

GATE DA

Machine Learning

50 Full-Length Sectional Mock Tests

MCQ · MSQ · NAT | 1-mark & 2-mark | Answer keys & detailed solutions

Aligned to GATE DA Section 6 (Machine Learning) syllabus

What is inside

- **50 mock tests**, each with 35 questions worth 55 marks (15 one-mark + 20 two-mark), the same weight the subject part of a GATE DA paper carries.
- **1750 questions** in total (266 MCQ, 391 MSQ, 1093 NAT), 2750 marks, with **696 questions built around a diagram**: scatter plots, decision boundaries, trees, network graphs, dendrograms, ROC curves and loss curves.
- Every test has an **answer key**, a **self-evaluation sheet** and **step-by-step solutions**. MSQ solutions explain every option.
- A **concept and formula revision section** with diagrams, and a **master answer key** at the end.

Syllabus coverage (GATE DA, Section 6: Machine Learning)

Supervised learning: regression and classification problems, simple linear regression, multiple linear regression, ridge regression, logistic regression, k-nearest neighbour, naive Bayes classifier, linear discriminant analysis, support vector machine, decision trees, bias-variance trade-off, cross-validation methods such as leave-one-out (LOO) cross-validation and k-fold cross-validation, multi-layer perceptron, feed-forward neural network.

Unsupervised learning: clustering algorithms, k-means/k-medoid, hierarchical clustering, top-down, bottom-up: single-linkage, multiple-linkage, dimensionality reduction, principal component analysis.

Every question is generated from a parameterised template whose answer is **computed in code**, so the numerical keys are exact. NAT keys are given as an accepted range around the exact value.

How to use this book

The GATE DA exam

GATE Data Science and Artificial Intelligence (DA) is a 3-hour computer-based test of 65 questions worth 100 marks: 10 General Aptitude questions (15 marks) and 55 subject questions (85 marks). Questions are of three kinds:

Type	What you do	Marks	Negative marking
MCQ: multiple choice	Pick the one correct option out of four	1 or 2	$-\frac{1}{3}$ (1-mark), $-\frac{2}{3}$ (2-mark)
MSQ: multiple select	Pick ALL correct options (one or more out of four). There is no partial credit.	1 or 2	None
NAT: numerical answer	Type a number; it is accepted if it lies in the given range	1 or 2	None

Machine Learning is the largest single block of the DA paper, so these tests concentrate on it. Each mock test is a **sectional test** of 35 questions worth 55 marks (15×1 mark + 20×2 marks), to be done in **100 minutes**. That is the same marks-per-minute pace as the real exam.

Suggested routine

1. Read the revision notes once and mark the formulas you are unsure of.
2. Take one test at a time under exam conditions: timer on, phone away, and a basic calculator only (GATE gives you an on-screen scientific calculator).
3. Mark your answers on a separate sheet. Use the answer key and the negative-marking rules to work out your score.
4. Read the detailed solution for **every** question, including the ones you got right. The MSQ explanations also say why each wrong statement is wrong, which is usually the trap.
5. Fill in the self-evaluation table. After every 5 tests, look across them for weak topics and re-read those parts of the notes.
6. In the last few weeks, re-attempt the tests you scored lowest on.

Exam-day strategy

- **NAT and MSQ have no negative marks**, so never leave them blank. For a NAT, an educated estimate costs nothing.
- Answer an MCQ only if you can rule out at least two options. A blind guess loses marks on average (expected value $\frac{1}{4} \cdot 1 - \frac{3}{4} \cdot \frac{1}{3} = 0$).
- For MSQ, judge each option on its own as true or false. Questions often have 2 or 3 correct options.
- Watch the rounding instruction on NAT questions. Do the arithmetic at full precision and round only at the end.
- In computation questions, watch these common slips: n vs $n-1$ (variance), log base (entropy uses $\log_2$ unless told otherwise), squared vs non-squared distance, and whether the bias term is included in parameter counts.

Coverage across all tests

Topic area	Questions	Marks
Probabilistic & Instance-based Classifiers	350	550
Unsupervised Learning	350	550
SVM & Decision Trees	350	550
Neural Networks (MLP / Feed-forward)	350	550
Regression & Model Selection	350	550

Sub-topic frequency

Topic area	Sub-topic	Questions
Neural Networks (MLP / Feed-forward)	Activation functions	51
Neural Networks (MLP / Feed-forward)	Backpropagation	42
Neural Networks (MLP / Feed-forward)	Expressiveness	27
Neural Networks (MLP / Feed-forward)	Forward pass	34
Neural Networks (MLP / Feed-forward)	Parameter counting	18
Neural Networks (MLP / Feed-forward)	Perceptron	51
Neural Networks (MLP / Feed-forward)	Softmax + cross-entropy	33
Neural Networks (MLP / Feed-forward)	Training & regularisation	94
Probabilistic & Instance-based Classifiers	Bayes optimal classifier	40
Probabilistic & Instance-based Classifiers	Evaluation metrics	68
Probabilistic & Instance-based Classifiers	Linear discriminant analysis	53
Probabilistic & Instance-based Classifiers	Logistic regression	71
Probabilistic & Instance-based Classifiers	Naive Bayes	47
Probabilistic & Instance-based Classifiers	k-nearest neighbours	71
Regression & Model Selection	Basis functions & feature scaling	31
Regression & Model Selection	Bias-variance trade-off	40
Regression & Model Selection	Cross-validation	48
Regression & Model Selection	Gradient descent	40
Regression & Model Selection	MLE & MAP view	23
Regression & Model Selection	Multiple linear regression	56
Regression & Model Selection	Regression vs classification	16
Regression & Model Selection	Ridge & Lasso	48
Regression & Model Selection	Simple linear regression	48
SVM & Decision Trees	Decision trees: impurity measures	25
SVM & Decision Trees	Decision trees: overfitting & pruning	26
SVM & Decision Trees	Decision trees: split selection	50
SVM & Decision Trees	Decision trees: structure & traversal	26
SVM & Decision Trees	Ensembles: bagging & random forests	26
SVM & Decision Trees	Ensembles: boosting	18
SVM & Decision Trees	Perceptron (linear separators)	10
SVM & Decision Trees	Regression trees	10
SVM & Decision Trees	SVM: dual & support vectors	36
SVM & Decision Trees	SVM: kernels	52
SVM & Decision Trees	SVM: margin & geometry	35

SVM & Decision Trees	SVM: soft margin & hinge loss	36
Unsupervised Learning	Cluster evaluation	34
Unsupervised Learning	Distance measures	41
Unsupervised Learning	Hierarchical clustering	72
Unsupervised Learning	PCA & dimensionality reduction	93
Unsupervised Learning	k-means	83
Unsupervised Learning	k-medoids	27

Concept and Formula Revision

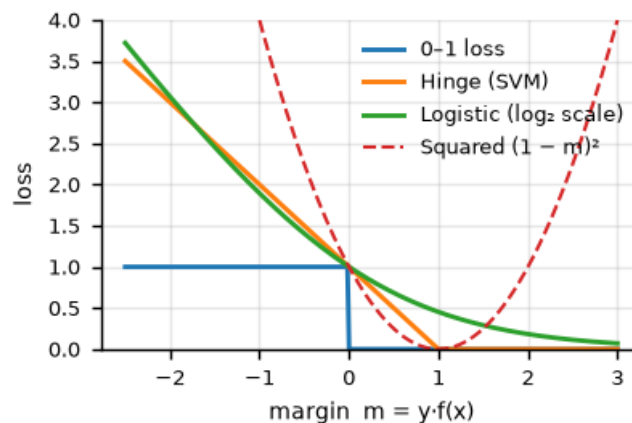
This is a short summary of the results the mock tests use most. Notation: n = number of samples, p = number of features, $\hat{y}$ = prediction, $1[\cdot]$ = indicator function.

1. Regression vs classification; loss functions

Regression: the target is continuous, and the usual loss is squared error $(\hat{y} - y)^2$ (or absolute error). **Classification:** the target is a discrete label; the usual losses are 0-1 loss, cross-entropy (log loss) and hinge loss.

Binary cross-entropy: $L = -[y \ln p + (1 - y) \ln(1 - p)]$. Hinge loss: $\max(0, 1 - y \cdot f(x))$ with $y \in \{-1, +1\}$.

Hinge, logistic and squared losses are convex surrogates for the 0-1 loss, which cannot be optimised directly. All of them are plotted against the margin $m = y \cdot f(x)$.



Classification losses as functions of the margin

2. Simple linear regression

Model $y = \beta_0 + \beta_1 x + \varepsilon$. $\beta_1 = S_{xy}/S_{xx} = r \cdot (s_y/s_x)$, $\beta_0 = \bar{y} - \beta_1 \bar{x}$, where $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$ and $S_{xx} = \sum (x_i - \bar{x})^2$.

The fitted line passes through $(\bar{x}, \bar{y})$. Residuals $e_i = y_i - \hat{y}_i$ satisfy $\sum e_i = 0$ and $\sum x_i e_i = 0$.

$SST = \sum (y_i - \bar{y})^2 = SSR + SSE$; $R^2 = SSR/SST = 1 - SSE/SST$, and $R^2 = r^2$ for simple regression.

Unbiased noise variance estimate: $\hat{\sigma}^2 = SSE/(n - 2)$ (in general $SSE/(n - p - 1)$). $\text{Var}(\beta_1) = \sigma^2/S_{xx}$.

3. Multiple linear regression

Normal equations $X^T X \beta = X^T y \Rightarrow \beta = (X^T X)^{-1} X^T y$ (X includes a column of 1s for the intercept).

Hat matrix $H = X(X^T X)^{-1} X^T$, $\hat{y} = Hy$. H is symmetric and idempotent ($H^2 = H$), and $\text{trace}(H) = \text{rank}(H) = \text{number of fitted parameters}$. $I - H$ is also idempotent, and the residuals are $e = (I - H)y$.

Adjusted $R^2 = 1 - (1 - R^2)(n - 1)/(n - p - 1)$. Adding a feature never decreases R^2 , but adjusted R^2 can go down.

Multicollinearity makes $X^T X$ close to singular, so the coefficient variances blow up. A categorical variable with K levels needs $K - 1$ dummy columns when an intercept is present (otherwise the dummy-variable trap makes X rank-deficient).

4. Ridge, lasso and the MAP view

Ridge: minimise $\|y - X\beta\|^2 + \lambda\|\beta\|^2$, with solution $\beta = (X^T X + \lambda I)^{-1} X^T y$. $X^T X + \lambda I$ is invertible for every $\lambda > 0$. In one dimension with centred data, $\beta_{\text{ridge}} = S_{xy}/(S_{xx} + \lambda)$. Ridge shrinks coefficients towards 0 but never exactly to 0. Larger λ means more bias and less variance.

Lasso: an L1 penalty $\lambda\|\beta\|_1$. Its constraint region is a diamond with corners on the axes, so the solution can set coefficients exactly to zero (feature selection). It has no closed form in general.

Least squares is the MLE under Gaussian noise. MAP with a Gaussian prior on β gives ridge ($\lambda = \sigma^2/\tau^2$); MAP with a Laplace prior gives lasso.

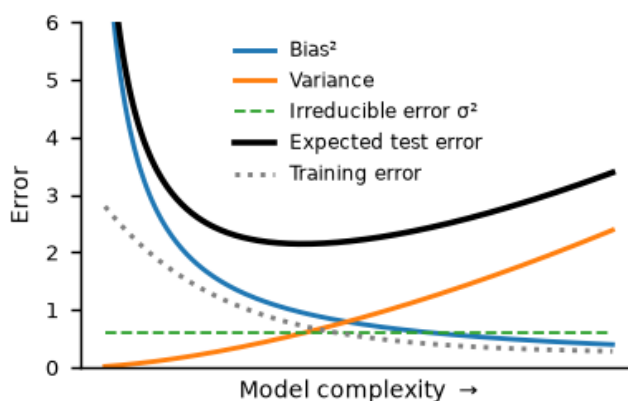
The intercept is usually not penalised, and features should be standardised before regularising, because the penalty is not scale-invariant.

5. Bias-variance trade-off

For squared loss: $E[(y - f(x))^2] = \text{Bias}[f(x)]^2 + \text{Var}[f(x)] + \sigma^2$ (irreducible noise).

Low complexity gives high bias (underfitting): training and test errors are both high. High complexity gives high variance (overfitting): training error is low and test error is high.

More training data lowers variance. Regularisation, bagging and larger k in k -NN also lower variance. Boosting and adding features mainly lower bias.



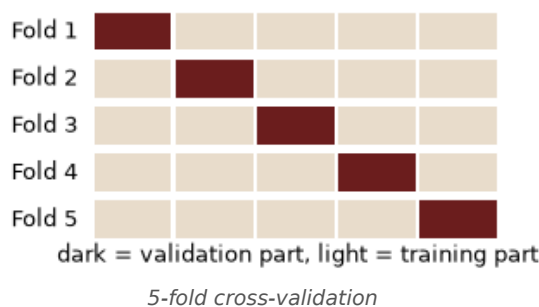
U-shaped test error: the sum of bias², variance and noise

6. Cross-validation

k-fold CV: split the data into k folds; train on $k - 1$ folds and validate on the remaining one; repeat k times. CV error = $(1/k) \sum E_j$ (weight by fold size if folds are unequal). This takes k model fits.

LOOCV is $k = n$: it needs n fits, has low bias and high variance, and involves no randomness. For least squares (and any linear smoother) there is a shortcut: $CV_{(n)} = (1/n) \sum [e_i / (1 - h_{ii})]^2$.

Stratified k -fold keeps the class proportions in every fold. Every preprocessing step (scaling, feature selection) must be fitted inside the training folds, or information leaks from the validation data. Choosing hyper-parameters on the test set biases the error estimate; use nested CV instead.



7. Gradient descent

$\theta \leftarrow \theta - \eta \nabla J(\theta)$. For MSE, $J = (1/n) \|X\theta - y\|^2$ and $\nabla J = (2/n) X^T(X\theta - y)$ (with the $\frac{1}{2}$ convention, $\nabla J = (1/n) X^T(X\theta - y)$).

A learning rate that is too large makes the loss oscillate or diverge; one that is too small makes convergence slow. For a convex quadratic, GD converges if $\eta < 2/L$, where L is the largest eigenvalue of the Hessian.

Updates per epoch: N/B (rounded up) for mini-batch size B . Batch GD makes 1 update per epoch; SGD makes N .

8. Logistic regression and softmax

$P(y = 1 | x) = \sigma(w^T x + b)$, with $\sigma(z) = 1/(1 + e^{-z})$, $\sigma(-z) = 1 - \sigma(z)$ and $\sigma'(z) = \sigma(z)(1 - \sigma(z)) \leq \frac{1}{4}$.

Log-odds are linear: $\ln[p/(1 - p)] = w^T x + b$. Increasing x_j by 1 multiplies the odds by e^{w_j} .

The decision boundary at threshold 0.5 is the hyperplane $w^T x + b = 0$. The gradient of the log loss is $(\sigma(w^T x) - y)x$. There is no closed-form solution, but the loss is convex. On linearly separable data, $\|w\| \rightarrow \infty$ unless the model is regularised.

Softmax: $p_k = e^{z_k} / \sum_j e^{z_j}$. Adding the same constant to all z_k leaves p unchanged. The cross-entropy gradient with respect to z is $p - y$.

9. k-nearest neighbours

Predict the majority class (classification) or the mean target (regression) of the k nearest training points. The method is non-parametric and lazy: there is no training step, and prediction costs $O(nd)$ per query.

Small k gives low bias and high variance (1-NN has zero training error). Large k gives a smoother boundary, with higher bias and lower variance. Use an odd k for binary problems to avoid ties.

k -NN needs feature scaling and suffers from the curse of dimensionality: in high dimensions, distances concentrate and all points look roughly equally far away.

Distances: Minkowski $d_p = (\sum |x_i - z_i|^p)^{1/p}$; $p = 1$ is Manhattan, $p = 2$ is Euclidean, $p \rightarrow \infty$ is Chebyshev (max).

10. Naive Bayes

$\hat{y} = \operatorname{argmax}_c P(c) \prod_j P(x_j | c)$. The model assumes the features are conditionally independent given the class.

Laplace (add- α) smoothing: $P(x_j = v | c) = (\text{count}(v, c) + \alpha) / (\text{count}(c) + \alpha \cdot |V_j|)$. Without smoothing, a single zero count makes the whole product zero.

Gaussian NB: $P(x_j | c) = N(x_j; \mu_{jc}, \sigma_{jc}^2)$. Parameters for d binary features and K classes: $(K - 1) + Kd$, compared with $K(2^d - 1) + (K - 1)$ for the full joint distribution.

11. Linear discriminant analysis (LDA)

LDA models each class as a Gaussian with its own mean and a shared covariance Σ . The discriminant $\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln \pi_k$ is linear in x , so the decision boundaries are linear.

In 1-D with two classes, equal variance and equal priors, the threshold is $(\mu_0 + \mu_1)/2$. With unequal priors it is $(\mu_0 + \mu_1)/2 + \sigma^2 \ln(\pi_0/\pi_1)/(\mu_1 - \mu_0)$.

QDA gives each class its own covariance Σ_k , so the boundaries are quadratic. Fisher's direction is $w \propto S_W^{-1}(\mu_1 - \mu_0)$. With K classes there are at most $K - 1$ discriminant directions.

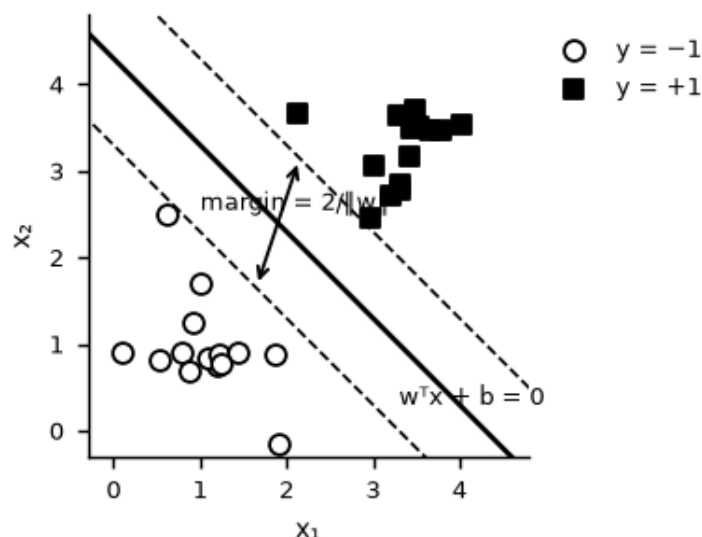
12. Support vector machines

Hard margin: minimise $\frac{1}{2}\|w\|^2$ subject to $y_i(w^T x_i + b) \geq 1$. The geometric margin (the width of the street) is $2/\|w\|$, and the distance from a point to the hyperplane is $|w^T x + b|/\|w\|$.

Dual solution: $w = \sum \alpha_i y_i x_i$, with $\sum \alpha_i y_i = 0$ and $\alpha_i \geq 0$. Only the support vectors have $\alpha_i > 0$.

Soft margin: $\frac{1}{2}\|w\|^2 + C \sum \xi_i$. A large C penalises violations heavily (narrower margin, lower bias); a small C allows a wider margin (more regularisation). $\xi_i = 0$ means outside the margin; $0 < \xi_i \leq 1$ means inside the margin but correctly classified; $\xi_i > 1$ means misclassified.

Kernels: polynomial $(x^T z + c)^d$; RBF $\exp(-\gamma\|x - z\|^2)$. A kernel is valid exactly when every Gram matrix is positive semi-definite. Sums, products and positive multiples of valid kernels are valid.



Maximum-margin hyperplane; the dashed lines pass through the support vectors

13. Decision trees

Entropy $H = -\sum p_k \log_2 p_k$ (maximum $\log_2 K$). Gini $= 1 - \sum p_k^2$ (maximum $1 - 1/K$).

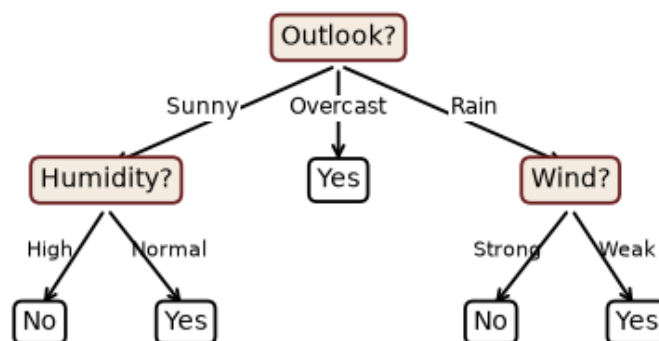
Misclassification error $= 1 - \max p_k$.

Information gain $IG = H(\text{parent}) - \sum (n_v/n) H(\text{child}_v)$. Gain ratio $= IG/\text{SplitInfo}$, where $\text{SplitInfo} = -\sum (n_v/n) \log_2 (n_v/n)$; it penalises attributes with many values.

For a continuous attribute, candidate thresholds are the midpoints between consecutive sorted values where the class changes. Regression trees choose splits that minimise SSE (that is, reduce variance) and predict the leaf mean.

Trees are greedy, non-parametric and unaffected by monotone feature transforms. A binary tree of depth d has at most 2^d leaves. Pre-pruning and post-pruning (cost-complexity $R_\alpha(T) = R(T) + \alpha|T|$) control overfitting.

Bagging and random forests reduce variance; each bootstrap sample leaves out about $(1 - 1/n)^n \approx e^{-1} \approx 36.8\%$ of the points (out-of-bag). AdaBoost gives a weak learner with error ϵ the weight $\alpha = \frac{1}{2} \ln((1 - \epsilon)/\epsilon)$.



A decision tree for the PlayTennis data

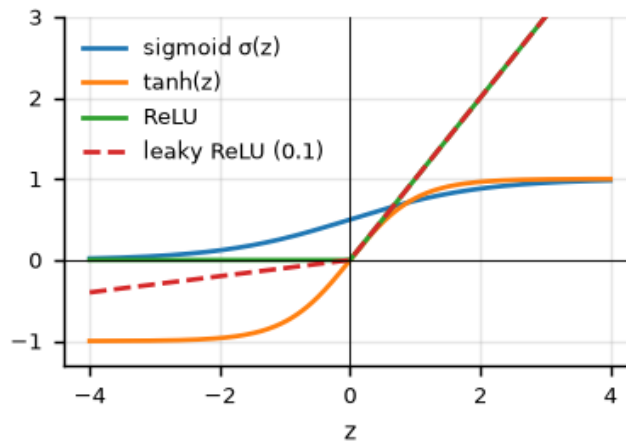
14. Perceptron and multi-layer perceptrons

Perceptron: $\hat{y} = \text{sign}(w^T x + b)$. Update on a mistake: $w \leftarrow w + \eta y_i x_i$, $b \leftarrow b + \eta y_i$. It converges in finitely many steps if the data are linearly separable. A single perceptron cannot represent XOR. Forward pass: $a^{(l)} = g(W^{(l)} a^{(l-1)} + b^{(l)})$. With linear activations, any number of layers collapses to one linear map. One hidden layer with enough units is a universal approximator.

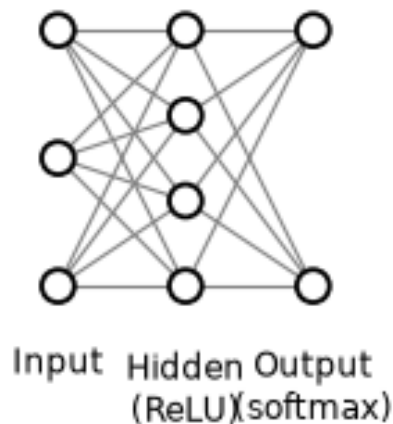
Parameter count for layer sizes $n_0 \rightarrow n_1 \rightarrow \dots \rightarrow n_L$: $\sum (n_{l-1} + 1) \cdot n_l$ (weights plus biases).

Backpropagation: $\delta^{(L)} = \partial L / \partial z^{(L)}$; $\delta^{(l)} = (W^{(l+1)^T} \delta^{(l+1)}) \odot g'(z^{(l)})$; $\partial L / \partial W^{(l)} = \delta^{(l)} a^{(l-1)^T}$. If all weights start at the same value, symmetry is never broken, so initialise randomly.

Inverted dropout with keep probability p scales the kept activations by $1/p$ during training. An L2 penalty gives weight decay: $w \leftarrow (1 - \eta \lambda) w - \eta \nabla L$.



Common activation functions



A 3-4-2 feed-forward network: $(3+1) \cdot 4 + (4+1) \cdot 2 = 26$ parameters

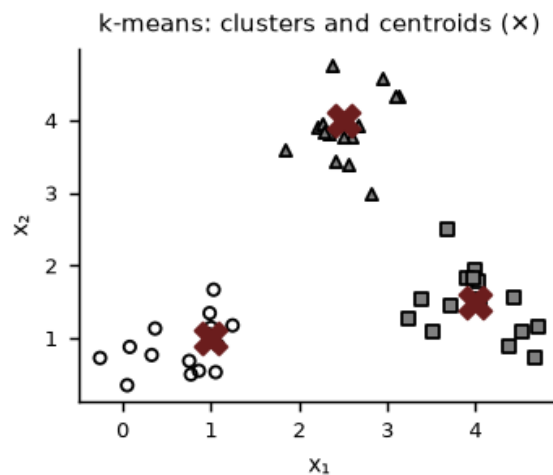
15. Clustering: k-means and k-medoids

k-means minimises $J = \sum_k \sum_{x \in C_k} \|x - \mu_k\|^2$. Lloyd's algorithm alternates (i) assigning each point to its nearest centroid and (ii) recomputing each centroid as its cluster mean. J never increases, the algorithm converges in finitely many steps, but it may stop at a local minimum.

k-means is sensitive to initialisation (k-means++ helps), to outliers and to feature scale, and it assumes convex, roughly spherical clusters. Use the elbow method or the silhouette to choose k .

k-medoids (PAM): every centre must be an actual data point (the medoid minimises the total distance to the other points in its cluster). It works with any dissimilarity and is more robust to outliers.

Silhouette $s(i) = (b - a) / \max(a, b) \in [-1, 1]$. Here a is the mean distance to the point's own cluster and b is the smallest mean distance to another cluster.

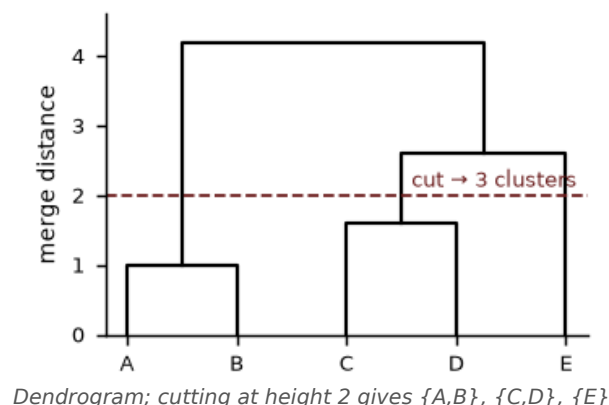


16. Hierarchical clustering

Agglomerative (bottom-up): start with n singleton clusters and repeatedly merge the closest pair, giving $n - 1$ merges in total. Divisive (top-down): start with one cluster and split recursively.

Single linkage: min distance between members (tends to chain; equivalent to building a minimum spanning tree). Complete linkage: max distance (compact clusters). Average linkage (UPGMA): mean of all pairwise distances. Ward: the merge that least increases the total within-cluster SSE.

Cutting the dendrogram at height h gives the clusters whose merge heights are below h . With single and complete linkage the merge heights never decrease (no inversions).



Dendrogram; cutting at height 2 gives $\{A, B\}$, $\{C, D\}$, $\{E\}$

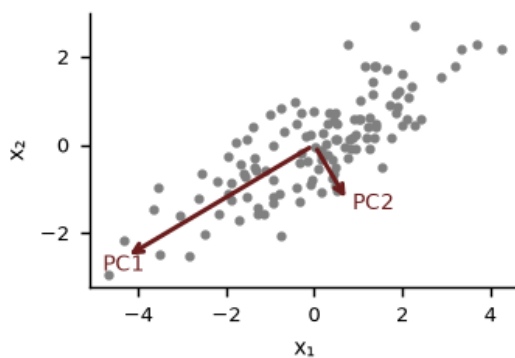
17. Principal component analysis

Centre the data (and standardise it if the units differ). The covariance matrix is $S = X^T X / (n - 1)$. Its eigenvectors are the principal directions, and its eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ are the variances along them.

Proportion of variance explained by PC k : $\lambda_k / \sum \lambda_j$. Squared reconstruction error when keeping the top m components: the sum of the discarded eigenvalues (times $n - 1$ for the total squared error).

SVD view: $X = U \Sigma V^T$. The principal directions are the columns of V , the scores are $U \Sigma = X V$, and $\lambda_k = \sigma_k^2 / (n - 1)$. PCs are orthogonal and their scores are uncorrelated.

PCA is unsupervised and linear and keeps directions of maximum variance. LDA is supervised and keeps directions of maximum class separation.



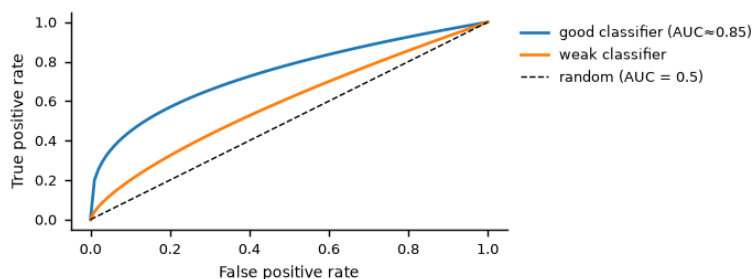
Principal directions of a correlated 2-D cloud

18. Classification metrics

Precision = $TP/(TP + FP)$. Recall (sensitivity, TPR) = $TP/(TP + FN)$. Specificity = $TN/(TN + FP)$. $FPR = 1 - \text{specificity}$.

$F1 = 2PR/(P + R)$. $F_\beta = (1 + \beta^2)PR/(\beta^2P + R)$. Accuracy = $(TP + TN)/N$ can be misleading when the classes are imbalanced.

The ROC curve plots TPR against FPR as the threshold varies. AUC is the probability that a random positive is scored higher than a random negative (ties count $\frac{1}{2}$). Macro-averaging takes the unweighted mean of the per-class metric; micro-averaging pools the counts first.



Mock Test 01

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	9	9	17

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Probabilistic & Instance-based Classifiers	7	11
Unsupervised Learning	7	11
SVM & Decision Trees	7	11
Neural Networks (MLP / Feed-forward)	7	11
Regression & Model Selection	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

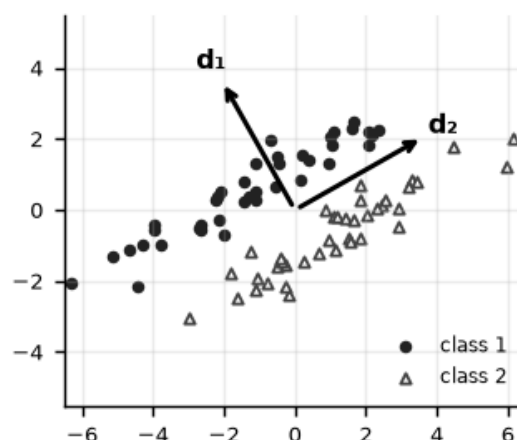
Q.1 [1 mark · MSQ (one or more correct)]

Which of the following statements about the k-nearest-neighbour method is/are CORRECT?

- (A) Manhattan and Euclidean distances can give different nearest neighbours for the same query.
- (B) k-NN can produce highly non-linear decision boundaries.
- (C) Increasing k generally increases bias and decreases variance of the k-NN classifier.
- (D) The number of neighbours k is a hyperparameter usually chosen by cross-validation.

Q.2 [1 mark · MCQ]

Two labelled classes are shown below, together with two orthogonal directions d_1 and d_2 . A single direction is to be selected by (i) PCA (first principal component, ignoring labels) and (ii) Fisher's LDA. Which statement is correct?



- (A) PCA (1st PC) chooses d_1 ; LDA chooses d_2
- (B) Both PCA and LDA choose d_2
- (C) PCA (1st PC) chooses d_2 ; LDA chooses d_1
- (D) Both PCA and LDA choose d_1

Q.3 [1 mark · NAT (numerical answer)]

The confusion matrix of a binary classifier on a test set is shown below (note which of rows/columns are actual and predicted). The balanced accuracy of the classifier is _____ (rounded off to two decimal places)

	Predicted +	Predicted –
Actual +	12	20
Actual –	42	45

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about decision-tree learning is/are CORRECT?

- (A) Small changes in the training data can produce a very different tree.
- (B) With squared-error loss, a regression-tree leaf predicts the mean response of its training samples.
- (C) A tree with axis-parallel splits predicts a constant within each axis-aligned rectangular region.
- (D) Growing a tree until all leaves are pure typically reduces variance.

Q.5 [1 mark · NAT (numerical answer)]

A binary decision tree must have at least 76 leaves to separate 76 distinct classes. The minimum possible depth of such a tree is _____ (Answer in integer)

Q.6 [1 mark · NAT (numerical answer)]

LDA is fitted with $K = 4$ classes and $p = 3$ features. Counting class priors, class means and the shared covariance matrix, the number of independent parameters is _____ (Answer in integer)

Q.7 [1 mark · NAT (numerical answer)]

A fully connected MLP has layer sizes 10-16-64-256-5 (input-...-output), every layer having a bias vector. The number of weights (excluding biases) is _____ (Answer in integer)

Q.8 [1 mark · NAT (numerical answer)]

A bagged ensemble is trained on $N = 8400$ points. Using the large- N approximation, the expected number of training points that are out-of-bag for a single bootstrap sample of size N is _____ (Answer in integer)

Q.9 [1 mark · MSQ (one or more correct)]

Regarding the bias-variance trade-off in supervised regression with squared-error loss, which of the following statements is/are CORRECT?

- (A) Increasing model flexibility typically decreases bias.
- (B) The bias of $\hat{f}$ at x_0 is $E[\hat{f}(x_0)] - f(x_0)$, the expectation being over training sets.
- (C) For a fixed model class, more training data tends to reduce variance.
- (D) The irreducible error cannot be reduced by choosing a better model.

Q.10 [1 mark · MSQ (one or more correct)]

Which of the following statements about cross-validation (CV) is/are CORRECT? (n denotes the number of observations.)

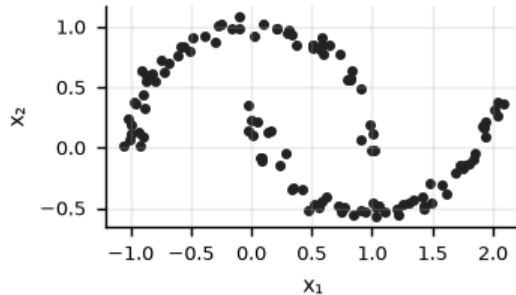
- (A) LOOCV estimates of test error have low bias but can have relatively high variance.
- (B) A single train/validation split gives an error estimate that can vary substantially with the split.
- (C) For least-squares linear regression, LOOCV error can be computed from a single fit using the leverages h_{ii} .
- (D) Nested CV gives a nearly unbiased estimate of generalisation error when hyperparameters are tuned by CV.

Q.11 [1 mark · NAT (numerical answer)]

Mini-batch SGD is run on $N = 60000$ examples with batch size $B = 32$ for $E = 11$ epochs, and any incomplete final batch in an epoch is **dropped**. The total number of parameter updates is _____ (Answer in integer)

Q.12 [1 mark · MCQ]

k-means with $k = 2$ fails to separate the two interleaving half-moons shown below, even with many random restarts. The main reason is that k-means



- (A) converges only when the data are standardized
- (B) produces convex clusters separated by linear (Voronoi) boundaries
- (C) requires class labels to place the centroids
- (D) cannot be applied to two-dimensional data

Q.13 [1 mark · MSQ (one or more correct)]

Let $p = \text{softmax}(z)$ for logits $z \in \mathbb{R}^K$ and $L = -\ln p_y$ the cross-entropy loss for true class y . Which of the following is/are CORRECT?

- (A) The cross-entropy loss $-\ln p_y$ can be negative.
- (B) Softmax with two classes is equivalent to a sigmoid applied to the logit difference.
- (C) Adding the same constant to all logits leaves the softmax output and the cross-entropy loss unchanged.
- (D) For softmax outputs with cross-entropy loss and one-hot target y , $\partial L / \partial z = p - y$.

Q.14 [1 mark · NAT (numerical answer)]

For $x = (3, 7, 1, 7, -4)$ and $y = (1, 2, 1, 3, 1)$, the cosine distance ($1 - \text{cosine similarity}$) between x and y is _____ (rounded off to two decimal places)

Q.15 [1 mark · MCQ]

For the Gaussian prior $w \sim N(0, \tau^2 I)$ and Gaussian noise, as $\tau^2 \rightarrow \infty$ the MAP estimate tends to:

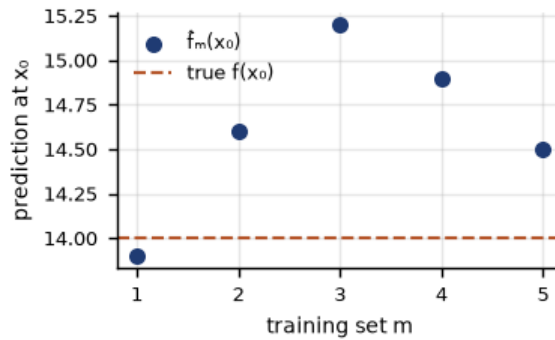
- (A) the zero vector
- (B) an undefined limit
- (C) the least-squares (ML) estimate
- (D) the lasso estimate

Q.16 - Q.35 : Two-mark questions

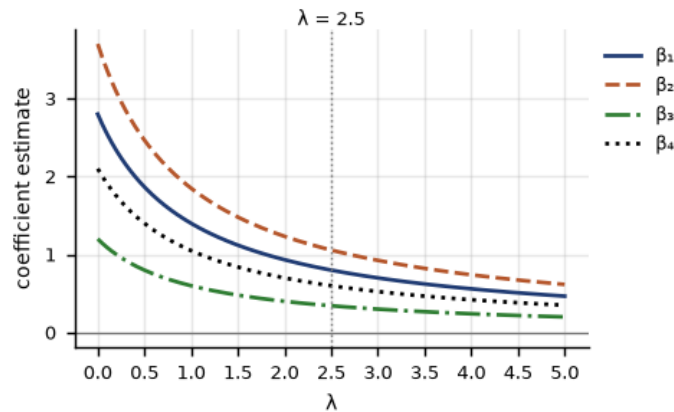
Q.16 [2 marks · NAT (numerical answer)]

Data are generated as $y = f(x) + \varepsilon$ with $E[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2 = 2$. At a test point x_0 the true value is $f(x_0) = 14$. A learning method is trained on 5 independent training sets, giving the predictions $\hat{f}_m(x_0)$ in the table (also plotted). Treating these as the full distribution of the method's prediction (variance with divisor 5), the variance of the method's prediction at x_0 is _____ (rounded off to three decimal places)

m	1	2	3	4	5
$\hat{f}_m(x_0)$	13.9	14.6	15.2	14.9	14.5

**Q.17 [2 marks · MSQ (one or more correct)]**

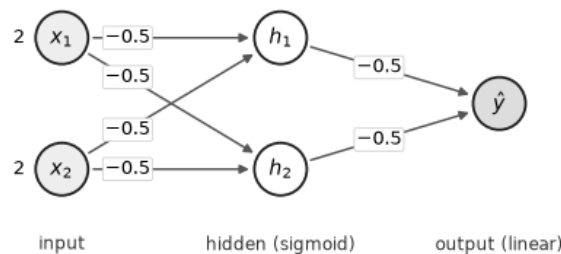
For a design with orthonormal columns ($X^T X = I$), the figure shows the coefficient paths $\beta_j(\lambda)$ of a penalised least-squares estimator that minimises $\frac{1}{2}\|y - X\beta\|^2 + P_\lambda(\beta)$, where the penalty P_λ is either the ridge penalty $(\lambda/2)\|\beta\|_2^2$ or the lasso penalty $\lambda\|\beta\|_1$. Which of the following statements is/are CORRECT?



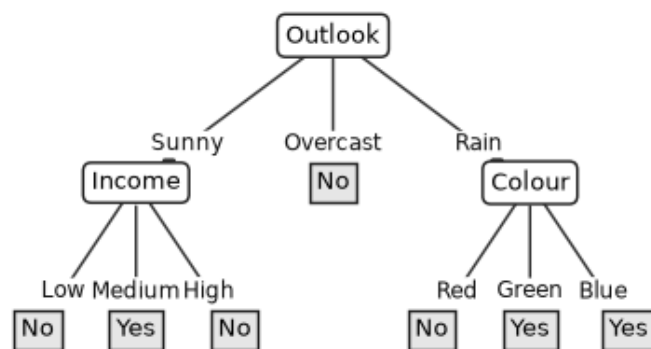
- (A) As λ increases, $|\beta_j(\lambda)|$ is non-increasing for every j .
- (B) The plotted paths are those of ridge regression (L2 penalty).
- (C) At $\lambda = 0$ the coefficients equal the least-squares estimates.
- (D) At $\lambda = 2.5$, every coefficient equals its least-squares value divided by 3.5.

Q.18 [2 marks · NAT (numerical answer)]

In the 2-2-1 network below (sigmoid hidden units, linear output $\hat{y} = v_1 h_1 + v_2 h_2$, all biases 0), **every** weight is initialised to -0.5 . One gradient-descent step with $\eta = 0.5$ is taken on $L = \frac{1}{2}(\hat{y} - y)^2$ for the input $x = (2, 2)$ with target $y = -1$. Then $w_{12} - w_{22}$ after the update (difference between the weights from x_2 into the two hidden units) is _____ (rounded off to 4 decimal places)

**Q.19 [2 marks · NAT (numerical answer)]**

The decision tree shown is applied to the 8 labelled records in the table (attributes not tested on a path are ignored). Find the precision of the tree for the class “Yes”. (rounded off to two decimal places)



#	Outlook	Colour	Income	Actual
1	Sunny	Blue	High	No
2	Overcast	Blue	Low	No
3	Overcast	Green	Low	Yes
4	Rain	Blue	Low	Yes
5	Sunny	Red	Low	No
6	Rain	Red	High	Yes
7	Sunny	Red	Low	Yes
8	Sunny	Red	Low	Yes

Q.20 [2 marks · MCQ]

A naive Bayes classifier is trained on the 12 examples below. The new instance (Fever = None, Cough = No, Fatigue = Yes) is classified (i) with maximum-likelihood (unsmoothed) estimates and (ii) with Laplace (add-one) smoothing of the class-conditional probabilities (class priors unsmoothed). The predicted classes are

#	Fever	Cough	Fatigue	Flu
1	Mild	Yes	Yes	Pos
2	None	Yes	Yes	Pos
3	None	Yes	No	Neg
4	None	Yes	No	Neg
5	None	Yes	No	Neg
6	Mild	Yes	No	Pos
7	Mild	No	Yes	Pos
8	Mild	Yes	No	Pos
9	None	Yes	No	Neg
10	None	No	No	Neg
11	Mild	Yes	Yes	Pos
12	None	Yes	No	Neg

- (A) Without smoothing: Pos; with Laplace smoothing: Pos
 (B) Without smoothing: Neg; with Laplace smoothing: Pos
 (C) Without smoothing: Neg; with Laplace smoothing: Neg
 (D) Without smoothing: Pos; with Laplace smoothing: Neg

Q.21 [2 marks · NAT (numerical answer)]

A 3-class softmax (multinomial logistic) regression model computes scores $\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}$ and $P(y = k | \mathbf{x}) = \frac{e^{\mathbf{z}_k}}{\sum_j e^{\mathbf{z}_j}}$, where row k of $\mathbf{W}$ holds the weights of class k . For the parameters below and $\mathbf{x} = (0, 1)^T$, the predicted probability $P(y = 3 | \mathbf{x})$ is _____ (rounded off to two decimal places)

$$\mathbf{W} = \begin{bmatrix} 2 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$$

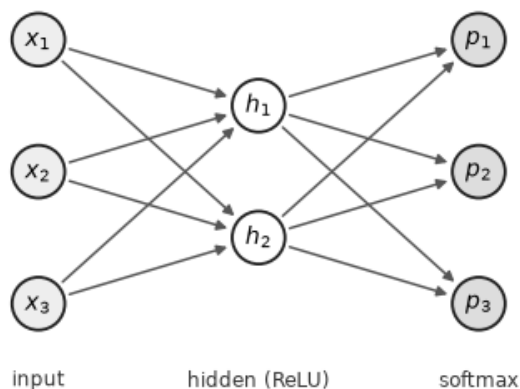
$$\mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Q.22 [2 marks · NAT (numerical answer)]

The eigenvalues of the sample covariance matrix (divisor $n - 1$) of a centered 7-dimensional dataset are 19, 18, 12, 6, 3, 2.5, 2. Then the mean squared reconstruction error $(1/(n - 1)) \sum_i \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2$, when each point is projected onto the top 4 principal components, is _____ (rounded off to two decimal places)

Q.23 [2 marks · NAT (numerical answer)]

A 3-2-3 network computes $h = \text{ReLU}(W_1 x + b_1)$, $z = W_2 h + b_2$, $p = \text{softmax}(z)$, with the parameters below. For $x = (-1, -1, 0)^T$, the predicted probability p_1 of class 1 is _____ (rounded off to three decimal places)



$$W_1 = \begin{bmatrix} -1 & -0.5 & 1 \\ 0.5 & 0.5 & 0 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0.5 \\ -1 \end{bmatrix}$$

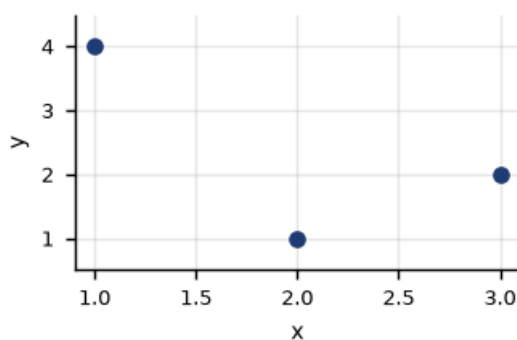
$$W_2 = \begin{bmatrix} 2 & -2 \\ 1 & -2 \\ 2 & 0 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 0.5 \\ 1 \\ 0.5 \end{bmatrix}$$

Q.24 [2 marks · NAT (numerical answer)]

A linear model $\hat{y} = w_0 + w_1 x$ is trained on the 3 points below by batch gradient descent on the loss $J(w) = (1/2n)\sum (w_0 + w_1 x_i - y_i)^2$, starting from $(w_0, w_1) = (-1, -1)$ with learning rate $\eta = 0.1$. Then the value of w_1 after two iterations is _____ (rounded off to three decimal places)

x	1	2	3
y	4	1	2



Q.25 [2 marks · MCQ]

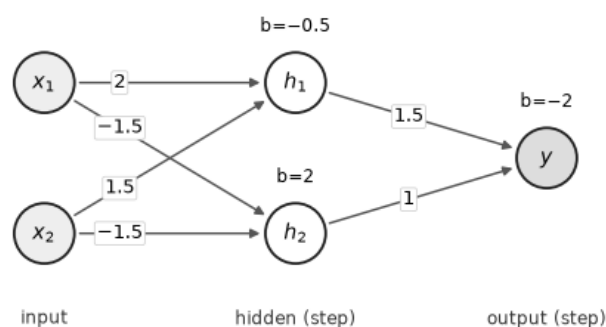
Six points have the pairwise distances below. Agglomerative clustering with **single linkage** is run and the dendrogram is cut to give exactly **2** clusters. The resulting clusters are

	P1	P2	P3	P4	P5	P6
P1	0	3	4	7	8	2
P2		0	7	8	9	3
P3			0	9	6	4
P4				0	15	5
P5					0	10
P6						0

- (A) {P1, P2, P4, P6}, {P3, P5} (B) {P1, P2, P3, P6}, {P4}, {P5}
 (C) {P1, P2, P3, P4, P6}, {P5} (D) {P1, P4}, {P2, P3, P5, P6}

Q.26 [2 marks · MCQ]

In the network below every unit (hidden and output) is a threshold unit that outputs 1 if its weighted input plus bias is > 0 , and 0 otherwise. Edge labels are weights. For binary inputs $x_1, x_2 \in \{0, 1\}$, the network computes



- (A) NOR (B) (NOT x_1) AND x_2
 (C) XOR (D) XNOR

Q.27 [2 marks · MSQ (one or more correct)]

A linear regression of annual salary (₹ lakh) on Exp (years of experience), Female (1 if the employee is female, 0 otherwise) and city $\in \{A, B, C\}$ encoded by dummies CityB and CityC (reference level A) gives the fitted model

$$\hat{y} = 10 + 2.6 \cdot \text{Exp} + 3.1 \cdot \text{Female} - 1.9 \cdot \text{CityB} - 3.2 \cdot \text{CityC}$$

Which of the following is/are CORRECT?

- (A) Holding other variables fixed, the predicted response for city B minus that for city C is 1.3.
 (B) Increasing Exp by 5 years, other variables fixed, changes the prediction by 7.6.
 (C) The coefficient of CityB equals the average response of all observations in city B.
 (D) The predicted annual salary for a female employee in city C with 6 years of experience is 25.5.

Q.28 [2 marks · MCQ]

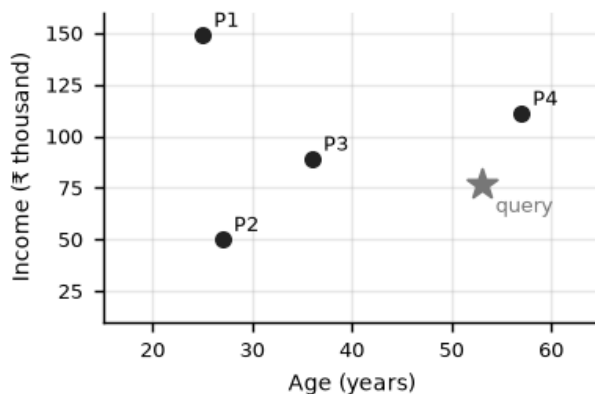
For logits $z = (0, -1, 2)$, softmax probabilities p and true class 3, the gradient of the cross-entropy loss $L = -\ln p_3$ with respect to z (rounded to two decimals) is

- (A) (0.11, 0.04, 0.84) (B) (0.11, 0.04, -0.16)
 (C) (0.01, 0, -0.02) (D) (-0.89, -0.96, -0.16)

Q.29 [2 marks · MCQ]

A 1-NN classifier uses the two features below. Before computing Euclidean distances, each feature is min-max scaled to $[0, 1]$ using the minimum and maximum of that feature over the four training points P1-P4 (the query is scaled with the same constants). The nearest neighbour of the query after scaling is

Point	Age (years)	Monthly income (₹ thousand)
P1	25	149
P2	27	50
P3	36	89
P4	57	111
Query	53	77



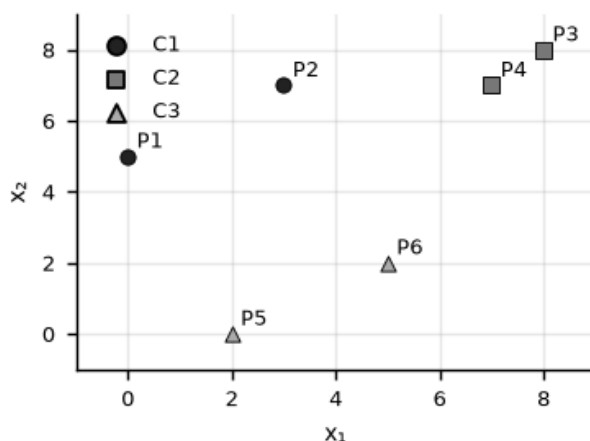
- (A) P1
(C) P3

- (B) P2
(D) P4

Q.30 [2 marks · NAT (numerical answer)]

A clustering of 6 points into 3 clusters is shown below. Using Euclidean distance, the silhouette coefficient $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$ of point P2 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5	P6
x_1	0	3	8	7	2	5
x_2	5	7	8	7	0	2
cluster	C1	C1	C2	C2	C3	C3



Q.31 [2 marks · MSQ (one or more correct)]

A node with class counts [60+, 60−] can be split by either of two binary splits S1 or S2 (table). For each impurity measure, the “gain” is the parent impurity minus the weighted child impurity. Which of the following statements is/are CORRECT?

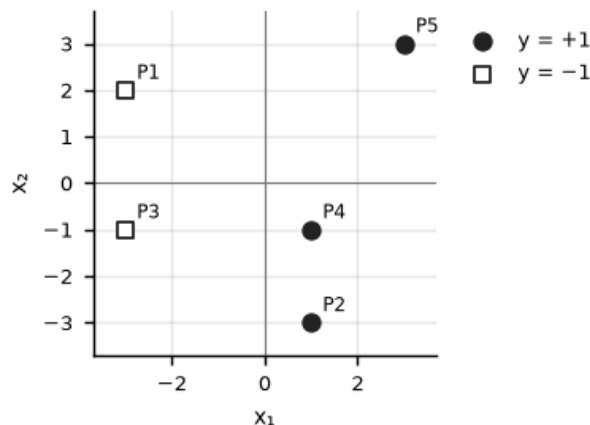
Split	Left child [+, −]	Right child [+, −]
S1	[48+, 27−]	[12+, 33−]
S2	[1+, 29−]	[59+, 31−]

- (A) The information gain of S2 is 0.2 bits (to two decimals).
- (B) Gini gain prefers S2 over S1.
- (C) Information gain (entropy) prefers S1 over S2.
- (D) Information gain (entropy) prefers S2 over S1.

Q.32 [2 marks · NAT (numerical answer)]

A perceptron with augmented weight vector $w = (w_0, w_1, w_2)$ (w_0 is the bias, $\tilde{x} = (1, x_1, x_2)$) starts from $w = (-1, 1, 1)$ and uses learning rate $\eta = 1$. The points P1, P2, ... are presented once, in order; whenever $y \cdot w^T \tilde{x} \leq 0$ the update $w \leftarrow w + \eta y \tilde{x}$ is applied. Find the value of w_2 after this epoch. (Answer in integer)

Point	x_1	x_2	y
P1	−3	2	−1
P2	1	−3	+1
P3	−3	−1	−1
P4	1	−1	+1
P5	3	3	+1

**Q.33 [2 marks · MSQ (one or more correct)]**

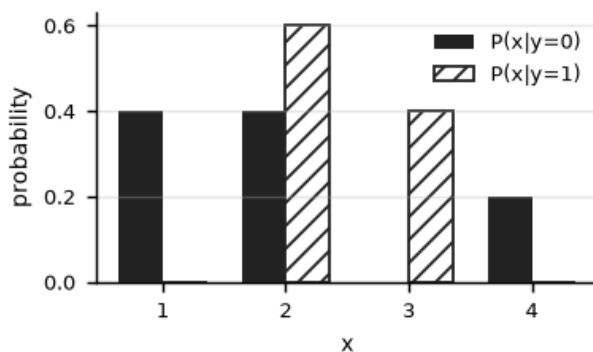
Consider k-means (Lloyd's algorithm) and k-medoids (PAM). Which of the following statements is/are CORRECT?

- (A) For mean centroids, total sum of squares = within-cluster SS – between-cluster SS.
- (B) k-medoids minimizes the sum of squared Euclidean distances to the cluster means.
- (C) Within a cluster C , $\sum_{x \in C} \|x - \mu\|^2 = (1/(2|C|)) \sum_{x \in C} \sum_{y \in C} \|x - y\|^2$.
- (D) The number of ways to split n points into 2 non-empty clusters is $n(n-1)/2$.

Q.34 [2 marks · NAT (numerical answer)]

A discrete feature $x \in \{1, \dots, 4\}$ has the class-conditional distributions below (also plotted), with priors $P(y = 0) = 0.5$ and $P(y = 1) = 0.5$. Then the Bayes error rate of the optimal classifier is _____ (rounded off to three decimal places)

x	1	2	3	4
$P(x y=0)$	0.4	0.4	0	0.2
$P(x y=1)$	0	0.6	0.4	0

**Q.35 [2 marks · MCQ]**

A decision tree is to be grown on the 10 training records below using information gain as the splitting criterion (multi-way split on each categorical attribute). Which attribute is chosen at the root?

#	Temp	Wind	Weekend	Outlook	Class
1	Hot	Weak	No	Sunny	Yes
2	Hot	Weak	Yes	Overcast	No
3	Cool	Strong	Yes	Overcast	Yes
4	Mild	Weak	Yes	Overcast	No
5	Cool	Weak	Yes	Sunny	No
6	Mild	Strong	Yes	Sunny	Yes
7	Mild	Strong	No	Rain	No
8	Cool	Weak	No	Sunny	No
9	Mild	Weak	Yes	Sunny	No
10	Cool	Strong	No	Overcast	Yes

(A) Temp**(B)** Wind**(C)** Weekend**(D)** Outlook

— END OF QUESTION PAPER —

Mock Test 01 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MSQ	1	(A), (B), (C), (D)	19	NAT	2	0.99 to 1.01
2	MCQ	1	(C)	20	MCQ	2	(A)
3	NAT	1	0.4361 to 0.4561	21	NAT	2	0.8568 to 0.8768
4	MSQ	1	(A), (B), (C)	22	NAT	2	7.49 to 7.51
5	NAT	1	7	23	NAT	2	0.4478 to 0.4518
6	NAT	1	21	24	NAT	2	0.5269 to 0.5309
7	NAT	1	18848	25	MCQ	2	(C)
8	NAT	1	3089 to 3092	26	MCQ	2	(C)
9	MSQ	1	(A), (B), (C), (D)	27	MSQ	2	(A), (D)
10	MSQ	1	(A), (B), (C), (D)	28	MCQ	2	(B)
11	NAT	1	20625	29	MCQ	2	(D)
12	MCQ	1	(B)	30	NAT	2	0.1975 to 0.2175
13	MSQ	1	(B), (C), (D)	31	MSQ	2	(B), (D)
14	NAT	1	0.2042 to 0.2242	32	NAT	2	1
15	MCQ	1	(C)	33	MSQ	2	(C)
16	NAT	2	0.1866 to 0.1926	34	NAT	2	0.198 to 0.202
17	MSQ	2	(A), (B), (C), (D)	35	MCQ	2	(B)
18	NAT	2	−0.0002 to 0.0002				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Probabilistic & Instance-based Classifiers					11
Unsupervised Learning					11
SVM & Decision Trees					11
Neural Networks (MLP / Feed-forward)					11
Regression & Model Selection					11
Total					55

Mock Test 01 — Detailed Solutions

Q.1 — Answer: (A), (B), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers › k-nearest neighbours]

- (A) **TRUE**. Their orderings of points differ in general.
- (B) **TRUE**. It is non-parametric.
- (C) **TRUE**. Averaging more neighbours smooths the decision boundary.
- (D) **TRUE**. Training error cannot select k (it always favours $k = 1$).

Q.2 — Answer: (C) [MCQ, 1M · Unsupervised Learning › PCA & dimensionality reduction]

PCA is unsupervised: it picks the direction of maximum total variance, which is along the long axis of the clouds (d_2).

LDA is supervised: it maximizes between-class separation relative to within-class scatter, so it picks the direction across which the class means differ (d_1).

Answer: **PCA (1st PC) chooses d_2 ; LDA chooses d_1** . Projecting on the PCA direction would mix the two classes completely.

Q.3 — Answer: 0.4361 to 0.4561 [NAT, 1M · Probabilistic & Instance-based Classifiers › Evaluation metrics]

TP = 12, FN = 20, FP = 42, TN = 45 ($N = 119$).

Balanced accuracy = $(\text{TPR} + \text{TNR})/2$.

TPR = $12/32 = 0.375$, TNR = $45/87 = 0.5172$; balanced accuracy = **0.45**.

Watch the orientation: swapping FP and FN is the most common error (it swaps precision and recall).

Q.4 — Answer: (A), (B), (C) [MSQ, 1M · SVM & Decision Trees › Decision trees: overfitting & pruning]

- (A) **TRUE**. Trees are high-variance learners.
- (B) **TRUE**. The mean minimises SSE.
- (C) **TRUE**. Each leaf corresponds to a hyper-rectangle.
- (D) **FALSE**. It increases variance (overfits).

Q.5 — Answer: 7 [NAT, 1M · SVM & Decision Trees › Decision trees: structure & traversal]

Depth d allows at most 2^d leaves, so we need $2^d \geq 76 \Rightarrow d \geq \log_2 76 = 6.248 \Rightarrow d = 7$.

Q.6 — Answer: 21 [NAT, 1M · Probabilistic & Instance-based Classifiers › Linear discriminant analysis]

Priors $K - 1 = 3$; means $K_p = 12$; one symmetric covariance $p(p+1)/2 = 6$.

Total = **21**.

Q.7 — Answer: 18848 [NAT, 1M · Neural Networks (MLP / Feed-forward) › Parameter counting]

Fully connected layer with n_{in} inputs and n_{out} outputs: $n_{\text{in}} \cdot n_{\text{out}}$ weights + n_{out} biases.

Layer 1: $10 \times 16 = 160$ weights, 16 biases.

Layer 2: $16 \times 64 = 1024$ weights, 64 biases.

Layer 3: $64 \times 256 = 16384$ weights, 256 biases.

Layer 4: $256 \times 5 = 1280$ weights, 5 biases.

Weights = 18848, biases = 341. Required: **18848**.

Q.8 — Answer: 3089 to 3092 [NAT, 1M · SVM & Decision Trees › Ensembles: bagging & random forests]

$P(\text{out-of-bag}) = (1 - 1/N)^N \approx e^{-1} \approx 0.3679$; expected count $\approx 8400 \times 0.3679 = \mathbf{3090}$.

Q.9 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection › Bias-variance trade-off]

- (A) **TRUE**. Flexible models can approximate the true f better.
- (B) **TRUE**. Definition of bias.
- (C) **TRUE**. Estimates stabilise as n grows.

(D) TRUE. It is the variance of the noise ε .

Q.10 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Cross-validation]

(A) TRUE. The n training sets overlap almost completely, so the fold errors are highly correlated.

(B) TRUE. That is why k -fold CV averages over several splits.

(C) TRUE. $CV_{(n)} = (1/n)\sum(e_i/(1 - h_{ii}))^2$.

(D) TRUE. The outer loop evaluates the whole tuning procedure on untouched data.

Q.11 — Answer: 20625 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Updates per epoch = $\lfloor N/B \rfloor = \lfloor 60000/32 \rfloor = 1875$.

Total = $1875 \times 11 = 20625$.

Q.12 — Answer: (B) [MCQ, 1M · Unsupervised Learning > k-means]

Points are assigned to the nearest centroid, so each k -means cluster is a convex Voronoi cell and two clusters are separated by the perpendicular bisector of their centroids. Non-convex shapes such as rings or moons cannot be separated by such a boundary.

k -means converges on raw data, works in any dimension, and is unsupervised (no labels).

Q.13 — Answer: (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

(A) FALSE. $p_y \leq 1 \Rightarrow -\ln p_y \geq 0$.

(B) TRUE. $p_1 = \sigma(z_1 - z_2)$.

(C) TRUE. Shift invariance.

(D) TRUE. Standard result.

Q.14 — Answer: 0.2042 to 0.2242 [NAT, 1M · Unsupervised Learning > Distance measures]

$x \cdot y = 35$, $\|x\| = \sqrt{124}$, $\|y\| = \sqrt{16}$; $\cos \theta = x \cdot y / (\|x\| \|y\|) = 0.7858$; distance = $1 - 0.7858 = 0.21$.

For reference: $L_1 = 16$, $L_2 = 8.37$, $L_\infty = 5$ (always $L_\infty \leq L_2 \leq L_1$).

Q.15 — Answer: (C) [MCQ, 1M · Regression & Model Selection > MLE & MAP view]

$\lambda = \sigma^2/\tau^2 \rightarrow 0$, so the penalty disappears.

Answer: **the least-squares (ML) estimate**.

Q.16 — Answer: 0.1866 to 0.1926 [NAT, 2M · Regression & Model Selection > Bias-variance trade-off]

Mean prediction $\bar{f} = (1/5)\sum f_m = 14.62$.

Bias² = $(\bar{f} - f(x_0))^2 = (14.62 - 14)^2 = 0.3844$.

Variance = $(1/5)\sum (f_m - \bar{f})^2 = 0.1896$.

Expected test error = Bias² + Variance + $\sigma^2 = 0.3844 + 0.1896 + 2 = 2.574$.

Answer: **0.19**.

Q.17 — Answer: (A), (B), (C), (D) [MSQ, 2M · Regression & Model Selection > Ridge & Lasso]

With $X^T X = I$ the ridge path is $b/(1 + \lambda)$ and the lasso path is $\text{sign}(b)(|b| - \lambda)_+$. The plot shows paths reaching zero at finite $\lambda \Leftrightarrow$ lasso; here it is **ridge**.

(A) TRUE. Both $b/(1 + \lambda)$ and soft-thresholding shrink monotonically.

(B) TRUE. Smooth decay $b/(1 + \lambda)$, no exact zeros.

(C) TRUE. Penalty vanishes at $\lambda = 0$.

(D) TRUE. $\beta(\lambda) = b/(1 + \lambda)$.

Q.18 — Answer: -0.0002 to 0.0002 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

All weights = -0.5 , biases 0 $\Rightarrow$ both hidden units get $a = -0.5 \times (2 + 2) = -2$, $h_1 = h_2 = \sigma(a) = 0.1192$; $\hat{y} = 2 \times -0.5 \times 0.1192 = -0.1192$.

$\hat{y} - y = 0.8808$. $\partial L / \partial v_j = (\hat{y} - y) h_j = 0.105$ (same for both j).

$\partial L / \partial w_{jk} = (\hat{y} - y) v_j h_j (1 - h_j) x_k = 0.8808 \times -0.5 \times 0.105 \times 2 = -0.0925$ (identical for $j = 1, 2$).

Both hidden units receive identical gradients, so the difference stays **0**.

Because the two hidden units start identical and get identical updates, they remain identical forever (symmetry problem) – hence weights must be initialised randomly.

Q.19 — Answer: 0.99 to 1.01 [NAT, 2M · SVM & Decision Trees > Decision trees: structure & traversal]

Route each record from the root to a leaf:

#	Predicted	Actual	Correct?
1	No	No	yes
2	No	No	yes
3	No	Yes	no
4	Yes	Yes	yes
5	No	No	yes
6	No	Yes	no
7	No	Yes	no
8	No	Yes	no

Misclassified: 4 of 8 (accuracy 50%). Answer: **1**.

Q.20 — Answer: (A) [MCQ, 2M · Probabilistic & Instance-based Classifiers > Naive Bayes]

(i) No smoothing.

Pos: $P(\text{Flu}=\text{Pos}) = 6/12$; $P(\text{Fever}=\text{None}|\text{Pos}) = 1/6$; $P(\text{Cough}=\text{No}|\text{Pos}) = 1/6$; $P(\text{Fatigue}=\text{Yes}|\text{Pos}) = 4/6 \Rightarrow \text{score} = 0.00926$.

Neg: $P(\text{Flu}=\text{Neg}) = 6/12$; $P(\text{Fever}=\text{None}|\text{Neg}) = 6/6$; $P(\text{Cough}=\text{No}|\text{Neg}) = 1/6$;

$P(\text{Fatigue}=\text{Yes}|\text{Neg}) = 0/6 \Rightarrow \text{score} = 0$.

A zero count makes the whole product for Neg equal to 0 (zero-frequency problem), so the prediction is Pos.

(ii) Laplace smoothing.

Pos: $P(\text{Flu}=\text{Pos}) = 6/12$; $P(\text{Fever}=\text{None}|\text{Pos}) = (1+1)/(6+3) = 2/9$; $P(\text{Cough}=\text{No}|\text{Pos}) = (1+1)/(6+2) = 1/4$; $P(\text{Fatigue}=\text{Yes}|\text{Pos}) = (4+1)/(6+2) = 5/8 \Rightarrow \text{score} = 0.01736$.

Neg: $P(\text{Flu}=\text{Neg}) = 6/12$; $P(\text{Fever}=\text{None}|\text{Neg}) = (6+1)/(6+3) = 7/9$; $P(\text{Cough}=\text{No}|\text{Neg}) = (1+1)/(6+2) = 1/4$; $P(\text{Fatigue}=\text{Yes}|\text{Neg}) = (0+1)/(6+2) = 1/8 \Rightarrow \text{score} = 0.01215$.

Larger score: Pos. Answer: **Without smoothing: Pos; with Laplace smoothing: Pos.**

Q.21 — Answer: 0.8568 to 0.8768 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b} = (-1, 1, 3)$.

$e^{\mathbf{z}} = (0.3679, 2.7183, 20.0855)$, sum = 23.1717.

Softmax probabilities = (0.0159, 0.1173, 0.8668).

$P(y = 3 | \mathbf{x}) = \mathbf{0.87}$.

Q.22 — Answer: 7.49 to 7.51 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

For centered data, projecting onto the top-k PCs and reconstructing leaves the components along the discarded eigenvectors; their average squared length (divisor $n - 1$) is the sum of the discarded eigenvalues.

Discarded eigenvalues: 3, 2.5, 2; sum = 7.5; total = 62.5.

Answer: **7.5**. (A common mistake is to add the *retained* eigenvalues 55.)

Q.23 — Answer: 0.4478 to 0.4518 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

$\mathbf{a} = \mathbf{W}_1\mathbf{x} + \mathbf{b}_1 = (2, -2)$; $\mathbf{h} = \text{ReLU}(\mathbf{a}) = (2, 0)$.

$\mathbf{z} = \mathbf{W}_2\mathbf{h} + \mathbf{b}_2 = (4.5, 3, 4.5)$.

$\exp(\mathbf{z}) = (90.0171, 20.0855, 90.0171)$, sum = 200.1198.

$p_1 = \exp(z_1)/\sum \exp(z_j) = \mathbf{0.4498}$. (All probabilities: (0.45, 0.1, 0.45).)

Q.24 — Answer: 0.5269 to 0.5309 [NAT, 2M · Regression & Model Selection > Gradient descent]

$\partial J/\partial w_0 = (1/n)\sum r_i$, $\partial J/\partial w_1 = (1/n)\sum r_i x_i$, with residual $r_i = w_0 + w_1 x_i - y_i$.

Step 1: $w = (-1, -1)$, $r = (-6, -4, -6)$, $\nabla J = (-5.3333, -10.6667) \Rightarrow w \leftarrow w - 0.1\nabla J = (-0.4667, 0.0667)$.

Step 2: $w = (-0.4667, 0.0667)$, $r = (-4.4, -1.3333, -2.2667)$, $\nabla J = (-2.6667, -4.6222) \Rightarrow w \leftarrow w - 0.1\nabla J = (-0.2, 0.5289)$.

Answer: **0.529**.

Q.25 — Answer: (C) [MCQ, 2M · Unsupervised Learning > Hierarchical clustering]

Use single linkage: $d(A, B) = \min_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P1\} + \{P6\}$ at height **2**. Updated: $d(\{P1, P6\}, \{P2\}) = 3$, $d(\{P1, P6\}, \{P3\}) = 4$, $d(\{P1, P6\}, \{P4\}) = 5$, $d(\{P1, P6\}, \{P5\}) = 8$.

Merge 2: $\{P2\} + \{P1, P6\}$ at height **3**. Updated: $d(\{P1, P2, P6\}, \{P3\}) = 4$, $d(\{P1, P2, P6\}, \{P4\}) = 5$, $d(\{P1, P2, P6\}, \{P5\}) = 8$.

Merge 3: $\{P3\} + \{P1, P2, P6\}$ at height **4**. Updated: $d(\{P1, P2, P3, P6\}, \{P4\}) = 5$, $d(\{P1, P2, P3, P6\}, \{P5\}) = 6$.

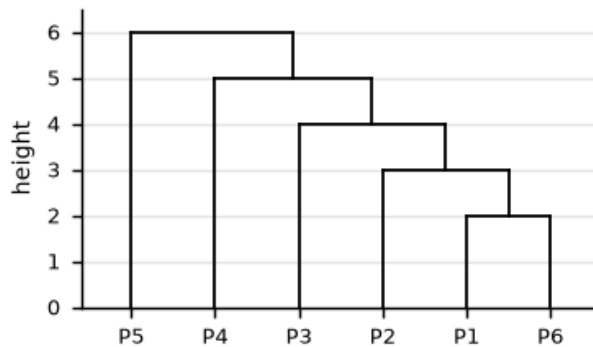
Merge 4: $\{P4\} + \{P1, P2, P3, P6\}$ at height **5**. Updated: $d(\{P1, P2, P3, P4, P6\}, \{P5\}) = 6$.

Merge 5: $\{P5\} + \{P1, P2, P3, P4, P6\}$ at height **6**.

Stopping when 2 clusters remain (i.e. after 4 merges): **$\{P1, P2, P3, P4, P6\}, \{P5\}$** .

$\{P1, P2, P4, P6\}, \{P3, P5\}$ is what *complete* linkage would give.

$\{P1, P2, P3, P6\}, \{P4\}, \{P5\}$ has 3 clusters (cut at the wrong level).



Q.26 — Answer: (C) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Expressiveness]

Each unit outputs 1 iff its net input (weighted sum + bias) is > 0 . Evaluate on all four inputs:

x_1	x_2	h_1	h_2	y
0	0	0	1	0
0	1	1	1	1
1	0	1	1	1
1	1	1	0	0

h_1 computes OR, h_2 computes NAND; the output column $(0,0) \rightarrow (1,1)$ is 0, 1, 1, 0, i.e. **XOR**.

This shows that one hidden layer of two threshold units suffices for the non-linearly-separable XOR, which a single perceptron cannot represent.

Q.27 — Answer: (A), (D) [MSQ, 2M · Regression & Model Selection > Multiple linear regression]

(A) **TRUE**. Difference of the two dummy coefficients: $-1.9 - (-3.2)$.

(B) **FALSE**. The change is $5 \times 2.6 = 13$.

(C) **FALSE**. It is a difference relative to the reference level A, adjusted for the other variables.

(D) **TRUE**. $\hat{y} = 10 + 2.6 \times 6 + (3.1) \times 1 + (-3.2) = 25.5$.

Q.28 — Answer: (B) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$p = \text{softmax}((0, -1, 2)) = (0.1142, 0.042, 0.8438)$; $y = (0, 0, 1)$.

$\partial L / \partial z = p - y = \mathbf{(0.11, 0.04, -0.16)}$.

Distractors: $y - p = (-0.11, -0.04, 0.16)$ (wrong sign); p alone = $(0.11, 0.04, 0.84)$ (forgot the target); $(p - y) \odot p \odot (1 - p) = (0.01, 0, -0.02)$ (spurious σ' -type factor, which cancels for cross-entropy); $-y/p = (0, 0, -1.19)$ is $\partial L / \partial p$, not $\partial L / \partial z$.

Q.29 — Answer: (D) [MCQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Ranges: age [25, 57], income [50, 149]. Scaled query = (0.875, 0.273).

Point	scaled age	scaled income	scaled distance	raw distance
P1	0	1	1.138	77.25
P2	0.062	0	0.857	37.48
P3	0.344	0.394	0.545	20.81
P4	1	0.616	0.365	34.23

Nearest after scaling: **P4**.

Without scaling the nearest point would be P3, because income differences (tens of thousands) dominate the age differences — exactly why k-NN needs feature scaling.

Q.30 — Answer: 0.1975 to 0.2175 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

a(i) = mean Euclidean distance from P2 to the other members of its own cluster C1: $d(P2, P1) = 3.606 \rightarrow a = 3.6056$.

b(i) = smallest mean distance to the points of another cluster: C2: 4.5495; C3: 6.2281 $\rightarrow b = 4.5495$.

s(i) = $(b - a) / \max(a, b) = (4.5495 - 3.6056) / 4.5495 = \mathbf{0.21}$.

s ranges in $[-1, 1]$; values near 1 mean well clustered, negative values suggest a wrong assignment.

Q.31 — Answer: (B), (D) [MSQ, 2M · SVM & Decision Trees > Decision trees: impurity measures]

Parent [60+, 60-]: $H = 1$, Gini = 0.5, error = 0.5.

S1: children $H = 0.9427$, 0.8366; Gini = 0.4608, 0.3911; error = 0.36, 0.2667. Gains: IG = 0.0971, Gini = 0.0653, error = 0.175.

S2: children $H = 0.2108$, 0.929; Gini = 0.0644, 0.4516; error = 0.0333, 0.3444. Gains: IG = 0.2505, Gini = 0.1452, error = 0.2333.

(A) FALSE. IG(S2) = 0.2505.

(B) TRUE. S1: 0.0653, S2: 0.1452.

(C) FALSE. S1: 0.0971, S2: 0.2505.

(D) TRUE. S1: 0.0971, S2: 0.2505.

Q.32 — Answer: 1 [NAT, 2M · SVM & Decision Trees > Perceptron (linear separators)]

Trace the epoch (a point with $y \cdot w^T \tilde{x} \leq 0$, including 0, triggers an update):

Step	$\tilde{x} = (1, x_1, x_2)$	y	$w^T \tilde{x}$	$y \cdot w^T \tilde{x} \leq 0$?	w after step
1	(1, -3, 2)	-1	-2	no	(-1, 1, 1)
2	(1, 1, -3)	+1	-3	yes $\rightarrow$ update	(0, 2, -2)
3	(1, -3, -1)	-1	-4	no	(0, 2, -2)
4	(1, 1, -1)	+1	4	no	(0, 2, -2)
5	(1, 3, 3)	+1	0	yes $\rightarrow$ update	(1, 5, 1)

Final $w = (1, 5, 1)$, number of updates = 2. Answer: **1**.

Q.33 — Answer: (C) [MSQ, 2M · Unsupervised Learning > k-means]

(A) FALSE. The decomposition is $TSS = WSS + BSS$.

(B) FALSE. It minimizes the sum of dissimilarities to medoids (data points).

(C) TRUE. A standard identity; the SSE can be written via pairwise distances.

(D) FALSE. It is $2^{n-1} - 1$.

Q.34 — Answer: 0.198 to 0.202 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

Bayes rule: choose the class with larger $\pi_k P(x|k)$. Bayes error = $\sum_x \min(\pi_0 P(x|0), \pi_1 P(x|1))$.

x	$\pi_0 P(x 0)$	$\pi_1 P(x 1)$	decision	min
1	0.2	0	0	0

2	0.2	0.3	1	0.2
3	0	0.2	1	0
4	0.1	0	0	0

Bayes error = **0.2**.

Q.35 — Answer: (B) [MCQ, 2M · SVM & Decision Trees › Decision trees: split selection]

Parent: 4 Yes, 6 No $\Rightarrow H = 0.971$.

Temp — Hot: [1+, 1-], Mild: [1+, 3-], Cool: [2+, 2-]; gain = 0.0464.

Wind — Weak: [1+, 5-], Strong: [3+, 1-]; gain = 0.2564.

Weekend — Yes: [2+, 4-], No: [2+, 2-]; gain = 0.02.

Outlook — Sunny: [2+, 3-], Overcast: [2+, 2-], Rain: [0+, 1-]; gain = 0.0855.

The largest information gain is for **Wind**, so it is chosen at the root.

Mock Test 02

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	7	11	17

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Unsupervised Learning	7	11
Neural Networks (MLP / Feed-forward)	7	11
Regression & Model Selection	7	11
Probabilistic & Instance-based Classifiers	7	11
SVM & Decision Trees	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

A complete (binary) dendrogram over $n = 23$ points has how many nodes in total (leaves plus internal nodes)? (Answer in integer)

Q.2 [1 mark · NAT (numerical answer)]

The value of the derivative $\sigma'(z)$ of the logistic sigmoid at $z = 2$ is _____ (rounded off to three decimal places)

Q.3 [1 mark · MSQ (one or more correct)]

Which of the following statements about distance / similarity measures is/are CORRECT?

- (A) Minkowski distance with $p = 1$ is the Manhattan distance and with $p = 2$ the Euclidean distance.
- (B) Cosine distance $1 - \cos \theta$ satisfies the triangle inequality for all vectors.
- (C) Cosine similarity changes when both vectors are multiplied by the same positive constant.
- (D) Euclidean distance is unaffected by changing the unit of one feature (e.g. metres to centimetres).

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about basis-function (polynomial) regression and feature scaling is/are CORRECT?

- (A) Polynomial regression $y = w_0 + w_1x + w_2x^2$ is linear in the parameters, so it can be fitted by ordinary least squares.
- (B) Standardising two features changes the correlation between them.
- (C) High-degree polynomial features can make $X^T X$ badly ill-conditioned; centring/scaling x helps numerically.
- (D) Gradient descent for linear regression usually converges faster when features are on comparable scales.

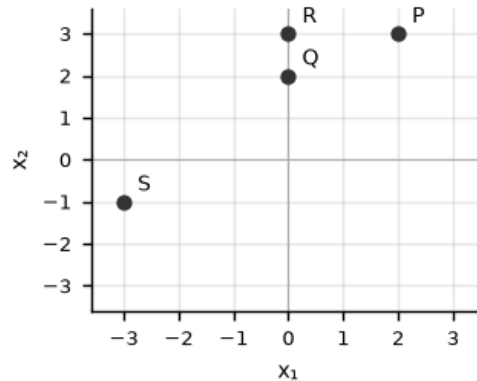
Q.5 [1 mark · MSQ (one or more correct)]

Which of the following statements about naive Bayes classifiers is/are CORRECT?

- (A) In practice, naive Bayes sums log-probabilities to avoid numerical underflow.
- (B) If a feature is duplicated, naive Bayes counts its evidence twice.
- (C) The number of naive Bayes parameters grows linearly with the number of features d .
- (D) Maximum-likelihood training of categorical naive Bayes requires only counting (closed form).

Q.6 [1 mark · MSQ (one or more correct)]

A perceptron with weights $w = (-2, -3)$ and bias $b = -3$ outputs $\hat{y} = +1$ if $w \cdot x + b \geq 0$ and $\hat{y} = -1$ otherwise. Which of the points shown below are classified as $\hat{y} = +1$?



- (A) Point P = (2, 3)
- (B) Point Q = (0, 2)
- (C) Point R = (0, 3)
- (D) Point S = (-3, -1)

Q.7 [1 mark · NAT (numerical answer)]

A linear regression model with an intercept uses 2 numeric features and a categorical feature *education level* with 3 levels (School, UG, PG). Each categorical feature is encoded by dummy (indicator) variables using a reference level, so that the design matrix has full column rank. The model is fitted to $n = 100$ observations. The residual degrees of freedom used in the unbiased estimate $\hat{\sigma}^2 = \text{SSE}/(\text{degrees of freedom})$ is _____ (Answer in integer)

Q.8 [1 mark · NAT (numerical answer)]

In a defect detector, a false negative costs 18 units and a false positive costs 5 units (correct decisions cost 0). For an input x the model gives $P(\text{positive} | x) = 0.8$. The expected cost of the Bayes-optimal decision for this x is _____ (rounded off to two decimal places)

Q.9 [1 mark · MCQ]

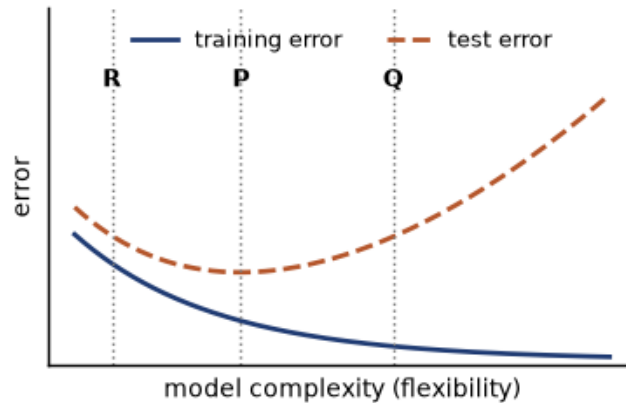
Cost-complexity pruning of a classification tree produced the nested subtrees $T1 \supset T2 \supset T3 \supset T4$ below. For complexity parameter $\alpha = 0.008$, which subtree minimises $R_\alpha(T) = R(T) + \alpha|T|$?

Subtree	T1	T2	T3	T4
Leaves $ T $	12	11	3	2
Training error $R(T)$	0.055	0.105	0.115	0.19

- (A) T1
- (B) T2
- (C) T3
- (D) T4

Q.10 [1 mark · MCQ]

The figure shows training and test error as a function of model complexity, with three complexity levels marked P, Q and R. Which marked level corresponds to underfitting (high bias, low variance)?



- (A) R (B) Q
(C) None of P, Q, R (D) P

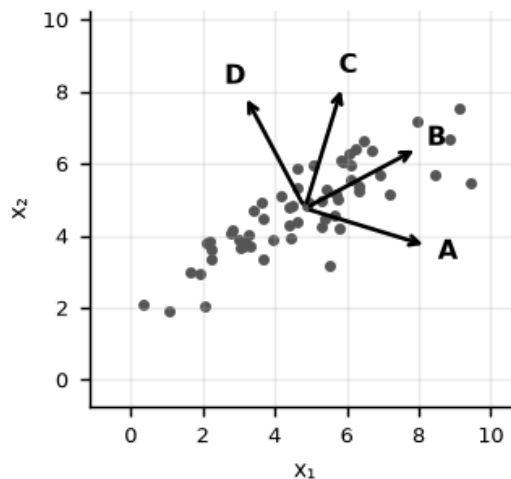
Q.11 [1 mark · MSQ (one or more correct)]

Which of the following statements about bagging, random forests and boosting is/are CORRECT?

- (A) AdaBoost requires weak learners whose error is strictly greater than 0.5.
(B) AdaBoost is sensitive to label noise because the weights of persistently misclassified points grow exponentially.
(C) After an AdaBoost reweighting step, the misclassified examples carry exactly half of the total weight.
(D) Bagged trees can be trained in parallel, whereas boosting is inherently sequential.

Q.12 [1 mark · MCQ]

The scatter plot shows a 2-D dataset together with four candidate directions A-D drawn from the data mean. Which direction best represents the first principal component (PC1)?



- (A) Direction A (B) Direction B
(C) Direction C (D) Direction D

Q.13 [1 mark · NAT (numerical answer)]

A 3-class softmax classifier gives the predicted probabilities below for a mini-batch of 3 examples. The mean categorical cross-entropy loss (natural log) over the mini-batch is _____ (rounded off to three decimal places)

example	p(class 1)	p(class 2)	p(class 3)	true class
1	0.05	0.9	0.05	3
2	0.1	0.8	0.1	2
3	0.3	0.2	0.5	3

Q.14 [1 mark · NAT (numerical answer)]

A binary logistic regression model is given by $\ln[P(y=1|\mathbf{x}) / P(y=0|\mathbf{x})] = -1.5 + (1.5)x_1 + (0.5)x_2$. For the input $\mathbf{x} = (x_1, x_2) = (1, 2)$, the value of $P(y = 0 | \mathbf{x})$ is _____ (rounded off to two decimal places)

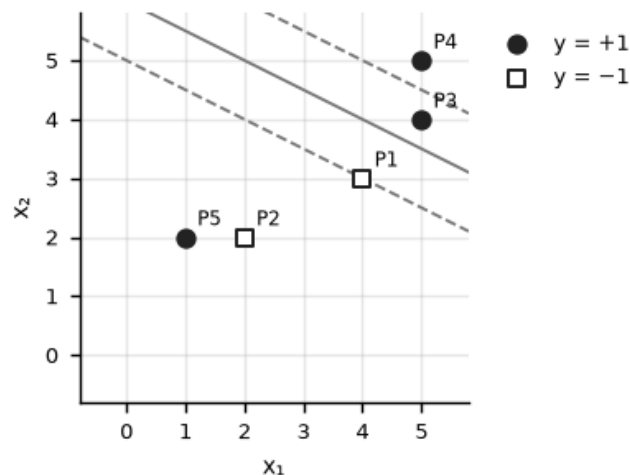
Q.15 [1 mark · NAT (numerical answer)]

A node contains 12 records spread equally over 4 classes (3 records per class). Its entropy (in bits) is _____ (rounded off to two decimal places)

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · MSQ (one or more correct)]**

A soft-margin linear SVM has learned $f(\mathbf{x}) = 0.5x_1 + x_2 - 6$. The training points are listed below and plotted (boundary solid, margins $f = \pm 1$ dashed). With ξ_i the optimal slack of point i , which of the following statements is/are CORRECT?

Point	x_1	x_2	y
P1	4	3	-1
P2	2	2	-1
P3	5	4	+1
P4	5	5	+1
P5	1	2	+1



- (A) P1 has slack $\xi_1 = 0$.
- (B) P5 is misclassified ($\xi_5 > 1$).
- (C) $\sum \xi_i = 5$.
- (D) P3 is correctly classified but lies strictly inside the margin.

Q.17 [2 marks · MSQ (one or more correct)]

For the confusion matrix below, which of the following statements is/are CORRECT? (values rounded to 2 decimals)

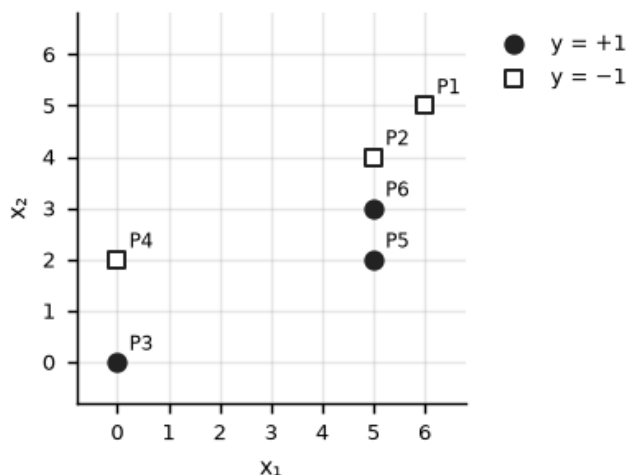
	Actual +	Actual –
Predicted +	43	13
Predicted –	32	76

- (A) The negative predictive value is 0.85.
- (B) The F1-score is 0.67.
- (C) The specificity (TNR) is 0.7.
- (D) The F1-score is less than the arithmetic mean of precision and recall.

Q.18 [2 marks · MSQ (one or more correct)]

A hard-margin linear SVM is trained on the 6 labelled points shown. Which of the following points is/are support vectors of the resulting maximum-margin classifier?

Point	x_1	x_2	y
P1	6	5	–1
P2	5	4	–1
P3	0	0	+1
P4	0	2	–1
P5	5	2	+1
P6	5	3	+1

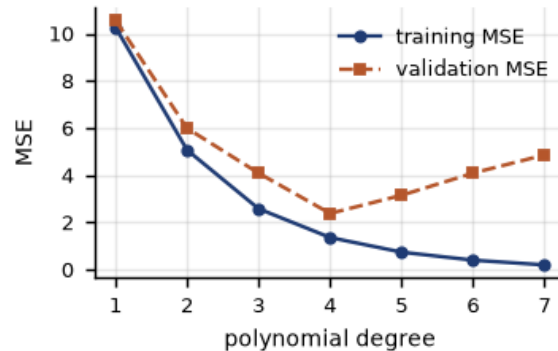


- (A) P1
- (B) P2
- (C) P4
- (D) P6

Q.19 [2 marks · MSQ (one or more correct)]

Polynomial regression models of degree 1 to 7 are fitted by least squares on a training set and evaluated on a separate validation set. The resulting mean squared errors are given below. Which of the following statements is/are CORRECT?

Degree	1	2	3	4	5	6	7
Training MSE	10.27	5.07	2.58	1.36	0.74	0.4	0.2
Validation MSE	10.57	5.99	4.09	2.38	3.15	4.09	4.86



- (A) The degree-7 model has higher variance than the degree-4 model.
- (B) The validation MSE of the selected degree is an unbiased estimate of its test error, even though it was used for the selection.
- (C) The generalisation gap (validation MSE – training MSE) is largest for degree 7.
- (D) Based on the validation MSE, degree 4 should be selected.

Q.20 [2 marks · MCQ]

A network computes $\hat{y} = W_2 \text{ReLU}(W_1 x + b_1) + b_2$ with the parameters below. For $x = (2, 2, -1)^T$, the output $\hat{y}$ is

$$W_1 = \begin{bmatrix} -2 & 2 & 1 \\ 2 & -2 & -1 \\ 2 & 0 & 2 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 2 & 2 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

- (A) $\hat{y} = (2, 1)^T$
- (B) $\hat{y} = (2, 5)^T$
- (C) $\hat{y} = (2, 6)^T$
- (D) $\hat{y} = (2, 4)^T$

Q.21 [2 marks · NAT (numerical answer)]

A 2-2-1 network uses the identity (linear) activation in its hidden layer: $\hat{y} = W_2(W_1 x + b_1) + b_2$, with the parameters below. The network is equivalent to a single linear model $\hat{y} = \tilde{w}_1 x_1 + \tilde{w}_2 x_2 + \tilde{b}$. Then the intercept $\tilde{b}$ is _____ (Answer in integer)

$$W_1 = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} -2 & -2 \end{bmatrix}$$

$$b_2 = 0$$

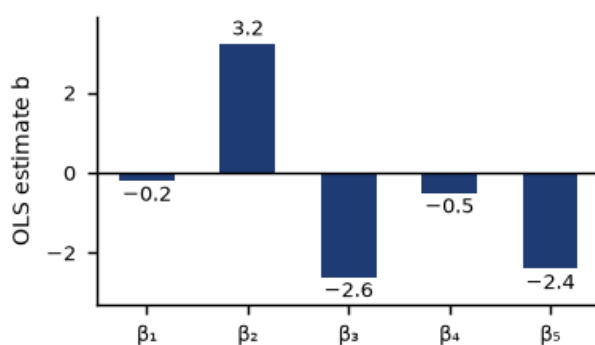
Q.22 [2 marks · NAT (numerical answer)]

The predicted probabilities $\hat{p}_i = P(y_i = 1 | x_i)$ of a binary classifier on 4 examples, and their true labels, are given below. Using natural logarithms, the **average** binary cross-entropy loss $-[y \ln \hat{p} + (1 - y) \ln(1 - \hat{p})]$ over these examples is _____ (rounded off to two decimal places)

i	1	2	3	4
y_i	0	1	0	1
$\hat{p}_i = P(y_i=1)$	0.6	0.8	0.45	0.45

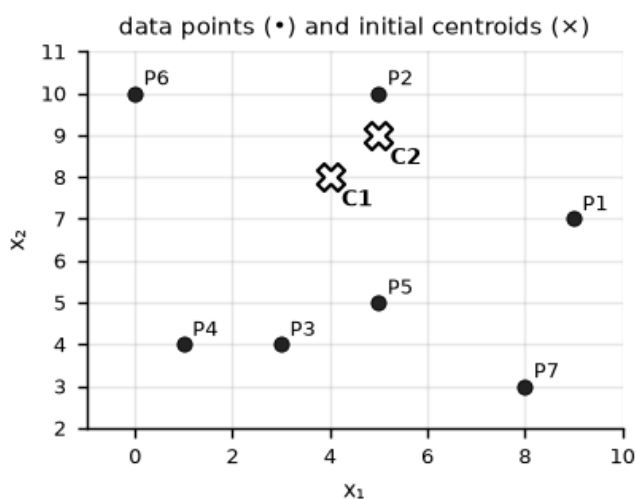
Q.23 [2 marks · NAT (numerical answer)]

In a regression problem the design matrix X has orthonormal columns ($X^T X = I$). The ordinary least-squares estimates $b = X^T y$ are shown below. Ridge minimises $\|y - X\beta\|^2 + \lambda \|\beta\|_2^2$ and the lasso minimises $\frac{1}{2} \|y - X\beta\|^2 + \lambda \|\beta\|_1$, each with $\lambda = 1$. Then the number of non-zero coefficients in the lasso solution is _____ (Answer in integer)

**Q.24 [2 marks · NAT (numerical answer)]**

k-means (Lloyd's algorithm, Euclidean distance) with $k = 2$ is run on the 7 points below, starting from the initial centroids $C1 = (4, 8)$, $C2 = (5, 9)$. After **two** iterations, the x_1 -coordinate of the centroid of cluster C1 is _____ (rounded off to two decimal places)

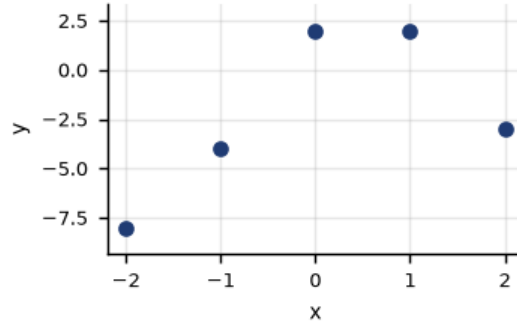
Point	P1	P2	P3	P4	P5	P6	P7
x_1	9	5	3	1	5	0	8
x_2	7	10	4	4	5	10	3



Q.25 [2 marks · NAT (numerical answer)]

The quadratic model $y = a + bx + cx^2$ is fitted by least squares to the 5 points below (a polynomial / basis-function regression with basis $1, x, x^2$). Then the intercept a is _____ (rounded off to two decimal places)

x	-2	-1	0	1	2
y	-8	-4	2	2	-3

**Q.26 [2 marks · MSQ (one or more correct)]**

Which of the following statements about loss functions and optimisation for training neural networks is/are CORRECT?

- (A) Too small a learning rate leads to very slow convergence.
- (B) Minimising the mean squared error corresponds to maximum likelihood under additive Gaussian noise with fixed variance.
- (C) Minimising cross-entropy loss is equivalent to maximising the log-likelihood of the labels under the model.
- (D) With momentum β ($v \leftarrow \beta v + g$, $w \leftarrow w - \eta v$) and a constant gradient g , the step size approaches $\eta g / (1 - \beta)$.

Q.27 [2 marks · NAT (numerical answer)]

For a two-class problem, the class means are $\mu_0 = (0, 1)^T$ and $\mu_1 = (-1, 1)^T$ and the pooled within-class scatter matrix S_W is given below. Fisher's linear discriminant uses $\mathbf{w} \propto S_W^{-1}(\mu_1 - \mu_0)$. Then the ratio w_2/w_1 of the components of the Fisher discriminant direction $\mathbf{w}$ is _____ (rounded off to two decimal places)

$$S_W = \begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix}$$

Q.28 [2 marks · NAT (numerical answer)]

Two clusterings U and V of 8 points are given below (cluster names are arbitrary). Then the number of point pairs placed in the same cluster by **both** clusterings is _____ (Answer in integer)

Point	P1	P2	P3	P4	P5	P6	P7	P8
Clustering U	γ	α	α	α	α	β	β	γ
Clustering V	3	1	1	1	1	1	2	3

Q.29 [2 marks · MSQ (one or more correct)]

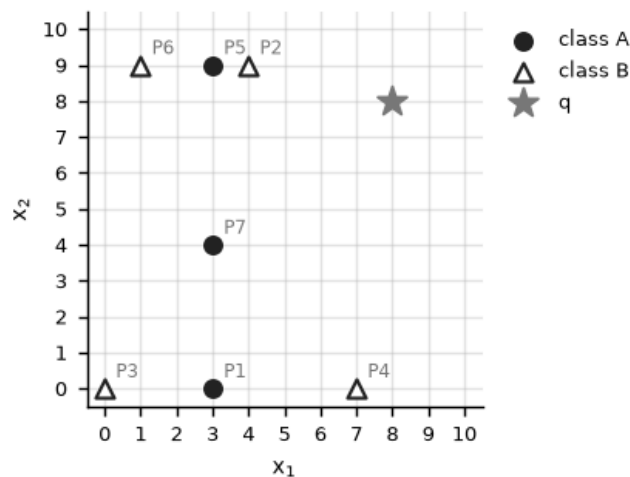
Which of the following statements about PCA, SVD and dimensionality reduction is/are CORRECT?

- (A) Multiplying all features by the same non-zero constant c leaves the proportion of variance explained by each PC unchanged.
- (B) PCA always preserves the class separability present in the original feature space.
- (C) If $X = U\Sigma V^T$, the PC scores are given by $V\Sigma$.
- (D) To capture a fraction r of uniform data in a d -dimensional unit hypercube, a sub-cube needs edge length $r^{1/d}$, which approaches 1 as d grows.

Q.30 [2 marks · MCQ]

Consider the training set below. A 3-NN classifier (majority vote) is applied to the query point $q = (8, 8)$, once with Euclidean distance and once with Manhattan distance. The predicted classes are

Point	x_1	x_2	class
P1	3	0	A
P2	4	9	B
P3	0	0	B
P4	7	0	B
P5	3	9	A
P6	1	9	B
P7	3	4	A



- (A) Euclidean: B; Manhattan: B
 (C) Euclidean: A; Manhattan: A

- (B) Euclidean: B; Manhattan: A
 (D) Euclidean: A; Manhattan: B

Q.31 [2 marks · MCQ]

For $n = 6$ paired observations: $\sum x = 43$, $\sum y = -66$, $\sum x^2 = 347$, $\sum y^2 = 804$, $\sum xy = -509$. The least-squares regression line of x on y is (coefficients rounded to two decimals)

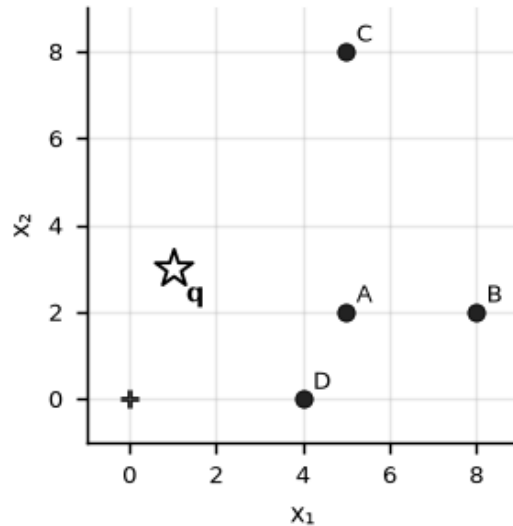
- (A) $\hat{x} = 2.09 - 0.46y$
 (C) $\hat{x} = -4.7 - 1.08y$

- (B) $\hat{x} = -7.69 - 0.46y$
 (D) $\hat{x} = 12.24 - 0.46y$

Q.32 [2 marks · MCQ]

A query point $q = (1, 3)$ and four points A-D are shown below. Using **cosine similarity**, which point is most similar to q ?

Point	A	B	C	D
x_1	5	8	5	4
x_2	2	2	8	0



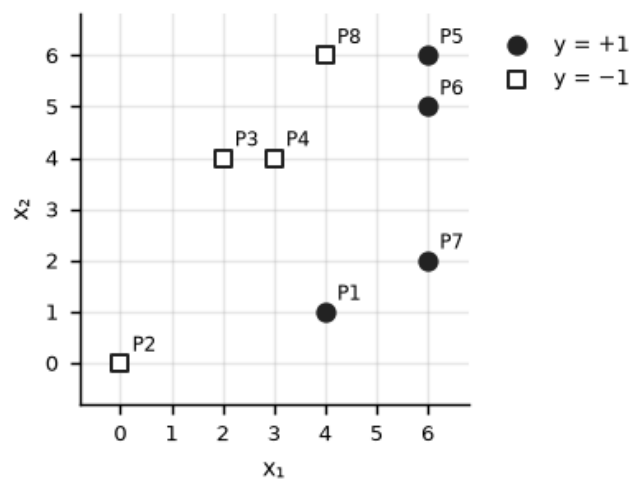
- (A) Point A
(C) Point C

- (B) Point B
(D) Point D

Q.33 [2 marks · NAT (numerical answer)]

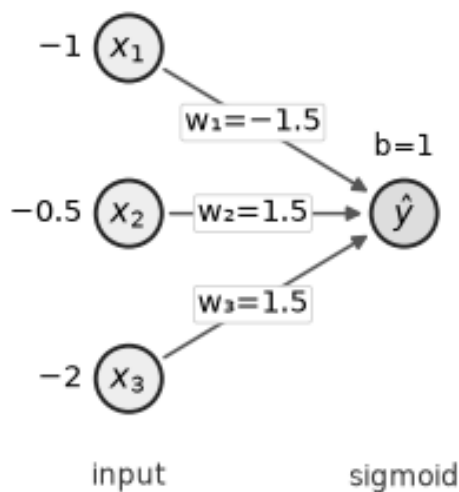
The labelled training points shown below are linearly separable. A hard-margin linear SVM is trained on them. Compute the width of the margin (distance between the two margin hyperplanes). (rounded off to two decimal places)

Point	x_1	x_2	y
P1	4	1	+1
P2	0	0	-1
P3	2	4	-1
P4	3	4	-1
P5	6	6	+1
P6	6	5	+1
P7	6	2	+1
P8	4	6	-1



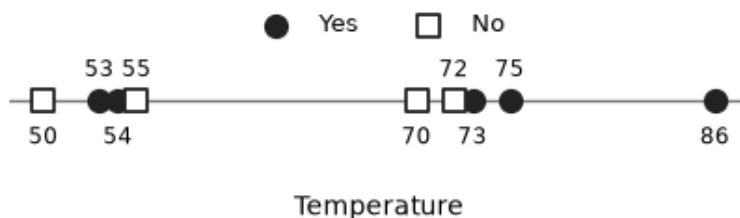
Q.34 [2 marks · NAT (numerical answer)]

A single sigmoid neuron (shown below) receives $x = (-1, -0.5, -2)$ and has target $y = 1$. The loss is $L = \frac{1}{2}(\hat{y} - y)^2$. The value of $\partial L / \partial w_1$ is _____ (rounded off to three decimal places)

**Q.35 [2 marks · NAT (numerical answer)]**

The continuous attribute Temperature and the class label of 9 training records are given below (unsorted). A binary split “Temperature $\leq t$ ” is chosen to maximise information gain, with t restricted to midpoints between consecutive sorted values. Find the threshold value t chosen for the split “Temperature $\leq t$ ”. (rounded off to one decimal place)

Temperature	53	72	50	70	75	86	55	73	54
Class	Yes	No	No	No	Yes	Yes	No	Yes	Yes



— END OF QUESTION PAPER —

Mock Test 02 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	45	19	MSQ	2	(A), (C), (D)
2	NAT	1	0.103 to 0.107	20	MCQ	2	(B)
3	MSQ	1	(A)	21	NAT	2	0
4	MSQ	1	(A), (C), (D)	22	NAT	2	0.6239 to 0.6439
5	MSQ	1	(A), (B), (C), (D)	23	NAT	2	3
6	MSQ	1	(D)	24	NAT	2	3.39 to 3.41
7	NAT	1	95	25	NAT	2	1.2086 to 1.2486
8	NAT	1	0.99 to 1.01	26	MSQ	2	(A), (B), (C), (D)
9	MCQ	1	(C)	27	NAT	2	0.49 to 0.51
10	MCQ	1	(A)	28	NAT	2	7
11	MSQ	1	(B), (C), (D)	29	MSQ	2	(A), (D)
12	MCQ	1	(B)	30	MCQ	2	(D)
13	NAT	1	1.302 to 1.306	31	MCQ	2	(A)
14	NAT	1	0.2589 to 0.2789	32	MCQ	2	(C)
15	NAT	1	1.99 to 2.01	33	NAT	2	1.847 to 1.867
16	MSQ	2	(A), (B), (C), (D)	34	NAT	2	0.1326 to 0.1366
17	MSQ	2	(D)	35	NAT	2	72.49 to 72.51
18	MSQ	2	(B), (C), (D)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Unsupervised Learning					11
Neural Networks (MLP / Feed-forward)					11
Regression & Model Selection					11
Probabilistic & Instance-based Classifiers					11
SVM & Decision Trees					11
Total					55

Mock Test 02 — Detailed Solutions

Q.1 — Answer: 45 [NAT, 1M · Unsupervised Learning › Hierarchical clustering]

n leaves + $(n - 1)$ internal (merge) nodes = $2n - 1 = 45$.

Q.2 — Answer: 0.103 to 0.107 [NAT, 1M · Neural Networks (MLP / Feed-forward) › Activation functions]

$\sigma(2) = 0.8808$; $\sigma'(z) = \sigma(z)(1 - \sigma(z)) = 0.8808 \times 0.1192 = 0.105$.

Note $\sigma' \leq 0.25$, with equality only at $z = 0$.

Q.3 — Answer: (A) [MSQ, 1M · Unsupervised Learning › Distance measures]

(A) TRUE. Definitions.

(B) FALSE. It does not in general.

(C) FALSE. Scaling does not change the angle.

(D) FALSE. Rescaling one coordinate changes distances.

Q.4 — Answer: (A), (C), (D) [MSQ, 1M · Regression & Model Selection › Basis functions & feature scaling]

(A) TRUE. Treat 1, x , x^2 as columns of the design matrix.

(B) FALSE. Correlation is invariant to affine maps.

(C) TRUE. Powers of large x differ by orders of magnitude and are highly correlated.

(D) TRUE. Better conditioned Hessian.

Q.5 — Answer: (A), (B), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers › Naive Bayes]

(A) TRUE. Products of many small probabilities underflow.

(B) TRUE. The duplicated factor appears twice in the product, overstating confidence.

(C) TRUE. Each feature adds $K(m - 1)$ parameters.

(D) TRUE. Estimates are relative frequencies.

Q.6 — Answer: (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) › Perceptron]

Net input $a = w \cdot x + b = -2x_1 + (-3)x_2 + (-3)$; output +1 iff $a \geq 0$.

(P) $a = -2 \cdot 2 + (-3) \cdot 3 + (-3) = -16 \rightarrow -1$.

(Q) $a = -2 \cdot 0 + (-3) \cdot 2 + (-3) = -9 \rightarrow -1$.

(R) $a = -2 \cdot 0 + (-3) \cdot 3 + (-3) = -12 \rightarrow -1$.

(S) $a = -2 \cdot -3 + (-3) \cdot -1 + (-3) = 6 \rightarrow +1$ (selected).

Answer: (D).

Q.7 — Answer: 95 [NAT, 1M · Regression & Model Selection › Multiple linear regression]

A categorical feature with k levels needs $k - 1$ dummies (one level is the reference absorbed by the intercept; using all k dummies with an intercept gives perfect multicollinearity).

Parameters = 1 (intercept) + 2 (numeric) + $(3 - 1) = 5$.

Residual degrees of freedom = $n - (\text{number of coefficients}) = 100 - 5 = 95$.

Answer: 95.

Q.8 — Answer: 0.99 to 1.01 [NAT, 1M · Probabilistic & Instance-based Classifiers › Bayes optimal classifier]

Predict positive: $(1 - 0.8) \times 5 = 1$. Predict negative: $0.8 \times 18 = 14.4$.

Optimal decision: positive; expected cost = 1.

Q.9 — Answer: (C) [MCQ, 1M · SVM & Decision Trees › Decision trees: overfitting & pruning]

Subtree	$R(T) + \alpha T $
T1	$0.055 + 0.008 \times 12 = 0.151$
T2	$0.105 + 0.008 \times 11 = 0.193$
T3	$0.115 + 0.008 \times 3 = 0.139$
T4	$0.19 + 0.008 \times 2 = 0.206$

Minimum at **T3**. Larger α penalises leaves more and favours smaller trees; $\alpha = 0$ would pick the largest tree T1.

Q.10 — Answer: (A) [MCQ, 1M · Regression & Model Selection > Bias-variance trade-off]

At R: both training and test errors are high (left side) → underfitting.

At P: the test error is at its minimum → best trade-off.

At Q: training error is very low but test error has risen well above its minimum (right side) → overfitting.

Answer: **R**.

Q.11 — Answer: (B), (C), (D) [MSQ, 1M · SVM & Decision Trees > Ensembles: bagging & random forests]

(A) **FALSE**. It needs error below 0.5.

(B) **TRUE**. Outliers get dominant weight.

(C) **TRUE**. $w' = w/(2\epsilon)$ for misclassified points.

(D) **TRUE**. Boosting needs the previous round's weights.

Q.12 — Answer: (B) [MCQ, 1M · Unsupervised Learning > PCA & dimensionality reduction]

PC1 is the direction of maximum variance — along the long axis of the elongated cloud (direction B). PC2 must be orthogonal to PC1 and captures the remaining variance (direction D).

Answer: **Direction B**. Directions C and A are at 45° to the main axis and capture intermediate variance.

Q.13 — Answer: 1.302 to 1.306 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

CE for one example = $-\ln p_{\text{true class}}$.

Example 1: $-\ln 0.05 = 2.9957$.

Example 2: $-\ln 0.8 = 0.2231$.

Example 3: $-\ln 0.5 = 0.6931$.

Mean = **1.304**.

Q.14 — Answer: 0.2589 to 0.2789 [NAT, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

$z = w_0 + w_1x_1 + w_2x_2 = -1.5 + (1.5)(1) + (0.5)(2) = 1$ (this is the log-odds).

$P(y = 1 | \mathbf{x}) = \sigma(z) = 1/(1 + e^{-1}) = 0.7311$.

$P(y = 0 | \mathbf{x}) = 1 - \sigma(z) = \sigma(-z) = \mathbf{0.27}$.

Q.15 — Answer: 1.99 to 2.01 [NAT, 1M · SVM & Decision Trees > Decision trees: impurity measures]

$p_k = 3/12 = 1/4$ for every class $\Rightarrow H = -4 \cdot (1/4) \log_2(1/4) = \log_2 4 = \mathbf{2}$ — the maximum possible for 4 classes.

Q.16 — Answer: (A), (B), (C), (D) [MSQ, 2M · SVM & Decision Trees > SVM: soft margin & hinge loss]

$\xi_i = \max(0, 1 - y_i f(x_i))$.

Point	$y f(x)$	ξ
P1	1	0
P2	3	0
P3	0.5	0.5
P4	1.5	0
P5	-3.5	4.5

(A) **TRUE**. $y f = 1 \Rightarrow \xi = 0$.

(B) **TRUE**. $y f = -3.5 \Rightarrow \xi = 4.5$.

(C) **TRUE**. Sum of the individual slacks.

(D) **TRUE**. $y f = 0.5 \Rightarrow \xi = 0.5$.

Q.17 — Answer: (D) [MSQ, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

TP = 43, FN = 32, FP = 13, TN = 76.

(A) **FALSE**. negative predictive value = $TN/(TN + FN) = 0.7037$, not 0.85.

(B) **FALSE**. F1-score = $2PR/(P + R) = 0.6565$, not 0.67.

(C) **FALSE**. specificity (TNR) = $TN/(TN + FP) = 0.8539$, not 0.7.

(D) **TRUE**. The harmonic mean is $\leq$ the arithmetic mean, with equality only if $P = R$ (not the case here).

Q.18 — Answer: (B), (C), (D) [MSQ, 2M · SVM & Decision Trees > SVM: dual & support vectors]

P2 and P4 (same class) lie on one margin line and P6 on the other; the optimal boundary is parallel to the line through P2, P4, midway to P6. Support vectors: P2, P4, P6.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (4/5, -2)$, $b = 3$; $\|w\| = 2.1541$, margin $2/\|w\| = 0.9285$.

Point	P1	P2	P3	P4	P5	P6
$y(w^T x + b)$	2.2	1	3	1	3	1

Support vectors are exactly the points with $y_i(w^T x_i + b) = 1$ ($\alpha_i > 0$): **P2, P4, P6**.

Dual check: $\alpha = (\alpha_2 = 54/25, \alpha_4 = 4/25, \alpha_6 = 58/25)$ satisfy $\sum \alpha_i y_i = 0$ and $\sum \alpha_i y_i x_i = w$.

Q.19 — Answer: (A), (C), (D) [MSQ, 2M · Regression & Model Selection > Bias-variance trade-off]

(A) **TRUE**. Higher-degree polynomials are more flexible $\rightarrow$ higher variance.

(B) **FALSE**. Choosing the minimum makes it optimistically biased; a separate test set is needed.

(C) **TRUE**. Gaps: 0.3, 0.92, 1.51, 1.02, 2.41, 3.69, 4.66.

(D) **TRUE**. Validation MSE is minimum (2.38) at degree 4.

Q.20 — Answer: (B) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

$a = W_1 x + b_1 = (-1, 2, 1)^T$; $h = \text{ReLU}(a) = (0, 2, 1)^T$.

$\hat{y} = W_2 h + b_2 = (2, 5)^T$.

Distractors: skipping the ReLU gives $(2, 6)^T$; using W_1^T instead of W_1 gives $(2, 1)^T$; dropping the biases gives $(-1, 6)^T$; using $|a|$ instead of ReLU gives $(2, 4)^T$.

Q.21 — Answer: 0 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Expressiveness]

With identity activations, $\hat{y} = W_2(W_1 x + b_1) + b_2 = (W_2 W_1)x + (W_2 b_1 + b_2)$: a single affine map.

$\tilde{w}^T = W_2 W_1 = (-6, 0)$; $\tilde{b} = W_2 b_1 + b_2 = 0$.

Answer: **0**.

Hence stacking linear layers adds no expressive power – a non-linear activation is essential.

Q.22 — Answer: 0.6239 to 0.6439 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

Per-example loss $L_i = -\ln \hat{p}_i$ if $y_i = 1$ and $-\ln(1 - \hat{p}_i)$ if $y_i = 0$:

i	y	$\hat{p}$	L_i
1	0	0.6	0.9163
2	1	0.8	0.2231
3	0	0.45	0.5978
4	1	0.45	0.7985

$\sum L_i = 2.5358$; average = $2.5358/4 = \mathbf{0.63}$.

Common mistakes: using $\ln \hat{p}$ for $y = 0$ examples, or $\log_{10}$ instead of $\ln$.

Q.23 — Answer: 3 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

With $X^T X = I$: $\beta_{\text{OLS}} = X^T y = b$.

Ridge: $\nabla[\|y - X\beta\|^2 + \lambda\|\beta\|^2] = 0 \Rightarrow (1 + \lambda)\beta = b \Rightarrow \beta_{\text{ridge}} = b/(1 + \lambda)$.

Lasso: the problem separates per coordinate, $\min \frac{1}{2}(\beta_j - b_j)^2 + \lambda|\beta_j| \Rightarrow$ soft-thresholding $\beta_j = \text{sign}(b_j) \cdot \max(|b_j| - \lambda, 0)$.

j	1	2	3	4	5
---	---	---	---	---	---

b_j	-0.2	3.2	-2.6	-0.5	-2.4
ridge	-0.1	1.6	-1.3	-0.25	-1.2
lasso	0	2.2	-1.6	0	-1.4

Answer: **3**.

Q.24 — Answer: 3.39 to 3.41 [NAT, 2M · Unsupervised Learning › k-means]

Iteration 1 - assignment (nearest initial centroid, squared Euclidean distance):

Point	$d^2(\cdot, C1)$	$d^2(\cdot, C2)$	cluster
P1 (9, 7)	26	20	C2
P2 (5, 10)	5	1	C2
P3 (3, 4)	17	29	C1
P4 (1, 4)	25	41	C1
P5 (5, 5)	10	16	C1
P6 (0, 10)	20	26	C1
P7 (8, 3)	41	45	C1

Update: each centroid becomes the mean of its assigned points: $C1 \rightarrow (3.4, 5.2)$; $C2 \rightarrow (7, 8.5)$.

Iteration 2 - assignment with the updated centroids:

Point	$d^2(\cdot, C1)$	$d^2(\cdot, C2)$	cluster
P1 (9, 7)	34.6	6.25	C2
P2 (5, 10)	25.6	6.25	C2
P3 (3, 4)	1.6	36.25	C1
P4 (1, 4)	7.2	56.25	C1
P5 (5, 5)	2.6	16.25	C1
P6 (0, 10)	34.6	51.25	C1
P7 (8, 3)	26	31.25	C1

Update: $C1 \rightarrow (3.4, 5.2)$; $C2 \rightarrow (7, 8.5)$. Required coordinate = **3.4**.

Note: each step of Lloyd's algorithm can only decrease (or keep) J , which is why it converges.

Q.25 — Answer: 1.2086 to 1.2486 [NAT, 2M · Regression & Model Selection › Basis functions & feature scaling]

Design columns 1, x , x^2 . Because the x -values are symmetric about 0, $\sum x = \sum x^3 = 0$, so the normal equations decouple:

$$\sum x^2 = 10, \sum x^4 = 34, \sum y = -11, \sum xy = 16, \sum x^2y = -46.$$

$$b = \sum xy / \sum x^2 = 16/10 = 1.6.$$

$$[n, \sum x^2, \sum x^4][a; c] = [\sum y; \sum x^2y] = 5a + 10c = -11, 10a + 34c = -46.$$

$$\text{Solving: } c = (n\sum x^2y - \sum x^2\sum y) / (n\sum x^4 - (\sum x^2)^2) = -1.7143, a = (\sum y - c\sum x^2) / n = 1.2286.$$

Answer: **1.23**.

Q.26 — Answer: (A), (B), (C), (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

(A) TRUE. Tiny steps.

(B) TRUE. $-\log N(y; \hat{y}, \sigma^2) \propto (y - \hat{y})^2$.

(C) TRUE. CE = negative log-likelihood of a categorical/Bernoulli model.

(D) TRUE. Geometric series $1 + \beta + \beta^2 + \dots$

Q.27 — Answer: 0.49 to 0.51 [NAT, 2M · Probabilistic & Instance-based Classifiers › Linear discriminant analysis]

$$\mu_1 - \mu_0 = (-1, 0)^T; \det S_W = 12.$$

$$S_W^{-1} = \begin{bmatrix} 0.3333 & 0.1667 \\ 0.1667 & 0.3333 \end{bmatrix}$$

$$\mathbf{w} = S_W^{-1}(\mu_1 - \mu_0) = (-0.3333, -0.1667)^T.$$

$$w_2/w_1 = -0.1667/-0.3333 = \mathbf{0.5}. \text{ (The direction, not the length, matters.)}$$

Note: using $\mu_1 - \mu_0$ without S_W^{-1} is correct only when $S_W \propto I$.

Q.28 — Answer: 7 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

Total pairs = $C(8,2) = 28$.

a = pairs together in both = 7; b = pairs separated in both = 16; pairs together in U only = 1; together in V only = 4.

Rand index = $(a + b)/C(n,2) = (7 + 16)/28 = 0.8214$.

Answer: 7. (Cluster names do not matter: only whether pairs are together.)

Q.29 — Answer: (A), (D) [MSQ, 2M · Unsupervised Learning > PCA & dimensionality reduction]

(A) TRUE. All eigenvalues scale by c^2 .

(B) FALSE. Discarded low-variance directions may be the discriminative ones.

(C) FALSE. Scores are $U\Sigma = XV$.

(D) TRUE. Neighbourhoods stop being local.

Q.30 — Answer: (D) [MCQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Distances from **q**:

Point	class	Euclidean d	Manhattan d
P1	A	9.434	13
P2	B	4.123	5
P3	B	11.314	16
P4	B	8.062	9
P5	A	5.099	6
P6	B	7.071	8
P7	A	6.403	9

Euclidean 3 nearest: P2(B), P5(A), P7(A) = **A**.

Manhattan 3 nearest: P2(B), P5(A), P6(B) = **B**.

Answer: **Euclidean: A; Manhattan: B**. The two metrics disagree because L_1 penalises diagonal offsets more than L_2 , changing the neighbour set.

Q.31 — Answer: (A) [MCQ, 2M · Regression & Model Selection > Simple linear regression]

$\bar{x} = 7.1667$, $\bar{y} = -11$.

$S_{xx} = \Sigma x^2 - (\Sigma x)^2/n = 38.8333$, $S_{yy} = \Sigma y^2 - (\Sigma y)^2/n = 78$, $S_{xy} = \Sigma xy - \Sigma x \Sigma y/n = -36$.

Regress x on y: $b' = S_{xy}/S_{yy} = -0.4615$; $a' = \bar{x} - b'\bar{y} = 2.0897$.

Line: $\hat{x} = 2.09 - 0.46y$.

A common error is to invert the y-on-x line (slope $1/b$), which minimises the wrong (vertical) errors.

Q.32 — Answer: (C) [MCQ, 2M · Unsupervised Learning > Distance measures]

Compute the measure from **q** = (1, 3) to every point (cosine column shows similarity, larger = more similar):

Point	coordinates	Manhattan	Euclidean	Chebyshev	cosine similarity
A	(5, 2)	5	4.123	4	0.646
B	(8, 2)	8	7.071	7	0.537
C	(5, 8)	9	6.403	5	0.972
D	(4, 0)	6	4.243	3	0.316

Under cosine similarity the most similar to **q** is **point C**.

Note how the answer depends on the measure: Manhattan (L_1) distance → A; Euclidean (L_2) distance → A; Chebyshev (L_∞) distance → D; cosine similarity → C. Cosine similarity looks only at the angle from the origin, L_1 sums coordinate gaps, L_∞ takes the largest gap.

Q.33 — Answer: 1.847 to 1.867 [NAT, 2M · SVM & Decision Trees > SVM: margin & geometry]

P1 and P5 (same class) lie on one margin line and P8 on the other; the optimal boundary is parallel to the line through P1, P5, midway to P8. Support vectors: P1, P5, P8.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (1, -2/5)$, $b = -13/5$; $\|w\| = 1.077$, margin $2/\|w\| = 1.857$.

Point	P1	P2	P3	P4	P5	P6	P7	P8
$y(w^T x + b)$	1	2.6	2.2	1.2	1	1.4	2.6	1

All values are ≥ 1 (feasible), with equality exactly at the support vectors.

Margin width = $2/\|w\| = \mathbf{1.86}$.

Q.34 — Answer: 0.1326 to 0.1366 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Backpropagation]

$a = w \cdot x + b = -1.25$; $\hat{y} = \sigma(a) = 0.2227$.

$L = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial L / \partial \hat{y} = \hat{y} - y = -0.7773$; $\partial \hat{y} / \partial a = \hat{y}(1 - \hat{y}) = 0.1731$.

$\delta = \partial L / \partial a = (\hat{y} - y)\hat{y}(1 - \hat{y}) = -0.1346$ ($= \partial L / \partial b$).

$\partial L / \partial w_1 = \delta \cdot x_1 = -0.1346 \times -1 = 0.1346$.

Answer: **0.1346**.

Forgetting the factor $\hat{y}(1 - \hat{y})$ (i.e. treating the output as linear) is the most common error.

Q.35 — Answer: 72.49 to 72.51 [NAT, 2M · SVM & Decision Trees › Decision trees: split selection]

Sort the values: 50(N), 53(Y), 54(Y), 55(N), 70(N), 72(N), 73(Y), 75(Y), 86(Y).

Parent [5 Yes, 4 No], $H = 0.9911$. The optimal threshold always lies where the class changes between neighbours (Fayyad-Irani), giving 3 candidates:

Threshold	Left ($\leq t$) [Yes, No]	Right [Yes, No]	IG
51.5	[0, 1]	[5, 3]	0.1427
54.5	[2, 1]	[3, 3]	0.0183
72.5	[2, 4]	[3, 0]	0.3789

Best: $t = 72.5$ with $IG = 0.3789$. Answer: **72.5**.

Mock Test 03

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	3	11	21

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
SVM & Decision Trees	7	11
Unsupervised Learning	7	11
Neural Networks (MLP / Feed-forward)	7	11
Probabilistic & Instance-based Classifiers	7	11
Regression & Model Selection	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

A node of a classification tree contains 15 training records: 7 of class C_1 , 5 of class C_2 , 3 of class C_3 . The entropy (in bits) of this node is _____ (rounded off to two decimal places)

Q.2 [1 mark · NAT (numerical answer)]

A clustering algorithm groups labelled data into three clusters; the contingency table of cluster versus true class counts is given below. If every point in a cluster is labelled with the cluster's majority class, the number of misclassified points is _____ (Answer in integer)

	Class P	Class Q	Class R
Cluster 1	10	0	19
Cluster 2	8	16	2
Cluster 3	25	9	3

Q.3 [1 mark · MSQ (one or more correct)]

Which of the following statements about the perceptron is/are CORRECT?

- (A) The decision boundary of a perceptron, $w \cdot x + b = 0$, is a hyperplane.
- (B) The perceptron is not trained by gradient descent on the 0-1 loss, because that loss has zero gradient almost everywhere.
- (C) XOR becomes linearly separable if the product feature $x_1 x_2$ is added as an extra input.
- (D) The perceptron mistake bound grows linearly with the input dimension d .

Q.4 [1 mark · NAT (numerical answer)]

The perpendicular distance of the point $x = (-1, 3, -2)$ from the hyperplane $2x_1 + 3x_2 + 6x_3 - 4 = 0$ is _____ (rounded off to two decimal places)

Q.5 [1 mark · MSQ (one or more correct)]

Which of the following statements about Bayes-optimal classification and generative vs. discriminative models is/are CORRECT?

- (A) Increasing the prior of class 1 shrinks the region assigned to class 1 by the Bayes rule.
- (B) A sufficiently flexible classifier can achieve expected test error strictly below the Bayes error.
- (C) Adding a new informative feature can reduce the Bayes error rate.
- (D) If the class-conditional densities have non-overlapping supports, the Bayes error rate is zero.

Q.6 [1 mark · NAT (numerical answer)]

In standard (non-inverted) dropout, each input h_i of a unit is kept with probability $p = 0.75$ during training (and not rescaled). For $h = (3, 3, 0.5)$ and weights $w = (-0.5, 1, -0.5)$, the expected value of the unit's pre-activation $\sum w_i m_i h_i$ ($m_i \sim \text{Bernoulli}(p)$) during training is _____ (rounded off to two decimal places)

Q.7 [1 mark · NAT (numerical answer)]

How many distinct monomials of total degree **exactly** 3 can be formed from 2 variables (e.g. for $p = 2, d = 2$: $x_1^2, x_1 x_2, x_2^2$)? _____ (Answer in integer)

Q.8 [1 mark · MSQ (one or more correct)]

Which of the following statements about the representational power of feed-forward neural networks is/are CORRECT?

- (A) The universal approximation theorem also holds with ReLU activations.
- (B) The universal approximation theorem fails for sigmoid activations; it holds only for ReLU.
- (C) Every Boolean function of n binary inputs can be represented by an MLP with a single hidden layer of threshold units (possibly with exponentially many hidden units).
- (D) A single sigmoid neuron (logistic regression) has a linear decision boundary.

Q.9 [1 mark · MSQ (one or more correct)]

Which of the following statements about principal component analysis (PCA) is/are CORRECT?

- (A) When features are in different units, PCA is usually applied to standardized features (correlation matrix).
- (B) The first PC is always the best single direction for classification.
- (C) The proportion of variance explained by the first k PCs is $(\lambda_1 + \dots + \lambda_k)/(\lambda_1 + \dots + \lambda_d)$.
- (D) LDA, unlike PCA, uses class labels to find discriminative directions.

Q.10 [1 mark · NAT (numerical answer)]

Data are uniformly distributed in the unit hypercube $[0, 1]^5$. To capture a fraction 0.05 of the data, a k -NN method uses a sub-cube neighbourhood. The required edge length of this sub-cube is _____ (rounded off to two decimal places)

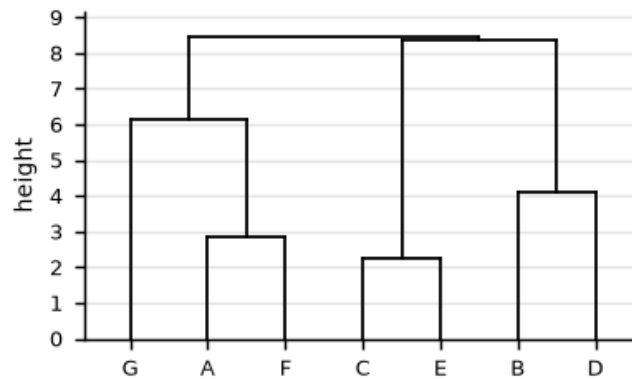
Q.11 [1 mark · MSQ (one or more correct)]

Which of the following statements about logistic regression is/are CORRECT?

- (A) Changing the probability threshold from 0.5 to 0.8 moves the logistic-regression decision boundary to a parallel hyperplane.
- (B) The average cross-entropy loss of logistic regression is a convex function of (w, b) .
- (C) Logistic regression cannot be used when some input features are categorical.
- (D) If the training data are linearly separable, the norm of the unregularised logistic-regression weights grows without bound during gradient descent.

Q.12 [1 mark · NAT (numerical answer)]

The dendrogram below was obtained by agglomerative clustering of 7 points A-G. If it is cut to obtain exactly 3 clusters, the number of points in the cluster that contains point C is _____
(Answer in integer)

**Q.13 [1 mark · MSQ (one or more correct)]**

Linear regression is trained by minimising the squared-error loss. Which of the following statements about gradient-based training is/are CORRECT?

- (A) Stochastic gradient descent updates the parameters using the gradient on one (or a few) randomly chosen example(s).
- (B) Standardising features usually speeds up gradient descent for linear regression.
- (C) With a sufficiently small constant learning rate, batch gradient descent on the least-squares loss converges to a global minimiser.
- (D) Batch gradient descent performs one parameter update per training example in each epoch.

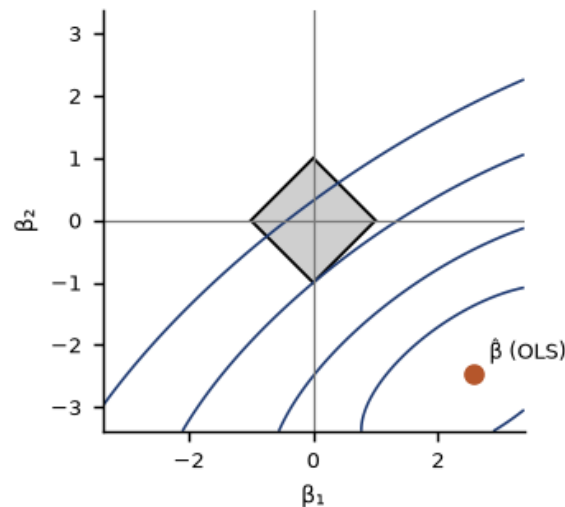
Q.14 [1 mark · MSQ (one or more correct)]

Consider the soft-margin SVM: minimise $\frac{1}{2}\|w\|^2 + C \sum_i \xi_i$ subject to $y_i(w^T x_i + b) \geq 1 - \xi_i$, $\xi_i \geq 0$. Which of the following statements is/are CORRECT?

- (A) A point with $\xi_i > 1$ is misclassified.
- (B) A point with $\xi_i = 0.5$ is misclassified.
- (C) A small C allows more margin violations and typically yields a wider margin.
- (D) $\sum \xi_i$ equals the number of misclassified training points.

Q.15 [1 mark · MCQ]

For a two-predictor penalised least-squares problem, the figure shows the constraint region (shaded) of the penalty together with elliptical contours of the residual sum of squares centred at the least-squares estimate $\hat{\beta}$ (OLS). The penalised estimate $(\hat{\beta}_1, \hat{\beta}_2)$ satisfies



- (A) $\hat{\beta}_1 = 0$ exactly and $\hat{\beta}_2 \neq 0$
- (B) $\hat{\beta}_1 \neq 0$ and $\hat{\beta}_2 \neq 0$ (both shrunk, neither exactly zero)
- (C) $\hat{\beta}_2 = 0$ exactly and $\hat{\beta}_1 \neq 0$
- (D) $\hat{\beta}_1 = \hat{\beta}_2 = 0$

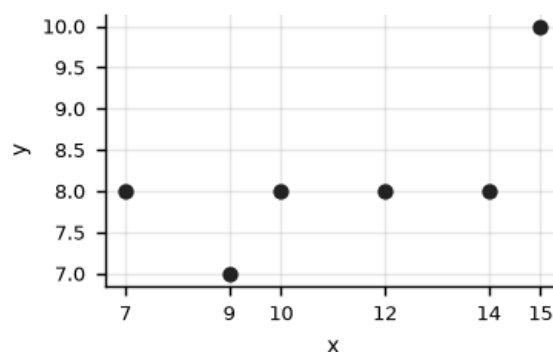
Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

A centered data matrix X with $n = 26$ rows (samples) and $d = 4$ columns has singular values 21, 18, 12, 4. Using the sample covariance with divisor $n - 1$, the minimum number of principal components needed to retain at least 80% of the variance is ____ (Answer in integer)

Q.17 [2 marks · NAT (numerical answer)]

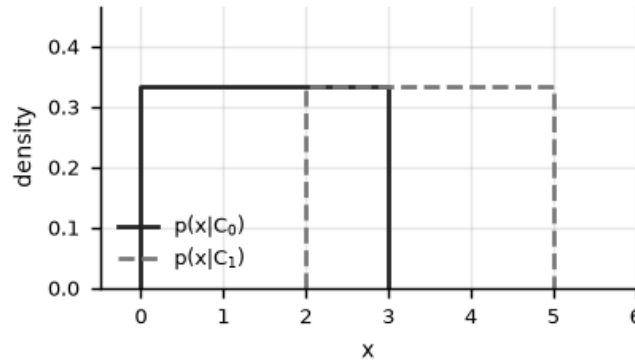
A regression tree (squared-error criterion) is fitted to the 6 points below. The root split " $x \leq t$ " is chosen among midpoints of consecutive x values to minimise the total SSE of the two children (each child predicts its mean). Find the prediction of the resulting depth-1 regression tree (stump) at $x = 5$. (rounded off to two decimal places)

x	7	9	10	12	14	15
y	8	7	8	8	8	10



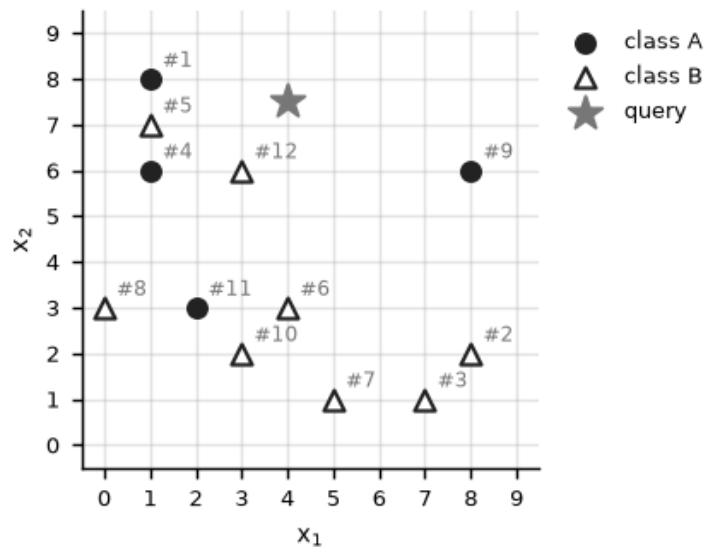
Q.18 [2 marks · NAT (numerical answer)]

For a binary problem with scalar x , $p(x | C_0)$ is uniform on $[0, 3]$ and $p(x | C_1)$ is uniform on $[2, 5]$, with priors $P(C_0) = 0.3$ and $P(C_1) = 0.7$. Then the Bayes (minimum achievable) error rate is _____ (rounded off to three decimal places)

**Q.19 [2 marks · MSQ (one or more correct)]**

The figure and table show 12 labelled training points (class A: filled circles, class B: open triangles) and a query point (star) at $(4, 7.5)$. A k -NN classifier with **Euclidean** distance and majority vote is used. Which of the following statements is/are CORRECT?

#	1	2	3	4	5	6	7	8	9	10	11	12
x_1	1	8	7	1	1	4	5	0	8	3	2	3
x_2	8	2	1	6	7	3	1	3	6	2	3	6
class	A	B	B	A	B	B	B	B	A	B	A	B



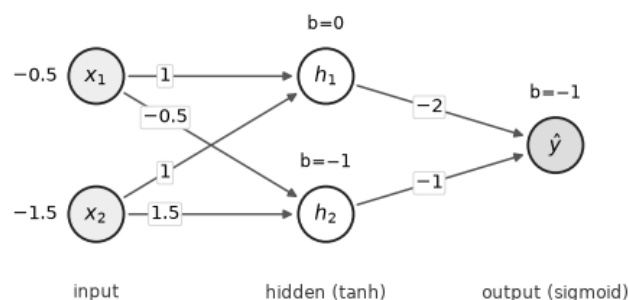
- (A) With $k = 7$, the query is classified as class B.
- (B) With $k = 3$, the query is classified as class B.
- (C) With $k = 1$, the query is classified as class A.
- (D) With $k = 5$, the query is classified as class A.

Q.20 [2 marks · NAT (numerical answer)]

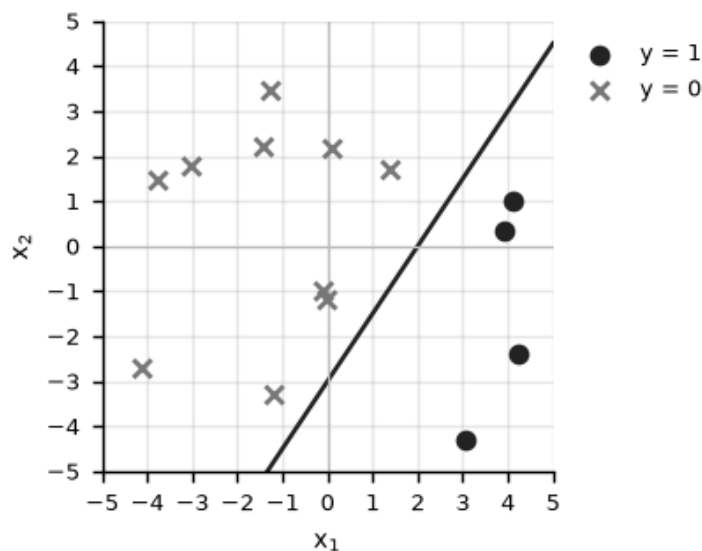
A 4-class classifier outputs logits $z = (1, -0.5, -0.5, 0.5)$ and probabilities $p = \text{softmax}(z)$. The true class is 1, and the loss is the cross-entropy $L = -\ln p_1$. Then the value of z_3 after one gradient step on the logits with $\eta = 0.5$ is _____ (rounded off to three decimal places)

Q.21 [2 marks · NAT (numerical answer)]

The feed-forward network below has tanh hidden units and a sigmoid output unit. The number on each edge is its weight and 'b' above a node is that node's bias. For input $\mathbf{x} = (-0.5, -1.5)$ (values shown left of the inputs), the network output $\hat{y}$ is _____ (rounded off to three decimal places)

**Q.22 [2 marks · MCQ]**

A logistic regression model $P(y = 1 | \mathbf{x}) = \sigma(w_0 + w_1x_1 + w_2x_2)$ perfectly classifies the training points shown below using threshold 0.5. Its decision boundary is the solid line (it crosses the axes at grid points). Which of the following parameter vectors is consistent with the figure?



- (A) $(w_0, w_1, w_2) = (-12, -4, 6)$
 (B) $(w_0, w_1, w_2) = (12, -4, 6)$
 (C) $(w_0, w_1, w_2) = (-12, 4, -6)$
 (D) $(w_0, w_1, w_2) = (-12, 6, -4)$

Q.23 [2 marks · NAT (numerical answer)]

Three customers are described by age and income. Age ranges over [20, 60] years and income over [10, 210] thousand rupees in the full dataset. After min-max scaling of each feature to [0, 1] (using the stated ranges), the Euclidean distance from A to its **nearest** neighbour among B and C is _____ (rounded off to two decimal places)

Customer	Age (years)	Income (₹ thousand)
A	46	100
B	20	55
C	41	195

Q.24 [2 marks · MSQ (one or more correct)]

Each option shows a 2×2 matrix $[[K(x_1, x_1), K(x_1, x_2)], [K(x_2, x_1), K(x_2, x_2)]]$. Which of them can be the Gram matrix of some valid (Mercer) kernel K on two points x_1, x_2 ?

- (A) $[[7, 0], [0, 8]]$
 (B) $[[9, 3], [3, 3]]$
 (C) $[[2, 0], [0, 2]]$
 (D) $[[4, 0], [1, 8]]$

Q.25 [2 marks · MSQ (one or more correct)]

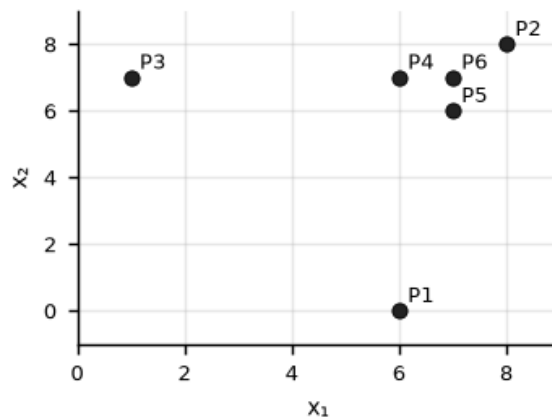
A threshold unit outputs 1 if $w_1 x_1 + w_2 x_2 + b > 0$ and 0 otherwise, for binary inputs $x_1, x_2 \in \{0, 1\}$. Which of the following parameter settings make the unit implement the Boolean **AND** function?

- (A) $w_1 = 1.5, w_2 = 0.5, b = 3$
 (B) $w_1 = 1.5, w_2 = 1, b = -2$
 (C) $w_1 = -1, w_2 = 0, b = -2.5$
 (D) $w_1 = -2.5, w_2 = -2.5, b = 2$

Q.26 [2 marks · NAT (numerical answer)]

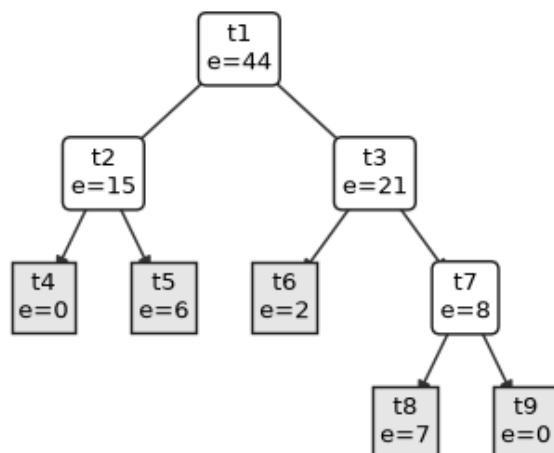
Single-linkage agglomerative clustering with Euclidean distance is applied to the six points below. Then the height at which the **last** merge (forming a single cluster) occurs is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5	P6
x_1	6	8	1	6	7	7
x_2	0	8	7	7	6	7

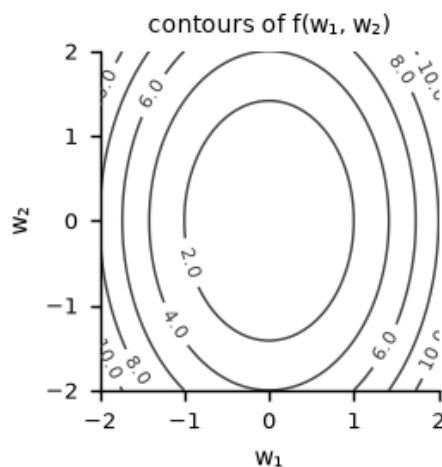


Q.27 [2 marks · NAT (numerical answer)]

A classification tree was grown on $N = 100$ training records. Each node shows $e(t)$, the number of training records that would be misclassified if t were a leaf; $R(t) = e(t)/N$. For $\alpha = 0.02$, the cost-complexity $R_\alpha(T) = R(T) + \alpha|T|$ of the full tree T is _____ (rounded off to three decimal places)

**Q.28 [2 marks · MSQ (one or more correct)]**

Gradient descent $w \leftarrow w - \eta \nabla f(w)$ is applied to $f(w_1, w_2) = \frac{1}{2}(4w_1^2 + 2w_2^2)$ from a generic starting point (both coordinates non-zero). For which of the following learning rates does GD **diverge**?

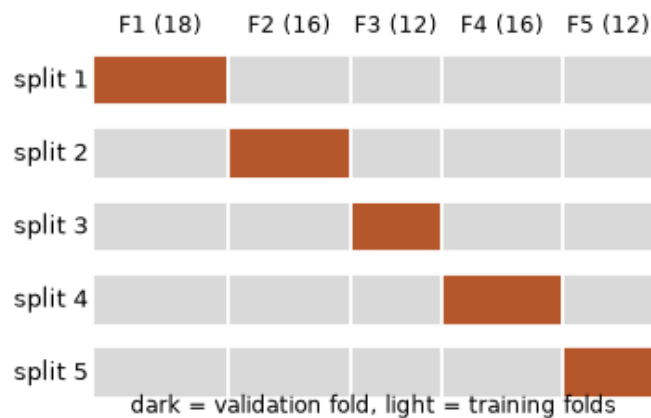


- (A) $\eta = 0.3$
(C) $\eta = 0.125$

- (B) $\eta = 0.75$
(D) $\eta = 0.475$

Q.29 [2 marks · NAT (numerical answer)]

A regression model is evaluated by 5-fold cross-validation on $n = 74$ observations. The folds (F1 ... F5) have unequal sizes, as shown in the diagram and table. The k-fold CV estimate is defined as the mean squared error over all n out-of-fold predictions (i.e. pooled over observations). Its value is _____ (rounded off to two decimal places)

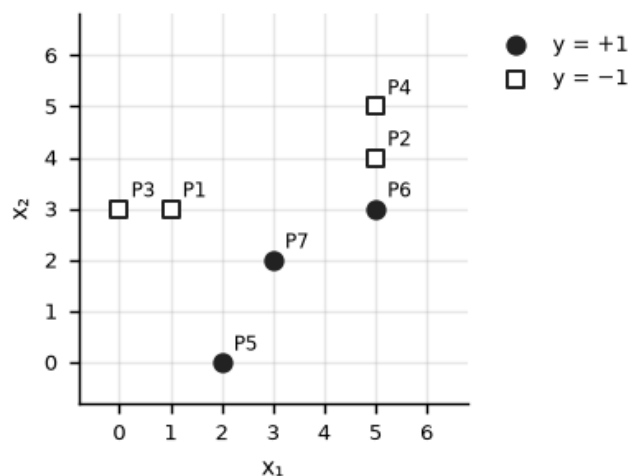


Fold	F1	F2	F3	F4	F5
size	18	16	12	16	12
validation MSE	2.2	9	5.9	8.1	7.9

Q.30 [2 marks · MCQ]

Four candidate linear decision boundaries are proposed for the labelled training points shown. Which boundary separates the two classes with the LARGEST geometric margin (minimum distance from the boundary to any training point, all points correctly classified)?

Point	x_1	x_2	y
P1	1	3	-1
P2	5	4	-1
P3	0	3	-1
P4	5	5	-1
P5	2	0	+1
P6	5	3	+1
P7	3	2	+1

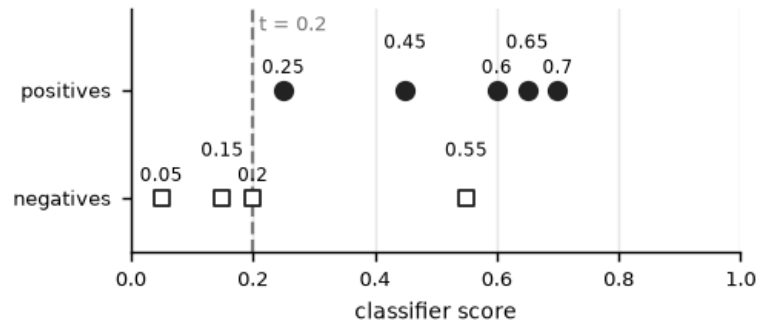


- (A) $x_1 - 2x_2 + 1.5 = 0$
 (B) $x_1 - 2x_2 + 2.5 = 0$
 (C) $x_1 - 2x_2 + 2 = 0$
 (D) $2x_1 - x_2 + 2.5 = 0$

Q.31 [2 marks · NAT (numerical answer)]

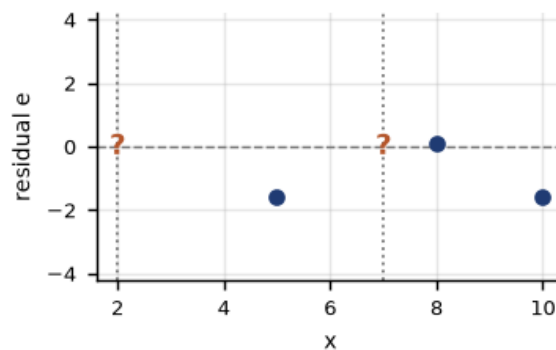
A probabilistic classifier produces the scores below on 9 labelled examples (1 = positive). An example is predicted positive if its score is $\geq t$, with $t = 0.2$. The false positive rate at this threshold is _____ (rounded off to two decimal places)

example	1	2	3	4	5	6	7	8	9
score	0.7	0.65	0.6	0.55	0.45	0.15	0.05	0.25	0.2
label	1	1	1	0	1	0	0	1	0

**Q.32 [2 marks · NAT (numerical answer)]**

A simple linear regression with intercept is fitted by least squares. The table gives the x_i -values and the residuals $e_i = y_i - \hat{y}_i$, but two residuals are missing. The fitted line is $\hat{y} = 1.8 - 1.9x$. The observed value y at $x = 7$ is _____ (rounded off to two decimal places)

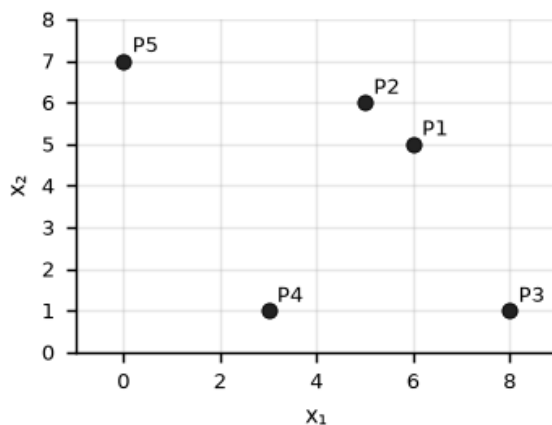
x_i	2	5	7	8	10
e_i	?	-1.6	?	0.1	-1.6



Q.33 [2 marks · NAT (numerical answer)]

A cluster contains the 5 points below. Distances are Euclidean (L_2). The medoid is the point that minimizes the sum of distances to all points of the cluster. The minimum total distance (cost of the medoid) is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5
x_1	6	5	8	3	0
x_2	5	6	1	1	7

**Q.34 [2 marks · NAT (numerical answer)]**

A regression with intercept and $p = 4$ predictors on $n = 23$ observations has $R^2 = 0.69$. Its adjusted R^2 is _____ (rounded off to three decimal places)

Q.35 [2 marks · NAT (numerical answer)]

Consider the model $y = wx + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$, $\sigma^2 = 2$, and $w \sim N(0, \tau^2)$ with $\tau^2 = 0.5$. Given the four observations below, the MAP estimate of w is _____ (rounded off to three decimal places)

x	-1	2	1	4
y	2	0	-3	-1

— END OF QUESTION PAPER —

Mock Test 03 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	1.4958 to 1.5158	19	MSQ	2	(B), (D)
2	NAT	1	32	20	NAT	2	-0.5564 to -0.5524
3	MSQ	1	(A), (B), (C)	21	NAT	2	0.8705 to 0.8745
4	NAT	1	1.2757 to 1.2957	22	MCQ	2	(D)
5	MSQ	1	(C), (D)	23	NAT	2	0.4812 to 0.5012
6	NAT	1	0.9275 to 0.9475	24	MSQ	2	(A), (B), (C)
7	NAT	1	4	25	MSQ	2	(B)
8	MSQ	1	(A), (C), (D)	26	NAT	2	6.0728 to 6.0928
9	MSQ	1	(A), (C), (D)	27	NAT	2	0.2485 to 0.2515
10	NAT	1	0.5393 to 0.5593	28	MSQ	2	(B)
11	MSQ	1	(A), (B), (D)	29	NAT	2	6.4603 to 6.4803
12	NAT	1	2	30	MCQ	2	(C)
13	MSQ	1	(A), (B), (C)	31	NAT	2	0.49 to 0.51
14	MSQ	1	(A), (C)	32	NAT	2	-8.1405 to -8.0595
15	MCQ	1	(A)	33	NAT	2	17.2009 to 17.2209
16	NAT	2	2	34	NAT	2	0.6191 to 0.6231

Q	Type	M	Key	Q	Type	M	Key
17	NAT	2	7.79 to 7.81	35	NAT	2	−0.3482 to −0.3442
18	NAT	2	0.098 to 0.102				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
SVM & Decision Trees					11
Unsupervised Learning					11
Neural Networks (MLP / Feed-forward)					11
Probabilistic & Instance-based Classifiers					11
Regression & Model Selection					11
Total					55

Mock Test 03 — Detailed Solutions

Q.1 — Answer: 1.4958 to 1.5158 [NAT, 1M · SVM & Decision Trees › Decision trees: impurity measures]

$H = -\sum p_k \log_2 p_k = -(7/15)\log_2(7/15) - (5/15)\log_2(5/15) - (3/15)\log_2(3/15) = \mathbf{1.506}$ bits.
(Maximum possible with 3 classes: $\log_2 3 = 1.585$.)

Q.2 — Answer: 32 [NAT, 1M · Unsupervised Learning › Cluster evaluation]

Majority count in each cluster: 19, 16, 25 (sum 60); $N = 92$.

Purity = $(1/N) \sum_k \max_j n_{kj} = 60/92 = 0.6522$.

Misclassified = $N - \sum \max = 32$.

Answer: **32**.

Purity is an external measure (needs class labels) and is trivially 1 if each point forms its own cluster, so it cannot be used to choose k .

Q.3 — Answer: (A), (B), (C) [MSQ, 1M · Neural Networks (MLP / Feed-forward) › Perceptron]

(A) **TRUE**. It is an affine set of dimension $d-1$.

(B) **TRUE**. The 0-1 loss is piecewise constant in w .

(C) **TRUE**. $XOR = x_1 + x_2 - 2x_1x_2$ is linear in (x_1, x_2, x_1x_2) .

(D) **FALSE**. The bound $(R/\gamma)^2$ has no explicit dependence on d .

Q.4 — Answer: 1.2757 to 1.2957 [NAT, 1M · SVM & Decision Trees › SVM: margin & geometry]

Distance = $|w^T x + b| / \|w\|$, with $w = (2, 3, 6)$, $b = -4$.

$w^T x + b = -9$, $\|w\| = 7$.

Distance = $9/7 = \mathbf{1.29}$.

Q.5 — Answer: (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers › Bayes optimal classifier]

(A) **FALSE**. It enlarges it.

(B) **FALSE**. The Bayes error is a lower bound.

(C) **TRUE**. The classes may overlap less in the larger space.

(D) **TRUE**. Every x can be assigned to its only possible class.

Q.6 — Answer: 0.9275 to 0.9475 [NAT, 1M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

$E[m_i] = p \Rightarrow E[\sum w_i m_i h_i] = p \sum w_i h_i = 0.75 \times 1.25 = \mathbf{0.9375}$.

This is why, at test time, standard dropout scales the weights by p (inverted dropout instead scales by $1/p$ during training).

Q.7 — Answer: 4 [NAT, 1M · Regression & Model Selection › Basis functions & feature scaling]

Multisets of size d from p variables: $C(p + d - 1, d) = C(4, 3) = \mathbf{4}$.

Q.8 — Answer: (A), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) › Expressiveness]

(A) **TRUE**. ReLU is non-polynomial, so the Leshno et al. theorem applies.

(B) **FALSE**. The original result (Cybenko) was proved for sigmoids.

(C) **TRUE**. One hidden unit per positive input pattern (DNF).

(D) **TRUE**. $\sigma(w \cdot x + b) = 0.5 \Leftrightarrow w \cdot x + b = 0$.

Q.9 — Answer: (A), (C), (D) [MSQ, 1M · Unsupervised Learning › PCA & dimensionality reduction]

(A) **TRUE**. Otherwise large-scale features dominate.

(B) **FALSE**. High variance $\neq$ discriminative.

(C) **TRUE**. Eigenvalues are variances along PCs.

(D) **TRUE**. LDA is supervised.

Q.10 — Answer: 0.5393 to 0.5593 [NAT, 1M · Probabilistic & Instance-based Classifiers › k-nearest neighbours]

Volume of a cube with edge e is e^5 ; set $e^5 = 0.05$.

$$e = 0.05^{1/5} = \mathbf{0.55}.$$

In high dimensions even a small fraction of data needs an edge spanning most of each coordinate's range, so 'nearest' neighbours are not local — the curse of dimensionality.

Q.11 — Answer: (A), (B), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers · Logistic regression]

(A) **TRUE**. The boundary becomes $w^T x + b = \ln 4$, same normal vector w .

(B) **TRUE**. Its Hessian $X^T S X$ with $S = \text{diag}(p_i(1-p_i))$ is positive semidefinite.

(C) **FALSE**. Categorical features are simply one-hot encoded.

(D) **TRUE**. Scaling a separating w up always lowers the loss, so the MLE does not exist (it diverges).

Q.12 — Answer: 2 [NAT, 1M · Unsupervised Learning · Hierarchical clustering]

Undo the top 2 merge(s) (equivalently cut just below height 8.34). Clusters: $\{A, F, G\}$, $\{B, D\}$, $\{C, E\}$.

C lies in a cluster of size 2.

Q.13 — Answer: (A), (B), (C) [MSQ, 1M · Regression & Model Selection · Gradient descent]

(A) **TRUE**. Definition of SGD / mini-batch GD.

(B) **TRUE**. It reduces the condition number of $X^T X$, making contours rounder.

(C) **TRUE**. Convex quadratic with Lipschitz gradient; $\eta < 2/\lambda_{\max}(\text{Hessian})$ suffices.

(D) **FALSE**. Batch GD makes one update per pass over the data.

Q.14 — Answer: (A), (C) [MSQ, 1M · SVM & Decision Trees · SVM: soft margin & hinge loss]

(A) **TRUE**. $y f = 1 - \xi < 0$.

(B) **FALSE**. It has $y f = 0.5 > 0$ — correct but inside the margin.

(C) **TRUE**. Small $C \Rightarrow$ stronger regularisation.

(D) **FALSE**. It is only an upper bound.

Q.15 — Answer: (A) [MCQ, 1M · Regression & Model Selection · Ridge & Lasso]

The shaded region is the constraint set of the lasso ($L1$: $|\beta_1| + |\beta_2| \leq t$, a diamond). The penalised estimate is the first point at which the RSS contours (ellipses centred at β_{OLS} , which lies outside the region) touch the region.

From the figure the tangency point is approximately $(0, -1)$.

The contour first touches the diamond at a corner on the β_2 -axis, so $\beta_1 = 0$ exactly — this is why the lasso performs variable selection.

Answer: $\beta_1 = 0$ exactly and $\beta_2 \neq 0$. Both coefficients cannot be zero since the OLS point lies outside the region (the constraint is active but $t > 0$).

Q.16 — Answer: 2 [NAT, 2M · Unsupervised Learning · PCA & dimensionality reduction]

With $X = U \Sigma V^T$ (X centered, $n \times d$), the sample covariance is $S = X^T X / (n - 1) = V(\Sigma^2 / (n - 1))V^T$, so the PC directions are the columns of V and the eigenvalues are $\lambda_j = \sigma_j^2 / (n - 1)$.

$$\lambda = 21^2/25 = 17.64, 18^2/25 = 12.96, 12^2/25 = 5.76, 4^2/25 = 0.64.$$

Cumulative % variance: 47.68, 82.7, 98.27, 100 (note the $(n - 1)$ factor cancels in proportions: $\text{PVE} = \sigma_j^2 / \Sigma \sigma^2$).

Answer: 2. (Using σ instead of σ^2 is the classic mistake.)

Q.17 — Answer: 7.79 to 7.81 [NAT, 2M · SVM & Decision Trees · Regression trees]

Parent mean $\bar{y} = 8.167$, $\text{SSE}_0 = \Sigma(y - \bar{y})^2 = 4.833$.

Evaluate every split:

Split t	Left mean	Right mean	$\text{SSE}_L + \text{SSE}_R$
8	8	8.2	4.8
9.5	7.5	8.5	3.5
11	7.667	8.667	3.333
13	7.75	9	2.75

14.5	7.8	10	0.8
------	-----	----	-----

Best split $t = 14.5$: left mean 7.8, right mean 10, SSE = 0.8, reduction = 4.033.

Answer: **7.8**.

Q.18 — Answer: 0.098 to 0.102 [NAT, 2M · Probabilistic & Instance-based Classifiers · Bayes optimal classifier]

$\pi_0 p(x|C_0) = 0.3/3 = 0.1$ on $[0, 3]$; $\pi_1 p(x|C_1) = 0.7/3 = 0.2333$ on $[2, 5]$.

Outside the overlap $[2, 3]$ only one class has non-zero density $\Rightarrow$ no error. Inside it the Bayes rule chooses C_1 (larger weighted density) and errs with density 0.1.

Bayes error = (overlap length) $\times$ min = $1 \times 0.1 = 0.1$.

Q.19 — Answer: (B), (D) [MSQ, 2M · Probabilistic & Instance-based Classifiers · k-nearest neighbours]

Distances from the query, sorted:

rank	#	(x_1, x_2)	distance	class
1	12	(3, 6)	$\sqrt{3.25} = 1.803$	B
2	1	(1, 8)	$\sqrt{9.25} = 3.041$	A
3	5	(1, 7)	$\sqrt{9.25} = 3.041$	B
4	4	(1, 6)	$\sqrt{11.25} = 3.354$	A
5	9	(8, 6)	$\sqrt{18.25} = 4.272$	A
6	6	(4, 3)	$\sqrt{20.25} = 4.5$	B
7	11	(2, 3)	$\sqrt{24.25} = 4.924$	A
8	10	(3, 2)	$\sqrt{31.25} = 5.59$	B

(A) FALSE. 7 nearest: #12, #1, #5, #4, #9, #6, #11 $\rightarrow$ 4 A vs 3 B $\Rightarrow$ class A.

(B) TRUE. 3 nearest: #12, #1, #5 $\rightarrow$ 1 A vs 2 B $\Rightarrow$ class B.

(C) FALSE. 1 nearest: #12 $\rightarrow$ 0 A vs 1 B $\Rightarrow$ class B.

(D) TRUE. 5 nearest: #12, #1, #5, #4, #9 $\rightarrow$ 3 A vs 2 B $\Rightarrow$ class A.

Q.20 — Answer: -0.5564 to -0.5524 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Softmax + cross-entropy]

$p = \text{softmax}(z)$: $\exp(z) = (2.7183, 0.6065, 0.6065, 1.6487)$, sum = 5.5801, so $p = (0.4871, 0.1087, 0.1087, 0.2955)$.

For $L = -\ln p_y$ with softmax, $\partial L / \partial z = p - y$ (y one-hot, true class 1): $\partial L / \partial z = (-0.5129, 0.1087, 0.1087, 0.2955)$.

$z_3 \leftarrow -0.5 - 0.5 \times (0.1087) = -0.5543$.

Note $\sum_k \partial L / \partial z_k = \sum p_k - 1 = 0$.

Q.21 — Answer: 0.8705 to 0.8745 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Forward pass]

Hidden pre-activations: $a_1 = 1 \cdot -0.5 + (1) \cdot -1.5 + (0) = -2$; $a_2 = -0.5 \cdot -0.5 + (1.5) \cdot -1.5 + (-1) = -3$.

Hidden activations ($\tanh(a)$): $h_1 = -0.964$, $h_2 = -0.9951$.

Output pre-activation: $o = -2 \cdot -0.964 + (-1) \cdot -0.9951 + (-1) = 1.9231$.

$\hat{y} = \sigma(o) = 0.872$.

Q.22 — Answer: (D) [MCQ, 2M · Probabilistic & Instance-based Classifiers · Logistic regression]

The line passes through (2, 0) and (0, -3): $x_1/2 + x_2/-3 = 1$, i.e. $-3x_1 + 2x_2 - 6 = 0$. Hence (w_0, w_1, w_2) must be a non-zero multiple of (6, -3, 2).

Orientation: the class $y = 1$ region must have $w_0 + w_1 x_1 + w_2 x_2 > 0$. At the origin the $y = 1$ side has sign $-(w_0 = -12)$, which matches the figure (the origin lies on the $y = 0$ side).

So the answer is $(w_0, w_1, w_2) = (-12, 6, -4)$ (any positive multiple gives the same classifier).

Other options either describe a different line (swapped/negated coefficients move the intercepts) or the same line with the sign flipped, which would label every training point incorrectly.

Q.23 — Answer: 0.4812 to 0.5012 [NAT, 2M · Unsupervised Learning > Distance measures]

Min-max scaling: $x' = (x - \min)/(\max - \min)$; age range [20, 60], income range [10, 210].

Customer	age'	income'
A	0.65	0.45
B	0	0.225
C	0.525	0.925

Raw distances: $d(A,B) = 51.97$, $d(A,C) = 95.13 \rightarrow$ nearest is B (income dominates because of its larger range).

Scaled distances: $d(A,B) = 0.6878$, $d(A,C) = 0.4912 \rightarrow$ nearest is C.

Answer: **0.49**. Scaling changes which neighbour is nearest — distance-based methods (k-means, k-NN, hierarchical clustering) should be run on standardized features.

Q.24 — Answer: (A), (B), (C) [MSQ, 2M · SVM & Decision Trees > SVM: kernels]

A Gram matrix of a valid kernel must be symmetric and positive semidefinite. For a symmetric 2×2 matrix this is equivalent to: both diagonal entries ≥ 0 and determinant ≥ 0 .

(A) TRUE. symmetric, diagonal entries ≥ 0 and $\det = 56 > 0 \Rightarrow$ positive definite.

(B) TRUE. symmetric, diagonal entries ≥ 0 and $\det = 18 > 0 \Rightarrow$ positive definite.

(C) TRUE. symmetric, diagonal entries ≥ 0 and $\det = 4 > 0 \Rightarrow$ positive definite.

(D) FALSE. not symmetric, but $K(x, z) = K(z, x)$ for every kernel.

Q.25 — Answer: (B) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Target AND truth table for $(x_1, x_2) = (0,0), (0,1), (1,0), (1,1)$: 0, 0, 0, 1. For each option compute $a = w_1x_1 + w_2x_2 + b$ at the four inputs and output 1 iff $a > 0$.

(A) FALSE. $a = 3, 3.5, 4.5, 5 \rightarrow$ outputs 1, 1, 1, 1 - constant 1.

(B) TRUE. $a = -2, -1, -0.5, 0.5 \rightarrow$ outputs 0, 0, 0, 1 - AND.

(C) FALSE. $a = -2.5, -2.5, -3.5, -3.5 \rightarrow$ outputs 0, 0, 0, 0 - constant 0.

(D) FALSE. $a = 2, -0.5, -0.5, -3 \rightarrow$ outputs 1, 0, 0, 0 - NOR.

Q.26 — Answer: 6.0728 to 6.0928 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Single-linkage merge heights are exactly the edge weights of a minimum spanning tree (Kruskal's algorithm adds the shortest edge joining two different components = single-linkage merge).

MST edges (Prim's algorithm): P1-P5 (6.08); P5-P6 (1); P6-P4 (1); P6-P2 (1.41); P4-P3 (5).

Sorted merge heights: 1, 1, 1.41, 5, 6.08.

The last single-linkage merge happens at the largest MST edge: **6.08**.

Q.27 — Answer: 0.2485 to 0.2515 [NAT, 2M · SVM & Decision Trees > Decision trees: overfitting & pruning]

$R(T) = \text{sum of leaf errors} / N = (0 + 6 + 2 + 7 + 0)/100 = 0.15$; $|T| = 5$ leaves.

$R_\alpha(T) = 0.15 + 0.02 \times 5 = \mathbf{0.25}$.

Q.28 — Answer: (B) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$\nabla f = (4w_1, 2w_2) = w_i \leftarrow (1 - \eta a_i) w_i$. GD converges (from every start) iff $|1 - \eta a_i| < 1$ for both i , i.e. $0 < \eta < 2/\max(a_i) = 2/4 = 0.5$. The steep direction ($a = 4$) limits the learning rate.

(A) FALSE. $\eta = 0.3$: $|1 - \eta \cdot 4| = 0.2$, $|1 - \eta \cdot 2| = 0.4 \rightarrow$ converges.

(B) TRUE. $\eta = 0.75$: $|1 - \eta \cdot 4| = 2$, $|1 - \eta \cdot 2| = 0.5 \rightarrow$ diverges.

(C) FALSE. $\eta = 0.125$: $|1 - \eta \cdot 4| = 0.5$, $|1 - \eta \cdot 2| = 0.75 \rightarrow$ converges.

(D) FALSE. $\eta = 0.475$: $|1 - \eta \cdot 4| = 0.9$, $|1 - \eta \cdot 2| = 0.05 \rightarrow$ converges.

Q.29 — Answer: 6.4603 to 6.4803 [NAT, 2M · Regression & Model Selection > Cross-validation]

Pooled CV MSE $= \sum_k n_k \cdot \text{MSE}_k / n$ (a size-weighted average, not the plain average of fold MSEs, since folds are unequal).

$\sum n_k \text{MSE}_k = 18 \times 2.2 + 16 \times 9 + 12 \times 5.9 + 16 \times 8.1 + 12 \times 7.9 = 478.8$; $n = 74$.

CV $= 478.8/74 = 6.4703$ (plain average of fold MSEs $= 6.62$ — a tempting but wrong value).

Answer: **6.47**.

Q.30 — Answer: (C) [MCQ, 2M · SVM & Decision Trees > SVM: margin & geometry]

For $n^T x - c = 0$, the geometric margin is $\min_i y_i s(n^T x_i - c)/\|n\|$ with orientation $s = \pm 1$ chosen so that the positive class is on the positive side; a negative value means some point is misclassified.

Hyperplane	$\ n\ $	$\min_i$ distance (signed, best orientation)
$x_1 - 2x_2 + 1.5 = 0$	2.236	0.224
$x_1 - 2x_2 + 2.5 = 0$	2.236	0.224
$x_1 - 2x_2 + 2 = 0$	2.236	0.447
$2x_1 - x_2 + 2.5 = 0$	2.236	-3.801 (does not separate)

The largest margin is 0.447, for $x_1 - 2x_2 + 2 = 0$. Note that a boundary can separate the data perfectly yet pass close to some point — only the minimum distance matters.

Q.31 — Answer: 0.49 to 0.51 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Predicted positive (score ≥ 0.2): examples 1, 2, 3, 4, 5, 8, 9.

TP = 5, FP = 2, FN = 0, TN = 2.

False positive rate = $FP/(FP + TN) = 0.5$.

Raising t typically raises precision and lowers recall/FPR; lowering t does the opposite (threshold moving).

Q.32 — Answer: -8.1405 to -8.0595 [NAT, 2M · Regression & Model Selection > Simple linear regression]

For the least-squares fit with intercept the normal equations give $\Sigma e_i = 0$ and $\Sigma x_i e_i = 0$.

Known: $\Sigma e = -3.1$, $\Sigma x e = -23.2$. With unknowns $u = e$ at $x = 2$ and $v = e$ at $x = 7$:

$u + v = 3.1$, $2u + 7v = 23.2$.

Solving: $u = -0.3$, $v = 3.4$.

$y = \hat{y} + e = (1.8 + (-1.9)(7)) + (3.4) = -8.1$.

Answer: **-8.1**.

Q.33 — Answer: 17.2009 to 17.2209 [NAT, 2M · Unsupervised Learning > k-medoids]

Compute the Euclidean (L_2) distance matrix and its row sums:

	P1	P2	P3	P4	P5	Σ
P1	0	1.41	4.47	5	6.32	17.21
P2	1.41	0	5.83	5.39	5.1	17.73
P3	4.47	5.83	0	5	10	25.3
P4	5	5.39	5	0	6.71	22.09
P5	6.32	5.1	10	6.71	0	28.13

The smallest row sum is 17.21 for P1, so the medoid is P1.

Answer: **17.21**.

Unlike a k-means centroid, the medoid is always an actual data point.

Q.34 — Answer: 0.6191 to 0.6231 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

Adjusted $R^2 = 1 - [SSE/(n - p - 1)]/[SST/(n - 1)] = 1 - (1 - R^2)(n - 1)/(n - p - 1)$, where p is the number of predictors (excluding the intercept).

$= 1 - (1 - 0.69) \times 22/18 = 0.6211$.

Answer: **0.621**.

Q.35 — Answer: -0.3482 to -0.3442 [NAT, 2M · Regression & Model Selection > MLE & MAP view]

$-\ln p(w | \text{data}) = (1/(2\sigma^2))\Sigma(y_i - wx_i)^2 + w^2/(2\tau^2) + \text{const.}$

Multiplying by $2\sigma^2$: $\Sigma(y_i - wx_i)^2 + (\sigma^2/\tau^2)w^2$ — ridge with $\lambda = \sigma^2/\tau^2 = 4$.

$\Sigma x_i^2 = 22$, $\Sigma x_i y_i = -9$; $\hat{w}_{\text{MAP}} = \Sigma xy / (\Sigma x^2 + \lambda) = -9/(22 + 4) = -0.3462$.

Answer: **-0.346**.

Mock Test 04

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	7	6	22

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Regression & Model Selection	7	11
Probabilistic & Instance-based Classifiers	7	11
Neural Networks (MLP / Feed-forward)	7	11
Unsupervised Learning	7	11
SVM & Decision Trees	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

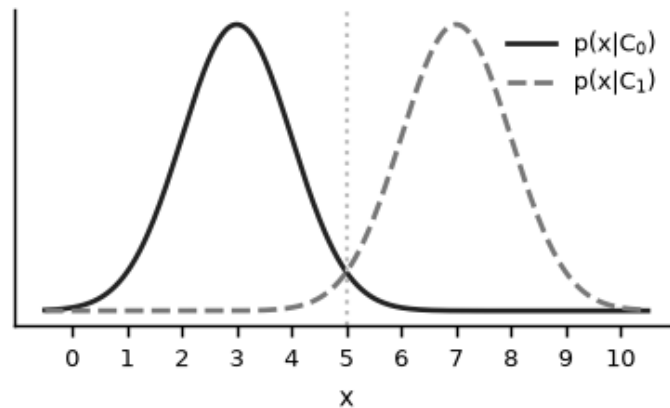
Q.1 [1 mark · MSQ (one or more correct)]

Consider ridge regression (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_2^2$) and the lasso (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_1$) with $\lambda \geq 0$. Which of the following statements is/are CORRECT?

- (A) Among a group of highly correlated relevant predictors, the lasso tends to keep all of them with similar coefficients.
- (B) Ridge regression typically sets many coefficients exactly to zero.
- (C) Predictors are usually standardised before fitting ridge or lasso.
- (D) Increasing λ decreases the bias of the ridge estimator.

Q.2 [1 mark · MCQ]

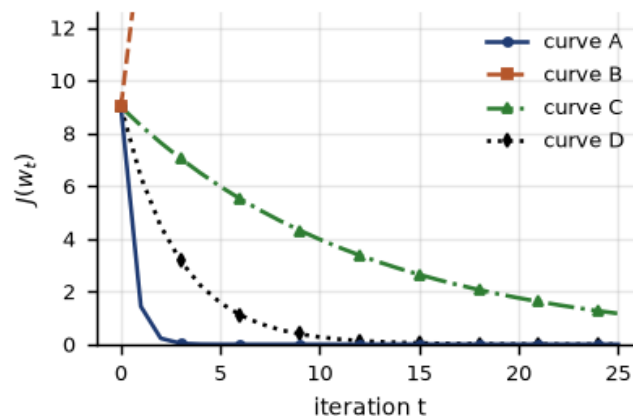
Two classes have unit-variance Gaussian class-conditional densities with means 3 (C_0) and 7 (C_1), as plotted. With equal priors the Bayes decision threshold is the dotted line at $x = 5$. If the prior of C_0 is increased to 0.7, the threshold



- (A) It splits into two thresholds, giving a quadratic boundary
- (B) It moves towards $x = 7$, enlarging the region assigned to C_0
- (C) It stays at $x = 5$ because the two variances are equal
- (D) It moves towards $x = 3$, shrinking the region assigned to C_0

Q.3 [1 mark · MCQ]

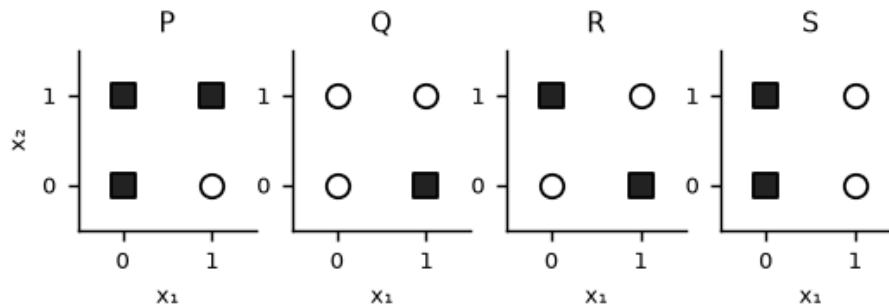
Gradient descent $w_{t+1} = w_t - \eta J'(w_t)$ is run on $J(w) = (w - 3)^2$ from $w_0 = 6$ with four learning rates $\eta \in \{0.02, 0.08, 0.3, 1.15\}$. The loss curves are shown below (curves that leave the plotted range are cut off). Which curve corresponds to $\eta = 0.08$?



- (A) Curve D
- (B) Curve C
- (C) Curve A
- (D) Curve B

Q.4 [1 mark · MCQ]

Each panel shows the four binary inputs (x_1, x_2) labelled as class 1 (filled square) or class 0 (open circle). Which dataset can NOT be classified perfectly by a single perceptron (one linear threshold unit)?



- (A) Dataset P
(B) Dataset Q
(C) Dataset R
(D) Dataset S

Q.5 [1 mark · NAT (numerical answer)]

For logits $z = (0.5, 2, 0, -0.5)$, the softmax probability $p_1 = \exp(z_1) / \sum_j \exp(z_j)$ is _____ (rounded off to three decimal places)

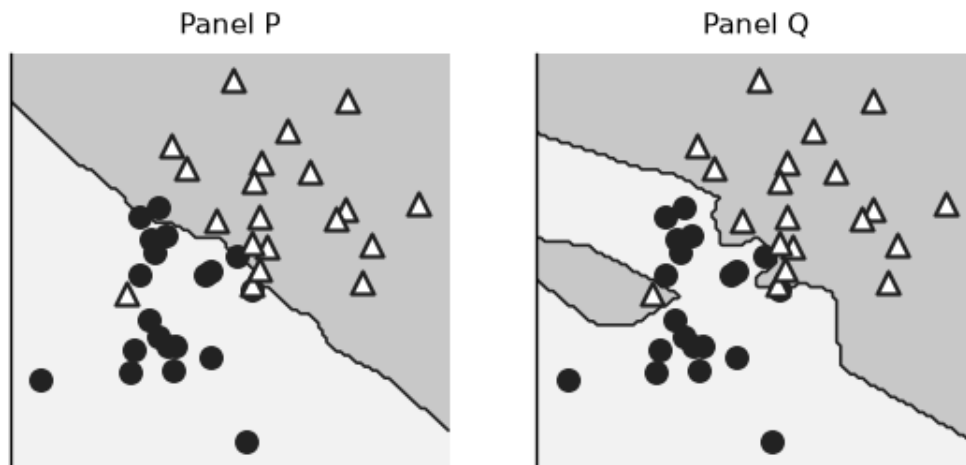
Q.6 [1 mark · MSQ (one or more correct)]

Let X be an $n \times p$ design matrix of full column rank p ($p < n$), and let $H = X(X^T X)^{-1} X^T$ be the hat matrix of least-squares regression of y on X . Which of the following statements is/are CORRECT?

- (A) The residual vector is $e = (I - H)y$.
(B) H is symmetric ($H^T = H$).
(C) The vector of fitted values is $\hat{y} = Hy$.
(D) $\text{trace}(H) = p$.

Q.7 [1 mark · MCQ]

The two panels show the decision regions of k -NN classifiers trained on the same 2-D data set, one with $k = 1$ and the other with $k = 21$. Which of the following statements is CORRECT?



- (A) Panel Q uses $k = 1$; it has lower bias and higher variance
(B) Panel P uses $k = 21$; it has zero training error
(C) Panel Q uses $k = 1$; it has higher bias and lower variance
(D) Panel P uses $k = 1$; it has lower bias and higher variance

Q.8 [1 mark · NAT (numerical answer)]

The 5 two-dimensional observations below have zero mean in each feature. Then the sample variance of x_1 (divisor $n - 1$) is _____ (rounded off to two decimal places)

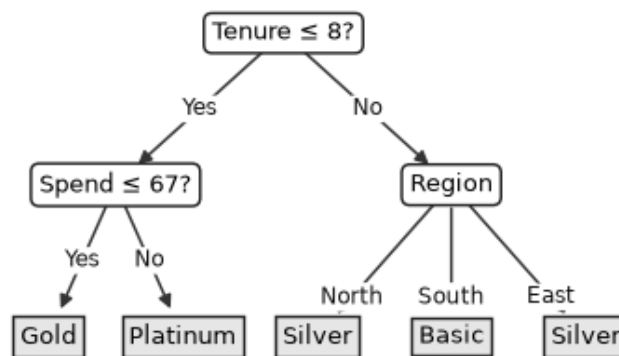
Point	x_1	x_2	x_3	x_4	x_5
x_1	-1	4	2	-2	-3
x_2	3	4	0	-4	-3

Q.9 [1 mark · NAT (numerical answer)]

In AdaBoost ($\alpha_t = \frac{1}{2} \ln((1 - \epsilon_t)/\epsilon_t)$), a weak learner received weight $\alpha_t = 0.7$. Its weighted error ϵ_t was _____ (rounded off to two decimal places)

Q.10 [1 mark · MCQ]

The decision tree below predicts the membership tier offered to a customer. Edges labelled Yes/No answer the test in the node. What is the prediction for the record shown in the table?



Tenure	Spend	Visits	Region
7	67	17	South

- (A) Gold (B) Silver
(C) Platinum (D) Basic

Q.11 [1 mark · NAT (numerical answer)]

A Gaussian naive Bayes classifier has $d = 7$ continuous features and $K = 4$ classes, with a separate mean and variance for every (feature, class) pair. The number of independent parameters, including class priors, is _____ (Answer in integer)

Q.12 [1 mark · NAT (numerical answer)]

Two shopping baskets are encoded as binary vectors (1 = item bought). The difference (simple matching coefficient – Jaccard similarity) between basket A and basket B is _____ (rounded off to two decimal places)

Item	milk	bread	eggs	butter	jam	tea	rice	salt
Basket A	0	0	1	0	0	1	0	0
Basket B	1	0	0	1	0	1	1	1

Q.13 [1 mark · MCQ]

A soft-margin linear SVM has learned $f(x) = x_1 + 2x_2 - 4$. For the training example $x = (0, 2)$ with label $y = -1$, which statement is CORRECT?

- (A) Exactly on the decision boundary ($\xi = 1$)
(B) Misclassified ($\xi > 1$)
(C) Correctly classified, on or outside the margin ($\xi = 0$)
(D) Correctly classified but strictly inside the margin ($0 < \xi < 1$)

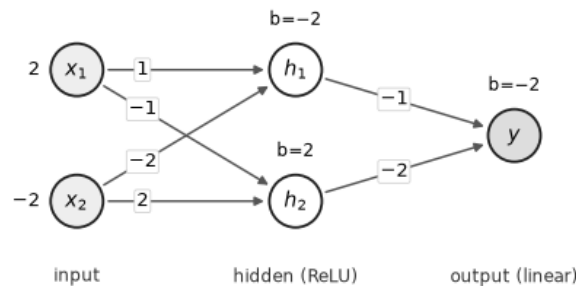
Q.14 [1 mark · MSQ (one or more correct)]

Which of the following statements about clustering evaluation measures is/are CORRECT?

- (A) The Rand index changes if the cluster labels are renamed.
- (B) Purity can be made equal to 1 by putting every point in its own cluster.
- (C) In the silhouette, $b(i)$ is the mean distance to the farthest other cluster.
- (D) The average silhouette width can be used to choose k by picking the k with the largest value.

Q.15 [1 mark · NAT (numerical answer)]

For the ReLU network shown (edge labels are weights, 'b' denotes a bias) and input $x = (2, -2)$, the output y is _____. (Answer in integer)

**Q.16 - Q.35 : Two-mark questions****Q.16 [2 marks · MSQ (one or more correct)]**

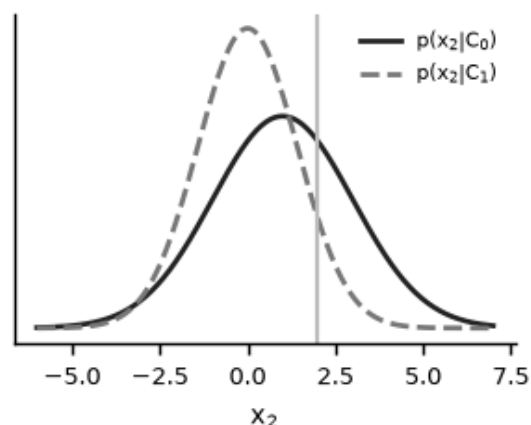
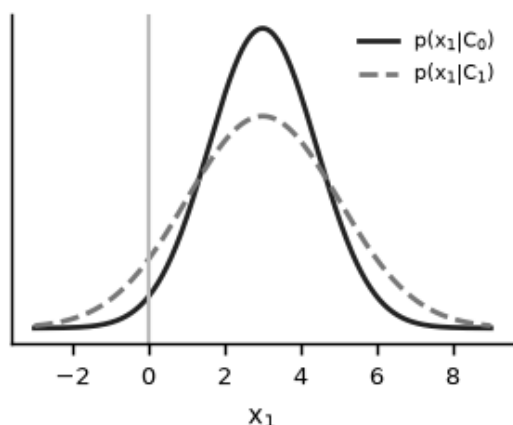
Consider agglomerative hierarchical clustering on n points with a dissimilarity matrix (no ties). Which of the following statements is/are CORRECT?

- (A) Cophenetic distances from single, complete or average linkage satisfy the ultrametric inequality $d(x, z) \leq \max(d(x, y), d(y, z))$.
- (B) Complete-linkage merge heights equal the edge weights of a minimum spanning tree.
- (C) The single-linkage merge heights are exactly the edge weights of a minimum spanning tree of the complete distance graph.
- (D) For any two points, the single-linkage cophenetic distance is $\leq$ their original distance.

Q.17 [2 marks · NAT (numerical answer)]

A Gaussian naive Bayes classifier with features x_1, x_2 has the class priors and per-class Gaussian parameters below (σ^2 = variance). For the input $x = (0, 2)$, the posterior probability $P(C_1 | x)$ is _____ (rounded off to two decimal places)

class	prior	$\mu(x_1)$	$\sigma^2(x_1)$	$\mu(x_2)$	$\sigma^2(x_2)$
C_0	0.4	3	2	1	4
C_1	0.6	3	4	0	2



Q.18 [2 marks · NAT (numerical answer)]

Least-squares linear regression on $n = 2$ observations minimises $J(w) = (1/2n)\|Xw - y\|^2$ by batch gradient descent with constant learning rate η . The data give the matrix $X^T X$ below. Then the constant learning rate η that minimises the worst-case contraction factor $\max_i |1 - \eta \lambda_i|$ is _____ (rounded off to three decimal places)

$$X^T X = \begin{bmatrix} 13 & 7 \\ 7 & 13 \end{bmatrix}$$

Q.19 [2 marks · MSQ (one or more correct)]

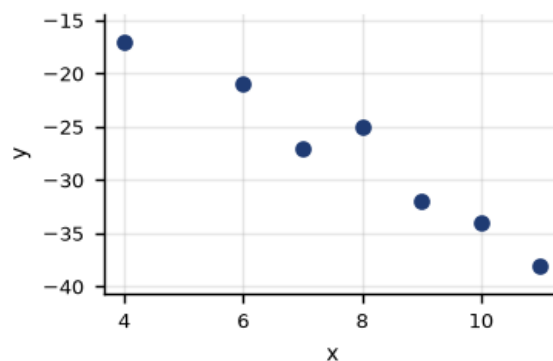
Let σ be the logistic sigmoid, and $'$ denote the derivative. At $z = 2.5$ (and $-z = -2.5$), which of the following is/are CORRECT? ($\text{softplus}(z) = \ln(1 + e^z)$.)

- (A) $\sigma(2.5) + \sigma(-2.5) = 1$ (B) $\text{ReLU}'(2.5) = 1$
 (C) $\text{softplus}(2.5) > \text{ReLU}(2.5)$ (D) $\tanh(2.5) = 2\sigma(5) - 1$

Q.20 [2 marks · NAT (numerical answer)]

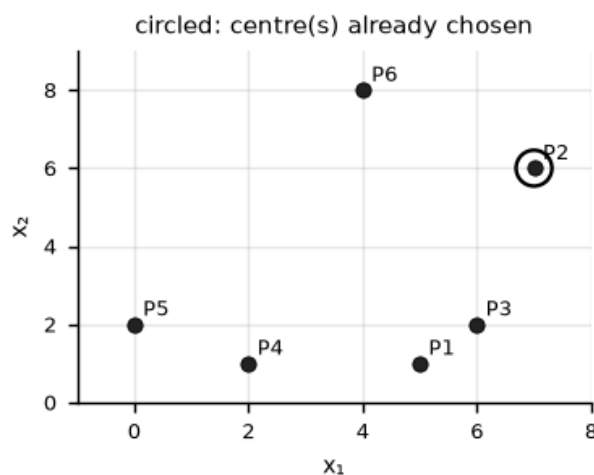
The 7 observations shown in the table are plotted below. For the simple linear regression of y on x (with intercept), the value predicted by the least-squares line at $x = 12$ is _____ (rounded off to two decimal places)

x	4	6	7	8	9	10	11
y	-17	-21	-27	-25	-32	-34	-38

**Q.21 [2 marks · NAT (numerical answer)]**

k-means++ initialization (squared Euclidean seeding) is applied to the six points below. P2 has already been chosen as centre. The probability that P3 is selected as the second centre is _____ (rounded off to three decimal places)

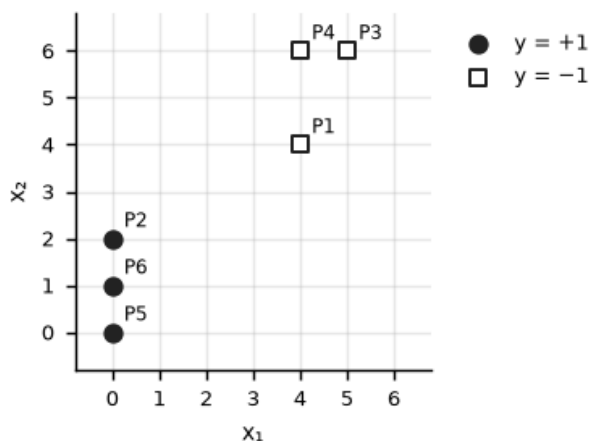
Point	P1	P2	P3	P4	P5	P6
x_1	5	7	6	2	0	4
x_2	1	6	2	1	2	8



Q.22 [2 marks · NAT (numerical answer)]

A hard-margin linear SVM is trained on the points below. Solving the dual problem gives the Lagrange multipliers α_i listed in the table. Using the KKT conditions, compute the margin width $2/\|w\|$. (rounded off to two decimal places)

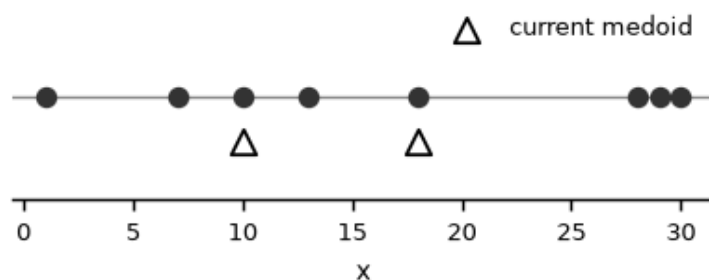
Point	x_1	x_2	y	α_i
P1	4	4	-1	0.1
P2	0	2	+1	0.1
P3	5	6	-1	0
P4	4	6	-1	0
P5	0	0	+1	0
P6	0	1	+1	0

**Q.23 [2 marks · NAT (numerical answer)]**

For the kernel $K(x, z) = (x^T z + 1)^2$ on $\mathbb{R}^2$ with feature map ϕ , compute the cosine of the angle between $\phi(x)$ and $\phi(z)$, where $x = (1, 2)$ and $z = (-1, -2)$. (rounded off to two decimal places)

Q.24 [2 marks · NAT (numerical answer)]

PAM (k-medoids, $k = 2$, absolute-difference dissimilarity) is applied to the 1-D data $\{1, 7, 10, 13, 18, 28, 29, 30\}$. The current medoids are 10 and 18. The change in total cost, (cost after swap) – (cost before swap), when medoid 18 is swapped with non-medoid 7 is _____. (Answer in integer)



Q.25 [2 marks · MCQ]

Ridge regression is tuned by 10-fold cross-validation. For each λ the table gives the mean CV MSE and its standard error (SE) across folds. (Larger λ means a simpler, more regularised model.) Using the **one-standard-error rule** (choose the most regularised model whose CV error is within one standard error of the minimum CV error), the selected value is

λ	0.01	0.1	0.3	1	3	10
CV MSE	3.72	3.46	3.6	4.12	5.15	5.76
SE	0.37	0.37	0.46	0.34	0.27	0.35

(A) $\lambda = 0.01$ **(B)** $\lambda = 0.3$ **(C)** $\lambda = 3$ **(D)** $\lambda = 0.1$ **Q.26 [2 marks · NAT (numerical answer)]**

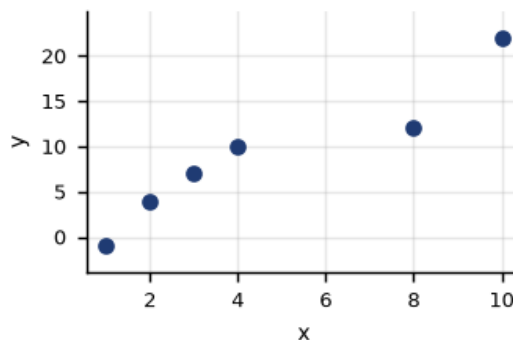
The sample covariance matrix of a two-feature dataset is S given below. In principal component analysis, if the first principal direction is written as $(1, m)$ (up to scaling), the value of m is _____ (rounded off to two decimal places)

$$S = \begin{bmatrix} 6 & -6 \\ -6 & 10 \end{bmatrix}$$

Q.27 [2 marks · NAT (numerical answer)]

Assume $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with ε_i i.i.d. $N(0, \sigma^2)$. For the 6 observations below, with β estimated by least squares (= maximum likelihood), the maximised log-likelihood $\ln L(\beta, \hat{\sigma}^2)$ (natural log, with $\hat{\sigma}^2$ the maximum-likelihood estimate) is _____ (rounded off to two decimal places)

x	1	2	3	4	8	10
y	-1	4	7	10	12	22

**Q.28 [2 marks · MSQ (one or more correct)]**

Which of the following statements about LDA, QDA and logistic regression is/are CORRECT?

- (A)** Fisher's LDA chooses w to maximise the within-class scatter of the projected data.
- (B)** QDA allows each class its own covariance matrix, giving quadratic decision boundaries in general.
- (C)** Fisher's LDA for K classes yields at most $K - 1$ discriminant directions.
- (D)** Changing the class priors in LDA shifts the decision boundary but does not change its orientation.

Q.29 [2 marks · NAT (numerical answer)]

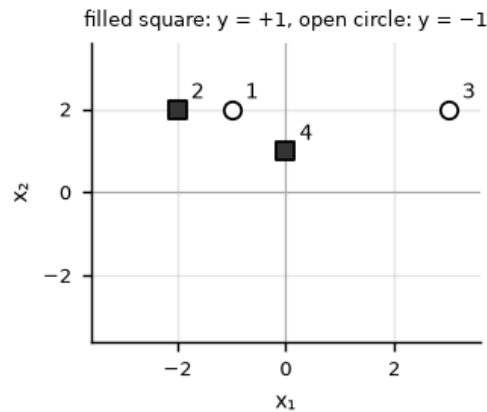
In one AdaBoost round the current example weights and whether the chosen weak classifier h classifies each example correctly are given. Using $\alpha = \frac{1}{2} \ln((1 - \varepsilon)/\varepsilon)$ and the update $w_i \leftarrow w_i \cdot \exp(-\alpha y_i h(x_i))/Z$, find the normalised weight of example 1 for the next round. (rounded off to three decimal places)

Example i	1	2	3	4	5	6	7
Weight w_i	1/7	1/7	1/7	1/7	1/7	1/7	1/7
$h(x_i)$ correct?	No	Yes	Yes	No	Yes	Yes	Yes

Q.30 [2 marks · NAT (numerical answer)]

A perceptron with parameters $w \in \mathbb{R}^2$, b is trained on the data below, starting from $w = (0, 0)$, $b = 0$, with learning rate $\eta = 2$. Examples are presented once in the order 1, 2, 3, 4 (one epoch). An update $w \leftarrow w + \eta y x$, $b \leftarrow b + \eta y$ is made whenever $y(w \cdot x + b) \leq 0$. After the epoch, the bias b is _____ (Answer in integer)

i	x_1	x_2	y
1	-1	2	-1
2	-2	2	1
3	3	2	-1
4	0	1	1

**Q.31 [2 marks · NAT (numerical answer)]**

In C4.5, the attribute Region splits a node of 27 records as tabulated (class Yes/No counts in each branch). Compute the gain ratio of Region (all logarithms base 2). (rounded off to two decimal places)

Region	#Yes	#No
v1	1	8
v2	8	2
v3	4	2
v4	2	0

Q.32 [2 marks · NAT (numerical answer)]

Logistic regression $P(y = 1 | \mathbf{x}) = \sigma(w_1 x_1 + w_2 x_2 + b)$ is trained by **batch** gradient descent on the **average** cross-entropy loss over the 3 examples below, with learning rate $\eta = 0.5$. Starting from $w_1 = 0.5$, $w_2 = -0.5$, $b = 0.5$, the value of b after ONE gradient-descent update is _____ (rounded off to three decimal places)

i	x_1	x_2	y
1	1	1	0
2	-1	-1	1
3	-1	2	0

Q.33 [2 marks · NAT (numerical answer)]

A network with one hidden layer maps 50 inputs to 10 outputs; all layers are fully connected and have biases. It has exactly 986 trainable parameters. The number of hidden units is _____ (Answer in integer)

Q.34 [2 marks · NAT (numerical answer)]

A softmax-regression output layer computes $z = Wx + b$, $p = \text{softmax}(z)$ with W and b below, and is trained with cross-entropy loss $L = -\ln p_y$. For $x = (-1, -2)^T$ with true class $y = 2$, $\partial L / \partial W_{31}$ is _____ (rounded off to 4 decimal places)

$$W = \begin{bmatrix} 1 & -1 \\ -1 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

$$b = \begin{bmatrix} 0 \\ 0.5 \\ 0 \end{bmatrix}$$

Q.35 [2 marks · NAT (numerical answer)]

The actual and predicted labels of a binary classifier (1 = positive) on 15 test examples are given below. The F_β -score with $\beta = 0.5$, $F_\beta = (1 + \beta^2)PR / (\beta^2P + R)$, is _____ (rounded off to two decimal places)

#	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
actual y	1	1	1	0	0	1	0	0	1	1	1	1	0	0	0
predicted $\hat{y}$	1	0	1	0	0	1	1	0	0	0	1	1	0	1	0

— END OF QUESTION PAPER —

Mock Test 04 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MSQ	1	(C)	19	MSQ	2	(A), (B), (C), (D)
2	MCQ	1	(B)	20	NAT	2	-40.3606 to -39.959
3	MCQ	1	(A)	21	NAT	2	0.0957 to 0.0997
4	MCQ	1	(C)	22	NAT	2	4.4621 to 4.4821
5	NAT	1	0.1529 to 0.1569	23	NAT	2	0.4344 to 0.4544
6	MSQ	1	(A), (B), (C), (D)	24	NAT	2	26
7	MCQ	1	(A)	25	MCQ	2	(B)
8	NAT	1	8.49 to 8.51	26	NAT	2	-1.3974 to -1.3774
9	NAT	1	0.1878 to 0.2078	27	NAT	2	-13.699 to -13.5356
10	MCQ	1	(A)	28	MSQ	2	(B), (C), (D)
11	NAT	1	59	29	NAT	2	0.248 to 0.252
12	NAT	1	0.1983 to 0.2183	30	NAT	2	2
13	MCQ	1	(A)	31	NAT	2	0.1834 to 0.2034
14	MSQ	1	(B), (D)	32	NAT	2	0.4124 to 0.4164
15	NAT	1	-6	33	NAT	2	16
16	MSQ	2	(A), (C), (D)	34	NAT	2	-0.0399 to -0.0383
17	NAT	2	0.6482 to 0.6682	35	NAT	2	0.6844 to 0.7044
18	NAT	2	0.1519 to 0.1558				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Regression & Model Selection					11
Probabilistic & Instance-based Classifiers					11
Neural Networks (MLP / Feed-forward)					11
Unsupervised Learning					11
SVM & Decision Trees					11

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Total					55

Mock Test 04 — Detailed Solutions

Q.1 — Answer: (C) [MSQ, 1M · Regression & Model Selection > Ridge & Lasso]

(A) **FALSE**. Lasso tends to pick one and zero the others; ridge spreads weight among them.

(B) **FALSE**. The L2 ball is smooth; ridge shrinks coefficients but rarely makes them exactly zero.

(C) **TRUE**. The penalty treats all coefficients alike, so the solution depends on the scale of each predictor.

(D) **FALSE**. More shrinkage → more bias.

Q.2 — Answer: (B) [MCQ, 1M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$$x^* = (\mu_0 + \mu_1)/2 + \sigma^2 \ln(\pi_0/\pi_1)/(\mu_1 - \mu_0) = 5.212.$$

A larger prior for C_0 means more of the x-axis is assigned to it, so the threshold moves away from μ_0 towards $\mu_1 = 7$. Answer: **It moves towards $x = 7$, enlarging the region assigned to C_0 .**

Equal variances keep the boundary a single point (linear); priors shift it but never make it quadratic.

Q.3 — Answer: (A) [MCQ, 1M · Regression & Model Selection > Gradient descent]

$$J(w) = (w - 3)^2 \text{ has } J' = 2. \text{ GD: } w_{t+1} - 3 = (1 - 2\eta)(w_t - 3), \text{ so } J(w_t) = (1 - 2\eta)^{2t} J(w_0).$$

Ratios $(1 - 2\eta)^2$ per step: $\eta = 0.02 \rightarrow 0.9216$ (curve C); $\eta = 0.08 \rightarrow 0.7056$ (curve D); $\eta = 0.3 \rightarrow 0.16$ (curve A); $\eta = 1.15 \rightarrow 1.69$ (curve B).

Ratio $> 1 \Rightarrow$ divergence ($\eta > 1 = 2/J'$); among convergent rates (all < 0.5 here), a smaller η gives a ratio closer to 1, i.e. slower decrease.

Hence $\eta = 0.08$ corresponds to **Curve D**.

Q.4 — Answer: (C) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

A single perceptron can classify a dataset perfectly iff the two classes are linearly separable.

Dataset P: labels for $(0,0),(0,1),(1,0),(1,1) = 1, 1, 0, 1 \rightarrow (\text{NOT } x_1) \text{ OR } x_2$ – linearly separable.

Dataset Q: labels for $(0,0),(0,1),(1,0),(1,1) = 0, 0, 1, 0 \rightarrow x_1 \text{ AND } (\text{NOT } x_2)$ – linearly separable.

Dataset R: labels for $(0,0),(0,1),(1,0),(1,1) = 0, 1, 1, 0 \rightarrow \text{XOR}$ – **not** linearly separable (diagonal pattern).

Dataset S: labels for $(0,0),(0,1),(1,0),(1,1) = 1, 1, 0, 0 \rightarrow \text{NOT } x_1$ – linearly separable.

Answer: **Dataset R**. Of the 14 non-constant Boolean functions of two inputs, only XOR and XNOR are not linearly separable.

Q.5 — Answer: 0.1529 to 0.1569 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

exp values: 1.6487, 7.3891, 1, 0.6065; sum = 10.6443.

$$p_1 = 1.6487/10.6443 = \mathbf{0.1549}.$$

Q.6 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Multiple linear regression]

(A) **TRUE**. $e = y - \hat{y} = (I - H)y$.

(B) **TRUE**. $H^T = X((X^T X)^{-1})^T X^T = H$ since $X^T X$ is symmetric.

(C) **TRUE**. $\hat{y} = X\beta = X(X^T X)^{-1} X^T y$.

(D) **TRUE**. $\text{trace}(X(X^T X)^{-1} X^T) = \text{trace}((X^T X)^{-1} X^T X) = \text{trace}(I_p) = p$.

Q.7 — Answer: (A) [MCQ, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

The jagged boundary with small islands around individual points (Panel Q) is produced by $k = 1$, which fits every training point (zero training error): low bias, high variance.

The smooth boundary (Panel P) comes from $k = 21$: averaging many neighbours raises bias and lowers variance; it misclassifies some training points, so its training error is not zero.

Answer: **Panel Q uses $k = 1$; it has lower bias and higher variance.**

Q.8 — Answer: 8.49 to 8.51 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

The data are already centered (column means are 0), so $S = X^T X / (n - 1)$.

$$\Sigma_{x_1^2} = 34, \Sigma_{x_2^2} = 50, \Sigma_{x_1 x_2} = 30.$$

$S = [[8.5, 7.5], [7.5, 12.5]]$.

Required value = **8.5**. (Dividing by n instead of $n - 1$ would give 6.8.)

Q.9 — Answer: 0.1878 to 0.2078 [NAT, 1M · SVM & Decision Trees > Ensembles: boosting]

$(1 - \varepsilon)/\varepsilon = e^{2\alpha} = e^{1.4} = 4.0552 \Rightarrow \varepsilon = 1/(1 + e^{2\alpha}) = \mathbf{0.2}$.

$\varepsilon < 0.5 \Rightarrow \alpha > 0$; $\varepsilon = 0.5 \Rightarrow \alpha = 0$ (no better than chance).

Q.10 — Answer: (A) [MCQ, 1M · SVM & Decision Trees > Decision trees: structure & traversal]

Follow the record from the root, taking the branch that matches each test:

Tenure = $7 \leq 8 \rightarrow$ Yes.

Spend = $67 \leq 67 \rightarrow$ Yes.

(Only the tests on the actual path matter.) The record reaches the leaf **Gold**.

Note: " $\leq$ " includes equality, so a value equal to the threshold goes to the Yes branch.

Q.11 — Answer: 59 [NAT, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Means: $K_d = 28$; variances: $K_d = 28$; priors: $K - 1 = 3$.

Total = $2K_d + K - 1 = \mathbf{59}$.

Q.12 — Answer: 0.1983 to 0.2183 [NAT, 1M · Unsupervised Learning > Distance measures]

Counts: $f_{11} = 1, f_{10} = 1, f_{01} = 4, f_{00} = 2$.

Jaccard $J = f_{11}/(f_{11} + f_{10} + f_{01}) = 1/6 = 0.1667$ (0-0 matches are ignored).

SMC = $(f_{11} + f_{00})/m = 3/8 = 0.375$.

Required difference (simple matching coefficient – Jaccard similarity) = **0.21**.

Q.13 — Answer: (A) [MCQ, 1M · SVM & Decision Trees > SVM: soft margin & hinge loss]

$y \cdot f(x) = (-1) \cdot (0) = 0$.

Slack $\xi = \max(0, 1 - y f(x)) = 1$.

Rules: $y f \geq 1 \Rightarrow \xi = 0$ (on/outside margin); $0 < y f < 1 \Rightarrow$ inside margin, still correct; $y f = 0 \Rightarrow$ on the boundary ($\xi = 1$); $y f < 0 \Rightarrow$ misclassified ($\xi > 1$).

Hence: **Exactly on the decision boundary ($\xi = 1$)**.

Q.14 — Answer: (B), (D) [MSQ, 1M · Unsupervised Learning > Cluster evaluation]

(A) **FALSE**. Renaming does not change co-membership.

(B) **TRUE**. Hence purity alone cannot choose k .

(C) **FALSE**. $b(i)$ uses the nearest other cluster.

(D) **TRUE**. A common model-selection heuristic.

Q.15 — Answer: -6 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Forward pass]

$a = Wx + b = (4, -4)$; $h = \text{ReLU}(a) = (4, 0)$ – a negative pre-activation is clipped to 0.

$y = -1 \cdot 4 + (-2) \cdot 0 + (-2) = \mathbf{-6}$.

A common slip is to forget the ReLU and use the raw pre-activation (which would give 2).

Q.16 — Answer: (A), (C), (D) [MSQ, 2M · Unsupervised Learning > Hierarchical clustering]

(A) **TRUE**. Monotone dendrograms define ultrametrics.

(B) **FALSE**. That holds for single linkage.

(C) **TRUE**. Kruskal's algorithm performs the same merges in the same order.

(D) **TRUE**. They are joined no later than when their own edge would merge them.

Q.17 — Answer: 0.6482 to 0.6682 [NAT, 2M · Probabilistic & Instance-based Classifiers > Naive Bayes]

$N(x; \mu, \sigma^2) = \exp(-(x-\mu)^2/(2\sigma^2))/\sqrt{(2\pi\sigma^2)}$. Naive Bayes multiplies the per-feature densities.

C_0 : $N(0; 3, 2) = 0.02973 \times N(2; 1, 4) = 0.17603$; $\times$ prior $0.4 \Rightarrow 0.002094$.

C_1 : $N(0; 3, 4) = 0.06476 \times N(2; 0, 2) = 0.10378$; $\times$ prior $0.6 \Rightarrow 0.004032$.

$P(C_1 | x) = 0.004032/(0.002094 + 0.004032) = \mathbf{0.66}$.

Q.18 — Answer: 0.1519 to 0.1558 [NAT, 2M · Regression & Model Selection > Gradient descent]

Hessian of J : $\nabla^2 J = (1/n)X^T X$. Eigenvalues of $X^T X$: $13 \pm 7 = 20, 6$ (eigenvectors $(1, 1), (1, -1)$).

So the Hessian eigenvalues are $L = 10$ and $\mu = 3$.

GD error evolves as $w_{t+1} - w^* = (I - \eta \nabla^2 J)(w_t - w^*)$, so it converges iff $|1 - \eta \lambda| < 1$ for every Hessian eigenvalue λ .

Balance $|1 - \eta L| = |1 - \eta \mu| \Rightarrow \eta^* = 2/(L + \mu) = 2/(10 + 3) = 0.1538$.

Answer: **0.154**.

Q.19 — Answer: (A), (B), (C), (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Activation functions]

$\sigma(2.5) = 0.9241$, $\tanh(2.5) = 0.9866$.

(A) TRUE. $\sigma(-z) = 1 - \sigma(z)$.

(B) TRUE. $\text{ReLU}' = 1$ for $z > 0$ and 0 for $z < 0$.

(C) TRUE. $\text{softplus} = 2.5789 > 2.5$; $\ln(1+e^z) > \max(0, z)$ always.

(D) TRUE. Both equal 0.9866 .

Q.20 — Answer: -40.3606 to -39.959 [NAT, 2M · Regression & Model Selection > Simple linear regression]

$\bar{x} = 7.857$, $\bar{y} = -27.714$.

i	1	2	3	4	5	6	7
$x_i - \bar{x}$	-3.86	-1.86	-0.86	0.14	1.14	2.14	3.14
$y_i - \bar{y}$	10.71	6.71	0.71	2.71	-4.29	-6.29	-10.29
$(x_i - \bar{x})(y_i - \bar{y})$	-41.33	-12.47	-0.61	0.39	-4.9	-13.47	-32.33

$S_{xx} = \sum (x_i - \bar{x})^2 = 34.857$, $S_{xy} = -104.714$.

$\beta_1 = S_{xy}/S_{xx} = -3.0041$, $\beta_0 = \bar{y} - \beta_1 \bar{x} = -4.1107$.

$\hat{y}(12) = -4.1107 + (-3.0041)(12) = -40.1598$.

Answer: **-40.16**.

Q.21 — Answer: 0.0957 to 0.0997 [NAT, 2M · Unsupervised Learning > k-means]

k-means++ picks the next centre x with probability $D(x)^2 / \sum_x D(x)^2$, where $D(x)$ is the distance from x to the **nearest** centre already chosen (chosen centres have $D = 0$).

Point	P1	P2	P3	P4	P5	P6
$D(x)^2$	29	0	17	50	65	13

$\sum D^2 = 174$, so $P(P3) = 17/174 = \mathbf{0.098}$.

Common mistakes: using D instead of D^2 (that is not k-means++), or using the distance to the *farthest*/first centre instead of the nearest chosen centre.

Q.22 — Answer: 4.4621 to 4.4821 [NAT, 2M · SVM & Decision Trees > SVM: dual & support vectors]

$w = \sum \alpha_i y_i x_i$ over points with $\alpha_i > 0$:

$w = (0.1)(-1)(4, 4) + (0.1)(+1)(0, 2) = (-2/5, -1/5)$.

b from any support vector, e.g. P1: $y = w^T x + b \Rightarrow b = -1 - (-2.4) = 7/5$.

Check: $\sum \alpha_i y_i = 0$ ($= 0$, as required).

$\|w\| = 0.4472 \Rightarrow \text{margin} = \mathbf{4.47}$.

Q.23 — Answer: 0.4344 to 0.4544 [NAT, 2M · SVM & Decision Trees > SVM: kernels]

$\cos \theta = \varphi(x)^T \varphi(z) / (\|\varphi(x)\| \|\varphi(z)\|) = K(x, z) / \sqrt{K(x, x) K(z, z)}$.

$K(x, z) = 16$, $K(x, x) = 36$, $K(z, z) = 36$.

$\cos \theta = 16 / \sqrt{36 \times 36} = \mathbf{0.44}$.

Q.24 — Answer: 26 [NAT, 2M · Unsupervised Learning > k-medoids]

PAM cost = $\sum$ over all points of the absolute distance to the nearest medoid.

Current medoids $\{10, 18\}$: distances $1 \rightarrow 9$, $7 \rightarrow 3$, $10 \rightarrow 0$, $13 \rightarrow 3$, $18 \rightarrow 0$, $28 \rightarrow 10$, $29 \rightarrow 11$, $30 \rightarrow 12$; cost = **48**.

After swap, medoids $\{7, 10\}$: distances $1 \rightarrow 6$, $7 \rightarrow 0$, $10 \rightarrow 0$, $13 \rightarrow 3$, $18 \rightarrow 8$, $28 \rightarrow 18$, $29 \rightarrow 19$, $30 \rightarrow 20$; cost = 74.

Change = $74 - 48 = 26$ (positive $\rightarrow$ PAM rejects the swap).

Q.25 — Answer: (B) [MCQ, 2M · Regression & Model Selection > Cross-validation]

Minimum CV MSE = 3.46 at $\lambda = 0.1$, with SE = 0.37.

One-SE threshold = $3.46 + 0.37 = 3.83$.

Models within the threshold: $\lambda = 0.01$ (3.72), $\lambda = 0.1$ (3.46), $\lambda = 0.3$ (3.6).

The one-SE rule picks the largest such λ (simplest model): $\lambda = 0.3$; the minimum-CV rule picks $\lambda = 0.1$.

Answer: $\lambda = 0.3$.

Q.26 — Answer: -1.3974 to -1.3774 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

Eigenvalues of $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$: $\lambda = (a + c)/2 \pm \sqrt{((a - c)/2)^2 + b^2}$.

$\lambda = 8 \pm \sqrt{(4 + 36)} = 8 \pm 6.3246 \rightarrow \lambda_1 = 14.3246, \lambda_2 = 1.6754$ (check: $\lambda_1 + \lambda_2 = \text{trace} = 16, \lambda_1 \lambda_2 = \text{det} = 24$).

Eigenvector for λ_1 : $(S - \lambda_1 I)v = 0 \rightarrow (-8.3246)v_1 + (-6)v_2 = 0 \rightarrow v \propto (1, -1.3874)$.

$\text{PVE}(\text{PC1}) = \lambda_1/(\lambda_1 + \lambda_2) = 89.53\%$.

Answer: **-1.39**.

Q.27 — Answer: -13.699 to -13.5356 [NAT, 2M · Regression & Model Selection > MLE & MAP view]

$\bar{x} = 4.667, \bar{y} = 9$.

i	1	2	3	4	5	6
$x_i - \bar{x}$	-3.67	-2.67	-1.67	-0.67	3.33	5.33
$y_i - \bar{y}$	-10	-5	-2	1	3	13
$(x_i - \bar{x})(y_i - \bar{y})$	36.67	13.33	3.33	-0.67	10	69.33

$S_{xx} = \sum (x_i - \bar{x})^2 = 63.333, S_{xy} = 132$.

$\beta_1 = S_{xy}/S_{xx} = 2.0842, \beta_0 = \bar{y} - \beta_1 \bar{x} = -0.7263$.

$\text{SSE} = S_{yy} - S_{xy}^2/S_{xx} = 308 - 275.1158 = 32.8842$.

$\ln L = -(n/2)\ln(2\pi\sigma^2) - \text{SSE}/(2\sigma^2)$. At $\hat{\sigma}^2 = \text{SSE}/n$: $\ln L = -(n/2)[\ln(2\pi\hat{\sigma}^2) + 1]$.

$\hat{\sigma}^2 = 5.4807 \Rightarrow \ln L = -(6/2)[\ln(2\pi \times 5.4807) + 1] = -13.6173$.

Answer: **-13.62**.

Q.28 — Answer: (B), (C), (D) [MSQ, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

(A) **FALSE**. It minimises within-class scatter relative to between-class scatter.

(B) **TRUE**. The $x^T \Sigma_k^{-1} x$ terms no longer cancel.

(C) **TRUE**. $\text{rank}(S_B) \leq K - 1$.

(D) **TRUE**. Priors only enter the constant term $\ln \pi_k$.

Q.29 — Answer: 0.248 to 0.252 [NAT, 2M · SVM & Decision Trees > Ensembles: boosting]

$\varepsilon = \sum_{\text{misclassified}} w_i = 0.2857; \alpha = \frac{1}{2} \ln(0.7143/0.2857) = 0.4581$.

Misclassified weights $\times e^\alpha = 1.5811$; correct weights $\times e^{-\alpha} = 0.6325$.

$Z = \varepsilon e^\alpha + (1 - \varepsilon) e^{-\alpha} = 2\sqrt{\varepsilon(1 - \varepsilon)} = 0.9035$.

New weights: 0.25, 0.1, 0.1, 0.25, 0.1, 0.1, 0.1.

Shortcut: after normalisation the misclassified set has total weight exactly $\frac{1}{2}$: $w_i' = w_i/(2\varepsilon)$ if misclassified, $w_i/(2(1 - \varepsilon))$ otherwise.

Answer: **0.25**.

Q.30 — Answer: 2 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Rule: if $y_i(w \cdot x_i + b) \leq 0$, then $w \leftarrow w + \eta y_i x_i, b \leftarrow b + \eta y_i$ ($\eta = 2$). Start $w = (0, 0), b = 0$.

Example 1: $a = 0, y \cdot a = 0 \rightarrow$ **mistake**, update $\rightarrow w = (2, -4), b = -2$.

Example 2: $a = -14, y \cdot a = -14 \rightarrow$ **mistake**, update $\rightarrow w = (-2, 0), b = 0$.

Example 3: $a = -6, y \cdot a = 6 \rightarrow$ correct, no change.

Example 4: $a = 0$, $y \cdot a = 0 \rightarrow$ **mistake**, update $\rightarrow w = (-2, 2)$, $b = 2$.

Required value = **2**.

Note: a point with net input exactly 0 counts as a mistake ($y \cdot a \leq 0$) – this is why the first example always triggers an update when starting from zero.

Q.31 — Answer: 0.1834 to 0.2034 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [15 Yes, 12 No]: $H = 0.9911$.

Child entropies: $v1: 0.5033$, $v2: 0.7219$, $v3: 0.9183$, $v4: 0$.

$IG = 0.9911 - \sum (n_v/27)H_v = 0.3519$.

$SplitInfo = -\sum (n_v/N) \log_2(n_v/N) = -(9/27)\log_2(9/27) - (10/27)\log_2(10/27) - (6/27)\log_2(6/27) - (2/27)\log_2(2/27) = 1.8194$.

Gain ratio = $IG / SplitInfo = 0.3519/1.8194 = \mathbf{0.19}$.

C4.5 uses the gain ratio to penalise attributes that split the data into many small branches.

Q.32 — Answer: 0.4124 to 0.4164 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

For the average cross-entropy, $\partial L/\partial w_j = (1/n)\sum (\hat{p}_i - y_i)x_{ij}$ and $\partial L/\partial b = (1/n)\sum (\hat{p}_i - y_i)$, with $\hat{p}_i = \sigma(z_i)$.

i	z_i	$\hat{p}_i$	$\hat{p}_i - y_i$
1	0.5	0.6225	0.6225
2	0.5	0.6225	-0.3775
3	-1	0.2689	0.2689

Gradient: $\partial L/\partial w_1 = 0.2437$, $\partial L/\partial w_2 = 0.5126$, $\partial L/\partial b = 0.1713$.

Update $\theta \leftarrow \theta - \eta \nabla L$: $w_1 = 0.5 - 0.5 \times (0.2437) = 0.3782$, $w_2 = -0.5 - 0.5 \times (0.5126) = -0.7563$, $b = 0.5 - 0.5 \times (0.1713) = 0.4144$.

Required: $b = \mathbf{0.414}$.

Q.33 — Answer: 16 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Parameter counting]

$P = (d + 1)h + (h + 1)k = h(d + k + 1) + k \Rightarrow h = (P - k)/(d + k + 1) = (986 - 10)/(50 + 10 + 1) = \mathbf{16}$.

Q.34 — Answer: -0.0399 to -0.0383 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$z = Wx + b = (1, 2.5, -0.5)$; $p = \text{softmax}(z) = (0.1753, 0.7856, 0.0391)$.

$\delta = \partial L/\partial z = p - y = (0.1753, -0.2144, 0.0391)$ (true class 2).

Since $z_k = \sum_j W_{kj}x_j + b_k$: $\partial L/\partial W_{kj} = (p_k - y_k)x_j$ and $\partial L/\partial b_k = p_k - y_k$ (gradient matrix = $(p - y)x^T$).

$\partial L/\partial W_{31} = 0.0391 \times -1 = \mathbf{-0.0391}$.

Q.35 — Answer: 0.6844 to 0.7044 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Counting: TP = 5, FP = 2, FN = 3.

$P = 5/7 = 0.7143$, $R = 5/8 = 0.625$.

$F_{0.5} = (1 + 0.25) \times 0.7143 \times 0.625 / (0.25 \times 0.7143 + 0.625) = \mathbf{0.69}$.

$\beta < 1$ weights precision more heavily (F1 here would be 0.667).

Mock Test 05

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	4	3	28

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Neural Networks (MLP / Feed-forward)	7	11
Regression & Model Selection	7	11
Probabilistic & Instance-based Classifiers	7	11
SVM & Decision Trees	7	11
Unsupervised Learning	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

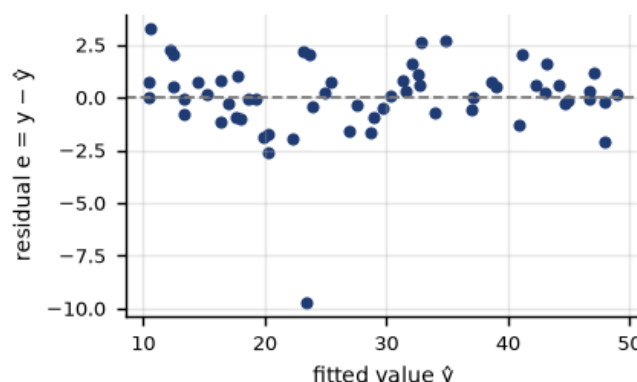
Q.1 [1 mark · MSQ (one or more correct)]

Which of the following statements about backpropagation is/are CORRECT?

- (A) If all weights of an MLP are initialised to the same value, backpropagation gives different gradients to the hidden units of a layer, breaking the symmetry automatically.
- (B) In backpropagation, ReLU units with negative pre-activation pass the gradient unchanged.
- (C) Backpropagation computes the gradient of the loss with respect to every weight using the chain rule, at a cost of the same order as one forward pass.
- (D) With $L = \frac{1}{2}(\hat{y} - y)^2$ and a sigmoid output unit, $\delta_{\text{out}} = (\hat{y} - y) \hat{y} (1 - \hat{y})$.

Q.2 [1 mark · MCQ]

After fitting a linear regression model, the residuals are plotted against the fitted values as shown. The plot most strongly suggests:

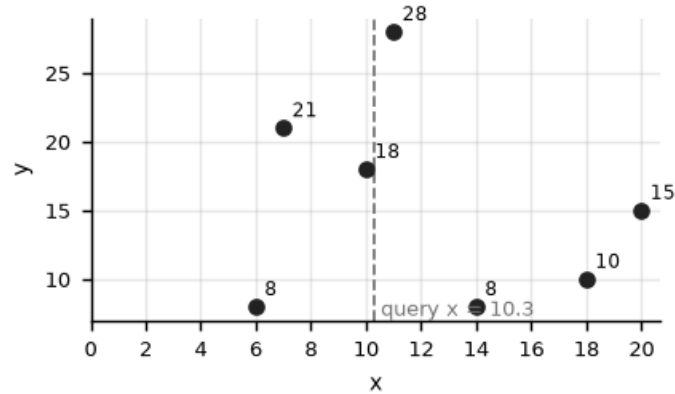


- (A) No evident violation of the linear-model assumptions
- (B) A non-linear relationship not captured by the model (e.g. a missing x^2 term)
- (C) A single outlying observation, with the rest of the fit looking adequate
- (D) Non-constant error variance (heteroscedasticity)

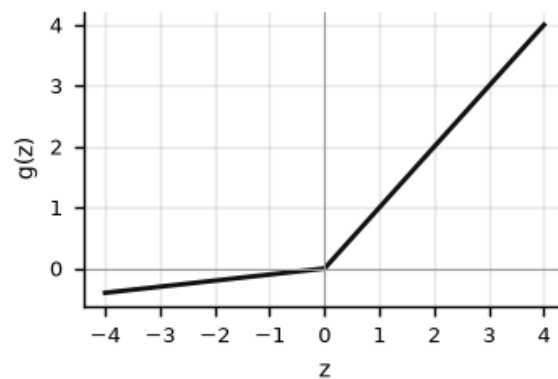
Q.3 [1 mark · NAT (numerical answer)]

The training data for a k-NN regressor are given below (also plotted). Using $k = 3$ and absolute distance $|x - x_i|$, the predicted value (unweighted average of neighbours) at $x = 10.3$ is _____ (rounded off to two decimal places)

x	6	7	10	11	14	18	20
y	8	21	18	28	8	10	15

**Q.4 [1 mark · MCQ]**

The figure shows the graph of a function $g(z)$ used in neural networks. Which function is it?



- (A) Softplus: $\ln(1 + e^z)$
- (B) ReLU: $\max(0, z)$
- (C) Leaky ReLU with $\alpha = 0.1$
- (D) ELU with $\alpha = 1$

Q.5 [1 mark · MSQ (one or more correct)]

Which of the following statements about support vector machines is/are CORRECT?

- (A) The optimal weight vector of an SVM can be written as $w = \sum \alpha_i y_i x_i$.
- (B) Hinge loss is zero for a correctly classified point with $y f(x) \geq 1$, so such points do not affect the SVM solution.
- (C) The geometric margin of a point equals its functional margin divided by $\|w\|$.
- (D) A kernel SVM predicts with $f(x) = \sum \alpha_i y_i K(x_i, x) + b$, where only support vectors contribute.

Q.6 [1 mark · MSQ (one or more correct)]

Which of the following statements about k-means clustering is/are CORRECT?

- (A) k-means is well suited to clusters of very different sizes and densities.
- (B) k-means is guaranteed to find the partition with the globally minimum SSE.
- (C) k-means is a soft clustering method that gives each point a membership probability for each cluster.
- (D) Rescaling one feature (e.g. changing its units) can change the clusters found by k-means.

Q.7 [1 mark · MCQ]

Among all functions g , the expected squared loss $E[(y - g(x))^2]$ is minimised by:

- (A) the conditional variance $\text{Var}(y | x)$
- (B) the conditional median of y given x
- (C) the conditional mean $E[y | x]$
- (D) the marginal mean $E[y]$

Q.8 [1 mark · NAT (numerical answer)]

In a training set, the 9 examples of class Buys = No have the following counts for the feature Age. Using Laplace (add-one) smoothing, the naive Bayes estimate of $P(\text{Age} = \text{Senior} | \text{Buys} = \text{No})$ is _____ (rounded off to three decimal places)

Age	Young	Middle	Senior
count in Buys = No	5	4	0

Q.9 [1 mark · NAT (numerical answer)]

For a hard-margin linear SVM, the support vectors $x_+ = (4, 0)$ (class +1) and $x_- = (1, 3)$ (class -1) lie on the hyperplanes $w^T x + b = +1$ and $w^T x + b = -1$ respectively, and the segment joining them is perpendicular to the decision boundary. The value of $\|w\|$ is _____ (rounded off to two decimal places)

Q.10 [1 mark · MCQ]

In a logistic regression model for loan approval, the coefficient of *credit score (in 10s)* is $\beta = 0.5$. A one-unit increase in *credit score (in 10s)* (others fixed) changes the odds of loan approval by approximately

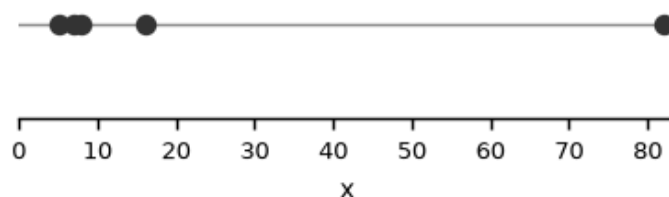
- (A) 164.9%
- (B) 50%
- (C) 39.3%
- (D) 64.9%

Q.11 [1 mark · NAT (numerical answer)]

Data are uniformly distributed in the d -dimensional unit hypercube with $d = 50$. A sub-cube is to capture a fraction $r = 0.01$ of the data. The required edge length of the sub-cube is _____ (rounded off to two decimal places)

Q.12 [1 mark · NAT (numerical answer)]

A cluster contains the 1-D values $\{5, 7, 8, 16, 82\}$. Treating them as one cluster, the absolute difference between the k-means centroid (mean) and the k-medoids representative (absolute-difference dissimilarity) is _____ (rounded off to two decimal places)



Q.13 [1 mark · NAT (numerical answer)]

A data set has $n = 41$ observations. Leave-one-out cross-validation is used to compare 5 candidate models (no final refit is counted). The total number of model fits is _____ (Answer in integer)

Q.14 [1 mark · NAT (numerical answer)]

A weight $w = 0.5$ is trained on the regularised objective $L(w) + (\lambda/2)w^2$ with $\lambda = 1$. The data-loss gradient is $\partial L/\partial w = 0.5$. After one gradient-descent step with $\eta = 0.01$, the new weight is _____ (rounded off to three decimal places)

Q.15 [1 mark · NAT (numerical answer)]

Consider the polynomial kernel $K(x, z) = (x^T z)^3$ with $x = (1, -1, 2)$ and $z = (1, -1, 2)$. The value of $K(x, x)$ is _____ (Answer in integer)

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

The 3×3 matrix $X^T X$ of a (centred) design matrix X has eigenvalues 25, 12, 1. For ridge regression with penalty $\lambda \|\beta\|_2^2$, $df(4) - df(12)$ is _____ (rounded off to two decimal places)

Q.17 [2 marks · NAT (numerical answer)]

Agglomerative clustering with **complete linkage** is applied to the distance matrix below. The cophenetic distance between P2 and P3 (the dendrogram height at which they are first joined) is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5	P6
P1	0	10	10	8	7	9
P2		0	8	6	5	9
P3			0	2	3	5
P4				0	1	5
P5					0	4
P6						0

Q.18 [2 marks · NAT (numerical answer)]

A naive Bayes classifier is trained on the 10 examples below, using maximum-likelihood estimates (no smoothing). For the new instance (Outlook = Overcast, Temp = Hot, Wind = Strong), the posterior probability $P(\text{Play} = \text{Yes} | x)$ is _____ (rounded off to two decimal places)

#	Outlook	Temp	Wind	Play
1	Sunny	Mild	Strong	No
2	Sunny	Mild	Strong	No
3	Overcast	Cool	Strong	No
4	Rain	Cool	Strong	No
5	Rain	Mild	Strong	No
6	Rain	Cool	Weak	Yes
7	Overcast	Hot	Strong	No
8	Overcast	Hot	Strong	Yes
9	Rain	Cool	Weak	Yes
10	Rain	Cool	Strong	Yes

Q.19 [2 marks · NAT (numerical answer)]

A simple linear regression fitted by ordinary least squares gives $\hat{y} = 9 - 5.5x$. The feature x was measured in kilometres. If x is re-expressed in metres (so $x' = 1000x$) and the model is refitted, the new slope is _____ (rounded off to 4 decimal places)

Q.20 [2 marks · NAT (numerical answer)]

The labelled data below are linearly separable through the origin by the unit vector $u = (1, -1)/\sqrt{2}$ (i.e. $y_i = \text{sign}(u \cdot x_i)$). Take $\gamma = \min_i y_i(u \cdot x_i)$ and $R = \max_i \|x_i\|$. The perceptron convergence theorem (no bias, $w_0 = 0$, $\eta = 1$) bounds the number of mistakes by $(R/\gamma)^2$. The value of this bound is _____ (rounded off to two decimal places)

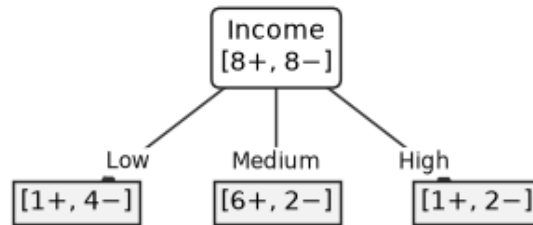
i	1	2	3	4	5
x_i	$(-1, 1)$	$(3, -3)$	$(3, 4)$	$(-1, -4)$	$(1, -3)$
y_i	-1	1	-1	1	1

Q.21 [2 marks · NAT (numerical answer)]

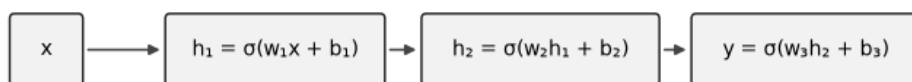
Gradient descent with (heavy-ball) momentum minimises $f(w) = (2/2)w^2$ using $v_t = \beta v_{t-1} + \nabla f(w_{t-1})$ and $w_t = w_{t-1} - \eta v_t$, with $w_0 = -2$, $v_0 = 0$, $\eta = 0.1$, $\beta = 0.9$. The value of w_3 is _____ (rounded off to three decimal places)

Q.22 [2 marks · NAT (numerical answer)]

A node with 16 training records (class counts [positive+, negative-]) is split on the attribute Income as shown. Compute the information gain (in bits) of splitting on Income. (rounded off to two decimal places)

**Q.23 [2 marks · NAT (numerical answer)]**

Consider the scalar chain shown below with weights $w_1 = -1$, $w_2 = -2$, $w_3 = 2$, biases $b_1 = 0.5$, $b_2 = 0$, $b_3 = -0.5$ and activation the logistic sigmoid σ at every stage. For input $x = 1$, the value of $\partial y / \partial w_1$ is _____ (rounded off to 4 decimal places)

**Q.24 [2 marks · NAT (numerical answer)]**

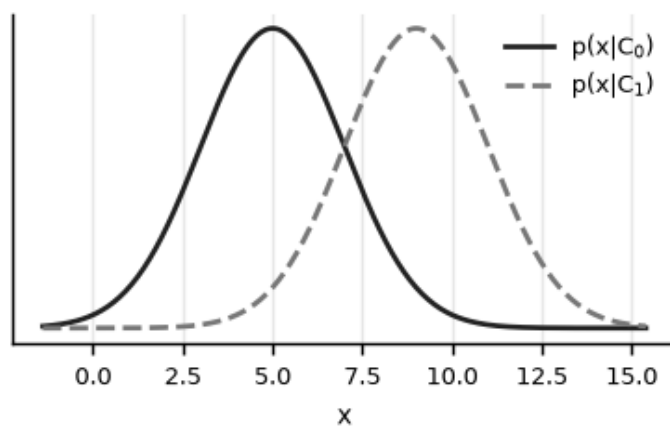
A logistic regression classifier has $P(y = 1 | \mathbf{x}) = \sigma(-2 + (-2)x_1 + (1)x_2)$. An input is labelled $y = 1$ if and only if $P(y = 1 | \mathbf{x}) \geq 0.1$. For this rule, the perpendicular distance of the decision boundary from the origin is _____ (rounded off to two decimal places)

Q.25 [2 marks · NAT (numerical answer)]

An ensemble combines 3 binary classifiers by majority vote. Each classifier is correct with probability 0.7, independently of the others. Compute the error rate of the majority-vote ensemble. (rounded off to three decimal places)

Q.26 [2 marks · NAT (numerical answer)]

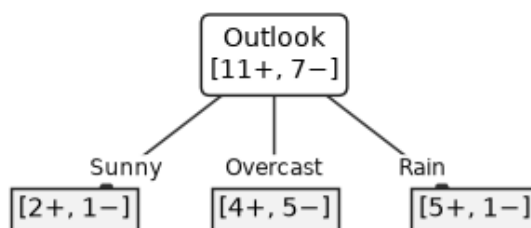
In a two-class problem with one feature, $p(x | C_0) = N(5, 4)$ and $p(x | C_1) = N(9, 4)$ (second argument = variance), with priors $P(C_0) = 0.7$ and $P(C_1) = 0.3$. Under 0-1 loss, the Bayes-optimal (minimum-error) decision threshold x^* is _____ (rounded off to two decimal places)



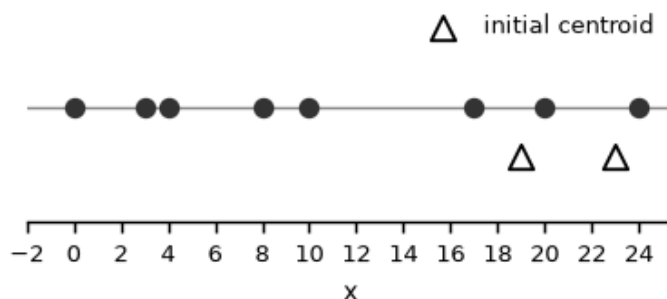
Class-conditional densities (not weighted by priors).

Q.27 [2 marks · NAT (numerical answer)]

A CART node with 18 training records (class counts [positive+, negative−]) is split on Outlook as shown. Compute the reduction in Gini index (Gini gain) obtained by splitting on Outlook. (rounded off to two decimal places)

**Q.28 [2 marks · NAT (numerical answer)]**

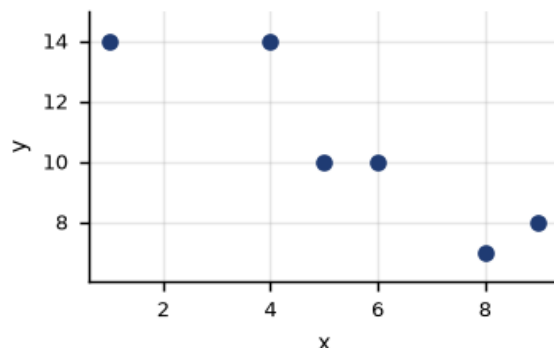
The 1-D dataset $\{0, 3, 4, 8, 10, 17, 20, 24\}$ is clustered with k-means ($k = 2$) using initial centroids 19 and 23 (shown as triangles). Lloyd's iterations (assign to nearest centroid, then recompute means) are run until convergence. Then the **larger** of the two final centroids is _____ (rounded off to two decimal places)



Q.29 [2 marks • NAT (numerical answer)]

A simple linear regression (with intercept) is fitted by least squares to the 6 points below. Using leave-one-out cross-validation, the leave-one-out residual $y_i - \hat{y}_{(-i)}(x_i)$ for the point (4, 14) — where $\hat{y}_{(-i)}$ is the line fitted without that point — is _____ (rounded off to two decimal places)

x	1	4	5	6	8	9
y	14	14	10	10	7	8

**Q.30 [2 marks • NAT (numerical answer)]**

A linear model $\hat{y} = w_1x_1 + w_2x_2 + b$ is trained by stochastic gradient descent (batch size 1) on the per-example loss $\frac{1}{2}(\hat{y} - y)^2$ with learning rate $\eta = 0.2$, starting from $w = (-1, 0)$, $b = 1$. The two examples below are processed once, in the given order. After both updates, the value of w_2 is _____ (rounded off to three decimal places)

example	x_1	x_2	y
1	0	2	6
2	2	0	-1

Q.31 [2 marks • NAT (numerical answer)]

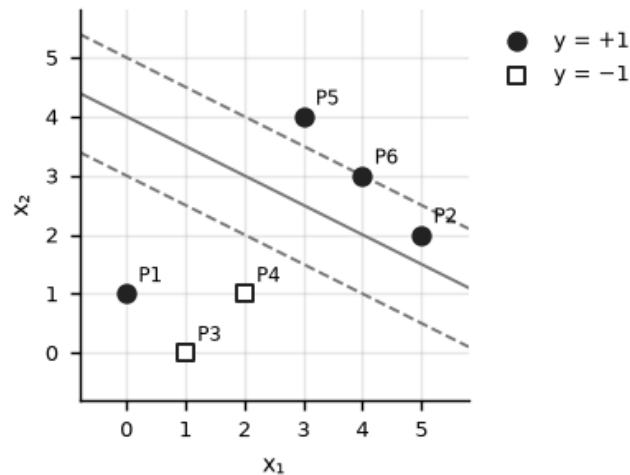
Agglomerative hierarchical clustering with **complete linkage** is applied to 5 points whose pairwise distance matrix is given below (upper triangle shown). The height (linkage distance) at which the **second** merge takes place is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5
P1	0	3	8	5	6
P2		0	5	8	7
P3			0	5	4
P4				0	1
P5					0

Q.32 [2 marks · NAT (numerical answer)]

A linear SVM with $f(x) = 0.5x_1 + x_2 - 4$ is evaluated on the training points in the table (also plotted, with the boundary $f = 0$ solid and $f = \pm 1$ dashed). Compute the total hinge loss $\sum_i \max(0, 1 - y_i f(x_i))$. (rounded off to two decimal places)

Point	x_1	x_2	y
P1	0	1	+1
P2	5	2	+1
P3	1	0	-1
P4	2	1	-1
P5	3	4	+1
P6	4	3	+1

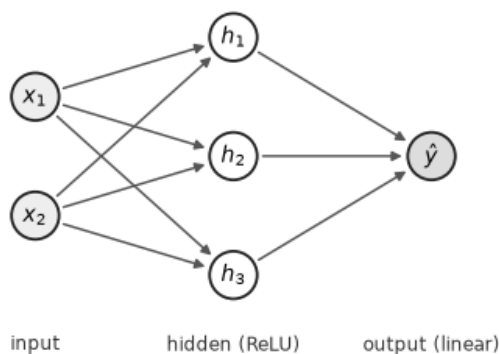
**Q.33 [2 marks · NAT (numerical answer)]**

The confusion matrix of a 3-class classifier is given below (rows = actual class, columns = predicted class). The micro-averaged F1-score is _____ (rounded off to two decimal places)

actual \ predicted	A	B	C
A	7	0	5
B	2	12	3
C	4	3	6

Q.34 [2 marks · NAT (numerical answer)]

The 2-3-1 network below computes $\hat{y} = v \cdot \text{ReLU}(Wx + b) + c$ with W , b , v given below and $c = -1$. With $x = (2, -2)$, target $y = -2$ and loss $L = \frac{1}{2}(\hat{y} - y)^2$, the value of $\partial L / \partial W_{22}$ is _____ (Answer in integer)



$$W = \begin{bmatrix} -2 & 2 \\ 0 & -1 \\ -2 & -1 \end{bmatrix}$$

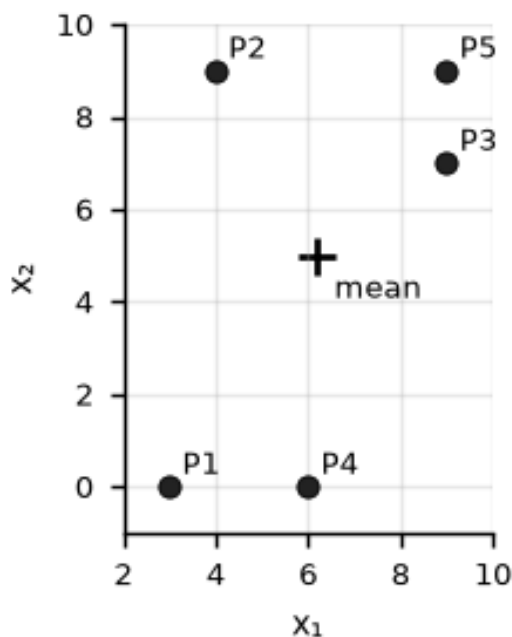
$$b = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

$$v^T = \begin{bmatrix} -2 & 1 & -1 \end{bmatrix}$$

Q.35 [2 marks · NAT (numerical answer)]

PCA is applied to the five 2-D points below (data are mean-centered first; sample covariance uses divisor $n - 1$). If each point is reconstructed from its PC1 score only ($\hat{x} = \bar{x} + z_1 v_1$), the squared reconstruction error $\|x - \hat{x}\|^2$ of P5 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5
x_1	3	4	9	6	9
x_2	0	9	7	0	9



— END OF QUESTION PAPER —

Mock Test 05 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MSQ	1	(C), (D)	19	NAT	2	−0.0056 to −0.0054
2	MCQ	1	(C)	20	NAT	2	49.9 to 50.1
3	NAT	1	22.3233 to 22.3433	21	NAT	2	−0.126 to −0.122
4	MCQ	1	(C)	22	NAT	2	0.1866 to 0.2066
5	MSQ	1	(A), (B), (C), (D)	23	NAT	2	−0.0514 to −0.0504
6	MSQ	1	(D)	24	NAT	2	0.0782 to 0.0982
7	MCQ	1	(C)	25	NAT	2	0.214 to 0.218
8	NAT	1	0.0813 to 0.0853	26	NAT	2	7.8373 to 7.8573
9	NAT	1	0.4614 to 0.4814	27	NAT	2	0.0517 to 0.0717
10	MCQ	1	(D)	28	NAT	2	20.3233 to 20.3433
11	NAT	1	0.902 to 0.922	29	NAT	2	2.6791 to 2.7332
12	NAT	1	15.59 to 15.61	30	NAT	2	1.994 to 2.006
13	NAT	1	205	31	NAT	2	2.99 to 3.01
14	NAT	1	0.488 to 0.492	32	NAT	2	4.49 to 4.51
15	NAT	1	216	33	NAT	2	0.5852 to 0.6052
16	NAT	2	0.5495 to 0.5695	34	NAT	2	−6
17	NAT	2	8.99 to 9.01	35	NAT	2	1.3671 to 1.3871
18	NAT	2	0.2627 to 0.2827				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Neural Networks (MLP / Feed-forward)					11
Regression & Model Selection					11
Probabilistic & Instance-based Classifiers					11
SVM & Decision Trees					11
Unsupervised Learning					11
Total					55

Mock Test 05 — Detailed Solutions

Q.1 — Answer: (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Backpropagation]

(A) FALSE. Identical units receive identical gradients; symmetry is never broken.

(B) FALSE. Their local derivative is 0, blocking the gradient.

(C) TRUE. Reverse-mode differentiation reuses intermediate results.

(D) TRUE. Chain rule through σ .

Q.2 — Answer: (C) [MCQ, 1M · Regression & Model Selection > Simple linear regression]

All residuals lie in a narrow band except one far from 0 → an outlier that should be investigated.

Answer: **A single outlying observation, with the rest of the fit looking adequate.**

Other patterns: funnel → heteroscedasticity; curvature → missing non-linearity; isolated point → outlier; random band → adequate fit.

Q.3 — Answer: 22.3233 to 22.3433 [NAT, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Distances $|x_i - 10.3|$: $x=10$: 0.3, $x=11$: 0.7, $x=7$: 3.3, $x=14$: 3.7, $x=6$: 4.3, $x=18$: 7.7, $x=20$: 9.7.

The 3 nearest neighbours are $x = 10, 11, 7$ with $y = 18, 28, 21$.

$\hat{y} = (18 + 28 + 21)/3 = \mathbf{22.33}$.

Q.4 — Answer: (C) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

Key features of the plotted curve: Small negative linear part (slope 0.1) for $z < 0$, slope 1 for $z > 0$.

Hence $g = \mathbf{Leaky ReLU with \alpha = 0.1}$.

Softplus: $\ln(1 + e^z)$: Smooth, always positive, value $\ln 2 \approx 0.69$ at $z = 0$, $\sim z$ for large z . – does not match.

ReLU: $\max(0, z)$: Exactly 0 for $z \leq 0$, slope 1 for $z > 0$, kink at origin. – does not match.

ELU with $\alpha = 1$: Smoothly saturates to -1 for $z \rightarrow -\infty$, linear for $z > 0$, passes through origin. – does not match.

Q.5 — Answer: (A), (B), (C), (D) [MSQ, 1M · SVM & Decision Trees > SVM: dual & support vectors]

(A) TRUE. From $\partial L / \partial w = 0$ in the Lagrangian.

(B) TRUE. They contribute neither loss nor gradient.

(C) TRUE. That normalisation makes it invariant to scaling (w, b).

(D) TRUE. $\alpha_i = 0$ for all other points.

Q.6 — Answer: (D) [MSQ, 1M · Unsupervised Learning > k-means]

(A) FALSE. It tends to split large clusters and merge small ones.

(B) FALSE. It only reaches a local optimum.

(C) FALSE. That describes Gaussian mixture models fitted with EM.

(D) TRUE. Euclidean distances change non-uniformly, so assignments can change.

Q.7 — Answer: (C) [MCQ, 1M · Regression & Model Selection > Regression vs classification]

$E[(y-g)^2|x] = \text{Var}(y|x) + (E[y|x] - g)^2$, minimised at $g = E[y|x]$.

Hence the answer is **the conditional mean $E[y | x]$** .

Q.8 — Answer: 0.0813 to 0.0853 [NAT, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Add- α estimate: $(\text{count} + \alpha) / (N_c + \alpha V)$ with $V = 3$ possible values.

$= (0 + 1) / (9 + 1 \times 3) = 1/12 = \mathbf{0.083}$.

(The unsmoothed estimate would be $0/9$, i.e. zero, which would wipe out the whole posterior product.)

Q.9 — Answer: 0.4614 to 0.4814 [NAT, 1M · SVM & Decision Trees > SVM: margin & geometry]

Since $x_+ - x_- = (3, -3)$ is parallel to w , the margin width equals $\|x_+ - x_-\| = 4.2426$.

Margin $= 2/\|w\| \Rightarrow \|w\| = 2/4.2426 = \mathbf{0.47}$.

(Check: $w^T(x_+ - x_-) = 2$ and $w \parallel (x_+ - x_-)$ give the same result.)

Q.10 — Answer: (D) [MCQ, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

Odds multiply by $e^\beta = e^{0.5} = 1.6487$.

Percentage change = $100(e^\beta - 1)\% = 64.9\%$.

$100\beta\%$ is only a small- β approximation; $100e^\beta\%$ is the new odds as a percentage of the old odds, not the change.

Q.11 — Answer: 0.902 to 0.922 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

Volume of sub-cube = $e^d = r \Rightarrow e = r^{1/d} = 0.01^{1/50} = 0.91$.

Even to capture a small fraction of data, the 'local' neighbourhood must span most of each feature's range — the curse of dimensionality.

Q.12 — Answer: 15.59 to 15.61 [NAT, 1M · Unsupervised Learning > k-medoids]

Mean = $118/5 = 23.6$ (pulled towards the outlier 82).

Medoid = data point minimizing $\sum |x_i - m|$. Costs: $m=5$: 93, $m=7$: 87, $m=8$: 86, $m=16$: 94, $m=82$: 292.

The minimum is at $m = 8$ (the median), cost 86.

Answer: **15.6**. The medoid is barely affected by the outlier, unlike the mean.

Q.13 — Answer: 205 [NAT, 1M · Regression & Model Selection > Cross-validation]

LOOCV = n-fold CV: 41 fits per model $\times$ 5 models = **205**.

Q.14 — Answer: 0.488 to 0.492 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$\nabla = \partial L / \partial w + \lambda w = 0.5 + 1 \times 0.5 = 1$.

$w \leftarrow 0.5 - 0.01 \times 1 = 0.49$ (equivalently $(1 - \eta\lambda)w - \eta \partial L / \partial w$).

Q.15 — Answer: 216 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

$x^T x = (1)(1) + (-1)(-1) + (2)(2) = 6$.

$K(x, x) = (6 + 0)^3 = 6^3 = 216$.

The kernel equals the inner product $\phi(x)^T \phi(z)$ of the explicit (all-monomials-up-to-degree-d) feature maps, without forming ϕ .

Q.16 — Answer: 0.5495 to 0.5695 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

Ridge fitted values $\hat{y} = H_\lambda y$ with $H_\lambda = X(X^T X + \lambda I)^{-1} X^T$. Writing $X^T X = V D V^T$ with eigenvalues d_j^2 gives $\text{trace}(H_\lambda) = \sum d_j^2 / (d_j^2 + \lambda)$ (effective degrees of freedom).

$25/(25 + 4) + 12/(12 + 4) + 1/(1 + 4) = 1.8121$.

$25/(25 + 12) + 12/(12 + 12) + 1/(1 + 12) = 1.2526$.

Difference = $1.8121 - 1.2526 = 0.5595$ (df decreases as λ grows).

Answer: **0.56**.

Q.17 — Answer: 8.99 to 9.01 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Use complete linkage: $d(A, B) = \max_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P4\} + \{P5\}$ at height **1**. Updated: $d(\{P4, P5\}, \{P1\}) = 8$, $d(\{P4, P5\}, \{P2\}) = 6$, $d(\{P4, P5\}, \{P3\}) = 3$, $d(\{P4, P5\}, \{P6\}) = 5$.

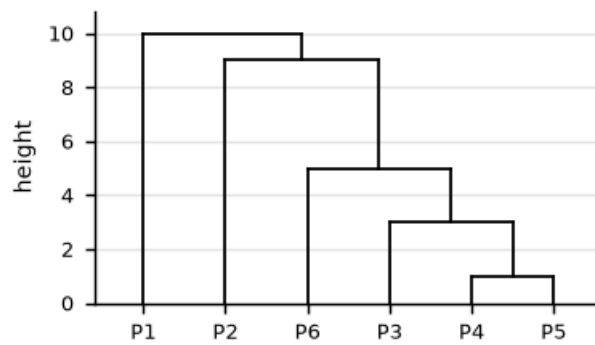
Merge 2: $\{P3\} + \{P4, P5\}$ at height **3**. Updated: $d(\{P3, P4, P5\}, \{P1\}) = 10$, $d(\{P3, P4, P5\}, \{P2\}) = 8$, $d(\{P3, P4, P5\}, \{P6\}) = 5$.

Merge 3: $\{P6\} + \{P3, P4, P5\}$ at height **5**. Updated: $d(\{P3, P4, P5, P6\}, \{P1\}) = 10$, $d(\{P3, P4, P5, P6\}, \{P2\}) = 9$.

Merge 4: $\{P2\} + \{P3, P4, P5, P6\}$ at height **9**. Updated: $d(\{P2, P3, P4, P5, P6\}, \{P1\}) = 10$.

Merge 5: $\{P1\} + \{P2, P3, P4, P5, P6\}$ at height **10**.

$P2$ and $P3$ first lie in the same cluster at the merge of height 9, so their cophenetic distance is **9** (their original distance was 8).



Q.18 — Answer: 0.2627 to 0.2827 [NAT, 2M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Naive Bayes: $P(c | x) \propto P(c) \prod_j P(x_j | c)$.

Yes: $P(\text{Play}=\text{Yes}) = 4/10$; $P(\text{Outlook}=\text{Overcast}|\text{Yes}) = 1/4$; $P(\text{Temp}=\text{Hot}|\text{Yes}) = 1/4$; $P(\text{Wind}=\text{Strong}|\text{Yes}) = 2/4$. Product = 0.0125.

No: $P(\text{Play}=\text{No}) = 6/10$; $P(\text{Outlook}=\text{Overcast}|\text{No}) = 2/6$; $P(\text{Temp}=\text{Hot}|\text{No}) = 1/6$; $P(\text{Wind}=\text{Strong}|\text{No}) = 6/6$. Product = 0.03333.

$P(\text{Yes} | x) = 0.0125 / (0.0125 + 0.03333) = \mathbf{0.27}$.

Q.19 — Answer: -0.0056 to -0.0054 [NAT, 2M · Regression & Model Selection > Basis functions & feature scaling]

$x = x'/1000 \Rightarrow \hat{y} = \beta_0 + (\beta_1/1000)x'$. Fitted values are unchanged; the slope becomes $-5.5/1000 = -0.0055$.

Answer: **-0.0055**.

Q.20 — Answer: 49.9 to 50.1 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Unit separator $u = (1, -1)/\sqrt{2}$. Margins $y_i(u \cdot x_i)$: 1.414, 4.243, 0.707, 2.121, 2.828 $\rightarrow \gamma = \min = 0.7071$.

Norms $\|x_i\|$: 1.414, 4.243, 5, 4.123, 3.162 $\rightarrow R = \max = 5$.

Novikoff bound: number of mistakes $\leq (R/\gamma)^2 = (5/0.7071)^2 = \mathbf{50}$.

Common mistakes: forgetting to normalise u (γ must be measured with $\|u\| = 1$), or using R/γ without squaring.

Q.21 — Answer: -0.126 to -0.122 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$f(w) = (2/2)w^2 \Rightarrow \nabla f(w) = 2w$. Update: $v_t = \beta v_{t-1} + \nabla f(w_{t-1})$, $w_t = w_{t-1} - \eta v_t$, with $v_0 = 0$, $\beta = 0.9$, $\eta = 0.1$.

$t = 1$: $g = 2 \times (-2) = -4$; $v_1 = 0.9 \times 0 + (-4) = -4$; $w_1 = -2 - 0.1 \times (-4) = -1.6$.

$t = 2$: $g = 2 \times (-1.6) = -3.2$; $v_2 = 0.9 \times (-4) + (-3.2) = -6.8$; $w_2 = -1.6 - 0.1 \times (-6.8) = -0.92$.

$t = 3$: $g = 2 \times (-0.92) = -1.84$; $v_3 = 0.9 \times (-6.8) + (-1.84) = -7.96$; $w_3 = -0.92 - 0.1 \times (-7.96) = -0.124$.

Answer: **-0.124**. (Plain gradient descent without momentum would give $w_3 = -1.024$.)

Q.22 — Answer: 0.1866 to 0.2066 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent $[8+, 8-]$: $H = -(8/16)\log_2(8/16) - (8/16)\log_2(8/16) = 1$.

Income = Low: $[1+, 4-]$, $H = -(1/5)\log_2(1/5) - (4/5)\log_2(4/5) = 0.7219$.

Income = Medium: $[6+, 2-]$, $H = -(6/8)\log_2(6/8) - (2/8)\log_2(2/8) = 0.8113$.

Income = High: $[1+, 2-]$, $H = -(1/3)\log_2(1/3) - (2/3)\log_2(2/3) = 0.9183$.

Weighted child entropy = $(5/16)(0.7219) + (8/16)(0.8113) + (3/16)(0.9183) = 0.8034$.

$IG = 1 - 0.8034 = 0.1966$.

Answer: **0.2**.

Q.23 — Answer: -0.0514 to -0.0504 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Activation functions]

Forward: $a_1 = -1 \cdot 1 + 0.5 = -0.5$, $h_1 = 0.3775$; $a_2 = -2 \cdot 0.3775 + 0 = -0.7551$, $h_2 = 0.3197$; $a_3 = 2 \cdot 0.3197 + -0.5 = 0.1394$, $y = 0.5348$.

Local derivatives $\sigma'(a)$: $\sigma'(a) = \sigma(a)(1 - \sigma(a))$: $\sigma'(a_1) = 0.235$, $\sigma'(a_2) = 0.2175$, $\sigma'(a_3) = 0.2488$.

Chain rule: $\partial y / \partial w_1 = \sigma'(a_3) \cdot w_3 \cdot \sigma'(a_2) \cdot w_2 \cdot \sigma'(a_1) \cdot x = -0.0509$.

Each layer contributes a factor $\sigma' \leq 0.25$ – the product shrinks with depth (vanishing gradient).

Q.24 — Answer: 0.0782 to 0.0982 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

$P(y=1|x) \geq 0.1 \Leftrightarrow z \geq c$, where $c = \ln(0.1/0.9) = -2.1972$.

Decision boundary: $-2 + (-2)x_1 + (1)x_2 = -2.1972$ (a straight line; changing the threshold only shifts it parallel to the 0.5 boundary).

distance = $|w_0 - c| / \|(w_1, w_2)\| = |-2 - -2.1972| / \sqrt{5} = 0.09$.

Q.25 — Answer: 0.214 to 0.218 [NAT, 2M · SVM & Decision Trees > Ensembles: bagging & random forests]

The vote is correct when at least 2 of 3 classifiers are correct (Binomial(3, 0.7)).

$P(\text{correct}) = C(3,2)(0.7)^2(0.3)^1 + C(3,3)(0.7)^3(0.3)^0 = 0.784$.

Answer: **0.216**. Independence is crucial — correlated classifiers gain much less.

Q.26 — Answer: 7.8373 to 7.8573 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

With equal variances, $\ln[\pi_1 p(x|C_1) / (\pi_0 p(x|C_0))] = (\mu_1 - \mu_0)x / \sigma^2 - (\mu_1^2 - \mu_0^2) / (2\sigma^2) + \ln(\pi_1 / \pi_0)$ — linear in x (LDA).

Setting it to 0: $x^* = (\mu_0 + \mu_1) / 2 + \sigma^2 \ln(\pi_0 / \pi_1) / (\mu_1 - \mu_0) = 7 + 4 \times \ln(0.7/0.3) / 4 = 7 + (0.8473) = 7.85$.

The threshold moves away from the more probable class C_0 's mean, enlarging its decision region.

Q.27 — Answer: 0.0517 to 0.0717 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [11+, 7-]: Gini = $1 - [(11/18)^2 + (7/18)^2] = 0.4753$.

Outlook = Sunny: [2+, 1-], Gini = $1 - [(2/3)^2 + (1/3)^2] = 0.4444$.

Outlook = Overcast: [4+, 5-], Gini = $1 - [(4/9)^2 + (5/9)^2] = 0.4938$.

Outlook = Rain: [5+, 1-], Gini = $1 - [(5/6)^2 + (1/6)^2] = 0.2778$.

Weighted Gini = $(3/18)(0.4444) + (9/18)(0.4938) + (6/18)(0.2778) = 0.4136$.

Gini gain = $0.4753 - 0.4136 = 0.0617$.

Answer: **0.06**.

Q.28 — Answer: 20.3233 to 20.3433 [NAT, 2M · Unsupervised Learning > k-means]

Data: {0, 3, 4, 8, 10, 17, 20, 24}; initial centroids $c_1 = 19$, $c_2 = 23$.

Iter	centroids used	cluster 1	cluster 2	new centroids
1	19, 23	{0, 3, 4, 8, 10, 17, 20}	{24}	8.86, 24
2	8.86, 24	{0, 3, 4, 8, 10}	{17, 20, 24}	5, 20.33
3	5, 20.33	{0, 3, 4, 8, 10}	{17, 20, 24}	5, 20.33

In the last row the centroids do not change, so the algorithm has converged: final centroids 5 and 20.33.

Larger final centroid = **20.33**.

Q.29 — Answer: 2.6791 to 2.7332 [NAT, 2M · Regression & Model Selection > Cross-validation]

$\bar{x} = 5.5$, $\bar{y} = 10.5$.

i	1	2	3	4	5	6
$x_i - \bar{x}$	-4.5	-1.5	-0.5	0.5	2.5	3.5
$y_i - \bar{y}$	3.5	3.5	-0.5	-0.5	-3.5	-2.5
$(x_i - \bar{x})(y_i - \bar{y})$	-15.75	-5.25	0.25	-0.25	-8.75	-8.75

$S_{xx} = \sum (x_i - \bar{x})^2 = 41.5$, $S_{xy} = -38.5$.

$$\beta_1 = S_{xy}/S_{xx} = -0.9277, \beta_0 = \bar{y} - \beta_1 \bar{x} = 15.6024.$$

For a linear smoother, the leave-one-out residual is $e_{(i)} = e_i/(1 - h_{ii})$, with $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$.

x_i	1	4	5	6	8	9
e_i	-0.675	2.108	-0.964	-0.036	-1.181	0.747
h_{ii}	0.655	0.221	0.173	0.173	0.317	0.462
$e_i/(1-h_{ii})$	-1.953	2.706	-1.165	-0.044	-1.729	1.388

Answer: **2.71** (verified by explicitly refitting without the point).

Q.30 — Answer: 1.994 to 2.006 [NAT, 2M · Regression & Model Selection > Gradient descent]

Per-example loss $\square = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial \square / \partial w = (\hat{y} - y)x$, $\partial \square / \partial b = (\hat{y} - y)$. Update after each example.

Example 1: $\hat{y} = 1$, error $\hat{y} - y = -5$; $\nabla_w = (0, -10)$, $\nabla_b = -5 \Rightarrow w = (-1, 2)$, $b = 2$.

Example 2: $\hat{y} = 0$, error $\hat{y} - y = 1$; $\nabla_w = (2, 0)$, $\nabla_b = 1 \Rightarrow w = (-1.4, 2)$, $b = 1.8$.

Answer: **2**.

Q.31 — Answer: 2.99 to 3.01 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Use complete linkage: $d(A, B) = \max_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P4\} + \{P5\}$ at height **1**. Updated: $d(\{P4, P5\}, \{P1\}) = 6$, $d(\{P4, P5\}, \{P2\}) = 8$, $d(\{P4, P5\}, \{P3\}) = 5$.

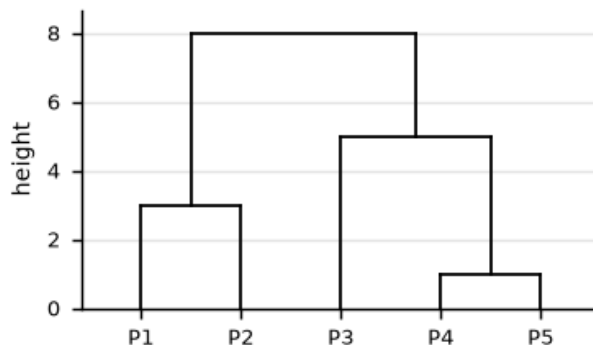
Merge 2: $\{P1\} + \{P2\}$ at height **3**. Updated: $d(\{P1, P2\}, \{P3\}) = 8$, $d(\{P1, P2\}, \{P4, P5\}) = 8$.

Merge 3: $\{P3\} + \{P4, P5\}$ at height **5**. Updated: $d(\{P3, P4, P5\}, \{P1, P2\}) = 8$.

Merge 4: $\{P1, P2\} + \{P3, P4, P5\}$ at height **8**.

The second merge occurs at height **3**.

Dendrogram:



Q.32 — Answer: 4.49 to 4.51 [NAT, 2M · SVM & Decision Trees > SVM: soft margin & hinge loss]

Compute $f(x_i)$, the functional margin $y_i f(x_i)$ and the slack/hinge loss for each point:

Point	x	y	f(x)	$y f(x)$	$\xi = \max(0, 1 - y f)$
P1	(0, 1)	+1	-3	-3	4
P2	(5, 2)	+1	0.5	0.5	0.5
P3	(1, 0)	-1	-3.5	3.5	0
P4	(2, 1)	-1	-2	2	0
P5	(3, 4)	+1	1.5	1.5	0
P6	(4, 3)	+1	1	1	0

$$\sum \xi_i = 4.5.$$

Points with $y f \geq 1$ contribute 0; points inside the margin contribute between 0 and 1; misclassified points contribute more than 1.

Q.33 — Answer: 0.5852 to 0.6052 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

class	TP	predicted total	actual total	precision	recall	F1
A	7	13	12	0.5385	0.5833	0.56

B	12	15	17	0.8	0.7059	0.75
C	6	14	13	0.4286	0.4615	0.4444

Precision of class $k = C_{kk}/(\text{column sum})$; recall $= C_{kk}/(\text{row sum})$.

Micro-averaging pools TP/FP/FN over classes; for single-label multiclass, micro-P = micro-R = micro-F1 = accuracy = $25/42 = \mathbf{0.6}$.

Q.34 — Answer: -6 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Backpropagation]

$a = Wx + b = (-8, 2, -3)$, $h = \text{ReLU}(a) = (0, 2, 0)$, $\hat{y} = v \cdot h + c = 1$; $\partial L / \partial \hat{y} = \hat{y} - y = 3$.

$\text{ReLU}'(a) = (0, 1, 0)$ (unit j is active iff $a_j > 0$).

$\partial L / \partial W_{22} = (\hat{y} - y) \cdot v_2 \cdot \text{ReLU}'(a_2) \cdot x_2 = 3 \times 1 \times 1 \times -2 = \mathbf{-6}$.

Q.35 — Answer: 1.3671 to 1.3871 [NAT, 2M · Unsupervised Learning › PCA & dimensionality reduction]

Mean $\bar{x} = (6.2, 5)$. Centered data rows: P1: $(-3.2, -5)$, P2: $(-2.2, 4)$, P3: $(2.8, 2)$, P4: $(-0.2, -5)$, P5: $(2.8, 4)$.

Sample covariance $S = (1/(n - 1)) \Sigma(x - \bar{x})(x - \bar{x})^T = [[7.7, 6.25], [6.25, 21.5]]$.

Eigenvalues: $\lambda_1 = 23.9098$, $\lambda_2 = 5.2902$; unit PC1 direction $v_1 = (0.3598, 0.933)$.

Score $z_1 = v_1 \cdot (x - \bar{x}) = 0.3598 \cdot 2.8 + 0.933 \cdot 4 = 4.7395$.

$\|x - \bar{x}\|^2 = 23.84$; error $= \|x - \bar{x}\|^2 - z_1^2 = 23.84 - 22.4629 = \mathbf{1.38}$ (Pythagoras: the residual is the PC2 component).

Mock Test 06

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	3	9	23

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
SVM & Decision Trees	7	11
Neural Networks (MLP / Feed-forward)	7	11
Regression & Model Selection	7	11
Probabilistic & Instance-based Classifiers	7	11
Unsupervised Learning	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MSQ (one or more correct)]

Which of the following statements about kernel functions (Mercer kernels) is/are CORRECT?

- (A) $k(x, z) = (xz - 1)^2$ is a valid kernel on $\mathbb{R}$.
- (B) Any function of the form $k(x, z) = \phi(x)^T \phi(z)$ is a valid kernel.
- (C) $k(x, z) = f(x)f(z)$ is valid for any real-valued f .
- (D) Mercer's condition requires every entry of the Gram matrix to be positive.

Q.2 [1 mark · MSQ (one or more correct)]

Which of the following statements about initialisation and regularisation of neural networks is/are CORRECT?

- (A) Dropout is applied at test time with freshly sampled random masks, exactly as during training.
- (B) L2 regularisation tends to increase the magnitude of the weights.
- (C) Increasing the dropout rate always reduces the training error.
- (D) Increasing the L2 penalty λ generally increases bias and reduces variance.

Q.3 [1 mark · NAT (numerical answer)]

For a data set, $\bar{x} = 7$, $\bar{y} = 30$, $s_x = 7$, $s_y = 2$ and the sample correlation between x and y is $r = -0.85$. Using the least-squares regression line of y on x , the predicted value of y at $x = 13$ is _____ (rounded off to two decimal places)

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following tasks is/are most naturally formulated as **regression** problems?

- (A) Estimating a patient's systolic blood pressure from age and BMI
- (B) Identifying the language (Hindi, Tamil, English, ...) of a sentence
- (C) Assigning a news article to one of: sports, politics, business
- (D) Predicting the number of minutes a user will spend on an app tomorrow

Q.5 [1 mark · NAT (numerical answer)]

A hard-margin SVM has exactly three support vectors with the optimal dual variables shown. The component w_1 of the optimal weight vector is _____ (rounded off to two decimal places)

i	x_i	y_i	α_i
1	(4, -2)	-1	0.25
2	(0, -1)	-1	1.25
3	(3, 1)	+1	1.5

Q.6 [1 mark · NAT (numerical answer)]

Fisher's multi-class LDA is applied to data with $K = 8$ classes and $p = 3$ features. The maximum number of discriminant directions (non-zero generalised eigenvalues of $S_W^{-1}S_B$) is _____ (Answer in integer)

Q.7 [1 mark · NAT (numerical answer)]

Regressing predictor x_3 on all the other predictors gives $R^2 = 0.61$. The VIF of x_3 is _____ (rounded off to two decimal places)

Q.8 [1 mark · MSQ (one or more correct)]

Which of the following statements about classifier evaluation is/are CORRECT?

- (A) If 99% of test examples are negative, a classifier that always predicts negative has 99% accuracy but zero recall.
- (B) Macro-averaging weights each class by its number of examples.
- (C) Applying the transformation $s \rightarrow s^3$ to the scores changes the ROC AUC.
- (D) At fixed TPR and FPR, precision does not depend on the class prevalence.

Q.9 [1 mark · NAT (numerical answer)]

A 4-class softmax classifier produces the logits $z_1 = 3, z_2 = -1, z_3 = 1, z_4 = -1$ for an input $\mathbf{x}$. The predicted probability of class 1, $P(y = 1 | \mathbf{x})$, is _____ (rounded off to two decimal places)

Q.10 [1 mark · NAT (numerical answer)]

The Gaussian kernel is $K(\mathbf{x}, \mathbf{z}) = \exp(-\|\mathbf{x} - \mathbf{z}\|^2 / (2\sigma^2))$ with $\sigma = 2$. For $\mathbf{x} = (-2, 1, 0)$ and $\mathbf{z} = (-1, 2, -1)$, the value of $K(\mathbf{x}, \mathbf{z})$ is _____ (rounded off to three decimal places)

Q.11 [1 mark · MSQ (one or more correct)]

Which of the following statements about activation functions is/are CORRECT? (σ denotes the logistic sigmoid.)

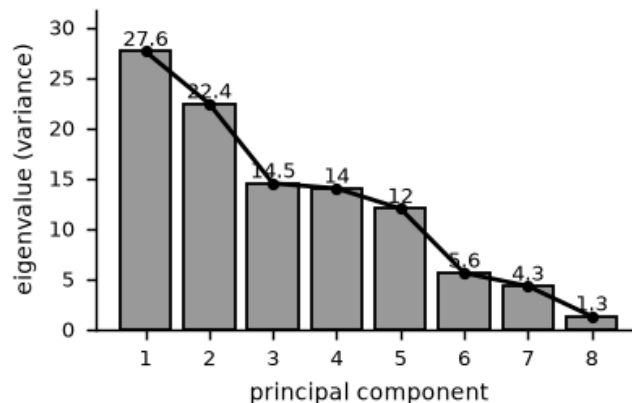
- (A) Softmax is invariant to multiplying all logits by the same positive constant.
- (B) $\tanh$ outputs lie in the interval $(0, 1)$.
- (C) Dividing the logits by a temperature $T > 1$ makes the softmax distribution more uniform.
- (D) $\tanh$ outputs lie in $(-1, 1)$ and are zero-centred.

Q.12 [1 mark · NAT (numerical answer)]

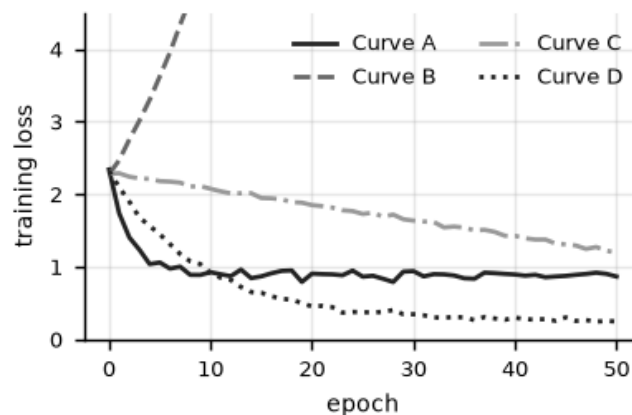
In one iteration of PAM (k-medoids) on $n = 98$ points with $k = 6$, every (medoid, non-medoid) pair is considered as a candidate swap. The number of candidate swaps is _____ (Answer in integer)

Q.13 [1 mark · NAT (numerical answer)]

The scree plot below shows the eigenvalues of the covariance matrix of a 8-dimensional dataset. The minimum number of principal components needed to explain at least 70% of the total variance is _____ (Answer in integer)

**Q.14 [1 mark · MCQ]**

The training-loss curves below were obtained with four different learning rates (everything else fixed). Which curve corresponds to a learning rate that is far too high (training diverges)?

**(A)** Curve A**(B)** Curve B**(C)** Curve C**(D)** Curve D**Q.15 [1 mark · MSQ (one or more correct)]**

Which of the following statements about hierarchical clustering is/are CORRECT?

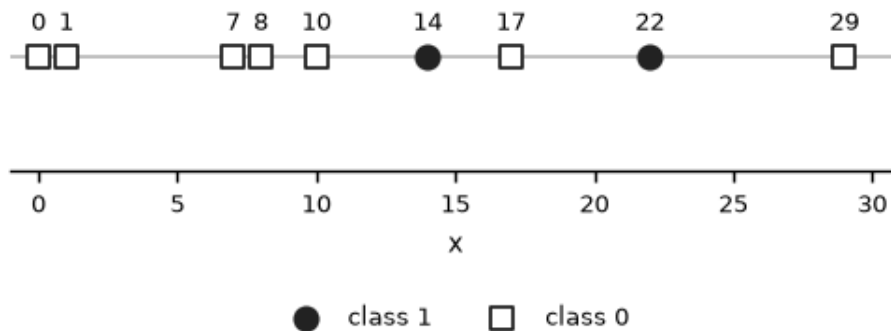
- (A)** Average linkage defines the distance between clusters as the distance between their medoids.
- (B)** A merge made in agglomerative clustering can be undone in a later step.
- (C)** For single, complete and average linkage, successive merge heights are non-decreasing.
- (D)** The agglomerative dendrogram depends on a random initialization.

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

$X_1, \dots, X_n$ are i.i.d. with mean $\mu = 4$ and variance $\sigma^2 = 4$, with $n = 5$. Consider the shrinkage estimator $\hat{\mu} = c \cdot \bar{x}$ of μ , where $\bar{x}$ is the sample mean. Using $\text{MSE} = \text{Bias}^2 + \text{Variance}$, $\text{MSE}(\bar{x}) - \text{MSE}(0.5 \cdot \bar{x})$ is (a negative value means shrinkage is worse) _____ (rounded off to three decimal places)

Q.17 [2 marks · NAT (numerical answer)]

9 one-dimensional points with binary labels are shown on the number line (filled circle = class 1, open square = class 0). A 1-NN classifier (absolute distance, majority vote) is evaluated by leave-one-out cross-validation: each point is classified using all the *other* points. The number of training points misclassified in leave-one-out cross-validation is _____ (Answer in integer)

**Q.18 [2 marks · MSQ (one or more correct)]**

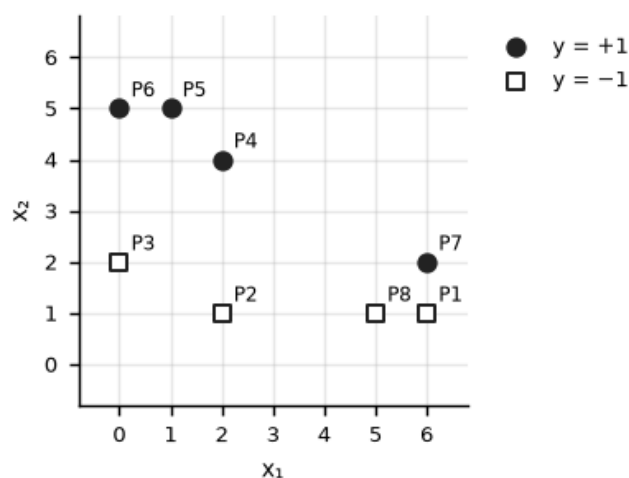
Each option shows a 2×2 matrix $[[K(x_1, x_1), K(x_1, x_2)], [K(x_2, x_1), K(x_2, x_2)]]$. Which of them can be the Gram matrix of some valid (Mercer) kernel $\tilde{K}$ on two points x_1, x_2 ?

- (A) $[[8, -10], [-10, 7]]$
- (B) $[[4, -2], [-2, 1]]$
- (C) $[[6, 1], [1, 3]]$
- (D) $[[2, 3], [3, 5]]$

Q.19 [2 marks • MCQ]

Four candidate linear decision boundaries are proposed for the labelled training points shown. Which boundary separates the two classes with the LARGEST geometric margin (minimum distance from the boundary to any training point, all points correctly classified)?

Point	x_1	x_2	y
P1	6	1	-1
P2	2	1	-1
P3	0	2	-1
P4	2	4	+1
P5	1	5	+1
P6	0	5	+1
P7	6	2	+1
P8	5	1	-1



- (A) $x_1 + 2x_2 - 9 = 0$
 (B) $x_1 + 2x_2 - 9.5 = 0$
 (C) $x_1 + 2x_2 - 8.5 = 0$
 (D) $2x_1 - x_2 - 3 = 0$

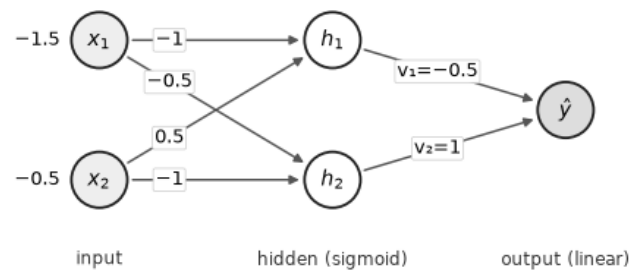
Q.20 [2 marks • NAT (numerical answer)]

The sample covariance matrix of a two-feature dataset is S given below. In principal component analysis, the variance of the data along the second principal component is _____ (rounded off to two decimal places)

$$S = \begin{bmatrix} 12 & -6 \\ -6 & 5 \end{bmatrix}$$

Q.21 [2 marks · NAT (numerical answer)]

In the network below, hidden units are sigmoid, the output is linear ($\hat{y} = v_1 h_1 + v_2 h_2$), all biases are 0, and the loss is $L = \frac{1}{2}(\hat{y} - y)^2$. For input $x = (-1.5, -0.5)$ and target $y = 1.5$, the value of $\partial L / \partial w_{11}$ (the weight from x_1 to h_1) is _____ (rounded off to three decimal places)

**Q.22 [2 marks · NAT (numerical answer)]**

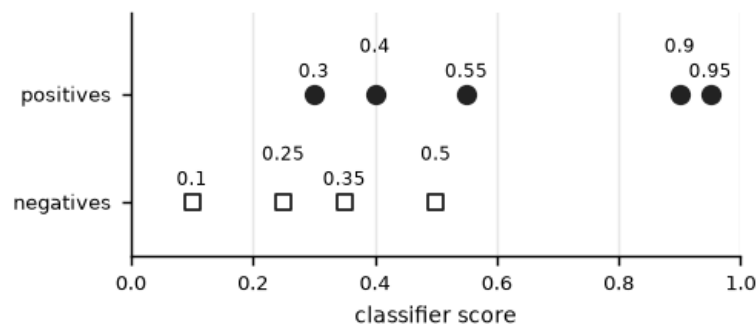
A logistic-regression neuron $p = \sigma(w_1 x_1 + w_2 x_2 + b)$ has $w = (-1, 0.5)$, $b = 0$ and is trained with binary cross-entropy loss. After one gradient-descent step using the **average** loss over both examples (a mini-batch of size 2) with learning rate $\eta = 0.1$, the new value of b is _____ (rounded off to 4 decimal places)

example	x_1	x_2	y
1	1	-1.5	1
2	-0.5	0.5	1

Q.23 [2 marks · NAT (numerical answer)]

A classifier assigns the scores below to 9 test examples (5 positives, label 1; 4 negatives, label 0). The area under the ROC curve (AUC) is _____ (rounded off to two decimal places)

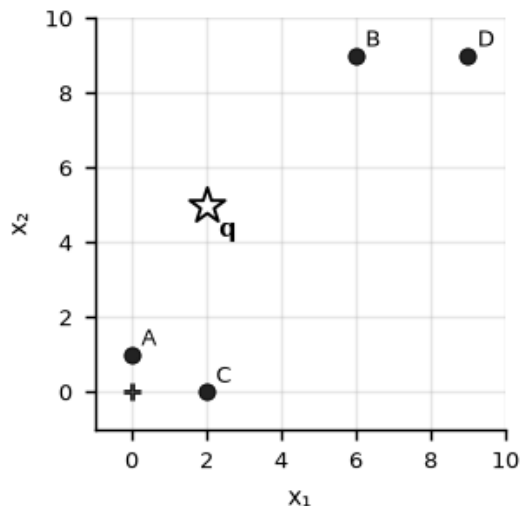
example	1	2	3	4	5	6	7	8	9
score	0.35	0.55	0.4	0.25	0.1	0.5	0.9	0.3	0.95
label	0	1	1	0	0	0	1	1	1



Q.24 [2 marks · MCQ]

A query point $q = (2, 5)$ and four points A-D are shown below. Using **Manhattan (L_1) distance**, which point is nearest to q ?

Point	A	B	C	D
x_1	0	6	2	9
x_2	1	9	0	9



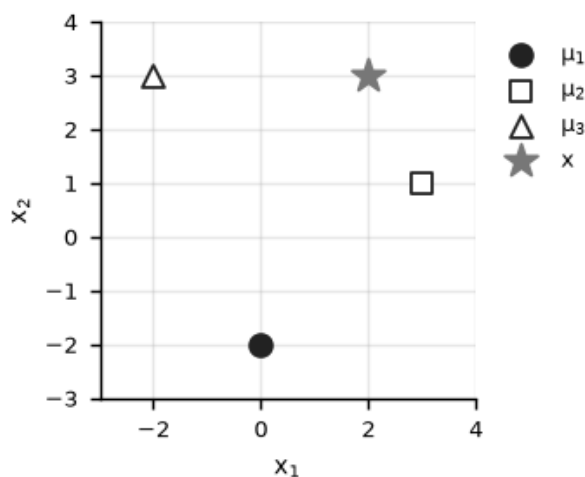
- (A) Point A
(C) Point C

- (B) Point B
(D) Point D

Q.25 [2 marks · NAT (numerical answer)]

An LDA classifier for 3 classes has shared covariance matrix Σ (below), class means $\mu_1 = (0, -2)^T$; $\mu_2 = (3, 1)^T$; $\mu_3 = (-2, 3)^T$, and priors $\pi_1 = 0.25$, $\pi_2 = 0.25$, $\pi_3 = 0.5$. The linear discriminant is $\delta_k(\mathbf{x}) = \mathbf{x}^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln \pi_k$ (natural log). For $\mathbf{x} = (2, 3)^T$, the value of the discriminant function $\delta_1(\mathbf{x})$ is _____ (rounded off to two decimal places)

$$\Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$



Q.26 [2 marks · NAT (numerical answer)]

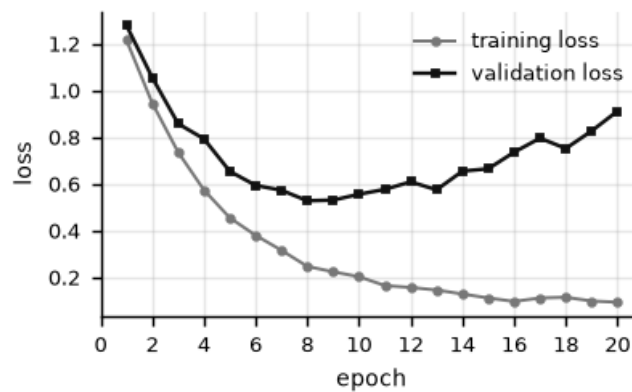
Ridge regression without intercept minimises $\|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2$ with $\lambda = 3$, for the data below ($\beta = (\beta_1, \beta_2)^T$). The value of $\|\beta\|_2^2$ is _____ (rounded off to three decimal places)

$$X = \begin{bmatrix} 3 & -1 \\ 3 & 3 \\ 1 & -1 \end{bmatrix}$$

$$y = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}$$

Q.27 [2 marks · NAT (numerical answer)]

A network is trained with early stopping on validation loss using patience $P = 4$: training stops as soon as the validation loss has failed to improve on its best value for 4 consecutive epochs, and the weights from the best epoch are restored. Using the validation losses below (also plotted), the epoch whose weights are returned (restored) as the final model is _____ (Answer in integer)

**Q.28 [2 marks · NAT (numerical answer)]**

Consider the model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ (first column of X is the intercept column), fitted by least squares using $\beta = (X^T X)^{-1} X^T y$ with the design matrix X and response y below. Then the predicted response at $(x_1, x_2) = (-0.5, 2)$ is _____ (rounded off to two decimal places)

$$X = \begin{bmatrix} 1 & -2 & 2 \\ 1 & -1 & -1 \\ 1 & 0 & -2 \\ 1 & 1 & -1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$y = \begin{bmatrix} -4 \\ 1 \\ 5 \\ 7 \\ 2 \end{bmatrix}$$

Q.29 [2 marks · MSQ (one or more correct)]

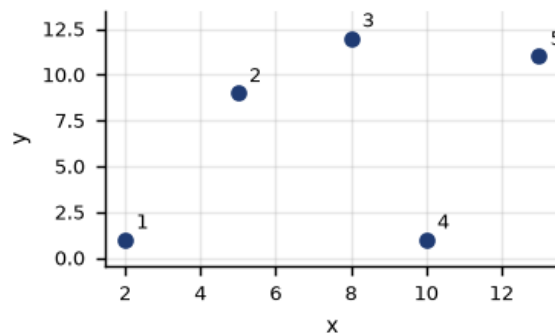
A threshold unit outputs 1 if $w_1x_1 + w_2x_2 + b > 0$ and 0 otherwise, for binary inputs $x_1, x_2 \in \{0, 1\}$. Which of the following parameter settings make the unit implement the Boolean **OR** function?

- (A) $w_1 = 2, w_2 = 3, b = -1$
- (B) $w_1 = 2.5, w_2 = 2, b = -0.5$
- (C) $w_1 = 1, w_2 = 2.5, b = -0.5$
- (D) $w_1 = 1.5, w_2 = 2, b = 0.5$

Q.30 [2 marks · NAT (numerical answer)]

Consider the 5 training points (x_i, y_i) shown below. Two-fold cross-validation is used with fold 1 = {points 1, 3, 5, ...} and fold 2 = {points 2, 4, ...} (points numbered in increasing x). The model predicts the mean of the training-fold targets. The pooled 2-fold CV MSE (over all n points) is _____ (rounded off to two decimal places)

x	2	5	8	10	13
y	1	9	12	1	11

**Q.31 [2 marks · MSQ (one or more correct)]**

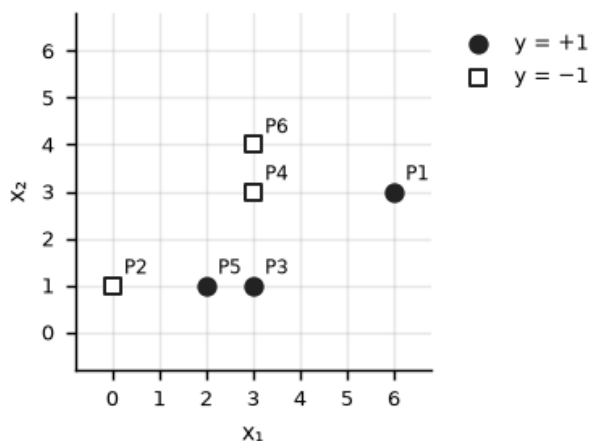
Consider agglomerative hierarchical clustering on n points with a dissimilarity matrix (no ties). Which of the following statements is/are CORRECT?

- (A) Complete-linkage merge heights equal the edge weights of a minimum spanning tree.
- (B) Cutting the single-linkage dendrogram into k clusters is equivalent to deleting the $k - 1$ heaviest edges of an MST.
- (C) Replacing every dissimilarity by its square does not change the single-linkage merge order.
- (D) The single-linkage merge heights are exactly the edge weights of a minimum spanning tree of the complete distance graph.

Q.32 [2 marks · NAT (numerical answer)]

A hard-margin linear SVM is trained on the points below. Solving the dual problem gives the Lagrange multipliers α_i listed in the table. Using the KKT conditions, compute the margin width $2/\|w\|$. (rounded off to two decimal places)

Point	x_1	x_2	y	α_i
P1	6	3	+1	0
P2	0	1	-1	0.875
P3	3	1	+1	0
P4	3	3	-1	0.75
P5	2	1	+1	1.625
P6	3	4	-1	0

**Q.33 [2 marks · NAT (numerical answer)]**

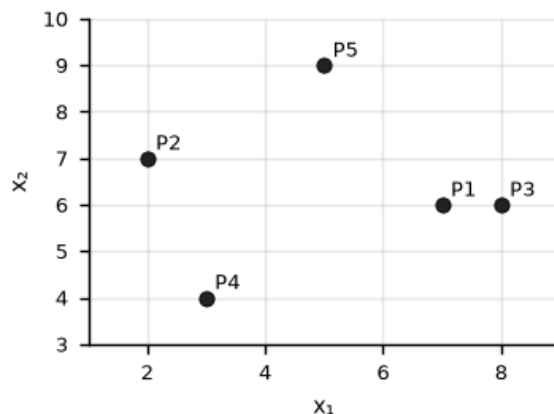
For an input x , a classifier's posterior probabilities are $P(C_1|x) = 0.2$, $P(C_2|x) = 0.3$, $P(C_3|x) = 0.5$. The loss of taking action α_i (deciding C_i) when the true class is C_j is given below. Then the index i of the Bayes-optimal (minimum-risk) action α_i is ____ (Answer in integer)

loss $L(\alpha_i, C_j)$	true C_1	true C_2	true C_3
α_1 : decide C_1	0	9	4
α_2 : decide C_2	6	0	2
α_3 : decide C_3	8	6	0

Q.34 [2 marks · NAT (numerical answer)]

A cluster contains the 5 points below. Distances are Manhattan (L_1). The medoid is the point that minimizes the sum of distances to all points of the cluster. The minimum total distance (cost of the medoid) is _____ (Answer in integer)

Point	P1	P2	P3	P4	P5
x_1	7	2	8	3	5
x_2	6	7	6	4	9

**Q.35 [2 marks · NAT (numerical answer)]**

An ensemble combines 3 binary classifiers by majority vote. Each classifier is correct with probability 0.8, independently of the others. Compute the probability that the majority vote is correct. (rounded off to three decimal places)

— END OF QUESTION PAPER —

Mock Test 06 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MSQ	1	(B), (C)	19	MCQ	2	(A)
2	MSQ	1	(D)	20	NAT	2	1.5438 to 1.5638
3	NAT	1	28.4001 to 28.6856	21	NAT	2	-0.1463 to -0.1423
4	MSQ	1	(A), (D)	22	NAT	2	0.0575 to 0.0598
5	NAT	1	3.49 to 3.51	23	NAT	2	0.84 to 0.86
6	NAT	1	3	24	MCQ	2	(C)
7	NAT	1	2.5513 to 2.5769	25	NAT	2	-9.3963 to -9.3763
8	MSQ	1	(A)	26	NAT	2	1.4332 to 1.4418
9	NAT	1	0.8433 to 0.8633	27	NAT	2	8
10	NAT	1	0.6853 to 0.6893	28	NAT	2	-1.8629 to -1.8229
11	MSQ	1	(C), (D)	29	MSQ	2	(A), (B), (C)
12	NAT	1	552	30	NAT	2	30.049 to 30.351
13	NAT	1	4	31	MSQ	2	(B), (C), (D)
14	MCQ	1	(B)	32	NAT	2	1.0994 to 1.1194
15	MSQ	1	(C)	33	NAT	2	2
16	NAT	2	-3.403 to -3.397	34	NAT	2	18
17	NAT	2	4	35	NAT	2	0.894 to 0.898
18	MSQ	2	(B), (C), (D)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
SVM & Decision Trees					11
Neural Networks (MLP / Feed-forward)					11
Regression & Model Selection					11
Probabilistic & Instance-based Classifiers					11
Unsupervised Learning					11
Total					55

Mock Test 06 — Detailed Solutions

Q.1 — Answer: (B), (C) [MSQ, 1M · SVM & Decision Trees > SVM: kernels]

- (A) **FALSE**. For points 0 and 1 the Gram matrix $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ has determinant -1 .
 (B) **TRUE**. Gram matrix $= \Phi\Phi^T$ is PSD.
 (C) **TRUE**. Rank-one PSD Gram matrix.
 (D) **FALSE**. It requires PSD; entries may be negative (e.g. linear kernel).

Q.2 — Answer: (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

- (A) **FALSE**. Dropout is disabled at test time (deterministic prediction).
 (B) **FALSE**. It shrinks weights towards 0.
 (C) **FALSE**. More dropout adds noise and usually increases training error.
 (D) **TRUE**. Stronger shrinkage.

Q.3 — Answer: 28.4001 to 28.6856 [NAT, 1M · Regression & Model Selection > Simple linear regression]

$\beta_1 = r \cdot s_y / s_x = -0.85 \times 2/7 = -0.2429$.
 $\hat{y} = \bar{y} + \beta_1(x_0 - \bar{x}) = 30 + (-0.2429)(6) = 28.5429$ (the line passes through $(\bar{x}, \bar{y})$).
 Answer: **28.54**.

Q.4 — Answer: (A), (D) [MSQ, 1M · Regression & Model Selection > Regression vs classification]

A supervised task is *regression* when the target y is real-valued (loss such as squared error), and *classification* when y takes values in a finite set of labels (loss such as cross-entropy or 0-1 loss).

- (A) **TRUE**. The output (blood pressure in mm Hg) is a continuous real-valued quantity $\rightarrow$ regression.
 (B) **FALSE**. The output (language label) is a discrete class label $\rightarrow$ classification.
 (C) **FALSE**. The output (one of 3 categories) is a discrete class label $\rightarrow$ classification.
 (D) **TRUE**. The output (time in minutes) is a continuous real-valued quantity $\rightarrow$ regression.

Q.5 — Answer: 3.49 to 3.51 [NAT, 1M · SVM & Decision Trees > SVM: dual & support vectors]

$w = \sum_i \alpha_i y_i x_i$ (sum over support vectors).
 $w_1 = (0.25)(-1)(4) + (1.25)(-1)(0) + (1.5)(+1)(3) = \mathbf{3.5}$.
 (Full vector $w = (3.5, 3.25)$; note $\sum_i \alpha_i y_i = 0$ holds.)

Q.6 — Answer: 3 [NAT, 1M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

S_B is a sum of K outer products of $(\mu_k - \mu)$ which are linearly dependent (weighted sum is 0), so $\text{rank}(S_B) \leq K - 1 = 7$; also $\leq p = 3$.
 Maximum $= \min(K - 1, p) = \mathbf{3}$.

Q.7 — Answer: 2.5513 to 2.5769 [NAT, 1M · Regression & Model Selection > Multiple linear regression]

Variance inflation factor of predictor j : $\text{VIF}_j = 1/(1 - R_j^2)$, where R_j^2 is the R^2 from regressing x_j on all other predictors. It is the factor by which $\text{Var}(\beta_j)$ is inflated by collinearity; $\sqrt{\text{VIF}}$ inflates the standard error.

$\text{VIF} = 1/(1 - 0.61) = 2.5641$.
 Answer: **2.56**.

Q.8 — Answer: (A) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

- (A) **TRUE**. $\text{TP} = 0 \Rightarrow \text{recall } 0$.
 (B) **FALSE**. That is weighted averaging.
 (C) **FALSE**. Monotone transforms preserve the ranking.
 (D) **FALSE**. It does.

Q.9 — Answer: 0.8433 to 0.8633 [NAT, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

softmax: $P(y = k | \mathbf{x}) = e^{z_k} / \sum_j e^{z_j}$.

e^z values: 20.0855, 0.3679, 2.7183, 0.3679; sum = 23.5396.

$P(y = 1 \mid \mathbf{x}) = 20.0855/23.5396 = \mathbf{0.85}$.

Q.10 — Answer: 0.6853 to 0.6893 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

$\|\mathbf{x} - \mathbf{z}\|^2 = 3$; $2\sigma^2 = 8$.

$K = \exp(-3/8) = \mathbf{0.687}$.

Note: $K \in (0, 1]$, $K(\mathbf{x}, \mathbf{x}) = 1$, and K decreases as the points move apart or as γ grows.

Q.11 — Answer: (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

(A) FALSE. Scaling changes the sharpness (temperature); only additive shifts leave it unchanged.

(B) FALSE. $\tanh$ ranges over $(-1, 1)$.

(C) TRUE. Differences between logits shrink.

(D) TRUE. $\tanh$ is odd and bounded by ± 1 .

Q.12 — Answer: 552 [NAT, 1M · Unsupervised Learning > k-means]

There are k medoids and $n - k$ non-medoids, so $k(n - k) = 6 \cdot 92 = \mathbf{552}$ swaps (each evaluated in $O(n - k)$, giving $O(k(n - k)^2)$ per iteration, far costlier than k-means' $O(nkd)$).

Q.13 — Answer: 4 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

Total variance = $\sum \lambda = 101.7$. Cumulative proportion of variance explained:

k	1	2	3	4	5	6	7	8
λ_k	27.6	22.4	14.5	14	12	5.6	4.3	1.3
cumulative %	27.14	49.16	63.42	77.19	88.99	94.49	98.72	100

The cumulative share first reaches 70% at $k = \mathbf{4}$.

Q.14 — Answer: (B) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Typical signatures: far-too-high $\eta \rightarrow$ loss blows up; somewhat-too-high $\eta \rightarrow$ quick drop then plateau at a high loss (bouncing around the minimum); too-low $\eta \rightarrow$ slow, nearly linear decrease; good $\eta \rightarrow$ fast decrease to the lowest loss.

Curve A: a learning rate that is somewhat too high (fast initial drop, then the loss plateaus at a high value).

Curve B: a learning rate that is far too high (training diverges).

Curve C: a learning rate that is too low (very slow, almost linear decrease).

Curve D: a well-chosen learning rate.

Hence the answer is **Curve B**.

Q.15 — Answer: (C) [MSQ, 1M · Unsupervised Learning > Hierarchical clustering]

(A) FALSE. It averages all pairwise distances.

(B) FALSE. Merges are greedy and final.

(C) TRUE. These linkages are monotone (no inversions).

(D) FALSE. The procedure is deterministic (apart from tie-breaking).

Q.16 — Answer: -3.403 to -3.397 [NAT, 2M · Regression & Model Selection > Bias-variance trade-off]

$E[c\bar{x}] = c\mu = \text{Bias} = (c - 1)\mu$; $\text{Var}(c\bar{x}) = c^2\sigma^2/n$. $\text{MSE}(c) = (c - 1)^2\mu^2 + c^2\sigma^2/n$.

Here $\mu = 4$, $\sigma^2/n = 4/5 = 0.8$.

$\text{MSE}(\bar{x}) = \sigma^2/n = 0.8$ (unbiased). $\text{MSE}(0.5\bar{x}) = 4.2$. Difference = -3.4 .

Answer: $\mathbf{-3.4}$.

Q.17 — Answer: 4 [NAT, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Classify each point by its nearest neighbour(s) among the remaining points:

x	true	1 nearest (excluding itself)	LOO prediction	
0	0	1(0)	0	
1	0	0(0)	0	
7	0	8(0)	0	

8	0	7(0)	0	
10	0	8(0)	0	
14	1	17(0)	0	wrong
17	0	14(1)	1	wrong
22	1	17(0)	0	wrong
29	0	22(1)	1	wrong

Misclassified: 4 of 9. Answer: **4**.

Note: the *training* error of 1-NN is 0, but the LOOCV error is not — a point cannot be its own neighbour.

Q.18 — Answer: (B), (C), (D) [MSQ, 2M · SVM & Decision Trees · SVM: kernels]

A Gram matrix of a valid kernel must be symmetric and positive semidefinite. For a symmetric 2×2 matrix this is equivalent to: both diagonal entries ≥ 0 and determinant ≥ 0 .

(A) **FALSE**. $\det = -44 < 0 \Rightarrow$ one negative eigenvalue, not PSD.

(B) **TRUE**. symmetric, diagonal ≥ 0 , $\det = 0 \Rightarrow$ PSD (singular) — still a valid Gram matrix.

(C) **TRUE**. symmetric, diagonal entries ≥ 0 and $\det = 17 > 0 \Rightarrow$ positive definite.

(D) **TRUE**. symmetric, diagonal entries ≥ 0 and $\det = 1 > 0 \Rightarrow$ positive definite.

Q.19 — Answer: (A) [MCQ, 2M · SVM & Decision Trees · SVM: margin & geometry]

For $\mathbf{n}^T \mathbf{x} - c = 0$, the geometric margin is $\min_i y_i s(\mathbf{n}^T \mathbf{x}_i - c) / \|\mathbf{n}\|$ with orientation $s = \pm 1$ chosen so that the positive class is on the positive side; a negative value means some point is misclassified.

Hyperplane	$\ \mathbf{n}\ $	$\min_i$ distance (signed, best orientation)
$\mathbf{x}_1 + 2\mathbf{x}_2 - 9 = 0$	2.236	0.447
$\mathbf{x}_1 + 2\mathbf{x}_2 - 9.5 = 0$	2.236	0.224
$\mathbf{x}_1 + 2\mathbf{x}_2 - 8.5 = 0$	2.236	0.224
$2\mathbf{x}_1 - \mathbf{x}_2 - 3 = 0$	2.236	-3.13 (does not separate)

The largest margin is 0.447, for $\mathbf{x}_1 + 2\mathbf{x}_2 - 9 = 0$. Note that a boundary can separate the data perfectly yet pass close to some point — only the minimum distance matters.

Q.20 — Answer: 1.5438 to 1.5638 [NAT, 2M · Unsupervised Learning · PCA & dimensionality reduction]

Eigenvalues of $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$: $\lambda = (a + c)/2 \pm \sqrt{((a - c)/2)^2 + b^2}$.

$\lambda = 8.5 \pm \sqrt{(12.25 + 36)} = 8.5 \pm 6.9462 \rightarrow \lambda_1 = 15.4462, \lambda_2 = 1.5538$ (check: $\lambda_1 + \lambda_2 = \text{trace} = 17$, $\lambda_1 \lambda_2 = \det = 24$).

Eigenvector for λ_1 : $(S - \lambda_1 I)\mathbf{v} = 0 \rightarrow (-3.4462)v_1 + (-6)v_2 = 0 \rightarrow \mathbf{v} \propto (1, -0.5744)$.

$\text{PVE}(\text{PC1}) = \lambda_1 / (\lambda_1 + \lambda_2) = 90.86\%$.

Answer: **1.55**.

Q.21 — Answer: -0.1463 to -0.1423 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Backpropagation]

Forward: $\mathbf{a} = \mathbf{W}\mathbf{x} = (1.25, 1.25)$, $\mathbf{h} = \sigma(\mathbf{a}) = (0.7773, 0.7773)$, $\hat{y} = \mathbf{v} \cdot \mathbf{h} = 0.3886$.

Output error $\delta_o = \partial L / \partial \hat{y} = \hat{y} - y = -1.1114$ (linear output, $L = \frac{1}{2}(\hat{y} - y)^2$).

Hidden delta $\delta_1 = \delta_o \cdot \mathbf{v}_1 \cdot \mathbf{h}_1(1 - \mathbf{h}_1) = -1.1114 \times -0.5 \times 0.1731 = 0.0962$.

$\partial L / \partial \mathbf{w}_{11} = \delta_1 \cdot \mathbf{x}_1 = 0.0962 \times -1.5 = \mathbf{-0.1443}$.

Q.22 — Answer: 0.0575 to 0.0598 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Backpropagation]

For σ output with binary cross-entropy $L = -[y \ln p + (1 - y) \ln(1 - p)]$, $\partial L / \partial a = p - y$, so $\partial L / \partial \mathbf{w}_j = (p - y)\mathbf{x}_j$ and $\partial L / \partial b = p - y$.

Example 1: $a = -1.75$, $p = 0.148$, $p - y = -0.852$.

Example 2: $a = 0.75$, $p = 0.6792$, $p - y = -0.3208$.

Mini-batch (mean) gradient for b : -0.5864 .

$b \leftarrow 0 - 0.1 \times (-0.5864) = \mathbf{0.0586}$.

Note there is no σ' factor: it cancels with the derivative of the log loss.

Q.23 — Answer: 0.84 to 0.86 [NAT, 2M · Probabilistic & Instance-based Classifiers · Evaluation metrics]

AUC = fraction of (positive, negative) pairs in which the positive has the higher score (Mann-Whitney statistic).

For each positive, count negatives with lower score: $0.95 \rightarrow 4$; $0.9 \rightarrow 4$; $0.55 \rightarrow 4$; $0.4 \rightarrow 3$; $0.3 \rightarrow 2$.

Concordant pairs = 17 out of $5 \times 4 = 20$.

AUC = $17/20 = 0.85$.

(Equivalently, trace the ROC staircase by lowering the threshold and add the rectangle areas.)

Q.24 — Answer: (C) [MCQ, 2M · Unsupervised Learning · Distance measures]

Compute the measure from $q = (2, 5)$ to every point (cosine column shows similarity, larger = more similar):

Point	coordinates	Manhattan	Euclidean	Chebyshev	cosine similarity
A	(0, 1)	6	4.472	4	0.928
B	(6, 9)	8	5.657	4	0.979
C	(2, 0)	5	5	5	0.371
D	(9, 9)	11	8.062	7	0.919

Under Manhattan (L_1) distance the nearest to q is **point C**.

Note how the answer depends on the measure: Manhattan (L_1) distance $\rightarrow$ C; Euclidean (L_2) distance $\rightarrow$ A; Chebyshev (L_∞) distance $\rightarrow$ A; cosine similarity $\rightarrow$ B. Cosine similarity looks only at the angle from the origin, L_1 sums coordinate gaps, L_∞ takes the largest gap.

Q.25 — Answer: -9.3963 to -9.3763 [NAT, 2M · Probabilistic & Instance-based Classifiers · Linear discriminant analysis]

$$\Sigma^{-1} = \begin{bmatrix} 0.5 & 0 \\ 0 & 1 \end{bmatrix}$$

$k = 1$: $\Sigma^{-1}\mu_1 = (0, -2)$; $\mathbf{x}^T \Sigma^{-1}\mu = -6$; $\frac{1}{2}\mu^T \Sigma^{-1}\mu = 2$; $\ln \pi = -1.3863 \Rightarrow \delta_1 = -9.3863$.

$k = 2$: $\Sigma^{-1}\mu_2 = (1.5, 1)$; $\mathbf{x}^T \Sigma^{-1}\mu = 6$; $\frac{1}{2}\mu^T \Sigma^{-1}\mu = 2.75$; $\ln \pi = -1.3863 \Rightarrow \delta_2 = 1.8637$.

$k = 3$: $\Sigma^{-1}\mu_3 = (-1, 3)$; $\mathbf{x}^T \Sigma^{-1}\mu = 7$; $\frac{1}{2}\mu^T \Sigma^{-1}\mu = 5.5$; $\ln \pi = -0.6931 \Rightarrow \delta_3 = 0.8069$.

Required $\delta_3(\mathbf{x}) = -9.39$.

Q.26 — Answer: 1.4332 to 1.4418 [NAT, 2M · Regression & Model Selection · Ridge & Lasso]

$\beta_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y$ with $\lambda = 3$.

$$X^T X = \begin{bmatrix} 19 & 5 \\ 5 & 11 \end{bmatrix}$$

$$X^T X + \lambda I = \begin{bmatrix} 22 & 5 \\ 5 & 14 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 9 \\ -11 \end{bmatrix}$$

$\det = 22 \times 14 - (5)^2 = 283$; $(X^T X + \lambda I)^{-1} = (1/283)[[14, -5], [-5, 22]]$.

$\beta = (0.6396, -1.0141)$.

$\|\beta\|^2 = (0.6396)^2 + (-1.0141)^2 = 1.4375$.

Answer: **1.438**.

Q.27 — Answer: 8 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Training & regularisation]

Rule: keep track of the best (lowest) validation loss; an epoch that does not strictly improve on it increments a counter, which is reset on improvement; stop when the counter reaches patience $P = 4$.

Trace - 1: 1.279 (new best); 2: 1.055 (new best); 3: 0.858 (new best); 4: 0.793 (new best); 5: 0.654 (new best); 6: 0.594 (new best); 7: 0.573 (new best); 8: 0.528 (new best); 9: 0.531 (no improvement, count = 1); 10: 0.556 (no improvement, count = 2); 11: 0.577 (no improvement, count = 3); 12: 0.609 (no improvement, count = 4).

Training stops after epoch 12; the best validation loss 0.528 occurred at epoch 8, whose weights are restored.

Answer: **8**. The widening train/validation gap after this point is the signature of overfitting.

Q.28 — Answer: -1.8629 to -1.8229 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

The columns of X are mutually orthogonal, so $X^T X$ is diagonal and each coefficient is $\beta_j = (x_j^T y) / (x_j^T x_j)$.

$$X^T X = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 14 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 11 \\ 18 \\ -22 \end{bmatrix}$$

$$\beta = (11/5, 18/10, -22/14) = (2.2, 1.8, -1.5714).$$

$$\hat{y} = 2.2 + (1.8)(-0.5) + (-1.5714)(2) = -1.8429.$$

Answer: **-1.84**.

Q.29 — Answer: (A), (B), (C) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Target OR truth table for $(x_1, x_2) = (0,0), (0,1), (1,0), (1,1)$: 0, 1, 1, 1. For each option compute $a = w_1 x_1 + w_2 x_2 + b$ at the four inputs and output 1 iff $a > 0$.

(A) TRUE. $a = -1, 2, 1, 4 \rightarrow$ outputs 0, 1, 1, 1 - OR.

(B) TRUE. $a = -0.5, 1.5, 2, 4 \rightarrow$ outputs 0, 1, 1, 1 - OR.

(C) TRUE. $a = -0.5, 2, 0.5, 3 \rightarrow$ outputs 0, 1, 1, 1 - OR.

(D) FALSE. $a = 0.5, 2.5, 2, 4 \rightarrow$ outputs 1, 1, 1, 1 - constant 1.

Q.30 — Answer: 30.049 to 30.351 [NAT, 2M · Regression & Model Selection > Cross-validation]

Fold 1 validated with mean of fold-2 targets = 5; fold 2 validated with mean of fold-1 targets = 8.

Σ squared errors = 151; pooled MSE = $151/5 = 30.2$.

Answer: **30.2**.

Q.31 — Answer: (B), (C), (D) [MSQ, 2M · Unsupervised Learning > Hierarchical clustering]

(A) FALSE. That holds for single linkage.

(B) TRUE. The remaining forest's components are the clusters.

(C) TRUE. Single (and complete) linkage depend only on the ranks of the distances.

(D) TRUE. Kruskal's algorithm performs the same merges in the same order.

Q.32 — Answer: 1.0994 to 1.1194 [NAT, 2M · SVM & Decision Trees > SVM: dual & support vectors]

$w = \Sigma \alpha_i y_i x_i$ over points with $\alpha_i > 0$:

$$w = (0.875)(-1)(0, 1) + (0.75)(-1)(3, 3) + (1.625)(+1)(2, 1) = (1, -3/2).$$

b from any support vector, e.g. P2: $y = w^T x + b \Rightarrow b = -1 - (-1.5) = 1/2$.

Check: $\Sigma \alpha_i y_i = 0$ ($= 0$, as required).

$$\|w\| = 1.8028 \Rightarrow \text{margin} = \mathbf{1.11}.$$

Q.33 — Answer: 2 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

Conditional risk $R(\alpha_i | x) = \Sigma_j L(\alpha_i, C_j) P(C_j | x)$:

$$R(\alpha_1) = 0 \times 0.2 + 9 \times 0.3 + 4 \times 0.5 = 4.7$$

$$R(\alpha_2) = 6 \times 0.2 + 0 \times 0.3 + 2 \times 0.5 = 2.2$$

$$R(\alpha_3) = 8 \times 0.2 + 6 \times 0.3 + 0 \times 0.5 = 3.4$$

Minimum risk: α_2 with $R = 2.2$. The MAP rule picks C_3 (risk 3.4) — different, because the losses are asymmetric.

Answer: **2**.

Q.34 — Answer: 18 [NAT, 2M · Unsupervised Learning > k-medoids]

Compute the Manhattan (L_1) distance matrix and its row sums:

	P1	P2	P3	P4	P5	Σ
P1	0	6	1	6	5	18
P2	6	0	7	4	5	22
P3	1	7	0	7	6	21
P4	6	4	7	0	7	24
P5	5	5	6	7	0	23

The smallest row sum is 18 for P1, so the medoid is P1.

Answer: **18**.

Unlike a k-means centroid, the medoid is always an actual data point.

Q.35 — Answer: 0.894 to 0.898 [NAT, 2M · SVM & Decision Trees > Ensembles: bagging & random forests]

The vote is correct when at least 2 of 3 classifiers are correct (Binomial(3, 0.8)).

$$P(\text{correct}) = C(3,2)(0.8)^2(0.2)^1 + C(3,3)(0.8)^3(0.2)^0 = 0.896.$$

Answer: **0.896**. Independence is crucial — correlated classifiers gain much less.

Mock Test 07

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	5	11	19

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Unsupervised Learning	7	11
Probabilistic & Instance-based Classifiers	7	11
SVM & Decision Trees	7	11
Regression & Model Selection	7	11
Neural Networks (MLP / Feed-forward)	7	11

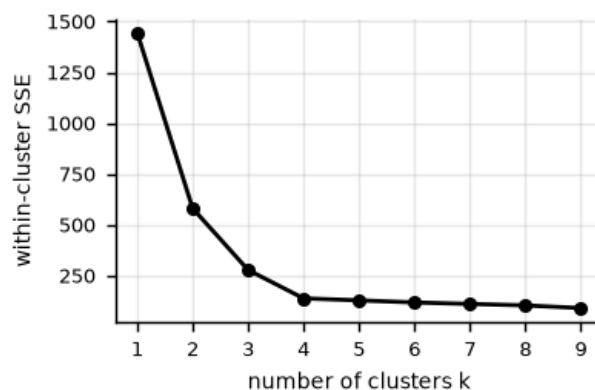
Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MCQ]

k-means is run for $k = 1, 2, \dots, 9$ on a dataset and the final within-cluster SSE is plotted below. Using the elbow method, the most appropriate number of clusters is



- (A) 9 (B) 3
(C) 5 (D) 4

Q.2 [1 mark · MSQ (one or more correct)]

Which of the following statements about Bayes-optimal classification and generative vs. discriminative models is/are CORRECT?

- (A) Increasing the prior of one class enlarges the region of input space assigned to that class by the Bayes rule.
 (B) When the misclassification costs are asymmetric, the Bayes-optimal rule need not pick the most probable class.
 (C) For binary classification with 0-1 loss, the Bayes error at x equals $\min(P(y=0|x), P(y=1|x))$.
 (D) If the class-conditional densities have non-overlapping supports, the Bayes error rate is zero.

Q.3 [1 mark · NAT (numerical answer)]

For a canonical hard-margin SVM (support vectors satisfy $y_i(w^T x_i + b) = 1$), the distance between the two margin hyperplanes is found to be 2.5. The value of $\|w\|^2$ is _____ (rounded off to two decimal places)

Q.4 [1 mark · MCQ]

In a logistic regression model for the probability of the email being spam, the coefficient of *number of links in an email* is $\beta = 0.3$. Holding all other features fixed, increasing *number of links in an email* by 2 units multiplies the **odds** of the email being spam by (approximately)

- (A) 0.6 (B) 1.82
(C) 1.35 (D) 1.6

Q.5 [1 mark · MSQ (one or more correct)]

Consider the soft-margin SVM: minimise $\frac{1}{2}\|w\|^2 + C \sum_i \xi_i$ subject to $y_i(w^T x_i + b) \geq 1 - \xi_i$, $\xi_i \geq 0$. Which of the following statements is/are CORRECT?

- (A) $\sum \xi_i$ is an upper bound on the number of misclassified training points.
(B) Increasing C penalises slack more heavily and typically gives a narrower margin with fewer margin violations.
(C) The soft-margin SVM has a solution even when the data are not linearly separable.
(D) A small C allows more margin violations and typically yields a wider margin.

Q.6 [1 mark · NAT (numerical answer)]

Nested cross-validation is performed with an outer 10-fold loop and an inner 4-fold loop. In each outer fold, the inner loop evaluates 2 hyperparameter values, after which the model with the best value is refitted once on that outer fold's training part and evaluated on its test part. The total number of model fits (no further refit at the end) is _____ (Answer in integer)

Q.7 [1 mark · MSQ (one or more correct)]

Which of the following tasks is/are most naturally formulated as **regression** problems?

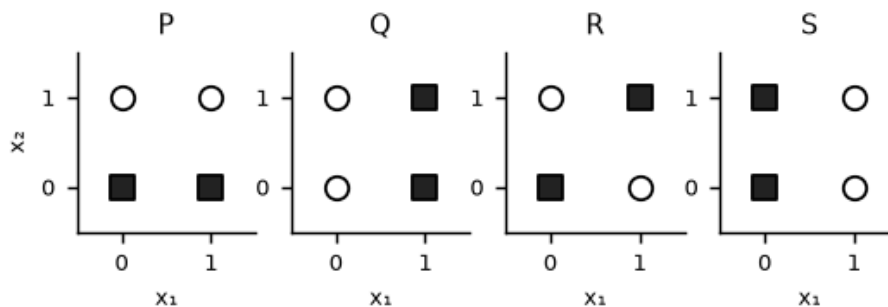
- (A) Deciding whether an X-ray shows pneumonia or not
(B) Predicting the number of minutes a user will spend on an app tomorrow
(C) Estimating the fuel efficiency (km per litre) of a car from engine specifications
(D) Estimating the PM2.5 concentration ($\mu\text{g}/\text{m}^3$) in the air for tomorrow

Q.8 [1 mark · NAT (numerical answer)]

The number of ways to partition 5 distinct points into **at most** 3 non-empty (unlabeled) clusters is _____ (Answer in integer)

Q.9 [1 mark · MCQ]

Each panel shows the four binary inputs (x_1, x_2) labelled as class 1 (filled square) or class 0 (open circle). Which dataset can NOT be classified perfectly by a single perceptron (one linear threshold unit)?

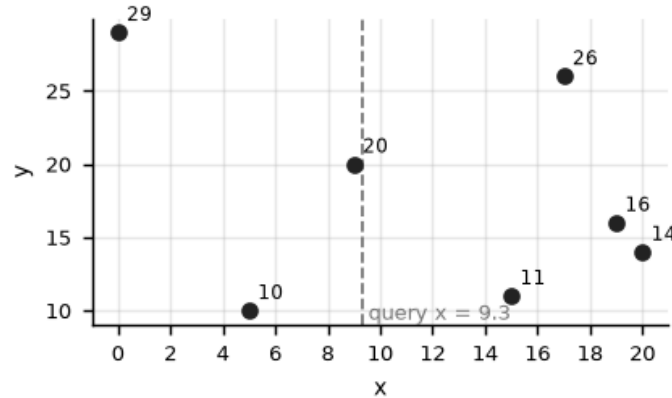


- (A) Dataset P (B) Dataset Q
(C) Dataset R (D) Dataset S

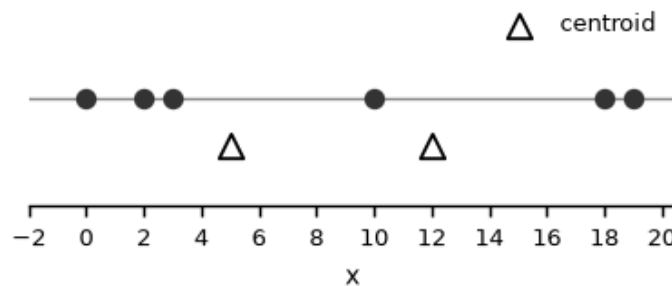
Q.10 [1 mark · NAT (numerical answer)]

The training data for a k-NN regressor are given below (also plotted). Using $k = 4$ and absolute distance $|x - x_i|$, the predicted value (unweighted average of neighbours) at $x = 9.3$ is _____ (rounded off to two decimal places)

x	0	5	9	15	17	19	20
y	29	10	20	11	26	16	14

**Q.11 [1 mark · NAT (numerical answer)]**

One-dimensional data $\{0, 2, 3, 10, 18, 19\}$ and $k = 2$ centroids at 5, 12 are given. Each point is assigned to its nearest centroid (centroids are **not** updated). The resulting within-cluster sum of squared errors $\sum (x_i - c_{c(i)})^2$ is _____ (rounded off to two decimal places)

**Q.12 [1 mark · MSQ (one or more correct)]**

A simple linear regression $\hat{y} = \beta_0 + \beta_1 x$ is fitted by ordinary least squares (with intercept) to data (x_i, y_i) , $i = 1, \dots, n$, with residuals $e_i = y_i - \hat{y}_i$ and sample correlation r . Which of the following statements is/are CORRECT?

- (A) The least-squares slope has the same sign as the sample correlation r .
- (B) The least-squares line passes through the point $(\bar{x}, \bar{y})$.
- (C) $SST = SSR + SSE$ for the least-squares fit with intercept.
- (D) For the least-squares line, $\sum e_i^2 = 0$.

Q.13 [1 mark · MSQ (one or more correct)]

Let $p = \text{softmax}(z)$ for logits $z \in \mathbb{R}^K$ and $L = -\ln p_y$ the cross-entropy loss for true class y . Which of the following is/are CORRECT?

- (A) The Jacobian of softmax is $\partial p_i / \partial z_j = p_i (\delta_{ij} - p_j)$.
- (B) For softmax + cross-entropy, $\partial L / \partial z_k$ for a wrong class k equals p_k (which is positive).
- (C) For softmax outputs with cross-entropy loss and one-hot target y , $\partial L / \partial z = p - y$.
- (D) The components of $\partial L / \partial z$ for softmax + cross-entropy sum to zero.

Q.14 [1 mark · NAT (numerical answer)]

The homogeneous polynomial kernel $K(x, z) = (x^T z)^4$ on $x \in \mathbb{R}^2$ corresponds to a feature map of all (scaled) monomials of degree exactly 4. The number of distinct such features is _____
(Answer in integer)

Q.15 [1 mark · MSQ (one or more correct)]

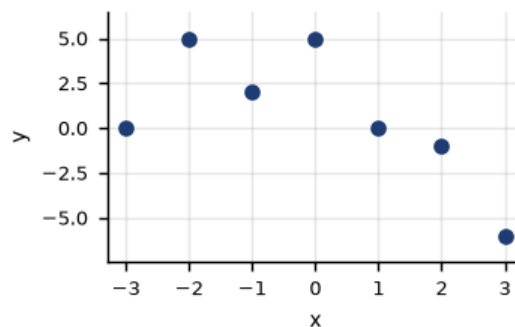
Which of the following statements about initialisation and regularisation of neural networks is/are CORRECT?

- (A) L2 regularisation tends to increase the magnitude of the weights.
- (B) Adding more (representative) training data typically increases overfitting.
- (C) An L1 penalty tends to drive some weights exactly to zero, unlike an L2 penalty.
- (D) Initialising sigmoid networks with very large random weights helps avoid vanishing gradients.

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

The quadratic model $y = a + bx + cx^2$ is fitted by least squares to the 7 points below (a polynomial / basis-function regression with basis 1, x , x^2). Then the intercept a is _____
(rounded off to two decimal places)

x	-3	-2	-1	0	1	2	3
y	0	5	2	5	0	-1	-6

**Q.17 [2 marks · MSQ (one or more correct)]**

A node with class counts [40+, 50-] can be split by either of two binary splits S1 or S2 (table). For each impurity measure, the “gain” is the parent impurity minus the weighted child impurity. Which of the following statements is/are CORRECT?

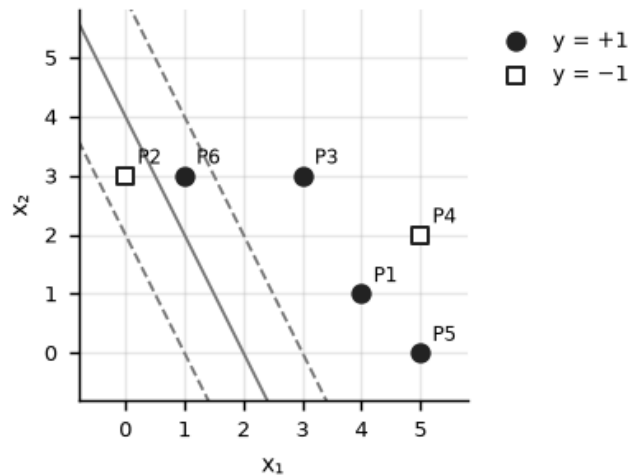
Split	Left child [+, -]	Right child [+, -]
S1	[39+, 19-]	[1+, 31-]
S2	[11+, 43-]	[29+, 7-]

- (A) Misclassification-error reduction is the same for S1 and S2.
- (B) Gini gain prefers S1 over S2.
- (C) Information gain (entropy) prefers S2 over S1.
- (D) Gini gain prefers S2 over S1.

Q.18 [2 marks · MSQ (one or more correct)]

A soft-margin linear SVM has learned $f(x) = x_1 + 0.5x_2 - 2$. The training points are listed below and plotted (boundary solid, margins $f = \pm 1$ dashed). With ξ_i the optimal slack of point i , which of the following statements is/are CORRECT?

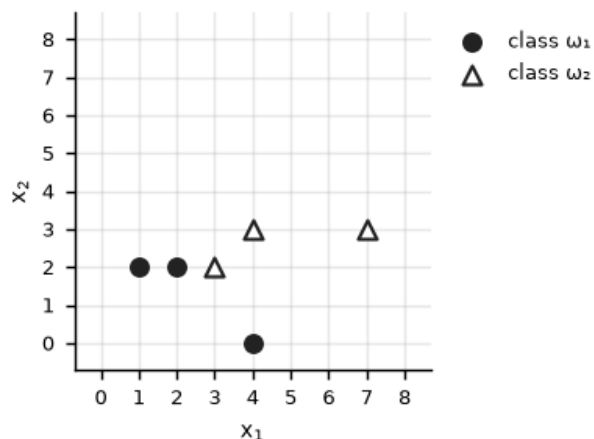
Point	x_1	x_2	y
P1	4	1	+1
P2	0	3	-1
P3	3	3	+1
P4	5	2	-1
P5	5	0	+1
P6	1	3	+1



- (A) $\sum \xi_i = 7$.
 (B) P5 is correctly classified but lies strictly inside the margin.
 (C) P4 is misclassified ($\xi_4 > 1$).
 (D) Exactly 4 of the points have $\xi_i > 0$.

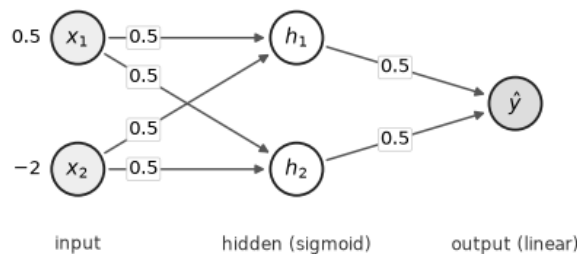
Q.19 [2 marks · NAT (numerical answer)]

Class ω_1 has training points (1, 2), (4, 0), (2, 2) and class ω_2 has points (3, 2), (4, 3), (7, 3). Let $\mathbf{m}_1, \mathbf{m}_2$ be the class means and $S_W = \sum_{k=1}^K \sum_{x \in \omega_k} (x - \mathbf{m}_k)(x - \mathbf{m}_k)^T$ the within-class scatter matrix (no division by n). Then $(\mathbf{m}_2 - \mathbf{m}_1)^T S_W (\mathbf{m}_2 - \mathbf{m}_1)$ is _____ (rounded off to two decimal places)



Q.20 [2 marks · NAT (numerical answer)]

In the 2-2-1 network below (sigmoid hidden units, linear output $\hat{y} = v_1 h_1 + v_2 h_2$, all biases 0), **every** weight is initialised to 0.5. One gradient-descent step with $\eta = 1$ is taken on $L = \frac{1}{2}(\hat{y} - y)^2$ for the input $x = (0.5, -2)$ with target $y = 2$. Then $w_{11} - w_{21}$ after the update (difference between the weights from x_1 into the two hidden units) is _____ (rounded off to 4 decimal places)

**Q.21 [2 marks · NAT (numerical answer)]**

The labelled data below are linearly separable through the origin by the unit vector $u = (4, 3)/5$ (i.e. $y_i = \text{sign}(u \cdot x_i)$). Take $\gamma = \min_i y_i(u \cdot x_i)$ and $R = \max_i \|x_i\|$. The perceptron convergence theorem (no bias, $w_0 = 0$, $\eta = 1$) bounds the number of mistakes by $(R/\gamma)^2$. The value of this bound is _____ (rounded off to two decimal places)

i	1	2	3	4	5
x_i	(0, 1)	(-3, -4)	(1, 3)	(4, -3)	(3, 4)
y_i	1	-1	1	1	1

Q.22 [2 marks · MSQ (one or more correct)]

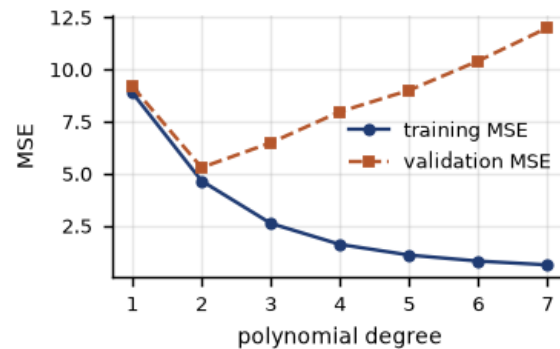
Let σ be the logistic sigmoid, and $'$ denote the derivative. At $z = 2$ (and $-z = -2$), which of the following is/are CORRECT? ($\text{softplus}(z) = \ln(1 + e^z)$.)

- (A) $\tanh(2) = 2\sigma(4) - 1$ (B) $\text{ReLU}'(2) = 0$
 (C) $\sigma'(2) \geq 0.25$ (D) $\tanh(2) + \tanh(-2) = 1$

Q.23 [2 marks · MSQ (one or more correct)]

Polynomial regression models of degree 1 to 7 are fitted by least squares on a training set and evaluated on a separate validation set. The resulting mean squared errors are given below. Which of the following statements is/are CORRECT?

Degree	1	2	3	4	5	6	7
Training MSE	8.89	4.66	2.62	1.61	1.1	0.81	0.63
Validation MSE	9.19	5.3	6.49	7.98	9.01	10.42	12.01



- (A) The degree-7 model overfits: it has the lowest training MSE but its validation MSE exceeds the minimum.
- (B) The generalisation gap (validation MSE – training MSE) is largest for degree 5.
- (C) The degree-2 model has higher variance than the degree-7 model.
- (D) Based on the validation MSE, degree 7 should be selected.

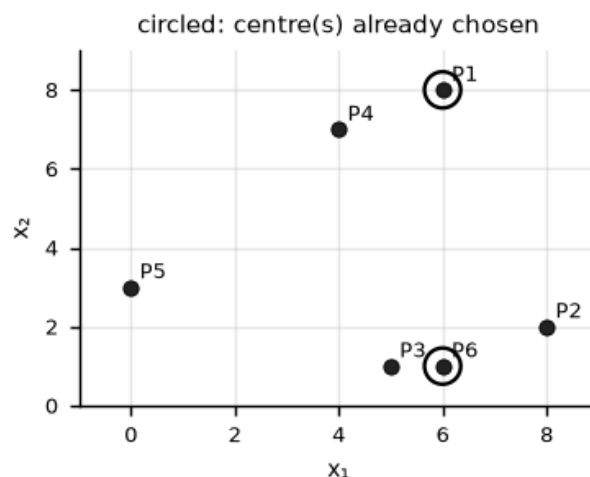
Q.24 [2 marks · NAT (numerical answer)]

A classifier for a rare condition has sensitivity (recall) 0.95 and specificity 0.98. It is deployed on a population where the prevalence of the positive class is 0.05. The expected F1-score in this population is _____ (rounded off to three decimal places)

Q.25 [2 marks · NAT (numerical answer)]

k-means++ initialization (squared Euclidean seeding) is applied to the six points below. P6 and P1 have already been chosen as centres. The probability that P4 is selected as the third centre is _____ (rounded off to three decimal places)

Point	P1	P2	P3	P4	P5	P6
x ₁	6	8	5	4	0	6
x ₂	8	2	1	7	3	1



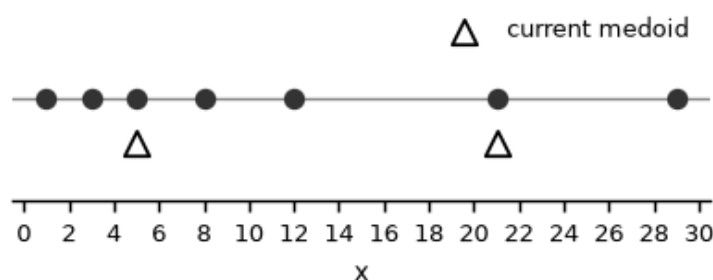
Q.26 [2 marks · MSQ (one or more correct)]

Which of the following statements about PCA, SVD and dimensionality reduction is/are CORRECT?

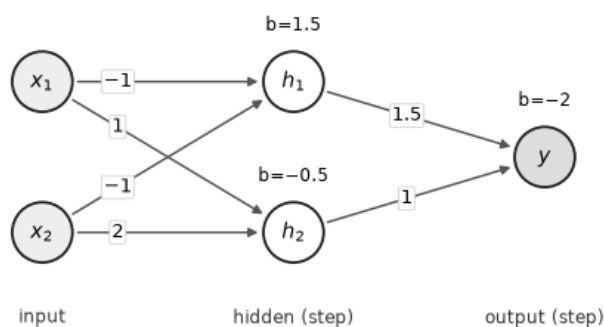
- (A) Rotating the data by an orthogonal matrix rotates the PC directions but leaves the eigenvalues unchanged.
- (B) If two features are perfectly correlated, their 2×2 covariance matrix has a zero eigenvalue.
- (C) Multiplying all features by the same non-zero constant c leaves the proportion of variance explained by each PC unchanged.
- (D) Fisher LDA for C classes yields at most $C - 1$ discriminant directions.

Q.27 [2 marks · NAT (numerical answer)]

PAM (k-medoids, $k = 2$, absolute-difference dissimilarity) is applied to the 1-D data $\{1, 3, 5, 8, 12, 21, 29\}$. The current medoids are 5 and 21. The total cost $\sum_i \min_m |x_i - m|$ of the current configuration is ____ (Answer in integer)

**Q.28 [2 marks · MCQ]**

In the network below every unit (hidden and output) is a threshold unit that outputs 1 if its weighted input plus bias is > 0 , and 0 otherwise. Edge labels are weights. For binary inputs $x_1, x_2 \in \{0, 1\}$, the network computes

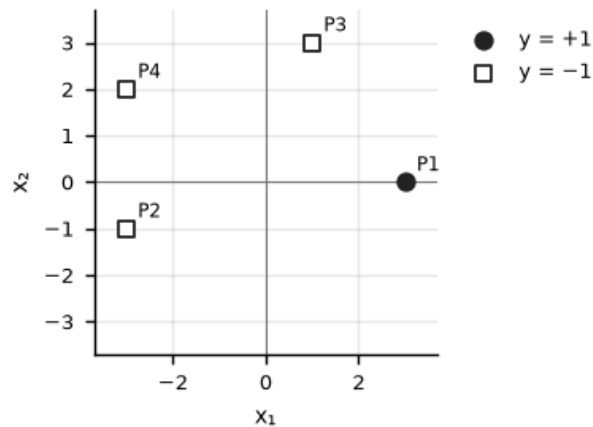


- | | |
|----------|----------|
| (A) NAND | (B) XNOR |
| (C) XOR | (D) OR |

Q.29 [2 marks · NAT (numerical answer)]

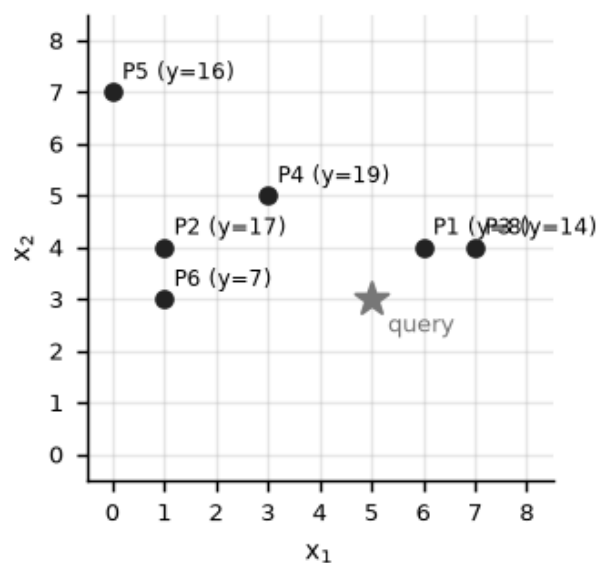
A perceptron with augmented weight vector $w = (w_0, w_1, w_2)$ (w_0 is the bias, $\tilde{x} = (1, x_1, x_2)$) starts from $w = (0, 0, 0)$ and uses learning rate $\eta = 1$. The points P1, P2, ... are presented once, in order; whenever $y \cdot w^T \tilde{x} \leq 0$ the update $w \leftarrow w + \eta y \tilde{x}$ is applied. Find the value of w_2 after this epoch. (Answer in integer)

Point	x_1	x_2	y
P1	3	0	+1
P2	-3	-1	-1
P3	1	3	-1
P4	-3	2	-1

**Q.30 [2 marks · NAT (numerical answer)]**

A distance-weighted k-NN regressor with $k = 3$ predicts $\hat{y} = \sum w_i y_i / \sum w_i$ over the 3 nearest neighbours, with weights $w_i = 1/d_i^2$. For the training data below, the prediction at $\mathbf{x} = (5, 3)$ (Euclidean distance) is _____ (rounded off to two decimal places)

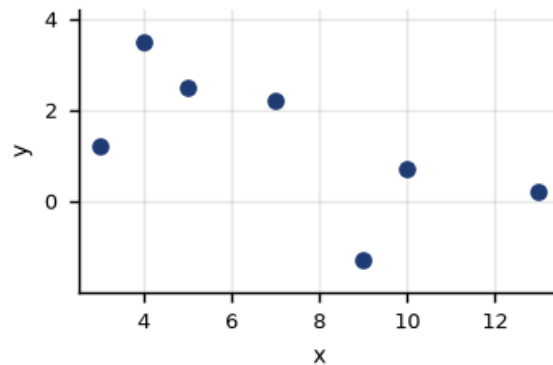
Point	x_1	x_2	y
P1	6	4	8
P2	1	4	17
P3	7	4	14
P4	3	5	19
P5	0	7	16
P6	1	3	7



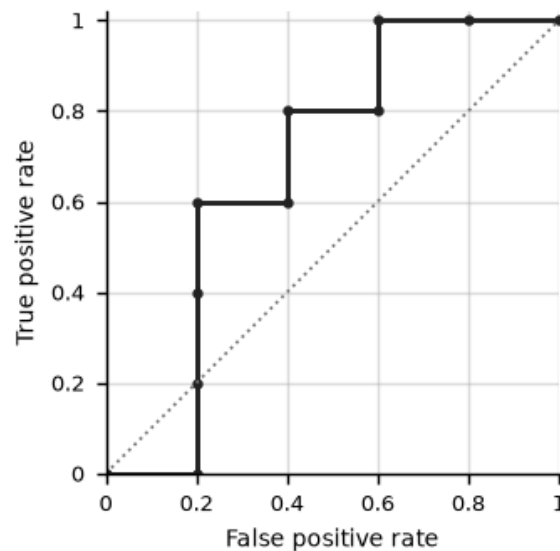
Q.31 [2 marks · NAT (numerical answer)]

A simple linear regression (with intercept) is fitted by least squares to the 7 points shown. With $H = X(X^T X)^{-1} X^T$, the largest leverage among the observations is _____ (rounded off to three decimal places)

x	3	4	5	7	9	10	13
y	1.2	3.5	2.5	2.2	-1.3	0.7	0.2

**Q.32 [2 marks · NAT (numerical answer)]**

The ROC curve below was obtained from a test set with 5 positives and 5 negatives (all scores distinct, so the curve moves in steps of 1/5 vertically and 1/5 horizontally). From the figure, the highest TPR attainable at $FPR \leq 0.4$ is _____ (rounded off to two decimal places)

**Q.33 [2 marks · MCQ]**

Ridge regression is tuned by 10-fold cross-validation. For each λ the table gives the mean CV MSE and its standard error (SE) across folds. (Larger λ means a simpler, more regularised model.) Using the **one-standard-error rule** (choose the most regularised model whose CV error is within one standard error of the minimum CV error), the selected value is

λ	0.01	0.1	0.3	1	3	10
CV MSE	5.21	4.75	4.63	4.78	5.87	6.61
SE	0.16	0.23	0.44	0.32	0.19	0.26

(A) $\lambda = 0.01$ **(B)** $\lambda = 0.1$ **(C)** $\lambda = 10$ **(D)** $\lambda = 1$

Q.34 [2 marks · NAT (numerical answer)]

In C4.5, the attribute Day splits a node of 22 records as tabulated (class Yes/No counts in each branch). Compute the split information of Day (all logarithms base 2). (rounded off to two decimal places)

Day	#Yes	#No
v1	0	5
v2	5	5
v3	1	6

Q.35 [2 marks · NAT (numerical answer)]

Agglomerative clustering with **average linkage** is applied to the distance matrix below. The cophenetic distance between P3 and P5 (the dendrogram height at which they are first joined) is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5
P1	0	4	3	7	7
P2		0	5	3	5
P3			0	8	8
P4				0	2
P5					0

— END OF QUESTION PAPER —

Mock Test 07 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MCQ	1	(D)	19	NAT	2	1.2433 to 1.2633
2	MSQ	1	(A), (B), (C), (D)	20	NAT	2	−0.0002 to 0.0002
3	NAT	1	0.63 to 0.65	21	NAT	2	69.3056 to 69.5833
4	MCQ	1	(B)	22	MSQ	2	(A)
5	MSQ	1	(A), (B), (C), (D)	23	MSQ	2	(A)
6	NAT	1	90	24	NAT	2	0.8135 to 0.8175
7	MSQ	1	(B), (C), (D)	25	NAT	2	0.096 to 0.1
8	NAT	1	41	26	MSQ	2	(A), (B), (C), (D)
9	MCQ	1	(C)	27	NAT	2	24
10	NAT	1	16.74 to 16.76	28	MCQ	2	(C)
11	NAT	1	126.99 to 127.01	29	NAT	2	−3
12	MSQ	1	(A), (B), (C)	30	NAT	2	11.1112 to 11.1312
13	MSQ	1	(A), (B), (C), (D)	31	NAT	2	0.5626 to 0.5666
14	NAT	1	5	32	NAT	2	0.79 to 0.81
15	MSQ	1	(C)	33	MCQ	2	(D)
16	NAT	2	3.3607 to 3.4012	34	NAT	2	1.5185 to 1.5385
17	MSQ	2	(B)	35	NAT	2	6.49 to 6.51
18	MSQ	2	(C)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Unsupervised Learning					11
Probabilistic & Instance-based Classifiers					11
SVM & Decision Trees					11

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Regression & Model Selection					11
Neural Networks (MLP / Feed-forward)					11
Total					55

Mock Test 07 — Detailed Solutions

Q.1 — Answer: (D) [MCQ, 1M · Unsupervised Learning > k-means]

The SSE drops steeply up to $k = 4$ and then decreases only slowly (relative drops: $k=2$: 60%, $k=3$: 52%, $k=4$: 50%, $k=5$: 7%, $k=6$: 8%, $k=7$: 6%, $k=8$: 7%, $k=9$: 13%). The bend ('elbow') is at $k = 4$. $k = 9$ gives the lowest SSE but SSE always (weakly) decreases with k , so minimizing it would pick the largest k ; it is not a model-selection criterion by itself.

Q.2 — Answer: (A), (B), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

(A) TRUE. π_k multiplies its weighted density.

(B) TRUE. A costly error type shifts the decision threshold.

(C) TRUE. The Bayes rule errs with the probability of the less probable class.

(D) TRUE. Every x can be assigned to its only possible class.

Q.3 — Answer: 0.63 to 0.65 [NAT, 1M · SVM & Decision Trees > SVM: margin & geometry]

Margin width $2/\|w\| = 2.5 \Rightarrow \|w\| = 2/2.5 = 0.8$.

$\|w\|^2 = 0.64$.

Q.4 — Answer: (B) [MCQ, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

$\log\text{-odds} = \beta_0 + \beta x + \dots$, so $\Delta(\log\text{-odds}) = \beta \cdot \Delta x = 0.3 \times 2 = 0.6$.

Odds are multiplied by $e^{\beta \Delta x} = e^{0.6} = 1.82$.

Distractors: $\beta \cdot \Delta x$ is the change in log-odds (not a multiplier); e^β ignores Δx ; $\sigma(\beta \Delta x)$ is a probability, not an odds ratio.

Q.5 — Answer: (A), (B), (C), (D) [MSQ, 1M · SVM & Decision Trees > SVM: soft margin & hinge loss]

(A) TRUE. Each misclassified point has $\xi_i > 1$.

(B) TRUE. Large C trades margin width for training fit.

(C) TRUE. Slack variables make the problem always feasible.

(D) TRUE. Small $C \Rightarrow$ stronger regularisation.

Q.6 — Answer: 90 [NAT, 1M · Regression & Model Selection > Cross-validation]

Per outer fold: inner CV $2 \times 4 = 8$ fits + 1 refit = 9.

Total = $10 \times 9 = 90$.

Q.7 — Answer: (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Regression vs classification]

A supervised task is *regression* when the target y is real-valued (loss such as squared error), and *classification* when y takes values in a finite set of labels (loss such as cross-entropy or 0-1 loss).

(A) FALSE. The output (pneumonia / normal) is a discrete class label $\rightarrow$ classification.

(B) TRUE. The output (time in minutes) is a continuous real-valued quantity $\rightarrow$ regression.

(C) TRUE. The output (km per litre) is a continuous real-valued quantity $\rightarrow$ regression.

(D) TRUE. The output (concentration) is a continuous real-valued quantity $\rightarrow$ regression.

Q.8 — Answer: 41 [NAT, 1M · Unsupervised Learning > k-means]

Sum Stirling numbers of the second kind: $S(5,1) = 1 + S(5,2) = 15 + S(5,3) = 25 \rightarrow$ total **41**.

Q.9 — Answer: (C) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

A single perceptron can classify a dataset perfectly iff the two classes are linearly separable.

Dataset P: labels for $(0,0),(0,1),(1,0),(1,1) = 1, 0, 1, 0 \rightarrow$ NOT x_2 - linearly separable.

Dataset Q: labels for $(0,0),(0,1),(1,0),(1,1) = 0, 0, 1, 1 \rightarrow x_1$ (identity of x_1) - linearly separable.

Dataset R: labels for $(0,0),(0,1),(1,0),(1,1) = 1, 0, 0, 1 \rightarrow$ XNOR - **not** linearly separable (diagonal pattern).

Dataset S: labels for $(0,0),(0,1),(1,0),(1,1) = 1, 1, 0, 0 \rightarrow$ NOT x_1 - linearly separable.

Answer: **Dataset R**. Of the 14 non-constant Boolean functions of two inputs, only XOR and XNOR are not linearly separable.

Q.10 — Answer: 16.74 to 16.76 [NAT, 1M · Probabilistic & Instance-based Classifiers › k-nearest neighbours]

Distances $|x_i - 9.3|$: $x=9$: 0.3, $x=5$: 4.3, $x=15$: 5.7, $x=17$: 7.7, $x=0$: 9.3, $x=19$: 9.7, $x=20$: 10.7.

The 4 nearest neighbours are $x = 9, 5, 15, 17$ with $y = 20, 10, 11, 26$.

$\hat{y} = (20 + 10 + 11 + 26)/4 = \mathbf{16.75}$.

Q.11 — Answer: 126.99 to 127.01 [NAT, 1M · Unsupervised Learning › k-means]

Assignment (nearest centroid): $c = 5$: $\{0, 2, 3\}$; $c = 12$: $\{10, 18, 19\}$.

$SSE = (0-5)^2 + (2-5)^2 + (3-5)^2 + (10-12)^2 + (18-12)^2 + (19-12)^2 = \mathbf{127}$.

Q.12 — Answer: (A), (B), (C) [MSQ, 1M · Regression & Model Selection › Simple linear regression]

(A) TRUE. $\beta_1 = r \cdot s_y / s_x$ with $s_x, s_y > 0$.

(B) TRUE. $\beta_0 = \bar{y} - \beta_1 \bar{x} = \hat{y}(\bar{x}) = \bar{y}$.

(C) TRUE. Cross term $2\sum(\hat{y}_i - \bar{y})e_i$ vanishes by the normal equations.

(D) FALSE. Only for a perfect fit; in general $SSE > 0$.

Q.13 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) › Softmax + cross-entropy]

(A) TRUE. Standard derivative.

(B) TRUE. $y_k = 0$.

(C) TRUE. Standard result.

(D) TRUE. $\sum(p_k - y_k) = 1 - 1$.

Q.14 — Answer: 5 [NAT, 1M · SVM & Decision Trees › SVM: kernels]

Monomials of degree exactly d in p variables = $C(p + d - 1, d)$ (stars and bars).

$C(5, 4) = \mathbf{5}$.

(Including all lower degrees, as for $(x^T z + 1)^4$, would give $C(6, 4) = 15$.)

Q.15 — Answer: (C) [MSQ, 1M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

(A) FALSE. It shrinks weights towards 0.

(B) FALSE. More data reduces variance.

(C) TRUE. The L1 subgradient has constant magnitude λ .

(D) FALSE. Large weights saturate the sigmoids, making gradients vanish.

Q.16 — Answer: 3.3607 to 3.4012 [NAT, 2M · Regression & Model Selection › Basis functions & feature scaling]

Design columns $1, x, x^2$. Because the x -values are symmetric about 0, $\sum x = \sum x^3 = 0$, so the normal equations decouple:

$\sum x^2 = 28, \sum x^4 = 196, \sum y = 5, \sum xy = -32, \sum x^2 y = -36$.

$b = \sum xy / \sum x^2 = -32/28 = -1.1429$.

$[n, \sum x^2; \sum x^2, \sum x^4][a; c] = [\sum y; \sum x^2 y] = 7a + 28c = 5, 28a + 196c = -36$.

Solving: $c = (n\sum x^2 y - \sum x^2 \sum y) / (n\sum x^4 - (\sum x^2)^2) = -0.6667, a = (\sum y - c\sum x^2) / n = 3.381$.

Answer: **3.38**.

Q.17 — Answer: (B) [MSQ, 2M · SVM & Decision Trees › Decision trees: impurity measures]

Parent $[40+, 50-]$: $H = 0.9911, \text{Gini} = 0.4938, \text{error} = 0.4444$.

S1: children $H = 0.9124, 0.2006$; $\text{Gini} = 0.4405, 0.0605$; $\text{error} = 0.3276, 0.0312$. Gains: $\text{IG} = 0.3317, \text{Gini} = 0.1884, \text{error} = 0.2222$.

S2: children $H = 0.7293, 0.7107$; $\text{Gini} = 0.3244, 0.3133$; $\text{error} = 0.2037, 0.1944$. Gains: $\text{IG} = 0.2692, \text{Gini} = 0.1739, \text{error} = 0.2444$.

(A) FALSE. S1: 0.2222, S2: 0.2444.

(B) TRUE. S1: 0.1884, S2: 0.1739.

(C) FALSE. S1: 0.3317, S2: 0.2692.

(D) FALSE. S1: 0.1884, S2: 0.1739.

Q.18 — Answer: (C) [MSQ, 2M · SVM & Decision Trees › SVM: soft margin & hinge loss]

$$\xi_i = \max(0, 1 - y_i f(x_i)).$$

Point	y f(x)	ξ
P1	2.5	0
P2	0.5	0.5
P3	2.5	0
P4	-4	5
P5	3	0
P6	0.5	0.5

- (A) FALSE. $\sum \xi_i = 6$.
 (B) FALSE. $y f = 3 \Rightarrow \xi = 0$.
 (C) TRUE. $y f = -4 \Rightarrow \xi = 5$.
 (D) FALSE. The correct count is 3.

Q.19 — Answer: 1.2433 to 1.2633 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$$\mathbf{m}_1 = (2.333, 1.333), \mathbf{m}_2 = (4.667, 2.667).$$

$$S_1 = \begin{bmatrix} 4.667 & -3.333 \\ -3.333 & 2.667 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 8.667 & 1.667 \\ 1.667 & 0.667 \end{bmatrix}$$

$$S_W = \begin{bmatrix} 13.333 & -1.667 \\ -1.667 & 3.333 \end{bmatrix}$$

$$\mathbf{m}_2 - \mathbf{m}_1 = (2.333, 1.333).$$

$$\mathbf{w} = S_W^{-1}(\mathbf{m}_2 - \mathbf{m}_1) = (0.24, 0.52).$$

$$(\mathbf{m}_2 - \mathbf{m}_1)^T \mathbf{w} = 1.25.$$

Q.20 — Answer: -0.0002 to 0.0002 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

All weights = 0.5, biases 0 $\Rightarrow$ both hidden units get $a = 0.5 \times (0.5 + -2) = -0.75$, $h_1 = h_2 = \sigma(a) = 0.3208$; $\hat{y} = 2 \times 0.5 \times 0.3208 = 0.3208$.

$$\hat{y} - y = -1.6792. \partial L / \partial v_j = (\hat{y} - y) h_j = -0.5387 \text{ (same for both } j).$$

$$\partial L / \partial w_{jk} = (\hat{y} - y) v_j h_j (1 - h_j) x_k = -1.6792 \times 0.5 \times 0.2179 \times 0.5 = -0.0915 \text{ (identical for } j = 1, 2).$$

Both hidden units receive identical gradients, so the difference stays 0.

Because the two hidden units start identical and get identical updates, they remain identical forever (symmetry problem) - hence weights must be initialised randomly.

Q.21 — Answer: 69.3056 to 69.5833 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Unit separator $u = (4, 3)/5$. Margins $y_i(u \cdot x_i)$: 0.6, 4.8, 2.6, 1.4, 4.8 $\rightarrow \gamma = \min = 0.6$.

Norms $\|x_i\|$: 1, 5, 3.162, 5, 5 $\rightarrow R = \max = 5$.

Novikoff bound: number of mistakes $\leq (R/\gamma)^2 = (5/0.6)^2 = 69.44$.

Common mistakes: forgetting to normalise u (γ must be measured with $\|u\| = 1$), or using R/γ without squaring.

Q.22 — Answer: (A) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Activation functions]

$$\sigma(2) = 0.8808, \tanh(2) = 0.964.$$

- (A) TRUE. Both equal 0.964.
 (B) FALSE. $\text{ReLU}' = 1$ for $z > 0$ and 0 for $z < 0$.
 (C) FALSE. $\sigma'(2) = 0.105 < 0.25$ (max only at 0).
 (D) FALSE. $\tanh$ is odd: the sum is 0.

Q.23 — Answer: (A) [MSQ, 2M · Regression & Model Selection > Bias-variance trade-off]

(A) TRUE. Train 0.63 (lowest), validation 12.01 $>$ 5.3.

(B) FALSE. Gaps: 0.3, 0.64, 3.87, 6.37, 7.91, 9.61, 11.38; the largest is at degree 7.

(C) FALSE. Lower-degree models are less flexible and have lower variance.

(D) FALSE. The minimum validation MSE is at degree 2, not 7.

Q.24 — Answer: 0.8135 to 0.8175 [NAT, 2M · Probabilistic & Instance-based Classifiers · Evaluation metrics]

Per unit population: $TP = Se \cdot \pi$, $FN = (1 - Se) \cdot \pi$, $FP = (1 - Sp)(1 - \pi)$, $TN = Sp(1 - \pi)$.

$TP = 0.0475$, $FN = 0.0025$, $FP = 0.019$, $TN = 0.931$.

$F1\text{-score} = 2TP / (2TP + FP + FN) = \mathbf{0.815}$.

Q.25 — Answer: 0.096 to 0.1 [NAT, 2M · Unsupervised Learning · k-means]

k-means++ picks the next centre x with probability $D(x)^2 / \sum_x D(x)^2$, where $D(x)$ is the distance from x to the **nearest** centre already chosen (chosen centres have $D = 0$).

Point	P1	P2	P3	P4	P5	P6
$D(x)^2$	0	5	1	5	40	0

$\sum D^2 = 51$, so $P(P4) = 5/51 = \mathbf{0.098}$.

Common mistakes: using D instead of D^2 (that is not k-means++), or using the distance to the *farthest*/first centre instead of the nearest chosen centre.

Q.26 — Answer: (A), (B), (C), (D) [MSQ, 2M · Unsupervised Learning · PCA & dimensionality reduction]

(A) TRUE. $S \rightarrow RSR^T$ is a similarity transform.

(B) TRUE. The matrix is singular.

(C) TRUE. All eigenvalues scale by c^2 .

(D) TRUE. The between-class scatter has rank $\leq C - 1$.

Q.27 — Answer: 24 [NAT, 2M · Unsupervised Learning · k-medoids]

PAM cost = $\sum$ over all points of the absolute distance to the nearest medoid.

Current medoids {5, 21}: distances 1→4, 3→2, 5→0, 8→3, 12→7, 21→0, 29→8; cost = **24**.

Q.28 — Answer: (C) [MCQ, 2M · Neural Networks (MLP / Feed-forward) · Expressiveness]

Each unit outputs 1 iff its net input (weighted sum + bias) is > 0 . Evaluate on all four inputs:

x_1	x_2	h_1	h_2	y
0	0	1	0	0
0	1	1	1	1
1	0	1	1	1
1	1	0	1	0

h_1 computes NAND, h_2 computes OR; the output column (0,0)→(1,1) is 0, 1, 1, 0, i.e. **XOR**.

This shows that one hidden layer of two threshold units suffices for the non-linearly-separable XOR, which a single perceptron cannot represent.

Q.29 — Answer: -3 [NAT, 2M · SVM & Decision Trees · Perceptron (linear separators)]

Trace the epoch (a point with $y \cdot w^T \tilde{x} \leq 0$, including 0, triggers an update):

Step	$\tilde{x} = (1, x_1, x_2)$	y	$w^T \tilde{x}$	$y \cdot w^T \tilde{x} \leq 0$?	w after step
1	(1, 3, 0)	+1	0	yes → update	(1, 3, 0)
2	(1, -3, -1)	-1	-8	no	(1, 3, 0)
3	(1, 1, 3)	-1	4	yes → update	(0, 2, -3)
4	(1, -3, 2)	-1	-12	no	(0, 2, -3)

Final $w = (0, 2, -3)$, number of updates = 2. Answer: **-3**.

Q.30 — Answer: 11.1112 to 11.1312 [NAT, 2M · Probabilistic & Instance-based Classifiers · k-nearest neighbours]

Three nearest neighbours: P1 ($d^2=2$, $y=8$), P3 ($d^2=5$, $y=14$), P4 ($d^2=8$, $y=19$).

Weights $1/d^2$: 0.5, 0.2, 0.125; $\sum w = 0.825$.

$$\sum w_i y_i = 9.175 \Rightarrow \hat{y} = 9.175/0.825 = \mathbf{11.12}.$$

(The unweighted 3-NN average would be 13.67; weighting pulls $\hat{y}$ towards the closest point.)

Q.31 — Answer: 0.5626 to 0.5666 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

For simple linear regression with intercept, the leverage (diagonal of the hat matrix) is $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$.

$$\bar{x} = 7.2857, S_{xx} = 77.4286.$$

The largest leverage is at the x farthest from $\bar{x}$: $x = 13$, $h = 1/7 + (5.7143)^2/77.4286 = 0.5646$.

(Check: $\sum h_{ii} = 2 = 2 = \text{number of parameters.}$)

Answer: **0.565**.

Q.32 — Answer: 0.79 to 0.81 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Among the operating points with $\text{FPR} \leq 0.4$, take the highest point of the curve (the top of the vertical segment at $\text{FPR} = 0.4$).

$\text{TPR} = \mathbf{0.8}$.

Q.33 — Answer: (D) [MCQ, 2M · Regression & Model Selection > Cross-validation]

Minimum CV MSE = 4.63 at $\lambda = 0.3$, with SE = 0.44.

One-SE threshold = $4.63 + 0.44 = 5.07$.

Models within the threshold: $\lambda = 0.1$ (4.75), $\lambda = 0.3$ (4.63), $\lambda = 1$ (4.78).

The one-SE rule picks the largest such λ (simplest model): $\lambda = 1$; the minimum-CV rule picks $\lambda = 0.3$.

Answer: $\lambda = \mathbf{1}$.

Q.34 — Answer: 1.5185 to 1.5385 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [6 Yes, 16 No]: $H = 0.8454$.

Child entropies: $v1: 0$, $v2: 1$, $v3: 0.5917$.

$$\text{IG} = 0.8454 - \sum (n_v/22) H_v = 0.2025.$$

$$\text{SplitInfo} = -\sum (n_v/N) \log_2(n_v/N) = -(5/22)\log_2(5/22) - (10/22)\log_2(10/22) - (7/22)\log_2(7/22) = 1.5285.$$

Answer: **1.53**.

C4.5 uses the gain ratio to penalise attributes that split the data into many small branches.

Q.35 — Answer: 6.49 to 6.51 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Use average linkage (UPGMA): $d(A,B) = (1/|A||B|) \sum_{a \in A} \sum_{b \in B} d(a,b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

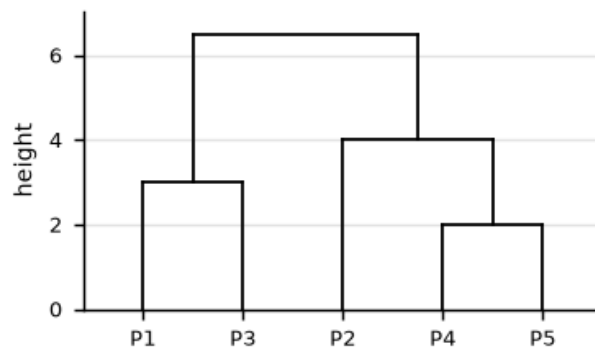
Merge 1: $\{P4\} + \{P5\}$ at height **2**. Updated: $d(\{P4, P5\}, \{P1\}) = 7$, $d(\{P4, P5\}, \{P2\}) = 4$, $d(\{P4, P5\}, \{P3\}) = 8$.

Merge 2: $\{P1\} + \{P3\}$ at height **3**. Updated: $d(\{P1, P3\}, \{P2\}) = 4.5$, $d(\{P1, P3\}, \{P4, P5\}) = 7.5$.

Merge 3: $\{P2\} + \{P4, P5\}$ at height **4**. Updated: $d(\{P2, P4, P5\}, \{P1, P3\}) = 6.5$.

Merge 4: $\{P1, P3\} + \{P2, P4, P5\}$ at height **6.5**.

$P3$ and $P5$ first lie in the same cluster at the merge of height 6.5, so their cophenetic distance is **6.5** (their original distance was 8).



Mock Test 08

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	8	8	19

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Unsupervised Learning	7	11
Neural Networks (MLP / Feed-forward)	7	11
Probabilistic & Instance-based Classifiers	7	11
Regression & Model Selection	7	11
SVM & Decision Trees	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

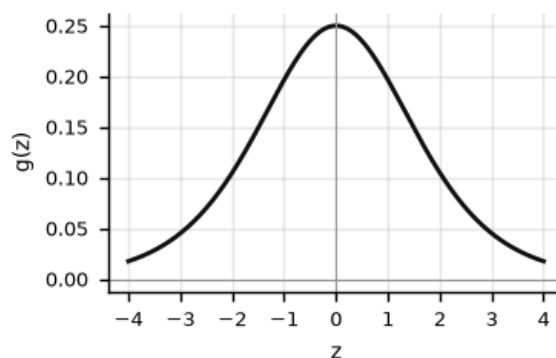
Q.1 [1 mark · MSQ (one or more correct)]

Which of the following statements about principal component analysis (PCA) is/are CORRECT?

- (A) The first PC is the eigenvector of the covariance matrix with the smallest eigenvalue.
- (B) PCA is a supervised method that uses class labels.
- (C) The principal component directions are mutually orthogonal.
- (D) The variance of the projected data along the j -th PC equals the j -th largest covariance eigenvalue.

Q.2 [1 mark · MCQ]

The figure shows the graph of a function $g(z)$ used in neural networks. Which function is it?



- (A) Sigmoid $\sigma(z) = 1/(1+e^{-z})$
- (B) $\tanh(z)$
- (C) $1 - \tanh^2(z)$
- (D) $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Q.3 [1 mark · NAT (numerical answer)]

Two shopping baskets are encoded as binary vectors (1 = item bought). The Jaccard similarity between basket A and basket B is _____ (rounded off to two decimal places)

Item	milk	bread	eggs	butter	jam	tea	rice	salt
Basket A	1	0	0	0	1	1	0	1
Basket B	1	0	1	0	1	1	1	1

Q.4 [1 mark · NAT (numerical answer)]

In a training set, the 6 examples of class Play = No have the following counts for the feature Outlook. Using Laplace (add-one) smoothing, the naive Bayes estimate of $P(\text{Outlook} = \text{Rain} \mid \text{Play} = \text{No})$ is _____ (rounded off to three decimal places)

Outlook	Sunny	Overcast	Rain
count in Play = No	3	0	3

Q.5 [1 mark · MSQ (one or more correct)]

Which of the following statements about the representational power of feed-forward neural networks is/are CORRECT?

- (A) The universal approximation theorem requires at least two hidden layers.
- (B) The universal approximation theorem states that the number of hidden units needed grows at most linearly with the input dimension.
- (C) Logistic regression has a non-linear decision boundary because the sigmoid is non-linear.
- (D) A single sigmoid neuron (logistic regression) has a linear decision boundary.

Q.6 [1 mark · MSQ (one or more correct)]

A simple linear regression $\hat{y} = \beta_0 + \beta_1 x$ is fitted by ordinary least squares (with intercept) to data (x_i, y_i) , $i = 1, \dots, n$, with residuals $e_i = y_i - \hat{y}_i$ and sample correlation r . Which of the following statements is/are CORRECT?

- (A) If $r = 0$ the least-squares line is $\hat{y} = 0$.
- (B) Least squares minimises the sum of perpendicular distances from the points to the line.
- (C) R^2 is unchanged if y is replaced by $a + by$ for any constants a and $b \neq 0$.
- (D) In simple linear regression, $R^2 = r$ (the correlation itself).

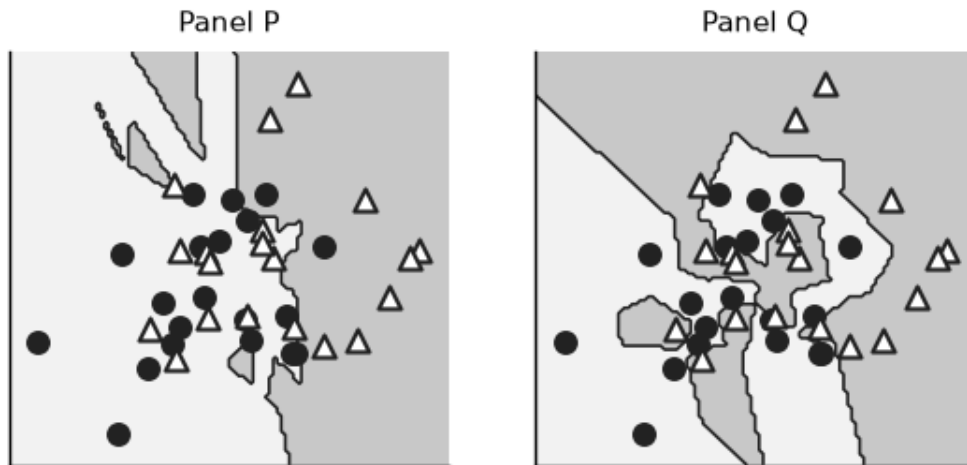
Q.7 [1 mark · MSQ (one or more correct)]

Consider ridge regression (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_2^2$) and the lasso (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_1$) with $\lambda \geq 0$. Which of the following statements is/are CORRECT?

- (A) $\|\beta_{\text{ridge}}(\lambda)\|_2$ is a non-increasing function of λ .
- (B) The training error of ridge regression decreases as λ increases.
- (C) λ should be chosen to minimise the training error.
- (D) The ridge objective $\|y - X\beta\|^2 + \lambda\|\beta\|^2$ ($\lambda > 0$) is non-convex.

Q.8 [1 mark · MCQ]

The two panels show the decision regions of k-NN classifiers trained on the same 2-D data set, one with $k = 1$ and the other with $k = 25$. Which of the following statements is CORRECT?



- (A) Panel P uses $k = 1$; it has higher bias and lower variance
- (B) Panel P uses $k = 1$; it has lower bias and higher variance
- (C) Panel Q uses $k = 1$; it has higher bias and lower variance
- (D) Panel Q uses $k = 1$; it has lower bias and higher variance

Q.9 [1 mark · NAT (numerical answer)]

For the RBF kernel $K(x, z) = \exp(-\gamma \|x - z\|^2)$, with $x = (-1, 1, 1)$ and $z = (-3, 2, 2)$, the value of γ for which $K(x, z) = 0.8$ is _____ (rounded off to three decimal places)

Q.10 [1 mark · NAT (numerical answer)]

A node of a classification tree contains 24 training records: 12 of class C_1 , 12 of class C_2 . The entropy (in bits) of this node is _____ (rounded off to two decimal places)

Q.11 [1 mark · NAT (numerical answer)]

In one iteration of PAM (k-medoids) on $n = 53$ points with $k = 8$, every (medoid, non-medoid) pair is considered as a candidate swap. The number of candidate swaps is _____ (Answer in integer)

Q.12 [1 mark · NAT (numerical answer)]

The confusion matrix of a binary classifier on a test set is shown below (note which of rows/columns are actual and predicted). The balanced accuracy of the classifier is _____ (rounded off to two decimal places)

	Predicted +	Predicted -
Actual +	27	9
Actual -	34	6

Q.13 [1 mark · NAT (numerical answer)]

A model has weights $w = (-2, 2, -0.5)$. The L2 penalty term $(\lambda/2)\|w\|^2$ with $\lambda = 0.01$ added to the loss equals _____ (rounded off to 4 decimal places)

Q.14 [1 mark · NAT (numerical answer)]

Regressing predictor x_3 on all the other predictors gives $R^2 = 0.67$. The VIF of x_3 is _____ (rounded off to two decimal places)

Q.15 [1 mark · MSQ (one or more correct)]

Which of the following statements about bagging, random forests and boosting is/are CORRECT?

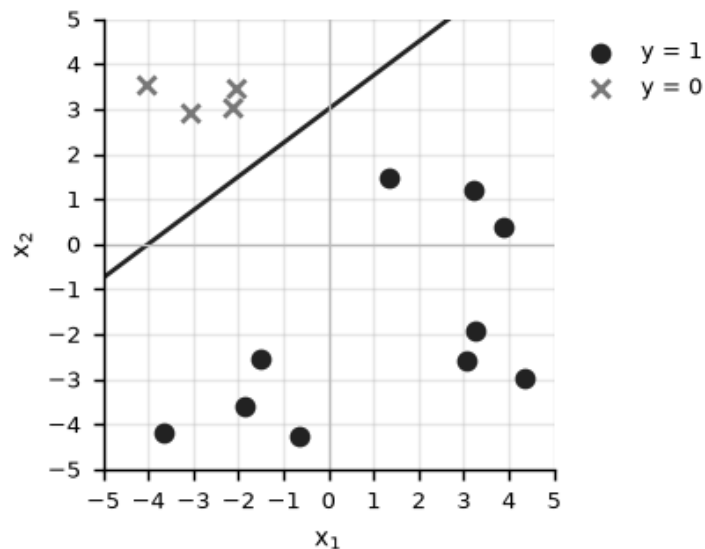
- (A) In AdaBoost, the weights of misclassified examples increase for the next round.
- (B) A weak learner with weighted error $\varepsilon_t = 0.5$ receives $\alpha_t = 0$.
- (C) A common default for random-forest classification is $m \approx \sqrt{p}$ features per split.
- (D) Random forests decorrelate trees by considering a random subset of features at each split.

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

Gradient descent with (heavy-ball) momentum minimises $f(w) = (1/2)w^2$ using $v_t = \beta v_{t-1} + \nabla f(w_{t-1})$ and $w_t = w_{t-1} - \eta v_t$, with $w_0 = 3$, $v_0 = 0$, $\eta = 0.2$, $\beta = 0.5$. The value of v_2 is _____ (rounded off to three decimal places)

Q.17 [2 marks · MCQ]

A logistic regression model $P(y = 1 | \mathbf{x}) = \sigma(w_0 + w_1 x_1 + w_2 x_2)$ perfectly classifies the training points shown below using threshold 0.5. Its decision boundary is the solid line (it crosses the axes at grid points). Which of the following parameter vectors is consistent with the figure?

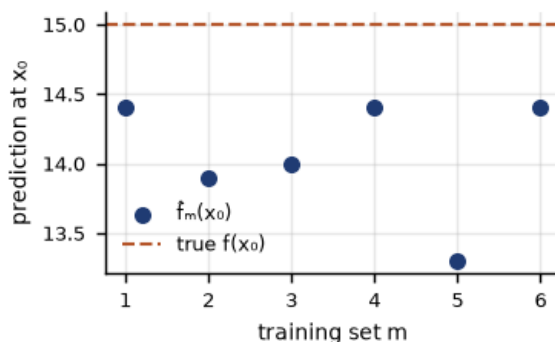


- (A) $(w_0, w_1, w_2) = (12, -4, 3)$
- (B) $(w_0, w_1, w_2) = (12, 3, -4)$
- (C) $(w_0, w_1, w_2) = (-12, -3, 4)$
- (D) $(w_0, w_1, w_2) = (12, 4, -3)$

Q.18 [2 marks · NAT (numerical answer)]

Data are generated as $y = f(x) + \varepsilon$ with $E[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2 = 1$. At a test point x_0 the true value is $f(x_0) = 15$. A learning method is trained on 6 independent training sets, giving the predictions $\hat{f}_m(x_0)$ in the table (also plotted). Treating these as the full distribution of the method's prediction (variance with divisor 6), the variance of the method's prediction at x_0 is _____ (rounded off to three decimal places)

m	1	2	3	4	5	6
$\hat{f}_m(x_0)$	14.4	13.9	14	14.4	13.3	14.4

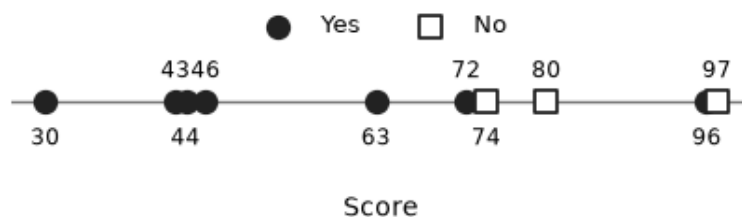
**Q.19 [2 marks · NAT (numerical answer)]**

The eigenvalues of the sample covariance matrix (divisor $n - 1$) of a centered 7-dimensional dataset are 20, 18.5, 16.5, 16, 12.5, 10, 2.5. Then with $n = 11$ samples, the total squared reconstruction error $\sum_i \|x_i - \hat{x}_i\|^2$ when keeping the top 4 principal components is _____ (rounded off to two decimal places)

Q.20 [2 marks · NAT (numerical answer)]

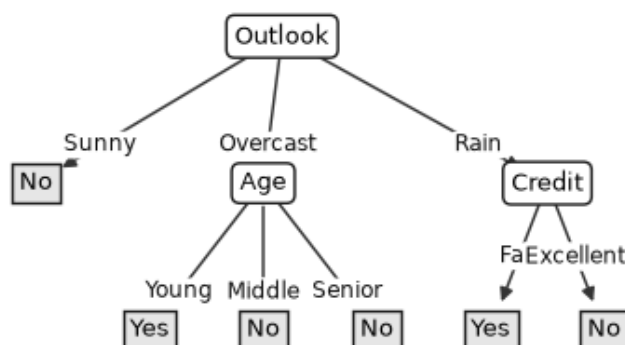
The continuous attribute Score and the class label of 10 training records are given below (unsorted). A binary split “Score $\leq t$ ” is chosen to maximise information gain, with t restricted to midpoints between consecutive sorted values. Find the threshold value t chosen for the split “Score $\leq t$ ”. (rounded off to one decimal place)

Score	43	97	74	63	30	46	80	72	44	96
Class	Yes	No	No	Yes	Yes	Yes	No	Yes	Yes	Yes



Q.21 [2 marks · NAT (numerical answer)]

The decision tree shown is applied to the 10 labelled records in the table (attributes not tested on a path are ignored). Find the training accuracy of the tree (as a percentage). (rounded off to one decimal place)



#	Outlook	Age	Credit	Actual
1	Rain	Middle	Excellent	No
2	Overcast	Young	Fair	Yes
3	Rain	Middle	Fair	No
4	Rain	Young	Fair	Yes
5	Overcast	Young	Excellent	No
6	Overcast	Young	Excellent	No
7	Sunny	Senior	Fair	No
8	Sunny	Young	Fair	No
9	Overcast	Young	Excellent	Yes
10	Rain	Middle	Excellent	No

Q.22 [2 marks · NAT (numerical answer)]

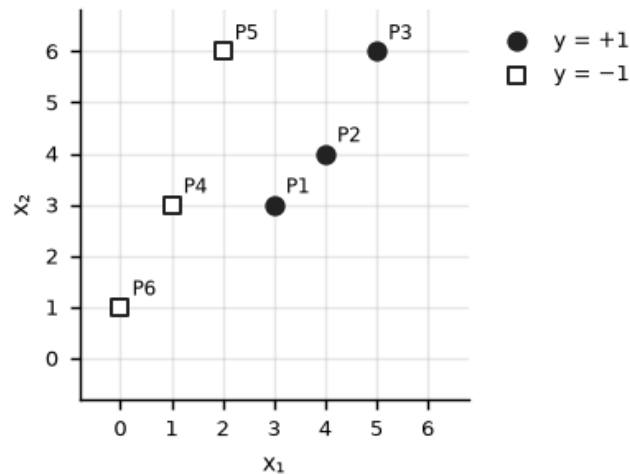
Three customers are described by age and income. Age ranges over [20, 60] years and income over [10, 210] thousand rupees in the full dataset. After min-max scaling of each feature to [0, 1] (using the stated ranges), the Euclidean distance from A to its **nearest** neighbour among B and C is _____ (rounded off to two decimal places)

Customer	Age (years)	Income (₹ thousand)
A	44	25
B	60	195
C	41	200

Q.23 [2 marks · MSQ (one or more correct)]

A hard-margin linear SVM is trained on the 6 labelled points shown. Which of the following points is/are support vectors of the resulting maximum-margin classifier?

Point	x_1	x_2	y
P1	3	3	+1
P2	4	4	+1
P3	5	6	+1
P4	1	3	-1
P5	2	6	-1
P6	0	1	-1



- (A)** P1 **(B)** P4
(C) P5 **(D)** P6

Q.24 [2 marks · NAT (numerical answer)]

For the kernel $K(x, z) = (x^T z + 1)^2$ on $\mathbb{R}^2$ with feature map ϕ , compute the cosine of the angle between $\phi(x)$ and $\phi(z)$, where $x = (-1, -2)$ and $z = (1, 1)$. (rounded off to two decimal places)

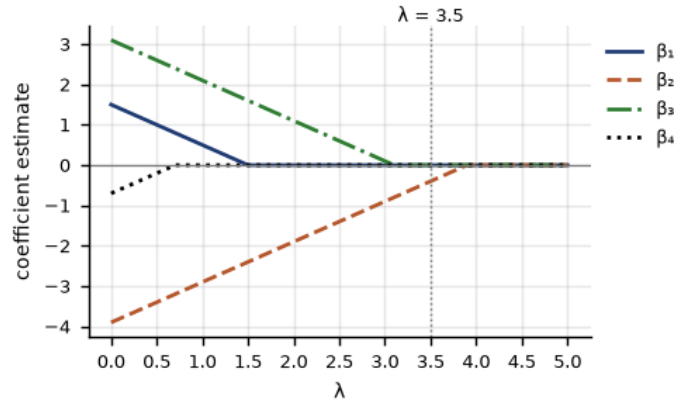
Q.25 [2 marks · MCQ]

For logits $z = (2, -1, 0)$, softmax probabilities p and true class 3, the gradient of the cross-entropy loss $L = -\ln p_3$ with respect to z (rounded to two decimals) is

- (A) $(0.84, 0.04, -0.89)$ (B) $(-0.84, -0.04, 0.89)$
(C) $(-0.16, -0.96, -0.89)$ (D) $(0, 0, -8.76)$

Q.26 [2 marks · MSQ (one or more correct)]

For a design with orthonormal columns ($X^T X = I$), the figure shows the coefficient paths $\beta_i(\lambda)$ of a penalised least-squares estimator that minimises $\frac{1}{2}\|y - X\beta\|^2 + P_\lambda(\beta)$, where the penalty P_λ is either the ridge penalty $(\lambda/2)\|\beta\|_2^2$ or the lasso penalty $\lambda\|\beta\|_1$. Which of the following statements is/are CORRECT?

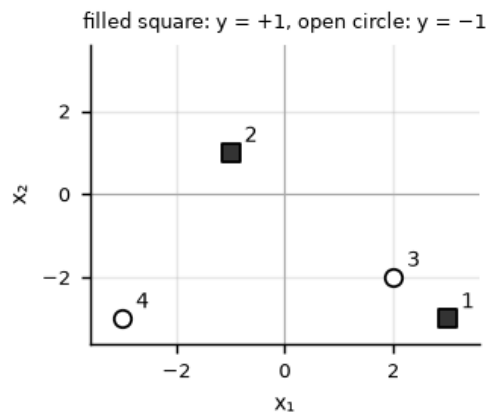


- (A) At $\lambda = 3.5$, exactly 4 coefficient(s) are zero.
- (B) At $\lambda = 0$ the coefficients equal the least-squares estimates.
- (C) No coefficient becomes exactly zero for any finite λ .
- (D) The plotted paths are those of the lasso (L1 penalty).

Q.27 [2 marks · NAT (numerical answer)]

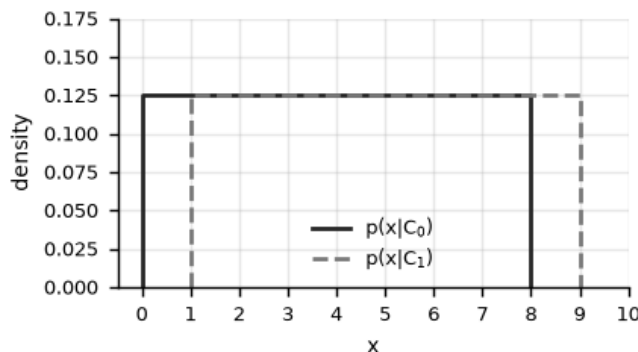
A perceptron with parameters $w \in \mathbb{R}^2$, b is trained on the data below, starting from $w = (0, 0)$, $b = 0$, with learning rate $\eta = 1$. Examples are presented once in the order 1, 2, 3, 4 (one epoch). An update $w \leftarrow w + \eta y x$, $b \leftarrow b + \eta y$ is made whenever $y(w \cdot x + b) \leq 0$. After the epoch, the weight w_2 is _____ (Answer in integer)

i	x_1	x_2	y
1	3	-3	1
2	-1	1	1
3	2	-2	-1
4	-3	-3	-1



Q.28 [2 marks · NAT (numerical answer)]

For a binary problem with scalar x , $p(x | C_0)$ is uniform on $[0, 8]$ and $p(x | C_1)$ is uniform on $[1, 9]$, with priors $P(C_0) = 0.6$ and $P(C_1) = 0.4$. Then the Bayes (minimum achievable) error rate is _____ (rounded off to three decimal places)

**Q.29 [2 marks · MSQ (one or more correct)]**

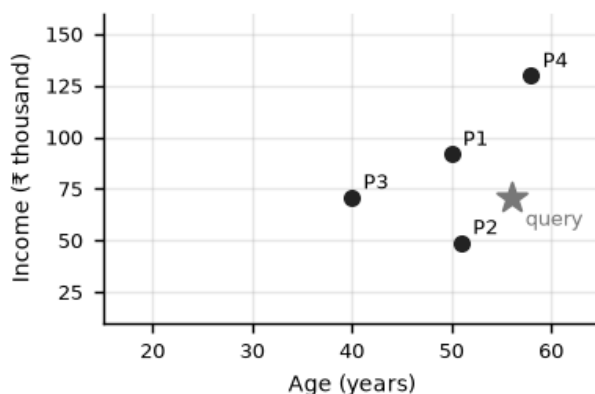
Consider k-means (Lloyd's algorithm) and k-medoids (PAM). Which of the following statements is/are CORRECT?

- (A) k-means++ initialization guarantees the globally optimal clustering.
- (B) k-medoids is generally more robust to outliers than k-means.
- (C) For mean centroids, total sum of squares = within-cluster SS + between-cluster SS.
- (D) Multiplying one feature by 10 never changes the k-means partition.

Q.30 [2 marks · MCQ]

A 1-NN classifier uses the two features below. Before computing Euclidean distances, each feature is min-max scaled to $[0, 1]$ using the minimum and maximum of that feature over the four training points P1-P4 (the query is scaled with the same constants). The nearest neighbour of the query after scaling is

Point	Age (years)	Monthly income (₹ thousand)
P1	50	92
P2	51	49
P3	40	71
P4	58	130
Query	56	71



- (A) P1
- (B) P2
- (C) P3
- (D) P4

Q.31 [2 marks · MCQ]

Six points have the pairwise distances below. Agglomerative clustering with **average linkage** is run and the dendrogram is cut to give exactly **2** clusters. The resulting clusters are

	P1	P2	P3	P4	P5	P6
P1	0	1	2	11	8	15
P2		0	3	12	7	16
P3			0	11	10	15
P4				0	9	4
P5					0	9
P6						0

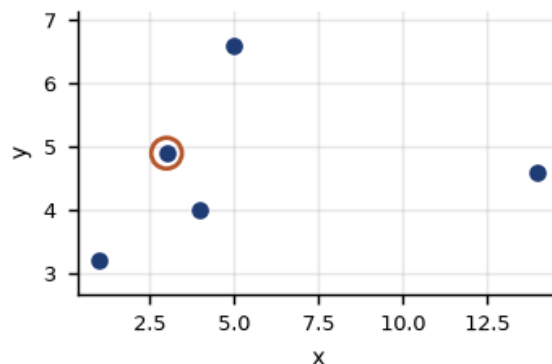
(A) {P1}, {P2, P3, P4, P5, P6}**(B)** {P1, P2, P3}, {P4, P5, P6}**(C)** {P1, P2, P3, P5}, {P4, P6}**(D)** {P1, P2, P3}, {P4, P6}, {P5}**Q.32 [2 marks · MCQ]**

For $n = 8$ paired observations: $\Sigma x = 66$, $\Sigma y = -30$, $\Sigma x^2 = 766$, $\Sigma y^2 = 256$, $\Sigma xy = -404$. The least-squares regression line of y on x is (coefficients rounded to two decimals)

(A) $\hat{y} = 0.6 - 0.53x$ **(B)** $\hat{y} = 2.08 - 0.71x$ **(C)** $\hat{y} = -9.58 - 0.71x$ **(D)** $\hat{y} = 5.25 - 1.09x$ **Q.33 [2 marks · NAT (numerical answer)]**

A simple linear regression (with intercept) is fitted by least squares to the 5 points shown. With $H = X(X^T X)^{-1} X^T$, if the errors have variance $\sigma^2 = 1$, the variance of the residual e_i of the observation with $x = 3$ (circled) is _____ (rounded off to three decimal places)

x	1	3	4	5	14
y	3.2	4.9	4	6.6	4.6



Q.34 [2 marks · MCQ]

A network computes $\hat{y} = W_2 \text{ReLU}(W_1 x + b_1) + b_2$ with the parameters below. For $x = (1, 2, -1)^T$, the output $\hat{y}$ is

$$W_1 = \begin{bmatrix} -1 & 1 & -2 \\ 1 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 1 & -1 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

(A) $\hat{y} = (-2, -5)^T$

(B) $\hat{y} = (-3, -4)^T$

(C) $\hat{y} = (-2, -4)^T$

(D) $\hat{y} = (-1, -4)^T$

Q.35 [2 marks · NAT (numerical answer)]

A classifier for a rare condition has sensitivity (recall) 0.9 and specificity 0.99. It is deployed on a population where the prevalence of the positive class is 0.01. The expected negative predictive value in this population is _____ (rounded off to three decimal places)

— END OF QUESTION PAPER —

Mock Test 08 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MSQ	1	(C), (D)	19	NAT	2	249.99 to 250.01
2	MCQ	1	(D)	20	NAT	2	72.99 to 73.01
3	NAT	1	0.6567 to 0.6767	21	NAT	2	69.95 to 70.05
4	NAT	1	0.4424 to 0.4464	22	NAT	2	0.8682 to 0.8882
5	MSQ	1	(D)	23	MSQ	2	(A), (B), (C)
6	MSQ	1	(C)	24	NAT	2	0.2122 to 0.2322
7	MSQ	1	(A)	25	MCQ	2	(A)
8	MCQ	1	(D)	26	MSQ	2	(B), (D)
9	NAT	1	0.0352 to 0.0392	27	NAT	2	3
10	NAT	1	0.99 to 1.01	28	NAT	2	0.348 to 0.352
11	NAT	1	360	29	MSQ	2	(B), (C)
12	NAT	1	0.44 to 0.46	30	MCQ	2	(B)
13	NAT	1	0.0404 to 0.0421	31	MCQ	2	(C)
14	NAT	1	3.0152 to 3.0455	32	MCQ	2	(B)
15	MSQ	1	(A), (B), (C), (D)	33	NAT	2	0.7409 to 0.7453
16	NAT	2	3.898 to 3.902	34	MCQ	2	(C)
17	MCQ	2	(B)	35	NAT	2	0.997 to 1.001
18	NAT	2	0.1559 to 0.1619				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Unsupervised Learning					11
Neural Networks (MLP / Feed-forward)					11
Probabilistic & Instance-based Classifiers					11

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Regression & Model Selection					11
SVM & Decision Trees					11
Total					55

Mock Test 08 — Detailed Solutions

Q.1 — Answer: (C), (D) [MSQ, 1M · Unsupervised Learning > PCA & dimensionality reduction]

(A) FALSE. It has the largest eigenvalue.

(B) FALSE. PCA ignores labels.

(C) TRUE. Eigenvectors of a symmetric matrix.

(D) TRUE. $v^T S v = \lambda$ for a unit eigenvector v .

Q.2 — Answer: (D) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

Key features of the plotted curve: Bell-shaped with peak 0.25 at $z = 0$.

Hence $g = \sigma'(z) = \sigma(z)(1 - \sigma(z))$.

Sigmoid $\sigma(z) = 1/(1+e^{-z})$: S-shaped, range (0, 1), value 0.5 at $z = 0$. – does not match.

$\tanh(z)$: S-shaped, range (-1, 1), passes through the origin. – does not match.

$1 - \tanh^2(z)$: Bell-shaped with peak 1 at $z = 0$. – does not match.

Q.3 — Answer: 0.6567 to 0.6767 [NAT, 1M · Unsupervised Learning > Distance measures]

Counts: $f_{11} = 4$, $f_{10} = 0$, $f_{01} = 2$, $f_{00} = 2$.

Jaccard $J = f_{11}/(f_{11}+f_{10}+f_{01}) = 4/6 = 0.6667$ (0-0 matches are ignored).

SMC = $(f_{11}+f_{00})/m = 6/8 = 0.75$.

Required Jaccard similarity = **0.67**.

Q.4 — Answer: 0.4424 to 0.4464 [NAT, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Add- α estimate: $(\text{count} + \alpha)/(N_c + \alpha V)$ with $V = 3$ possible values.

$= (3 + 1)/(6 + 1 \times 3) = 4/9 = \mathbf{0.444}$.

(The unsmoothed estimate would be $3/6$.)

Q.5 — Answer: (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Expressiveness]

(A) FALSE. One hidden layer suffices.

(B) FALSE. It gives no such bound; the width can be very large.

(C) FALSE. Thresholding $\sigma(w \cdot x + b)$ at 0.5 gives the hyperplane $w \cdot x + b = 0$.

(D) TRUE. $\sigma(w \cdot x + b) = 0.5 \Leftrightarrow w \cdot x + b = 0$.

Q.6 — Answer: (C) [MSQ, 1M · Regression & Model Selection > Simple linear regression]

(A) FALSE. It is $\hat{y} = \bar{y}$.

(B) FALSE. OLS minimises vertical distances; perpendicular distances give total least squares.

(C) TRUE. r^2 is invariant to affine changes of either variable.

(D) FALSE. $R^2 = r^2$, which is non-negative.

Q.7 — Answer: (A) [MSQ, 1M · Regression & Model Selection > Ridge & Lasso]

(A) TRUE. Each component in the SVD basis is $d_j^2/(d_j^2 + \lambda)$ times the OLS component.

(B) FALSE. $\lambda = 0$ (OLS) minimises training RSS; training error is non-decreasing in λ .

(C) FALSE. That always picks $\lambda = 0$; use validation / CV.

(D) FALSE. It is strictly convex.

Q.8 — Answer: (D) [MCQ, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

The jagged boundary with small islands around individual points (Panel Q) is produced by $k = 1$, which fits every training point (zero training error): low bias, high variance.

The smooth boundary (Panel P) comes from $k = 25$: averaging many neighbours raises bias and lowers variance; it misclassifies some training points, so its training error is not zero.

Answer: **Panel Q uses $k = 1$; it has lower bias and higher variance.**

Q.9 — Answer: 0.0352 to 0.0392 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

$\|x - z\|^2 = 6$.

$\exp(-\gamma \cdot 6) = 0.8 \Rightarrow \gamma = -\ln(0.8)/6 = 0.2231/6 = \mathbf{0.037}$.

Note: $K \in (0, 1]$, $K(x, x) = 1$, and K decreases as the points move apart or as γ grows.

Q.10 — Answer: 0.99 to 1.01 [NAT, 1M · SVM & Decision Trees · Decision trees: impurity measures]

$H = -\sum p_k \log_2 p_k = -(12/24)\log_2(12/24) - (12/24)\log_2(12/24) = 1$ bits.
(Maximum possible with 2 classes: $\log_2 2 = 1$.)

Q.11 — Answer: 360 [NAT, 1M · Unsupervised Learning · k-means]

There are k medoids and $n - k$ non-medoids, so $k(n - k) = 8 \cdot 45 = 360$ swaps (each evaluated in $O(n - k)$, giving $O(k(n - k)^2)$ per iteration, far costlier than k-means' $O(nkd)$).

Q.12 — Answer: 0.44 to 0.46 [NAT, 1M · Probabilistic & Instance-based Classifiers · Evaluation metrics]

TP = 27, FN = 9, FP = 34, TN = 6 ($N = 76$).

Balanced accuracy = $(\text{TPR} + \text{TNR})/2$.

TPR = $27/36 = 0.75$, TNR = $6/40 = 0.15$; balanced accuracy = **0.45**.

Watch the orientation: swapping FP and FN is the most common error (it swaps precision and recall).

Q.13 — Answer: 0.0404 to 0.0421 [NAT, 1M · Neural Networks (MLP / Feed-forward) · Training & regularisation]

$\|w\|^2 = 4 + 4 + 0.25 = 8.25$; $(\lambda/2)\|w\|^2 = 0.01 \times 8.25 = \mathbf{0.0413}$.

Q.14 — Answer: 3.0152 to 3.0455 [NAT, 1M · Regression & Model Selection · Multiple linear regression]

Variance inflation factor of predictor j : $\text{VIF}_j = 1/(1 - R_j^2)$, where R_j^2 is the R^2 from regressing x_j on all other predictors. It is the factor by which $\text{Var}(\beta_j)$ is inflated by collinearity; $\sqrt{\text{VIF}}$ inflates the standard error.

$\text{VIF} = 1/(1 - 0.67) = 3.0303$.

Answer: **3.03**.

Q.15 — Answer: (A), (B), (C), (D) [MSQ, 1M · SVM & Decision Trees · Ensembles: bagging & random forests]

(A) TRUE. Multiplied by e^α .

(B) TRUE. $\ln(1) = 0$.

(C) TRUE. Breiman's recommendation.

(D) TRUE. Feature subsampling reduces correlation between trees.

Q.16 — Answer: 3.898 to 3.902 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Training & regularisation]

$f(w) = (1/2)w^2 \Rightarrow \nabla f(w) = 1w$. Update: $v_t = \beta v_{t-1} + \nabla f(w_{t-1})$, $w_t = w_{t-1} - \eta v_t$, with $v_0 = 0$, $\beta = 0.5$, $\eta = 0.2$.

$t = 1$: $g = 1 \times 3 = 3$; $v_1 = 0.5 \times 0 + 3 = 3$; $w_1 = 3 - 0.2 \times 3 = 2.4$.

$t = 2$: $g = 1 \times 2.4 = 2.4$; $v_2 = 0.5 \times 3 + 2.4 = 3.9$; $w_2 = 2.4 - 0.2 \times 3.9 = 1.62$.

Answer: **3.9**.

Q.17 — Answer: (B) [MCQ, 2M · Probabilistic & Instance-based Classifiers · Logistic regression]

The line passes through $(-4, 0)$ and $(0, 3)$: $x_1/-4 + x_2/3 = 1$, i.e. $3x_1 + -4x_2 - -12 = 0$. Hence (w_0, w_1, w_2) must be a non-zero multiple of $(12, 3, -4)$.

Orientation: the class $y = 1$ region must have $w_0 + w_1x_1 + w_2x_2 > 0$. At the origin the $y = 1$ side has sign $+$ ($w_0 = 12$), which matches the figure (the origin lies on the $y = 1$ side).

So the answer is $(w_0, w_1, w_2) = (12, 3, -4)$ (any positive multiple gives the same classifier).

Other options either describe a different line (swapped/negated coefficients move the intercepts) or the same line with the sign flipped, which would label every training point incorrectly.

Q.18 — Answer: 0.1559 to 0.1619 [NAT, 2M · Regression & Model Selection · Bias-variance trade-off]

Mean prediction $\bar{f} = (1/6)\sum \hat{f}_m = 14.0667$.

$\text{Bias}^2 = (\bar{f} - f(x_0))^2 = (14.0667 - 15)^2 = 0.8711$.

Variance = $(1/6)\Sigma(\hat{f}_m - \bar{f})^2 = 0.1589$.

Expected test error = Bias² + Variance + $\sigma^2 = 0.8711 + 0.1589 + 1 = 2.03$.

Answer: **0.159**.

Q.19 — Answer: 249.99 to 250.01 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

For centered data, projecting onto the top-k PCs and reconstructing leaves the components along the discarded eigenvectors; their average squared length (divisor $n - 1$) is the sum of the discarded eigenvalues.

Discarded eigenvalues: 12.5, 10, 2.5; sum = 25; total = 96.

Total error = $(n - 1) \times 25 = 10 \times 25 = 250$.

Q.20 — Answer: 72.99 to 73.01 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Sort the values: 30(Y), 43(Y), 44(Y), 46(Y), 63(Y), 72(Y), 74(N), 80(N), 96(Y), 97(N).

Parent [7 Yes, 3 No], $H = 0.8813$. The optimal threshold always lies where the class changes between neighbours (Fayyad-Irani), giving 3 candidates:

Threshold	Left ($\leq t$) [Yes, No]	Right [Yes, No]	IG
73	[6, 0]	[1, 3]	0.5568
88	[6, 2]	[1, 1]	0.0323
96.5	[7, 2]	[0, 1]	0.1935

Best: $t = 73$ with IG = 0.5568. Answer: **73**.

Q.21 — Answer: 69.95 to 70.05 [NAT, 2M · SVM & Decision Trees > Decision trees: structure & traversal]

Route each record from the root to a leaf:

#	Predicted	Actual	Correct?
1	No	No	yes
2	Yes	Yes	yes
3	Yes	No	no
4	Yes	Yes	yes
5	Yes	No	no
6	Yes	No	no
7	No	No	yes
8	No	No	yes
9	Yes	Yes	yes
10	No	No	yes

Misclassified: 3 of 10 (accuracy 70%). Answer: **70**.

Q.22 — Answer: 0.8682 to 0.8882 [NAT, 2M · Unsupervised Learning > Distance measures]

Min-max scaling: $x' = (x - \min)/(\max - \min)$; age range [20, 60], income range [10, 210].

Customer	age'	income'
A	0.6	0.075
B	1	0.925
C	0.525	0.95

Raw distances: $d(A,B) = 170.75$, $d(A,C) = 175.03 \rightarrow$ nearest is B (income dominates because of its larger range).

Scaled distances: $d(A,B) = 0.9394$, $d(A,C) = 0.8782 \rightarrow$ nearest is C.

Answer: **0.88**. Scaling changes which neighbour is nearest — distance-based methods (k-means, k-NN, hierarchical clustering) should be run on standardized features.

Q.23 — Answer: (A), (B), (C) [MSQ, 2M · SVM & Decision Trees > SVM: dual & support vectors]

P4 and P5 (same class) lie on one margin line and P1 on the other; the optimal boundary is parallel to the line through P4, P5, midway to P1. Support vectors: P1, P4, P5.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (1, -1/3)$, $b = -1$; $\|w\| = 1.0541$, margin $2/\|w\| = 1.8974$.

Point	P1	P2	P3	P4	P5	P6
$y(w^T x + b)$	1	1.667	2	1	1	1.333

Support vectors are exactly the points with $y_i(w^T x_i + b) = 1$ ($\alpha_i > 0$): **P1, P4, P5**.

Dual check: $\alpha = (\alpha_1 = 5/9, \alpha_4 = 4/9, \alpha_5 = 1/9)$ satisfy $\sum \alpha_i y_i = 0$ and $\sum \alpha_i y_i x_i = w$.

Q.24 — Answer: 0.2122 to 0.2322 [NAT, 2M · SVM & Decision Trees > SVM: kernels]

$\cos \theta = \varphi(x)^T \varphi(z) / (\|\varphi(x)\| \|\varphi(z)\|) = K(x, z) / \sqrt{K(x, x) K(z, z)}$.

$K(x, z) = 4$, $K(x, x) = 36$, $K(z, z) = 9$.

$\cos \theta = 4 / \sqrt{36 \times 9} = \mathbf{0.22}$.

Q.25 — Answer: (A) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$p = \text{softmax}((2, -1, 0)) = (0.8438, 0.042, 0.1142)$; $y = (0, 0, 1)$.

$\partial L / \partial z = p - y = \mathbf{(0.84, 0.04, -0.89)}$.

Distractors: $y - p = (-0.84, -0.04, 0.89)$ (wrong sign); p alone = $(0.84, 0.04, 0.11)$ (forgot the target); $(p - y) \odot p \odot (1 - p) = (0.11, 0, -0.09)$ (spurious σ' -type factor, which cancels for cross-entropy); $-y/p = (0, 0, -8.76)$ is $\partial L / \partial p$, not $\partial L / \partial z$.

Q.26 — Answer: (B), (D) [MSQ, 2M · Regression & Model Selection > Ridge & Lasso]

With $X^T X = I$ the ridge path is $b/(1 + \lambda)$ and the lasso path is $\text{sign}(b)(|b| - \lambda)_+$. The plot shows paths reaching zero at finite $\lambda \Rightarrow$ lasso; here it is **lasso**.

(A) FALSE. The correct count is 3.

(B) TRUE. Penalty vanishes at $\lambda = 0$.

(C) FALSE. Under lasso $\beta_j = 0$ for $\lambda \geq |b_j|$.

(D) TRUE. Coefficients hit exactly zero at finite λ .

Q.27 — Answer: 3 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Rule: if $y_i(w \cdot x_i + b) \leq 0$, then $w \leftarrow w + \eta y_i x_i$, $b \leftarrow b + \eta y_i$ ($\eta = 1$). Start $w = (0, 0)$, $b = 0$.

Example 1: $a = 0$, $y \cdot a = 0 \rightarrow$ **mistake**, update $\rightarrow w = (3, -3)$, $b = 1$.

Example 2: $a = -5$, $y \cdot a = -5 \rightarrow$ **mistake**, update $\rightarrow w = (2, -2)$, $b = 2$.

Example 3: $a = 10$, $y \cdot a = -10 \rightarrow$ **mistake**, update $\rightarrow w = (0, 0)$, $b = 1$.

Example 4: $a = 1$, $y \cdot a = -1 \rightarrow$ **mistake**, update $\rightarrow w = (3, 3)$, $b = 0$.

Required value = **3**.

Note: a point with net input exactly 0 counts as a mistake ($y \cdot a \leq 0$) - this is why the first example always triggers an update when starting from zero.

Q.28 — Answer: 0.348 to 0.352 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

$\pi_0 p(x|C_0) = 0.6/8 = 0.075$ on $[0, 8]$; $\pi_1 p(x|C_1) = 0.4/8 = 0.05$ on $[1, 9]$.

Outside the overlap $[1, 8]$ only one class has non-zero density $\Rightarrow$ no error. Inside it the Bayes rule chooses C_0 (larger weighted density) and errs with density 0.05.

Bayes error = (overlap length) $\times$ min = $7 \times 0.05 = \mathbf{0.35}$.

Q.29 — Answer: (B), (C) [MSQ, 2M · Unsupervised Learning > k-means]

(A) FALSE. It only gives an $O(\log k)$ approximation in expectation.

(B) TRUE. A medoid is an actual point and uses unsquared dissimilarities, so an outlier pulls it far less.

(C) TRUE. The standard ANOVA decomposition; reducing WSS increases BSS.

(D) FALSE. Non-uniform scaling changes distances and can change the partition.

Q.30 — Answer: (B) [MCQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Ranges: age $[40, 58]$, income $[49, 130]$. Scaled query = $(0.889, 0.272)$.

Point	scaled age	scaled income	scaled distance	raw distance
P1	0.556	0.531	0.422	21.84

P2	0.611	0	0.388	22.56
P3	0	0.272	0.889	16
P4	1	1	0.737	59.03

Nearest after scaling: **P2**.

Without scaling the nearest point would be P3, because income differences (tens of thousands) dominate the age differences — exactly why k-NN needs feature scaling.

Q.31 — Answer: (C) [MCQ, 2M · Unsupervised Learning > Hierarchical clustering]

Use average linkage (UPGMA): $d(A,B) = (1/|A||B|) \sum_{a \in A} \sum_{b \in B} d(a,b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: {P1} + {P2} at height **1**. Updated: $d(\{P1, P2\}, \{P3\}) = 2.5$, $d(\{P1, P2\}, \{P4\}) = 11.5$, $d(\{P1, P2\}, \{P5\}) = 7.5$, $d(\{P1, P2\}, \{P6\}) = 15.5$.

Merge 2: {P3} + {P1, P2} at height **2.5**. Updated: $d(\{P1, P2, P3\}, \{P4\}) = 11.33$, $d(\{P1, P2, P3\}, \{P5\}) = 8.33$, $d(\{P1, P2, P3\}, \{P6\}) = 15.33$.

Merge 3: {P4} + {P6} at height **4**. Updated: $d(\{P4, P6\}, \{P5\}) = 9$, $d(\{P4, P6\}, \{P1, P2, P3\}) = 13.33$.

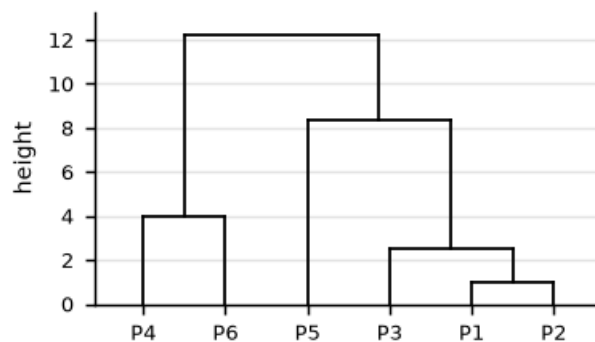
Merge 4: {P5} + {P1, P2, P3} at height **8.33**. Updated: $d(\{P1, P2, P3, P5\}, \{P4, P6\}) = 12.25$.

Merge 5: {P4, P6} + {P1, P2, P3, P5} at height **12.25**.

Stopping when 2 clusters remain (i.e. after 4 merges): **{P1, P2, P3, P5}, {P4, P6}**.

{P1, P2, P3}, {P4, P5, P6} is what *complete* linkage would give.

{P1, P2, P3}, {P4, P6}, {P5} has 3 clusters (cut at the wrong level).



Q.32 — Answer: (B) [MCQ, 2M · Regression & Model Selection > Simple linear regression]

$\bar{x} = 8.25$, $\bar{y} = -3.75$.

$S_{xx} = \sum x^2 - (\sum x)^2/n = 221.5$, $S_{yy} = \sum y^2 - (\sum y)^2/n = 143.5$, $S_{xy} = \sum xy - \sum x \sum y/n = -156.5$.

$b = S_{xy}/S_{xx} = -0.7065$; $a = \bar{y} - b \bar{x} = 2.079$.

Line: $\hat{y} = 2.08 - 0.71x$.

Distractors: using S_{xy}/S_{yy} (x-on-y slope), using uncentred sums $\sum xy/\sum x^2$, or a sign error in the intercept.

Q.33 — Answer: 0.7409 to 0.7453 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

For simple linear regression with intercept, the leverage (diagonal of the hat matrix) is $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$.

$\bar{x} = 5.4$, $S_{xx} = 101.2$.

$h = 0.2569$; $\text{Var}(e_i) = \sigma^2(1 - h_{ii}) = 1 \times 0.7431 = 0.7431$.

Answer: **0.743**.

Q.34 — Answer: (C) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

$a = W_1 x + b_1 = (3, 4, -1)^T$; $h = \text{ReLU}(a) = (3, 4, 0)^T$.

$\hat{y} = W_2 h + b_2 = (-2, -4)^T$.

Distractors: skipping the ReLU gives $(-3, -4)^T$; using W_1^T instead of W_1 gives $(-3, -4)^T$; dropping the biases gives $(-2, -5)^T$; using $|a|$ instead of ReLU gives $(-1, -4)^T$.

Q.35 — Answer: 0.997 to 1.001 [NAT, 2M · Probabilistic & Instance-based Classifiers › Evaluation metrics]

Per unit population: $TP = Se \cdot \pi$, $FN = (1 - Se) \cdot \pi$, $FP = (1 - Sp)(1 - \pi)$, $TN = Sp(1 - \pi)$.

$TP = 0.009$, $FN = 0.001$, $FP = 0.0099$, $TN = 0.9801$.

Negative predictive value = $TN / (TN + FN) = \mathbf{0.999}$.

Mock Test 09

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	5	8	22

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
SVM & Decision Trees	7	11
Probabilistic & Instance-based Classifiers	7	11
Regression & Model Selection	7	11
Neural Networks (MLP / Feed-forward)	7	11
Unsupervised Learning	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

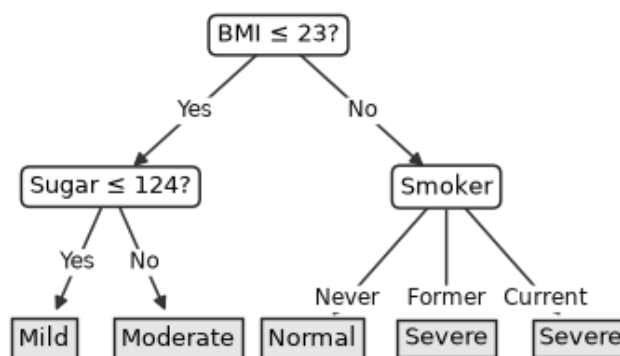
Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

A linear classifier has decision boundary $8x_1 + 15x_2 - 4 = 0$ (predict +1 when the left side is positive). The *signed geometric margin* $\gamma = y(w^T x + b)/\|w\|$ of the labelled example $x = (1, 4)$, $y = -1$ is _____ (rounded off to two decimal places)

Q.2 [1 mark · MCQ]

The decision tree below predicts the diagnosis category of a patient. Edges labelled Yes/No answer the test in the node. What is the prediction for the record shown in the table?



BMI	Sugar	BP	Smoker
24	134	95	Current

- (A) Normal
(C) Severe

- (B) Mild
(D) Moderate

Q.3 [1 mark · MSQ (one or more correct)]

Which of the following statements about naive Bayes classifiers is/are CORRECT?

- (A) Gaussian naive Bayes with class-specific variances generally yields a quadratic decision boundary.
- (B) Laplace smoothing is used mainly to reduce the number of parameters of naive Bayes.
- (C) Naive Bayes posterior probabilities are always well calibrated.
- (D) The number of parameters of naive Bayes grows exponentially with the number of features.

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about cross-validation (CV) is/are CORRECT? (n denotes the number of observations.)

- (A) A single train/validation split gives an error estimate that can vary substantially with the split.
- (B) In k -fold CV each observation is used for validation k times.
- (C) 2-fold CV gives a less (pessimistically) biased estimate of test error than LOOCV.
- (D) LOOCV on n observations requires n^2 model fits.

Q.5 [1 mark · NAT (numerical answer)]

For logits $z = (-0.5, 3, -2, 0)$, the temperature-scaled softmax $p_k = \exp(z_k/T) / \sum_j \exp(z_j/T)$ with $T = 2$ gives $p_2 = \underline{\hspace{2cm}}$ (rounded off to three decimal places)

Q.6 [1 mark · MCQ]

With Gaussian noise $N(0, \sigma^2)$ and prior $w \sim N(0, \tau^2 I)$, the MAP estimate of w equals:

- (A) ridge regression with $\lambda = \sigma^2/\tau^2$
- (B) ordinary least squares (the prior has no effect)
- (C) ridge regression with $\lambda = \sigma\tau$
- (D) lasso with $\lambda = \sigma^2/\tau^2$

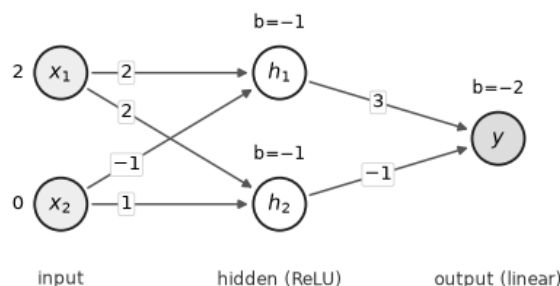
Q.7 [1 mark · MSQ (one or more correct)]

Which of the following statements about the k -nearest-neighbour method is/are CORRECT?

- (A) In distance-weighted k -NN with weights $1/d$, farther neighbours get larger weights.
- (B) The decision boundary of k -NN is always a straight line (hyperplane).
- (C) As $n \rightarrow \infty$, the error rate of 1-NN is at most twice the Bayes error rate (binary case).
- (D) The training error of k -NN is zero for every value of k .

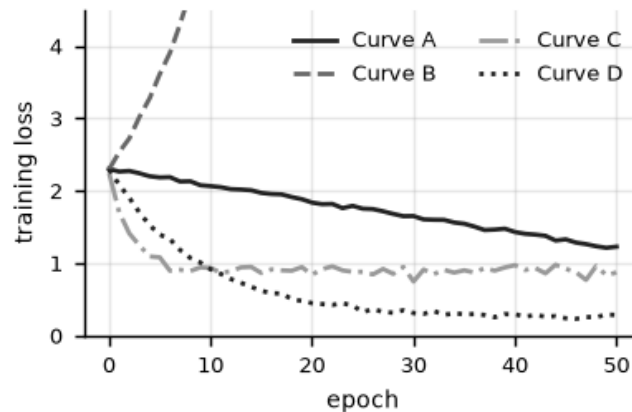
Q.8 [1 mark · NAT (numerical answer)]

For the ReLU network shown (edge labels are weights, 'b' denotes a bias) and input $x = (2, 0)$, the output y is (Answer in integer)



Q.9 [1 mark · MCQ]

The training-loss curves below were obtained with four different learning rates (everything else fixed). Which curve corresponds to a learning rate that is too low (very slow, almost linear decrease)?

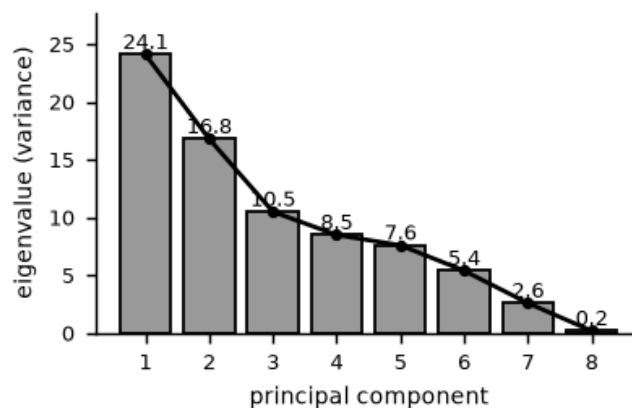


- (A) Curve A
(C) Curve C

- (B) Curve B
(D) Curve D

Q.10 [1 mark · NAT (numerical answer)]

The scree plot below shows the eigenvalues of the covariance matrix of a 8-dimensional dataset. The percentage of total variance explained by the first two principal components together is _____ (rounded off to one decimal place)

**Q.11 [1 mark · MCQ]**

A soft-margin linear SVM has learned $f(x) = 0.5x_1 + 0.5x_2 - 2$. For the training example $x = (-1, 4)$ with label $y = -1$, which statement is CORRECT?

- (A) Exactly on the decision boundary ($\xi = 1$)
(B) Correctly classified but strictly inside the margin ($0 < \xi < 1$)
(C) Misclassified ($\xi > 1$)
(D) Correctly classified, on or outside the margin ($\xi = 0$)

Q.12 [1 mark · MSQ (one or more correct)]

Which of the following statements about distance / similarity measures is/are CORRECT?

- (A) Euclidean distance is invariant to rotations and translations of the coordinate system.
(B) Euclidean distance is unaffected by changing the unit of one feature (e.g. metres to centimetres).
(C) Minkowski 'distance' with $0 < p < 1$ violates the triangle inequality.
(D) Two vectors with cosine similarity 1 must be identical.

Q.13 [1 mark · NAT (numerical answer)]

A 4-class softmax classifier produces the logits $z_1 = -1$, $z_2 = -1$, $z_3 = 1$, $z_4 = 0$ for an input $\mathbf{x}$. The predicted probability of class 4, $P(y = 4 | \mathbf{x})$, is _____ (rounded off to two decimal places)

Q.14 [1 mark · MSQ (one or more correct)]

Regarding the bias-variance trade-off in supervised regression with squared-error loss, which of the following statements is/are CORRECT?

- (A) A model with zero training error must have zero test error.
- (B) Expected test MSE = Bias + Variance + noise (bias not squared).
- (C) The irreducible error cannot be reduced by choosing a better model.
- (D) Regularisation reduces the bias of a model.

Q.15 [1 mark · NAT (numerical answer)]

A clustering algorithm groups labelled data into three clusters; the contingency table of cluster versus true class counts is given below. If every point in a cluster is labelled with the cluster's majority class, the number of misclassified points is _____ (Answer in integer)

	Class P	Class Q	Class R
Cluster 1	1	5	27
Cluster 2	11	8	5
Cluster 3	0	4	12

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · MCQ]**

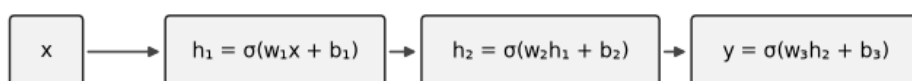
A decision tree is to be grown on the 10 training records below using Gini gain as the splitting criterion (multi-way split on each categorical attribute). Which attribute is chosen at the root?

#	Temp	Colour	Outlook	Wind	Class
1	Hot	Green	Sunny	Weak	No
2	Cool	Red	Sunny	Strong	No
3	Cool	Green	Sunny	Weak	No
4	Cool	Blue	Sunny	Weak	No
5	Hot	Blue	Rain	Strong	No
6	Cool	Blue	Rain	Strong	No
7	Mild	Green	Rain	Weak	Yes
8	Hot	Blue	Overcast	Strong	Yes
9	Cool	Green	Rain	Strong	Yes
10	Cool	Blue	Sunny	Weak	No

- (A) Temp
- (B) Colour
- (C) Outlook
- (D) Wind

Q.17 [2 marks · NAT (numerical answer)]

Consider the scalar chain shown below with weights $w_1 = -3$, $w_2 = -1$, $w_3 = -1$, biases $b_1 = -0.5$, $b_2 = 0.5$, $b_3 = -0.5$ and activation the logistic sigmoid σ at every stage. For input $x = -0.5$, the value of $\partial y / \partial w_1$ is _____ (rounded off to 4 decimal places)



Q.18 [2 marks · NAT (numerical answer)]

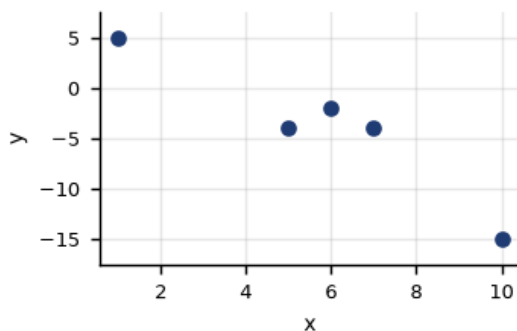
Agglomerative hierarchical clustering with **single linkage** is applied to 5 points whose pairwise distance matrix is given below (upper triangle shown). The height (linkage distance) at which the **fourth** merge takes place is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5
P1	0	5	8	4	2
P2		0	9	1	5
P3			0	10	10
P4				0	4
P5					0

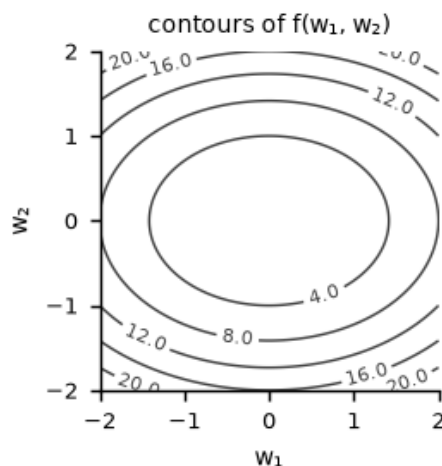
Q.19 [2 marks · NAT (numerical answer)]

Assume $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with ε_i i.i.d. $N(0, \sigma^2)$. For the 5 observations below, with β estimated by least squares (= maximum likelihood), the maximum-likelihood estimate of σ^2 is _____ (rounded off to two decimal places)

x	1	5	6	7	10
y	5	-4	-2	-4	-15

**Q.20 [2 marks · MSQ (one or more correct)]**

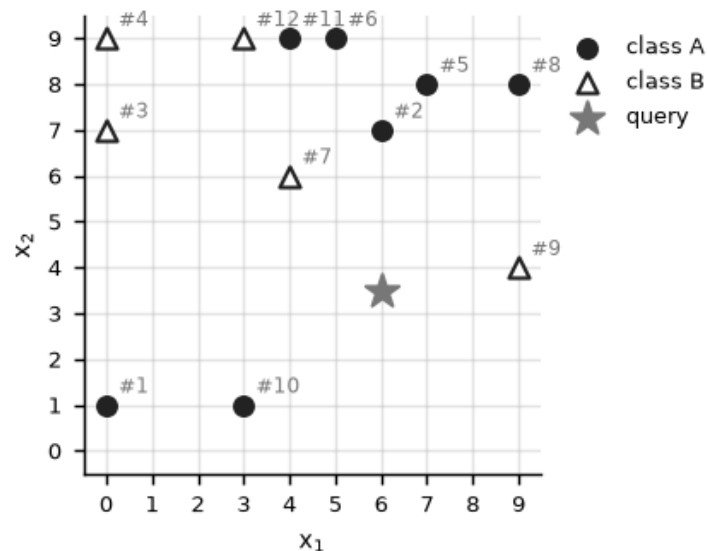
Gradient descent $w \leftarrow w - \eta \nabla f(w)$ is applied to $f(w_1, w_2) = \frac{1}{2}(4w_1^2 + 8w_2^2)$ from a generic starting point (both coordinates non-zero). For which of the following learning rates does GD **diverge**?

**(A)** $\eta = 0.0375$ **(B)** $\eta = 0.375$ **(C)** $\eta = 0.2375$ **(D)** $\eta = 0.1$

Q.21 [2 marks · MSQ (one or more correct)]

The figure and table show 12 labelled training points (class A: filled circles, class B: open triangles) and a query point (star) at (6, 3.5). A k-NN classifier with **Euclidean** distance and majority vote is used. Which of the following statements is/are CORRECT?

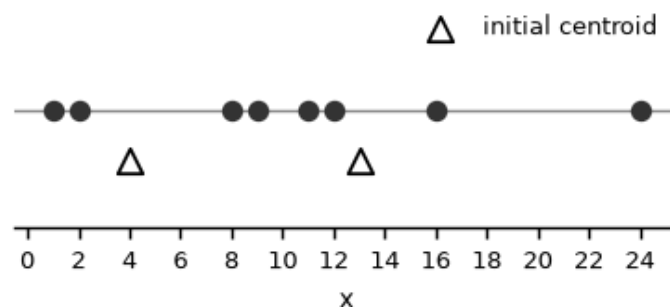
#	1	2	3	4	5	6	7	8	9	10	11	12
x_1	0	6	0	0	7	5	4	9	9	3	4	3
x_2	1	7	7	9	8	9	6	8	4	1	9	9
class	A	A	B	B	A	A	B	A	B	A	A	B



- (A) With $k = 3$, the query is classified as class B.
- (B) With $k = 7$, the query is classified as class A.
- (C) With $k = 1$, the query is classified as class B.
- (D) With $k = 5$, the query is classified as class B.

Q.22 [2 marks · NAT (numerical answer)]

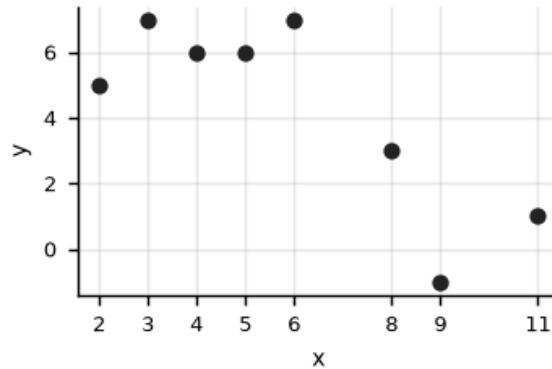
The 1-D dataset $\{1, 2, 8, 9, 11, 12, 16, 24\}$ is clustered with k-means ($k = 2$) using initial centroids 4 and 13 (shown as triangles). Lloyd's iterations (assign to nearest centroid, then recompute means) are run until convergence. Then the **larger** of the two final centroids is _____ (rounded off to two decimal places)



Q.23 [2 marks · NAT (numerical answer)]

A regression tree (squared-error criterion) is fitted to the 8 points below. The root split " $x \leq t$ " is chosen among midpoints of consecutive x values to minimise the total SSE of the two children (each child predicts its mean). Find the threshold t of the best split " $x \leq t$ ". (rounded off to one decimal place)

x	2	3	4	5	6	8	9	11
y	5	7	6	6	7	3	-1	1

**Q.24 [2 marks · NAT (numerical answer)]**

A naive Bayes classifier is trained on the 12 examples below, using **Laplace (add-one) smoothing** for the class-conditional feature probabilities (the class prior is not smoothed). For the new instance (Fever = Mild, Cough = No, Fatigue = Yes), the posterior probability $P(\text{Flu} = \text{Pos} | x)$ is _____ (rounded off to two decimal places)

#	Fever	Cough	Fatigue	Flu
1	High	Yes	No	Neg
2	High	No	No	Pos
3	High	Yes	No	Neg
4	Mild	No	Yes	Pos
5	None	Yes	No	Neg
6	High	No	No	Neg
7	High	Yes	No	Neg
8	High	Yes	Yes	Pos
9	Mild	No	No	Pos
10	High	No	No	Neg
11	High	No	No	Pos
12	High	No	No	Neg

Q.25 [2 marks · NAT (numerical answer)]

Consider the model $y = \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ (no intercept), fitted by least squares using $\beta = (X^T X)^{-1} X^T y$ with the design matrix X and response y below. Then the predicted response at $(x_1, x_2) = (1, 2)$ is _____ (rounded off to two decimal places)

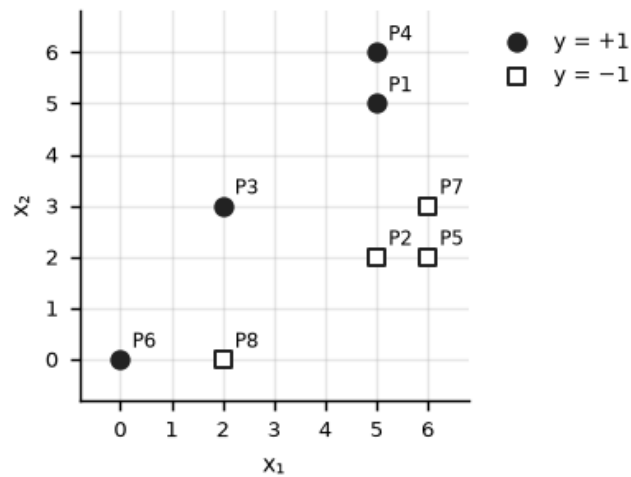
$$X = \begin{bmatrix} 0 & 1 \\ 3 & 3 \\ 2 & -1 \\ -1 & 3 \end{bmatrix}$$

$$y = \begin{bmatrix} 2 \\ -1 \\ 8 \\ -1 \end{bmatrix}$$

Q.26 [2 marks · NAT (numerical answer)]

The labelled training points shown below are linearly separable. A hard-margin linear SVM is trained on them. Compute the value of the canonical decision function $f(x) = w^T x + b$ (support vectors satisfy $y_i f(x_i) = 1$) at the test point $x = (1, 3)$. (rounded off to two decimal places)

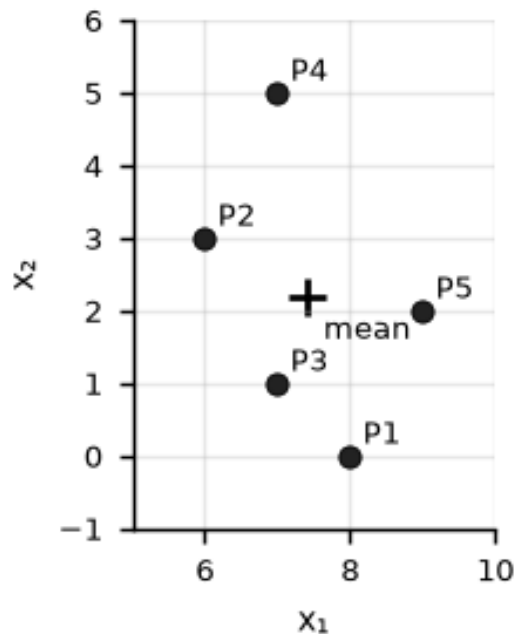
Point	x_1	x_2	y
P1	5	5	+1
P2	5	2	-1
P3	2	3	+1
P4	5	6	+1
P5	6	2	-1
P6	0	0	+1
P7	6	3	-1
P8	2	0	-1



Q.27 [2 marks · NAT (numerical answer)]

PCA is applied to the five 2-D points below (data are mean-centered first; sample covariance uses divisor $n - 1$). If each point is reconstructed from its PC1 score only ($\hat{x} = \bar{x} + z_1 v_1$), the squared reconstruction error $\|x - \hat{x}\|^2$ of P4 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5
x_1	8	6	7	7	9
x_2	0	3	1	5	2

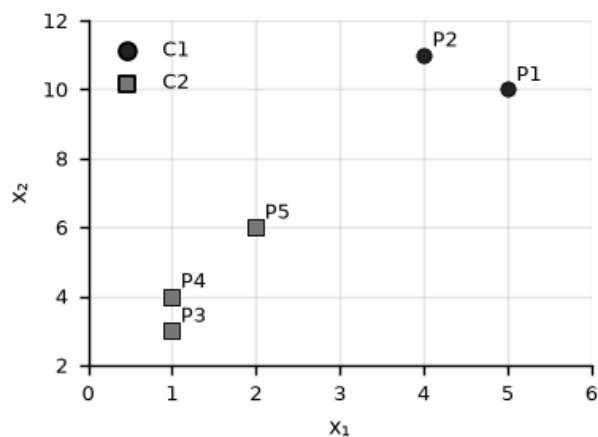
**Q.28 [2 marks · NAT (numerical answer)]**

$X_1, \dots, X_n$ are i.i.d. with mean $\mu = 3$ and variance $\sigma^2 = 9$, with $n = 20$. Consider the shrinkage estimator $\hat{\mu} = c \cdot \bar{x}$ of μ , where $\bar{x}$ is the sample mean. Using $\text{MSE} = \text{Bias}^2 + \text{Variance}$, the value of c that minimises the MSE of $\hat{\mu} = c \cdot \bar{x}$ is _____ (rounded off to three decimal places)

Q.29 [2 marks · NAT (numerical answer)]

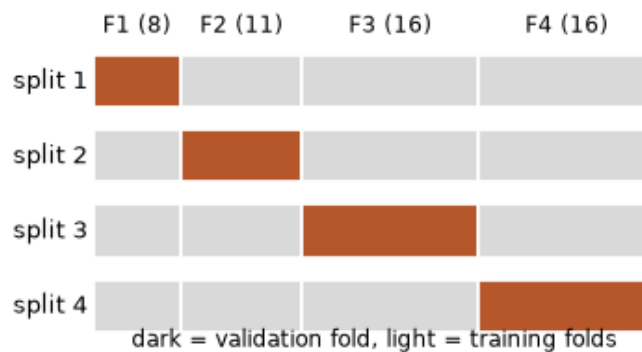
A clustering of 5 points into 2 clusters is shown below. Using Manhattan distance, the silhouette coefficient $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$ of point P2 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5
x_1	5	4	1	1	2
x_2	10	11	3	4	6
cluster	C1	C1	C2	C2	C2



Q.30 [2 marks · NAT (numerical answer)]

A regression model is evaluated by 4-fold cross-validation on $n = 51$ observations. The folds (F1 ... F4) have unequal sizes, as shown in the diagram and table. The cross-validated root mean squared error (pooled over all n out-of-fold predictions) is _____ (rounded off to two decimal places)



Fold	F1	F2	F3	F4
size	8	11	16	16
sum of squared errors	22.4	89.1	80	52.8

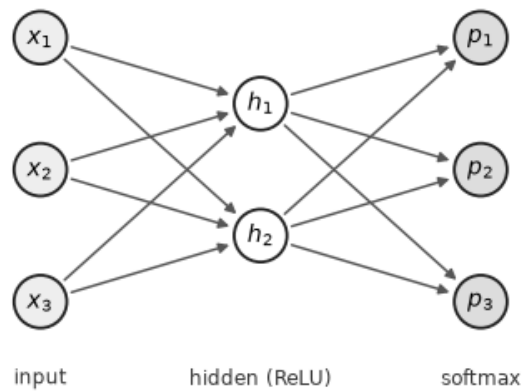
Q.31 [2 marks · MSQ (one or more correct)]

Which of the following statements about LDA, QDA and logistic regression is/are CORRECT?

- (A) Fisher's LDA directions coincide with the directions of maximum total variance found by PCA.
- (B) QDA has fewer parameters than LDA.
- (C) Logistic regression assumes Gaussian class-conditional densities.
- (D) QDA allows each class its own covariance matrix, giving quadratic decision boundaries in general.

Q.32 [2 marks · NAT (numerical answer)]

A 3-2-3 network computes $h = \text{ReLU}(W_1 x + b_1)$, $z = W_2 h + b_2$, $p = \text{softmax}(z)$, with the parameters below. For $x = (0, 2, -1)^T$, the predicted probability p_2 of class 2 is _____ (rounded off to three decimal places)



$$W_1 = \begin{bmatrix} 1 & -0.5 & 0.5 \\ -1 & 0.5 & -0.5 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

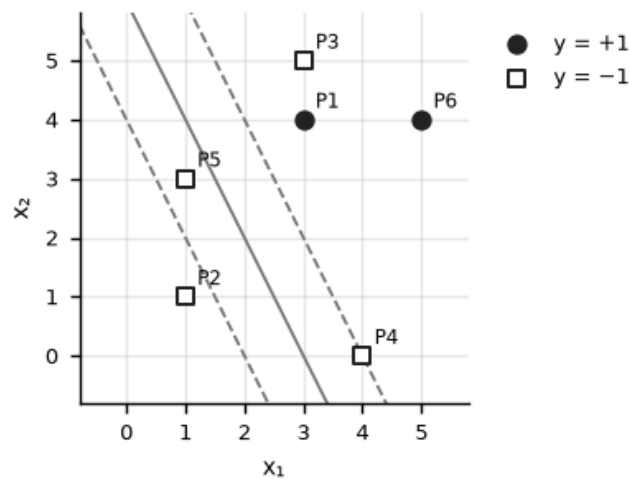
$$W_2 = \begin{bmatrix} 0 & -1 \\ 2 & 2 \\ -2 & -2 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

Q.33 [2 marks · NAT (numerical answer)]

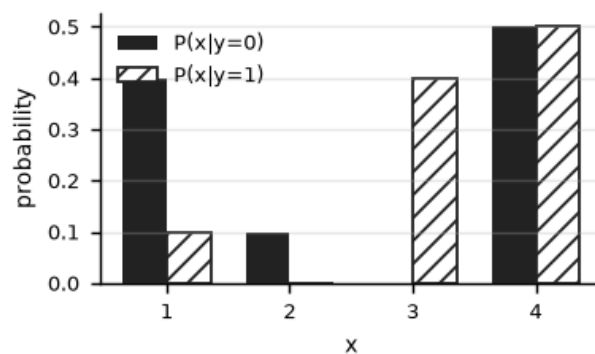
A linear SVM with $f(x) = x_1 + 0.5x_2 - 3$ is evaluated on the training points in the table (also plotted, with the boundary $f = 0$ solid and $f = \pm 1$ dashed). Compute the value of the soft-margin primal objective $\frac{1}{2}\|w\|^2 + C \sum_i \xi_i$ with $C = 5$ (using the optimal slacks for this (w, b)). (rounded off to two decimal places)

Point	x_1	x_2	y
P1	3	4	+1
P2	1	1	-1
P3	3	5	-1
P4	4	0	-1
P5	1	3	-1
P6	5	4	+1

**Q.34 [2 marks · NAT (numerical answer)]**

A discrete feature $x \in \{1, \dots, 4\}$ has the class-conditional distributions below (also plotted), with priors $P(y = 0) = 0.6$ and $P(y = 1) = 0.4$. Then the posterior probability $P(y = 1 \mid x = 2)$ is _____ (rounded off to three decimal places)

x	1	2	3	4
$P(x \mid y=0)$	0.4	0.1	0	0.5
$P(x \mid y=1)$	0.1	0	0.4	0.5



Q.35 [2 marks · NAT (numerical answer)]

A 2-2-1 network uses the identity (linear) activation in its hidden layer: $\hat{y} = W_2(W_1x + b_1) + b_2$, with the parameters below. The network is equivalent to a single linear model $\hat{y} = \tilde{w}_1x_1 + \tilde{w}_2x_2 + \tilde{b}$. Then the coefficient $\tilde{w}_1$ is _____ (Answer in integer)

$$W_1 = \begin{bmatrix} -2 & -2 \\ 2 & 2 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} -2 & -1 \end{bmatrix}$$

$$b_2 = 2$$

— END OF QUESTION PAPER —

Mock Test 09 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	−3.7747 to −3.7547	19	NAT	2	4.1609 to 4.2111
2	MCQ	1	(C)	20	MSQ	2	(B)
3	MSQ	1	(A)	21	MSQ	2	(A), (B), (C)
4	MSQ	1	(A)	22	NAT	2	15.74 to 15.76
5	NAT	1	0.6741 to 0.6781	23	NAT	2	6.99 to 7.01
6	MCQ	1	(A)	24	NAT	2	0.9332 to 0.9532
7	MSQ	1	(C)	25	NAT	2	−0.4745 to −0.4345
8	NAT	1	4	26	NAT	2	2.99 to 3.01
9	MCQ	1	(A)	27	NAT	2	0.2091 to 0.2291
10	NAT	1	53.829 to 54.229	28	NAT	2	0.9494 to 0.9554
11	MCQ	1	(B)	29	NAT	2	0.7757 to 0.7957
12	MSQ	1	(A), (C)	30	NAT	2	2.1787 to 2.1987
13	NAT	1	0.2145 to 0.2345	31	MSQ	2	(D)
14	MSQ	1	(C)	32	NAT	2	0.994 to 0.998
15	NAT	1	23	33	NAT	2	30.615 to 30.635
16	MCQ	2	(C)	34	NAT	2	−0.002 to 0.002
17	NAT	2	−0.0052 to −0.0046	35	NAT	2	2
18	NAT	2	7.99 to 8.01				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
SVM & Decision Trees					11
Probabilistic & Instance-based Classifiers					11
Regression & Model Selection					11
Neural Networks (MLP / Feed-forward)					11
Unsupervised Learning					11
Total					55

Mock Test 09 — Detailed Solutions

Q.1 — Answer: **-3.7747 to -3.7547** [NAT, 1M · SVM & Decision Trees > SVM: margin & geometry]

Functional margin $y(w^T x + b) = (-1)(64) = -64$.

$\|w\| = 17$, so $\gamma = -64/17 = -3.76$.

A negative value means the example is on the wrong side of the boundary (misclassified).

Q.2 — Answer: **(C)** [MCQ, 1M · SVM & Decision Trees > Decision trees: structure & traversal]

Follow the record from the root, taking the branch that matches each test:

BMI = 24 > 23 → No.

Smoker = Current → branch Current.

(Only the tests on the actual path matter.) The record reaches the leaf **Severe**.

Note: “≤” includes equality, so a value equal to the threshold goes to the Yes branch.

Q.3 — Answer: **(A)** [MSQ, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

(A) TRUE. Log-ratio contains x_j^2 terms with coefficients $(1/\sigma_{j0}^2 - 1/\sigma_{j1}^2)/2$.

(B) FALSE. It avoids zero probability estimates; the parameter count is unchanged.

(C) FALSE. They are often over-confident.

(D) FALSE. That is the full joint model; naive Bayes is linear in d .

Q.4 — Answer: **(A)** [MSQ, 1M · Regression & Model Selection > Cross-validation]

(A) TRUE. That is why k-fold CV averages over several splits.

(B) FALSE. Each observation is validated exactly once.

(C) FALSE. 2-fold trains on only half the data, so its estimate is more pessimistically biased.

(D) FALSE. It requires n fits.

Q.5 — Answer: **0.6741 to 0.6781** [NAT, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

Scaled logits $z/T = (-0.25, 1.5, -1, 0)$.

exp values: 0.7788, 4.4817, 0.3679, 1; sum = 6.6284.

$p_2 = 4.4817/6.6284 = \mathbf{0.6761}$.

Q.6 — Answer: **(A)** [MCQ, 1M · Regression & Model Selection > MLE & MAP view]

$-\log \text{posterior} \propto (1/2\sigma^2)\|y - Xw\|^2 + (1/2\tau^2)\|w\|^2$; multiplying by $2\sigma^2$ gives $\|y - Xw\|^2 + (\sigma^2/\tau^2)\|w\|^2$.

Answer: **ridge regression with $\lambda = \sigma^2/\tau^2$** .

Q.7 — Answer: **(C)** [MSQ, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

(A) FALSE. Weights decrease with distance.

(B) FALSE. It is generally piecewise and non-linear.

(C) TRUE. Cover-Hart bound: $R^* \leq R_{1NN} \leq 2R^*(1 - R^*)$.

(D) FALSE. Only 1-NN is guaranteed zero training error (without duplicates).

Q.8 — Answer: **4** [NAT, 1M · Neural Networks (MLP / Feed-forward) > Forward pass]

$a = Wx + b = (3, 3)$; $h = \text{ReLU}(a) = (3, 3)$.

$y = 3 \cdot 3 + (-1) \cdot 3 + (-2) = \mathbf{4}$.

A common slip is to forget the ReLU and use the raw pre-activation (which would give 4).

Q.9 — Answer: **(A)** [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Typical signatures: far-too-high $\eta \rightarrow$ loss blows up; somewhat-too-high $\eta \rightarrow$ quick drop then plateau at a high loss (bouncing around the minimum); too-low $\eta \rightarrow$ slow, nearly linear decrease; good $\eta \rightarrow$ fast decrease to the lowest loss.

Curve A: a learning rate that is too low (very slow, almost linear decrease).

Curve B: a learning rate that is far too high (training diverges).

Curve C: a learning rate that is somewhat too high (fast initial drop, then the loss plateaus at a high value).

Curve D: a well-chosen learning rate.

Hence the answer is **Curve A**.

Q.10 — Answer: 53.829 to 54.229 [NAT, 1M · Unsupervised Learning › PCA & dimensionality reduction]

Total variance = $\Sigma \lambda = 75.7$. Cumulative proportion of variance explained:

k	1	2	3	4	5	6	7	8
λ_k	24.1	16.8	10.5	8.5	7.6	5.4	2.6	0.2
cumulative %	31.84	54.03	67.9	79.13	89.17	96.3	99.74	100

$$(\lambda_1 + \lambda_2)/\Sigma \lambda = 40.9/75.7 = \mathbf{54\%}.$$

Q.11 — Answer: (B) [MCQ, 1M · SVM & Decision Trees › SVM: soft margin & hinge loss]

$$y \cdot f(x) = (-1) \cdot (-0.5) = 0.5.$$

$$\text{Slack } \xi = \max(0, 1 - y f(x)) = 0.5.$$

Rules: $y f \geq 1 \Rightarrow \xi = 0$ (on/outside margin); $0 < y f < 1 \Rightarrow$ inside margin, still correct; $y f = 0 \Rightarrow$ on the boundary ($\xi = 1$); $y f < 0 \Rightarrow$ misclassified ($\xi > 1$).

Hence: **Correctly classified but strictly inside the margin ($0 < \xi < 1$)**.

Q.12 — Answer: (A), (C) [MSQ, 1M · Unsupervised Learning › Distance measures]

(A) **TRUE**. Orthogonal transforms preserve L_2 norms.

(B) **FALSE**. Rescaling one coordinate changes distances.

(C) **TRUE**. e.g. $p = 0.5$: $d((0,0),(1,1)) = 4$, which exceeds $d((0,0),(1,0)) + d((1,0),(1,1)) = 2$.

(D) **FALSE**. They need only point in the same direction.

Q.13 — Answer: 0.2145 to 0.2345 [NAT, 1M · Probabilistic & Instance-based Classifiers › Logistic regression]

$$\text{softmax: } P(y = k | \mathbf{x}) = e^{z_k} / \sum_j e^{z_j}.$$

$$e^z \text{ values: } 0.3679, 0.3679, 2.7183, 1; \text{ sum} = 4.454.$$

$$P(y = 4 | \mathbf{x}) = 1/4.454 = \mathbf{0.22}.$$

Q.14 — Answer: (C) [MSQ, 1M · Regression & Model Selection › Bias-variance trade-off]

(A) **FALSE**. It may have badly overfit.

(B) **FALSE**. It is the squared bias that appears.

(C) **TRUE**. It is the variance of the noise ϵ .

(D) **FALSE**. It increases bias, reduces variance.

Q.15 — Answer: 23 [NAT, 1M · Unsupervised Learning › Cluster evaluation]

Majority count in each cluster: 27, 11, 12 (sum 50); $N = 73$.

$$\text{Purity} = (1/N) \sum_k \max_j n_{kj} = 50/73 = 0.6849.$$

$$\text{Misclassified} = N - \sum \max = 23.$$

Answer: **23**.

Purity is an external measure (needs class labels) and is trivially 1 if each point forms its own cluster, so it cannot be used to choose k .

Q.16 — Answer: (C) [MCQ, 2M · SVM & Decision Trees › Decision trees: split selection]

Parent: 3 Yes, 7 No $\Rightarrow$ Gini = 0.42.

Temp — Hot: [1+, 2-], Mild: [1+, 0-], Cool: [1+, 5-]; gain = 0.12.

Colour — Red: [0+, 1-], Green: [2+, 2-], Blue: [1+, 4-]; gain = 0.06.

Outlook — Sunny: [0+, 5-], Overcast: [1+, 0-], Rain: [2+, 2-]; gain = 0.22.

Wind — Weak: [1+, 4-], Strong: [2+, 3-]; gain = 0.02.

The largest Gini gain is for **Outlook**, so it is chosen at the root.

Q.17 — Answer: -0.0052 to -0.0046 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Activation functions]

$$\text{Forward: } a_1 = -3 \cdot -0.5 + -0.5 = 1, h_1 = 0.7311; a_2 = -1 \cdot 0.7311 + 0.5 = -0.2311, h_2 = 0.4425; a_3 = -1 \cdot 0.4425 + -0.5 = -0.9425, y = 0.2804.$$

$$\text{Local derivatives } \sigma'(a): \sigma'(a) = \sigma(a)(1 - \sigma(a)): \sigma'(a_1) = 0.1966, \sigma'(a_2) = 0.2467, \sigma'(a_3) = 0.2018.$$

Chain rule: $\partial y / \partial w_1 = \sigma'(a_3) \cdot w_3 \cdot \sigma'(a_2) \cdot w_2 \cdot \sigma'(a_1) \cdot x = -0.0049$.

Each layer contributes a factor $\sigma' \leq 0.25$ – the product shrinks with depth (vanishing gradient).

Q.18 — Answer: 7.99 to 8.01 [NAT, 2M · Unsupervised Learning · Hierarchical clustering]

Use single linkage: $d(A, B) = \min_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P2\} + \{P4\}$ at height **1**. Updated: $d(\{P2, P4\}, \{P1\}) = 4$, $d(\{P2, P4\}, \{P3\}) = 9$, $d(\{P2, P4\}, \{P5\}) = 4$.

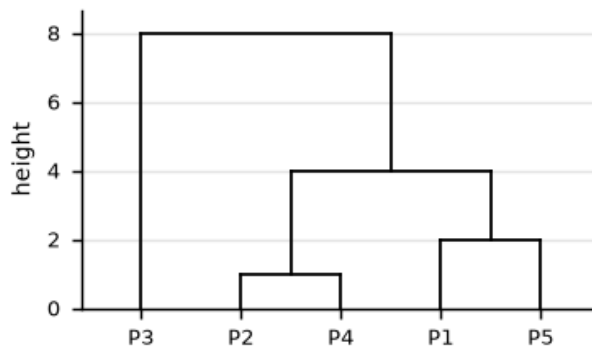
Merge 2: $\{P1\} + \{P5\}$ at height **2**. Updated: $d(\{P1, P5\}, \{P3\}) = 8$, $d(\{P1, P5\}, \{P2, P4\}) = 4$.

Merge 3: $\{P2, P4\} + \{P1, P5\}$ at height **4**. Updated: $d(\{P1, P2, P4, P5\}, \{P3\}) = 8$.

Merge 4: $\{P3\} + \{P1, P2, P4, P5\}$ at height **8**.

The fourth merge occurs at height **8**.

Dendrogram:



Q.19 — Answer: 4.1609 to 4.2111 [NAT, 2M · Regression & Model Selection · MLE & MAP view]

$\bar{x} = 5.8$, $\bar{y} = -4$.

i	1	2	3	4	5
$x_i - \bar{x}$	-4.8	-0.8	0.2	1.2	4.2
$y_i - \bar{y}$	9	0	2	0	-11
$(x_i - \bar{x})(y_i - \bar{y})$	-43.2	0	0.4	0	-46.2

$$S_{xx} = \sum (x_i - \bar{x})^2 = 42.8, S_{xy} = -89.$$

$$\beta_1 = S_{xy} / S_{xx} = -2.0794, \beta_0 = \bar{y} - \beta_1 \bar{x} = 8.0607.$$

$$SSE = S_{yy} - S_{xy}^2 / S_{xx} = 206 - 185.0701 = 20.9299.$$

Maximising the Gaussian likelihood over σ^2 gives $\hat{\sigma}_{ML}^2 = SSE/n = 20.9299/5 = 4.186$.

Answer: **4.19**.

Q.20 — Answer: (B) [MSQ, 2M · Neural Networks (MLP / Feed-forward) · Training & regularisation]

$\nabla f = (4w_1, 8w_2) = w_i \leftarrow (1 - \eta a_i) w_i$. GD converges (from every start) iff $|1 - \eta a_i| < 1$ for both i , i.e. $0 < \eta < 2/\max(a_i) = 2/8 = 0.25$. The steep direction ($a = 8$) limits the learning rate.

(A) FALSE. $\eta = 0.0375$: $|1 - \eta \cdot 4| = 0.85$, $|1 - \eta \cdot 8| = 0.7 \rightarrow$ converges.

(B) TRUE. $\eta = 0.375$: $|1 - \eta \cdot 4| = 0.5$, $|1 - \eta \cdot 8| = 2 \rightarrow$ diverges.

(C) FALSE. $\eta = 0.2375$: $|1 - \eta \cdot 4| = 0.05$, $|1 - \eta \cdot 8| = 0.9 \rightarrow$ converges.

(D) FALSE. $\eta = 0.1$: $|1 - \eta \cdot 4| = 0.6$, $|1 - \eta \cdot 8| = 0.2 \rightarrow$ converges.

Q.21 — Answer: (A), (B), (C) [MSQ, 2M · Probabilistic & Instance-based Classifiers · k-nearest neighbours]

Distances from the query, sorted:

rank	#	(x_1, x_2)	distance	class
1	9	(9, 4)	$\sqrt{9.25} = 3.041$	B
2	7	(4, 6)	$\sqrt{10.25} = 3.202$	B
3	2	(6, 7)	$\sqrt{12.25} = 3.5$	A
4	10	(3, 1)	$\sqrt{15.25} = 3.905$	A

5	5	(7, 8)	$\sqrt{21.25} = 4.61$	A
6	8	(9, 8)	$\sqrt{29.25} = 5.408$	A
7	6	(5, 9)	$\sqrt{31.25} = 5.59$	A
8	11	(4, 9)	$\sqrt{34.25} = 5.852$	A

(A) TRUE. 3 nearest: #9, #7, #2 $\rightarrow$ 1 A vs 2 B $\Rightarrow$ class B.

(B) TRUE. 7 nearest: #9, #7, #2, #10, #5, #8, #6 $\rightarrow$ 5 A vs 2 B $\Rightarrow$ class A.

(C) TRUE. 1 nearest: #9 $\rightarrow$ 0 A vs 1 B $\Rightarrow$ class B.

(D) FALSE. 5 nearest: #9, #7, #2, #10, #5 $\rightarrow$ 3 A vs 2 B $\Rightarrow$ class A.

Q.22 — Answer: 15.74 to 15.76 [NAT, 2M · Unsupervised Learning > k-means]

Data: {1, 2, 8, 9, 11, 12, 16, 24}; initial centroids $c_1 = 4, c_2 = 13$.

Iter	centroids used	cluster 1	cluster 2	new centroids
1	4, 13	{1, 2, 8}	{9, 11, 12, 16, 24}	3.67, 14.4
2	3.67, 14.4	{1, 2, 8, 9}	{11, 12, 16, 24}	5, 15.75
3	5, 15.75	{1, 2, 8, 9}	{11, 12, 16, 24}	5, 15.75

In the last row the centroids do not change, so the algorithm has converged: final centroids 5 and 15.75.

Larger final centroid = **15.75**.

Q.23 — Answer: 6.99 to 7.01 [NAT, 2M · SVM & Decision Trees > Regression trees]

Parent mean $\bar{y} = 4.25$, $SSE_0 = \sum (y - \bar{y})^2 = 61.5$.

Evaluate every split:

Split t	Left mean	Right mean	$SSE_L + SSE_R$
2.5	5	4.143	60.857
3.5	6	3.667	53.333
4.5	6	3.2	46.8
5.5	6	2.5	37
7	6.2	1	10.8
8.5	5.667	0	13.333
10	4.714	1	49.429

Best split $t = 7$: left mean 6.2, right mean 1, $SSE = 10.8$, reduction = 50.7.

Answer: **7**.

Q.24 — Answer: 0.9332 to 0.9532 [NAT, 2M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Naive Bayes: $P(c | x) \propto P(c) \prod_j P(x_j | c)$. With Laplace smoothing, $P(x_j = v | c) = (\text{count} + 1) / (n_c + V_j)$, V_j = number of values of feature j .

Pos: $P(\text{Flu}=\text{Pos}) = 5/12$; $P(\text{Fever}=\text{Mild}|\text{Pos}) = (2+1)/(5+3) = 3/8$; $P(\text{Cough}=\text{No}|\text{Pos}) = (4+1)/(5+2) = 5/7$; $P(\text{Fatigue}=\text{Yes}|\text{Pos}) = (2+1)/(5+2) = 3/7$. Product = 0.04783.

Neg: $P(\text{Flu}=\text{Neg}) = 7/12$; $P(\text{Fever}=\text{Mild}|\text{Neg}) = (0+1)/(7+3) = 1/10$; $P(\text{Cough}=\text{No}|\text{Neg}) = (3+1)/(7+2) = 4/9$; $P(\text{Fatigue}=\text{Yes}|\text{Neg}) = (0+1)/(7+2) = 1/9$. Product = 0.00288.

$P(\text{Pos} | x) = 0.04783 / (0.04783 + 0.00288) = \mathbf{0.94}$.

Q.25 — Answer: -0.4745 to -0.4345 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

$$X^T X = \begin{bmatrix} 14 & 4 \\ 4 & 20 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 14 \\ -12 \end{bmatrix}$$

$$\det(X^T X) = 264; (X^T X)^{-1} = (1/264) \cdot \begin{bmatrix} 20 & -4 \\ -4 & 14 \end{bmatrix}.$$

$$\beta = (X^T X)^{-1} X^T y = (1.2424, -0.8485).$$

$$\hat{y} = (1.2424)(1) + (-0.8485)(2) = -0.4545.$$

Answer: **-0.45**.

Q.26 — Answer: 2.99 to 3.01 [NAT, 2M · SVM & Decision Trees > SVM: margin & geometry]

P1 and P6 (same class) lie on one margin line and P8 on the other; the optimal boundary is parallel to the line through P1, P6, midway to P8. Support vectors: P1, P6, P8.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (-1, 1)$, $b = 1$; $\|w\| = 1.4142$, margin $2/\|w\| = 1.4142$.

Point	P1	P2	P3	P4	P5	P6	P7	P8
$y(w^T x + b)$	1	2	2	2	3	1	2	1

All values are ≥ 1 (feasible), with equality exactly at the support vectors.

$$f((1, 3)) = (-1)(1) + (1)(3) + 1 = 3.$$

Q.27 — Answer: 0.2091 to 0.2291 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

Mean $\bar{x} = (7.4, 2.2)$. Centered data rows: P1: (0.6, -2.2), P2: (-1.4, 0.8), P3: (-0.4, -1.2), P4: (-0.4, 2.8), P5: (1.6, -0.2).

Sample covariance $S = (1/(n-1)) \sum (x - \bar{x})(x - \bar{x})^T = [[1.3, -0.85], [-0.85, 3.7]]$.

Eigenvalues: $\lambda_1 = 3.9705$, $\lambda_2 = 1.0295$; unit PC1 direction $v_1 = (0.3033, -0.9529)$.

Score $z_1 = v_1 \cdot (x - \bar{x}) = 0.3033 \cdot -0.4 + -0.9529 \cdot 2.8 = -2.7894$.

$\|x - \bar{x}\|^2 = 8$; error $= \|x - \bar{x}\|^2 - z_1^2 = 8 - 7.7809 = 0.22$ (Pythagoras: the residual is the PC2 component).

Q.28 — Answer: 0.9494 to 0.9554 [NAT, 2M · Regression & Model Selection > Bias-variance trade-off]

$E[c\bar{x}] = c\mu = \text{Bias} = (c-1)\mu$; $\text{Var}(c\bar{x}) = c^2\sigma^2/n$. $\text{MSE}(c) = (c-1)^2\mu^2 + c^2\sigma^2/n$.

Here $\mu = 3$, $\sigma^2/n = 9/20 = 0.45$.

$d\text{MSE}/dc = 2(c-1)\mu^2 + 2c\sigma^2/n = 0 \Rightarrow c^* = \mu^2/(\mu^2 + \sigma^2/n) = 9/(9 + 0.45) = 0.9524$.

$c^* < 1$: accepting some bias reduces variance enough to lower MSE.

Answer: **0.952**.

Q.29 — Answer: 0.7757 to 0.7957 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

a(i) = mean Manhattan distance from P2 to the other members of its own cluster C1: $d(P2, P1) = 2 \rightarrow a = 2$.

b(i) = smallest mean distance to the points of another cluster: C2: $9.3333 \rightarrow b = 9.3333$.

s(i) = $(b - a)/\max(a, b) = (9.3333 - 2)/9.3333 = 0.79$.

s ranges in $[-1, 1]$; values near 1 mean well clustered, negative values suggest a wrong assignment.

Q.30 — Answer: 2.1787 to 2.1987 [NAT, 2M · Regression & Model Selection > Cross-validation]

Total SSE = $22.4 + 89.1 + 80 + 52.8 = 244.3$; $n = 51$.

CV MSE = $244.3/51 = 4.7902$; RMSE = $\sqrt{4.7902} = 2.1887$.

Answer: **2.19**.

Q.31 — Answer: (D) [MSQ, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

(A) FALSE. PCA ignores labels; LDA uses them.

(B) FALSE. It has $(K-1)p(p+1)/2$ more.

(C) FALSE. It makes no assumption on $P(x|y)$.

(D) TRUE. The $x^T \Sigma_k^{-1} x$ terms no longer cancel.

Q.32 — Answer: 0.994 to 0.998 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

$a = W_1 x + b_1 = (-2.5, 1.5)$; $h = \text{ReLU}(a) = (0, 1.5)$.

$z = W_2 h + b_2 = (-2.5, 4, -2)$.

$\exp(z) = (0.0821, 54.5982, 0.1353)$, sum = 54.8156.

$p_2 = \exp(z_2)/\sum \exp(z_j) = 0.996$. (All probabilities: (0.001, 0.996, 0.002).)

Q.33 — Answer: 30.615 to 30.635 [NAT, 2M · SVM & Decision Trees › SVM: soft margin & hinge loss]

Compute $f(x_i)$, the functional margin $y_i f(x_i)$ and the slack/hinge loss for each point:

Point	x	y	f(x)	y f(x)	$\xi = \max(0, 1 - y f)$
P1	(3, 4)	+1	2	2	0
P2	(1, 1)	-1	-1.5	1.5	0
P3	(3, 5)	-1	2.5	-2.5	3.5
P4	(4, 0)	-1	1	-1	2
P5	(1, 3)	-1	-0.5	0.5	0.5
P6	(5, 4)	+1	4	4	0

$\frac{1}{2}\|w\|^2 = 0.625$; objective = $0.625 + 5 \times 6 = \mathbf{30.62}$.

Points with $y f \geq 1$ contribute 0; points inside the margin contribute between 0 and 1; misclassified points contribute more than 1.

Q.34 — Answer: -0.002 to 0.002 [NAT, 2M · Probabilistic & Instance-based Classifiers › Bayes optimal classifier]

$P(y=1|x=2) = \pi_1 P(x|1) / (\pi_0 P(x|0) + \pi_1 P(x|1)) = 0 / (0.06 + 0) = \mathbf{0}$.

Q.35 — Answer: 2 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Expressiveness]

With identity activations, $\hat{y} = W_2(W_1x + b_1) + b_2 = (W_2W_1)x + (W_2b_1 + b_2)$: a single affine map.

$\tilde{w}^T = W_2W_1 = (2, 2)$; $\tilde{b} = W_2b_1 + b_2 = 2$.

Answer: **2**.

Hence stacking linear layers adds no expressive power – a non-linear activation is essential.

Mock Test 10

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	3	7	25

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Unsupervised Learning	7	11
SVM & Decision Trees	7	11
Regression & Model Selection	7	11
Probabilistic & Instance-based Classifiers	7	11
Neural Networks (MLP / Feed-forward)	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

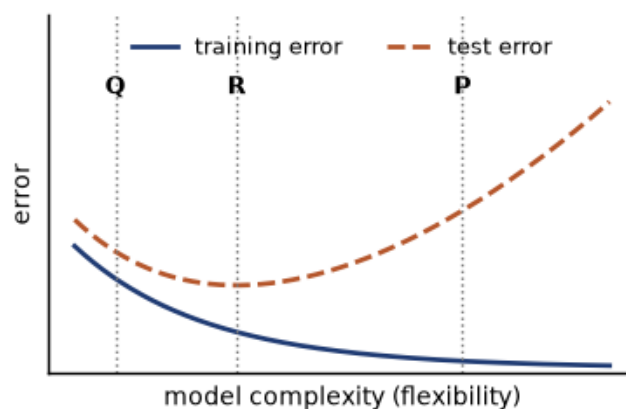
A complete (binary) dendrogram over $n = 45$ points has how many nodes in total (leaves plus internal nodes)? (Answer in integer)

Q.2 [1 mark · NAT (numerical answer)]

In AdaBoost with $\epsilon_t = 0.4$, before normalisation each misclassified example's weight is multiplied by e^{α_t} , where $\alpha_t = \frac{1}{2} \ln((1 - \epsilon_t)/\epsilon_t)$. This multiplying factor is _____ (rounded off to two decimal places)

Q.3 [1 mark · MCQ]

The figure shows training and test error as a function of model complexity, with three complexity levels marked P, Q and R. Which marked level corresponds to overfitting (low bias, high variance)?



- (A) None of P, Q, R
(C) Q

- (B) P
(D) R

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about clustering evaluation measures is/are CORRECT?

- (A) Purity can be made equal to 1 by putting every point in its own cluster.
- (B) The Rand index can compare two clusterings with different numbers of clusters.
- (C) In the silhouette, $b(i)$ is the mean distance to the farthest other cluster.
- (D) The silhouette coefficient always lies in $[0, 1]$.

Q.5 [1 mark · NAT (numerical answer)]

The homogeneous polynomial kernel $K(x, z) = (x^T z)^3$ on $x \in \mathbb{R}^6$ corresponds to a feature map of all (scaled) monomials of degree exactly 3. The number of distinct such features is _____
(Answer in integer)

Q.6 [1 mark · NAT (numerical answer)]

Data are uniformly distributed in the unit hypercube $[0, 1]^5$. To capture a fraction 0.01 of the data, a k-NN method uses a sub-cube neighbourhood. The required edge length of this sub-cube is _____ (rounded off to two decimal places)

Q.7 [1 mark · NAT (numerical answer)]

For a data set, $\bar{x} = 24$, $\bar{y} = 91$, $s_x = 2$, $s_y = 3$ and the sample correlation between x and y is $r = 0.45$. The slope of the least-squares regression line of y on x is _____ (rounded off to two decimal places)

Q.8 [1 mark · NAT (numerical answer)]

A fully connected MLP has layer sizes 12-10-3 (input-...-output), every layer having a bias vector. The number of bias parameters is _____ (Answer in integer)

Q.9 [1 mark · MSQ (one or more correct)]

Which of the following statements about classifier evaluation is/are CORRECT?

- (A) For a classifier with random scores, the expected precision at any threshold equals the prevalence of the positive class.
- (B) At fixed TPR and FPR, precision does not depend on the class prevalence.
- (C) ROC AUC is unchanged by any strictly increasing transformation of the scores.
- (D) The point $(FPR, TPR) = (0, 1)$ on the ROC plane corresponds to a perfect classifier.

Q.10 [1 mark · MSQ (one or more correct)]

Which of the following statements about support vector machines is/are CORRECT?

- (A) Scaling (w, b) by a positive constant does not change the decision boundary.
- (B) In the hard-margin SVM dual, some α_i may be negative at the optimum.
- (C) Moving a non-support-vector point (keeping it strictly outside the margin) changes the SVM boundary.
- (D) The hard-margin SVM primal is a convex quadratic program, so its optimal w is unique.

Q.11 [1 mark · NAT (numerical answer)]

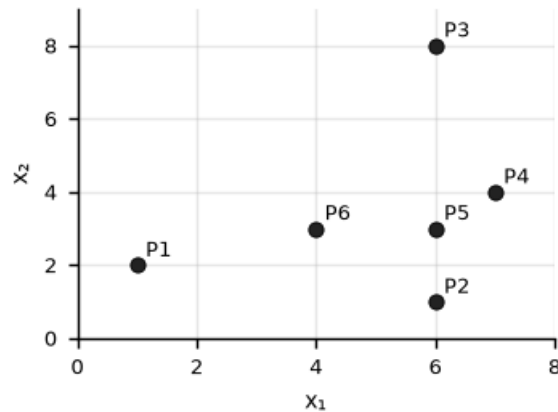
A 3-class softmax classifier gives the predicted probabilities below for a mini-batch of 3 examples. The mean categorical cross-entropy loss (natural log) over the mini-batch is _____ (rounded off to three decimal places)

example	p(class 1)	p(class 2)	p(class 3)	true class
1	0.25	0.25	0.5	1
2	0.1	0.2	0.7	3
3	0.7	0.2	0.1	1

Q.16 [2 marks · NAT (numerical answer)]

Single-linkage agglomerative clustering with Manhattan distance is applied to the six points below. Then the **sum of the heights** of all merges in the dendrogram is ____ (Answer in integer)

Point	P1	P2	P3	P4	P5	P6
x_1	1	6	6	7	6	4
x_2	2	1	8	4	3	3

**Q.17 [2 marks · MSQ (one or more correct)]**

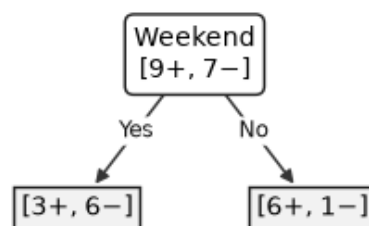
For the confusion matrix below, which of the following statements is/are CORRECT? (values rounded to 2 decimals)

	Actual +	Actual -
Predicted +	65	27
Predicted -	33	34

- (A) The accuracy is 0.52.
- (B) The F1-score is 0.68.
- (C) The negative predictive value is 0.56.
- (D) The F1-score is less than the arithmetic mean of precision and recall.

Q.18 [2 marks · NAT (numerical answer)]

A node with 16 training records (class counts [positive+, negative-]) is split on the attribute Weekend as shown. Compute the information gain (in bits) of splitting on Weekend. (rounded off to two decimal places)

**Q.19 [2 marks · NAT (numerical answer)]**

For the fully connected network 32-128-16-1, a forward pass on a mini-batch of 16 inputs is computed as dense matrix products. Counting only the scalar multiplications in the weight-matrix products (ignore biases and activations), the number of multiplications is ____ (Answer in integer)

Q.20 [2 marks · NAT (numerical answer)]

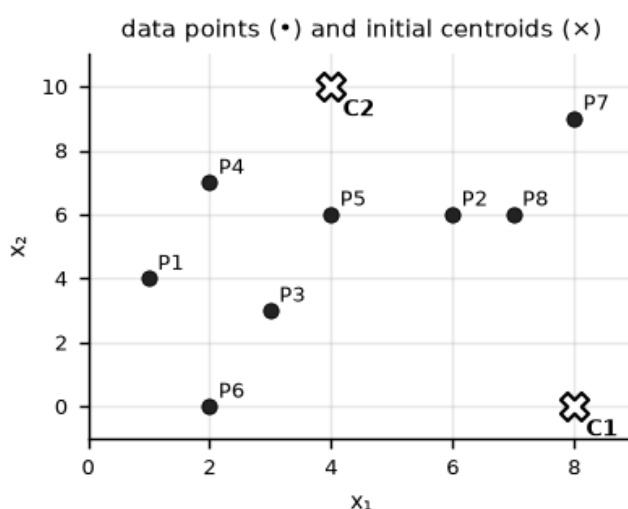
Two clusterings U and V of 6 points are given below (cluster names are arbitrary). Then the number of point pairs on which the two clusterings **disagree** is ____ (Answer in integer)

Point	P1	P2	P3	P4	P5	P6
Clustering U	α	γ	β	β	α	γ
Clustering V	1	3	2	2	3	3

Q.21 [2 marks · NAT (numerical answer)]

k-means (Lloyd's algorithm, Euclidean distance) with $k = 2$ is run on the 8 points below, starting from the initial centroids $C1 = (8, 0)$, $C2 = (4, 10)$. After **one** iteration (assignment step followed by update step), the x_2 -coordinate of the updated centroid of cluster C1 is ____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5	P6	P7	P8
x_1	1	6	3	2	4	2	8	7
x_2	4	6	3	7	6	0	9	6

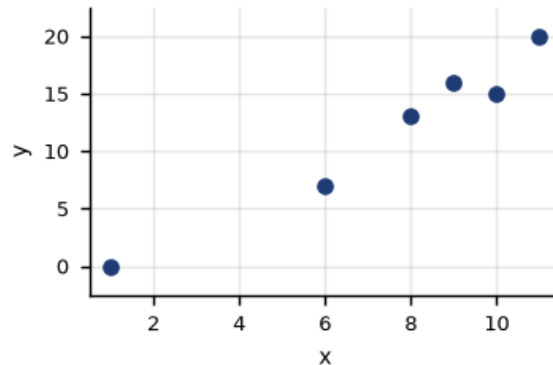
**Q.22 [2 marks · NAT (numerical answer)]**

A simple linear regression fitted by ordinary least squares gives $\hat{y} = 6.3 - 20.1x$. The feature x was measured in thousands of rupees. If x is re-expressed in rupees (so $x' = 1000x$) and the model is refitted, the new slope is ____ (rounded off to 4 decimal places)

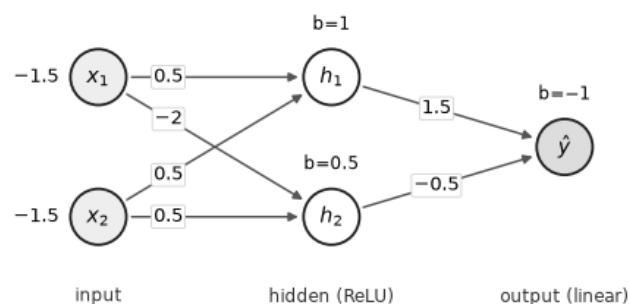
Q.23 [2 marks • NAT (numerical answer)]

The 6 observations shown in the table are plotted below. For the simple linear regression of y on x (with intercept), the coefficient of determination R^2 of the least-squares fit is _____ (rounded off to three decimal places)

x	1	6	8	9	10	11
y	0	7	13	16	15	20

**Q.24 [2 marks • NAT (numerical answer)]**

The feed-forward network below has ReLU hidden units and a linear output unit. The number on each edge is its weight and 'b' above a node is that node's bias. For input $x = (-1.5, -1.5)$ (values shown left of the inputs), the network output $\hat{y}$ is _____ (rounded off to three decimal places)

**Q.25 [2 marks • NAT (numerical answer)]**

An AdaBoost ensemble has three weak classifiers with the weighted errors below, and their predictions on a test point x are listed. The ensemble score $F(x) = \sum \alpha_t h_t(x)$, with $\alpha_t = \frac{1}{2} \ln((1 - \varepsilon_t)/\varepsilon_t)$, is _____ (rounded off to two decimal places)

Round t	ε_t	$h_t(x)$
1	0.15	-1
2	0.4	+1
3	0.25	-1

Q.26 [2 marks • NAT (numerical answer)]

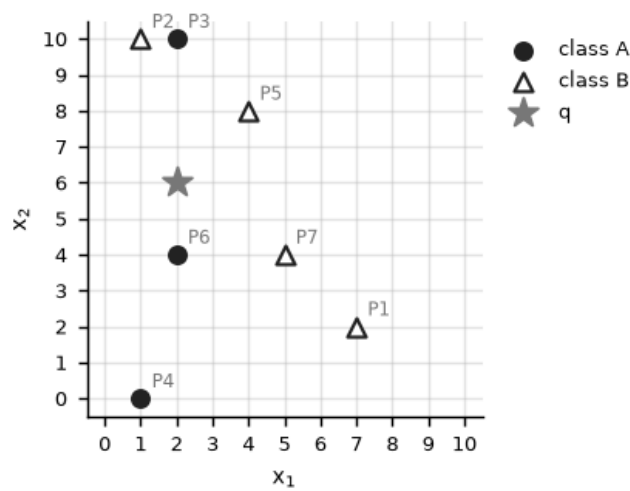
For a two-class problem, the class means are $\mu_0 = (2, 1)^T$ and $\mu_1 = (2, 4)^T$ and the pooled within-class scatter matrix S_W is given below. Fisher's linear discriminant uses $\mathbf{w} \propto S_W^{-1}(\mu_1 - \mu_0)$. Then the ratio w_2/w_1 of the components of the Fisher discriminant direction $\mathbf{w}$ is _____ (rounded off to two decimal places)

$$S_W = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Q.27 [2 marks · MCQ]

Consider the training set below. A 3-NN classifier (majority vote) is applied to the query point $\mathbf{q} = (2, 6)$, once with Euclidean distance and once with Manhattan distance. The predicted classes are

Point	x_1	x_2	class
P1	7	2	B
P2	1	10	B
P3	2	10	A
P4	1	0	A
P5	4	8	B
P6	2	4	A
P7	5	4	B



- (A) Euclidean: B; Manhattan: A
(C) Euclidean: A; Manhattan: A

- (B) Euclidean: A; Manhattan: B
(D) Euclidean: B; Manhattan: B

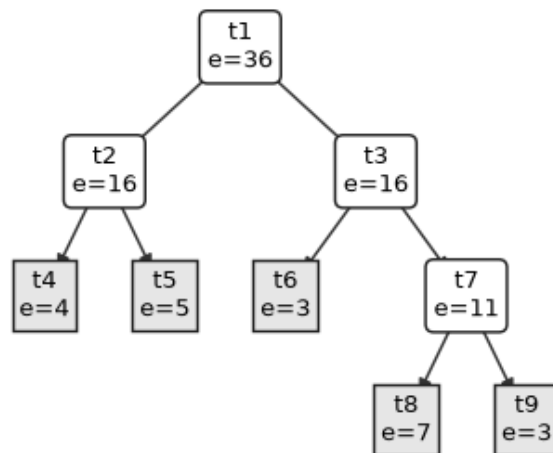
Q.28 [2 marks · NAT (numerical answer)]

A logistic-regression neuron $p = \sigma(w_1x_1 + w_2x_2 + b)$ has $w = (1, -1)$, $b = 0$ and is trained with binary cross-entropy loss. After one gradient-descent step using the **average** loss over both examples (a mini-batch of size 2) with learning rate $\eta = 0.1$, the new value of w_1 is _____ (rounded off to three decimal places)

example	x_1	x_2	y
1	1.5	2	1
2	1.5	-1	1

Q.29 [2 marks · NAT (numerical answer)]

A classification tree was grown on $N = 100$ training records. Each node shows $e(t)$, the number of training records that would be misclassified if t were a leaf; $R(t) = e(t)/N$. For $\alpha = 0.02$, the cost-complexity $R_\alpha(T) = R(T) + \alpha|T|$ of the full tree T is _____ (rounded off to three decimal places)

**Q.30 [2 marks · MSQ (one or more correct)]**

A linear regression of annual salary (₹ lakh) on Exp (years of experience), Female (1 if the employee is female, 0 otherwise) and city $\in \{A, B, C\}$ encoded by dummies CityB and CityC (reference level A) gives the fitted model

$$\hat{y} = 16 + 2.6 \cdot \text{Exp} + 1.5 \cdot \text{Female} + 5.6 \cdot \text{CityB} - 3.5 \cdot \text{CityC}$$

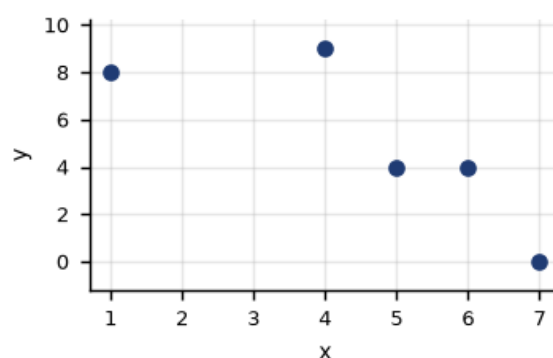
Which of the following is/are CORRECT?

- (A) Increasing Exp by 2 years, other variables fixed, changes the prediction by 4.6.
- (B) Increasing Exp by 2 years, other variables fixed, changes the prediction by 5.2.
- (C) The coefficient of CityB equals the average response of all observations in city B.
- (D) The predicted annual salary for a female employee in city C with 10 years of experience is 43.5.

Q.31 [2 marks · NAT (numerical answer)]

A simple linear regression (with intercept) is fitted by least squares to the 5 points below. Using leave-one-out cross-validation, the LOOCV estimate of test MSE, $(1/n)\sum(y_i - \hat{y}_{(-i)}(x_i))^2$, is _____ (rounded off to two decimal places)

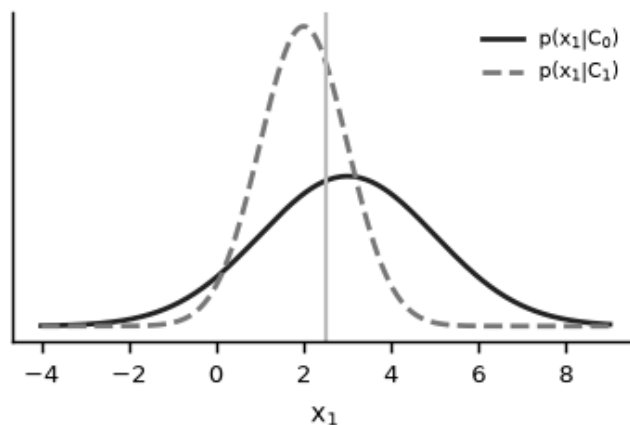
x	1	4	5	6	7
y	8	9	4	4	0



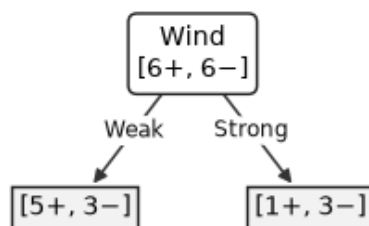
Q.32 [2 marks · NAT (numerical answer)]

A Gaussian naive Bayes classifier with feature x_1 has the class priors and per-class Gaussian parameters below (σ^2 = variance). For the input $x = 2.5$, the posterior probability $P(C_1 | x)$ is _____ (rounded off to two decimal places)

class	prior	$\mu(x_1)$	$\sigma^2(x_1)$
C_0	0.7	3	4
C_1	0.3	2	1

**Q.33 [2 marks · NAT (numerical answer)]**

A CART node with 12 training records (class counts [positive+, negative–]) is split on Wind as shown. Compute the weighted Gini index of the split on Wind. (rounded off to two decimal places)

**Q.34 [2 marks · MSQ (one or more correct)]**

Which of the following statements about loss functions and optimisation for training neural networks is/are CORRECT?

- (A) Decaying the learning rate during training can help SGD settle closer to a minimum.
- (B) With momentum $\beta = 0.9$ and a constant gradient g , the step size eventually becomes $0.1\eta g$.
- (C) Minimising cross-entropy loss is equivalent to maximising the log-likelihood of the labels under the model.
- (D) Using MSE loss for a classification network makes training by backpropagation impossible.

Q.35 [2 marks · NAT (numerical answer)]

A centered data matrix X with $n = 21$ rows (samples) and $d = 3$ columns has singular values 17, 5, 2. Using the sample covariance with divisor $n - 1$, the percentage of total variance explained by the first principal component is _____ (rounded off to two decimal places)

— END OF QUESTION PAPER —

Mock Test 10 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	89	19	NAT	2	98560
2	NAT	1	1.2147 to 1.2347	20	NAT	2	3
3	MCQ	1	(B)	21	NAT	2	1.49 to 1.51
4	MSQ	1	(A), (B)	22	NAT	2	−0.0202 to −0.02
5	NAT	1	56	23	NAT	2	0.9541 to 0.9637
6	NAT	1	0.3881 to 0.4081	24	NAT	2	−2.377 to −2.373
7	NAT	1	0.665 to 0.685	25	NAT	2	−1.2239 to −1.2039
8	NAT	1	13	26	NAT	2	1.99 to 2.01
9	MSQ	1	(A), (C), (D)	27	MCQ	2	(A)
10	MSQ	1	(A), (D)	28	NAT	2	1.0504 to 1.0544
11	NAT	1	0.6979 to 0.7019	29	NAT	2	0.3185 to 0.3215
12	NAT	1	0.6567 to 0.6767	30	MSQ	2	(B)
13	NAT	1	0.1708 to 0.1908	31	NAT	2	20.35 to 20.7611
14	MSQ	1	(B), (C)	32	NAT	2	0.4283 to 0.4483
15	MCQ	1	(D)	33	NAT	2	0.4275 to 0.4475
16	NAT	2	15	34	MSQ	2	(A), (C)
17	MSQ	2	(B), (D)	35	NAT	2	90.8705 to 90.8905
18	NAT	2	0.2033 to 0.2233				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Unsupervised Learning					11
SVM & Decision Trees					11
Regression & Model Selection					11
Probabilistic & Instance-based Classifiers					11
Neural Networks (MLP / Feed-forward)					11
Total					55

Mock Test 10 — Detailed Solutions

Q.1 — Answer: 89 [NAT, 1M · Unsupervised Learning > Hierarchical clustering]

n leaves + $(n - 1)$ internal (merge) nodes = $2n - 1 = 89$.

Q.2 — Answer: 1.2147 to 1.2347 [NAT, 1M · SVM & Decision Trees > Ensembles: boosting]

$$e^\alpha = e^{\frac{1}{2} \ln((1-\epsilon)/\epsilon)} = \sqrt{((1-\epsilon)/\epsilon)} = \sqrt{1.5} = 1.22.$$

Correctly classified examples are multiplied by $e^{-\alpha} = \sqrt{(\epsilon/(1-\epsilon))}$.

$\epsilon < 0.5 \Rightarrow \alpha > 0$; $\epsilon = 0.5 \Rightarrow \alpha = 0$ (no better than chance).

Q.3 — Answer: (B) [MCQ, 1M · Regression & Model Selection > Bias-variance trade-off]

At Q: both training and test errors are high (left side) → underfitting.

At R: the test error is at its minimum → best trade-off.

At P: training error is very low but test error has risen well above its minimum (right side) → overfitting.

Answer: **P**.

Q.4 — Answer: (A), (B) [MSQ, 1M · Unsupervised Learning > Cluster evaluation]

(A) TRUE. Hence purity alone cannot choose k .

(B) TRUE. Only pairs are compared.

(C) FALSE. $b(i)$ uses the nearest other cluster.

(D) FALSE. It can be negative; range is $[-1, 1]$.

Q.5 — Answer: 56 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

Monomials of degree exactly d in p variables = $C(p + d - 1, d)$ (stars and bars).

$$C(8, 3) = 56.$$

(Including all lower degrees, as for $(x^T z + 1)^3$, would give $C(9, 3) = 84$.)

Q.6 — Answer: 0.3881 to 0.4081 [NAT, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Volume of a cube with edge e is e^5 ; set $e^5 = 0.01$.

$$e = 0.01^{1/5} = 0.4.$$

In high dimensions even a small fraction of data needs an edge spanning most of each coordinate's range, so 'nearest' neighbours are not local — the curse of dimensionality.

Q.7 — Answer: 0.665 to 0.685 [NAT, 1M · Regression & Model Selection > Simple linear regression]

$$\beta_1 = r \cdot s_y / s_x = 0.45 \times 3/2 = 0.675.$$

Answer: **0.68**.

Q.8 — Answer: 13 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Parameter counting]

Fully connected layer with n_{in} inputs and n_{out} outputs: $n_{in} \cdot n_{out}$ weights + n_{out} biases.

Layer 1: $12 \times 10 = 120$ weights, 10 biases.

Layer 2: $10 \times 3 = 30$ weights, 3 biases.

Weights = 150, biases = 13. Required: **13**.

Q.9 — Answer: (A), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

(A) TRUE. The PR-curve baseline is horizontal at π .

(B) FALSE. It does.

(C) TRUE. Only the ranking matters.

(D) TRUE. No FP and no FN.

Q.10 — Answer: (A), (D) [MSQ, 1M · SVM & Decision Trees > SVM: dual & support vectors]

(A) TRUE. $\{x : w^T x + b = 0\}$ is unchanged by positive scaling.

(B) FALSE. Lagrange multipliers of inequality constraints satisfy $\alpha_i \geq 0$.

(C) FALSE. Its constraint stays inactive, so the solution is unchanged.

(D) TRUE. Strictly convex objective $\frac{1}{2}\|w\|^2$ over a convex feasible set.

Q.11 — Answer: 0.6979 to 0.7019 [NAT, 1M · Neural Networks (MLP / Feed-forward) · Training & regularisation]

CE for one example = $-\ln p_{\text{true class}}$.

Example 1: $-\ln 0.25 = 1.3863$.

Example 2: $-\ln 0.7 = 0.3567$.

Example 3: $-\ln 0.7 = 0.3567$.

Mean = **0.6999**.

Q.12 — Answer: 0.6567 to 0.6767 [NAT, 1M · Probabilistic & Instance-based Classifiers · Bayes optimal classifier]

Expected cost of predicting positive = $(1 - p) \cdot C_{FP}$; of predicting negative = $p \cdot C_{FN}$.

Predict positive iff $p \cdot 2 > (1 - p) \cdot 4 \Leftrightarrow p > C_{FP} / (C_{FP} + C_{FN}) = 4/6 = \mathbf{0.67}$.

Q.13 — Answer: 0.1708 to 0.1908 [NAT, 1M · Unsupervised Learning · Distance measures]

$x \cdot y = 11$, $\|x\| = \sqrt{50}$, $\|y\| = \sqrt{74}$; $\cos \theta = x \cdot y / (\|x\| \|y\|) = 0.1808 = \mathbf{0.18}$.

For reference: $L_1 = 12$, $L_2 = 10.1$, $L_\infty = 10$ (always $L_\infty \leq L_2 \leq L_1$).

Q.14 — Answer: (B), (C) [MSQ, 1M · Neural Networks (MLP / Feed-forward) · Perceptron]

Net input $a = w \cdot x + b = -1x_1 + (1)x_2 + (0)$; output +1 iff $a \geq 0$.

(P) $a = -1 \cdot 3 + (1) \cdot 1 + (0) = -2 \rightarrow -1$.

(Q) $a = -1 \cdot -3 + (1) \cdot -1 + (0) = 2 \rightarrow +1$ (selected).

(R) $a = -1 \cdot -1 + (1) \cdot 2 + (0) = 3 \rightarrow +1$ (selected).

(S) $a = -1 \cdot 0 + (1) \cdot -1 + (0) = -1 \rightarrow -1$.

Answer: **(B), (C)**.

Q.15 — Answer: (D) [MCQ, 1M · Regression & Model Selection · Regression vs classification]

$E[(y - g)^2 | x] = \text{Var}(y | x) + (E[y | x] - g)^2$, minimised at $g = E[y | x]$.

Hence the answer is **the conditional mean $E[y | x]$** .

Q.16 — Answer: 15 [NAT, 2M · Unsupervised Learning · Hierarchical clustering]

Single-linkage merge heights are exactly the edge weights of a minimum spanning tree (Kruskal's algorithm adds the shortest edge joining two different components = single-linkage merge).

MST edges (Prim's algorithm): P1-P6 (4); P6-P5 (2); P5-P2 (2); P5-P4 (2); P4-P3 (5).

Sorted merge heights: 2, 2, 2, 4, 5.

Sum of merge heights = total MST weight = $2 + 2 + 2 + 4 + 5 = \mathbf{15}$.

Q.17 — Answer: (B), (D) [MSQ, 2M · Probabilistic & Instance-based Classifiers · Evaluation metrics]

TP = 65, FN = 33, FP = 27, TN = 34.

(A) FALSE. accuracy = $(TP + TN) / N = 0.6226$, not 0.52.

(B) TRUE. F1-score = $2PR / (P + R) = 0.6842$.

(C) FALSE. negative predictive value = $TN / (TN + FN) = 0.5075$, not 0.56.

(D) TRUE. The harmonic mean is $\leq$ the arithmetic mean, with equality only if $P = R$ (not the case here).

Q.18 — Answer: 0.2033 to 0.2233 [NAT, 2M · SVM & Decision Trees · Decision trees: split selection]

Parent [9+, 7-]: $H = -(9/16)\log_2(9/16) - (7/16)\log_2(7/16) = 0.9887$.

Weekend = Yes: [3+, 6-], $H = -(3/9)\log_2(3/9) - (6/9)\log_2(6/9) = 0.9183$.

Weekend = No: [6+, 1-], $H = -(6/7)\log_2(6/7) - (1/7)\log_2(1/7) = 0.5917$.

Weighted child entropy = $(9/16)(0.9183) + (7/16)(0.5917) = 0.7754$.

IG = $0.9887 - 0.7754 = 0.2133$.

Answer: **0.21**.

Q.19 — Answer: 98560 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Parameter counting]

Each weight participates in exactly one multiplication per example.

Layer 1: $32 \times 128 = 4096$.

Layer 2: $128 \times 16 = 2048$.

Layer 3: $16 \times 1 = 16$.

Per example: 6160; for 16 examples: $6160 \times 16 = \mathbf{98560}$.

Q.20 — Answer: 3 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

Total pairs = $C(6,2) = 15$.

a = pairs together in both = 2; b = pairs separated in both = 10; pairs together in U only = 1; together in V only = 2.

Rand index = $(a + b)/C(n,2) = (2 + 10)/15 = 0.8$.

Answer: **3**. (Cluster names do not matter: only whether pairs are together.)

Q.21 — Answer: 1.49 to 1.51 [NAT, 2M · Unsupervised Learning > k-means]

Iteration 1 - assignment (nearest initial centroid, squared Euclidean distance):

Point	$d^2(\cdot, C1)$	$d^2(\cdot, C2)$	cluster
P1 (1, 4)	65	45	C2
P2 (6, 6)	40	20	C2
P3 (3, 3)	34	50	C1
P4 (2, 7)	85	13	C2
P5 (4, 6)	52	16	C2
P6 (2, 0)	36	104	C1
P7 (8, 9)	81	17	C2
P8 (7, 6)	37	25	C2

Update: each centroid becomes the mean of its assigned points: $C1 \rightarrow (2.5, 1.5)$; $C2 \rightarrow (4.67, 6.33)$.

Required coordinate = **1.5**.

Note: each step of Lloyd's algorithm can only decrease (or keep) J , which is why it converges.

Q.22 — Answer: -0.0202 to -0.02 [NAT, 2M · Regression & Model Selection > Basis functions & feature scaling]

$x = x'/1000 = \hat{y} = \beta_0 + (\beta_1/1000)x'$. Fitted values are unchanged; the slope becomes $-20.1/1000 = -0.0201$.

Answer: **-0.0201**.

Q.23 — Answer: 0.9541 to 0.9637 [NAT, 2M · Regression & Model Selection > Simple linear regression]

$\bar{x} = 7.5$, $\bar{y} = 11.833$.

i	1	2	3	4	5	6
$x_i - \bar{x}$	-6.5	-1.5	0.5	1.5	2.5	3.5
$y_i - \bar{y}$	-11.83	-4.83	1.17	4.17	3.17	8.17
$(x_i - \bar{x})(y_i - \bar{y})$	76.92	7.25	0.58	6.25	7.92	28.58

$S_{xx} = \sum (x_i - \bar{x})^2 = 65.5$, $S_{xy} = 127.5$.

$\beta_1 = S_{xy}/S_{xx} = 1.9466$, $\beta_0 = \bar{y} - \beta_1 \bar{x} = -2.7659$.

$S_{yy} = SST = 258.8333$; $SSR = \beta_1 S_{xy} = 248.187$.

$R^2 = SSR/SST = S_{xy}^2/(S_{xx} S_{yy}) = 0.9589$.

Answer: **0.959**.

Q.24 — Answer: -2.377 to -2.373 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

Hidden pre-activations: $a_1 = 0.5 \cdot -1.5 + (0.5) \cdot -1.5 + (1) = -0.5$; $a_2 = -2 \cdot -1.5 + (0.5) \cdot -1.5 + (0.5) = 2.75$.

Hidden activations ($\text{ReLU}(a) = \max(0, a)$): $h_1 = 0$, $h_2 = 2.75$.

Output pre-activation: $o = 1.5 \cdot 0 + (-0.5) \cdot 2.75 + (-1) = -2.375$.

$$\hat{y} = 0 = -2.375.$$

Q.25 — Answer: -1.2239 to -1.2039 [NAT, 2M · SVM & Decision Trees › Ensembles: boosting]

$$\alpha_t: \alpha_1 = 0.8673, \alpha_2 = 0.2027, \alpha_3 = 0.5493.$$

$$F(x) = -0.8673 + 0.2027 - 0.5493 = -1.21; \text{prediction sign}(F) = -1.$$

Note that a majority of weak votes need not win: a low-error learner carries a larger α .

Q.26 — Answer: 1.99 to 2.01 [NAT, 2M · Probabilistic & Instance-based Classifiers › Linear discriminant analysis]

$$\mu_1 - \mu_0 = (0, 3)^T; \det S_W = 3.$$

$$S_W^{-1} = \begin{bmatrix} 0.6667 & 0.3333 \\ 0.3333 & 0.6667 \end{bmatrix}$$

$$\mathbf{w} = S_W^{-1}(\mu_1 - \mu_0) = (1, 2)^T.$$

$$w_2/w_1 = 2/1 = 2. \text{ (The direction, not the length, matters.)}$$

Note: using $\mu_1 - \mu_0$ without S_W^{-1} is correct only when $S_W \propto I$.

Q.27 — Answer: (A) [MCQ, 2M · Probabilistic & Instance-based Classifiers › k-nearest neighbours]

Distances from $\mathbf{q}$:

Point	class	Euclidean d	Manhattan d
P1	B	6.403	9
P2	B	4.123	5
P3	A	4	4
P4	A	6.083	7
P5	B	2.828	4
P6	A	2	2
P7	B	3.606	5

Euclidean 3 nearest: P6(A), P5(B), P7(B) $\Rightarrow$ **B**.

Manhattan 3 nearest: P6(A), P3(A), P5(B) $\Rightarrow$ **A**.

Answer: **Euclidean: B; Manhattan: A**. The two metrics disagree because L_1 penalises diagonal offsets more than L_2 , changing the neighbour set.

Q.28 — Answer: 1.0504 to 1.0544 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Backpropagation]

For σ output with binary cross-entropy $L = -[y \ln p + (1 - y) \ln(1 - p)]$, $\partial L / \partial a = p - y$, so $\partial L / \partial w_j = (p - y)x_j$ and $\partial L / \partial b = p - y$.

$$\text{Example 1: } a = -0.5, p = 0.3775, p - y = -0.6225.$$

$$\text{Example 2: } a = 2.5, p = 0.9241, p - y = -0.0759.$$

$$\text{Mini-batch (mean) gradient for } w_1: -0.5237.$$

$$w_1 \leftarrow 1 - 0.1 \times (-0.5237) = \mathbf{1.0524}.$$

Note there is no σ' factor: it cancels with the derivative of the log loss.

Q.29 — Answer: 0.3185 to 0.3215 [NAT, 2M · SVM & Decision Trees › Decision trees: overfitting & pruning]

$$R(T) = \text{sum of leaf errors} / N = (4 + 5 + 3 + 7 + 3)/100 = 0.22; |T| = 5 \text{ leaves.}$$

$$R_\alpha(T) = 0.22 + 0.02 \times 5 = \mathbf{0.32}.$$

Q.30 — Answer: (B) [MSQ, 2M · Regression & Model Selection › Multiple linear regression]

(A) **FALSE**. The change is $2 \times 2.6 = 5.2$.

(B) **TRUE**. $2 \times 2.6 = 5.2$.

(C) **FALSE**. It is a difference relative to the reference level A, adjusted for the other variables.

(D) **FALSE**. This omits the city dummy; the correct value is 40.

Q.31 — Answer: 20.35 to 20.7611 [NAT, 2M · Regression & Model Selection › Cross-validation]

$$\bar{x} = 4.6, \bar{y} = 5.$$

i	1	2	3	4	5
$x_i - \bar{x}$	-3.6	-0.6	0.4	1.4	2.4
$y_i - \bar{y}$	3	4	-1	-1	-5
$(x_i - \bar{x})(y_i - \bar{y})$	-10.8	-2.4	-0.4	-1.4	-12

$$S_{xx} = \sum (x_i - \bar{x})^2 = 21.2, S_{xy} = -27.$$

$$\beta_1 = S_{xy}/S_{xx} = -1.2736, \beta_0 = \bar{y} - \beta_1 \bar{x} = 10.8585.$$

For a linear smoother, the leave-one-out residual is $e_{(i)} = e_i/(1 - h_{ii})$, with $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$.

x_i	1	4	5	6	7
e_i	-1.585	3.236	-0.491	0.783	-1.943
h_{ii}	0.811	0.217	0.208	0.292	0.472
$e_i/(1-h_{ii})$	-8.4	4.133	-0.619	1.107	-3.679

$$CV_{(n)} = (1/5) \sum (e_i/(1-h_{ii}))^2 = 20.5555.$$

Answer: **20.56** (verified by explicitly refitting without the point).

Q.32 — Answer: 0.4283 to 0.4483 [NAT, 2M · Probabilistic & Instance-based Classifiers > Naive Bayes]

$N(x; \mu, \sigma^2) = \exp(-(x-\mu)^2/(2\sigma^2))/\sqrt{(2\pi\sigma^2)}$. Naive Bayes multiplies the per-feature densities.

$$C_0: N(2.5; 3, 4) = 0.19333; \times \text{prior } 0.7 \Rightarrow 0.135334.$$

$$C_1: N(2.5; 2, 1) = 0.35207; \times \text{prior } 0.3 \Rightarrow 0.10562.$$

$$P(C_1 | x) = 0.10562/(0.135334 + 0.10562) = \mathbf{0.44}.$$

Q.33 — Answer: 0.4275 to 0.4475 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

$$\text{Parent } [6+, 6-]: \text{Gini} = 1 - [(6/12)^2 + (6/12)^2] = 0.5.$$

$$\text{Wind} = \text{Weak}: [5+, 3-], \text{Gini} = 1 - [(5/8)^2 + (3/8)^2] = 0.4688.$$

$$\text{Wind} = \text{Strong}: [1+, 3-], \text{Gini} = 1 - [(1/4)^2 + (3/4)^2] = 0.375.$$

$$\text{Weighted Gini} = (8/12)(0.4688) + (4/12)(0.375) = 0.4375.$$

$$\text{Gini gain} = 0.5 - 0.4375 = 0.0625.$$

Answer: **0.44**.

Q.34 — Answer: (A), (C) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

(A) **TRUE**. Reduces the noise floor of SGD.

(B) **FALSE**. It approaches $\eta g/(1 - \beta) = 10\eta g$.

(C) **TRUE**. CE = negative log-likelihood of a categorical/Bernoulli model.

(D) **FALSE**. It is possible, just usually slower/worse.

Q.35 — Answer: 90.8705 to 90.8905 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

With $X = U\Sigma V^T$ (X centered, $n \times d$), the sample covariance is $S = X^T X/(n - 1) = V(\Sigma^2/(n - 1))V^T$, so the PC directions are the columns of V and the eigenvalues are $\lambda_j = \sigma_j^2/(n - 1)$.

$$\lambda = 17^2/20 = 14.45, 5^2/20 = 1.25, 2^2/20 = 0.2.$$

Cumulative % variance: 90.88, 98.74, 100 (note the $(n - 1)$ factor cancels in proportions: $PVE = \sigma_j^2/\Sigma\sigma^2$).

Answer: **90.88**. (Using σ instead of σ^2 is the classic mistake.)

Mock Test 11

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	7	7	21

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Neural Networks (MLP / Feed-forward)	7	11
SVM & Decision Trees	7	11
Regression & Model Selection	7	11
Probabilistic & Instance-based Classifiers	7	11
Unsupervised Learning	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MSQ (one or more correct)]

Which of the following statements about backpropagation is/are CORRECT?

- (A) For a sigmoid output with binary cross-entropy (or softmax with cross-entropy), the error at the output pre-activation is $\hat{y} - y$.
- (B) Backpropagation only computes gradients; the parameter update rule (SGD, momentum, Adam) is a separate choice.
- (C) The gradient with respect to a bias b_j equals δ_j .
- (D) A finite-difference (central difference) check can be used to verify a backprop implementation.

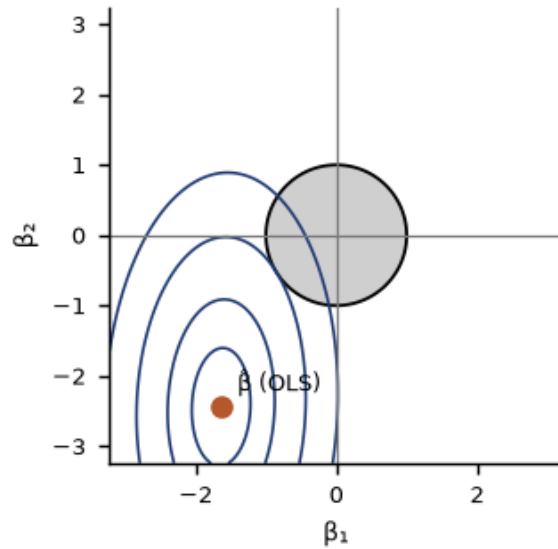
Q.2 [1 mark · NAT (numerical answer)]

In the dual solution of a linear SVM, the non-zero Lagrange multipliers α_i and labels y_i of the support vectors are given below (all other $\alpha_i = 0$). The value of α_5 is _____ (rounded off to one decimal place)

i	1	2	3	4	5
y_i	+1	+1	+1	-1	-1
α_i	0.2	0.5	0.2	0.2	?

Q.3 [1 mark · MCQ]

For a two-predictor penalised least-squares problem, the figure shows the constraint region (shaded) of the penalty together with elliptical contours of the residual sum of squares centred at the least-squares estimate β (OLS). The penalised estimate (β_1, β_2) satisfies



- (A) $\beta_1 = 0$ exactly and $\beta_2 \neq 0$
- (B) $\beta_2 = 0$ exactly and $\beta_1 \neq 0$
- (C) $\beta_1 = \beta_2 = 0$
- (D) $\beta_1 \neq 0$ and $\beta_2 \neq 0$ (both shrunk, neither exactly zero)

Q.4 [1 mark · NAT (numerical answer)]

Inverted dropout with keep probability $p = 0.75$ is applied during training. A hidden unit with activation $h = 3.5$ that is **kept** passes on the value _____ (rounded off to two decimal places)

Q.5 [1 mark · NAT (numerical answer)]

The maximum number of leaf nodes in a binary decision tree of depth 8 (root at depth 0) is _____ (Answer in integer)

Q.6 [1 mark · NAT (numerical answer)]

A naive Bayes classifier has $d = 10$ binary features and $K = 2$ classes. The number of independent parameters (class priors plus class-conditional feature probabilities) to be estimated is _____ (Answer in integer)

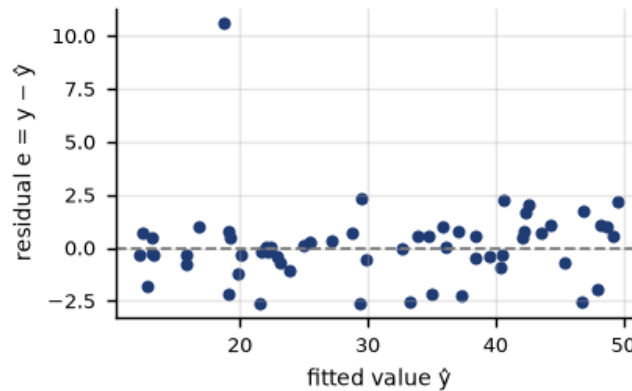
Q.7 [1 mark · MSQ (one or more correct)]

Which of the following statements about decision-tree learning is/are CORRECT?

- (A) Pruning a subtree can never increase the training error.
- (B) A binary tree of depth d has at most 2^d leaves.
- (C) Information gain and Gini gain always select the same attribute.
- (D) The maximum Gini index for a binary classification node is 1.

Q.8 [1 mark · MCQ]

After fitting a linear regression model, the residuals are plotted against the fitted values as shown. The plot most strongly suggests:



- (A) Non-constant error variance (heteroscedasticity)
- (B) No evident violation of the linear-model assumptions
- (C) A non-linear relationship not captured by the model (e.g. a missing x^2 term)
- (D) A single outlying observation, with the rest of the fit looking adequate

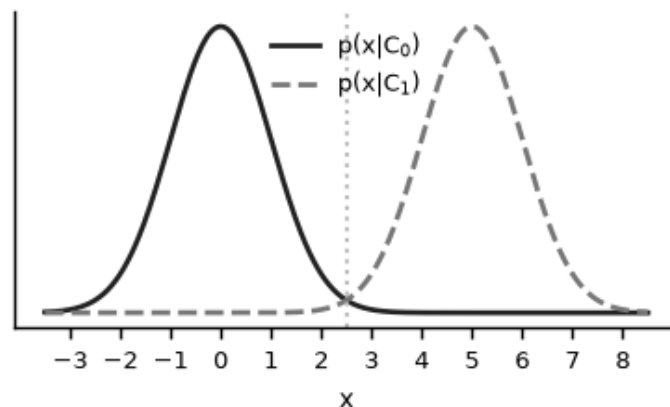
Q.9 [1 mark · MSQ (one or more correct)]

Which of the following statements about activation functions is/are CORRECT? (σ denotes the logistic sigmoid.)

- (A) Leaky ReLU has a non-zero gradient for $z < 0$, which helps avoid 'dead' units.
- (B) The derivative of $\tanh(z)$ has maximum value 1, attained at $z = 0$.
- (C) ReLU is not differentiable at $z = 0$; in practice a subgradient value (0 or 1) is used there.
- (D) The output of ReLU lies in $[0, \infty)$.

Q.10 [1 mark · MCQ]

Two classes have unit-variance Gaussian class-conditional densities with means 0 (C_0) and 5 (C_1), as plotted. With equal priors the Bayes decision threshold is the dotted line at $x = 2.5$. If the prior of C_0 is increased to 0.9, the threshold



- (A) It moves towards $x = 0$, enlarging the region assigned to C_0
- (B) It splits into two thresholds, giving a quadratic boundary
- (C) It moves towards $x = 5$, enlarging the region assigned to C_0
- (D) It moves towards $x = 0$, shrinking the region assigned to C_0

Q.11 [1 mark · MSQ (one or more correct)]

Which of the following statements about k-means clustering is/are CORRECT?

- (A) k-means requires labelled training data.
- (B) Each k-means cluster region (Voronoi cell) is convex.
- (C) k-means results are invariant to rescaling individual features.
- (D) The assignment step partitions the feature space into the Voronoi cells of the centroids.

Q.12 [1 mark · NAT (numerical answer)]

The 5 two-dimensional observations below have zero mean in each feature. Then the correlation coefficient between x_1 and x_2 is _____ (rounded off to two decimal places)

Point	x_1	x_2	x_3	x_4	x_5
x_1	-4	3	-3	-2	6
x_2	-1	-1	4	0	-2

Q.13 [1 mark · MSQ (one or more correct)]

Which of the following statements about basis-function (polynomial) regression and feature scaling is/are CORRECT?

- (A) Gaussian radial basis functions $\phi_j(x) = \exp(-(x - \mu_j)^2/(2s^2))$ can be used in a model that is linear in its weights.
- (B) For least-squares polynomial fits on the same data, increasing the degree never increases the training RSS.
- (C) Ridge and lasso solutions depend on the scales of the features.
- (D) k-NN regression predictions can change if one feature is rescaled.

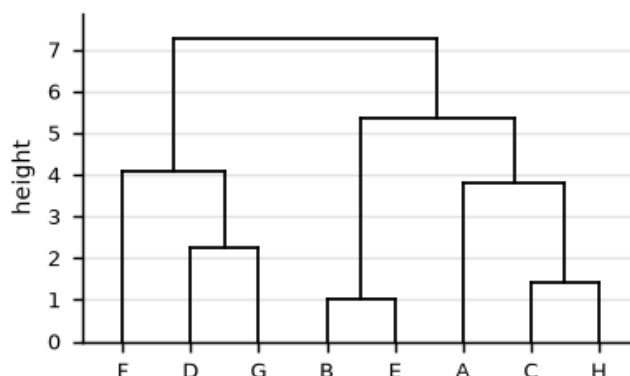
Q.14 [1 mark · MSQ (one or more correct)]

Which of the following statements about logistic regression is/are CORRECT?

- (A) Increasing x_j by one unit (others fixed) multiplies the odds of $y = 1$ by e^{β_j} .
- (B) Raising the classification threshold from 0.5 to 0.9 can increase recall.
- (C) Multiplying all weights and the bias by 2 shifts the 0.5-threshold decision boundary.
- (D) In softmax regression with K classes, the boundary between two classes is in general a curved surface in x .

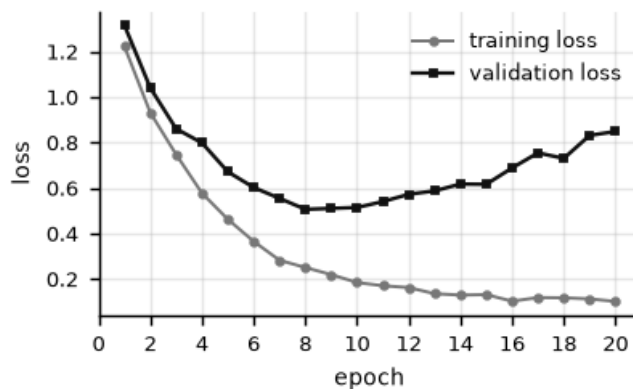
Q.15 [1 mark · NAT (numerical answer)]

The dendrogram below was obtained by agglomerative clustering of 8 points A-H. If the dendrogram is cut by a horizontal line at height 3, the number of clusters obtained is _____ (Answer in integer)



Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

A network is trained with early stopping on validation loss using patience $P = 2$: training stops as soon as the validation loss has failed to improve on its best value for 2 consecutive epochs, and the weights from the best epoch are restored. Using the validation losses below (also plotted), the epoch after which training is stopped is ____ (Answer in integer)



epoch	1	2	3	4	5	6	7	8	9	10
val. loss	1.316	1.041	0.861	0.799	0.672	0.603	0.555	0.507	0.511	0.515

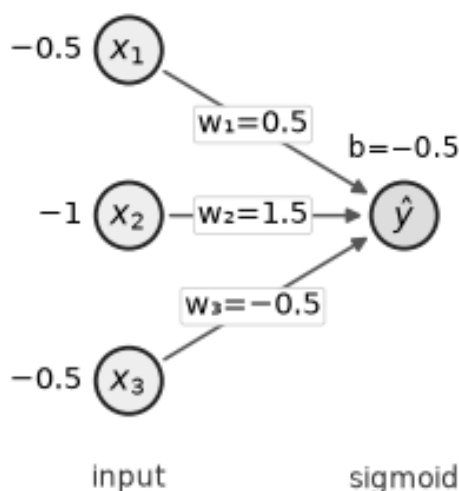
epoch	11	12	13	14	15	16	17	18	19	20
val. loss	0.541	0.573	0.588	0.618	0.617	0.689	0.755	0.731	0.832	0.849

Q.17 [2 marks · NAT (numerical answer)]

$B = 50$ regression trees are averaged. Each tree's prediction has variance $\sigma^2 = 4$, and every pair of trees has correlation $\rho = 0.2$. The variance of the averaged prediction is ____ (rounded off to two decimal places)

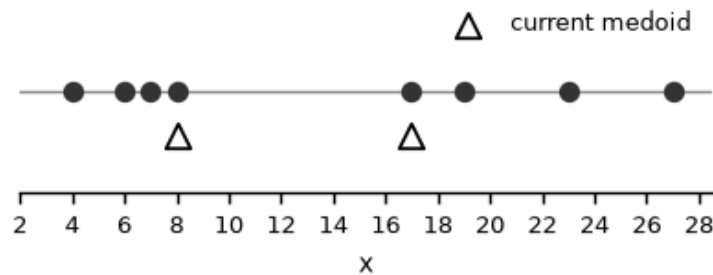
Q.18 [2 marks · NAT (numerical answer)]

A single sigmoid neuron (shown below) receives $x = (-0.5, -1, -0.5)$ and has target $y = 1$. The loss is $L = \frac{1}{2}(\hat{y} - y)^2$. The value of $\partial L / \partial b$ is ____ (rounded off to 4 decimal places)



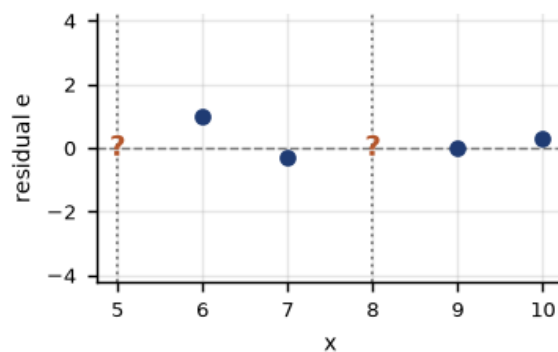
Q.19 [2 marks · NAT (numerical answer)]

PAM (k-medoids, $k = 2$, absolute-difference dissimilarity) is applied to the 1-D data $\{4, 6, 7, 8, 17, 19, 23, 27\}$. The current medoids are 8 and 17. The change in total cost, (cost after swap) – (cost before swap), when medoid 8 is swapped with non-medoid 19 is _____ (Answer in integer)

**Q.20 [2 marks · NAT (numerical answer)]**

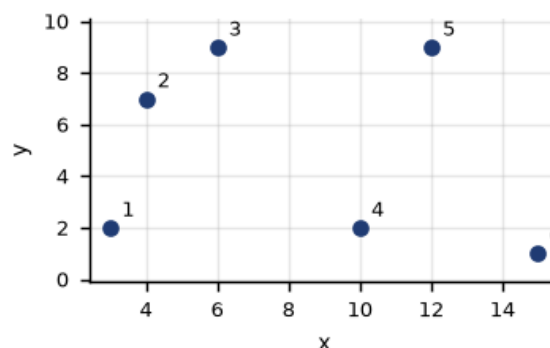
A simple linear regression with intercept is fitted by least squares. The table gives the x -values and the residuals $e_i = y_i - \hat{y}_i$, but two residuals are missing. The fitted line is $\hat{y} = 2.8 - 0.6x$. The observed value y at $x = 5$ is _____ (rounded off to two decimal places)

x_i	5	6	7	8	9	10
e_i	?	1	-0.3	?	0	0.3

**Q.21 [2 marks · NAT (numerical answer)]**

Consider the 6 training points (x_i, y_i) shown below. The model always predicts the mean of the training targets (ignoring x). Its leave-one-out cross-validation MSE is _____ (rounded off to two decimal places)

x	3	4	6	10	12	15
y	2	7	9	2	9	1

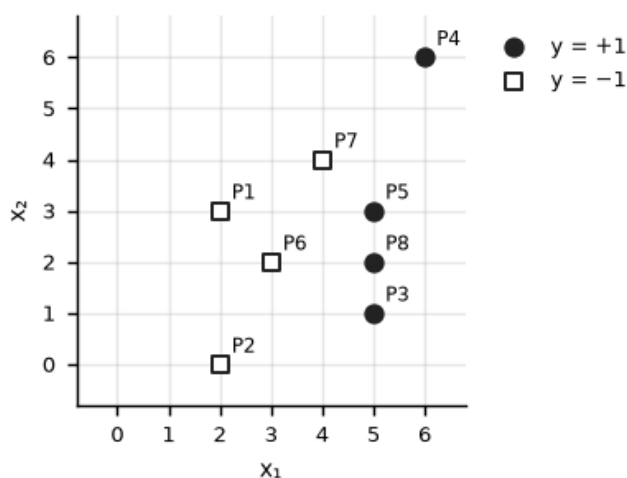
**Q.22 [2 marks · NAT (numerical answer)]**

A regression with intercept and $p = 2$ predictors on $n = 47$ observations has $R^2 = 0.47$. Its adjusted R^2 is _____ (rounded off to three decimal places)

Q.23 [2 marks · MCQ]

Four candidate linear decision boundaries are proposed for the labelled training points shown. Which boundary separates the two classes with the LARGEST geometric margin (minimum distance from the boundary to any training point, all points correctly classified)?

Point	x_1	x_2	y
P1	2	3	-1
P2	2	0	-1
P3	5	1	+1
P4	6	6	+1
P5	5	3	+1
P6	3	2	-1
P7	4	4	-1
P8	5	2	+1



- (A) $2x_1 - x_2 - 4.5 = 0$
 (B) $2x_1 - x_2 - 5 = 0$
 (C) $2x_1 - x_2 - 5.5 = 0$
 (D) $x_1 + x_2 - 6.5 = 0$

Q.24 [2 marks · NAT (numerical answer)]

Two clusterings U and V of 8 points are given below (cluster names are arbitrary). Then the number of point pairs placed in the same cluster by **both** clusterings is _____ (Answer in integer)

Point	P1	P2	P3	P4	P5	P6	P7	P8
Clustering U	α	α	β	α	α	α	α	α
Clustering V	1	3	2	1	3	1	1	1

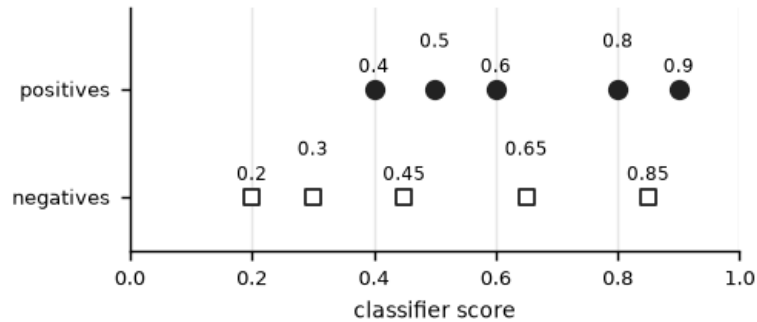
Q.25 [2 marks · NAT (numerical answer)]

For the kernel $K(x, z) = (x^T z + 1)^2$ on $\mathbb{R}^2$ with feature map ϕ , compute the cosine of the angle between $\phi(x)$ and $\phi(z)$, where $x = (-1, 0)$ and $z = (-2, 1)$. (rounded off to two decimal places)

Q.26 [2 marks · NAT (numerical answer)]

A classifier assigns the scores below to 10 test examples (5 positives, label 1; 5 negatives, label 0). The area under the ROC curve (AUC) is _____ (rounded off to two decimal places)

example	1	2	3	4	5	6	7	8	9	10
score	0.5	0.6	0.65	0.3	0.45	0.85	0.2	0.9	0.8	0.4
label	1	1	0	0	0	0	0	1	1	1

**Q.27 [2 marks · MCQ]**

A decision tree is to be grown on the 10 training records below using Gini gain as the splitting criterion (multi-way split on each categorical attribute). Which attribute is chosen at the root?

#	Size	Income	Outlook	Shape	Class
1	Small	Medium	Overcast	Square	Yes
2	Large	Low	Rain	Oval	Yes
3	Small	High	Sunny	Square	Yes
4	Small	Low	Sunny	Square	Yes
5	Large	High	Sunny	Oval	No
6	Small	Low	Sunny	Square	Yes
7	Large	High	Overcast	Square	No
8	Large	Low	Overcast	Round	No
9	Large	Medium	Overcast	Oval	No
10	Large	Medium	Overcast	Round	Yes

(A) Size

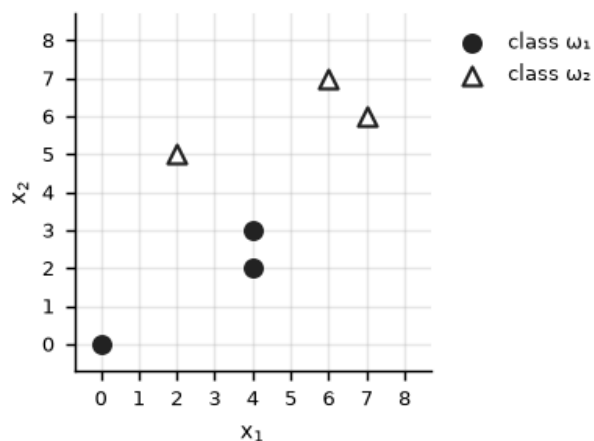
(B) Income

(C) Outlook

(D) Shape

Q.28 [2 marks · NAT (numerical answer)]

Class ω_1 has training points (4, 3), (0, 0), (4, 2) and class ω_2 has points (7, 6), (6, 7), (2, 5). Let $\mathbf{m}_1$, $\mathbf{m}_2$ be the class means and $S_W = \sum_k \sum_{x \in \omega_k} (x - \mathbf{m}_k)(x - \mathbf{m}_k)^T$ the within-class scatter matrix (no division by n). Then $(\mathbf{m}_2 - \mathbf{m}_1)^T S_W^{-1} (\mathbf{m}_2 - \mathbf{m}_1)$ is _____ (rounded off to two decimal places)



Q.29 [2 marks · MSQ (one or more correct)]

Consider k-means (Lloyd's algorithm) and k-medoids (PAM). Which of the following statements is/are CORRECT?

- (A) The medoid of a cluster is the member that minimizes the sum of dissimilarities to the other members.
- (B) The number of ways to split n points into 2 non-empty clusters is $n(n-1)/2$.
- (C) k-medoids minimizes the sum of squared Euclidean distances to the cluster means.
- (D) For a 1-D cluster with an odd number of points, the medoid under absolute distance is the mean.

Q.30 [2 marks · MCQ]

A naive Bayes classifier is trained on the 11 examples below. The new instance (Age = Middle, Student = No, Rating = Excellent) is classified (i) with maximum-likelihood (unsmoothed) estimates and (ii) with Laplace (add-one) smoothing of the class-conditional probabilities (class priors unsmoothed). The predicted classes are

#	Age	Student	Rating	Buys
1	Senior	Yes	Fair	No
2	Middle	Yes	Fair	Yes
3	Senior	No	Fair	Yes
4	Young	No	Fair	Yes
5	Middle	No	Fair	Yes
6	Young	Yes	Excellent	No
7	Middle	Yes	Fair	No
8	Young	Yes	Excellent	No
9	Middle	No	Fair	No
10	Young	Yes	Fair	No
11	Middle	No	Fair	Yes

- (A) Without smoothing: Yes; with Laplace smoothing: Yes
- (B) Without smoothing: Yes; with Laplace smoothing: No
- (C) Without smoothing: No; with Laplace smoothing: No
- (D) Without smoothing: No; with Laplace smoothing: Yes

Q.31 [2 marks · MCQ]

Six points have the pairwise distances below. Agglomerative clustering with **complete linkage** is run and the dendrogram is cut to give exactly **2** clusters. The resulting clusters are

	P1	P2	P3	P4	P5	P6
P1	0	5	4	6	6	8
P2		0	1	11	7	9
P3			0	10	6	8
P4				0	4	14
P5					0	10
P6						0

- (A) {P1, P2, P3}, {P4, P5}, {P6}
- (B) {P1, P5, P6}, {P2, P3, P4}
- (C) {P1, P2, P3, P4, P5}, {P6}
- (D) {P1, P2, P3, P6}, {P4, P5}

Q.32 [2 marks · NAT (numerical answer)]

Least-squares linear regression on $n = 5$ observations minimises $J(w) = (1/2n)\|Xw - y\|^2$ by batch gradient descent with constant learning rate η . The data give the matrix $X^T X$ below. Then with $\eta = 0.248$, the contraction factor $\rho = \max_i |1 - \eta \lambda_i|$ (so that $\|w_{t+1} - w^*\| \leq \rho \|w_t - w^*\|$) is _____ (rounded off to three decimal places)

$$X^T X = \begin{bmatrix} 12 & 4 \\ 4 & 12 \end{bmatrix}$$

Q.33 [2 marks · NAT (numerical answer)]

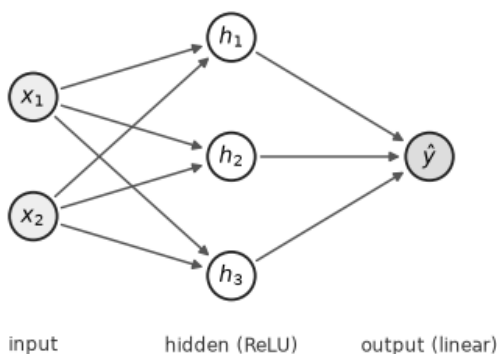
A softmax-regression output layer computes $z = Wx + b$, $p = \text{softmax}(z)$ with W and b below, and is trained with cross-entropy loss $L = -\ln p_y$. For $x = (1, -1)^T$ with true class $y = 1$, $\partial L / \partial b_3$ is _____ (rounded off to three decimal places)

$$W = \begin{bmatrix} 0 & 1 \\ -0.5 & 0.5 \\ -1 & -1 \end{bmatrix}$$

$$b = \begin{bmatrix} -0.5 \\ 0 \\ 0.5 \end{bmatrix}$$

Q.34 [2 marks · NAT (numerical answer)]

The 2-3-1 network below computes $\hat{y} = v \cdot \text{ReLU}(Wx + b) + c$ with W , b , v given below and $c = -1$. With $x = (-2, -2)$, target $y = 3$ and loss $L = \frac{1}{2}(\hat{y} - y)^2$, the value of $\partial L / \partial W_{22}$ is _____ (Answer in integer)



$$W = \begin{bmatrix} -1 & 2 \\ -2 & -1 \\ 0 & -1 \end{bmatrix}$$

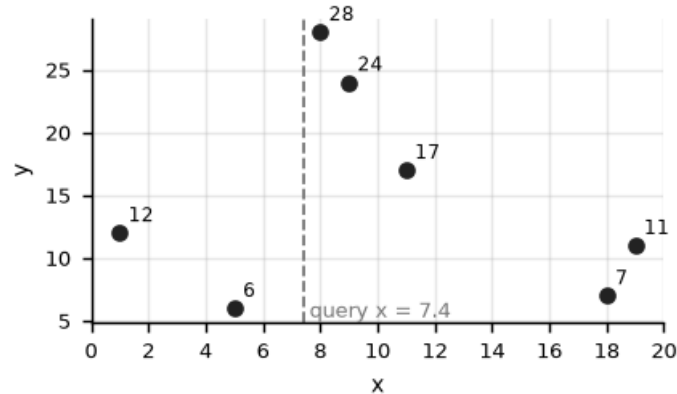
$$b = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$v^T = \begin{bmatrix} -2 & -1 & -2 \end{bmatrix}$$

Q.35 [2 marks · NAT (numerical answer)]

A distance-weighted k-NN regressor with $k = 3$ predicts $\hat{y} = \sum w_i y_i / \sum w_i$ over the 3 nearest neighbours, with weights $w_i = 1/d_i^2$. For the training data below, the prediction at $x = 7.4$ is _____ (rounded off to two decimal places)

x	1	5	8	9	11	18	19
y	12	6	28	24	17	7	11



— END OF QUESTION PAPER —

Mock Test 11 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MSQ	1	(A), (B), (C), (D)	19	NAT	2	30
2	NAT	1	0.69 to 0.71	20	NAT	2	−0.5767 to −0.5567
3	MCQ	1	(D)	21	NAT	2	16.716 to 16.884
4	NAT	1	4.6567 to 4.6767	22	NAT	2	0.4439 to 0.4479
5	NAT	1	256	23	MCQ	2	(B)
6	NAT	1	21	24	NAT	2	11
7	MSQ	1	(B)	25	NAT	2	0.74 to 0.76
8	MCQ	1	(D)	26	NAT	2	0.67 to 0.69
9	MSQ	1	(A), (B), (C), (D)	27	MCQ	2	(A)
10	MCQ	1	(C)	28	NAT	2	5.5909 to 5.6109
11	MSQ	1	(B), (D)	29	MSQ	2	(A)
12	NAT	1	−0.58 to −0.56	30	MCQ	2	(D)
13	MSQ	1	(A), (B), (C), (D)	31	MCQ	2	(D)
14	MSQ	1	(A)	32	NAT	2	0.6012 to 0.6052
15	NAT	1	5	33	NAT	2	0.7341 to 0.7381
16	NAT	2	10	34	NAT	2	−32
17	NAT	2	0.854 to 0.874	35	NAT	2	26.3796 to 26.3996
18	NAT	2	−0.0943 to −0.0906				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Neural Networks (MLP / Feed-forward)					11
SVM & Decision Trees					11
Regression & Model Selection					11
Probabilistic & Instance-based Classifiers					11
Unsupervised Learning					11

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Total					55

Mock Test 11 — Detailed Solutions

Q.1 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Backpropagation]

- (A) **TRUE**. The σ' factor cancels with the log-loss derivative.
 (B) **TRUE**. Backprop $\neq$ optimiser.
 (C) **TRUE**. a_j depends on b_j with coefficient 1.
 (D) **TRUE**. $[L(w+\varepsilon) - L(w-\varepsilon)]/2\varepsilon \approx \partial L/\partial w$.

Q.2 — Answer: 0.69 to 0.71 [NAT, 1M · SVM & Decision Trees > SVM: dual & support vectors]

Stationarity w.r.t. b gives the dual constraint $\sum_i \alpha_i y_i = 0$.

$$\sum_{i \neq 5} \alpha_i y_i = +0.2 + 0.5 + 0.2 - 0.2 = 0.7.$$

So $\alpha_5(-1) = -0.7 \Rightarrow \alpha_5 = \mathbf{0.7}$ (it must be ≥ 0).

Q.3 — Answer: (D) [MCQ, 1M · Regression & Model Selection > Ridge & Lasso]

The shaded region is the constraint set of the ridge (L_2 : $\beta_1^2 + \beta_2^2 \leq t^2$, a disc). The penalised estimate is the first point at which the RSS contours (ellipses centred at β_{OLS} , which lies outside the region) touch the region.

From the figure the tangency point is approximately $(-0.85, -0.53)$.

The contour touches the boundary at a point that is not on an axis, so both coefficients are non-zero. For the disc this is the generic case — ridge shrinks but does not zero out.

Answer: **$\beta_1 \neq 0$ and $\beta_2 \neq 0$ (both shrunk, neither exactly zero)**. Both coefficients cannot be zero since the OLS point lies outside the region (the constraint is active but $t > 0$).

Q.4 — Answer: 4.6567 to 4.6767 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Inverted dropout multiplies kept activations by $1/p$ so that $E[\text{output}] = h$: $3.5/0.75 = \mathbf{4.6667}$. No scaling is applied at test time.

Q.5 — Answer: 256 [NAT, 1M · SVM & Decision Trees > Decision trees: structure & traversal]

Each level at most doubles the nodes: leaves $\leq 2^d = 2^8 = \mathbf{256}$.

Q.6 — Answer: 21 [NAT, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Priors: $K - 1 = 1$. Each binary feature needs 1 parameter $P(x_j=1|c)$ per class: $Kd = 20$.

Total = $20 + 1 = \mathbf{21}$.

Q.7 — Answer: (B) [MSQ, 1M · SVM & Decision Trees > Decision trees: overfitting & pruning]

- (A) **FALSE**. Pruning usually increases (or keeps) training error.
 (B) **TRUE**. Doubling per level.
 (C) **FALSE**. They usually agree but can differ.
 (D) **FALSE**. It is 0.5.

Q.8 — Answer: (D) [MCQ, 1M · Regression & Model Selection > Simple linear regression]

All residuals lie in a narrow band except one far from 0 $\rightarrow$ an outlier that should be investigated.

Answer: **A single outlying observation, with the rest of the fit looking adequate**.

Other patterns: funnel $\rightarrow$ heteroscedasticity; curvature $\rightarrow$ missing non-linearity; isolated point $\rightarrow$ outlier; random band $\rightarrow$ adequate fit.

Q.9 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

- (A) **TRUE**. Its gradient is $\alpha > 0$ for negative inputs.
 (B) **TRUE**. $\tanh' = 1 - \tanh^2$, and $\tanh(0) = 0$.
 (C) **TRUE**. Left and right derivatives are 0 and 1.
 (D) **TRUE**. $\max(0, z)$ is never negative and is unbounded above.

Q.10 — Answer: (C) [MCQ, 1M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$$x^* = (\mu_0 + \mu_1)/2 + \sigma^2 \ln(\pi_0/\pi_1)/(\mu_1 - \mu_0) = 2.939.$$

A larger prior for C_0 means more of the x-axis is assigned to it, so the threshold moves away from μ_0 towards $\mu_1 = 5$. Answer: **It moves towards $x = 5$, enlarging the region assigned to C_0 .**
Equal variances keep the boundary a single point (linear); priors shift it but never make it quadratic.

Q.11 — Answer: (B), (D) [MSQ, 1M · Unsupervised Learning > k-means]

(A) FALSE. It is unsupervised.

(B) TRUE. A Voronoi cell is an intersection of half-spaces.

(C) FALSE. Rescaling changes Euclidean distances and hence assignments.

(D) TRUE. Each point goes to its nearest centroid.

Q.12 — Answer: -0.58 to -0.56 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

The data are already centered (column means are 0), so $S = X^T X / (n - 1)$.

$\Sigma_{X_1^2} = 74$, $\Sigma_{X_2^2} = 22$, $\Sigma_{X_1 X_2} = -23$.

$S = [[18.5, -5.75], [-5.75, 5.5]]$.

Required value = **-0.57**.

Q.13 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Basis functions & feature scaling]

(A) TRUE. $y = \sum w_j \phi_j(x)$ is linear in w .

(B) TRUE. Nested models.

(C) TRUE. The penalty is not scale-invariant.

(D) TRUE. Distances change.

Q.14 — Answer: (A) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

(A) TRUE. The log-odds increase by β_j .

(B) FALSE. Fewer examples are predicted positive, so TP can only stay the same or fall.

(C) FALSE. $w^T x + b = 0 \Leftrightarrow 2w^T x + 2b = 0$; the boundary is the same.

(D) FALSE. It is the hyperplane $(w_k - w_l)^T x + b_k - b_l = 0$.

Q.15 — Answer: 5 [NAT, 1M · Unsupervised Learning > Hierarchical clustering]

Merges below the cut height 3: 3 (heights 1, 1.41, 2.24).

Each merge reduces the number of clusters by one: $8 - 3 = 5$. Equivalently, count the vertical lines crossed by the horizontal cut.

Q.16 — Answer: 10 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Rule: keep track of the best (lowest) validation loss; an epoch that does not strictly improve on it increments a counter, which is reset on improvement; stop when the counter reaches patience $P = 2$.

Trace - 1: 1.316 (new best); 2: 1.041 (new best); 3: 0.861 (new best); 4: 0.799 (new best); 5: 0.672 (new best); 6: 0.603 (new best); 7: 0.555 (new best); 8: 0.507 (new best); 9: 0.511 (no improvement, count = 1); 10: 0.515 (no improvement, count = 2).

Training stops after epoch 10; the best validation loss 0.507 occurred at epoch 8, whose weights are restored.

Answer: **10**. The widening train/validation gap after this point is the signature of overfitting.

Q.17 — Answer: 0.854 to 0.874 [NAT, 2M · SVM & Decision Trees > Ensembles: bagging & random forests]

$\text{Var}((1/B)\sum_b f_b) = (1/B^2)[B\sigma^2 + B(B-1)\rho\sigma^2] = \rho\sigma^2 + (1-\rho)\sigma^2/B$.

$= 0.2 \times 4 + 0.8 \times 4/50 = 0.8 + 0.064 = \mathbf{0.86}$.

Even with $B \rightarrow \infty$ the variance cannot fall below $\rho\sigma^2 = 0.8$ — which is why random forests try to reduce ρ by feature subsampling.

Q.18 — Answer: -0.0943 to -0.0906 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

$a = w \cdot x + b = -2$; $\hat{y} = \sigma(a) = 0.1192$.

$$L = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial L / \partial \hat{y} = \hat{y} - y = -0.8808; \partial \hat{y} / \partial a = \hat{y}(1 - \hat{y}) = 0.105.$$

$$\delta = \partial L / \partial a = (\hat{y} - y)\hat{y}(1 - \hat{y}) = -0.0925 (= \partial L / \partial b).$$

Answer: **-0.0925**.

Forgetting the factor $\hat{y}(1 - \hat{y})$ (i.e. treating the output as linear) is the most common error.

Q.19 — Answer: 30 [NAT, 2M · Unsupervised Learning > k-medoids]

PAM cost = Σ over all points of the absolute distance to the nearest medoid.

Current medoids {8, 17}: distances 4→4, 6→2, 7→1, 8→0, 17→0, 19→2, 23→6, 27→10; cost = **25**.

After swap, medoids {17, 19}: distances 4→13, 6→11, 7→10, 8→9, 17→0, 19→0, 23→4, 27→8; cost = 55.

Change = 55 - 25 = **30** (positive → PAM rejects the swap).

Q.20 — Answer: -0.5767 to -0.5567 [NAT, 2M · Regression & Model Selection > Simple linear regression]

For the least-squares fit with intercept the normal equations give $\Sigma e_i = 0$ and $\Sigma x_i e_i = 0$.

Known: $\Sigma e = 1$, $\Sigma x e = 6.9$. With unknowns $u = e$ at $x = 5$ and $v = e$ at $x = 8$:

$$u + v = -1, 5u + 8v = -6.9.$$

$$\text{Solving: } u = -0.3667, v = -0.6333.$$

$$y = \hat{y} + e = (2.8 + (-0.6)(5)) + (-0.3667) = -0.5667.$$

Answer: **-0.57**.

Q.21 — Answer: 16.716 to 16.884 [NAT, 2M · Regression & Model Selection > Cross-validation]

Leaving out point i , the prediction is $(\Sigma y - y_i)/(n - 1)$ with $\Sigma y = 30$.

y_i	2	7	9	2	9	1
prediction	5.6	4.6	4.2	5.6	4.2	5.8
error	-3.6	2.4	4.8	-3.6	4.8	-4.8

(Shortcut: error = $n(y_i - \bar{y})/(n - 1)$, $\bar{y} = 5$.) LOOCV MSE = mean of squared errors = 16.8.

Answer: **16.8**.

Q.22 — Answer: 0.4439 to 0.4479 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

Adjusted $R^2 = 1 - [\text{SSE}/(n - p - 1)]/[\text{SST}/(n - 1)] = 1 - (1 - R^2)(n - 1)/(n - p - 1)$, where p is the number of predictors (excluding the intercept).

$$= 1 - (1 - 0.47) \times 46/44 = 0.4459.$$

Answer: **0.446**.

Q.23 — Answer: (B) [MCQ, 2M · SVM & Decision Trees > SVM: margin & geometry]

For $n^T x - c = 0$, the geometric margin is $\min_i y_i s(n^T x_i - c)/\|n\|$ with orientation $s = \pm 1$ chosen so that the positive class is on the positive side; a negative value means some point is misclassified.

Hyperplane	$\ n\ $	$\min_i$ distance (signed, best orientation)
$2x_1 - x_2 - 4.5 = 0$	2.236	0.224
$2x_1 - x_2 - 5 = 0$	2.236	0.447
$2x_1 - x_2 - 5.5 = 0$	2.236	0.224
$x_1 + x_2 - 6.5 = 0$	1.414	-1.061 (does not separate)

The largest margin is 0.447, for $2x_1 - x_2 - 5 = 0$. Note that a boundary can separate the data perfectly yet pass close to some point — only the minimum distance matters.

Q.24 — Answer: 11 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

Total pairs = $C(8, 2) = 28$.

a = pairs together in both = 11; b = pairs separated in both = 7; pairs together in U only = 10; together in V only = 0.

$$\text{Rand index} = (a + b)/C(n, 2) = (11 + 7)/28 = 0.6429.$$

Answer: **11**. (Cluster names do not matter: only whether pairs are together.)

Q.25 — Answer: 0.74 to 0.76 [NAT, 2M · SVM & Decision Trees · SVM: kernels]

$$\cos \theta = \varphi(x)^T \varphi(z) / (\|\varphi(x)\| \|\varphi(z)\|) = K(x, z) / \sqrt{K(x, x)K(z, z)}.$$

$$K(x, z) = 9, K(x, x) = 4, K(z, z) = 36.$$

$$\cos \theta = 9 / \sqrt{4 \times 36} = \mathbf{0.75}.$$

Q.26 — Answer: 0.67 to 0.69 [NAT, 2M · Probabilistic & Instance-based Classifiers · Evaluation metrics]

AUC = fraction of (positive, negative) pairs in which the positive has the higher score (Mann-Whitney statistic).

For each positive, count negatives with lower score: $0.9 \rightarrow 5$; $0.8 \rightarrow 4$; $0.6 \rightarrow 3$; $0.5 \rightarrow 3$; $0.4 \rightarrow 2$.

Concordant pairs = 17 out of $5 \times 5 = 25$.

$$\text{AUC} = 17/25 = \mathbf{0.68}.$$

(Equivalently, trace the ROC staircase by lowering the threshold and add the rectangle areas.)

Q.27 — Answer: (A) [MCQ, 2M · SVM & Decision Trees · Decision trees: split selection]

Parent: 6 Yes, 4 No $\Rightarrow$ Gini = 0.48.

Size — Small: [4+, 0-], Large: [2+, 4-]; gain = 0.2133.

Income — Low: [3+, 1-], Medium: [2+, 1-], High: [1+, 2-]; gain = 0.0633.

Outlook — Sunny: [3+, 1-], Overcast: [2+, 3-], Rain: [1+, 0-]; gain = 0.09.

Shape — Round: [1+, 1-], Square: [4+, 1-], Oval: [1+, 2-]; gain = 0.0867.

The largest Gini gain is for **Size**, so it is chosen at the root.

Q.28 — Answer: 5.5909 to 5.6109 [NAT, 2M · Probabilistic & Instance-based Classifiers · Linear discriminant analysis]

$$\mathbf{m}_1 = (2.667, 1.667), \mathbf{m}_2 = (5, 6).$$

$$S_1 = \begin{bmatrix} 10.667 & 6.667 \\ 6.667 & 4.667 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 14 & 4 \\ 4 & 2 \end{bmatrix}$$

$$S_w = \begin{bmatrix} 24.667 & 10.667 \\ 10.667 & 6.667 \end{bmatrix}$$

$$\mathbf{m}_2 - \mathbf{m}_1 = (2.333, 4.333).$$

$$\mathbf{w} = S_w^{-1}(\mathbf{m}_2 - \mathbf{m}_1) = (-0.6053, 1.6184).$$

$$(\mathbf{m}_2 - \mathbf{m}_1)^T \mathbf{w} = \mathbf{5.6}.$$

Q.29 — Answer: (A) [MSQ, 2M · Unsupervised Learning · k-means]

(A) TRUE. This is the definition.

(B) FALSE. It is $2^{n-1} - 1$.

(C) FALSE. It minimizes the sum of dissimilarities to medoids (data points).

(D) FALSE. It is the median, which is a data point.

Q.30 — Answer: (D) [MCQ, 2M · Probabilistic & Instance-based Classifiers · Naive Bayes]

(i) No smoothing.

Yes: $P(\text{Buys}=\text{Yes}) = 5/11$; $P(\text{Age}=\text{Middle}|\text{Yes}) = 3/5$; $P(\text{Student}=\text{No}|\text{Yes}) = 4/5$;
 $P(\text{Rating}=\text{Excellent}|\text{Yes}) = 0/5 \Rightarrow \text{score} = 0$.

No: $P(\text{Buys}=\text{No}) = 6/11$; $P(\text{Age}=\text{Middle}|\text{No}) = 2/6$; $P(\text{Student}=\text{No}|\text{No}) = 1/6$;
 $P(\text{Rating}=\text{Excellent}|\text{No}) = 2/6 \Rightarrow \text{score} = 0.0101$.

A zero count makes the whole product for Yes equal to 0 (zero-frequency problem), so the prediction is No.

(ii) Laplace smoothing.

Yes: $P(\text{Buys}=\text{Yes}) = 5/11$; $P(\text{Age}=\text{Middle}|\text{Yes}) = (3+1)/(5+3) = 1/2$; $P(\text{Student}=\text{No}|\text{Yes}) = (4+1)/(5+2) = 5/7$; $P(\text{Rating}=\text{Excellent}|\text{Yes}) = (0+1)/(5+2) = 1/7 \Rightarrow \text{score} = 0.02319$.

No: $P(\text{Buys}=\text{No}) = 6/11$; $P(\text{Age}=\text{Middle}|\text{No}) = (2+1)/(6+3) = 1/3$; $P(\text{Student}=\text{No}|\text{No}) = (1+1)/(6+2) = 1/4$; $P(\text{Rating}=\text{Excellent}|\text{No}) = (2+1)/(6+2) = 3/8 \Rightarrow \text{score} = 0.01705$.

Larger score: Yes. Answer: **Without smoothing: No; with Laplace smoothing: Yes.**

Q.31 — Answer: (D) [MCQ, 2M · Unsupervised Learning › Hierarchical clustering]

Use complete linkage: $d(A, B) = \max_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P2\} + \{P3\}$ at height **1**. Updated: $d(\{P2, P3\}, \{P1\}) = 5$, $d(\{P2, P3\}, \{P4\}) = 11$, $d(\{P2, P3\}, \{P5\}) = 7$, $d(\{P2, P3\}, \{P6\}) = 9$.

Merge 2: $\{P4\} + \{P5\}$ at height **4**. Updated: $d(\{P4, P5\}, \{P1\}) = 6$, $d(\{P4, P5\}, \{P6\}) = 14$, $d(\{P4, P5\}, \{P2, P3\}) = 11$.

Merge 3: $\{P1\} + \{P2, P3\}$ at height **5**. Updated: $d(\{P1, P2, P3\}, \{P6\}) = 9$, $d(\{P1, P2, P3\}, \{P4, P5\}) = 11$.

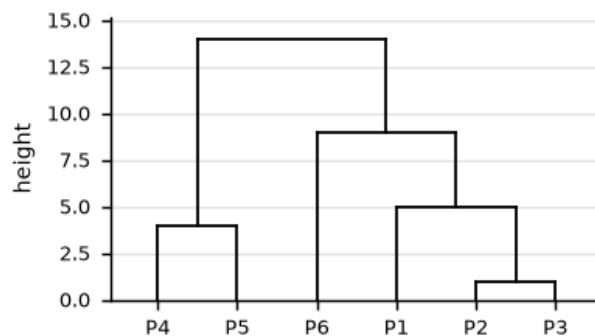
Merge 4: $\{P6\} + \{P1, P2, P3\}$ at height **9**. Updated: $d(\{P1, P2, P3, P6\}, \{P4, P5\}) = 14$.

Merge 5: $\{P4, P5\} + \{P1, P2, P3, P6\}$ at height **14**.

Stopping when 2 clusters remain (i.e. after 4 merges): **$\{P1, P2, P3, P6\}, \{P4, P5\}$** .

$\{P1, P2, P3, P4, P5\}, \{P6\}$ is what *average* linkage would give.

$\{P1, P2, P3\}, \{P4, P5\}, \{P6\}$ has 3 clusters (cut at the wrong level).



Q.32 — Answer: 0.6012 to 0.6052 [NAT, 2M · Regression & Model Selection › Gradient descent]

Hessian of J : $\nabla^2 J = (1/n)X^T X$. Eigenvalues of $X^T X$: $12 \pm 4 = 16, 8$ (eigenvectors $(1, 1), (1, -1)$).

So the Hessian eigenvalues are $L = 3.2$ and $\mu = 1.6$.

GD error evolves as $w_{t+1} - w^* = (I - \eta \nabla^2 J)(w_t - w^*)$, so it converges iff $|1 - \eta \lambda| < 1$ for every Hessian eigenvalue λ .

$|1 - 0.248 \times 3.2| = 0.2064$, $|1 - 0.248 \times 1.6| = 0.6032$; $\rho = 0.6032$.

Answer: **0.603**.

Q.33 — Answer: 0.7341 to 0.7381 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Softmax + cross-entropy]

$z = Wx + b = (-1.5, -1, 0.5)$; $p = \text{softmax}(z) = (0.0996, 0.1643, 0.7361)$.

$\delta = \partial L / \partial z = p - y = (-0.9004, 0.1643, 0.7361)$ (true class 1).

Since $z_k = \sum_j W_{kj} x_j + b_k$: $\partial L / \partial W_{kj} = (p_k - y_k) x_j$ and $\partial L / \partial b_k = p_k - y_k$ (gradient matrix = $(p - y)x^T$). $\partial L / \partial b_3 = \mathbf{0.7361}$.

Q.34 — Answer: -32 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Backpropagation]

$a = Wx + b = (-2, 6, 3)$, $h = \text{ReLU}(a) = (0, 6, 3)$, $\hat{y} = v \cdot h + c = -13$; $\partial L / \partial \hat{y} = \hat{y} - y = -16$.

$\text{ReLU}'(a) = (0, 1, 1)$ (unit j is active iff $a_j > 0$).

$\partial L / \partial W_{22} = (\hat{y} - y) \cdot v_2 \cdot \text{ReLU}'(a_2) \cdot x_2 = -16 \times -1 \times 1 \times -2 = \mathbf{-32}$.

Q.35 — Answer: 26.3796 to 26.3996 [NAT, 2M · Probabilistic & Instance-based Classifiers › k-nearest neighbours]

Three nearest neighbours: $x=8$ ($d=0.6, y=28$), $x=9$ ($d=1.6, y=24$), $x=5$ ($d=2.4, y=6$).

Weights $1/d^2$: 2.7778, 0.3906, 0.1736; $\Sigma w = 3.342$.

$\Sigma w_i y_i = 88.1944 \Rightarrow \hat{y} = 88.1944 / 3.342 = \mathbf{26.39}$.

(The unweighted 3-NN average would be 19.33; weighting pulls $\hat{y}$ towards the closest point.)

Mock Test 12

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	4	7	24

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
SVM & Decision Trees	7	11
Probabilistic & Instance-based Classifiers	7	11
Neural Networks (MLP / Feed-forward)	7	11
Regression & Model Selection	7	11
Unsupervised Learning	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MCQ]

Cost-complexity pruning of a classification tree produced the nested subtrees $T1 \supset T2 \supset T3 \supset T4$ below. For complexity parameter $\alpha = 0.02$, which subtree minimises $R_\alpha(T) = R(T) + \alpha|T|$?

Subtree	T1	T2	T3	T4
Leaves $ T $	7	6	5	4
Training error $R(T)$	0.015	0.06	0.065	0.11

- (A) T1 (B) T2
(C) T3 (D) T4

Q.2 [1 mark · NAT (numerical answer)]

A bagged ensemble is trained on $N = 8900$ points. Using the large- N approximation, the expected number of training points that are out-of-bag for a single bootstrap sample of size N is _____ (Answer in integer)

Q.3 [1 mark · MSQ (one or more correct)]

Which of the following statements about classifier evaluation is/are CORRECT?

- (A) The F1-score is the harmonic mean of precision and recall.
(B) Micro-averaged and macro-averaged F1 are always equal for multiclass problems.
(C) A classifier that assigns random scores has an expected ROC AUC of 0.5.
(D) ROC AUC depends on the particular decision threshold chosen.

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about the perceptron is/are CORRECT?

- (A) The perceptron mistake bound becomes smaller when the margin γ decreases.
(B) A two-layer network of threshold units can compute XOR.
(C) The perceptron update rule changes the weights even on correctly classified examples.
(D) The perceptron mistake bound $(R/\gamma)^2$ does not depend explicitly on the input dimension d .

Q.5 [1 mark · MSQ (one or more correct)]

Linear regression is trained by minimising the squared-error loss. Which of the following statements about gradient-based training is/are CORRECT?

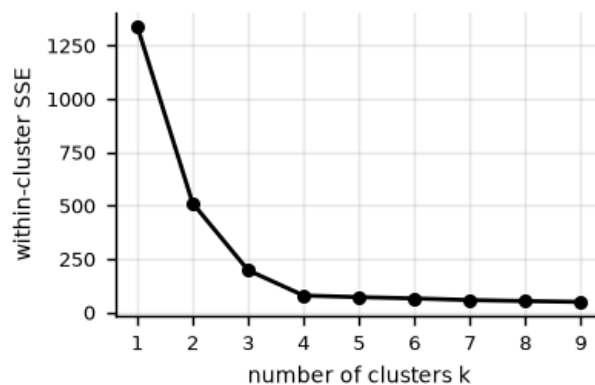
- (A) One epoch of SGD with batch size 1 on n examples performs n parameter updates.
- (B) Gradient descent on the MSE loss converges for every learning rate $\eta > 0$.
- (C) Solving the normal equations requires choosing a learning rate.
- (D) To minimise the loss, gradient descent updates $w \leftarrow w + \eta \nabla J(w)$.

Q.6 [1 mark · NAT (numerical answer)]

Polynomial feature expansion of degree 2 is applied to 4 input variables (all monomials of total degree 1 to 2, including interaction terms, but excluding the constant). The number of generated features is _____ (Answer in integer)

Q.7 [1 mark · MCQ]

k-means is run for $k = 1, 2, \dots, 9$ on a dataset and the final within-cluster SSE is plotted below. Using the elbow method, the most appropriate number of clusters is



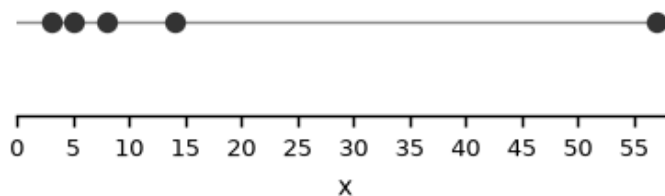
- (A) 4
- (B) 3
- (C) 9
- (D) 5

Q.8 [1 mark · NAT (numerical answer)]

The value of the derivative $\sigma'(z)$ of the logistic sigmoid at $z = -0.5$ is _____ (rounded off to three decimal places)

Q.9 [1 mark · NAT (numerical answer)]

A cluster contains the 1-D values $\{3, 5, 8, 14, 57\}$. Treating them as one cluster, the medoid of this cluster (absolute-difference dissimilarity) is _____ (rounded off to two decimal places)

**Q.10 [1 mark · NAT (numerical answer)]**

A node contains 12 records spread equally over 4 classes (3 records per class). Its entropy (in bits) is _____ (rounded off to two decimal places)

Q.11 [1 mark · NAT (numerical answer)]

To keep the same sampling density as 20 points per axis in 1-D, the number of points needed in $d = 3$ dimensions (a grid with 20 points per axis) is _____ (Answer in integer)

Q.12 [1 mark · NAT (numerical answer)]

A binary logistic regression model is given by $P(y = 1 | \mathbf{x}) = \sigma(w_0 + w_1x_1 + w_2x_2)$ with $w_0 = 2$, $w_1 = -1$, $w_2 = -2$, where $\sigma(z) = 1/(1 + e^{-z})$. For the input $\mathbf{x} = (x_1, x_2) = (-2, 3)$, the value of $P(y = 1 | \mathbf{x})$ is _____ (rounded off to two decimal places)

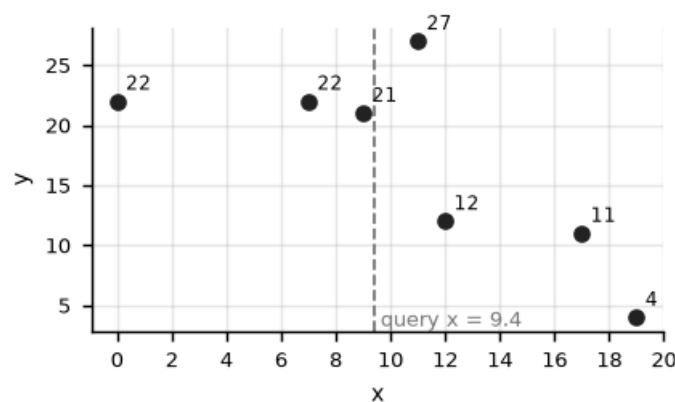
Q.13 [1 mark · NAT (numerical answer)]

Mini-batch SGD is run on $N = 5000$ examples with batch size $B = 16$ for $E = 20$ epochs, and any incomplete final batch in an epoch is **dropped**. The total number of parameter updates is _____ (Answer in integer)

Q.14 [1 mark · NAT (numerical answer)]

The training data for a k-NN regressor are given below (also plotted). Using $k = 3$ and absolute distance $|x - x_i|$, the predicted value (unweighted average of neighbours) at $x = 9.4$ is _____ (rounded off to two decimal places)

x	0	7	9	11	12	17	19
y	22	22	21	27	12	11	4

**Q.15 [1 mark · NAT (numerical answer)]**

A linear regression model with an intercept uses 3 numeric features and a categorical feature *day type* with 3 levels (Weekday, Saturday, Sunday). Each categorical feature is encoded by dummy (indicator) variables using a reference level, so that the design matrix has full column rank. The total number of regression coefficients (including the intercept) to be estimated is _____ (Answer in integer)

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · MSQ (one or more correct)]**

A threshold unit outputs 1 if $w_1x_1 + w_2x_2 + b > 0$ and 0 otherwise, for binary inputs $x_1, x_2 \in \{0, 1\}$. Which of the following parameter settings make the unit implement the Boolean **AND** function?

- (A) $w_1 = 1, w_2 = 2, b = -2.5$
- (B) $w_1 = 1.5, w_2 = 1, b = -2$
- (C) $w_1 = 1.5, w_2 = 2, b = -2.5$
- (D) $w_1 = -2, w_2 = 0, b = 1.5$

Q.17 [2 marks · NAT (numerical answer)]

A 3-class classifier outputs logits $z = (0, -0.5, 2)$ and probabilities $p = \text{softmax}(z)$. The true class is 2, and the loss is the cross-entropy $L = -\ln p_2$. Then the cross-entropy loss $L = -\ln p_y$ (natural log) is _____ (rounded off to three decimal places)

Q.18 [2 marks · NAT (numerical answer)]

A linear model $\hat{y} = w_1 x_1 + w_2 x_2 + b$ is trained by stochastic gradient descent (batch size 1) on the per-example loss $\frac{1}{2}(\hat{y} - y)^2$ with learning rate $\eta = 0.1$, starting from $w = (0, 1)$, $b = 1$. The two examples below are processed once, in the given order. After both updates, the value of w_2 is _____ (rounded off to three decimal places)

example	x_1	x_2	y
1	2	-1	-3
2	1	1	-3

Q.19 [2 marks · NAT (numerical answer)]

Ridge regression without intercept minimises $\|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2$ with $\lambda = 1$, for the data below ($\beta = (\beta_1, \beta_2)^T$). The value of β_1 is _____ (rounded off to three decimal places)

$$X = \begin{bmatrix} 2 & 1 \\ -2 & 0 \\ 2 & 2 \\ 0 & 0 \end{bmatrix}$$

$$y = \begin{bmatrix} 7 \\ 1 \\ 3 \\ 3 \end{bmatrix}$$

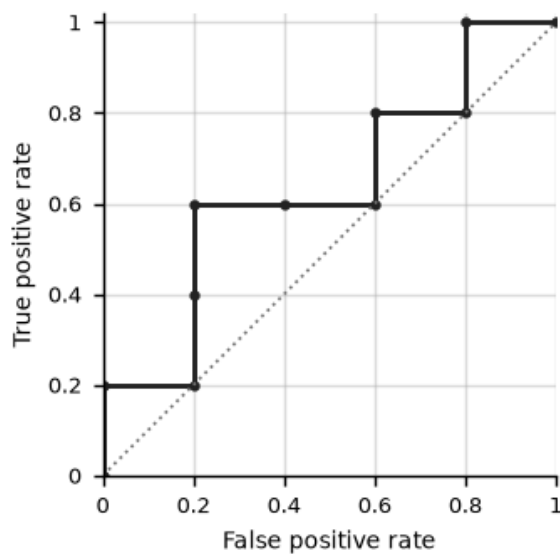
Q.20 [2 marks · MSQ (one or more correct)]

Each option shows a 2×2 matrix $[[K(x_1, x_1), K(x_1, x_2)], [K(x_2, x_1), K(x_2, x_2)]]$. Which of them can be the Gram matrix of some valid (Mercer) kernel K on two points x_1, x_2 ?

- (A) $[[7, -5], [-5, 1]]$
- (B) $[[-6, 0], [0, 1]]$
- (C) $[[8, 6], [6, 6]]$
- (D) $[[5, 1], [3, 3]]$

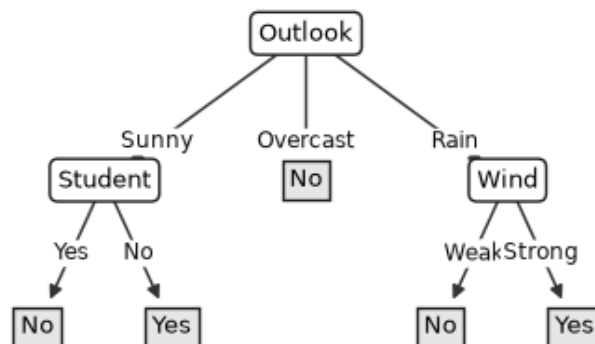
Q.21 [2 marks · NAT (numerical answer)]

The ROC curve below was obtained from a test set with 5 positives and 5 negatives (all scores distinct, so the curve moves in steps of $1/5$ vertically and $1/5$ horizontally). From the figure, the highest TPR attainable at $\text{FPR} \leq 0.2$ is _____ (rounded off to two decimal places)



Q.22 [2 marks · NAT (numerical answer)]

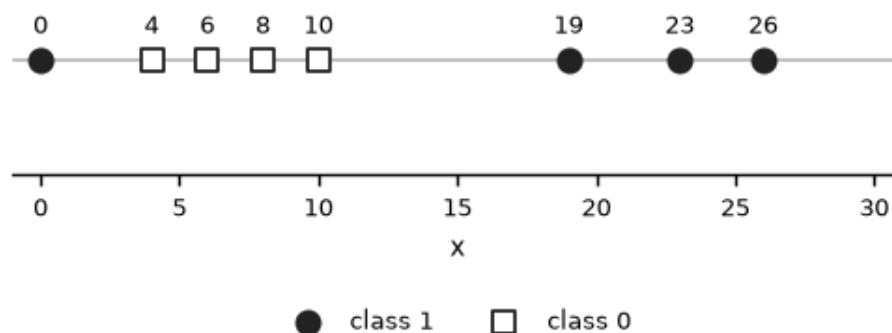
The decision tree shown is applied to the 10 labelled records in the table (attributes not tested on a path are ignored). Find the precision of the tree for the class “Yes”. (rounded off to two decimal places)



#	Outlook	Wind	Student	Actual
1	Overcast	Strong	No	Yes
2	Sunny	Strong	No	No
3	Rain	Strong	Yes	No
4	Sunny	Weak	Yes	No
5	Overcast	Strong	No	Yes
6	Overcast	Strong	Yes	No
7	Sunny	Weak	No	Yes
8	Rain	Weak	Yes	No
9	Overcast	Strong	Yes	Yes
10	Rain	Strong	No	No

Q.23 [2 marks · NAT (numerical answer)]

8 one-dimensional points with binary labels are shown on the number line (filled circle = class 1, open square = class 0). A 3-NN classifier (absolute distance, majority vote) is evaluated by leave-one-out cross-validation: each point is classified using all the *other* points. The number of training points misclassified in leave-one-out cross-validation is ____ (Answer in integer)



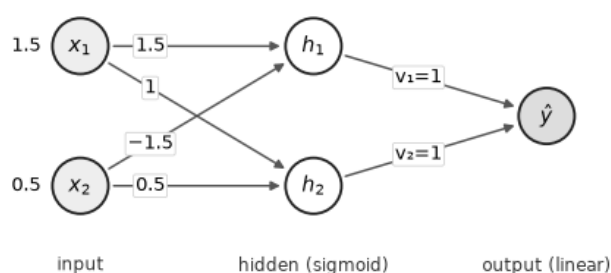
Q.24 [2 marks · NAT (numerical answer)]

For an input x , a classifier's posterior probabilities are $P(C_1|x) = 0.6$, $P(C_2|x) = 0.3$, $P(C_3|x) = 0.1$. The loss of taking action α_i (deciding C_i) when the true class is C_j is given below. Then the conditional risk of the action chosen by the MAP rule (i.e. choosing the class with highest posterior) is _____ (rounded off to two decimal places)

loss $L(\alpha_i, C_j)$	true C_1	true C_2	true C_3
α_1 : decide C_1	0	10	10
α_2 : decide C_2	1	0	9
α_3 : decide C_3	8	5	0

Q.25 [2 marks · NAT (numerical answer)]

In the network below, hidden units are sigmoid, the output is linear ($\hat{y} = v_1 h_1 + v_2 h_2$), all biases are 0, and the loss is $L = \frac{1}{2}(\hat{y} - y)^2$. For input $x = (1.5, 0.5)$ and target $y = 1$, the value of $\partial L / \partial w_{12}$ (the weight from x_2 to h_1) is _____ (rounded off to 4 decimal places)

**Q.26 [2 marks · NAT (numerical answer)]**

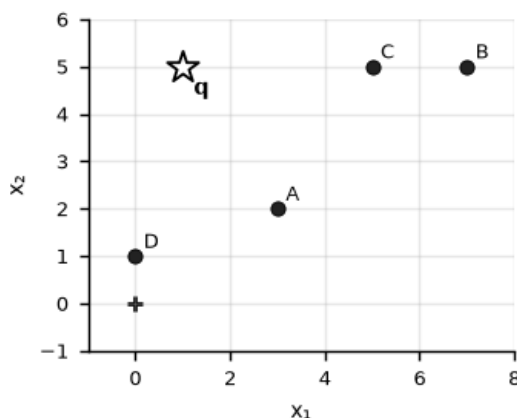
Consider the model $y = wx + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$, $\sigma^2 = 1$, and $w \sim N(0, \tau^2)$ with $\tau^2 = 0.5$. Given the four observations below, $\hat{w}_{ML} - \hat{w}_{MAP}$ is _____ (rounded off to three decimal places)

x	2	3	-1	-1
y	-2	0	0	5

Q.27 [2 marks · MCQ]

A query point $q = (1, 5)$ and four points A-D are shown below. Using **Euclidean (L_2) distance**, which point is nearest to q ?

Point	A	B	C	D
x_1	3	7	5	0
x_2	2	5	5	1



- (A) Point A
(C) Point C

- (B) Point B
(D) Point D

Q.28 [2 marks · NAT (numerical answer)]

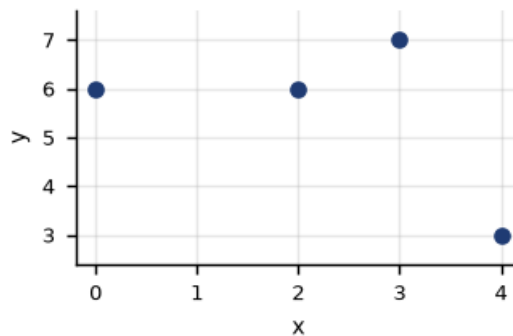
Agglomerative hierarchical clustering with **complete linkage** is applied to 5 points whose pairwise distance matrix is given below (upper triangle shown). The height (linkage distance) at which the **third** merge takes place is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5
P1	0	10	5	11	11
P2		0	5	1	9
P3			0	6	8
P4				0	10
P5					0

Q.29 [2 marks · NAT (numerical answer)]

A linear model $\hat{y} = w_0 + w_1 x$ is trained on the 4 points below by batch gradient descent on the loss $J(w) = (1/2n)\sum(w_0 + w_1 x_i - y_i)^2$, starting from $(w_0, w_1) = (0, 0)$ with learning rate $\eta = 0.02$. Then the value of w_1 after two iterations is _____ (rounded off to three decimal places)

x	0	2	3	4
y	6	6	7	3

**Q.30 [2 marks · MSQ (one or more correct)]**

Which of the following statements about PCA, SVD and dimensionality reduction is/are CORRECT?

- (A) Multiplying all features by the same non-zero constant changes the proportion of variance explained by each PC.
- (B) The reconstruction error when keeping k PCs equals the sum of the retained eigenvalues.
- (C) With n samples and $d \geq n$ features, the centered data matrix has rank at most $n - 1$.
- (D) PCA always preserves the class separability present in the original feature space.

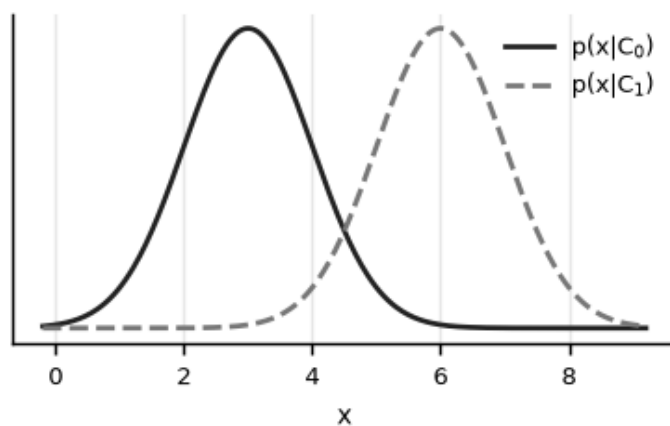
Q.31 [2 marks · NAT (numerical answer)]

In C4.5, the attribute Day splits a node of 20 records as tabulated (class Yes/No counts in each branch). Compute the gain ratio of Day (all logarithms base 2). (rounded off to two decimal places)

Day	#Yes	#No
v1	4	3
v2	0	6
v3	0	3
v4	1	3

Q.32 [2 marks · NAT (numerical answer)]

In a two-class problem with one feature, $p(x | C_0) = N(3, 1)$ and $p(x | C_1) = N(6, 1)$ (second argument = variance), with priors $P(C_0) = 0.8$ and $P(C_1) = 0.2$. Under 0-1 loss, the Bayes-optimal (minimum-error) decision threshold x^* is _____ (rounded off to two decimal places)

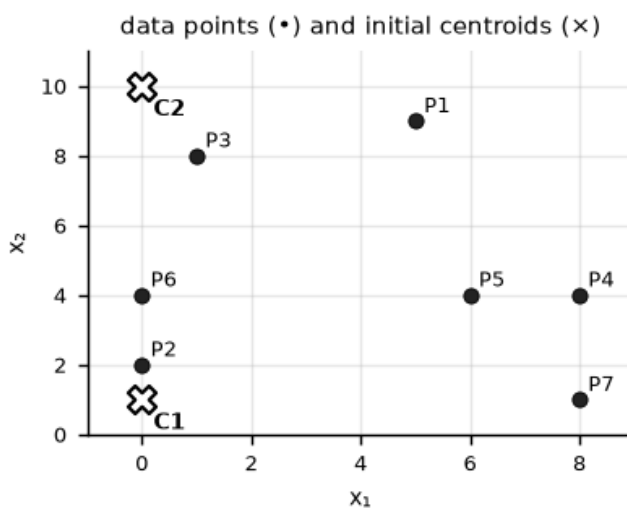


Class-conditional densities (not weighted by priors).

Q.33 [2 marks · NAT (numerical answer)]

k-means (Lloyd's algorithm, Euclidean distance) with $k = 2$ is run on the 7 points below, starting from the initial centroids $C1 = (0, 1)$, $C2 = (0, 10)$. After **one** iteration (assignment step followed by update step), the within-cluster sum of squared Euclidean distances $J = \sum_i \|x_i - \mu_{c(i)}\|^2$, computed with the *updated* centroids and the assignments of that iteration, is _____ (rounded off to two decimal places)

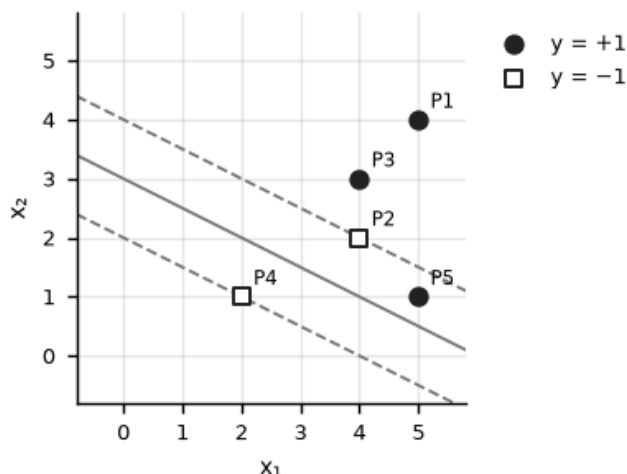
Point	P1	P2	P3	P4	P5	P6	P7
x_1	5	0	1	8	6	0	8
x_2	9	2	8	4	4	4	1



Q.34 [2 marks · MSQ (one or more correct)]

A soft-margin linear SVM has learned $f(x) = 0.5x_1 + x_2 - 3$. The training points are listed below and plotted (boundary solid, margins $f = \pm 1$ dashed). With ξ_i the optimal slack of point i , which of the following statements is/are CORRECT?

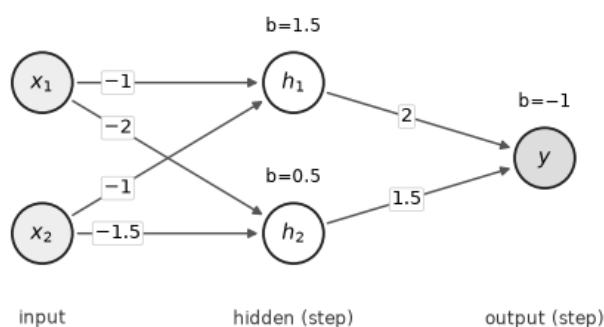
Point	x_1	x_2	y
P1	5	4	+1
P2	4	2	-1
P3	4	3	+1
P4	2	1	-1
P5	5	1	+1



- (A) P1 has slack $\xi_1 = 0$.
- (B) P2 is misclassified ($\xi_2 > 1$).
- (C) P3 has slack $\xi_3 = 0$.
- (D) $\sum \xi_i = 2.5$.

Q.35 [2 marks · MCQ]

In the network below every unit (hidden and output) is a threshold unit that outputs 1 if its weighted input plus bias is > 0 , and 0 otherwise. Edge labels are weights. For binary inputs $x_1, x_2 \in \{0, 1\}$, the network computes



- (A) NAND
- (B) x_1 AND (NOT x_2)
- (C) XNOR
- (D) OR

— END OF QUESTION PAPER —

Mock Test 12 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MCQ	1	(A)	19	NAT	2	0.7113 to 0.7173
2	NAT	1	3273 to 3276	20	MSQ	2	(C)
3	MSQ	1	(A), (C)	21	NAT	2	0.59 to 0.61
4	MSQ	1	(B), (D)	22	NAT	2	0.24 to 0.26
5	MSQ	1	(A)	23	NAT	2	1
6	NAT	1	14	24	NAT	2	3.99 to 4.01
7	MCQ	1	(A)	25	NAT	2	0.0489 to 0.0509
8	NAT	1	0.233 to 0.237	26	NAT	2	−0.0726 to −0.0686
9	NAT	1	7.99 to 8.01	27	MCQ	2	(A)
10	NAT	1	1.99 to 2.01	28	NAT	2	9.99 to 10.01
11	NAT	1	8000	29	NAT	2	0.4104 to 0.4144
12	NAT	1	0.1092 to 0.1292	30	MSQ	2	(C)
13	NAT	1	6240	31	NAT	2	0.1479 to 0.1679
14	NAT	1	23.3233 to 23.3433	32	NAT	2	4.9521 to 4.9721
15	NAT	1	6	33	NAT	2	83.69 to 83.71
16	MSQ	2	(A), (B), (C)	34	MSQ	2	(A), (B), (C), (D)
17	NAT	2	2.6947 to 2.6987	35	MCQ	2	(A)
18	NAT	2	0.8574 to 0.8626				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
SVM & Decision Trees					11
Probabilistic & Instance-based Classifiers					11
Neural Networks (MLP / Feed-forward)					11
Regression & Model Selection					11
Unsupervised Learning					11
Total					55

Mock Test 12 — Detailed Solutions

Q.1 — Answer: (A) [MCQ, 1M · SVM & Decision Trees > Decision trees: overfitting & pruning]

Subtree	$R(T) + \alpha T $
T1	$0.015 + 0.02 \times 7 = 0.155$
T2	$0.06 + 0.02 \times 6 = 0.18$
T3	$0.065 + 0.02 \times 5 = 0.165$
T4	$0.11 + 0.02 \times 4 = 0.19$

Minimum at **T1**. Larger α penalises leaves more and favours smaller trees; $\alpha = 0$ would pick the largest tree T1.

Q.2 — Answer: 3273 to 3276 [NAT, 1M · SVM & Decision Trees > Ensembles: bagging & random forests]

$P(\text{out-of-bag}) = (1 - 1/N)^N \approx e^{-1} \approx 0.3679$; expected count $\approx 8900 \times 0.3679 = \mathbf{3274}$.

Q.3 — Answer: (A), (C) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

(A) **TRUE**. $F1 = 2PR/(P + R)$.

(B) **FALSE**. They differ when classes have different sizes/performance.

(C) **TRUE**. Its ROC curve is the diagonal.

(D) **FALSE**. It summarises all thresholds.

Q.4 — Answer: (B), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

(A) **FALSE**. $(R/\gamma)^2$ grows as γ shrinks.

(B) **TRUE**. E.g. $\text{XOR} = \text{AND}(\text{OR}(x_1, x_2), \text{NAND}(x_1, x_2))$.

(C) **FALSE**. Correctly classified points ($y \cdot a > 0$) cause no update.

(D) **TRUE**. It depends only on the radius R of the data and the margin γ .

Q.5 — Answer: (A) [MSQ, 1M · Regression & Model Selection > Gradient descent]

(A) **TRUE**. One update per example.

(B) **FALSE**. It diverges if η is too large.

(C) **FALSE**. It is a direct (closed-form) solve.

(D) **FALSE**. That is gradient ascent.

Q.6 — Answer: 14 [NAT, 1M · Regression & Model Selection > Basis functions & feature scaling]

All monomials of degree ≤ 2 : $C(6, 2) = 15$; excluding the constant: **14**.

Q.7 — Answer: (A) [MCQ, 1M · Unsupervised Learning > k-means]

The SSE drops steeply up to $k = 4$ and then decreases only slowly (relative drops: $k=2$: 62%, $k=3$: 61%, $k=4$: 60%, $k=5$: 9%, $k=6$: 8%, $k=7$: 12%, $k=8$: 8%, $k=9$: 7%). The bend ('elbow') is at **$k = 4$** . $k = 9$ gives the lowest SSE but SSE always (weakly) decreases with k , so minimizing it would pick the largest k ; it is not a model-selection criterion by itself.

Q.8 — Answer: 0.233 to 0.237 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

$\sigma(-0.5) = 0.3775$; $\sigma'(z) = \sigma(z)(1 - \sigma(z)) = 0.3775 \times 0.6225 = \mathbf{0.235}$.

Note $\sigma' \leq 0.25$, with equality only at $z = 0$.

Q.9 — Answer: 7.99 to 8.01 [NAT, 1M · Unsupervised Learning > k-medoids]

Mean = $87/5 = 17.4$ (pulled towards the outlier 57).

Medoid = data point minimizing $\sum |x_i - m|$. Costs: $m=3$: 72, $m=5$: 66, $m=8$: 63, $m=14$: 69, $m=57$: 198.

The minimum is at $m = 8$ (the median), cost 63.

Answer: **8**. The medoid is barely affected by the outlier, unlike the mean.

Q.10 — Answer: 1.99 to 2.01 [NAT, 1M · SVM & Decision Trees > Decision trees: impurity measures]

$p_k = 3/12 = 1/4$ for every class $\Rightarrow H = -4 \cdot (1/4) \log_2(1/4) = \log_2 4 = 2$ — the maximum possible for 4 classes.

Q.11 — Answer: 8000 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

A grid with 20 points per axis needs $20^3 = 8000$ points — exponential growth with d .

Q.12 — Answer: 0.1092 to 0.1292 [NAT, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

$z = w_0 + w_1 x_1 + w_2 x_2 = 2 + (-1)(-2) + (-2)(3) = -2$ (this is the log-odds).

$P(y = 1 | \mathbf{x}) = \sigma(z) = 1/(1 + e^2) = 0.1192$.

Answer: **0.12**.

Q.13 — Answer: 6240 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Updates per epoch = $\lfloor N/B \rfloor = \lfloor 5000/16 \rfloor = 312$.

Total = $312 \times 20 = 6240$.

Q.14 — Answer: 23.3233 to 23.3433 [NAT, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Distances $|x_i - 9.4|$: $x=9$: 0.4, $x=11$: 1.6, $x=7$: 2.4, $x=12$: 2.6, $x=17$: 7.6, $x=0$: 9.4, $x=19$: 9.6.

The 3 nearest neighbours are $x = 9, 11, 7$ with $y = 21, 27, 22$.

$\hat{y} = (21 + 27 + 22)/3 = 23.33$.

Q.15 — Answer: 6 [NAT, 1M · Regression & Model Selection > Multiple linear regression]

A categorical feature with k levels needs $k - 1$ dummies (one level is the reference absorbed by the intercept; using all k dummies with an intercept gives perfect multicollinearity).

Parameters = 1 (intercept) + 3 (numeric) + (3 - 1) = 6.

Answer: **6**.

Q.16 — Answer: (A), (B), (C) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Target AND truth table for $(x_1, x_2) = (0,0), (0,1), (1,0), (1,1)$: 0, 0, 0, 1. For each option compute $a = w_1 x_1 + w_2 x_2 + b$ at the four inputs and output 1 iff $a > 0$.

(A) TRUE. $a = -2.5, -0.5, -1.5, 0.5 \rightarrow$ outputs 0, 0, 0, 1 - AND.

(B) TRUE. $a = -2, -1, -0.5, 0.5 \rightarrow$ outputs 0, 0, 0, 1 - AND.

(C) TRUE. $a = -2.5, -0.5, -1, 1 \rightarrow$ outputs 0, 0, 0, 1 - AND.

(D) FALSE. $a = 1.5, 1.5, -0.5, -0.5 \rightarrow$ outputs 1, 1, 0, 0 - NOT x_1 .

Q.17 — Answer: 2.6947 to 2.6987 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$p = \text{softmax}(z)$: $\exp(z) = (1, 0.6065, 7.3891)$, $\text{sum} = 8.9956$, so $p = (0.1112, 0.0674, 0.8214)$.

For $L = -\ln p_y$ with softmax, $\partial L / \partial z = p - y$ (y one-hot, true class 2): $\partial L / \partial z = (0.1112, -0.9326, 0.8214)$.

$L = -\ln 0.0674 = 2.6967$.

Note $\sum_k \partial L / \partial z_k = \sum p_k - 1 = 0$.

Q.18 — Answer: 0.8574 to 0.8626 [NAT, 2M · Regression & Model Selection > Gradient descent]

Per-example loss $\square = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial \square / \partial w = (\hat{y} - y)x$, $\partial \square / \partial b = (\hat{y} - y)$. Update after each example.

Example 1: $\hat{y} = 0$, error $\hat{y} - y = 3$; $\nabla_w = (6, -3)$, $\nabla_b = 3 \Rightarrow w = (-0.6, 1.3)$, $b = 0.7$.

Example 2: $\hat{y} = 1.4$, error $\hat{y} - y = 4.4$; $\nabla_w = (4.4, 4.4)$, $\nabla_b = 4.4 \Rightarrow w = (-1.04, 0.86)$, $b = 0.26$.

Answer: **0.86**.

Q.19 — Answer: 0.7113 to 0.7173 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

$\beta_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y$ with $\lambda = 1$.

$$X^T X = \begin{bmatrix} 12 & 6 \\ 6 & 5 \end{bmatrix}$$

$$X^T X + \lambda I = \begin{bmatrix} 13 & 6 \\ 6 & 6 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 18 \\ 13 \end{bmatrix}$$

$$\det = 13 \times 6 - (6)^2 = 42; (X^T X + \lambda I)^{-1} = (1/42)[[6, -6], [-6, 13]].$$

$$\beta = (0.7143, 1.4524).$$

Answer: **0.714**.

Q.20 — Answer: (C) [MSQ, 2M · SVM & Decision Trees > SVM: kernels]

A Gram matrix of a valid kernel must be symmetric and positive semidefinite. For a symmetric 2×2 matrix this is equivalent to: both diagonal entries ≥ 0 and determinant ≥ 0 .

(A) **FALSE**. $\det = -18 < 0 \Rightarrow$ one negative eigenvalue, not PSD.

(B) **FALSE**. a diagonal entry $K(x, x) = \|\phi(x)\|^2$ is negative — impossible.

(C) **TRUE**. symmetric, diagonal entries ≥ 0 and $\det = 12 > 0 \Rightarrow$ positive definite.

(D) **FALSE**. not symmetric, but $K(x, z) = K(z, x)$ for every kernel.

Q.21 — Answer: 0.59 to 0.61 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Among the operating points with $FPR \leq 0.2$, take the highest point of the curve (the top of the vertical segment at $FPR = 0.2$).

TPR = **0.6**.

Q.22 — Answer: 0.24 to 0.26 [NAT, 2M · SVM & Decision Trees > Decision trees: structure & traversal]

Route each record from the root to a leaf:

#	Predicted	Actual	Correct?
1	No	Yes	no
2	Yes	No	no
3	Yes	No	no
4	No	No	yes
5	No	Yes	no
6	No	No	yes
7	Yes	Yes	yes
8	No	No	yes
9	No	Yes	no
10	Yes	No	no

Misclassified: 6 of 10 (accuracy 40%). Answer: **0.25**.

Q.23 — Answer: 1 [NAT, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Classify each point by its nearest neighbour(s) among the remaining points:

x	true	3 nearest (excluding itself)	LOO prediction	
0	1	4(0), 6(0), 8(0)	0	wrong
4	0	6(0), 0(1), 8(0)	0	
6	0	4(0), 8(0), 10(0)	0	
8	0	6(0), 10(0), 4(0)	0	
10	0	8(0), 6(0), 4(0)	0	
19	1	23(1), 26(1), 10(0)	1	
23	1	26(1), 19(1), 10(0)	1	
26	1	23(1), 19(1), 10(0)	1	

Misclassified: 1 of 8. Answer: **1**.

Note: the *training* error of 1-NN is 0, but the LOOCV error is not — a point cannot be its own neighbour.

Q.24 — Answer: 3.99 to 4.01 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

Conditional risk $R(\alpha_i|x) = \sum_j L(\alpha_i, C_j)P(C_j|x)$:

$$R(\alpha_1) = 0 \times 0.6 + 10 \times 0.3 + 10 \times 0.1 = 4$$

$$R(\alpha_2) = 1 \times 0.6 + 0 \times 0.3 + 9 \times 0.1 = 1.5$$

$$R(\alpha_3) = 8 \times 0.6 + 5 \times 0.3 + 0 \times 0.1 = 6.3$$

Minimum risk: α_2 with $R = 1.5$. The MAP rule picks C_1 (risk 4) — different, because the losses are asymmetric.

Answer: **4**.

Q.25 — Answer: 0.0489 to 0.0509 [NAT, 2M · Neural Networks (MLP / Feed-forward) >

Backpropagation]

Forward: $a = Wx = (1.5, 1.75)$, $h = \sigma(a) = (0.8176, 0.852)$, $\hat{y} = v \cdot h = 1.6695$.

Output error $\delta_o = \partial L / \partial \hat{y} = \hat{y} - y = 0.6695$ (linear output, $L = \frac{1}{2}(\hat{y} - y)^2$).

Hidden delta $\delta_1 = \delta_o \cdot v_1 \cdot h_1(1 - h_1) = 0.6695 \times 1 \times 0.1491 = 0.0999$.

$$\partial L / \partial w_{12} = \delta_1 \cdot x_2 = 0.0999 \times 0.5 = \mathbf{0.0499}.$$

Q.26 — Answer: -0.0726 to -0.0686 [NAT, 2M · Regression & Model Selection > MLE & MAP view]

$$-\ln p(w | \text{data}) = (1/(2\sigma^2))\sum(y_i - wx_i)^2 + w^2/(2\tau^2) + \text{const.}$$

Multiplying by $2\sigma^2$: $\sum(y_i - wx_i)^2 + (\sigma^2/\tau^2)w^2$ — ridge with $\lambda = \sigma^2/\tau^2 = 2$.

$$\sum x_i^2 = 15, \sum x_i y_i = -9; \hat{w}_{\text{MAP}} = \sum xy / (\sum x^2 + \lambda) = -9 / (15 + 2) = -0.5294.$$

$$\hat{w}_{\text{ML}} = \sum xy / \sum x^2 = -0.6; \text{ difference} = -0.0706.$$

Answer: **-0.071**.

Q.27 — Answer: (A) [MCQ, 2M · Unsupervised Learning > Distance measures]

Compute the measure from $q = (1, 5)$ to every point (cosine column shows similarity, larger = more similar):

Point	coordinates	Manhattan	Euclidean	Chebyshev	cosine similarity
A	(3, 2)	5	3.606	3	0.707
B	(7, 5)	6	6	6	0.73
C	(5, 5)	4	4	4	0.832
D	(0, 1)	5	4.123	4	0.981

Under Euclidean (L_2) distance the nearest to q is **point A**.

Note how the answer depends on the measure: Manhattan (L_1) distance $\rightarrow$ C; Euclidean (L_2) distance $\rightarrow$ A; Chebyshev (L_∞) distance $\rightarrow$ A; cosine similarity $\rightarrow$ D. Cosine similarity looks only at the angle from the origin, L_1 sums coordinate gaps, L_∞ takes the largest gap.

Q.28 — Answer: 9.99 to 10.01 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Use complete linkage: $d(A, B) = \max_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P2\} + \{P4\}$ at height **1**. Updated: $d(\{P2, P4\}, \{P1\}) = 11$, $d(\{P2, P4\}, \{P3\}) = 6$, $d(\{P2, P4\}, \{P5\}) = 10$.

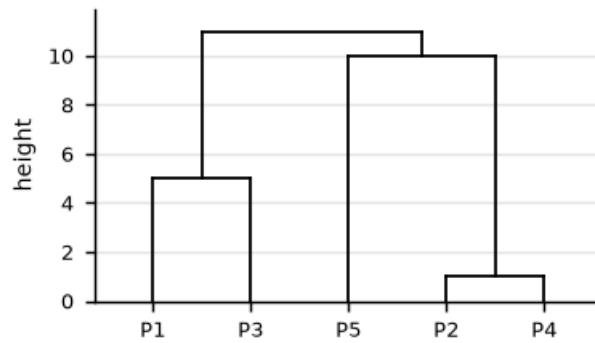
Merge 2: $\{P1\} + \{P3\}$ at height **5**. Updated: $d(\{P1, P3\}, \{P5\}) = 11$, $d(\{P1, P3\}, \{P2, P4\}) = 11$.

Merge 3: $\{P5\} + \{P2, P4\}$ at height **10**. Updated: $d(\{P2, P4, P5\}, \{P1, P3\}) = 11$.

Merge 4: $\{P1, P3\} + \{P2, P4, P5\}$ at height **11**.

The third merge occurs at height **10**.

Dendrogram:



Q.29 — Answer: 0.4104 to 0.4144 [NAT, 2M · Regression & Model Selection > Gradient descent]

$\partial J / \partial w_0 = (1/n) \sum r_i$, $\partial J / \partial w_1 = (1/n) \sum r_i x_i$, with residual $r_i = w_0 + w_1 x_i - y_i$.

Step 1: $w = (0, 0)$, $r = (-6, -6, -7, -3)$, $\nabla J = (-5.5, -11.25) \Rightarrow w \leftarrow w - 0.02 \nabla J = (0.11, 0.225)$.

Step 2: $w = (0.11, 0.225)$, $r = (-5.89, -5.44, -6.215, -1.99)$, $\nabla J = (-4.8837, -9.3712) \Rightarrow w \leftarrow w - 0.02 \nabla J = (0.2077, 0.4124)$.

Answer: **0.412**.

Q.30 — Answer: (C) [MSQ, 2M · Unsupervised Learning > PCA & dimensionality reduction]

(A) **FALSE**. Every eigenvalue is scaled by c^2 , so proportions are unchanged.

(B) **FALSE**. It equals the sum of the discarded eigenvalues.

(C) **TRUE**. Centering removes one degree of freedom.

(D) **FALSE**. Discarded low-variance directions may be the discriminative ones.

Q.31 — Answer: 0.1479 to 0.1679 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [5 Yes, 15 No]: $H = 0.8113$.

Child entropies: $v1: 0.9852$, $v2: 0$, $v3: 0$, $v4: 0.8113$.

$IG = 0.8113 - \sum (n_v/20) H_v = 0.3042$.

$\text{SplitInfo} = -\sum (n_v/N) \log_2(n_v/N) = -(7/20) \log_2(7/20) - (6/20) \log_2(6/20) - (3/20) \log_2(3/20) - (4/20) \log_2(4/20) = 1.9261$.

Gain ratio = $IG / \text{SplitInfo} = 0.3042/1.9261 = \mathbf{0.16}$.

C4.5 uses the gain ratio to penalise attributes that split the data into many small branches.

Q.32 — Answer: 4.9521 to 4.9721 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

With equal variances, $\ln[\pi_1 p(x|C_1)/(\pi_0 p(x|C_0))] = (\mu_1 - \mu_0)x/\sigma^2 - (\mu_1^2 - \mu_0^2)/(2\sigma^2) + \ln(\pi_1/\pi_0)$ — linear in x (LDA).

Setting it to 0: $x^* = (\mu_0 + \mu_1)/2 + \sigma^2 \ln(\pi_0/\pi_1)/(\mu_1 - \mu_0) = 4.5 + 1 \times \ln(0.8/0.2)/3$.

$= 4.5 + (0.4621) = \mathbf{4.96}$.

The threshold moves away from the more probable class C_0 's mean, enlarging its decision region.

Q.33 — Answer: 83.69 to 83.71 [NAT, 2M · Unsupervised Learning > k-means]

Iteration 1 - assignment (nearest initial centroid, squared Euclidean distance):

Point	$d^2(\cdot, C1)$	$d^2(\cdot, C2)$	cluster
P1 (5, 9)	89	26	C2
P2 (0, 2)	1	64	C1
P3 (1, 8)	50	5	C2
P4 (8, 4)	73	100	C1
P5 (6, 4)	45	72	C1
P6 (0, 4)	9	36	C1
P7 (8, 1)	64	145	C1

Update: each centroid becomes the mean of its assigned points: $C1 \rightarrow (4.4, 3)$; $C2 \rightarrow (3, 8.5)$.

$J = \sum \|x_i - \mu_{c(i)}\|^2$ with the new centroids = **83.7**.

Note: each step of Lloyd's algorithm can only decrease (or keep) J , which is why it converges.

Q.34 — Answer: (A), (B), (C), (D) [MSQ, 2M · SVM & Decision Trees › SVM: soft margin & hinge loss]

$$\xi_i = \max(0, 1 - y_i f(x_i)).$$

Point	$y f(x)$	ξ
P1	3.5	0
P2	-1	2
P3	2	0
P4	1	0
P5	0.5	0.5

(A) TRUE. $y f = 3.5 \Rightarrow \xi = 0$.

(B) TRUE. $y f = -1 \Rightarrow \xi = 2$.

(C) TRUE. $y f = 2 \Rightarrow \xi = 0$.

(D) TRUE. Sum of the individual slacks.

Q.35 — Answer: (A) [MCQ, 2M · Neural Networks (MLP / Feed-forward) › Expressiveness]

Each unit outputs 1 iff its net input (weighted sum + bias) is > 0 . Evaluate on all four inputs:

x_1	x_2	h_1	h_2	y
0	0	1	1	1
0	1	1	0	1
1	0	1	0	1
1	1	0	0	0

h_1 computes NAND, h_2 computes NOR; the output column $(0,0) \rightarrow (1,1)$ is 1, 1, 1, 0, i.e. **NAND**.

Mock Test 13

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	5	10	20

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
SVM & Decision Trees	7	11
Unsupervised Learning	7	11
Probabilistic & Instance-based Classifiers	7	11
Neural Networks (MLP / Feed-forward)	7	11
Regression & Model Selection	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

Consider the polynomial kernel $K(x, z) = (x^T z)^2$ with $x = (3, 2, 3)$ and $z = (3, 2, 3)$. The value of $K(x, x)$ is _____ (Answer in integer)

Q.2 [1 mark · MSQ (one or more correct)]

Which of the following statements about hierarchical clustering is/are CORRECT?

- (A) Complete linkage tends to produce compact clusters with small diameters.
- (B) Hierarchical clustering does not need the number of clusters in advance; it can be chosen afterwards by cutting the dendrogram.
- (C) Storing the full distance matrix for agglomerative clustering needs $O(n^2)$ memory.
- (D) Single and complete linkage always produce the same dendrogram.

Q.3 [1 mark · MSQ (one or more correct)]

Which of the following statements about the k-nearest-neighbour method is/are CORRECT?

- (A) Increasing k always reduces the test error of k-NN.
- (B) In distance-weighted k-NN with weights $1/d$, farther neighbours get larger weights.
- (C) The cost of a brute-force k-NN prediction grows linearly with the number of training points n.
- (D) k-NN is a parametric model whose number of parameters does not depend on n.

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about kernel functions (Mercer kernels) is/are CORRECT?

- (A) $k(x, z) = x^T A z$ is a valid kernel when A is symmetric positive semidefinite.
- (B) $p(k(x, z))$ is valid for any polynomial p with non-negative coefficients when k is valid.
- (C) If k is valid, $\exp(k(x, z))$ is valid.
- (D) $k(x, z) = x^T z + c$ with $c \geq 0$ is a valid kernel.

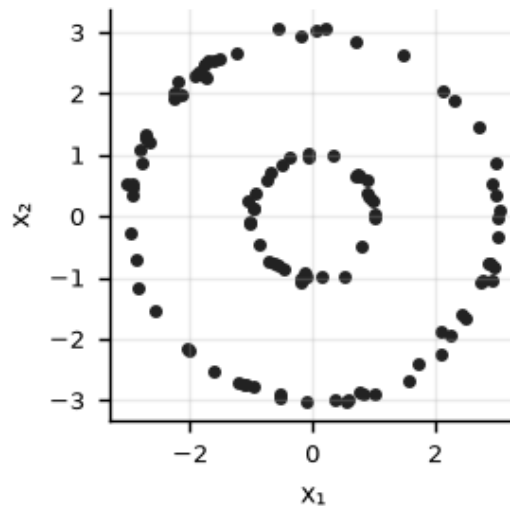
Q.5 [1 mark · NAT (numerical answer)]

The confusion matrix of a binary classifier on a test set is shown below (note which of rows/columns are actual and predicted). The false positive rate of the classifier is _____ (rounded off to two decimal places)

	Predicted +	Predicted -
Actual +	41	18
Actual -	8	55

Q.6 [1 mark · MCQ]

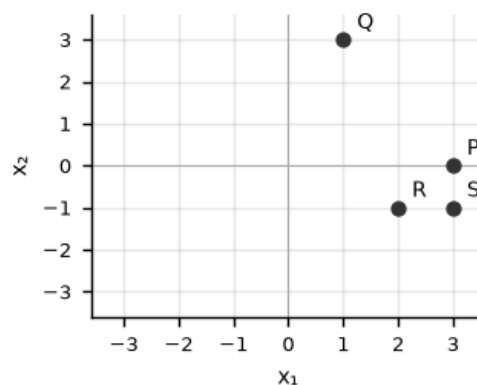
k-means with $k = 2$ fails to separate the two concentric rings shown below, even with many random restarts. The main reason is that k-means



- (A) produces convex clusters separated by linear (Voronoi) boundaries
- (B) cannot be applied to two-dimensional data
- (C) requires class labels to place the centroids
- (D) converges only when the data are standardized

Q.7 [1 mark · MSQ (one or more correct)]

A perceptron with weights $w = (2, 1)$ and bias $b = 1$ outputs $\hat{y} = +1$ if $w \cdot x + b \geq 0$ and $\hat{y} = -1$ otherwise. Which of the points shown below are classified as $\hat{y} = +1$?



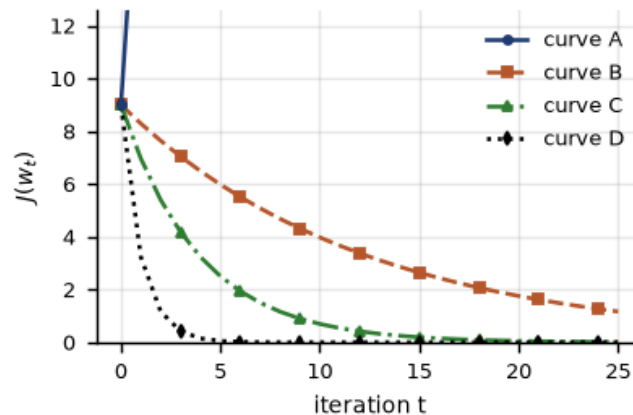
- (A) Point P = (3, 0)
- (B) Point Q = (1, 3)
- (C) Point R = (2, -1)
- (D) Point S = (3, -1)

Q.8 [1 mark · NAT (numerical answer)]

A network is trained with standard (non-inverted) dropout, keep probability $p = 0.6$, on the inputs of a layer. A trained outgoing weight equals 1.5. To match expected activations, the weight used at test time is _____ (rounded off to two decimal places)

Q.9 [1 mark · MCQ]

Gradient descent $w_{t+1} = w_t - \eta J'(w_t)$ is run on $J(w) = (w - 4)^2$ from $w_0 = 1$ with four learning rates $\eta \in \{0.02, 0.06, 0.2, 1.25\}$. The loss curves are shown below (curves that leave the plotted range are cut off). Which curve corresponds to $\eta = 0.02$?



- (A) Curve B
(C) Curve C

- (B) Curve D
(D) Curve A

Q.10 [1 mark · NAT (numerical answer)]

In a spam filter, a false negative costs 2 units and a false positive costs 8 units (correct decisions cost 0). The model outputs $p = P(\text{positive} | x)$, i.e. the probability that the email is spam. The Bayes-optimal rule predicts positive iff p exceeds the threshold _____ (rounded off to two decimal places)

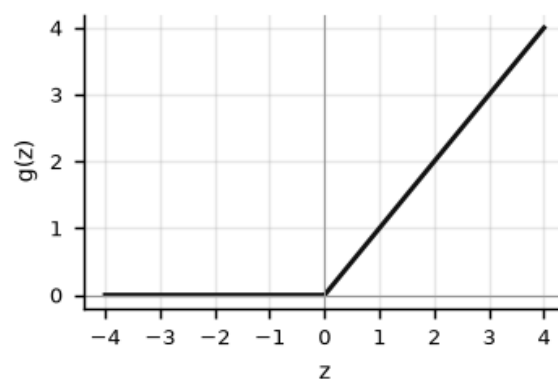
Q.11 [1 mark · MSQ (one or more correct)]

Which of the following tasks is/are most naturally formulated as **regression** problems?

- (A) Estimating the remaining useful life (in hours) of a machine from sensor data
(B) Estimating the electricity demand (MW) of a city for the next hour
(C) Predicting whether a customer will cancel a subscription next month
(D) Estimating the time (in seconds) a runner will take to finish 400 m

Q.12 [1 mark · MCQ]

The figure shows the graph of a function $g(z)$ used in neural networks. Which function is it?



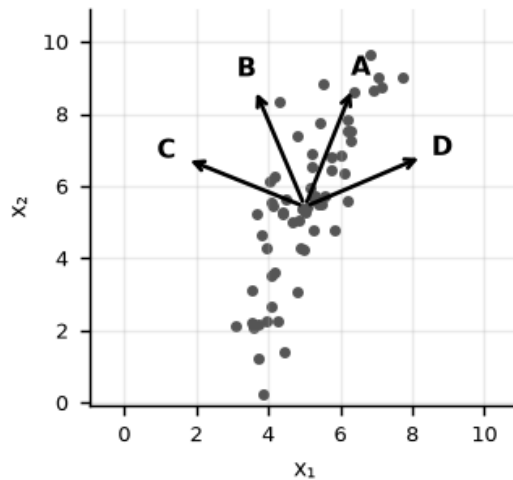
- (A) Leaky ReLU with $\alpha = 0.1$
(B) ReLU: $\max(0, z)$
(C) Softplus: $\ln(1 + e^z)$
(D) ELU with $\alpha = 1$

Q.13 [1 mark · NAT (numerical answer)]

The maximum number of leaf nodes in a binary decision tree of depth 5 (root at depth 0) is _____ (Answer in integer)

Q.14 [1 mark · MCQ]

The scatter plot shows a 2-D dataset together with four candidate directions A-D drawn from the data mean. Which direction best represents the first principal component (PC1)?



- (A)** Direction A **(B)** Direction B
(C) Direction C **(D)** Direction D

Q.15 [1 mark · MSQ (one or more correct)]

Let X be an $n \times p$ design matrix of full column rank p ($p < n$), and let $H = X(X^T X)^{-1} X^T$ be the hat matrix of least-squares regression of y on X . Which of the following statements is/are CORRECT?

- (A) $\text{rank}(H) = p$.
 (B) If X contains a column of ones, each row of H sums to 1.
 (C) Under $\text{Var}(y) = \sigma^2 I$, $\text{Var}(e) = \sigma^2(I - H)$.
 (D) The average leverage $(1/n)\sum h_{ii}$ equals p/n .

Q.16 - Q.35 : Two-mark questions

Q.16 [2 marks · MCQ]

Ridge regression is tuned by 10-fold cross-validation. For each λ the table gives the mean CV MSE and its standard error (SE) across folds. (Larger λ means a simpler, more regularised model.) Using the **one-standard-error rule** (choose the most regularised model whose CV error is within one standard error of the minimum CV error), the selected value is

λ	0.01	0.1	0.3	1	3	10
CV MSE	5.41	5.11	4.83	5.09	5.74	5.96
SE	0.2	0.21	0.47	0.16	0.28	0.29

- (A) $\lambda = 0.3$ (B) $\lambda = 0.01$
(C) $\lambda = 3$ (D) $\lambda = 1$

Q.17 [2 marks · NAT (numerical answer)]

A logistic regression classifier has $P(y = 1 \mid \mathbf{x}) = \sigma(-1 + (-2)x_1 + (-2)x_2)$. An input is labelled $y = 1$ if and only if $P(y = 1 \mid \mathbf{x}) \geq 0.4$. For this rule, the perpendicular distance of the decision boundary from the origin is _____ (rounded off to two decimal places)

Q.18 [2 marks · NAT (numerical answer)]

Agglomerative clustering with **complete linkage** is applied to the distance matrix below. The cophenetic distance between P1 and P4 (the dendrogram height at which they are first joined) is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5
P1	0	3	6	8	9
P2		0	5	5	6
P3			0	6	7
P4				0	1
P5					0

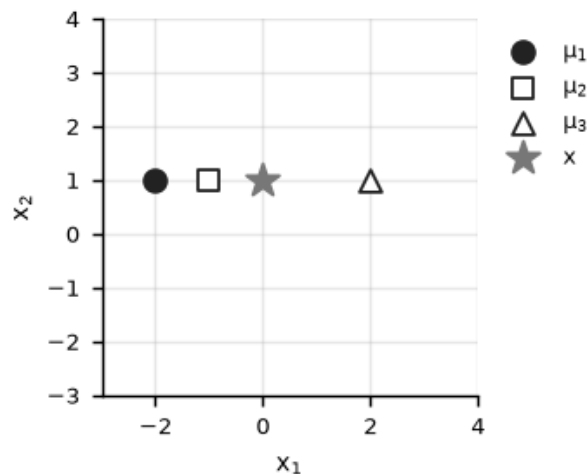
Q.19 [2 marks · NAT (numerical answer)]

The 2×2 matrix $X^T X$ of a (centred) design matrix X has eigenvalues 16, 9. For ridge regression with penalty $\lambda \|\beta\|_2^2$, the effective degrees of freedom $df(\lambda) = \text{trace}[X(X^T X + \lambda I)^{-1} X^T]$ for $\lambda = 2$ is _____ (rounded off to two decimal places)

Q.20 [2 marks · NAT (numerical answer)]

An LDA classifier for 3 classes has shared covariance matrix Σ (below), class means $\mu_1 = (-2, 1)^T$; $\mu_2 = (-1, 1)^T$; $\mu_3 = (2, 1)^T$, and priors $\pi_1 = 0.5$, $\pi_2 = 0.3$, $\pi_3 = 0.2$. The linear discriminant is $\delta_k(\mathbf{x}) = \mathbf{x}^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln \pi_k$ (natural log). For $\mathbf{x} = (0, 1)^T$, the value of the discriminant function $\delta_3(\mathbf{x})$ is _____ (rounded off to two decimal places)

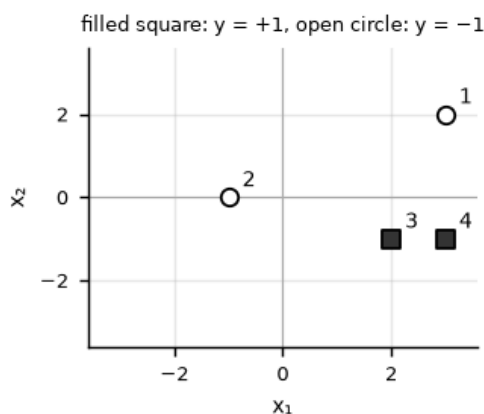
$$\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$



Q.21 [2 marks · NAT (numerical answer)]

A perceptron with parameters $w \in \mathbb{R}^2$, b is trained on the data below, starting from $w = (0, 0)$, $b = 0$, with learning rate $\eta = 1$. Examples are presented once in the order 1, 2, 3, 4 (one epoch). An update $w \leftarrow w + \eta y x$, $b \leftarrow b + \eta y$ is made whenever $y(w \cdot x + b) \leq 0$. After the epoch, the weight w_2 is _____ (Answer in integer)

i	x_1	x_2	y
1	3	2	-1
2	-1	0	-1
3	2	-1	1
4	3	-1	1

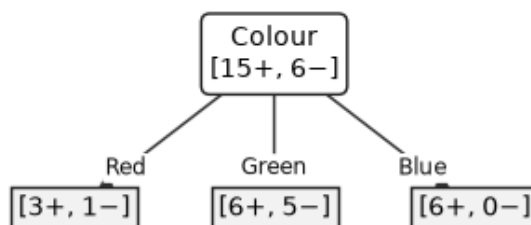
**Q.22 [2 marks · NAT (numerical answer)]**

A one-feature logistic regression model $P(y = 1 | x) = \sigma(wx + b)$ has $w = -1.5$ and $b = -0.5$. It is evaluated on the 4 examples below. Using natural logarithms, the **average** binary cross-entropy loss $-[y \ln \hat{p} + (1 - y) \ln(1 - \hat{p})]$ over these examples is _____ (rounded off to two decimal places)

i	1	2	3	4
x_i	2	2	1	-2
y_i	1	1	0	0

Q.23 [2 marks · NAT (numerical answer)]

A CART node with 21 training records (class counts [positive+, negative-]) is split on Colour as shown. Compute the weighted Gini index of the split on Colour. (rounded off to two decimal places)



Q.24 [2 marks · NAT (numerical answer)]

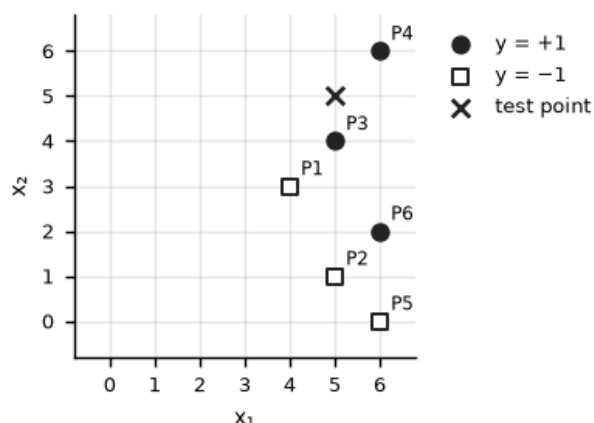
An AdaBoost ensemble has three weak classifiers with the weighted errors below, and their predictions on a test point x are listed. The ensemble score $F(x) = \sum \alpha_t h_t(x)$, with $\alpha_t = \frac{1}{2} \ln((1 - \varepsilon_t)/\varepsilon_t)$, is _____ (rounded off to two decimal places)

Round t	ε_t	$h_t(x)$
1	0.15	+1
2	0.15	+1
3	0.3	-1

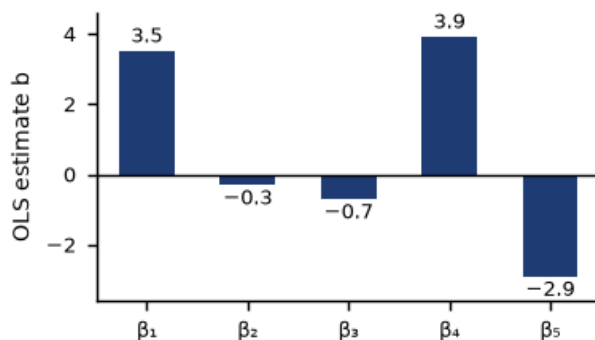
Q.25 [2 marks · NAT (numerical answer)]

A hard-margin linear SVM is trained on the points below. Solving the dual problem gives the Lagrange multipliers α_i listed in the table. Using the KKT conditions, compute the decision value $f(x) = w^T x + b$ at $x = (5, 5)$. (rounded off to two decimal places)

Point	x_1	x_2	y	α_i
P1	4	3	-1	0.75
P2	5	1	-1	0
P3	5	4	+1	0
P4	6	6	+1	0
P5	6	0	-1	0.875
P6	6	2	+1	1.625

**Q.26 [2 marks · NAT (numerical answer)]**

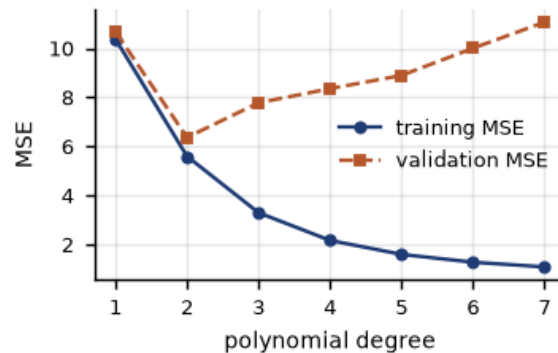
In a regression problem the design matrix X has orthonormal columns ($X^T X = I$). The ordinary least-squares estimates $b = X^T y$ are shown below. Ridge minimises $\|y - X\beta\|^2 + \lambda \|\beta\|_2^2$ and the lasso minimises $\frac{1}{2} \|y - X\beta\|^2 + \lambda \|\beta\|_1$, each with $\lambda = 2$. Then (ignoring the given λ) the smallest value of λ for which the lasso sets at least 2 coefficients exactly to zero is _____ (rounded off to two decimal places)



Q.27 [2 marks · MSQ (one or more correct)]

Polynomial regression models of degree 1 to 7 are fitted by least squares on a training set and evaluated on a separate validation set. The resulting mean squared errors are given below. Which of the following statements is/are CORRECT?

Degree	1	2	3	4	5	6	7
Training MSE	10.38	5.6	3.3	2.17	1.6	1.28	1.09
Validation MSE	10.68	6.37	7.78	8.36	8.9	10.01	11.07

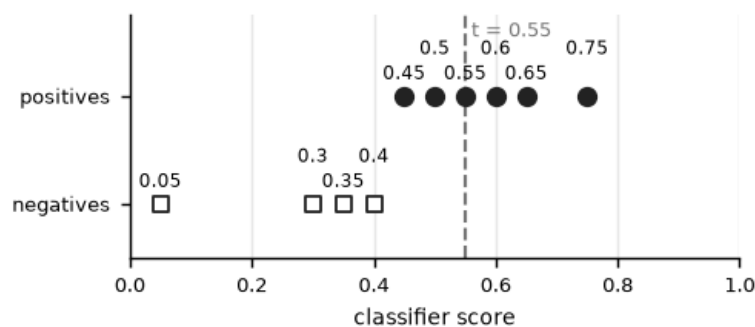


- (A) Since training MSE keeps decreasing, the degree-7 model generalises best.
- (B) Based on the validation MSE, degree 6 should be selected.
- (C) Based on the validation MSE, degree 2 should be selected.
- (D) The degree-2 model has higher variance than the degree-7 model.

Q.28 [2 marks · NAT (numerical answer)]

A probabilistic classifier produces the scores below on 10 labelled examples (1 = positive). An example is predicted positive if its score is $\geq t$, with $t = 0.55$. The precision at this threshold is _____ (rounded off to two decimal places)

example	1	2	3	4	5	6	7	8	9	10
score	0.75	0.35	0.6	0.65	0.05	0.3	0.4	0.55	0.5	0.45
label	1	0	1	1	0	0	0	1	1	1

**Q.29 [2 marks · NAT (numerical answer)]**

A softmax-regression output layer computes $z = Wx + b$, $p = \text{softmax}(z)$ with W and b below, and is trained with cross-entropy loss $L = -\ln p_y$. For $x = (2, 2)^T$ with true class $y = 1$, $\partial L / \partial W_{32}$ is _____ (rounded off to three decimal places)

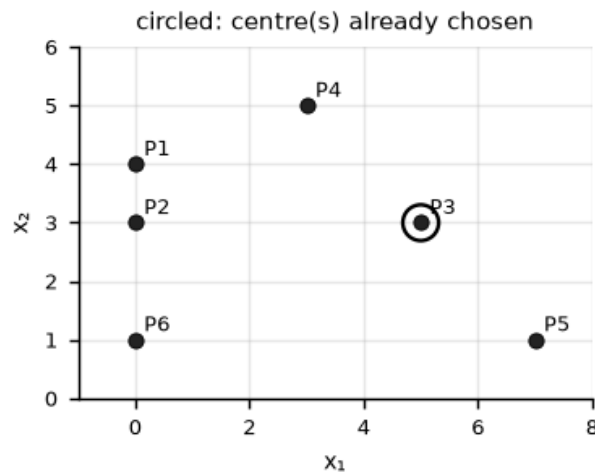
$$W = \begin{bmatrix} 0 & 0.5 \\ 0 & -1 \\ 1 & -1 \end{bmatrix}$$

$$b = \begin{bmatrix} -0.5 \\ -0.5 \\ 0.5 \end{bmatrix}$$

Q.30 [2 marks · NAT (numerical answer)]

k-means++ initialization (squared Euclidean seeding) is applied to the six points below. P3 has already been chosen as centre. The probability that P4 is selected as the second centre is _____ (rounded off to three decimal places)

Point	P1	P2	P3	P4	P5	P6
x_1	0	0	5	3	7	0
x_2	4	3	3	5	1	1

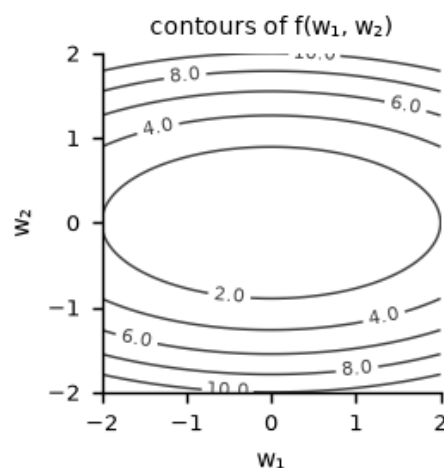
**Q.31 [2 marks · NAT (numerical answer)]**

The sample covariance matrix of a two-feature dataset is S given below. In principal component analysis, the variance of the data along the first principal component is _____ (rounded off to two decimal places)

$$S = \begin{bmatrix} 10 & 5 \\ 5 & 4 \end{bmatrix}$$

Q.32 [2 marks · MSQ (one or more correct)]

Gradient descent $w \leftarrow w - \eta \nabla f(w)$ is applied to $f(w_1, w_2) = \frac{1}{2}(1w_1^2 + 5w_2^2)$ from a generic starting point (both coordinates non-zero). For which of the following learning rates does GD converge to the minimum?



- (A) $\eta = 0.52$
(C) $\eta = 0.3$

- (B) $\eta = 0.1$
(D) $\eta = 0.44$

Q.33 [2 marks · MSQ (one or more correct)]

Let σ be the logistic sigmoid, and $'$ denote the derivative. At $z = -0.5$ (and $-z = 0.5$), which of the following is/are CORRECT? ($\text{softplus}(z) = \ln(1 + e^z)$.)

- (A) $\sigma(-0.5) + \sigma(0.5) = 1$ (B) $\text{ReLU}'(-0.5) = 0$
 (C) $\sigma(-0.5) > 0.5$ (D) $\text{softplus}(-0.5) > \text{ReLU}(-0.5)$

Q.34 [2 marks · NAT (numerical answer)]

Three customers are described by age and income. Age ranges over [20, 60] years and income over [10, 210] thousand rupees in the full dataset. On the **raw** (unscaled) features, the Euclidean distance from A to its nearest neighbour among B and C is _____ (rounded off to two decimal places)

Customer	Age (years)	Income (₹ thousand)
A	23	30
B	58	50
C	31	190

Q.35 [2 marks · MSQ (one or more correct)]

A node with class counts [40+, 20−] can be split by either of two binary splits S1 or S2 (table). For each impurity measure, the “gain” is the parent impurity minus the weighted child impurity. Which of the following statements is/are CORRECT?

Split	Left child [+, −]	Right child [+, −]
S1	[32+, 6−]	[8+, 14−]
S2	[6+, 12−]	[34+, 8−]

- (A) Gini gain prefers S1 over S2.
 (B) Misclassification-error reduction is the same for S1 and S2.
 (C) The information gain of S1 is 0.17 bits (to two decimals).
 (D) Information gain (entropy) prefers S1 over S2.

— END OF QUESTION PAPER —

Mock Test 13 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	484	19	NAT	2	1.6971 to 1.7171
2	MSQ	1	(A), (B), (C)	20	NAT	2	−2.6194 to −2.5994
3	MSQ	1	(C)	21	NAT	2	−3
4	MSQ	1	(A), (B), (C), (D)	22	NAT	2	2.4313 to 2.4513
5	NAT	1	0.117 to 0.137	23	NAT	2	0.3212 to 0.3412
6	MCQ	1	(A)	24	NAT	2	1.301 to 1.321
7	MSQ	1	(A), (B), (C), (D)	25	NAT	2	2.49 to 2.51
8	NAT	1	0.89 to 0.91	26	NAT	2	0.69 to 0.71
9	MCQ	1	(A)	27	MSQ	2	(C)
10	NAT	1	0.79 to 0.81	28	NAT	2	0.99 to 1.01
11	MSQ	1	(A), (B), (D)	29	NAT	2	0.9737 to 0.9777
12	MCQ	1	(B)	30	NAT	2	0.0813 to 0.0853
13	NAT	1	32	31	NAT	2	12.821 to 12.841
14	MCQ	1	(A)	32	MSQ	2	(B), (C)
15	MSQ	1	(A), (B), (C), (D)	33	MSQ	2	(A), (B), (D)
16	MCQ	2	(D)	34	NAT	2	40.3013 to 40.3213
17	NAT	2	0.2002 to 0.2202	35	MSQ	2	(A), (B), (C), (D)
18	NAT	2	8.99 to 9.01				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
SVM & Decision Trees					11
Unsupervised Learning					11
Probabilistic & Instance-based Classifiers					11
Neural Networks (MLP / Feed-forward)					11
Regression & Model Selection					11
Total					55

Mock Test 13 — Detailed Solutions

Q.1 — Answer: 484 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

$$\mathbf{x}^T \mathbf{x} = (3)(3) + (2)(2) + (3)(3) = 22.$$

$$K(\mathbf{x}, \mathbf{x}) = (22 + 0)^2 = 22^2 = \mathbf{484}.$$

The kernel equals the inner product $\phi(\mathbf{x})^T \phi(\mathbf{z})$ of the explicit (all-monomials-up-to-degree-d) feature maps, without forming ϕ .

Q.2 — Answer: (A), (B), (C) [MSQ, 1M · Unsupervised Learning > Hierarchical clustering]

(A) TRUE. It merges on the largest within-pair distance.

(B) TRUE. One run gives all levels.

(C) TRUE. There are $n(n-1)/2$ pairs.

(D) FALSE. They generally differ.

Q.3 — Answer: (C) [MSQ, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

(A) FALSE. Too large k underfits; test error is typically U-shaped in k .

(B) FALSE. Weights decrease with distance.

(C) TRUE. Distances to all n points are computed for each query.

(D) FALSE. It is non-parametric.

Q.4 — Answer: (A), (B), (C), (D) [MSQ, 1M · SVM & Decision Trees > SVM: kernels]

(A) TRUE. $\phi(\mathbf{x}) = \mathbf{A}^{1/2} \mathbf{x}$.

(B) TRUE. Sums, products and positive scalings.

(C) TRUE. Limit of polynomials with non-negative coefficients.

(D) TRUE. Sum of the linear kernel and a constant kernel.

Q.5 — Answer: 0.117 to 0.137 [NAT, 1M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

$$TP = 41, FN = 18, FP = 8, TN = 55 \text{ (} N = 122 \text{)}.$$

$$\text{False positive rate} = FP / (FP + TN).$$

$$\text{Value} = \mathbf{0.13}.$$

Watch the orientation: swapping FP and FN is the most common error (it swaps precision and recall).

Q.6 — Answer: (A) [MCQ, 1M · Unsupervised Learning > k-means]

Points are assigned to the nearest centroid, so each k -means cluster is a convex Voronoi cell and two clusters are separated by the perpendicular bisector of their centroids. Non-convex shapes such as rings or moons cannot be separated by such a boundary.

k -means converges on raw data, works in any dimension, and is unsupervised (no labels).

Q.7 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

Net input $a = \mathbf{w} \cdot \mathbf{x} + b = 2x_1 + (1)x_2 + (1)$; output $+1$ iff $a \geq 0$.

(P) $a = 2 \cdot 3 + (1) \cdot 0 + (1) = 7 \rightarrow +1$ (selected).

(Q) $a = 2 \cdot 1 + (1) \cdot 3 + (1) = 6 \rightarrow +1$ (selected).

(R) $a = 2 \cdot 2 + (1) \cdot -1 + (1) = 4 \rightarrow +1$ (selected).

(S) $a = 2 \cdot 3 + (1) \cdot -1 + (1) = 6 \rightarrow +1$ (selected).

Answer: **(A), (B), (C), (D)**.

Q.8 — Answer: 0.89 to 0.91 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

During training each input is present with probability p , so at test time weights are scaled by p :

$$0.6 \times 1.5 = \mathbf{0.9}.$$

Q.9 — Answer: (A) [MCQ, 1M · Regression & Model Selection > Gradient descent]

$$J(\mathbf{w}) = (\mathbf{w} - 4)^2 \text{ has } J' = 2. \text{ GD: } \mathbf{w}_{t+1} - 4 = (1 - 2\eta)(\mathbf{w}_t - 4), \text{ so } J(\mathbf{w}_t) = (1 - 2\eta)^{2t} J(\mathbf{w}_0).$$

Ratios $(1 - 2\eta)^2$ per step: $\eta = 0.02 \rightarrow 0.9216$ (curve B); $\eta = 0.06 \rightarrow 0.7744$ (curve C); $\eta = 0.2 \rightarrow 0.36$ (curve D); $\eta = 1.25 \rightarrow 2.25$ (curve A).

Ratio $> 1 \Rightarrow$ divergence ($\eta > 1 = 2/J''$); among convergent rates (all < 0.5 here), a smaller η gives a ratio closer to 1, i.e. slower decrease.

Hence $\eta = 0.02$ corresponds to **Curve B**.

Q.10 — Answer: 0.79 to 0.81 [NAT, 1M · Probabilistic & Instance-based Classifiers · Bayes optimal classifier]

Expected cost of predicting positive $= (1 - p) \cdot C_{FP}$; of predicting negative $= p \cdot C_{FN}$.

Predict positive iff $p \cdot 2 > (1 - p) \cdot 8 \Leftrightarrow p > C_{FP} / (C_{FP} + C_{FN}) = 8/10 = \mathbf{0.8}$.

Q.11 — Answer: (A), (B), (D) [MSQ, 1M · Regression & Model Selection · Regression vs classification]

A supervised task is *regression* when the target y is real-valued (loss such as squared error), and *classification* when y takes values in a finite set of labels (loss such as cross-entropy or 0-1 loss).

(A) TRUE. The output (life in hours) is a continuous real-valued quantity $\rightarrow$ regression.

(B) TRUE. The output (demand in MW) is a continuous real-valued quantity $\rightarrow$ regression.

(C) FALSE. The output (churn / stay) is a discrete class label $\rightarrow$ classification.

(D) TRUE. The output (time in seconds) is a continuous real-valued quantity $\rightarrow$ regression.

Q.12 — Answer: (B) [MCQ, 1M · Neural Networks (MLP / Feed-forward) · Activation functions]

Key features of the plotted curve: Exactly 0 for $z \leq 0$, slope 1 for $z > 0$, kink at origin.

Hence $g = \mathbf{ReLU: \max(0, z)}$.

Leaky ReLU with $\alpha = 0.1$: Small negative linear part (slope 0.1) for $z < 0$, slope 1 for $z > 0$. – does not match.

Softplus: $\ln(1 + e^z)$: Smooth, always positive, value $\ln 2 \approx 0.69$ at $z = 0$, $\sim z$ for large z . – does not match.

ELU with $\alpha = 1$: Smoothly saturates to -1 for $z \rightarrow -\infty$, linear for $z > 0$, passes through origin. – does not match.

Q.13 — Answer: 32 [NAT, 1M · SVM & Decision Trees · Decision trees: structure & traversal]

Each level at most doubles the nodes: leaves $\leq 2^d = 2^5 = \mathbf{32}$.

Q.14 — Answer: (A) [MCQ, 1M · Unsupervised Learning · PCA & dimensionality reduction]

PC1 is the direction of maximum variance — along the long axis of the elongated cloud (direction A). PC2 must be orthogonal to PC1 and captures the remaining variance (direction C).

Answer: **Direction A**. Directions B and D are at 45° to the main axis and capture intermediate variance.

Q.15 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection · Multiple linear regression]

(A) TRUE. For an idempotent matrix rank = trace = p .

(B) TRUE. $H1 = 1$ because 1 is in the column space of X .

(C) TRUE. $\text{Var}((I-H)y) = (I-H)\sigma^2 I(I-H)^T = \sigma^2(I-H)$.

(D) TRUE. $\sum_{ii} h_{ii} = \text{trace}(H) = p$.

Q.16 — Answer: (D) [MCQ, 2M · Regression & Model Selection · Cross-validation]

Minimum CV MSE = 4.83 at $\lambda = 0.3$, with SE = 0.47.

One-SE threshold = $4.83 + 0.47 = 5.3$.

Models within the threshold: $\lambda = 0.1$ (5.11), $\lambda = 0.3$ (4.83), $\lambda = 1$ (5.09).

The one-SE rule picks the largest such λ (simplest model): $\lambda = 1$; the minimum-CV rule picks $\lambda = 0.3$.

Answer: $\lambda = \mathbf{1}$.

Q.17 — Answer: 0.2002 to 0.2202 [NAT, 2M · Probabilistic & Instance-based Classifiers · Logistic regression]

$P(y=1|x) \geq 0.4 \Leftrightarrow z \geq c$, where $c = \ln(0.4/0.6) = -0.4055$.

Decision boundary: $-1 + (-2)x_1 + (-2)x_2 = -0.4055$ (a straight line; changing the threshold only shifts it parallel to the 0.5 boundary).

$$\text{distance} = |w_0 - c|/\|(w_1, w_2)\| = |-1 - -0.4055|/\sqrt{8} = \mathbf{0.21}.$$

Q.18 — Answer: 8.99 to 9.01 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Use complete linkage: $d(A, B) = \max_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

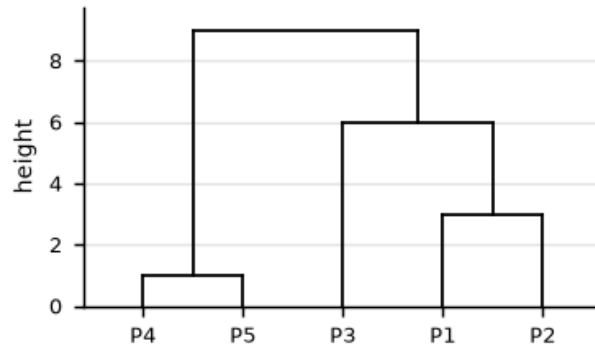
Merge 1: $\{P4\} + \{P5\}$ at height **1**. Updated: $d(\{P4, P5\}, \{P1\}) = 9$, $d(\{P4, P5\}, \{P2\}) = 6$, $d(\{P4, P5\}, \{P3\}) = 7$.

Merge 2: $\{P1\} + \{P2\}$ at height **3**. Updated: $d(\{P1, P2\}, \{P3\}) = 6$, $d(\{P1, P2\}, \{P4, P5\}) = 9$.

Merge 3: $\{P3\} + \{P1, P2\}$ at height **6**. Updated: $d(\{P1, P2, P3\}, \{P4, P5\}) = 9$.

Merge 4: $\{P4, P5\} + \{P1, P2, P3\}$ at height **9**.

P1 and P4 first lie in the same cluster at the merge of height 9, so their cophenetic distance is **9** (their original distance was 8).



Q.19 — Answer: 1.6971 to 1.7171 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

Ridge fitted values $\hat{y} = H_{\lambda} y$ with $H_{\lambda} = X(X^T X + \lambda I)^{-1} X^T$. Writing $X^T X = V D V^T$ with eigenvalues d_j^2 gives $\text{trace}(H_{\lambda}) = \sum d_j^2 / (d_j^2 + \lambda)$ (effective degrees of freedom).

$$16/(16 + 2) + 9/(9 + 2) = 1.7071.$$

Answer: **1.71**.

Q.20 — Answer: -2.6194 to -2.5994 [NAT, 2M · Probabilistic & Instance-based Classifiers >

Linear discriminant analysis]

$$\Sigma^{-1} = \begin{bmatrix} 0.6 & -0.2 \\ -0.2 & 0.4 \end{bmatrix}$$

$$k = 1: \Sigma^{-1} \mu_1 = (-1.4, 0.8); \mathbf{x}^T \Sigma^{-1} \mu = 0.8; \frac{1}{2} \mu^T \Sigma^{-1} \mu = 1.8; \ln \pi = -0.6931 \Rightarrow \delta_1 = -1.6931.$$

$$k = 2: \Sigma^{-1} \mu_2 = (-0.8, 0.6); \mathbf{x}^T \Sigma^{-1} \mu = 0.6; \frac{1}{2} \mu^T \Sigma^{-1} \mu = 0.7; \ln \pi = -1.204 \Rightarrow \delta_2 = -1.304.$$

$$k = 3: \Sigma^{-1} \mu_3 = (1, 0); \mathbf{x}^T \Sigma^{-1} \mu = 0; \frac{1}{2} \mu^T \Sigma^{-1} \mu = 1; \ln \pi = -1.6094 \Rightarrow \delta_3 = -2.6094.$$

$$\text{Required } \delta_3(\mathbf{x}) = \mathbf{-2.61}.$$

Q.21 — Answer: -3 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Rule: if $y_i(w \cdot x_i + b) \leq 0$, then $w \leftarrow w + \eta y_i x_i$, $b \leftarrow b + \eta y_i$ ($\eta = 1$). Start $w = (0, 0)$, $b = 0$.

Example 1: $a = 0$, $y \cdot a = 0 \rightarrow$ **mistake**, update $\rightarrow w = (-3, -2)$, $b = -1$.

Example 2: $a = 2$, $y \cdot a = -2 \rightarrow$ **mistake**, update $\rightarrow w = (-2, -2)$, $b = -2$.

Example 3: $a = -4$, $y \cdot a = -4 \rightarrow$ **mistake**, update $\rightarrow w = (0, -3)$, $b = -1$.

Example 4: $a = 2$, $y \cdot a = 2 \rightarrow$ correct, no change.

Required value = **-3**.

Note: a point with net input exactly 0 counts as a mistake ($y \cdot a \leq 0$) - this is why the first example always triggers an update when starting from zero.

Q.22 — Answer: 2.4313 to 2.4513 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

$$z_i = -1.5x_i + -0.5; \hat{p}_i = \sigma(z_i): 0.0293, 0.0293, 0.1192, 0.9241.$$

Per-example loss $L_i = -\ln \hat{p}_i$ if $y_i = 1$ and $-\ln(1 - \hat{p}_i)$ if $y_i = 0$:

i	y	$\hat{p}$	L_i
1	1	0.0293	3.5298

2	1	0.0293	3.5298
3	0	0.1192	0.1269
4	0	0.9241	2.5789

$\Sigma L_i = 9.7653$; average = $9.7653/4 = \mathbf{2.44}$.

Common mistakes: using $\ln \hat{p}$ for $y = 0$ examples, or $\log_{10}$ instead of $\ln$.

Q.23 — Answer: 0.3212 to 0.3412 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [15+, 6-]: Gini = $1 - [(15/21)^2 + (6/21)^2] = 0.4082$.

Colour = Red: [3+, 1-], Gini = $1 - [(3/4)^2 + (1/4)^2] = 0.375$.

Colour = Green: [6+, 5-], Gini = $1 - [(6/11)^2 + (5/11)^2] = 0.4959$.

Colour = Blue: [6+, 0-], Gini = $1 - [(6/6)^2 + (0/6)^2] = 0$.

Weighted Gini = $(4/21)(0.375) + (11/21)(0.4959) + (6/21)(0) = 0.3312$.

Gini gain = $0.4082 - 0.3312 = 0.077$.

Answer: **0.33**.

Q.24 — Answer: 1.301 to 1.321 [NAT, 2M · SVM & Decision Trees > Ensembles: boosting]

α_t : $\alpha_1 = 0.8673$, $\alpha_2 = 0.8673$, $\alpha_3 = 0.4236$.

$F(x) = +0.8673 + 0.8673 - 0.4236 = \mathbf{1.31}$; prediction $\text{sign}(F) = +1$.

Note that a majority of weak votes need not win: a low-error learner carries a larger α .

Q.25 — Answer: 2.49 to 2.51 [NAT, 2M · SVM & Decision Trees > SVM: dual & support vectors]

$w = \sum \alpha_i y_i x_i$ over points with $\alpha_i > 0$:

$w = (0.75)(-1)(4, 3) + (0.875)(-1)(6, 0) + (1.625)(+1)(6, 2) = (3/2, 1)$.

b from any support vector, e.g. P1: $y = w^T x + b \Rightarrow b = -1 - (9) = -10$.

Check: $\sum \alpha_i y_i = 0$ ($= 0$, as required).

$f((5, 5)) = (3/2)(5) + (1)(5) - 10 = \mathbf{2.5}$.

Q.26 — Answer: 0.69 to 0.71 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

With $X^T X = I$: $\beta_{OLS} = X^T y = b$.

Ridge: $\nabla[\|y - X\beta\|^2 + \lambda\|\beta\|^2] = 0 \Rightarrow (1 + \lambda)\beta = b \Rightarrow \beta_{\text{ridge}} = b/(1 + \lambda)$.

Lasso: the problem separates per coordinate, $\min \frac{1}{2}(\beta_j - b_j)^2 + \lambda|\beta_j| \Rightarrow \text{soft-thresholding } \beta_j = \text{sign}(b_j) \cdot \max(|b_j| - \lambda, 0)$.

Coefficient j becomes 0 once $\lambda \geq |b_j|$. Sorted $|b|$: 0.3, 0.7, 2.9, 3.5, 3.9; the 2-th smallest is 0.7.

Answer: **0.7**.

Q.27 — Answer: (C) [MSQ, 2M · Regression & Model Selection > Bias-variance trade-off]

(A) FALSE. Training error always falls with degree for nested fits; it says nothing about generalisation.

(B) FALSE. The minimum validation MSE is at degree 2, not 6.

(C) TRUE. Validation MSE is minimum (6.37) at degree 2.

(D) FALSE. Lower-degree models are less flexible and have lower variance.

Q.28 — Answer: 0.99 to 1.01 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Predicted positive (score ≥ 0.55): examples 1, 3, 4, 8.

TP = 4, FP = 0, FN = 2, TN = 4.

Precision = $TP/(TP + FP) = \mathbf{1}$.

Raising t typically raises precision and lowers recall/FPR; lowering t does the opposite (threshold moving).

Q.29 — Answer: 0.9737 to 0.9777 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$z = Wx + b = (0.5, -2.5, 0.5)$; $p = \text{softmax}(z) = (0.4879, 0.0243, 0.4879)$.

$\delta = \partial L / \partial z = p - y = (-0.5121, 0.0243, 0.4879)$ (true class 1).

Since $z_k = \sum_j W_{kj} x_j + b_k$: $\partial L / \partial W_{kj} = (p_k - y_k) x_j$ and $\partial L / \partial b_k = p_k - y_k$ (gradient matrix = $(p - y)x^T$).
 $\partial L / \partial W_{32} = 0.4879 \times 2 = \mathbf{0.9757}$.

Q.30 — Answer: 0.0813 to 0.0853 [NAT, 2M · Unsupervised Learning > k-means]

k-means++ picks the next centre x with probability $D(x)^2 / \sum_{x'} D(x')^2$, where $D(x)$ is the distance from x to the **nearest** centre already chosen (chosen centres have $D = 0$).

Point	P1	P2	P3	P4	P5	P6
$D(x)^2$	26	25	0	8	8	29

$\sum D^2 = 96$, so $P(P4) = 8/96 = \mathbf{0.083}$.

Common mistakes: using D instead of D^2 (that is not k-means++), or using the distance to the *farthest*/first centre instead of the nearest chosen centre.

Q.31 — Answer: 12.821 to 12.841 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

Eigenvalues of $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$: $\lambda = (a + c)/2 \pm \sqrt{((a - c)/2)^2 + b^2}$.

$\lambda = 7 \pm \sqrt{(9 + 25)} = 7 \pm 5.831 \rightarrow \lambda_1 = 12.831, \lambda_2 = 1.169$ (check: $\lambda_1 + \lambda_2 = \text{trace} = 14$, $\lambda_1 \lambda_2 = \det = 15$).

Eigenvector for λ_1 : $(S - \lambda_1 I)v = 0 \rightarrow (-2.831)v_1 + (5)v_2 = 0 \rightarrow v \propto (1, 0.5662)$.

$\text{PVE}(\text{PC1}) = \lambda_1 / (\lambda_1 + \lambda_2) = 91.65\%$.

Answer: **12.83**.

Q.32 — Answer: (B), (C) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$\nabla f = (1w_1, 5w_2) = w_i \leftarrow (1 - \eta a_i) w_i$. GD converges (from every start) iff $|1 - \eta a_i| < 1$ for both i , i.e. $0 < \eta < 2/\max(a_i) = 2/5 = 0.4$. The steep direction ($a = 5$) limits the learning rate.

(A) **FALSE**. $\eta = 0.52$: $|1 - \eta \cdot 1| = 0.48$, $|1 - \eta \cdot 5| = 1.6 \rightarrow$ diverges.

(B) **TRUE**. $\eta = 0.1$: $|1 - \eta \cdot 1| = 0.9$, $|1 - \eta \cdot 5| = 0.5 \rightarrow$ converges.

(C) **TRUE**. $\eta = 0.3$: $|1 - \eta \cdot 1| = 0.7$, $|1 - \eta \cdot 5| = 0.5 \rightarrow$ converges.

(D) **FALSE**. $\eta = 0.44$: $|1 - \eta \cdot 1| = 0.56$, $|1 - \eta \cdot 5| = 1.2 \rightarrow$ diverges.

Q.33 — Answer: (A), (B), (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Activation functions]

$\sigma(-0.5) = 0.3775$, $\tanh(-0.5) = -0.4621$.

(A) **TRUE**. $\sigma(-z) = 1 - \sigma(z)$.

(B) **TRUE**. $\text{ReLU}' = 1$ for $z > 0$ and 0 for $z < 0$.

(C) **FALSE**. $\sigma(-0.5) = 0.3775$.

(D) **TRUE**. $\text{softplus} = 0.4741 > 0$; $\ln(1 + e^z) > \max(0, z)$ always.

Q.34 — Answer: 40.3013 to 40.3213 [NAT, 2M · Unsupervised Learning > Distance measures]

Min-max scaling: $x' = (x - \min)/(\max - \min)$; age range $[20, 60]$, income range $[10, 210]$.

Customer	age'	income'
A	0.075	0.1
B	0.95	0.2
C	0.275	0.9

Raw distances: $d(A, B) = 40.31$, $d(A, C) = 160.2 \rightarrow$ nearest is B (income dominates because of its larger range).

Scaled distances: $d(A, B) = 0.8807$, $d(A, C) = 0.8246 \rightarrow$ nearest is C.

Answer: **40.31**. Scaling changes which neighbour is nearest — distance-based methods (k-means, k-NN, hierarchical clustering) should be run on standardized features.

Q.35 — Answer: (A), (B), (C), (D) [MSQ, 2M · SVM & Decision Trees > Decision trees: impurity measures]

Parent $[40+, 20-]$: $H = 0.9183$, $\text{Gini} = 0.4444$, $\text{error} = 0.3333$.

S1: children H = 0.6292, 0.9457; Gini = 0.2659, 0.4628; error = 0.1579, 0.3636. Gains: IG = 0.173, Gini = 0.1063, error = 0.1.

S2: children H = 0.9183, 0.7025; Gini = 0.4444, 0.3084; error = 0.3333, 0.1905. Gains: IG = 0.1511, Gini = 0.0952, error = 0.1.

(A) TRUE. S1: 0.1063, S2: 0.0952.

(B) TRUE. S1: 0.1, S2: 0.1.

(C) TRUE. IG(S1) = 0.173.

(D) TRUE. S1: 0.173, S2: 0.1511.

Mock Test 14

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	7	7	21

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Unsupervised Learning	7	11
Regression & Model Selection	7	11
Neural Networks (MLP / Feed-forward)	7	11
SVM & Decision Trees	7	11
Probabilistic & Instance-based Classifiers	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

k-means with $k = 2$ searches over partitions of the data into k non-empty, unlabeled clusters. The number of distinct such partitions of $n = 4$ points is _____ (Answer in integer)

Q.2 [1 mark · MSQ (one or more correct)]

Let X be an $n \times p$ design matrix of full column rank p ($p < n$), and let $H = X(X^T X)^{-1} X^T$ be the hat matrix of least-squares regression of y on X . Which of the following statements is/are CORRECT?

- (A) $I - H$ is also symmetric and idempotent.
- (B) $\text{rank}(H) = p$.
- (C) $\text{trace}(H) = p$.
- (D) H is symmetric ($H^T = H$).

Q.3 [1 mark · MSQ (one or more correct)]

Let $p = \text{softmax}(z)$ for logits $z \in \mathbb{R}^K$ and $L = -\ln p_y$ the cross-entropy loss for true class y . Which of the following is/are CORRECT?

- (A) For softmax + cross-entropy, the gradient with respect to the true-class logit is negative unless $p_y = 1$.
- (B) Multiplying all logits by 2 leaves the softmax probabilities unchanged.
- (C) Cross-entropy loss $-\ln p_y$ is 0 only when $p_y = 1$.
- (D) Cross-entropy loss is maximised when the model predicts the correct class with probability 1.

Q.4 [1 mark · NAT (numerical answer)]

The Gaussian kernel is $K(x, z) = \exp(-\|x - z\|^2 / (2\sigma^2))$ with $\sigma = 3$. For $x = (0, 2)$ and $z = (-1, 2)$, the value of $K(x, z)$ is _____ (rounded off to three decimal places)

Q.5 [1 mark · NAT (numerical answer)]

A sigmoid-output classifier gives the probabilities $p = P(y = 1 | x)$ below. The mean binary cross-entropy $-[y \ln p + (1 - y) \ln(1 - p)]$ over the 4 examples is _____ (rounded off to three decimal places)

example	$p = P(y = 1)$	y
1	0.1	1
2	0.1	0
3	0.3	0
4	0.3	0

Q.6 [1 mark · MCQ]

For the Gaussian prior $w \sim N(0, \tau^2 I)$ and Gaussian noise, as $\tau^2 \rightarrow \infty$ the MAP estimate tends to:

- (A) a vector with all components equal (B) the least-squares (ML) estimate
(C) an undefined limit (D) the zero vector

Q.7 [1 mark · NAT (numerical answer)]

In a training set, the 7 examples of class Buys = Yes have the following counts for the feature Age. Using Laplace (add-one) smoothing, the naive Bayes estimate of $P(\text{Age} = \text{Middle} | \text{Buys} = \text{Yes})$ is _____ (rounded off to three decimal places)

Age	Young	Middle	Senior
count in Buys = Yes	4	0	3

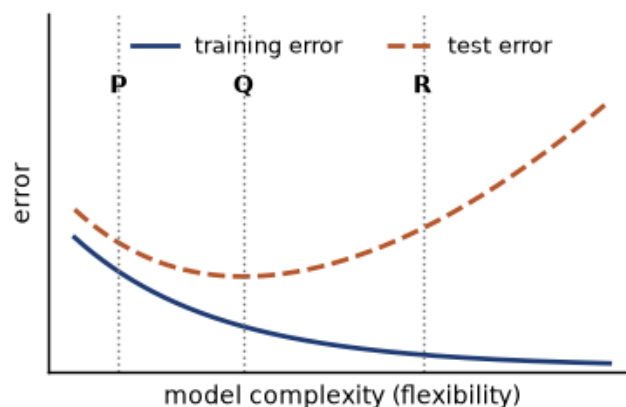
Q.8 [1 mark · MCQ]

A soft-margin linear SVM has learned $f(x) = x_1 + 2x_2$. For the training example $x = (-1, 1)$ with label $y = +1$, which statement is CORRECT?

- (A) Correctly classified but strictly inside the margin ($0 < \xi < 1$)
(B) Misclassified ($\xi > 1$)
(C) Exactly on the decision boundary ($\xi = 1$)
(D) Correctly classified, on or outside the margin ($\xi = 0$)

Q.9 [1 mark · MCQ]

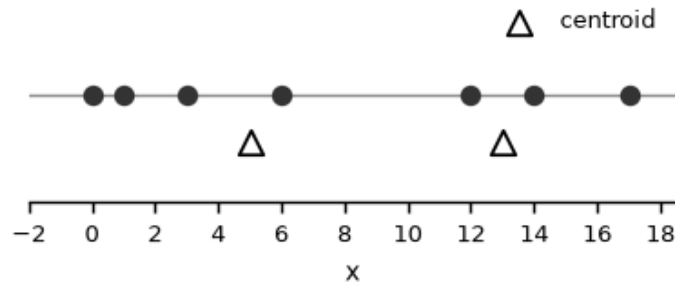
The figure shows training and test error as a function of model complexity, with three complexity levels marked P, Q and R. Which marked level corresponds to the best generalisation (lowest expected test error)?



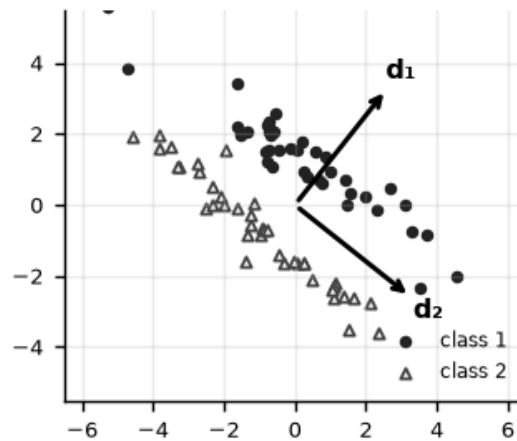
- (A) Q (B) R
(C) None of P, Q, R (D) P

Q.10 [1 mark · NAT (numerical answer)]

One-dimensional data $\{0, 1, 3, 6, 12, 14, 17\}$ and $k = 2$ centroids at 5, 13 are given. Each point is assigned to its nearest centroid (centroids are **not** updated). The resulting within-cluster sum of squared errors $\sum (x_i - c_{c(i)})^2$ is _____ (rounded off to two decimal places)

**Q.11 [1 mark · MCQ]**

Two labelled classes are shown below, together with two orthogonal directions d_1 and d_2 . A single direction is to be selected by (i) PCA (first principal component, ignoring labels) and (ii) Fisher's LDA. Which statement is correct?



- (A) Both PCA and LDA choose d_1
- (B) Both PCA and LDA choose d_2
- (C) PCA (1st PC) chooses d_1 ; LDA chooses d_2
- (D) PCA (1st PC) chooses d_2 ; LDA chooses d_1

Q.12 [1 mark · MSQ (one or more correct)]

Which of the following statements about bagging, random forests and boosting is/are CORRECT?

- (A) Bagging is mainly used to reduce the bias of high-bias learners such as decision stumps.
- (B) Bagged trees can be trained in parallel, whereas boosting is inherently sequential.
- (C) AdaBoost trains its weak learners independently and in parallel.
- (D) A weak learner with weighted error 0.3 receives a negative α in AdaBoost.

Q.13 [1 mark · MSQ (one or more correct)]

Which of the following statements about activation functions is/are CORRECT? (σ denotes the logistic sigmoid.)

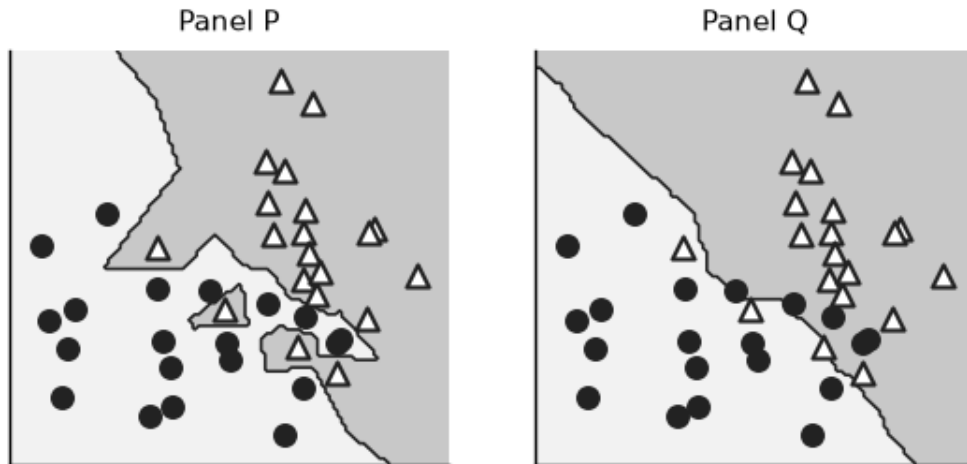
- (A) ReLU is not differentiable at $z = 0$; in practice a subgradient value (0 or 1) is used there.
- (B) $\sigma(-z) = 1 - \sigma(z)$.
- (C) $\tanh(z) = \sigma(z) - 0.5$ for all z .
- (D) ReLU is a bounded function.

Q.14 [1 mark · NAT (numerical answer)]

Fisher's multi-class LDA is applied to data with $K = 2$ classes and $p = 3$ features. The maximum number of discriminant directions (non-zero generalised eigenvalues of $S_W^{-1}S_B$) is _____.
(Answer in integer)

Q.15 [1 mark · MCQ]

The two panels show the decision regions of k-NN classifiers trained on the same 2-D data set, one with $k = 1$ and the other with $k = 21$. Which of the following statements is CORRECT?



- (A) Panel P uses $k = 1$; it has higher bias and lower variance
- (B) Panel P uses $k = 1$; it has lower bias and higher variance
- (C) Panel Q uses $k = 21$; it has zero training error
- (D) Panel Q uses $k = 1$; it has higher bias and lower variance

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · MCQ]**

A network computes $\hat{y} = W_2 \text{ReLU}(W_1 x + b_1) + b_2$ with the parameters below. For $x = (-1, 2, 2)^T$, the output $\hat{y}$ is

$$W_1 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

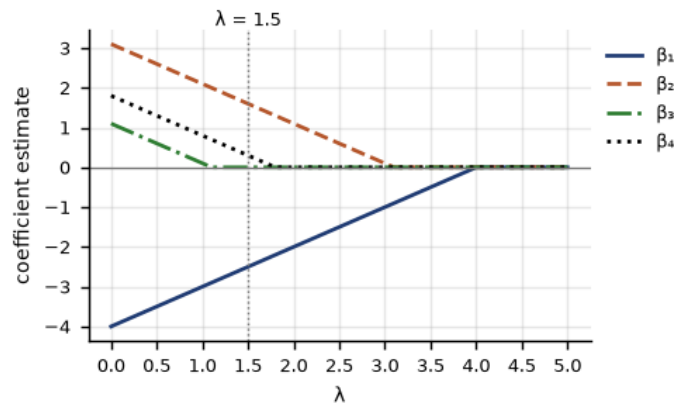
$$W_2 = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- (A) $\hat{y} = (5, 1)^T$
- (B) $\hat{y} = (8, -3)^T$
- (C) $\hat{y} = (-2, 9)^T$
- (D) $\hat{y} = (2, 5)^T$

Q.17 [2 marks · MSQ (one or more correct)]

For a design with orthonormal columns ($X^T X = I$), the figure shows the coefficient paths $\beta_j(\lambda)$ of a penalised least-squares estimator that minimises $\frac{1}{2}\|y - X\beta\|^2 + P_\lambda(\beta)$, where the penalty P_λ is either the ridge penalty $(\lambda/2)\|\beta\|_2^2$ or the lasso penalty $\lambda\|\beta\|_1$. Which of the following statements is/are CORRECT?



- (A) As λ increases, $|\beta_j(\lambda)|$ is non-increasing for every j .
- (B) At $\lambda = 0$ the coefficients equal the least-squares estimates.
- (C) At $\lambda = 1.5$, exactly 1 coefficient(s) are zero.
- (D) Each coefficient path is piecewise linear in λ .

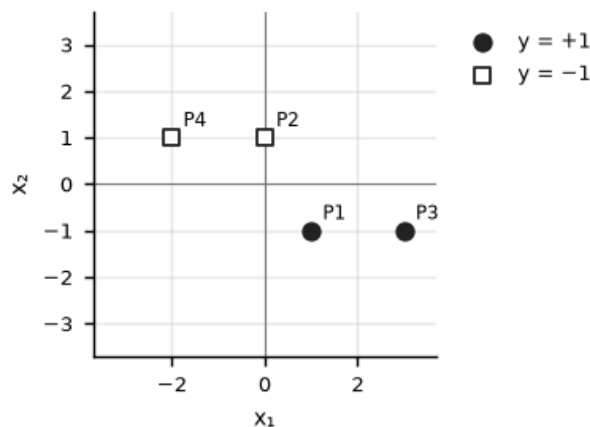
Q.18 [2 marks · NAT (numerical answer)]

A 64-64-10 fully connected network (with biases) is modified by inserting a new hidden layer of 8 units between the 64-unit layer and the output, giving 64-64-8-10. The change in the number of trainable parameters (new minus old) is _____. (Answer in integer)

Q.19 [2 marks · NAT (numerical answer)]

A perceptron with augmented weight vector $w = (w_0, w_1, w_2)$ (w_0 is the bias, $\tilde{x} = (1, x_1, x_2)$) starts from $w = (0, 0, 0)$ and uses learning rate $\eta = 0.5$. The points P1, P2, ... are presented once, in order; whenever $y \cdot w^T \tilde{x} \leq 0$ the update $w \leftarrow w + \eta y \tilde{x}$ is applied. Find the value of w_0 (bias) after this epoch. (rounded off to one decimal place)

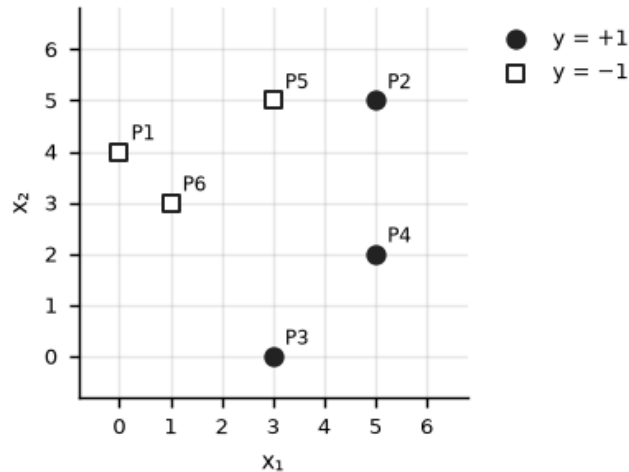
Point	x_1	x_2	y
P1	1	-1	+1
P2	0	1	-1
P3	3	-1	+1
P4	-2	1	-1



Q.20 [2 marks · MSQ (one or more correct)]

A hard-margin linear SVM is trained on the 6 labelled points shown. Which of the following points is/are support vectors of the resulting maximum-margin classifier?

Point	x_1	x_2	y
P1	0	4	-1
P2	5	5	+1
P3	3	0	+1
P4	5	2	+1
P5	3	5	-1
P6	1	3	-1



- (A)** P2 **(B)** P3
(C) P4 **(D)** P5

Q.21 [2 marks · NAT (numerical answer)]

The actual and predicted labels of a binary classifier (1 = positive) on 12 test examples are given below. The F_β -score with $\beta = 0.5$, $F_\beta = (1 + \beta^2)PR/(\beta^2P + R)$, is _____ (rounded off to two decimal places)

#	1	2	3	4	5	6	7	8	9	10	11	12
actual y	1	1	0	0	0	0	1	0	0	1	1	0
predicted $\hat{y}$	0	0	1	0	0	0	0	1	0	1	1	0

Q.22 [2 marks · NAT (numerical answer)]

Gradient descent with (heavy-ball) momentum minimises $f(w) = (0.5/2)w^2$ using $v_t = \beta v_{t-1} + \nabla f(w_{t-1})$ and $w_t = w_{t-1} - \eta v_t$, with $w_0 = 2$, $v_0 = 0$, $\eta = 0.25$, $\beta = 0.9$. The value of v_3 is _____ (rounded off to three decimal places)

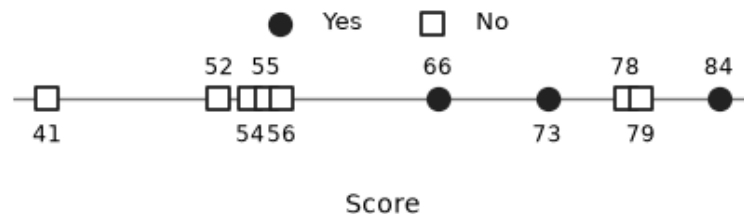
Q.23 [2 marks · NAT (numerical answer)]

A 4-class classifier outputs logits $z = (-2, -1, 1.5, 1)$ and probabilities $p = \text{softmax}(z)$. The true class is 1, and the loss is the cross-entropy $L = -\ln p_1$. Then the cross-entropy loss $L = -\ln p_y$ (natural log) is _____ (rounded off to three decimal places)

Q.24 [2 marks · NAT (numerical answer)]

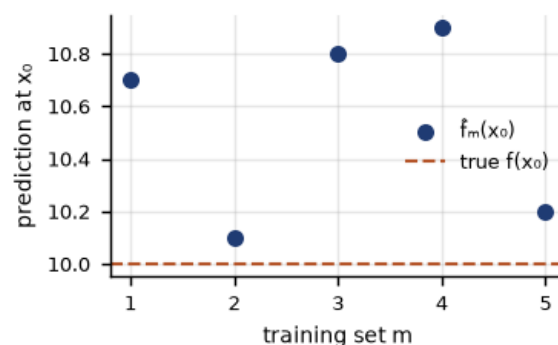
The continuous attribute Score and the class label of 10 training records are given below (unsorted). A binary split “Score $\leq t$ ” is chosen to maximise information gain, with t restricted to midpoints between consecutive sorted values. Find the threshold value t chosen for the split “Score $\leq t$ ”. (rounded off to one decimal place)

Score	55	52	41	54	73	56	79	84	78	66
Class	No	No	No	No	Yes	No	No	Yes	No	Yes

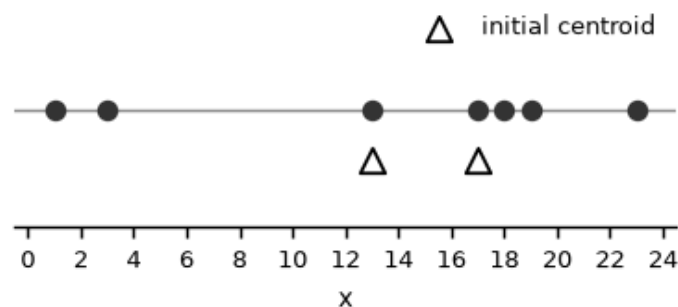
**Q.25 [2 marks · NAT (numerical answer)]**

Data are generated as $y = f(x) + \varepsilon$ with $E[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2 = 1.5$. At a test point x_0 the true value is $f(x_0) = 10$. A learning method is trained on 5 independent training sets, giving the predictions $\hat{f}_m(x_0)$ in the table (also plotted). Treating these as the full distribution of the method's prediction (variance with divisor 5), the expected squared prediction error $E[(y_0 - \hat{f}(x_0))^2]$ at x_0 is _____ (rounded off to three decimal places)

m	1	2	3	4	5
$\hat{f}_m(x_0)$	10.7	10.1	10.8	10.9	10.2

**Q.26 [2 marks · NAT (numerical answer)]**

The 1-D dataset $\{1, 3, 13, 17, 18, 19, 23\}$ is clustered with k-means ($k = 2$) using initial centroids 13 and 17 (shown as triangles). Lloyd's iterations (assign to nearest centroid, then recompute means) are run until convergence. Then the number of iterations in which at least one centroid **changes** its value is _____ (Answer in integer)



Q.27 [2 marks · NAT (numerical answer)]

Least-squares linear regression on $n = 5$ observations minimises $J(w) = (1/2n)\|Xw - y\|^2$ by batch gradient descent with constant learning rate η . The data give the matrix $X^T X$ below. Then the supremum of learning rates η for which batch gradient descent converges from every starting point is _____ (rounded off to three decimal places)

$$X^T X = \begin{bmatrix} 7 & 1 \\ 1 & 7 \end{bmatrix}$$

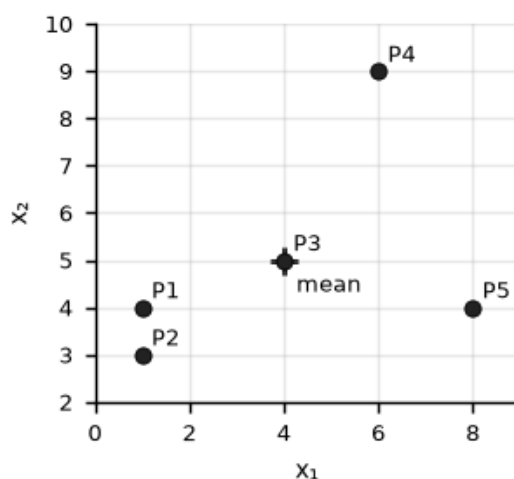
Q.28 [2 marks · MCQ]

For $n = 10$ paired observations: $\sum x = 80$, $\sum y = 47$, $\sum x^2 = 756$, $\sum y^2 = 315$, $\sum xy = 454$. The sample correlation coefficient r between x and y is

(A) -0.747**(B)** 0.672**(C)** 0.829**(D)** 0.747**Q.29 [2 marks · NAT (numerical answer)]**

PCA is applied to the five 2-D points below (data are mean-centered first; sample covariance uses divisor $n - 1$). The sample variance (divisor $n - 1$) of the PC1 scores of the five points is _____ (rounded off to two decimal places)

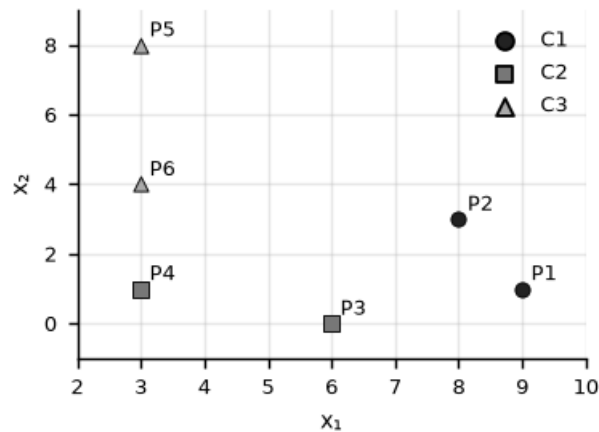
Point	P1	P2	P3	P4	P5
x_1	1	1	4	6	8
x_2	4	3	5	9	4



Q.30 [2 marks · NAT (numerical answer)]

A clustering of 6 points into 3 clusters is shown below. Using Euclidean distance, the silhouette coefficient $s(i) = (b(i) - a(i))/\max(a(i), b(i))$ of point P6 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5	P6
x_1	9	8	6	3	3	3
x_2	1	3	0	1	8	4
cluster	C1	C1	C2	C2	C3	C3

**Q.31 [2 marks · NAT (numerical answer)]**

The confusion matrix of a 3-class classifier is given below (rows = actual class, columns = predicted class). The micro-averaged F1-score is _____ (rounded off to two decimal places)

actual \ predicted	A	B	C
A	16	5	2
B	5	18	1
C	4	3	10

Q.32 [2 marks · NAT (numerical answer)]

A 3-class softmax (multinomial logistic) regression model computes scores $\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}$ and $P(y = k | \mathbf{x}) = e^{z_k} / \sum_j e^{z_j}$, where row k of $\mathbf{W}$ holds the weights of class k . For the parameters below and $\mathbf{x} = (-1, -2)^T$, the cross-entropy loss $-\ln P(y = 1 | \mathbf{x})$ incurred if the true class is 1 (natural log) is _____ (rounded off to two decimal places)

$$\mathbf{W} = \begin{bmatrix} -2 & 1 \\ -1 & 2 \\ 0 & 0 \end{bmatrix}$$

$$\mathbf{b} = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

Q.33 [2 marks · MSQ (one or more correct)]

Consider agglomerative hierarchical clustering on n points with a dissimilarity matrix (no ties). Which of the following statements is/are CORRECT?

- (A) For any two points, the complete-linkage cophenetic distance is $\leq$ their original distance.
- (B) Replacing every dissimilarity by its square never changes the average-linkage dendrogram topology.
- (C) Different linkages applied to the same distance matrix must yield the same 2-cluster partition.
- (D) The Lance-Williams formula updates inter-cluster distances after a merge without going back to the raw points.

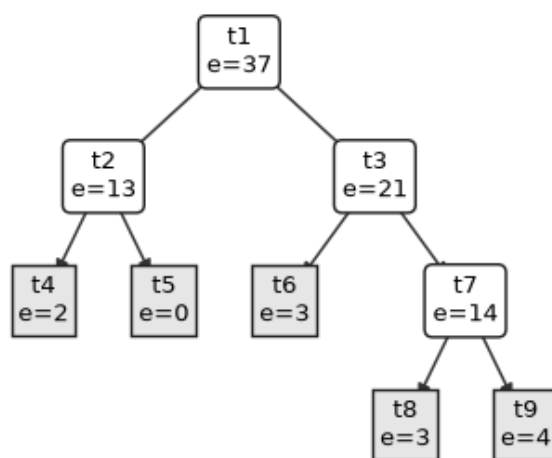
Q.34 [2 marks · NAT (numerical answer)]

Logistic regression $P(y = 1 | \mathbf{x}) = \sigma(w_1 x + b)$ is trained by **batch** gradient descent on the **average** cross-entropy loss over the 2 examples below, with learning rate $\eta = 1$. Starting from $w_1 = 0$, $b = 0.5$, the value of b after ONE gradient-descent update is _____ (rounded off to three decimal places)

i	x	y
1	1	0
2	-2	1

Q.35 [2 marks · NAT (numerical answer)]

A classification tree was grown on $N = 100$ training records. Each node shows $e(t)$, the number of training records that would be misclassified if t were a leaf; $R(t) = e(t)/N$. For $\alpha = 0.025$, the cost-complexity $R_\alpha(T) = R(T) + \alpha|T|$ of the full tree T is _____ (rounded off to three decimal places)



— END OF QUESTION PAPER —

Mock Test 14 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	7	19	NAT	2	-0.01 to 0.01
2	MSQ	1	(A), (B), (C), (D)	20	MSQ	2	(A), (B), (D)
3	MSQ	1	(A), (C)	21	NAT	2	0.4662 to 0.4862
4	NAT	1	0.944 to 0.948	22	NAT	2	2.2486 to 2.2526
5	NAT	1	0.7783 to 0.7823	23	NAT	2	4.0396 to 4.0436
6	MCQ	1	(B)	24	NAT	2	60.99 to 61.01
7	NAT	1	0.098 to 0.102	25	NAT	2	1.8923 to 1.9037
8	MCQ	1	(D)	26	NAT	2	2
9	MCQ	1	(A)	27	NAT	2	1.2463 to 1.2537
10	NAT	1	63.99 to 64.01	28	MCQ	2	(D)
11	MCQ	1	(D)	29	NAT	2	11.3061 to 11.3261
12	MSQ	1	(B)	30	NAT	2	-0.01 to 0.01
13	MSQ	1	(A), (B)	31	NAT	2	0.6775 to 0.6975
14	NAT	1	1	32	NAT	2	0.1228 to 0.1428
15	MCQ	1	(B)	33	MSQ	2	(D)
16	MCQ	2	(A)	34	NAT	2	0.3755 to 0.3795
17	MSQ	2	(A), (B), (C), (D)	35	NAT	2	0.2435 to 0.2465
18	NAT	2	-40				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Unsupervised Learning					11
Regression & Model Selection					11
Neural Networks (MLP / Feed-forward)					11
SVM & Decision Trees					11
Probabilistic & Instance-based Classifiers					11
Total					55

Mock Test 14 — Detailed Solutions

Q.1 — Answer: 7 [NAT, 1M · Unsupervised Learning > k-means]

This is the Stirling number of the second kind $S(n,k)$, with $S(n,k) = k \cdot S(n-1,k) + S(n-1,k-1)$.
 $S(4,2) = 7$. (For $k = 2$ it equals $2^{n-1} - 1 = 7$.)

Q.2 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Multiple linear regression]

(A) TRUE. $(I-H)^2 = I - 2H + H^2 = I - H$.

(B) TRUE. For an idempotent matrix $\text{rank} = \text{trace} = p$.

(C) TRUE. $\text{trace}(X(X^T X)^{-1} X^T) = \text{trace}((X^T X)^{-1} X^T X) = \text{trace}(I_p) = p$.

(D) TRUE. $H^T = X((X^T X)^{-1})^T X^T = H$ since $X^T X$ is symmetric.

Q.3 — Answer: (A), (C) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

(A) TRUE. $p_y - 1 < 0$.

(B) FALSE. It sharpens the distribution.

(C) TRUE. $-\ln 1 = 0$.

(D) FALSE. It is minimised ($= 0$) there.

Q.4 — Answer: 0.944 to 0.948 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

$\|x - z\|^2 = 1$; $2\sigma^2 = 18$.

$K = \exp(-1/18) = \mathbf{0.946}$.

Note: $K \in (0, 1]$, $K(x, x) = 1$, and K decreases as the points move apart or as γ grows.

Q.5 — Answer: 0.7783 to 0.7823 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Example 1: $y = 1 \rightarrow -\ln 0.1 = 2.3026$.

Example 2: $y = 0 \rightarrow -\ln(1 - 0.1) = 0.1054$.

Example 3: $y = 0 \rightarrow -\ln(1 - 0.3) = 0.3567$.

Example 4: $y = 0 \rightarrow -\ln(1 - 0.3) = 0.3567$.

Mean = **0.7803**.

Q.6 — Answer: (B) [MCQ, 1M · Regression & Model Selection > MLE & MAP view]

$\lambda = \sigma^2/\tau^2 \rightarrow 0$, so the penalty disappears.

Answer: **the least-squares (ML) estimate**.

Q.7 — Answer: 0.098 to 0.102 [NAT, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Add- α estimate: $(\text{count} + \alpha)/(\mathcal{N}_c + \alpha V)$ with $V = 3$ possible values.

$= (0 + 1)/(7 + 1 \times 3) = 1/10 = \mathbf{0.1}$.

(The unsmoothed estimate would be $0/7$, i.e. zero, which would wipe out the whole posterior product.)

Q.8 — Answer: (D) [MCQ, 1M · SVM & Decision Trees > SVM: soft margin & hinge loss]

$y \cdot f(x) = (+1) \cdot (1) = 1$.

Slack $\xi = \max(0, 1 - y f(x)) = 0$.

Rules: $y f \geq 1 \Rightarrow \xi = 0$ (on/outside margin); $0 < y f < 1 \Rightarrow$ inside margin, still correct; $y f = 0 \Rightarrow$ on the boundary ($\xi = 1$); $y f < 0 \Rightarrow$ misclassified ($\xi > 1$).

Hence: **Correctly classified, on or outside the margin ($\xi = 0$)**.

Q.9 — Answer: (A) [MCQ, 1M · Regression & Model Selection > Bias-variance trade-off]

At P: both training and test errors are high (left side) $\rightarrow$ underfitting.

At Q: the test error is at its minimum $\rightarrow$ best trade-off.

At R: training error is very low but test error has risen well above its minimum (right side) $\rightarrow$ overfitting.

Answer: **Q**.

Q.10 — Answer: 63.99 to 64.01 [NAT, 1M · Unsupervised Learning > k-means]

Assignment (nearest centroid): $c = 5$: {0, 1, 3, 6}; $c = 13$: {12, 14, 17}.

$$\text{SSE} = (0-5)^2 + (1-5)^2 + (3-5)^2 + (6-5)^2 + (12-13)^2 + (14-13)^2 + (17-13)^2 = \mathbf{64}.$$

Q.11 — Answer: (D) [MCQ, 1M · Unsupervised Learning > PCA & dimensionality reduction]

PCA is unsupervised: it picks the direction of maximum total variance, which is along the long axis of the clouds (d_2).

LDA is supervised: it maximizes between-class separation relative to within-class scatter, so it picks the direction across which the class means differ (d_1).

Answer: **PCA (1st PC) chooses d_2 ; LDA chooses d_1 .** Projecting on the PCA direction would mix the two classes completely.

Q.12 — Answer: (B) [MSQ, 1M · SVM & Decision Trees > Ensembles: bagging & random forests]

(A) **FALSE.** Bagging reduces variance; boosting reduces bias.

(B) **TRUE.** Boosting needs the previous round's weights.

(C) **FALSE.** It is sequential.

(D) **FALSE.** $\alpha = \frac{1}{2} \ln(0.7/0.3) > 0$.

Q.13 — Answer: (A), (B) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

(A) **TRUE.** Left and right derivatives are 0 and 1.

(B) **TRUE.** $1/(1+e^z) = e^{-z}/(1+e^{-z})$.

(C) **FALSE.** The correct identity is $\tanh(z) = 2\sigma(2z) - 1$.

(D) **FALSE.** It is unbounded above.

Q.14 — Answer: 1 [NAT, 1M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

S_B is a sum of K outer products of $(\mu_k - \mu)$ which are linearly dependent (weighted sum is 0), so $\text{rank}(S_B) \leq K - 1 = 1$; also $\leq p = 3$.

Maximum = $\min(K - 1, p) = \mathbf{1}$.

Q.15 — Answer: (B) [MCQ, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

The jagged boundary with small islands around individual points (Panel P) is produced by $k = 1$, which fits every training point (zero training error): low bias, high variance.

The smooth boundary (Panel Q) comes from $k = 21$: averaging many neighbours raises bias and lowers variance; it misclassifies some training points, so its training error is not zero.

Answer: **Panel P uses $k = 1$; it has lower bias and higher variance.**

Q.16 — Answer: (A) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

$a = W_1 x + b_1 = (-2, -1, 4)^T$; $h = \text{ReLU}(a) = (0, 0, 4)^T$.

$\hat{y} = W_2 h + b_2 = \mathbf{(5, 1)^s}$.

Distractors: skipping the ReLU gives $(8, -3)^T$; using W_1^T instead of W_1 gives $(-2, 9)^T$; dropping the biases gives $(4, 0)^T$; using $|a|$ instead of ReLU gives $(2, 5)^T$.

Q.17 — Answer: (A), (B), (C), (D) [MSQ, 2M · Regression & Model Selection > Ridge & Lasso]

With $X^T X = I$ the ridge path is $b/(1 + \lambda)$ and the lasso path is $\text{sign}(b)(|b| - \lambda)_+$. The plot shows paths reaching zero at finite $\lambda \Leftrightarrow$ lasso; here it is **lasso**.

(A) **TRUE.** Both $b/(1+\lambda)$ and soft-thresholding shrink monotonically.

(B) **TRUE.** Penalty vanishes at $\lambda = 0$.

(C) **TRUE.** Coefficients with $|b_j| \leq 1.5$ are zero: count = 1.

(D) **TRUE.** $\text{sign}(b)(|b| - \lambda)_+$ is piecewise linear.

Q.18 — Answer: -40 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Parameter counting]

Old: $(64+1) \cdot 64 + (64+1) \cdot 10 = 4810$.

New: $(64+1) \cdot 64 + (64+1) \cdot 8 + (8+1) \cdot 10 = 4770$.

Change = **-40** (negative: the bottleneck reduces parameters).

Q.19 — Answer: -0.01 to 0.01 [NAT, 2M · SVM & Decision Trees > Perceptron (linear separators)]

Trace the epoch (a point with $y \cdot w^T \tilde{x} \leq 0$, including 0, triggers an update):

Step	$\tilde{x} = (1, x_1, x_2)$	y	$w^T \tilde{x}$	$y \cdot w^T \tilde{x} \leq 0$?	w after step
1	(1, 1, -1)	+1	0	yes → update	(0.5, 0.5, -0.5)
2	(1, 0, 1)	-1	0	yes → update	(0, 0.5, -1)
3	(1, 3, -1)	+1	2.5	no	(0, 0.5, -1)
4	(1, -2, 1)	-1	-2	no	(0, 0.5, -1)

Final $w = (0, 0.5, -1)$, number of updates = 2. Answer: **0**.

Q.20 — Answer: (A), (B), (D) [MSQ, 2M · SVM & Decision Trees > SVM: dual & support vectors]

P2 and P3 (same class) lie on one margin line and P5 on the other; the optimal boundary is parallel to the line through P2, P3, midway to P5. Support vectors: P2, P3, P5.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (1, -2/5)$, $b = -2$; $\|w\| = 1.077$, margin $2/\|w\| = 1.857$.

Point	P1	P2	P3	P4	P5	P6
$y(w^T x + b)$	3.6	1	1	2.2	1	2.2

Support vectors are exactly the points with $y_i(w^T x_i + b) = 1$ ($\alpha_i > 0$): **P2, P3, P5**.

Dual check: $\alpha = (\alpha_2 = 1/2, \alpha_3 = 2/25, \alpha_5 = 29/50)$ satisfy $\sum \alpha_i y_i = 0$ and $\sum \alpha_i y_i x_i = w$.

Q.21 — Answer: 0.4662 to 0.4862 [NAT, 2M · Probabilistic & Instance-based Classifiers >

Evaluation metrics]

Counting: TP = 2, FP = 2, FN = 3.

$P = 2/4 = 0.5$, $R = 2/5 = 0.4$.

$F_{0.5} = (1 + 0.25) \times 0.5 \times 0.4 / (0.25 \times 0.5 + 0.4) = \mathbf{0.48}$.

$\beta < 1$ weights precision more heavily (F1 here would be 0.444).

Q.22 — Answer: 2.2486 to 2.2526 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$f(w) = (0.5/2)w^2 \Rightarrow \nabla f(w) = 0.5w$. Update: $v_t = \beta v_{t-1} + \nabla f(w_{t-1})$, $w_t = w_{t-1} - \eta v_t$, with $v_0 = 0$, $\beta = 0.9$, $\eta = 0.25$.

$t = 1$: $g = 0.5 \times 2 = 1$; $v_1 = 0.9 \times 0 + 1 = 1$; $w_1 = 2 - 0.25 \times 1 = 1.75$.

$t = 2$: $g = 0.5 \times 1.75 = 0.875$; $v_2 = 0.9 \times 1 + 0.875 = 1.775$; $w_2 = 1.75 - 0.25 \times 1.775 = 1.3062$.

$t = 3$: $g = 0.5 \times 1.3062 = 0.6531$; $v_3 = 0.9 \times 1.775 + 0.6531 = 2.2506$; $w_3 = 1.3062 - 0.25 \times 2.2506 = 0.7436$.

Answer: **2.2506**.

Q.23 — Answer: 4.0396 to 4.0436 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$p = \text{softmax}(z)$: $\exp(z) = (0.1353, 0.3679, 4.4817, 2.7183)$, $\text{sum} = 7.7032$, so $p = (0.0176, 0.0478, 0.5818, 0.3529)$.

For $L = -\ln p_y$ with softmax, $\partial L / \partial z = p - y$ (y one-hot, true class 1): $\partial L / \partial z = (-0.9824, 0.0478, 0.5818, 0.3529)$.

$L = -\ln 0.0176 = \mathbf{4.0416}$.

Note $\sum_k \partial L / \partial z_k = \sum p_k - 1 = 0$.

Q.24 — Answer: 60.99 to 61.01 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Sort the values: 41(N), 52(N), 54(N), 55(N), 56(N), 66(Y), 73(Y), 78(N), 79(N), 84(Y).

Parent [3 Yes, 7 No], $H = 0.8813$. The optimal threshold always lies where the class changes between neighbours (Fayyad-Irani), giving 3 candidates:

Threshold	Left ($\leq t$) [Yes, No]	Right [Yes, No]	IG
61	[0, 5]	[3, 2]	0.3958
75.5	[2, 5]	[1, 2]	0.0016
81.5	[2, 7]	[1, 0]	0.1935

Best: $t = 61$ with $IG = 0.3958$. Answer: **61**.

Q.25 — Answer: 1.8923 to 1.9037 [NAT, 2M · Regression & Model Selection > Bias-variance trade-off]

Mean prediction $\bar{f} = (1/5)\sum f_m = 10.54$.

Bias² = $(\bar{f} - f(x_0))^2 = (10.54 - 10)^2 = 0.2916$.

Variance = $(1/5)\sum (f_m - \bar{f})^2 = 0.1064$.

Expected test error = Bias² + Variance + $\sigma^2 = 0.2916 + 0.1064 + 1.5 = 1.898$.

Answer: **1.898**.

Q.26 — Answer: 2 [NAT, 2M · Unsupervised Learning > k-means]

Data: {1, 3, 13, 17, 18, 19, 23}; initial centroids $c_1 = 13, c_2 = 17$.

Iter	centroids used	cluster 1	cluster 2	new centroids
1	13, 17	{1, 3, 13}	{17, 18, 19, 23}	5.67, 19.25
2	5.67, 19.25	{1, 3}	{13, 17, 18, 19, 23}	2, 18
3	2, 18	{1, 3}	{13, 17, 18, 19, 23}	2, 18

In the last row the centroids do not change, so the algorithm has converged: final centroids 2 and 18.

Centroids changed in 2 iteration(s); the final iteration only confirms convergence. Answer **2**.

Q.27 — Answer: 1.2463 to 1.2537 [NAT, 2M · Regression & Model Selection > Gradient descent]

Hessian of J: $\nabla^2 J = (1/n)X^T X$. Eigenvalues of $X^T X$: $7 \pm 1 = 8, 6$ (eigenvectors (1, 1), (1, -1)).

So the Hessian eigenvalues are $L = 1.6$ and $\mu = 1.2$.

GD error evolves as $w_{t+1} - w^* = (I - \eta \nabla^2 J)(w_t - w^*)$, so it converges iff $|1 - \eta \lambda| < 1$ for every Hessian eigenvalue λ .

Need $\eta < 2/L = 2/1.6 = 1.25$.

Answer: **1.25**.

Q.28 — Answer: (D) [MCQ, 2M · Regression & Model Selection > Simple linear regression]

$\bar{x} = 8, \bar{y} = 4.7$.

$S_{xx} = \sum x^2 - (\sum x)^2/n = 116, S_{yy} = \sum y^2 - (\sum y)^2/n = 94.1, S_{xy} = \sum xy - \sum x \sum y/n = 78$.

$r = S_{xy}/\sqrt{(S_{xx} S_{yy})} = 78/\sqrt{(116 \times 94.1)} = \mathbf{0.747}$.

($r^2 = 0.557$ would be R^2 ; the slopes $b_{yx} = 0.672, b_{xy} = 0.829$ satisfy $b_{yx} b_{xy} = r^2$.)

Q.29 — Answer: 11.3061 to 11.3261 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

Mean $\bar{x} = (4, 5)$. Centered data rows: P1: (-3, -1), P2: (-3, -2), P3: (0, 0), P4: (2, 4), P5: (4, -1).

Sample covariance $S = (1/(n-1)) \sum (x - \bar{x})(x - \bar{x})^T = [[9.5, 3.25], [3.25, 5.5]]$.

Eigenvalues: $\lambda_1 = 11.3161, \lambda_2 = 3.6839$; unit PC1 direction $v_1 = (0.873, 0.4878)$.

Score $z_1 = v_1 \cdot (x - \bar{x}) = 0.873 \cdot 4 + 0.4878 \cdot -1 = 3.004$.

The variance of the PC1 scores equals the top eigenvalue: **11.32**.

Q.30 — Answer: -0.01 to 0.01 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

a(i) = mean Euclidean distance from P6 to the other members of its own cluster C3: $d(P6, P5) = 4 \rightarrow a = 4$.

b(i) = smallest mean distance to the points of another cluster: C1: 5.9036; C2: 4 $\rightarrow b = 4$.

s(i) = $(b - a)/\max(a, b) = (4 - 4)/4 = \mathbf{0}$.

s ranges in $[-1, 1]$; values near 1 mean well clustered, negative values suggest a wrong assignment.

Q.31 — Answer: 0.6775 to 0.6975 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

class	TP	predicted total	actual total	precision	recall	F1
A	16	25	23	0.64	0.6957	0.6667
B	18	26	24	0.6923	0.75	0.72
C	10	13	17	0.7692	0.5882	0.6667

Precision of class $k = C_{kk}/(\text{column sum})$; recall = $C_{kk}/(\text{row sum})$.

Micro-averaging pools TP/FP/FN over classes; for single-label multiclass, micro-P = micro-R = micro-F1 = accuracy = $44/64 = \mathbf{0.69}$.

Q.32 — Answer: 0.1228 to 0.1428 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b} = (1, -4, -1)$.

$e^{\mathbf{z}} = (2.7183, 0.0183, 0.3679)$, sum = 3.1045.

Softmax probabilities = (0.8756, 0.0059, 0.1185).

Loss = $-\ln(0.8756) = \mathbf{0.13}$.

Q.33 — Answer: (D) [MSQ, 2M · Unsupervised Learning > Hierarchical clustering]

(A) FALSE. It is $\geq$ the original distance.

(B) FALSE. Average linkage depends on actual values, not only on ranks.

(C) FALSE. Single and complete linkage often disagree.

(D) TRUE. It covers single, complete, average, Ward and others.

Q.34 — Answer: 0.3755 to 0.3795 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

For the average cross-entropy, $\partial L/\partial w_j = (1/n)\sum(\hat{p}_i - y_i)x_{ij}$, and $\partial L/\partial b = (1/n)\sum(\hat{p}_i - y_i)$, with $\hat{p}_i = \sigma(z_i)$.

i	z_i	$\hat{p}_i$	$\hat{p}_i - y_i$
1	0.5	0.6225	0.6225
2	0.5	0.6225	-0.3775

Gradient: $\partial L/\partial w_1 = 0.6888$, $\partial L/\partial b = 0.1225$.

Update $\theta \leftarrow \theta - \eta \nabla L$: $w_1 = 0 - 1 \times (0.6888) = -0.6888$, $b = 0.5 - 1 \times (0.1225) = 0.3775$.

Required: $b = \mathbf{0.378}$.

Q.35 — Answer: 0.2435 to 0.2465 [NAT, 2M · SVM & Decision Trees > Decision trees: overfitting & pruning]

$R(T) = \text{sum of leaf errors} / N = (2 + 0 + 3 + 3 + 4)/100 = 0.12$; $|T| = 5$ leaves.

$R_\alpha(T) = 0.12 + 0.025 \times 5 = \mathbf{0.245}$.

Mock Test 15

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	5	9	21

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Probabilistic & Instance-based Classifiers	7	11
Unsupervised Learning	7	11
SVM & Decision Trees	7	11
Regression & Model Selection	7	11
Neural Networks (MLP / Feed-forward)	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

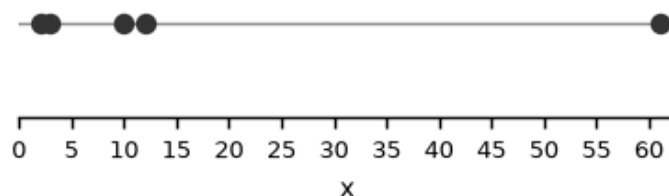
Q.1 [1 mark · MCQ]

In a logistic regression model for a positive response, the coefficient of *dose (mg)* is $\beta = 0.3$. A one-unit increase in *dose (mg)* (others fixed) changes the odds of a positive response by approximately

- (A) 25.9% (B) 57.4%
(C) 30% (D) 35%

Q.2 [1 mark · NAT (numerical answer)]

A cluster contains the 1-D values $\{2, 3, 10, 12, 61\}$. Treating them as one cluster, the absolute difference between the k-means centroid (mean) and the k-medoids representative (absolute-difference dissimilarity) is _____ (rounded off to two decimal places)



Q.3 [1 mark · NAT (numerical answer)]

To keep the same sampling density as 5 points per axis in 1-D, the number of points needed in $d = 3$ dimensions (a grid with 5 points per axis) is _____ (Answer in integer)

Q.4 [1 mark · NAT (numerical answer)]

For $x \in \mathbb{R}^5$, the quadratic kernel $K(x, z) = (1 + x^T z)^2$ equals $\phi(x)^T \phi(z)$ for a feature map ϕ containing a constant, the linear terms, the squares and the pairwise cross terms (with $\sqrt{2}$ scaling). The length of $\phi(x)$ is _____ (Answer in integer)

Q.5 [1 mark · MSQ (one or more correct)]

Regarding the bias–variance trade-off in supervised regression with squared-error loss, which of the following statements is/are CORRECT?

- (A) Regularisation reduces the bias of a model.
- (B) Regularisation trades a small increase in bias for a reduction in variance.
- (C) Averaging many (approximately) independent unbiased estimators reduces variance.
- (D) Expected test MSE = Bias + Variance + noise (bias not squared).

Q.6 [1 mark · MSQ (one or more correct)]

Linear regression is trained by minimising the squared-error loss. Which of the following statements about gradient-based training is/are CORRECT?

- (A) The Hessian of $\|Xw - y\|^2$ is $2X^T X$.
- (B) The MSE loss of linear regression, $J(w) = (1/n)\|Xw - y\|^2$, is a convex function of w .
- (C) One epoch of SGD with batch size 1 on n examples performs n parameter updates.
- (D) Standardising features changes the minimum training MSE achievable by OLS with an intercept.

Q.7 [1 mark · MSQ (one or more correct)]

Which of the following statements about clustering evaluation measures is/are CORRECT?

- (A) A silhouette value of -1 indicates a perfectly clustered point.
- (B) The Rand index is the fraction of point pairs on which two clusterings agree (both together or both apart).
- (C) The average silhouette width can be used to choose k by picking the k with the largest value.
- (D) The adjusted Rand index corrects the Rand index for chance; its expected value under random labelings is 0.

Q.8 [1 mark · NAT (numerical answer)]

For $K = 3$ classes and $d = 4$ binary features, a Bayes classifier that models the FULL joint class-conditional distribution $P(x_1, \dots, x_d | c)$ (no independence assumption) needs how many independent parameters, including the class priors? _____ (Answer in integer)

Q.9 [1 mark · MSQ (one or more correct)]

Which of the following statements about support vector machines is/are CORRECT?

- (A) A kernel SVM predicts with $f(x) = \sum \alpha_i y_i K(x_i, x) + b$, where only support vectors contribute.
- (B) Removing a training point that is not a support vector does not change the hard-margin SVM solution.
- (C) The optimal weight vector of an SVM can be written as $w = \sum \alpha_i y_i x_i$.
- (D) The functional margin $y_i(w^T x_i + b)$ is positive if and only if point i is correctly classified.

Q.10 [1 mark · NAT (numerical answer)]

For logits $z = (-1.5, -2, 3)$, the temperature-scaled softmax $p_k = \exp(z_k/T) / \sum_j \exp(z_j/T)$ with $T = 0.5$ gives $p_1 =$ _____ (rounded off to 4 decimal places)

Q.11 [1 mark · MSQ (one or more correct)]

Consider the soft-margin SVM: minimise $\frac{1}{2}\|w\|^2 + C \sum_i \xi_i$ subject to $y_i(w^T x_i + b) \geq 1 - \xi_i$, $\xi_i \geq 0$. Which of the following statements is/are CORRECT?

- (A) $\sum \xi_i$ equals the number of misclassified training points.
- (B) Points with $\alpha_i = C$ may lie inside the margin or be misclassified.
- (C) The soft-margin SVM is infeasible if the data are not linearly separable.
- (D) Every point strictly inside the margin has $\alpha_i = 0$.

Q.12 [1 mark · NAT (numerical answer)]

A linear regression model with an intercept uses 4 numeric features and a categorical feature *education level* with 4 levels (School, UG, PG, PhD). Each categorical feature is encoded by dummy (indicator) variables using a reference level, so that the design matrix has full column rank. An interaction between the first numeric feature x_1 and *education level* is also added (products of x_1 with each education level dummy). The total number of coefficients (including the intercept) is _____ (Answer in integer)

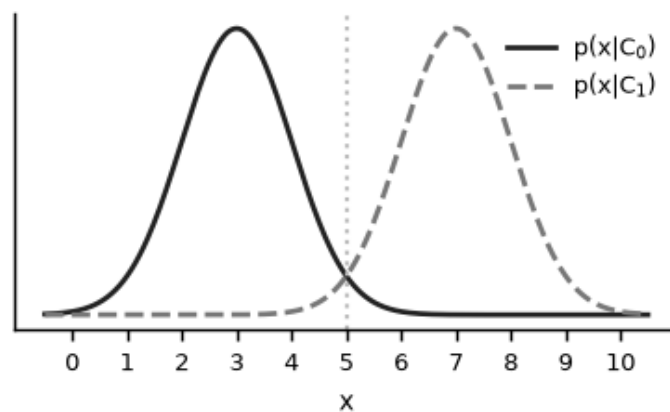
Q.13 [1 mark · MSQ (one or more correct)]

Which of the following statements about initialisation and regularisation of neural networks is/are CORRECT?

- (A) With inverted dropout (keep probability p), activations must additionally be multiplied by p at test time.
- (B) Initialising all weights of an MLP to the same value makes the hidden units of a layer compute identical functions and receive identical updates, so they stay identical.
- (C) High training loss and high validation loss together indicate overfitting.
- (D) Early stopping acts as a regulariser.

Q.14 [1 mark · MCQ]

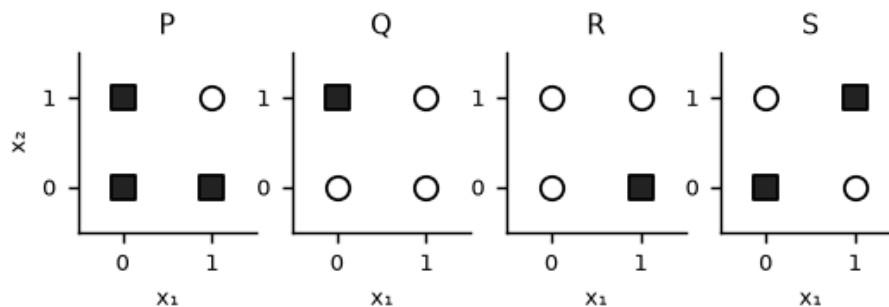
Two classes have unit-variance Gaussian class-conditional densities with means 3 (C_0) and 7 (C_1), as plotted. With equal priors the Bayes decision threshold is the dotted line at $x = 5$. If the prior of C_1 is increased to 0.9, the threshold



- (A) It moves towards $x = 3$, enlarging the region assigned to C_1
- (B) It moves towards $x = 7$, shrinking the region assigned to C_1
- (C) It moves towards $x = 7$, enlarging the region assigned to C_1
- (D) It splits into two thresholds, giving a quadratic boundary

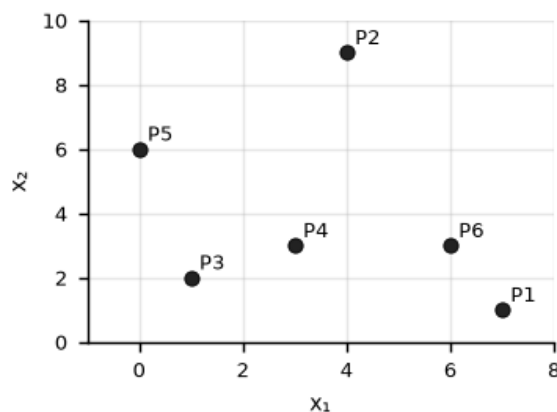
Q.15 [1 mark · MCQ]

Each panel shows the four binary inputs (x_1, x_2) labelled as class 1 (filled square) or class 0 (open circle). Which dataset can NOT be classified perfectly by a single perceptron (one linear threshold unit)?

**(A)** Dataset P**(B)** Dataset Q**(C)** Dataset R**(D)** Dataset S**Q.16 - Q.35 : Two-mark questions****Q.16 [2 marks · NAT (numerical answer)]**

A cluster contains the 6 points below. Distances are Manhattan (L_1). If P6 were used as the cluster representative instead of the medoid, the total distance (cost) would increase by _____ (Answer in integer)

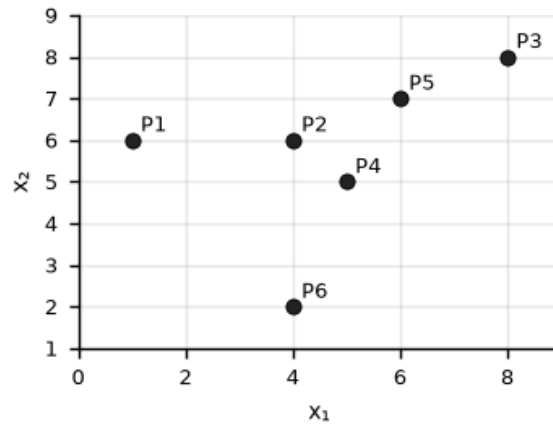
Point	P1	P2	P3	P4	P5	P6
x_1	7	4	1	3	0	6
x_2	1	9	2	3	6	3



Q.17 [2 marks · NAT (numerical answer)]

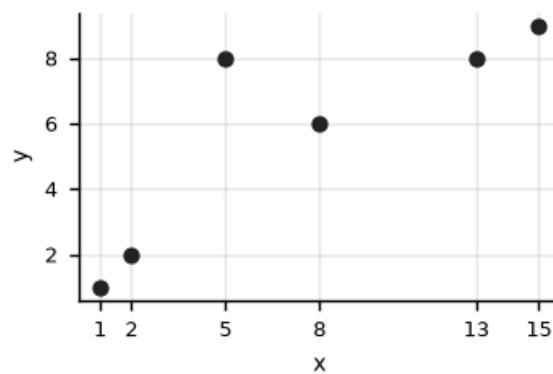
Single-linkage agglomerative clustering with Euclidean distance is applied to the six points below. Then the number of clusters obtained by cutting the dendrogram at height 1.83 is _____ (Answer in integer)

Point	P1	P2	P3	P4	P5	P6
x_1	1	4	8	5	6	4
x_2	6	6	8	5	7	2

**Q.18 [2 marks · NAT (numerical answer)]**

A regression tree (squared-error criterion) is fitted to the 6 points below. The root split " $x \leq t$ " is chosen among midpoints of consecutive x values to minimise the total SSE of the two children (each child predicts its mean). Find the threshold t of the best split " $x \leq t$ ". (rounded off to one decimal place)

x	1	2	5	8	13	15
y	1	2	8	6	8	9



Q.19 [2 marks · MCQ]

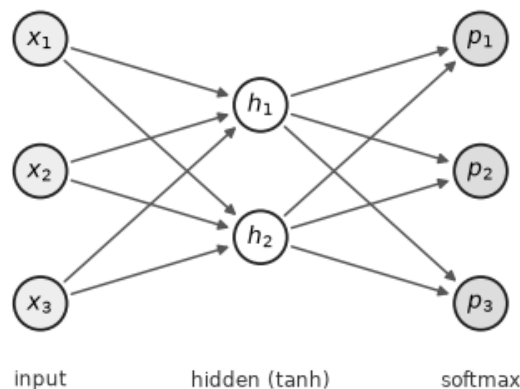
A naive Bayes classifier is trained on the 12 examples below. The new instance (Sender = Unknown, Length = Long, HasLink = No) is classified (i) with maximum-likelihood (unsmoothed) estimates and (ii) with Laplace (add-one) smoothing of the class-conditional probabilities (class priors unsmoothed). The predicted classes are

#	Sender	Length	HasLink	Class
1	Known	Long	Yes	Ham
2	Known	Medium	No	Ham
3	Unknown	Medium	No	Spam
4	Unknown	Medium	Yes	Spam
5	Unknown	Long	Yes	Spam
6	Unknown	Medium	Yes	Spam
7	Known	Long	Yes	Ham
8	Unknown	Long	Yes	Spam
9	Unknown	Short	Yes	Spam
10	Known	Short	Yes	Ham
11	Unknown	Medium	Yes	Spam
12	Unknown	Long	No	Spam

- (A) Without smoothing: Spam; with Laplace smoothing: Spam
 (B) Without smoothing: Ham; with Laplace smoothing: Ham
 (C) Without smoothing: Spam; with Laplace smoothing: Ham
 (D) Without smoothing: Ham; with Laplace smoothing: Spam

Q.20 [2 marks · NAT (numerical answer)]

A 3-2-3 network computes $h = \tanh(W_1 x + b_1)$, $z = W_2 h + b_2$, $p = \text{softmax}(z)$, with the parameters below. For $x = (0, 2, -1)^T$, the predicted probability p_1 of class 1 is _____ (rounded off to three decimal places)



$$W_1 = \begin{bmatrix} 0.5 & 1 & -0.5 \\ 1 & -1 & -1 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 2 & 1 \\ 0 & -2 \\ -2 & 2 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} -0.5 \\ -1 \\ 1 \end{bmatrix}$$

Q.21 [2 marks · MCQ]

For logits $z = (2, 0, 1)$, softmax probabilities p and true class 3, the gradient of the cross-entropy loss $L = -\ln p_3$ with respect to z (rounded to two decimals) is

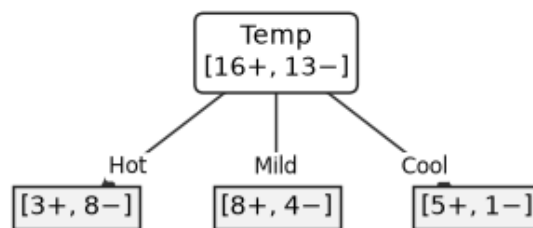
- (A) $(-0.33, -0.91, -0.76)$ (B) $(0.15, 0.01, -0.14)$
 (C) $(0.67, 0.09, 0.24)$ (D) $(0.67, 0.09, -0.76)$

Q.22 [2 marks · NAT (numerical answer)]

A centered data matrix X with $n = 17$ rows (samples) and $d = 3$ columns has singular values 15, 11, 8. Using the sample covariance with divisor $n - 1$, the variance of the data along the third principal component is _____ (rounded off to two decimal places)

Q.23 [2 marks · NAT (numerical answer)]

A node with 29 training records (class counts [positive+, negative−]) is split on the attribute Temp as shown. Compute the information gain (in bits) of splitting on Temp. (rounded off to two decimal places)

**Q.24 [2 marks · NAT (numerical answer)]**

A logistic regression classifier has $P(y = 1 | \mathbf{x}) = \sigma(-3 + (2)x_1 + (2)x_2)$. An input is labelled $y = 1$ if and only if $P(y = 1 | \mathbf{x}) \geq 0.5$. For this rule, the perpendicular distance of the decision boundary from the origin is _____ (rounded off to two decimal places)

Q.25 [2 marks · MSQ (one or more correct)]

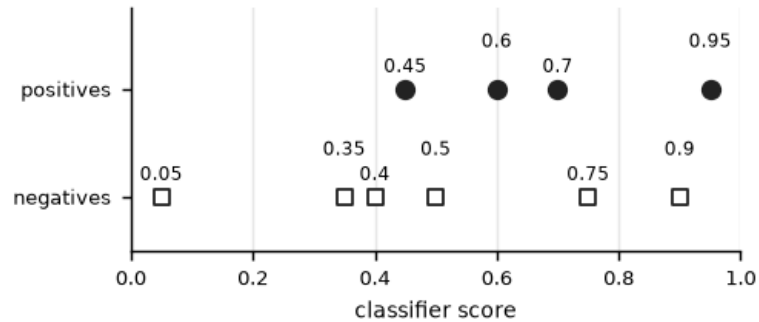
Which of the following statements about loss functions and optimisation for training neural networks is/are CORRECT?

- (A) Momentum accumulates an exponentially decaying average of past gradients, which damps oscillations across steep directions.
 (B) Decaying the learning rate during training can help SGD settle closer to a minimum.
 (C) Minimising cross-entropy loss is equivalent to maximising the log-likelihood of the labels under the model.
 (D) Too large a learning rate can make the training loss oscillate or diverge.

Q.26 [2 marks · NAT (numerical answer)]

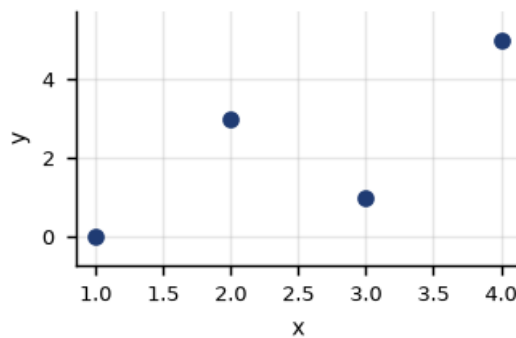
A classifier assigns the scores below to 10 test examples (4 positives, label 1; 6 negatives, label 0). The area under the ROC curve (AUC) is _____ (rounded off to two decimal places)

example	1	2	3	4	5	6	7	8	9	10
score	0.45	0.4	0.95	0.05	0.75	0.5	0.7	0.9	0.6	0.35
label	1	0	1	0	0	0	1	0	1	0

**Q.27 [2 marks · NAT (numerical answer)]**

A linear model $\hat{y} = w_0 + w_1 x$ is trained on the 4 points below by batch gradient descent on the loss $J(w) = (1/n)\sum (w_0 + w_1 x_i - y_i)^2$, starting from $(w_0, w_1) = (1, 0)$ with learning rate $\eta = 0.05$. Then the value of w_1 after two iterations is _____ (rounded off to three decimal places)

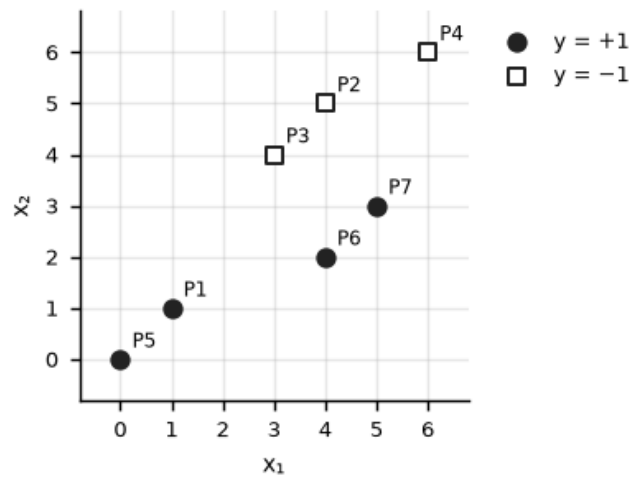
x	1	2	3	4
y	0	3	1	5



Q.28 [2 marks · NAT (numerical answer)]

The labelled training points shown below are linearly separable. A hard-margin linear SVM is trained on them. Compute $\|w\|$ for the canonical solution (support vectors satisfy $y_i(w^T x_i + b) = 1$). (rounded off to two decimal places)

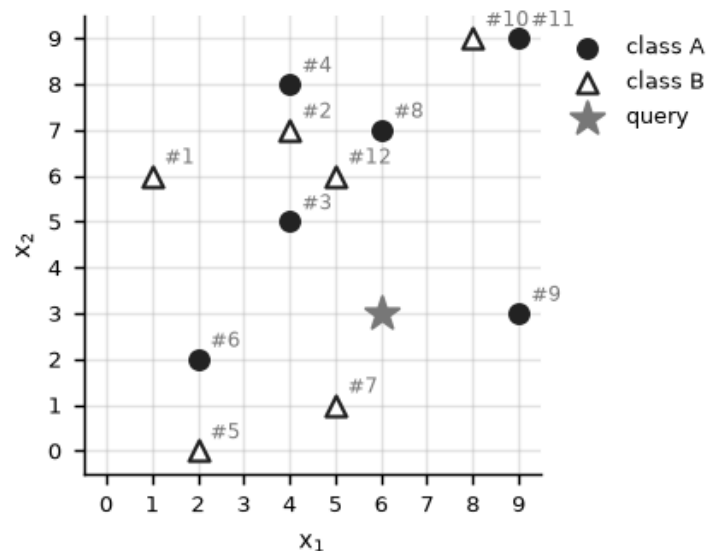
Point	x_1	x_2	y
P1	1	1	+1
P2	4	5	-1
P3	3	4	-1
P4	6	6	-1
P5	0	0	+1
P6	4	2	+1
P7	5	3	+1



Q.29 [2 marks · MSQ (one or more correct)]

The figure and table show 12 labelled training points (class A: filled circles, class B: open triangles) and a query point (star) at (6, 3). A k-NN classifier with **Euclidean** distance and majority vote is used. Which of the following statements is/are CORRECT?

#	1	2	3	4	5	6	7	8	9	10	11	12
x_1	1	4	4	4	2	2	5	6	9	8	9	5
x_2	6	7	5	8	0	2	1	7	3	9	9	6
class	B	B	A	A	B	A	B	A	A	B	A	B

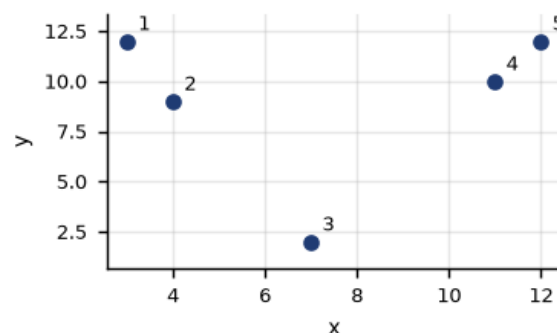


- (A) With $k = 5$, the query is classified as class B.
- (B) With $k = 1$, the query is classified as class B.
- (C) With $k = 7$, the query is classified as class B.
- (D) With $k = 3$, the query is classified as class B.

Q.30 [2 marks · NAT (numerical answer)]

Consider the 5 training points (x_i, y_i) shown below. Two-fold cross-validation is used with fold 1 = {points 1, 3, 5, ...} and fold 2 = {points 2, 4, ...} (points numbered in increasing x). The model predicts the mean of the training-fold targets. The pooled 2-fold CV MSE (over all n points) is _____ (rounded off to two decimal places)

x	3	4	7	11	12
y	12	9	2	10	12



Q.31 [2 marks · MSQ (one or more correct)]

A linear regression of house price (₹ lakh) on Area (area in hundreds of sq. ft), Furnished (1 if furnished, 0 otherwise) and zone $\in \{\text{North, South, East}\}$ encoded by dummies ZoneSouth and ZoneEast (reference level North) gives the fitted model

$$\hat{y} = 21 + 2.6 \cdot \text{Area} - 2.5 \cdot \text{Furnished} + 5.6 \cdot \text{ZoneSouth} + 4.1 \cdot \text{ZoneEast}$$

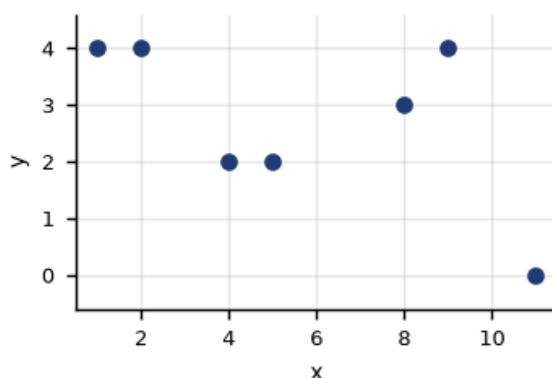
Which of the following is/are CORRECT?

- (A) Holding other variables fixed, the predicted response for zone South minus that for zone East is 9.7.
- (B) Increasing Area by 4 hundred sq. ft, other variables fixed, changes the prediction by 10.4.
- (C) Adding a third dummy for zone North while keeping the intercept would make $X^T X$ singular.
- (D) The coefficient of ZoneSouth is the expected difference in response between zone South and zone North, holding the other variables fixed.

Q.32 [2 marks · NAT (numerical answer)]

The 7 observations shown in the table are plotted below. For the simple linear regression of y on x (with intercept), the intercept β_0 of the least-squares line $\hat{y} = \beta_0 + \beta_1 x$ is _____ (rounded off to two decimal places)

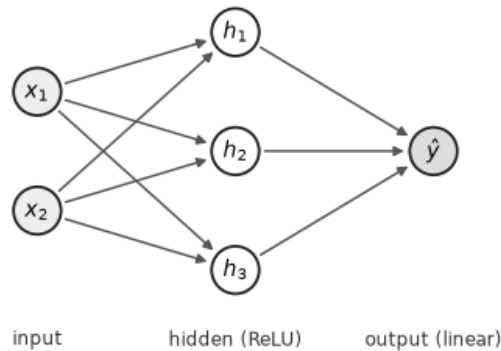
x	1	2	4	5	8	9	11
y	4	4	2	2	3	4	0

**Q.33 [2 marks · NAT (numerical answer)]**

The eigenvalues of the sample covariance matrix (divisor $n - 1$) of a centered 6-dimensional dataset are 19.5, 17.5, 14, 13.5, 12.5, 11.5. Then the percentage of total variance **lost** by keeping only the top 4 principal components is _____ (rounded off to two decimal places)

Q.34 [2 marks · NAT (numerical answer)]

The 2-3-1 network below computes $\hat{y} = v \cdot \text{ReLU}(Wx + b) + c$ with W , b , v given below and $c = 1$. With $x = (1, -2)$, target $y = -3$ and loss $L = \frac{1}{2}(\hat{y} - y)^2$, the value of $\partial L / \partial x_1$ (gradient with respect to the input) is _____ (Answer in integer)



$$W = \begin{bmatrix} 2 & 2 \\ 2 & 0 \\ -1 & 0 \end{bmatrix}$$

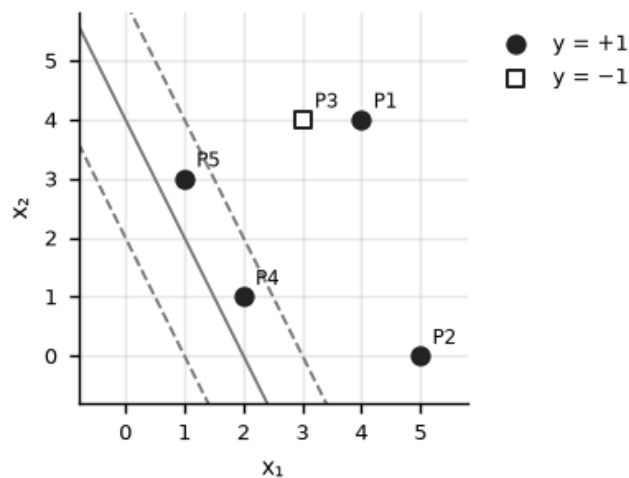
$$b = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}$$

$$v^T = \begin{bmatrix} -1 & 2 & 1 \end{bmatrix}$$

Q.35 [2 marks · NAT (numerical answer)]

A linear SVM with $f(x) = x_1 + 0.5x_2 - 2$ is evaluated on the training points in the table (also plotted, with the boundary $f = 0$ solid and $f = \pm 1$ dashed). Compute the total hinge loss $\sum_i \max(0, 1 - y_i f(x_i))$. (rounded off to two decimal places)

Point	x_1	x_2	y
P1	4	4	+1
P2	5	0	+1
P3	3	4	-1
P4	2	1	+1
P5	1	3	+1



— END OF QUESTION PAPER —

Mock Test 15 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MCQ	1	(D)	19	MCQ	2	(A)
2	NAT	1	7.59 to 7.61	20	NAT	2	0.3507 to 0.3547
3	NAT	1	125	21	MCQ	2	(D)
4	NAT	1	21	22	NAT	2	3.99 to 4.01
5	MSQ	1	(B), (C)	23	NAT	2	0.1471 to 0.1671
6	MSQ	1	(A), (B), (C)	24	NAT	2	1.0507 to 1.0707
7	MSQ	1	(B), (C), (D)	25	MSQ	2	(A), (B), (C), (D)
8	NAT	1	47	26	NAT	2	0.6983 to 0.7183
9	MSQ	1	(A), (B), (C), (D)	27	NAT	2	0.5605 to 0.5645
10	NAT	1	−0.0001 to 0.0003	28	NAT	2	1.108 to 1.128
11	MSQ	1	(B)	29	MSQ	2	(B)
12	NAT	1	11	30	NAT	2	14.0571 to 14.1984
13	MSQ	1	(B), (D)	31	MSQ	2	(B), (C), (D)
14	MCQ	1	(A)	32	NAT	2	3.9663 to 4.0063
15	MCQ	1	(D)	33	NAT	2	27.1086 to 27.1286
16	NAT	2	4	34	NAT	2	24
17	NAT	2	5	35	NAT	2	4.99 to 5.01
18	NAT	2	3.49 to 3.51				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Probabilistic & Instance-based Classifiers					11
Unsupervised Learning					11
SVM & Decision Trees					11
Regression & Model Selection					11
Neural Networks (MLP / Feed-forward)					11
Total					55

Mock Test 15 — Detailed Solutions

Q.1 — Answer: (D) [MCQ, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

Odds multiply by $e^\beta = e^{0.3} = 1.3499$.

Percentage change = $100(e^\beta - 1)\% = 35\%$.

$100\beta\%$ is only a small- β approximation; $100e^\beta\%$ is the new odds as a percentage of the old odds, not the change.

Q.2 — Answer: 7.59 to 7.61 [NAT, 1M · Unsupervised Learning > k-medoids]

Mean = $88/5 = 17.6$ (pulled towards the outlier 61).

Medoid = data point minimizing $\sum |x_i - m|$. Costs: $m=2$: 78, $m=3$: 75, $m=10$: 68, $m=12$: 70, $m=61$: 217.

The minimum is at $m = 10$ (the median), cost 68.

Answer: **7.6**. The medoid is barely affected by the outlier, unlike the mean.

Q.3 — Answer: 125 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

A grid with 5 points per axis needs $5^3 = 125$ points — exponential growth with d .

Q.4 — Answer: 21 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

Constant: 1; linear: 5; squares: 5; cross terms: $C(5, 2) = 10$.

Total = $1 + 5 + 5 + 10 = 21 = C(5 + 2, 2)$.

Q.5 — Answer: (B), (C) [MSQ, 1M · Regression & Model Selection > Bias-variance trade-off]

(A) FALSE. It increases bias, reduces variance.

(B) TRUE. Shrinkage stabilises estimates.

(C) TRUE. Var of mean = σ^2/B .

(D) FALSE. It is the squared bias that appears.

Q.6 — Answer: (A), (B), (C) [MSQ, 1M · Regression & Model Selection > Gradient descent]

(A) TRUE. Differentiate $2X^T(Xw - y)$ once more.

(B) TRUE. Its Hessian $(2/n)X^TX$ is positive semidefinite.

(C) TRUE. One update per example.

(D) FALSE. Affine rescaling of features does not change the span of the design, hence not the fitted values.

Q.7 — Answer: (B), (C), (D) [MSQ, 1M · Unsupervised Learning > Cluster evaluation]

(A) FALSE. +1 is ideal; -1 is worst.

(B) TRUE. $RI = (a + b)/C(n, 2)$.

(C) TRUE. A common model-selection heuristic.

(D) TRUE. That is the purpose of the adjustment.

Q.8 — Answer: 47 [NAT, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Each class-conditional joint over $2^4 = 16$ configurations needs $2^4 - 1 = 15$ parameters; for K classes: 45.

Priors: $K - 1 = 2$. Total = **47**.

Naive Bayes would need only $Kd + K - 1 = 14$ — linear rather than exponential in d .

Q.9 — Answer: (A), (B), (C), (D) [MSQ, 1M · SVM & Decision Trees > SVM: dual & support vectors]

(A) TRUE. $\alpha_i = 0$ for all other points.

(B) TRUE. Non-support vectors have $\alpha_i = 0$ and inactive constraints.

(C) TRUE. From $\partial L / \partial w = 0$ in the Lagrangian.

(D) TRUE. Its sign compares the prediction with the label.

Q.10 — Answer: -0.0001 to 0.0003 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

Scaled logits $z/T = (-3, -4, 6)$.

exp values: 0.0498, 0.0183, 403.4288; sum = 403.4969.

$$p_1 = 0.0498/403.4969 = \mathbf{0.0001}.$$

Q.11 — Answer: (B) [MSQ, 1M · SVM & Decision Trees > SVM: soft margin & hinge loss]

(A) FALSE. It is only an upper bound.

(B) TRUE. Bound support vectors.

(C) FALSE. Slack makes it always feasible.

(D) FALSE. Such points have $\alpha_i = C$.

Q.12 — Answer: 11 [NAT, 1M · Regression & Model Selection > Multiple linear regression]

A categorical feature with k levels needs $k - 1$ dummies (one level is the reference absorbed by the intercept; using all k dummies with an intercept gives perfect multicollinearity).

Parameters = 1 (intercept) + 4 (numeric) + (4 - 1) = 8.

The interaction adds one product term per education level dummy: 4 - 1 = 3. Total = 8 + 3 = 11.

Answer: **11**.

Q.13 — Answer: (B), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

(A) FALSE. The $1/p$ scaling during training already makes test-time scaling unnecessary.

(B) TRUE. The symmetry is never broken by GD.

(C) FALSE. That pattern is underfitting.

(D) TRUE. It limits how far the weights move from their initialisation.

Q.14 — Answer: (A) [MCQ, 1M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$$x^* = (\mu_0 + \mu_1)/2 + \sigma^2 \ln(\pi_0/\pi_1)/(\mu_1 - \mu_0) = 4.451.$$

A larger prior for C_1 means more of the x -axis is assigned to it, so the threshold moves away from μ_1 towards $\mu_0 = 3$. Answer: **It moves towards $x = 3$, enlarging the region assigned to C_1 .**

Equal variances keep the boundary a single point (linear); priors shift it but never make it quadratic.

Q.15 — Answer: (D) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

A single perceptron can classify a dataset perfectly iff the two classes are linearly separable.

Dataset P: labels for (0,0),(0,1),(1,0),(1,1) = 1, 1, 1, 0 → NAND - linearly separable.

Dataset Q: labels for (0,0),(0,1),(1,0),(1,1) = 0, 1, 0, 0 → (NOT x_1) AND x_2 - linearly separable.

Dataset R: labels for (0,0),(0,1),(1,0),(1,1) = 0, 0, 1, 0 → x_1 AND (NOT x_2) - linearly separable.

Dataset S: labels for (0,0),(0,1),(1,0),(1,1) = 1, 0, 0, 1 → XNOR - **not** linearly separable (diagonal pattern).

Answer: **Dataset S**. Of the 14 non-constant Boolean functions of two inputs, only XOR and XNOR are not linearly separable.

Q.16 — Answer: 4 [NAT, 2M · Unsupervised Learning > k-medoids]

Compute the Manhattan (L_1) distance matrix and its row sums:

	P1	P2	P3	P4	P5	P6	Σ
P1	0	11	7	6	12	3	39
P2	11	0	10	7	7	8	43
P3	7	10	0	3	5	6	31
P4	6	7	3	0	6	3	25
P5	12	7	5	6	0	9	39
P6	3	8	6	3	9	0	29

The smallest row sum is 25 for P4, so the medoid is P4.

Cost with P6: 29; increase = 29 - 25 = **4**.

Unlike a k-means centroid, the medoid is always an actual data point.

Q.17 — Answer: 5 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Single-linkage merge heights are exactly the edge weights of a minimum spanning tree (Kruskal's algorithm adds the shortest edge joining two different components = single-linkage merge).

MST edges (Prim's algorithm): P1-P2 (3); P2-P4 (1.41); P2-P5 (2.24); P5-P3 (2.24); P4-P6 (3.16).

Sorted merge heights: 1.41, 2.24, 2.24, 3, 3.16.

MST edges shorter than 1.83: 1; clusters = $n - (\text{number of such edges}) = 6 - 1 = 5$.

Q.18 — Answer: 3.49 to 3.51 [NAT, 2M · SVM & Decision Trees > Regression trees]

Parent mean $\bar{y} = 5.667$, $SSE_0 = \sum (y - \bar{y})^2 = 57.333$.

Evaluate every split:

Split t	Left mean	Right mean	$SSE_L + SSE_R$
1.5	1	6.6	31.2
3.5	1.5	7.75	5.25
6.5	3.667	7.667	33.333
10.5	4.25	8.5	33.25
14	5	9	44

Best split $t = 3.5$: left mean 1.5, right mean 7.75, $SSE = 5.25$, reduction = 52.083.

Answer: **3.5**.

Q.19 — Answer: (A) [MCQ, 2M · Probabilistic & Instance-based Classifiers > Naive Bayes]

(i) **No smoothing.**

Spam: $P(\text{Class}=\text{Spam}) = 8/12$; $P(\text{Sender}=\text{Unknown}|\text{Spam}) = 8/8$; $P(\text{Length}=\text{Long}|\text{Spam}) = 3/8$; $P(\text{HasLink}=\text{No}|\text{Spam}) = 2/8 \Rightarrow \text{score} = 0.0625$.

Ham: $P(\text{Class}=\text{Ham}) = 4/12$; $P(\text{Sender}=\text{Unknown}|\text{Ham}) = 0/4$; $P(\text{Length}=\text{Long}|\text{Ham}) = 2/4$; $P(\text{HasLink}=\text{No}|\text{Ham}) = 1/4 \Rightarrow \text{score} = 0$.

A zero count makes the whole product for Ham equal to 0 (zero-frequency problem), so the prediction is Spam.

(ii) **Laplace smoothing.**

Spam: $P(\text{Class}=\text{Spam}) = 8/12$; $P(\text{Sender}=\text{Unknown}|\text{Spam}) = (8+1)/(8+2) = 9/10$;

$P(\text{Length}=\text{Long}|\text{Spam}) = (3+1)/(8+3) = 4/11$; $P(\text{HasLink}=\text{No}|\text{Spam}) = (2+1)/(8+2) = 3/10 \Rightarrow \text{score} = 0.06545$.

Ham: $P(\text{Class}=\text{Ham}) = 4/12$; $P(\text{Sender}=\text{Unknown}|\text{Ham}) = (0+1)/(4+2) = 1/6$;

$P(\text{Length}=\text{Long}|\text{Ham}) = (2+1)/(4+3) = 3/7$; $P(\text{HasLink}=\text{No}|\text{Ham}) = (1+1)/(4+2) = 1/3 \Rightarrow \text{score} = 0.00794$.

Larger score: Spam. Answer: **Without smoothing: Spam; with Laplace smoothing: Spam.**

Q.20 — Answer: 0.3507 to 0.3547 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

$a = W_1 x + b_1 = (1.5, -2)$; $h = \tanh(a) = (0.9051, -0.964)$.

$z = W_2 h + b_2 = (0.3463, 0.9281, -2.7384)$.

$\exp(z) = (1.4138, 2.5296, 0.0647)$, sum = 4.008.

$p_1 = \exp(z_1)/\sum \exp(z_j) = \mathbf{0.3527}$. (All probabilities: (0.353, 0.631, 0.016).)

Q.21 — Answer: (D) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$p = \text{softmax}((2, 0, 1)) = (0.6652, 0.09, 0.2447)$; $y = (0, 0, 1)$.

$\partial L/\partial z = p - y = \mathbf{(0.67, 0.09, -0.76)}$.

Distractors: $y - p = (-0.67, -0.09, 0.76)$ (wrong sign); p alone = (0.67, 0.09, 0.24) (forgot the target); $(p - y) \odot p \odot (1 - p) = (0.15, 0.01, -0.14)$ (spurious σ' -type factor, which cancels for cross-entropy); $-y/p = (0, 0, -4.09)$ is $\partial L/\partial p$, not $\partial L/\partial z$.

Q.22 — Answer: 3.99 to 4.01 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

With $X = U \Sigma V^T$ (X centered, $n \times d$), the sample covariance is $S = X^T X / (n - 1) = V(\Sigma^2 / (n - 1))V^T$, so the PC directions are the columns of V and the eigenvalues are $\lambda_j = \sigma_j^2 / (n - 1)$.

$\lambda = 15^2/16 = 14.0625$, $11^2/16 = 7.5625$, $8^2/16 = 4$.

Cumulative % variance: 54.88, 84.39, 100 (note the $(n - 1)$ factor cancels in proportions: $PVE = \sigma_j^2 / \Sigma \sigma^2$).

Answer: **4**. (Using σ instead of σ^2 is the classic mistake.)

Q.23 — Answer: 0.1471 to 0.1671 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [16+, 13-]: $H = -(16/29)\log_2(16/29) - (13/29)\log_2(13/29) = 0.9923$.

Temp = Hot: [3+, 8-], $H = -(3/11)\log_2(3/11) - (8/11)\log_2(8/11) = 0.8454$.

Temp = Mild: [8+, 4-], $H = -(8/12)\log_2(8/12) - (4/12)\log_2(4/12) = 0.9183$.

Temp = Cool: [5+, 1-], $H = -(5/6)\log_2(5/6) - (1/6)\log_2(1/6) = 0.65$.

Weighted child entropy = $(11/29)(0.8454) + (12/29)(0.9183) + (6/29)(0.65) = 0.8351$.

IG = $0.9923 - 0.8351 = 0.1571$.

Answer: **0.16**.

Q.24 — Answer: 1.0507 to 1.0707 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

$P(y=1|x) \geq 0.5 \Leftrightarrow z \geq c$, where $c = \ln(0.5/0.5) = 0$.

Here $c = 0$ because the threshold is 0.5.

Decision boundary: $-3 + (2)x_1 + (2)x_2 = 0$ (a straight line; changing the threshold only shifts it parallel to the 0.5 boundary).

distance = $|w_0 - c| / \|(w_1, w_2)\| = |-3 - 0| / \sqrt{8} = \mathbf{1.06}$.

Q.25 — Answer: (A), (B), (C), (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

(A) TRUE. Opposite-sign components cancel.

(B) TRUE. Reduces the noise floor of SGD.

(C) TRUE. CE = negative log-likelihood of a categorical/Bernoulli model.

(D) TRUE. Steps overshoot the minimum.

Q.26 — Answer: 0.6983 to 0.7183 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

AUC = fraction of (positive, negative) pairs in which the positive has the higher score (Mann-Whitney statistic).

For each positive, count negatives with lower score: $0.95 \rightarrow 6$; $0.7 \rightarrow 4$; $0.6 \rightarrow 4$; $0.45 \rightarrow 3$.

Concordant pairs = 17 out of $4 \times 6 = 24$.

AUC = $17/24 = \mathbf{0.71}$.

(Equivalently, trace the ROC staircase by lowering the threshold and add the rectangle areas.)

Q.27 — Answer: 0.5605 to 0.5645 [NAT, 2M · Regression & Model Selection > Gradient descent]

$\partial J / \partial w_0 = (2/n) \sum r_i$, $\partial J / \partial w_1 = (2/n) \sum r_i x_i$, with residual $r_i = w_0 + w_1 x_i - y_i$.

Step 1: $w = (1, 0)$, $r = (1, -2, 0, -4)$, $\nabla J = (-2.5, -9.5) \Rightarrow w \leftarrow w - 0.05 \nabla J = (1.125, 0.475)$.

Step 2: $w = (1.125, 0.475)$, $r = (1.6, -0.925, 1.55, -1.975)$, $\nabla J = (0.125, -1.75) \Rightarrow w \leftarrow w - 0.05 \nabla J = (1.1187, 0.5625)$.

Answer: **0.562**.

Q.28 — Answer: 1.108 to 1.128 [NAT, 2M · SVM & Decision Trees > SVM: margin & geometry]

P1 and P7 (same class) lie on one margin line and P3 on the other; the optimal boundary is parallel to the line through P1, P7, midway to P3. Support vectors: P1, P3, P7.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (1/2, -1)$, $b = 3/2$; $\|w\| = 1.118$, margin $2/\|w\| = 1.7889$.

Point	P1	P2	P3	P4	P5	P6	P7
$y(w^T x + b)$	1	1.5	1	1.5	1.5	1.5	1

All values are ≥ 1 (feasible), with equality exactly at the support vectors.

$\|w\| = \mathbf{1.12}$.

Q.29 — Answer: (B) [MSQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Distances from the query, sorted:

rank	#	(x_1, x_2)	distance	class
1	7	(5, 1)	$\sqrt{5} = 2.236$	B
2	3	(4, 5)	$\sqrt{8} = 2.828$	A
3	9	(9, 3)	$\sqrt{9} = 3$	A
4	12	(5, 6)	$\sqrt{10} = 3.162$	B
5	8	(6, 7)	$\sqrt{16} = 4$	A
6	6	(2, 2)	$\sqrt{17} = 4.123$	A
7	2	(4, 7)	$\sqrt{20} = 4.472$	B
8	5	(2, 0)	$\sqrt{25} = 5$	B

(A) FALSE. 5 nearest: #7, #3, #9, #12, #8 → 3 A vs 2 B ⇒ class A.**(B) TRUE.** 1 nearest: #7 → 0 A vs 1 B ⇒ class B.**(C) FALSE.** 7 nearest: #7, #3, #9, #12, #8, #6, #2 → 4 A vs 3 B ⇒ class A.**(D) FALSE.** 3 nearest: #7, #3, #9 → 2 A vs 1 B ⇒ class A.**Q.30 — Answer: 14.0571 to 14.1984** [NAT, 2M · Regression & Model Selection > Cross-validation]

Fold 1 validated with mean of fold-2 targets = 9.5; fold 2 validated with mean of fold-1 targets = 8.6667.

 Σ squared errors = 70.6389; pooled MSE = 70.6389/5 = 14.1278.Answer: **14.13**.**Q.31 — Answer: (B), (C), (D)** [MSQ, 2M · Regression & Model Selection > Multiple linear regression]**(A) FALSE.** It is $5.6 - (4.1) = 1.5$, not the sum.**(B) TRUE.** $4 \times 2.6 = 10.4$.**(C) TRUE.** The three dummies would sum to the intercept column (dummy-variable trap).**(D) TRUE.** North is the reference level (all dummies 0).**Q.32 — Answer: 3.9663 to 4.0063** [NAT, 2M · Regression & Model Selection > Simple linear regression] $\bar{x} = 5.714, \bar{y} = 2.714$.

i	1	2	3	4	5	6	7
$x_i - \bar{x}$	-4.71	-3.71	-1.71	-0.71	2.29	3.29	5.29
$y_i - \bar{y}$	1.29	1.29	-0.71	-0.71	0.29	1.29	-2.71
$(x_i - \bar{x})(y_i - \bar{y})$	-6.06	-4.78	1.22	0.51	0.65	4.22	-14.35

 $S_{xx} = \Sigma(x_i - \bar{x})^2 = 83.429, S_{xy} = -18.571$. $\beta_1 = S_{xy}/S_{xx} = -0.2226, \beta_0 = \bar{y} - \beta_1 \bar{x} = 3.9863$.Answer: **3.99**.**Q.33 — Answer: 27.1086 to 27.1286** [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]For centered data, projecting onto the top-k PCs and reconstructing leaves the components along the discarded eigenvectors; their average squared length (divisor $n - 1$) is the sum of the discarded eigenvalues.

Discarded eigenvalues: 12.5, 11.5; sum = 24; total = 88.5.

Lost fraction = $24/88.5 = 27.12\%$.**Q.34 — Answer: 24** [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation] $a = Wx + b = (-3, 1, -2), h = \text{ReLU}(a) = (0, 1, 0), \hat{y} = v \cdot h + c = 3; \partial L / \partial \hat{y} = \hat{y} - y = 6$. $\text{ReLU}'(a) = (0, 1, 0)$ (unit j is active iff $a_j > 0$). $\partial L / \partial x_1 = (\hat{y} - y) \sum_j v_j \text{ReLU}'(a_j) W_{j1} = 6 \times (-1 \cdot 0 \cdot 2 + 2 \cdot 1 \cdot 2 + 1 \cdot 0 \cdot -1) = 24$.**Q.35 — Answer: 4.99 to 5.01** [NAT, 2M · SVM & Decision Trees > SVM: soft margin & hinge loss]

Compute $f(x_i)$, the functional margin $y_i f(x_i)$ and the slack/hinge loss for each point:

Point	x	y	$f(x)$	$y f(x)$	$\xi = \max(0, 1 - y f)$
P1	(4, 4)	+1	4	4	0
P2	(5, 0)	+1	3	3	0
P3	(3, 4)	-1	3	-3	4
P4	(2, 1)	+1	0.5	0.5	0.5
P5	(1, 3)	+1	0.5	0.5	0.5

$\sum \xi_i = 5$.

Points with $y f \geq 1$ contribute 0; points inside the margin contribute between 0 and 1; misclassified points contribute more than 1.

Mock Test 16

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	3	12	20

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
SVM & Decision Trees	7	11
Neural Networks (MLP / Feed-forward)	7	11
Unsupervised Learning	7	11
Regression & Model Selection	7	11
Probabilistic & Instance-based Classifiers	7	11

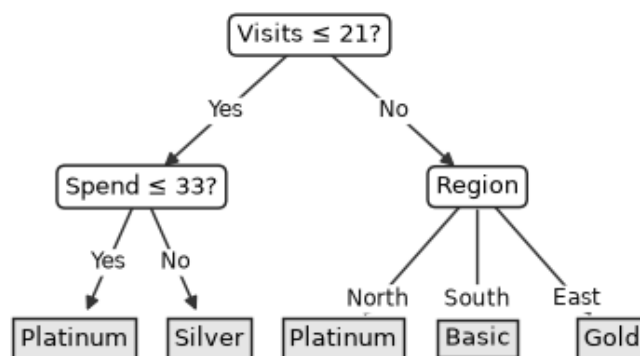
Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MCQ]

The decision tree below predicts the membership tier offered to a customer. Edges labelled Yes/No answer the test in the node. What is the prediction for the record shown in the table?



Visits	Spend	Tenure	Region
20	25	2	South

- (A) Basic
(C) Gold

- (B) Platinum
(D) Silver

Q.2 [1 mark · MSQ (one or more correct)]

Which of the following statements about the perceptron is/are CORRECT?

- (A) A perceptron with a bias term can only realise decision boundaries passing through the origin.
- (B) For separable data, the separating hyperplane found by the perceptron can depend on the order in which examples are presented.
- (C) A single perceptron cannot represent the XOR function of two binary inputs.
- (D) In the perceptron learning rule (labels ± 1), the weights change only on misclassified examples.

Q.3 [1 mark · NAT (numerical answer)]

A clustering algorithm groups labelled data into three clusters; the contingency table of cluster versus true class counts is given below. The purity of this clustering is _____ (rounded off to two decimal places)

	Class P	Class Q	Class R
Cluster 1	10	6	6
Cluster 2	18	4	6
Cluster 3	16	1	11

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about k-means clustering is/are CORRECT?

- (A) Running k-means from several random initializations and keeping the run with lowest SSE is a common remedy for poor local optima.
- (B) Finding the globally optimal k-means partition is NP-hard in general.
- (C) Rescaling one feature (e.g. changing its units) can change the clusters found by k-means.
- (D) With Euclidean distance, the boundary between two k-means clusters is the perpendicular bisector (hyperplane) of their centroids.

Q.5 [1 mark · MSQ (one or more correct)]

Which of the following statements about basis-function (polynomial) regression and feature scaling is/are CORRECT?

- (A) With n distinct x values, a polynomial of degree $n - 1$ can fit the n training points exactly.
- (B) Adding x^2 as an extra feature makes the least-squares objective non-convex.
- (C) Polynomial regression is non-linear in its parameters and therefore cannot be solved with the normal equations.
- (D) Ridge regression solutions are invariant to rescaling individual features.

Q.6 [1 mark · NAT (numerical answer)]

For a data set, $\bar{x} = 38$, $\bar{y} = 68$, $s_x = 8$, $s_y = 7$ and the sample correlation between x and y is $r = -0.85$. The slope of the least-squares regression line of y on x is _____ (rounded off to two decimal places)

Q.7 [1 mark · MSQ (one or more correct)]

Which of the following statements about decision-tree learning is/are CORRECT?

- (A) The information gain of a poorly chosen split can be negative.
- (B) Greedy ID3 always finds the smallest tree consistent with the training data.
- (C) A tree with finitely many axis-parallel splits can represent the diagonal boundary $x_1 = x_2$ exactly on continuous inputs.
- (D) A decision tree's training predictions are unchanged by a strictly increasing transformation of a feature (e.g. $x \rightarrow \log x$).

Q.8 [1 mark · NAT (numerical answer)]

A 4-class softmax classifier produces the logits $z_1 = 0, z_2 = 3, z_3 = 2, z_4 = 1$ for an input $\mathbf{x}$. The ratio $P(y = 1 | \mathbf{x}) / P(y = 4 | \mathbf{x})$ is _____ (rounded off to two decimal places)

Q.9 [1 mark · NAT (numerical answer)]

The distance between the parallel hyperplanes $3x_1 + 4x_3 - 5 = 0$ and $3x_1 + 4x_3 - 1 = 0$ is _____ (rounded off to two decimal places)

Q.10 [1 mark · NAT (numerical answer)]

Two shopping baskets are encoded as binary vectors (1 = item bought). The difference (simple matching coefficient – Jaccard similarity) between basket A and basket B is _____ (rounded off to two decimal places)

Item	milk	bread	eggs	butter	jam	tea	rice
Basket A	0	0	0	0	1	1	1
Basket B	0	0	0	0	0	1	0

Q.11 [1 mark · MSQ (one or more correct)]

Which of the following statements about naive Bayes classifiers is/are CORRECT?

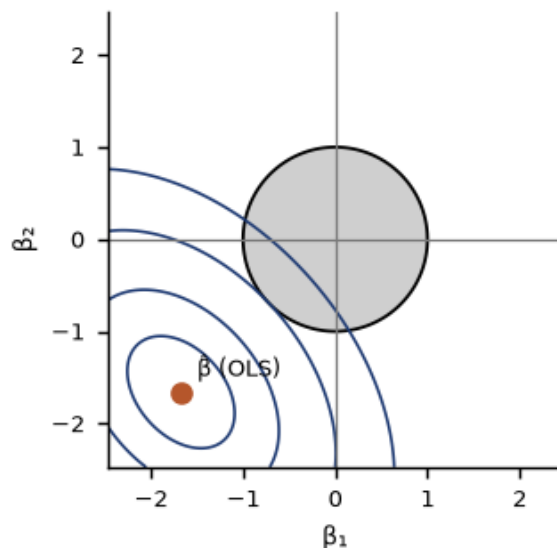
- (A) Naive Bayes is a discriminative classifier that models $P(y | \mathbf{x})$ directly.
- (B) The Laplace-smoothed estimate $(n_{v,c} + 1)/(n_c + V)$ does not depend on the number V of possible feature values.
- (C) Maximum-likelihood training of categorical naive Bayes requires only counting (closed form).
- (D) Naive Bayes cannot be applied to problems with more than two classes.

Q.12 [1 mark · NAT (numerical answer)]

Points are drawn uniformly from the unit hypercube $[0, 1]^5$. The fraction of points lying within distance 0.01 of the boundary (i.e. outside the inner cube $[0.01, 0.99]^5$) is _____ (rounded off to two decimal places)

Q.13 [1 mark · MCQ]

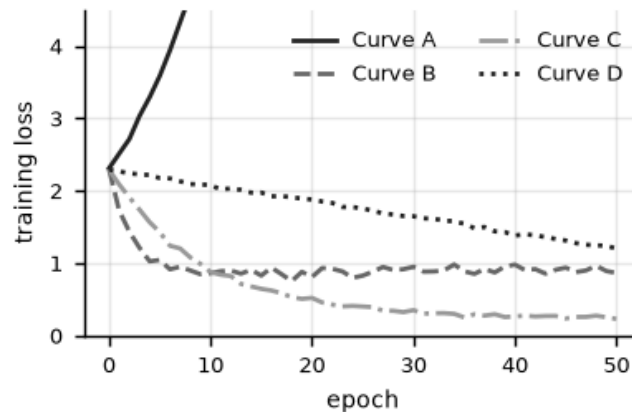
For a two-predictor penalised least-squares problem, the figure shows the constraint region (shaded) of the penalty together with elliptical contours of the residual sum of squares centred at the least-squares estimate β (OLS). The penalised estimate (β_1, β_2) satisfies



- (A) $\beta_1 \neq 0$ and $\beta_2 \neq 0$ (both shrunk, neither exactly zero)
- (B) $\beta_1 = 0$ exactly and $\beta_2 \neq 0$
- (C) $\beta_1 = \beta_2 = 0$
- (D) $\beta_2 = 0$ exactly and $\beta_1 \neq 0$

Q.14 [1 mark · MCQ]

The training-loss curves below were obtained with four different learning rates (everything else fixed). Which curve corresponds to a learning rate that is too low (very slow, almost linear decrease)?



- (A) Curve A
(C) Curve C

- (B) Curve B
(D) Curve D

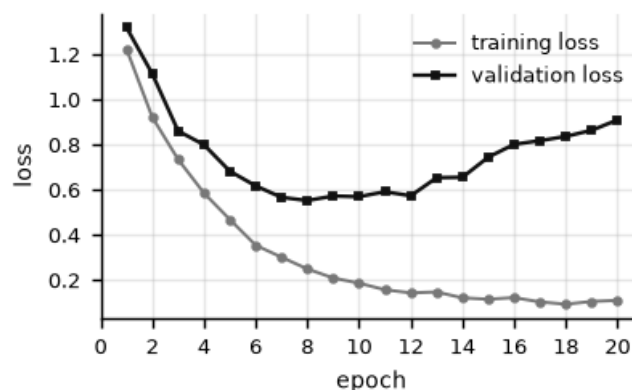
Q.15 [1 mark · MSQ (one or more correct)]

Which of the following statements about the representational power of feed-forward neural networks is/are CORRECT?

- (A) Every Boolean function of n binary inputs can be represented by an MLP with a single hidden layer of threshold units (possibly with exponentially many hidden units).
(B) XOR can be computed by an MLP with one hidden layer containing two threshold units.
(C) The universal approximation theorem is an existence result; it does not say that gradient descent will find the approximating weights.
(D) A single sigmoid neuron (logistic regression) has a linear decision boundary.

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

A network is trained with early stopping on validation loss using patience $P = 2$: training stops as soon as the validation loss has failed to improve on its best value for 2 consecutive epochs, and the weights from the best epoch are restored. Using the validation losses below (also plotted), the epoch whose weights are returned (restored) as the final model is ____ (Answer in integer)



epoch	1	2	3	4	5	6	7	8	9	10
val. loss	1.320	1.114	0.861	0.800	0.683	0.618	0.567	0.553	0.573	0.570

epoch	11	12	13	14	15	16	17	18	19	20
val. loss	0.592	0.574	0.654	0.656	0.745	0.802	0.818	0.837	0.864	0.910

Q.17 [2 marks · NAT (numerical answer)]

Agglomerative clustering with **single linkage** is applied to the distance matrix below. The cophenetic distance between P1 and P3 (the dendrogram height at which they are first joined) is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5
P1	0	5	10	8	9
P2		0	11	3	10
P3			0	8	1
P4				0	9
P5					0

Q.18 [2 marks · MSQ (one or more correct)]

For the confusion matrix below, which of the following statements is/are CORRECT? (values rounded to 2 decimals)

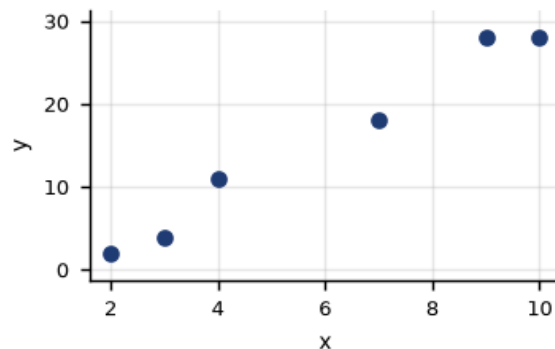
	Predicted +	Predicted –
Actual +	73	68
Actual –	8	66

- (A) The balanced accuracy is 0.7.
- (B) The precision is 0.52.
- (C) The F1-score is less than the arithmetic mean of precision and recall.
- (D) The specificity (TNR) is 0.89.

Q.19 [2 marks · NAT (numerical answer)]

A simple linear regression (with intercept) is fitted by least squares to the 6 points below. Using leave-one-out cross-validation, the leave-one-out residual $y_i - \hat{y}_{(-i)}(x_i)$ for the point (3, 4) — where $\hat{y}_{(-i)}$ is the line fitted without that point — is _____ (rounded off to two decimal places)

x	2	3	4	7	9	10
y	2	4	11	18	28	28

**Q.20 [2 marks · NAT (numerical answer)]**

The labelled data below are linearly separable through the origin by the unit vector $u = (0, 1)$ (i.e. $y_i = \text{sign}(u \cdot x_i)$). Take $\gamma = \min_i y_i(u \cdot x_i)$ and $R = \max_i \|x_i\|$. The perceptron convergence theorem (no bias, $w_0 = 0$, $\eta = 1$) bounds the number of mistakes by $(R/\gamma)^2$. The value of this bound is _____ (rounded off to two decimal places)

i	1	2	3	4	5
x_i	(-2, -1)	(2, 2)	(1, 2)	(-3, 4)	(-1, 2)
y_i	-1	1	1	1	1

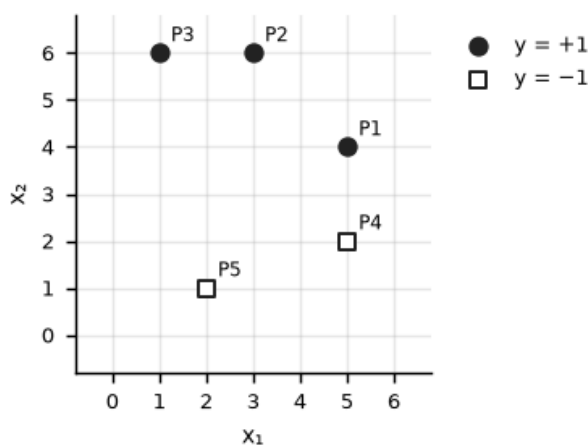
Q.21 [2 marks · NAT (numerical answer)]

A simple linear regression fitted by ordinary least squares gives $\hat{y} = 14.6 - 1x$. After standardising x as $z = (x - \bar{x})/s_x$, with $\bar{x} = 59$ and $s_x = 5$, the model is refitted as $\hat{y} = \gamma_0 + \gamma_1 z$. The value of γ_1 is _____ (rounded off to two decimal places)

Q.22 [2 marks · NAT (numerical answer)]

A hard-margin linear SVM is trained on the points below. Solving the dual problem gives the Lagrange multipliers α_i listed in the table. Using the KKT conditions, compute the weight component w_1 . (rounded off to two decimal places)

Point	x_1	x_2	y	α_i
P1	5	4	+1	0.5
P2	3	6	+1	0
P3	1	6	+1	0
P4	5	2	-1	0.5
P5	2	1	-1	0

**Q.23 [2 marks · NAT (numerical answer)]**

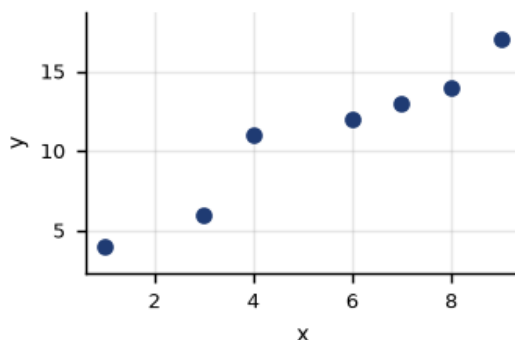
Two clusterings U and V of 7 points are given below (cluster names are arbitrary). Then the Rand index between the two clusterings is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5	P6	P7
Clustering U	β	α	β	α	β	β	α
Clustering V	2	1	1	1	2	2	1

Q.24 [2 marks · NAT (numerical answer)]

Assume $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with ε_i i.i.d. $N(0, \sigma^2)$. For the 7 observations below, with β estimated by least squares (= maximum likelihood), the difference between the unbiased estimate $SSE/(n - 2)$ and the maximum-likelihood estimate of σ^2 is _____ (rounded off to two decimal places)

x	1	3	4	6	7	8	9
y	4	6	11	12	13	14	17



Q.25 [2 marks · MSQ (one or more correct)]

Consider k-means (Lloyd's algorithm) and k-medoids (PAM). Which of the following statements is/are CORRECT?

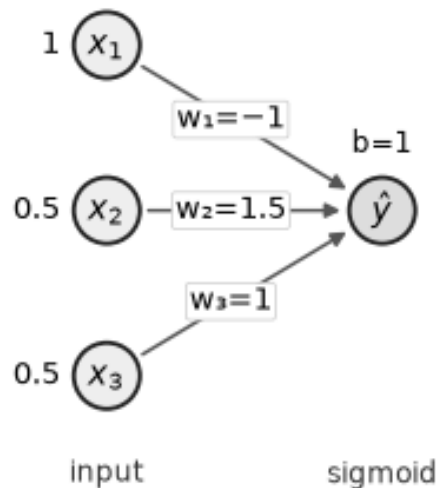
- (A) k-medoids is generally more robust to outliers than k-means.
- (B) For mean centroids, total sum of squares = within-cluster SS + between-cluster SS.
- (C) Multiplying one feature by 10 never changes the k-means partition.
- (D) k-means++ initialization guarantees the globally optimal clustering.

Q.26 [2 marks · NAT (numerical answer)]

The 2×2 matrix $X^T X$ of a (centred) design matrix X has eigenvalues 6, 4. For ridge regression with penalty $\lambda \|\beta\|_2^2$, the effective degrees of freedom $df(\lambda) = \text{trace}[X(X^T X + \lambda I)^{-1} X^T]$ for $\lambda = 4$ is _____ (rounded off to two decimal places)

Q.27 [2 marks · NAT (numerical answer)]

A single sigmoid neuron (shown below) receives $x = (1, 0.5, 0.5)$ and has target $y = 0$. The loss is $L = \frac{1}{2}(\hat{y} - y)^2$. The value of $\partial L / \partial w_1$ is _____ (rounded off to three decimal places)

**Q.28 [2 marks · MSQ (one or more correct)]**

Which of the following statements about LDA, QDA and logistic regression is/are CORRECT?

- (A) Changing the class priors in LDA shifts the decision boundary but does not change its orientation.
- (B) When the Gaussian shared-covariance assumption holds, LDA can estimate the boundary with lower variance than logistic regression.
- (C) LDA is a discriminative model that never models $P(x | y)$.
- (D) The LDA discriminant function $\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln \pi_k$ is linear in x .

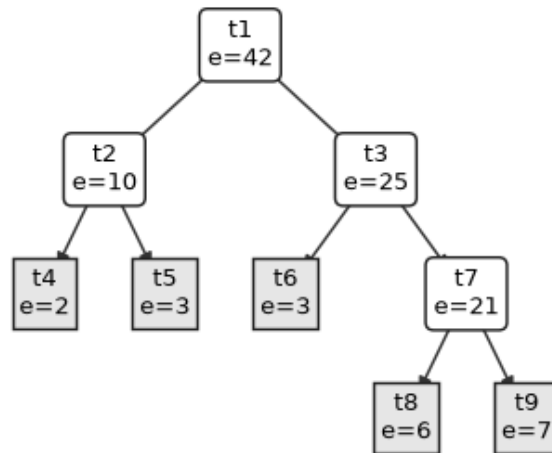
Q.29 [2 marks · NAT (numerical answer)]

A one-feature logistic regression model $P(y = 1 | x) = \sigma(wx + b)$ has $w = -1$ and $b = -1$. It is evaluated on the 5 examples below. Using natural logarithms, the **average** binary cross-entropy loss $-[y \ln \hat{p} + (1 - y) \ln(1 - \hat{p})]$ over these examples is _____ (rounded off to two decimal places)

i	1	2	3	4	5
x_i	-2	-1	0	0	0
y_i	1	0	1	0	0

Q.30 [2 marks · NAT (numerical answer)]

A classification tree was grown on $N = 100$ training records. Each node shows $e(t)$, the number of training records that would be misclassified if t were a leaf; $R(t) = e(t)/N$. The smallest value of α at which cost-complexity pruning first prunes the tree (the weakest link) is _____ (rounded off to three decimal places)

**Q.31 [2 marks · MSQ (one or more correct)]**

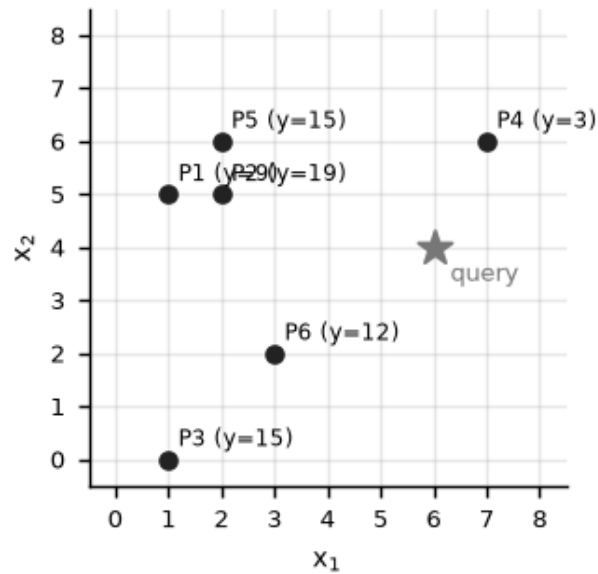
Which of the following statements about PCA, SVD and dimensionality reduction is/are CORRECT?

- (A) For a covariance matrix $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$ with $b > 0$, PC1 is along $(1, 1)/\sqrt{2}$.
- (B) In high dimensions, pairwise distances between random points concentrate, so nearest and farthest neighbours become similar in distance.
- (C) Fisher LDA for C classes can give up to $C + 1$ discriminant directions.
- (D) Rotating the data by an orthogonal matrix rotates the PC directions but leaves the eigenvalues unchanged.

Q.32 [2 marks · NAT (numerical answer)]

A distance-weighted k-NN regressor with $k = 3$ predicts $\hat{y} = \sum w_i y_i / \sum w_i$ over the 3 nearest neighbours, with weights $w_i = 1/d_i^2$. For the training data below, the prediction at $\mathbf{x} = (6, 4)$ (Euclidean distance) is _____ (rounded off to two decimal places)

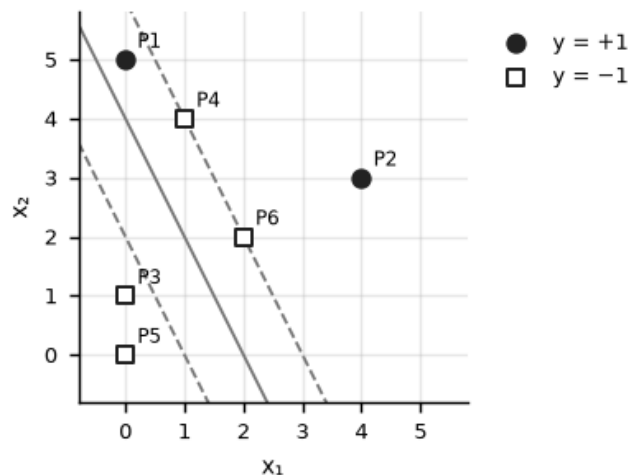
Point	x_1	x_2	y
P1	1	5	9
P2	2	5	19
P3	1	0	15
P4	7	6	3
P5	2	6	15
P6	3	2	12



Q.33 [2 marks · MSQ (one or more correct)]

A soft-margin linear SVM has learned $f(x) = x_1 + 0.5x_2 - 2$. The training points are listed below and plotted (boundary solid, margins $f = \pm 1$ dashed). With ξ_i the optimal slack of point i , which of the following statements is/are CORRECT?

Point	x_1	x_2	y
P1	0	5	+1
P2	4	3	+1
P3	0	1	-1
P4	1	4	-1
P5	0	0	-1
P6	2	2	-1



- (A) $\sum \xi_i = 4.5$.
 (B) P5 has slack $\xi_5 = 0$.
 (C) Exactly 3 of the points have $\xi_i > 0$.
 (D) P3 has slack $\xi_3 = 0$.

Q.34 [2 marks · MSQ (one or more correct)]

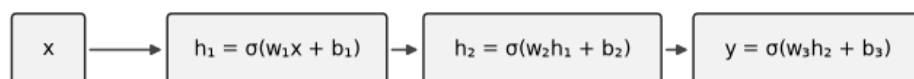
A node with class counts [40+, 60-] can be split by either of two binary splits S1 or S2 (table). For each impurity measure, the “gain” is the parent impurity minus the weighted child impurity. Which of the following statements is/are CORRECT?

Split	Left child [+, -]	Right child [+, -]
S1	[8+, 31-]	[32+, 29-]
S2	[13+, 32-]	[27+, 28-]

- (A) Misclassification-error reduction is the same for S1 and S2.
 (B) Gini gain prefers S1 over S2.
 (C) Misclassification-error reduction prefers S2 over S1.
 (D) Information gain (entropy) prefers S1 over S2.

Q.35 [2 marks · NAT (numerical answer)]

Consider the scalar chain shown below with weights $w_1 = 3$, $w_2 = -2$, $w_3 = 3$, biases $b_1 = 0$, $b_2 = 0$, $b_3 = 0.5$ and activation the logistic sigmoid σ at every stage. For input $x = 0.5$, the value of $\partial y / \partial x$ is _____ (rounded off to 4 decimal places)



— END OF QUESTION PAPER —

Mock Test 16 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MCQ	1	(B)	19	NAT	2	−2.1988 to −2.1552
2	MSQ	1	(B), (C), (D)	20	NAT	2	24.95 to 25.05
3	NAT	1	0.5541 to 0.5741	21	NAT	2	−5.01 to −4.99
4	MSQ	1	(A), (B), (C), (D)	22	NAT	2	−0.01 to 0.01
5	MSQ	1	(A)	23	NAT	2	0.7043 to 0.7243
6	NAT	1	−0.7538 to −0.7338	24	NAT	2	0.4266 to 0.4666
7	MSQ	1	(D)	25	MSQ	2	(A), (B)
8	NAT	1	0.3579 to 0.3779	26	NAT	2	1.09 to 1.11
9	NAT	1	0.79 to 0.81	27	NAT	2	0.1326 to 0.1366
10	NAT	1	0.371 to 0.391	28	MSQ	2	(A), (B), (D)
11	MSQ	1	(C)	29	NAT	2	0.5792 to 0.5992
12	NAT	1	0.0861 to 0.1061	30	NAT	2	0.0435 to 0.0465
13	MCQ	1	(A)	31	MSQ	2	(A), (B), (D)
14	MCQ	1	(D)	32	NAT	2	7.8552 to 7.8752
15	MSQ	1	(A), (B), (C), (D)	33	MSQ	2	(A), (B), (C), (D)
16	NAT	2	8	34	MSQ	2	(B), (D)
17	NAT	2	7.99 to 8.01	35	NAT	2	−0.0731 to −0.0717
18	MSQ	2	(A), (C), (D)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
SVM & Decision Trees					11
Neural Networks (MLP / Feed-forward)					11
Unsupervised Learning					11
Regression & Model Selection					11
Probabilistic & Instance-based Classifiers					11
Total					55

Mock Test 16 — Detailed Solutions

Q.1 — Answer: (B) [MCQ, 1M · SVM & Decision Trees > Decision trees: structure & traversal]

Follow the record from the root, taking the branch that matches each test:

Visits = $20 \leq 21 \rightarrow \text{Yes}$.

Spend = $25 \leq 33 \rightarrow \text{Yes}$.

(Only the tests on the actual path matter.) The record reaches the leaf **Platinum**.

Note: " $\leq$ " includes equality, so a value equal to the threshold goes to the Yes branch.

Q.2 — Answer: (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

(A) **FALSE**. The bias b shifts the hyperplane away from the origin.

(B) **TRUE**. Different orders lead to different update sequences and final weights.

(C) **TRUE**. The positive points (0,1),(1,0) and negative points (0,0),(1,1) are not linearly separable.

(D) **TRUE**. Update $w \leftarrow w + \eta yx$ is applied only when $y(w \cdot x + b) \leq 0$.

Q.3 — Answer: 0.5541 to 0.5741 [NAT, 1M · Unsupervised Learning > Cluster evaluation]

Majority count in each cluster: 10, 18, 16 (sum 44); $N = 78$.

Purity = $(1/N) \sum_k \max_j n_{kj} = 44/78 = 0.5641$.

Misclassified = $N - \sum \max = 34$.

Answer: **0.56**.

Purity is an external measure (needs class labels) and is trivially 1 if each point forms its own cluster, so it cannot be used to choose k .

Q.4 — Answer: (A), (B), (C), (D) [MSQ, 1M · Unsupervised Learning > k-means]

(A) **TRUE**. Multiple restarts reduce the chance of a bad local minimum.

(B) **TRUE**. Hence the use of the Lloyd heuristic.

(C) **TRUE**. Euclidean distances change non-uniformly, so assignments can change.

(D) **TRUE**. Points equidistant from two centroids form that hyperplane.

Q.5 — Answer: (A) [MSQ, 1M · Regression & Model Selection > Basis functions & feature scaling]

(A) **TRUE**. Lagrange interpolation; training RSS = 0.

(B) **FALSE**. It remains a convex quadratic in the weights.

(C) **FALSE**. It is linear in the parameters.

(D) **FALSE**. They are not.

Q.6 — Answer: -0.7538 to -0.7338 [NAT, 1M · Regression & Model Selection > Simple linear regression]

$\beta_1 = r \cdot s_y / s_x = -0.85 \times 7/8 = -0.7438$.

Answer: **-0.74**.

Q.7 — Answer: (D) [MSQ, 1M · SVM & Decision Trees > Decision trees: overfitting & pruning]

(A) **FALSE**. $IG \geq 0$ always.

(B) **FALSE**. Greedy $\neq$ optimal.

(C) **FALSE**. Only a staircase approximation is possible.

(D) **TRUE**. Splits depend only on the ordering of values.

Q.8 — Answer: 0.3579 to 0.3779 [NAT, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

The normalising constant $\sum e^{z_i}$ cancels in the ratio:

$P(1)/P(4) = e^{z_1 - z_4} = e^{-1} = \mathbf{0.37}$.

Q.9 — Answer: 0.79 to 0.81 [NAT, 1M · SVM & Decision Trees > SVM: margin & geometry]

For $w^T x + b_1 = 0$ and $w^T x + b_2 = 0$, distance = $|b_1 - b_2| / \|w\|$.

= $|-5 - (-1)|/5 = 4/5 = \mathbf{0.8}$.

Q.10 — Answer: 0.371 to 0.391 [NAT, 1M · Unsupervised Learning > Distance measures]

Counts: $f_{11} = 1$, $f_{10} = 2$, $f_{01} = 0$, $f_{00} = 4$.

Jaccard $J = f_{11}/(f_{11}+f_{10}+f_{01}) = 1/3 = 0.3333$ (0-0 matches are ignored).

SMC $= (f_{11}+f_{00})/m = 5/7 = 0.7143$.

Required difference (simple matching coefficient – Jaccard similarity) = **0.38**.

Q.11 — Answer: (C) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

(A) **FALSE**. It is generative.

(B) **FALSE**. V appears in the denominator.

(C) **TRUE**. Estimates are relative frequencies.

(D) **FALSE**. It handles any K .

Q.12 — Answer: 0.0861 to 0.1061 [NAT, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Inner cube volume $= (1 - 2 \times 0.01)^5 = 0.98^5 = 0.9039$.

Fraction near the boundary $= 1 - 0.9039 = \mathbf{0.1}$.

In high dimensions most points lie near the boundary, far from each other — a symptom of the curse of dimensionality that hurts k-NN.

Q.13 — Answer: (A) [MCQ, 1M · Regression & Model Selection > Ridge & Lasso]

The shaded region is the constraint set of the ridge (L2: $\beta_1^2 + \beta_2^2 \leq t^2$, a disc). The penalised estimate is the first point at which the RSS contours (ellipses centred at β_{OLS} , which lies outside the region) touch the region.

From the figure the tangency point is approximately $(-0.73, -0.69)$.

The contour touches the boundary at a point that is not on an axis, so both coefficients are non-zero. For the disc this is the generic case — ridge shrinks but does not zero out.

Answer: **$\beta_1 \neq 0$ and $\beta_2 \neq 0$ (both shrunk, neither exactly zero)**. Both coefficients cannot be zero since the OLS point lies outside the region (the constraint is active but $t > 0$).

Q.14 — Answer: (D) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Typical signatures: far-too-high $\eta \rightarrow$ loss blows up; somewhat-too-high $\eta \rightarrow$ quick drop then plateau at a high loss (bouncing around the minimum); too-low $\eta \rightarrow$ slow, nearly linear decrease; good $\eta \rightarrow$ fast decrease to the lowest loss.

Curve A: a learning rate that is far too high (training diverges).

Curve B: a learning rate that is somewhat too high (fast initial drop, then the loss plateaus at a high value).

Curve C: a well-chosen learning rate.

Curve D: a learning rate that is too low (very slow, almost linear decrease).

Hence the answer is **Curve D**.

Q.15 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Expressiveness]

(A) **TRUE**. One hidden unit per positive input pattern (DNF).

(B) **TRUE**. E.g. $h_1 = \text{OR}$, $h_2 = \text{NAND}$, $y = \text{AND}(h_1, h_2)$.

(C) **TRUE**. It says nothing about learnability or sample complexity.

(D) **TRUE**. $\sigma(w \cdot x + b) = 0.5 \Leftrightarrow w \cdot x + b = 0$.

Q.16 — Answer: 8 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Rule: keep track of the best (lowest) validation loss; an epoch that does not strictly improve on it increments a counter, which is reset on improvement; stop when the counter reaches patience $P = 2$.

Trace – 1: 1.320 (new best); 2: 1.114 (new best); 3: 0.861 (new best); 4: 0.800 (new best); 5: 0.683 (new best); 6: 0.618 (new best); 7: 0.567 (new best); 8: 0.553 (new best); 9: 0.573 (no improvement, count = 1); 10: 0.570 (no improvement, count = 2).

Training stops after epoch 10; the best validation loss 0.553 occurred at epoch 8, whose weights are restored.

Answer: **8**. The widening train/validation gap after this point is the signature of overfitting.

Q.17 — Answer: 7.99 to 8.01 [NAT, 2M · Unsupervised Learning · Hierarchical clustering]

Use single linkage: $d(A, B) = \min_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

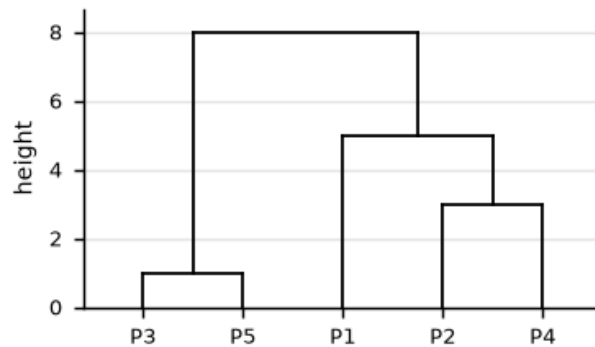
Merge 1: $\{P3\} + \{P5\}$ at height 1. Updated: $d(\{P3, P5\}, \{P1\}) = 9$, $d(\{P3, P5\}, \{P2\}) = 10$, $d(\{P3, P5\}, \{P4\}) = 8$.

Merge 2: $\{P2\} + \{P4\}$ at height 3. Updated: $d(\{P2, P4\}, \{P1\}) = 5$, $d(\{P2, P4\}, \{P3, P5\}) = 8$.

Merge 3: $\{P1\} + \{P2, P4\}$ at height 5. Updated: $d(\{P1, P2, P4\}, \{P3, P5\}) = 8$.

Merge 4: $\{P3, P5\} + \{P1, P2, P4\}$ at height 8.

P1 and P3 first lie in the same cluster at the merge of height 8, so their cophenetic distance is 8 (their original distance was 10).

**Q.18 — Answer: (A), (C), (D)** [MSQ, 2M · Probabilistic & Instance-based Classifiers · Evaluation metrics]

TP = 73, FN = 68, FP = 8, TN = 66.

(A) **TRUE**. balanced accuracy = $(\text{TPR} + \text{TNR})/2 = 0.7048$.

(B) **FALSE**. precision = $\text{TP}/(\text{TP} + \text{FP}) = 0.9012$, not 0.52.

(C) **TRUE**. The harmonic mean is $\leq$ the arithmetic mean, with equality only if $P = R$ (not the case here).

(D) **TRUE**. specificity (TNR) = $\text{TN}/(\text{TN} + \text{FP}) = 0.8919$.

Q.19 — Answer: -2.1988 to -2.1552 [NAT, 2M · Regression & Model Selection · Cross-validation]

$\bar{x} = 5.833$, $\bar{y} = 15.167$.

i	1	2	3	4	5	6
$x_i - \bar{x}$	-3.83	-2.83	-1.83	1.17	3.17	4.17
$y_i - \bar{y}$	-13.17	-11.17	-4.17	2.83	12.83	12.83
$(x_i - \bar{x})(y_i - \bar{y})$	50.47	31.64	7.64	3.31	40.64	53.47

$S_{xx} = \sum (x_i - \bar{x})^2 = 54.833$, $S_{xy} = 187.167$.

$\beta_1 = S_{xy}/S_{xx} = 3.4134$, $\beta_0 = \bar{y} - \beta_1 \bar{x} = -4.7447$.

For a linear smoother, the leave-one-out residual is $e_{(i)} = e_i/(1 - h_{ii})$, with $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$.

x_i	2	3	4	7	9	10
e_i	-0.082	-1.495	2.091	-1.149	2.024	-1.389
h_{ii}	0.435	0.313	0.228	0.191	0.35	0.483
$e_i/(1 - h_{ii})$	-0.145	-2.177	2.709	-1.421	3.112	-2.688

Answer: **-2.18** (verified by explicitly refitting without the point).

Q.20 — Answer: 24.95 to 25.05 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Perceptron]

Unit separator $u = (0, 1)$. Margins $y_i(u \cdot x_i)$: 1, 2, 2, 4, 2 $\rightarrow \gamma = \min = 1$.

Norms $\|x_i\|$: 2.236, 2.828, 2.236, 5, 2.236 $\rightarrow R = \max = 5$.

Novikoff bound: number of mistakes $\leq (R/\gamma)^2 = (5/1)^2 = 25$.

Common mistakes: forgetting to normalise u (γ must be measured with $\|u\| = 1$), or using R/γ without squaring.

Q.21 — Answer: -5.01 to -4.99 [NAT, 2M · Regression & Model Selection > Basis functions & feature scaling]

Since $x = \bar{x} + s_x z$, $\hat{y} = \beta_0 + \beta_1 \bar{x} + \beta_1 s_x z$, and OLS with intercept is equivariant under affine maps of x (same fitted values).

$$\gamma_0 = \beta_0 + \beta_1 \bar{x} = 14.6 + (-1)(59) = -44.4 (= \bar{y}, \text{ since } z \text{ is centred});$$

$$\gamma_1 = \beta_1 s_x = (-1)(5) = -5.$$

Answer: **-5**.

Q.22 — Answer: -0.01 to 0.01 [NAT, 2M · SVM & Decision Trees > SVM: dual & support vectors]

$w = \sum \alpha_i y_i x_i$ over points with $\alpha_i > 0$:

$$w = (0.5)(+1)(5, 4) + (0.5)(-1)(5, 2) = (0, 1).$$

b from any support vector, e.g. P1: $y = w^T x + b \Rightarrow b = +1 - (4) = -3$.

Check: $\sum \alpha_i y_i = 0$ ($= 0$, as required).

Answer: **0**.

Q.23 — Answer: 0.7043 to 0.7243 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

Total pairs = $C(7, 2) = 21$.

a = pairs together in both = 6; b = pairs separated in both = 9; pairs together in U only = 3; together in V only = 3.

$$\text{Rand index} = (a + b)/C(n, 2) = (6 + 9)/21 = 0.7143.$$

Answer: **0.71**. (Cluster names do not matter: only whether pairs are together.)

Q.24 — Answer: 0.4266 to 0.4666 [NAT, 2M · Regression & Model Selection > MLE & MAP view]

$\bar{x} = 5.429$, $\bar{y} = 11$.

i	1	2	3	4	5	6	7
$x_i - \bar{x}$	-4.43	-2.43	-1.43	0.57	1.57	2.57	3.57
$y_i - \bar{y}$	-7	-5	0	1	2	3	6
$(x_i - \bar{x})(y_i - \bar{y})$	31	12.14	0	0.57	3.14	7.71	21.43

$$S_{xx} = \sum (x_i - \bar{x})^2 = 49.714, S_{xy} = 76.$$

$$\beta_1 = S_{xy}/S_{xx} = 1.5287, \beta_0 = \bar{y} - \beta_1 \bar{x} = 2.7011.$$

$$SSE = S_{yy} - S_{xy}^2/S_{xx} = 124 - 116.1839 = 7.8161.$$

$$SSE/(n - 2) - SSE/n = 1.5632 - 1.1166 = 0.4466.$$

Answer: **0.45**.

Q.25 — Answer: (A), (B) [MSQ, 2M · Unsupervised Learning > k-means]

(A) TRUE. A medoid is an actual point and uses unsquared dissimilarities, so an outlier pulls it far less.

(B) TRUE. The standard ANOVA decomposition; reducing WSS increases BSS.

(C) FALSE. Non-uniform scaling changes distances and can change the partition.

(D) FALSE. It only gives an $O(\log k)$ approximation in expectation.

Q.26 — Answer: 1.09 to 1.11 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

Ridge fitted values $\hat{y} = H_\lambda y$ with $H_\lambda = X(X^T X + \lambda I)^{-1} X^T$. Writing $X^T X = V D V^T$ with eigenvalues d_j^2 gives $\text{trace}(H_\lambda) = \sum d_j^2 / (d_j^2 + \lambda)$ (effective degrees of freedom).

$$6/(6 + 4) + 4/(4 + 4) = 1.1.$$

Answer: **1.1**.

Q.27 — Answer: 0.1326 to 0.1366 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

$$a = w \cdot x + b = 1.25; \hat{y} = \sigma(a) = 0.7773.$$

$$L = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial L / \partial \hat{y} = \hat{y} - y = 0.7773; \partial \hat{y} / \partial a = \hat{y}(1 - \hat{y}) = 0.1731.$$

$$\delta = \partial L / \partial a = (\hat{y} - y)\hat{y}(1 - \hat{y}) = 0.1346 (= \partial L / \partial b).$$

$$\partial L / \partial w_1 = \delta \cdot x_1 = 0.1346 \times 1 = 0.1346.$$

Answer: **0.1346**.

Forgetting the factor $\hat{y}(1 - \hat{y})$ (i.e. treating the output as linear) is the most common error.

Q.28 — Answer: (A), (B), (D) [MSQ, 2M · Probabilistic & Instance-based Classifiers · Linear discriminant analysis]

(A) **TRUE**. Priors only enter the constant term $\ln \pi_k$.

(B) **TRUE**. It uses the extra (correct) distributional information.

(C) **FALSE**. LDA is generative.

(D) **TRUE**. Only first-order terms in x .

Q.29 — Answer: 0.5792 to 0.5992 [NAT, 2M · Probabilistic & Instance-based Classifiers · Logistic regression]

$z_i = -1x_i + -1$; $\hat{p}_i = \sigma(z_i)$: 0.7311, 0.5, 0.2689, 0.2689, 0.2689.

Per-example loss $L_i = -\ln \hat{p}_i$ if $y_i = 1$ and $-\ln(1 - \hat{p}_i)$ if $y_i = 0$:

i	y	$\hat{p}$	L_i
1	1	0.7311	0.3133
2	0	0.5	0.6931
3	1	0.2689	1.3133
4	0	0.2689	0.3133
5	0	0.2689	0.3133

$\Sigma L_i = 2.9462$; average = $2.9462/5 = \mathbf{0.59}$.

Common mistakes: using $\ln \hat{p}$ for $y = 0$ examples, or $\log_{10}$ instead of $\ln$.

Q.30 — Answer: 0.0435 to 0.0465 [NAT, 2M · SVM & Decision Trees · Decision trees: overfitting & pruning]

$R(t) = e(t)/N$ with $N = 100$. For an internal node t with subtree T_t : $g(t) = [R(t) - R(T_t)]/(|T_t| - 1)$, where $|T_t|$ = number of leaves.

Node	$R(t)$	$R(T_t)$	$ T_t $	$g(t)$
t1	0.42	0.21	5	0.0525
t2	0.1	0.05	2	0.05
t3	0.25	0.16	3	0.045
t7	0.21	0.13	2	0.08

The weakest link is t3 with the minimum $g = \mathbf{0.045}$; for $\alpha \geq$ this value, collapsing t3 into a leaf does not increase R_α .

Q.31 — Answer: (A), (B), (D) [MSQ, 2M · Unsupervised Learning · PCA & dimensionality reduction]

(A) **TRUE**. Eigenvalue $a + b$.

(B) **TRUE**. A symptom of the curse of dimensionality.

(C) **FALSE**. At most $C - 1$.

(D) **TRUE**. $S \rightarrow RSR^T$ is a similarity transform.

Q.32 — Answer: 7.8552 to 7.8752 [NAT, 2M · Probabilistic & Instance-based Classifiers · k-nearest neighbours]

Three nearest neighbours: P4 ($d^2=5$, $y=3$), P6 ($d^2=13$, $y=12$), P2 ($d^2=17$, $y=19$).

Weights $1/d^2$: 0.2, 0.0769, 0.0588; $\Sigma w = 0.3357$.

$\Sigma w_i y_i = 2.6407 \Rightarrow \hat{y} = 2.6407/0.3357 = \mathbf{7.87}$.

(The unweighted 3-NN average would be 11.33; weighting pulls $\hat{y}$ towards the closest point.)

Q.33 — Answer: (A), (B), (C), (D) [MSQ, 2M · SVM & Decision Trees · SVM: soft margin & hinge loss]

$\xi_i = \max(0, 1 - y_i f(x_i))$.

Point	$y f(x)$	ξ
P1	0.5	0.5
P2	3.5	0
P3	1.5	0

P4	-1	2
P5	2	0
P6	-1	2

(A) TRUE. Sum of the individual slacks.

(B) TRUE. $y f = 2 \Rightarrow \xi = 0$.

(C) TRUE. Count of points with $y f < 1$.

(D) TRUE. $y f = 1.5 \Rightarrow \xi = 0$.

Q.34 — Answer: (B), (D) [MSQ, 2M · SVM & Decision Trees › Decision trees: impurity measures]

Parent [40+, 60-]: $H = 0.971$, $Gini = 0.48$, $error = 0.4$.

S1: children $H = 0.7321, 0.9983$; $Gini = 0.3261, 0.4988$; $error = 0.2051, 0.4754$. Gains: $IG = 0.0765$, $Gini = 0.0486$, $error = 0.03$.

S2: children $H = 0.8673, 0.9998$; $Gini = 0.4109, 0.4998$; $error = 0.2889, 0.4909$. Gains: $IG = 0.0308$, $Gini = 0.0202$, $error = 0$.

(A) FALSE. S1: 0.03, S2: 0.

(B) TRUE. S1: 0.0486, S2: 0.0202.

(C) FALSE. S1: 0.03, S2: 0.

(D) TRUE. S1: 0.0765, S2: 0.0308.

Q.35 — Answer: -0.0731 to -0.0717 [NAT, 2M · Neural Networks (MLP / Feed-forward) ›

Activation functions]

Forward: $a_1 = 3 \cdot 0.5 + 0 = 1.5$, $h_1 = 0.8176$; $a_2 = -2 \cdot 0.8176 + 0 = -1.6351$, $h_2 = 0.1631$; $a_3 = 3 \cdot 0.1631 + 0.5 = 0.9894$, $y = 0.729$.

Local derivatives $\sigma'(a)$: $\sigma'(a) = \sigma(a)(1 - \sigma(a))$: $\sigma'(a_1) = 0.1491$, $\sigma'(a_2) = 0.1365$, $\sigma'(a_3) = 0.1976$.

Chain rule: $\partial y / \partial x = \sigma'(a_3) \cdot w_3 \cdot \sigma'(a_2) \cdot w_2 \cdot \sigma'(a_1) \cdot w_1 = -0.0724$.

Each layer contributes a factor $\sigma' \leq 0.25$ - the product shrinks with depth (vanishing gradient).

Mock Test 17

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	8	5	22

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Probabilistic & Instance-based Classifiers	7	11
Neural Networks (MLP / Feed-forward)	7	11
Regression & Model Selection	7	11
SVM & Decision Trees	7	11
Unsupervised Learning	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MSQ (one or more correct)]

Which of the following statements about Bayes-optimal classification and generative vs. discriminative models is/are CORRECT?

- (A) A generative model can be used to sample synthetic input vectors $\mathbf{x}$.
- (B) When the misclassification costs are asymmetric, the Bayes-optimal rule need not pick the most probable class.
- (C) Under 0-1 loss, the Bayes classifier predicts $\arg \max_k P(y = k | \mathbf{x})$ and minimises the probability of misclassification.
- (D) Adding a new informative feature can reduce the Bayes error rate.

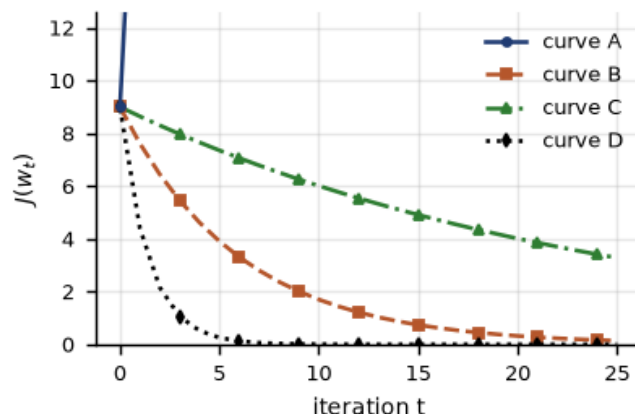
Q.2 [1 mark · MSQ (one or more correct)]

Which of the following statements about backpropagation is/are CORRECT?

- (A) In backpropagation, the deltas of a layer are computed after (from) the deltas of the next layer.
- (B) With $L = \frac{1}{2}(\hat{y} - y)^2$ and a sigmoid output unit, $\delta_{\text{out}} = (\hat{y} - y) \hat{y} (1 - \hat{y})$.
- (C) The gradient with respect to a bias b_j equals δ_j .
- (D) A finite-difference (central difference) check can be used to verify a backprop implementation.

Q.3 [1 mark · MCQ]

Gradient descent $w_{t+1} = w_t - \eta J'(w_t)$ is run on $J(w) = (w - 3)^2$ from $w_0 = 0$ with four learning rates $\eta \in \{0.01, 0.04, 0.15, 1.3\}$. The loss curves are shown below (curves that leave the plotted range are cut off). Which curve corresponds to $\eta = 1.3$?



- (A) Curve B
(C) Curve A

- (B) Curve D
(D) Curve C

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about logistic regression is/are CORRECT?

- (A) For finite z , the sigmoid output $\sigma(z)$ can be exactly 0 or exactly 1.
 (B) The decision boundaries between pairs of classes in softmax (multinomial logistic) regression are linear in x .
 (C) If the training data are linearly separable, the norm of the unregularised logistic-regression weights grows without bound during gradient descent.
 (D) A predicted probability of 0.8 corresponds to log-odds equal to 0.8.

Q.5 [1 mark · MCQ]

Cost-complexity pruning of a classification tree produced the nested subtrees $T1 \supset T2 \supset T3 \supset T4$ below. For complexity parameter $\alpha = 0.015$, which subtree minimises $R_\alpha(T) = R(T) + \alpha|T|$?

Subtree	T1	T2	T3	T4
Leaves $ T $	14	10	6	4
Training error $R(T)$	0.03	0.08	0.17	0.175

- (A) T1
(C) T3

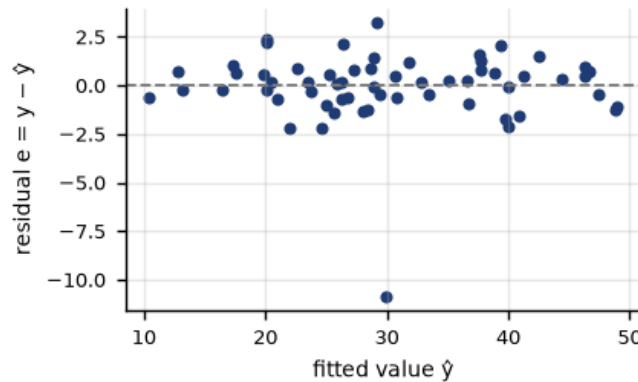
- (B) T2
(D) T4

Q.6 [1 mark · NAT (numerical answer)]

A node of a classification tree contains 18 training records: 8 of class C_1 , 10 of class C_2 . The Gini index of this node is _____ (rounded off to two decimal places)

Q.7 [1 mark · MCQ]

After fitting a linear regression model, the residuals are plotted against the fitted values as shown. The plot most strongly suggests:



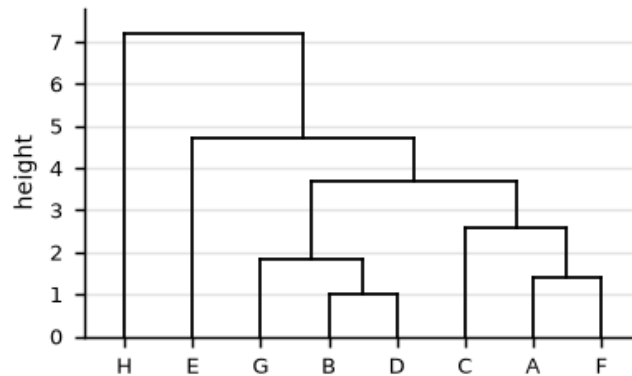
- (A) A single outlying observation, with the rest of the fit looking adequate
- (B) A non-linear relationship not captured by the model (e.g. a missing x^2 term)
- (C) No evident violation of the linear-model assumptions
- (D) Non-constant error variance (heteroscedasticity)

Q.8 [1 mark · NAT (numerical answer)]

k-means with $k = 2$ searches over partitions of the data into k non-empty, unlabeled clusters. The number of distinct such partitions of $n = 4$ points is _____ (Answer in integer)

Q.9 [1 mark · NAT (numerical answer)]

The dendrogram below was obtained by agglomerative clustering of 8 points A-H. If it is cut to obtain exactly 3 clusters, the number of points in the cluster that contains point E is _____ (Answer in integer)

**Q.10 [1 mark · NAT (numerical answer)]**

The VIF of a predictor x_j is 20. The R^2 obtained by regressing x_j on the other predictors is _____ (rounded off to two decimal places)

Q.11 [1 mark · NAT (numerical answer)]

A binary logistic regression model is given by $\ln[P(y=1|\mathbf{x}) / P(y=0|\mathbf{x})] = 1 + (2)x_1 + (-1.5)x_2$. For the input $\mathbf{x} = (x_1, x_2) = (-1, -2)$, the value of $P(y = 0 | \mathbf{x})$ is _____ (rounded off to two decimal places)

Q.12 [1 mark · NAT (numerical answer)]

A fully connected MLP has layer sizes 20-50-8-64-2 (input-...-output), every layer having a bias vector. The number of bias parameters is _____ (Answer in integer)

Q.13 [1 mark · NAT (numerical answer)]

In AdaBoost with $\varepsilon_t = 0.25$, before normalisation each misclassified example's weight is multiplied by e^{α_t} , where $\alpha_t = \frac{1}{2} \ln((1 - \varepsilon_t)/\varepsilon_t)$. This multiplying factor is _____ (rounded off to two decimal places)

Q.14 [1 mark · NAT (numerical answer)]

The value of $\text{LeakyReLU}_\alpha(z_1) + \text{LeakyReLU}_\alpha(z_2) + \text{ReLU}(z_1)$ with $\alpha = 0.01$, $z_1 = -3$, $z_2 = 2$ ($\text{LeakyReLU}_\alpha(z) = z$ for $z \geq 0$ and αz for $z < 0$) is _____ (rounded off to three decimal places)

Q.15 [1 mark · MSQ (one or more correct)]

Which of the following statements about distance / similarity measures is/are CORRECT?

- (A) Cosine distance $1 - \cos \theta$ does not satisfy the triangle inequality in general.
- (B) Cosine similarity between two vectors with non-negative entries can be negative.
- (C) The Hamming distance between two binary vectors equals their Manhattan distance.
- (D) Manhattan distance is always $\leq$ Euclidean distance.

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

The continuous attribute Age and the class label of 9 training records are given below (unsorted). A binary split "Age $\leq t$ " is chosen to maximise information gain, with t restricted to midpoints between consecutive sorted values. Find the threshold value t chosen for the split "Age $\leq t$ ". (rounded off to one decimal place)

Age	39	60	20	51	26	47	44	45	66
Class	Yes	No	Yes	No	No	Yes	Yes	Yes	No

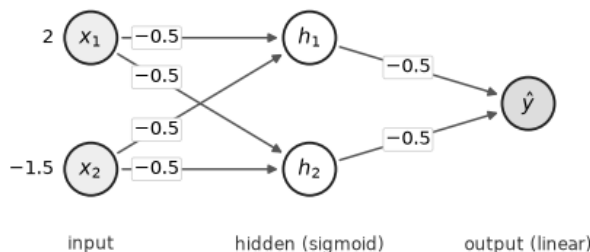
**Q.17 [2 marks · NAT (numerical answer)]**

The sample covariance matrix of a two-feature dataset is S given below. In principal component analysis, the variance of the data along the first principal component is _____ (rounded off to two decimal places)

$$S = \begin{bmatrix} 8 & 3 \\ 3 & 10 \end{bmatrix}$$

Q.18 [2 marks · NAT (numerical answer)]

In the 2-2-1 network below (sigmoid hidden units, linear output $\hat{y} = v_1 h_1 + v_2 h_2$, all biases 0), **every** weight is initialised to -0.5 . One gradient-descent step with $\eta = 1$ is taken on $L = \frac{1}{2}(\hat{y} - y)^2$ for the input $x = (2, -1.5)$ with target $y = 0$. Then $w_{12} - w_{22}$ after the update (difference between the weights from x_2 into the two hidden units) is _____ (rounded off to 4 decimal places)

**Q.19 [2 marks · NAT (numerical answer)]**

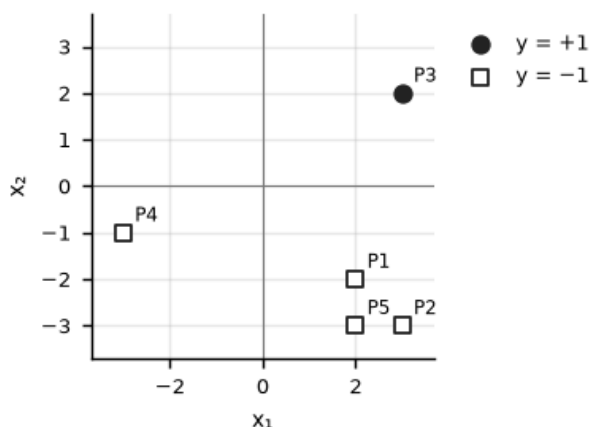
For a two-class problem, the class means are $\mu_0 = (2, -2)^T$ and $\mu_1 = (2, -1)^T$ and the pooled within-class scatter matrix S_W is given below. Fisher's linear discriminant uses $\mathbf{w} \propto S_W^{-1}(\mu_1 - \mu_0)$. Then the separation of the projected class means, $\mathbf{w}^T(\mu_1 - \mu_0)$, with $\mathbf{w} = S_W^{-1}(\mu_1 - \mu_0)$ is _____ (rounded off to two decimal places)

$$S_W = \begin{bmatrix} 20 & 8 \\ 8 & 8 \end{bmatrix}$$

Q.20 [2 marks · NAT (numerical answer)]

A perceptron with augmented weight vector $\mathbf{w} = (w_0, w_1, w_2)$ (w_0 is the bias, $\tilde{x} = (1, x_1, x_2)$) starts from $\mathbf{w} = (0, 0, 0)$ and uses learning rate $\eta = 0.5$. The points P1, P2, ... are presented once, in order; whenever $y \cdot \mathbf{w}^T \tilde{x} \leq 0$ the update $\mathbf{w} \leftarrow \mathbf{w} + \eta y \tilde{x}$ is applied. Find the value of w_0 (bias) after this epoch. (rounded off to one decimal place)

Point	x_1	x_2	y
P1	2	-2	-1
P2	3	-3	-1
P3	3	2	+1
P4	-3	-1	-1
P5	2	-3	-1



Q.21 [2 marks • MCQ]

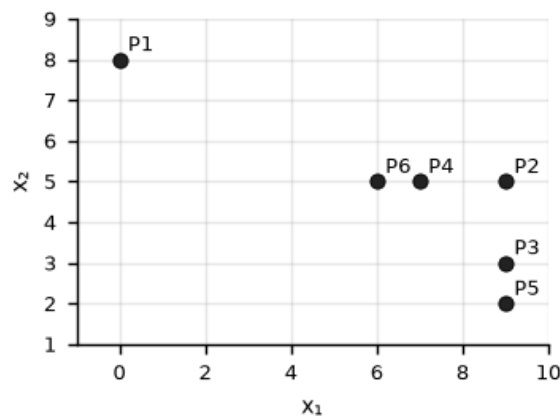
Six points have the pairwise distances below. Agglomerative clustering with **average linkage** is run and the dendrogram is cut to give exactly **2** clusters. The resulting clusters are

	P1	P2	P3	P4	P5	P6
P1	0	4	6	11	3	10
P2		0	8	13	1	6
P3			0	5	9	8
P4				0	14	9
P5					0	7
P6						0

(A) {P1, P2, P5, P6}, {P3, P4}**(B)** {P1, P2, P5}, {P3, P4, P6}**(C)** {P1, P2, P3, P4}, {P5, P6}**(D)** {P1, P2, P5}, {P3, P4}, {P6}**Q.22 [2 marks • NAT (numerical answer)]**

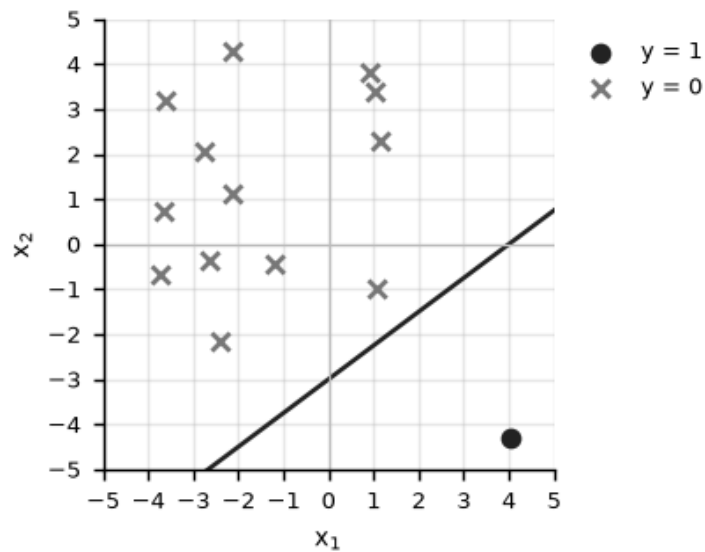
A cluster contains the 6 points below. Distances are Euclidean (L_2). If P6 were used as the cluster representative instead of the medoid, the total distance (cost) would increase by _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5	P6
x_1	0	9	9	7	9	6
x_2	8	5	3	5	2	5



Q.23 [2 marks · MCQ]

A logistic regression model $P(y = 1 | \mathbf{x}) = \sigma(w_0 + w_1x_1 + w_2x_2)$ perfectly classifies the training points shown below using threshold 0.5. Its decision boundary is the solid line (it crosses the axes at grid points). Which of the following parameter vectors is consistent with the figure?

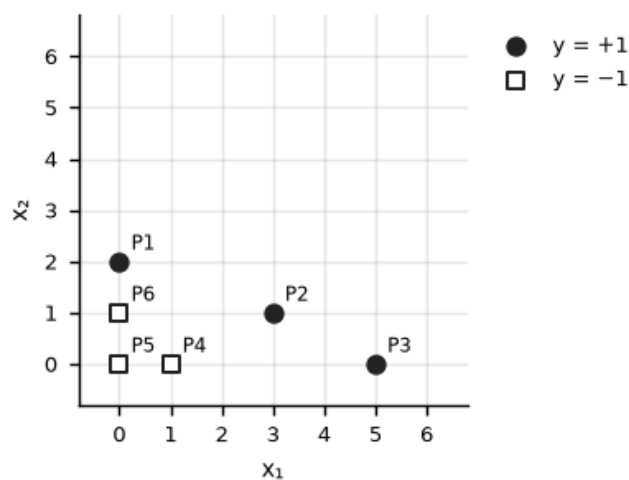


- (A) $(w_0, w_1, w_2) = (24, -8, 6)$
 (B) $(w_0, w_1, w_2) = (-24, 6, -8)$
 (C) $(w_0, w_1, w_2) = (-24, 8, -6)$
 (D) $(w_0, w_1, w_2) = (-24, -6, -8)$

Q.24 [2 marks · MSQ (one or more correct)]

A hard-margin linear SVM is trained on the 6 labelled points shown. Which of the following points is/are support vectors of the resulting maximum-margin classifier?

Point	x_1	x_2	y
P1	0	2	+1
P2	3	1	+1
P3	5	0	+1
P4	1	0	-1
P5	0	0	-1
P6	0	1	-1

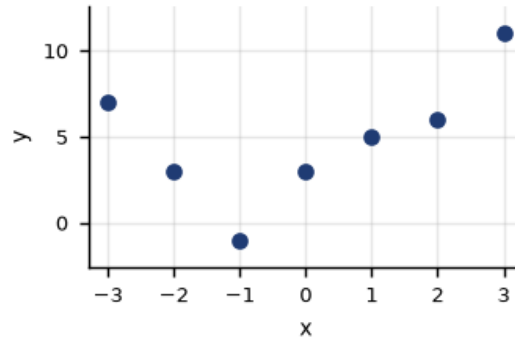


- (A) P1
 (B) P2
 (C) P3
 (D) P6

Q.25 [2 marks · NAT (numerical answer)]

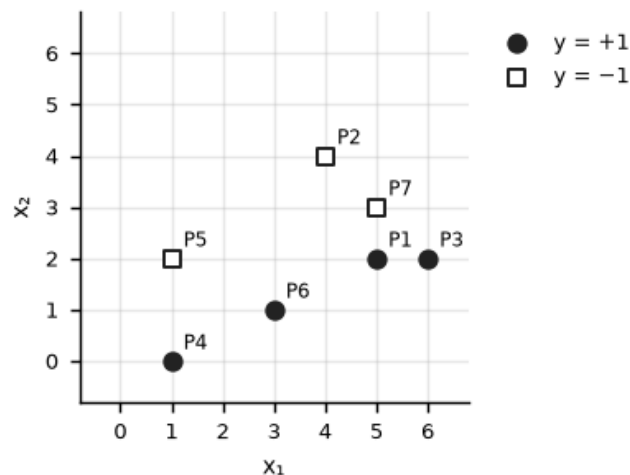
The quadratic model $y = a + bx + cx^2$ is fitted by least squares to the 7 points below (a polynomial / basis-function regression with basis $1, x, x^2$). Then the intercept a is _____ (rounded off to two decimal places)

x	-3	-2	-1	0	1	2	3
y	7	3	-1	3	5	6	11

**Q.26 [2 marks · MCQ]**

Four candidate linear decision boundaries are proposed for the labelled training points shown. Which boundary separates the two classes with the LARGEST geometric margin (minimum distance from the boundary to any training point, all points correctly classified)?

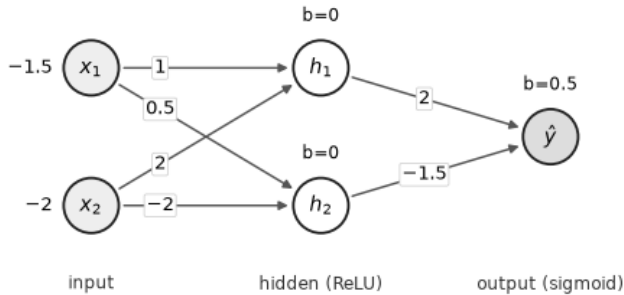
Point	x_1	x_2	y
P1	5	2	+1
P2	4	4	-1
P3	6	2	+1
P4	1	0	+1
P5	1	2	-1
P6	3	1	+1
P7	5	3	-1



- (A) $x_1 - 2x_2 = 0$
 (B) $x_1 - 2x_2 - 0.5 = 0$
 (C) $x_1 - 2x_2 + 0.5 = 0$
 (D) $2x_1 + x_2 - 9 = 0$

Q.27 [2 marks · NAT (numerical answer)]

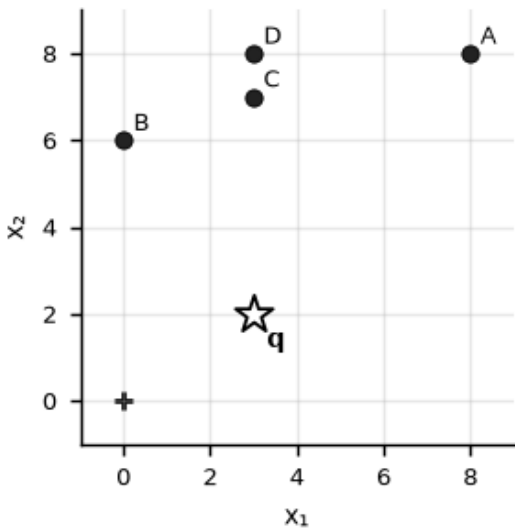
The feed-forward network below has ReLU hidden units and a sigmoid output unit. The number on each edge is its weight and 'b' above a node is that node's bias. For input $x = (-1.5, -2)$ (values shown left of the inputs), the network output $\hat{y}$ is _____ (rounded off to 4 decimal places)



Q.28 [2 marks · MCQ]

A query point $q = (3, 2)$ and four points A-D are shown below. Using **Chebyshev (L_∞) distance**, which point is nearest to q ?

Point	A	B	C	D
x_1	8	0	3	3
x_2	8	6	7	8



- (A) Point A
- (B) Point B
- (C) Point C
- (D) Point D

Q.29 [2 marks · NAT (numerical answer)]

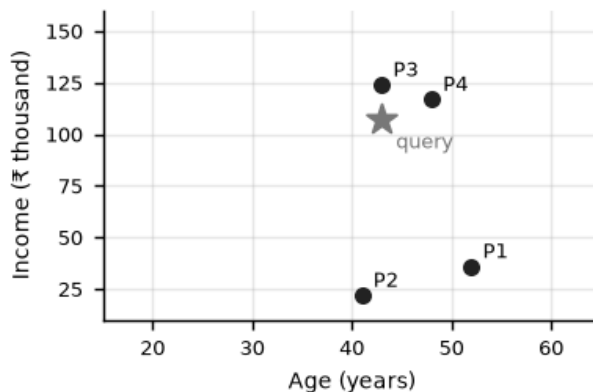
Consider the model $y = wx + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$, $\sigma^2 = 4$, and $w \sim N(0, \tau^2)$ with $\tau^2 = 0.25$. Given the four observations below, $\hat{w}_{ML} - \hat{w}_{MAP}$ is _____ (rounded off to three decimal places)

x	0	0	0	1
y	7	3	6	4

Q.30 [2 marks · MCQ]

A 1-NN classifier uses the two features below. Before computing Euclidean distances, each feature is min-max scaled to $[0, 1]$ using the minimum and maximum of that feature over the four training points P1-P4 (the query is scaled with the same constants). The nearest neighbour of the query after scaling is

Point	Age (years)	Monthly income (₹ thousand)
P1	52	36
P2	41	22
P3	43	124
P4	48	117
Query	43	107



- (A) P1
(C) P3

- (B) P2
(D) P4

Q.31 [2 marks · NAT (numerical answer)]

A logistic-regression neuron $p = \sigma(w_1x_1 + w_2x_2 + b)$ has $w = (-0.5, -1)$, $b = 1$ and is trained with binary cross-entropy loss. After one gradient-descent step using the **average** loss over both examples (a mini-batch of size 2) with learning rate $\eta = 0.1$, the new value of b is _____ (rounded off to three decimal places)

example	x_1	x_2	y
1	-1	1.5	1
2	2	2	0

Q.32 [2 marks · NAT (numerical answer)]

A 2-2-1 network uses the identity (linear) activation in its hidden layer: $\hat{y} = W_2(W_1x + b_1) + b_2$, with the parameters below. The network is equivalent to a single linear model $\hat{y} = \tilde{w}_1x_1 + \tilde{w}_2x_2 + \tilde{b}$. Then the output for $x = (-3, 2)$ is _____ (Answer in integer)

$$W_1 = \begin{bmatrix} 2 & -2 \\ 0 & -2 \end{bmatrix}$$

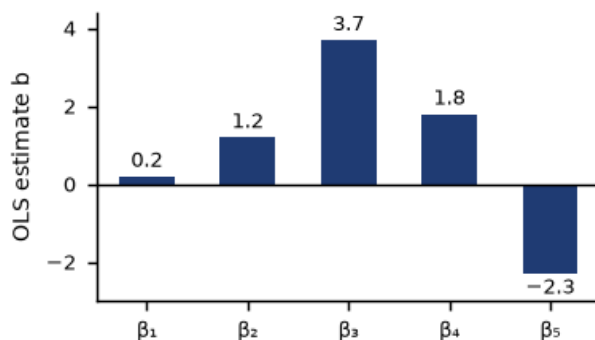
$$b_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 2 & 2 \end{bmatrix}$$

$$b_2 = 0$$

Q.33 [2 marks · NAT (numerical answer)]

In a regression problem the design matrix X has orthonormal columns ($X^T X = I$). The ordinary least-squares estimates $b = X^T y$ are shown below. Ridge minimises $\|y - X\beta\|^2 + \lambda\|\beta\|_2^2$ and the lasso minimises $\frac{1}{2}\|y - X\beta\|^2 + \lambda\|\beta\|_1$, each with $\lambda = 2$. Then the ridge estimate of β_1 is _____ (rounded off to two decimal places)

**Q.34 [2 marks · NAT (numerical answer)]**

On $n = 30$ observations, model A (intercept + 1 predictor) has $R^2 = 0.57$, and model B (intercept + 4 predictors, containing those of A) has $R^2 = 0.59$. The value of (adjusted R^2 of B) – (adjusted R^2 of A) is _____ (rounded off to three decimal places)

Q.35 [2 marks · NAT (numerical answer)]

The confusion matrix of a 3-class classifier is given below (rows = actual class, columns = predicted class). The macro-averaged precision is _____ (rounded off to two decimal places)

actual \ predicted	A	B	C
A	7	6	5
B	1	17	1
C	1	5	17

— END OF QUESTION PAPER —

Mock Test 17 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MSQ	1	(A), (B), (C), (D)	19	NAT	2	0.1983 to 0.2183
2	MSQ	1	(A), (B), (C), (D)	20	NAT	2	-0.01 to 0.01
3	MCQ	1	(C)	21	MCQ	2	(A)
4	MSQ	1	(B), (C)	22	NAT	2	1.4966 to 1.5166
5	MCQ	1	(B)	23	MCQ	2	(B)
6	NAT	1	0.4838 to 0.5038	24	MSQ	2	(A), (C), (D)
7	MCQ	1	(A)	25	NAT	2	1.6943 to 1.7343
8	NAT	1	7	26	MCQ	2	(A)
9	NAT	1	1	27	NAT	2	0.0122 to 0.0127
10	NAT	1	0.94 to 0.96	28	MCQ	2	(B)
11	NAT	1	0.1092 to 0.1292	29	NAT	2	3.7534 to 3.776
12	NAT	1	124	30	MCQ	2	(C)
13	NAT	1	1.7221 to 1.7421	31	NAT	2	1.017 to 1.021
14	NAT	1	1.968 to 1.972	32	NAT	2	-22
15	MSQ	1	(A), (C)	33	NAT	2	0.0567 to 0.0767
16	NAT	2	48.99 to 49.01	34	NAT	2	-0.0322 to -0.0282
17	NAT	2	12.1523 to 12.1723	35	NAT	2	0.698 to 0.718
18	NAT	2	-0.0002 to 0.0002				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Probabilistic & Instance-based Classifiers					11
Neural Networks (MLP / Feed-forward)					11
Regression & Model Selection					11
SVM & Decision Trees					11
Unsupervised Learning					11
Total					55

Mock Test 17 — Detailed Solutions

Q.1 — Answer: (A), (B), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

- (A) **TRUE**. It models the distribution of x .
 (B) **TRUE**. A costly error type shifts the decision threshold.
 (C) **TRUE**. This is the MAP rule.
 (D) **TRUE**. The classes may overlap less in the larger space.

Q.2 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Backpropagation]

- (A) **TRUE**. Errors propagate from output towards input.
 (B) **TRUE**. Chain rule through σ .
 (C) **TRUE**. a_j depends on b_j with coefficient 1.
 (D) **TRUE**. $[L(w+\varepsilon) - L(w-\varepsilon)]/2\varepsilon \approx \partial L/\partial w$.

Q.3 — Answer: (C) [MCQ, 1M · Regression & Model Selection > Gradient descent]

$J(w) = (w - 3)^2$ has $J' = 2$. GD: $w_{t+1} - 3 = (1 - 2\eta)(w_t - 3)$, so $J(w_t) = (1 - 2\eta)^{2t} J(w_0)$.

Ratios $(1 - 2\eta)^2$ per step: $\eta = 0.01 \rightarrow 0.9604$ (curve C); $\eta = 0.04 \rightarrow 0.8464$ (curve B); $\eta = 0.15 \rightarrow 0.49$ (curve D); $\eta = 1.3 \rightarrow 2.56$ (curve A).

Ratio $> 1 \Rightarrow$ divergence ($\eta > 1 = 2/J'$); among convergent rates (all < 0.5 here), a smaller η gives a ratio closer to 1, i.e. slower decrease.

Hence $\eta = 1.3$ corresponds to **Curve A**.

Q.4 — Answer: (B), (C) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

- (A) **FALSE**. $0 < \sigma(z) < 1$ for every finite z .
 (B) **TRUE**. Boundary between k and l : $(w_k - w_l)^T x + (b_k - b_l) = 0$.
 (C) **TRUE**. Scaling a separating w up always lowers the loss, so the MLE does not exist (it diverges).
 (D) **FALSE**. $\log\text{-odds} = \ln(0.8/0.2) = \ln 4 \approx 1.386$.

Q.5 — Answer: (B) [MCQ, 1M · SVM & Decision Trees > Decision trees: overfitting & pruning]

Subtree	$R(T) + \alpha T $
T1	$0.03 + 0.015 \times 14 = 0.24$
T2	$0.08 + 0.015 \times 10 = 0.23$
T3	$0.17 + 0.015 \times 6 = 0.26$
T4	$0.175 + 0.015 \times 4 = 0.235$

Minimum at **T2**. Larger α penalises leaves more and favours smaller trees; $\alpha = 0$ would pick the largest tree T1.

Q.6 — Answer: 0.4838 to 0.5038 [NAT, 1M · SVM & Decision Trees > Decision trees: impurity measures]

Gini $= 1 - \sum p_k^2 = 1 - [(8/18)^2 + (10/18)^2] = \mathbf{0.494}$.

(Maximum possible with 2 classes: $1 - 1/2 = 0.5$.)

Q.7 — Answer: (A) [MCQ, 1M · Regression & Model Selection > Simple linear regression]

All residuals lie in a narrow band except one far from 0 $\rightarrow$ an outlier that should be investigated.

Answer: **A single outlying observation, with the rest of the fit looking adequate.**

Other patterns: funnel $\rightarrow$ heteroscedasticity; curvature $\rightarrow$ missing non-linearity; isolated point $\rightarrow$ outlier; random band $\rightarrow$ adequate fit.

Q.8 — Answer: 7 [NAT, 1M · Unsupervised Learning > k-means]

This is the Stirling number of the second kind $S(n, k)$, with $S(n, k) = k \cdot S(n-1, k) + S(n-1, k-1)$.

$S(4, 2) = \mathbf{7}$. (For $k = 2$ it equals $2^{n-1} - 1 = 7$.)

Q.9 — Answer: 1 [NAT, 1M · Unsupervised Learning > Hierarchical clustering]

Undo the top 2 merge(s) (equivalently cut just below height 4.73). Clusters: {A, B, C, D, F, G}, {E}, {H}.

E lies in a cluster of size **1**.

Q.10 — Answer: 0.94 to 0.96 [NAT, 1M · Regression & Model Selection · Multiple linear regression]

Variance inflation factor of predictor j : $VIF_j = 1/(1 - R_j^2)$, where R_j^2 is the R^2 from regressing x_j on all other predictors. It is the factor by which $\text{Var}(\beta_j)$ is inflated by collinearity; $\sqrt{VIF}$ inflates the standard error.

$$R_j^2 = 1 - 1/VIF = 1 - 1/20 = 0.95.$$

Answer: **0.95**.

Q.11 — Answer: 0.1092 to 0.1292 [NAT, 1M · Probabilistic & Instance-based Classifiers · Logistic regression]

$$z = w_0 + w_1x_1 + w_2x_2 = 1 + (2)(-1) + (-1.5)(-2) = 2 \text{ (this is the log-odds).}$$

$$P(y = 1 | \mathbf{x}) = \sigma(z) = 1/(1 + e^{-2}) = 0.8808.$$

$$P(y = 0 | \mathbf{x}) = 1 - \sigma(z) = \sigma(-z) = \mathbf{0.12}.$$

Q.12 — Answer: 124 [NAT, 1M · Neural Networks (MLP / Feed-forward) · Parameter counting]

Fully connected layer with n_{in} inputs and n_{out} outputs: $n_{\text{in}} \cdot n_{\text{out}}$ weights + n_{out} biases.

Layer 1: $20 \times 50 = 1000$ weights, 50 biases.

Layer 2: $50 \times 8 = 400$ weights, 8 biases.

Layer 3: $8 \times 64 = 512$ weights, 64 biases.

Layer 4: $64 \times 2 = 128$ weights, 2 biases.

Weights = 2040, biases = 124. Required: **124**.

Q.13 — Answer: 1.7221 to 1.7421 [NAT, 1M · SVM & Decision Trees · Ensembles: boosting]

$$e^\alpha = e^{\frac{1}{2} \ln((1-\epsilon)/\epsilon)} = \sqrt{((1-\epsilon)/\epsilon)} = \sqrt{3} = \mathbf{1.73}.$$

Correctly classified examples are multiplied by $e^{-\alpha} = \sqrt{(\epsilon/(1-\epsilon))}$.

$\epsilon < 0.5 \Rightarrow \alpha > 0$; $\epsilon = 0.5 \Rightarrow \alpha = 0$ (no better than chance).

Q.14 — Answer: 1.968 to 1.972 [NAT, 1M · Neural Networks (MLP / Feed-forward) · Activation functions]

$$\text{LeakyReLU}(-3) = 0.01 \times -3 = -0.03; \text{LeakyReLU}(2) = 2; \text{ReLU}(-3) = 0. \text{ Sum} = \mathbf{1.97}.$$

Q.15 — Answer: (A), (C) [MSQ, 1M · Unsupervised Learning · Distance measures]

(A) **TRUE**. So it is not a true metric.

(B) **FALSE**. Their dot product is ≥ 0 , so $\cos \theta \geq 0$.

(C) **TRUE**. Each mismatched coordinate contributes 1.

(D) **FALSE**. It is the other way round: $L_2 \leq L_1$.

Q.16 — Answer: 48.99 to 49.01 [NAT, 2M · SVM & Decision Trees · Decision trees: split selection]

Sort the values: 20(Y), 26(N), 39(Y), 44(Y), 45(Y), 47(Y), 51(N), 60(N), 66(N).

Parent [5 Yes, 4 No], $H = 0.9911$. The optimal threshold always lies where the class changes between neighbours (Fayyad-Irani), giving 3 candidates:

Threshold	Left ($\leq t$) [Yes, No]	Right [Yes, No]	IG
23	[1, 0]	[4, 4]	0.1022
32.5	[1, 1]	[4, 3]	0.0026
49	[5, 1]	[0, 3]	0.5577

Best: $t = 49$ with $IG = 0.5577$. Answer: **49**.

Q.17 — Answer: 12.1523 to 12.1723 [NAT, 2M · Unsupervised Learning · PCA & dimensionality reduction]

Eigenvalues of $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$: $\lambda = (a + c)/2 \pm \sqrt{((a - c)/2)^2 + b^2}$.

$\lambda = 9 \pm \sqrt{(1 + 9)} = 9 \pm 3.1623 \rightarrow \lambda_1 = 12.1623, \lambda_2 = 5.8377$ (check: $\lambda_1 + \lambda_2 = \text{trace} = 18, \lambda_1\lambda_2 = \det = 71$).

Eigenvector for λ_1 : $(S - \lambda_1 I)v = 0 \rightarrow (-4.1623)v_1 + (3)v_2 = 0 \rightarrow v \propto (1, 1.3874)$.

$PVE(PC1) = \lambda_1/(\lambda_1 + \lambda_2) = 67.57\%$.

Answer: **12.16**.

Q.18 — Answer: -0.0002 to 0.0002 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

All weights = -0.5, biases 0 $\Rightarrow$ both hidden units get $a = -0.5 \times (2 + -1.5) = -0.25$, $h_1 = h_2 = \sigma(a) = 0.4378$; $\hat{y} = 2 \times -0.5 \times 0.4378 = -0.4378$.

$\hat{y} - y = -0.4378$. $\partial L / \partial v_j = (\hat{y} - y)h_j = -0.1917$ (same for both j).

$\partial L / \partial w_{jk} = (\hat{y} - y) v_j h_j (1 - h_j) x_k = -0.4378 \times -0.5 \times 0.2461 \times -1.5 = -0.0808$ (identical for $j = 1, 2$).

Both hidden units receive identical gradients, so the difference stays **0**.

Because the two hidden units start identical and get identical updates, they remain identical forever (symmetry problem) - hence weights must be initialised randomly.

Q.19 — Answer: 0.1983 to 0.2183 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$\mu_1 - \mu_0 = (0, 1)^T$; $\det S_W = 96$.

$$S_W^{-1} = \begin{bmatrix} 0.0833 & -0.0833 \\ -0.0833 & 0.2083 \end{bmatrix}$$

$w = S_W^{-1}(\mu_1 - \mu_0) = (-0.0833, 0.2083)^T$.

$w^T(\mu_1 - \mu_0) = (\mu_1 - \mu_0)^T S_W^{-1}(\mu_1 - \mu_0) = \mathbf{0.21}$ (always ≥ 0 since S_W is positive definite).

Note: using $\mu_1 - \mu_0$ without S_W^{-1} is correct only when $S_W \propto I$.

Q.20 — Answer: -0.01 to 0.01 [NAT, 2M · SVM & Decision Trees > Perceptron (linear separators)]

Trace the epoch (a point with $y \cdot w^T \tilde{x} \leq 0$, including 0, triggers an update):

Step	$\tilde{x} = (1, x_1, x_2)$	y	$w^T \tilde{x}$	$y \cdot w^T \tilde{x} \leq 0$?	w after step
1	(1, 2, -2)	-1	0	yes $\rightarrow$ update	(-0.5, -1, 1)
2	(1, 3, -3)	-1	-6.5	no	(-0.5, -1, 1)
3	(1, 3, 2)	+1	-1.5	yes $\rightarrow$ update	(0, 0.5, 2)
4	(1, -3, -1)	-1	-3.5	no	(0, 0.5, 2)
5	(1, 2, -3)	-1	-5	no	(0, 0.5, 2)

Final $w = (0, 0.5, 2)$, number of updates = 2. Answer: **0**.

Q.21 — Answer: (A) [MCQ, 2M · Unsupervised Learning > Hierarchical clustering]

Use average linkage (UPGMA): $d(A, B) = (1/|A||B|) \sum_{a \in A} \sum_{b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: {P2} + {P5} at height **1**. Updated: $d(\{P2, P5\}, \{P1\}) = 3.5$, $d(\{P2, P5\}, \{P3\}) = 8.5$, $d(\{P2, P5\}, \{P4\}) = 13.5$, $d(\{P2, P5\}, \{P6\}) = 6.5$.

Merge 2: {P1} + {P2, P5} at height **3.5**. Updated: $d(\{P1, P2, P5\}, \{P3\}) = 7.67$, $d(\{P1, P2, P5\}, \{P4\}) = 12.67$, $d(\{P1, P2, P5\}, \{P6\}) = 7.67$.

Merge 3: {P3} + {P4} at height **5**. Updated: $d(\{P3, P4\}, \{P6\}) = 8.5$, $d(\{P3, P4\}, \{P1, P2, P5\}) = 10.17$.

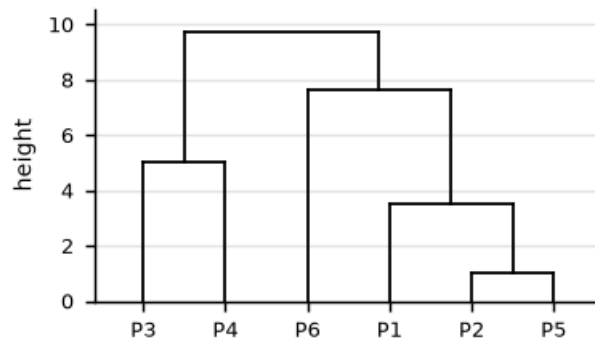
Merge 4: {P6} + {P1, P2, P5} at height **7.67**. Updated: $d(\{P1, P2, P5, P6\}, \{P3, P4\}) = 9.75$.

Merge 5: {P3, P4} + {P1, P2, P5, P6} at height **9.75**.

Stopping when 2 clusters remain (i.e. after 4 merges): **{P1, P2, P5, P6}, {P3, P4}**.

{P1, P2, P5}, {P3, P4, P6} is what *complete* linkage would give.

{P1, P2, P5}, {P3, P4}, {P6} has 3 clusters (cut at the wrong level).



Q.22 — Answer: 1.4966 to 1.5166 [NAT, 2M · Unsupervised Learning > k-medoids]

Compute the Euclidean (L_2) distance matrix and its row sums:

	P1	P2	P3	P4	P5	P6	Σ
P1	0	9.49	10.3	7.62	10.82	6.71	44.92
P2	9.49	0	2	2	3	3	19.49
P3	10.3	2	0	2.83	1	3.61	19.73
P4	7.62	2	2.83	0	3.61	1	17.05
P5	10.82	3	1	3.61	0	4.24	22.66
P6	6.71	3	3.61	1	4.24	0	18.56

The smallest row sum is 17.05 for P4, so the medoid is P4.

Cost with P6: 18.56; increase = $18.56 - 17.05 = 1.51$.

Unlike a k-means centroid, the medoid is always an actual data point.

Q.23 — Answer: (B) [MCQ, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

The line passes through (4, 0) and (0, -3): $x_1/4 + x_2/-3 = 1$, i.e. $-3x_1 + 4x_2 - 12 = 0$. Hence (w_0, w_1, w_2) must be a non-zero multiple of (12, -3, 4).

Orientation: the class $y = 1$ region must have $w_0 + w_1x_1 + w_2x_2 > 0$. At the origin the $y = 1$ side has sign $-(w_0 = -24)$, which matches the figure (the origin lies on the $y = 0$ side).

So the answer is $(w_0, w_1, w_2) = (-24, 6, -8)$ (any positive multiple gives the same classifier).

Other options either describe a different line (swapped/negated coefficients move the intercepts) or the same line with the sign flipped, which would label every training point incorrectly.

Q.24 — Answer: (A), (C), (D) [MSQ, 2M · SVM & Decision Trees > SVM: dual & support vectors]

P1 and P3 (same class) lie on one margin line and P6 on the other; the optimal boundary is parallel to the line through P1, P3, midway to P6. Support vectors: P1, P3, P6.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (4/5, 2)$, $b = -3$; $\|w\| = 2.1541$, margin $2/\|w\| = 0.9285$.

Point	P1	P2	P3	P4	P5	P6
$y(w^T x + b)$	1	1.4	1	2.2	3	1

Support vectors are exactly the points with $y_i(w^T x_i + b) = 1$ ($\alpha_i > 0$): **P1, P3, P6**.

Dual check: $\alpha = (\alpha_1 = 54/25, \alpha_3 = 4/25, \alpha_6 = 58/25)$ satisfy $\sum \alpha_i y_i = 0$ and $\sum \alpha_i y_i x_i = w$.

Q.25 — Answer: 1.6943 to 1.7343 [NAT, 2M · Regression & Model Selection > Basis functions & feature scaling]

Design columns 1, x , x^2 . Because the x -values are symmetric about 0, $\Sigma x = \Sigma x^3 = 0$, so the normal equations decouple:

$$\Sigma x^2 = 28, \Sigma x^4 = 196, \Sigma y = 34, \Sigma xy = 24, \Sigma x^2 y = 202.$$

$$b = \Sigma xy / \Sigma x^2 = 24/28 = 0.8571.$$

$$[n, \Sigma x^2; \Sigma x^2, \Sigma x^4][a; c] = [\Sigma y; \Sigma x^2 y] = 7a + 28c = 34, 28a + 196c = 202.$$

$$\text{Solving: } c = (n\Sigma x^2 y - \Sigma x^2 \Sigma y) / (n\Sigma x^4 - (\Sigma x^2)^2) = 0.7857, a = (\Sigma y - c\Sigma x^2) / n = 1.7143.$$

Answer: **1.71**.

Q.26 — Answer: (A) [MCQ, 2M · SVM & Decision Trees > SVM: margin & geometry]

For $n^T x - c = 0$, the geometric margin is $\min_i y_i s(n^T x_i - c)/\|n\|$ with orientation $s = \pm 1$ chosen so that the positive class is on the positive side; a negative value means some point is misclassified.

Hyperplane	$\ n\ $	$\min_i$ distance (signed, best orientation)
$x_1 - 2x_2 = 0$	2.236	0.447
$x_1 - 2x_2 - 0.5 = 0$	2.236	0.224
$x_1 - 2x_2 + 0.5 = 0$	2.236	0.224
$2x_1 + x_2 - 9 = 0$	2.236	-2.236 (does not separate)

The largest margin is 0.447, for $x_1 - 2x_2 = 0$. Note that a boundary can separate the data perfectly yet pass close to some point — only the minimum distance matters.

Q.27 — Answer: 0.0122 to 0.0127 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

Hidden pre-activations: $a_1 = 1 \cdot -1.5 + (2) \cdot -2 + (0) = -5.5$; $a_2 = 0.5 \cdot -1.5 + (-2) \cdot -2 + (0) = 3.25$.

Hidden activations ($\text{ReLU}(a) = \max(0, a)$): $h_1 = 0$, $h_2 = 3.25$.

Output pre-activation: $o = 2 \cdot 0 + (-1.5) \cdot 3.25 + (0.5) = -4.375$.

$\hat{y} = \sigma(o) = \mathbf{0.012}$.

Q.28 — Answer: (B) [MCQ, 2M · Unsupervised Learning > Distance measures]

Compute the measure from $q = (3, 2)$ to every point (cosine column shows similarity, larger = more similar):

Point	coordinates	Manhattan	Euclidean	Chebyshev	cosine similarity
A	(8, 8)	11	7.81	6	0.981
B	(0, 6)	7	5	4	0.555
C	(3, 7)	5	5	5	0.838
D	(3, 8)	6	6	6	0.812

Under Chebyshev (L_∞) distance the nearest to q is **point B**.

Note how the answer depends on the measure: Manhattan (L_1) distance $\rightarrow$ C; Euclidean (L_2) distance $\rightarrow$ B; Chebyshev (L_∞) distance $\rightarrow$ B; cosine similarity $\rightarrow$ A. Cosine similarity looks only at the angle from the origin, L_1 sums coordinate gaps, L_∞ takes the largest gap.

Q.29 — Answer: 3.7534 to 3.776 [NAT, 2M · Regression & Model Selection > MLE & MAP view]

$-\ln p(w \mid \text{data}) = (1/(2\sigma^2))\sum (y_i - wx_i)^2 + w^2/(2\tau^2) + \text{const.}$

Multiplying by $2\sigma^2$: $\sum (y_i - wx_i)^2 + (\sigma^2/\tau^2)w^2$ — ridge with $\lambda = \sigma^2/\tau^2 = 16$.

$\sum x_i^2 = 1$, $\sum x_i y_i = 4$; $\hat{w}_{\text{MAP}} = \sum xy / (\sum x^2 + \lambda) = 4/(1 + 16) = 0.2353$.

$\hat{w}_{\text{ML}} = \sum xy / \sum x^2 = 4$; difference = 3.7647.

Answer: **3.765**.

Q.30 — Answer: (C) [MCQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Ranges: age [41, 52], income [22, 124]. Scaled query = (0.182, 0.833).

Point	scaled age	scaled income	scaled distance	raw distance
P1	1	0.137	1.074	71.57
P2	0	0	0.853	85.02
P3	0.182	1	0.167	17
P4	0.636	0.931	0.465	11.18

Nearest after scaling: **P3**.

Without scaling the nearest point would be P4, because income differences (tens of thousands) dominate the age differences — exactly why k-NN needs feature scaling.

Q.31 — Answer: 1.017 to 1.021 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

For σ output with binary cross-entropy $L = -[y \ln p + (1 - y) \ln(1 - p)]$, $\partial L / \partial a = p - y$, so $\partial L / \partial w_j = (p - y)x_j$ and $\partial L / \partial b = p - y$.

Example 1: $a = 0$, $p = 0.5$, $p - y = -0.5$.

Example 2: $a = -2$, $p = 0.1192$, $p - y = 0.1192$.

Mini-batch (mean) gradient for b : -0.1904 .

$b \leftarrow 1 - 0.1 \times (-0.1904) = \mathbf{1.019}$.

Note there is no σ' factor: it cancels with the derivative of the log loss.

Q.32 — Answer: -22 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Expressiveness]

With identity activations, $\hat{y} = W_2(W_1x + b_1) + b_2 = (W_2W_1)x + (W_2b_1 + b_2)$: a single affine map.

$\tilde{w}^T = W_2W_1 = (4, -8)$; $\tilde{b} = W_2b_1 + b_2 = 6$.

$\hat{y} = 4 \cdot -3 + (-8) \cdot 2 + (6) = \mathbf{-22}$.

Hence stacking linear layers adds no expressive power – a non-linear activation is essential.

Q.33 — Answer: 0.0567 to 0.0767 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

With $X^TX = I$: $\beta_{OLS} = X^Ty = b$.

Ridge: $\nabla[\|y - X\beta\|^2 + \lambda\|\beta\|^2] = 0 \Rightarrow (1 + \lambda)\beta = b \Rightarrow \beta_{\text{ridge}} = b/(1 + \lambda)$.

Lasso: the problem separates per coordinate, $\min \frac{1}{2}(\beta_j - b_j)^2 + \lambda|\beta_j| \Rightarrow \text{soft-thresholding } \beta_j = \text{sign}(b_j) \cdot \max(|b_j| - \lambda, 0)$.

j	1	2	3	4	5
b_j	0.2	1.2	3.7	1.8	-2.3
ridge	0.0667	0.4	1.2333	0.6	-0.7667
lasso	0	0	1.7	0	-0.3

Answer: **0.07**.

Q.34 — Answer: -0.0322 to -0.0282 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

Adjusted $R^2 = 1 - [\text{SSE}/(n - p - 1)]/[\text{SST}/(n - 1)] = 1 - (1 - R^2)(n - 1)/(n - p - 1)$, where p is the number of predictors (excluding the intercept).

A: $1 - (1 - 0.57)(29)/(28) = 0.5546$.

B: $1 - (1 - 0.59)(29)/(25) = 0.5244$.

Difference = -0.0302 (negative: the extra predictors do not justify their cost even though R^2 increased).

Answer: **-0.03**.

Q.35 — Answer: 0.698 to 0.718 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

class	TP	predicted total	actual total	precision	recall	F1
A	7	9	18	0.7778	0.3889	0.5185
B	17	28	19	0.6071	0.8947	0.7234
C	17	23	23	0.7391	0.7391	0.7391

Precision of class $k = C_{kk}/(\text{column sum})$; recall = $C_{kk}/(\text{row sum})$.

Macro precision = mean of per-class precisions = **0.71**.

Mock Test 18

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	2	8	25

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Probabilistic & Instance-based Classifiers	7	11
Unsupervised Learning	7	11
SVM & Decision Trees	7	11
Neural Networks (MLP / Feed-forward)	7	11
Regression & Model Selection	7	11

Start time: _____ End time: _____ Score: _____ / 55

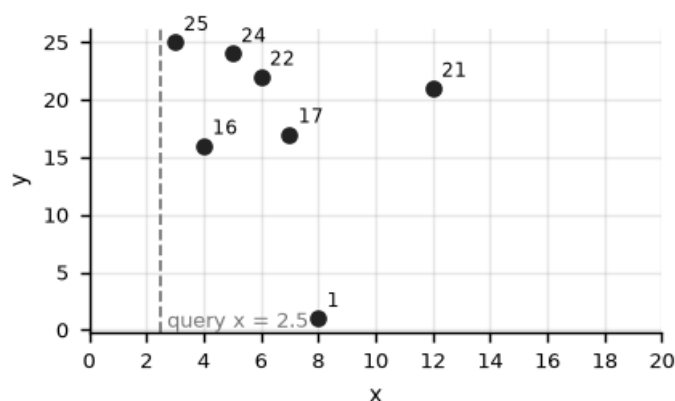
Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

The training data for a k-NN regressor are given below (also plotted). Using $k = 4$ and absolute distance $|x - x_i|$, the predicted value (unweighted average of neighbours) at $x = 2.5$ is _____ (rounded off to two decimal places)

x	3	4	5	6	7	8	12
y	25	16	24	22	17	1	21



Q.2 [1 mark · MSQ (one or more correct)]

Which of the following statements about Bayes-optimal classification and generative vs. discriminative models is/are CORRECT?

- (A) Given enough training data, any classifier can reach zero test error for every distribution.
- (B) When the misclassification costs are asymmetric, the Bayes-optimal rule need not pick the most probable class.
- (C) Under 0-1 loss, the Bayes classifier predicts $\arg \max_k P(y = k | x)$ and minimises the probability of misclassification.
- (D) Generative classifiers can handle missing input features by marginalising over them.

Q.3 [1 mark · MSQ (one or more correct)]

Which of the following statements about hierarchical clustering is/are CORRECT?

- (A) Storing the full distance matrix for agglomerative clustering needs $O(n^2)$ memory.
- (B) Divisive clustering starts with all points in one cluster and recursively splits clusters.
- (C) Ward's method merges the pair of clusters giving the smallest increase in total within-cluster sum of squares.
- (D) Naive agglomerative clustering runs in $O(n)$ time.

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about kernel functions (Mercer kernels) is/are CORRECT?

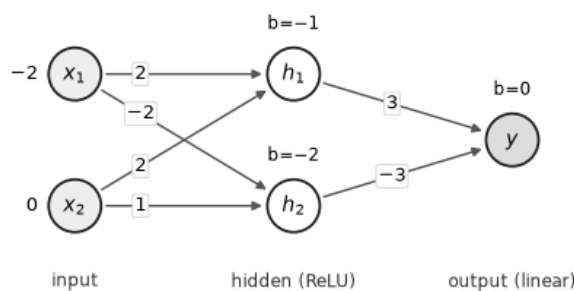
- (A) $\exp(-\gamma \|x - z\|^2)$ with $\gamma > 0$ is a valid kernel.
- (B) The Gram matrix of an RBF kernel can have diagonal entries larger than 1.
- (C) $k(x, z) = \exp(+\gamma \|x - z\|^2)$ with $\gamma > 0$ is a valid kernel.
- (D) $k(x, z) = x^T z + c$ with $c \geq 0$ is a valid kernel.

Q.5 [1 mark · NAT (numerical answer)]

A training run on $N = 50000$ examples with batch size $B = 50$ (last partial batch kept) performed 10000 parameter updates in total. The number of epochs completed is ____ (Answer in integer)

Q.6 [1 mark · NAT (numerical answer)]

For the ReLU network shown (edge labels are weights, 'b' denotes a bias) and input $x = (-2, 0)$, the output y is ____ (Answer in integer)

**Q.7 [1 mark · NAT (numerical answer)]**

The confusion matrix of a binary classifier on a test set is shown below (note which of rows/columns are actual and predicted). The specificity (TNR) of the classifier is ____ (rounded off to two decimal places)

	Predicted +	Predicted -
Actual +	57	56
Actual -	27	23

Q.8 [1 mark · MSQ (one or more correct)]

Consider ridge regression (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_2^2$) and the lasso (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_1$) with $\lambda \geq 0$. Which of the following statements is/are CORRECT?

- (A) The lasso constraint region $\sum |\beta_j| \leq t$ in two dimensions is a diamond (square rotated by 45°).
- (B) For $\lambda > 0$, $X^T X + \lambda I$ is invertible even when $p > n$.
- (C) With an orthonormal design ($X^T X = I$), minimising $\|y - X\beta\|^2 + \lambda\|\beta\|^2$ gives $\beta_{OLS}/(1 + \lambda)$.
- (D) The L1 penalty $|\beta|$ is not differentiable at $\beta = 0$.

Q.9 [1 mark · NAT (numerical answer)]

A weight $w = 1.5$ is trained on the regularised objective $L(w) + (\lambda/2)w^2$ with $\lambda = 0.01$. The data-loss gradient is $\partial L/\partial w = -1.5$. After one gradient-descent step with $\eta = 0.1$, the new weight is _____ (rounded off to three decimal places)

Q.10 [1 mark · NAT (numerical answer)]

For a canonical hard-margin SVM (support vectors satisfy $y_i(w^T x_i + b) = 1$), the distance between the two margin hyperplanes is found to be 0.5. The value of $\|w\|^2$ is _____ (rounded off to two decimal places)

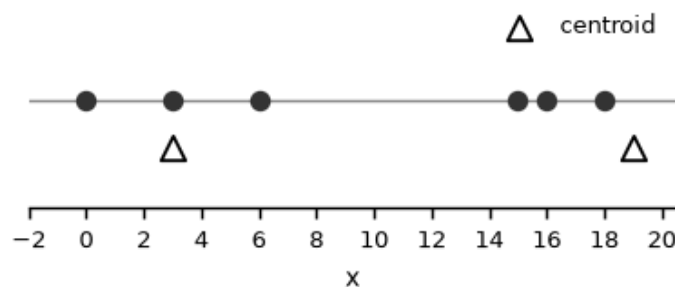
Q.11 [1 mark · MSQ (one or more correct)]

A simple linear regression $\hat{y} = \beta_0 + \beta_1 x$ is fitted by ordinary least squares (with intercept) to data (x_i, y_i) , $i = 1, \dots, n$, with residuals $e_i = y_i - \hat{y}_i$ and sample correlation r . Which of the following statements is/are CORRECT?

- (A) $SST = SSR - SSE$.
- (B) If every y_i is multiplied by a constant c , the least-squares slope is multiplied by c .
- (C) If $r = 0$ the least-squares line is $\hat{y} = 0$.
- (D) The slope of the x-on-y regression line is always the reciprocal of the slope of the y-on-x line.

Q.12 [1 mark · NAT (numerical answer)]

One-dimensional data $\{0, 3, 6, 15, 16, 18\}$ and $k = 2$ centroids at 3, 19 are given. Each point is assigned to its nearest centroid and then each centroid is recomputed as the mean of its points. The within-cluster SSE with the **updated** centroids is _____ (rounded off to two decimal places)

**Q.13 [1 mark · NAT (numerical answer)]**

The 5 two-dimensional observations below have zero mean in each feature. Then the correlation coefficient between x_1 and x_2 is _____ (rounded off to two decimal places)

Point	x_1	x_2	x_3	x_4	x_5
x_1	0	2	0	3	-5
x_2	-4	3	-1	-3	5

Q.14 [1 mark · NAT (numerical answer)]

A node contains 24 records spread equally over 8 classes (3 records per class). Its entropy (in bits) is _____ (rounded off to two decimal places)

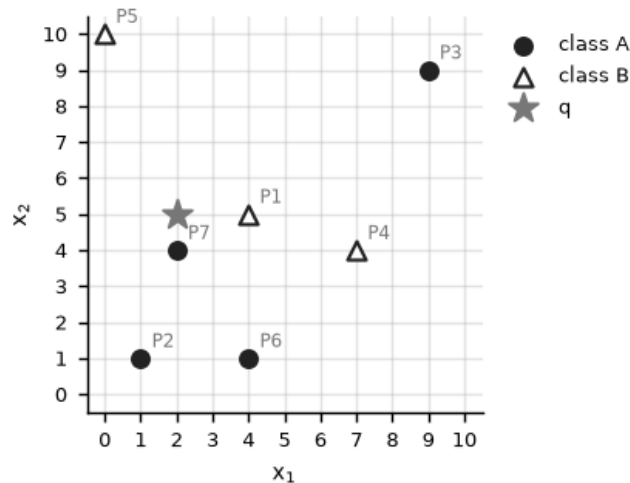
Q.15 [1 mark · NAT (numerical answer)]

A regularisation parameter is tuned over 7 candidate values using 10-fold cross-validation. After selecting the best value, the model is refitted once on the full training data. The total number of times the model is fitted is _____ (Answer in integer)

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · MCQ]**

Consider the training set below. A 3-NN classifier (majority vote) is applied to the query point $\mathbf{q} = (2, 5)$, once with Euclidean distance and once with Manhattan distance. The predicted classes are

Point	x_1	x_2	class
P1	4	5	B
P2	1	1	A
P3	9	9	A
P4	7	4	B
P5	0	10	B
P6	4	1	A
P7	2	4	A



- (A) Euclidean: A; Manhattan: A
(C) Euclidean: B; Manhattan: B

- (B) Euclidean: B; Manhattan: A
(D) Euclidean: A; Manhattan: B

Q.17 [2 marks · MSQ (one or more correct)]

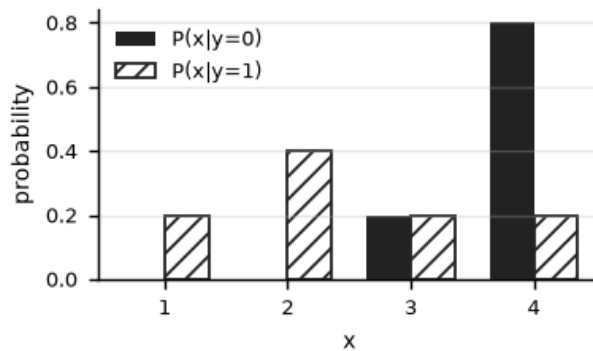
Each option shows a 2×2 matrix $[[K(x_1, x_1), K(x_1, x_2)], [K(x_2, x_1), K(x_2, x_2)]]$. Which of them can be the Gram matrix of some valid (Mercer) kernel $\tilde{K}$ on two points x_1, x_2 ?

- (A) $[[8, -6], [-6, 6]]$
(B) $[[4, 4], [4, 4]]$
(C) $[[4, 0], [2, 9]]$
(D) $[[8, -5], [-5, 5]]$

Q.18 [2 marks · NAT (numerical answer)]

A discrete feature $x \in \{1, \dots, 4\}$ has the class-conditional distributions below (also plotted), with priors $P(y = 0) = 0.4$ and $P(y = 1) = 0.6$. Then the posterior probability $P(y = 1 \mid x = 2)$ is _____ (rounded off to three decimal places)

x	1	2	3	4
$P(x \mid y=0)$	0	0	0.2	0.8
$P(x \mid y=1)$	0.2	0.4	0.2	0.2

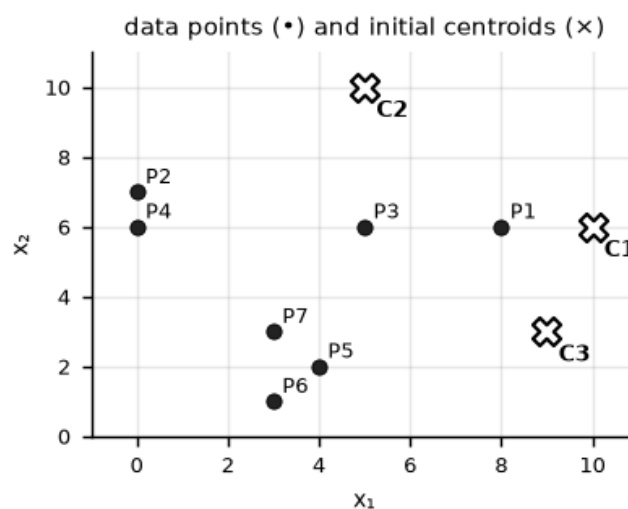
**Q.19 [2 marks · NAT (numerical answer)]**

A centered data matrix X with $n = 9$ rows (samples) and $d = 3$ columns has singular values 21, 10, 8. Using the sample covariance with divisor $n - 1$, the minimum number of principal components needed to retain at least 95% of the variance is _____ (Answer in integer)

Q.20 [2 marks · NAT (numerical answer)]

k-means (Lloyd's algorithm, Euclidean distance) with $k = 3$ is run on the 7 points below, starting from the initial centroids $C1 = (10, 6)$, $C2 = (5, 10)$, $C3 = (9, 3)$. After **one** iteration (assignment step followed by update step), the x_1 -coordinate of the updated centroid of cluster $C1$ is _____ (rounded off to two decimal places)

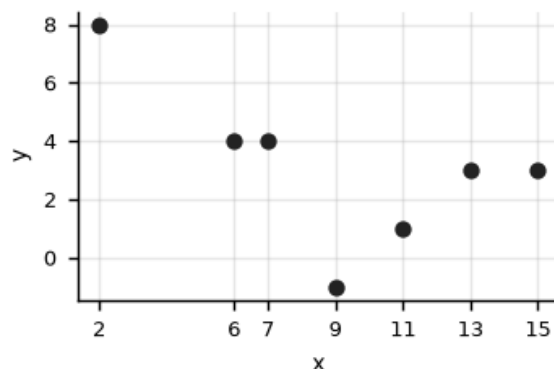
Point	P1	P2	P3	P4	P5	P6	P7
x_1	8	0	5	0	4	3	3
x_2	6	7	6	6	2	1	3



Q.21 [2 marks · NAT (numerical answer)]

A regression tree (squared-error criterion) is fitted to the 7 points below. The root split “ $x \leq t$ ” is chosen among midpoints of consecutive x values to minimise the total SSE of the two children (each child predicts its mean). Find the prediction of the resulting depth-1 regression tree (stump) at $x = 10$. (rounded off to two decimal places)

x	2	6	7	9	11	13	15
y	8	4	4	-1	1	3	3

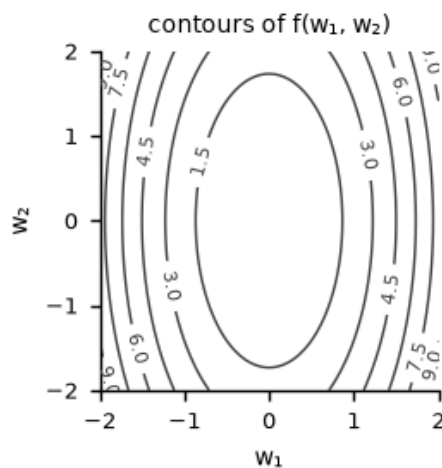
**Q.22 [2 marks · MCQ]**

For logits $z = (1, 0, 2)$, softmax probabilities p and true class 1, the gradient of the cross-entropy loss $L = -\ln p_1$ with respect to z (rounded to two decimals) is

- (A) $(-0.14, 0.01, 0.15)$ (B) $(0.76, -0.09, -0.67)$
 (C) $(-0.76, 0.09, 0.67)$ (D) $(-4.09, 0, 0)$

Q.23 [2 marks · MSQ (one or more correct)]

Gradient descent $w \leftarrow w - \eta \nabla f(w)$ is applied to $f(w_1, w_2) = \frac{1}{2}(4w_1^2 + 1w_2^2)$ from a generic starting point (both coordinates non-zero). For which of the following learning rates does GD **diverge**?



- (A) $\eta = 0.3$ (B) $\eta = 0.75$
 (C) $\eta = 0.65$ (D) $\eta = 0.2$

Q.24 [2 marks · MSQ (one or more correct)]

Consider agglomerative hierarchical clustering on n points with a dissimilarity matrix (no ties). Which of the following statements is/are CORRECT?

- (A) For any two points, the complete-linkage cophenetic distance is $\leq$ their original distance.
- (B) Replacing every dissimilarity by its square never changes the average-linkage dendrogram topology.
- (C) The single-linkage merge heights are exactly the edge weights of a minimum spanning tree of the complete distance graph.
- (D) For any two points, the single-linkage cophenetic distance is $\leq$ their original distance.

Q.25 [2 marks · NAT (numerical answer)]

Ridge regression without intercept minimises $\|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2$ with $\lambda = 2$, for the data below ($\beta = (\beta_1, \beta_2)^T$). The value of β_2 is _____ (rounded off to three decimal places)

$$X = \begin{bmatrix} 3 & -2 \\ 0 & 0 \\ 0 & -1 \\ 0 & -1 \end{bmatrix}$$

$$y = \begin{bmatrix} -3 \\ 5 \\ -2 \\ 5 \end{bmatrix}$$

Q.26 [2 marks · NAT (numerical answer)]

$X_1, \dots, X_n$ are i.i.d. with mean $\mu = 2$ and variance $\sigma^2 = 25$, with $n = 4$. Consider the shrinkage estimator $\hat{\mu} = c\bar{x}$ of μ , where $\bar{x}$ is the sample mean. Using $\text{MSE} = \text{Bias}^2 + \text{Variance}$, the value of c that minimises the MSE of $\hat{\mu} = c\bar{x}$ is _____ (rounded off to three decimal places)

Q.27 [2 marks · NAT (numerical answer)]

A 3-class softmax (multinomial logistic) regression model computes scores $\mathbf{z} = W\mathbf{x} + \mathbf{b}$ and $P(y = k | \mathbf{x}) = e^{z_k} / \sum_j e^{z_j}$, where row k of W holds the weights of class k . For the parameters below and $\mathbf{x} = (1, -1)^T$, the predicted probability $P(y = 1 | \mathbf{x})$ is _____ (rounded off to two decimal places)

$$W = \begin{bmatrix} 0 & 0 \\ 2 & -2 \\ 1 & -1 \end{bmatrix}$$

$$\mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Q.28 [2 marks · NAT (numerical answer)]

The labelled data below are linearly separable through the origin by the unit vector $\mathbf{u} = (1, 1)/\sqrt{2}$ (i.e. $y_i = \text{sign}(\mathbf{u} \cdot \mathbf{x}_i)$). Take $\gamma = \min_i y_i (\mathbf{u} \cdot \mathbf{x}_i)$ and $R = \max_i \|\mathbf{x}_i\|$. The perceptron convergence theorem (no bias, $w_0 = 0$, $\eta = 1$) bounds the number of mistakes by $(R/\gamma)^2$. The value of this bound is _____ (rounded off to two decimal places)

i	1	2	3	4	5
$\mathbf{x}_i$	(2, 4)	(0, 3)	(-3, -3)	(0, 2)	(0, -4)
y_i	1	1	-1	1	-1

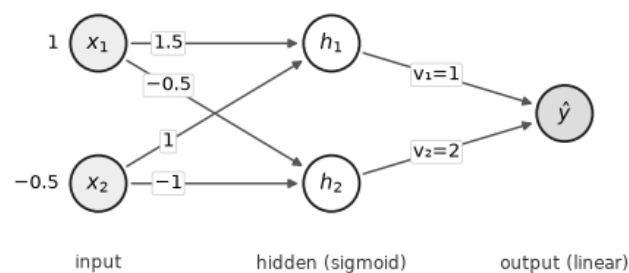
Q.29 [2 marks · NAT (numerical answer)]

In C4.5, the attribute Region splits a node of 17 records as tabulated (class Yes/No counts in each branch). Compute the split information of Region (all logarithms base 2). (rounded off to two decimal places)

Region	#Yes	#No
v1	3	6
v2	2	1
v3	4	1

Q.30 [2 marks · NAT (numerical answer)]

In the network below, hidden units are sigmoid, the output is linear ($\hat{y} = v_1 h_1 + v_2 h_2$), all biases are 0, and the loss is $L = \frac{1}{2}(\hat{y} - y)^2$. For input $x = (1, -0.5)$ and target $y = -1$, the value of $\partial L / \partial b_2$ (bias of hidden unit h_2 , currently 0) is _____ (rounded off to three decimal places)

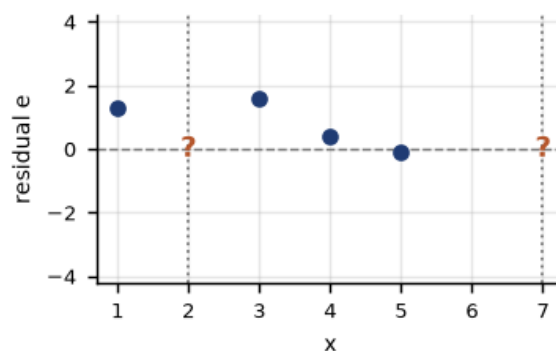
**Q.31 [2 marks · NAT (numerical answer)]**

A classifier for a rare condition has sensitivity (recall) 0.8 and specificity 0.9. It is deployed on a population where the prevalence of the positive class is 0.01. The expected precision (positive predictive value) in this population is _____ (rounded off to three decimal places)

Q.32 [2 marks · NAT (numerical answer)]

A simple linear regression with intercept is fitted by least squares. The table gives the x -values and the residuals $e_i = y_i - \hat{y}_i$, but two residuals are missing. The fitted line is $\hat{y} = 1.7 + 1.1x$. The observed value y at $x = 7$ is _____ (rounded off to two decimal places)

x_i	1	2	3	4	5	7
e_i	1.3	?	1.6	0.4	-0.1	?



Q.33 [2 marks · NAT (numerical answer)]

Consider the model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ (first column of X is the intercept column), fitted by least squares using $\beta = (X^T X)^{-1} X^T y$ with the design matrix X and response y below. Then the predicted response at $(x_1, x_2) = (2, 2)$ is _____ (rounded off to two decimal places)

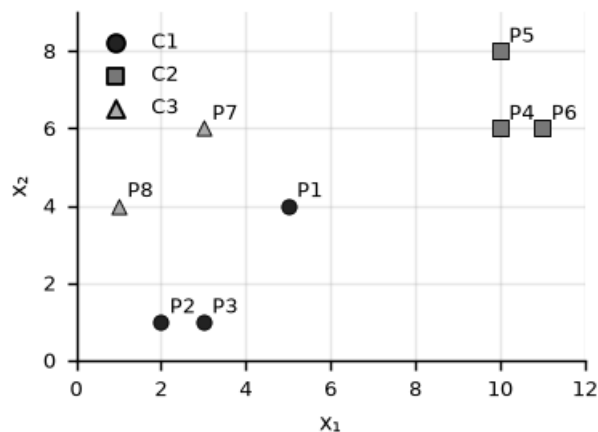
$$X = \begin{bmatrix} 1 & -2 & 2 \\ 1 & -1 & -1 \\ 1 & 0 & -2 \\ 1 & 1 & -1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$y = \begin{bmatrix} -1 \\ 6 \\ 12 \\ 14 \\ 13 \end{bmatrix}$$

Q.34 [2 marks · NAT (numerical answer)]

A clustering of 8 points into 3 clusters is shown below. Using Euclidean distance, the silhouette coefficient $s(i) = (b(i) - a(i))/\max(a(i), b(i))$ of point P7 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5	P6	P7	P8
x_1	5	2	3	10	10	11	3	1
x_2	4	1	1	6	8	6	6	4
cluster	C1	C1	C1	C2	C2	C2	C3	C3

**Q.35 [2 marks · NAT (numerical answer)]**

In one AdaBoost round the current example weights and whether the chosen weak classifier h classifies each example correctly are given. Using $\alpha = \frac{1}{2} \ln((1 - \varepsilon)/\varepsilon)$ and the update $w_i \leftarrow w_i \cdot \exp(-\alpha y_i h(x_i))/Z$, find the normalisation constant $Z = \sum_i w_i \cdot \exp(-\alpha y_i h(x_i))$. (rounded off to three decimal places)

Example i	1	2	3	4	5	6
Weight w_i	1/6	1/6	1/6	1/6	1/6	1/6
$h(x_i)$ correct?	No	Yes	Yes	Yes	Yes	No

— END OF QUESTION PAPER —

Mock Test 18 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	21.74 to 21.76	19	NAT	2	3
2	MSQ	1	(B), (C), (D)	20	NAT	2	7.99 to 8.01
3	MSQ	1	(A), (B), (C)	21	NAT	2	2.3233 to 2.3433
4	MSQ	1	(A), (D)	22	MCQ	2	(C)
5	NAT	1	10	23	MSQ	2	(B), (C)
6	NAT	1	−6	24	MSQ	2	(C), (D)
7	NAT	1	0.45 to 0.47	25	NAT	2	−0.4068 to −0.4008
8	MSQ	1	(A), (B), (C), (D)	26	NAT	2	0.3872 to 0.3932
9	NAT	1	1.6465 to 1.6505	27	NAT	2	0.0059 to 0.0259
10	NAT	1	15.99 to 16.01	28	NAT	2	9.98 to 10.02
11	MSQ	1	(B)	29	NAT	2	1.4366 to 1.4566
12	NAT	1	22.6567 to 22.6767	30	NAT	2	1.3635 to 1.3675
13	NAT	1	−0.5964 to −0.5764	31	NAT	2	0.0728 to 0.0768
14	NAT	1	2.99 to 3.01	32	NAT	2	9.1938 to 9.2862
15	NAT	1	71	33	NAT	2	13.0771 to 13.2086
16	MCQ	2	(A)	34	NAT	2	0.3336 to 0.3536
17	MSQ	2	(A), (B), (D)	35	NAT	2	0.9408 to 0.9448
18	NAT	2	0.998 to 1.002				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Probabilistic & Instance-based Classifiers					11
Unsupervised Learning					11
SVM & Decision Trees					11
Neural Networks (MLP / Feed-forward)					11
Regression & Model Selection					11
Total					55

Mock Test 18 — Detailed Solutions

Q.1 — Answer: 21.74 to 21.76 [NAT, 1M · Probabilistic & Instance-based Classifiers › k-nearest neighbours]

Distances $|x_i - 2.5|$: $x=3$: 0.5, $x=4$: 1.5, $x=5$: 2.5, $x=6$: 3.5, $x=7$: 4.5, $x=8$: 5.5, $x=12$: 9.5.

The 4 nearest neighbours are $x = 3, 4, 5, 6$ with $y = 25, 16, 24, 22$.

$\hat{y} = (25 + 16 + 24 + 22)/4 = \mathbf{21.75}$.

Q.2 — Answer: (B), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers › Bayes optimal classifier]

(A) **FALSE**. Error cannot go below the Bayes error, which can be positive.

(B) **TRUE**. A costly error type shifts the decision threshold.

(C) **TRUE**. This is the MAP rule.

(D) **TRUE**. $P(x_{\text{obs}}|y)$ is obtained from the joint model.

Q.3 — Answer: (A), (B), (C) [MSQ, 1M · Unsupervised Learning › Hierarchical clustering]

(A) **TRUE**. There are $n(n-1)/2$ pairs.

(B) **TRUE**. Top-down procedure.

(C) **TRUE**. Definition of Ward's criterion.

(D) **FALSE**. The naive algorithm is $O(n^3)$ ($O(n^2 \log n)$ with heaps).

Q.4 — Answer: (A), (D) [MSQ, 1M · SVM & Decision Trees › SVM: kernels]

(A) **TRUE**. The Gaussian/RBF kernel.

(B) **FALSE**. All diagonal entries equal 1.

(C) **FALSE**. For two distinct points $\det \begin{bmatrix} 1 & e^{y d^2} \\ e^{y d^2} & 1 \end{bmatrix} < 0$.

(D) **TRUE**. Sum of the linear kernel and a constant kernel.

Q.5 — Answer: 10 [NAT, 1M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

Updates per epoch = $\lceil 50000/50 \rceil = 1000$; epochs = $10000/1000 = \mathbf{10}$.

Q.6 — Answer: -6 [NAT, 1M · Neural Networks (MLP / Feed-forward) › Forward pass]

$a = Wx + b = (-5, 2)$; $h = \text{ReLU}(a) = (0, 2)$ - a negative pre-activation is clipped to 0.

$y = 3 \cdot 0 + (-3) \cdot 2 + (0) = \mathbf{-6}$.

A common slip is to forget the ReLU and use the raw pre-activation (which would give -21).

Q.7 — Answer: 0.45 to 0.47 [NAT, 1M · Probabilistic & Instance-based Classifiers › Evaluation metrics]

$TP = 57$, $FN = 56$, $FP = 27$, $TN = 23$ ($N = 163$).

Specificity (TNR) = $TN/(TN + FP)$.

Value = **0.46**.

Watch the orientation: swapping FP and FN is the most common error (it swaps precision and recall).

Q.8 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection › Ridge & Lasso]

(A) **TRUE**. Its vertices lie on the coordinate axes.

(B) **TRUE**. $X^T X$ is PSD, so $X^T X + \lambda I$ has eigenvalues $\geq \lambda > 0$.

(C) **TRUE**. $(I + \lambda I)\beta = X^T y = \beta_{\text{OLS}}$.

(D) **TRUE**. Its subgradient at 0 is the interval $[-1, 1]$.

Q.9 — Answer: 1.6465 to 1.6505 [NAT, 1M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

$\nabla = \partial L / \partial w + \lambda w = -1.5 + 0.01 \times 1.5 = -1.485$.

$w \leftarrow 1.5 - 0.1 \times -1.485 = \mathbf{1.6485}$ (equivalently $(1 - \eta \lambda)w - \eta \partial L / \partial w$).

Q.10 — Answer: 15.99 to 16.01 [NAT, 1M · SVM & Decision Trees › SVM: margin & geometry]

Margin width $2/\|w\| = 0.5 \Rightarrow \|w\| = 2/0.5 = 4$.

$\|w\|^2 = \mathbf{16}$.

Q.11 — Answer: (B) [MSQ, 1M · Regression & Model Selection > Simple linear regression]

(A) FALSE. The decomposition is $SST = SSR + SSE$.

(B) TRUE. S_{xy} scales by c while S_{xx} is unchanged.

(C) FALSE. It is $\hat{y} = \bar{y}$.

(D) FALSE. Their product is r^2 , so they are reciprocals only when $|r| = 1$.

Q.12 — Answer: 22.6567 to 22.6767 [NAT, 1M · Unsupervised Learning > k-means]

Assignment (nearest centroid): $c = 3$: {0, 3, 6}; $c = 19$: {15, 16, 18}.

Updated centroids (cluster means): 3, 16.33.

$SSE = (0-3)^2 + (3-3)^2 + (6-3)^2 + (15-16.33)^2 + (16-16.33)^2 + (18-16.33)^2 = 22.67$.

(With the old centroids the SSE was 44; the update step can only lower it.)

Q.13 — Answer: -0.5964 to -0.5764 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

The data are already centered (column means are 0), so $S = X^T X / (n - 1)$.

$\Sigma_{X_1^2} = 38$, $\Sigma_{X_2^2} = 60$, $\Sigma_{X_1 X_2} = -28$.

$S = \begin{bmatrix} 9.5 & -7 \\ -7 & 15 \end{bmatrix}$.

Required value = **-0.59**.

Q.14 — Answer: 2.99 to 3.01 [NAT, 1M · SVM & Decision Trees > Decision trees: impurity measures]

$p_k = 3/24 = 1/8$ for every class $\Rightarrow H = -8 \cdot (1/8) \log_2(1/8) = \log_2 8 = 3$ — the maximum possible for 8 classes.

Q.15 — Answer: 71 [NAT, 1M · Regression & Model Selection > Cross-validation]

Each candidate needs $k = 10$ fits: $7 \times 10 = 70$. Plus 1 final refit: **71**.

Q.16 — Answer: (A) [MCQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Distances from $\mathbf{q}$:

Point	class	Euclidean d	Manhattan d
P1	B	2	2
P2	A	4.123	5
P3	A	8.062	11
P4	B	5.099	6
P5	B	5.385	7
P6	A	4.472	6
P7	A	1	1

Euclidean 3 nearest: P7(A), P1(B), P2(A) $\Rightarrow$ **A**.

Manhattan 3 nearest: P7(A), P1(B), P2(A) $\Rightarrow$ **A**.

Answer: **Euclidean: A; Manhattan: A**. Here both metrics lead to the same vote.

Q.17 — Answer: (A), (B), (D) [MSQ, 2M · SVM & Decision Trees > SVM: kernels]

A Gram matrix of a valid kernel must be symmetric and positive semidefinite. For a symmetric 2×2 matrix this is equivalent to: both diagonal entries ≥ 0 and determinant ≥ 0 .

(A) TRUE. symmetric, diagonal ≥ 0 , $\det = 12 > 0 \Rightarrow$ PD (negative off-diagonal entries are fine).

(B) TRUE. symmetric, diagonal ≥ 0 , $\det = 0 \Rightarrow$ PSD (singular) — still a valid Gram matrix.

(C) FALSE. not symmetric, but $K(x, z) = K(z, x)$ for every kernel.

(D) TRUE. symmetric, diagonal ≥ 0 , $\det = 15 > 0 \Rightarrow$ PD (negative off-diagonal entries are fine).

Q.18 — Answer: 0.998 to 1.002 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

$P(y=1|x=2) = \pi_1 P(x|1) / (\pi_0 P(x|0) + \pi_1 P(x|1)) = 0.24 / (0 + 0.24) = 1$.

Q.19 — Answer: 3 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

With $X = U \Sigma V^T$ (X centered, $n \times d$), the sample covariance is $S = X^T X / (n - 1) = V(\Sigma^2 / (n - 1))V^T$, so the PC directions are the columns of V and the eigenvalues are $\lambda_j = \sigma_j^2 / (n - 1)$.

$$\lambda = 21^2/8 = 55.125, 10^2/8 = 12.5, 8^2/8 = 8.$$

Cumulative % variance: 72.89, 89.42, 100 (note the $(n - 1)$ factor cancels in proportions: $PVE = \sigma_j^2 / \Sigma \sigma^2$).

Answer: **3**. (Using σ instead of σ^2 is the classic mistake.)

Q.20 — Answer: 7.99 to 8.01 [NAT, 2M · Unsupervised Learning > k-means]

Iteration 1 - assignment (nearest initial centroid, squared Euclidean distance):

Point	$d^2(\cdot, C1)$	$d^2(\cdot, C2)$	$d^2(\cdot, C3)$	cluster
P1 (8, 6)	4	25	10	C1
P2 (0, 7)	101	34	97	C2
P3 (5, 6)	25	16	25	C2
P4 (0, 6)	100	41	90	C2
P5 (4, 2)	52	65	26	C3
P6 (3, 1)	74	85	40	C3
P7 (3, 3)	58	53	36	C3

Update: each centroid becomes the mean of its assigned points: $C1 \rightarrow (8, 6)$; $C2 \rightarrow (1.67, 6.33)$; $C3 \rightarrow (3.33, 2)$.

Required coordinate = **8**.

Note: each step of Lloyd's algorithm can only decrease (or keep) J , which is why it converges.

Q.21 — Answer: 2.3233 to 2.3433 [NAT, 2M · SVM & Decision Trees > Regression trees]

Parent mean $\bar{y} = 3.143$, $SSE_0 = \Sigma(y - \bar{y})^2 = 46.857$.

Evaluate every split:

Split t	Left mean	Right mean	$SSE_L + SSE_R$
4	8	2.333	19.333
6.5	6	2	24
8	5.333	1.5	21.667
10	3.75	2.333	43.417
12	3.2	3	46.8
14	3.167	3	46.833

Best split $t = 4$: left mean 8, right mean 2.333, $SSE = 19.333$, reduction = 27.524.

Answer: **2.33**.

Q.22 — Answer: (C) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$p = \text{softmax}((1, 0, 2)) = (0.2447, 0.09, 0.6652)$; $y = (1, 0, 0)$.

$\partial L / \partial z = p - y = (-0.76, 0.09, 0.67)$.

Distractors: $y - p = (0.76, -0.09, -0.67)$ (wrong sign); p alone = $(0.24, 0.09, 0.67)$ (forgot the target); $(p - y) \odot p \odot (1 - p) = (-0.14, 0.01, 0.15)$ (spurious σ' -type factor, which cancels for cross-entropy); $-y/p = (-4.09, 0, 0)$ is $\partial L / \partial p$, not $\partial L / \partial z$.

Q.23 — Answer: (B), (C) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$\nabla f = (4w_1, 1w_2) = w_i \leftarrow (1 - \eta a_i) w_i$. GD converges (from every start) iff $|1 - \eta a_i| < 1$ for both i , i.e. $0 < \eta < 2/\max(a_i) = 2/4 = 0.5$. The steep direction ($a = 4$) limits the learning rate.

(A) FALSE. $\eta = 0.3$: $|1 - \eta \cdot 4| = 0.2$, $|1 - \eta \cdot 1| = 0.7 \rightarrow$ converges.

(B) TRUE. $\eta = 0.75$: $|1 - \eta \cdot 4| = 2$, $|1 - \eta \cdot 1| = 0.25 \rightarrow$ diverges.

(C) TRUE. $\eta = 0.65$: $|1 - \eta \cdot 4| = 1.6$, $|1 - \eta \cdot 1| = 0.35 \rightarrow$ diverges.

(D) FALSE. $\eta = 0.2$: $|1 - \eta \cdot 4| = 0.2$, $|1 - \eta \cdot 1| = 0.8 \rightarrow$ converges.

Q.24 — Answer: (C), (D) [MSQ, 2M · Unsupervised Learning > Hierarchical clustering]

(A) FALSE. It is $\geq$ the original distance.

(B) FALSE. Average linkage depends on actual values, not only on ranks.

(C) TRUE. Kruskal's algorithm performs the same merges in the same order.

(D) TRUE. They are joined no later than when their own edge would merge them.

Q.25 — Answer: -0.4068 to -0.4008 [NAT, 2M · Regression & Model Selection › Ridge & Lasso]

$$\beta_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y \text{ with } \lambda = 2.$$

$$X^T X = \begin{bmatrix} 9 & -6 \\ -6 & 6 \end{bmatrix}$$

$$X^T X + \lambda I = \begin{bmatrix} 11 & -6 \\ -6 & 8 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} -9 \\ 3 \end{bmatrix}$$

$$\det = 11 \times 8 - (-6)^2 = 52; (X^T X + \lambda I)^{-1} = (1/52)[[8, 6], [6, 11]].$$

$$\beta = (-1.0385, -0.4038).$$

Answer: **-0.404**.

Q.26 — Answer: 0.3872 to 0.3932 [NAT, 2M · Regression & Model Selection › Bias-variance trade-off]

$$E[c \bar{x}] = c\mu = \text{Bias} = (c - 1)\mu; \text{Var}(c \bar{x}) = c^2 \sigma^2/n. \text{MSE}(c) = (c - 1)^2 \mu^2 + c^2 \sigma^2/n.$$

$$\text{Here } \mu = 2, \sigma^2/n = 25/4 = 6.25.$$

$$d\text{MSE}/dc = 2(c - 1)\mu^2 + 2c\sigma^2/n = 0 = c^* = \mu^2/(\mu^2 + \sigma^2/n) = 4/(4 + 6.25) = 0.3902.$$

$c^* < 1$: accepting some bias reduces variance enough to lower MSE.

Answer: **0.39**.

Q.27 — Answer: 0.0059 to 0.0259 [NAT, 2M · Probabilistic & Instance-based Classifiers › Logistic regression]

$$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b} = (1, 5, 3).$$

$$e^{\mathbf{z}} = (2.7183, 148.4132, 20.0855), \text{sum} = 171.217.$$

$$\text{Softmax probabilities} = (0.0159, 0.8668, 0.1173).$$

$$P(y = 1 | \mathbf{x}) = \mathbf{0.02}.$$

Q.28 — Answer: 9.98 to 10.02 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Perceptron]

$$\text{Unit separator } \mathbf{u} = (1, 1)/\sqrt{2}. \text{Margins } y_i(\mathbf{u} \cdot \mathbf{x}_i): 4.243, 2.121, 4.243, 1.414, 2.828 \rightarrow \gamma = \min = 1.4142.$$

$$\text{Norms } \|\mathbf{x}_i\|: 4.472, 3, 4.243, 2, 4 \rightarrow R = \max = 4.4721.$$

$$\text{Novikoff bound: number of mistakes} \leq (R/\gamma)^2 = (4.4721/1.4142)^2 = \mathbf{10}.$$

Common mistakes: forgetting to normalise $\mathbf{u}$ (γ must be measured with $\|\mathbf{u}\| = 1$), or using R/γ without squaring.

Q.29 — Answer: 1.4366 to 1.4566 [NAT, 2M · SVM & Decision Trees › Decision trees: split selection]

$$\text{Parent [9 Yes, 8 No]: } H = 0.9975.$$

$$\text{Child entropies: } v1: 0.9183, v2: 0.9183, v3: 0.7219.$$

$$IG = 0.9975 - \sum (n_v/17) H_v = 0.137.$$

$$\text{SplitInfo} = -\sum (n_v/N) \log_2(n_v/N) = -(9/17)\log_2(9/17) - (3/17)\log_2(3/17) - (5/17)\log_2(5/17) = 1.4466.$$

Answer: **1.45**.

C4.5 uses the gain ratio to penalise attributes that split the data into many small branches.

Q.30 — Answer: 1.3635 to 1.3675 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Backpropagation]

$$\text{Forward: } \mathbf{a} = \mathbf{W}\mathbf{x} = (1, 0), \mathbf{h} = \sigma(\mathbf{a}) = (0.7311, 0.5), \hat{y} = \mathbf{v} \cdot \mathbf{h} = 1.7311.$$

$$\text{Output error } \delta_o = \partial L / \partial \hat{y} = \hat{y} - y = 2.7311 \text{ (linear output, } L = \frac{1}{2}(\hat{y} - y)^2).$$

$$\text{Hidden delta } \delta_2 = \delta_o \cdot v_2 \cdot h_2(1 - h_2) = 2.7311 \times 2 \times 0.25 = 1.3655.$$

$$\partial L / \partial b_2 = \delta_2 = \mathbf{1.3655}.$$

Q.31 — Answer: 0.0728 to 0.0768 [NAT, 2M · Probabilistic & Instance-based Classifiers >**Evaluation metrics]**

Per unit population: $TP = Se \cdot \pi$, $FN = (1 - Se) \cdot \pi$, $FP = (1 - Sp)(1 - \pi)$, $TN = Sp(1 - \pi)$.

$TP = 0.008$, $FN = 0.002$, $FP = 0.099$, $TN = 0.891$.

Precision (positive predictive value) = $TP/(TP + FP) = \mathbf{0.075}$.

Even a good test has low precision when the positive class is rare: most positives predicted are false positives from the large negative class (class-imbalance effect; ROC is unaffected).

Q.32 — Answer: 9.1938 to 9.2862 [NAT, 2M · Regression & Model Selection > Simple linear regression]

For the least-squares fit with intercept the normal equations give $\sum e_i = 0$ and $\sum x_i e_i = 0$.

Known: $\sum e = 3.2$, $\sum xe = 7.2$. With unknowns $u = e$ at $x = 2$ and $v = e$ at $x = 7$:

$u + v = -3.2$, $2u + 7v = -7.2$.

Solving: $u = -3.04$, $v = -0.16$.

$y = \hat{y} + e = (1.7 + (1.1)(7)) + (-0.16) = 9.24$.

Answer: **9.24**.

Q.33 — Answer: 13.0771 to 13.2086 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

The columns of X are mutually orthogonal, so $X^T X$ is diagonal and each coefficient is $\beta_j = (x_j^T y)/(x_j^T x_j)$.

$$X^T X = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 14 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 44 \\ 36 \\ -20 \end{bmatrix}$$

$\beta = (44/5, 36/10, -20/14) = (8.8, 3.6, -1.4286)$.

$\hat{y} = 8.8 + (3.6)(2) + (-1.4286)(2) = 13.1429$.

Answer: **13.14**.

Q.34 — Answer: 0.3336 to 0.3536 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

a(i) = mean Euclidean distance from P_7 to the other members of its own cluster C_3 : $d(P_7, P_8) = 2.828 \rightarrow a = 2.8284$.

b(i) = smallest mean distance to the points of another cluster: $C_1: 4.3091$; $C_2: 7.4267 \rightarrow b = 4.3091$.

$s(i) = (b - a)/\max(a, b) = (4.3091 - 2.8284)/4.3091 = \mathbf{0.34}$.

s ranges in $[-1, 1]$; values near 1 mean well clustered, negative values suggest a wrong assignment.

Q.35 — Answer: 0.9408 to 0.9448 [NAT, 2M · SVM & Decision Trees > Ensembles: boosting]

$\varepsilon = \sum_{\text{misclassified}} w_i = 0.3333$; $\alpha = \frac{1}{2} \ln(0.6667/0.3333) = 0.3466$.

Misclassified weights $\times e^\alpha = 1.4142$; correct weights $\times e^{-\alpha} = 0.7071$.

$Z = \varepsilon e^\alpha + (1 - \varepsilon) e^{-\alpha} = 2\sqrt{\varepsilon(1 - \varepsilon)} = 0.9428$.

New weights: 0.25, 0.125, 0.125, 0.125, 0.125, 0.25.

Shortcut: after normalisation the misclassified set has total weight exactly $\frac{1}{2}$: $w_i' = w_i/(2\varepsilon)$ if misclassified, $w_i/(2(1 - \varepsilon))$ otherwise.

Answer: **0.943**.

Mock Test 19

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	4	5	26

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Neural Networks (MLP / Feed-forward)	7	11
Probabilistic & Instance-based Classifiers	7	11
Unsupervised Learning	7	11
Regression & Model Selection	7	11
SVM & Decision Trees	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MSQ (one or more correct)]

Which of the following statements about backpropagation is/are CORRECT?

- (A) The gradient with respect to a bias b_j equals δ_j .
- (B) Hidden-layer deltas are computed before the output-layer deltas.
- (C) Backpropagation can only be used with sigmoid activation functions.
- (D) With a sigmoid output unit and squared-error loss $\frac{1}{2}(\hat{y} - y)^2$, the output delta is simply $\hat{y} - y$.

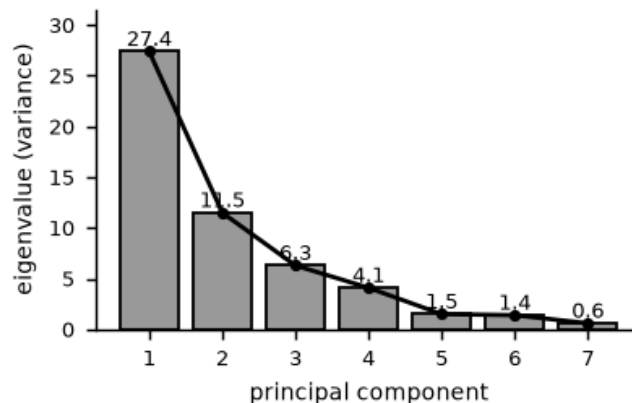
Q.2 [1 mark · NAT (numerical answer)]

In a training set, the 5 examples of class Buys = Yes have the following counts for the feature Age. Using additive (add- α) smoothing with $\alpha = 2$, the naive Bayes estimate of $P(\text{Age} = \text{Middle} \mid \text{Buys} = \text{Yes})$ is _____ (rounded off to three decimal places)

Age	Young	Middle	Senior
count in Buys = Yes	2	0	3

Q.3 [1 mark · NAT (numerical answer)]

The scree plot below shows the eigenvalues of the covariance matrix of a 7-dimensional dataset. The minimum number of principal components needed to explain at least 90% of the total variance is _____ (Answer in integer)

**Q.4 [1 mark · MCQ]**

Logistic regression, despite its name, is primarily used for:

- (A) classification (B) density estimation
(C) regression of real-valued targets (D) clustering

Q.5 [1 mark · NAT (numerical answer)]

In the dual solution of a linear SVM, the non-zero Lagrange multipliers α_i and labels y_i of the support vectors are given below (all other $\alpha_i = 0$). The value of α_5 is _____ (rounded off to one decimal place)

i	1	2	3	4	5
y_i	+1	+1	-1	-1	-1
α_i	1.2	1.2	1.2	0.9	?

Q.6 [1 mark · NAT (numerical answer)]

A bootstrap sample of size 6 is drawn with replacement from 6 distinct training points. The expected number of distinct points appearing in the sample is _____ (rounded off to two decimal places)

Q.7 [1 mark · MSQ (one or more correct)]

Which of the following statements about cross-validation (CV) is/are CORRECT? (n denotes the number of observations.)

- (A) For least-squares linear regression, LOOCV error can be computed from a single fit using the leverages h_{ii} .
(B) Compared with 2-fold CV, 10-fold CV uses larger training sets in each fit.
(C) 2-fold CV gives a less (pessimistically) biased estimate of test error than LOOCV.
(D) For any learning algorithm, LOOCV can be computed from a single model fit.

Q.8 [1 mark · NAT (numerical answer)]

Consider the polynomial kernel $K(x, z) = (x^T z + 1)^2$ with $x = (3, -2, -2)$ and $z = (0, 0, 3)$. The value of $K(x, z)$ is _____ (Answer in integer)

Q.9 [1 mark · NAT (numerical answer)]

How many distinct monomials of total degree **exactly** 2 can be formed from 2 variables (e.g. for $p = 2, d = 2: x_1^2, x_1 x_2, x_2^2$)? _____ (Answer in integer)

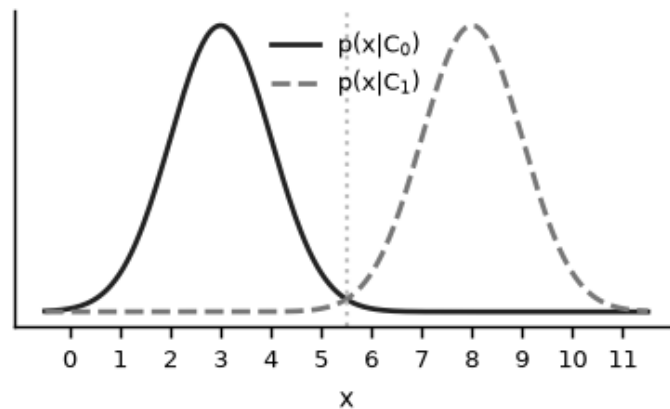
Q.10 [1 mark · MSQ (one or more correct)]

Which of the following statements about the perceptron is/are CORRECT?

- (A) XOR becomes linearly separable if the product feature x_1x_2 is added as an extra input.
- (B) If the training data are linearly separable, the perceptron learning algorithm makes only a finite number of updates.
- (C) The perceptron mistake bound $(R/\gamma)^2$ does not depend explicitly on the input dimension d .
- (D) In the perceptron learning rule (labels ± 1), the weights change only on misclassified examples.

Q.11 [1 mark · MCQ]

Two classes have unit-variance Gaussian class-conditional densities with means 3 (C_0) and 8 (C_1), as plotted. With equal priors the Bayes decision threshold is the dotted line at $x = 5.5$. If the prior of C_0 is increased to 0.8, the threshold



- (A) It moves towards $x = 8$, enlarging the region assigned to C_0
- (B) It stays at $x = 5.5$ because the two variances are equal
- (C) It splits into two thresholds, giving a quadratic boundary
- (D) It moves towards $x = 3$, shrinking the region assigned to C_0

Q.12 [1 mark · NAT (numerical answer)]

A training run on $N = 1200$ examples with batch size $B = 128$ (last partial batch kept) performed 130 parameter updates in total. The number of epochs completed is _____ (Answer in integer)

Q.13 [1 mark · NAT (numerical answer)]

Lloyd's k -means is run on $n = 1000$ points in $d = 7$ dimensions with $k = 7$ for $T = 15$ iterations. Counting one subtraction per coordinate per point-centroid pair in the assignment step, the total number of such subtractions is _____ (Answer in integer)

Q.14 [1 mark · NAT (numerical answer)]

For $\mathbf{x} = (1, 0, 0)$ and $\mathbf{y} = (1, 1, 7)$, the cosine similarity between $\mathbf{x}$ and $\mathbf{y}$ is _____ (rounded off to two decimal places)

Q.15 [1 mark · NAT (numerical answer)]

A 3-class softmax classifier produces the logits $z_1 = -1$, $z_2 = 2$, $z_3 = 1$ for an input $\mathbf{x}$. The predicted probability of class 1, $P(y = 1 | \mathbf{x})$, is _____ (rounded off to two decimal places)

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · MSQ (one or more correct)]**

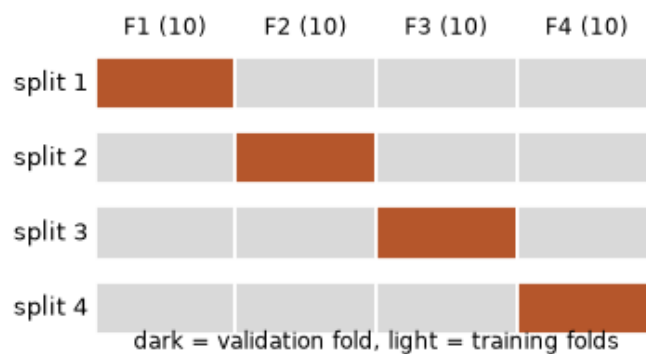
Let σ be the logistic sigmoid, and $'$ denote the derivative. At $z = -2$ (and $-z = 2$), which of the following is/are CORRECT? ($\text{softplus}(z) = \ln(1 + e^z)$.)

- (A) $\sigma'(-2) > \tanh'(-2)$ (B) $\sigma'(-2) = \sigma'(2)$
 (C) $\tanh(-2) + \tanh(2) = 1$ (D) $\text{ReLU}'(-2) = 0$

Q.17 [2 marks · NAT (numerical answer)]

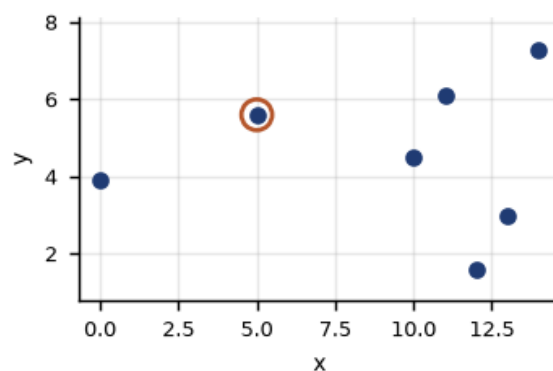
Ridge regression is tuned by 4-fold cross-validation (equal fold sizes) over $\lambda \in \{0.01, 2, 10\}$. The validation MSE on each fold is shown. The CV estimate (mean over folds) of the λ that would be selected is _____ (rounded off to two decimal places)

λ	Fold 1	Fold 2	Fold 3	Fold 4
0.01	2.5	3.7	2.4	5.1
2	5.3	4.5	2.3	2.2
10	3.8	2.3	2.9	3.3

**Q.18 [2 marks · NAT (numerical answer)]**

A simple linear regression (with intercept) is fitted by least squares to the 7 points shown. With $H = X(X^T X)^{-1} X^T$, the leverage h_{ii} of the observation with $x = 5$ (circled) is _____ (rounded off to three decimal places)

x	0	5	10	11	12	13	14
y	3.9	5.6	4.5	6.1	1.6	3	7.3



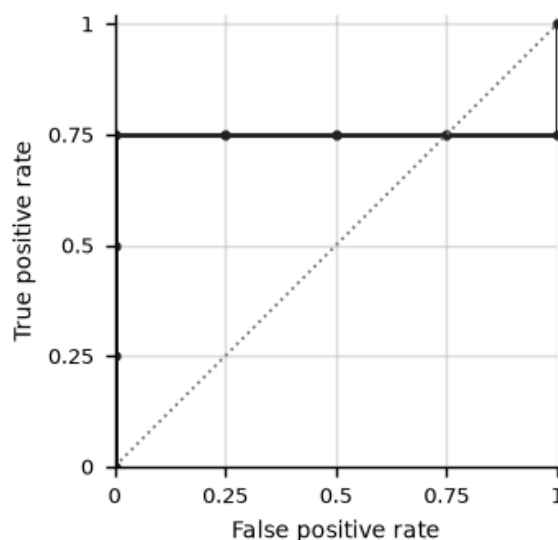
Q.19 [2 marks · NAT (numerical answer)]

Three customers are described by age and income. Age ranges over [20, 60] years and income over [10, 210] thousand rupees in the full dataset. On the **raw** (unscaled) features, the Euclidean distance from A to its nearest neighbour among B and C is _____ (rounded off to two decimal places)

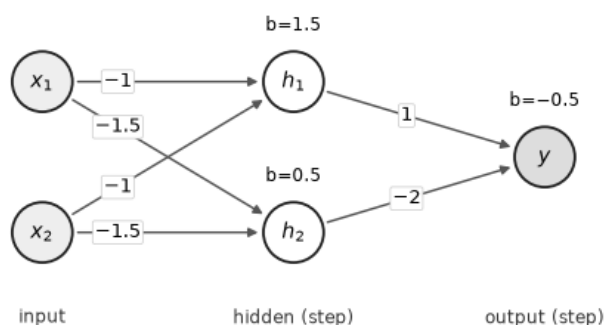
Customer	Age (years)	Income (₹ thousand)
A	29	55
B	34	165
C	58	135

Q.20 [2 marks · NAT (numerical answer)]

The ROC curve below was obtained from a test set with 4 positives and 4 negatives (all scores distinct, so the curve moves in steps of 1/4 vertically and 1/4 horizontally). From the figure, the area under this ROC curve (AUC) is _____ (rounded off to two decimal places)

**Q.21 [2 marks · MCQ]**

In the network below every unit (hidden and output) is a threshold unit that outputs 1 if its weighted input plus bias is > 0 , and 0 otherwise. Edge labels are weights. For binary inputs $x_1, x_2 \in \{0, 1\}$, the network computes



- (A) (NOT x_1) AND x_2
(C) XOR

- (B) XNOR
(D) NOR

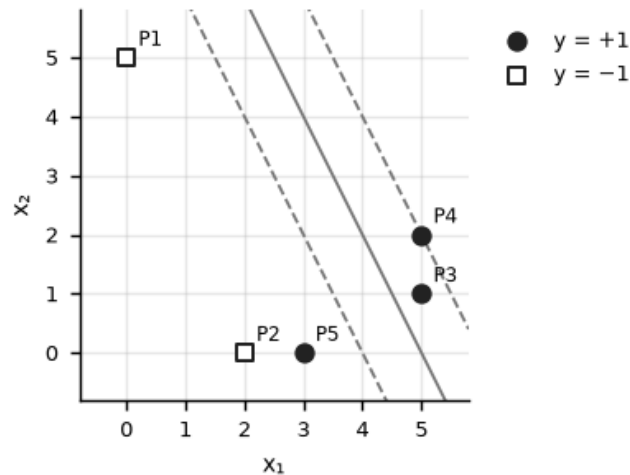
Q.22 [2 marks · NAT (numerical answer)]

The eigenvalues of the sample covariance matrix (divisor $n - 1$) of a centered 5-dimensional dataset are 20, 19.5, 9.5, 8, 5. Then the mean squared reconstruction error $(1/(n - 1)) \sum_i \|x_i - \hat{x}_i\|^2$, when each point is projected onto the top 2 principal components, is _____ (rounded off to two decimal places)

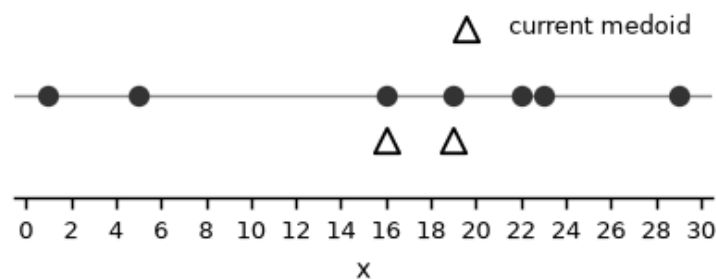
Q.23 [2 marks · NAT (numerical answer)]

A linear SVM with $f(x) = x_1 + 0.5x_2 - 5$ is evaluated on the training points in the table (also plotted, with the boundary $f = 0$ solid and $f = \pm 1$ dashed). Compute the value of the soft-margin primal objective $\frac{1}{2}\|w\|^2 + C \sum_i \xi_i$ with $C = 1$ (using the optimal slacks for this (w, b)). (rounded off to two decimal places)

Point	x_1	x_2	y
P1	0	5	-1
P2	2	0	-1
P3	5	1	+1
P4	5	2	+1
P5	3	0	+1

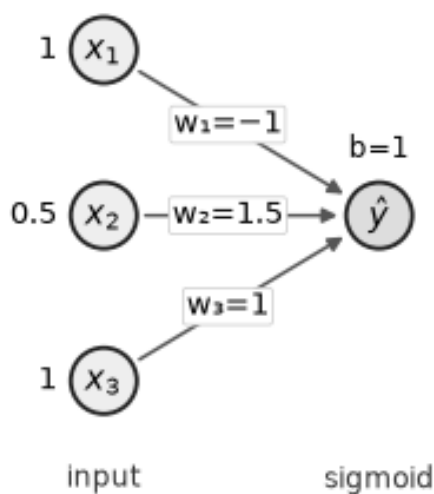
**Q.24 [2 marks · NAT (numerical answer)]**

PAM (k-medoids, $k = 2$, absolute-difference dissimilarity) is applied to the 1-D data $\{1, 5, 16, 19, 22, 23, 29\}$. The current medoids are 16 and 19. The total cost $\sum_i \min_m |x_i - m|$ of the current configuration is ____ (Answer in integer)



Q.25 [2 marks · NAT (numerical answer)]

A single sigmoid neuron (shown below) receives $\mathbf{x} = (1, 0.5, 1)$ and has target $y = 0$. The loss is $L = \frac{1}{2}(\hat{y} - y)^2$. The value of $\partial L / \partial w_2$ is _____ (rounded off to 4 decimal places)

**Q.26 [2 marks · NAT (numerical answer)]**

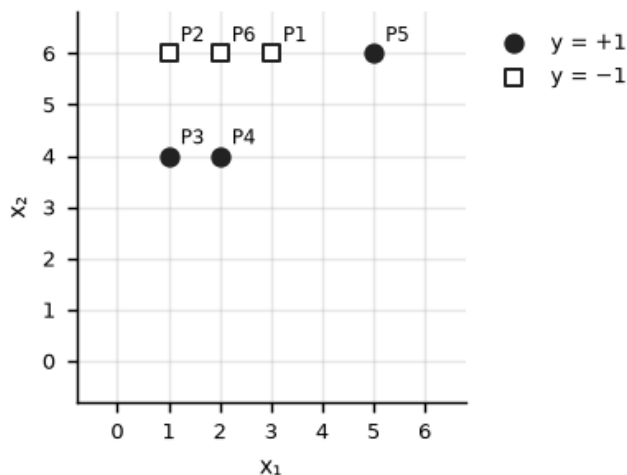
Logistic regression $P(y = 1 \mid \mathbf{x}) = \sigma(w_1 x + b)$ is trained by **batch** gradient descent on the **average** cross-entropy loss over the 3 examples below, with learning rate $\eta = 0.5$. Starting from $w_1 = 0.5$, $b = 0.5$, the value of w_1 after ONE gradient-descent update is _____ (rounded off to three decimal places)

i	x	y
1	2	0
2	-2	1
3	-1	0

Q.27 [2 marks · NAT (numerical answer)]

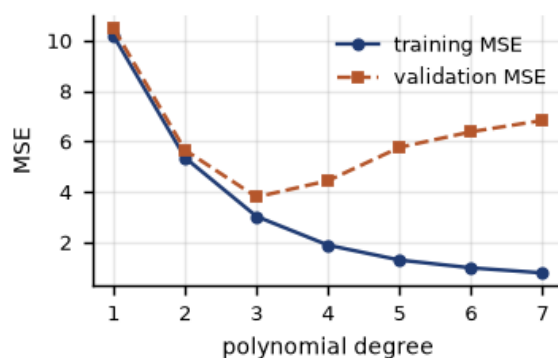
The labelled training points shown below are linearly separable. A hard-margin linear SVM is trained on them. Compute the width of the margin (distance between the two margin hyperplanes). (rounded off to two decimal places)

Point	x_1	x_2	y
P1	3	6	-1
P2	1	6	-1
P3	1	4	+1
P4	2	4	+1
P5	5	6	+1
P6	2	6	-1

**Q.28 [2 marks · MSQ (one or more correct)]**

Polynomial regression models of degree 1 to 7 are fitted by least squares on a training set and evaluated on a separate validation set. The resulting mean squared errors are given below. Which of the following statements is/are CORRECT?

Degree	1	2	3	4	5	6	7
Training MSE	10.2	5.36	3.03	1.89	1.3	0.99	0.79
Validation MSE	10.5	5.66	3.8	4.45	5.77	6.39	6.84



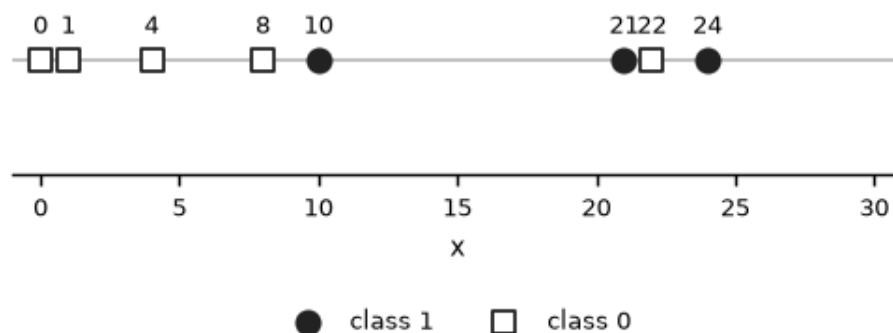
- (A) Based on the validation MSE, degree 1 should be selected.
- (B) Based on the validation MSE, degree 3 should be selected.
- (C) The degree-1 model has higher variance than the degree-7 model.
- (D) The generalisation gap (validation MSE – training MSE) is largest for degree 7.

Q.29 [2 marks · NAT (numerical answer)]

Let ϕ be the feature map of the kernel $K(x, z) = (x^T z)^2$ on $\mathbb{R}^2$, so that $K(x, z) = \phi(x)^T \phi(z)$. For $x = (2, -1)$ and $z = (2, 2)$, the squared distance $\|\phi(x) - \phi(z)\|^2$ in feature space is _____ (Answer in integer)

Q.30 [2 marks · NAT (numerical answer)]

8 one-dimensional points with binary labels are shown on the number line (filled circle = class 1, open square = class 0). A 1-NN classifier (absolute distance, majority vote) is evaluated by leave-one-out cross-validation: each point is classified using all the *other* points. The leave-one-out cross-validation error rate (as a fraction) is _____ (rounded off to two decimal places)

**Q.31 [2 marks · MCQ]**

A decision tree is to be grown on the 12 training records below using information gain as the splitting criterion (multi-way split on each categorical attribute). Which attribute is chosen at the root?

#	Shape	Student	Credit	Outlook	Class
1	Round	No	Fair	Rain	No
2	Round	No	Fair	Overcast	Yes
3	Square	No	Fair	Overcast	Yes
4	Oval	Yes	Excellent	Overcast	Yes
5	Oval	No	Excellent	Sunny	No
6	Oval	No	Excellent	Overcast	Yes
7	Oval	No	Excellent	Sunny	Yes
8	Round	Yes	Fair	Sunny	Yes
9	Round	No	Fair	Overcast	No
10	Square	No	Excellent	Sunny	Yes
11	Square	Yes	Excellent	Sunny	No
12	Round	No	Excellent	Rain	No

(A) Shape**(B)** Student**(C)** Credit**(D)** Outlook

Q.32 [2 marks · NAT (numerical answer)]

A linear model $\hat{y} = w_1 x_1 + w_2 x_2 + b$ is trained by stochastic gradient descent (batch size 1) on the per-example loss $\frac{1}{2}(\hat{y} - y)^2$ with learning rate $\eta = 0.1$, starting from $w = (0, 0)$, $b = 1$. The two examples below are processed once, in the given order. After both updates, the value of b is _____ (rounded off to three decimal places)

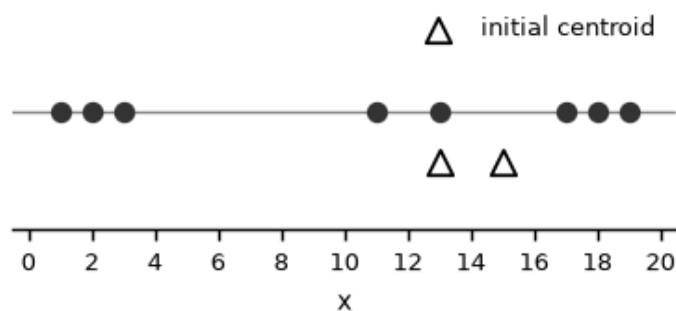
example	x_1	x_2	y
1	0	2	2
2	-1	-1	5

Q.33 [2 marks · NAT (numerical answer)]

A 3-class classifier outputs logits $z = (-1.5, 0.5, -1.5)$ and probabilities $p = \text{softmax}(z)$. The true class is 2, and the loss is the cross-entropy $L = -\ln p_2$. Then the cross-entropy loss $L = -\ln p_y$ (natural log) is _____ (rounded off to three decimal places)

Q.34 [2 marks · NAT (numerical answer)]

The 1-D dataset $\{1, 2, 3, 11, 13, 17, 18, 19\}$ is clustered with k-means ($k = 2$) using initial centroids 13 and 15 (shown as triangles). Lloyd's iterations (assign to nearest centroid, then recompute means) are run until convergence. Then the final within-cluster sum of squared errors (SSE) is _____ (rounded off to two decimal places)

**Q.35 [2 marks · NAT (numerical answer)]**

A naive Bayes classifier is trained on the 12 examples below, using **Laplace (add-one) smoothing** for the class-conditional feature probabilities (the class prior is not smoothed). For the new instance (Age = Young, Student = Yes, Rating = Excellent), the posterior probability $P(\text{Buys} = \text{Yes} \mid x)$ is _____ (rounded off to two decimal places)

#	Age	Student	Rating	Buys
1	Senior	No	Excellent	No
2	Senior	No	Fair	Yes
3	Senior	No	Fair	No
4	Senior	Yes	Fair	No
5	Middle	No	Fair	No
6	Senior	No	Fair	Yes
7	Senior	Yes	Fair	Yes
8	Senior	No	Fair	Yes
9	Middle	No	Fair	Yes
10	Senior	Yes	Fair	Yes
11	Senior	Yes	Fair	No
12	Young	No	Fair	Yes

— END OF QUESTION PAPER —

Mock Test 19 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MSQ	1	(A)	19	NAT	2	85.0841 to 85.1041
2	NAT	1	0.1798 to 0.1838	20	NAT	2	0.74 to 0.76
3	NAT	1	4	21	MCQ	2	(C)
4	MCQ	1	(A)	22	NAT	2	22.49 to 22.51
5	NAT	1	0.29 to 0.31	23	NAT	2	4.115 to 4.135
6	NAT	1	3.9806 to 4.0006	24	NAT	2	43
7	MSQ	1	(A), (B)	25	NAT	2	0.0527 to 0.0548
8	NAT	1	25	26	NAT	2	0.1013 to 0.1053
9	NAT	1	3	27	NAT	2	0.8844 to 0.9044
10	MSQ	1	(A), (B), (C), (D)	28	MSQ	2	(B), (D)
11	MCQ	1	(A)	29	NAT	2	81
12	NAT	1	13	30	NAT	2	0.615 to 0.635
13	NAT	1	735000	31	MCQ	2	(D)
14	NAT	1	0.13 to 0.15	32	NAT	2	1.5055 to 1.5145
15	NAT	1	0.0251 to 0.0451	33	NAT	2	0.2375 to 0.2415
16	MSQ	2	(A), (B), (D)	34	NAT	2	49.19 to 49.21
17	NAT	2	3.065 to 3.085	35	NAT	2	0.3939 to 0.4139
18	NAT	2	0.2621 to 0.2661				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Neural Networks (MLP / Feed-forward)					11
Probabilistic & Instance-based Classifiers					11
Unsupervised Learning					11
Regression & Model Selection					11
SVM & Decision Trees					11
Total					55

Mock Test 19 — Detailed Solutions

Q.1 — Answer: (A) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Backpropagation]

- (A) **TRUE**. a_j depends on b_j with coefficient 1.
 (B) **FALSE**. Order is output $\rightarrow$ input.
 (C) **FALSE**. Any (sub)differentiable activation works.
 (D) **FALSE**. It is $(\hat{y} - y)\hat{y}(1 - \hat{y})$; the simple form holds for cross-entropy.

Q.2 — Answer: 0.1798 to 0.1838 [NAT, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Add- α estimate: $(\text{count} + \alpha)/(N_c + \alpha V)$ with $V = 3$ possible values.
 $= (0 + 2)/(5 + 2 \times 3) = 2/11 = \mathbf{0.182}$.

(The unsmoothed estimate would be $0/5$, i.e. zero, which would wipe out the whole posterior product.)

Q.3 — Answer: 4 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

Total variance = $\Sigma \lambda = 52.8$. Cumulative proportion of variance explained:

k	1	2	3	4	5	6	7
λ_k	27.4	11.5	6.3	4.1	1.5	1.4	0.6
cumulative %	51.89	73.67	85.61	93.37	96.21	98.86	100

The cumulative share first reaches 90% at $k = \mathbf{4}$.

Q.4 — Answer: (A) [MCQ, 1M · Regression & Model Selection > Regression vs classification]

It models $P(y=1|x) = \sigma(w^T x + b)$ and predicts a class by thresholding.

Hence the answer is **classification**.

Q.5 — Answer: 0.29 to 0.31 [NAT, 1M · SVM & Decision Trees > SVM: dual & support vectors]

Stationarity w.r.t. b gives the dual constraint $\sum_i \alpha_i y_i = 0$.

$$\sum_{i \neq 5} \alpha_i y_i = +1.2 + 1.2 - 1.2 - 0.9 = 0.3.$$

So $\alpha_5(-1) = -0.3 \Rightarrow \alpha_5 = \mathbf{0.3}$ (it must be ≥ 0).

Q.6 — Answer: 3.9806 to 4.0006 [NAT, 1M · SVM & Decision Trees > Ensembles: bagging & random forests]

By linearity: $n \cdot P(\text{point appears}) = 6 \cdot [1 - (1 - 1/6)^6] = \mathbf{3.991}$.

Q.7 — Answer: (A), (B) [MSQ, 1M · Regression & Model Selection > Cross-validation]

(A) **TRUE**. $CV_{(n)} = (1/n) \sum (e_i / (1 - h_{ii}))^2$.

(B) **TRUE**. 90% vs 50% of the data.

(C) **FALSE**. 2-fold trains on only half the data, so its estimate is more pessimistically biased.

(D) **FALSE**. The shortcut holds for linear smoothers such as least squares, not in general.

Q.8 — Answer: 25 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

$$x^T z = (3)(0) + (-2)(0) + (-2)(3) = -6.$$

$$K(x, z) = (-6 + 1)^2 = -5^2 = \mathbf{25}.$$

The kernel equals the inner product $\phi(x)^T \phi(z)$ of the explicit (all-monomials-up-to-degree-d) feature maps, without forming ϕ .

Q.9 — Answer: 3 [NAT, 1M · Regression & Model Selection > Basis functions & feature scaling]

Multisets of size d from p variables: $C(p + d - 1, d) = C(3, 2) = \mathbf{3}$.

Q.10 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

(A) **TRUE**. $XOR = x_1 + x_2 - 2x_1x_2$ is linear in (x_1, x_2, x_1x_2) .

(B) **TRUE**. Perceptron convergence theorem (Novikoff): at most $(R/\gamma)^2$ mistakes.

(C) **TRUE**. It depends only on the radius R of the data and the margin γ .

(D) **TRUE**. Update $w \leftarrow w + \eta yx$ is applied only when $y(w \cdot x + b) \leq 0$.

Q.11 — Answer: (A) [MCQ, 1M · Probabilistic & Instance-based Classifiers › Linear discriminant analysis]

$$x^* = (\mu_0 + \mu_1)/2 + \sigma^2 \ln(\pi_0/\pi_1)/(\mu_1 - \mu_0) = 5.777.$$

A larger prior for C_0 means more of the x-axis is assigned to it, so the threshold moves away from μ_0 towards $\mu_1 = 8$. Answer: **It moves towards $x = 8$, enlarging the region assigned to C_0 .**

Equal variances keep the boundary a single point (linear); priors shift it but never make it quadratic.

Q.12 — Answer: 13 [NAT, 1M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

Updates per epoch = $\lceil 1200/128 \rceil = 10$; epochs = $130/10 = 13$.

Q.13 — Answer: 735000 [NAT, 1M · Unsupervised Learning › k-means]

Assignment cost per iteration = $n \cdot k \cdot d = 1000 \cdot 7 \cdot 7 = 49000$; over T iterations **735000**. This is the $O(nkdT)$ complexity of k-means.

Q.14 — Answer: 0.13 to 0.15 [NAT, 1M · Unsupervised Learning › Distance measures]

$$x \cdot y = 1, \|x\| = \sqrt{1}, \|y\| = \sqrt{51}; \cos \theta = x \cdot y / (\|x\| \|y\|) = 0.14 = \mathbf{0.14}.$$

For reference: $L_1 = 8$, $L_2 = 7.07$, $L_\infty = 7$ (always $L_\infty \leq L_2 \leq L_1$).

Q.15 — Answer: 0.0251 to 0.0451 [NAT, 1M · Probabilistic & Instance-based Classifiers › Logistic regression]

$$\text{softmax: } P(y = k | \mathbf{x}) = e^{z_k} / \sum_j e^{z_j}.$$

e^z values: 0.3679, 7.3891, 2.7183; sum = 10.4752.

$$P(y = 1 | \mathbf{x}) = 0.3679/10.4752 = \mathbf{0.04}.$$

Q.16 — Answer: (A), (B), (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) › Activation functions]

$$\sigma(-2) = 0.1192, \tanh(-2) = -0.964.$$

(A) TRUE. $\sigma' = 0.105$, $\tanh' = 0.0707$.

(B) TRUE. σ' is an even function.

(C) FALSE. $\tanh$ is odd: the sum is 0.

(D) TRUE. $\text{ReLU}' = 1$ for $z > 0$ and 0 for $z < 0$.

Q.17 — Answer: 3.065 to 3.085 [NAT, 2M · Regression & Model Selection › Cross-validation]

$\text{CV}(\lambda)$ = average of the fold MSEs: $\lambda = 0.01$: 3.425; $\lambda = 2$: 3.575; $\lambda = 10$: 3.075.

Smallest CV error at $\lambda = 10$: $\text{CV} = 3.075$.

Answer: **3.08**.

Q.18 — Answer: 0.2621 to 0.2661 [NAT, 2M · Regression & Model Selection › Multiple linear regression]

For simple linear regression with intercept, the leverage (diagonal of the hat matrix) is $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$.

$$\bar{x} = 9.2857, S_{xx} = 151.4286.$$

$$h = 1/7 + (-4.2857)^2/151.4286 = 0.2642.$$

Answer: **0.264**.

Q.19 — Answer: 85.0841 to 85.1041 [NAT, 2M · Unsupervised Learning › Distance measures]

Min-max scaling: $x' = (x - \min)/(\max - \min)$; age range [20, 60], income range [10, 210].

Customer	age'	income'
A	0.225	0.225
B	0.35	0.775
C	0.95	0.625

Raw distances: $d(A,B) = 110.11$, $d(A,C) = 85.09 \rightarrow$ nearest is C (income dominates because of its larger range).

Scaled distances: $d(A,B) = 0.564$, $d(A,C) = 0.828 \rightarrow$ nearest is B.

Answer: **85.09**. Scaling changes which neighbour is nearest — distance-based methods (k-means, k-NN, hierarchical clustering) should be run on standardized features.

Q.20 — Answer: 0.74 to 0.76 [NAT, 2M · Probabilistic & Instance-based Classifiers · Evaluation metrics]

The ROC curve is a staircase: each vertical step is a positive and each horizontal step a negative. Area = Σ over horizontal steps of (width $1/N$) $\times$ (current TPR): $0.75 + 0.75 + 0.75 + 0.75$, times $1/4$.

AUC = **0.75**.

Q.21 — Answer: (C) [MCQ, 2M · Neural Networks (MLP / Feed-forward) · Expressiveness]

Each unit outputs 1 iff its net input (weighted sum + bias) is > 0 . Evaluate on all four inputs:

x_1	x_2	h_1	h_2	y
0	0	1	1	0
0	1	1	0	1
1	0	1	0	1
1	1	0	0	0

h_1 computes NAND, h_2 computes NOR; the output column $(0,0) \rightarrow (1,1)$ is 0, 1, 1, 0, i.e. **XOR**.

This shows that one hidden layer of two threshold units suffices for the non-linearly-separable XOR, which a single perceptron cannot represent.

Q.22 — Answer: 22.49 to 22.51 [NAT, 2M · Unsupervised Learning · PCA & dimensionality reduction]

For centered data, projecting onto the top-k PCs and reconstructing leaves the components along the discarded eigenvectors; their average squared length (divisor $n - 1$) is the sum of the discarded eigenvalues.

Discarded eigenvalues: 9.5, 8, 5; sum = 22.5; total = 62.

Answer: **22.5**. (A common mistake is to add the *retained* eigenvalues 39.5.)

Q.23 — Answer: 4.115 to 4.135 [NAT, 2M · SVM & Decision Trees · SVM: soft margin & hinge loss]

Compute $f(x_i)$, the functional margin $y_i f(x_i)$ and the slack/hinge loss for each point:

Point	x	y	$f(x)$	$y f(x)$	$\xi = \max(0, 1 - y f)$
P1	(0, 5)	-1	-2.5	2.5	0
P2	(2, 0)	-1	-3	3	0
P3	(5, 1)	+1	0.5	0.5	0.5
P4	(5, 2)	+1	1	1	0
P5	(3, 0)	+1	-2	-2	3

$\frac{1}{2}\|w\|^2 = 0.625$; objective = $0.625 + 1 \times 3.5 = \mathbf{4.12}$.

Points with $y f \geq 1$ contribute 0; points inside the margin contribute between 0 and 1; misclassified points contribute more than 1.

Q.24 — Answer: 43 [NAT, 2M · Unsupervised Learning · k-medoids]

PAM cost = Σ over all points of the absolute distance to the nearest medoid.

Current medoids {16, 19}: distances 1 $\rightarrow$ 15, 5 $\rightarrow$ 11, 16 $\rightarrow$ 0, 19 $\rightarrow$ 0, 22 $\rightarrow$ 3, 23 $\rightarrow$ 4, 29 $\rightarrow$ 10; cost = **43**.

Q.25 — Answer: 0.0527 to 0.0548 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Backpropagation]

$a = w \cdot x + b = 1.75$; $\hat{y} = \sigma(a) = 0.852$.

$L = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial L / \partial \hat{y} = \hat{y} - y = 0.852$; $\partial \hat{y} / \partial a = \hat{y}(1 - \hat{y}) = 0.1261$.

$\delta = \partial L / \partial a = (\hat{y} - y)\hat{y}(1 - \hat{y}) = 0.1075$ ($= \partial L / \partial b$).

$\partial L / \partial w_2 = \delta \cdot x_2 = 0.1075 \times 0.5 = 0.0537$.

Answer: **0.0537**.

Forgetting the factor $\hat{y}(1 - \hat{y})$ (i.e. treating the output as linear) is the most common error.

Q.26 — Answer: 0.1013 to 0.1053 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

For the average cross-entropy, $\partial L / \partial w_j = (1/n) \sum (\hat{p}_i - y_i) x_{ij}$ and $\partial L / \partial b = (1/n) \sum (\hat{p}_i - y_i)$, with $\hat{p}_i = \sigma(z_i)$.

i	z_i	$\hat{p}_i$	$\hat{p}_i - y_i$
1	1.5	0.8176	0.8176
2	-0.5	0.3775	-0.6225
3	0	0.5	0.5

Gradient: $\partial L / \partial w_1 = 0.7934$, $\partial L / \partial b = 0.2317$.

Update $\theta \leftarrow \theta - \eta \nabla L$: $w_1 = 0.5 - 0.5 \times (0.7934) = 0.1033$, $b = 0.5 - 0.5 \times (0.2317) = 0.3841$.

Required: $w_1 = \mathbf{0.103}$.

Q.27 — Answer: 0.8844 to 0.9044 [NAT, 2M · SVM & Decision Trees > SVM: margin & geometry]

P3 and P5 (same class) lie on one margin line and P1 on the other; the optimal boundary is parallel to the line through P3, P5, midway to P1. Support vectors: P1, P3, P5.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (1, -2)$, $b = 8$; $\|w\| = 2.2361$, margin $2/\|w\| = 0.8944$.

Point	P1	P2	P3	P4	P5	P6
$y(w^T x + b)$	1	3	1	2	1	2

All values are ≥ 1 (feasible), with equality exactly at the support vectors.

Margin width = $2/\|w\| = \mathbf{0.89}$.

Q.28 — Answer: (B), (D) [MSQ, 2M · Regression & Model Selection > Bias-variance trade-off]

(A) FALSE. The minimum validation MSE is at degree 3, not 1.

(B) TRUE. Validation MSE is minimum (3.8) at degree 3.

(C) FALSE. Lower-degree models are less flexible and have lower variance.

(D) TRUE. Gaps: 0.3, 0.3, 0.77, 2.56, 4.47, 5.4, 6.05.

Q.29 — Answer: 81 [NAT, 2M · SVM & Decision Trees > SVM: kernels]

$\|\varphi(x) - \varphi(z)\|^2 = K(x, x) + K(z, z) - 2K(x, z)$.

$K(x, x) = (5 + 0)^2 = 25$, $K(z, z) = (8 + 0)^2 = 64$, $K(x, z) = (2 + 0)^2 = 4$.

Squared distance = $25 + 64 - 2(4) = \mathbf{81}$.

Q.30 — Answer: 0.615 to 0.635 [NAT, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Classify each point by its nearest neighbour(s) among the remaining points:

x	true	1 nearest (excluding itself)	LOO prediction	
0	0	1(0)	0	
1	0	0(0)	0	
4	0	1(0)	0	
8	0	10(1)	1	wrong
10	1	8(0)	0	wrong
21	1	22(0)	0	wrong
22	0	21(1)	1	wrong
24	1	22(0)	0	wrong

Misclassified: 5 of 8. Error rate = $5/8 = \mathbf{0.62}$.

Note: the *training* error of 1-NN is 0, but the LOOCV error is not — a point cannot be its own neighbour.

Q.31 — Answer: (D) [MCQ, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent: 7 Yes, 5 No $\Rightarrow H = 0.9799$.

Shape — Round: [2+, 3-], Square: [2+, 1-], Oval: [3+, 1-]; gain = 0.0753.

Student — Yes: [2+, 1-], No: [5+, 4-]; gain = 0.007.

Credit — Fair: [3+, 2–], Excellent: [4+, 3–]; gain = 0.0006.

Outlook — Sunny: [3+, 2–], Overcast: [4+, 1–], Rain: [0+, 2–]; gain = 0.2745.

The largest information gain is for **Outlook**, so it is chosen at the root.

Q.32 — Answer: 1.5055 to 1.5145 [NAT, 2M · Regression & Model Selection > Gradient descent]

Per-example loss $\mathcal{J} = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial \mathcal{J} / \partial w = (\hat{y} - y)x$, $\partial \mathcal{J} / \partial b = (\hat{y} - y)$. Update after each example.

Example 1: $\hat{y} = 1$, error $\hat{y} - y = -1$; $\nabla_w = (0, -2)$, $\nabla_b = -1 \Rightarrow w = (0, 0.2)$, $b = 1.1$.

Example 2: $\hat{y} = 0.9$, error $\hat{y} - y = -4.1$; $\nabla_w = (4.1, 4.1)$, $\nabla_b = -4.1 \Rightarrow w = (-0.41, -0.21)$, $b = 1.51$.

Answer: **1.51**.

Q.33 — Answer: 0.2375 to 0.2415 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$p = \text{softmax}(z)$: $\exp(z) = (0.2231, 1.6487, 0.2231)$, sum = 2.095, so $p = (0.1065, 0.787, 0.1065)$.

For $L = -\ln p_y$ with softmax, $\partial L / \partial z = p - y$ (y one-hot, true class 2): $\partial L / \partial z = (0.1065, -0.213, 0.1065)$.

$L = -\ln 0.787 = \mathbf{0.2395}$.

Note $\sum_k \partial L / \partial z_k = \sum p_k - 1 = 0$.

Q.34 — Answer: 49.19 to 49.21 [NAT, 2M · Unsupervised Learning > k-means]

Data: {1, 2, 3, 11, 13, 17, 18, 19}; initial centroids $c_1 = 13$, $c_2 = 15$.

Iter	centroids used	cluster 1	cluster 2	new centroids
1	13, 15	{1, 2, 3, 11, 13}	{17, 18, 19}	6, 18
2	6, 18	{1, 2, 3, 11}	{13, 17, 18, 19}	4.25, 16.75
3	4.25, 16.75	{1, 2, 3}	{11, 13, 17, 18, 19}	2, 15.6
4	2, 15.6	{1, 2, 3}	{11, 13, 17, 18, 19}	2, 15.6

In the last row the centroids do not change, so the algorithm has converged: final centroids 2 and 15.6.

$SSE = (1 - 2)^2 + (2 - 2)^2 + (3 - 2)^2 + (11 - 15.6)^2 + (13 - 15.6)^2 + (17 - 15.6)^2 + (18 - 15.6)^2 + (19 - 15.6)^2 = \mathbf{49.2}$.

Q.35 — Answer: 0.3939 to 0.4139 [NAT, 2M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Naive Bayes: $P(c | x) \propto P(c) \prod_j P(x_j | c)$. With Laplace smoothing, $P(x_j = v | c) = (\text{count} + 1) / (n_c + V_j)$, V_j = number of values of feature j .

Yes: $P(\text{Buys}=\text{Yes}) = 7/12$; $P(\text{Age}=\text{Young}|\text{Yes}) = (1+1)/(7+3) = 1/5$; $P(\text{Student}=\text{Yes}|\text{Yes}) = (2+1)/(7+2) = 1/3$; $P(\text{Rating}=\text{Excellent}|\text{Yes}) = (0+1)/(7+2) = 1/9$. Product = 0.00432.

No: $P(\text{Buys}=\text{No}) = 5/12$; $P(\text{Age}=\text{Young}|\text{No}) = (0+1)/(5+3) = 1/8$; $P(\text{Student}=\text{Yes}|\text{No}) = (2+1)/(5+2) = 3/7$; $P(\text{Rating}=\text{Excellent}|\text{No}) = (1+1)/(5+2) = 2/7$. Product = 0.00638.

$P(\text{Yes} | x) = 0.00432 / (0.00432 + 0.00638) = \mathbf{0.4}$.

Mock Test 20

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	6	8	21

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Neural Networks (MLP / Feed-forward)	7	11
SVM & Decision Trees	7	11
Regression & Model Selection	7	11
Unsupervised Learning	7	11
Probabilistic & Instance-based Classifiers	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MSQ (one or more correct)]

Let $p = \text{softmax}(z)$ for logits $z \in \mathbb{R}^K$ and $L = -\ln p_y$ the cross-entropy loss for true class y . Which of the following is/are CORRECT?

- (A) Adding the same constant to all logits leaves the softmax output and the cross-entropy loss unchanged.
- (B) For softmax + cross-entropy, the gradient with respect to the true-class logit is negative unless $p_y = 1$.
- (C) The cross-entropy loss $-\ln p_y$ can be negative.
- (D) The softmax Jacobian is diagonal, so each p_i depends only on z_i .

Q.2 [1 mark · NAT (numerical answer)]

In AdaBoost with $\varepsilon_t = 0.4$, before normalisation each misclassified example's weight is multiplied by e^{α_t} , where $\alpha_t = \frac{1}{2} \ln((1 - \varepsilon_t)/\varepsilon_t)$. This multiplying factor is _____ (rounded off to two decimal places)

Q.3 [1 mark · NAT (numerical answer)]

A node of a classification tree contains 17 training records: 11 of class C_1 , 6 of class C_2 . The misclassification error of this node is _____ (rounded off to two decimal places)

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about basis-function (polynomial) regression and feature scaling is/are CORRECT?

- (A) Adding x^2 as an extra feature makes the least-squares objective non-convex.
- (B) Increasing the polynomial degree always reduces test error.
- (C) For OLS with an intercept, rescaling a feature as $x' = ax + b$ ($a \neq 0$) leaves the fitted values unchanged.
- (D) Min-max scaling is robust to outliers.

Q.5 [1 mark · NAT (numerical answer)]

Agglomerative clustering of 32 points is stopped when 5 clusters remain. The number of merges performed is _____ (Answer in integer)

Q.6 [1 mark · MSQ (one or more correct)]

Which of the following statements about bagging, random forests and boosting is/are CORRECT?

- (A) A random forest with $m = p$ (all features considered at each split) is equivalent to bagged trees.
- (B) Random forests reduce variance by making the individual trees more correlated.
- (C) Bagging is mainly used to reduce the bias of high-bias learners such as decision stumps.
- (D) A common default for random-forest classification is $m \approx \sqrt{p}$ features per split.

Q.7 [1 mark · NAT (numerical answer)]

A fully connected MLP has layer sizes 64-16-8-64-2 (input-...-output), every layer having a bias vector. The number of bias parameters is _____ (Answer in integer)

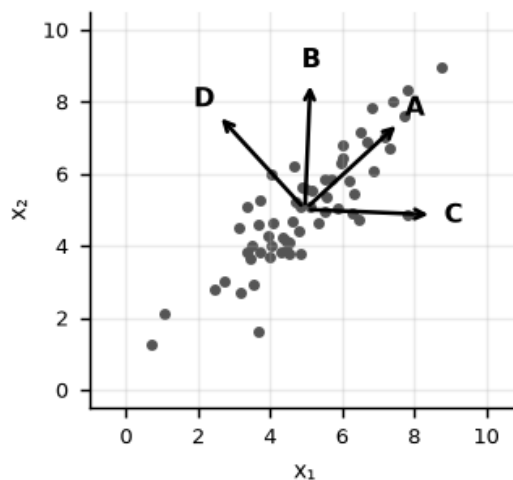
Q.8 [1 mark · MSQ (one or more correct)]

Which of the following statements about naive Bayes classifiers is/are CORRECT?

- (A) Whenever the independence assumption is violated, naive Bayes predicts the wrong class.
- (B) Under the naive Bayes assumption, $P(x_1, x_2) = P(x_1)P(x_2)$ for any two features.
- (C) Laplace smoothing adds a pseudo-count (typically 1) to every feature-value count within each class.
- (D) Without smoothing, a feature value never observed with class c in training forces the posterior of c to 0 for any input containing that value.

Q.9 [1 mark · MCQ]

The scatter plot shows a 2-D dataset together with four candidate directions A-D drawn from the data mean. Which direction best represents the second principal component (PC2)?



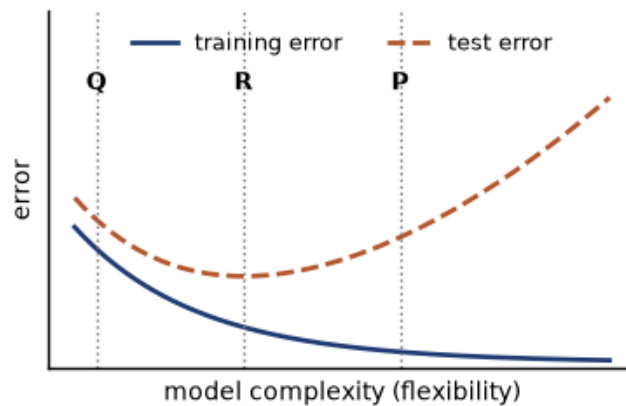
- (A) Direction A
- (B) Direction B
- (C) Direction C
- (D) Direction D

Q.10 [1 mark · NAT (numerical answer)]

LDA is fitted with $K = 4$ classes and $p = 9$ features. Counting class priors, class means and the shared covariance matrix, the number of independent parameters is _____ (Answer in integer)

Q.11 [1 mark · MCQ]

The figure shows training and test error as a function of model complexity, with three complexity levels marked P, Q and R. Which marked level corresponds to underfitting (high bias, low variance)?



- (A)** R
(B) Q
(C) P
(D) None of P, Q, R

Q.12 [1 mark · MSQ (one or more correct)]

Which of the following statements about the k-nearest-neighbour method is/are CORRECT?

- (A)** The number of neighbours k is a hyperparameter usually chosen by cross-validation.
- (B)** The training error of k -NN is zero for every value of k .
- (C)** Rescaling one feature (e.g. metres $\rightarrow$ millimetres) can change k -NN predictions.
- (D)** k -NN is a non-parametric method: its effective complexity grows with the amount of training data.

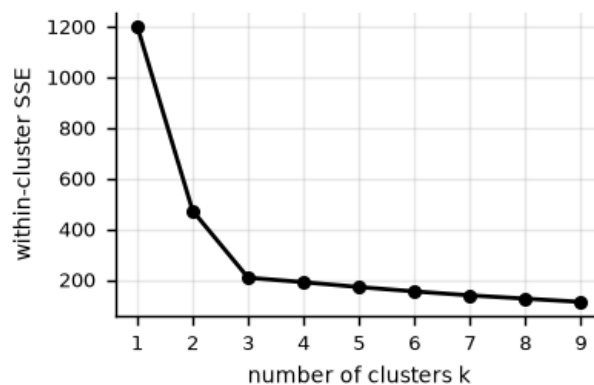
Q.13 [1 mark · MSQ (one or more correct)]

Which of the following statements about initialisation and regularisation of neural networks is/are CORRECT?

- (A)** An L1 penalty tends to drive some weights exactly to zero, unlike an L2 penalty.
- (B)** Increasing the dropout rate always reduces the training error.
- (C)** Dropout is applied at test time with freshly sampled random masks, exactly as during training.
- (D)** Xavier/Glorot initialisation chooses the weight variance based on the layer's fan-in and fan-out to keep activation and gradient variances roughly constant across layers.

Q.14 [1 mark · MCQ]

k-means is run for $k = 1, 2, \dots, 9$ on a dataset and the final within-cluster SSE is plotted below. Using the elbow method, the most appropriate number of clusters is



- (A) 9
(B) 2
(C) 4
(D) 3

Q.15 [1 mark · MCQ]

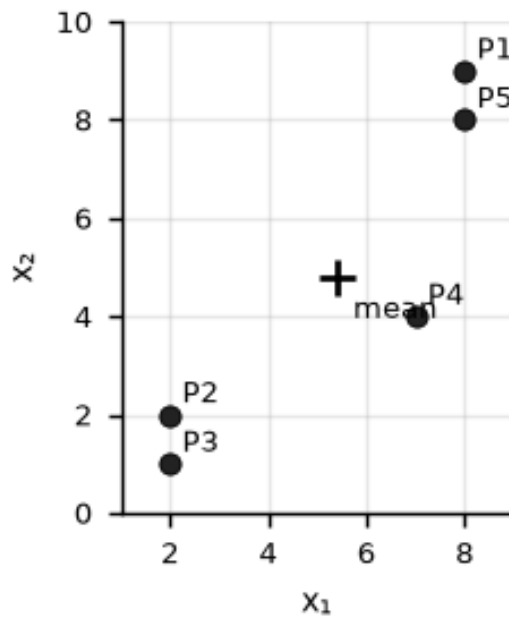
Which of the following is NOT an appropriate evaluation metric for a regression model?

- (A) Root mean squared error (B) Coefficient of determination R^2
 (C) F1-score (D) Mean squared error

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

PCA is applied to the five 2-D points below (data are mean-centered first; sample covariance uses divisor $n - 1$). If each point is reconstructed from its PC1 score only ($\hat{x} = \bar{x} + z_1 v_1$), the squared reconstruction error $\|x - \hat{x}\|^2$ of P5 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5
x_1	8	2	2	7	8
x_2	9	2	1	4	8

**Q.17 [2 marks · MCQ]**

Ridge regression is tuned by 10-fold cross-validation. For each λ the table gives the mean CV MSE and its standard error (SE) across folds. (Larger λ means a simpler, more regularised model.) Using the **one-standard-error rule** (choose the most regularised model whose CV error is within one standard error of the minimum CV error), the selected value is

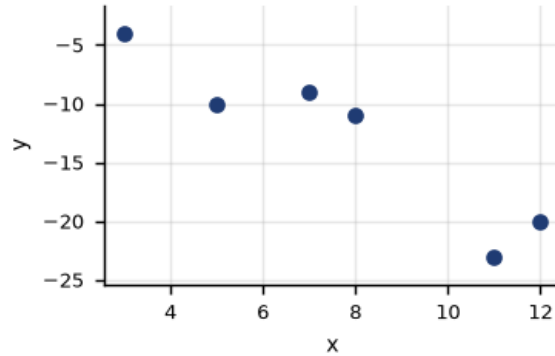
λ	0.01	0.1	0.3	1	3	10
CV MSE	4.68	4.21	4.1	4.19	5.18	5.78
SE	0.41	0.43	0.4	0.28	0.31	0.18

- (A) $\lambda = 10$ (B) $\lambda = 0.1$
 (C) $\lambda = 1$ (D) $\lambda = 0.3$

Q.18 [2 marks · NAT (numerical answer)]

The 6 observations shown in the table are plotted below. For the simple linear regression of y on x (with intercept), the value predicted by the least-squares line at $x = 10$ is _____ (rounded off to two decimal places)

x	3	5	7	8	11	12
y	-4	-10	-9	-11	-23	-20

**Q.19 [2 marks · NAT (numerical answer)]**

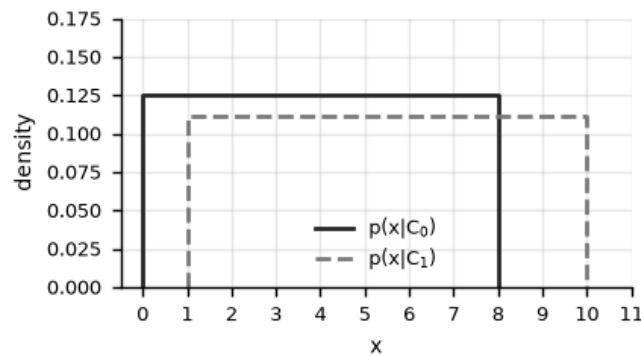
Ridge regression without intercept minimises $\|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2$ with $\lambda = 4$, for the data below ($\beta = (\beta_1, \beta_2)^T$). The value of β_1 is _____ (rounded off to three decimal places)

$$X = \begin{bmatrix} 0 & 1 \\ -2 & 0 \\ 1 & 1 \end{bmatrix}$$

$$y = \begin{bmatrix} 1 \\ 7 \\ 5 \end{bmatrix}$$

Q.20 [2 marks · NAT (numerical answer)]

For a binary problem with scalar x , $p(x | C_0)$ is uniform on $[0, 8]$ and $p(x | C_1)$ is uniform on $[1, 10]$, with priors $P(C_0) = 0.4$ and $P(C_1) = 0.6$. Then the Bayes (minimum achievable) error rate is _____ (rounded off to three decimal places)



Q.21 [2 marks · MSQ (one or more correct)]

A linear regression of annual salary (₹ lakh) on Exp (years of experience), Female (1 if the employee is female, 0 otherwise) and city $\in \{A, B, C\}$ encoded by dummies CityB and CityC (reference level A) gives the fitted model

$$\hat{y} = 14 + 2.7 \cdot \text{Exp} + 3.5 \cdot \text{Female} + 6 \cdot \text{CityB} - 4.1 \cdot \text{CityC}$$

Which of the following is/are CORRECT?

- (A) Adding a third dummy for city A while keeping the intercept would make $X^T X$ singular.
- (B) Increasing Exp by 3 years, other variables fixed, changes the prediction by 8.1.
- (C) The predicted annual salary for a female employee in city B with 9 years of experience is 47.8.
- (D) The coefficient of CityB is the expected difference in response between city B and city A, holding the other variables fixed.

Q.22 [2 marks · NAT (numerical answer)]

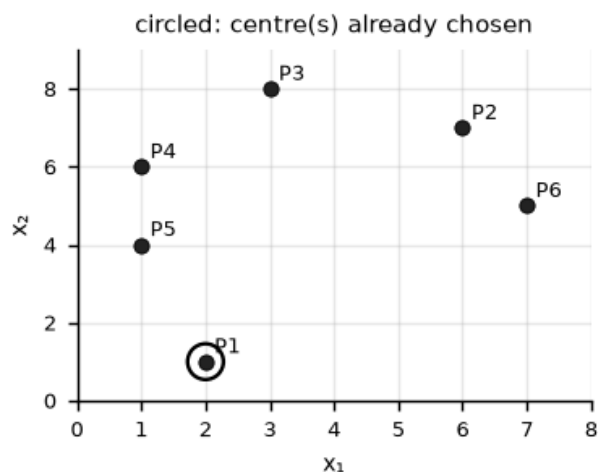
Agglomerative hierarchical clustering with **average linkage** is applied to 6 points whose pairwise distance matrix is given below (upper triangle shown). The height (linkage distance) at which the **third** merge takes place is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5	P6
P1	0	10	3	3	2	10
P2		0	11	9	10	2
P3			0	6	1	11
P4				0	5	7
P5					0	10
P6						0

Q.23 [2 marks · NAT (numerical answer)]

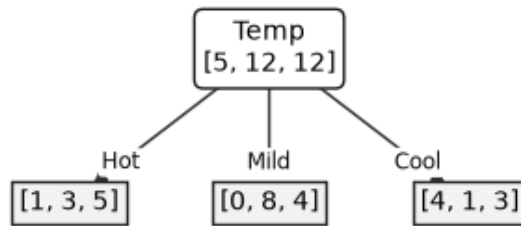
k-means++ initialization (squared Euclidean seeding) is applied to the six points below. P1 has already been chosen as centre. The probability that P2 is selected as the second centre is _____ (rounded off to three decimal places)

Point	P1	P2	P3	P4	P5	P6
x_1	2	6	3	1	1	7
x_2	1	7	8	6	4	5



Q.24 [2 marks · NAT (numerical answer)]

A CART node with 29 training records (class counts [C1, C2, C3]) is split on Temp as shown. Compute the reduction in Gini index (Gini gain) obtained by splitting on Temp. (rounded off to two decimal places)

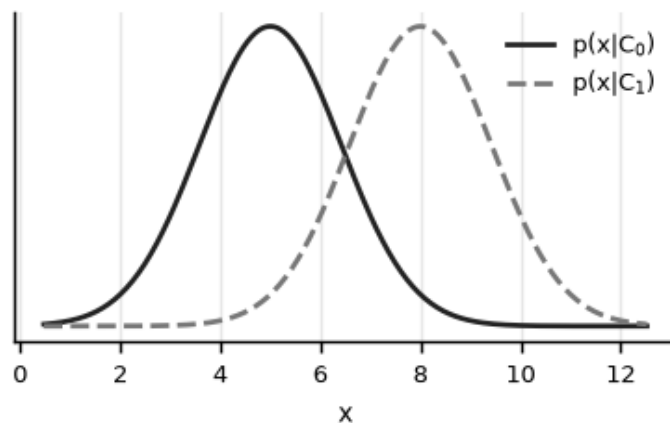
**Q.25 [2 marks · MSQ (one or more correct)]**

A threshold unit outputs 1 if $w_1x_1 + w_2x_2 + b > 0$ and 0 otherwise, for binary inputs $x_1, x_2 \in \{0, 1\}$. Which of the following parameter settings make the unit implement the Boolean **OR** function?

- (A) $w_1 = 2.5, w_2 = 2.5, b = -2$
- (B) $w_1 = 0, w_2 = -3, b = 1.5$
- (C) $w_1 = 2, w_2 = 2.5, b = -0.5$
- (D) $w_1 = 1.5, w_2 = 2, b = -1$

Q.26 [2 marks · NAT (numerical answer)]

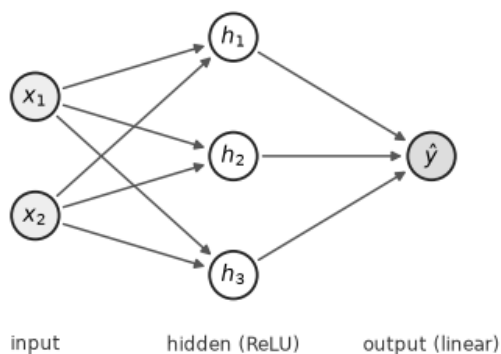
In a two-class problem with one feature, $p(x | C_0) = N(5, 2)$ and $p(x | C_1) = N(8, 2)$ (second argument = variance), with priors $P(C_0) = 0.4$ and $P(C_1) = 0.6$. Under 0-1 loss, the Bayes-optimal (minimum-error) decision threshold x^* is _____ (rounded off to two decimal places)



Class-conditional densities (not weighted by priors).

Q.27 [2 marks · NAT (numerical answer)]

The 2-3-1 network below computes $\hat{y} = v \cdot \text{ReLU}(Wx + b) + c$ with W , b , v given below and $c = -1$. With $x = (1, -1)$, target $y = -1$ and loss $L = \frac{1}{2}(\hat{y} - y)^2$, the value of $\partial L / \partial W_{11}$ is _____ (Answer in integer)



$$W = \begin{bmatrix} 2 & 1 \\ -2 & 0 \\ -1 & 2 \end{bmatrix}$$

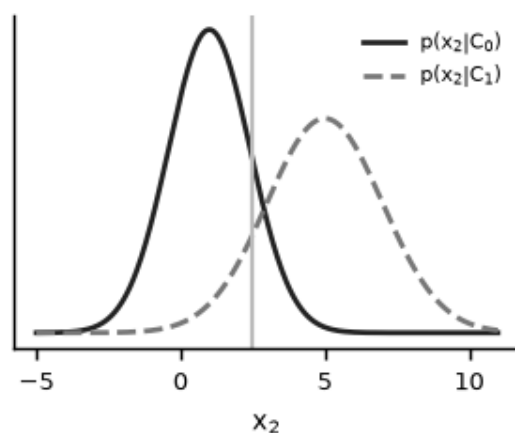
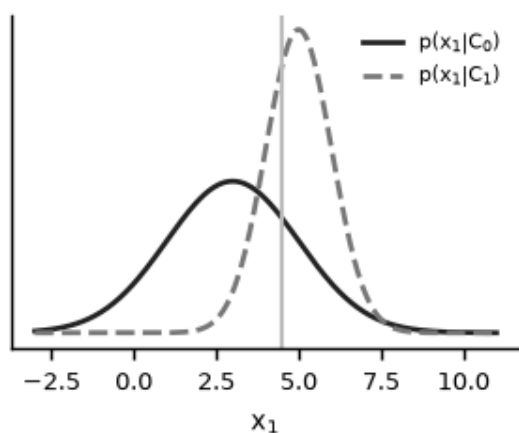
$$b = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$v^T = \begin{bmatrix} 2 & -1 & -1 \end{bmatrix}$$

Q.28 [2 marks · NAT (numerical answer)]

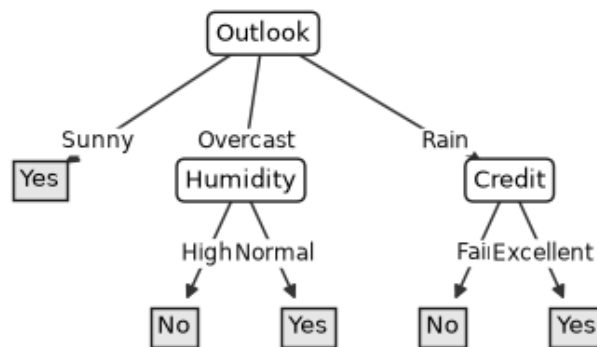
A Gaussian naive Bayes classifier with features x_1, x_2 has the class priors and per-class Gaussian parameters below (σ^2 = variance). For the input $x = (4.5, 2.5)$, the posterior probability $P(C_1 | x)$ is _____ (rounded off to two decimal places)

class	prior	$\mu(x_1)$	$\sigma^2(x_1)$	$\mu(x_2)$	$\sigma^2(x_2)$
C_0	0.5	3	4	1	2
C_1	0.5	5	1	5	4



Q.29 [2 marks · NAT (numerical answer)]

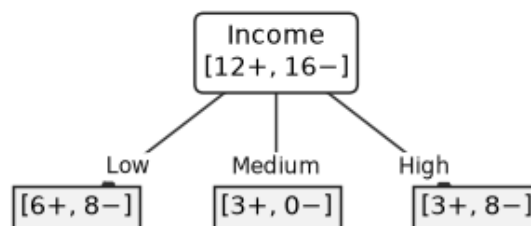
The decision tree shown is applied to the 8 labelled records in the table (attributes not tested on a path are ignored). Find the number of records misclassified by the tree. (Answer in integer)



#	Outlook	Humidity	Credit	Actual
1	Rain	Normal	Excellent	Yes
2	Rain	Normal	Excellent	Yes
3	Sunny	Normal	Fair	Yes
4	Sunny	High	Fair	No
5	Overcast	High	Excellent	No
6	Rain	Normal	Excellent	No
7	Sunny	Normal	Fair	Yes
8	Rain	High	Excellent	Yes

Q.30 [2 marks · NAT (numerical answer)]

A node with 28 training records (class counts [positive+, negative–]) is split on the attribute Income as shown. Compute the information gain (in bits) of splitting on Income. (rounded off to two decimal places)

**Q.31 [2 marks · NAT (numerical answer)]**

$B = 20$ regression trees are averaged. Each tree's prediction has variance $\sigma^2 = 4$, and every pair of trees has correlation $\rho = 0.5$. The variance of the averaged prediction is _____ (rounded off to two decimal places)

Q.32 [2 marks · MCQ]

A network computes $\hat{y} = W_2 \text{ReLU}(W_1 x + b_1) + b_2$ with the parameters below. For $x = (-2, 1, -1)^T$, the output $\hat{y}$ is

$$W_1 = \begin{bmatrix} -2 & 0 & 2 \\ 2 & -1 & 2 \\ 1 & 0 & -2 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} -1 & -1 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

(A) $\hat{y} = (-2, 4)^T$

(B) $\hat{y} = (-1, 4)^T$

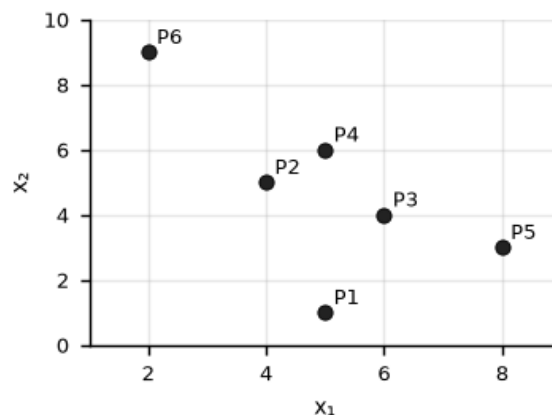
(C) $\hat{y} = (-4, 10)^T$

(D) $\hat{y} = (6, -3)^T$

Q.33 [2 marks · NAT (numerical answer)]

Single-linkage agglomerative clustering with Manhattan distance is applied to the six points below. Then the height at which the **last** merge (forming a single cluster) occurs is _____. (Answer in integer)

Point	P1	P2	P3	P4	P5	P6
x_1	5	4	6	5	8	2
x_2	1	5	4	6	3	9

**Q.34 [2 marks · NAT (numerical answer)]**

The actual and predicted labels of a binary classifier (1 = positive) on 15 test examples are given below. The F_β -score with $\beta = 0.5$, $F_\beta = (1 + \beta^2)PR/(\beta^2P + R)$, is _____ (rounded off to two decimal places)

#	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
actual y	1	0	0	1	1	0	1	1	1	1	1	1	1	0	0
predicted $\hat{y}$	1	0	1	0	1	0	1	0	1	0	0	1	1	0	0

Q.35 [2 marks · NAT (numerical answer)]

A softmax-regression output layer computes $z = Wx + b$, $p = \text{softmax}(z)$ with W and b below, and is trained with cross-entropy loss $L = -\ln p_y$. For $x = (1, -2)^T$ with true class $y = 1$, $\partial L / \partial W_{11}$ is _____ (rounded off to 4 decimal places)

$$W = \begin{bmatrix} 0 & -0.5 \\ -0.5 & 1 \\ -0.5 & 1 \end{bmatrix}$$

$$b = \begin{bmatrix} 0.5 \\ 0 \\ -0.5 \end{bmatrix}$$

— END OF QUESTION PAPER —

Mock Test 20 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MSQ	1	(A), (B)	19	NAT	2	−1.1355 to −1.1287
2	NAT	1	1.2147 to 1.2347	20	NAT	2	0.348 to 0.352
3	NAT	1	0.3429 to 0.3629	21	MSQ	2	(A), (B), (C), (D)
4	MSQ	1	(C)	22	NAT	2	2.49 to 2.51
5	NAT	1	27	23	NAT	2	0.2885 to 0.2925
6	MSQ	1	(A), (D)	24	NAT	2	0.0939 to 0.1139
7	NAT	1	90	25	MSQ	2	(A), (C), (D)
8	MSQ	1	(C), (D)	26	NAT	2	6.2197 to 6.2397
9	MCQ	1	(D)	27	NAT	2	8
10	NAT	1	84	28	NAT	2	0.5605 to 0.5805
11	MCQ	1	(B)	29	NAT	2	2
12	MSQ	1	(A), (C), (D)	30	NAT	2	0.1505 to 0.1705
13	MSQ	1	(A), (D)	31	NAT	2	2.09 to 2.11
14	MCQ	1	(D)	32	MCQ	2	(B)
15	MCQ	1	(C)	33	NAT	2	6
16	NAT	2	0.0072 to 0.0272	34	NAT	2	0.7795 to 0.7995
17	MCQ	2	(C)	35	NAT	2	−0.0292 to −0.028
18	NAT	2	−17.4689 to −17.2951				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Neural Networks (MLP / Feed-forward)					11
SVM & Decision Trees					11
Regression & Model Selection					11
Unsupervised Learning					11
Probabilistic & Instance-based Classifiers					11
Total					55

Mock Test 20 — Detailed Solutions

Q.1 — Answer: (A), (B) [MSQ, 1M · Neural Networks (MLP / Feed-forward) › Softmax + cross-entropy]

(A) TRUE. Shift invariance.

(B) TRUE. $p_y - 1 < 0$.

(C) FALSE. $p_y \leq 1 \Rightarrow -\ln p_y \geq 0$.

(D) FALSE. Off-diagonal terms $-p_i p_j$ are non-zero.

Q.2 — Answer: 1.2147 to 1.2347 [NAT, 1M · SVM & Decision Trees › Ensembles: boosting]

$$e^\alpha = e^{\frac{1}{2} \ln((1-\epsilon)/\epsilon)} = \sqrt{(1-\epsilon)/\epsilon} = \sqrt{1.5} = \mathbf{1.22}.$$

Correctly classified examples are multiplied by $e^{-\alpha} = \sqrt{\epsilon/(1-\epsilon)}$.

$\epsilon < 0.5 \Rightarrow \alpha > 0$; $\epsilon = 0.5 \Rightarrow \alpha = 0$ (no better than chance).

Q.3 — Answer: 0.3429 to 0.3629 [NAT, 1M · SVM & Decision Trees › Decision trees: impurity measures]

$$\text{Error} = 1 - \max_k p_k = 1 - 11/17 = \mathbf{0.353}.$$

Q.4 — Answer: (C) [MSQ, 1M · Regression & Model Selection › Basis functions & feature scaling]

(A) FALSE. It remains a convex quadratic in the weights.

(B) FALSE. It eventually overfits.

(C) TRUE. The column space of $[1, x]$ is unchanged; only the coefficients adjust.

(D) FALSE. A single extreme value compresses all others.

Q.5 — Answer: 27 [NAT, 1M · Unsupervised Learning › Hierarchical clustering]

$$32 - 5 = \mathbf{27} \text{ merges.}$$

Q.6 — Answer: (A), (D) [MSQ, 1M · SVM & Decision Trees › Ensembles: bagging & random forests]

(A) TRUE. No feature subsampling.

(B) FALSE. They de-correlate them.

(C) FALSE. Bagging reduces variance; boosting reduces bias.

(D) TRUE. Breiman's recommendation.

Q.7 — Answer: 90 [NAT, 1M · Neural Networks (MLP / Feed-forward) › Parameter counting]

Fully connected layer with n_{in} inputs and n_{out} outputs: $n_{\text{in}} \cdot n_{\text{out}}$ weights + n_{out} biases.

Layer 1: $64 \times 16 = 1024$ weights, 16 biases.

Layer 2: $16 \times 8 = 128$ weights, 8 biases.

Layer 3: $8 \times 64 = 512$ weights, 64 biases.

Layer 4: $64 \times 2 = 128$ weights, 2 biases.

Weights = 1792, biases = 90. Required: **90**.

Q.8 — Answer: (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers › Naive Bayes]

(A) FALSE. It often still predicts correctly.

(B) FALSE. Only the class-conditional factorisation holds; marginally the features are usually dependent.

(C) TRUE. Estimate becomes $(n_{v,c} + 1)/(n_c + V)$.

(D) TRUE. One zero factor makes the product zero (zero-frequency problem).

Q.9 — Answer: (D) [MCQ, 1M · Unsupervised Learning › PCA & dimensionality reduction]

PC1 is the direction of maximum variance — along the long axis of the elongated cloud (direction A). PC2 must be orthogonal to PC1 and captures the remaining variance (direction D).

Answer: **Direction D**. Directions B and C are at 45° to the main axis and capture intermediate variance.

Q.10 — Answer: 84 [NAT, 1M · Probabilistic & Instance-based Classifiers › Linear discriminant analysis]

Priors $K - 1 = 3$; means $Kp = 36$; one symmetric covariance $p(p+1)/2 = 45$.

Total = **84**.

Q.11 — Answer: (B) [MCQ, 1M · Regression & Model Selection > Bias-variance trade-off]

At Q: both training and test errors are high (left side) → underfitting.

At R: the test error is at its minimum → best trade-off.

At P: training error is very low but test error has risen well above its minimum (right side) → overfitting.

Answer: **Q**.

Q.12 — Answer: (A), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

(A) **TRUE**. Training error cannot select k (it always favours $k = 1$).

(B) **FALSE**. Only 1-NN is guaranteed zero training error (without duplicates).

(C) **TRUE**. Distances, and hence neighbour sets, depend on feature scales.

(D) **TRUE**. The whole training set is the model.

Q.13 — Answer: (A), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

(A) **TRUE**. The L1 subgradient has constant magnitude λ .

(B) **FALSE**. More dropout adds noise and usually increases training error.

(C) **FALSE**. Dropout is disabled at test time (deterministic prediction).

(D) **TRUE**. $\text{Var}(w) = 2/(\text{fan_in} + \text{fan_out})$.

Q.14 — Answer: (D) [MCQ, 1M · Unsupervised Learning > k-means]

The SSE drops steeply up to $k = 3$ and then decreases only slowly (relative drops: $k=2$: 61%, $k=3$: 56%, $k=4$: 8%, $k=5$: 10%, $k=6$: 10%, $k=7$: 10%, $k=8$: 9%, $k=9$: 10%). The bend ('elbow') is at **$k = 3$** .

$k = 9$ gives the lowest SSE but SSE always (weakly) decreases with k , so minimizing it would pick the largest k ; it is not a model-selection criterion by itself.

Q.15 — Answer: (C) [MCQ, 1M · Regression & Model Selection > Regression vs classification]

F1-score combines precision and recall of class predictions; it is a classification metric.

Hence the answer is **F1-score**.

Q.16 — Answer: 0.0072 to 0.0272 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

Mean $\bar{x} = (5.4, 4.8)$. Centered data rows: P1: (2.6, 4.2), P2: (-3.4, -2.8), P3: (-3.4, -3.8), P4: (1.6, -0.8), P5: (2.6, 3.2).

Sample covariance $S = (1/(n - 1)) \sum (x - \bar{x})(x - \bar{x})^T = [[9.8, 10.1], [10.1, 12.7]]$.

Eigenvalues: $\lambda_1 = 21.4536$, $\lambda_2 = 1.0464$; unit PC1 direction $v_1 = (0.6549, 0.7557)$.

Score $z_1 = v_1 \cdot (x - \bar{x}) = 0.6549 \cdot 2.6 + 0.7557 \cdot 3.2 = 4.121$.

$\|x - \bar{x}\|^2 = 17$; error = $\|x - \bar{x}\|^2 - z_1^2 = 17 - 16.9828 = \mathbf{0.02}$ (Pythagoras: the residual is the PC2 component).

Q.17 — Answer: (C) [MCQ, 2M · Regression & Model Selection > Cross-validation]

Minimum CV MSE = 4.1 at $\lambda = 0.3$, with SE = 0.4.

One-SE threshold = $4.1 + 0.4 = 4.5$.

Models within the threshold: $\lambda = 0.1$ (4.21), $\lambda = 0.3$ (4.1), $\lambda = 1$ (4.19).

The one-SE rule picks the largest such λ (simplest model): $\lambda = 1$; the minimum-CV rule picks $\lambda = 0.3$.

Answer: **$\lambda = 1$** .

Q.18 — Answer: -17.4689 to -17.2951 [NAT, 2M · Regression & Model Selection > Simple linear regression]

$\bar{x} = 7.667$, $\bar{y} = -12.833$.

i	1	2	3	4	5	6
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$x_i - \bar{x}$	-4.67	-2.67	-0.67	0.33	3.33	4.33
$y_i - \bar{y}$	8.83	2.83	3.83	1.83	-10.17	-7.17
$(x_i - \bar{x})(y_i - \bar{y})$	-41.22	-7.56	-2.56	0.61	-33.89	-31.06

$$S_{xx} = \sum (x_i - \bar{x})^2 = 59.333, S_{xy} = -115.667.$$

$$\beta_1 = S_{xy}/S_{xx} = -1.9494, \beta_0 = \bar{y} - \beta_1 \bar{x} = 2.1124.$$

$$\hat{y}(10) = 2.1124 + (-1.9494)(10) = -17.382.$$

Answer: **-17.38**.

Q.19 — Answer: -1.1355 to -1.1287 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

$$\beta_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y \text{ with } \lambda = 4.$$

$$X^T X = \begin{bmatrix} 5 & 1 \\ 1 & 2 \end{bmatrix}$$

$$X^T X + \lambda I = \begin{bmatrix} 9 & 1 \\ 1 & 6 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} -9 \\ 6 \end{bmatrix}$$

$$\det = 9 \times 6 - (1)^2 = 53; (X^T X + \lambda I)^{-1} = (1/53)[[6, -1], [-1, 9]].$$

$$\beta = (-1.1321, 1.1887).$$

Answer: **-1.132**.

Q.20 — Answer: 0.348 to 0.352 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

$$\pi_0 p(x|C_0) = 0.4/8 = 0.05 \text{ on } [0, 8]; \pi_1 p(x|C_1) = 0.6/9 = 0.0667 \text{ on } [1, 10].$$

Outside the overlap $[1, 8]$ only one class has non-zero density $\Rightarrow$ no error. Inside it the Bayes rule chooses C_1 (larger weighted density) and errs with density 0.05.

$$\text{Bayes error} = (\text{overlap length}) \times \min = 7 \times 0.05 = \mathbf{0.35}.$$

Q.21 — Answer: (A), (B), (C), (D) [MSQ, 2M · Regression & Model Selection > Multiple linear regression]

(A) TRUE. The three dummies would sum to the intercept column (dummy-variable trap).

(B) TRUE. $3 \times 2.7 = 8.1$.

(C) TRUE. $\hat{y} = 14 + 2.7 \times 9 + (3.5) \times 1 + (6) = 47.8$.

(D) TRUE. A is the reference level (all dummies 0).

Q.22 — Answer: 2.49 to 2.51 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Use average linkage (UPGMA): $d(A, B) = (1/|A||B|) \sum_{a \in A} \sum_{b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P3\} + \{P5\}$ at height **1**. Updated: $d(\{P3, P5\}, \{P1\}) = 2.5$, $d(\{P3, P5\}, \{P2\}) = 10.5$, $d(\{P3, P5\}, \{P4\}) = 5.5$, $d(\{P3, P5\}, \{P6\}) = 10.5$.

Merge 2: $\{P2\} + \{P6\}$ at height **2**. Updated: $d(\{P2, P6\}, \{P1\}) = 10$, $d(\{P2, P6\}, \{P4\}) = 8$, $d(\{P2, P6\}, \{P3, P5\}) = 10.5$.

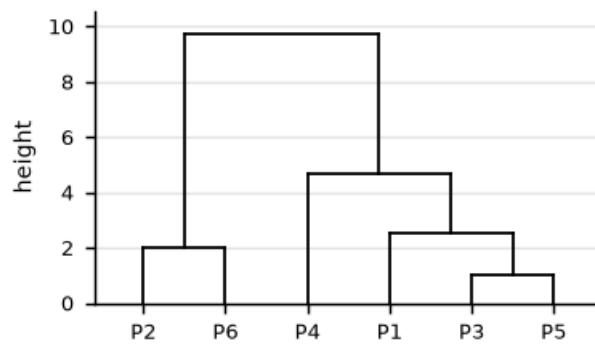
Merge 3: $\{P1\} + \{P3, P5\}$ at height **2.5**. Updated: $d(\{P1, P3, P5\}, \{P4\}) = 4.67$, $d(\{P1, P3, P5\}, \{P2, P6\}) = 10.33$.

Merge 4: $\{P4\} + \{P1, P3, P5\}$ at height **4.67**. Updated: $d(\{P1, P3, P4, P5\}, \{P2, P6\}) = 9.75$.

Merge 5: $\{P2, P6\} + \{P1, P3, P4, P5\}$ at height **9.75**.

The third merge occurs at height **2.5**.

Dendrogram:



Q.23 — Answer: 0.2885 to 0.2925 [NAT, 2M · Unsupervised Learning > k-means]

k-means++ picks the next centre x with probability $D(x)^2 / \sum_{x'} D(x')^2$, where $D(x)$ is the distance from x to the **nearest** centre already chosen (chosen centres have $D = 0$).

Point	P1	P2	P3	P4	P5	P6
$D(x)^2$	0	52	50	26	10	41

$\sum D^2 = 179$, so $P(P2) = 52/179 = \mathbf{0.291}$.

Common mistakes: using D instead of D^2 (that is not k-means++), or using the distance to the *farthest*/first centre instead of the nearest chosen centre.

Q.24 — Answer: 0.0939 to 0.1139 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [5, 12, 12]: Gini = $1 - [(5/29)^2 + (12/29)^2 + (12/29)^2] = 0.6278$.

Temp = Hot: [1, 3, 5], Gini = $1 - [(1/9)^2 + (3/9)^2 + (5/9)^2] = 0.5679$.

Temp = Mild: [0, 8, 4], Gini = $1 - [(0/12)^2 + (8/12)^2 + (4/12)^2] = 0.4444$.

Temp = Cool: [4, 1, 3], Gini = $1 - [(4/8)^2 + (1/8)^2 + (3/8)^2] = 0.5938$.

Weighted Gini = $(9/29)(0.5679) + (12/29)(0.4444) + (8/29)(0.5938) = 0.5239$.

Gini gain = $0.6278 - 0.5239 = 0.1039$.

Answer: **0.1**.

Q.25 — Answer: (A), (C), (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Target OR truth table for $(x_1, x_2) = (0,0), (0,1), (1,0), (1,1)$: 0, 1, 1, 1. For each option compute $a = w_1x_1 + w_2x_2 + b$ at the four inputs and output 1 iff $a > 0$.

(A) TRUE. $a = -2, 0.5, 0.5, 3 \rightarrow$ outputs 0, 1, 1, 1 - OR.

(B) FALSE. $a = 1.5, -1.5, 1.5, -1.5 \rightarrow$ outputs 1, 0, 1, 0 - NOT x_2 .

(C) TRUE. $a = -0.5, 2, 1.5, 4 \rightarrow$ outputs 0, 1, 1, 1 - OR.

(D) TRUE. $a = -1, 1, 0.5, 2.5 \rightarrow$ outputs 0, 1, 1, 1 - OR.

Q.26 — Answer: 6.2197 to 6.2397 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

With equal variances, $\ln[\pi_1 p(x|C_1)/(\pi_0 p(x|C_0))] = (\mu_1 - \mu_0)x/\sigma^2 - (\mu_1^2 - \mu_0^2)/(2\sigma^2) + \ln(\pi_1/\pi_0)$ — linear in x (LDA).

Setting it to 0: $x^* = (\mu_0 + \mu_1)/2 + \sigma^2 \ln(\pi_0/\pi_1)/(\mu_1 - \mu_0) = 6.5 + 2 \times \ln(0.4/0.6)/3$.

$= 6.5 + (-0.2703) = \mathbf{6.23}$.

The threshold moves away from the more probable class C_1 's mean, enlarging its decision region.

Q.27 — Answer: 8 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

$a = Wx + b = (2, -1, -3)$, $h = \text{ReLU}(a) = (2, 0, 0)$, $\hat{y} = v \cdot h + c = 3$; $\partial L/\partial \hat{y} = \hat{y} - y = 4$.

$\text{ReLU}'(a) = (1, 0, 0)$ (unit j is active iff $a_j > 0$).

$\partial L/\partial W_{11} = (\hat{y} - y) \cdot v_1 \cdot \text{ReLU}'(a_1) \cdot x_1 = 4 \times 2 \times 1 \times 1 = \mathbf{8}$.

Q.28 — Answer: 0.5605 to 0.5805 [NAT, 2M · Probabilistic & Instance-based Classifiers > Naive Bayes]

$N(x; \mu, \sigma^2) = \exp(-(x-\mu)^2/(2\sigma^2))/\sqrt{(2\pi\sigma^2)}$. Naive Bayes multiplies the per-feature densities.

C_0 : $N(4.5; 3, 4) = 0.15057 \times N(2.5; 1, 2) = 0.16073$; $\times$ prior $0.5 = 0.012101$.

$$C_1: N(4.5; 5, 1) = 0.35207 \times N(2.5; 5, 4) = 0.09132; \times \text{prior } 0.5 \Rightarrow 0.016076.$$

$$P(C_1 | x) = 0.016076 / (0.012101 + 0.016076) = \mathbf{0.57}.$$

Q.29 — Answer: 2 [NAT, 2M · SVM & Decision Trees › Decision trees: structure & traversal]

Route each record from the root to a leaf:

#	Predicted	Actual	Correct?
1	Yes	Yes	yes
2	Yes	Yes	yes
3	Yes	Yes	yes
4	Yes	No	no
5	No	No	yes
6	Yes	No	no
7	Yes	Yes	yes
8	Yes	Yes	yes

Misclassified: 2 of 8 (accuracy 75%). Answer: **2**.

Q.30 — Answer: 0.1505 to 0.1705 [NAT, 2M · SVM & Decision Trees › Decision trees: split selection]

$$\text{Parent } [12+, 16-]: H = -(12/28)\log_2(12/28) - (16/28)\log_2(16/28) = 0.9852.$$

$$\text{Income = Low: } [6+, 8-], H = -(6/14)\log_2(6/14) - (8/14)\log_2(8/14) = 0.9852.$$

$$\text{Income = Medium: } [3+, 0-], H = 0 \text{ (pure node)} = 0.$$

$$\text{Income = High: } [3+, 8-], H = -(3/11)\log_2(3/11) - (8/11)\log_2(8/11) = 0.8454.$$

$$\text{Weighted child entropy} = (14/28)(0.9852) + (3/28)(0) + (11/28)(0.8454) = 0.8247.$$

$$IG = 0.9852 - 0.8247 = 0.1605.$$

Answer: **0.16**.

Q.31 — Answer: 2.09 to 2.11 [NAT, 2M · SVM & Decision Trees › Ensembles: bagging & random forests]

$$\text{Var}((1/B)\sum_b) = (1/B^2)[B\sigma^2 + B(B-1)\rho\sigma^2] = \rho\sigma^2 + (1-\rho)\sigma^2/B.$$

$$= 0.5 \times 4 + 0.5 \times 4/20 = 2 + 0.1 = \mathbf{2.1}.$$

Even with $B \rightarrow \infty$ the variance cannot fall below $\rho\sigma^2 = 2$ — which is why random forests try to reduce ρ by feature subsampling.

Q.32 — Answer: (B) [MCQ, 2M · Neural Networks (MLP / Feed-forward) › Forward pass]

$$a = W_1x + b_1 = (1, -7, 1)^T; h = \text{ReLU}(a) = (1, 0, 1)^T.$$

$$\hat{y} = W_2h + b_2 = \mathbf{(-1, 4)^s}.$$

Distractors: skipping the ReLU gives $(6, -3)^T$; using W_1^T instead of W_1 gives $(-4, 10)^T$; dropping the biases gives $(-2, 4)^T$; using $|a|$ instead of ReLU gives $(-8, 11)^T$.

Q.33 — Answer: 6 [NAT, 2M · Unsupervised Learning › Hierarchical clustering]

Single-linkage merge heights are exactly the edge weights of a minimum spanning tree (Kruskal's algorithm adds the shortest edge joining two different components = single-linkage merge).

MST edges (Prim's algorithm): P1-P3 (4); P3-P2 (3); P2-P4 (2); P3-P5 (3); P2-P6 (6).

Sorted merge heights: 2, 3, 3, 4, 6.

The last single-linkage merge happens at the largest MST edge: **6**.

Q.34 — Answer: 0.7795 to 0.7995 [NAT, 2M · Probabilistic & Instance-based Classifiers › Evaluation metrics]

Counting: TP = 6, FP = 1, FN = 4.

$$P = 6/7 = 0.8571, R = 6/10 = 0.6.$$

$$F_{0.5} = (1 + 0.25) \times 0.8571 \times 0.6 / (0.25 \times 0.8571 + 0.6) = \mathbf{0.79}.$$

$\beta < 1$ weights precision more heavily (F1 here would be 0.706).

Q.35 — Answer: -0.0292 to -0.028 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Softmax + cross-entropy]

$z = Wx + b = (1.5, -2.5, -3)$; $p = \text{softmax}(z) = (0.9714, 0.0178, 0.0108)$.

$\delta = \partial L / \partial z = p - y = (-0.0286, 0.0178, 0.0108)$ (true class 1).

Since $z_k = \sum_j W_{kj} x_j + b_k$: $\partial L / \partial W_{kj} = (p_k - y_k) x_j$ and $\partial L / \partial b_k = p_k - y_k$ (gradient matrix = $(p - y)x^T$).
 $\partial L / \partial W_{11} = -0.0286 \times 1 = \mathbf{-0.0286}$.

Mock Test 21

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	8	7	20

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Probabilistic & Instance-based Classifiers	7	11
SVM & Decision Trees	7	11
Unsupervised Learning	7	11
Neural Networks (MLP / Feed-forward)	7	11
Regression & Model Selection	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

A categorical naive Bayes classifier has $d = 6$ features, each taking $m = 4$ possible values, and $K = 4$ classes. The number of independent parameters, including class priors, is _____ (Answer in integer)

Q.2 [1 mark · MCQ]

Cost-complexity pruning of a classification tree produced the nested subtrees $T1 \supset T2 \supset T3 \supset T4$ below. For complexity parameter $\alpha = 0.008$, which subtree minimises $R_\alpha(T) = R(T) + \alpha|T|$?

Subtree	T1	T2	T3	T4
Leaves $ T $	12	9	6	3
Training error $R(T)$	0.015	0.03	0.14	0.175

(A) T1

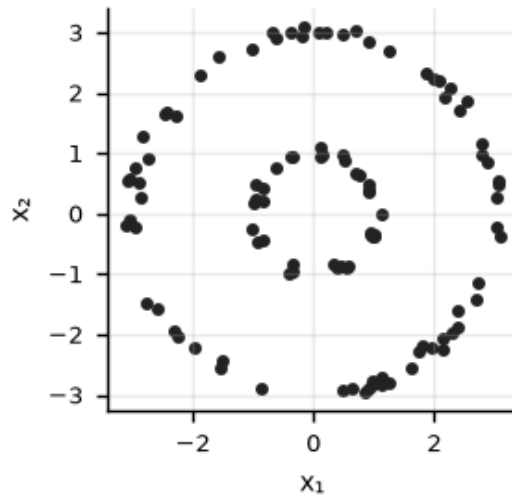
(B) T2

(C) T3

(D) T4

Q.3 [1 mark · MCQ]

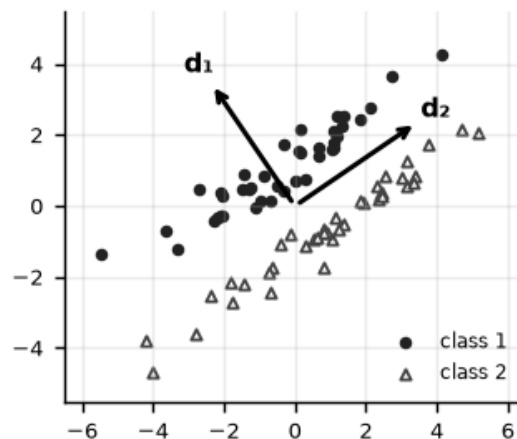
k-means with $k = 2$ fails to separate the two concentric rings shown below, even with many random restarts. The main reason is that k-means



- (A) produces convex clusters separated by linear (Voronoi) boundaries
- (B) cannot be applied to two-dimensional data
- (C) requires class labels to place the centroids
- (D) converges only when the data are standardized

Q.4 [1 mark · MCQ]

Two labelled classes are shown below, together with two orthogonal directions d_1 and d_2 . A single direction is to be selected by (i) PCA (first principal component, ignoring labels) and (ii) Fisher's LDA. Which statement is correct?



- (A) Both PCA and LDA choose d_1
- (B) PCA (1st PC) chooses d_1 ; LDA chooses d_2
- (C) PCA (1st PC) chooses d_2 ; LDA chooses d_1
- (D) Both PCA and LDA choose d_2

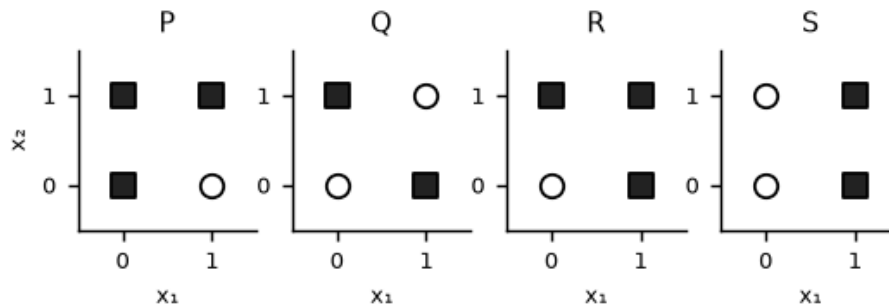
Q.5 [1 mark · MSQ (one or more correct)]

Consider the soft-margin SVM: minimise $\frac{1}{2}\|w\|^2 + C \sum_i \xi_i$ subject to $y_i(w^T x_i + b) \geq 1 - \xi_i$, $\xi_i \geq 0$. Which of the following statements is/are CORRECT?

- (A) Small C corresponds to higher bias/lower variance; large C to lower bias/higher variance.
- (B) The soft-margin SVM has a solution even when the data are not linearly separable.
- (C) Points with $0 < \alpha_i < C$ lie exactly on the margin ($\xi_i = 0$).
- (D) A small C allows more margin violations and typically yields a wider margin.

Q.6 [1 mark · MCQ]

Each panel shows the four binary inputs (x_1, x_2) labelled as class 1 (filled square) or class 0 (open circle). Which dataset can NOT be classified perfectly by a single perceptron (one linear threshold unit)?



- (A)** Dataset P **(B)** Dataset Q
(C) Dataset R **(D)** Dataset S

Q.7 [1 mark · NAT (numerical answer)]

A network is trained with standard (non-inverted) dropout, keep probability $p = 0.9$, on the inputs of a layer. A trained outgoing weight equals 3. To match expected activations, the weight used at test time is _____ (rounded off to two decimal places)

Q.8 [1 mark · NAT (numerical answer)]

A binary logistic regression model is given by $\ln[P(y=1|\mathbf{x}) / P(y=0|\mathbf{x})] = 0 + (2)x_1 + (1.5)x_2$. For the input $\mathbf{x} = (x_1, x_2) = (2, -1)$, the value of $P(y = 0 | \mathbf{x})$ is _____ (rounded off to two decimal places)

Q.9 [1 mark · MSQ (one or more correct)]

Which of the following statements about cross-validation (CV) is/are CORRECT? (n denotes the number of observations.)

- (A) Selecting features on the full data set and then cross-validating the selected model gives an optimistic error estimate.
- (B) For time-series data, validation folds should come after the training data in time.
- (C) The test set should be used to choose the hyperparameters.
- (D) The k-fold CV estimate does not depend on how the data are partitioned into folds.

Q.10 [1 mark · NAT (numerical answer)]

Regressing predictor x_3 on all the other predictors gives $R^2 = 0.58$. The VIF of x_3 is _____ (rounded off to two decimal places)

Q.11 [1 mark · NAT (numerical answer)]

For a data set, $\bar{x} = 23$, $\bar{y} = 47$, $s_x = 3$, $s_y = 15$ and the sample correlation between x and y is $r = 0.85$. Using the least-squares regression line of y on x , the predicted value of y at $x = 26$ is (rounded off to two decimal places)

Q.12 [1 mark · NAT (numerical answer)]

For $\mathbf{x} \in \mathbb{R}^3$, the quadratic kernel $K(\mathbf{x}, \mathbf{z}) = (1 + \mathbf{x}^T \mathbf{z})^2$ equals $\phi(\mathbf{x})^T \phi(\mathbf{z})$ for a feature map ϕ containing a constant, the linear terms, the squares and the pairwise cross terms (with $\sqrt{2}$ scaling). The length of $\phi(\mathbf{x})$ is (Answer in integer)

Q.13 [1 mark · NAT (numerical answer)]

The value of $\text{LeakyReLU}_\alpha(z_1) + \text{LeakyReLU}_\alpha(z_2) + \text{ReLU}(z_1)$ with $\alpha = 0.01$, $z_1 = -1$, $z_2 = 1$ ($\text{LeakyReLU}_\alpha(z) = z$ for $z \geq 0$ and αz for $z < 0$) is _____ (rounded off to three decimal places)

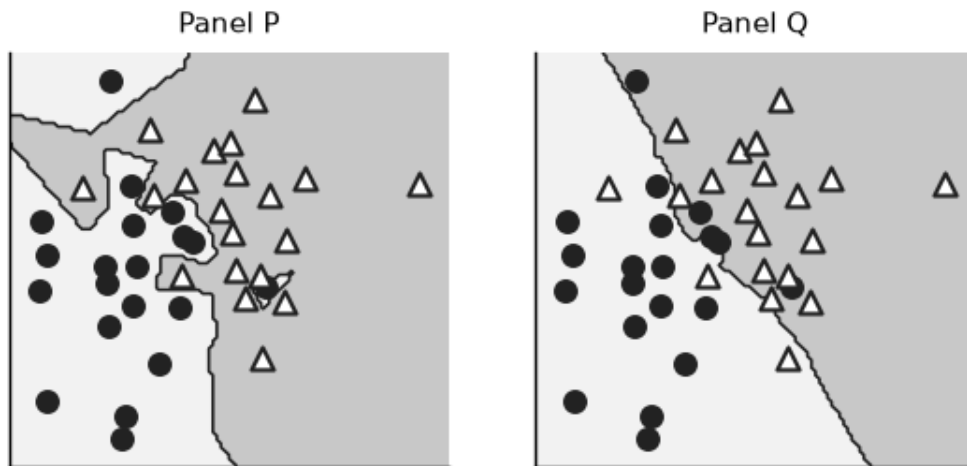
Q.14 [1 mark · MSQ (one or more correct)]

Which of the following statements about principal component analysis (PCA) is/are CORRECT?

- (A) When features are in different units, PCA is usually applied to standardized features (correlation matrix).
- (B) A scree plot shows the eigenvalues (or variance explained) in decreasing order against component number.
- (C) LDA, unlike PCA, uses class labels to find discriminative directions.
- (D) The variance of the projected data along the j -th PC equals the j -th largest covariance eigenvalue.

Q.15 [1 mark · MCQ]

The two panels show the decision regions of k -NN classifiers trained on the same 2-D data set, one with $k = 1$ and the other with $k = 21$. Which of the following statements is CORRECT?



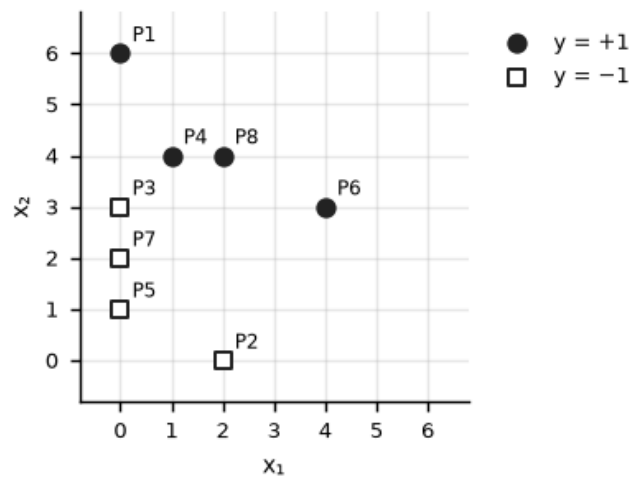
- (A) Panel P uses $k = 1$; it has lower bias and higher variance
- (B) Panel Q uses $k = 1$; it has lower bias and higher variance
- (C) Panel P uses $k = 1$; it has higher bias and lower variance
- (D) Panel Q uses $k = 21$; it has zero training error

Q.16 - Q.35 : Two-mark questions

Q.16 [2 marks · MSQ (one or more correct)]

A hard-margin linear SVM is trained on the 8 labelled points shown. Which of the following points is/are support vectors of the resulting maximum-margin classifier?

Point	x_1	x_2	y
P1	0	6	+1
P2	2	0	-1
P3	0	3	-1
P4	1	4	+1
P5	0	1	-1
P6	4	3	+1
P7	0	2	-1
P8	2	4	+1

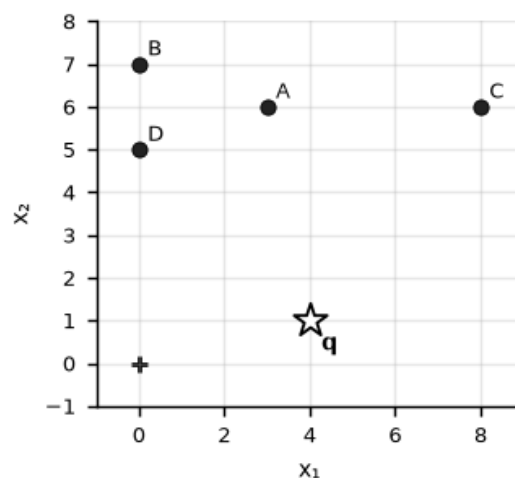


- (A) P1 (B) P2
 (C) P3 (D) P4

Q.17 [2 marks · MCQ]

A query point $q = (4, 1)$ and four points A-D are shown below. Using **Manhattan (L_1) distance**, which point is nearest to q ?

Point	A	B	C	D
x_1	3	0	8	0
x_2	6	7	6	5

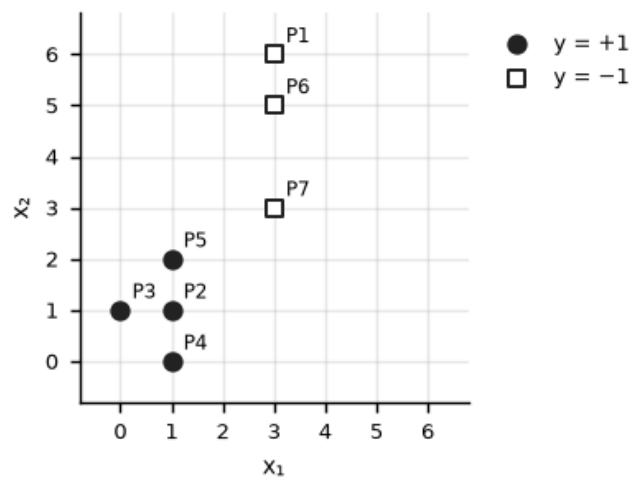


- (A) Point A (B) Point B
 (C) Point C (D) Point D

Q.18 [2 marks · MCQ]

Four candidate linear decision boundaries are proposed for the labelled training points shown. Which boundary separates the two classes with the LARGEST geometric margin (minimum distance from the boundary to any training point, all points correctly classified)?

Point	x_1	x_2	y
P1	3	6	-1
P2	1	1	+1
P3	0	1	+1
P4	1	0	+1
P5	1	2	+1
P6	3	5	-1
P7	3	3	-1

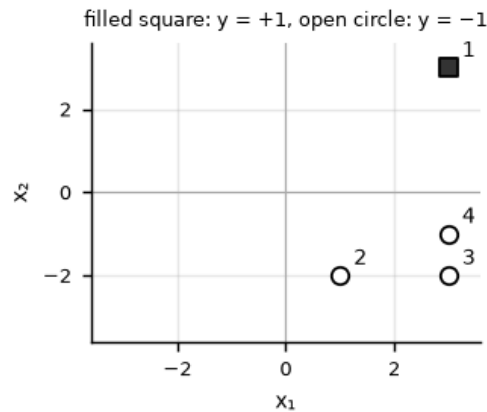


- (A) $x_1 + x_2 - 3.5 = 0$
- (B) $2x_1 + x_2 - 4.5 = 0$
- (C) $2x_1 + x_2 - 8.5 = 0$
- (D) $x_2 - 4.5 = 0$

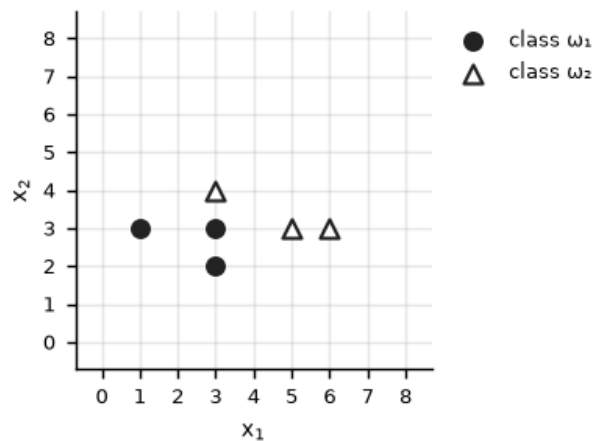
Q.19 [2 marks · NAT (numerical answer)]

A perceptron with parameters $w \in \mathbb{R}^2$, b is trained on the data below, starting from $w = (0, 0)$, $b = 0$, with learning rate $\eta = 0.5$. Examples are presented once in the order 1, 2, 3, 4 (one epoch). An update $w \leftarrow w + \eta y x$, $b \leftarrow b + \eta y$ is made whenever $y(w \cdot x + b) \leq 0$. After the epoch, the bias b is _____ (Answer in integer)

i	x_1	x_2	y
1	3	3	1
2	1	-2	-1
3	3	-2	-1
4	3	-1	-1

**Q.20 [2 marks · NAT (numerical answer)]**

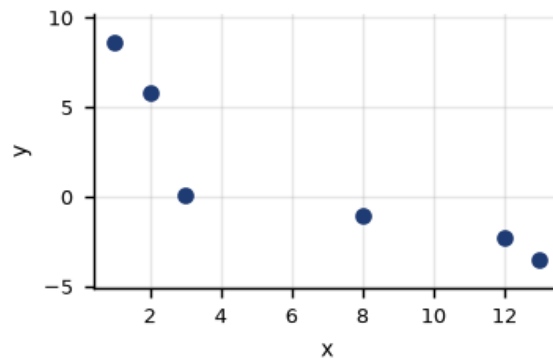
Class ω_1 has training points $(3, 2)$, $(1, 3)$, $(3, 3)$ and class ω_2 has points $(6, 3)$, $(5, 3)$, $(3, 4)$. Let $\mathbf{m}_1$, $\mathbf{m}_2$ be the class means and $S_W = \sum_k \sum_{x \in \omega_k} (x - \mathbf{m}_k)(x - \mathbf{m}_k)^T$ the within-class scatter matrix (no division by n). Then $(\mathbf{m}_2 - \mathbf{m}_1)^T S_W^{-1} (\mathbf{m}_2 - \mathbf{m}_1)$ is _____ (rounded off to two decimal places)



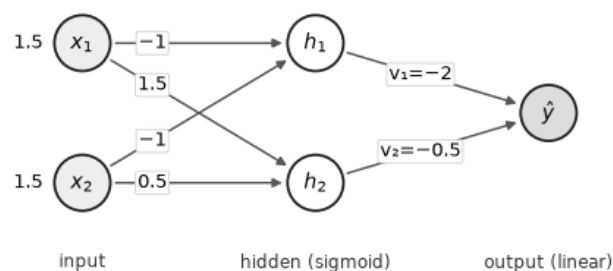
Q.21 [2 marks · NAT (numerical answer)]

A simple linear regression (with intercept) is fitted by least squares to the 6 points shown. With $H = X(X^T X)^{-1} X^T$, the largest leverage among the observations is _____ (rounded off to three decimal places)

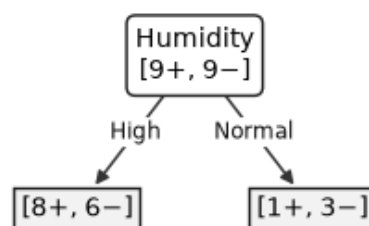
x	1	2	3	8	12	13
y	8.6	5.8	0.1	-1	-2.3	-3.5

**Q.22 [2 marks · NAT (numerical answer)]**

In the network below, hidden units are sigmoid, the output is linear ($\hat{y} = v_1 h_1 + v_2 h_2$), all biases are 0, and the loss is $L = \frac{1}{2}(\hat{y} - y)^2$. For input $x = (1.5, 1.5)$ and target $y = -1$, the value of $\partial L / \partial w_{22}$ (the weight from x_2 to h_2) is _____ (rounded off to 4 decimal places)

**Q.23 [2 marks · NAT (numerical answer)]**

A CART node with 18 training records (class counts [positive+, negative-]) is split on Humidity as shown. Compute the reduction in Gini index (Gini gain) obtained by splitting on Humidity. (rounded off to two decimal places)



Q.24 [2 marks · MSQ (one or more correct)]

Which of the following statements about loss functions and optimisation for training neural networks is/are CORRECT?

- (A) Momentum accumulates an exponentially decaying average of past gradients, which damps oscillations across steep directions.
- (B) With momentum β ($v \leftarrow \beta v + g$, $w \leftarrow w - \eta v$) and a constant gradient g , the step size approaches $\eta g / (1 - \beta)$.
- (C) Minimising the mean squared error corresponds to maximum likelihood under additive Gaussian noise with fixed variance.
- (D) With uniformly sampled mini-batches, the mini-batch gradient is an unbiased estimate of the full-batch gradient.

Q.25 [2 marks · NAT (numerical answer)]

Consider the model $y = wx + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$, $\sigma^2 = 2$, and w has a Laplace prior $p(w) \propto \exp(-|w|/b)$ with $b = 1$. Given the four observations below, the MAP estimate of w is _____ (rounded off to three decimal places)

x	4	4	2	1
y	6	7	-2	5

Q.26 [2 marks · NAT (numerical answer)]

For an input x , a classifier's posterior probabilities are $P(C_1|x) = 0.2$, $P(C_2|x) = 0.5$, $P(C_3|x) = 0.3$. The loss of taking action α_i (deciding C_i) when the true class is C_j is given below. Then the index i of the Bayes-optimal (minimum-risk) action α_i is _____. (Answer in integer)

loss $L(\alpha_i, C_j)$	true C_1	true C_2	true C_3
α_1 : decide C_1	0	2	9
α_2 : decide C_2	3	0	7
α_3 : decide C_3	3	3	0

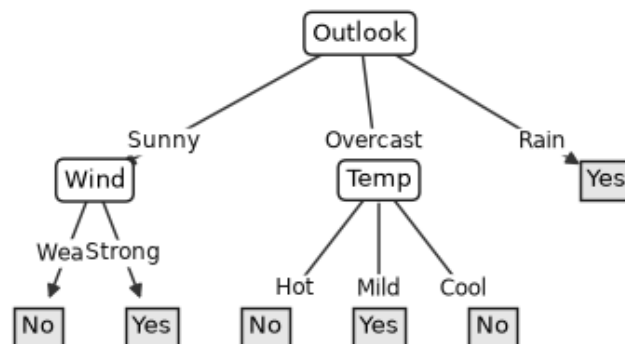
Q.27 [2 marks · MCQ]

For $n = 9$ paired observations: $\Sigma x = 86$, $\Sigma y = 21$, $\Sigma x^2 = 986$, $\Sigma y^2 = 159$, $\Sigma xy = 316$. The sample correlation coefficient r between x and y is

- (A) 1.048
- (B) 0.736
- (C) 0.858
- (D) 0.702

Q.28 [2 marks · NAT (numerical answer)]

The decision tree shown is applied to the 10 labelled records in the table (attributes not tested on a path are ignored). Find the precision of the tree for the class “Yes”. (rounded off to two decimal places)



#	Outlook	Wind	Temp	Actual
1	Overcast	Weak	Cool	Yes
2	Sunny	Weak	Hot	Yes
3	Sunny	Strong	Hot	Yes
4	Rain	Weak	Cool	Yes
5	Overcast	Weak	Cool	No
6	Overcast	Strong	Mild	Yes
7	Sunny	Weak	Mild	Yes
8	Sunny	Weak	Hot	No
9	Rain	Strong	Hot	No
10	Overcast	Strong	Hot	Yes

Q.29 [2 marks · NAT (numerical answer)]

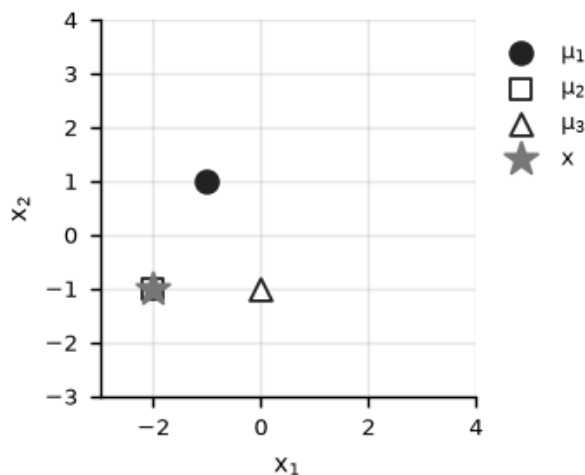
Agglomerative hierarchical clustering with **complete linkage** is applied to 6 points whose pairwise distance matrix is given below (upper triangle shown). The height (linkage distance) at which the **fifth** merge takes place is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5	P6
P1	0	7	4	5	5	5
P2		0	5	6	4	4
P3			0	3	9	9
P4				0	10	10
P5					0	2
P6						0

Q.30 [2 marks · NAT (numerical answer)]

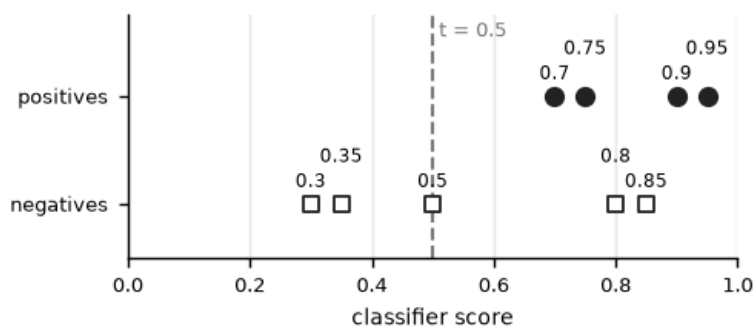
An LDA classifier for 3 classes has shared covariance matrix Σ (below), class means $\mu_1 = (-1, 1)^T$; $\mu_2 = (-2, -1)^T$; $\mu_3 = (0, -1)^T$, and priors $\pi_1 = 0.5$, $\pi_2 = 0.3$, $\pi_3 = 0.2$. The linear discriminant is $\delta_k(\mathbf{x}) = \mathbf{x}^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln \pi_k$ (natural log). For $\mathbf{x} = (-2, -1)^T$, the index $k \in \{1, 2, 3\}$ of the class predicted by LDA is _____ (Answer in integer)

$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

**Q.31 [2 marks · NAT (numerical answer)]**

A probabilistic classifier produces the scores below on 9 labelled examples (1 = positive). An example is predicted positive if its score is $\geq t$, with $t = 0.5$. The false positive rate at this threshold is _____ (rounded off to two decimal places)

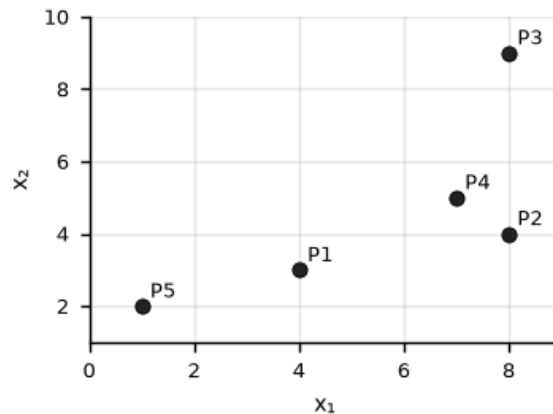
example	1	2	3	4	5	6	7	8	9
score	0.5	0.95	0.75	0.9	0.35	0.8	0.85	0.3	0.7
label	0	1	1	1	0	0	0	0	1



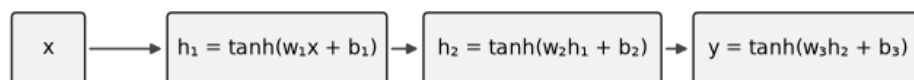
Q.32 [2 marks · NAT (numerical answer)]

A cluster contains the 5 points below. Distances are Euclidean (L_2). The medoid is the point that minimizes the sum of distances to all points of the cluster. The minimum total distance (cost of the medoid) is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5
x_1	4	8	8	7	1
x_2	3	4	9	5	2

**Q.33 [2 marks · NAT (numerical answer)]**

Consider the scalar chain shown below with weights $w_1 = 1$, $w_2 = 1$, $w_3 = 1$, biases $b_1 = -0.5$, $b_2 = 0$, $b_3 = 1$ and activation $\tanh$ at every stage. For input $x = 0.5$, the value of $\partial y / \partial w_1$ is _____ (rounded off to 4 decimal places)

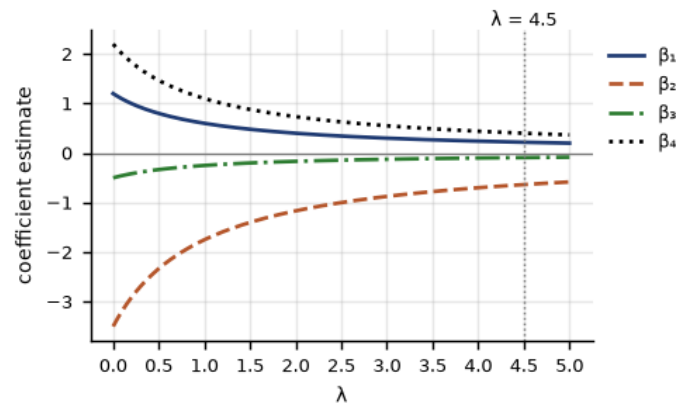
**Q.34 [2 marks · MSQ (one or more correct)]**

Which of the following statements about PCA, SVD and dimensionality reduction is/are CORRECT?

- (A) The PC score matrix equals $XV = U\Sigma$.
- (B) Kernel PCA can capture nonlinear structure by performing PCA in a feature space.
- (C) With n samples and $d \geq n$ features, the centered data matrix has rank at most $n - 1$.
- (D) The eigenvalues of the sample covariance equal the singular values σ_j of centered X .

Q.35 [2 marks · MSQ (one or more correct)]

For a design with orthonormal columns ($X^T X = I$), the figure shows the coefficient paths $\beta_i(\lambda)$ of a penalised least-squares estimator that minimises $\frac{1}{2}\|y - X\beta\|^2 + P_\lambda(\beta)$, where the penalty P_λ is either the ridge penalty $(\lambda/2)\|\beta\|_2^2$ or the lasso penalty $\lambda\|\beta\|_1$. Which of the following statements is/are CORRECT?



- (A) The plotted paths are those of ridge regression (L2 penalty).
- (B) As λ increases, the training residual sum of squares decreases.
- (C) Each coefficient path is piecewise linear in λ .
- (D) At $\lambda = 0$ all coefficients equal zero.

— END OF QUESTION PAPER —

Mock Test 21 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	75	19	NAT	2	0
2	MCQ	1	(B)	20	NAT	2	4.0926 to 4.1126
3	MCQ	1	(A)	21	NAT	2	0.4719 to 0.4759
4	MCQ	1	(C)	22	NAT	2	-0.0148 to -0.0142
5	MSQ	1	(A), (B), (C), (D)	23	NAT	2	0.0257 to 0.0457
6	MCQ	1	(B)	24	MSQ	2	(A), (B), (C), (D)
7	NAT	1	2.69 to 2.71	25	NAT	2	1.3742 to 1.3825
8	NAT	1	0.0659 to 0.0859	26	NAT	2	3
9	MSQ	1	(A), (B)	27	MCQ	2	(C)
10	NAT	1	2.369 to 2.3929	28	NAT	2	0.74 to 0.76
11	NAT	1	59.4513 to 60.0487	29	NAT	2	9.99 to 10.01
12	NAT	1	10	30	NAT	2	2
13	NAT	1	0.988 to 0.992	31	NAT	2	0.59 to 0.61
14	MSQ	1	(A), (B), (C), (D)	32	NAT	2	15.8411 to 15.8611
15	MCQ	1	(A)	33	NAT	2	0.2079 to 0.2121
16	MSQ	2	(C), (D)	34	MSQ	2	(A), (B), (C)
17	MCQ	2	(A)	35	MSQ	2	(A)
18	MCQ	2	(A)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Probabilistic & Instance-based Classifiers					11
SVM & Decision Trees					11
Unsupervised Learning					11

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Neural Networks (MLP / Feed-forward)					11
Regression & Model Selection					11
Total					55

Mock Test 21 — Detailed Solutions

Q.1 — Answer: 75 [NAT, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

For each class and feature, a distribution over m values has $m - 1 = 3$ free parameters: $Kd(m - 1) = 72$.

Priors: $K - 1 = 3$. Total = 75.

Q.2 — Answer: (B) [MCQ, 1M · SVM & Decision Trees > Decision trees: overfitting & pruning]

Subtree	$R(T) + \alpha T $
T1	$0.015 + 0.008 \times 12 = 0.111$
T2	$0.03 + 0.008 \times 9 = 0.102$
T3	$0.14 + 0.008 \times 6 = 0.188$
T4	$0.175 + 0.008 \times 3 = 0.199$

Minimum at **T2**. Larger α penalises leaves more and favours smaller trees; $\alpha = 0$ would pick the largest tree T1.

Q.3 — Answer: (A) [MCQ, 1M · Unsupervised Learning > k-means]

Points are assigned to the nearest centroid, so each k-means cluster is a convex Voronoi cell and two clusters are separated by the perpendicular bisector of their centroids. Non-convex shapes such as rings or moons cannot be separated by such a boundary.

k-means converges on raw data, works in any dimension, and is unsupervised (no labels).

Q.4 — Answer: (C) [MCQ, 1M · Unsupervised Learning > PCA & dimensionality reduction]

PCA is unsupervised: it picks the direction of maximum total variance, which is along the long axis of the clouds (d_2).

LDA is supervised: it maximizes between-class separation relative to within-class scatter, so it picks the direction across which the class means differ (d_1).

Answer: **PCA (1st PC) chooses d_2 ; LDA chooses d_1** . Projecting on the PCA direction would mix the two classes completely.

Q.5 — Answer: (A), (B), (C), (D) [MSQ, 1M · SVM & Decision Trees > SVM: soft margin & hinge loss]

(A) **TRUE**. C acts like an inverse regularisation strength.

(B) **TRUE**. Slack variables make the problem always feasible.

(C) **TRUE**. KKT: $\alpha_i < C \Rightarrow \xi_i = 0$, and $\alpha_i > 0 \Rightarrow y f = 1 - \xi_i$.

(D) **TRUE**. Small $C \Rightarrow$ stronger regularisation.

Q.6 — Answer: (B) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

A single perceptron can classify a dataset perfectly iff the two classes are linearly separable.

Dataset P: labels for $(0,0),(0,1),(1,0),(1,1) = 1, 1, 0, 1 \rightarrow (\text{NOT } x_1) \text{ OR } x_2$ – linearly separable.

Dataset Q: labels for $(0,0),(0,1),(1,0),(1,1) = 0, 1, 1, 0 \rightarrow \text{XOR}$ – **not** linearly separable (diagonal pattern).

Dataset R: labels for $(0,0),(0,1),(1,0),(1,1) = 0, 1, 1, 1 \rightarrow \text{OR}$ – linearly separable.

Dataset S: labels for $(0,0),(0,1),(1,0),(1,1) = 0, 0, 1, 1 \rightarrow x_1$ (identity of x_1) – linearly separable.

Answer: **Dataset Q**. Of the 14 non-constant Boolean functions of two inputs, only XOR and XNOR are not linearly separable.

Q.7 — Answer: 2.69 to 2.71 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

During training each input is present with probability p , so at test time weights are scaled by p : $0.9 \times 3 = 2.7$.

Q.8 — Answer: 0.0659 to 0.0859 [NAT, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

$z = w_0 + w_1 x_1 + w_2 x_2 = 0 + (2)(2) + (1.5)(-1) = 2.5$ (this is the log-odds).

$P(y = 1 | \mathbf{x}) = \sigma(z) = 1/(1 + e^{-2.5}) = 0.9241$.

$$P(y = 0 \mid \mathbf{x}) = 1 - \sigma(z) = \sigma(-z) = \mathbf{0.08}.$$

Q.9 — Answer: (A), (B) [MSQ, 1M · Regression & Model Selection > Cross-validation]

(A) **TRUE**. Classic data leakage — the selection already saw the validation folds.

(B) **TRUE**. Otherwise the model is trained on the future.

(C) **FALSE**. Then test error is no longer an unbiased estimate.

(D) **FALSE**. Different random partitions give different estimates.

Q.10 — Answer: 2.369 to 2.3929 [NAT, 1M · Regression & Model Selection > Multiple linear regression]

Variance inflation factor of predictor j : $VIF_j = 1/(1 - R_j^2)$, where R_j^2 is the R^2 from regressing x_j on all other predictors. It is the factor by which $\text{Var}(\beta_j)$ is inflated by collinearity; $\sqrt{VIF}$ inflates the standard error.

$$VIF = 1/(1 - 0.58) = 2.381.$$

Answer: **2.38**.

Q.11 — Answer: 59.4513 to 60.0487 [NAT, 1M · Regression & Model Selection > Simple linear regression]

$$\beta_1 = r \cdot s_y / s_x = 0.85 \times 15/3 = 4.25.$$

$$\hat{y} = \bar{y} + \beta_1(x_0 - \bar{x}) = 47 + (4.25)(3) = 59.75 \text{ (the line passes through } (\bar{x}, \bar{y})).$$

Answer: **59.75**.

Q.12 — Answer: 10 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

Constant: 1; linear: 3; squares: 3; cross terms: $C(3, 2) = 3$.

$$\text{Total} = 1 + 3 + 3 + 3 = \mathbf{10} = C(3 + 2, 2).$$

Q.13 — Answer: 0.988 to 0.992 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

$$\text{LeakyReLU}(-1) = 0.01 \times -1 = -0.01; \text{LeakyReLU}(1) = 1; \text{ReLU}(-1) = 0. \text{ Sum} = \mathbf{0.99}.$$

Q.14 — Answer: (A), (B), (C), (D) [MSQ, 1M · Unsupervised Learning > PCA & dimensionality reduction]

(A) **TRUE**. Otherwise large-scale features dominate.

(B) **TRUE**. Used to choose the number of PCs.

(C) **TRUE**. LDA is supervised.

(D) **TRUE**. $\mathbf{v}^T \mathbf{S} \mathbf{v} = \lambda$ for a unit eigenvector $\mathbf{v}$.

Q.15 — Answer: (A) [MCQ, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

The jagged boundary with small islands around individual points (Panel P) is produced by $k = 1$, which fits every training point (zero training error): low bias, high variance.

The smooth boundary (Panel Q) comes from $k = 21$: averaging many neighbours raises bias and lowers variance; it misclassifies some training points, so its training error is not zero.

Answer: **Panel P uses $k = 1$; it has lower bias and higher variance.**

Q.16 — Answer: (C), (D) [MSQ, 2M · SVM & Decision Trees > SVM: dual & support vectors]

The closest pair of opposite-class points is $P_3 = (0, 3)$ and $P_4 = (1, 4)$. Their perpendicular bisector separates the classes and no other point comes closer to it, so Hence $\mathbf{w} \parallel (\mathbf{x}_+ - \mathbf{x}_-)$ and the support vectors are P_3, P_4 .

Scaling so that $y_i(\mathbf{w}^T \mathbf{x}_i + b) = 1$ at the support vectors gives $\mathbf{w} = (1, 1)$, $b = -4$; $\|\mathbf{w}\| = 1.4142$, margin $2/\|\mathbf{w}\| = 1.4142$.

Point	P1	P2	P3	P4	P5	P6	P7	P8
$y(\mathbf{w}^T \mathbf{x} + b)$	2	2	1	1	3	3	2	2

Support vectors are exactly the points with $y_i(\mathbf{w}^T \mathbf{x}_i + b) = 1$ ($\alpha_i > 0$): **P3, P4**.

Dual check: $\alpha = (\alpha_3 = 1, \alpha_4 = 1)$ satisfy $\sum \alpha_i y_i = 0$ and $\sum \alpha_i y_i \mathbf{x}_i = \mathbf{w}$.

Q.17 — Answer: (A) [MCQ, 2M · Unsupervised Learning > Distance measures]

Compute the measure from $q = (4, 1)$ to every point (cosine column shows similarity, larger = more similar):

Point	coordinates	Manhattan	Euclidean	Chebyshev	cosine similarity
A	(3, 6)	6	5.099	5	0.651
B	(0, 7)	10	7.211	6	0.243
C	(8, 6)	9	6.403	5	0.922
D	(0, 5)	8	5.657	4	0.243

Under Manhattan (L_1) distance the nearest to q is **point A**.

Note how the answer depends on the measure: Manhattan (L_1) distance $\rightarrow$ A; Euclidean (L_2) distance $\rightarrow$ A; Chebyshev (L_∞) distance $\rightarrow$ D; cosine similarity $\rightarrow$ C. Cosine similarity looks only at the angle from the origin, L_1 sums coordinate gaps, L_∞ takes the largest gap.

Q.18 — Answer: (A) [MCQ, 2M · SVM & Decision Trees > SVM: margin & geometry]

For $n^T x - c = 0$, the geometric margin is $\min_i y_i (n^T x_i - c) / \|n\|$ with orientation $s = \pm 1$ chosen so that the positive class is on the positive side; a negative value means some point is misclassified.

Hyperplane	$\ n\ $	$\min_i$ distance (signed, best orientation)
$x_1 + x_2 - 3.5 = 0$	1.414	0.354
$2x_1 + x_2 - 4.5 = 0$	2.236	0.224
$2x_1 + x_2 - 8.5 = 0$	2.236	0.224
$x_2 - 4.5 = 0$	1	-1.5 (does not separate)

The largest margin is 0.354, for $x_1 + x_2 - 3.5 = 0$. Note that a boundary can separate the data perfectly yet pass close to some point — only the minimum distance matters.

Q.19 — Answer: 0 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Rule: if $y_i(w \cdot x_i + b) \leq 0$, then $w \leftarrow w + \eta y_i x_i$, $b \leftarrow b + \eta y_i$ ($\eta = 0.5$). Start $w = (0, 0)$, $b = 0$.

Example 1: $a = 0$, $y \cdot a = 0 \rightarrow$ **mistake**, update $\rightarrow w = (1.5, 1.5)$, $b = 0.5$.

Example 2: $a = -1$, $y \cdot a = 1 \rightarrow$ correct, no change.

Example 3: $a = 2$, $y \cdot a = -2 \rightarrow$ **mistake**, update $\rightarrow w = (0, 2.5)$, $b = 0$.

Example 4: $a = -2.5$, $y \cdot a = 2.5 \rightarrow$ correct, no change.

Required value = **0**.

Note: a point with net input exactly 0 counts as a mistake ($y \cdot a \leq 0$) — this is why the first example always triggers an update when starting from zero.

Q.20 — Answer: 4.0926 to 4.1126 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$m_1 = (2.333, 2.667)$, $m_2 = (4.667, 3.333)$.

$$S_1 = \begin{bmatrix} 2.667 & -0.667 \\ -0.667 & 0.667 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 4.667 & -1.667 \\ -1.667 & 0.667 \end{bmatrix}$$

$$S_W = \begin{bmatrix} 7.333 & -2.333 \\ -2.333 & 1.333 \end{bmatrix}$$

$m_2 - m_1 = (2.333, 0.667)$.

$w = S_W^{-1}(m_2 - m_1) = (1.0769, 2.3846)$.

$(m_2 - m_1)^T w = 4.1$.

Q.21 — Answer: 0.4719 to 0.4759 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

For simple linear regression with intercept, the leverage (diagonal of the hat matrix) is $h_{ii} = 1/n + (x_i - \bar{x})^2 / S_{xx}$.

$\bar{x} = 6.5$, $S_{xx} = 137.5$.

The largest leverage is at the x farthest from $\bar{x}$: $x = 13$, $h = 1/6 + (6.5)^2 / 137.5 = 0.4739$.

(Check: $\sum_{ii} = 2 = 2 = \text{number of parameters.}$)

Answer: **0.474**.

Q.22 — Answer: -0.0148 to -0.0142 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Backpropagation]

Forward: $a = Wx = (-3, 3)$, $h = \sigma(a) = (0.0474, 0.9526)$, $\hat{y} = v \cdot h = -0.5711$.

Output error $\delta_o = \partial L / \partial \hat{y} = \hat{y} - y = 0.4289$ (linear output, $L = \frac{1}{2}(\hat{y} - y)^2$).

Hidden delta $\delta_2 = \delta_o \cdot v_2 \cdot h_2(1 - h_2) = 0.4289 \times -0.5 \times 0.0452 = -0.0097$.

$\partial L / \partial w_{22} = \delta_2 \cdot x_2 = -0.0097 \times 1.5 = \mathbf{-0.0145}$.

Q.23 — Answer: 0.0257 to 0.0457 [NAT, 2M · SVM & Decision Trees › Decision trees: split selection]

Parent [9+, 9-]: Gini = $1 - [(9/18)^2 + (9/18)^2] = 0.5$.

Humidity = High: [8+, 6-], Gini = $1 - [(8/14)^2 + (6/14)^2] = 0.4898$.

Humidity = Normal: [1+, 3-], Gini = $1 - [(1/4)^2 + (3/4)^2] = 0.375$.

Weighted Gini = $(14/18)(0.4898) + (4/18)(0.375) = 0.4643$.

Gini gain = $0.5 - 0.4643 = 0.0357$.

Answer: **0.04**.

Q.24 — Answer: (A), (B), (C), (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

(A) TRUE. Opposite-sign components cancel.

(B) TRUE. Geometric series $1 + \beta + \beta^2 + \dots$

(C) TRUE. $-\log N(y; \hat{y}, \sigma^2) \propto (y - \hat{y})^2$.

(D) TRUE. Its expectation equals the average gradient.

Q.25 — Answer: 1.3742 to 1.3825 [NAT, 2M · Regression & Model Selection › MLE & MAP view]

$-\ln p(w | \text{data}) = (1/(2\sigma^2))\sum(y_i - wx_i)^2 + |w|/b + \text{const.}$ Multiply by $2\sigma^2$: $\sum(y_i - wx_i)^2 + (2\sigma^2/b)|w|$ (lasso).

$\sum x_i^2 = 37$, $\sum x_i y_i = 53$. Setting the subgradient to 0: $\sum x^2 \cdot w = \sum xy - (\sigma^2/b) \cdot \text{sign}(w) \Rightarrow \hat{w} = \text{sign}(\sum xy) \cdot \max(|\sum xy| - \sigma^2/b, 0) / \sum x^2$.

$\sigma^2/b = 2$; $\hat{w} = 1 \times \max(53 - 2, 0) / 37 = 1.3784$.

Answer: **1.378**.

Q.26 — Answer: 3 [NAT, 2M · Probabilistic & Instance-based Classifiers › Bayes optimal classifier]

Conditional risk $R(\alpha_i | x) = \sum_j L(\alpha_i, C_j) P(C_j | x)$:

$R(\alpha_1) = 0 \times 0.2 + 2 \times 0.5 + 9 \times 0.3 = 3.7$

$R(\alpha_2) = 3 \times 0.2 + 0 \times 0.5 + 7 \times 0.3 = 2.7$

$R(\alpha_3) = 3 \times 0.2 + 3 \times 0.5 + 0 \times 0.3 = 2.1$

Minimum risk: α_3 with $R = 2.1$. The MAP rule picks C_2 (risk 2.7) — different, because the losses are asymmetric.

Answer: **3**.

Q.27 — Answer: (C) [MCQ, 2M · Regression & Model Selection › Simple linear regression]

$\bar{x} = 9.5556$, $\bar{y} = 2.3333$.

$S_{xx} = \sum x^2 - (\sum x)^2/n = 164.2222$, $S_{yy} = \sum y^2 - (\sum y)^2/n = 110$, $S_{xy} = \sum xy - \sum x \sum y/n = 115.3333$.

$r = S_{xy} / \sqrt{(S_{xx} S_{yy})} = 115.3333 / \sqrt{(164.2222 \times 110)} = \mathbf{0.858}$.

($r^2 = 0.736$ would be R^2 ; the slopes $b_{yx} = 0.702$, $b_{xy} = 1.048$ satisfy $b_{yx} b_{xy} = r^2$.)

Q.28 — Answer: 0.74 to 0.76 [NAT, 2M · SVM & Decision Trees › Decision trees: structure & traversal]

Route each record from the root to a leaf:

#	Predicted	Actual	Correct?
1	No	Yes	no
2	No	Yes	no

3	Yes	Yes	yes
4	Yes	Yes	yes
5	No	No	yes
6	Yes	Yes	yes
7	No	Yes	no
8	No	No	yes
9	Yes	No	no
10	No	Yes	no

Misclassified: 5 of 10 (accuracy 50%). Answer: **0.75**.

Q.29 — Answer: 9.99 to 10.01 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Use complete linkage: $d(A, B) = \max_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P5\} + \{P6\}$ at height **2**. Updated: $d(\{P5, P6\}, \{P1\}) = 5$, $d(\{P5, P6\}, \{P2\}) = 4$, $d(\{P5, P6\}, \{P3\}) = 9$, $d(\{P5, P6\}, \{P4\}) = 10$.

Merge 2: $\{P3\} + \{P4\}$ at height **3**. Updated: $d(\{P3, P4\}, \{P1\}) = 5$, $d(\{P3, P4\}, \{P2\}) = 6$, $d(\{P3, P4\}, \{P5, P6\}) = 10$.

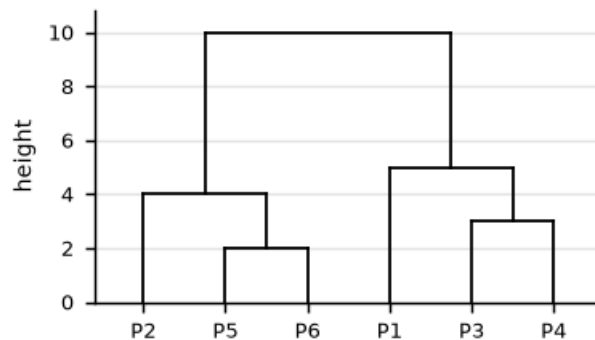
Merge 3: $\{P2\} + \{P5, P6\}$ at height **4**. Updated: $d(\{P2, P5, P6\}, \{P1\}) = 7$, $d(\{P2, P5, P6\}, \{P3, P4\}) = 10$.

Merge 4: $\{P1\} + \{P3, P4\}$ at height **5**. Updated: $d(\{P1, P3, P4\}, \{P2, P5, P6\}) = 10$.

Merge 5: $\{P2, P5, P6\} + \{P1, P3, P4\}$ at height **10**.

The fifth merge occurs at height **10**.

Dendrogram:



Q.30 — Answer: 2 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$$\Sigma^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 0.5 \end{bmatrix}$$

$k = 1$: $\Sigma^{-1}\mu_1 = (-1, 0.5)$; $\mathbf{x}^T \Sigma^{-1} \mu = 1.5$; $\frac{1}{2} \mu^T \Sigma^{-1} \mu = 0.75$; $\ln \pi = -0.6931 \Rightarrow \delta_1 = 0.0569$.

$k = 2$: $\Sigma^{-1}\mu_2 = (-2, -0.5)$; $\mathbf{x}^T \Sigma^{-1} \mu = 4.5$; $\frac{1}{2} \mu^T \Sigma^{-1} \mu = 2.25$; $\ln \pi = -1.204 \Rightarrow \delta_2 = 1.046$.

$k = 3$: $\Sigma^{-1}\mu_3 = (0, -0.5)$; $\mathbf{x}^T \Sigma^{-1} \mu = 0.5$; $\frac{1}{2} \mu^T \Sigma^{-1} \mu = 0.25$; $\ln \pi = -1.6094 \Rightarrow \delta_3 = -1.3594$.

The largest discriminant is $\delta_2 \Rightarrow$ predicted class **2**.

Q.31 — Answer: 0.59 to 0.61 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Predicted positive (score ≥ 0.5): examples 1, 2, 3, 4, 6, 7, 9.

TP = 4, FP = 3, FN = 0, TN = 2.

False positive rate = $FP / (FP + TN) = \mathbf{0.6}$.

Raising t typically raises precision and lowers recall/FPR; lowering t does the opposite (threshold moving).

Q.32 — Answer: 15.8411 to 15.8611 [NAT, 2M · Unsupervised Learning > k-medoids]

Compute the Euclidean (L_2) distance matrix and its row sums:

	P1	P2	P3	P4	P5	Σ
P1	0	4.12	7.21	3.61	3.16	18.1
P2	4.12	0	5	1.41	7.28	17.82
P3	7.21	5	0	4.12	9.9	26.23
P4	3.61	1.41	4.12	0	6.71	15.85
P5	3.16	7.28	9.9	6.71	0	27.05

The smallest row sum is 15.85 for P4, so the medoid is P4.

Answer: **15.85**.

Unlike a k-means centroid, the medoid is always an actual data point.

Q.33 — Answer: 0.2079 to 0.2121 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Activation functions]

Forward: $a_1 = 1 \cdot 0.5 + -0.5 = 0$, $h_1 = 0$; $a_2 = 1 \cdot 0 + 0 = 0$, $h_2 = 0$; $a_3 = 1 \cdot 0 + 1 = 1$, $y = 0.7616$.

Local derivatives $\tanh'(a)$: $\tanh'(a) = 1 - \tanh^2(a)$: $\tanh'(a_1) = 1$, $\tanh'(a_2) = 1$, $\tanh'(a_3) = 0.42$.

Chain rule: $\partial y / \partial w_1 = \tanh'(a_3) \cdot w_3 \cdot \tanh'(a_2) \cdot w_2 \cdot \tanh'(a_1) \cdot x = \mathbf{0.21}$.

Each layer contributes a factor $\tanh' \leq 1$ - the product shrinks with depth (vanishing gradient).

Q.34 — Answer: (A), (B), (C) [MSQ, 2M · Unsupervised Learning › PCA & dimensionality reduction]

(A) TRUE. Multiply $X = U\Sigma V^T$ by V .

(B) TRUE. Via the kernel trick.

(C) TRUE. Centering removes one degree of freedom.

(D) FALSE. They are $\sigma_j^2 / (n - 1)$.

Q.35 — Answer: (A) [MSQ, 2M · Regression & Model Selection › Ridge & Lasso]

With $X^T X = I$ the ridge path is $b / (1 + \lambda)$ and the lasso path is $\text{sign}(b)(|b| - \lambda)_+$. The plot shows paths reaching zero at finite $\lambda \Rightarrow$ lasso; here it is **ridge**.

(A) TRUE. Smooth decay $b / (1 + \lambda)$, no exact zeros.

(B) FALSE. Training RSS is minimised at $\lambda = 0$ and increases with λ .

(C) FALSE. $b / (1 + \lambda)$ is a smooth hyperbola, not piecewise linear.

(D) FALSE. At $\lambda = 0$ they are the OLS estimates.

Mock Test 22

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	4	8	23

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
SVM & Decision Trees	7	11
Unsupervised Learning	7	11
Neural Networks (MLP / Feed-forward)	7	11
Probabilistic & Instance-based Classifiers	7	11
Regression & Model Selection	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

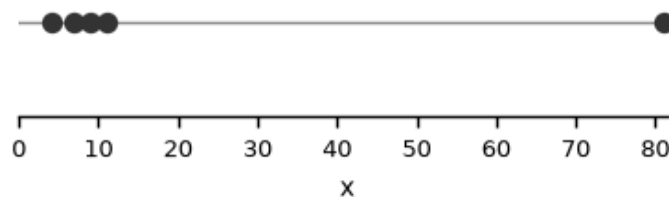
Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

A bootstrap sample of size $n = 3$ is drawn with replacement from a training set of 3 distinct points. The probability that a particular point is NOT in the sample (i.e. is out-of-bag) is _____ (rounded off to three decimal places)

Q.2 [1 mark · NAT (numerical answer)]

A cluster contains the 1-D values $\{4, 7, 9, 11, 81\}$. Treating them as one cluster, the absolute difference between the k-means centroid (mean) and the k-medoids representative (absolute-difference dissimilarity) is _____ (rounded off to two decimal places)



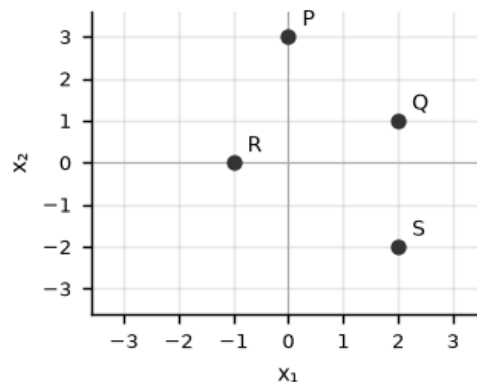
Q.3 [1 mark · MSQ (one or more correct)]

Which of the following statements about kernel functions (Mercer kernels) is/are CORRECT?

- (A) If k_1 and k_2 are valid kernels, the product $k_1 \cdot k_2$ is a valid kernel.
- (B) $k(x, z) = (xz - 1)^2$ is a valid kernel on $\mathbb{R}$.
- (C) $k(x, z) = x^T A z$ is a valid kernel when A is symmetric positive semidefinite.
- (D) Every valid kernel yields a symmetric PSD Gram matrix for any finite set of points.

Q.4 [1 mark · MSQ (one or more correct)]

A perceptron with weights $w = (-2, 3)$ and bias $b = 4$ outputs $\hat{y} = +1$ if $w \cdot x + b \geq 0$ and $\hat{y} = -1$ otherwise. Which of the points shown below are classified as $\hat{y} = +1$?

**(A)** Point P = (0, 3)**(B)** Point Q = (2, 1)**(C)** Point R = (-1, 0)**(D)** Point S = (2, -2)**Q.5 [1 mark · NAT (numerical answer)]**

In a spam filter, a false negative costs 13 units and a false positive costs 2 units (correct decisions cost 0). For an input x the model gives $P(\text{positive} \mid x) = 0.7$. The expected cost of the Bayes-optimal decision for this x is _____ (rounded off to two decimal places)

Q.6 [1 mark · NAT (numerical answer)]

For logits $z = (-2, -1, 2.5)$, the softmax probability $p_2 = \exp(z_2) / \sum_j \exp(z_j)$ is _____ (rounded off to 4 decimal places)

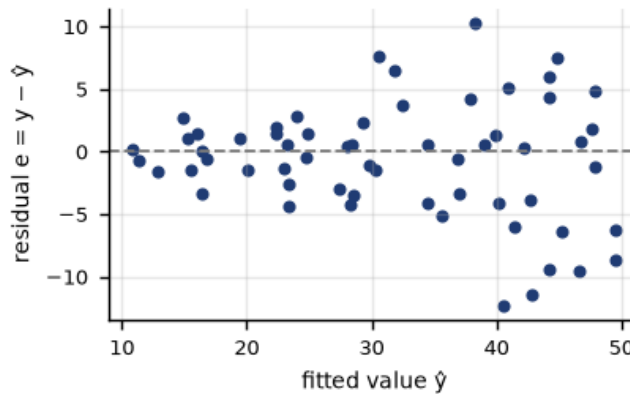
Q.7 [1 mark · NAT (numerical answer)]

A sigmoid-output classifier gives the probabilities $p = P(y = 1 \mid x)$ below. The mean binary cross-entropy $-[y \ln p + (1 - y) \ln(1 - p)]$ over the 4 examples is _____ (rounded off to three decimal places)

example	$p = P(y = 1)$	y
1	0.1	1
2	0.9	1
3	0.7	1
4	0.3	0

Q.8 [1 mark · MCQ]

After fitting a linear regression model, the residuals are plotted against the fitted values as shown. The plot most strongly suggests:



- (A) A single outlying observation, with the rest of the fit looking adequate
- (B) A non-linear relationship not captured by the model (e.g. a missing x^2 term)
- (C) Non-constant error variance (heteroscedasticity)
- (D) No evident violation of the linear-model assumptions

Q.9 [1 mark · NAT (numerical answer)]

A linear regression model with an intercept uses 5 numeric features and a categorical feature *day type* with 3 levels (Weekday, Saturday, Sunday); a categorical feature *education level* with 4 levels (School, UG, PG, PhD). Each categorical feature is encoded by dummy (indicator) variables using a reference level, so that the design matrix has full column rank. An interaction between the first numeric feature x_1 and *day type* is also added (products of x_1 with each day type dummy). The total number of coefficients (including the intercept) is _____. (Answer in integer)

Q.10 [1 mark · MCQ]

A fitted logistic regression model reports that each one-unit increase in *credit score (in 10s)* multiplies the odds of loan approval by 4 (other features fixed). The coefficient of *credit score (in 10s)* in the model is

- (A) 4
- (B) 0.6
- (C) 0.25
- (D) 1.39

Q.11 [1 mark · MSQ (one or more correct)]

Which of the following tasks is/are most naturally formulated as **classification** problems?

- (A) Predicting the monthly rent of a flat from its area and locality
- (B) Predicting the delivery time (in minutes) of a food order
- (C) Recognising the species of an iris flower from petal measurements
- (D) Predicting the blood group (A, B, AB or O) of a person from genetic markers

Q.12 [1 mark · MSQ (one or more correct)]

Which of the following statements about classifier evaluation is/are CORRECT?

- (A) Specificity equals $TN/(TN + FP)$.
- (B) ROC AUC is unchanged by any strictly increasing transformation of the scores.
- (C) Recall equals $TP/(TP + FP)$.
- (D) The precision of a random classifier is 0.5 regardless of the class prevalence.

Q.13 [1 mark · NAT (numerical answer)]

The distance between the parallel hyperplanes $2x_1 + x_2 + 2 = 0$ and $2x_1 + x_2 - 2 = 0$ is _____. (rounded off to two decimal places)

Q.14 [1 mark · NAT (numerical answer)]

A divisive (top-down) hierarchical algorithm considers **every** way of splitting a cluster of 7 points into two non-empty sub-clusters. The number of candidate splits is _____ (Answer in integer)

Q.15 [1 mark · MSQ (one or more correct)]

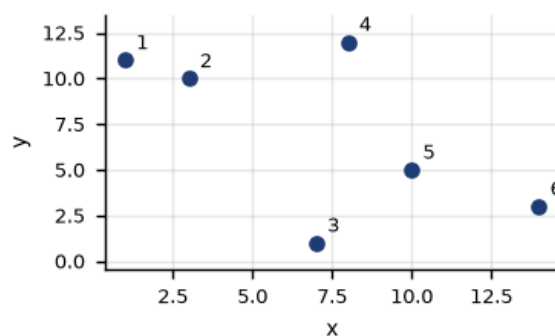
Which of the following statements about hierarchical clustering is/are CORRECT?

- (A) Single linkage is prone to the chaining effect, producing long, straggly clusters.
- (B) Complete linkage measures cluster distance by the maximum pairwise distance between their members.
- (C) Agglomerative clustering of n points performs exactly $n - 1$ merges.
- (D) For a fixed distance matrix without ties, agglomerative clustering is deterministic.

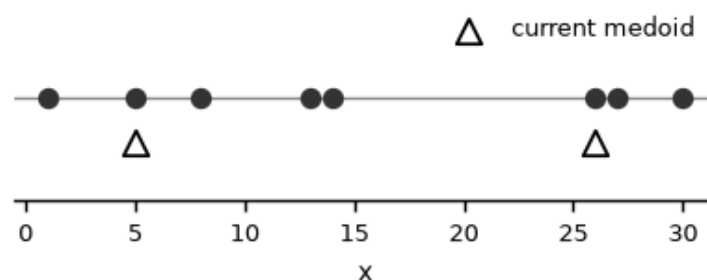
Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

Consider the 6 training points (x_i, y_i) shown below. A 1-nearest-neighbour regressor (prediction = y -value of the closest training x , $|\cdot|$ distance) is evaluated by leave-one-out cross-validation. The LOOCV MSE is _____ (rounded off to two decimal places)

x	1	3	7	8	10	14
y	11	10	1	12	5	3

**Q.17 [2 marks · NAT (numerical answer)]**

PAM (k-medoids, $k = 2$, absolute-difference dissimilarity) is applied to the 1-D data $\{1, 5, 8, 13, 14, 26, 27, 30\}$. The current medoids are 5 and 26. The total cost $\sum_i \min_m |x_i - m|$ of the current configuration is _____ (Answer in integer)



Q.18 [2 marks · NAT (numerical answer)]

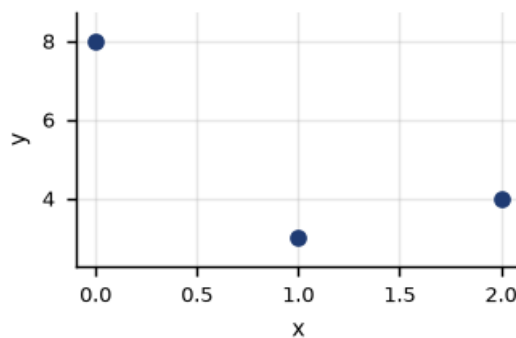
In one AdaBoost round the current example weights and whether the chosen weak classifier h classifies each example correctly are given. Using $\alpha = \frac{1}{2} \ln((1 - \epsilon)/\epsilon)$ and the update $w_i \leftarrow w_i \cdot \exp(-\alpha y_i h(x_i))/Z$, find the normalisation constant $Z = \sum_i w_i \cdot \exp(-\alpha y_i h(x_i))$. (rounded off to three decimal places)

Example i	1	2	3	4	5	6	7	8
Weight w_i	4/27	4/27	4/27	2/27	1/9	1/9	2/27	5/27
$h(x_i)$ correct?	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes

Q.19 [2 marks · NAT (numerical answer)]

A linear model $\hat{y} = w_0 + w_1 x$ is trained on the 3 points below by batch gradient descent on the loss $J(w) = (1/n) \sum (w_0 + w_1 x_i - y_i)^2$, starting from $(w_0, w_1) = (1, -1)$ with learning rate $\eta = 0.02$. Then the loss $J(w)$ after one iteration is _____ (rounded off to three decimal places)

x	0	1	2
y	8	3	4

**Q.20 [2 marks · MCQ]**

A naive Bayes classifier is trained on the 12 examples below. The new instance (Sender = Known, Length = Medium, HasLink = No) is classified (i) with maximum-likelihood (unsmoothed) estimates and (ii) with Laplace (add-one) smoothing of the class-conditional probabilities (class priors unsmoothed). The predicted classes are

#	Sender	Length	HasLink	Class
1	Known	Long	No	Ham
2	Known	Long	No	Ham
3	Known	Short	Yes	Spam
4	Known	Short	Yes	Spam
5	Unknown	Long	Yes	Spam
6	Known	Short	No	Ham
7	Known	Medium	Yes	Spam
8	Known	Medium	No	Spam
9	Unknown	Long	Yes	Spam
10	Known	Medium	Yes	Spam
11	Known	Long	No	Ham
12	Known	Long	No	Ham

- (A) Without smoothing: Ham; with Laplace smoothing: Ham
 (B) Without smoothing: Spam; with Laplace smoothing: Ham
 (C) Without smoothing: Ham; with Laplace smoothing: Spam
 (D) Without smoothing: Spam; with Laplace smoothing: Spam

Q.21 [2 marks · NAT (numerical answer)]

Consider the model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ (first column of X is the intercept column), fitted by least squares using $\beta = (X^T X)^{-1} X^T y$ with the design matrix X and response y below. Then the least-squares estimate of β_2 is _____ (rounded off to two decimal places)

$$X = \begin{bmatrix} 1 & -2 & 2 \\ 1 & -1 & -1 \\ 1 & 0 & -2 \\ 1 & 1 & -1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$y = \begin{bmatrix} 8 \\ 5 \\ 4 \\ 7 \\ 13 \end{bmatrix}$$

Q.22 [2 marks · MSQ (one or more correct)]

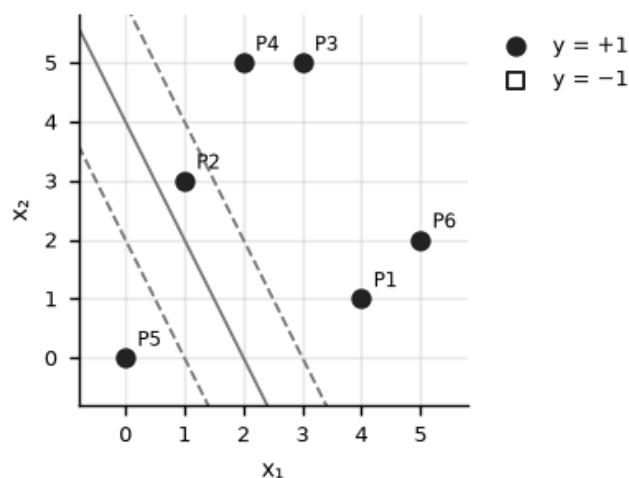
Consider k-means (Lloyd's algorithm) and k-medoids (PAM). Which of the following statements is/are CORRECT?

- (A) For a 1-D cluster with an odd number of points, the medoid under absolute distance is the mean.
- (B) k-medoids minimizes the sum of squared Euclidean distances to the cluster means.
- (C) Multiplying all coordinates by 2 multiplies the optimal k-means SSE by 4 and leaves the optimal partition unchanged.
- (D) k-medoids is generally more robust to outliers than k-means.

Q.23 [2 marks · NAT (numerical answer)]

A linear SVM with $f(x) = x_1 + 0.5x_2 - 2$ is evaluated on the training points in the table (also plotted, with the boundary $f = 0$ solid and $f = \pm 1$ dashed). Compute the average hinge loss $(1/n) \sum_i \max(0, 1 - y_i f(x_i))$. (rounded off to two decimal places)

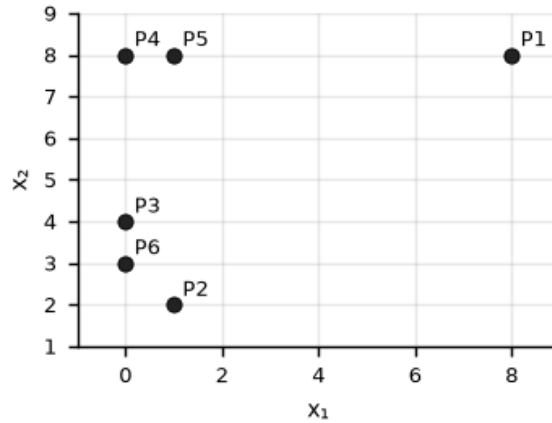
Point	x_1	x_2	y
P1	4	1	+1
P2	1	3	+1
P3	3	5	+1
P4	2	5	+1
P5	0	0	+1
P6	5	2	+1



Q.24 [2 marks · NAT (numerical answer)]

Single-linkage agglomerative clustering with Euclidean distance is applied to the six points below. Then the **sum of the heights** of all merges in the dendrogram is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5	P6
x_1	8	1	0	0	1	0
x_2	8	2	4	8	8	3

**Q.25 [2 marks · MSQ (one or more correct)]**

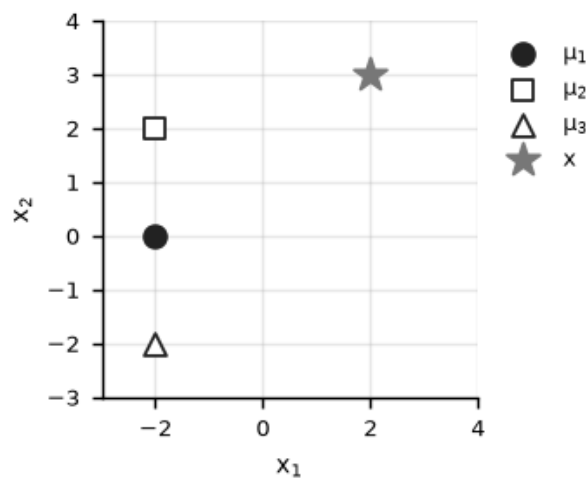
Each option shows a 2×2 matrix $[[K(x_1, x_1), K(x_1, x_2)], [K(x_2, x_1), K(x_2, x_2)]]$. Which of them can be the Gram matrix of some valid (Mercer) kernel K on two points x_1, x_2 ?

- (A) $[[7, 2], [3, 5]]$
 (B) $[[1, 2], [2, 4]]$
 (C) $[[1, -1], [-1, 1]]$
 (D) $[[8, -4], [-4, 2]]$

Q.26 [2 marks · NAT (numerical answer)]

An LDA classifier for 3 classes has shared covariance matrix Σ (below), class means $\mu_1 = (-2, 0)^T$; $\mu_2 = (-2, 2)^T$; $\mu_3 = (-2, -2)^T$, and priors $\pi_1 = 0.6$, $\pi_2 = 0.2$, $\pi_3 = 0.2$. The linear discriminant is $\delta_k(\mathbf{x}) = \mathbf{x}^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln \pi_k$ (natural log). For $\mathbf{x} = (2, 3)^T$, the index $k \in \{1, 2, 3\}$ of the class predicted by LDA is _____ (Answer in integer)

$$\Sigma = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$$



Q.27 [2 marks · NAT (numerical answer)]

The eigenvalues of the sample covariance matrix (divisor $n - 1$) of a centered 7-dimensional dataset are 20, 19.5, 14.5, 13, 12.5, 11, 3. Then the percentage of total variance **lost** by keeping only the top 5 principal components is _____ (rounded off to two decimal places)

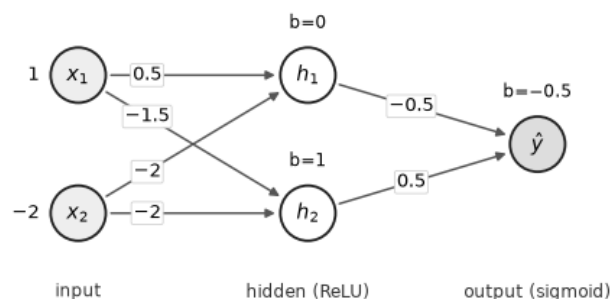
Q.28 [2 marks · NAT (numerical answer)]

A logistic-regression neuron $p = \sigma(w_1x_1 + w_2x_2 + b)$ has $w = (-0.5, -1)$, $b = 0$ and is trained with binary cross-entropy loss. After one stochastic gradient step on **example 1 only** with learning rate $\eta = 1$, the new value of w_1 is _____ (rounded off to three decimal places)

example	x_1	x_2	y
1	-1	1	1
2	-1.5	-1.5	0

Q.29 [2 marks · NAT (numerical answer)]

The feed-forward network below has ReLU hidden units and a sigmoid output unit. The number on each edge is its weight and 'b' above a node is that node's bias. For input $x = (1, -2)$ (values shown left of the inputs), the activation of hidden unit h_1 is _____ (rounded off to three decimal places)

**Q.30 [2 marks · MSQ (one or more correct)]**

For the confusion matrix below, which of the following statements is/are CORRECT? (values rounded to 2 decimals)

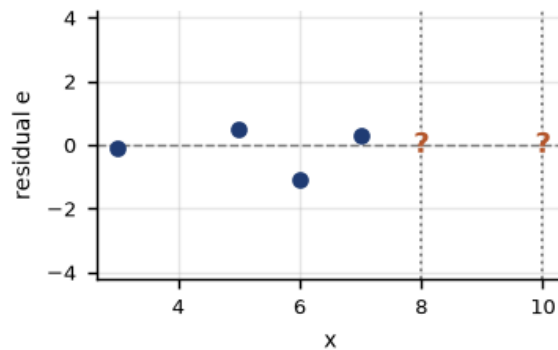
	Predicted +	Predicted -
Actual +	72	15
Actual -	34	42

- (A) The accuracy is 0.7.
- (B) The balanced accuracy is 0.7.
- (C) The false positive rate is 0.32.
- (D) The F1-score is less than the arithmetic mean of precision and recall.

Q.31 [2 marks · NAT (numerical answer)]

A simple linear regression with intercept is fitted by least squares. The table gives the x -values and the residuals $e_i = y_i - \hat{y}_i$, but two residuals are missing. The fitted line is $\hat{y} = 3.6 + 0.8x$. The observed value y at $x = 8$ is _____ (rounded off to two decimal places)

x_i	3	5	6	7	8	10
e_i	-0.1	0.5	-1.1	0.3	?	?

**Q.32 [2 marks · NAT (numerical answer)]**

A 2-2-1 network uses the identity (linear) activation in its hidden layer: $\hat{y} = W_2(W_1x + b_1) + b_2$, with the parameters below. The network is equivalent to a single linear model $\hat{y} = \tilde{w}_1x_1 + \tilde{w}_2x_2 + \tilde{b}$. Then the intercept $\tilde{b}$ is _____ (Answer in integer)

$$W_1 = \begin{bmatrix} -2 & -1 \\ 2 & 2 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 1 & 1 \end{bmatrix}$$

$$b_2 = -2$$

Q.33 [2 marks · MCQ]

A decision tree is to be grown on the 10 training records below using Gini gain as the splitting criterion (multi-way split on each categorical attribute). Which attribute is chosen at the root?

#	Student	Age	Income	Wind	Class
1	No	Middle	Low	Strong	No
2	No	Middle	Low	Weak	No
3	No	Young	Medium	Strong	No
4	Yes	Middle	Medium	Weak	Yes
5	No	Young	Medium	Strong	Yes
6	Yes	Middle	Low	Strong	Yes
7	No	Middle	High	Strong	No
8	No	Young	High	Weak	No
9	Yes	Young	Low	Strong	No
10	Yes	Senior	High	Weak	No

(A) Student**(B)** Age**(C)** Income**(D)** Wind

Q.34 [2 marks · NAT (numerical answer)]

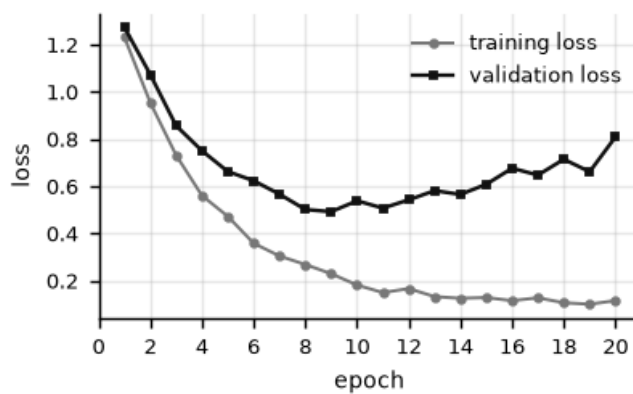
A 3-class softmax (multinomial logistic) regression model computes scores $\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}$ and $P(y = k | \mathbf{x}) = \frac{e^{z_k}}{\sum_j e^{z_j}}$, where row k of $\mathbf{W}$ holds the weights of class k . For the parameters below and $\mathbf{x} = (1, -1)^T$, the predicted probability $P(y = 1 | \mathbf{x})$ is _____ (rounded off to two decimal places)

$$\mathbf{W} = \begin{bmatrix} -2 & 1 \\ 1 & -1 \\ -2 & 0 \end{bmatrix}$$

$$\mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Q.35 [2 marks · NAT (numerical answer)]

A network is trained with early stopping on validation loss using patience $P = 3$: training stops as soon as the validation loss has failed to improve on its best value for 3 consecutive epochs, and the weights from the best epoch are restored. Using the validation losses below (also plotted), the epoch after which training is stopped is _____ (Answer in integer)



epoch	1	2	3	4	5	6	7	8	9	10
val. loss	1.274	1.074	0.857	0.751	0.663	0.624	0.567	0.502	0.493	0.539

epoch	11	12	13	14	15	16	17	18	19	20
val. loss	0.508	0.543	0.582	0.566	0.609	0.677	0.648	0.715	0.662	0.810

— END OF QUESTION PAPER —

Mock Test 22 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	0.2943 to 0.2983	19	NAT	2	24.2509 to 24.3968
2	NAT	1	13.39 to 13.41	20	MCQ	2	(B)
3	MSQ	1	(A), (C), (D)	21	NAT	2	1.5514 to 1.5914
4	MSQ	1	(A), (B), (C)	22	MSQ	2	(C), (D)
5	NAT	1	0.59 to 0.61	23	NAT	2	0.5733 to 0.5933
6	NAT	1	0.0284 to 0.0296	24	NAT	2	14.4042 to 14.4242
7	NAT	1	0.7783 to 0.7823	25	MSQ	2	(B), (C), (D)
8	MCQ	1	(C)	26	NAT	2	1
9	NAT	1	13	27	NAT	2	14.9633 to 14.9833
10	MCQ	1	(D)	28	NAT	2	-1.1245 to -1.1205
11	MSQ	1	(C), (D)	29	NAT	2	4.498 to 4.502
12	MSQ	1	(A), (B)	30	MSQ	2	(A), (D)
13	NAT	1	1.7789 to 1.7989	31	NAT	2	10.7958 to 10.9042
14	NAT	1	63	32	NAT	2	-6
15	MSQ	1	(A), (B), (C), (D)	33	MCQ	2	(C)

Q	Type	M	Key	Q	Type	M	Key
16	NAT	2	49.2525 to 49.7475	34	NAT	2	0.0071 to 0.0271
17	NAT	2	29	35	NAT	2	12
18	NAT	2	0.5218 to 0.5258				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
SVM & Decision Trees					11
Unsupervised Learning					11
Neural Networks (MLP / Feed-forward)					11
Probabilistic & Instance-based Classifiers					11
Regression & Model Selection					11
Total					55

Mock Test 22 — Detailed Solutions

Q.1 — Answer: 0.2943 to 0.2983 [NAT, 1M · SVM & Decision Trees > Ensembles: bagging & random forests]

Each draw misses the point with probability $1 - 1/3$; 3 independent draws: $(1 - 1/3)^3 = \mathbf{0.296}$ ($\rightarrow 1/e \approx 0.368$ as $n \rightarrow \infty$).

Q.2 — Answer: 13.39 to 13.41 [NAT, 1M · Unsupervised Learning > k-medoids]

Mean = $112/5 = 22.4$ (pulled towards the outlier 81).

Medoid = data point minimizing $\sum |x_i - m|$. Costs: $m=4$: 92, $m=7$: 83, $m=9$: 81, $m=11$: 83, $m=81$: 293.

The minimum is at $m = 9$ (the median), cost 81.

Answer: **13.4**. The medoid is barely affected by the outlier, unlike the mean.

Q.3 — Answer: (A), (C), (D) [MSQ, 1M · SVM & Decision Trees > SVM: kernels]

(A) TRUE. Schur product theorem.

(B) FALSE. For points 0 and 1 the Gram matrix $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ has determinant -1 .

(C) TRUE. $\phi(x) = A^{1/2}x$.

(D) TRUE. Mercer's condition.

Q.4 — Answer: (A), (B), (C) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

Net input $a = w \cdot x + b = -2x_1 + (3)x_2 + (4)$; output +1 iff $a \geq 0$.

(P) $a = -2 \cdot 0 + (3) \cdot 3 + (4) = 13 \rightarrow +1$ (selected).

(Q) $a = -2 \cdot 2 + (3) \cdot 1 + (4) = 3 \rightarrow +1$ (selected).

(R) $a = -2 \cdot -1 + (3) \cdot 0 + (4) = 6 \rightarrow +1$ (selected).

(S) $a = -2 \cdot 2 + (3) \cdot -2 + (4) = -6 \rightarrow -1$.

Answer: **(A), (B), (C)**.

Q.5 — Answer: 0.59 to 0.61 [NAT, 1M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

Predict positive: $(1 - 0.7) \times 2 = 0.6$. Predict negative: $0.7 \times 13 = 9.1$.

Optimal decision: positive; expected cost = **0.6**.

Q.6 — Answer: 0.0284 to 0.0296 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

exp values: 0.1353, 0.3679, 12.1825; sum = 12.6857.

$p_2 = 0.3679/12.6857 = \mathbf{0.029}$.

Q.7 — Answer: 0.7783 to 0.7823 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Example 1: $y = 1 \rightarrow -\ln 0.1 = 2.3026$.

Example 2: $y = 1 \rightarrow -\ln 0.9 = 0.1054$.

Example 3: $y = 1 \rightarrow -\ln 0.7 = 0.3567$.

Example 4: $y = 0 \rightarrow -\ln(1 - 0.3) = 0.3567$.

Mean = **0.7803**.

Q.8 — Answer: (C) [MCQ, 1M · Regression & Model Selection > Simple linear regression]

The spread of residuals grows with the fitted value (funnel shape) $\rightarrow$ the constant-variance assumption is violated; a transformation of y (e.g. log) or weighted least squares may help.

Answer: **Non-constant error variance (heteroscedasticity)**.

Other patterns: funnel $\rightarrow$ heteroscedasticity; curvature $\rightarrow$ missing non-linearity; isolated point $\rightarrow$ outlier; random band $\rightarrow$ adequate fit.

Q.9 — Answer: 13 [NAT, 1M · Regression & Model Selection > Multiple linear regression]

A categorical feature with k levels needs $k - 1$ dummies (one level is the reference absorbed by the intercept; using all k dummies with an intercept gives perfect multicollinearity).

Parameters = 1 (intercept) + 5 (numeric) + $(3 - 1)$ + $(4 - 1)$ = 11.

The interaction adds one product term per day type dummy: $3 - 1 = 2$. Total = $11 + 2 = 13$.

Answer: **13**.

Q.10 — Answer: (D) [MCQ, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

Odds ratio for a unit increase = $e^\beta = 4 \Rightarrow \beta = \ln 4 = \mathbf{1.39}$.

Distractors: the odds ratio itself, $\log_{10}$, or $1/R$ are common slips; the model is linear in the *natural* log-odds.

Q.11 — Answer: (C), (D) [MSQ, 1M · Regression & Model Selection > Regression vs classification]

A supervised task is *regression* when the target y is real-valued (loss such as squared error), and *classification* when y takes values in a finite set of labels (loss such as cross-entropy or 0-1 loss).

(A) **FALSE**. The output (rent) is a continuous real-valued quantity $\rightarrow$ regression.

(B) **FALSE**. The output (delivery time) is a continuous real-valued quantity $\rightarrow$ regression.

(C) **TRUE**. The output (species label) is a discrete class label $\rightarrow$ classification.

(D) **TRUE**. The output (one of 4 groups) is a discrete class label $\rightarrow$ classification.

Q.12 — Answer: (A), (B) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

(A) **TRUE**. True negative rate.

(B) **TRUE**. Only the ranking matters.

(C) **FALSE**. That is precision.

(D) **FALSE**. It equals the prevalence.

Q.13 — Answer: 1.7789 to 1.7989 [NAT, 1M · SVM & Decision Trees > SVM: margin & geometry]

For $w^T x + b_1 = 0$ and $w^T x + b_2 = 0$, distance = $|b_1 - b_2|/\|w\|$.

= $|2 - (-2)|/2.2361 = 4/2.2361 = \mathbf{1.79}$.

Q.14 — Answer: 63 [NAT, 1M · Unsupervised Learning > k-means]

Each split is an unordered pair of non-empty complementary subsets: $(2^7 - 2)/2 = 2^6 - 1 = \mathbf{63}$.

This exponential count is why divisive methods use heuristics (e.g. bisecting k-means).

Q.15 — Answer: (A), (B), (C), (D) [MSQ, 1M · Unsupervised Learning > Hierarchical clustering]

(A) **TRUE**. One close pair suffices to merge two clusters.

(B) **TRUE**. Definition of complete (farthest-neighbour) linkage.

(C) **TRUE**. Each merge removes one cluster.

(D) **TRUE**. No random initialization is involved.

Q.16 — Answer: 49.2525 to 49.7475 [NAT, 2M · Regression & Model Selection > Cross-validation]

For each left-out point, the prediction is the y of its nearest remaining neighbour (no ties here).

x_i	1	3	7	8	10	14
nearest x	3	1	8	7	8	10
prediction	10	11	12	1	12	5
error	1	-1	-11	11	-7	-2

LOOCV MSE = $(1/6)\sum \text{error}^2 = 297/6 = 49.5$.

Answer: **49.5**.

Q.17 — Answer: 29 [NAT, 2M · Unsupervised Learning > k-medoids]

PAM cost = $\sum$ over all points of the absolute distance to the nearest medoid.

Current medoids $\{5, 26\}$: distances $1 \rightarrow 4$, $5 \rightarrow 0$, $8 \rightarrow 3$, $13 \rightarrow 8$, $14 \rightarrow 9$, $26 \rightarrow 0$, $27 \rightarrow 1$, $30 \rightarrow 4$; cost = **29**.

Q.18 — Answer: 0.5218 to 0.5258 [NAT, 2M · SVM & Decision Trees > Ensembles: boosting]

$\varepsilon = \sum_{\text{misclassified}} w_i = 0.0741$; $\alpha = \frac{1}{2} \ln(0.9259/0.0741) = 1.2629$.

Misclassified weights $\times e^\alpha = 3.5355$; correct weights $\times e^{-\alpha} = 0.2828$.

$Z = \varepsilon e^\alpha + (1 - \varepsilon) e^{-\alpha} = 2\sqrt{\varepsilon(1 - \varepsilon)} = 0.5238$.

New weights: 0.08, 0.08, 0.08, 0.5, 0.06, 0.06, 0.04, 0.1.

Shortcut: after normalisation the misclassified set has total weight exactly $\frac{1}{2}$: $w_i' = w_i/(2\varepsilon)$ if misclassified, $w_i/(2(1 - \varepsilon))$ otherwise.

Answer: **0.524**.

Q.19 — Answer: 24.2509 to 24.3968 [NAT, 2M · Regression & Model Selection > Gradient descent]

$\partial J/\partial w_0 = (2/n)\sum r_i$, $\partial J/\partial w_1 = (2/n)\sum r_i x_i$, with residual $r_i = w_0 + w_1 x_i - y_i$.

Step 1: $w = (1, -1)$, $r = (-7, -3, -5)$, $\nabla J = (-10, -8.6667) \Rightarrow w \leftarrow w - 0.02\nabla J = (1.2, -0.8267)$.

$J = 24.3239$ (initial loss 27.6667).

Answer: **24.324**.

Q.20 — Answer: (B) [MCQ, 2M · Probabilistic & Instance-based Classifiers > Naive Bayes]

(i) No smoothing.

Spam: $P(\text{Class}=\text{Spam}) = 7/12$; $P(\text{Sender}=\text{Known}|\text{Spam}) = 5/7$; $P(\text{Length}=\text{Medium}|\text{Spam}) = 3/7$; $P(\text{HasLink}=\text{No}|\text{Spam}) = 1/7 \Rightarrow \text{score} = 0.02551$.

Ham: $P(\text{Class}=\text{Ham}) = 5/12$; $P(\text{Sender}=\text{Known}|\text{Ham}) = 5/5$; $P(\text{Length}=\text{Medium}|\text{Ham}) = 0/5$; $P(\text{HasLink}=\text{No}|\text{Ham}) = 5/5 \Rightarrow \text{score} = 0$.

A zero count makes the whole product for Ham equal to 0 (zero-frequency problem), so the prediction is Spam.

(ii) Laplace smoothing.

Spam: $P(\text{Class}=\text{Spam}) = 7/12$; $P(\text{Sender}=\text{Known}|\text{Spam}) = (5+1)/(7+2) = 2/3$; $P(\text{Length}=\text{Medium}|\text{Spam}) = (3+1)/(7+3) = 2/5$; $P(\text{HasLink}=\text{No}|\text{Spam}) = (1+1)/(7+2) = 2/9 \Rightarrow \text{score} = 0.03457$.

Ham: $P(\text{Class}=\text{Ham}) = 5/12$; $P(\text{Sender}=\text{Known}|\text{Ham}) = (5+1)/(5+2) = 6/7$; $P(\text{Length}=\text{Medium}|\text{Ham}) = (0+1)/(5+3) = 1/8$; $P(\text{HasLink}=\text{No}|\text{Ham}) = (5+1)/(5+2) = 6/7 \Rightarrow \text{score} = 0.03827$.

Larger score: Ham. Answer: **Without smoothing: Spam; with Laplace smoothing: Ham.**

Q.21 — Answer: 1.5514 to 1.5914 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

The columns of X are mutually orthogonal, so $X^T X$ is diagonal and each coefficient is $\beta_j = (x_j^T y)/(x_j^T x_j)$.

$$X^T X = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 14 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 37 \\ 12 \\ 22 \end{bmatrix}$$

$\beta = (37/5, 12/10, 22/14) = (7.4, 1.2, 1.5714)$.

Answer: **1.57**.

Q.22 — Answer: (C), (D) [MSQ, 2M · Unsupervised Learning > k-means]

(A) FALSE. It is the median, which is a data point.

(B) FALSE. It minimizes the sum of dissimilarities to medoids (data points).

(C) TRUE. Uniform scaling multiplies every squared distance by 4.

(D) TRUE. A medoid is an actual point and uses unsquared dissimilarities, so an outlier pulls it far less.

Q.23 — Answer: 0.5733 to 0.5933 [NAT, 2M · SVM & Decision Trees > SVM: soft margin & hinge loss]

Compute $f(x_i)$, the functional margin $y_i f(x_i)$ and the slack/hinge loss for each point:

Point	x	y	f(x)	y f(x)	$\xi = \max(0, 1 - y f)$
P1	(4, 1)	+1	2.5	2.5	0
P2	(1, 3)	+1	0.5	0.5	0.5
P3	(3, 5)	+1	3.5	3.5	0

P4	(2, 5)	+1	2.5	2.5	0
P5	(0, 0)	+1	-2	-2	3
P6	(5, 2)	+1	4	4	0

Average = $3.5/6 = 0.58$.

Points with $y \geq 1$ contribute 0; points inside the margin contribute between 0 and 1; misclassified points contribute more than 1.

Q.24 — Answer: 14.4042 to 14.4242 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Single-linkage merge heights are exactly the edge weights of a minimum spanning tree (Kruskal's algorithm adds the shortest edge joining two different components = single-linkage merge).

MST edges (Prim's algorithm): P1-P5 (7); P5-P4 (1); P4-P3 (4); P3-P6 (1); P6-P2 (1.41).

Sorted merge heights: 1, 1, 1.41, 4, 7.

Sum of merge heights = total MST weight = $1 + 1 + 1.41 + 4 + 7 = 14.41$.

Q.25 — Answer: (B), (C), (D) [MSQ, 2M · SVM & Decision Trees > SVM: kernels]

A Gram matrix of a valid kernel must be symmetric and positive semidefinite. For a symmetric 2×2 matrix this is equivalent to: both diagonal entries ≥ 0 and determinant ≥ 0 .

(A) FALSE. not symmetric, but $K(x, z) = K(z, x)$ for every kernel.

(B) TRUE. symmetric, diagonal ≥ 0 , $\det = 0 \Rightarrow$ PSD (singular) — still a valid Gram matrix.

(C) TRUE. symmetric, diagonal ≥ 0 , $\det = 0 \Rightarrow$ PSD (singular) — still a valid Gram matrix.

(D) TRUE. symmetric, diagonal ≥ 0 , $\det = 0 \Rightarrow$ PSD (singular) — still a valid Gram matrix.

Q.26 — Answer: 1 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$$\Sigma^{-1} = \begin{bmatrix} 0.333 & -0.333 \\ -0.333 & 0.833 \end{bmatrix}$$

$k = 1$: $\Sigma^{-1}\mu_1 = (-0.6667, 0.6667)$; $\mathbf{x}^T \Sigma^{-1} \mu = 0.6667$; $\frac{1}{2} \mu^T \Sigma^{-1} \mu = 0.6667$; $\ln \pi = -0.5108 \Rightarrow \delta_1 = -0.5108$.

$k = 2$: $\Sigma^{-1}\mu_2 = (-1.3333, 2.3333)$; $\mathbf{x}^T \Sigma^{-1} \mu = 4.3333$; $\frac{1}{2} \mu^T \Sigma^{-1} \mu = 3.6667$; $\ln \pi = -1.6094 \Rightarrow \delta_2 = -0.9428$.

$k = 3$: $\Sigma^{-1}\mu_3 = (0, -1)$; $\mathbf{x}^T \Sigma^{-1} \mu = -3$; $\frac{1}{2} \mu^T \Sigma^{-1} \mu = 1$; $\ln \pi = -1.6094 \Rightarrow \delta_3 = -5.6094$.

The largest discriminant is $\delta_1 \Rightarrow$ predicted class **1**.

Q.27 — Answer: 14.9633 to 14.9833 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

For centered data, projecting onto the top- k PCs and reconstructing leaves the components along the discarded eigenvectors; their average squared length (divisor $n - 1$) is the sum of the discarded eigenvalues.

Discarded eigenvalues: 11, 3; sum = 14; total = 93.5.

Lost fraction = $14/93.5 = 14.97\%$.

Q.28 — Answer: -1.1245 to -1.1205 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

For σ output with binary cross-entropy $L = -[y \ln p + (1 - y) \ln(1 - p)]$, $\partial L / \partial a = p - y$, so $\partial L / \partial w_j = (p - y)x_j$ and $\partial L / \partial b = p - y$.

Example 1: $a = -0.5$, $p = 0.3775$, $p - y = -0.6225$.

Gradient for w_1 : 0.6225.

$w_1 \leftarrow -0.5 - 1 \times (0.6225) = -1.1225$.

Note there is no σ' factor: it cancels with the derivative of the log loss.

Q.29 — Answer: 4.498 to 4.502 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

Hidden pre-activations: $a_1 = 0.5 \cdot 1 + (-2) \cdot -2 + (0) = 4.5$; $a_2 = -1.5 \cdot 1 + (-2) \cdot -2 + (1) = 3.5$.

Hidden activations ($\text{ReLU}(a) = \max(0, a)$): $h_1 = 4.5$, $h_2 = 3.5$.

Required: $h_1 = 4.5$.

Q.30 — Answer: (A), (D) [MSQ, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

TP = 72, FN = 15, FP = 34, TN = 42.

(A) TRUE. accuracy = $(TP + TN)/N = 0.6994$.

(B) FALSE. balanced accuracy = $(TPR + TNR)/2 = 0.6901$, not 0.7.

(C) FALSE. false positive rate = $FP/(FP + TN) = 0.4474$, not 0.32.

(D) TRUE. The harmonic mean is $\leq$ the arithmetic mean, with equality only if $P = R$ (not the case here).

Q.31 — Answer: 10.7958 to 10.9042 [NAT, 2M · Regression & Model Selection > Simple linear regression]

For the least-squares fit with intercept the normal equations give $\sum e_i = 0$ and $\sum x_i e_i = 0$.

Known: $\sum e = -0.4$, $\sum x e = -2.3$. With unknowns $u = e$ at $x = 8$ and $v = e$ at $x = 10$:

$u + v = 0.4$, $8u + 10v = 2.3$.

Solving: $u = 0.85$, $v = -0.45$.

$y = \hat{y} + e = (3.6 + (0.8)(8)) + (0.85) = 10.85$.

Answer: **10.85**.

Q.32 — Answer: -6 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Expressiveness]

With identity activations, $\hat{y} = W_2(W_1x + b_1) + b_2 = (W_2W_1)x + (W_2b_1 + b_2)$: a single affine map.

$\tilde{w}^T = W_2W_1 = (0, 1)$; $\tilde{b} = W_2b_1 + b_2 = -6$.

Answer: **-6**.

Hence stacking linear layers adds no expressive power – a non-linear activation is essential.

Q.33 — Answer: (C) [MCQ, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent: 3 Yes, 7 No $\Rightarrow$ Gini = 0.42.

Student — Yes: [2+, 2-], No: [1+, 5-]; gain = 0.0533.

Age — Young: [1+, 3-], Middle: [2+, 3-], Senior: [0+, 1-]; gain = 0.03.

Income — Low: [1+, 3-], Medium: [2+, 1-], High: [0+, 3-]; gain = 0.1367.

Wind — Weak: [1+, 3-], Strong: [2+, 4-]; gain = 0.0033.

The largest Gini gain is for **Income**, so it is chosen at the root.

Q.34 — Answer: 0.0071 to 0.0271 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

$z = Wx + b = (-2, 2, -1)$.

$e^z = (0.1353, 7.3891, 0.3679)$, sum = 7.8923.

Softmax probabilities = (0.0171, 0.9362, 0.0466).

$P(y = 1 | x) = \mathbf{0.02}$.

Q.35 — Answer: 12 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Rule: keep track of the best (lowest) validation loss; an epoch that does not strictly improve on it increments a counter, which is reset on improvement; stop when the counter reaches patience $P = 3$.

Trace – 1: 1.274 (new best); 2: 1.074 (new best); 3: 0.857 (new best); 4: 0.751 (new best); 5: 0.663 (new best); 6: 0.624 (new best); 7: 0.567 (new best); 8: 0.502 (new best); 9: 0.493 (new best); 10: 0.539 (no improvement, count = 1); 11: 0.508 (no improvement, count = 2); 12: 0.543 (no improvement, count = 3).

Training stops after epoch 12; the best validation loss 0.493 occurred at epoch 9, whose weights are restored.

Answer: **12**. The widening train/validation gap after this point is the signature of overfitting.

Mock Test 23

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	7	7	21

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
SVM & Decision Trees	7	11
Regression & Model Selection	7	11
Neural Networks (MLP / Feed-forward)	7	11
Probabilistic & Instance-based Classifiers	7	11
Unsupervised Learning	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

A node contains 4 records spread equally over 2 classes (2 records per class). Its entropy (in bits) is _____ (rounded off to two decimal places)

Q.2 [1 mark · MSQ (one or more correct)]

Linear regression is trained by minimising the squared-error loss. Which of the following statements about gradient-based training is/are CORRECT?

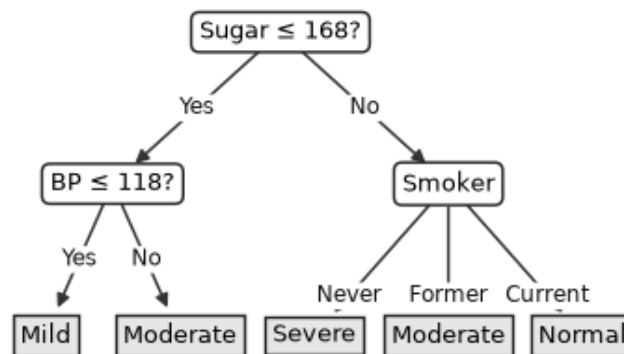
- (A) With a constant learning rate, SGD iterates typically keep fluctuating in a neighbourhood of the minimiser.
- (B) Solving the normal equations costs about $O(p^3)$ for the $p \times p$ system, which becomes expensive for very large p .
- (C) Stochastic gradient descent updates the parameters using the gradient on one (or a few) randomly chosen example(s).
- (D) With a sufficiently small constant learning rate, batch gradient descent on the least-squares loss converges to a global minimiser.

Q.3 [1 mark · NAT (numerical answer)]

A full polynomial regression of degree 2 in 5 input variables $x_1, \dots, x_5$ includes every monomial in these variables of total degree at most 2 (including the constant term and all interaction terms). The number of regression coefficients is _____ (Answer in integer)

Q.4 [1 mark · MCQ]

The decision tree below predicts the diagnosis category of a patient. Edges labelled Yes/No answer the test in the node. What is the prediction for the record shown in the table?



Sugar	BP	BMI	Smoker
74	117	26	Never

- (A) Severe (B) Normal
(C) Moderate (D) Mild

Q.5 [1 mark · MSQ (one or more correct)]

Which of the following statements about the representational power of feed-forward neural networks is/are CORRECT?

- (A) An MLP with 10 hidden layers and identity activations can represent XOR.
(B) XOR can be computed by an MLP with one hidden layer containing two threshold units.
(C) Removing all non-linear activations from a deep MLP reduces it to the hypothesis class of linear regression (or logistic regression if the output is sigmoid).
(D) An MLP of any depth with identity (linear) activations computes an affine function of its input.

Q.6 [1 mark · MSQ (one or more correct)]

Which of the following statements about logistic regression is/are CORRECT?

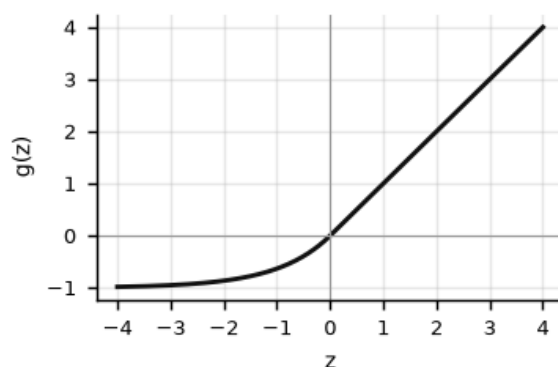
- (A) The squared-error loss composed with the sigmoid is the standard convex training loss of logistic regression.
(B) Softmax regression with $K = 2$ classes is equivalent to binary logistic regression.
(C) The derivative of the sigmoid is $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.
(D) A negative coefficient β_j means the predicted probability of $y = 1$ decreases as x_j increases (others fixed).

Q.7 [1 mark · NAT (numerical answer)]

Data are uniformly distributed in the unit hypercube $[0, 1]^{10}$. To capture a fraction 0.1 of the data, a k-NN method uses a sub-cube neighbourhood. The required edge length of this sub-cube is _____ (rounded off to two decimal places)

Q.8 [1 mark · MCQ]

The figure shows the graph of a function $g(z)$ used in neural networks. Which function is it?



- (A) $\tanh(z)$
- (B) Leaky ReLU with $\alpha = 0.1$
- (C) Softplus: $\ln(1 + e^z)$
- (D) ELU with $\alpha = 1$

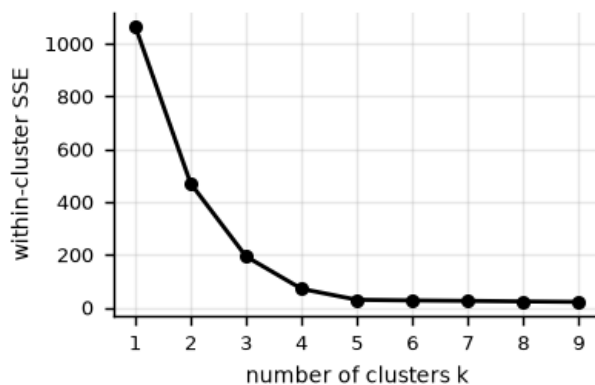
Q.9 [1 mark · NAT (numerical answer)]

A clustering algorithm groups labelled data into three clusters; the contingency table of cluster versus true class counts is given below. If every point in a cluster is labelled with the cluster's majority class, the number of misclassified points is ____ (Answer in integer)

	Class P	Class Q	Class R
Cluster 1	11	0	14
Cluster 2	6	7	24
Cluster 3	3	24	8

Q.10 [1 mark · MCQ]

k-means is run for $k = 1, 2, \dots, 9$ on a dataset and the final within-cluster SSE is plotted below. Using the elbow method, the most appropriate number of clusters is



- (A) 9
- (B) 5
- (C) 6
- (D) 4

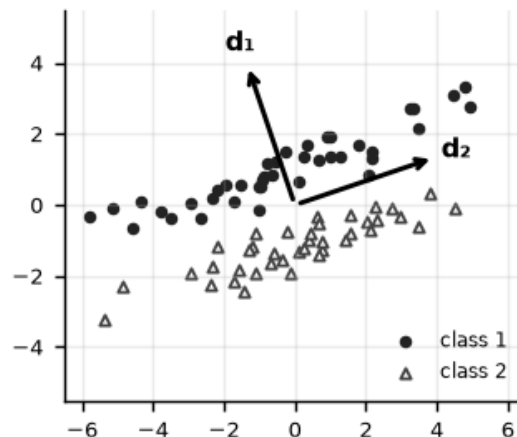
Q.11 [1 mark · MSQ (one or more correct)]

Let X be an $n \times p$ design matrix of full column rank p ($p < n$), and let $H = X(X^T X)^{-1} X^T$ be the hat matrix of least-squares regression of y on X . Which of the following statements is/are CORRECT?

- (A) H is symmetric ($H^T = H$).
- (B) $I - H$ is also symmetric and idempotent.
- (C) $\text{rank}(H) = p$.
- (D) H is idempotent ($H^2 = H$).

Q.12 [1 mark · MCQ]

Two labelled classes are shown below, together with two orthogonal directions d_1 and d_2 . A single direction is to be selected by (i) PCA (first principal component, ignoring labels) and (ii) Fisher's LDA. Which statement is correct?



- (A) PCA (1st PC) chooses d_2 ; LDA chooses d_1
- (B) PCA (1st PC) chooses d_1 ; LDA chooses d_2
- (C) Both PCA and LDA choose d_1
- (D) Both PCA and LDA choose d_2

Q.13 [1 mark · MCQ]

A soft-margin linear SVM has learned $f(x) = 0.5x_1 + 0.5x_2 - 2$. For the training example $x = (4, -1)$ with label $y = -1$, which statement is CORRECT?

- (A) Correctly classified but strictly inside the margin ($0 < \xi < 1$)
- (B) Exactly on the decision boundary ($\xi = 1$)
- (C) Misclassified ($\xi > 1$)
- (D) Correctly classified, on or outside the margin ($\xi = 0$)

Q.14 [1 mark · NAT (numerical answer)]

With L2 weight decay (penalty $(\lambda/2)w^2$, $\lambda = 2$) and learning rate $\eta = 0.1$, a weight $w = 0.5$ receives zero data-loss gradient for 5 consecutive gradient-descent steps. Its value after these 5 steps is _____ (rounded off to three decimal places)

Q.15 [1 mark · NAT (numerical answer)]

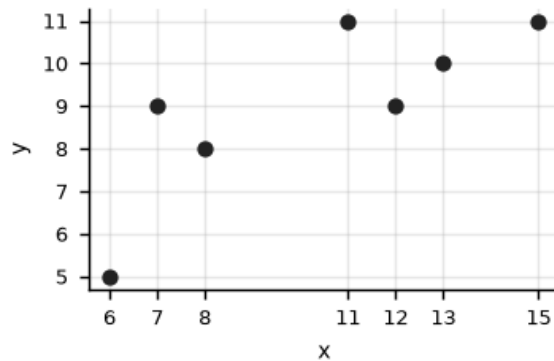
In a training set, the 4 examples of class Class = Ham have the following counts for the feature *Length*. Using Laplace (add-one) smoothing, the naive Bayes estimate of $P(\text{Length} = \text{Short} \mid \text{Class} = \text{Ham})$ is _____ (rounded off to three decimal places)

Length	Short	Medium	Long
count in Class = Ham	3	0	1

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

A regression tree (squared-error criterion) is fitted to the 7 points below. The root split " $x \leq t$ " is chosen among midpoints of consecutive x values to minimise the total SSE of the two children (each child predicts its mean). Find the reduction in SSE (sum of squared errors) achieved by the best split. (rounded off to two decimal places)

x	6	7	8	11	12	13	15
y	5	9	8	11	9	10	11

**Q.17 [2 marks · MCQ]**

Six points have the pairwise distances below. Agglomerative clustering with **single linkage** is run and the dendrogram is cut to give exactly **3** clusters. The resulting clusters are

	P1	P2	P3	P4	P5	P6
P1	0	1	7	7	5	11
P2		0	8	8	6	10
P3			0	12	2	6
P4				0	12	18
P5					0	6
P6						0

(A) {P1, P2, P3, P5}, {P4}, {P6}

(B) {P1, P2}, {P3, P5}, {P4}, {P6}

(C) {P1, P2}, {P3, P5, P6}, {P4}

(D) {P1, P2, P3, P5, P6}, {P4}

Q.18 [2 marks · NAT (numerical answer)]

Three customers are described by age and income. Age ranges over [20, 60] years and income over [10, 210] thousand rupees in the full dataset. After min-max scaling of each feature to [0, 1] (using the stated ranges), the Euclidean distance from A to its **nearest** neighbour among B and C is _____ (rounded off to two decimal places)

Customer	Age (years)	Income (₹ thousand)
A	53	50
B	50	140
C	21	20

Q.19 [2 marks · MSQ (one or more correct)]

Which of the following statements about LDA, QDA and logistic regression is/are CORRECT?

(A) Fisher's LDA for K classes yields at most $K - 1$ discriminant directions.

(B) When the Gaussian shared-covariance assumption holds, LDA can estimate the boundary with lower variance than logistic regression.

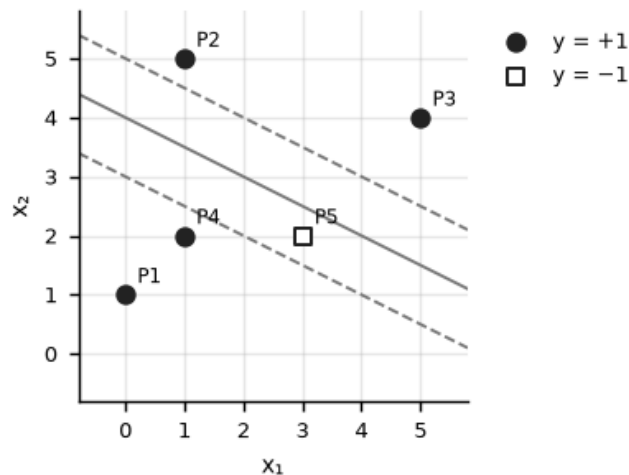
(C) The LDA discriminant function $\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln \pi_k$ is linear in x .

(D) Fisher's criterion $J(w)$ is unchanged if w is multiplied by a non-zero scalar.

Q.20 [2 marks · MSQ (one or more correct)]

A soft-margin linear SVM has learned $f(x) = 0.5x_1 + x_2 - 4$. The training points are listed below and plotted (boundary solid, margins $f = \pm 1$ dashed). With ξ_i the optimal slack of point i , which of the following statements is/are CORRECT?

Point	x_1	x_2	y
P1	0	1	+1
P2	1	5	+1
P3	5	4	+1
P4	1	2	+1
P5	3	2	-1



- (A) $\sum \xi_i = 7$.
- (B) Exactly 3 of the points have $\xi_i > 0$.
- (C) P1 is misclassified ($\xi_1 > 1$).
- (D) P2 has slack $\xi_2 = 0$.

Q.21 [2 marks · MSQ (one or more correct)]

A node with class counts [60+, 40−] can be split by either of two binary splits S1 or S2 (table). For each impurity measure, the “gain” is the parent impurity minus the weighted child impurity. Which of the following statements is/are CORRECT?

Split	Left child [+ , −]	Right child [+ , −]
S1	[15+, 25−]	[45+, 15−]
S2	[39+, 16−]	[21+, 24−]

- (A) Misclassification-error reduction prefers S2 over S1.
- (B) Information gain (entropy) prefers S2 over S1.
- (C) The information gain of S1 is 0.1 bits (to two decimals).
- (D) Misclassification-error reduction is the same for S1 and S2.

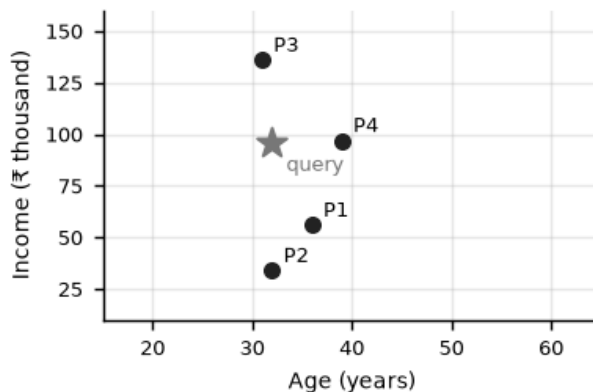
Q.22 [2 marks · NAT (numerical answer)]

A regression with intercept and $p = 5$ predictors on $n = 12$ observations has $R^2 = 0.66$. Its adjusted R^2 is _____ (rounded off to three decimal places)

Q.23 [2 marks · MCQ]

A 1-NN classifier uses the two features below. Before computing Euclidean distances, each feature is min-max scaled to $[0, 1]$ using the minimum and maximum of that feature over the four training points P1-P4 (the query is scaled with the same constants). The nearest neighbour of the query after scaling is

Point	Age (years)	Monthly income (₹ thousand)
P1	36	56
P2	32	34
P3	31	136
P4	39	97
Query	32	96

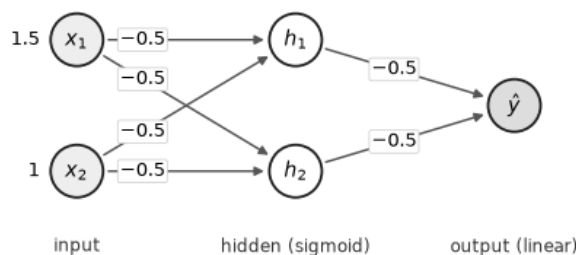


- (A) P1
(C) P3

- (B) P2
(D) P4

Q.24 [2 marks · NAT (numerical answer)]

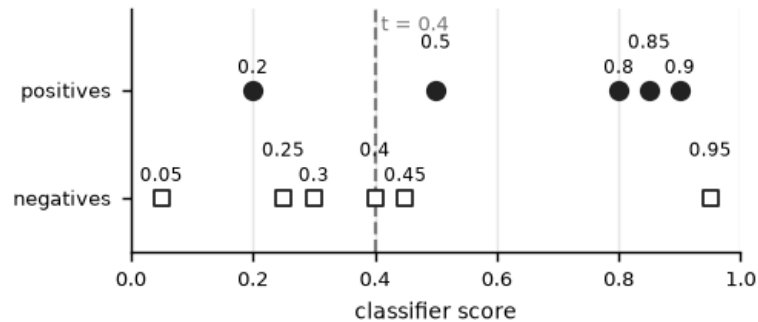
In the 2-2-1 network below (sigmoid hidden units, linear output $\hat{y} = v_1 h_1 + v_2 h_2$, all biases 0), **every** weight is initialised to -0.5 . One gradient-descent step with $\eta = 0.5$ is taken on $L = \frac{1}{2}(\hat{y} - y)^2$ for the input $x = (1.5, 1)$ with target $y = 2$. Then the updated value of the output weight v_1 is _____ (rounded off to three decimal places)



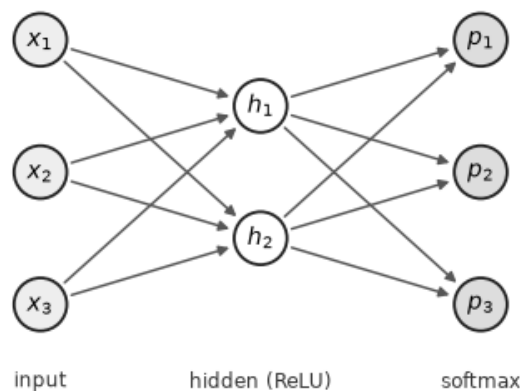
Q.25 [2 marks · NAT (numerical answer)]

A probabilistic classifier produces the scores below on 11 labelled examples (1 = positive). An example is predicted positive if its score is $\geq t$, with $t = 0.4$. The precision at this threshold is _____ (rounded off to two decimal places)

example	1	2	3	4	5	6	7	8	9	10	11
score	0.05	0.95	0.5	0.8	0.2	0.25	0.4	0.3	0.85	0.45	0.9
label	0	0	1	1	1	0	0	0	1	0	1

**Q.26 [2 marks · NAT (numerical answer)]**

A 3-2-3 network computes $h = \text{ReLU}(W_1 x + b_1)$, $z = W_2 h + b_2$, $p = \text{softmax}(z)$, with the parameters below. For $x = (1, 0, -1)^T$, the predicted probability p_3 of class 3 is _____ (rounded off to three decimal places)



$$W_1 = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -0.5 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

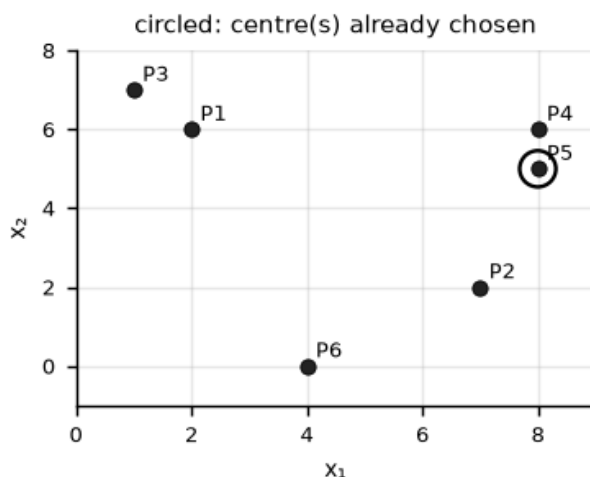
$$W_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

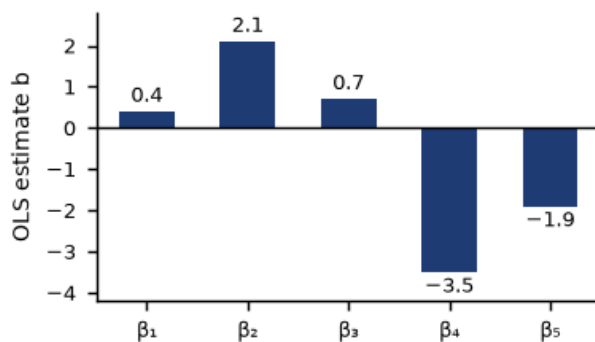
Q.27 [2 marks · NAT (numerical answer)]

k-means++ initialization (squared Euclidean seeding) is applied to the six points below. P5 has already been chosen as centre. The probability that P6 is selected as the second centre is _____ (rounded off to three decimal places)

Point	P1	P2	P3	P4	P5	P6
x_1	2	7	1	8	8	4
x_2	6	2	7	6	5	0

**Q.28 [2 marks · NAT (numerical answer)]**

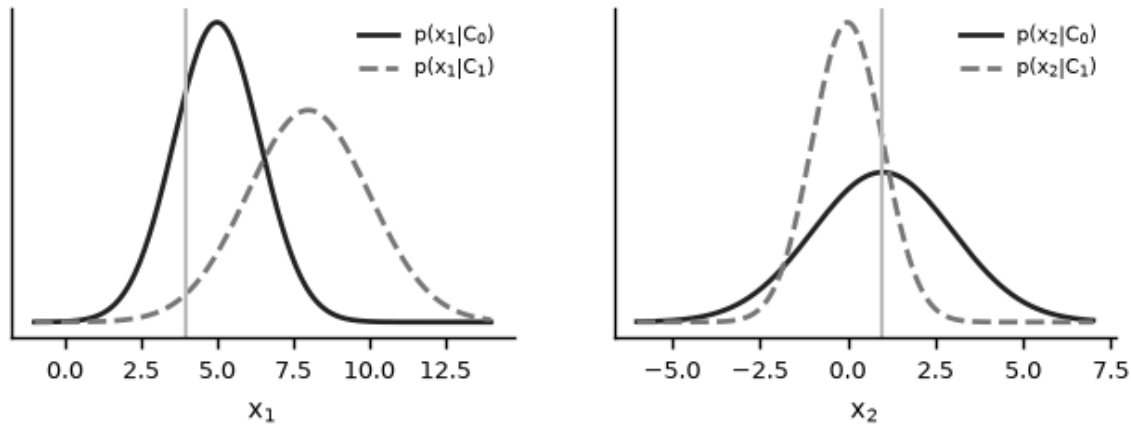
In a regression problem the design matrix X has orthonormal columns ($X^T X = I$). The ordinary least-squares estimates $b = X^T y$ are shown below. Ridge minimises $\|y - X\beta\|^2 + \lambda\|\beta\|_2^2$ and the lasso minimises $\frac{1}{2}\|y - X\beta\|^2 + \lambda\|\beta\|_1$, each with $\lambda = 1.5$. Then $\|\beta_{\text{lasso}}\|_1$ is _____ (rounded off to two decimal places)



Q.29 [2 marks · NAT (numerical answer)]

A Gaussian naive Bayes classifier with features x_1, x_2 has the class priors and per-class Gaussian parameters below (σ^2 = variance). For the input $x = (4, 1)$, the posterior probability $P(C_1 | x)$ is _____ (rounded off to two decimal places)

class	prior	$\mu(x_1)$	$\sigma^2(x_1)$	$\mu(x_2)$	$\sigma^2(x_2)$
C_0	0.5	5	2	1	4
C_1	0.5	8	4	0	1

**Q.30 [2 marks · NAT (numerical answer)]**

A centered data matrix X with $n = 21$ rows (samples) and $d = 4$ columns has singular values 26, 24, 16, 5. Using the sample covariance with divisor $n - 1$, the percentage of total variance explained by the first principal component is _____ (rounded off to two decimal places)

Q.31 [2 marks · NAT (numerical answer)]

A simple linear regression fitted by ordinary least squares gives $\hat{y} = 11.7 + 0.6x$. The sample standard deviations are $s_x = 3$ and $s_y = 41$. If both x and y are z-score standardised and the regression is refitted, the new slope (the standardised coefficient) is _____ (rounded off to three decimal places)

Q.32 [2 marks · NAT (numerical answer)]

Gradient descent with (heavy-ball) momentum minimises $f(w) = (0.5/2)w^2$ using $v_t = \beta v_{t-1} + \nabla f(w_{t-1})$ and $w_t = w_{t-1} - \eta v_t$, with $w_0 = 2$, $v_0 = 0$, $\eta = 0.2$, $\beta = 0.5$. The value of v_2 is _____ (rounded off to three decimal places)

Q.33 [2 marks · NAT (numerical answer)]

$B = 5$ regression trees are averaged. Each tree's prediction has variance $\sigma^2 = 4$, and every pair of trees has correlation $\rho = 0.2$. The variance of the averaged prediction is _____ (rounded off to two decimal places)

Q.34 [2 marks · NAT (numerical answer)]

A linear model $\hat{y} = w_1 x_1 + w_2 x_2 + b$ is trained by stochastic gradient descent (batch size 1) on the per-example loss $\frac{1}{2}(\hat{y} - y)^2$ with learning rate $\eta = 0.1$, starting from $w = (1, 1)$, $b = -1$. The two examples below are processed once, in the given order. After both updates, the value of b is _____ (rounded off to three decimal places)

example	x_1	x_2	y
1	1	-2	2
2	3	1	0

Q.35 [2 marks · NAT (numerical answer)]

For the fully connected network 64-16-64-10, a forward pass on a mini-batch of 32 inputs is computed as dense matrix products. Counting only the scalar multiplications in the weight-matrix products (ignore biases and activations), the number of multiplications is _____ (Answer in integer)

— END OF QUESTION PAPER —

Mock Test 23 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	0.99 to 1.01	19	MSQ	2	(A), (B), (C), (D)
2	MSQ	1	(A), (B), (C), (D)	20	MSQ	2	(A), (B), (C), (D)
3	NAT	1	21	21	MSQ	2	(C)
4	MCQ	1	(D)	22	NAT	2	0.3747 to 0.3787
5	MSQ	1	(B), (C), (D)	23	MCQ	2	(C)
6	MSQ	1	(B), (C), (D)	24	NAT	2	−0.2545 to −0.2505
7	NAT	1	0.7843 to 0.8043	25	NAT	2	0.5614 to 0.5814
8	MCQ	1	(D)	26	NAT	2	0.1843 to 0.1883
9	NAT	1	35	27	NAT	2	0.2867 to 0.2907
10	MCQ	1	(B)	28	NAT	2	2.988 to 3.012
11	MSQ	1	(A), (B), (C), (D)	29	NAT	2	0.1197 to 0.1397
12	MCQ	1	(A)	30	NAT	2	44.0865 to 44.1065
13	MCQ	1	(A)	31	NAT	2	0.0429 to 0.0449
14	NAT	1	0.1618 to 0.1658	32	NAT	2	1.398 to 1.402
15	NAT	1	0.5694 to 0.5734	33	NAT	2	1.43 to 1.45
16	NAT	2	18.6567 to 18.6767	34	NAT	2	−0.9829 to −0.9771
17	MCQ	2	(A)	35	NAT	2	86016
18	NAT	2	0.4462 to 0.4662				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
SVM & Decision Trees					11
Regression & Model Selection					11
Neural Networks (MLP / Feed-forward)					11
Probabilistic & Instance-based Classifiers					11
Unsupervised Learning					11
Total					55

Mock Test 23 — Detailed Solutions

Q.1 — Answer: 0.99 to 1.01 [NAT, 1M · SVM & Decision Trees › Decision trees: impurity measures]
 $p_k = 2/4 = 1/2$ for every class $\Rightarrow H = -2 \cdot (1/2) \log_2(1/2) = \log_2 2 = 1$ — the maximum possible for 2 classes.

Q.2 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection › Gradient descent]
(A) TRUE. Gradient noise does not vanish; decaying step sizes are needed for exact convergence.
(B) TRUE. Matrix factorisation/inversion is cubic in p .
(C) TRUE. Definition of SGD / mini-batch GD.
(D) TRUE. Convex quadratic with Lipschitz gradient; $\eta < 2/\lambda_{\max}$ (Hessian) suffices.

Q.3 — Answer: 21 [NAT, 1M · Regression & Model Selection › Basis functions & feature scaling]
 Number of monomials of degree $\leq d$ in p variables $= C(p + d, d) = C(7, 2) = 21$.

Q.4 — Answer: (D) [MCQ, 1M · SVM & Decision Trees › Decision trees: structure & traversal]
 Follow the record from the root, taking the branch that matches each test:
 Sugar $= 74 \leq 168 \rightarrow$ Yes.
 BP $= 117 \leq 118 \rightarrow$ Yes.
 (Only the tests on the actual path matter.) The record reaches the leaf **Mild**.
 Note: " $\leq$ " includes equality, so a value equal to the threshold goes to the Yes branch.

Q.5 — Answer: (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) › Expressiveness]
(A) FALSE. It is still a linear model; XOR is not linearly separable.
(B) TRUE. E.g. $h_1 = \text{OR}$, $h_2 = \text{NAND}$, $y = \text{AND}(h_1, h_2)$.
(C) TRUE. The pre-output computation is affine.
(D) TRUE. A composition of affine maps is affine.

Q.6 — Answer: (B), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers › Logistic regression]
(A) FALSE. The standard loss is cross-entropy; squared error with a sigmoid is non-convex in w .
(B) TRUE. $P(1) = e^{z_1} / (e^{z_1} + e^{z_2}) = \sigma(z_1 - z_2)$.
(C) TRUE. Standard identity; maximal value 0.25 at $z = 0$.
(D) TRUE. σ is increasing and the log-odds decrease with x_j .

Q.7 — Answer: 0.7843 to 0.8043 [NAT, 1M · Probabilistic & Instance-based Classifiers › k-nearest neighbours]
 Volume of a cube with edge e is e^{10} ; set $e^{10} = 0.1$.
 $e = 0.1^{1/10} = 0.79$.

In high dimensions even a small fraction of data needs an edge spanning most of each coordinate's range, so 'nearest' neighbours are not local — the curse of dimensionality.

Q.8 — Answer: (D) [MCQ, 1M · Neural Networks (MLP / Feed-forward) › Activation functions]
 Key features of the plotted curve: Smoothly saturates to -1 for $z \rightarrow -\infty$, linear for $z > 0$, passes through origin.
 Hence $g = \text{ELU with } \alpha = 1$.
 $\tanh(z)$: S-shaped, range $(-1, 1)$, passes through the origin. — does not match.
 Leaky ReLU with $\alpha = 0.1$: Small negative linear part (slope 0.1) for $z < 0$, slope 1 for $z > 0$. — does not match.
 Softplus: $\ln(1 + e^z)$: Smooth, always positive, value $\ln 2 \approx 0.69$ at $z = 0$, $\sim z$ for large z . — does not match.

Q.9 — Answer: 35 [NAT, 1M · Unsupervised Learning › Cluster evaluation]
 Majority count in each cluster: 14, 24, 24 (sum 62); $N = 97$.
 Purity $= (1/N) \sum_k \max_j n_{kj} = 62/97 = 0.6392$.
 Misclassified $= N - \sum \max = 35$.
 Answer: **35**.

Purity is an external measure (needs class labels) and is trivially 1 if each point forms its own cluster, so it cannot be used to choose k .

Q.10 — Answer: (B) [MCQ, 1M · Unsupervised Learning > k-means]

The SSE drops steeply up to $k = 5$ and then decreases only slowly (relative drops: $k=2$: 56%, $k=3$: 59%, $k=4$: 63%, $k=5$: 58%, $k=6$: 8%, $k=7$: 6%, $k=8$: 8%, $k=9$: 7%). The bend ('elbow') is at **$k = 5$** . $k = 9$ gives the lowest SSE but SSE always (weakly) decreases with k , so minimizing it would pick the largest k ; it is not a model-selection criterion by itself.

Q.11 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Multiple linear regression]

(A) TRUE. $H^T = X((X^T X)^{-1})^T X^T = H$ since $X^T X$ is symmetric.

(B) TRUE. $(I-H)^2 = I - 2H + H^2 = I - H$.

(C) TRUE. For an idempotent matrix $\text{rank} = \text{trace} = p$.

(D) TRUE. $H^2 = X(X^T X)^{-1} X^T X (X^T X)^{-1} X^T = H$.

Q.12 — Answer: (A) [MCQ, 1M · Unsupervised Learning > PCA & dimensionality reduction]

PCA is unsupervised: it picks the direction of maximum total variance, which is along the long axis of the clouds (d_2).

LDA is supervised: it maximizes between-class separation relative to within-class scatter, so it picks the direction across which the class means differ (d_1).

Answer: **PCA (1st PC) chooses d_2 ; LDA chooses d_1** . Projecting on the PCA direction would mix the two classes completely.

Q.13 — Answer: (A) [MCQ, 1M · SVM & Decision Trees > SVM: soft margin & hinge loss]

$y \cdot f(x) = (-1) \cdot (-0.5) = 0.5$.

Slack $\xi = \max(0, 1 - y f(x)) = 0.5$.

Rules: $y f \geq 1 \Rightarrow \xi = 0$ (on/outside margin); $0 < y f < 1 \Rightarrow$ inside margin, still correct; $y f = 0 \Rightarrow$ on the boundary ($\xi = 1$); $y f < 0 \Rightarrow$ misclassified ($\xi > 1$).

Hence: **Correctly classified but strictly inside the margin ($0 < \xi < 1$)**.

Q.14 — Answer: 0.1618 to 0.1658 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Each step: $w \leftarrow w - \eta \lambda w = (1 - \eta \lambda) w = 0.8 \cdot w$.

After 5 steps: $0.5 \times 0.8^5 = \mathbf{0.1638}$.

Q.15 — Answer: 0.5694 to 0.5734 [NAT, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Add- α estimate: $(\text{count} + \alpha) / (N_c + \alpha V)$ with $V = 3$ possible values.

$= (3 + 1) / (4 + 1 \times 3) = 4/7 = \mathbf{0.571}$.

(The unsmoothed estimate would be $3/4$.)

Q.16 — Answer: 18.6567 to 18.6767 [NAT, 2M · SVM & Decision Trees > Regression trees]

Parent mean $\hat{y} = 9$, $\text{SSE}_0 = \sum (y - \hat{y})^2 = 26$.

Evaluate every split:

Split t	Left mean	Right mean	$\text{SSE}_L + \text{SSE}_R$
6.5	5	9.667	7.333
7.5	7	9.8	14.8
9.5	7.333	10.25	11.417
11.5	8.25	10	20.75
12.5	8.4	10.5	19.7
14	8.667	11	21.333

Best split $t = 6.5$: left mean 5, right mean 9.667, $\text{SSE} = 7.333$, reduction = 18.667.

Answer: **18.67**.

Q.17 — Answer: (A) [MCQ, 2M · Unsupervised Learning > Hierarchical clustering]

Use single linkage: $d(A,B) = \min_{a \in A, b \in B} d(a,b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P1\} + \{P2\}$ at height **1**. Updated: $d(\{P1, P2\}, \{P3\}) = 7$, $d(\{P1, P2\}, \{P4\}) = 7$, $d(\{P1, P2\}, \{P5\}) = 5$, $d(\{P1, P2\}, \{P6\}) = 10$.

Merge 2: $\{P3\} + \{P5\}$ at height **2**. Updated: $d(\{P3, P5\}, \{P4\}) = 12$, $d(\{P3, P5\}, \{P6\}) = 6$, $d(\{P3, P5\}, \{P1, P2\}) = 5$.

Merge 3: $\{P1, P2\} + \{P3, P5\}$ at height **5**. Updated: $d(\{P1, P2, P3, P5\}, \{P4\}) = 7$, $d(\{P1, P2, P3, P5\}, \{P6\}) = 6$.

Merge 4: $\{P6\} + \{P1, P2, P3, P5\}$ at height **6**. Updated: $d(\{P1, P2, P3, P5, P6\}, \{P4\}) = 7$.

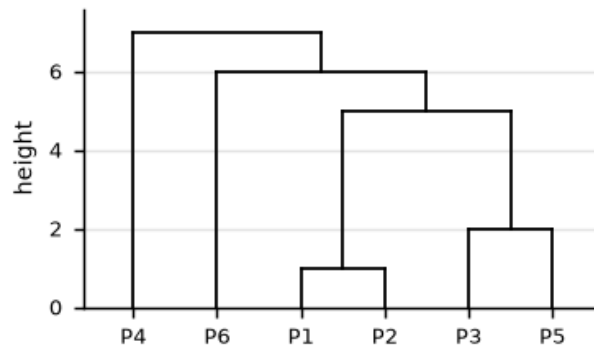
Merge 5: $\{P4\} + \{P1, P2, P3, P5, P6\}$ at height **7**.

Stopping when 3 clusters remain (i.e. after 3 merges): **$\{P1, P2, P3, P5\}$, $\{P4\}$, $\{P6\}$** .

$\{P1, P2\}$, $\{P3, P5, P6\}$, $\{P4\}$ is what *complete* linkage would give.

$\{P1, P2, P3, P5, P6\}$, $\{P4\}$ has 2 clusters (cut at the wrong level).

$\{P1, P2\}$, $\{P3, P5\}$, $\{P4\}$, $\{P6\}$ has 4 clusters (cut at the wrong level).



Q.18 — Answer: 0.4462 to 0.4662 [NAT, 2M · Unsupervised Learning > Distance measures]

Min-max scaling: $x' = (x - \min)/(\max - \min)$; age range [20, 60], income range [10, 210].

Customer	age'	income'
A	0.825	0.2
B	0.75	0.65
C	0.025	0.05

Raw distances: $d(A,B) = 90.05$, $d(A,C) = 43.86 \rightarrow$ nearest is C (income dominates because of its larger range).

Scaled distances: $d(A,B) = 0.4562$, $d(A,C) = 0.8139 \rightarrow$ nearest is B.

Answer: **0.46**. Scaling changes which neighbour is nearest — distance-based methods (k-means, k-NN, hierarchical clustering) should be run on standardized features.

Q.19 — Answer: (A), (B), (C), (D) [MSQ, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

(A) TRUE. $\text{rank}(S_B) \leq K - 1$.

(B) TRUE. It uses the extra (correct) distributional information.

(C) TRUE. Only first-order terms in x .

(D) TRUE. Numerator and denominator scale by c^2 .

Q.20 — Answer: (A), (B), (C), (D) [MSQ, 2M · SVM & Decision Trees > SVM: soft margin & hinge loss]

$$\xi_i = \max(0, 1 - y_i f(x_i)).$$

Point	$y f(x)$	ξ
P1	-3	4
P2	1.5	0
P3	2.5	0
P4	-1.5	2.5

P5	0.5	0.5
----	-----	-----

(A) **TRUE**. Sum of the individual slacks.

(B) **TRUE**. Count of points with $y f < 1$.

(C) **TRUE**. $y f = -3 \Rightarrow \xi = 4$.

(D) **TRUE**. $y f = 1.5 \Rightarrow \xi = 0$.

Q.21 — Answer: (C) [MSQ, 2M · SVM & Decision Trees > Decision trees: impurity measures]

Parent [60+, 40-]: $H = 0.971$, Gini = 0.48, error = 0.4.

S1: children $H = 0.9544$, 0.8113; Gini = 0.4688, 0.375; error = 0.375, 0.25. Gains: IG = 0.1024, Gini = 0.0675, error = 0.1.

S2: children $H = 0.8699$, 0.9968; Gini = 0.4126, 0.4978; error = 0.2909, 0.4667. Gains: IG = 0.044, Gini = 0.0291, error = 0.03.

(A) **FALSE**. S1: 0.1, S2: 0.03.

(B) **FALSE**. S1: 0.1024, S2: 0.044.

(C) **TRUE**. IG(S1) = 0.1024.

(D) **FALSE**. S1: 0.1, S2: 0.03.

Q.22 — Answer: 0.3747 to 0.3787 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

Adjusted $R^2 = 1 - [SSE/(n - p - 1)]/[SST/(n - 1)] = 1 - (1 - R^2)(n - 1)/(n - p - 1)$, where p is the number of predictors (excluding the intercept).

$= 1 - (1 - 0.66) \times 11/6 = 0.3767$.

Answer: **0.377**.

Q.23 — Answer: (C) [MCQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Ranges: age [31, 39], income [34, 136]. Scaled query = (0.125, 0.608).

Point	scaled age	scaled income	scaled distance	raw distance
P1	0.625	0.216	0.635	40.2
P2	0.125	0	0.608	62
P3	0	1	0.412	40.01
P4	1	0.618	0.875	7.07

Nearest after scaling: **P3**.

Without scaling the nearest point would be P4, because income differences (tens of thousands) dominate the age differences — exactly why k-NN needs feature scaling.

Q.24 — Answer: -0.2545 to -0.2505 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

All weights = -0.5, biases 0 $\Rightarrow$ both hidden units get $a = -0.5 \times (1.5 + 1) = -1.25$, $h_1 = h_2 = \sigma(a) = 0.2227$; $\hat{y} = 2 \times -0.5 \times 0.2227 = -0.2227$.

$\hat{y} - y = -2.2227$. $\partial L / \partial v_j = (\hat{y} - y) h_j = -0.495$ (same for both j).

$\partial L / \partial w_{jk} = (\hat{y} - y) v_j h_j (1 - h_j) x_k = -2.2227 \times -0.5 \times 0.1731 \times 1.5 = 0.2886$ (identical for $j = 1, 2$).

$v_1 \leftarrow -0.5 - 0.5 \times (-0.495) = \mathbf{-0.2525}$.

Because the two hidden units start identical and get identical updates, they remain identical forever (symmetry problem) — hence weights must be initialised randomly.

Q.25 — Answer: 0.5614 to 0.5814 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Predicted positive (score ≥ 0.4): examples 2, 3, 4, 7, 9, 10, 11.

TP = 4, FP = 3, FN = 1, TN = 3.

Precision = $TP / (TP + FP) = \mathbf{0.57}$.

Raising t typically raises precision and lowers recall/FPR; lowering t does the opposite (threshold moving).

Q.26 — Answer: 0.1843 to 0.1883 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Forward pass]

$$a = W_1 x + b_1 = (1, 0.5); h = \text{ReLU}(a) = (1, 0.5).$$

$$z = W_2 h + b_2 = (0, 0.5, -0.5).$$

$$\exp(z) = (1, 1.6487, 0.6065), \text{sum} = 3.2553.$$

$$p_3 = \exp(z_3) / \sum \exp(z_j) = \mathbf{0.1863}.$$
 (All probabilities: (0.307, 0.506, 0.186).)

Q.27 — Answer: 0.2867 to 0.2907 [NAT, 2M · Unsupervised Learning › k-means]

k-means++ picks the next centre x with probability $D(x)^2 / \sum_x D(x)^2$, where $D(x)$ is the distance from x to the **nearest** centre already chosen (chosen centres have $D = 0$).

Point	P1	P2	P3	P4	P5	P6
$D(x)^2$	37	10	53	1	0	41

$$\sum D^2 = 142, \text{ so } P(P6) = 41/142 = \mathbf{0.289}.$$

Common mistakes: using D instead of D^2 (that is not k-means++), or using the distance to the *farthest*/first centre instead of the nearest chosen centre.

Q.28 — Answer: 2.988 to 3.012 [NAT, 2M · Regression & Model Selection › Ridge & Lasso]

$$\text{With } X^T X = I: \beta_{\text{OLS}} = X^T y = b.$$

$$\text{Ridge: } \nabla[\|y - X\beta\|^2 + \lambda\|\beta\|^2] = 0 \Rightarrow (1 + \lambda)\beta = b \Rightarrow \beta_{\text{ridge}} = b/(1 + \lambda).$$

$$\text{Lasso: the problem separates per coordinate, } \min \frac{1}{2}(\beta_j - b_j)^2 + \lambda|\beta_j| \Rightarrow \text{soft-thresholding } \beta_j = \text{sign}(b_j) \cdot \max(|b_j| - \lambda, 0).$$

j	1	2	3	4	5
b_j	0.4	2.1	0.7	-3.5	-1.9
ridge	0.16	0.84	0.28	-1.4	-0.76
lasso	0	0.6	0	-2	-0.4

Answer: **3**.

Q.29 — Answer: 0.1197 to 0.1397 [NAT, 2M · Probabilistic & Instance-based Classifiers › Naive Bayes]

$$N(x; \mu, \sigma^2) = \exp(-(x-\mu)^2/(2\sigma^2))/\sqrt{(2\pi\sigma^2)}.$$
 Naive Bayes multiplies the per-feature densities.

$$C_0: N(4; 5, 2) = 0.2197 \times N(1; 1, 4) = 0.19947; \times \text{prior } 0.5 \Rightarrow 0.021911.$$

$$C_1: N(4; 8, 4) = 0.027 \times N(1; 0, 1) = 0.24197; \times \text{prior } 0.5 \Rightarrow 0.003266.$$

$$P(C_1 | x) = 0.003266/(0.021911 + 0.003266) = \mathbf{0.13}.$$

Q.30 — Answer: 44.0865 to 44.1065 [NAT, 2M · Unsupervised Learning › PCA & dimensionality reduction]

With $X = U\Sigma V^T$ (X centered, $n \times d$), the sample covariance is $S = X^T X/(n-1) = V(\Sigma^2/(n-1))V^T$, so the PC directions are the columns of V and the eigenvalues are $\lambda_j = \sigma_j^2/(n-1)$.

$$\lambda = 26^2/20 = 33.8, 24^2/20 = 28.8, 16^2/20 = 12.8, 5^2/20 = 1.25.$$

$$\text{Cumulative \% variance: } 44.1, 81.67, 98.37, 100 \text{ (note the } (n-1) \text{ factor cancels in proportions: } PVE = \sigma_j^2/\Sigma\sigma^2).$$

Answer: **44.1**. (Using σ instead of σ^2 is the classic mistake.)

Q.31 — Answer: 0.0429 to 0.0449 [NAT, 2M · Regression & Model Selection › Basis functions & feature scaling]

Standardising y divides the slope by s_y ; standardising x multiplies it by s_x : $\beta^* = \beta_1 s_x/s_y = 0.6 \times 3/41 = 0.0439$ (this equals the correlation r in simple regression).

Answer: **0.044**.

Q.32 — Answer: 1.398 to 1.402 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

$$f(w) = (0.5/2)w^2 \Rightarrow \nabla f(w) = 0.5w. \text{ Update: } v_t = \beta v_{t-1} + \nabla f(w_{t-1}), w_t = w_{t-1} - \eta v_t, \text{ with } v_0 = 0, \beta = 0.5, \eta = 0.2.$$

$$t = 1: g = 0.5 \times 2 = 1; v_1 = 0.5 \times 0 + 1 = 1; w_1 = 2 - 0.2 \times 1 = 1.8.$$

$$t = 2: g = 0.5 \times 1.8 = 0.9; v_2 = 0.5 \times 1 + 0.9 = 1.4; w_2 = 1.8 - 0.2 \times 1.4 = 1.52.$$

Answer: **1.4**.

Q.33 — Answer: 1.43 to 1.45 [NAT, 2M · SVM & Decision Trees › Ensembles: bagging & random forests]

$$\text{Var}((1/B)\Sigma f_p) = (1/B^2)[B\sigma^2 + B(B-1)\rho\sigma^2] = \rho\sigma^2 + (1-\rho)\sigma^2/B.$$

$$= 0.2 \times 4 + 0.8 \times 4/5 = 0.8 + 0.64 = \mathbf{1.44}.$$

Even with $B \rightarrow \infty$ the variance cannot fall below $\rho\sigma^2 = 0.8$ — which is why random forests try to reduce ρ by feature subsampling.

Q.34 — Answer: -0.9829 to -0.9771 [NAT, 2M · Regression & Model Selection › Gradient descent]

Per-example loss $\square = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial\square/\partial w = (\hat{y} - y)x$, $\partial\square/\partial b = (\hat{y} - y)$. Update after each example.

Example 1: $\hat{y} = -2$, error $\hat{y} - y = -4$; $\nabla_w = (-4, 8)$, $\nabla_b = -4 \Rightarrow w = (1.4, 0.2)$, $b = -0.6$.

Example 2: $\hat{y} = 3.8$, error $\hat{y} - y = 3.8$; $\nabla_w = (11.4, 3.8)$, $\nabla_b = 3.8 \Rightarrow w = (0.26, -0.18)$, $b = -0.98$.

Answer: **-0.98**.

Q.35 — Answer: 86016 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Parameter counting]

Each weight participates in exactly one multiplication per example.

Layer 1: $64 \times 16 = 1024$.

Layer 2: $16 \times 64 = 1024$.

Layer 3: $64 \times 10 = 640$.

Per example: 2688; for 32 examples: $2688 \times 32 = \mathbf{86016}$.

Mock Test 24

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	5	8	22

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Unsupervised Learning	7	11
Neural Networks (MLP / Feed-forward)	7	11
SVM & Decision Trees	7	11
Regression & Model Selection	7	11
Probabilistic & Instance-based Classifiers	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

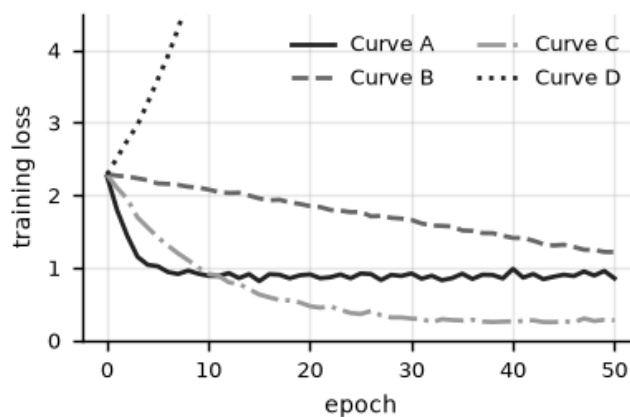
Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

For $x = (1, 7, -4, 5, -4)$ and $y = (0, 6, -4, 5, -1)$, the Manhattan (L_1) distance between x and y is _____ (rounded off to two decimal places)

Q.2 [1 mark · MCQ]

The training-loss curves below were obtained with four different learning rates (everything else fixed). Which curve corresponds to a well-chosen learning rate?



(A) Curve A

(B) Curve B

(C) Curve C

(D) Curve D

Q.3 [1 mark · MSQ (one or more correct)]

Which of the following statements about k-means clustering is/are CORRECT?

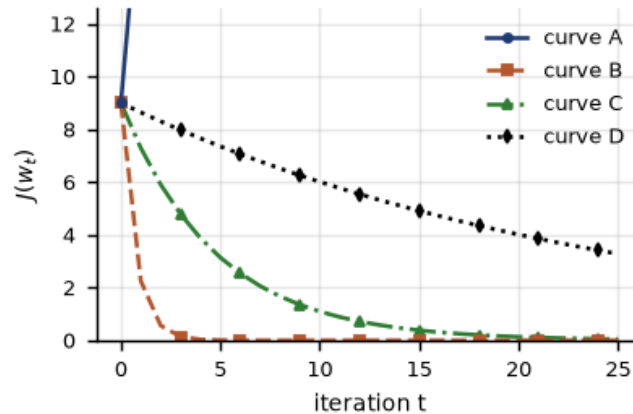
- (A) The solution found by k-means can depend on the choice of initial centroids.
- (B) k-means requires the number of clusters k to be specified in advance.
- (C) Rescaling one feature (e.g. changing its units) can change the clusters found by k-means.
- (D) k-means is an unsupervised method: it uses no class labels.

Q.4 [1 mark · NAT (numerical answer)]

The RBF (Gaussian) kernel is $K(x, z) = \exp(-\gamma\|x - z\|^2)$ with $\gamma = 0.2$. For $x = (2, 1, 1)$ and $z = (1, 3, 1)$, the value of $K(x, z)$ is _____ (rounded off to three decimal places)

Q.5 [1 mark · MCQ]

Gradient descent $w_{t+1} = w_t - \eta J'(w_t)$ is run on $J(w) = (w - 3)^2$ from $w_0 = 0$ with four learning rates $\eta \in \{0.01, 0.05, 0.25, 1.2\}$. The loss curves are shown below (curves that leave the plotted range are cut off). Which curve corresponds to $\eta = 0.05$?



- (A) Curve B
(C) Curve D

- (B) Curve C
(D) Curve A

Q.6 [1 mark · MSQ (one or more correct)]

Regarding the bias-variance trade-off in supervised regression with squared-error loss, which of the following statements is/are CORRECT?

- (A) Increasing model flexibility typically increases variance.
(B) A predictor that outputs a fixed constant regardless of the training data has zero variance.
(C) Expected test MSE at a point $x_0 = \text{Bias}^2 + \text{Variance} + \text{irreducible noise variance } \sigma^2$.
(D) k-NN regression with $k = 1$ has low bias and high variance.

Q.7 [1 mark · MCQ]

In a logistic regression model for a positive response, the coefficient of *dose (mg)* is $\beta = 0.4$. A one-unit increase in *dose (mg)* (others fixed) changes the odds of a positive response by approximately

- (A) 40%
(C) 33%
- (B) 59.9%
(D) 49.2%

Q.8 [1 mark · MSQ (one or more correct)]

Consider ridge regression (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_2^2$) and the lasso (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_1$) with $\lambda \geq 0$. Which of the following statements is/are CORRECT?

- (A) With an orthonormal design, the lasso estimate is a soft-thresholded version of the OLS estimate.
(B) The intercept is usually not penalised in ridge regression.
(C) Increasing λ generally increases bias and decreases variance.
(D) Ridge regression can be used to obtain a unique solution when predictors are perfectly collinear.

Q.9 [1 mark · NAT (numerical answer)]

The 5 two-dimensional observations below have zero mean in each feature. Then the sample variance of x_1 (divisor $n - 1$) is _____ (rounded off to two decimal places)

Point	x_1	x_2	x_3	x_4	x_5
x_1	-2	-2	3	-4	5
x_2	3	1	3	-2	-5

Q.10 [1 mark · NAT (numerical answer)]

The confusion matrix of a binary classifier on a test set is shown below (note which of rows/columns are actual and predicted). The specificity (TNR) of the classifier is _____ (rounded off to two decimal places)

	Predicted +	Predicted -
Actual +	53	19
Actual -	45	38

Q.11 [1 mark · MSQ (one or more correct)]

Which of the following statements about activation functions is/are CORRECT? (σ denotes the logistic sigmoid.)

- (A) ReLU is a bounded function.
- (B) Softmax outputs are strictly positive and sum to 1.
- (C) Replacing sigmoid by tanh in the hidden layers completely removes the vanishing-gradient problem.
- (D) tanh outputs lie in $(-1, 1)$ and are zero-centred.

Q.12 [1 mark · NAT (numerical answer)]

A hard-margin SVM has exactly three support vectors with the optimal dual variables shown. The component w_2 of the optimal weight vector is _____ (rounded off to two decimal places)

i	x_i	y_i	α_i
1	(4, 0)	+1	2
2	(2, -2)	+1	0.75
3	(0, -2)	-1	2.75

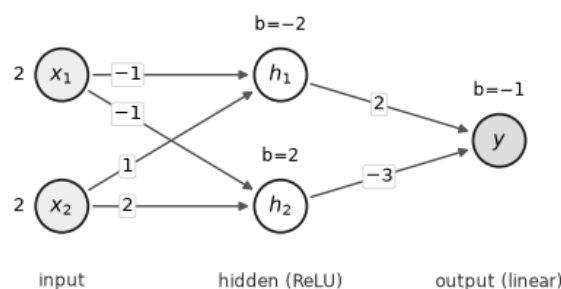
Q.13 [1 mark · MSQ (one or more correct)]

Which of the following statements about decision-tree learning is/are CORRECT?

- (A) The information gain of any split is non-negative.
- (B) Any Boolean function of n Boolean attributes can be represented by a decision tree.
- (C) With squared-error loss, a regression-tree leaf predicts the mean response of its training samples.
- (D) Deeper trees always generalise better because they have lower training error.

Q.14 [1 mark · NAT (numerical answer)]

For the ReLU network shown (edge labels are weights, 'b' denotes a bias) and input $x = (2, 2)$, the output y is _____ (Answer in integer)



Q.15 [1 mark · MSQ (one or more correct)]

Which of the following statements about Bayes-optimal classification and generative vs. discriminative models is/are CORRECT?

- (A) Logistic regression is a generative classifier.
- (B) Generative classifiers can handle missing input features by marginalising over them.
- (C) A generative model can be used to sample synthetic input vectors x .
- (D) The MAP decision rule ignores the class priors.

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

The sample covariance matrix of a two-feature dataset is S given below. In principal component analysis, the variance of the data along the first principal component is _____ (rounded off to two decimal places)

$$S = \begin{bmatrix} 11 & -1 \\ -1 & 9 \end{bmatrix}$$

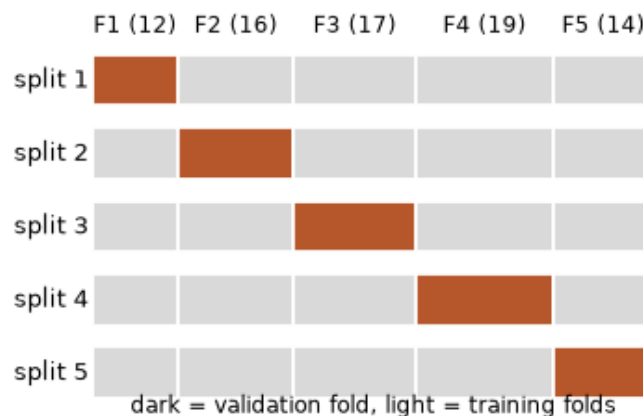
Q.17 [2 marks · NAT (numerical answer)]

The confusion matrix of a 3-class classifier is given below (rows = actual class, columns = predicted class). The macro-averaged recall is _____ (rounded off to two decimal places)

actual \ predicted	A	B	C
A	18	0	0
B	6	13	1
C	2	1	18

Q.18 [2 marks · NAT (numerical answer)]

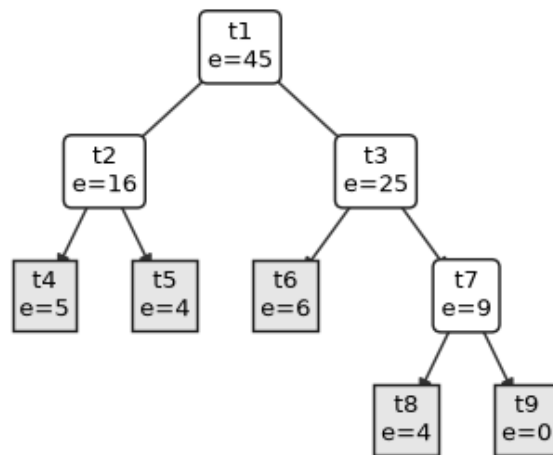
A regression model is evaluated by 5-fold cross-validation on $n = 78$ observations. The folds (F1 ... F5) have unequal sizes, as shown in the diagram and table. The cross-validated root mean squared error (pooled over all n out-of-fold predictions) is _____ (rounded off to two decimal places)



Fold	F1	F2	F3	F4	F5
size	12	16	17	19	14
sum of squared errors	102	108.8	79.9	167.2	112

Q.19 [2 marks · NAT (numerical answer)]

A classification tree was grown on $N = 100$ training records. Each node shows $e(t)$, the number of training records that would be misclassified if t were a leaf; $R(t) = e(t)/N$. The smallest value of α at which cost-complexity pruning first prunes the tree (the weakest link) is _____ (rounded off to three decimal places)

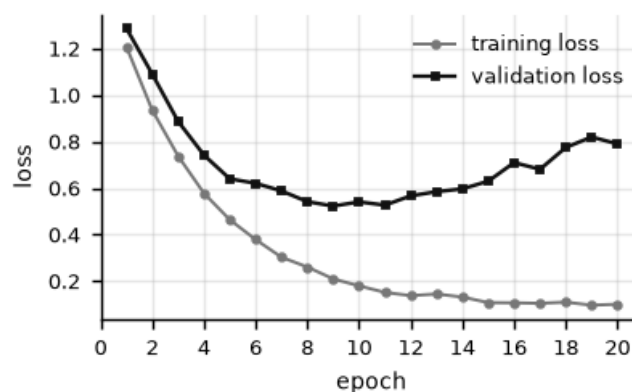
**Q.20 [2 marks · NAT (numerical answer)]**

Least-squares linear regression on $n = 5$ observations minimises $J(w) = (1/2n)\|Xw - y\|^2$ by batch gradient descent with constant learning rate η . The data give the matrix $X^T X$ below. Then with $\eta = 0.867$, the contraction factor $\rho = \max_i |1 - \eta \lambda_i|$ (so that $\|w_{t+1} - w^*\| \leq \rho \|w_t - w^*\|$) is _____ (rounded off to three decimal places)

$$X^T X = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$$

Q.21 [2 marks · NAT (numerical answer)]

A network is trained with early stopping on validation loss using patience $P = 5$: training stops as soon as the validation loss has failed to improve on its best value for 5 consecutive epochs, and the weights from the best epoch are restored. Using the validation losses below (also plotted), the epoch whose weights are returned (restored) as the final model is _____ (Answer in integer)



epoch	1	2	3	4	5	6	7	8	9	10
val. loss	1.287	1.090	0.887	0.742	0.640	0.621	0.588	0.542	0.523	0.541

epoch	11	12	13	14	15	16	17	18	19	20
val. loss	0.527	0.567	0.585	0.597	0.630	0.710	0.681	0.775	0.820	0.790

Q.22 [2 marks · NAT (numerical answer)]

For a two-class problem, the class means are $\mu_0 = (3, -1)^T$ and $\mu_1 = (-2, 3)^T$ and the pooled within-class scatter matrix S_W is given below. Fisher's linear discriminant uses $\mathbf{w} \propto S_W^{-1}(\mu_1 - \mu_0)$. Then the projection threshold $c = \mathbf{w}^T(\mu_0 + \mu_1)/2$, where $\mathbf{w} = S_W^{-1}(\mu_1 - \mu_0)$ (unnormalised) is _____ (rounded off to two decimal places)

$$S_W = \begin{bmatrix} 8 & 4 \\ 4 & 6 \end{bmatrix}$$

Q.23 [2 marks · MCQ]

A network computes $\hat{y} = W_2 \text{ReLU}(W_1 x + b_1) + b_2$ with the parameters below. For $x = (1, -1, 1)^T$, the output $\hat{y}$ is

$$W_1 = \begin{bmatrix} -2 & 1 & 1 \\ 1 & 0 & 1 \\ -1 & 2 & -1 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 2 & 2 & 2 \\ 1 & 1 & 0 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

(A) $\hat{y} = (3, 0)^T$

(B) $\hat{y} = (-13, -3)^T$

(C) $\hat{y} = (5, 1)^T$

(D) $\hat{y} = (19, 3)^T$

Q.24 [2 marks · NAT (numerical answer)]

A logistic-regression neuron $p = \sigma(w_1 x_1 + w_2 x_2 + b)$ has $w = (0, -1)$, $b = 1$ and is trained with binary cross-entropy loss. After one gradient-descent step using the **average** loss over both examples (a mini-batch of size 2) with learning rate $\eta = 1$, the new value of w_1 is _____ (rounded off to three decimal places)

example	x_1	x_2	y
1	-1	-1.5	0
2	-1	-1	0

Q.25 [2 marks · MSQ (one or more correct)]

Let σ be the logistic sigmoid, and $'$ denote the derivative. At $z = -1$ (and $-z = 1$), which of the following is/are CORRECT? (softplus(z) = $\ln(1 + e^z)$.)

(A) $\sigma(-1) > 0.5$

(B) $\text{ReLU}'(-1) = 1$

(C) $\sigma'(-1) = \sigma(-1)^2 = 0.072$

(D) $\tanh(-1) = 2\sigma(-2) - 1$

Q.26 [2 marks · MSQ (one or more correct)]

Consider agglomerative hierarchical clustering on n points with a dissimilarity matrix (no ties). Which of the following statements is/are CORRECT?

(A) For any two points, the complete-linkage cophenetic distance is $\leq$ their original distance.

(B) Different linkages applied to the same distance matrix must yield the same 2-cluster partition.

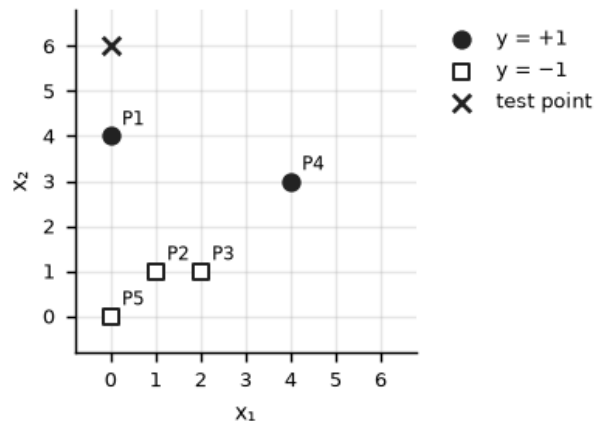
(C) Cutting the single-linkage dendrogram into k clusters is equivalent to deleting the $k - 1$ heaviest edges of an MST.

(D) A dendrogram over n leaves has n internal nodes.

Q.27 [2 marks · NAT (numerical answer)]

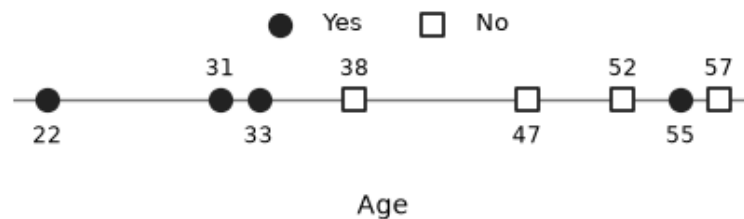
A hard-margin linear SVM is trained on the points below. Solving the dual problem gives the Lagrange multipliers α_i listed in the table. Using the KKT conditions, compute the decision value $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$ at $\mathbf{x} = (0, 6)$. (rounded off to two decimal places)

Point	x_1	x_2	y	α_i
P1	0	4	+1	0.12
P2	1	1	-1	0
P3	2	1	-1	0.34
P4	4	3	+1	0.22
P5	0	0	-1	0

**Q.28 [2 marks · NAT (numerical answer)]**

The continuous attribute Age and the class label of 8 training records are given below (unsorted). A binary split “Age $\leq t$ ” is chosen to maximise information gain, with t restricted to midpoints between consecutive sorted values. Find the threshold value t chosen for the split “Age $\leq t$ ”. (rounded off to one decimal place)

Age	31	22	52	55	57	33	47	38
Class	Yes	Yes	No	Yes	No	Yes	No	No

**Q.29 [2 marks · NAT (numerical answer)]**

The 4×4 matrix $X^T X$ of a (centred) design matrix X has eigenvalues 8, 6, 4, 3. For ridge regression with penalty $\lambda \|\beta\|_2^2$, $df(5) - df(15)$ is _____ (rounded off to two decimal places)

Q.30 [2 marks · NAT (numerical answer)]

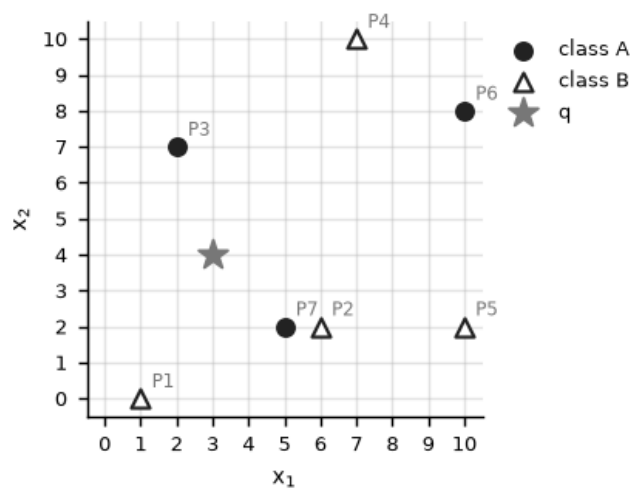
Logistic regression $P(y = 1 | \mathbf{x}) = \sigma(\mathbf{w}_1 \mathbf{x} + b)$ is trained by **batch** gradient descent on the **average** cross-entropy loss over the 2 examples below, with learning rate $\eta = 0.1$. Starting from $\mathbf{w}_1 = -0.5$, $b = 0.5$, the value of $\mathbf{w}_1$ after ONE gradient-descent update is _____ (rounded off to three decimal places)

i	$\mathbf{x}$	y
1	-2	0
2	0	1

Q.31 [2 marks · MCQ]

Consider the training set below. A 3-NN classifier (majority vote) is applied to the query point $\mathbf{q} = (3, 4)$, once with Euclidean distance and once with Manhattan distance. The predicted classes are

Point	x_1	x_2	class
P1	1	0	B
P2	6	2	B
P3	2	7	A
P4	7	10	B
P5	10	2	B
P6	10	8	A
P7	5	2	A



- (A) Euclidean: B; Manhattan: A
(C) Euclidean: A; Manhattan: A

- (B) Euclidean: A; Manhattan: B
(D) Euclidean: B; Manhattan: B

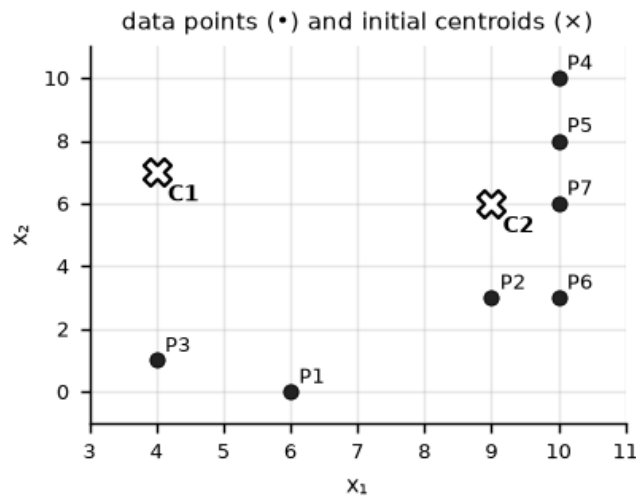
Q.32 [2 marks · NAT (numerical answer)]

$X_1, \dots, X_n$ are i.i.d. with mean $\mu = 1$ and variance $\sigma^2 = 9$, with $n = 20$. Consider the shrinkage estimator $\hat{\mu} = c \cdot \bar{x}$ of μ , where $\bar{x}$ is the sample mean. Using $\text{MSE} = \text{Bias}^2 + \text{Variance}$, $\text{MSE}(\bar{x}) - \text{MSE}(0.7 \cdot \bar{x})$ is (a negative value means shrinkage is worse) _____ (rounded off to three decimal places)

Q.33 [2 marks · NAT (numerical answer)]

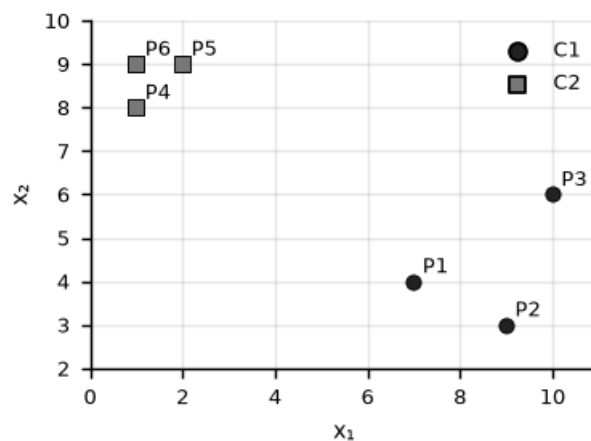
k-means (Lloyd's algorithm, Euclidean distance) with $k = 2$ is run on the 7 points below, starting from the initial centroids $C1 = (4, 7)$, $C2 = (9, 6)$. After **two** iterations, the x_1 -coordinate of the centroid of cluster C2 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5	P6	P7
x_1	6	9	4	10	10	10	10
x_2	0	3	1	10	8	3	6

**Q.34 [2 marks · NAT (numerical answer)]**

A clustering of 6 points into 2 clusters is shown below. Using Manhattan distance, the silhouette coefficient $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$ of point P3 is _____ (rounded off to two decimal places)

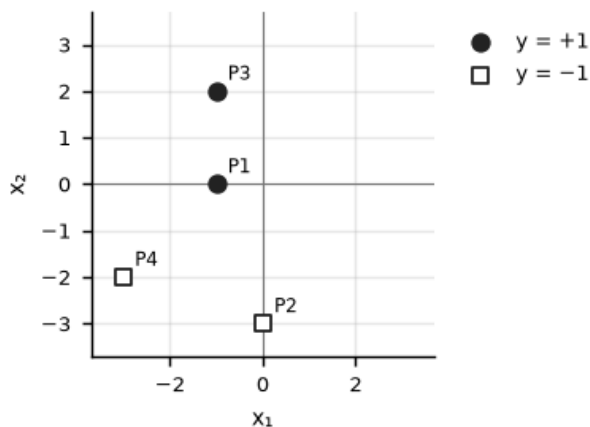
Point	P1	P2	P3	P4	P5	P6
x_1	7	9	10	1	2	1
x_2	4	3	6	8	9	9
cluster	C1	C1	C1	C2	C2	C2



Q.35 [2 marks · NAT (numerical answer)]

A perceptron with augmented weight vector $w = (w_0, w_1, w_2)$ (w_0 is the bias, $\tilde{x} = (1, x_1, x_2)$) starts from $w = (0, 0, 0)$ and uses learning rate $\eta = 1$. The points P1, P2, ... are presented once, in order; whenever $y \cdot w^T \tilde{x} \leq 0$ the update $w \leftarrow w + \eta y \tilde{x}$ is applied. Find the value of w_0 (bias) after this epoch. (Answer in integer)

Point	x_1	x_2	y
P1	-1	0	+1
P2	0	-3	-1
P3	-1	2	+1
P4	-3	-2	-1



— END OF QUESTION PAPER —

Mock Test 24 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	4.99 to 5.01	19	NAT	2	0.0485 to 0.0515
2	MCQ	1	(C)	20	NAT	2	0.6512 to 0.6552
3	MSQ	1	(A), (B), (C), (D)	21	NAT	2	9
4	NAT	1	0.3659 to 0.3699	22	NAT	2	0.8962 to 0.9162
5	MCQ	1	(B)	23	MCQ	2	(A)
6	MSQ	1	(A), (B), (C), (D)	24	NAT	2	0.9005 to 0.9045
7	MCQ	1	(D)	25	MSQ	2	(D)
8	MSQ	1	(A), (B), (C), (D)	26	MSQ	2	(C)
9	NAT	1	14.49 to 14.51	27	NAT	2	2.59 to 2.61
10	NAT	1	0.4478 to 0.4678	28	NAT	2	35.49 to 35.51
11	MSQ	1	(B), (D)	29	NAT	2	0.9596 to 0.9796
12	NAT	1	3.99 to 4.01	30	NAT	2	-0.4202 to -0.4162
13	MSQ	1	(A), (B), (C)	31	MCQ	2	(C)
14	NAT	1	-13	32	NAT	2	0.1365 to 0.1425
15	MSQ	1	(B), (C)	33	NAT	2	9.79 to 9.81
16	NAT	2	11.4042 to 11.4242	34	NAT	2	0.5929 to 0.6129
17	NAT	2	0.8257 to 0.8457	35	NAT	2	0
18	NAT	2	2.693 to 2.713				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Unsupervised Learning					11

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Neural Networks (MLP / Feed-forward)					11
SVM & Decision Trees					11
Regression & Model Selection					11
Probabilistic & Instance-based Classifiers					11
Total					55

Mock Test 24 — Detailed Solutions

Q.1 — Answer: 4.99 to 5.01 [NAT, 1M · Unsupervised Learning › Distance measures]

$$\sum |x_i - y_i| = 1 + 1 + 0 + 0 + 3 = 5.$$

For reference: $L_1 = 5$, $L_2 = 3.32$, $L_\infty = 3$ (always $L_\infty \leq L_2 \leq L_1$).

Q.2 — Answer: (C) [MCQ, 1M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

Typical signatures: far-too-high $\eta \rightarrow$ loss blows up; somewhat-too-high $\eta \rightarrow$ quick drop then plateau at a high loss (bouncing around the minimum); too-low $\eta \rightarrow$ slow, nearly linear decrease; good $\eta \rightarrow$ fast decrease to the lowest loss.

Curve A: a learning rate that is somewhat too high (fast initial drop, then the loss plateaus at a high value).

Curve B: a learning rate that is too low (very slow, almost linear decrease).

Curve C: a well-chosen learning rate.

Curve D: a learning rate that is far too high (training diverges).

Hence the answer is **Curve C**.

Q.3 — Answer: (A), (B), (C), (D) [MSQ, 1M · Unsupervised Learning › k-means]

(A) **TRUE**. Lloyd's algorithm converges to a local optimum determined by the initialization.

(B) **TRUE**. k is an input hyper-parameter.

(C) **TRUE**. Euclidean distances change non-uniformly, so assignments can change.

(D) **TRUE**. Only the feature vectors are used.

Q.4 — Answer: 0.3659 to 0.3699 [NAT, 1M · SVM & Decision Trees › SVM: kernels]

$$\|x - z\|^2 = 1 + 4 + 0 = 5.$$

$$K = \exp(-0.2 \times 5) = \exp(-1) = \mathbf{0.368}.$$

Note: $K \in (0, 1]$, $K(x, x) = 1$, and K decreases as the points move apart or as γ grows.

Q.5 — Answer: (B) [MCQ, 1M · Regression & Model Selection › Gradient descent]

$$J(w) = (w - 3)^2 \text{ has } J'' = 2. \text{ GD: } w_{t+1} - 3 = (1 - 2\eta)(w_t - 3), \text{ so } J(w_t) = (1 - 2\eta)^{2t} J(w_0).$$

Ratios $(1 - 2\eta)^2$ per step: $\eta = 0.01 \rightarrow 0.9604$ (curve D); $\eta = 0.05 \rightarrow 0.81$ (curve C); $\eta = 0.25 \rightarrow 0.25$ (curve B); $\eta = 1.2 \rightarrow 1.96$ (curve A).

Ratio $> 1 \Rightarrow$ divergence ($\eta > 1 = 2/J''$); among convergent rates (all < 0.5 here), a smaller η gives a ratio closer to 1, i.e. slower decrease.

Hence $\eta = 0.05$ corresponds to **Curve C**.

Q.6 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection › Bias-variance trade-off]

(A) **TRUE**. Flexible fits change more with the training sample.

(B) **TRUE**. Its prediction does not change across training sets.

(C) **TRUE**. Standard decomposition of $E[(y_0 - \hat{f}(x_0))^2]$.

(D) **TRUE**. It reproduces each training point.

Q.7 — Answer: (D) [MCQ, 1M · Probabilistic & Instance-based Classifiers › Logistic regression]

$$\text{Odds multiply by } e^\beta = e^{0.4} = 1.4918.$$

$$\text{Percentage change} = 100(e^\beta - 1)\% = \mathbf{49.2\%}.$$

$100\beta\%$ is only a small- β approximation; $100e^\beta\%$ is the new odds as a percentage of the old odds, not the change.

Q.8 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection › Ridge & Lasso]

(A) **TRUE**. $\beta_j = \text{sign}(b_j)(|b_j| - c)_+$ for a threshold c depending on λ .

(B) **TRUE**. Penalising it would make the fit depend on the origin chosen for y .

(C) **TRUE**. Stronger shrinkage = less flexible fit.

(D) **TRUE**. $X^T X + \lambda I$ is invertible for $\lambda > 0$.

Q.9 — Answer: 14.49 to 14.51 [NAT, 1M · Unsupervised Learning › PCA & dimensionality reduction]

The data are already centered (column means are 0), so $S = X^T X / (n - 1)$.

$$\Sigma x_1^2 = 58, \Sigma x_2^2 = 48, \Sigma x_1 x_2 = -16.$$

$$S = [[14.5, -4], [-4, 12]].$$

Required value = **14.5**. (Dividing by n instead of $n - 1$ would give 11.6.)

Q.10 — Answer: 0.4478 to 0.4678 [NAT, 1M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

$$TP = 53, FN = 19, FP = 45, TN = 38 \text{ (} N = 155 \text{)}.$$

$$\text{Specificity (TNR)} = TN / (TN + FP).$$

Value = **0.46**.

Watch the orientation: swapping FP and FN is the most common error (it swaps precision and recall).

Q.11 — Answer: (B), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

(A) **FALSE**. It is unbounded above.

(B) **TRUE**. $\exp(\cdot) > 0$ and the outputs are normalised.

(C) **FALSE**. $\tanh$ also saturates; it only helps (zero-centred, larger max slope).

(D) **TRUE**. $\tanh$ is odd and bounded by ± 1 .

Q.12 — Answer: 3.99 to 4.01 [NAT, 1M · SVM & Decision Trees > SVM: dual & support vectors]

$$w = \sum_i \alpha_i y_i x_i \text{ (sum over support vectors).}$$

$$w_2 = (2)(+1)(0) + (0.75)(+1)(-2) + (2.75)(-1)(-2) = 4.$$

(Full vector $w = (9.5, 4)$; note $\sum_i \alpha_i y_i = 0$ holds.)

Q.13 — Answer: (A), (B), (C) [MSQ, 1M · SVM & Decision Trees > Decision trees: overfitting & pruning]

(A) **TRUE**. Conditioning cannot increase entropy on average.

(B) **TRUE**. Up to 2^n leaves.

(C) **TRUE**. The mean minimises SSE.

(D) **FALSE**. They tend to overfit.

Q.14 — Answer: -13 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Forward pass]

$$a = Wx + b = (-2, 4); h = \text{ReLU}(a) = (0, 4) - \text{a negative pre-activation is clipped to 0.}$$

$$y = 2 \cdot 0 + (-3) \cdot 4 + (-1) = -13.$$

A common slip is to forget the ReLU and use the raw pre-activation (which would give -17).

Q.15 — Answer: (B), (C) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

(A) **FALSE**. It is discriminative.

(B) **TRUE**. $P(x_{\text{obs}}|y)$ is obtained from the joint model.

(C) **TRUE**. It models the distribution of x .

(D) **FALSE**. MAP uses the posterior $\propto$ likelihood $\times$ prior; ML ignores priors.

Q.16 — Answer: 11.4042 to 11.4242 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

$$\text{Eigenvalues of } [[a, b], [b, c]]: \lambda = (a + c)/2 \pm \sqrt{((a - c)/2)^2 + b^2}.$$

$$\lambda = 10 \pm \sqrt{1 + 1} = 10 \pm 1.4142 \rightarrow \lambda_1 = 11.4142, \lambda_2 = 8.5858 \text{ (check: } \lambda_1 + \lambda_2 = \text{trace} = 20, \lambda_1 \lambda_2 = \text{det} = 98 \text{)}.$$

$$\text{Eigenvector for } \lambda_1: (S - \lambda_1 I)v = 0 \rightarrow (-0.4142)v_1 + (-1)v_2 = 0 \rightarrow v \propto (1, -0.4142).$$

$$\text{PVE(PC1)} = \lambda_1 / (\lambda_1 + \lambda_2) = 57.07\%.$$

Answer: **11.41**.

Q.17 — Answer: 0.8257 to 0.8457 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

class	TP	predicted total	actual total	precision	recall	F1
A	18	26	18	0.6923	1	0.8182

B	13	14	20	0.9286	0.65	0.7647
C	18	19	21	0.9474	0.8571	0.9

Precision of class $k = C_{kk}/(\text{column sum})$; recall $= C_{kk}/(\text{row sum})$.

Macro recall = mean of per-class recalls = **0.84** (= balanced accuracy).

Q.18 — Answer: 2.693 to 2.713 [NAT, 2M · Regression & Model Selection > Cross-validation]

Total SSE = 102 + 108.8 + 79.9 + 167.2 + 112 = 569.9; $n = 78$.

CV MSE = 569.9/78 = 7.3064; RMSE = $\sqrt{7.3064} = 2.703$.

Answer: **2.7**.

Q.19 — Answer: 0.0485 to 0.0515 [NAT, 2M · SVM & Decision Trees > Decision trees: overfitting & pruning]

$R(t) = e(t)/N$ with $N = 100$. For an internal node t with subtree T_t : $g(t) = [R(t) - R(T_t)]/(|T_t| - 1)$, where $|T_t|$ = number of leaves.

Node	$R(t)$	$R(T_t)$	$ T_t $	$g(t)$
t1	0.45	0.19	5	0.065
t2	0.16	0.09	2	0.07
t3	0.25	0.1	3	0.075
t7	0.09	0.04	2	0.05

The weakest link is t7 with the minimum $g = 0.05$; for $\alpha \geq$ this value, collapsing t7 into a leaf does not increase R_α .

Q.20 — Answer: 0.6512 to 0.6552 [NAT, 2M · Regression & Model Selection > Gradient descent]

Hessian of J : $\nabla^2 J = (1/n)X^T X$. Eigenvalues of $X^T X$: $4 \pm 2 = 6, 2$ (eigenvectors $(1, 1), (1, -1)$).

So the Hessian eigenvalues are $L = 1.2$ and $\mu = 0.4$.

GD error evolves as $w_{t+1} - w^* = (I - \eta \nabla^2 J)(w_t - w^*)$, so it converges iff $|1 - \eta \lambda| < 1$ for every Hessian eigenvalue λ .

$|1 - 0.867 \times 1.2| = 0.0404$, $|1 - 0.867 \times 0.4| = 0.6532$; $\rho = 0.6532$.

Answer: **0.653**.

Q.21 — Answer: 9 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Rule: keep track of the best (lowest) validation loss; an epoch that does not strictly improve on it increments a counter, which is reset on improvement; stop when the counter reaches patience $P = 5$.

Trace - 1: 1.287 (new best); 2: 1.090 (new best); 3: 0.887 (new best); 4: 0.742 (new best); 5: 0.640 (new best); 6: 0.621 (new best); 7: 0.588 (new best); 8: 0.542 (new best); 9: 0.523 (new best); 10: 0.541 (no improvement, count = 1); 11: 0.527 (no improvement, count = 2); 12: 0.567 (no improvement, count = 3); 13: 0.585 (no improvement, count = 4); 14: 0.597 (no improvement, count = 5).

Training stops after epoch 14; the best validation loss 0.523 occurred at epoch 9, whose weights are restored.

Answer: **9**. The widening train/validation gap after this point is the signature of overfitting.

Q.22 — Answer: 0.8962 to 0.9162 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$\mu_1 - \mu_0 = (-5, 4)^T$; $\det S_W = 32$.

$$S_W^{-1} = \begin{bmatrix} 0.1875 & -0.125 \\ -0.125 & 0.25 \end{bmatrix}$$

$w = S_W^{-1}(\mu_1 - \mu_0) = (-1.4375, 1.625)^T$.

$(\mu_0 + \mu_1)/2 = (0.5, 1)$; $c = 0.91$.

Note: using $\mu_1 - \mu_0$ without S_W^{-1} is correct only when $S_W \propto I$.

Q.23 — Answer: (A) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

$a = W_1 x + b_1 = (-3, 1, -5)^T$; $h = \text{ReLU}(a) = (0, 1, 0)^T$.

$\hat{y} = W_2 h + b_2 = (3, 0)^s$.

Distractors: skipping the ReLU gives $(-13, -3)^T$; using W_1^T instead of W_1 gives $(5, 1)^T$; dropping the biases gives $(4, 2)^T$; using $|a|$ instead of ReLU gives $(19, 3)^T$.

Q.24 — Answer: 0.9005 to 0.9045 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

For σ output with binary cross-entropy $L = -[y \ln p + (1 - y) \ln(1 - p)]$, $\partial L / \partial a = p - y$, so $\partial L / \partial w_j = (p - y)x_j$ and $\partial L / \partial b = p - y$.

Example 1: $a = 2.5$, $p = 0.9241$, $p - y = 0.9241$.

Example 2: $a = 2$, $p = 0.8808$, $p - y = 0.8808$.

Mini-batch (mean) gradient for w_1 : -0.9025 .

$w_1 \leftarrow 0 - 1 \times (-0.9025) = \mathbf{0.9025}$.

Note there is no σ' factor: it cancels with the derivative of the log loss.

Q.25 — Answer: (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Activation functions]

$\sigma(-1) = 0.2689$, $\tanh(-1) = -0.7616$.

(A) FALSE. $\sigma(-1) = 0.2689$.

(B) FALSE. $\text{ReLU}' = 1$ for $z > 0$ and 0 for $z < 0$.

(C) FALSE. $\sigma' = \sigma(1 - \sigma) = 0.1966$.

(D) TRUE. Both equal -0.7616 .

Q.26 — Answer: (C) [MSQ, 2M · Unsupervised Learning > Hierarchical clustering]

(A) FALSE. It is $\geq$ the original distance.

(B) FALSE. Single and complete linkage often disagree.

(C) TRUE. The remaining forest's components are the clusters.

(D) FALSE. It has $n - 1$ internal nodes.

Q.27 — Answer: 2.59 to 2.61 [NAT, 2M · SVM & Decision Trees > SVM: dual & support vectors]

$w = \sum \alpha_i y_i x_i$ over points with $\alpha_i > 0$:

$w = (0.12)(+1)(0, 4) + (0.34)(-1)(2, 1) + (0.22)(+1)(4, 3) = (1/5, 4/5)$.

b from any support vector, e.g. P_1 : $y = w^T x + b \Rightarrow b = +1 - (3.2) = -11/5$.

Check: $\sum \alpha_i y_i = 0$ ($= 0$, as required).

$f((0, 6)) = (1/5)(0) + (4/5)(6) - 11/5 = \mathbf{2.6}$.

Q.28 — Answer: 35.49 to 35.51 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Sort the values: 22(Y), 31(Y), 33(Y), 38(N), 47(N), 52(N), 55(Y), 57(N).

Parent [4 Yes, 4 No], $H = 1$. The optimal threshold always lies where the class changes between neighbours (Fayyad-Irani), giving 3 candidates:

Threshold	Left ($\leq t$) [Yes, No]	Right [Yes, No]	IG
35.5	[3, 0]	[1, 4]	0.5488
53.5	[3, 3]	[1, 1]	0
56	[4, 3]	[0, 1]	0.1379

Best: $t = 35.5$ with $IG = 0.5488$. Answer: **35.5**.

Q.29 — Answer: 0.9596 to 0.9796 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

Ridge fitted values $\hat{y} = H_\lambda y$ with $H_\lambda = X(X^T X + \lambda I)^{-1} X^T$. Writing $X^T X = V D V^T$ with eigenvalues d_j^2 gives $\text{trace}(H_\lambda) = \sum d_j^2 / (d_j^2 + \lambda)$ (effective degrees of freedom).

$8/(8 + 5) + 6/(6 + 5) + 4/(4 + 5) + 3/(3 + 5) = 1.9803$.

$8/(8 + 15) + 6/(6 + 15) + 4/(4 + 15) + 3/(3 + 15) = 1.0107$.

Difference = $1.9803 - 1.0107 = 0.9696$ (df decreases as λ grows).

Answer: **0.97**.

Q.30 — Answer: -0.4202 to -0.4162 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

Logistic regression]

For the average cross-entropy, $\partial L / \partial w_j = (1/n) \sum (\hat{p}_i - y_i) x_{ij}$ and $\partial L / \partial b = (1/n) \sum (\hat{p}_i - y_i)$, with $\hat{p}_i = \sigma(z_i)$.

i	z_i	$\hat{p}_i$	$\hat{p}_i - y_i$
1	1.5	0.8176	0.8176
2	0.5	0.6225	-0.3775

Gradient: $\partial L / \partial w_1 = -0.8176$, $\partial L / \partial b = 0.22$.

Update $\theta \leftarrow \theta - \eta \nabla L$: $w_1 = -0.5 - 0.1 \times (-0.8176) = -0.4182$, $b = 0.5 - 0.1 \times (0.22) = 0.478$.

Required: $w_1 = \mathbf{-0.418}$.

Q.31 — Answer: (C) [MCQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Distances from $\mathbf{q}$:

Point	class	Euclidean d	Manhattan d
P1	B	4.472	6
P2	B	3.606	5
P3	A	3.162	4
P4	B	7.211	10
P5	B	7.28	9
P6	A	8.062	11
P7	A	2.828	4

Euclidean 3 nearest: P7(A), P3(A), P2(B) $\Rightarrow$ **A**.

Manhattan 3 nearest: P3(A), P7(A), P2(B) $\Rightarrow$ **A**.

Answer: **Euclidean: A; Manhattan: A**. Here both metrics lead to the same vote.

Q.32 — Answer: 0.1365 to 0.1425 [NAT, 2M · Regression & Model Selection > Bias-variance trade-off]

$E[c\bar{x}] = c\mu \Rightarrow \text{Bias} = (c - 1)\mu$; $\text{Var}(c\bar{x}) = c^2\sigma^2/n$. $\text{MSE}(c) = (c - 1)^2\mu^2 + c^2\sigma^2/n$.

Here $\mu = 1$, $\sigma^2/n = 9/20 = 0.45$.

$\text{MSE}(\bar{x}) = \sigma^2/n = 0.45$ (unbiased). $\text{MSE}(0.7\bar{x}) = 0.3105$. Difference = 0.1395.

Answer: **0.14**.

Q.33 — Answer: 9.79 to 9.81 [NAT, 2M · Unsupervised Learning > k-means]

Iteration 1 - assignment (nearest initial centroid, squared Euclidean distance):

Point	$d^2(\cdot, C1)$	$d^2(\cdot, C2)$	cluster
P1 (6, 0)	53	45	C2
P2 (9, 3)	41	9	C2
P3 (4, 1)	36	50	C1
P4 (10, 10)	45	17	C2
P5 (10, 8)	37	5	C2
P6 (10, 3)	52	10	C2
P7 (10, 6)	37	1	C2

Update: each centroid becomes the mean of its assigned points: $C1 \rightarrow (4, 1)$; $C2 \rightarrow (9.17, 5)$.

Iteration 2 - assignment with the updated centroids:

Point	$d^2(\cdot, C1)$	$d^2(\cdot, C2)$	cluster
P1 (6, 0)	5	35.03	C1
P2 (9, 3)	29	4.03	C2
P3 (4, 1)	0	42.69	C1
P4 (10, 10)	117	25.69	C2
P5 (10, 8)	85	9.69	C2
P6 (10, 3)	40	4.69	C2
P7 (10, 6)	61	1.69	C2

Update: $C1 \rightarrow (5, 0.5)$; $C2 \rightarrow (9.8, 6)$. Required coordinate = **9.8**.

Note: each step of Lloyd's algorithm can only decrease (or keep) J , which is why it converges.

Q.34 — Answer: 0.5929 to 0.6129 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

a(i) = mean Manhattan distance from P3 to the other members of its own cluster C1: $d(P3, P1) = 5$, $d(P3, P2) = 4 \rightarrow a = 4.5$.

b(i) = smallest mean distance to the points of another cluster: C2: $11.3333 \rightarrow b = 11.3333$.

s(i) = $(b - a) / \max(a, b) = (11.3333 - 4.5) / 11.3333 = \mathbf{0.6}$.

s ranges in $[-1, 1]$; values near 1 mean well clustered, negative values suggest a wrong assignment.

Q.35 — Answer: 0 [NAT, 2M · SVM & Decision Trees > Perceptron (linear separators)]

Trace the epoch (a point with $y \cdot w^T \tilde{x} \leq 0$, including 0, triggers an update):

Step	$\tilde{x} = (1, x_1, x_2)$	y	$w^T \tilde{x}$	$y \cdot w^T \tilde{x} \leq 0$?	w after step
1	(1, -1, 0)	+1	0	yes $\rightarrow$ update	(1, -1, 0)
2	(1, 0, -3)	-1	1	yes $\rightarrow$ update	(0, -1, 3)
3	(1, -1, 2)	+1	7	no	(0, -1, 3)
4	(1, -3, -2)	-1	-3	no	(0, -1, 3)

Final $w = (0, -1, 3)$, number of updates = 2. Answer: **0**.

Mock Test 25

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	6	6	23

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Unsupervised Learning	7	11
SVM & Decision Trees	7	11
Regression & Model Selection	7	11
Probabilistic & Instance-based Classifiers	7	11
Neural Networks (MLP / Feed-forward)	7	11

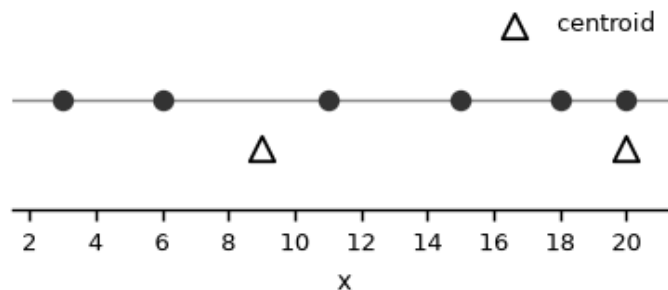
Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

One-dimensional data $\{3, 6, 11, 15, 18, 20\}$ and $k = 2$ centroids at 9, 20 are given. Each point is assigned to its nearest centroid and then each centroid is recomputed as the mean of its points. The within-cluster SSE with the **updated** centroids is _____ (rounded off to two decimal places)



Q.2 [1 mark · MSQ (one or more correct)]

Which of the following statements about support vector machines is/are CORRECT?

- (A) Logistic regression can never produce a linear decision boundary.
- (B) A kernel SVM predicts with $f(x) = \sum \alpha_i y_i K(x_i, x) + b$, where only support vectors contribute.
- (C) The logistic loss $\log(1 + e^{-y f(x)})$ is strictly positive for every finite $f(x)$.
- (D) Doubling both w and b doubles the geometric margin of every point.

Q.3 [1 mark · MCQ]

With Gaussian noise and independent Laplace (double-exponential) priors on each w_j , the MAP estimate corresponds to:

- (A) ordinary least squares
- (B) lasso (L1-penalised least squares)
- (C) principal component regression
- (D) ridge (L2-penalised least squares)

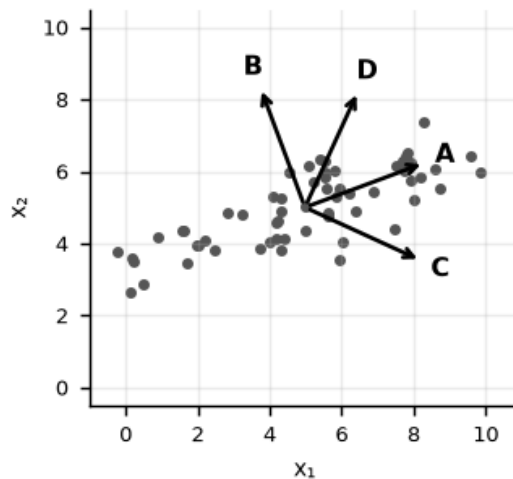
Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about classifier evaluation is/are CORRECT?

- (A) Balanced accuracy for binary classification is the mean of sensitivity and specificity.
- (B) The F_β -score with $\beta = 2$ weights precision more heavily than recall.
- (C) For fixed TPR and FPR, precision decreases as the positive class becomes rarer.
- (D) Moving the decision threshold trades precision against recall without retraining the model.

Q.5 [1 mark · MCQ]

The scatter plot shows a 2-D dataset together with four candidate directions A-D drawn from the data mean. Which direction best represents the second principal component (PC2)?



- (A) Direction A
- (B) Direction B
- (C) Direction C
- (D) Direction D

Q.6 [1 mark · NAT (numerical answer)]

Two shopping baskets are encoded as binary vectors (1 = item bought). The Jaccard distance between basket A and basket B is _____ (rounded off to two decimal places)

Item	milk	bread	eggs	butter	jam	tea	rice	salt
Basket A	1	1	0	0	0	1	0	0
Basket B	0	0	0	1	0	1	1	1

Q.7 [1 mark · MSQ (one or more correct)]

Which of the following statements about the representational power of feed-forward neural networks is/are CORRECT?

- (A) A single sigmoid neuron can represent XOR because the sigmoid is non-linear.
- (B) A network with a single hidden layer of two ReLU units can approximate any continuous function on $[0, 1]$ to arbitrary accuracy.
- (C) The universal approximation theorem is an existence result; it does not say that gradient descent will find the approximating weights.
- (D) The universal approximation theorem requires at least two hidden layers.

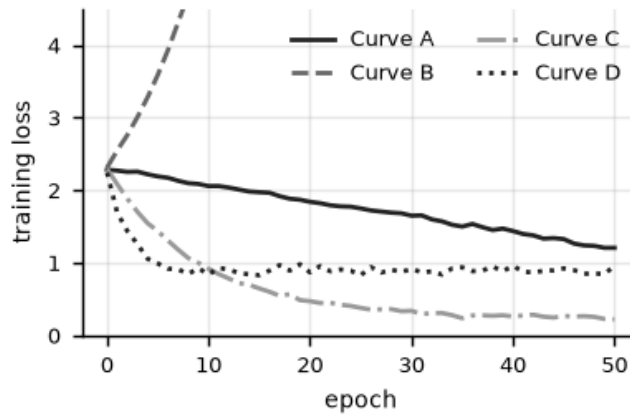
Q.8 [1 mark · MSQ (one or more correct)]

A simple linear regression $\hat{y} = \beta_0 + \beta_1 x$ is fitted by ordinary least squares (with intercept) to data (x_i, y_i) , $i = 1, \dots, n$, with residuals $e_i = y_i - \hat{y}_i$ and sample correlation r . Which of the following statements is/are CORRECT?

- (A) R^2 is unchanged if y is replaced by $a + by$ for any constants a and $b \neq 0$.
- (B) The product of the slope of the y -on- x line and the slope of the x -on- y line equals r^2 .
- (C) $SST = SSR - SSE$.
- (D) $\sum \hat{y}_i e_i = 0$ for the least-squares fit.

Q.9 [1 mark · MCQ]

The training-loss curves below were obtained with four different learning rates (everything else fixed). Which curve corresponds to a well-chosen learning rate?



- (A) Curve A
(C) Curve C

- (B) Curve B
(D) Curve D

Q.10 [1 mark · NAT (numerical answer)]

For a hard-margin linear SVM, the support vectors $x_+ = (1, 2)$ (class +1) and $x_- = (-2, 2)$ (class -1) lie on the hyperplanes $w^T x + b = +1$ and $w^T x + b = -1$ respectively, and the segment joining them is perpendicular to the decision boundary. The value of $\|w\|$ is _____ (rounded off to two decimal places)

Q.11 [1 mark · NAT (numerical answer)]

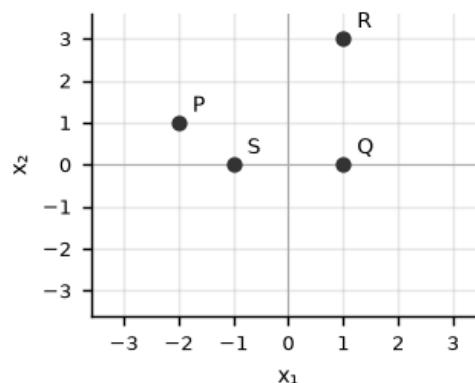
In a defect detector, a false negative costs 1 units and a false positive costs 8 units (correct decisions cost 0). The model outputs $p = P(\text{positive} | x)$, i.e. the probability that the part is defective. The Bayes-optimal rule predicts positive iff p exceeds the threshold _____ (rounded off to two decimal places)

Q.12 [1 mark · NAT (numerical answer)]

Nested cross-validation is performed with an outer 5-fold loop and an inner 3-fold loop. In each outer fold, the inner loop evaluates 4 hyperparameter values, after which the model with the best value is refitted once on that outer fold's training part and evaluated on its test part. The total number of model fits (no further refit at the end) is _____ (Answer in integer)

Q.13 [1 mark · MSQ (one or more correct)]

A perceptron with weights $w = (-1, 3)$ and bias $b = -4$ outputs $\hat{y} = +1$ if $w \cdot x + b \geq 0$ and $\hat{y} = -1$ otherwise. Which of the points shown below are classified as $\hat{y} = +1$?



- (A) Point P = (-2, 1)
(C) Point R = (1, 3)

- (B) Point Q = (1, 0)
(D) Point S = (-1, 0)

Q.14 [1 mark · NAT (numerical answer)]

Consider the polynomial kernel $K(x, z) = (x^T z + 2)^2$ with $x = (2, 2, -1)$ and $z = (1, 3, 2)$. The value of $K(x, z)$ is ____ (Answer in integer)

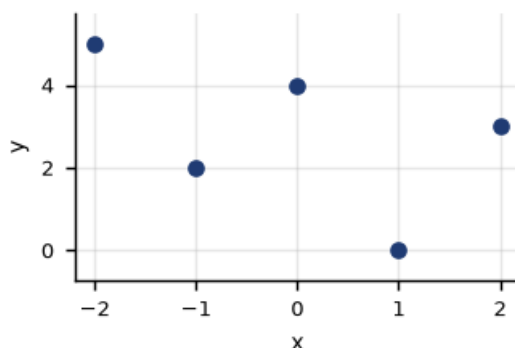
Q.15 [1 mark · NAT (numerical answer)]

Data are uniformly distributed in the unit hypercube $[0, 1]^5$. To capture a fraction 0.5 of the data, a k-NN method uses a sub-cube neighbourhood. The required edge length of this sub-cube is ____ (rounded off to two decimal places)

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

The quadratic model $y = a + bx + cx^2$ is fitted by least squares to the 5 points below (a polynomial / basis-function regression with basis $1, x, x^2$). Then the fitted value $\hat{y}$ at $x = 3$ is ____ (rounded off to two decimal places)

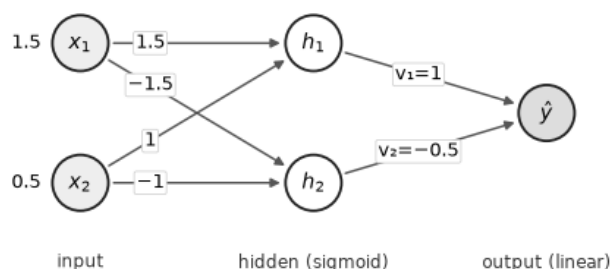
x	-2	-1	0	1	2
y	5	2	4	0	3

**Q.17 [2 marks · NAT (numerical answer)]**

Let ϕ be the feature map of the kernel $K(x, z) = (x^T z)^2$ on $\mathbb{R}^2$, so that $K(x, z) = \phi(x)^T \phi(z)$. For $x = (1, 0)$ and $z = (-2, 1)$, the squared distance $\|\phi(x) - \phi(z)\|^2$ in feature space is ____ (Answer in integer)

Q.18 [2 marks · NAT (numerical answer)]

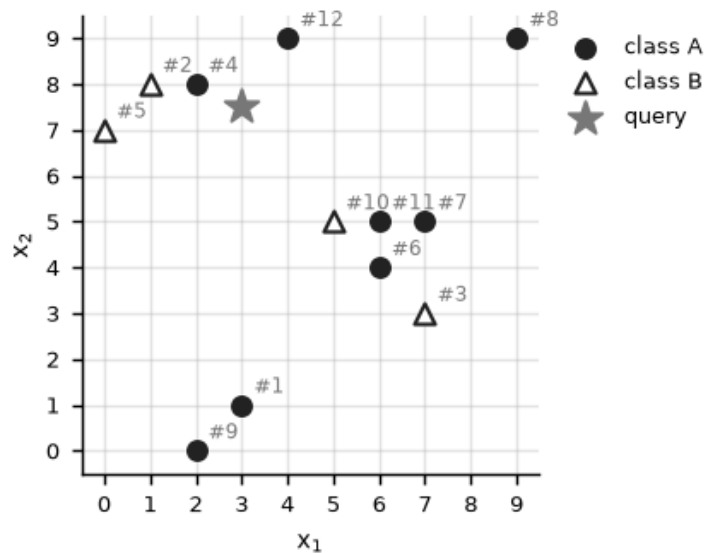
In the network below, hidden units are sigmoid, the output is linear ($\hat{y} = v_1 h_1 + v_2 h_2$), all biases are 0, and the loss is $L = \frac{1}{2}(\hat{y} - y)^2$. For input $x = (1.5, 0.5)$ and target $y = -0.5$, the value of $\partial L / \partial b_1$ (bias of hidden unit h_1 , currently 0) is ____ (rounded off to 4 decimal places)



Q.19 [2 marks · MSQ (one or more correct)]

The figure and table show 12 labelled training points (class A: filled circles, class B: open triangles) and a query point (star) at (3, 7.5). A k-NN classifier with **Euclidean** distance and majority vote is used. Which of the following statements is/are CORRECT?

#	1	2	3	4	5	6	7	8	9	10	11	12
x_1	3	1	7	2	0	6	7	9	2	5	6	4
x_2	1	8	3	8	7	4	5	9	0	5	5	9
class	A	B	B	A	B	A	A	A	A	B	A	A

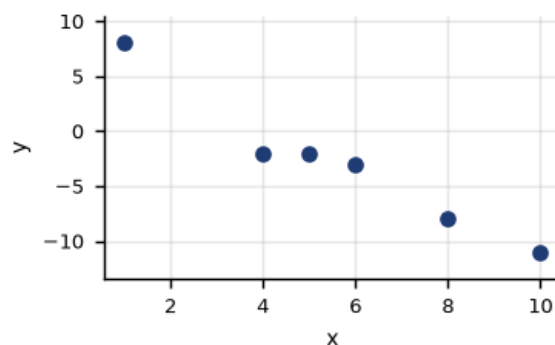


- (A) With $k = 1$, the query is classified as class A.
 (B) With $k = 3$, the query is classified as class B.
 (C) With $k = 5$, the query is classified as class B.
 (D) With $k = 7$, the query is classified as class A.

Q.20 [2 marks · NAT (numerical answer)]

A simple linear regression (with intercept) is fitted by least squares to the 6 points below. Using leave-one-out cross-validation, the LOOCV estimate of test MSE, $(1/n)\sum(y_i - \hat{y}_{(-i)}(x_i))^2$, is _____ (rounded off to two decimal places)

x	1	4	5	6	8	10
y	8	-2	-2	-3	-8	-11

**Q.21 [2 marks · MCQ]**

For logits $z = (0, 1, -1)$, softmax probabilities p and true class 2, the gradient of the cross-entropy loss $L = -\ln p_2$ with respect to z (rounded to two decimals) is

- (A) (0.05, -0.07, 0.01) (B) (-0.24, 0.33, -0.09)
 (C) (-0.76, -0.33, -0.91) (D) (0.24, -0.33, 0.09)

Q.22 [2 marks · NAT (numerical answer)]

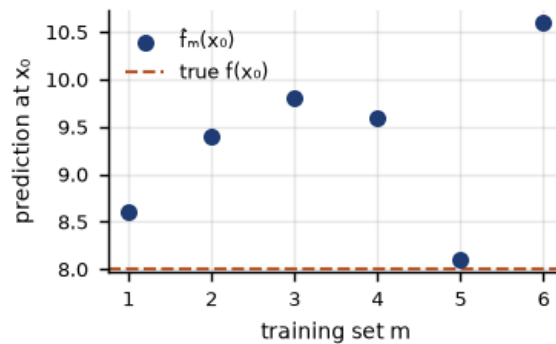
In C4.5, the attribute Product splits a node of 13 records as tabulated (class Yes/No counts in each branch). Compute the gain ratio of Product (all logarithms base 2). (rounded off to two decimal places)

Product	#Yes	#No
v1	4	0
v2	7	2

Q.23 [2 marks · NAT (numerical answer)]

Data are generated as $y = f(x) + \varepsilon$ with $E[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2 = 0.25$. At a test point x_0 the true value is $f(x_0) = 8$. A learning method is trained on 6 independent training sets, giving the predictions $\hat{f}_m(x_0)$ in the table (also plotted). Treating these as the full distribution of the method's prediction (variance with divisor 6), the expected squared prediction error $E[(y_0 - \hat{f}(x_0))^2]$ at x_0 is _____ (rounded off to three decimal places)

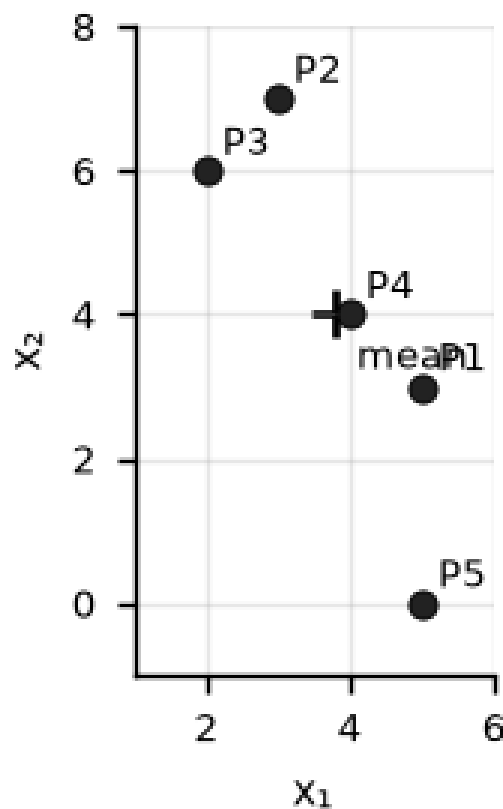
m	1	2	3	4	5	6
$\hat{f}_m(x_0)$	8.6	9.4	9.8	9.6	8.1	10.6



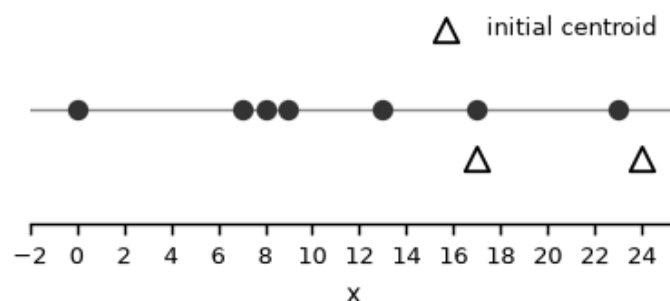
Q.24 [2 marks · NAT (numerical answer)]

PCA is applied to the five 2-D points below (data are mean-centered first; sample covariance uses divisor $n - 1$). Taking the unit-length first principal direction with a **positive first component**, the score (projection) of P5 on PC1 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5
x_1	5	3	2	4	5
x_2	3	7	6	4	0

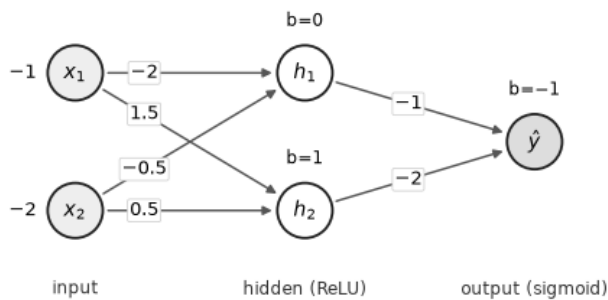
**Q.25 [2 marks · NAT (numerical answer)]**

The 1-D dataset $\{0, 7, 8, 9, 13, 17, 23\}$ is clustered with k-means ($k = 2$) using initial centroids 17 and 24 (shown as triangles). Lloyd's iterations (assign to nearest centroid, then recompute means) are run until convergence. Then the number of iterations in which at least one centroid **changes** its value is _____ (Answer in integer)



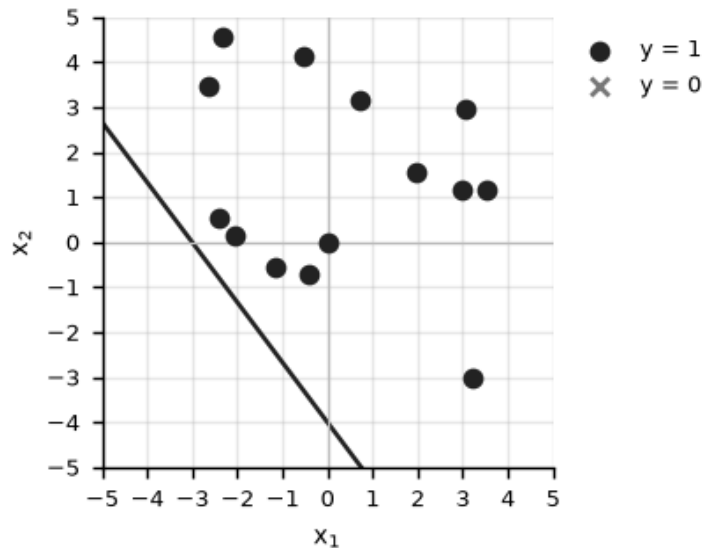
Q.26 [2 marks · NAT (numerical answer)]

The feed-forward network below has ReLU hidden units and a sigmoid output unit. The number on each edge is its weight and 'b' above a node is that node's bias. For input $\mathbf{x} = (-1, -2)$ (values shown left of the inputs), the network output $\hat{y}$ is _____ (rounded off to 4 decimal places)



Q.27 [2 marks · MCQ]

A logistic regression model $P(y = 1 \mid \mathbf{x}) = \sigma(w_0 + w_1x_1 + w_2x_2)$ perfectly classifies the training points shown below using threshold 0.5. Its decision boundary is the solid line (it crosses the axes at grid points). Which of the following parameter vectors is consistent with the figure?



- (A) $(w_0, w_1, w_2) = (24, -8, 6)$
- (B) $(w_0, w_1, w_2) = (24, 8, 6)$
- (C) $(w_0, w_1, w_2) = (-24, 6, 8)$
- (D) $(w_0, w_1, w_2) = (24, 8, -6)$

Q.28 [2 marks · NAT (numerical answer)]

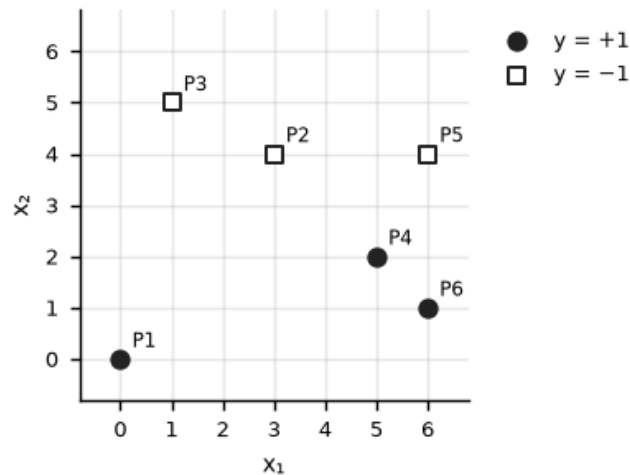
The actual and predicted labels of a binary classifier (1 = positive) on 13 test examples are given below. The F_β -score with $\beta = 0.5$, $F_\beta = (1 + \beta^2)PR/(\beta^2P + R)$, is _____ (rounded off to two decimal places)

#	1	2	3	4	5	6	7	8	9	10	11	12	13
actual y	1	0	0	1	1	0	0	0	0	0	1	0	0
predicted $\hat{y}$	1	1	0	1	0	1	0	0	1	0	0	0	1

Q.29 [2 marks · NAT (numerical answer)]

The labelled training points shown below are linearly separable. A hard-margin linear SVM is trained on them. Compute the width of the margin (distance between the two margin hyperplanes). (rounded off to two decimal places)

Point	x_1	x_2	y
P1	0	0	+1
P2	3	4	-1
P3	1	5	-1
P4	5	2	+1
P5	6	4	-1
P6	6	1	+1

**Q.30 [2 marks · NAT (numerical answer)]**

A naive Bayes classifier is trained on the 11 examples below, using maximum-likelihood estimates (no smoothing). For the new instance (Fever = None, Cough = Yes, Fatigue = No), the posterior probability $P(\text{Flu} = \text{Pos} \mid x)$ is _____ (rounded off to two decimal places)

#	Fever	Cough	Fatigue	Flu
1	None	Yes	Yes	Neg
2	None	Yes	No	Neg
3	None	Yes	Yes	Neg
4	None	Yes	No	Neg
5	None	No	Yes	Pos
6	None	No	No	Pos
7	Mild	Yes	No	Neg
8	None	Yes	Yes	Pos
9	None	No	No	Pos
10	Mild	No	Yes	Pos
11	None	No	Yes	Pos

Q.31 [2 marks · NAT (numerical answer)]

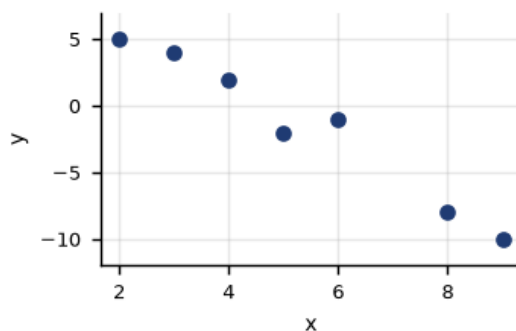
Agglomerative clustering with **complete linkage** is applied to the distance matrix below. The cophenetic distance between P3 and P6 (the dendrogram height at which they are first joined) is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5	P6
P1	0	9	6	2	8	6
P2		0	5	7	3	15
P3			0	4	4	10
P4				0	6	8
P5					0	14
P6						0

Q.32 [2 marks · NAT (numerical answer)]

Assume $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with ε_i i.i.d. $N(0, \sigma^2)$. For the 7 observations below, with β estimated by least squares (= maximum likelihood), the unbiased estimate $s^2 = \text{SSE}/(n - 2)$ of σ^2 is _____ (rounded off to two decimal places)

x	2	3	4	5	6	8	9
y	5	4	2	-2	-1	-8	-10

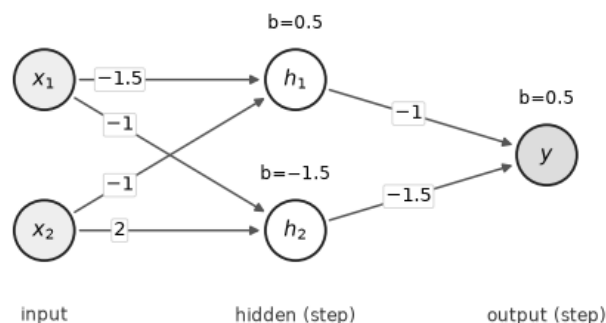
**Q.33 [2 marks · NAT (numerical answer)]**

Two clusterings U and V of 7 points are given below (cluster names are arbitrary). Then the number of point pairs on which the two clusterings **disagree** is _____ (Answer in integer)

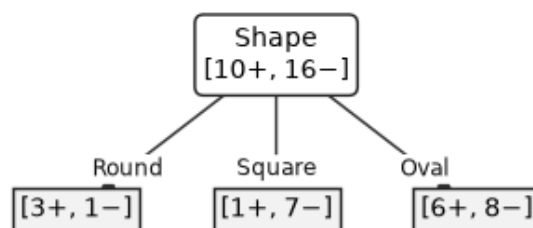
Point	P1	P2	P3	P4	P5	P6	P7
Clustering U	β	α	α	γ	β	γ	γ
Clustering V	2	1	1	3	2	1	3

Q.34 [2 marks · MCQ]

In the network below every unit (hidden and output) is a threshold unit that outputs 1 if its weighted input plus bias is > 0 , and 0 otherwise. Edge labels are weights. For binary inputs $x_1, x_2 \in \{0, 1\}$, the network computes

**(A)** x_1 (identity of x_1)**(B)** (NOT x_1) AND x_2 **(C)** NAND**(D)** NOR**Q.35 [2 marks · NAT (numerical answer)]**

A node with 26 training records (class counts [positive+, negative–]) is split on the attribute Shape as shown. Compute the weighted average entropy (in bits) of the children after splitting on Shape. (rounded off to two decimal places)



— END OF QUESTION PAPER —

Mock Test 25 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	45.3233 to 45.3433	19	MSQ	2	(A), (C), (D)
2	MSQ	1	(B), (C)	20	NAT	2	4.7523 to 4.8483
3	MCQ	1	(B)	21	MCQ	2	(D)
4	MSQ	1	(A), (C), (D)	22	NAT	2	0.0914 to 0.1114
5	MCQ	1	(B)	23	NAT	2	2.7235 to 2.7399
6	NAT	1	0.8233 to 0.8433	24	NAT	2	4.141 to 4.161
7	MSQ	1	(C)	25	NAT	2	2
8	MSQ	1	(A), (B), (D)	26	NAT	2	0.0176 to 0.0183
9	MCQ	1	(C)	27	MCQ	2	(B)
10	NAT	1	0.6567 to 0.6767	28	NAT	2	0.3471 to 0.3671
11	NAT	1	0.8789 to 0.8989	29	NAT	2	1.99 to 2.01
12	NAT	1	65	30	NAT	2	0.0937 to 0.1137
13	MSQ	1	(A), (C)	31	NAT	2	14.99 to 15.01
14	NAT	1	64	32	NAT	2	1.4032 to 1.4432
15	NAT	1	0.8606 to 0.8806	33	NAT	2	4
16	NAT	2	3.976 to 4.024	34	MCQ	2	(A)
17	NAT	2	18	35	NAT	2	0.8126 to 0.8326

Q	Type	M	Key	Q	Type	M	Key
18	NAT	2	0.078 to 0.0812				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Unsupervised Learning					11
SVM & Decision Trees					11
Regression & Model Selection					11
Probabilistic & Instance-based Classifiers					11
Neural Networks (MLP / Feed-forward)					11
Total					55

Mock Test 25 — Detailed Solutions

Q.1 — Answer: 45.3233 to 45.3433 [NAT, 1M · Unsupervised Learning > k-means]

Assignment (nearest centroid): $c = 9$: {3, 6, 11}; $c = 20$: {15, 18, 20}.

Updated centroids (cluster means): 6.67, 17.67.

$SSE = (3-6.67)^2 + (6-6.67)^2 + (11-6.67)^2 + (15-17.67)^2 + (18-17.67)^2 + (20-17.67)^2 = 45.33$.

(With the old centroids the SSE was 78; the update step can only lower it.)

Q.2 — Answer: (B), (C) [MSQ, 1M · SVM & Decision Trees > SVM: dual & support vectors]

(A) FALSE. Its boundary $w^T x + b = 0$ is linear.

(B) TRUE. $\alpha_i = 0$ for all other points.

(C) TRUE. Unlike hinge loss, it never becomes exactly zero.

(D) FALSE. Functional margin doubles; geometric margin is unchanged.

Q.3 — Answer: (B) [MCQ, 1M · Regression & Model Selection > MLE & MAP view]

$-\log$ Laplace prior $\propto \sum |w_j|/b$, an L1 penalty.

Answer: **lasso (L1-penalised least squares)**.

Q.4 — Answer: (A), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

(A) TRUE. $(TPR + TNR)/2$.

(B) FALSE. $\beta > 1$ favours recall.

(C) TRUE. Precision = $TPR \cdot \pi / (TPR \cdot \pi + FPR(1 - \pi))$.

(D) TRUE. Threshold moving is a common remedy for class imbalance.

Q.5 — Answer: (B) [MCQ, 1M · Unsupervised Learning > PCA & dimensionality reduction]

PC1 is the direction of maximum variance — along the long axis of the elongated cloud (direction A). PC2 must be orthogonal to PC1 and captures the remaining variance (direction B).

Answer: **Direction B**. Directions D and C are at 45° to the main axis and capture intermediate variance.

Q.6 — Answer: 0.8233 to 0.8433 [NAT, 1M · Unsupervised Learning > Distance measures]

Counts: $f_{11} = 1$, $f_{10} = 2$, $f_{01} = 3$, $f_{00} = 2$.

Jaccard $J = f_{11} / (f_{11} + f_{10} + f_{01}) = 1/6 = 0.1667$ (0-0 matches are ignored).

SMC = $(f_{11} + f_{00})/m = 3/8 = 0.375$.

Required Jaccard distance = **0.83**.

Q.7 — Answer: (C) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Expressiveness]

(A) FALSE. Its decision boundary $w \cdot x + b = 0$ is still linear.

(B) FALSE. Width must grow with the required accuracy.

(C) TRUE. It says nothing about learnability or sample complexity.

(D) FALSE. One hidden layer suffices.

Q.8 — Answer: (A), (B), (D) [MSQ, 1M · Regression & Model Selection > Simple linear regression]

(A) TRUE. r^2 is invariant to affine changes of either variable.

(B) TRUE. $(S_{xy}/S_{xx})(S_{xy}/S_{yy}) = r^2$.

(C) FALSE. The decomposition is $SST = SSR + SSE$.

(D) TRUE. $\hat{y}$ is a linear combination of 1 and x , both orthogonal to e .

Q.9 — Answer: (C) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Typical signatures: far-too-high $\eta \rightarrow$ loss blows up; somewhat-too-high $\eta \rightarrow$ quick drop then plateau at a high loss (bouncing around the minimum); too-low $\eta \rightarrow$ slow, nearly linear decrease; good $\eta \rightarrow$ fast decrease to the lowest loss.

Curve A: a learning rate that is too low (very slow, almost linear decrease).

Curve B: a learning rate that is far too high (training diverges).

Curve C: a well-chosen learning rate.

Curve D: a learning rate that is somewhat too high (fast initial drop, then the loss plateaus at a high value).

Hence the answer is **Curve C**.

Q.10 — Answer: 0.6567 to 0.6767 [NAT, 1M · SVM & Decision Trees > SVM: margin & geometry]

Since $x_+ - x_- = (3, 0)$ is parallel to w , the margin width equals $\|x_+ - x_-\| = 3$.

Margin = $2/\|w\| \Rightarrow \|w\| = 2/3 = \mathbf{0.67}$.

(Check: $w^T(x_+ - x_-) = 2$ and $w \parallel (x_+ - x_-)$ give the same result.)

Q.11 — Answer: 0.8789 to 0.8989 [NAT, 1M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

Expected cost of predicting positive = $(1 - p) \cdot C_{FP}$; of predicting negative = $p \cdot C_{FN}$.

Predict positive iff $p \cdot 1 > (1 - p) \cdot 8 \Leftrightarrow p > C_{FP}/(C_{FP} + C_{FN}) = 8/9 = \mathbf{0.89}$.

Q.12 — Answer: 65 [NAT, 1M · Regression & Model Selection > Cross-validation]

Per outer fold: inner CV $4 \times 3 = 12$ fits + 1 refit = 13.

Total = $5 \times 13 = \mathbf{65}$.

Q.13 — Answer: (A), (C) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

Net input $a = w \cdot x + b = -1x_1 + (3)x_2 + (-4)$; output +1 iff $a \geq 0$.

(P) $a = -1 \cdot -2 + (3) \cdot 1 + (-4) = 1 \rightarrow +1$ (selected).

(Q) $a = -1 \cdot 1 + (3) \cdot 0 + (-4) = -5 \rightarrow -1$.

(R) $a = -1 \cdot 1 + (3) \cdot 3 + (-4) = 4 \rightarrow +1$ (selected).

(S) $a = -1 \cdot -1 + (3) \cdot 0 + (-4) = -3 \rightarrow -1$.

Answer: **(A), (C)**.

Q.14 — Answer: 64 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

$x^T z = (2)(1) + (2)(3) + (-1)(2) = 6$.

$K(x, z) = (6 + 2)^2 = 8^2 = \mathbf{64}$.

The kernel equals the inner product $\phi(x)^T \phi(z)$ of the explicit (all-monomials-up-to-degree-d) feature maps, without forming ϕ .

Q.15 — Answer: 0.8606 to 0.8806 [NAT, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Volume of a cube with edge e is e^5 ; set $e^5 = 0.5$.

$e = 0.5^{1/5} = \mathbf{0.87}$.

In high dimensions even a small fraction of data needs an edge spanning most of each coordinate's range, so 'nearest' neighbours are not local — the curse of dimensionality.

Q.16 — Answer: 3.976 to 4.024 [NAT, 2M · Regression & Model Selection > Basis functions & feature scaling]

Design columns 1, x , x^2 . Because the x -values are symmetric about 0, $\Sigma x = \Sigma x^3 = 0$, so the normal equations decouple:

$\Sigma x^2 = 10$, $\Sigma x^4 = 34$, $\Sigma y = 14$, $\Sigma xy = -6$, $\Sigma x^2 y = 34$.

$b = \Sigma xy / \Sigma x^2 = -6/10 = -0.6$.

$[n, \Sigma x^2, \Sigma x^4][a; c] = [\Sigma y, \Sigma x^2 y] = 5a + 10c = 14$, $10a + 34c = 34$.

Solving: $c = (n \Sigma x^2 y - \Sigma x^2 \Sigma y) / (n \Sigma x^4 - (\Sigma x^2)^2) = 0.4286$, $a = (\Sigma y - c \Sigma x^2) / n = 1.9429$.

$\hat{y}(3) = 1.9429 + (-0.6)(3) + (0.4286)(3)^2 = 4$.

Answer: **4**.

Q.17 — Answer: 18 [NAT, 2M · SVM & Decision Trees > SVM: kernels]

$\|\phi(x) - \phi(z)\|^2 = K(x, x) + K(z, z) - 2K(x, z)$.

$K(x, x) = (1 + 0)^2 = 1$, $K(z, z) = (5 + 0)^2 = 25$, $K(x, z) = (-2 + 0)^2 = 4$.

Squared distance = $1 + 25 - 2(4) = \mathbf{18}$.

Q.18 — Answer: 0.078 to 0.0812 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

Forward: $a = Wx = (2.75, -2.75)$, $h = \sigma(a) = (0.9399, 0.0601)$, $\hat{y} = v \cdot h = 0.9099$.

Output error $\delta_o = \partial L / \partial \hat{y} = \hat{y} - y = 1.4099$ (linear output, $L = \frac{1}{2}(\hat{y} - y)^2$).

Hidden delta $\delta_1 = \delta_o \cdot v_1 \cdot h_1(1 - h_1) = 1.4099 \times 1 \times 0.0565 = 0.0796$.

$\partial L / \partial b_1 = \delta_1 = \mathbf{0.0796}$.

Q.19 — Answer: (A), (C), (D) [MSQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Distances from the query, sorted:

rank	#	(x_1, x_2)	distance	class
1	4	(2, 8)	$\sqrt{1.25} = 1.118$	A
2	12	(4, 9)	$\sqrt{3.25} = 1.803$	A
3	2	(1, 8)	$\sqrt{4.25} = 2.062$	B
4	5	(0, 7)	$\sqrt{9.25} = 3.041$	B
5	10	(5, 5)	$\sqrt{10.25} = 3.202$	B
6	11	(6, 5)	$\sqrt{15.25} = 3.905$	A
7	6	(6, 4)	$\sqrt{21.25} = 4.61$	A
8	7	(7, 5)	$\sqrt{22.25} = 4.717$	A

(A) TRUE. 1 nearest: #4 $\rightarrow$ 1 A vs 0 B $\Rightarrow$ class A.

(B) FALSE. 3 nearest: #4, #12, #2 $\rightarrow$ 2 A vs 1 B $\Rightarrow$ class A.

(C) TRUE. 5 nearest: #4, #12, #2, #5, #10 $\rightarrow$ 2 A vs 3 B $\Rightarrow$ class B.

(D) TRUE. 7 nearest: #4, #12, #2, #5, #10, #11, #6 $\rightarrow$ 4 A vs 3 B $\Rightarrow$ class A.

Q.20 — Answer: 4.7523 to 4.8483 [NAT, 2M · Regression & Model Selection > Cross-validation]

$\bar{x} = 5.667$, $\bar{y} = -3$.

i	1	2	3	4	5	6
$x_i - \bar{x}$	-4.67	-1.67	-0.67	0.33	2.33	4.33
$y_i - \bar{y}$	11	1	1	0	-5	-8
$(x_i - \bar{x})(y_i - \bar{y})$	-51.33	-1.67	-0.67	0	-11.67	-34.67

$S_{xx} = \Sigma(x_i - \bar{x})^2 = 49.333$, $S_{xy} = -100$.

$\beta_1 = S_{xy} / S_{xx} = -2.027$, $\beta_0 = \bar{y} - \beta_1 \bar{x} = 8.4865$.

For a linear smoother, the leave-one-out residual is $e_{(i)} = e_i / (1 - h_{ii})$, with $h_{ii} = 1/n + (x_i - \bar{x})^2 / S_{xx}$.

x_i	1	4	5	6	8	10
e_i	1.541	-2.378	-0.351	0.676	-0.27	0.784
h_{ii}	0.608	0.223	0.176	0.169	0.277	0.547
$e_i / (1 - h_{ii})$	3.931	-3.061	-0.426	0.813	-0.374	1.731

$CV_{(n)} = (1/6) \Sigma(e_i / (1 - h_{ii}))^2 = 4.8003$.

Answer: **4.8** (verified by explicitly refitting without the point).

Q.21 — Answer: (D) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$p = \text{softmax}((0, 1, -1)) = (0.2447, 0.6652, 0.09)$; $y = (0, 1, 0)$.

$\partial L / \partial z = p - y = \mathbf{(0.24, -0.33, 0.09)}$.

Distractors: $y - p = (-0.24, 0.33, -0.09)$ (wrong sign); p alone = $(0.24, 0.67, 0.09)$ (forgot the target); $(p - y) \odot p \odot (1 - p) = (0.05, -0.07, 0.01)$ (spurious σ' -type factor, which cancels for cross-entropy); $-y/p = (0, -1.5, 0)$ is $\partial L / \partial p$, not $\partial L / \partial z$.

Q.22 — Answer: 0.0914 to 0.1114 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [11 Yes, 2 No]: $H = 0.6194$.

Child entropies: $v1: 0$, $v2: 0.7642$.

$IG = 0.6194 - \Sigma(n_v/13)H_v = 0.0903$.

SplitInfo = $-\Sigma(n_v/N) \log_2(n_v/N) = -(4/13) \log_2(4/13) - (9/13) \log_2(9/13) = 0.8905$.

Gain ratio = IG / SplitInfo = 0.0903/0.8905 = **0.1**.

C4.5 uses the gain ratio to penalise attributes that split the data into many small branches.

Q.23 — Answer: 2.7235 to 2.7399 [NAT, 2M · Regression & Model Selection > Bias-variance trade-off]

Mean prediction $\bar{f} = (1/6)\sum f_m = 9.35$.

Bias² = $(\bar{f} - f(x_0))^2 = (9.35 - 8)^2 = 1.8225$.

Variance = $(1/6)\sum (f_m - \bar{f})^2 = 0.6592$.

Expected test error = Bias² + Variance + $\sigma^2 = 1.8225 + 0.6592 + 0.25 = 2.7317$.

Answer: **2.732**.

Q.24 — Answer: 4.141 to 4.161 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

Mean $\bar{x} = (3.8, 4)$. Centered data rows: P1: (1.2, -1), P2: (-0.8, 3), P3: (-1.8, 2), P4: (0.2, 0), P5: (1.2, -4).

Sample covariance $S = (1/(n-1)) \sum (x - \bar{x})(x - \bar{x})^T = [[1.7, -3], [-3, 7.5]]$.

Eigenvalues: $\lambda_1 = 8.7725$, $\lambda_2 = 0.4275$; unit PC1 direction $v_1 = (0.3905, -0.9206)$.

Score $z_1 = v_1 \cdot (x - \bar{x}) = 0.3905 \cdot 1.2 + -0.9206 \cdot -4 = 4.151$.

Answer: **4.15** (the sign depends on the orientation of v_1 ; with the stated convention it is 4.15).

Q.25 — Answer: 2 [NAT, 2M · Unsupervised Learning > k-means]

Data: {0, 7, 8, 9, 13, 17, 23}; initial centroids $c_1 = 17$, $c_2 = 24$.

Iter	centroids used	cluster 1	cluster 2	new centroids
1	17, 24	{0, 7, 8, 9, 13, 17}	{23}	9, 23
2	9, 23	{0, 7, 8, 9, 13}	{17, 23}	7.4, 20
3	7.4, 20	{0, 7, 8, 9, 13}	{17, 23}	7.4, 20

In the last row the centroids do not change, so the algorithm has converged: final centroids 7.4 and 20.

Centroids changed in 2 iteration(s); the final iteration only confirms convergence. Answer **2**.

Q.26 — Answer: 0.0176 to 0.0183 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

Hidden pre-activations: $a_1 = -2 \cdot -1 + (-0.5) \cdot -2 + (0) = 3$; $a_2 = 1.5 \cdot -1 + (0.5) \cdot -2 + (1) = -1.5$.

Hidden activations ($\text{ReLU}(a) = \max(0, a)$): $h_1 = 3$, $h_2 = 0$.

Output pre-activation: $o = -1 \cdot 3 + (-2) \cdot 0 + (-1) = -4$.

$\hat{y} = \sigma(o) = \mathbf{0.018}$.

Q.27 — Answer: (B) [MCQ, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

The line passes through (-3, 0) and (0, -4): $x_1/-3 + x_2/-4 = 1$, i.e. $-4x_1 + -3x_2 - 12 = 0$. Hence (w_0, w_1, w_2) must be a non-zero multiple of $(-12, -4, -3)$.

Orientation: the class $y = 1$ region must have $w_0 + w_1x_1 + w_2x_2 > 0$. At the origin the $y = 1$ side has sign + ($w_0 = 24$), which matches the figure (the origin lies on the $y = 1$ side).

So the answer is $(w_0, w_1, w_2) = (24, 8, 6)$ (any positive multiple gives the same classifier).

Other options either describe a different line (swapped/negated coefficients move the intercepts) or the same line with the sign flipped, which would label every training point incorrectly.

Q.28 — Answer: 0.3471 to 0.3671 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Counting: TP = 2, FP = 4, FN = 2.

$P = 2/6 = 0.3333$, $R = 2/4 = 0.5$.

$F_{0.5} = (1 + 0.25) \times 0.3333 \times 0.5 / (0.25 \times 0.3333 + 0.5) = \mathbf{0.36}$.

$\beta < 1$ weights precision more heavily (F1 here would be 0.4).

Q.29 — Answer: 1.99 to 2.01 [NAT, 2M · SVM & Decision Trees > SVM: margin & geometry]

P2 and P5 (same class) lie on one margin line and P4 on the other; the optimal boundary is parallel to the line through P2, P5, midway to P4. Support vectors: P2, P4, P5.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (0, -1)$, $b = 3$; $\|w\| = 1$, margin $2/\|w\| = 2$.

Point	P1	P2	P3	P4	P5	P6
$y(w^T x + b)$	3	1	2	1	1	2

All values are ≥ 1 (feasible), with equality exactly at the support vectors.

Margin width = $2/\|w\| = 2$.

Q.30 — Answer: 0.0937 to 0.1137 [NAT, 2M · Probabilistic & Instance-based Classifiers · Naive Bayes]

Naive Bayes: $P(c | x) \propto P(c) \prod_j P(x_j | c)$.

Pos: $P(\text{Flu}=\text{Pos}) = 6/11$; $P(\text{Fever}=\text{None}|\text{Pos}) = 5/6$; $P(\text{Cough}=\text{Yes}|\text{Pos}) = 1/6$; $P(\text{Fatigue}=\text{No}|\text{Pos}) = 2/6$. Product = 0.02525.

Neg: $P(\text{Flu}=\text{Neg}) = 5/11$; $P(\text{Fever}=\text{None}|\text{Neg}) = 4/5$; $P(\text{Cough}=\text{Yes}|\text{Neg}) = 5/5$; $P(\text{Fatigue}=\text{No}|\text{Neg}) = 3/5$. Product = 0.21818.

$P(\text{Pos} | x) = 0.02525 / (0.02525 + 0.21818) = \mathbf{0.1}$.

Q.31 — Answer: 14.99 to 15.01 [NAT, 2M · Unsupervised Learning · Hierarchical clustering]

Use complete linkage: $d(A, B) = \max_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P1\} + \{P4\}$ at height **2**. Updated: $d(\{P1, P4\}, \{P2\}) = 9$, $d(\{P1, P4\}, \{P3\}) = 6$, $d(\{P1, P4\}, \{P5\}) = 8$, $d(\{P1, P4\}, \{P6\}) = 8$.

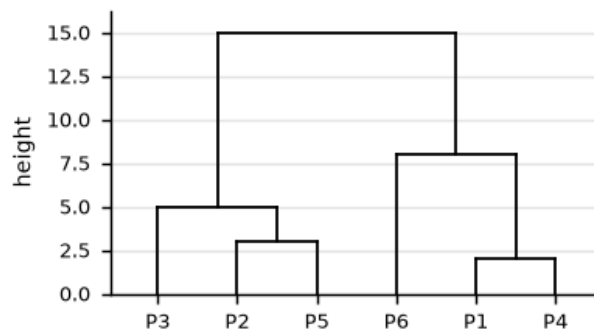
Merge 2: $\{P2\} + \{P5\}$ at height **3**. Updated: $d(\{P2, P5\}, \{P3\}) = 5$, $d(\{P2, P5\}, \{P6\}) = 15$, $d(\{P2, P5\}, \{P1, P4\}) = 9$.

Merge 3: $\{P3\} + \{P2, P5\}$ at height **5**. Updated: $d(\{P2, P3, P5\}, \{P6\}) = 15$, $d(\{P2, P3, P5\}, \{P1, P4\}) = 9$.

Merge 4: $\{P6\} + \{P1, P4\}$ at height **8**. Updated: $d(\{P1, P4, P6\}, \{P2, P3, P5\}) = 15$.

Merge 5: $\{P2, P3, P5\} + \{P1, P4, P6\}$ at height **15**.

P3 and P6 first lie in the same cluster at the merge of height 15, so their cophenetic distance is **15** (their original distance was 10).



Q.32 — Answer: 1.4032 to 1.4432 [NAT, 2M · Regression & Model Selection · MLE & MAP view]

$\bar{x} = 5.286$, $\bar{y} = -1.429$.

i	1	2	3	4	5	6	7
$x_i - \bar{x}$	-3.29	-2.29	-1.29	-0.29	0.71	2.71	3.71
$y_i - \bar{y}$	6.43	5.43	3.43	-0.57	0.43	-6.57	-8.57
$(x_i - \bar{x})(y_i - \bar{y})$	-21.12	-12.41	-4.41	0.16	0.31	-17.84	-31.84

$S_{xx} = \sum (x_i - \bar{x})^2 = 39.429$, $S_{xy} = -87.143$.

$\beta_1 = S_{xy} / S_{xx} = -2.2101$, $\beta_0 = \bar{y} - \beta_1 \bar{x} = 10.2536$.

$SSE = S_{yy} - S_{xy}^2 / S_{xx} = 199.7143 - 192.5983 = 7.1159$.

Two parameters are estimated, so $s^2 = SSE / (n - 2) = 7.1159 / 5 = 1.4232$.

Answer: **1.42**.

Q.33 — Answer: 4 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

Total pairs = $C(7,2) = 21$.

a = pairs together in both = 3; b = pairs separated in both = 14; pairs together in U only = 2; together in V only = 2.

Rand index = $(a + b)/C(n,2) = (3 + 14)/21 = 0.8095$.

Answer: **4**. (Cluster names do not matter: only whether pairs are together.)

Q.34 — Answer: (A) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Expressiveness]

Each unit outputs 1 iff its net input (weighted sum + bias) is > 0 . Evaluate on all four inputs:

x_1	x_2	h_1	h_2	y
0	0	1	0	0
0	1	0	1	0
1	0	0	0	1
1	1	0	0	1

h_1 computes NOR, h_2 computes (NOT x_1) AND x_2 ; the output column (0,0)→(1,1) is 0, 0, 1, 1, i.e. x_1 (**identity of x_1**).

Q.35 — Answer: 0.8126 to 0.8326 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [10+, 16-]: $H = -(10/26)\log_2(10/26) - (16/26)\log_2(16/26) = 0.9612$.

Shape = Round: [3+, 1-], $H = -(3/4)\log_2(3/4) - (1/4)\log_2(1/4) = 0.8113$.

Shape = Square: [1+, 7-], $H = -(1/8)\log_2(1/8) - (7/8)\log_2(7/8) = 0.5436$.

Shape = Oval: [6+, 8-], $H = -(6/14)\log_2(6/14) - (8/14)\log_2(8/14) = 0.9852$.

Weighted child entropy = $(4/26)(0.8113) + (8/26)(0.5436) + (14/26)(0.9852) = 0.8226$.

IG = $0.9612 - 0.8226 = 0.1387$.

Answer: **0.82**.

Mock Test 26

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	5	12	18

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Neural Networks (MLP / Feed-forward)	7	11
SVM & Decision Trees	7	11
Regression & Model Selection	7	11
Unsupervised Learning	7	11
Probabilistic & Instance-based Classifiers	7	11

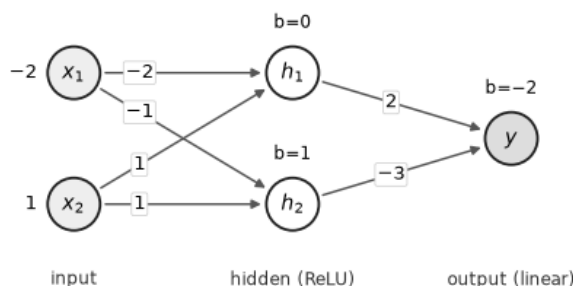
Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

For the ReLU network shown (edge labels are weights, 'b' denotes a bias) and input $x = (-2, 1)$, the output y is _____ (Answer in integer)



Q.2 [1 mark · MSQ (one or more correct)]

Let $p = \text{softmax}(z)$ for logits $z \in \mathbb{R}^K$ and $L = -\ln p_y$ the cross-entropy loss for true class y . Which of the following is/are CORRECT?

- (A) The sum of the softmax probabilities depends on the scale of the logits.
- (B) The cross-entropy loss $-\ln p_y$ can be negative.
- (C) The Jacobian of softmax is $\partial p_i / \partial z_j = p_i (\delta_{ij} - p_j)$.
- (D) For softmax + cross-entropy, $\partial L / \partial z_k = (p_k - y_k) p_k (1 - p_k)$.

Q.3 [1 mark · NAT (numerical answer)]

The Gaussian kernel is $K(x, z) = \exp(-\|x - z\|^2 / (2\sigma^2))$ with $\sigma = 1$. For $x = (1, -1)$ and $z = (1, 1)$, the value of $K(x, z)$ is _____ (rounded off to three decimal places)

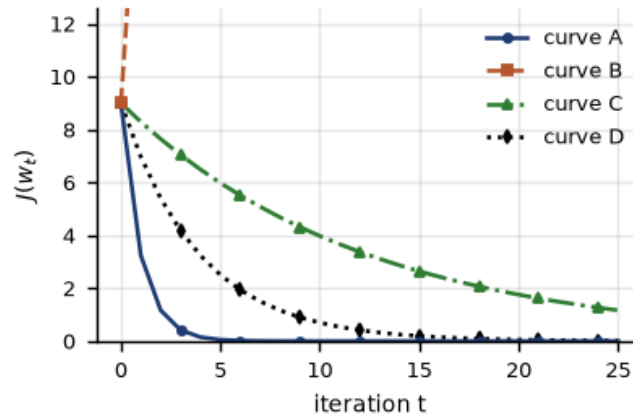
Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about initialisation and regularisation of neural networks is/are CORRECT?

- (A) Data augmentation can reduce overfitting.
- (B) Training loss decreasing while validation loss increases is a sign of overfitting.
- (C) Dropout is switched off at inference (test) time.
- (D) High training loss and high validation loss together indicate overfitting.

Q.5 [1 mark · MCQ]

Gradient descent $w_{t+1} = w_t - \eta J'(w_t)$ is run on $J(w) = (w - 4)^2$ from $w_0 = 1$ with four learning rates $\eta \in \{0.02, 0.06, 0.2, 1.25\}$. The loss curves are shown below (curves that leave the plotted range are cut off). Which curve corresponds to $\eta = 0.02$?



- (A) Curve A
- (B) Curve D
- (C) Curve B
- (D) Curve C

Q.6 [1 mark · MSQ (one or more correct)]

Which of the following statements about principal component analysis (PCA) is/are CORRECT?

- (A) The sample covariance matrix of n points has at most $\min(n - 1, d)$ non-zero eigenvalues.
- (B) PCA on the uncentered data matrix gives the same directions as PCA on centered data for every dataset.
- (C) Projecting onto the top- k PCs minimizes the mean squared reconstruction error over all k -dimensional linear subspaces (through the mean).
- (D) Scores on different principal components are uncorrelated.

Q.7 [1 mark · NAT (numerical answer)]

A binary decision tree must have at least 61 leaves to separate 61 distinct classes. The minimum possible depth of such a tree is _____ (Answer in integer)

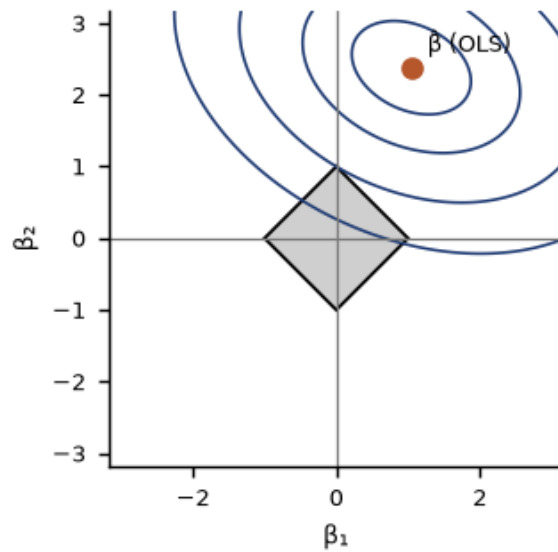
Q.8 [1 mark · MSQ (one or more correct)]

Which of the following statements about support vector machines is/are CORRECT?

- (A) The kernel trick computes inner products in a feature space without explicitly constructing $\phi(x)$.
- (B) A point with functional margin $y(w^T x + b) = 0.5$ is misclassified.
- (C) Removing a training point that is not a support vector does not change the hard-margin SVM solution.
- (D) The geometric margin of a point equals its functional margin divided by $\|w\|$.

Q.9 [1 mark · MCQ]

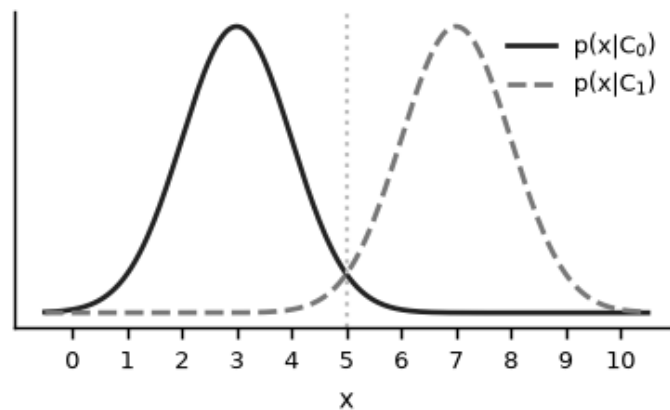
For a two-predictor penalised least-squares problem, the figure shows the constraint region (shaded) of the penalty together with elliptical contours of the residual sum of squares centred at the least-squares estimate β (OLS). The penalised estimate (β_1, β_2) satisfies



- (A) $\beta_1 = \beta_2 = 0$
- (B) $\beta_2 = 0$ exactly and $\beta_1 \neq 0$
- (C) $\beta_1 = 0$ exactly and $\beta_2 \neq 0$
- (D) $\beta_1 \neq 0$ and $\beta_2 \neq 0$ (both shrunk, neither exactly zero)

Q.10 [1 mark · MCQ]

Two classes have unit-variance Gaussian class-conditional densities with means 3 (C_0) and 7 (C_1), as plotted. With equal priors the Bayes decision threshold is the dotted line at $x = 5$. If the prior of C_0 is increased to 0.7, the threshold



- (A) It stays at $x = 5$ because the two variances are equal
- (B) It moves towards $x = 3$, shrinking the region assigned to C_0
- (C) It splits into two thresholds, giving a quadratic boundary
- (D) It moves towards $x = 7$, enlarging the region assigned to C_0

Q.11 [1 mark · NAT (numerical answer)]

A 4-class softmax classifier produces the logits $z_1 = 2, z_2 = 1, z_3 = -2, z_4 = -1$ for an input $\mathbf{x}$. The predicted probability of class 1, $P(y = 1 | \mathbf{x})$, is _____ (rounded off to two decimal places)

Q.12 [1 mark · MSQ (one or more correct)]

Which of the following statements about clustering evaluation measures is/are CORRECT?

- (A) The Rand index can compare two clusterings with different numbers of clusters.
- (B) A silhouette close to 1 means the point is much closer to its own cluster than to the nearest other cluster.
- (C) A negative silhouette suggests that the point may be assigned to the wrong cluster.
- (D) The Rand index is unchanged if the cluster labels are renamed (e.g. $1 \leftrightarrow 2$).

Q.13 [1 mark · MSQ (one or more correct)]

Regarding the bias-variance trade-off in supervised regression with squared-error loss, which of the following statements is/are CORRECT?

- (A) A model with zero training error must have zero test error.
- (B) For a fixed model class, more training data tends to reduce variance.
- (C) Ridge regression with a larger λ has higher variance.
- (D) Increasing k in k -NN regression increases its variance.

Q.14 [1 mark · MSQ (one or more correct)]

Which of the following statements about naive Bayes classifiers is/are CORRECT?

- (A) Without smoothing, a feature value never observed with class c in training forces the posterior of c to 0 for any input containing that value.
- (B) Gaussian naive Bayes with class-specific variances generally yields a quadratic decision boundary.
- (C) Naive Bayes posterior probabilities are often over-confident (close to 0 or 1) when features are correlated.
- (D) In practice, naive Bayes sums log-probabilities to avoid numerical underflow.

Q.15 [1 mark · NAT (numerical answer)]

Agglomerative hierarchical clustering of $n = 55$ points (until one cluster remains) performs how many merges? (Answer in integer)

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

A classifier for a rare condition has sensitivity (recall) 0.8 and specificity 0.98. It is deployed on a population where the prevalence of the positive class is 0.01. The expected precision (positive predictive value) in this population is _____ (rounded off to three decimal places)

Q.17 [2 marks · NAT (numerical answer)]

Consider the model $y = \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ (no intercept), fitted by least squares using $\beta = (X^T X)^{-1} X^T y$ with the design matrix X and response y below. Then the least-squares estimate of β_2 is _____ (rounded off to two decimal places)

$$X = \begin{bmatrix} 3 & 2 \\ 3 & -2 \\ -1 & 1 \end{bmatrix}$$

$$y = \begin{bmatrix} 7 \\ 1 \\ -3 \end{bmatrix}$$

Q.18 [2 marks · MSQ (one or more correct)]

Consider agglomerative hierarchical clustering on n points with a dissimilarity matrix (no ties). Which of the following statements is/are CORRECT?

- (A) Different linkages applied to the same distance matrix must yield the same 2-cluster partition.
- (B) Ward's merge cost for clusters A and B is $(|A||B|/(|A|+|B|)) \cdot \|\mu_A - \mu_B\|^2$.
- (C) Replacing every dissimilarity by its square never changes the average-linkage dendrogram topology.
- (D) Cutting the single-linkage dendrogram into k clusters is equivalent to deleting the $k - 1$ heaviest edges of an MST.

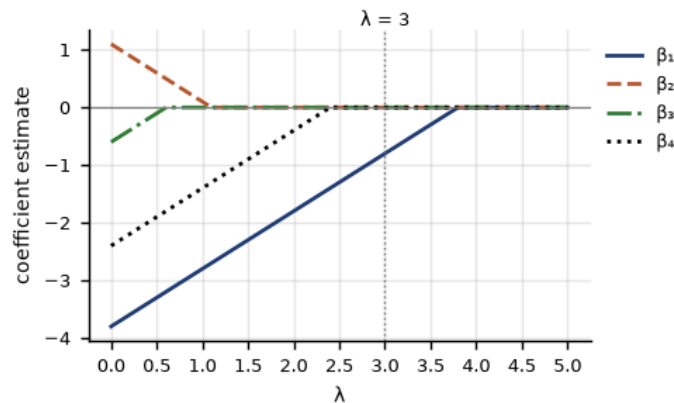
Q.19 [2 marks · NAT (numerical answer)]

The predicted probabilities $\hat{p}_i = P(y_i = 1 | x_i)$ of a binary classifier on 4 examples, and their true labels, are given below. Using natural logarithms, the **average** binary cross-entropy loss $-\{y \ln \hat{p} + (1 - y) \ln(1 - \hat{p})\}$ over these examples is _____ (rounded off to two decimal places)

i	1	2	3	4
y_i	0	0	0	1
$\hat{p}_i = P(y_i=1)$	0.65	0.95	0.35	0.9

Q.20 [2 marks · MSQ (one or more correct)]

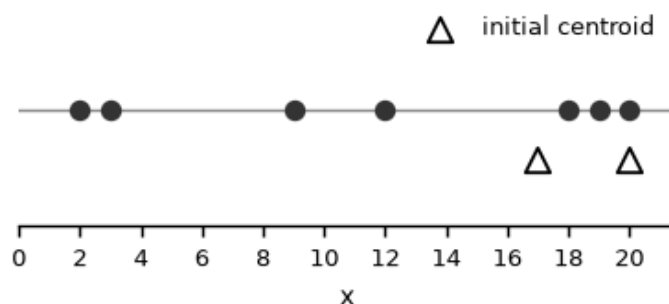
For a design with orthonormal columns ($X^T X = I$), the figure shows the coefficient paths $\beta_j(\lambda)$ of a penalised least-squares estimator that minimises $\frac{1}{2}\|y - X\beta\|^2 + P_\lambda(\beta)$, where the penalty P_λ is either the ridge penalty $(\lambda/2)\|\beta\|_2^2$ or the lasso penalty $\lambda\|\beta\|_1$. Which of the following statements is/are CORRECT?



- (A) At $\lambda = 0$ the coefficients equal the least-squares estimates.
- (B) Each coefficient path is piecewise linear in λ .
- (C) At $\lambda = 3$, exactly 3 coefficient(s) are zero.
- (D) As λ increases, $|\beta_j(\lambda)|$ is non-increasing for every j .

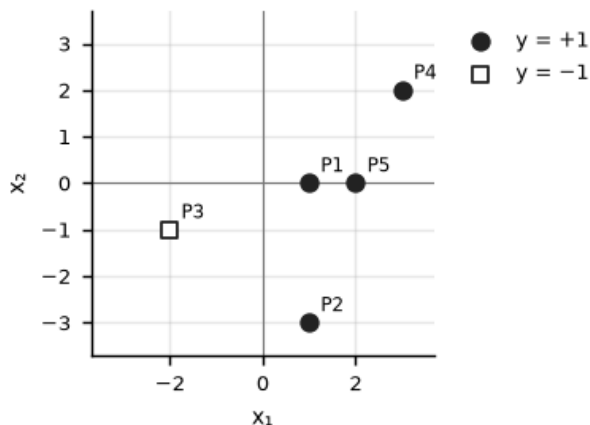
Q.21 [2 marks · NAT (numerical answer)]

The 1-D dataset $\{2, 3, 9, 12, 18, 19, 20\}$ is clustered with k-means ($k = 2$) using initial centroids 17 and 20 (shown as triangles). Lloyd's iterations (assign to nearest centroid, then recompute means) are run until convergence. Then the number of iterations in which at least one centroid **changes** its value is _____ (Answer in integer)

**Q.22 [2 marks · NAT (numerical answer)]**

A perceptron with augmented weight vector $w = (w_0, w_1, w_2)$ (w_0 is the bias, $\tilde{x} = (1, x_1, x_2)$) starts from $w = (1, 0, 1)$ and uses learning rate $\eta = 0.5$. The points $P_1, P_2, \dots$ are presented once, in order; whenever $y \cdot w^T \tilde{x} \leq 0$ the update $w \leftarrow w + \eta y \tilde{x}$ is applied. Find the value of w_0 (bias) after this epoch. (rounded off to one decimal place)

Point	x_1	x_2	y
P1	1	0	+1
P2	1	-3	+1
P3	-2	-1	-1
P4	3	2	+1
P5	2	0	+1

**Q.23 [2 marks · NAT (numerical answer)]**

For an input x , a classifier's posterior probabilities are $P(C_1|x) = 0.2$, $P(C_2|x) = 0.5$, $P(C_3|x) = 0.3$. The loss of taking action α_i (deciding C_i) when the true class is C_j is given below. Then the conditional risk of the action chosen by the MAP rule (i.e. choosing the class with highest posterior) is _____ (rounded off to two decimal places)

loss $L(\alpha_i, C_j)$	true C_1	true C_2	true C_3
α_1 : decide C_1	0	2	9
α_2 : decide C_2	7	0	10
α_3 : decide C_3	9	7	0

Q.24 [2 marks • MCQ]

For $n = 9$ paired observations: $\Sigma x = 67$, $\Sigma y = -77$, $\Sigma x^2 = 665$, $\Sigma y^2 = 857$, $\Sigma xy = -735$. The least-squares regression line of y on x is (coefficients rounded to two decimals)

(A) $\hat{y} = -1.31 - 0.97x$

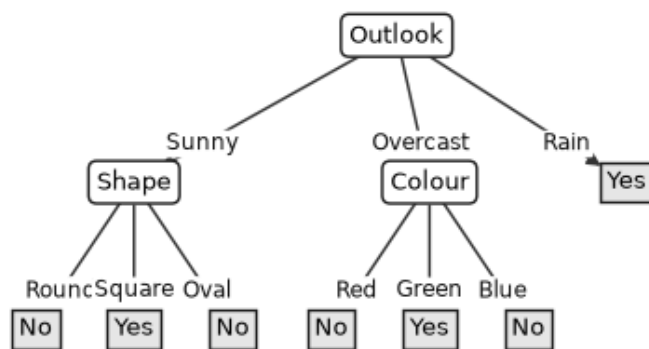
(B) $\hat{y} = -15.8 - 0.97x$

(C) $\hat{y} = -0.33 - 1.11x$

(D) $\hat{y} = 0.57 - 1.23x$

Q.25 [2 marks • NAT (numerical answer)]

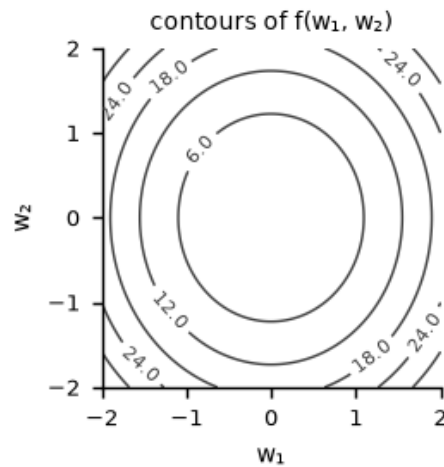
The decision tree shown is applied to the 9 labelled records in the table (attributes not tested on a path are ignored). Find the number of records misclassified by the tree. (Answer in integer)



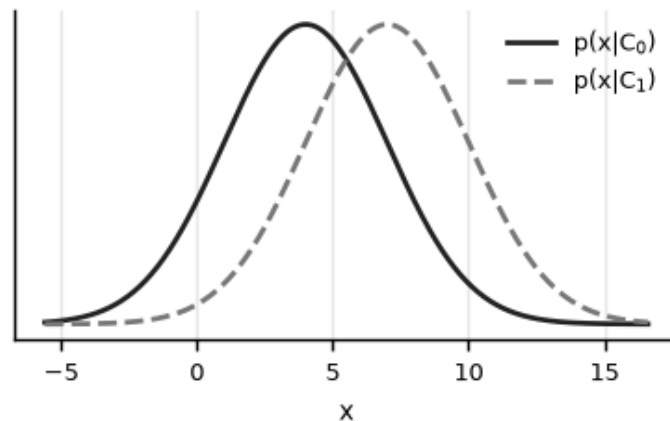
#	Outlook	Shape	Colour	Actual
1	Overcast	Oval	Blue	No
2	Sunny	Oval	Green	No
3	Overcast	Square	Blue	Yes
4	Rain	Round	Blue	Yes
5	Sunny	Square	Green	Yes
6	Sunny	Oval	Green	No
7	Sunny	Oval	Blue	Yes
8	Sunny	Square	Red	Yes
9	Overcast	Round	Blue	Yes

Q.26 [2 marks · MSQ (one or more correct)]

Gradient descent $w \leftarrow w - \eta \nabla f(w)$ is applied to $f(w_1, w_2) = \frac{1}{2}(10w_1^2 + 8w_2^2)$ from a generic starting point (both coordinates non-zero). For which of the following learning rates does GD **diverge**?

**(A)** $\eta = 0.22$ **(B)** $\eta = 0.12$ **(C)** $\eta = 0.08$ **(D)** $\eta = 0.05$ **Q.27 [2 marks · NAT (numerical answer)]**

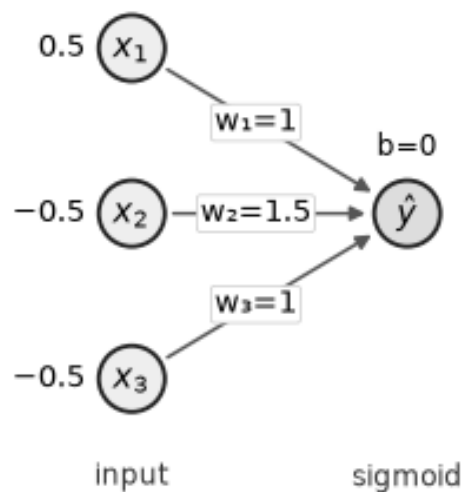
In a two-class problem with one feature, $p(x | C_0) = N(4, 9)$ and $p(x | C_1) = N(7, 9)$ (second argument = variance), with priors $P(C_0) = 0.25$ and $P(C_1) = 0.75$. Under 0-1 loss, the posterior probability $P(C_1 | x = 6)$ is _____ (rounded off to two decimal places)



Class-conditional densities (not weighted by priors).

Q.28 [2 marks · NAT (numerical answer)]

A single sigmoid neuron (shown below) receives $x = (0.5, -0.5, -0.5)$ and has target $y = 0$. The loss is $L = \frac{1}{2}(\hat{y} - y)^2$. The value of $\partial L / \partial w_2$ is _____ (rounded off to 4 decimal places)

**Q.29 [2 marks · MSQ (one or more correct)]**

A node with class counts [60+, 20-] can be split by either of two binary splits S1 or S2 (table). For each impurity measure, the “gain” is the parent impurity minus the weighted child impurity. Which of the following statements is/are CORRECT?

Split	Left child [+, -]	Right child [+, -]
S1	[13+, 7-]	[47+, 13-]
S2	[40+, 4-]	[20+, 16-]

- (A) Gini gain prefers S1 over S2.
- (B) The information gain of S2 is 0.12 bits (to two decimals).
- (C) Gini gain prefers S2 over S1.
- (D) Misclassification-error reduction is the same for S1 and S2.

Q.30 [2 marks · MSQ (one or more correct)]

Which of the following statements about PCA, SVD and dimensionality reduction is/are CORRECT?

- (A) For a covariance matrix $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$ with $b > 0$, PC1 is along $(1, 1)/\sqrt{2}$.
- (B) Rotating the data by an orthogonal matrix rotates the PC directions but leaves the eigenvalues unchanged.
- (C) With n samples and $d \geq n$ features, the centered data matrix has rank at most $n - 1$.
- (D) PCA on standardized features is equivalent to eigen-decomposition of the correlation matrix.

Q.31 [2 marks · NAT (numerical answer)]

A simple linear regression fitted by ordinary least squares gives $\hat{y} = 7.9 + 2.9x$. After standardising x as $z = (x - \bar{x})/s_x$, with $\bar{x} = 20$ and $s_x = 9$, the model is refitted as $\hat{y} = \gamma_0 + \gamma_1 z$. The value of γ_0 is _____ (rounded off to two decimal places)

Q.32 [2 marks · NAT (numerical answer)]

A 32-20-10 fully connected network (with biases) is modified by inserting a new hidden layer of 8 units between the 20-unit layer and the output, giving 32-20-8-10. The change in the number of trainable parameters (new minus old) is _____ (Answer in integer)

Q.33 [2 marks · NAT (numerical answer)]

Three customers are described by age and income. Age ranges over [20, 60] years and income over [10, 210] thousand rupees in the full dataset. After min-max scaling of each feature to [0, 1] (using the stated ranges), the Euclidean distance from A to its **nearest** neighbour among B and C is _____ (rounded off to two decimal places)

Customer	Age (years)	Income (₹ thousand)
A	51	165
B	29	185
C	40	90

Q.34 [2 marks · MCQ]

A decision tree is to be grown on the 10 training records below using Gini gain as the splitting criterion (multi-way split on each categorical attribute). Which attribute is chosen at the root?

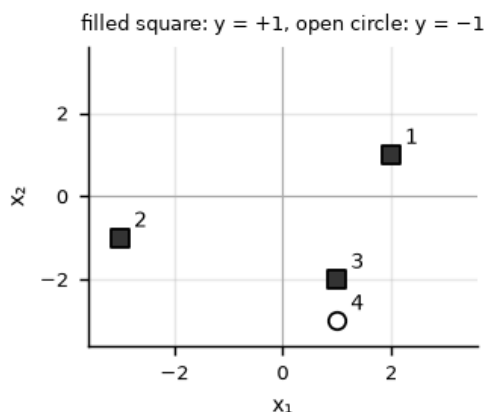
#	Outlook	Colour	Size	Shape	Class
1	Rain	Blue	Small	Round	Yes
2	Sunny	Blue	Small	Oval	Yes
3	Overcast	Blue	Large	Square	No
4	Rain	Red	Small	Square	Yes
5	Overcast	Green	Small	Round	No
6	Overcast	Green	Small	Round	Yes
7	Overcast	Red	Large	Round	No
8	Sunny	Blue	Small	Square	No
9	Rain	Blue	Large	Square	No
10	Sunny	Green	Large	Square	No

(A) Outlook**(B)** Colour**(C)** Size**(D)** Shape**Q.35 [2 marks · NAT (numerical answer)]**

A perceptron with parameters $w \in \mathbb{R}^2$, b is trained on the data below, starting from $w = (0, 0)$, $b = 0$, with learning rate $\eta = 1$. Examples are presented once in the order 1, 2, 3, 4 (one epoch).

An update $w \leftarrow w + \eta y x$, $b \leftarrow b + \eta y$ is made whenever $y(w \cdot x + b) \leq 0$. After the epoch, the weight w_1 is _____ (Answer in integer)

i	x_1	x_2	y
1	2	1	1
2	-3	-1	1
3	1	-2	1
4	1	-3	-1



— END OF QUESTION PAPER —

Mock Test 26 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	−4	19	NAT	2	1.1354 to 1.1554
2	MSQ	1	(C)	20	MSQ	2	(A), (B), (C), (D)
3	NAT	1	0.1333 to 0.1373	21	NAT	2	2
4	MSQ	1	(A), (B), (C)	22	NAT	2	0.99 to 1.01
5	MCQ	1	(D)	23	NAT	2	4.39 to 4.41
6	MSQ	1	(A), (C), (D)	24	MCQ	2	(A)
7	NAT	1	6	25	NAT	2	3
8	MSQ	1	(A), (C), (D)	26	MSQ	2	(A)
9	MCQ	1	(C)	27	NAT	2	0.7699 to 0.7899
10	MCQ	1	(D)	28	NAT	2	−0.0357 to −0.0343
11	NAT	1	0.6864 to 0.7064	29	MSQ	2	(B), (C), (D)
12	MSQ	1	(A), (B), (C), (D)	30	MSQ	2	(A), (B), (C), (D)
13	MSQ	1	(B)	31	NAT	2	65.7682 to 66.0318
14	MSQ	1	(A), (B), (C), (D)	32	NAT	2	48
15	NAT	1	54	33	NAT	2	0.455 to 0.475
16	NAT	2	0.2858 to 0.2898	34	MCQ	2	(C)
17	NAT	2	1.1447 to 1.1847	35	NAT	2	−2
18	MSQ	2	(B), (D)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Neural Networks (MLP / Feed-forward)					11
SVM & Decision Trees					11
Regression & Model Selection					11
Unsupervised Learning					11
Probabilistic & Instance-based Classifiers					11
Total					55

Mock Test 26 — Detailed Solutions

Q.1 — Answer: -4 [NAT, 1M · Neural Networks (MLP / Feed-forward) › Forward pass]

$a = Wx + b = (5, 4)$; $h = \text{ReLU}(a) = (5, 4)$.

$y = 2 \cdot 5 + (-3) \cdot 4 + (-2) = -4$.

A common slip is to forget the ReLU and use the raw pre-activation (which would give -4).

Q.2 — Answer: (C) [MSQ, 1M · Neural Networks (MLP / Feed-forward) › Softmax + cross-entropy]

(A) FALSE. It always equals 1.

(B) FALSE. $p_y \leq 1 \Rightarrow -\ln p_y \geq 0$.

(C) TRUE. Standard derivative.

(D) FALSE. The extra factor would appear only for an MSE-type loss on a sigmoid; it cancels here.

Q.3 — Answer: 0.1333 to 0.1373 [NAT, 1M · SVM & Decision Trees › SVM: kernels]

$\|x - z\|^2 = 4$; $2\sigma^2 = 2$.

$K = \exp(-4/2) = \mathbf{0.135}$.

Note: $K \in (0, 1]$, $K(x, x) = 1$, and K decreases as the points move apart or as γ grows.

Q.4 — Answer: (A), (B), (C) [MSQ, 1M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

(A) TRUE. It effectively enlarges the training set.

(B) TRUE. Generalisation gap grows.

(C) TRUE. The full network (appropriately scaled) is used.

(D) FALSE. That pattern is underfitting.

Q.5 — Answer: (D) [MCQ, 1M · Regression & Model Selection › Gradient descent]

$J(w) = (w - 4)^2$ has $J' = 2$. GD: $w_{t+1} - 4 = (1 - 2\eta)(w_t - 4)$, so $J(w_t) = (1 - 2\eta)^{2t} J(w_0)$.

Ratios $(1 - 2\eta)^2$ per step: $\eta = 0.02 \rightarrow 0.9216$ (curve C); $\eta = 0.06 \rightarrow 0.7744$ (curve D); $\eta = 0.2 \rightarrow 0.36$ (curve A); $\eta = 1.25 \rightarrow 2.25$ (curve B).

Ratio $> 1 \Rightarrow$ divergence ($\eta > 1 = 2/J'$); among convergent rates (all < 0.5 here), a smaller η gives a ratio closer to 1, i.e. slower decrease.

Hence $\eta = 0.02$ corresponds to **Curve C**.

Q.6 — Answer: (A), (C), (D) [MSQ, 1M · Unsupervised Learning › PCA & dimensionality reduction]

(A) TRUE. Centered data has rank $\leq n - 1$.

(B) FALSE. Without centering, the first direction tends to point towards the mean.

(C) TRUE. Equivalent formulation of PCA.

(D) TRUE. The covariance of the scores is diagonal.

Q.7 — Answer: 6 [NAT, 1M · SVM & Decision Trees › Decision trees: structure & traversal]

Depth d allows at most 2^d leaves, so we need $2^d \geq 61 \Rightarrow d \geq \log_2 61 = 5.931 \Rightarrow d = \mathbf{6}$.

Q.8 — Answer: (A), (C), (D) [MSQ, 1M · SVM & Decision Trees › SVM: dual & support vectors]

(A) TRUE. $K(x, z) = \phi(x)^T \phi(z)$ is evaluated directly.

(B) FALSE. It is on the correct side (positive), only inside the margin.

(C) TRUE. Non-support vectors have $\alpha_i = 0$ and inactive constraints.

(D) TRUE. That normalisation makes it invariant to scaling (w, b).

Q.9 — Answer: (C) [MCQ, 1M · Regression & Model Selection › Ridge & Lasso]

The shaded region is the constraint set of the lasso (L1: $|\beta_1| + |\beta_2| \leq t$, a diamond). The penalised estimate is the first point at which the RSS contours (ellipses centred at β_{OLS} , which lies outside the region) touch the region.

From the figure the tangency point is approximately $(0, 1)$.

The contour first touches the diamond at a corner on the β_2 -axis, so $\beta_1 = 0$ exactly — this is why the lasso performs variable selection.

Answer: $\beta_1 = \mathbf{0}$ exactly and $\beta_2 \neq \mathbf{0}$. Both coefficients cannot be zero since the OLS point lies outside the region (the constraint is active but $t > 0$).

Q.10 — Answer: (D) [MCQ, 1M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$$x^* = (\mu_0 + \mu_1)/2 + \sigma^2 \ln(\pi_0/\pi_1)/(\mu_1 - \mu_0) = 5.212.$$

A larger prior for C_0 means more of the x-axis is assigned to it, so the threshold moves away from μ_0 towards $\mu_1 = 7$. Answer: **It moves towards $x = 7$, enlarging the region assigned to C_0 .**

Equal variances keep the boundary a single point (linear); priors shift it but never make it quadratic.

Q.11 — Answer: 0.6864 to 0.7064 [NAT, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

$$\text{softmax: } P(y = k | \mathbf{x}) = e^{z_k} / \sum_j e^{z_j}.$$

e^z values: 7.3891, 2.7183, 0.1353, 0.3679; sum = 10.6106.

$$P(y = 1 | \mathbf{x}) = 7.3891/10.6106 = \mathbf{0.7}.$$

Q.12 — Answer: (A), (B), (C), (D) [MSQ, 1M · Unsupervised Learning > Cluster evaluation]

(A) TRUE. Only pairs are compared.

(B) TRUE. $a \sqcap b$ gives $s \approx 1$.

(C) TRUE. It is on average closer to another cluster ($b < a$).

(D) TRUE. It depends only on pair co-membership.

Q.13 — Answer: (B) [MSQ, 1M · Regression & Model Selection > Bias-variance trade-off]

(A) FALSE. It may have badly overfit.

(B) TRUE. Estimates stabilise as n grows.

(C) FALSE. Larger $\lambda \rightarrow$ lower variance.

(D) FALSE. Larger k averages more points $\rightarrow$ lower variance.

Q.14 — Answer: (A), (B), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

(A) TRUE. One zero factor makes the product zero (zero-frequency problem).

(B) TRUE. Log-ratio contains x_j^2 terms with coefficients $(1/\sigma_{j0}^2 - 1/\sigma_{j1}^2)/2$.

(C) TRUE. Correlated evidence is counted multiple times.

(D) TRUE. Products of many small probabilities underflow.

Q.15 — Answer: 54 [NAT, 1M · Unsupervised Learning > Hierarchical clustering]

Each merge reduces the number of clusters by one: from 55 to 1 needs $n - 1 = \mathbf{54}$ merges.

Q.16 — Answer: 0.2858 to 0.2898 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Per unit population: $TP = Se \cdot \pi$, $FN = (1 - Se) \cdot \pi$, $FP = (1 - Sp)(1 - \pi)$, $TN = Sp(1 - \pi)$.

$TP = 0.008$, $FN = 0.002$, $FP = 0.0198$, $TN = 0.9702$.

Precision (positive predictive value) = $TP/(TP + FP) = \mathbf{0.288}$.

Even a good test has low precision when the positive class is rare: most positives predicted are false positives from the large negative class (class-imbalance effect; ROC is unaffected).

Q.17 — Answer: 1.1447 to 1.1847 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

$$X^T X = \begin{bmatrix} 19 & -1 \\ -1 & 9 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 27 \\ 9 \end{bmatrix}$$

$$\det(X^T X) = 170; (X^T X)^{-1} = (1/170) \cdot \begin{bmatrix} 9 & 1 \\ 1 & 19 \end{bmatrix}.$$

$$\beta = (X^T X)^{-1} X^T y = (1.4824, 1.1647).$$

Answer: **1.16**.

Q.18 — Answer: (B), (D) [MSQ, 2M · Unsupervised Learning > Hierarchical clustering]

(A) FALSE. Single and complete linkage often disagree.

(B) TRUE. This is the increase in within-cluster SS when A and B merge.

(C) FALSE. Average linkage depends on actual values, not only on ranks.

(D) TRUE. The remaining forest's components are the clusters.

Q.19 — Answer: 1.1354 to 1.1554 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

Per-example loss $L_i = -\ln \hat{p}_i$ if $y_i = 1$ and $-\ln(1 - \hat{p}_i)$ if $y_i = 0$:

i	y	$\hat{p}$	L_i
1	0	0.65	1.0498
2	0	0.95	2.9957
3	0	0.35	0.4308
4	1	0.9	0.1054

$\sum L_i = 4.5817$; average = $4.5817/4 = 1.15$.

Common mistakes: using $\ln \hat{p}$ for $y = 0$ examples, or $\log_{10}$ instead of $\ln$.

Q.20 — Answer: (A), (B), (C), (D) [MSQ, 2M · Regression & Model Selection > Ridge & Lasso]

With $X^T X = I$ the ridge path is $b/(1 + \lambda)$ and the lasso path is $\text{sign}(b)(|b| - \lambda)_+$. The plot shows paths reaching zero at finite $\lambda \Rightarrow$ lasso; here it is **lasso**.

(A) TRUE. Penalty vanishes at $\lambda = 0$.

(B) TRUE. $\text{sign}(b)(|b| - \lambda)_+$ is piecewise linear.

(C) TRUE. Coefficients with $|b_j| \leq 3$ are zero: count = 3.

(D) TRUE. Both $b/(1 + \lambda)$ and soft-thresholding shrink monotonically.

Q.21 — Answer: 2 [NAT, 2M · Unsupervised Learning > k-means]

Data: {2, 3, 9, 12, 18, 19, 20}; initial centroids $c_1 = 17$, $c_2 = 20$.

Iter	centroids used	cluster 1	cluster 2	new centroids
1	17, 20	{2, 3, 9, 12, 18}	{19, 20}	8.8, 19.5
2	8.8, 19.5	{2, 3, 9, 12}	{18, 19, 20}	6.5, 19
3	6.5, 19	{2, 3, 9, 12}	{18, 19, 20}	6.5, 19

In the last row the centroids do not change, so the algorithm has converged: final centroids 6.5 and 19.

Centroids changed in 2 iteration(s); the final iteration only confirms convergence. Answer **2**.

Q.22 — Answer: 0.99 to 1.01 [NAT, 2M · SVM & Decision Trees > Perceptron (linear separators)]

Trace the epoch (a point with $y \cdot w^T \tilde{x} \leq 0$, including 0, triggers an update):

Step	$\tilde{x} = (1, x_1, x_2)$	y	$w^T \tilde{x}$	$y \cdot w^T \tilde{x} \leq 0$?	w after step
1	(1, 1, 0)	+1	1	no	(1, 0, 1)
2	(1, 1, -3)	+1	-2	yes $\rightarrow$ update	(1.5, 0.5, -0.5)
3	(1, -2, -1)	-1	1	yes $\rightarrow$ update	(1, 1.5, 0)
4	(1, 3, 2)	+1	5.5	no	(1, 1.5, 0)
5	(1, 2, 0)	+1	4	no	(1, 1.5, 0)

Final $w = (1, 1.5, 0)$, number of updates = 2. Answer: **1**.

Q.23 — Answer: 4.39 to 4.41 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

Conditional risk $R(\alpha_i | x) = \sum_j L(\alpha_i, C_j) P(C_j | x)$:

$$R(\alpha_1) = 0 \times 0.2 + 2 \times 0.5 + 9 \times 0.3 = 3.7$$

$$R(\alpha_2) = 7 \times 0.2 + 0 \times 0.5 + 10 \times 0.3 = 4.4$$

$$R(\alpha_3) = 9 \times 0.2 + 7 \times 0.5 + 0 \times 0.3 = 5.3$$

Minimum risk: α_1 with $R = 3.7$. The MAP rule picks C_2 (risk 4.4) — different, because the losses are asymmetric.

Answer: **4.4**.

Q.24 — Answer: (A) [MCQ, 2M · Regression & Model Selection › Simple linear regression]

$$\bar{x} = 7.4444, \bar{y} = -8.5556.$$

$$S_{xx} = \sum x^2 - (\sum x)^2/n = 166.2222, S_{yy} = \sum y^2 - (\sum y)^2/n = 198.2222, S_{xy} = \sum xy - \sum x \sum y/n = -161.7778.$$

$$b = S_{xy}/S_{xx} = -0.9733; a = \bar{y} - b\bar{x} = -1.3102.$$

$$\text{Line: } \hat{y} = -1.31 - 0.97x.$$

Distractors: using S_{xy}/S_{yy} (x-on-y slope), using uncentred sums $\sum xy/\sum x^2$, or a sign error in the intercept.

Q.25 — Answer: 3 [NAT, 2M · SVM & Decision Trees › Decision trees: structure & traversal]

Route each record from the root to a leaf:

#	Predicted	Actual	Correct?
1	No	No	yes
2	No	No	yes
3	No	Yes	no
4	Yes	Yes	yes
5	Yes	Yes	yes
6	No	No	yes
7	No	Yes	no
8	Yes	Yes	yes
9	No	Yes	no

Misclassified: 3 of 9 (accuracy 66.67%). Answer: **3**.

Q.26 — Answer: (A) [MSQ, 2M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

$\nabla f = (10w_1, 8w_2) \Rightarrow w_i \leftarrow (1 - \eta a_i) w_i$. GD converges (from every start) iff $|1 - \eta a_i| < 1$ for both i , i.e. $0 < \eta < 2/\max(a_i) = 2/10 = 0.2$. The steep direction ($a = 10$) limits the learning rate.

(A) TRUE. $\eta = 0.22$: $|1 - \eta \cdot 10| = 1.2$, $|1 - \eta \cdot 8| = 0.76 \rightarrow$ diverges.

(B) FALSE. $\eta = 0.12$: $|1 - \eta \cdot 10| = 0.2$, $|1 - \eta \cdot 8| = 0.04 \rightarrow$ converges.

(C) FALSE. $\eta = 0.08$: $|1 - \eta \cdot 10| = 0.2$, $|1 - \eta \cdot 8| = 0.36 \rightarrow$ converges.

(D) FALSE. $\eta = 0.05$: $|1 - \eta \cdot 10| = 0.5$, $|1 - \eta \cdot 8| = 0.6 \rightarrow$ converges.

Q.27 — Answer: 0.7699 to 0.7899 [NAT, 2M · Probabilistic & Instance-based Classifiers › Linear discriminant analysis]

With equal variances, $\ln[\pi_1 p(x|C_1)/(\pi_0 p(x|C_0))] = (\mu_1 - \mu_0)x/\sigma^2 - (\mu_1^2 - \mu_0^2)/(2\sigma^2) + \ln(\pi_1/\pi_0)$ — linear in x (LDA).

$$\text{Slope } a = (\mu_1 - \mu_0)/\sigma^2 = 0.3333; \text{ intercept } c = -(\mu_1^2 - \mu_0^2)/(2\sigma^2) + \ln(\pi_1/\pi_0) = -0.7347.$$

$$P(C_1|x) = \sigma(a x + c) = \sigma(1.2653) = \mathbf{0.78}.$$

Q.28 — Answer: -0.0357 to -0.0343 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Backpropagation]

$$a = w \cdot x + b = -0.75; \hat{y} = \sigma(a) = 0.3208.$$

$$L = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial L/\partial \hat{y} = \hat{y} - y = 0.3208; \partial \hat{y}/\partial a = \hat{y}(1 - \hat{y}) = 0.2179.$$

$$\delta = \partial L/\partial a = (\hat{y} - y)\hat{y}(1 - \hat{y}) = 0.0699 (= \partial L/\partial b).$$

$$\partial L/\partial w_2 = \delta \cdot x_2 = 0.0699 \times -0.5 = -0.035.$$

Answer: **-0.035**.

Forgetting the factor $\hat{y}(1 - \hat{y})$ (i.e. treating the output as linear) is the most common error.

Q.29 — Answer: (B), (C), (D) [MSQ, 2M · SVM & Decision Trees › Decision trees: impurity measures]

Parent [60+, 20-]: $H = 0.8113$, $Gini = 0.375$, $\text{error} = 0.25$.

S1: children $H = 0.9341$, 0.754 ; $Gini = 0.455$, 0.3394 ; $\text{error} = 0.35$, 0.2167 . Gains: $IG = 0.0122$, $Gini = 0.0067$, $\text{error} = 0$.

S2: children $H = 0.4395$, 0.9911 ; $Gini = 0.1653$, 0.4938 ; $\text{error} = 0.0909$, 0.4444 . Gains: $IG = 0.1236$, $Gini = 0.0619$, $\text{error} = 0$.

(A) FALSE. S1: 0.0067 , S2: 0.0619 .

(B) TRUE. $IG(S_2) = 0.1236$.

(C) TRUE. $S_1: 0.0067, S_2: 0.0619$.

(D) TRUE. $S_1: 0, S_2: 0$.

Q.30 — Answer: (A), (B), (C), (D) [MSQ, 2M · Unsupervised Learning > PCA & dimensionality reduction]

(A) TRUE. Eigenvalue $a + b$.

(B) TRUE. $S \rightarrow RS R^T$ is a similarity transform.

(C) TRUE. Centering removes one degree of freedom.

(D) TRUE. The covariance of z-scores is the correlation matrix.

Q.31 — Answer: 65.7682 to 66.0318 [NAT, 2M · Regression & Model Selection > Basis functions & feature scaling]

Since $x = \bar{x} + s_x z$, $\hat{y} = \beta_0 + \beta_1 \bar{x} + \beta_1 s_x z$, and OLS with intercept is equivariant under affine maps of x (same fitted values).

$y_0 = \beta_0 + \beta_1 \bar{x} = 7.9 + (2.9)(20) = 65.9$ ($= \bar{y}$, since z is centred);

$y_1 = \beta_1 s_x = (2.9)(9) = 26.1$.

Answer: **65.9**.

Q.32 — Answer: 48 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Parameter counting]

Old: $(32+1) \cdot 20 + (20+1) \cdot 10 = 870$.

New: $(32+1) \cdot 20 + (20+1) \cdot 8 + (8+1) \cdot 10 = 918$.

Change = **48**.

Q.33 — Answer: 0.455 to 0.475 [NAT, 2M · Unsupervised Learning > Distance measures]

Min-max scaling: $x' = (x - \min) / (\max - \min)$; age range [20, 60], income range [10, 210].

Customer	age'	income'
A	0.775	0.775
B	0.225	0.875
C	0.5	0.4

Raw distances: $d(A,B) = 29.73$, $d(A,C) = 75.8 \rightarrow$ nearest is B (income dominates because of its larger range).

Scaled distances: $d(A,B) = 0.559$, $d(A,C) = 0.465 \rightarrow$ nearest is C.

Answer: **0.47**. Scaling changes which neighbour is nearest — distance-based methods (k-means, k-NN, hierarchical clustering) should be run on standardized features.

Q.34 — Answer: (C) [MCQ, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent: 4 Yes, 6 No $\Rightarrow$ Gini = 0.48.

Outlook — Sunny: [1+, 2-], Overcast: [1+, 3-], Rain: [2+, 1-]; gain = 0.0633.

Colour — Red: [1+, 1-], Green: [1+, 2-], Blue: [2+, 3-]; gain = 0.0067.

Size — Small: [4+, 2-], Large: [0+, 4-]; gain = 0.2133.

Shape — Round: [2+, 2-], Square: [1+, 4-], Oval: [1+, 0-]; gain = 0.12.

The largest Gini gain is for **Size**, so it is chosen at the root.

Q.35 — Answer: -2 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Rule: if $y_i(w \cdot x_i + b) \leq 0$, then $w \leftarrow w + \eta y_i x_i$, $b \leftarrow b + \eta y_i$ ($\eta = 1$). Start $w = (0, 0)$, $b = 0$.

Example 1: $a = 0$, $y \cdot a = 0 \rightarrow$ **mistake**, update $\rightarrow w = (2, 1)$, $b = 1$.

Example 2: $a = -6$, $y \cdot a = -6 \rightarrow$ **mistake**, update $\rightarrow w = (-1, 0)$, $b = 2$.

Example 3: $a = 1$, $y \cdot a = 1 \rightarrow$ correct, no change.

Example 4: $a = 1$, $y \cdot a = -1 \rightarrow$ **mistake**, update $\rightarrow w = (-2, 3)$, $b = 1$.

Required value = **-2**.

Note: a point with net input exactly 0 counts as a mistake ($y \cdot a \leq 0$) — this is why the first example always triggers an update when starting from zero.

Mock Test 27

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	7	9	19

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Neural Networks (MLP / Feed-forward)	7	11
SVM & Decision Trees	7	11
Probabilistic & Instance-based Classifiers	7	11
Regression & Model Selection	7	11
Unsupervised Learning	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

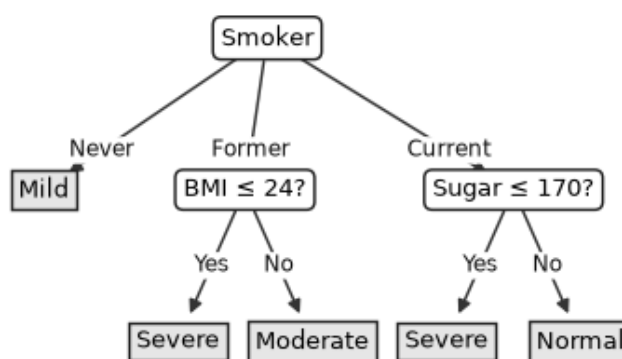
Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

A fully connected MLP has layer sizes 32-100-256-3 (input-...-output), every layer having a bias vector. The total number of trainable parameters (weights and biases) is _____ (Answer in integer)

Q.2 [1 mark · MCQ]

The decision tree below predicts the diagnosis category of a patient. Edges labelled Yes/No answer the test in the node. What is the prediction for the record shown in the table?



BMI	Sugar	BP	Smoker
22	239	146	Never

(A) Normal

(C) Severe

(B) Mild

(D) Moderate

Q.3 [1 mark · MSQ (one or more correct)]

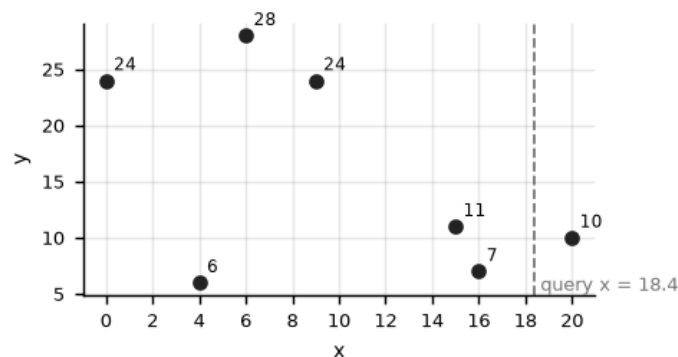
Which of the following statements about logistic regression is/are CORRECT?

- (A) The derivative of the sigmoid is $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.
- (B) If the training data are linearly separable, the norm of the unregularised logistic-regression weights grows without bound during gradient descent.
- (C) Adding an L2 penalty $\lambda \|w\|^2$ ($\lambda > 0$) keeps the logistic-regression weights finite even on separable data.
- (D) There is no closed-form expression for the maximum-likelihood weights of logistic regression; they are found iteratively (gradient descent, Newton/IRLS).

Q.4 [1 mark · NAT (numerical answer)]

The training data for a k-NN regressor are given below (also plotted). Using $k = 4$ and absolute distance $|x - x_i|$, the predicted value (unweighted average of neighbours) at $x = 18.4$ is _____ (rounded off to two decimal places)

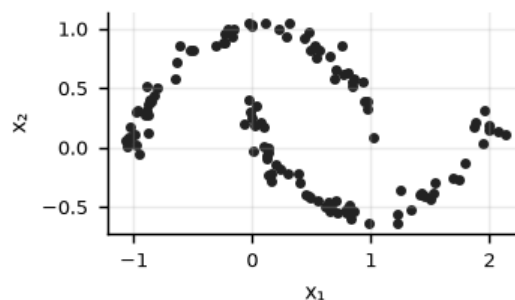
x	0	4	6	9	15	16	20
y	24	6	28	24	11	7	10

**Q.5 [1 mark · NAT (numerical answer)]**

For a data set, $\bar{x} = 20$, $\bar{y} = 75$, $s_x = 3$, $s_y = 14$ and the sample correlation between x and y is $r = 0.65$. The intercept of the least-squares regression line of y on x is _____ (rounded off to two decimal places)

Q.6 [1 mark · MCQ]

k-means with $k = 2$ fails to separate the two interleaving half-moons shown below, even with many random restarts. The main reason is that k-means



- (A) produces convex clusters separated by linear (Voronoi) boundaries
- (B) cannot be applied to two-dimensional data
- (C) requires class labels to place the centroids
- (D) converges only when the data are standardized

Q.7 [1 mark · MSQ (one or more correct)]

Which of the following statements about activation functions is/are CORRECT? (σ denotes the logistic sigmoid.)

- (A) For two classes, the softmax probability of class 1 equals $\sigma(z_1 - z_2)$.
- (B) $\sigma(0) = 0$ for the logistic sigmoid.
- (C) $\tanh(z) = 2\sigma(2z) - 1$.
- (D) $\tanh$ outputs lie in the interval $(0, 1)$.

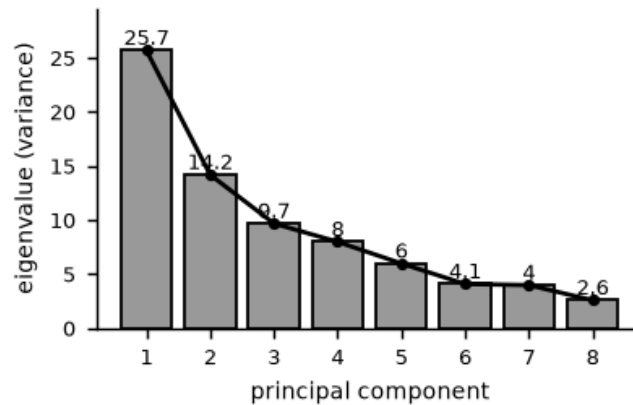
Q.8 [1 mark · MSQ (one or more correct)]

Which of the following statements about distance / similarity measures is/are CORRECT?

- (A) Mahalanobis distance accounts for feature correlations through the inverse covariance matrix.
- (B) The Jaccard similarity of sets A and B is $|A \cap B| / |A \cup B|$.
- (C) Cosine similarity is unchanged if one of the vectors is multiplied by a positive scalar.
- (D) Minkowski 'distance' with $0 < p < 1$ violates the triangle inequality.

Q.9 [1 mark · NAT (numerical answer)]

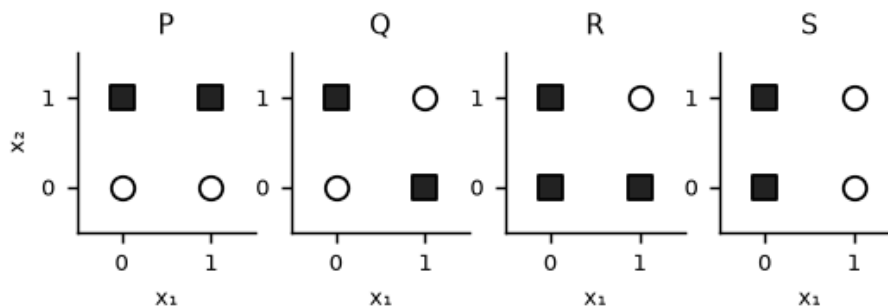
The scree plot below shows the eigenvalues of the covariance matrix of a 8-dimensional dataset. The percentage of total variance explained by the first two principal components together is _____ (rounded off to one decimal place)

**Q.10 [1 mark · NAT (numerical answer)]**

A bagged ensemble is trained on $N = 2400$ points. Using the large- N approximation, the expected number of training points that are out-of-bag for a single bootstrap sample of size N is _____ (Answer in integer)

Q.11 [1 mark · MCQ]

Each panel shows the four binary inputs (x_1, x_2) labelled as class 1 (filled square) or class 0 (open circle). Which dataset can NOT be classified perfectly by a single perceptron (one linear threshold unit)?



- (A) Dataset P
- (B) Dataset Q
- (C) Dataset R
- (D) Dataset S

Q.12 [1 mark · MSQ (one or more correct)]

Which of the following tasks is/are most naturally formulated as **classification** problems?

- (A) Estimating the electricity demand (MW) of a city for the next hour
- (B) Detecting whether a credit-card transaction is fraudulent
- (C) Predicting whether a student will pass or fail a course
- (D) Estimating the remaining useful life (in hours) of a machine from sensor data

Q.13 [1 mark · NAT (numerical answer)]

A linear regression model with an intercept uses 1 numeric feature and a categorical feature *education level* with 3 levels (School, UG, PG); a categorical feature *payment mode* with 3 levels (Cash, Card, UPI). Each categorical feature is encoded by dummy (indicator) variables using a reference level, so that the design matrix has full column rank. The model is fitted to $n = 110$ observations. The residual degrees of freedom used in the unbiased estimate $\hat{\sigma}^2 = \text{SSE}/(\text{degrees of freedom})$ is _____ (Answer in integer)

Q.14 [1 mark · NAT (numerical answer)]

For $K = 4$ classes and $p = 5$ features, how many MORE covariance parameters does QDA estimate than LDA? _____ (Answer in integer)

Q.15 [1 mark · MCQ]

A soft-margin linear SVM has learned $f(x) = 2x_1 + x_2 - 4$. For the training example $x = (1, -1)$ with label $y = +1$, which statement is CORRECT?

- (A) Exactly on the decision boundary ($\xi = 1$)
- (B) Correctly classified, on or outside the margin ($\xi = 0$)
- (C) Correctly classified but strictly inside the margin ($0 < \xi < 1$)
- (D) Misclassified ($\xi > 1$)

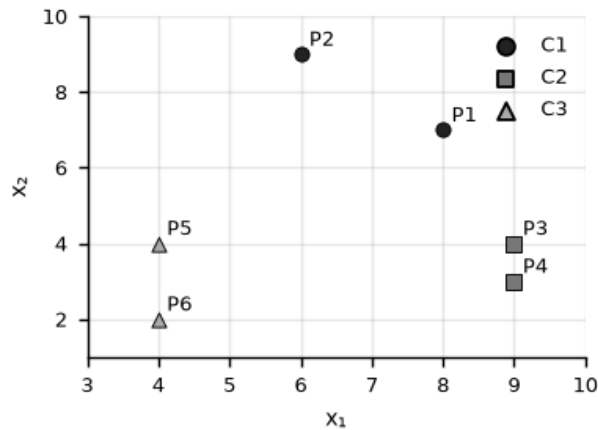
Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

$X_1, \dots, X_n$ are i.i.d. with mean $\mu = 3$ and variance $\sigma^2 = 25$, with $n = 8$. Consider the shrinkage estimator $\hat{\mu} = c \cdot \bar{x}$ of μ , where $\bar{x}$ is the sample mean. Using $\text{MSE} = \text{Bias}^2 + \text{Variance}$, the mean squared error of $\hat{\mu} = 0.7 \cdot \bar{x}$ is _____ (rounded off to three decimal places)

Q.17 [2 marks • NAT (numerical answer)]

A clustering of 6 points into 3 clusters is shown below. Using Euclidean distance, the silhouette coefficient $s(i) = (b(i) - a(i))/\max(a(i), b(i))$ of point P4 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5	P6
x_1	8	6	9	9	4	4
x_2	7	9	4	3	4	2
cluster	C1	C1	C2	C2	C3	C3

**Q.18 [2 marks • MCQ]**

Ridge regression is tuned by 10-fold cross-validation. For each λ the table gives the mean CV MSE and its standard error (SE) across folds. (Larger λ means a simpler, more regularised model.) Using the **one-standard-error rule** (choose the most regularised model whose CV error is within one standard error of the minimum CV error), the selected value is

λ	0.01	0.1	0.3	1	3	10
CV MSE	5.92	5.87	5.69	5.84	6.54	7.93
SE	0.45	0.25	0.4	0.35	0.37	0.16

(A) $\lambda = 0.3$ **(B)** $\lambda = 3$ **(C)** $\lambda = 0.01$ **(D)** $\lambda = 1$ **Q.19 [2 marks • NAT (numerical answer)]**

A softmax-regression output layer computes $z = Wx + b$, $p = \text{softmax}(z)$ with W and b below, and is trained with cross-entropy loss $L = -\ln p_y$. For $x = (2, 2)^T$ with true class $y = 1$, the updated W_{31} after one gradient-descent step with $\eta = 0.1$ is _____ (rounded off to three decimal places)

$$W = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ -1 & 0 \end{bmatrix}$$

$$b = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

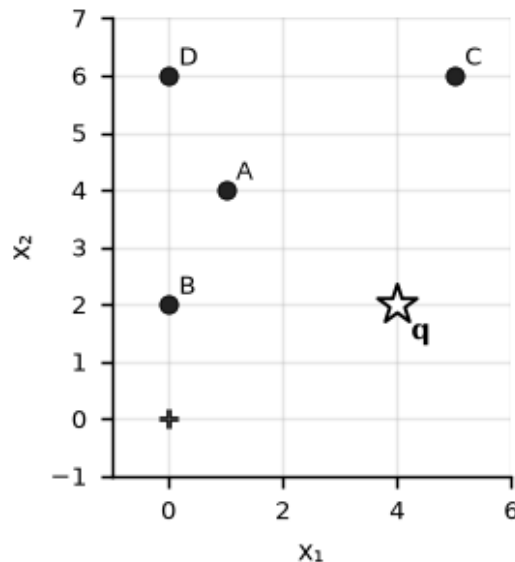
Q.20 [2 marks • NAT (numerical answer)]

A logistic regression classifier has $P(y = 1 | \mathbf{x}) = \sigma(0 + (-1)x_1 + (2)x_2)$. An input is labelled $y = 1$ if and only if $P(y = 1 | \mathbf{x}) \geq 0.9$. For this rule, the x_1 -intercept of the decision boundary (the value of x_1 where it meets $x_2 = 0$) is _____ (rounded off to two decimal places)

Q.21 [2 marks · MCQ]

A query point $q = (4, 2)$ and four points A-D are shown below. Using **Euclidean (L_2) distance**, which point is nearest to q ?

Point	A	B	C	D
x_1	1	0	5	0
x_2	4	2	6	6



- (A) Point A
(C) Point C

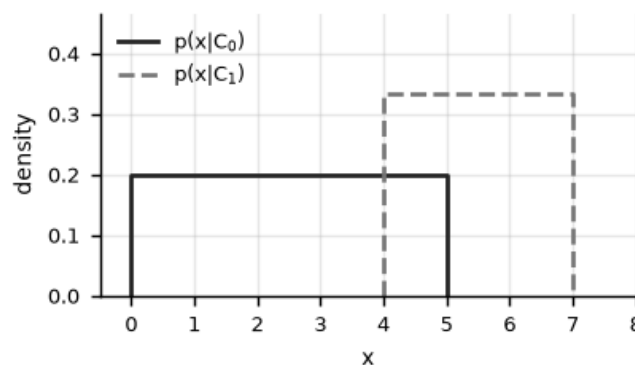
- (B) Point B
(D) Point D

Q.22 [2 marks · NAT (numerical answer)]

A centered data matrix X with $n = 11$ rows (samples) and $d = 3$ columns has singular values 24, 21, 7. Using the sample covariance with divisor $n - 1$, the variance of the data along the second principal component is _____ (rounded off to two decimal places)

Q.23 [2 marks · NAT (numerical answer)]

For a binary problem with scalar x , $p(x | C_0)$ is uniform on $[0, 5]$ and $p(x | C_1)$ is uniform on $[4, 7]$, with priors $P(C_0) = 0.4$ and $P(C_1) = 0.6$. Then the error rate of the rule "predict C_0 if $x < 4.5$, else C_1 " is _____ (rounded off to three decimal places)

**Q.24 [2 marks · MSQ (one or more correct)]**

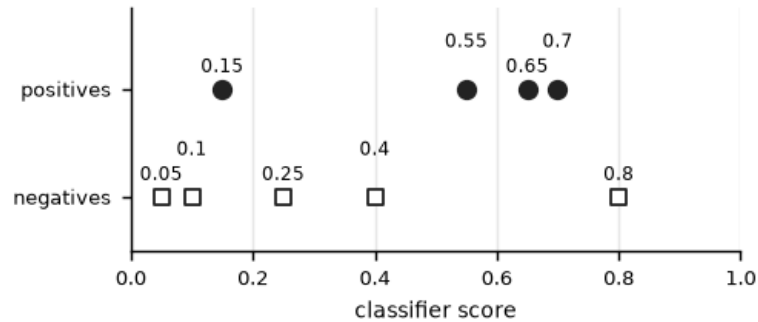
Each option shows a 2×2 matrix $[[K(x_1, x_1), K(x_1, x_2)], [K(x_2, x_1), K(x_2, x_2)]]$. Which of them can be the Gram matrix of some valid (Mercer) kernel $\tilde{K}$ on two points x_1, x_2 ?

- (A) $\begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$
 (B) $\begin{bmatrix} 1 & 4 \\ 4 & 9 \end{bmatrix}$
 (C) $\begin{bmatrix} -6 & 1 \\ 1 & 8 \end{bmatrix}$
 (D) $\begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}$

Q.25 [2 marks · NAT (numerical answer)]

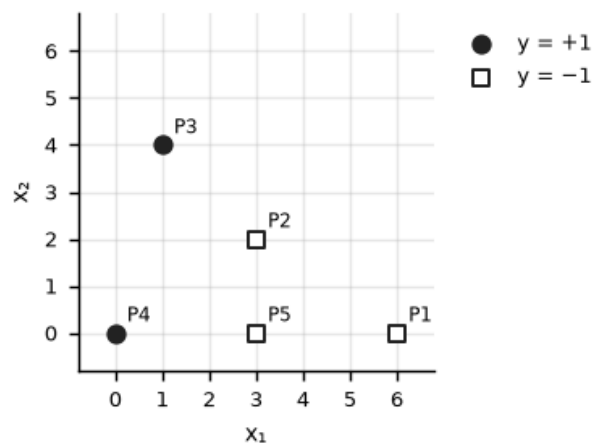
A classifier assigns the scores below to 9 test examples (4 positives, label 1; 5 negatives, label 0). The area under the ROC curve (AUC) is _____ (rounded off to two decimal places)

example	1	2	3	4	5	6	7	8	9
score	0.8	0.4	0.25	0.1	0.05	0.15	0.65	0.7	0.55
label	0	0	0	0	0	1	1	1	1

**Q.26 [2 marks · NAT (numerical answer)]**

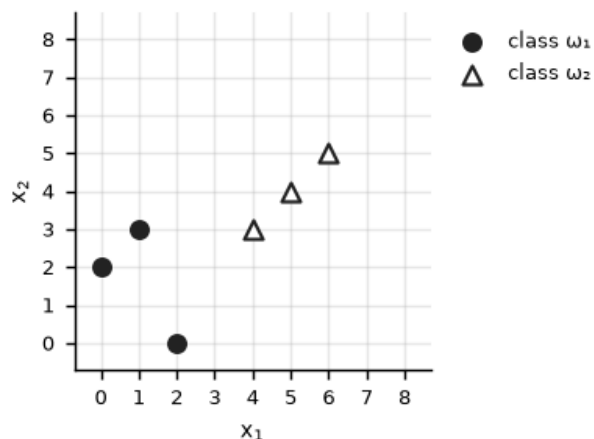
A hard-margin linear SVM is trained on the points below. Solving the dual problem gives the Lagrange multipliers α_i listed in the table. Using the KKT conditions, compute the bias b . (rounded off to two decimal places)

Point	x_1	x_2	y	α_i
P1	6	0	-1	0
P2	3	2	-1	0.34
P3	1	4	+1	0.22
P4	0	0	+1	0.12
P5	3	0	-1	0



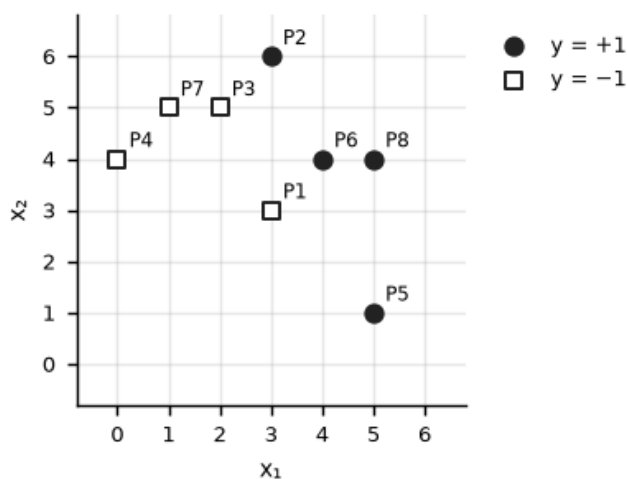
Q.27 [2 marks · NAT (numerical answer)]

Class ω_1 has training points (0, 2), (1, 3), (2, 0) and class ω_2 has points (4, 3), (6, 5), (5, 4). Let $\mathbf{m}_1, \mathbf{m}_2$ be the class means and $S_W = \sum_k \sum_{x \in \omega_k} (x - \mathbf{m}_k)(x - \mathbf{m}_k)^T$ the within-class scatter matrix (no division by n). Then $(\mathbf{m}_2 - \mathbf{m}_1)^T S_W (\mathbf{m}_2 - \mathbf{m}_1)$ is _____ (rounded off to two decimal places)

**Q.28 [2 marks · NAT (numerical answer)]**

The labelled training points shown below are linearly separable. A hard-margin linear SVM is trained on them. Compute $\|w\|$ for the canonical solution (support vectors satisfy $y_i(w^T x_i + b) = 1$). (rounded off to two decimal places)

Point	x_1	x_2	y
P1	3	3	-1
P2	3	6	+1
P3	2	5	-1
P4	0	4	-1
P5	5	1	+1
P6	4	4	+1
P7	1	5	-1
P8	5	4	+1



Q.29 [2 marks · MSQ (one or more correct)]

A threshold unit outputs 1 if $w_1x_1 + w_2x_2 + b > 0$ and 0 otherwise, for binary inputs $x_1, x_2 \in \{0, 1\}$. Which of the following parameter settings make the unit implement the Boolean **AND** function?

- (A) $w_1 = 1, w_2 = 0.5, b = 3$
- (B) $w_1 = 2, w_2 = 2, b = -2.5$
- (C) $w_1 = 2.5, w_2 = 1.5, b = -3$
- (D) $w_1 = 1, w_2 = 3, b = -1.5$

Q.30 [2 marks · MSQ (one or more correct)]

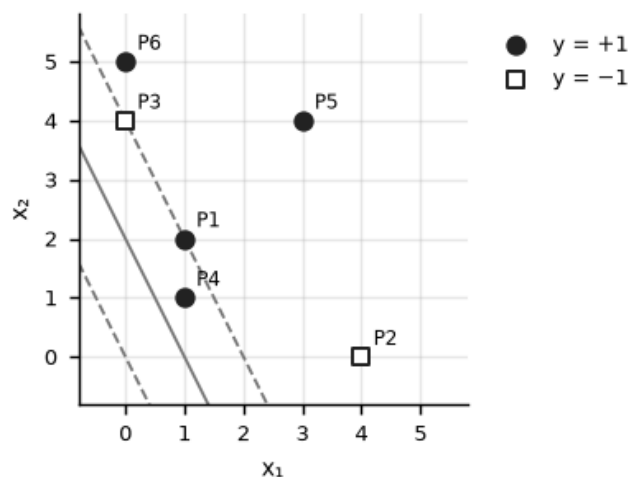
Which of the following statements about loss functions and optimisation for training neural networks is/are CORRECT?

- (A) Minimising cross-entropy loss is equivalent to maximising the log-likelihood of the labels under the model.
- (B) Decaying the learning rate during training can help SGD settle closer to a minimum.
- (C) With uniformly sampled mini-batches, the mini-batch gradient is an unbiased estimate of the full-batch gradient.
- (D) Increasing the batch size reduces the variance of the gradient estimate.

Q.31 [2 marks · MSQ (one or more correct)]

A soft-margin linear SVM has learned $f(x) = x_1 + 0.5x_2 - 1$. The training points are listed below and plotted (boundary solid, margins $f = \pm 1$ dashed). With ξ_i the optimal slack of point i , which of the following statements is/are CORRECT?

Point	x_1	x_2	y
P1	1	2	+1
P2	4	0	-1
P3	0	4	-1
P4	1	1	+1
P5	3	4	+1
P6	0	5	+1



- (A) P3 is misclassified ($\xi_3 > 1$).
- (B) P3 has slack $\xi_3 = 0$.
- (C) P5 has slack $\xi_5 = 0$.
- (D) P5 is misclassified ($\xi_5 > 1$).

Q.32 [2 marks · MCQ]

Six points have the pairwise distances below. Agglomerative clustering with **single linkage** is run and the dendrogram is cut to give exactly **2** clusters. The resulting clusters are

	P1	P2	P3	P4	P5	P6
P1	0	9	11	8	3	8
P2		0	2	7	6	5
P3			0	9	8	7
P4				0	7	4
P5					0	7
P6						0

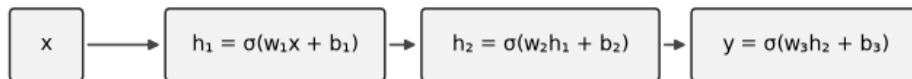
- (A) {P1, P5}, {P2, P3}, {P4, P6} (B) {P1, P5}, {P2, P3, P4, P6}
 (C) {P1, P4, P5, P6}, {P2, P3} (D) {P1, P3, P6}, {P2, P4, P5}

Q.33 [2 marks · NAT (numerical answer)]

The 3×3 matrix $X^T X$ of a (centred) design matrix X has eigenvalues 25, 16, 6. For ridge regression with penalty $\lambda \|\beta\|_2^2$, the effective degrees of freedom $df(\lambda) = \text{trace}[X(X^T X + \lambda I)^{-1} X^T]$ for $\lambda = 5$ is _____ (rounded off to two decimal places)

Q.34 [2 marks · NAT (numerical answer)]

Consider the scalar chain shown below with weights $w_1 = 1$, $w_2 = -2$, $w_3 = 3$, biases $b_1 = -0.5$, $b_2 = -1$, $b_3 = 0$ and activation the logistic sigmoid σ at every stage. For input $x = -0.5$, the value of $\partial y / \partial x$ is _____ (rounded off to 4 decimal places)

**Q.35 [2 marks · MSQ (one or more correct)]**

A linear regression of annual salary (₹ lakh) on Exp (years of experience), Female (1 if the employee is female, 0 otherwise) and city $\in \{A, B, C\}$ encoded by dummies CityB and CityC (reference level A) gives the fitted model

$$\hat{y} = 5 + 3 \cdot \text{Exp} + 3.8 \cdot \text{Female} + 2 \cdot \text{CityB} + 3.9 \cdot \text{CityC}$$

Which of the following is/are CORRECT?

- (A) Increasing Exp by 2 years, other variables fixed, changes the prediction by 6.
 (B) The coefficient of CityB is the expected difference in response between city B and city A, holding the other variables fixed.
 (C) Adding a third dummy for city A while keeping the intercept would make $X^T X$ singular.
 (D) Holding other variables fixed, the predicted response for city B minus that for city C is -1.9 .

— END OF QUESTION PAPER —

Mock Test 27 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	29927	19	NAT	2	−1.0038 to −0.9998
2	MCQ	1	(B)	20	NAT	2	−2.2072 to −2.1872
3	MSQ	1	(A), (B), (C), (D)	21	MCQ	2	(A)
4	NAT	1	12.99 to 13.01	22	NAT	2	44.09 to 44.11
5	NAT	1	14.2617 to 14.405	23	NAT	2	0.138 to 0.142
6	MCQ	1	(A)	24	MSQ	2	(D)
7	MSQ	1	(A), (C)	25	NAT	2	0.69 to 0.71
8	MSQ	1	(A), (B), (C), (D)	26	NAT	2	0.99 to 1.01
9	NAT	1	53.501 to 53.901	27	NAT	2	4.8067 to 4.8267
10	NAT	1	881 to 884	28	NAT	2	1.7851 to 1.8051
11	MCQ	1	(B)	29	MSQ	2	(B), (C)
12	MSQ	1	(B), (C)	30	MSQ	2	(A), (B), (C), (D)
13	NAT	1	104	31	MSQ	2	(A), (C)
14	NAT	1	45	32	MCQ	2	(B)
15	MCQ	1	(D)	33	NAT	2	2.1307 to 2.1507
16	NAT	2	2.3382 to 2.3443	34	NAT	2	−0.0404 to −0.0396
17	NAT	2	0.7939 to 0.8139	35	MSQ	2	(A), (B), (C), (D)
18	MCQ	2	(D)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Neural Networks (MLP / Feed-forward)					11
SVM & Decision Trees					11
Probabilistic & Instance-based Classifiers					11
Regression & Model Selection					11
Unsupervised Learning					11
Total					55

Mock Test 27 — Detailed Solutions

Q.1 — Answer: 29927 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Parameter counting]

Fully connected layer with n_{in} inputs and n_{out} outputs: $n_{in} \cdot n_{out}$ weights + n_{out} biases.

Layer 1: $32 \times 100 = 3200$ weights, 100 biases.

Layer 2: $100 \times 256 = 25600$ weights, 256 biases.

Layer 3: $256 \times 3 = 768$ weights, 3 biases.

Weights = 29568, biases = 359. Required: **29927**.

Q.2 — Answer: (B) [MCQ, 1M · SVM & Decision Trees > Decision trees: structure & traversal]

Follow the record from the root, taking the branch that matches each test:

Smoker = Never $\rightarrow$ branch Never.

(Only the tests on the actual path matter.) The record reaches the leaf **Mild**.

Note: " $\leq$ " includes equality, so a value equal to the threshold goes to the Yes branch.

Q.3 — Answer: (A), (B), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

(A) **TRUE**. Standard identity; maximal value 0.25 at $z = 0$.

(B) **TRUE**. Scaling a separating w up always lowers the loss, so the MLE does not exist (it diverges).

(C) **TRUE**. The penalty grows with $\|w\|$ and makes the objective strictly convex and coercive.

(D) **TRUE**. Setting the gradient $\Sigma(\sigma(w^T x_i) - y_i)x_i = 0$ gives non-linear equations.

Q.4 — Answer: 12.99 to 13.01 [NAT, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Distances $|x_i - 18.4|$: $x=20$: 1.6, $x=16$: 2.4, $x=15$: 3.4, $x=9$: 9.4, $x=6$: 12.4, $x=4$: 14.4, $x=0$: 18.4.

The 4 nearest neighbours are $x = 20, 16, 15, 9$ with $y = 10, 7, 11, 24$.

$\hat{y} = (10 + 7 + 11 + 24)/4 = \mathbf{13}$.

Q.5 — Answer: 14.2617 to 14.405 [NAT, 1M · Regression & Model Selection > Simple linear regression]

$\beta_1 = r \cdot s_y / s_x = 0.65 \times 14/3 = 3.0333$.

$\beta_0 = \bar{y} - \beta_1 \bar{x} = 75 - (3.0333)(20) = 14.3333$.

Answer: **14.33**.

Q.6 — Answer: (A) [MCQ, 1M · Unsupervised Learning > k-means]

Points are assigned to the nearest centroid, so each k-means cluster is a convex Voronoi cell and two clusters are separated by the perpendicular bisector of their centroids. Non-convex shapes such as rings or moons cannot be separated by such a boundary.

k-means converges on raw data, works in any dimension, and is unsupervised (no labels).

Q.7 — Answer: (A), (C) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

(A) **TRUE**. $e^{z_1}/(e^{z_1} + e^{z_2}) = 1/(1 + e^{-(z_1 - z_2)})$.

(B) **FALSE**. $\sigma(0) = 1/2$.

(C) **TRUE**. Standard identity relating tanh and the logistic sigmoid.

(D) **FALSE**. tanh ranges over $(-1, 1)$.

Q.8 — Answer: (A), (B), (C), (D) [MSQ, 1M · Unsupervised Learning > Distance measures]

(A) **TRUE**. $d^2 = (x - y)^T \Sigma^{-1} (x - y)$.

(B) **TRUE**. Definition.

(C) **TRUE**. It depends only on the angle.

(D) **TRUE**. e.g. $p = 0.5$: $d((0,0),(1,1)) = 4$, which exceeds $d((0,0),(1,0)) + d((1,0),(1,1)) = 2$.

Q.9 — Answer: 53.501 to 53.901 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

Total variance = $\Sigma \lambda = 74.3$. Cumulative proportion of variance explained:

k	1	2	3	4	5	6	7	8
λ_k	25.7	14.2	9.7	8	6	4.1	4	2.6
cumulative %	34.59	53.7	66.76	77.52	85.6	91.12	96.5	100

$$(\lambda_1 + \lambda_2)/\Sigma\lambda = 39.9/74.3 = \mathbf{53.7\%}.$$

Q.10 — Answer: 881 to 884 [NAT, 1M · SVM & Decision Trees › Ensembles: bagging & random forests]

$$P(\text{out-of-bag}) = (1 - 1/N)^N \approx e^{-1} \approx 0.3679; \text{ expected count} \approx 2400 \times 0.3679 = \mathbf{883}.$$

Q.11 — Answer: (B) [MCQ, 1M · Neural Networks (MLP / Feed-forward) › Perceptron]

A single perceptron can classify a dataset perfectly iff the two classes are linearly separable.

Dataset P: labels for (0,0),(0,1),(1,0),(1,1) = 0, 1, 0, 1 $\rightarrow x_2$ (identity of x_2) – linearly separable.

Dataset Q: labels for (0,0),(0,1),(1,0),(1,1) = 0, 1, 1, 0 $\rightarrow$ XOR – **not** linearly separable (diagonal pattern).

Dataset R: labels for (0,0),(0,1),(1,0),(1,1) = 1, 1, 1, 0 $\rightarrow$ NAND – linearly separable.

Dataset S: labels for (0,0),(0,1),(1,0),(1,1) = 1, 1, 0, 0 $\rightarrow$ NOT x_1 – linearly separable.

Answer: **Dataset Q**. Of the 14 non-constant Boolean functions of two inputs, only XOR and XNOR are not linearly separable.

Q.12 — Answer: (B), (C) [MSQ, 1M · Regression & Model Selection › Regression vs classification]

A supervised task is *regression* when the target y is real-valued (loss such as squared error), and *classification* when y takes values in a finite set of labels (loss such as cross-entropy or 0-1 loss).

(A) FALSE. The output (demand in MW) is a continuous real-valued quantity $\rightarrow$ regression.

(B) TRUE. The output (fraud / genuine) is a discrete class label $\rightarrow$ classification.

(C) TRUE. The output (pass / fail) is a discrete class label $\rightarrow$ classification.

(D) FALSE. The output (life in hours) is a continuous real-valued quantity $\rightarrow$ regression.

Q.13 — Answer: 104 [NAT, 1M · Regression & Model Selection › Multiple linear regression]

A categorical feature with k levels needs $k - 1$ dummies (one level is the reference absorbed by the intercept; using all k dummies with an intercept gives perfect multicollinearity).

$$\text{Parameters} = 1 (\text{intercept}) + 1 (\text{numeric}) + (3 - 1) + (3 - 1) = 6.$$

$$\text{Residual degrees of freedom} = n - (\text{number of coefficients}) = 110 - 6 = 104.$$

Answer: **104**.

Q.14 — Answer: 45 [NAT, 1M · Probabilistic & Instance-based Classifiers › Linear discriminant analysis]

$$\text{QDA: } K \cdot p(p+1)/2 = 60; \text{ LDA: } p(p+1)/2 = 15.$$

$$\text{Difference} = (K - 1)p(p+1)/2 = \mathbf{45}.$$

This extra variance is why LDA often beats QDA when n is small.

Q.15 — Answer: (D) [MCQ, 1M · SVM & Decision Trees › SVM: soft margin & hinge loss]

$$y \cdot f(x) = (+1) \cdot (-3) = -3.$$

$$\text{Slack } \xi = \max(0, 1 - y f(x)) = 4.$$

Rules: $y f \geq 1 \Rightarrow \xi = 0$ (on/outside margin); $0 < y f < 1 \Rightarrow$ inside margin, still correct; $y f = 0 \Rightarrow$ on the boundary ($\xi = 1$); $y f < 0 \Rightarrow$ misclassified ($\xi > 1$).

Hence: **Misclassified ($\xi > 1$)**.

Q.16 — Answer: 2.3382 to 2.3443 [NAT, 2M · Regression & Model Selection › Bias-variance trade-off]

$$E[c \bar{x}] = c\mu \Rightarrow \text{Bias} = (c - 1)\mu; \text{Var}(c \bar{x}) = c^2\sigma^2/n. \text{MSE}(c) = (c - 1)^2\mu^2 + c^2\sigma^2/n.$$

$$\text{Here } \mu = 3, \sigma^2/n = 25/8 = 3.125.$$

$$\text{MSE} = (-0.3)^2(3)^2 + (0.7)^2(3.125) = 0.81 + 1.5312 = 2.3413.$$

Answer: **2.341**.

Q.17 — Answer: 0.7939 to 0.8139 [NAT, 2M · Unsupervised Learning › Cluster evaluation]

a(i) = mean Euclidean distance from P4 to the other members of its own cluster C2: $d(P4, P3) = 1 \rightarrow a = 1$.

b(i) = smallest mean distance to the points of another cluster: C1: 5.4157; C3: 5.099 $\rightarrow b = 5.099$.

s(i) = $(b - a)/\max(a, b) = (5.099 - 1)/5.099 = \mathbf{0.8}$.

s ranges in $[-1, 1]$; values near 1 mean well clustered, negative values suggest a wrong assignment.

Q.18 — Answer: (D) [MCQ, 2M · Regression & Model Selection > Cross-validation]

Minimum CV MSE = 5.69 at $\lambda = 0.3$, with SE = 0.4.

One-SE threshold = $5.69 + 0.4 = 6.09$.

Models within the threshold: $\lambda = 0.01$ (5.92), $\lambda = 0.1$ (5.87), $\lambda = 0.3$ (5.69), $\lambda = 1$ (5.84).

The one-SE rule picks the largest such λ (simplest model): $\lambda = 1$; the minimum-CV rule picks $\lambda = 0.3$.

Answer: $\lambda = 1$.

Q.19 — Answer: -1.0038 to -0.9998 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$z = Wx + b = (2.5, 2.5, -1.5)$; $p = \text{softmax}(z) = (0.4955, 0.4955, 0.0091)$.

$\delta = \partial L / \partial z = p - y = (-0.5045, 0.4955, 0.0091)$ (true class 1).

Since $z_k = \sum_j W_{kj} x_j + b_k$: $\partial L / \partial W_{kj} = (p_k - y_k) x_j$ and $\partial L / \partial b_k = p_k - y_k$ (gradient matrix = $(p - y)x^T$).

$W_{31} \leftarrow -1 - 0.1 \times 0.0091 \times 2 = \mathbf{-1.0018}$.

Q.20 — Answer: -2.2072 to -2.1872 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

$P(y=1|x) \geq 0.9 \Leftrightarrow z \geq c$, where $c = \ln(0.9/0.1) = 2.1972$.

Decision boundary: $0 + (-1)x_1 + (2)x_2 = 2.1972$ (a straight line; changing the threshold only shifts it parallel to the 0.5 boundary).

$x_1 = (c - w_0)/w_1 = (2.1972 - (0))/(-1) = \mathbf{-2.2}$.

Q.21 — Answer: (A) [MCQ, 2M · Unsupervised Learning > Distance measures]

Compute the measure from $q = (4, 2)$ to every point (cosine column shows similarity, larger = more similar):

Point	coordinates	Manhattan	Euclidean	Chebyshev	cosine similarity
A	(1, 4)	5	3.606	3	0.651
B	(0, 2)	4	4	4	0.447
C	(5, 6)	5	4.123	4	0.916
D	(0, 6)	8	5.657	4	0.447

Under Euclidean (L_2) distance the nearest to q is **point A**.

Note how the answer depends on the measure: Manhattan (L_1) distance $\rightarrow$ B; Euclidean (L_2) distance $\rightarrow$ A; Chebyshev (L_∞) distance $\rightarrow$ A; cosine similarity $\rightarrow$ C. Cosine similarity looks only at the angle from the origin, L_1 sums coordinate gaps, L_∞ takes the largest gap.

Q.22 — Answer: 44.09 to 44.11 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

With $X = U\Sigma V^T$ (X centered, $n \times d$), the sample covariance is $S = X^T X / (n - 1) = V(\Sigma^2 / (n - 1))V^T$, so the PC directions are the columns of V and the eigenvalues are $\lambda_j = \sigma_j^2 / (n - 1)$.

$\lambda = 24^2/10 = 57.6$, $21^2/10 = 44.1$, $7^2/10 = 4.9$.

Cumulative % variance: 54.03, 95.4, 100 (note the $(n - 1)$ factor cancels in proportions: $PVE = \sigma_j^2 / \Sigma \sigma^2$).

Answer: **44.1**. (Using σ instead of σ^2 is the classic mistake.)

Q.23 — Answer: 0.138 to 0.142 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

$P(\text{error}) = \pi_0 P(x \geq 4.5 | C_0) + \pi_1 P(x < 4.5 | C_1)$.

$= 0.4 \times (5 - 4.5)/5 + 0.6 \times (4.5 - 4)/3 = 0.04 + 0.1 = \mathbf{0.14}$.

(For comparison, the Bayes error is 0.08: the Bayes rule assigns the whole overlap to the class with the larger $\pi_k p(x|C_k)$.)

Q.24 — Answer: (D) [MSQ, 2M · SVM & Decision Trees > SVM: kernels]

A Gram matrix of a valid kernel must be symmetric and positive semidefinite. For a symmetric 2×2 matrix this is equivalent to: both diagonal entries ≥ 0 and determinant ≥ 0 .

(A) FALSE. $\det = -2 < 0 \Rightarrow$ one negative eigenvalue, not PSD.

(B) FALSE. $\det = -7 < 0 \Rightarrow$ one negative eigenvalue, not PSD.

(C) FALSE. a diagonal entry $K(x, x) = \|\phi(x)\|^2$ is negative — impossible.

(D) TRUE. symmetric, diagonal ≥ 0 , $\det = 0 \Rightarrow$ PSD (singular) — still a valid Gram matrix.

Q.25 — Answer: 0.69 to 0.71 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

AUC = fraction of (positive, negative) pairs in which the positive has the higher score (Mann-Whitney statistic).

For each positive, count negatives with lower score: $0.7 \rightarrow 4$; $0.65 \rightarrow 4$; $0.55 \rightarrow 4$; $0.15 \rightarrow 2$.

Concordant pairs = 14 out of $4 \times 5 = 20$.

AUC = $14/20 = 0.7$.

(Equivalently, trace the ROC staircase by lowering the threshold and add the rectangle areas.)

Q.26 — Answer: 0.99 to 1.01 [NAT, 2M · SVM & Decision Trees > SVM: dual & support vectors]

$w = \sum \alpha_i y_i x_i$ over points with $\alpha_i > 0$:

$w = (0.34)(-1)(3, 2) + (0.22)(+1)(1, 4) + (0.12)(+1)(0, 0) = (-4/5, 1/5)$.

b from any support vector, e.g. P2: $y = w^T x + b = b = -1 - (-2) = 1$.

Check: $\sum \alpha_i y_i = 0$ ($= 0$, as required).

Answer: **1**.

Q.27 — Answer: 4.8067 to 4.8267 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$m_1 = (1, 1.667)$, $m_2 = (5, 4)$.

$$S_1 = \begin{bmatrix} 2 & -2 \\ -2 & 4.667 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

$$S_W = \begin{bmatrix} 4 & 0 \\ 0 & 6.667 \end{bmatrix}$$

$m_2 - m_1 = (4, 2.333)$.

$w = S_W^{-1}(m_2 - m_1) = (1, 0.35)$.

$(m_2 - m_1)^T w = 4.82$.

Q.28 — Answer: 1.7851 to 1.8051 [NAT, 2M · SVM & Decision Trees > SVM: margin & geometry]

P2 and P5 (same class) lie on one margin line and P1 on the other; the optimal boundary is parallel to the line through P2, P5, midway to P1. Support vectors: P1, P2, P5.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (5/3, 2/3)$, $b = -8$; $\|w\| = 1.7951$, margin $2/\|w\| = 1.1142$.

Point	P1	P2	P3	P4	P5	P6	P7	P8
$y(w^T x + b)$	1	1	1.333	5.333	1	1.333	3	3

All values are ≥ 1 (feasible), with equality exactly at the support vectors.

$\|w\| = 1.8$.

Q.29 — Answer: (B), (C) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Target AND truth table for $(x_1, x_2) = (0,0), (0,1), (1,0), (1,1)$: 0, 0, 0, 1. For each option compute $a = w_1 x_1 + w_2 x_2 + b$ at the four inputs and output 1 iff $a > 0$.

(A) FALSE. $a = 3, 3.5, 4, 4.5 \rightarrow$ outputs 1, 1, 1, 1 — constant 1.

(B) TRUE. $a = -2.5, -0.5, -0.5, 1.5 \rightarrow$ outputs 0, 0, 0, 1 - AND.

(C) TRUE. $a = -3, -1.5, -0.5, 1 \rightarrow$ outputs 0, 0, 0, 1 - AND.

(D) FALSE. $a = -1.5, 1.5, -0.5, 2.5 \rightarrow$ outputs 0, 1, 0, 1 - x_2 (identity of x_2).

Q.30 — Answer: (A), (B), (C), (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

(A) TRUE. CE = negative log-likelihood of a categorical/Bernoulli model.

(B) TRUE. Reduces the noise floor of SGD.

(C) TRUE. Its expectation equals the average gradient.

(D) TRUE. Variance $\propto 1/B$.

Q.31 — Answer: (A), (C) [MSQ, 2M · SVM & Decision Trees > SVM: soft margin & hinge loss]

$$\xi_i = \max(0, 1 - y_i f(x_i)).$$

Point	$y f(x)$	ξ
P1	1	0
P2	-3	4
P3	-1	2
P4	0.5	0.5
P5	4	0
P6	1.5	0

(A) TRUE. $y f = -1 \Rightarrow \xi = 2$.

(B) FALSE. $y f = -1 \Rightarrow \xi = 2$.

(C) TRUE. $y f = 4 \Rightarrow \xi = 0$.

(D) FALSE. $y f = 4 \Rightarrow \xi = 0$.

Q.32 — Answer: (B) [MCQ, 2M · Unsupervised Learning > Hierarchical clustering]

Use single linkage: $d(A, B) = \min_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P2\} + \{P3\}$ at height 2. Updated: $d(\{P2, P3\}, \{P1\}) = 9$, $d(\{P2, P3\}, \{P4\}) = 7$, $d(\{P2, P3\}, \{P5\}) = 6$, $d(\{P2, P3\}, \{P6\}) = 5$.

Merge 2: $\{P1\} + \{P5\}$ at height 3. Updated: $d(\{P1, P5\}, \{P4\}) = 7$, $d(\{P1, P5\}, \{P6\}) = 7$, $d(\{P1, P5\}, \{P2, P3\}) = 6$.

Merge 3: $\{P4\} + \{P6\}$ at height 4. Updated: $d(\{P4, P6\}, \{P2, P3\}) = 5$, $d(\{P4, P6\}, \{P1, P5\}) = 7$.

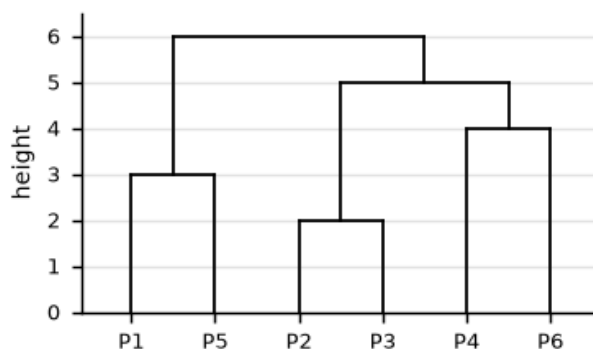
Merge 4: $\{P2, P3\} + \{P4, P6\}$ at height 5. Updated: $d(\{P2, P3, P4, P6\}, \{P1, P5\}) = 6$.

Merge 5: $\{P1, P5\} + \{P2, P3, P4, P6\}$ at height 6.

Stopping when 2 clusters remain (i.e. after 4 merges): **$\{P1, P5\}, \{P2, P3, P4, P6\}$** .

$\{P1, P4, P5, P6\}, \{P2, P3\}$ is what *complete* linkage would give.

$\{P1, P5\}, \{P2, P3\}, \{P4, P6\}$ has 3 clusters (cut at the wrong level).



Q.33 — Answer: 2.1307 to 2.1507 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

Ridge fitted values $\hat{y} = H_\lambda y$ with $H_\lambda = X(X^T X + \lambda I)^{-1} X^T$. Writing $X^T X = V D V^T$ with eigenvalues d_j^2 gives $\text{trace}(H_\lambda) = \sum d_j^2 / (d_j^2 + \lambda)$ (effective degrees of freedom).

$$25/(25 + 5) + 16/(16 + 5) + 6/(6 + 5) = 2.1407.$$

Answer: **2.14**.

Q.34 — Answer: -0.0404 to -0.0396 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Activation functions]

Forward: $a_1 = 1 \cdot -0.5 + -0.5 = -1$, $h_1 = 0.2689$; $a_2 = -2 \cdot 0.2689 + -1 = -1.5379$, $h_2 = 0.1768$; $a_3 = 3 \cdot 0.1768 + 0 = 0.5305$, $y = 0.6296$.

Local derivatives $\sigma'(a)$: $\sigma'(a) = \sigma(a)(1 - \sigma(a))$: $\sigma'(a_1) = 0.1966$, $\sigma'(a_2) = 0.1456$, $\sigma'(a_3) = 0.2332$.

Chain rule: $\partial y / \partial x = \sigma'(a_3) \cdot w_3 \cdot \sigma'(a_2) \cdot w_2 \cdot \sigma'(a_1) \cdot w_1 = -0.04$.

Each layer contributes a factor $\sigma' \leq 0.25$ – the product shrinks with depth (vanishing gradient).

Q.35 — Answer: (A), (B), (C), (D) [MSQ, 2M · Regression & Model Selection › Multiple linear regression]

(A) TRUE. $2 \times 3 = 6$.

(B) TRUE. A is the reference level (all dummies 0).

(C) TRUE. The three dummies would sum to the intercept column (dummy-variable trap).

(D) TRUE. Difference of the two dummy coefficients: $2 - (3.9)$.

Mock Test 28

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	2	8	25

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
SVM & Decision Trees	7	11
Regression & Model Selection	7	11
Neural Networks (MLP / Feed-forward)	7	11
Unsupervised Learning	7	11
Probabilistic & Instance-based Classifiers	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MSQ (one or more correct)]

Which of the following statements about kernel functions (Mercer kernels) is/are CORRECT?

- (A) If k is valid, $c \cdot k$ is valid for every $c < 0$.
- (B) Every valid kernel yields a symmetric PSD Gram matrix for any finite set of points.
- (C) Any function of the form $k(x, z) = \phi(x)^T \phi(z)$ is a valid kernel.
- (D) $p(k(x, z))$ is valid for any polynomial p with non-negative coefficients when k is valid.

Q.2 [1 mark · MSQ (one or more correct)]

Which of the following statements about cross-validation (CV) is/are CORRECT? (n denotes the number of observations.)

- (A) For least-squares linear regression, LOOCV error can be computed from a single fit using the leverages h_{ii} .
- (B) The LOOCV estimate involves no randomness in the choice of splits.
- (C) In k -fold CV every observation is used for validation exactly once.
- (D) Selecting features on the full data set and then cross-validating the selected model gives an optimistic error estimate.

Q.3 [1 mark · NAT (numerical answer)]

In AdaBoost with $\varepsilon_t = 0.45$, before normalisation each misclassified example's weight is multiplied by e^{α_t} , where $\alpha_t = \frac{1}{2} \ln((1 - \varepsilon_t)/\varepsilon_t)$. This multiplying factor is _____ (rounded off to two decimal places)

Q.4 [1 mark · NAT (numerical answer)]

For a canonical hard-margin SVM (support vectors satisfy $y_i(w^T x_i + b) = 1$), the distance between the two margin hyperplanes is found to be 4. The value of $\|w\|^2$ is _____ (rounded off to two decimal places)

Q.5 [1 mark · MSQ (one or more correct)]

Consider ridge regression (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_2^2$) and the lasso (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_1$) with $\lambda \geq 0$. Which of the following statements is/are CORRECT?

- (A) The ridge objective $\|y - X\beta\|^2 + \lambda\|\beta\|^2$ ($\lambda > 0$) is non-convex.
- (B) The L1 penalty $|\beta|$ is not differentiable at $\beta = 0$.
- (C) Ridge regression requires $X^T X$ to be invertible.
- (D) The training error of ridge regression decreases as λ increases.

Q.6 [1 mark · MSQ (one or more correct)]

Which of the following statements about backpropagation is/are CORRECT?

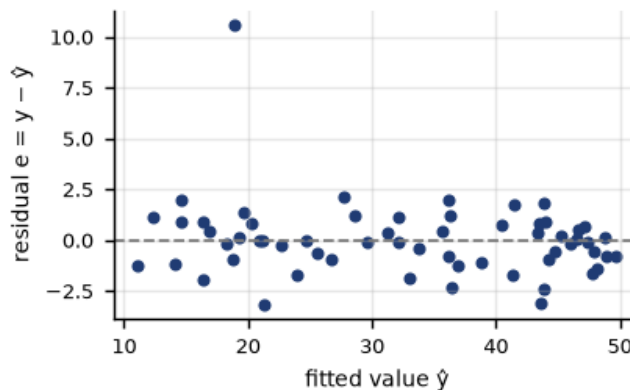
- (A) Gradients with respect to biases are always zero, so biases need not be trained.
- (B) The gradient of a sum of per-example losses equals the sum of the per-example gradients.
- (C) If a ReLU unit's pre-activation is negative for an input, no gradient flows through that unit for that input.
- (D) Backpropagation only computes gradients; the parameter update rule (SGD, momentum, Adam) is a separate choice.

Q.7 [1 mark · NAT (numerical answer)]

Inverted dropout with keep probability $p = 0.5$ is applied during training. A hidden unit with activation $h = 3.5$ that is **kept** passes on the value _____ (rounded off to two decimal places)

Q.8 [1 mark · MCQ]

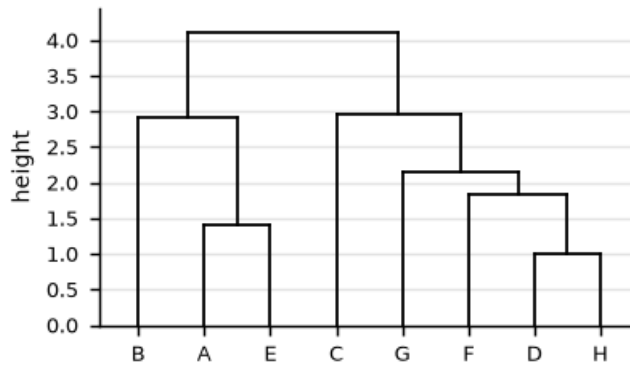
After fitting a linear regression model, the residuals are plotted against the fitted values as shown. The plot most strongly suggests:



- (A) No evident violation of the linear-model assumptions
- (B) A single outlying observation, with the rest of the fit looking adequate
- (C) Non-constant error variance (heteroscedasticity)
- (D) A non-linear relationship not captured by the model (e.g. a missing x^2 term)

Q.9 [1 mark · NAT (numerical answer)]

The dendrogram below was obtained by agglomerative clustering of 8 points A-H. If it is cut to obtain exactly 4 clusters, the number of points in the cluster that contains point C is ____ (Answer in integer)

**Q.10 [1 mark · MSQ (one or more correct)]**

Which of the following statements about the k-nearest-neighbour method is/are CORRECT?

- (A) k-NN training requires solving an optimisation problem by gradient descent.
- (B) The decision boundary of k-NN is always a straight line (hyperplane).
- (C) 1-NN has high bias and low variance.
- (D) The cost of a brute-force k-NN prediction grows linearly with the number of training points n .

Q.11 [1 mark · NAT (numerical answer)]

To keep the same sampling density as 10 points per axis in 1-D, the number of points needed in $d = 5$ dimensions (a grid with 10 points per axis) is ____ (Answer in integer)

Q.12 [1 mark · NAT (numerical answer)]

Lloyd's k-means is run on $n = 1000$ points in $d = 9$ dimensions with $k = 3$ for $T = 19$ iterations. Counting one subtraction per coordinate per point-centroid pair in the assignment step, the total number of such subtractions is ____ (Answer in integer)

Q.13 [1 mark · NAT (numerical answer)]

A binary logistic regression model is given by $P(y = 1 | \mathbf{x}) = \sigma(w_0 + w_1x_1 + w_2x_2)$ with $w_0 = -1$, $w_1 = 1.5$, $w_2 = -1$, where $\sigma(z) = 1/(1 + e^{-z})$. For the input $\mathbf{x} = (x_1, x_2) = (0, -3)$, the value of $P(y = 1 | \mathbf{x})$ is ____ (rounded off to two decimal places)

Q.14 [1 mark · NAT (numerical answer)]

A categorical naive Bayes classifier has $d = 4$ features, each taking $m = 5$ possible values, and $K = 2$ classes. The number of independent parameters, including class priors, is ____ (Answer in integer)

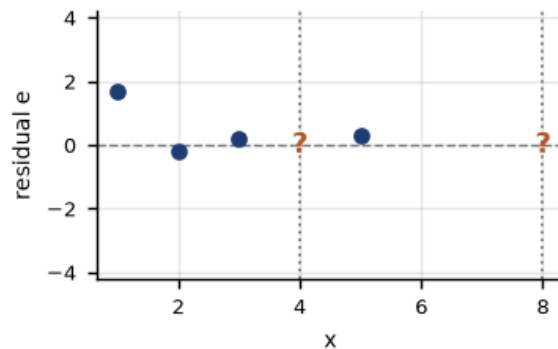
Q.15 [1 mark · NAT (numerical answer)]

A network produces logits $\mathbf{z} = (0, -1, -2, 0.5)$. A constant 100 is added to every logit. The softmax probability of class 4 computed from the shifted logits is ____ (rounded off to three decimal places)

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

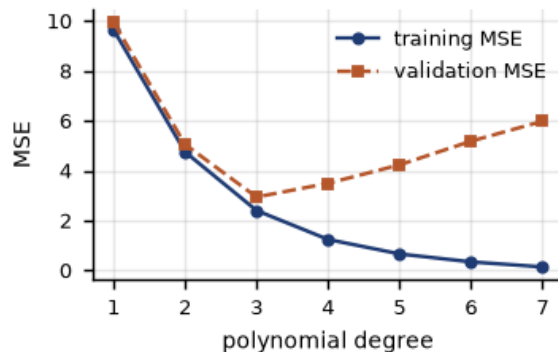
A simple linear regression with intercept is fitted by least squares. The table gives the x -values and the residuals $e_i = y_i - \hat{y}_i$, but two residuals are missing. The missing residual at $x = 4$ is _____ (rounded off to two decimal places)

x_i	1	2	3	4	5	8
e_i	1.7	-0.2	0.2	?	0.3	?

**Q.17 [2 marks · MSQ (one or more correct)]**

Polynomial regression models of degree 1 to 7 are fitted by least squares on a training set and evaluated on a separate validation set. The resulting mean squared errors are given below. Which of the following statements is/are CORRECT?

Degree	1	2	3	4	5	6	7
Training MSE	9.65	4.75	2.4	1.24	0.66	0.34	0.14
Validation MSE	9.95	5.05	2.94	3.49	4.23	5.18	5.98

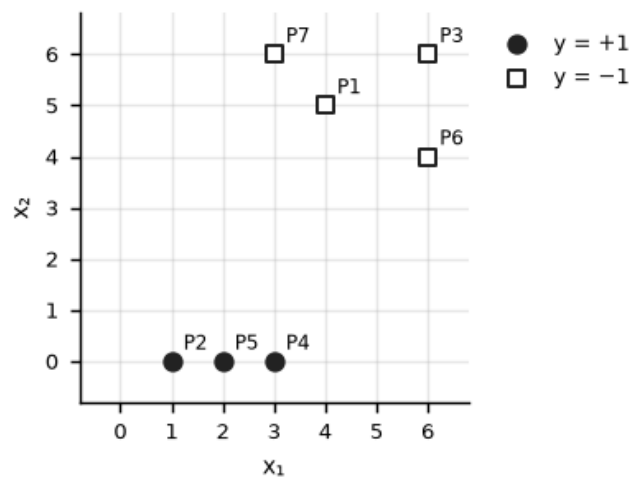


- (A) The degree-7 model has higher variance than the degree-1 model.
- (B) The degree-7 model overfits: it has the lowest training MSE but its validation MSE exceeds the minimum.
- (C) The generalisation gap (validation MSE – training MSE) is largest for degree 7.
- (D) The degree-1 model underfits: its training and validation errors are both high.

Q.18 [2 marks • MCQ]

Four candidate linear decision boundaries are proposed for the labelled training points shown. Which boundary separates the two classes with the LARGEST geometric margin (minimum distance from the boundary to any training point, all points correctly classified)?

Point	x_1	x_2	y
P1	4	5	-1
P2	1	0	+1
P3	6	6	-1
P4	3	0	+1
P5	2	0	+1
P6	6	4	-1
P7	3	6	-1

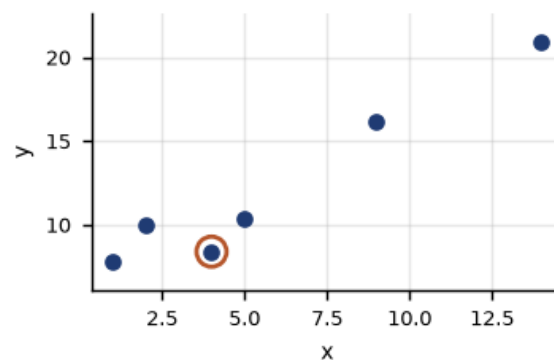


- (A) $x_2 - 2.5 = 0$
 (B) $x_1 + 2x_2 - 11 = 0$
 (C) $x_1 - 2x_2 - 0.5 = 0$
 (D) $x_1 + x_2 - 6 = 0$

Q.19 [2 marks • NAT (numerical answer)]

A simple linear regression (with intercept) is fitted by least squares to the 6 points shown. With $H = X(X^T X)^{-1} X^T$, the leverage h_{ii} of the observation with $x = 4$ (circled) is _____ (rounded off to three decimal places)

x	1	2	4	5	9	14
y	7.8	10	8.4	10.4	16.2	20.9



Q.20 [2 marks · NAT (numerical answer)]

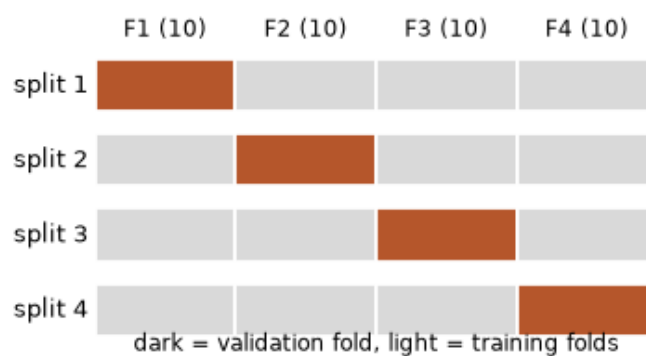
The sample covariance matrix of a two-feature dataset is S given below. In principal component analysis, the percentage of total variance explained by the first principal component is _____ (rounded off to two decimal places)

$$S = \begin{bmatrix} 7 & 3 \\ 3 & 9 \end{bmatrix}$$

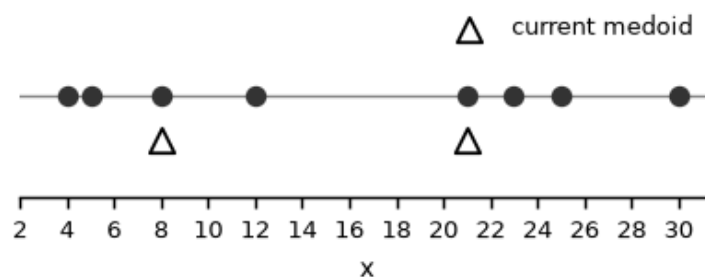
Q.21 [2 marks · NAT (numerical answer)]

Ridge regression is tuned by 4-fold cross-validation (equal fold sizes) over $\lambda \in \{0.5, 2, 10\}$. The validation MSE on each fold is shown. The CV estimate (mean over folds) of the λ that would be selected is _____ (rounded off to two decimal places)

λ	Fold 1	Fold 2	Fold 3	Fold 4
0.5	2.2	5.1	3.5	3.9
2	4.8	2.6	3	3.5
10	5.2	3.6	4.4	3.8

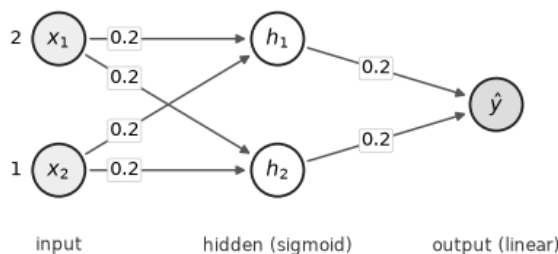
**Q.22 [2 marks · NAT (numerical answer)]**

PAM (k-medoids, $k = 2$, absolute-difference dissimilarity) is applied to the 1-D data $\{4, 5, 8, 12, 21, 23, 25, 30\}$. The current medoids are 8 and 21. The total cost $\sum_i \min_m |x_i - m|$ of the current configuration is _____ (Answer in integer)

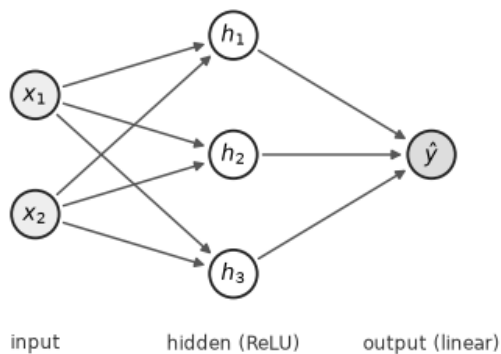


Q.23 [2 marks • NAT (numerical answer)]

In the 2-2-1 network below (sigmoid hidden units, linear output $\hat{y} = v_1 h_1 + v_2 h_2$, all biases 0), **every** weight is initialised to 0.2. One gradient-descent step with $\eta = 0.1$ is taken on $L = \frac{1}{2}(\hat{y} - y)^2$ for the input $x = (2, 1)$ with target $y = 0$. Then the updated value of w_{11} (weight from x_1 to h_1) is _____ (rounded off to three decimal places)

**Q.24 [2 marks • NAT (numerical answer)]**

The 2-3-1 network below computes $\hat{y} = v \cdot \text{ReLU}(Wx + b) + c$ with W, b, v given below and $c = 1$. With $x = (2, -2)$, target $y = 2$ and loss $L = \frac{1}{2}(\hat{y} - y)^2$, the value of $\partial L / \partial v_1$ is _____ (Answer in integer)



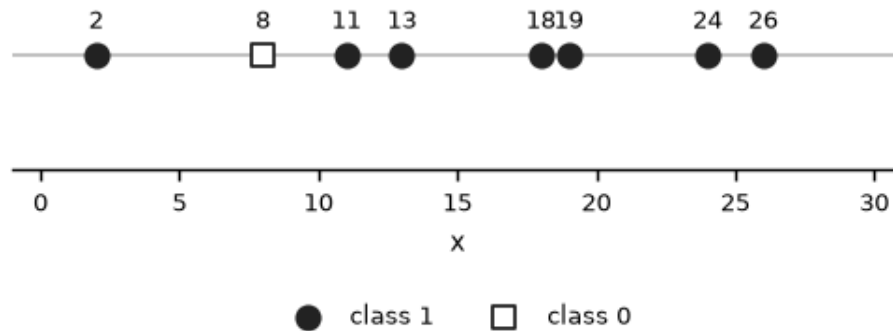
$$W = \begin{bmatrix} -1 & -2 \\ 1 & 2 \\ -2 & 2 \end{bmatrix}$$

$$b = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$v^T = \begin{bmatrix} 1 & -1 & -1 \end{bmatrix}$$

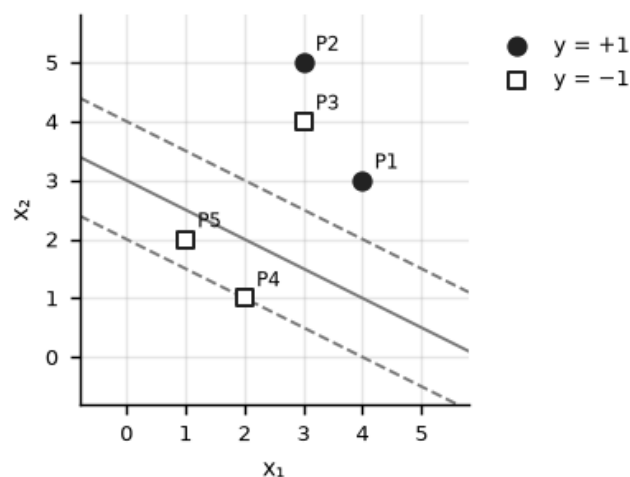
Q.25 [2 marks · NAT (numerical answer)]

8 one-dimensional points with binary labels are shown on the number line (filled circle = class 1, open square = class 0). A 1-NN classifier (absolute distance, majority vote) is evaluated by leave-one-out cross-validation: each point is classified using all the *other* points. The leave-one-out cross-validation error rate (as a fraction) is _____ (rounded off to two decimal places)

**Q.26 [2 marks · NAT (numerical answer)]**

A linear SVM with $f(x) = 0.5x_1 + x_2 - 3$ is evaluated on the training points in the table (also plotted, with the boundary $f = 0$ solid and $f = \pm 1$ dashed). Compute the value of the soft-margin primal objective $\frac{1}{2}\|w\|^2 + C \sum_i \xi_i$ with $C = 10$ (using the optimal slacks for this (w, b)). (rounded off to two decimal places)

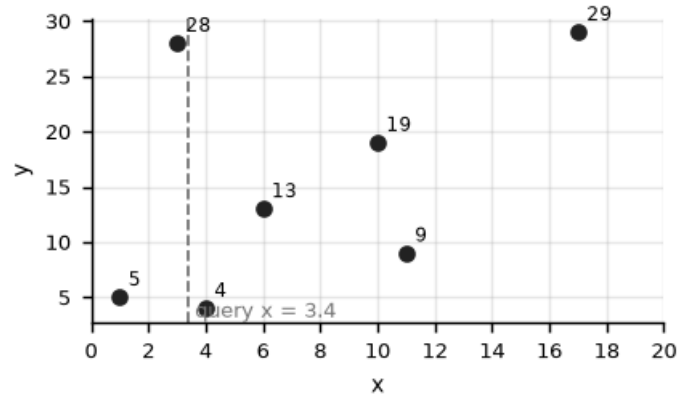
Point	x_1	x_2	y
P1	4	3	+1
P2	3	5	+1
P3	3	4	-1
P4	2	1	-1
P5	1	2	-1



Q.27 [2 marks · NAT (numerical answer)]

A distance-weighted k-NN regressor with $k = 3$ predicts $\hat{y} = \sum w_i y_i / \sum w_i$ over the 3 nearest neighbours, with weights $w_i = 1/d_i^2$. For the training data below, the prediction at $x = 3.4$ is _____ (rounded off to two decimal places)

x	1	3	4	6	10	11	17
y	5	28	4	13	19	9	29

**Q.28 [2 marks · NAT (numerical answer)]**

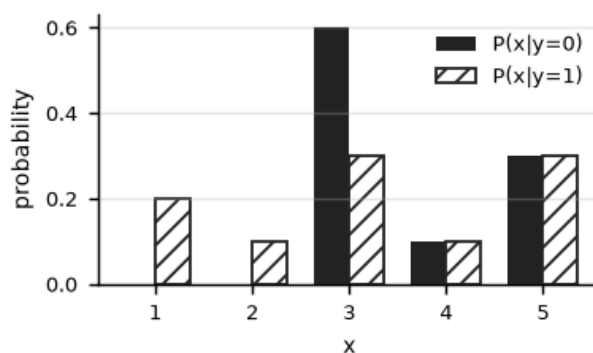
Agglomerative hierarchical clustering with **single linkage** is applied to 6 points whose pairwise distance matrix is given below (upper triangle shown). The height (linkage distance) at which the **second** merge takes place is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5	P6
P1	0	7	2	8	12	13
P2		0	7	5	7	8
P3			0	10	14	15
P4				0	4	5
P5					0	1
P6						0

Q.29 [2 marks · NAT (numerical answer)]

A discrete feature $x \in \{1, \dots, 5\}$ has the class-conditional distributions below (also plotted), with priors $P(y = 0) = 0.6$ and $P(y = 1) = 0.4$. Then the Bayes error rate of the optimal classifier is _____ (rounded off to three decimal places)

x	1	2	3	4	5
$P(x y=0)$	0	0	0.6	0.1	0.3
$P(x y=1)$	0.2	0.1	0.3	0.1	0.3



Q.30 [2 marks · NAT (numerical answer)]

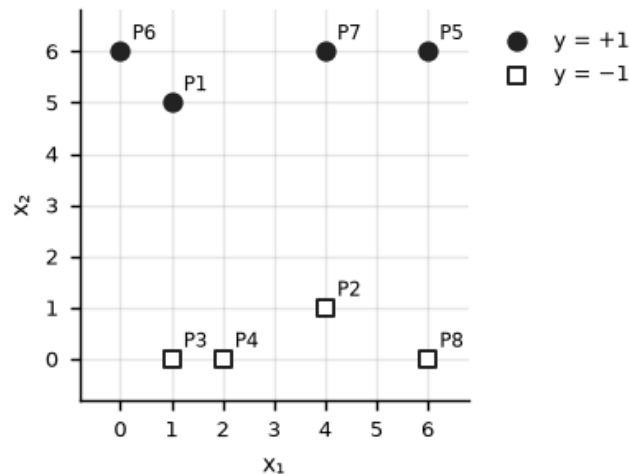
The continuous attribute Income and the class label of 9 training records are given below (unsorted). A binary split “Income $\leq t$ ” is chosen to maximise information gain, with t restricted to midpoints between consecutive sorted values. Find the number of candidate thresholds that need to be evaluated (midpoints between consecutive sorted values at which the class label changes). (Answer in integer)

Income	25	49	74	63	15	56	72	65	67
Class	Yes	Yes	No	No	Yes	No	Yes	No	No

**Q.31 [2 marks · MSQ (one or more correct)]**

A hard-margin linear SVM is trained on the 8 labelled points shown. Which of the following points is/are support vectors of the resulting maximum-margin classifier?

Point	x_1	x_2	y
P1	1	5	+1
P2	4	1	-1
P3	1	0	-1
P4	2	0	-1
P5	6	6	+1
P6	0	6	+1
P7	4	6	+1
P8	6	0	-1



- (A) P1
(C) P3

- (B) P2
(D) P5

Q.32 [2 marks · NAT (numerical answer)]

A 2-2-1 network uses the identity (linear) activation in its hidden layer: $\hat{y} = W_2(W_1x + b_1) + b_2$, with the parameters below. The network is equivalent to a single linear model $\hat{y} = \tilde{w}_1x_1 + \tilde{w}_2x_2 + \tilde{b}$. Then the intercept $\tilde{b}$ is _____ (Answer in integer)

$$W_1 = \begin{bmatrix} 1 & -2 \\ -1 & 1 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} -1 & -1 \end{bmatrix}$$

$$b_2 = -2$$

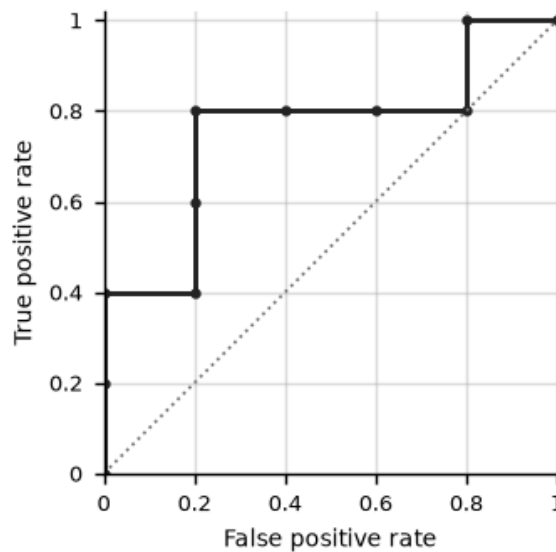
Q.33 [2 marks · MSQ (one or more correct)]

Consider k-means (Lloyd's algorithm) and k-medoids (PAM). Which of the following statements is/are CORRECT?

- (A) The medoid of a cluster is the member that minimizes the sum of dissimilarities to the other members.
- (B) Within a cluster C , $\sum_{x \in C} \|x - \mu\|^2 = (1/(2|C|)) \sum_{x \in C} \sum_{y \in C} \|x - y\|^2$.
- (C) For 1-D data and a single cluster, the k-medoids representative under absolute distance is a median data point.
- (D) k-medoids is generally more robust to outliers than k-means.

Q.34 [2 marks · NAT (numerical answer)]

The ROC curve below was obtained from a test set with 5 positives and 5 negatives (all scores distinct, so the curve moves in steps of 1/5 vertically and 1/5 horizontally). From the figure, the highest TPR attainable at FPR ≤ 0.4 is _____ (rounded off to two decimal places)

**Q.35 [2 marks · NAT (numerical answer)]**

A 3-class classifier outputs logits $z = (0, 0.5, 2)$ and probabilities $p = \text{softmax}(z)$. The true class is 1, and the loss is the cross-entropy $L = -\ln p_1$. Then the cross-entropy loss $L = -\ln p_y$ (natural log) is _____ (rounded off to three decimal places)

— END OF QUESTION PAPER —

Mock Test 28 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MSQ	1	(B), (C), (D)	19	NAT	2	0.193 to 0.1969
2	MSQ	1	(A), (B), (C), (D)	20	NAT	2	69.7542 to 69.7742
3	NAT	1	1.0955 to 1.1155	21	NAT	2	3.465 to 3.485
4	NAT	1	0.24 to 0.26	22	NAT	2	26
5	MSQ	1	(B)	23	NAT	2	0.1956 to 0.1996
6	MSQ	1	(B), (C), (D)	24	NAT	2	6
7	NAT	1	6.99 to 7.01	25	NAT	2	0.24 to 0.26
8	MCQ	1	(B)	26	NAT	2	40.615 to 40.635
9	NAT	1	1	27	NAT	2	20.3108 to 20.3308
10	MSQ	1	(D)	28	NAT	2	1.99 to 2.01
11	NAT	1	100000	29	NAT	2	0.278 to 0.282
12	NAT	1	513000	30	NAT	2	3
13	NAT	1	0.8708 to 0.8908	31	MSQ	2	(A), (B), (D)
14	NAT	1	33	32	NAT	2	−4
15	NAT	1	0.5211 to 0.5251	33	MSQ	2	(A), (B), (C), (D)
16	NAT	2	−3.1658 to −3.1342	34	NAT	2	0.79 to 0.81
17	MSQ	2	(A), (B), (C), (D)	35	NAT	2	2.3044 to 2.3084
18	MCQ	2	(D)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
SVM & Decision Trees					11
Regression & Model Selection					11
Neural Networks (MLP / Feed-forward)					11
Unsupervised Learning					11
Probabilistic & Instance-based Classifiers					11
Total					55

Mock Test 28 — Detailed Solutions

Q.1 — Answer: (B), (C), (D) [MSQ, 1M · SVM & Decision Trees > SVM: kernels]

- (A) FALSE. Negative scaling breaks PSD.
- (B) TRUE. Mercer's condition.
- (C) TRUE. Gram matrix = $\Phi\Phi^T$ is PSD.
- (D) TRUE. Sums, products and positive scalings.

Q.2 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Cross-validation]

- (A) TRUE. $CV_{(n)} = (1/n)\sum(e_i/(1 - h_{ii}))^2$.
- (B) TRUE. There is exactly one way to leave one out.
- (C) TRUE. Folds partition the data.
- (D) TRUE. Classic data leakage — the selection already saw the validation folds.

Q.3 — Answer: 1.0955 to 1.1155 [NAT, 1M · SVM & Decision Trees > Ensembles: boosting]

$$e^\alpha = e^{\frac{1}{2} \ln((1-\epsilon)/\epsilon)} = \sqrt{((1-\epsilon)/\epsilon)} = \sqrt{1.2222} = 1.11.$$

Correctly classified examples are multiplied by $e^{-\alpha} = \sqrt{(\epsilon/(1-\epsilon))}$.

$\epsilon < 0.5 \Rightarrow \alpha > 0$; $\epsilon = 0.5 \Rightarrow \alpha = 0$ (no better than chance).

Q.4 — Answer: 0.24 to 0.26 [NAT, 1M · SVM & Decision Trees > SVM: margin & geometry]

Margin width $2/\|w\| = 4 \Rightarrow \|w\| = 2/4 = 0.5$.

$$\|w\|^2 = 0.25.$$

Q.5 — Answer: (B) [MSQ, 1M · Regression & Model Selection > Ridge & Lasso]

- (A) FALSE. It is strictly convex.
- (B) TRUE. Its subgradient at 0 is the interval $[-1, 1]$.
- (C) FALSE. $X^T X + \lambda I$ is invertible for any $\lambda > 0$.
- (D) FALSE. $\lambda = 0$ (OLS) minimises training RSS; training error is non-decreasing in λ .

Q.6 — Answer: (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Backpropagation]

- (A) FALSE. $\partial L / \partial b_j = \delta_j$, generally non-zero.
- (B) TRUE. Differentiation is linear.
- (C) TRUE. $\text{ReLU}'(a) = 0$ for $a < 0$.
- (D) TRUE. Backprop $\neq$ optimiser.

Q.7 — Answer: 6.99 to 7.01 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Inverted dropout multiplies kept activations by $1/p$ so that $E[\text{output}] = h$: $3.5/0.5 = 7$. No scaling is applied at test time.

Q.8 — Answer: (B) [MCQ, 1M · Regression & Model Selection > Simple linear regression]

All residuals lie in a narrow band except one far from 0 $\rightarrow$ an outlier that should be investigated.

Answer: **A single outlying observation, with the rest of the fit looking adequate.**

Other patterns: funnel $\rightarrow$ heteroscedasticity; curvature $\rightarrow$ missing non-linearity; isolated point $\rightarrow$ outlier; random band $\rightarrow$ adequate fit.

Q.9 — Answer: 1 [NAT, 1M · Unsupervised Learning > Hierarchical clustering]

Undo the top 3 merge(s) (equivalently cut just below height 2.92). Clusters: $\{A, E\}$, $\{B\}$, $\{C\}$, $\{D, F, G, H\}$.

C lies in a cluster of size 1.

Q.10 — Answer: (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

- (A) FALSE. It simply stores the data.
- (B) FALSE. It is generally piecewise and non-linear.
- (C) FALSE. 1-NN has low bias and high variance.
- (D) TRUE. Distances to all n points are computed for each query.

Q.11 — Answer: 100000 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

A grid with 10 points per axis needs $10^5 = \mathbf{100000}$ points — exponential growth with d .

Q.12 — Answer: 513000 [NAT, 1M · Unsupervised Learning > k-means]

Assignment cost per iteration = $n \cdot k \cdot d = 1000 \cdot 3 \cdot 9 = 27000$; over T iterations **513000**. This is the $O(nkdT)$ complexity of k-means.

Q.13 — Answer: 0.8708 to 0.8908 [NAT, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

$z = w_0 + w_1x_1 + w_2x_2 = -1 + (1.5)(0) + (-1)(-3) = 2$ (this is the log-odds).

$P(y = 1 | \mathbf{x}) = \sigma(z) = 1/(1 + e^{-2}) = 0.8808$.

Answer: **0.88**.

Q.14 — Answer: 33 [NAT, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

For each class and feature, a distribution over m values has $m - 1 = 4$ free parameters: $Kd(m - 1) = 32$.

Priors: $K - 1 = 1$. Total = **33**.

Q.15 — Answer: 0.5211 to 0.5251 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

Adding a constant to all logits does not change softmax (e^c cancels).

exp values: 1, 0.3679, 0.1353, 1.6487; sum = 3.1519.

$p_4 = 1.6487/3.1519 = \mathbf{0.5231}$.

Q.16 — Answer: -3.1658 to -3.1342 [NAT, 2M · Regression & Model Selection > Simple linear regression]

For the least-squares fit with intercept the normal equations give $\Sigma e_i = 0$ and $\Sigma x_i e_i = 0$.

Known: $\Sigma e = 2$, $\Sigma xe = 3.4$. With unknowns $u = e$ at $x = 4$ and $v = e$ at $x = 8$:

$u + v = -2$, $4u + 8v = -3.4$.

Solving: $u = -3.15$, $v = 1.15$.

Answer: **-3.15**.

Q.17 — Answer: (A), (B), (C), (D) [MSQ, 2M · Regression & Model Selection > Bias-variance trade-off]

(A) TRUE. Higher-degree polynomials are more flexible $\rightarrow$ higher variance.

(B) TRUE. Train 0.14 (lowest), validation 5.98 > 2.94.

(C) TRUE. Gaps: 0.3, 0.3, 0.54, 2.25, 3.57, 4.84, 5.84.

(D) TRUE. Training 9.65 and validation 9.95 are both large — high bias.

Q.18 — Answer: (D) [MCQ, 2M · SVM & Decision Trees > SVM: margin & geometry]

For $n^T x - c = 0$, the geometric margin is $\min_i y_i s(n^T x_i - c)/\|n\|$ with orientation $s = \pm 1$ chosen so that the positive class is on the positive side; a negative value means some point is misclassified.

Hyperplane	$\ n\ $	$\min_i$ distance (signed, best orientation)
$x_2 - 2.5 = 0$	1	1.5
$x_1 + 2x_2 - 11 = 0$	2.236	1.342
$x_1 - 2x_2 - 0.5 = 0$	2.236	0.224
$x_1 + x_2 - 6 = 0$	1.414	2.121

The largest margin is 2.121, for $\mathbf{x_1 + x_2 - 6 = 0}$. Note that a boundary can separate the data perfectly yet pass close to some point — only the minimum distance matters.

Q.19 — Answer: 0.193 to 0.1969 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

For simple linear regression with intercept, the leverage (diagonal of the hat matrix) is $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$.

$\bar{x} = 5.8333$, $S_{xx} = 118.8333$.

$h = 1/6 + (-1.8333)^2/118.8333 = 0.195$.

Answer: **0.195**.

Q.20 — Answer: 69.7542 to 69.7742 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

Eigenvalues of $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$: $\lambda = (a + c)/2 \pm \sqrt{((a - c)/2)^2 + b^2}$.

$\lambda = 8 \pm \sqrt{(1 + 9)} = 8 \pm 3.1623 \rightarrow \lambda_1 = 11.1623, \lambda_2 = 4.8377$ (check: $\lambda_1 + \lambda_2 = \text{trace} = 16, \lambda_1\lambda_2 = \det = 54$).

Eigenvector for λ_1 : $(S - \lambda_1 I)v = 0 \rightarrow (-4.1623)v_1 + (3)v_2 = 0 \rightarrow v \propto (1, 1.3874)$.

$\text{PVE}(\text{PC1}) = \lambda_1/(\lambda_1 + \lambda_2) = 69.76\%$.

Answer: **69.76**.

Q.21 — Answer: 3.465 to 3.485 [NAT, 2M · Regression & Model Selection > Cross-validation]

$\text{CV}(\lambda) = \text{average of the fold MSEs}$: $\lambda = 0.5$: 3.675; $\lambda = 2$: 3.475; $\lambda = 10$: 4.25.

Smallest CV error at $\lambda = 2$: $\text{CV} = 3.475$.

Answer: **3.48**.

Q.22 — Answer: 26 [NAT, 2M · Unsupervised Learning > k-medoids]

PAM cost = Σ over all points of the absolute distance to the nearest medoid.

Current medoids $\{8, 21\}$: distances $4 \rightarrow 4, 5 \rightarrow 3, 8 \rightarrow 0, 12 \rightarrow 4, 21 \rightarrow 0, 23 \rightarrow 2, 25 \rightarrow 4, 30 \rightarrow 9$; cost = **26**.

Q.23 — Answer: 0.1956 to 0.1996 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

All weights = 0.2, biases 0 $\Rightarrow$ both hidden units get $a = 0.2 \times (2 + 1) = 0.6, h_1 = h_2 = \sigma(a) = 0.6457$; $\hat{y} = 2 \times 0.2 \times 0.6457 = 0.2583$.

$\hat{y} - y = 0.2583, \partial L / \partial v_j = (\hat{y} - y)h_j = 0.1667$ (same for both j).

$\partial L / \partial w_{jk} = (\hat{y} - y) v_j h_j (1 - h_j) x_k = 0.2583 \times 0.2 \times 0.2288 \times 2 = 0.0236$ (identical for $j = 1, 2$).

$w_{11} \leftarrow 0.2 - 0.1 \times (0.0236) = \mathbf{0.1976}$.

Because the two hidden units start identical and get identical updates, they remain identical forever (symmetry problem) – hence weights must be initialised randomly.

Q.24 — Answer: 6 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

$a = Wx + b = (3, -2, -9), h = \text{ReLU}(a) = (3, 0, 0), \hat{y} = v \cdot h + c = 4; \partial L / \partial \hat{y} = \hat{y} - y = 2$.

$\text{ReLU}'(a) = (1, 0, 0)$ (unit j is active iff $a_j > 0$).

$\partial L / \partial v_1 = (\hat{y} - y) \cdot h_1 = 2 \times 3 = \mathbf{6}$.

Q.25 — Answer: 0.24 to 0.26 [NAT, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Classify each point by its nearest neighbour(s) among the remaining points:

x	true	1 nearest (excluding itself)	LOO prediction	
2	1	8(0)	0	wrong
8	0	11(1)	1	wrong
11	1	13(1)	1	
13	1	11(1)	1	
18	1	19(1)	1	
19	1	18(1)	1	
24	1	26(1)	1	
26	1	24(1)	1	

Misclassified: 2 of 8. Error rate = $2/8 = \mathbf{0.25}$.

Note: the *training* error of 1-NN is 0, but the LOOCV error is not — a point cannot be its own neighbour.

Q.26 — Answer: 40.615 to 40.635 [NAT, 2M · SVM & Decision Trees > SVM: soft margin & hinge loss]

Compute $f(x_i)$, the functional margin $y_i f(x_i)$ and the slack/hinge loss for each point:

Point	x	y	$f(x)$	$y f(x)$	$\xi = \max(0, 1 - y f)$
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P1	(4, 3)	+1	2	2	0
P2	(3, 5)	+1	3.5	3.5	0
P3	(3, 4)	-1	2.5	-2.5	3.5
P4	(2, 1)	-1	-1	1	0
P5	(1, 2)	-1	-0.5	0.5	0.5

$\frac{1}{2}\|w\|^2 = 0.625$; objective = $0.625 + 10 \times 4 = \mathbf{40.62}$.

Points with $y f \geq 1$ contribute 0; points inside the margin contribute between 0 and 1; misclassified points contribute more than 1.

Q.27 — Answer: 20.3108 to 20.3308 [NAT, 2M · Probabilistic & Instance-based Classifiers · k-nearest neighbours]

Three nearest neighbours: $x=3$ ($d=0.4$, $y=28$), $x=4$ ($d=0.6$, $y=4$), $x=1$ ($d=2.4$, $y=5$).

Weights $1/d^2$: 6.25, 2.7778, 0.1736; $\Sigma w = 9.2014$.

$\Sigma w_i y_i = 186.9792 = \hat{y} = 186.9792/9.2014 = \mathbf{20.32}$.

(The unweighted 3-NN average would be 12.33; weighting pulls $\hat{y}$ towards the closest point.)

Q.28 — Answer: 1.99 to 2.01 [NAT, 2M · Unsupervised Learning · Hierarchical clustering]

Use single linkage: $d(A,B) = \min_{a \in A, b \in B} d(a,b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P5\} + \{P6\}$ at height **1**. Updated: $d(\{P5, P6\}, \{P1\}) = 12$, $d(\{P5, P6\}, \{P2\}) = 7$, $d(\{P5, P6\}, \{P3\}) = 14$, $d(\{P5, P6\}, \{P4\}) = 4$.

Merge 2: $\{P1\} + \{P3\}$ at height **2**. Updated: $d(\{P1, P3\}, \{P2\}) = 7$, $d(\{P1, P3\}, \{P4\}) = 8$, $d(\{P1, P3\}, \{P5, P6\}) = 12$.

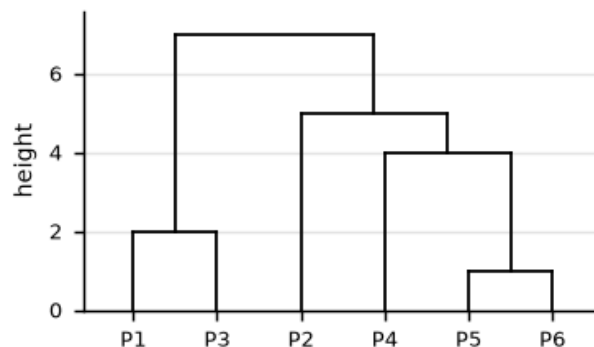
Merge 3: $\{P4\} + \{P5, P6\}$ at height **4**. Updated: $d(\{P4, P5, P6\}, \{P2\}) = 5$, $d(\{P4, P5, P6\}, \{P1, P3\}) = 8$.

Merge 4: $\{P2\} + \{P4, P5, P6\}$ at height **5**. Updated: $d(\{P2, P4, P5, P6\}, \{P1, P3\}) = 7$.

Merge 5: $\{P1, P3\} + \{P2, P4, P5, P6\}$ at height **7**.

The second merge occurs at height **2**.

Dendrogram:



Q.29 — Answer: 0.278 to 0.282 [NAT, 2M · Probabilistic & Instance-based Classifiers · Bayes optimal classifier]

Bayes rule: choose the class with larger $\pi_k P(x|k)$. Bayes error = $\sum_x \min(\pi_0 P(x|0), \pi_1 P(x|1))$.

x	$\pi_0 P(x 0)$	$\pi_1 P(x 1)$	decision	min
1	0	0.08	1	0
2	0	0.04	1	0
3	0.36	0.12	0	0.12
4	0.06	0.04	0	0.04
5	0.18	0.12	0	0.12

Bayes error = **0.28**.

Q.30 — Answer: 3 [NAT, 2M · SVM & Decision Trees · Decision trees: split selection]

Sort the values: 15(Y), 25(Y), 49(Y), 56(N), 63(N), 65(N), 67(N), 72(Y), 74(N).

Parent [4 Yes, 5 No], $H = 0.9911$. The optimal threshold always lies where the class changes between neighbours (Fayyad-Irani), giving 3 candidates:

Threshold	Left ($\leq t$) [Yes, No]	Right [Yes, No]	IG
52.5	[3, 0]	[1, 5]	0.5577
69.5	[3, 4]	[1, 1]	0.0026
73	[4, 4]	[0, 1]	0.1022

Best: $t = 52.5$ with $IG = 0.5577$. Answer: **3**.

Q.31 — Answer: (A), (B), (D) [MSQ, 2M · SVM & Decision Trees > SVM: dual & support vectors]

P1 and P5 (same class) lie on one margin line and P2 on the other; the optimal boundary is parallel to the line through P1, P5, midway to P2. Support vectors: P1, P2, P5.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (-2/23, 10/23)$, $b = -25/23$; $\|w\| = 0.4434$, margin $2/\|w\| = 4.5107$.

Point	P1	P2	P3	P4	P5	P6	P7	P8
$y(w^T x + b)$	1	1	1.174	1.261	1	1.522	1.174	1.609

Support vectors are exactly the points with $y_i(w^T x_i + b) = 1$ ($\alpha_i > 0$): **P1, P2, P5**.

Dual check: $\alpha = (\alpha_1 = 19/335, \alpha_2 = 29/295, \alpha_5 = 16/385)$ satisfy $\sum \alpha_i y_i = 0$ and $\sum \alpha_i y_i x_i = w$.

Q.32 — Answer: -4 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Expressiveness]

With identity activations, $\hat{y} = W_2(W_1 x + b_1) + b_2 = (W_2 W_1)x + (W_2 b_1 + b_2)$: a single affine map.

$\tilde{w}^T = W_2 W_1 = (0, 1)$; $\tilde{b} = W_2 b_1 + b_2 = -4$.

Answer: **-4**.

Hence stacking linear layers adds no expressive power – a non-linear activation is essential.

Q.33 — Answer: (A), (B), (C), (D) [MSQ, 2M · Unsupervised Learning > k-means]

(A) TRUE. This is the definition.

(B) TRUE. A standard identity; the SSE can be written via pairwise distances.

(C) TRUE. The median minimizes the sum of absolute deviations.

(D) TRUE. A medoid is an actual point and uses unsquared dissimilarities, so an outlier pulls it far less.

Q.34 — Answer: 0.79 to 0.81 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Among the operating points with $FPR \leq 0.4$, take the highest point of the curve (the top of the vertical segment at $FPR = 0.4$).

$TPR = \mathbf{0.8}$.

Q.35 — Answer: 2.3044 to 2.3084 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$p = \text{softmax}(z)$: $\exp(z) = (1, 1.6487, 7.3891)$, $\text{sum} = 10.0378$, so $p = (0.0996, 0.1643, 0.7361)$.

For $L = -\ln p_y$ with softmax, $\partial L / \partial z = p - y$ (y one-hot, true class 1): $\partial L / \partial z = (-0.9004, 0.1643, 0.7361)$.

$L = -\ln 0.0996 = \mathbf{2.3064}$.

Note $\sum_k \partial L / \partial z_k = \sum p_k - 1 = 0$.

Mock Test 29

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	7	4	24

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Regression & Model Selection	7	11
Neural Networks (MLP / Feed-forward)	7	11
Probabilistic & Instance-based Classifiers	7	11
SVM & Decision Trees	7	11
Unsupervised Learning	7	11

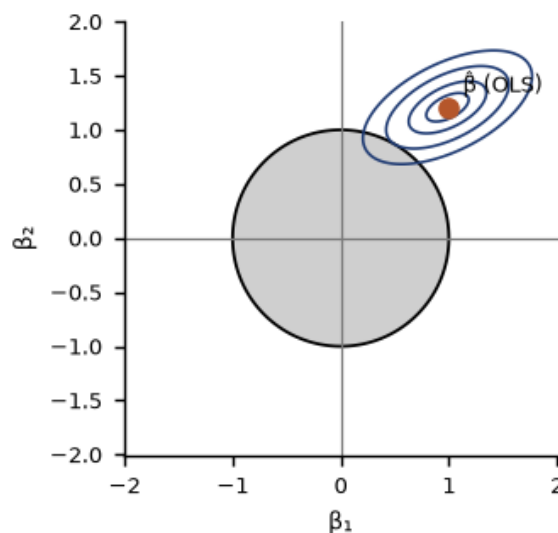
Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MCQ]

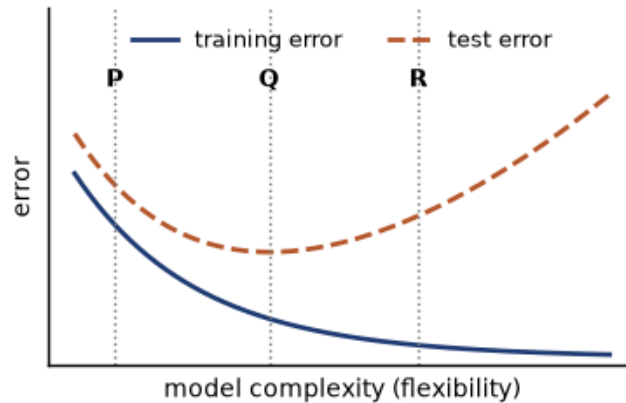
For a two-predictor penalised least-squares problem, the figure shows the constraint region (shaded) of the penalty together with elliptical contours of the residual sum of squares centred at the least-squares estimate $\hat{\beta}$ (OLS). The penalised estimate (β_1, β_2) satisfies



- (A) $\beta_1 = 0$ exactly and $\beta_2 \neq 0$
- (B) $\beta_2 = 0$ exactly and $\beta_1 \neq 0$
- (C) $\beta_1 \neq 0$ and $\beta_2 \neq 0$ (both shrunk, neither exactly zero)
- (D) $\beta_1 = \beta_2 = 0$

Q.2 [1 mark · MCQ]

The figure shows training and test error as a function of model complexity, with three complexity levels marked P, Q and R. Which marked level corresponds to overfitting (low bias, high variance)?



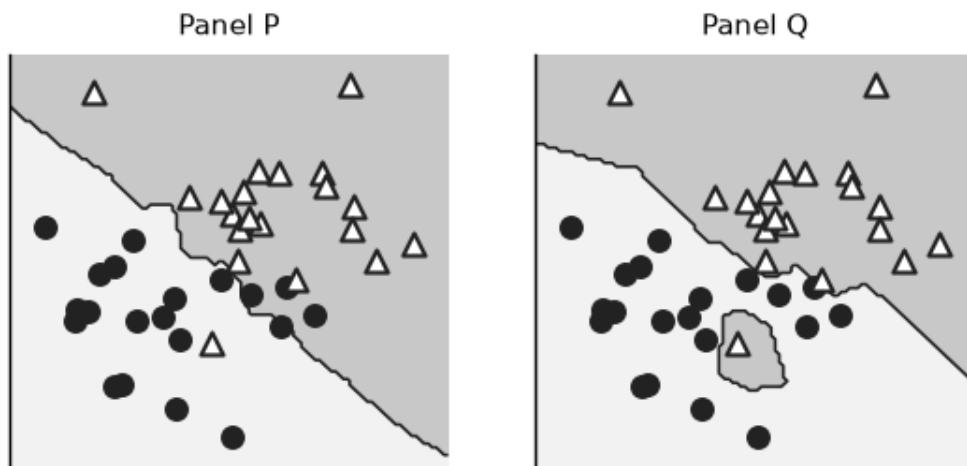
- (A) R (B) Q
(C) None of P, Q, R (D) P

Q.3 [1 mark · NAT (numerical answer)]

A network is trained with mini-batch SGD on $N = 50000$ examples with batch size $B = 100$ for $E = 20$ epochs. The last, smaller batch of each epoch is kept (not dropped). The total number of parameter updates is _____ (Answer in integer)

Q.4 [1 mark · MCQ]

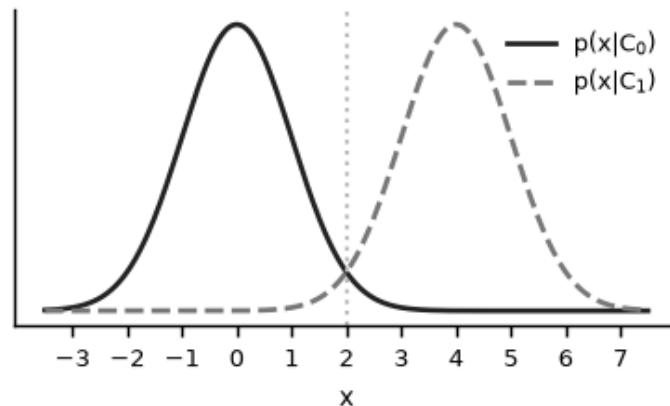
The two panels show the decision regions of k-NN classifiers trained on the same 2-D data set, one with $k = 1$ and the other with $k = 21$. Which of the following statements is CORRECT?



- (A)** Panel Q uses $k = 1$; it has higher bias and lower variance
(B) Panel Q uses $k = 1$; it has lower bias and higher variance
(C) Panel P uses $k = 1$; it has lower bias and higher variance
(D) Panel P uses $k = 1$; it has higher bias and lower variance

Q.5 [1 mark · MCQ]

Two classes have unit-variance Gaussian class-conditional densities with means 0 (C_0) and 4 (C_1), as plotted. With equal priors the Bayes decision threshold is the dotted line at $x = 2$. If the prior of C_1 is increased to 0.7, the threshold



- (A) It splits into two thresholds, giving a quadratic boundary
- (B) It moves towards $x = 0$, enlarging the region assigned to C_1
- (C) It stays at $x = 2$ because the two variances are equal
- (D) It moves towards $x = 4$, shrinking the region assigned to C_1

Q.6 [1 mark · NAT (numerical answer)]

A 4-class softmax classifier produces the logits $z_1 = 1, z_2 = 2, z_3 = 2, z_4 = -2$ for an input $\mathbf{x}$. The ratio $P(y = 2 | \mathbf{x}) / P(y = 4 | \mathbf{x})$ is _____ (rounded off to two decimal places)

Q.7 [1 mark · NAT (numerical answer)]

A node contains 8 records spread equally over 4 classes (2 records per class). Its entropy (in bits) is _____ (rounded off to two decimal places)

Q.8 [1 mark · NAT (numerical answer)]

The value of the softplus function $\ln(1 + e^z)$ at $z = -0.5$ is _____ (rounded off to three decimal places)

Q.9 [1 mark · MSQ (one or more correct)]

Which of the following statements about distance / similarity measures is/are CORRECT?

- (A) For any two vectors, $L_\infty \leq L_2 \leq L_1$ distance.
- (B) For unit-length vectors, squared Euclidean distance equals $2(1 - \cos \theta)$.
- (C) Cosine similarity is unchanged if one of the vectors is multiplied by a positive scalar.
- (D) As $p \rightarrow \infty$ the Minkowski distance tends to the Chebyshev distance $\max_i |x_i - y_i|$.

Q.10 [1 mark · MSQ (one or more correct)]

Which of the following statements about basis-function (polynomial) regression and feature scaling is/are CORRECT?

- (A) z-score standardisation $z = (x - \bar{x})/s$ gives the feature sample mean 0 and sample standard deviation 1.
- (B) Ridge and lasso solutions depend on the scales of the features.
- (C) Gaussian radial basis functions $\phi_j(\mathbf{x}) = \exp(-(x - \mu_j)^2/(2s^2))$ can be used in a model that is linear in its weights.
- (D) For OLS with an intercept, rescaling a feature as $x' = ax + b$ ($a \neq 0$) leaves the fitted values unchanged.

Q.11 [1 mark · MCQ]

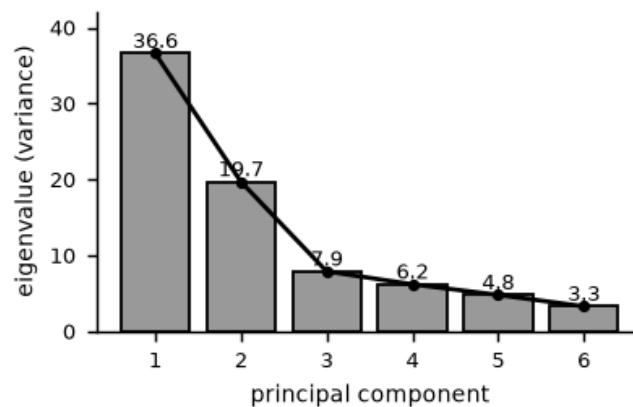
Cost-complexity pruning of a classification tree produced the nested subtrees $T1 \supset T2 \supset T3 \supset T4$ below. For complexity parameter $\alpha = 0.008$, which subtree minimises $R_\alpha(T) = R(T) + \alpha|T|$?

Subtree	T1	T2	T3	T4
Leaves $ T $	13	11	10	3
Training error $R(T)$	0.06	0.07	0.135	0.16

- (A) T1 (B) T2
(C) T3 (D) T4

Q.12 [1 mark · NAT (numerical answer)]

The scree plot below shows the eigenvalues of the covariance matrix of a 6-dimensional dataset. The percentage of total variance explained by the first two principal components together is _____ (rounded off to one decimal place)

**Q.13 [1 mark · NAT (numerical answer)]**

The inhomogeneous polynomial kernel $K(x, z) = (x^T z + 1)^3$ on $x \in \mathbb{R}^4$ corresponds to an explicit feature map containing all monomials of degree at most 3. The dimension of this feature space is _____ (Answer in integer)

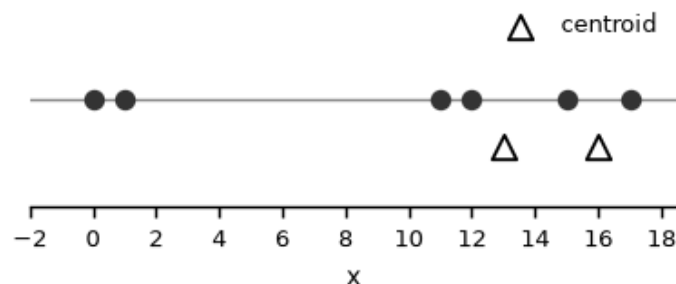
Q.14 [1 mark · MSQ (one or more correct)]

Which of the following statements about the perceptron is/are CORRECT?

- (A) XOR becomes linearly separable if the product feature $x_1 x_2$ is added as an extra input.
 (B) A two-layer network of threshold units can compute XOR.
 (C) The decision boundary of a perceptron, $w \cdot x + b = 0$, is a hyperplane.
 (D) Each mistake of the perceptron increases $w \cdot w^*$ by at least γ , where w^* is a unit-norm separator with margin γ ($\eta = 1$).

Q.15 [1 mark · NAT (numerical answer)]

One-dimensional data $\{0, 1, 11, 12, 15, 17\}$ and $k = 2$ centroids at 13, 16 are given. Each point is assigned to its nearest centroid (centroids are **not** updated). The resulting within-cluster sum of squared errors $\sum (x_i - c_{c(i)})^2$ is _____ (rounded off to two decimal places)



Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

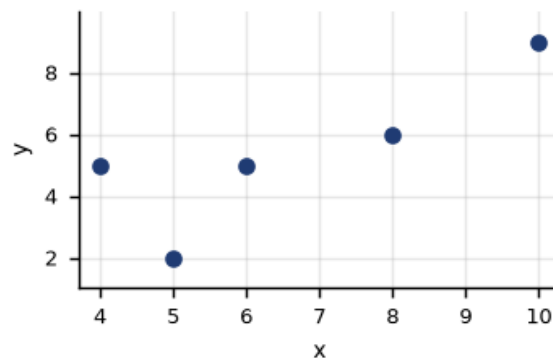
In one AdaBoost round the current example weights and whether the chosen weak classifier h classifies each example correctly are given. Using $\alpha = \frac{1}{2} \ln((1 - \epsilon)/\epsilon)$ and the update $w_i \leftarrow w_i \cdot \exp(-\alpha y_i h(x_i))/Z$, find the normalised weight of example 4 for the next round. (rounded off to three decimal places)

Example i	1	2	3	4	5	6
Weight w_i	1/6	1/6	1/6	1/6	1/6	1/6
$h(x_i)$ correct?	Yes	Yes	Yes	No	Yes	Yes

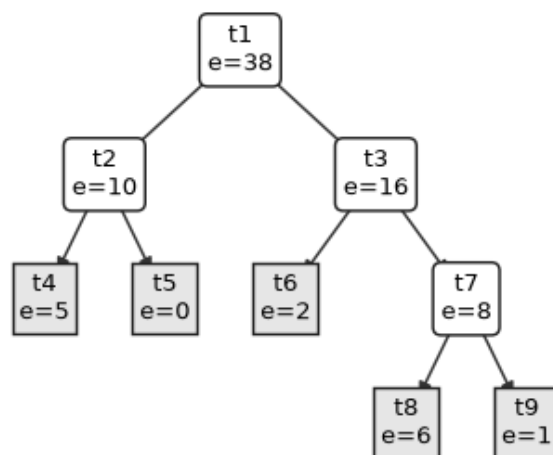
Q.17 [2 marks · NAT (numerical answer)]

A simple linear regression (with intercept) is fitted by least squares to the 5 points below. Using leave-one-out cross-validation, the prediction at $x = 8$ made by the least-squares line fitted to the other 4 points (i.e. with the point at that x left out) is _____ (rounded off to two decimal places)

x	4	5	6	8	10
y	5	2	5	6	9

**Q.18 [2 marks · NAT (numerical answer)]**

A classification tree was grown on $N = 100$ training records. Each node shows $e(t)$, the number of training records that would be misclassified if t were a leaf; $R(t) = e(t)/N$. The smallest value of α at which cost-complexity pruning first prunes the tree (the weakest link) is _____ (rounded off to three decimal places)

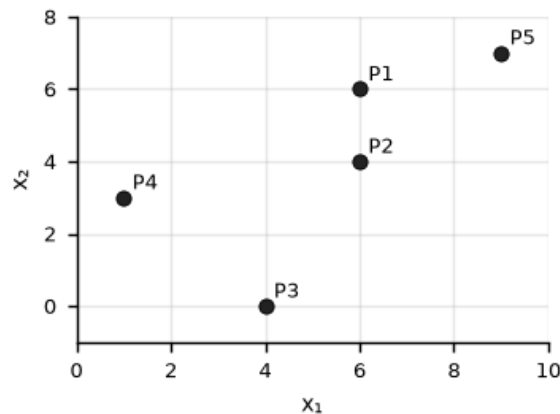
**Q.19 [2 marks · NAT (numerical answer)]**

A regression with intercept and $p = 5$ predictors on $n = 42$ observations has $R^2 = 0.5$. Its adjusted R^2 is _____ (rounded off to three decimal places)

Q.20 [2 marks · NAT (numerical answer)]

A cluster contains the 5 points below. Distances are Euclidean (L_2). If P1 were used as the cluster representative instead of the medoid, the total distance (cost) would increase by _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5
x_1	6	6	4	1	9
x_2	6	4	0	3	7

**Q.21 [2 marks · MSQ (one or more correct)]**

Which of the following statements about LDA, QDA and logistic regression is/are CORRECT?

- (A) Logistic regression assumes Gaussian class-conditional densities.
- (B) When the Gaussian shared-covariance assumption holds, LDA can estimate the boundary with lower variance than logistic regression.
- (C) Both LDA and logistic regression yield log-posterior-odds that are linear in x ; they differ in how the parameters are estimated.
- (D) Unlike unregularised logistic regression, LDA's parameter estimates remain finite when the classes are perfectly separable (if the pooled covariance is non-singular).

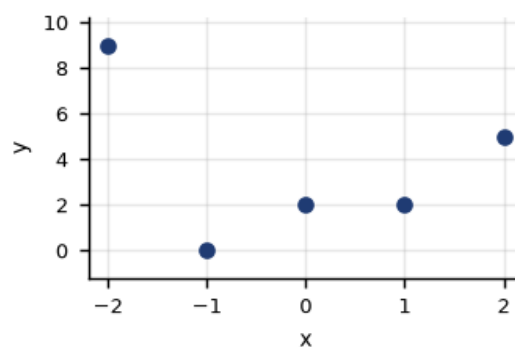
Q.22 [2 marks · NAT (numerical answer)]

Gradient descent with (heavy-ball) momentum minimises $f(w) = (1/2)w^2$ using $v_t = \beta v_{t-1} + \nabla f(w_{t-1})$ and $w_t = w_{t-1} - \eta v_t$, with $w_0 = 1$, $v_0 = 0$, $\eta = 0.2$, $\beta = 0.9$. The value of w_2 is _____ (rounded off to three decimal places)

Q.23 [2 marks · NAT (numerical answer)]

The quadratic model $y = a + bx + cx^2$ is fitted by least squares to the 5 points below (a polynomial / basis-function regression with basis $1, x, x^2$). Then the coefficient c of x^2 is _____ (rounded off to two decimal places)

x	-2	-1	0	1	2
y	9	0	2	2	5



Q.24 [2 marks · NAT (numerical answer)]

The labelled data below are linearly separable through the origin by the unit vector $u = (1, -1)/\sqrt{2}$ (i.e. $y_i = \text{sign}(u \cdot x_i)$). Take $\gamma = \min_i y_i(u \cdot x_i)$ and $R = \max_i \|x_i\|$. The perceptron convergence theorem (no bias, $w_0 = 0$, $\eta = 1$) bounds the number of mistakes by $(R/\gamma)^2$. The value of this bound is _____ (rounded off to two decimal places)

i	1	2	3	4	5
x_i	(3, -3)	(3, 1)	(-4, 1)	(3, 0)	(2, -4)
y_i	1	1	-1	1	1

Q.25 [2 marks · MCQ]

A naive Bayes classifier is trained on the 10 examples below. The new instance (Fever = None, Cough = Yes, Fatigue = No) is classified (i) with maximum-likelihood (unsmoothed) estimates and (ii) with Laplace (add-one) smoothing of the class-conditional probabilities (class priors unsmoothed). The predicted classes are

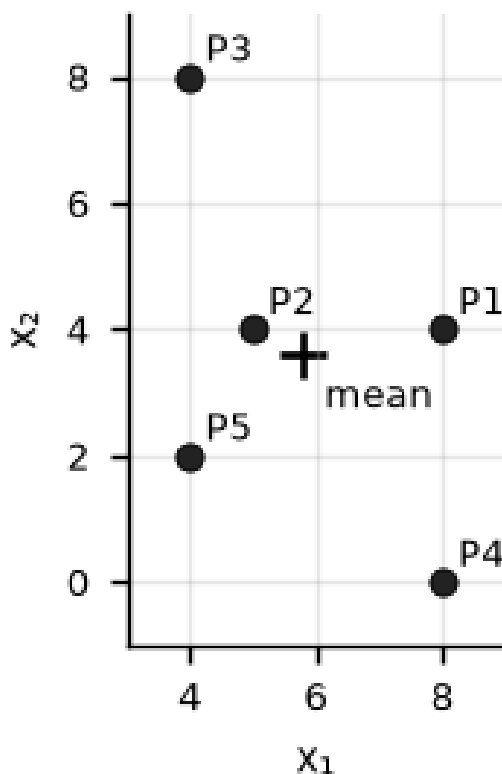
#	Fever	Cough	Fatigue	Flu
1	High	Yes	No	Neg
2	Mild	No	No	Pos
3	Mild	Yes	No	Pos
4	None	No	Yes	Neg
5	High	No	Yes	Neg
6	High	No	No	Pos
7	Mild	No	Yes	Neg
8	Mild	Yes	No	Pos
9	Mild	Yes	No	Pos
10	High	No	Yes	Neg

- (A) Without smoothing: Neg; with Laplace smoothing: Neg
 (B) Without smoothing: Pos; with Laplace smoothing: Neg
 (C) Without smoothing: Neg; with Laplace smoothing: Pos
 (D) Without smoothing: Pos; with Laplace smoothing: Pos

Q.26 [2 marks · NAT (numerical answer)]

PCA is applied to the five 2-D points below (data are mean-centered first; sample covariance uses divisor $n - 1$). Taking the unit-length first principal direction with a **positive first component**, the score (projection) of P5 on PC1 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5
x_1	8	5	4	8	4
x_2	4	4	8	0	2

**Q.27 [2 marks · NAT (numerical answer)]**

Let ϕ be the feature map of the kernel $K(x, z) = (x^T z + 1)^2$ on $\mathbb{R}^2$, so that $K(x, z) = \phi(x)^T \phi(z)$. For $x = (-2, -1)$ and $z = (0, -1)$, the squared distance $\|\phi(x) - \phi(z)\|^2$ in feature space is _____ (Answer in integer)

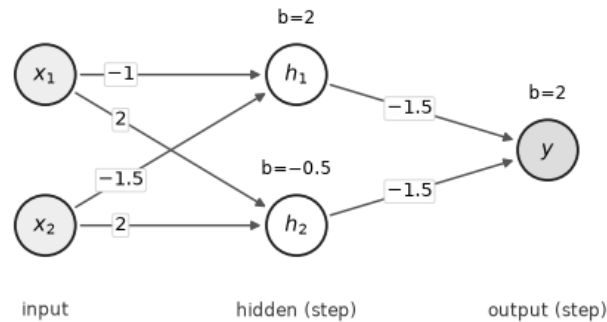
Q.28 [2 marks · NAT (numerical answer)]

A linear model $\hat{y} = w_1 x_1 + w_2 x_2 + b$ is trained by stochastic gradient descent (batch size 1) on the per-example loss $\frac{1}{2}(\hat{y} - y)^2$ with learning rate $\eta = 0.1$, starting from $w = (-1, -1)$, $b = -1$. The two examples below are processed once, in the given order. After both updates, the value of b is _____ (rounded off to three decimal places)

example	x_1	x_2	y
1	0	1	-2
2	0	-1	4

Q.29 [2 marks · MCQ]

In the network below every unit (hidden and output) is a threshold unit that outputs 1 if its weighted input plus bias is > 0 , and 0 otherwise. Edge labels are weights. For binary inputs $x_1, x_2 \in \{0, 1\}$, the network computes

**(A)** AND**(B)** OR**(C)** XNOR**(D)** NAND**Q.30 [2 marks · NAT (numerical answer)]**

An ensemble combines 5 binary classifiers by majority vote. Each classifier is correct with probability 0.8, independently of the others. Compute the probability that the majority vote is correct. (rounded off to three decimal places)

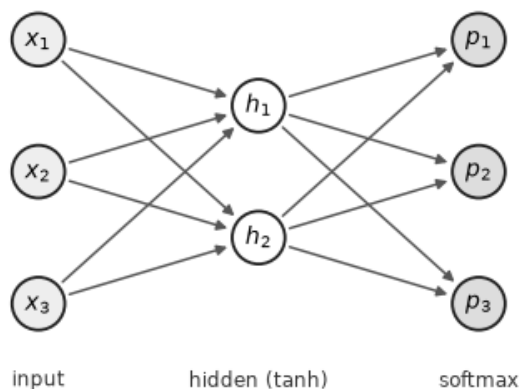
Q.31 [2 marks · NAT (numerical answer)]

The confusion matrix of a 3-class classifier is given below (rows = actual class, columns = predicted class). The macro-averaged recall is _____ (rounded off to two decimal places)

actual \ predicted	A	B	C
A	10	4	0
B	6	12	0
C	0	2	11

Q.32 [2 marks · NAT (numerical answer)]

A 3-2-3 network computes $h = \tanh(W_1 x + b_1)$, $z = W_2 h + b_2$, $p = \text{softmax}(z)$, with the parameters below. For $x = (-1, 1, 2)^T$, the predicted probability p_3 of class 3 is _____ (rounded off to three decimal places)



$$W_1 = \begin{bmatrix} 0.5 & 1 & 1 \\ -0.5 & -1 & -1 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$$

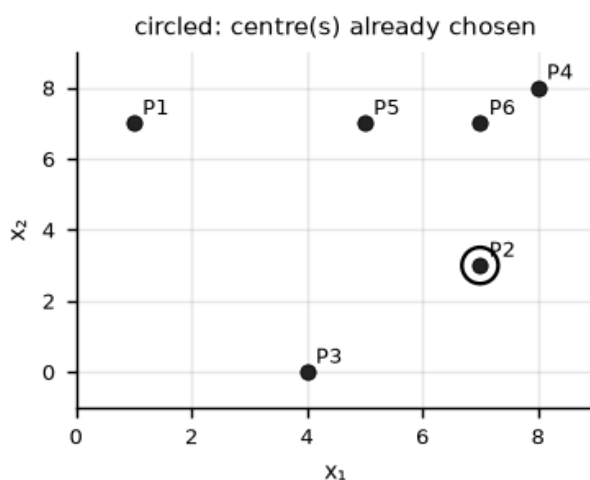
$$W_2 = \begin{bmatrix} -1 & 2 \\ -1 & 2 \\ 2 & -1 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 1 \\ 0.5 \\ 0.5 \end{bmatrix}$$

Q.33 [2 marks · NAT (numerical answer)]

k-means++ initialization (squared Euclidean seeding) is applied to the six points below. P2 has already been chosen as centre. The probability that P6 is selected as the second centre is _____ (rounded off to three decimal places)

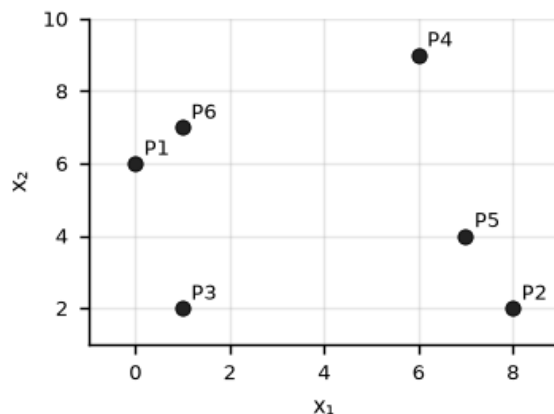
Point	P1	P2	P3	P4	P5	P6
x_1	1	7	4	8	5	7
x_2	7	3	0	8	7	7



Q.34 [2 marks · NAT (numerical answer)]

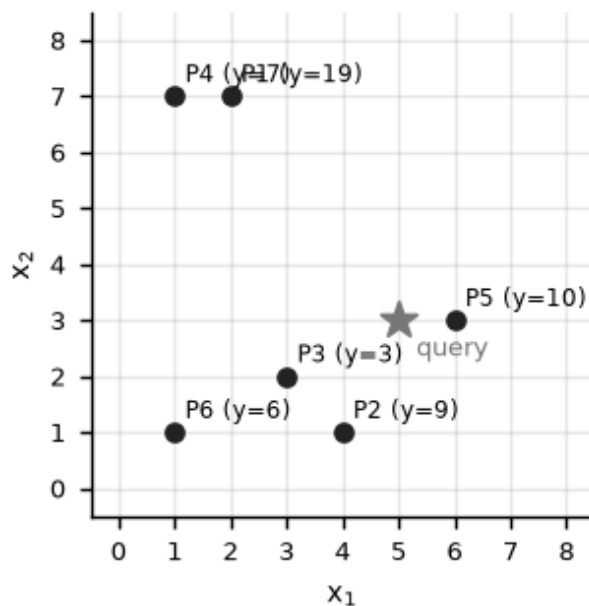
Single-linkage agglomerative clustering with Manhattan distance is applied to the six points below. Then the number of clusters obtained by cutting the dendrogram at height 2.5 is _____ (Answer in integer)

Point	P1	P2	P3	P4	P5	P6
x_1	0	8	1	6	7	1
x_2	6	2	2	9	4	7

**Q.35 [2 marks · NAT (numerical answer)]**

A distance-weighted k-NN regressor with $k = 3$ predicts $\hat{y} = \sum w_i y_i / \sum w_i$ over the 3 nearest neighbours, with weights $w_i = 1/d_i^2$. For the training data below, the prediction at $\mathbf{x} = (5, 3)$ (Euclidean distance) is _____ (rounded off to two decimal places)

Point	x_1	x_2	y
P1	2	7	19
P2	4	1	9
P3	3	2	3
P4	1	7	7
P5	6	3	10
P6	1	1	6



— END OF QUESTION PAPER —

Mock Test 29 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MCQ	1	(C)	19	NAT	2	0.4286 to 0.4326
2	MCQ	1	(A)	20	NAT	2	1.494 to 1.514
3	NAT	1	10000	21	MSQ	2	(B), (C), (D)
4	MCQ	1	(B)	22	NAT	2	0.458 to 0.462
5	MCQ	1	(B)	23	NAT	2	1.5514 to 1.5914
6	NAT	1	54.5882 to 54.6082	24	NAT	2	9.98 to 10.02
7	NAT	1	1.99 to 2.01	25	MCQ	2	(C)
8	NAT	1	0.4721 to 0.4761	26	NAT	2	0.6101 to 0.6301
9	MSQ	1	(A), (B), (C), (D)	27	NAT	2	32
10	MSQ	1	(A), (B), (C), (D)	28	NAT	2	−0.602 to −0.598
11	MCQ	1	(B)	29	MCQ	2	(C)
12	NAT	1	71.52 to 71.92	30	NAT	2	0.9401 to 0.9441
13	NAT	1	35	31	NAT	2	0.7324 to 0.7524
14	MSQ	1	(A), (B), (C), (D)	32	NAT	2	0.9906 to 0.9946
15	NAT	1	319.99 to 320.01	33	NAT	2	0.1192 to 0.1232
16	NAT	2	0.498 to 0.502	34	NAT	2	5
17	NAT	2	6.763 to 6.8996	35	NAT	2	8.8471 to 8.8671
18	NAT	2	0.0085 to 0.0115				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Regression & Model Selection					11
Neural Networks (MLP / Feed-forward)					11
Probabilistic & Instance-based Classifiers					11
SVM & Decision Trees					11
Unsupervised Learning					11
Total					55

Mock Test 29 — Detailed Solutions

Q.1 — Answer: (C) [MCQ, 1M · Regression & Model Selection > Ridge & Lasso]

The shaded region is the constraint set of the ridge (L2: $\beta_1^2 + \beta_2^2 \leq t^2$, a disc). The penalised estimate is the first point at which the RSS contours (ellipses centred at β_{OLS} , which lies outside the region) touch the region.

From the figure the tangency point is approximately (0.5, 0.87).

The contour touches the boundary at a point that is not on an axis, so both coefficients are non-zero. For the disc this is the generic case — ridge shrinks but does not zero out.

Answer: **$\beta_1 \neq 0$ and $\beta_2 \neq 0$ (both shrunk, neither exactly zero)**. Both coefficients cannot be zero since the OLS point lies outside the region (the constraint is active but $t > 0$).

Q.2 — Answer: (A) [MCQ, 1M · Regression & Model Selection > Bias-variance trade-off]

At P: both training and test errors are high (left side) → underfitting.

At Q: the test error is at its minimum → best trade-off.

At R: training error is very low but test error has risen well above its minimum (right side) → overfitting.

Answer: **R**.

Q.3 — Answer: 10000 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Updates per epoch = $\lceil N/B \rceil = \lceil 50000/100 \rceil = 500$.

Total = $500 \times 20 = 10000$.

Q.4 — Answer: (B) [MCQ, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

The jagged boundary with small islands around individual points (Panel Q) is produced by $k = 1$, which fits every training point (zero training error): low bias, high variance.

The smooth boundary (Panel P) comes from $k = 21$: averaging many neighbours raises bias and lowers variance; it misclassifies some training points, so its training error is not zero.

Answer: **Panel Q uses $k = 1$; it has lower bias and higher variance.**

Q.5 — Answer: (B) [MCQ, 1M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$x^* = (\mu_0 + \mu_1)/2 + \sigma^2 \ln(\pi_0/\pi_1)/(\mu_1 - \mu_0) = 1.788$.

A larger prior for C_1 means more of the x-axis is assigned to it, so the threshold moves away from μ_1 towards $\mu_0 = 0$. Answer: **It moves towards $x = 0$, enlarging the region assigned to C_1 .**

Equal variances keep the boundary a single point (linear); priors shift it but never make it quadratic.

Q.6 — Answer: 54.5882 to 54.6082 [NAT, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

The normalising constant $\sum e^{z_j}$ cancels in the ratio:

$P(2)/P(4) = e^{z_2 - z_4} = e^4 = 54.6$.

Q.7 — Answer: 1.99 to 2.01 [NAT, 1M · SVM & Decision Trees > Decision trees: impurity measures]

$p_k = 2/8 = 1/4$ for every class $\Rightarrow H = -4 \cdot (1/4) \log_2(1/4) = \log_2 4 = 2$ — the maximum possible for 4 classes.

Q.8 — Answer: 0.4721 to 0.4761 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

$\ln(1 + e^{-0.5}) = \ln(1 + 0.6065) = 0.4741$. (Its derivative is $\sigma(z) = 0.3775$.)

Q.9 — Answer: (A), (B), (C), (D) [MSQ, 1M · Unsupervised Learning > Distance measures]

(A) **TRUE**. Standard norm inequalities.

(B) **TRUE**. $\|x - y\|^2 = 2 - 2x \cdot y$.

(C) **TRUE**. It depends only on the angle.

(D) **TRUE**. The largest coordinate gap dominates.

Q.10 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Basis functions & feature scaling]

(A) TRUE. By construction.

(B) TRUE. The penalty is not scale-invariant.

(C) TRUE. $y = \sum_j w_j \phi_j(x)$ is linear in w .

(D) TRUE. The column space of $[1, x]$ is unchanged; only the coefficients adjust.

Q.11 — Answer: (B) [MCQ, 1M · SVM & Decision Trees > Decision trees: overfitting & pruning]

Subtree	$R(T) + \alpha T $
T1	$0.06 + 0.008 \times 13 = 0.164$
T2	$0.07 + 0.008 \times 11 = 0.158$
T3	$0.135 + 0.008 \times 10 = 0.215$
T4	$0.16 + 0.008 \times 3 = 0.184$

Minimum at **T2**. Larger α penalises leaves more and favours smaller trees; $\alpha = 0$ would pick the largest tree T1.

Q.12 — Answer: 71.52 to 71.92 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

Total variance = $\sum \lambda = 78.5$. Cumulative proportion of variance explained:

k	1	2	3	4	5	6
λ_k	36.6	19.7	7.9	6.2	4.8	3.3
cumulative %	46.62	71.72	81.78	89.68	95.8	100

$(\lambda_1 + \lambda_2)/\sum \lambda = 56.3/78.5 = \mathbf{71.7\%}$.

Q.13 — Answer: 35 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

Number of monomials of degree $\leq d$ in p variables = $C(p + d, d)$.

$C(4 + 3, 3) = C(7, 3) = \mathbf{35}$.

Q.14 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

(A) TRUE. $XOR = x_1 + x_2 - 2x_1x_2$ is linear in (x_1, x_2, x_1x_2) .

(B) TRUE. E.g. $XOR = AND(OR(x_1, x_2), NAND(x_1, x_2))$.

(C) TRUE. It is an affine set of dimension $d-1$.

(D) TRUE. Key step of the convergence proof: $w \cdot w^*$ grows linearly in the number of mistakes.

Q.15 — Answer: 319.99 to 320.01 [NAT, 1M · Unsupervised Learning > k-means]

Assignment (nearest centroid): $c = 13$: $\{0, 1, 11, 12\}$; $c = 16$: $\{15, 17\}$.

$SSE = (0-13)^2 + (1-13)^2 + (11-13)^2 + (12-13)^2 + (15-16)^2 + (17-16)^2 = \mathbf{320}$.

Q.16 — Answer: 0.498 to 0.502 [NAT, 2M · SVM & Decision Trees > Ensembles: boosting]

$\varepsilon = \sum_{\text{misclassified}} w_i = 0.1667$; $\alpha = \frac{1}{2} \ln(0.8333/0.1667) = 0.8047$.

Misclassified weights $\times e^\alpha = 2.2361$; correct weights $\times e^{-\alpha} = 0.4472$.

$Z = \varepsilon e^\alpha + (1 - \varepsilon) e^{-\alpha} = 2\sqrt{\varepsilon(1 - \varepsilon)} = 0.7454$.

New weights: 0.1, 0.1, 0.1, 0.5, 0.1, 0.1.

Shortcut: after normalisation the misclassified set has total weight exactly $\frac{1}{2}$: $w_i' = w_i/(2\varepsilon)$ if misclassified, $w_i/(2(1 - \varepsilon))$ otherwise.

Answer: **0.5**.

Q.17 — Answer: 6.763 to 6.8996 [NAT, 2M · Regression & Model Selection > Cross-validation]

$\bar{x} = 6.6$, $\bar{y} = 5.4$.

i	1	2	3	4	5
$x_i - \bar{x}$	-2.6	-1.6	-0.6	1.4	3.4
$y_i - \bar{y}$	-0.4	-3.4	-0.4	0.6	3.6
$(x_i - \bar{x})(y_i - \bar{y})$	1.04	5.44	0.24	0.84	12.24

$S_{xx} = \sum (x_i - \bar{x})^2 = 23.2$, $S_{xy} = 19.8$.

$$\beta_1 = S_{xy}/S_{xx} = 0.8534, \beta_0 = \bar{y} - \beta_1 \bar{x} = -0.2328.$$

For a linear smoother, the leave-one-out residual is $e_{(i)} = e_i/(1 - h_{ii})$, with $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$.

x_i	4	5	6	8	10
e_i	1.819	-2.034	0.112	-0.595	0.698
h_{ii}	0.491	0.31	0.216	0.284	0.698
$e_i/(1-h_{ii})$	3.576	-2.95	0.143	-0.831	2.314

$$\hat{y}_{(-i)} = y_i - e_{(i)} = 6 - (-0.8313) = 6.8313.$$

Answer: **6.83** (verified by explicitly refitting without the point).

Q.18 — Answer: 0.0085 to 0.0115 [NAT, 2M · SVM & Decision Trees > Decision trees: overfitting & pruning]

$R(t) = e(t)/N$ with $N = 100$. For an internal node t with subtree T_t : $g(t) = [R(t) - R(T_t)]/(|T_t| - 1)$, where $|T_t|$ = number of leaves.

Node	$R(t)$	$R(T_t)$	$ T_t $	$g(t)$
t1	0.38	0.14	5	0.06
t2	0.1	0.05	2	0.05
t3	0.16	0.09	3	0.035
t7	0.08	0.07	2	0.01

The weakest link is t7 with the minimum $g = 0.01$; for $\alpha \geq$ this value, collapsing t7 into a leaf does not increase R_α .

Q.19 — Answer: 0.4286 to 0.4326 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

Adjusted $R^2 = 1 - [SSE/(n - p - 1)]/[SST/(n - 1)] = 1 - (1 - R^2)(n - 1)/(n - p - 1)$, where p is the number of predictors (excluding the intercept).

$$= 1 - (1 - 0.5) \times 41/36 = 0.4306.$$

Answer: **0.431**.

Q.20 — Answer: 1.494 to 1.514 [NAT, 2M · Unsupervised Learning > k-medoids]

Compute the Euclidean (L_2) distance matrix and its row sums:

	P1	P2	P3	P4	P5	Σ
P1	0	2	6.32	5.83	3.16	17.32
P2	2	0	4.47	5.1	4.24	15.81
P3	6.32	4.47	0	4.24	8.6	23.64
P4	5.83	5.1	4.24	0	8.94	24.12
P5	3.16	4.24	8.6	8.94	0	24.95

The smallest row sum is 15.81 for P2, so the medoid is P2.

Cost with P1: 17.32; increase = $17.32 - 15.81 = 1.5$.

Unlike a k-means centroid, the medoid is always an actual data point.

Q.21 — Answer: (B), (C), (D) [MSQ, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

(A) FALSE. It makes no assumption on $P(x|y)$.

(B) TRUE. It uses the extra (correct) distributional information.

(C) TRUE. LDA maximises the joint likelihood, logistic regression the conditional likelihood.

(D) TRUE. LDA estimates are sample means and covariances.

Q.22 — Answer: 0.458 to 0.462 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$f(w) = (1/2)w^2 = \nabla f(w) = 1w$. Update: $v_t = \beta v_{t-1} + \nabla f(w_{t-1})$, $w_t = w_{t-1} - \eta v_t$, with $v_0 = 0$, $\beta = 0.9$, $\eta = 0.2$.

$t = 1$: $g = 1 \times 1 = 1$; $v_1 = 0.9 \times 0 + 1 = 1$; $w_1 = 1 - 0.2 \times 1 = 0.8$.

$t = 2$: $g = 1 \times 0.8 = 0.8$; $v_2 = 0.9 \times 1 + 0.8 = 1.7$; $w_2 = 0.8 - 0.2 \times 1.7 = 0.46$.

Answer: **0.46**. (Plain gradient descent without momentum would give $w_2 = 0.64$.)

Q.23 — Answer: 1.5514 to 1.5914 [NAT, 2M · Regression & Model Selection > Basis functions & feature scaling]

Design columns 1, x , x^2 . Because the x -values are symmetric about 0, $\sum x = \sum x^3 = 0$, so the normal equations decouple:

$$\sum x^2 = 10, \sum x^4 = 34, \sum y = 18, \sum xy = -6, \sum x^2y = 58.$$

$$b = \sum xy / \sum x^2 = -6/10 = -0.6.$$

$$[n, \sum x^2; \sum x^2, \sum x^4][a; c] = [\sum y; \sum x^2y] = 5a + 10c = 18, 10a + 34c = 58.$$

$$\text{Solving: } c = (n\sum x^2y - \sum x^2\sum y) / (n\sum x^4 - (\sum x^2)^2) = 1.5714, a = (\sum y - c\sum x^2) / n = 0.4571.$$

Answer: **1.57**.

Q.24 — Answer: 9.98 to 10.02 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Unit separator $u = (1, -1)/\sqrt{2}$. Margins $y_i(u \cdot x_i)$: 4.243, 1.414, 3.536, 2.121, 4.243 $\rightarrow \gamma = \min = 1.4142$.

$$\text{Norms } \|x_i\|: 4.243, 3.162, 4.123, 3, 4.472 \rightarrow R = \max = 4.4721.$$

$$\text{Novikoff bound: number of mistakes} \leq (R/\gamma)^2 = (4.4721/1.4142)^2 = \mathbf{10}.$$

Common mistakes: forgetting to normalise u (γ must be measured with $\|u\| = 1$), or using R/γ without squaring.

Q.25 — Answer: (C) [MCQ, 2M · Probabilistic & Instance-based Classifiers > Naive Bayes]

(i) No smoothing.

Pos: $P(\text{Flu}=\text{Pos}) = 5/10$; $P(\text{Fever}=\text{None}|\text{Pos}) = 0/5$; $P(\text{Cough}=\text{Yes}|\text{Pos}) = 3/5$; $P(\text{Fatigue}=\text{No}|\text{Pos}) = 5/5 \Rightarrow \text{score} = 0$.

Neg: $P(\text{Flu}=\text{Neg}) = 5/10$; $P(\text{Fever}=\text{None}|\text{Neg}) = 1/5$; $P(\text{Cough}=\text{Yes}|\text{Neg}) = 1/5$;

$P(\text{Fatigue}=\text{No}|\text{Neg}) = 1/5 \Rightarrow \text{score} = 0.004$.

A zero count makes the whole product for Pos equal to 0 (zero-frequency problem), so the prediction is Neg.

(ii) Laplace smoothing.

Pos: $P(\text{Flu}=\text{Pos}) = 5/10$; $P(\text{Fever}=\text{None}|\text{Pos}) = (0+1)/(5+3) = 1/8$; $P(\text{Cough}=\text{Yes}|\text{Pos}) = (3+1)/(5+2) = 4/7$; $P(\text{Fatigue}=\text{No}|\text{Pos}) = (5+1)/(5+2) = 6/7 \Rightarrow \text{score} = 0.03061$.

Neg: $P(\text{Flu}=\text{Neg}) = 5/10$; $P(\text{Fever}=\text{None}|\text{Neg}) = (1+1)/(5+3) = 1/4$; $P(\text{Cough}=\text{Yes}|\text{Neg}) = (1+1)/(5+2) = 2/7$; $P(\text{Fatigue}=\text{No}|\text{Neg}) = (1+1)/(5+2) = 2/7 \Rightarrow \text{score} = 0.0102$.

Larger score: Pos. Answer: **Without smoothing: Neg; with Laplace smoothing: Pos.**

Q.26 — Answer: 0.6101 to 0.6301 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

Mean $\bar{x} = (5.8, 3.6)$. Centered data rows: P1: (2.2, 0.4), P2: (-0.8, 0.4), P3: (-1.8, 4.4), P4: (2.2, -3.6), P5: (-1.8, -1.6).

$$\text{Sample covariance } S = (1/(n-1)) \sum (x - \bar{x})(x - \bar{x})^T = [[4.2, -3.1], [-3.1, 8.8]].$$

Eigenvalues: $\lambda_1 = 10.3601$, $\lambda_2 = 2.6399$; unit PC1 direction $v_1 = (0.4495, -0.8933)$.

$$\text{Score } z_1 = v_1 \cdot (x - \bar{x}) = 0.4495 \cdot -1.8 + -0.8933 \cdot -1.6 = 0.6201.$$

Answer: **0.62** (the sign depends on the orientation of v_1 ; with the stated convention it is 0.62).

Q.27 — Answer: 32 [NAT, 2M · SVM & Decision Trees > SVM: kernels]

$$\|\varphi(x) - \varphi(z)\|^2 = K(x, x) + K(z, z) - 2K(x, z).$$

$$K(x, x) = (5 + 1)^2 = 36, K(z, z) = (1 + 1)^2 = 4, K(x, z) = (1 + 1)^2 = 4.$$

$$\text{Squared distance} = 36 + 4 - 2(4) = \mathbf{32}.$$

Q.28 — Answer: -0.602 to -0.598 [NAT, 2M · Regression & Model Selection > Gradient descent]

Per-example loss $\square = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial \square / \partial w = (\hat{y} - y)x$, $\partial \square / \partial b = (\hat{y} - y)$. Update after each example.

Example 1: $\hat{y} = -2$, error $\hat{y} - y = 0$; $\nabla_w = (0, 0)$, $\nabla_b = 0 \Rightarrow w = (-1, -1)$, $b = -1$.

Example 2: $\hat{y} = 0$, error $\hat{y} - y = -4$; $\nabla_w = (0, 4)$, $\nabla_b = -4 \Rightarrow w = (-1, -1.4)$, $b = -0.6$.

Answer: **-0.6**.

Q.29 — Answer: (C) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Expressiveness]

Each unit outputs 1 iff its net input (weighted sum + bias) is > 0 . Evaluate on all four inputs:

x_1	x_2	h_1	h_2	y
0	0	1	0	1
0	1	1	1	0
1	0	1	1	0
1	1	0	1	1

h_1 computes NAND, h_2 computes OR; the output column (0,0)→(1,1) is 1, 0, 0, 1, i.e. **XNOR**.

This shows that one hidden layer of two threshold units suffices for the non-linearly-separable XNOR, which a single perceptron cannot represent.

Q.30 — Answer: 0.9401 to 0.9441 [NAT, 2M · SVM & Decision Trees > Ensembles: bagging & random forests]

The vote is correct when at least 3 of 5 classifiers are correct (Binomial(5, 0.8)).

$$P(\text{correct}) = C(5,3)(0.8)^3(0.2)^2 + C(5,4)(0.8)^4(0.2)^1 + C(5,5)(0.8)^5(0.2)^0 = 0.9421.$$

Answer: **0.942**. Independence is crucial — correlated classifiers gain much less.

Q.31 — Answer: 0.7324 to 0.7524 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

class	TP	predicted total	actual total	precision	recall	F1
A	10	16	14	0.625	0.7143	0.6667
B	12	18	18	0.6667	0.6667	0.6667
C	11	11	13	1	0.8462	0.9167

Precision of class $k = C_{kk}/(\text{column sum})$; recall = $C_{kk}/(\text{row sum})$.

Macro recall = mean of per-class recalls = **0.74** (= balanced accuracy).

Q.32 — Answer: 0.9906 to 0.9946 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

$$a = W_1x + b_1 = (3, -2); h = \tanh(a) = (0.9951, -0.964).$$

$$z = W_2h + b_2 = (-1.9231, -2.4231, 3.4541).$$

$$\exp(z) = (0.1462, 0.0886, 31.631), \text{ sum} = 31.8658.$$

$$p_3 = \exp(z_3)/\sum \exp(z_j) = \mathbf{0.9926}. \text{ (All probabilities: (0.005, 0.003, 0.993).)}$$

Q.33 — Answer: 0.1192 to 0.1232 [NAT, 2M · Unsupervised Learning > k-means]

k-means++ picks the next centre x with probability $D(x)^2 / \sum_x D(x)^2$, where $D(x)$ is the distance from x to the **nearest** centre already chosen (chosen centres have $D = 0$).

Point	P1	P2	P3	P4	P5	P6
$D(x)^2$	52	0	18	26	20	16

$$\sum D^2 = 132, \text{ so } P(P6) = 16/132 = \mathbf{0.121}.$$

Common mistakes: using D instead of D^2 (that is not k-means++), or using the distance to the *farthest*/first centre instead of the nearest chosen centre.

Q.34 — Answer: 5 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Single-linkage merge heights are exactly the edge weights of a minimum spanning tree (Kruskal's algorithm adds the shortest edge joining two different components = single-linkage merge).

MST edges (Prim's algorithm): P1-P6 (2); P1-P3 (5); P3-P2 (7); P2-P5 (3); P5-P4 (6).

Sorted merge heights: 2, 3, 5, 6, 7.

MST edges shorter than 2.5: 1; clusters = $n - (\text{number of such edges}) = 6 - 1 = \mathbf{5}$.

Q.35 — Answer: 8.8471 to 8.8671 [NAT, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Three nearest neighbours: P5 ($d^2=1$, $y=10$), P2 ($d^2=5$, $y=9$), P3 ($d^2=5$, $y=3$).

Weights $1/d^2$: 1, 0.2, 0.2; $\sum w = 1.4$.

$$\sum w_i y_i = 12.4 = \hat{y} = 12.4/1.4 = \mathbf{8.86}.$$

(The unweighted 3-NN average would be 7.33; weighting pulls $\hat{y}$ towards the closest point.)

Mock Test 30

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	6	5	24

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Unsupervised Learning	7	11
Probabilistic & Instance-based Classifiers	7	11
Regression & Model Selection	7	11
SVM & Decision Trees	7	11
Neural Networks (MLP / Feed-forward)	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

Two shopping baskets are encoded as binary vectors (1 = item bought). The difference (simple matching coefficient – Jaccard similarity) between basket A and basket B is _____ (rounded off to two decimal places)

Item	milk	bread	eggs	butter	jam	tea	rice	salt	sugar
Basket A	1	1	1	0	1	1	1	1	1
Basket B	1	1	0	0	1	0	1	0	0

Q.2 [1 mark · NAT (numerical answer)]

The confusion matrix of a binary classifier on a test set is shown below (note which of rows/columns are actual and predicted). The false positive rate of the classifier is _____ (rounded off to two decimal places)

	Actual +	Actual –
Predicted +	9	42
Predicted –	54	14

Q.3 [1 mark · NAT (numerical answer)]

A full polynomial regression of degree 3 in 5 input variables $x_1, \dots, x_5$ includes every monomial in these variables of total degree at most 3 (including the constant term and all interaction terms). The number of regression coefficients is _____ (Answer in integer)

Q.4 [1 mark · MCQ]

In Gaussian linear regression with n observations and p coefficients, an unbiased estimator of σ^2 is:

- (A) $SSE/(n - p)$ (B) $SSE/(n + p)$
 (C) $\sqrt{SSE/(n - p)}$ (D) SSE/n

Q.5 [1 mark · MSQ (one or more correct)]

Which of the following statements about Bayes-optimal classification and generative vs. discriminative models is/are CORRECT?

- (A) For binary classification under 0-1 loss, the Bayes error at x equals $\max(P(y=0|x), P(y=1|x))$.
- (B) Increasing the prior of class 1 shrinks the region assigned to class 1 by the Bayes rule.
- (C) The Bayes error rate is the lowest error rate achievable by any classifier for the given joint distribution of (x, y) .
- (D) Under 0-1 loss, the Bayes classifier predicts $\arg \max_k P(y = k | x)$ and minimises the probability of misclassification.

Q.6 [1 mark · NAT (numerical answer)]

The perpendicular distance of the point $x = (0, 3, 4)$ from the hyperplane $2x_1 + 2x_2 + x_3 - 1 = 0$ is _____ (rounded off to two decimal places)

Q.7 [1 mark · MSQ (one or more correct)]

Which of the following statements about bagging, random forests and boosting is/are CORRECT?

- (A) Averaging B independent predictors, each with variance σ^2 , gives variance σ^2/B .
- (B) About 63.2% of the training points are out-of-bag for a given bootstrap sample.
- (C) In AdaBoost, α_t increases as ε_t increases (for $\varepsilon_t < 0.5$).
- (D) Every bootstrap sample contains each training point exactly once.

Q.8 [1 mark · MSQ (one or more correct)]

Let X be an $n \times p$ design matrix of full column rank p ($p < n$), and let $H = X(X^T X)^{-1} X^T$ be the hat matrix of least-squares regression of y on X . Which of the following statements is/are CORRECT?

- (A) The residual vector is $e = (I - H)y$.
- (B) Under $\text{Var}(y) = \sigma^2 I$, the residuals are uncorrelated with common variance σ^2 .
- (C) H depends on the response vector y .
- (D) Some leverage h_{ii} can exceed 1.

Q.9 [1 mark · MSQ (one or more correct)]

Which of the following statements about the k -nearest-neighbour method is/are CORRECT?

- (A) The cost of a brute-force k -NN prediction grows linearly with the number of training points n .
- (B) k -NN can produce highly non-linear decision boundaries.
- (C) 1-NN has high bias and low variance.
- (D) As $n \rightarrow \infty$, the error rate of 1-NN is at most twice the Bayes error rate (binary case).

Q.10 [1 mark · NAT (numerical answer)]

A weight $w = 2$ is trained on the regularised objective $L(w) + (\lambda/2)w^2$ with $\lambda = 2$. The data-loss gradient is $\partial L/\partial w = 0$. After one gradient-descent step with $\eta = 0.1$, the new weight is _____ (rounded off to three decimal places)

Q.11 [1 mark · NAT (numerical answer)]

A divisive (top-down) hierarchical algorithm considers **every** way of splitting a cluster of 7 points into two non-empty sub-clusters. The number of candidate splits is _____ (Answer in integer)

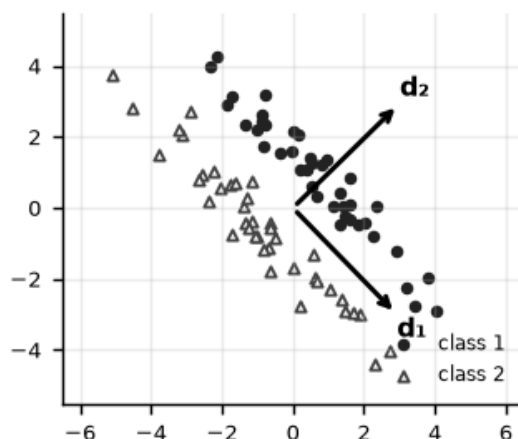
Q.12 [1 mark · NAT (numerical answer)]

A 3-class softmax classifier gives the predicted probabilities below for a mini-batch of 3 examples. The mean categorical cross-entropy loss (natural log) over the mini-batch is _____ (rounded off to three decimal places)

example	p(class 1)	p(class 2)	p(class 3)	true class
1	0.1	0.7	0.2	1
2	0.1	0.2	0.7	1
3	0.05	0.05	0.9	3

Q.13 [1 mark · MCQ]

Two labelled classes are shown below, together with two orthogonal directions d_1 and d_2 . A single direction is to be selected by (i) PCA (first principal component, ignoring labels) and (ii) Fisher's LDA. Which statement is correct?



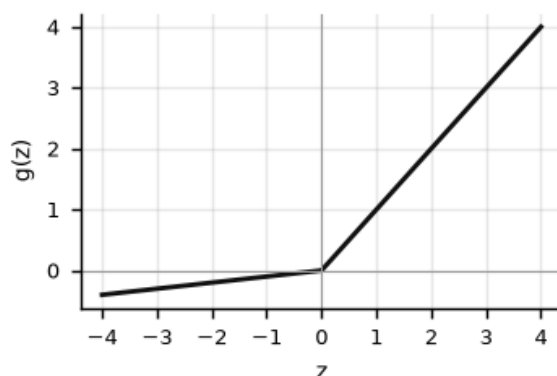
- (A) PCA (1st PC) chooses d_1 ; LDA chooses d_2
- (B) Both PCA and LDA choose d_2
- (C) Both PCA and LDA choose d_1
- (D) PCA (1st PC) chooses d_2 ; LDA chooses d_1

Q.14 [1 mark · NAT (numerical answer)]

Consider the polynomial kernel $K(x, z) = (x^T z + 2)^2$ with $x = (0, 3, 2)$ and $z = (0, 3, 2)$. The value of $K(x, x)$ is _____ (Answer in integer)

Q.15 [1 mark · MCQ]

The figure shows the graph of a function $g(z)$ used in neural networks. Which function is it?



- (A) Softplus: $\ln(1 + e^z)$
- (B) ReLU: $\max(0, z)$
- (C) Leaky ReLU with $\alpha = 0.1$
- (D) ELU with $\alpha = 1$

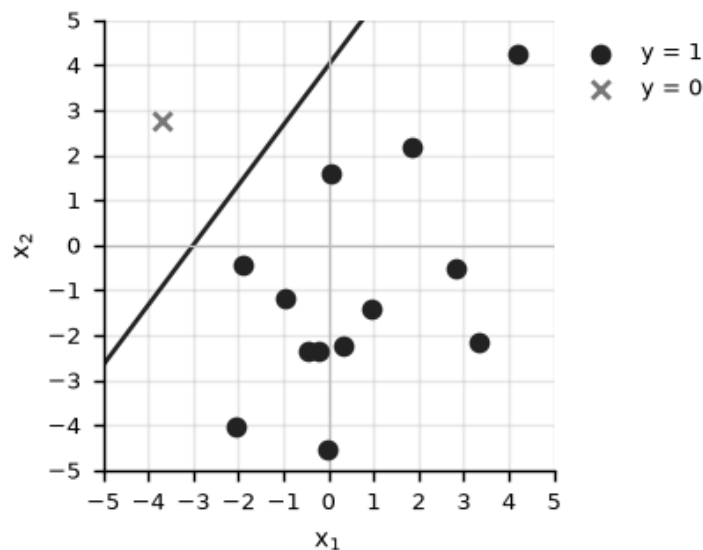
Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

A logistic-regression neuron $p = \sigma(w_1x_1 + w_2x_2 + b)$ has $w = (-1, 0.5)$, $b = 0.5$ and is trained with binary cross-entropy loss. After one stochastic gradient step on **example 1 only** with learning rate $\eta = 1$, the new value of w_1 is _____ (rounded off to three decimal places)

example	x_1	x_2	y
1	-0.5	-1	0
2	0.5	1	0

Q.17 [2 marks · MCQ]

A logistic regression model $P(y = 1 | \mathbf{x}) = \sigma(w_0 + w_1x_1 + w_2x_2)$ perfectly classifies the training points shown below using threshold 0.5. Its decision boundary is the solid line (it crosses the axes at grid points). Which of the following parameter vectors is consistent with the figure?



- (A) $(w_0, w_1, w_2) = (12, 4, -3)$
 (B) $(w_0, w_1, w_2) = (-12, -3, 4)$
 (C) $(w_0, w_1, w_2) = (-12, 4, -3)$
 (D) $(w_0, w_1, w_2) = (-12, -4, 3)$

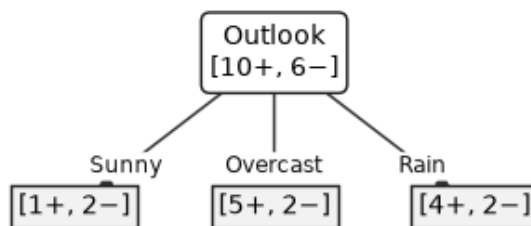
Q.18 [2 marks · NAT (numerical answer)]

In C4.5, the attribute Day splits a node of 19 records as tabulated (class Yes/No counts in each branch). Compute the split information of Day (all logarithms base 2). (rounded off to two decimal places)

Day	#Yes	#No
v1	6	8
v2	4	1

Q.19 [2 marks · NAT (numerical answer)]

A CART node with 16 training records (class counts [positive+, negative−]) is split on Outlook as shown. Compute the reduction in Gini index (Gini gain) obtained by splitting on Outlook. (rounded off to two decimal places)

**Q.20 [2 marks · NAT (numerical answer)]**

Consider the model $y = wx + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$, $\sigma^2 = 4$, and w has a Laplace prior $p(w) \propto \exp(-|w|/b)$ with $b = 1$. Given the four observations below, the MAP estimate of w is _____ (rounded off to three decimal places)

x	-1	2	1	1
y	6	6	8	8

Q.21 [2 marks · NAT (numerical answer)]

Two clusterings U and V of 6 points are given below (cluster names are arbitrary). Then the number of point pairs on which the two clusterings **disagree** is _____ (Answer in integer)

Point	P1	P2	P3	P4	P5	P6
Clustering U	α	β	β	γ	β	γ
Clustering V	1	2	2	3	3	1

Q.22 [2 marks · MCQ]

A network computes $\hat{y} = W_2 \text{ReLU}(W_1 x + b_1) + b_2$ with the parameters below. For $x = (0, 1, -1)^T$, the output $\hat{y}$ is

$$W_1 = \begin{bmatrix} 1 & 2 & -2 \\ 0 & -2 & 0 \\ -1 & 2 & 1 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 0 & 2 & 0 \\ 2 & 1 & 0 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

(A) $\hat{y} = (-1, 4)^T$

(C) $\hat{y} = (5, 13)^T$

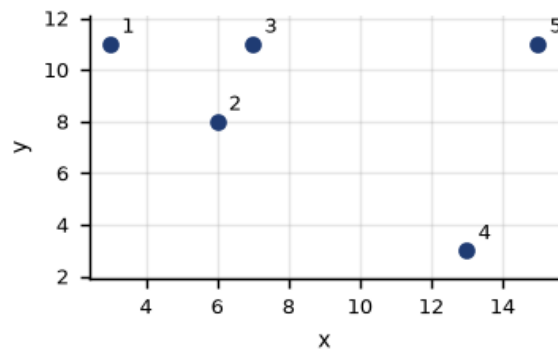
(B) $\hat{y} = (-1, 10)^T$

(D) $\hat{y} = (-7, 7)^T$

Q.23 [2 marks · NAT (numerical answer)]

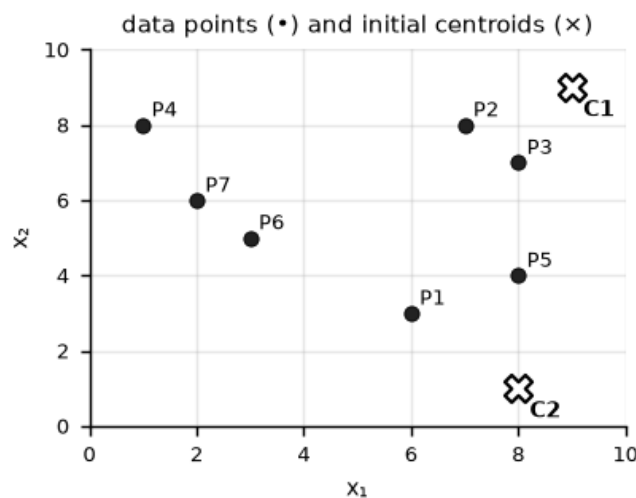
Consider the 5 training points (x_i, y_i) shown below. A 1-nearest-neighbour regressor (prediction = y-value of the closest training x , $|\cdot|$ distance) is evaluated by leave-one-out cross-validation. The LOOCV MSE is _____ (rounded off to two decimal places)

x	3	6	7	13	15
y	11	8	11	3	11

**Q.24 [2 marks · NAT (numerical answer)]**

k-means (Lloyd's algorithm, Euclidean distance) with $k = 2$ is run on the 7 points below, starting from the initial centroids $C1 = (9, 9)$, $C2 = (8, 1)$. The number of points whose cluster membership **changes** in the second iteration (compared with the first-iteration assignment) is _____ (Answer in integer)

Point	P1	P2	P3	P4	P5	P6	P7
x_1	6	7	8	1	8	3	2
x_2	3	8	7	8	4	5	6

**Q.25 [2 marks · NAT (numerical answer)]**

Agglomerative clustering with **single linkage** is applied to the distance matrix below. The cophenetic distance between P4 and P5 (the dendrogram height at which they are first joined) is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5
P1	0	11	9	14	8
P2		0	2	3	7
P3			0	5	7
P4				0	8
P5					0

Q.26 [2 marks · NAT (numerical answer)]

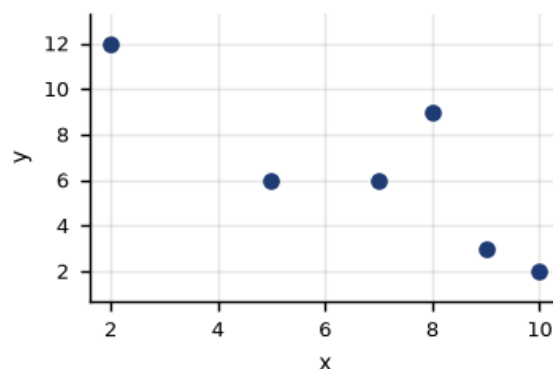
For an input x , a classifier's posterior probabilities are $P(C_1|x) = 0.5$, $P(C_2|x) = 0.2$, $P(C_3|x) = 0.3$. The loss of taking action α_i (deciding C_i) when the true class is C_j is given below. Then the minimum conditional risk (expected loss) achievable for this x is _____ (rounded off to two decimal places)

loss $L(\alpha_i, C_j)$	true C_1	true C_2	true C_3
α_1 : decide C_1	0	10	8
α_2 : decide C_2	10	0	1
α_3 : decide C_3	7	9	0

Q.27 [2 marks · NAT (numerical answer)]

The 6 observations shown in the table are plotted below. For the simple linear regression of y on x (with intercept), the residual $e = y - \hat{y}$ of the observation $(x, y) = (8, 9)$ under the least-squares line is _____ (rounded off to two decimal places)

x	2	5	7	8	9	10
y	12	6	6	9	3	2

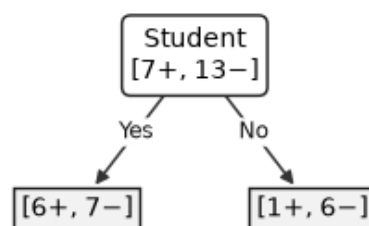
**Q.28 [2 marks · MSQ (one or more correct)]**

Which of the following statements about loss functions and optimisation for training neural networks is/are CORRECT?

- (A) Decaying the learning rate during training can help SGD settle closer to a minimum.
- (B) With uniformly sampled mini-batches, the mini-batch gradient is an unbiased estimate of the full-batch gradient.
- (C) Minimising cross-entropy loss is equivalent to maximising the log-likelihood of the labels under the model.
- (D) Too large a learning rate can make the training loss oscillate or diverge.

Q.29 [2 marks · NAT (numerical answer)]

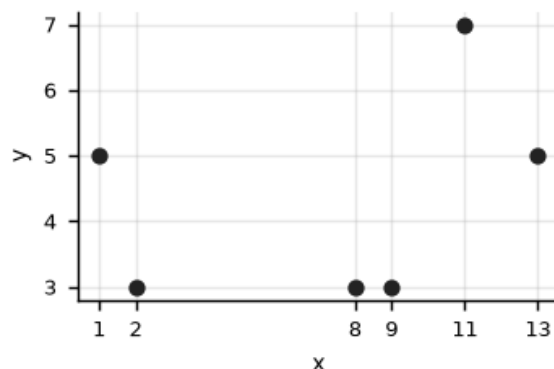
A node with 20 training records (class counts [positive+, negative−]) is split on the attribute Student as shown. Compute the information gain (in bits) of splitting on Student. (rounded off to two decimal places)



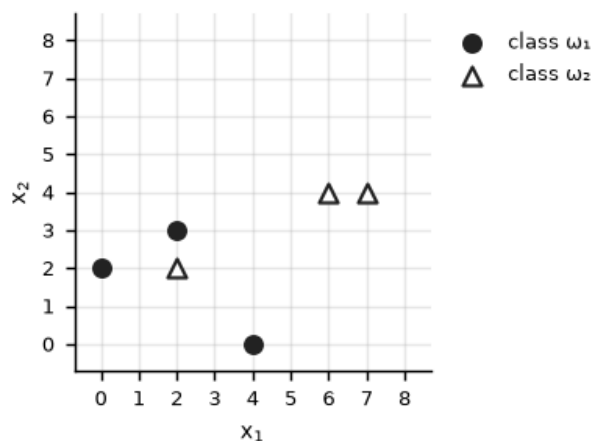
Q.30 [2 marks · NAT (numerical answer)]

A regression tree (squared-error criterion) is fitted to the 6 points below. The root split “ $x \leq t$ ” is chosen among midpoints of consecutive x values to minimise the total SSE of the two children (each child predicts its mean). Find the prediction of the resulting depth-1 regression tree (stump) at $x = 15$. (rounded off to two decimal places)

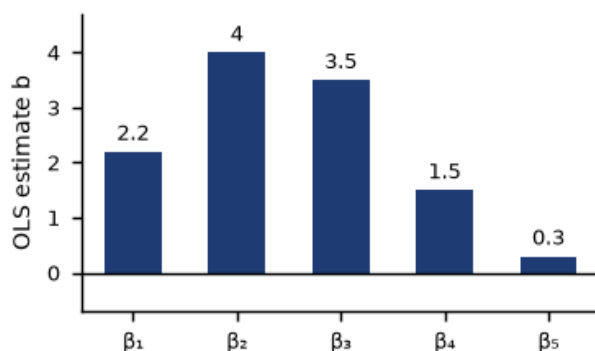
x	1	2	8	9	11	13
y	5	3	3	3	7	5

**Q.31 [2 marks · NAT (numerical answer)]**

Class ω_1 has training points $(0, 2)$, $(2, 3)$, $(4, 0)$ and class ω_2 has points $(7, 4)$, $(2, 2)$, $(6, 4)$. Let $\mathbf{m}_1$, $\mathbf{m}_2$ be the class means and $S_W = \sum_k \sum_{x \in \omega_k} (x - \mathbf{m}_k)(x - \mathbf{m}_k)^T$ the within-class scatter matrix (no division by n). Then $(\mathbf{m}_2 - \mathbf{m}_1)^T S_W (\mathbf{m}_2 - \mathbf{m}_1)$ is _____ (rounded off to two decimal places)

**Q.32 [2 marks · NAT (numerical answer)]**

In a regression problem the design matrix X has orthonormal columns ($X^T X = I$). The ordinary least-squares estimates $\mathbf{b} = X^T \mathbf{y}$ are shown below. Ridge minimises $\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|_2^2$ and the lasso minimises $\frac{1}{2} \|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|_1$, each with $\lambda = 2.5$. Then (ignoring the given λ) the smallest value of λ for which the lasso sets at least 4 coefficients exactly to zero is _____ (rounded off to two decimal places)



Q.33 [2 marks · NAT (numerical answer)]

An autoencoder with layer sizes 64-8-64 (fully connected, separate encoder and decoder weights, biases in both layers) has _____ trainable parameters. (Answer in integer)

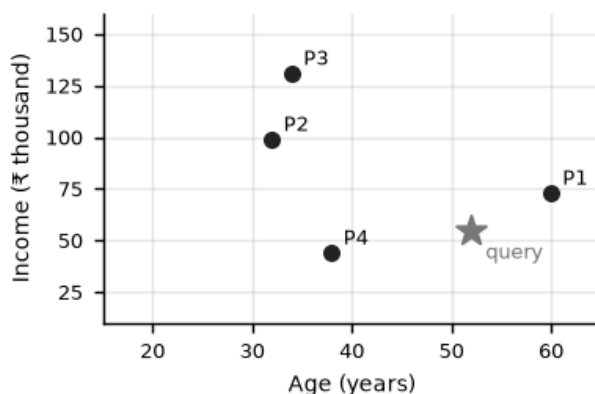
Q.34 [2 marks · NAT (numerical answer)]

The eigenvalues of the sample covariance matrix (divisor $n - 1$) of a centered 6-dimensional dataset are 19.5, 18.5, 18, 13, 4, 1. Then with $n = 51$ samples, the total squared reconstruction error $\sum_i \|x_i - \hat{x}_i\|^2$ when keeping the top 2 principal components is _____ (rounded off to two decimal places)

Q.35 [2 marks · MCQ]

A 1-NN classifier uses the two features below. Before computing Euclidean distances, each feature is min-max scaled to $[0, 1]$ using the minimum and maximum of that feature over the four training points P1-P4 (the query is scaled with the same constants). The nearest neighbour of the query after scaling is

Point	Age (years)	Monthly income (₹ thousand)
P1	60	73
P2	32	99
P3	34	131
P4	38	44
Query	52	55



- (A) P1
(C) P3

- (B) P2
(D) P4

— END OF QUESTION PAPER —

Mock Test 30 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	0.0456 to 0.0656	19	NAT	2	0.0302 to 0.0502
2	NAT	1	0.74 to 0.76	20	NAT	2	2.5637 to 2.5791
3	NAT	1	56	21	NAT	2	5
4	MCQ	1	(A)	22	MCQ	2	(B)
5	MSQ	1	(C), (D)	23	NAT	2	30.845 to 31.155
6	NAT	1	2.99 to 3.01	24	NAT	2	1
7	MSQ	1	(A)	25	NAT	2	6.99 to 7.01
8	MSQ	1	(A)	26	NAT	2	4.39 to 4.41
9	MSQ	1	(A), (B), (D)	27	NAT	2	3.8633 to 3.9033
10	NAT	1	1.598 to 1.602	28	MSQ	2	(A), (B), (C), (D)
11	NAT	1	63	29	NAT	2	0.0698 to 0.0898
12	NAT	1	1.5682 to 1.5722	30	NAT	2	5.99 to 6.01
13	MCQ	1	(A)	31	NAT	2	0.6708 to 0.6908
14	NAT	1	225	32	NAT	2	3.486 to 3.514
15	MCQ	1	(C)	33	NAT	2	1096
16	NAT	2	−0.6908 to −0.6868	34	NAT	2	1799.99 to 1800.01
17	MCQ	2	(A)	35	MCQ	2	(A)
18	NAT	2	0.8215 to 0.8415				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Unsupervised Learning					11
Probabilistic & Instance-based Classifiers					11
Regression & Model Selection					11
SVM & Decision Trees					11
Neural Networks (MLP / Feed-forward)					11
Total					55

Mock Test 30 — Detailed Solutions

Q.1 — Answer: 0.0456 to 0.0656 [NAT, 1M · Unsupervised Learning › Distance measures]

Counts: $f_{11} = 4$, $f_{10} = 4$, $f_{01} = 0$, $f_{00} = 1$.

Jaccard $J = f_{11}/(f_{11}+f_{10}+f_{01}) = 4/8 = 0.5$ (0-0 matches are ignored).

SMC = $(f_{11}+f_{00})/m = 5/9 = 0.5556$.

Required difference (simple matching coefficient – Jaccard similarity) = **0.06**.

Q.2 — Answer: 0.74 to 0.76 [NAT, 1M · Probabilistic & Instance-based Classifiers › Evaluation metrics]

TP = 9, FN = 54, FP = 42, TN = 14 (N = 119).

False positive rate = $FP/(FP + TN)$.

Value = **0.75**.

Watch the orientation: swapping FP and FN is the most common error (it swaps precision and recall).

Q.3 — Answer: 56 [NAT, 1M · Regression & Model Selection › Basis functions & feature scaling]

Number of monomials of degree $\leq d$ in p variables = $C(p + d, d) = C(8, 3) = \mathbf{56}$.

Q.4 — Answer: (A) [MCQ, 1M · Regression & Model Selection › MLE & MAP view]

$E[SSE] = \sigma^2(n - p)$ because the residuals live in an $(n - p)$ -dimensional space.

Answer: **SSE/(n - p)**.

Q.5 — Answer: (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers › Bayes optimal classifier]

(A) FALSE. It is the minimum of the two posteriors.

(B) FALSE. It enlarges it.

(C) TRUE. It is the error of the Bayes classifier.

(D) TRUE. This is the MAP rule.

Q.6 — Answer: 2.99 to 3.01 [NAT, 1M · SVM & Decision Trees › SVM: margin & geometry]

Distance = $|w^T x + b| / \|w\|$, with $w = (2, 2, 1)$, $b = -1$.

$w^T x + b = 9$, $\|w\| = 3$.

Distance = $9/3 = \mathbf{3}$.

Q.7 — Answer: (A) [MSQ, 1M · SVM & Decision Trees › Ensembles: bagging & random forests]

(A) TRUE. Variance of a mean.

(B) FALSE. $\approx 63.2\%$ are in-bag; $\approx 36.8\%$ are out-of-bag.

(C) FALSE. α decreases as ε increases.

(D) FALSE. Sampling is with replacement.

Q.8 — Answer: (A) [MSQ, 1M · Regression & Model Selection › Multiple linear regression]

(A) TRUE. $e = y - \hat{y} = (I - H)y$.

(B) FALSE. $\text{Var}(e) = \sigma^2(I - H)$, which is not diagonal in general.

(C) FALSE. $H = X(X^T X)^{-1} X^T$ depends only on X .

(D) FALSE. $0 \leq h_{ii} \leq 1$ always.

Q.9 — Answer: (A), (B), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers › k-nearest neighbours]

(A) TRUE. Distances to all n points are computed for each query.

(B) TRUE. It is non-parametric.

(C) FALSE. 1-NN has low bias and high variance.

(D) TRUE. Cover-Hart bound: $R^* \leq R_{1NN} \leq 2R^*(1 - R^*)$.

Q.10 — Answer: 1.598 to 1.602 [NAT, 1M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

$\nabla = \partial L / \partial w + \lambda w = 0 + 2 \times 2 = 4$.

$w \leftarrow 2 - 0.1 \times 4 = \mathbf{1.6}$ (equivalently $(1 - \eta\lambda)w - \eta\partial L/\partial w$).

Q.11 — Answer: 63 [NAT, 1M · Unsupervised Learning > k-means]

Each split is an unordered pair of non-empty complementary subsets: $(2^7 - 2)/2 = 2^6 - 1 = \mathbf{63}$.

This exponential count is why divisive methods use heuristics (e.g. bisecting k-means).

Q.12 — Answer: 1.5682 to 1.5722 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

CE for one example = $-\ln p_{\text{true class}}$.

Example 1: $-\ln 0.1 = 2.3026$.

Example 2: $-\ln 0.1 = 2.3026$.

Example 3: $-\ln 0.9 = 0.1054$.

Mean = **1.5702**.

Q.13 — Answer: (A) [MCQ, 1M · Unsupervised Learning > PCA & dimensionality reduction]

PCA is unsupervised: it picks the direction of maximum total variance, which is along the long axis of the clouds (d_1).

LDA is supervised: it maximizes between-class separation relative to within-class scatter, so it picks the direction across which the class means differ (d_2).

Answer: **PCA (1st PC) chooses d_1 ; LDA chooses d_2** . Projecting on the PCA direction would mix the two classes completely.

Q.14 — Answer: 225 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

$x^T x = (0)(0) + (3)(3) + (2)(2) = 13$.

$K(x, x) = (13 + 2)^2 = 15^2 = \mathbf{225}$.

The kernel equals the inner product $\phi(x)^T \phi(z)$ of the explicit (all-monomials-up-to-degree-d) feature maps, without forming ϕ .

Q.15 — Answer: (C) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

Key features of the plotted curve: Small negative linear part (slope 0.1) for $z < 0$, slope 1 for $z > 0$.

Hence $g = \mathbf{Leaky ReLU}$ with $\alpha = \mathbf{0.1}$.

Softplus: $\ln(1 + e^z)$: Smooth, always positive, value $\ln 2 \approx 0.69$ at $z = 0$, $\sim z$ for large z . - does not match.

ReLU: $\max(0, z)$: Exactly 0 for $z \leq 0$, slope 1 for $z > 0$, kink at origin. - does not match.

ELU with $\alpha = 1$: Smoothly saturates to -1 for $z \rightarrow -\infty$, linear for $z > 0$, passes through origin. - does not match.

Q.16 — Answer: -0.6908 to -0.6868 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

For σ output with binary cross-entropy $L = -[y \ln p + (1 - y) \ln(1 - p)]$, $\partial L/\partial a = p - y$, so $\partial L/\partial w_j = (p - y)x_j$ and $\partial L/\partial b = p - y$.

Example 1: $a = 0.5$, $p = 0.6225$, $p - y = 0.6225$.

Gradient for w_1 : -0.3112 .

$w_1 \leftarrow -1 - 1 \times (-0.3112) = \mathbf{-0.6888}$.

Note there is no σ' factor: it cancels with the derivative of the log loss.

Q.17 — Answer: (A) [MCQ, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

The line passes through $(-3, 0)$ and $(0, 4)$: $x_1/-3 + x_2/4 = 1$, i.e. $4x_1 + -3x_2 - -12 = 0$. Hence (w_0, w_1, w_2) must be a non-zero multiple of $(12, 4, -3)$.

Orientation: the class $y = 1$ region must have $w_0 + w_1 x_1 + w_2 x_2 > 0$. At the origin the $y = 1$ side has sign $+$ ($w_0 = 12$), which matches the figure (the origin lies on the $y = 1$ side).

So the answer is **$(w_0, w_1, w_2) = (12, 4, -3)$** (any positive multiple gives the same classifier).

Other options either describe a different line (swapped/negated coefficients move the intercepts) or the same line with the sign flipped, which would label every training point incorrectly.

Q.18 — Answer: 0.8215 to 0.8415 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [10 Yes, 9 No]: $H = 0.998$.

Child entropies: $v_1: 0.9852, v_2: 0.7219$.

$IG = 0.998 - \sum (n_v/N) H_v = 0.0821$.

$\text{SplitInfo} = -\sum (n_v/N) \log_2(n_v/N) = -(14/19)\log_2(14/19) - (5/19)\log_2(5/19) = 0.8315$.

Answer: **0.83**.

C4.5 uses the gain ratio to penalise attributes that split the data into many small branches.

Q.19 — Answer: 0.0302 to 0.0502 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [10+, 6-]: $\text{Gini} = 1 - [(10/16)^2 + (6/16)^2] = 0.4688$.

Outlook = Sunny: [1+, 2-], $\text{Gini} = 1 - [(1/3)^2 + (2/3)^2] = 0.4444$.

Outlook = Overcast: [5+, 2-], $\text{Gini} = 1 - [(5/7)^2 + (2/7)^2] = 0.4082$.

Outlook = Rain: [4+, 2-], $\text{Gini} = 1 - [(4/6)^2 + (2/6)^2] = 0.4444$.

Weighted Gini = $(3/16)(0.4444) + (7/16)(0.4082) + (6/16)(0.4444) = 0.4286$.

Gini gain = $0.4688 - 0.4286 = 0.0402$.

Answer: **0.04**.

Q.20 — Answer: 2.5637 to 2.5791 [NAT, 2M · Regression & Model Selection > MLE & MAP view]

$-\ln p(w | \text{data}) = (1/(2\sigma^2))\sum (y_i - wx_i)^2 + |w|/b + \text{const}$. Multiply by $2\sigma^2$: $\sum (y_i - wx_i)^2 + (2\sigma^2/b)|w|$ (lasso).

$\sum x_i^2 = 7, \sum x_i y_i = 22$. Setting the subgradient to 0: $\sum x_i^2 \cdot w = \sum x_i y_i - (\sigma^2/b) \cdot \text{sign}(w) = \hat{w} = \text{sign}(\sum x_i y_i) \cdot \max(|\sum x_i y_i| - \sigma^2/b, 0) / \sum x_i^2$.

$\sigma^2/b = 4$; $\hat{w} = 1 \times \max(22 - 4, 0) / 7 = 2.5714$.

Answer: **2.571**.

Q.21 — Answer: 5 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

Total pairs = $C(6,2) = 15$.

a = pairs together in both = 1; b = pairs separated in both = 9; pairs together in U only = 3; together in V only = 2.

Rand index = $(a + b)/C(n,2) = (1 + 9)/15 = 0.6667$.

Answer: **5**. (Cluster names do not matter: only whether pairs are together.)

Q.22 — Answer: (B) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

$a = W_1 x + b_1 = (5, -3, 1)^T$; $h = \text{ReLU}(a) = (5, 0, 1)^T$.

$\hat{y} = W_2 h + b_2 = (-1, 10)^T$.

Distractors: skipping the ReLU gives $(-7, 7)^T$; using W_1^T instead of W_1 gives $(-1, 4)^T$; dropping the biases gives $(0, 8)^T$; using $|a|$ instead of ReLU gives $(5, 13)^T$.

Q.23 — Answer: 30.845 to 31.155 [NAT, 2M · Regression & Model Selection > Cross-validation]

For each left-out point, the prediction is the y of its nearest remaining neighbour (no ties here).

x_i	3	6	7	13	15
nearest x	6	7	6	15	13
prediction	8	11	8	11	3
error	3	-3	3	-8	8

LOOCV MSE = $(1/5)\sum \text{error}^2 = 155/5 = 31$.

Answer: **31**.

Q.24 — Answer: 1 [NAT, 2M · Unsupervised Learning > k-means]

Iteration 1 - assignment (nearest initial centroid, squared Euclidean distance):

Point	$d^2(\cdot, C1)$	$d^2(\cdot, C2)$	cluster
P1 (6, 3)	45	8	C2
P2 (7, 8)	5	50	C1
P3 (8, 7)	5	36	C1
P4 (1, 8)	65	98	C1

P5 (8, 4)	26	9	C2
P6 (3, 5)	52	41	C2
P7 (2, 6)	58	61	C1

Update: each centroid becomes the mean of its assigned points: $C1 \rightarrow (4.5, 7.25)$; $C2 \rightarrow (5.67, 4)$.

Iteration 2 - assignment with the updated centroids:

Point	$d^2(\cdot, C1)$	$d^2(\cdot, C2)$	cluster
P1 (6, 3)	20.31	1.11	C2
P2 (7, 8)	6.81	17.78	C1
P3 (8, 7)	12.31	14.44	C1
P4 (1, 8)	12.81	37.78	C1
P5 (8, 4)	22.81	5.44	C2
P6 (3, 5)	7.31	8.11	C1
P7 (2, 6)	7.81	17.44	C1

Points that changed cluster: P6 $\rightarrow$ answer **1**.

Note: each step of Lloyd's algorithm can only decrease (or keep) J, which is why it converges.

Q.25 — Answer: 6.99 to 7.01 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Use single linkage: $d(A, B) = \min_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

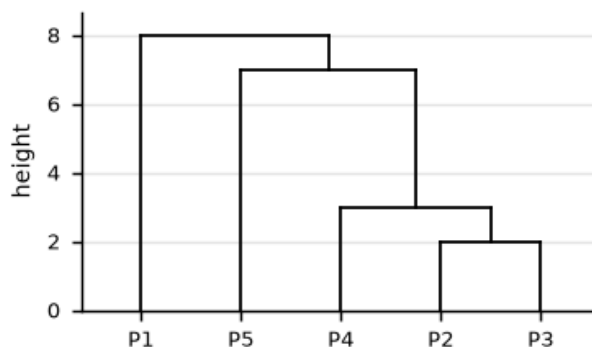
Merge 1: $\{P2\} + \{P3\}$ at height **2**. Updated: $d(\{P2, P3\}, \{P1\}) = 9$, $d(\{P2, P3\}, \{P4\}) = 3$, $d(\{P2, P3\}, \{P5\}) = 7$.

Merge 2: $\{P4\} + \{P2, P3\}$ at height **3**. Updated: $d(\{P2, P3, P4\}, \{P1\}) = 9$, $d(\{P2, P3, P4\}, \{P5\}) = 7$.

Merge 3: $\{P5\} + \{P2, P3, P4\}$ at height **7**. Updated: $d(\{P2, P3, P4, P5\}, \{P1\}) = 8$.

Merge 4: $\{P1\} + \{P2, P3, P4, P5\}$ at height **8**.

P4 and P5 first lie in the same cluster at the merge of height 7, so their cophenetic distance is 7 (their original distance was 8).



Q.26 — Answer: 4.39 to 4.41 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

Conditional risk $R(\alpha_i | x) = \sum_j L(\alpha_i, C_j) P(C_j | x)$:

$$R(\alpha_1) = 0 \times 0.5 + 10 \times 0.2 + 8 \times 0.3 = 4.4$$

$$R(\alpha_2) = 10 \times 0.5 + 0 \times 0.2 + 1 \times 0.3 = 5.3$$

$$R(\alpha_3) = 7 \times 0.5 + 9 \times 0.2 + 0 \times 0.3 = 5.3$$

Minimum risk: α_1 with $R = 4.4$. The MAP rule picks C_1 (risk 4.4) — the same action here.

Answer: **4.4**.

Q.27 — Answer: 3.8633 to 3.9033 [NAT, 2M · Regression & Model Selection > Simple linear regression]

$$\bar{x} = 6.833, \bar{y} = 6.333.$$

i	1	2	3	4	5	6
$x_i - \bar{x}$	-4.83	-1.83	0.17	1.17	2.17	3.17

$y_i - \bar{y}$	5.67	-0.33	-0.33	2.67	-3.33	-4.33
$(x_i - \bar{x})(y_i - \bar{y})$	-27.39	0.61	-0.06	3.11	-7.22	-13.72

$$S_{xx} = \sum (x_i - \bar{x})^2 = 42.833, S_{xy} = -44.667.$$

$$\beta_1 = S_{xy}/S_{xx} = -1.0428, \beta_0 = \bar{y} - \beta_1 \bar{x} = 13.4591.$$

$$\hat{y} = 13.4591 + (-1.0428)(8) = 5.1167; e = 9 - 5.1167 = 3.8833.$$

Answer: **3.88**.

Q.28 — Answer: (A), (B), (C), (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

(A) TRUE. Reduces the noise floor of SGD.

(B) TRUE. Its expectation equals the average gradient.

(C) TRUE. CE = negative log-likelihood of a categorical/Bernoulli model.

(D) TRUE. Steps overshoot the minimum.

Q.29 — Answer: 0.0698 to 0.0898 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

$$\text{Parent } [7+, 13-]: H = -(7/20)\log_2(7/20) - (13/20)\log_2(13/20) = 0.9341.$$

$$\text{Student} = \text{Yes}: [6+, 7-], H = -(6/13)\log_2(6/13) - (7/13)\log_2(7/13) = 0.9957.$$

$$\text{Student} = \text{No}: [1+, 6-], H = -(1/7)\log_2(1/7) - (6/7)\log_2(6/7) = 0.5917.$$

$$\text{Weighted child entropy} = (13/20)(0.9957) + (7/20)(0.5917) = 0.8543.$$

$$IG = 0.9341 - 0.8543 = 0.0798.$$

Answer: **0.08**.

Q.30 — Answer: 5.99 to 6.01 [NAT, 2M · SVM & Decision Trees > Regression trees]

$$\text{Parent mean } \bar{y} = 4.333, \text{SSE}_0 = \sum (y - \bar{y})^2 = 13.333.$$

Evaluate every split:

Split t	Left mean	Right mean	$\text{SSE}_L + \text{SSE}_R$
1.5	5	4.2	12.8
5	4	4.5	13
8.5	3.667	5	10.667
10	3.5	6	5
12	4.2	5	12.8

Best split $t = 10$: left mean 3.5, right mean 6, $\text{SSE} = 5$, reduction = 8.333.

Answer: **6**.

Q.31 — Answer: 0.6708 to 0.6908 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$$\mathbf{m}_1 = (2, 1.667), \mathbf{m}_2 = (5, 3.333).$$

$$S_1 = \begin{bmatrix} 8 & -4 \\ -4 & 4.667 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 14 & 6 \\ 6 & 2.667 \end{bmatrix}$$

$$S_W = \begin{bmatrix} 22 & 2 \\ 2 & 7.333 \end{bmatrix}$$

$$\mathbf{m}_2 - \mathbf{m}_1 = (3, 1.667).$$

$$\mathbf{w} = S_W^{-1}(\mathbf{m}_2 - \mathbf{m}_1) = (0.1186, 0.1949).$$

$$(\mathbf{m}_2 - \mathbf{m}_1)^T \mathbf{w} = \mathbf{0.68}.$$

Q.32 — Answer: 3.486 to 3.514 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

$$\text{With } X^T X = I: \beta_{\text{OLS}} = X^T y = b.$$

$$\text{Ridge: } \nabla [\|y - X\beta\|^2 + \lambda \|\beta\|^2] = 0 \Rightarrow (1 + \lambda)\beta = b \Rightarrow \beta_{\text{ridge}} = b/(1 + \lambda).$$

$$\text{Lasso: the problem separates per coordinate, } \min \frac{1}{2}(\beta_j - b_j)^2 + \lambda |\beta_j| \Rightarrow \text{soft-thresholding } \beta_j = \text{sign}(b_j) \cdot \max(|b_j| - \lambda, 0).$$

Coefficient j becomes 0 once $\lambda \geq |b_j|$. Sorted $|b|$: 0.3, 1.5, 2.2, 3.5, 4; the 4-th smallest is 3.5.
 Answer: **3.5**.

Q.33 — Answer: 1096 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Parameter counting]
 Encoder: $64 \times 8 + 8 = 520$; decoder: $8 \times 64 + 64 = 576$. Total = **1096**.

Q.34 — Answer: 1799.99 to 1800.01 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

For centered data, projecting onto the top- k PCs and reconstructing leaves the components along the discarded eigenvectors; their average squared length (divisor $n - 1$) is the sum of the discarded eigenvalues.

Discarded eigenvalues: 18, 13, 4, 1; sum = 36; total = 74.

Total error = $(n - 1) \times 36 = 50 \times 36 = \mathbf{1800}$.

Q.35 — Answer: (A) [MCQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]
 Ranges: age [32, 60], income [44, 131]. Scaled query = (0.714, 0.126).

Point	scaled age	scaled income	scaled distance	raw distance
P1	1	0.333	0.353	19.7
P2	0	0.632	0.875	48.33
P3	0.071	1	1.085	78.1
P4	0.214	0	0.516	17.8

Nearest after scaling: **P1**.

Without scaling the nearest point would be P4, because income differences (tens of thousands) dominate the age differences — exactly why k-NN needs feature scaling.

Mock Test 31

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	5	11	19

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Regression & Model Selection	7	11
Probabilistic & Instance-based Classifiers	7	11
SVM & Decision Trees	7	11
Unsupervised Learning	7	11
Neural Networks (MLP / Feed-forward)	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

A regularisation parameter is tuned over 6 candidate values using 4-fold cross-validation. After selecting the best value, the model is refitted once on the full training data. The total number of times the model is fitted is _____ (Answer in integer)

Q.2 [1 mark · NAT (numerical answer)]

Points are drawn uniformly from the unit hypercube $[0, 1]^5$. The fraction of points lying within distance 0.05 of the boundary (i.e. outside the inner cube $[0.05, 0.95]^5$) is _____ (rounded off to two decimal places)

Q.3 [1 mark · MSQ (one or more correct)]

Which of the following statements about decision-tree learning is/are CORRECT?

- (A) A binary tree of depth d has at most 2^d leaves.
- (B) A decision tree is a linear classifier.
- (C) Pruning a subtree can never increase the training error.
- (D) Post-pruning grows a full tree and then removes subtrees that do not help validation performance.

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about clustering evaluation measures is/are CORRECT?

- (A) A larger within-cluster SSE indicates more compact clusters.
- (B) The (unadjusted) Rand index of two random labelings has expected value 0.
- (C) The silhouette coefficient is an internal measure: it uses only the data and the cluster assignments.
- (D) In the silhouette, $b(i)$ is the mean distance to the farthest other cluster.

Q.5 [1 mark · MSQ (one or more correct)]

Consider the soft-margin SVM: minimise $\frac{1}{2}\|w\|^2 + C \sum_i \xi_i$ subject to $y_i(w^T x_i + b) \geq 1 - \xi_i$, $\xi_i \geq 0$. Which of the following statements is/are CORRECT?

- (A) Slack variables ξ_i can be negative for points far from the boundary.
- (B) A point lying exactly on the margin ($y f(x) = 1$) incurs zero hinge loss.
- (C) The soft-margin SVM objective is non-convex because of the slack variables.
- (D) Increasing C increases the regularisation strength.

Q.6 [1 mark · MSQ (one or more correct)]

A simple linear regression $\hat{y} = \beta_0 + \beta_1 x$ is fitted by ordinary least squares (with intercept) to data (x_i, y_i) , $i = 1, \dots, n$, with residuals $e_i = y_i - \hat{y}_i$ and sample correlation r . Which of the following statements is/are CORRECT?

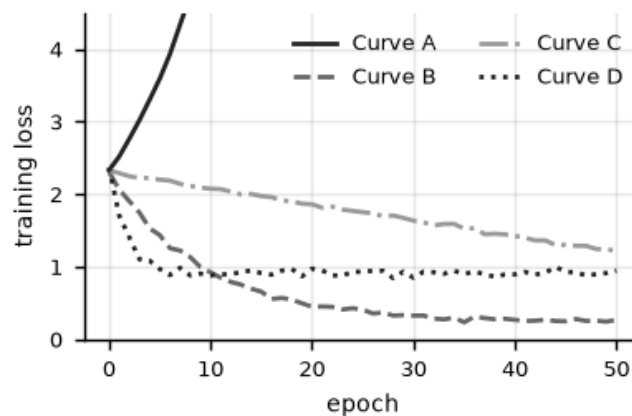
- (A) In simple linear regression with intercept, $R^2 = r^2$.
- (B) $SST = SSR + SSE$ for the least-squares fit with intercept.
- (C) If every y_i is multiplied by a constant c , the least-squares slope is multiplied by c .
- (D) The product of the slope of the y-on-x line and the slope of the x-on-y line equals r^2 .

Q.7 [1 mark · NAT (numerical answer)]

To keep the same sampling density as 20 points per axis in 1-D, the number of points needed in $d = 4$ dimensions (a grid with 20 points per axis) is _____. (Answer in integer)

Q.8 [1 mark · MCQ]

The training-loss curves below were obtained with four different learning rates (everything else fixed). Which curve corresponds to a well-chosen learning rate?



- (A) Curve A
- (B) Curve B
- (C) Curve C
- (D) Curve D

Q.9 [1 mark · NAT (numerical answer)]

Regressing x_2 on the remaining predictors gives $R^2 = 0.88$. By what factor is the standard error of β_2 inflated relative to the case of x_2 being uncorrelated with the other predictors? _____ (rounded off to two decimal places)

Q.10 [1 mark · NAT (numerical answer)]

For $K = 5$ classes and $p = 5$ features, how many MORE covariance parameters does QDA estimate than LDA? _____ (Answer in integer)

Q.11 [1 mark · MSQ (one or more correct)]

Which of the following statements about backpropagation is/are CORRECT?

- (A) A finite-difference (central difference) check can be used to verify a backprop implementation.
- (B) For a sigmoid output with binary cross-entropy (or softmax with cross-entropy), the error at the output pre-activation is $\hat{y} - y$.
- (C) The gradient with respect to a bias b_j equals δ_j .
- (D) With $L = \frac{1}{2}(\hat{y} - y)^2$ and a sigmoid output unit, $\delta_{\text{out}} = (\hat{y} - y) \hat{y} (1 - \hat{y})$.

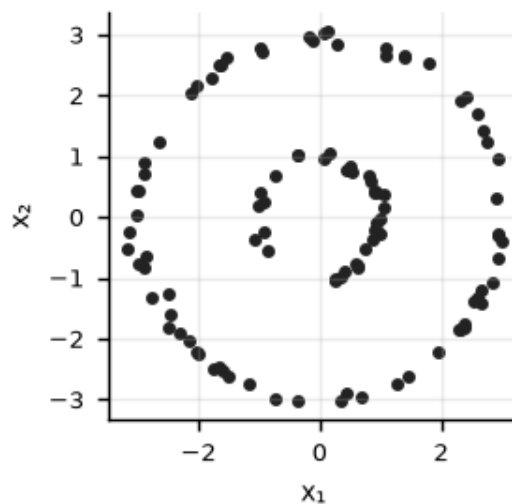
Q.12 [1 mark · NAT (numerical answer)]

In the dual solution of a linear SVM, the non-zero Lagrange multipliers α_i and labels y_i of the support vectors are given below (all other $\alpha_i = 0$). The value of α_1 is _____ (rounded off to one decimal place)

i	1	2	3	4
y_i	+1	+1	-1	-1
α_i	?	0.1	1.4	0.5

Q.13 [1 mark · MCQ]

The unlabeled data below visibly consists of two concentric rings (well separated, dense). Which method is most likely to recover these two groups exactly?



- (A) Complete-linkage agglomerative clustering, cut into 2 clusters
- (B) k-medoids (PAM) with $k = 2$
- (C) Single-linkage agglomerative clustering, cut into 2 clusters
- (D) k-means with $k = 2$ (best of many restarts)

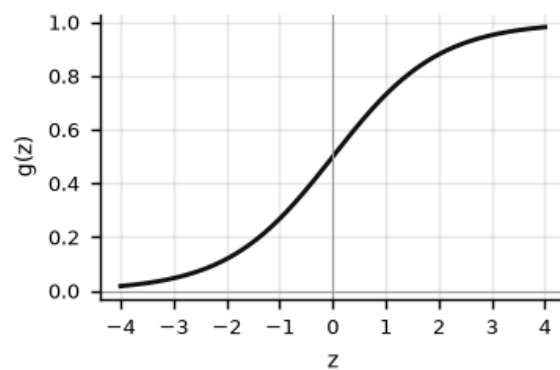
Q.14 [1 mark · MSQ (one or more correct)]

Which of the following statements about classifier evaluation is/are CORRECT?

- (A) ROC AUC is unchanged by any strictly increasing transformation of the scores.
- (B) Specificity equals $TN/(TN + FP)$.
- (C) For fixed TPR and FPR, precision decreases as the positive class becomes rarer.
- (D) Under heavy class imbalance, precision-recall curves are usually more informative than ROC curves.

Q.15 [1 mark · MCQ]

The figure shows the graph of a function $g(z)$ used in neural networks. Which function is it?



- (A) $\sigma'(z) = \sigma(z)(1 - \sigma(z))$
- (B) $\tanh(z)$
- (C) Sigmoid $\sigma(z) = 1/(1+e^{-z})$
- (D) Softplus: $\ln(1 + e^z)$

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · MSQ (one or more correct)]**

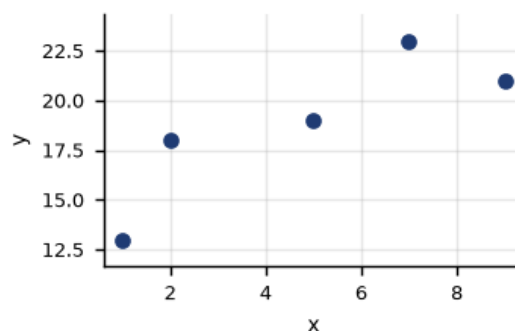
A threshold unit outputs 1 if $w_1x_1 + w_2x_2 + b > 0$ and 0 otherwise, for binary inputs $x_1, x_2 \in \{0, 1\}$. Which of the following parameter settings make the unit implement the Boolean **AND** function?

- (A) $w_1 = 2, w_2 = 0, b = -3$
- (B) $w_1 = 2, w_2 = 2, b = -2.5$
- (C) $w_1 = 2, w_2 = -1.5, b = 0.5$
- (D) $w_1 = 1.5, w_2 = 3, b = 1$

Q.17 [2 marks · NAT (numerical answer)]

Assume $y_i = \beta_0 + \beta_1x_i + \varepsilon_i$ with ε_i i.i.d. $N(0, \sigma^2)$. For the 5 observations below, with β estimated by least squares (= maximum likelihood), the maximum-likelihood estimate of σ^2 is _____ (rounded off to two decimal places)

x	1	2	5	7	9
y	13	18	19	23	21



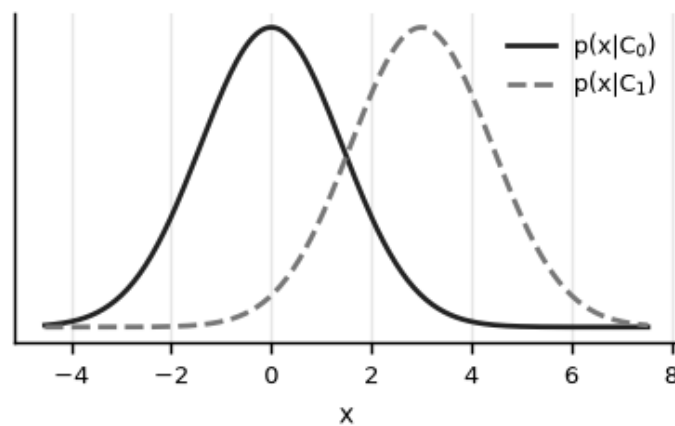
Q.18 [2 marks · NAT (numerical answer)]

Agglomerative hierarchical clustering with **complete linkage** is applied to 5 points whose pairwise distance matrix is given below (upper triangle shown). The height (linkage distance) at which the **third** merge takes place is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5
P1	0	6	15	3	13
P2		0	9	7	7
P3			0	12	2
P4				0	10
P5					0

Q.19 [2 marks · NAT (numerical answer)]

In a two-class problem with one feature, $p(x | C_0) = N(0, 2)$ and $p(x | C_1) = N(3, 2)$ (second argument = variance), with priors $P(C_0) = 0.7$ and $P(C_1) = 0.3$. Under 0-1 loss, the posterior probability $P(C_1 | x = 3)$ is _____ (rounded off to two decimal places)

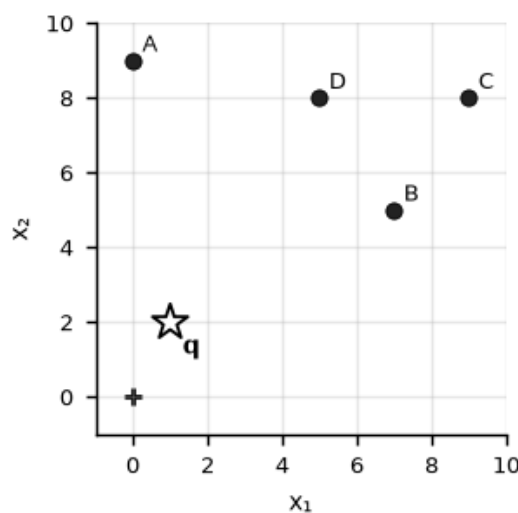


Class-conditional densities (not weighted by priors).

Q.20 [2 marks · MCQ]

A query point $q = (1, 2)$ and four points A-D are shown below. Using **cosine similarity**, which point is most similar to q ?

Point	A	B	C	D
x_1	0	7	9	5
x_2	9	5	8	8



- (A) Point A
(C) Point C

- (B) Point B
(D) Point D

Q.21 [2 marks · NAT (numerical answer)]

Ridge regression without intercept minimises $\|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2$ with $\lambda = 3$, for the data below ($\beta = (\beta_1, \beta_2)^T$). The value of β_2 is _____ (rounded off to three decimal places)

$$X = \begin{bmatrix} -2 & -1 \\ 2 & 2 \\ 0 & 0 \end{bmatrix}$$

$$y = \begin{bmatrix} 5 \\ 0 \\ -2 \end{bmatrix}$$

Q.22 [2 marks · NAT (numerical answer)]

Logistic regression $P(y = 1 | \mathbf{x}) = \sigma(w_1x_1 + w_2x_2 + b)$ is trained by **batch** gradient descent on the **average** cross-entropy loss over the 3 examples below, with learning rate $\eta = 0.2$. Starting from $w_1 = 0.5$, $w_2 = -0.5$, $b = 0$, the value of b after ONE gradient-descent update is _____ (rounded off to three decimal places)

i	x_1	x_2	y
1	0	0	1
2	1	2	0
3	1	2	0

Q.23 [2 marks · MSQ (one or more correct)]

For the confusion matrix below, which of the following statements is/are CORRECT? (values rounded to 2 decimals)

	Actual +	Actual –
Predicted +	30	64
Predicted –	58	43

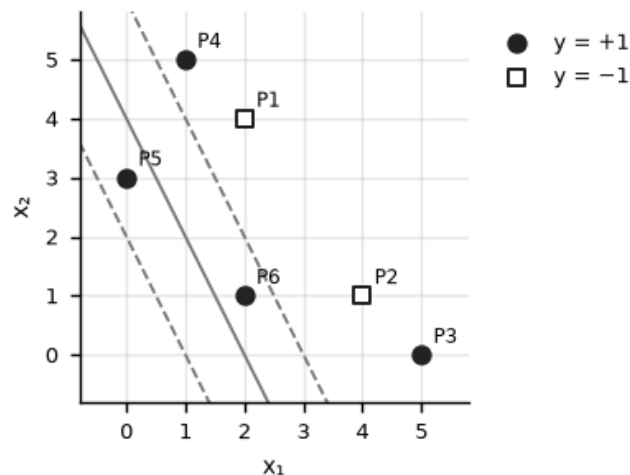
- (A) Precision is greater than recall.
(C) The F1-score is 0.33.

- (B) The specificity (TNR) is 0.4.
(D) The false positive rate is 0.68.

Q.24 [2 marks · MSQ (one or more correct)]

A soft-margin linear SVM has learned $f(x) = x_1 + 0.5x_2 - 2$. The training points are listed below and plotted (boundary solid, margins $f = \pm 1$ dashed). With ξ_i the optimal slack of point i , which of the following statements is/are CORRECT?

Point	x_1	x_2	y
P1	2	4	-1
P2	4	1	-1
P3	5	0	+1
P4	1	5	+1
P5	0	3	+1
P6	2	1	+1

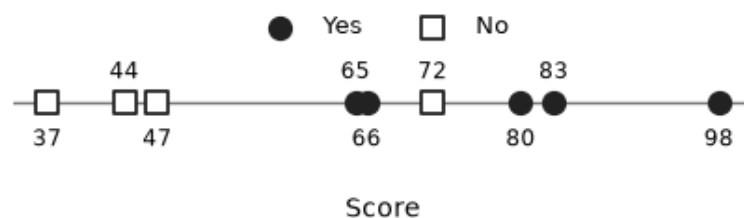


- (A) P3 is misclassified ($\xi_3 > 1$).
 (B) P6 is correctly classified but lies strictly inside the margin.
 (C) P5 has slack $\xi_5 = 0$.
 (D) Exactly 4 of the points have $\xi_i > 0$.

Q.25 [2 marks · NAT (numerical answer)]

The continuous attribute Score and the class label of 9 training records are given below (unsorted). A binary split “Score $\leq t$ ” is chosen to maximise information gain, with t restricted to midpoints between consecutive sorted values. Find the information gain (in bits) of the best threshold split. (rounded off to two decimal places)

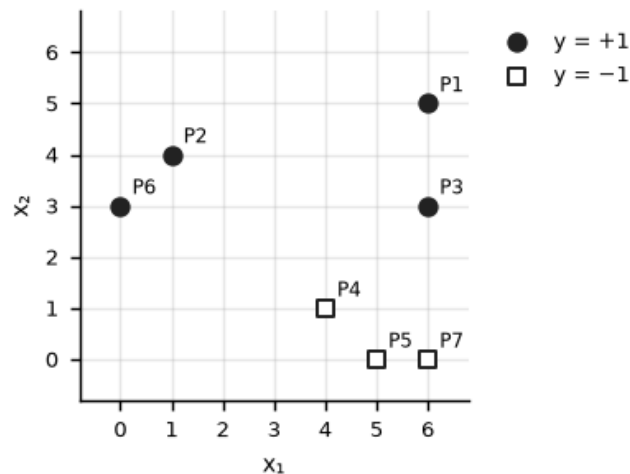
Score	37	83	65	47	66	72	80	44	98
Class	No	Yes	Yes	No	Yes	No	Yes	No	Yes



Q.26 [2 marks · MSQ (one or more correct)]

A hard-margin linear SVM is trained on the 7 labelled points shown. Which of the following points is/are support vectors of the resulting maximum-margin classifier?

Point	x ₁	x ₂	y
P1	6	5	+1
P2	1	4	+1
P3	6	3	+1
P4	4	1	-1
P5	5	0	-1
P6	0	3	+1
P7	6	0	-1

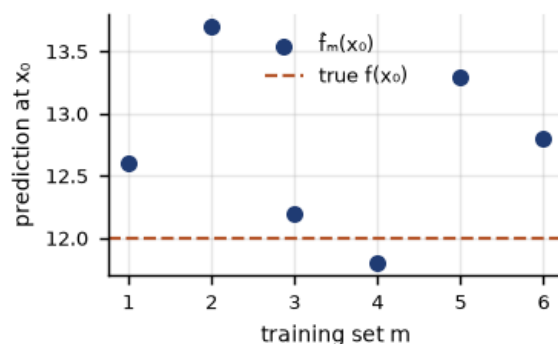


- (A)** P2 **(B)** P3
(C) P4 **(D)** P6

Q.27 [2 marks · NAT (numerical answer)]

Data are generated as $y = f(x) + \varepsilon$ with $E[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2 = 1.5$. At a test point x_0 the true value is $f(x_0) = 12$. A learning method is trained on 6 independent training sets, giving the predictions $\hat{f}_m(x_0)$ in the table (also plotted). Treating these as the full distribution of the method's prediction (variance with divisor 6), the squared bias of the method at x_0 is _____ (rounded off to three decimal places)

m	1	2	3	4	5	6
$\hat{f}_m(x_0)$	12.6	13.7	12.2	11.8	13.3	12.8



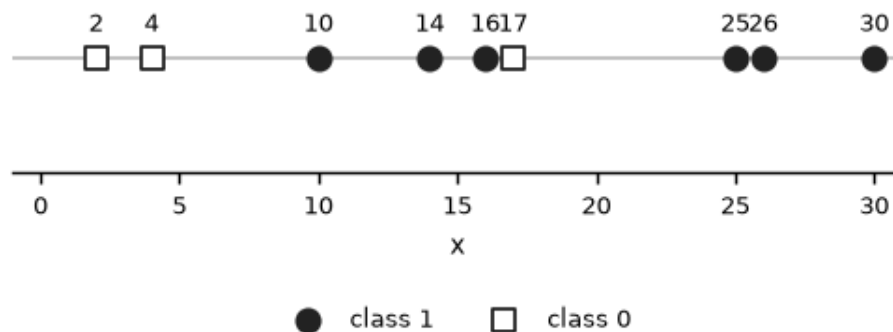
Q.28 [2 marks · MCQ]

For logits $z = (-1, 0, 2)$, softmax probabilities p and true class 3, the gradient of the cross-entropy loss $L = -\ln p_3$ with respect to z (rounded to two decimals) is

- (A) (0.04, 0.11, 0.84) (B) (0.04, 0.11, -0.16)
(C) (-0.96, -0.89, -0.16) (D) (0, 0, -1.19)

Q.29 [2 marks · NAT (numerical answer)]

9 one-dimensional points with binary labels are shown on the number line (filled circle = class 1, open square = class 0). A 3-NN classifier (absolute distance, majority vote) is evaluated by leave-one-out cross-validation: each point is classified using all the *other* points. The leave-one-out cross-validation error rate (as a fraction) is _____ (rounded off to two decimal places)

**Q.30 [2 marks · MSQ (one or more correct)]**

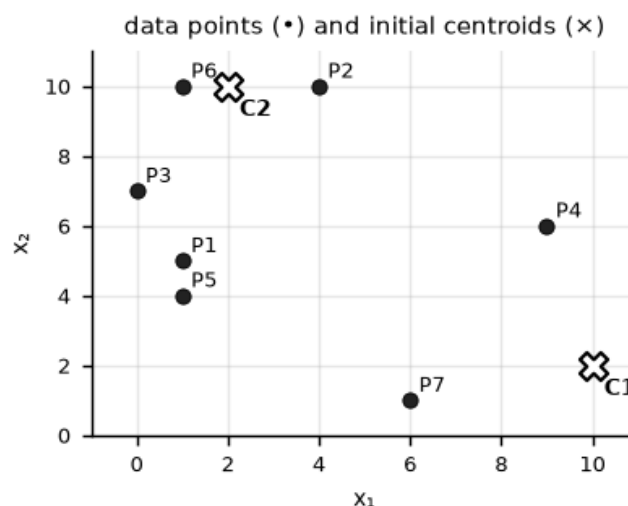
Which of the following statements about PCA, SVD and dimensionality reduction is/are CORRECT?

- (A) If two features are perfectly correlated, their 2×2 covariance matrix has a zero eigenvalue.
- (B) Multiplying all features by the same non-zero constant c leaves the proportion of variance explained by each PC unchanged.
- (C) Rotating the data by an orthogonal matrix rotates the PC directions but leaves the eigenvalues unchanged.
- (D) If centered X ($n \times d$) has SVD $X = U\Sigma V^T$, the eigenvalues of the sample covariance are $\sigma_j^2/(n-1)$.

Q.31 [2 marks · NAT (numerical answer)]

k-means (Lloyd's algorithm, Euclidean distance) with $k = 2$ is run on the 7 points below, starting from the initial centroids $C1 = (10, 2)$, $C2 = (2, 10)$. After **two** iterations, the x_2 -coordinate of the centroid of cluster $C2$ is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5	P6	P7
x_1	1	4	0	9	1	1	6
x_2	5	10	7	6	4	10	1



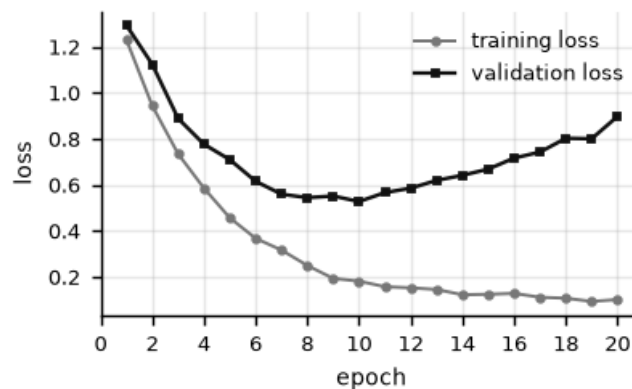
Q.32 [2 marks · NAT (numerical answer)]

Least-squares linear regression on $n = 10$ observations minimises $J(w) = (1/n)\|Xw - y\|^2$ by batch gradient descent with constant learning rate η . The data give the matrix $X^T X$ below. Then the supremum of learning rates η for which batch gradient descent converges from every starting point is _____ (rounded off to three decimal places)

$$X^T X = \begin{bmatrix} 9 & 7 \\ 7 & 9 \end{bmatrix}$$

Q.33 [2 marks · NAT (numerical answer)]

A network is trained with early stopping on validation loss using patience $P = 3$: training stops as soon as the validation loss has failed to improve on its best value for 3 consecutive epochs, and the weights from the best epoch are restored. Using the validation losses below (also plotted), the epoch after which training is stopped is _____ (Answer in integer)

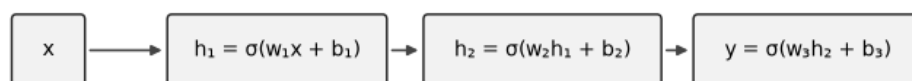


epoch	1	2	3	4	5	6	7	8	9	10
val. loss	1.294	1.124	0.889	0.777	0.710	0.616	0.560	0.545	0.552	0.528

epoch	11	12	13	14	15	16	17	18	19	20
val. loss	0.567	0.586	0.620	0.642	0.669	0.717	0.743	0.803	0.801	0.898

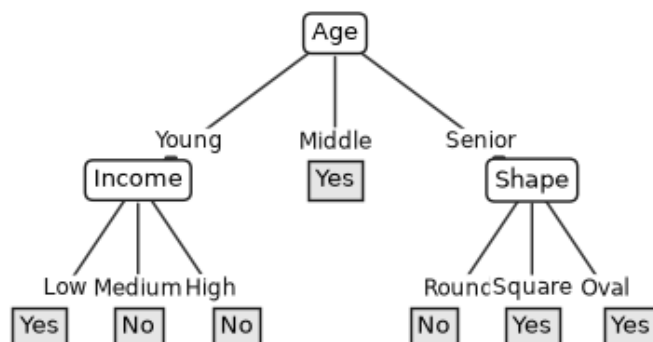
Q.34 [2 marks · NAT (numerical answer)]

Consider the scalar chain shown below with weights $w_1 = 2$, $w_2 = -3$, $w_3 = -2$, biases $b_1 = -0.5$, $b_2 = 1$, $b_3 = 0$ and activation the logistic sigmoid σ at every stage. For input $x = 1$, the value of $\partial y / \partial x$ is _____ (rounded off to 4 decimal places)



Q.35 [2 marks · NAT (numerical answer)]

The decision tree shown is applied to the 9 labelled records in the table (attributes not tested on a path are ignored). Find the training accuracy of the tree (as a percentage). (rounded off to one decimal place)



#	Age	Shape	Income	Actual
1	Senior	Square	Low	Yes
2	Middle	Oval	Low	No
3	Young	Square	Medium	No
4	Senior	Round	High	No
5	Senior	Square	Low	Yes
6	Senior	Oval	Low	No
7	Middle	Oval	Low	Yes
8	Senior	Square	Low	Yes
9	Young	Oval	Low	Yes

— END OF QUESTION PAPER —

Mock Test 31 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	25	19	NAT	2	0.7926 to 0.8126
2	NAT	1	0.3995 to 0.4195	20	MCQ	2	(D)
3	MSQ	1	(A), (D)	21	NAT	2	0.0931 to 0.0992
4	MSQ	1	(C)	22	NAT	2	−0.019 to −0.015
5	MSQ	1	(B)	23	MSQ	2	(B), (C)
6	MSQ	1	(A), (B), (C), (D)	24	MSQ	2	(B), (D)
7	NAT	1	160000	25	NAT	2	0.5477 to 0.5677
8	MCQ	1	(B)	26	MSQ	2	(B), (C), (D)
9	NAT	1	2.8723 to 2.9012	27	NAT	2	0.5348 to 0.5408
10	NAT	1	60	28	MCQ	2	(B)
11	MSQ	1	(A), (B), (C), (D)	29	NAT	2	0.3233 to 0.3433
12	NAT	1	1.79 to 1.81	30	MSQ	2	(A), (B), (C), (D)
13	MCQ	1	(C)	31	NAT	2	7.19 to 7.21
14	MSQ	1	(A), (B), (C), (D)	32	NAT	2	0.623 to 0.627
15	MCQ	1	(C)	33	NAT	2	13
16	MSQ	2	(B)	34	NAT	2	0.0657 to 0.067
17	NAT	2	3.1621 to 3.2021	35	NAT	2	77.728 to 77.828
18	NAT	2	6.99 to 7.01				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Regression & Model Selection					11
Probabilistic & Instance-based Classifiers					11
SVM & Decision Trees					11
Unsupervised Learning					11
Neural Networks (MLP / Feed-forward)					11
Total					55

Mock Test 31 — Detailed Solutions

Q.1 — Answer: 25 [NAT, 1M · Regression & Model Selection > Cross-validation]

Each candidate needs $k = 4$ fits: $6 \times 4 = 24$. Plus 1 final refit: **25**.

Q.2 — Answer: 0.3995 to 0.4195 [NAT, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Inner cube volume $= (1 - 2 \times 0.05)^5 = 0.9^5 = 0.5905$.

Fraction near the boundary $= 1 - 0.5905 = \mathbf{0.41}$.

In high dimensions most points lie near the boundary, far from each other — a symptom of the curse of dimensionality that hurts k-NN.

Q.3 — Answer: (A), (D) [MSQ, 1M · SVM & Decision Trees > Decision trees: overfitting & pruning]

(A) **TRUE**. Doubling per level.

(B) **FALSE**. Its boundary is piecewise axis-parallel, generally non-linear.

(C) **FALSE**. Pruning usually increases (or keeps) training error.

(D) **TRUE**. E.g. reduced-error or cost-complexity pruning.

Q.4 — Answer: (C) [MSQ, 1M · Unsupervised Learning > Cluster evaluation]

(A) **FALSE**. Compact clusters have smaller SSE.

(B) **FALSE**. It is typically well above 0; the ARI is the chance-corrected version.

(C) **TRUE**. No ground-truth labels are needed.

(D) **FALSE**. b(i) uses the nearest other cluster.

Q.5 — Answer: (B) [MSQ, 1M · SVM & Decision Trees > SVM: soft margin & hinge loss]

(A) **FALSE**. $\xi_i \geq 0$ by definition.

(B) **TRUE**. $\max(0, 1 - 1) = 0$.

(C) **FALSE**. Objective and constraints remain convex (a QP).

(D) **FALSE**. Larger C means weaker regularisation.

Q.6 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Simple linear regression]

(A) **TRUE**. $SSR/SST = S_{xy}^2 / (S_{xx} S_{yy}) = r^2$.

(B) **TRUE**. Cross term $2\sum(\hat{y}_i - \bar{y})e_i$ vanishes by the normal equations.

(C) **TRUE**. S_{xy} scales by c while S_{xx} is unchanged.

(D) **TRUE**. $(S_{xy}/S_{xx})(S_{xy}/S_{yy}) = r^2$.

Q.7 — Answer: 160000 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

A grid with 20 points per axis needs $20^4 = \mathbf{160000}$ points — exponential growth with d .

Q.8 — Answer: (B) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Typical signatures: far-too-high $\eta \rightarrow$ loss blows up; somewhat-too-high $\eta \rightarrow$ quick drop then plateau at a high loss (bouncing around the minimum); too-low $\eta \rightarrow$ slow, nearly linear decrease; good $\eta \rightarrow$ fast decrease to the lowest loss.

Curve A: a learning rate that is far too high (training diverges).

Curve B: a well-chosen learning rate.

Curve C: a learning rate that is too low (very slow, almost linear decrease).

Curve D: a learning rate that is somewhat too high (fast initial drop, then the loss plateaus at a high value).

Hence the answer is **Curve B**.

Q.9 — Answer: 2.8723 to 2.9012 [NAT, 1M · Regression & Model Selection > Multiple linear regression]

Variance inflation factor of predictor j : $VIF_j = 1/(1 - R_j^2)$, where R_j^2 is the R^2 from regressing x_j on all other predictors. It is the factor by which $\text{Var}(\beta_j)$ is inflated by collinearity; $\sqrt{VIF}$ inflates the standard error.

$VIF = 1/(1 - 0.88) = 8.3333$; SE inflation $= \sqrt{VIF} = 2.8868$.

Answer: **2.89**.

Q.10 — Answer: 60 [NAT, 1M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

QDA: $K \cdot p(p+1)/2 = 75$; LDA: $p(p+1)/2 = 15$.

Difference = $(K - 1)p(p+1)/2 = 60$.

This extra variance is why LDA often beats QDA when n is small.

Q.11 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Backpropagation]

(A) TRUE. $[L(w+\varepsilon) - L(w-\varepsilon)]/2\varepsilon \approx \partial L/\partial w$.

(B) TRUE. The σ' factor cancels with the log-loss derivative.

(C) TRUE. a_j depends on b_j with coefficient 1.

(D) TRUE. Chain rule through σ .

Q.12 — Answer: 1.79 to 1.81 [NAT, 1M · SVM & Decision Trees > SVM: dual & support vectors]

Stationarity w.r.t. b gives the dual constraint $\sum_i \alpha_i y_i = 0$.

$\sum_{i \neq 1} \alpha_i y_i = +0.1 - 1.4 - 0.5 = -1.8$.

So $\alpha_1(+1) = 1.8 \Rightarrow \alpha_1 = 1.8$ (it must be ≥ 0).

Q.13 — Answer: (C) [MCQ, 1M · Unsupervised Learning > k-means]

Each group is connected by chains of small gaps, while the gap between groups is larger. Single linkage merges via nearest neighbours (MST edges) and therefore follows these chains: cutting the largest MST edge separates the two concentric rings.

k-means and k-medoids produce convex (Voronoi) clusters around representatives, so they split the shapes by a straight boundary. Complete linkage favours compact, small-diameter clusters and also cuts across the non-convex shapes.

Q.14 — Answer: (A), (B), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

(A) TRUE. Only the ranking matters.

(B) TRUE. True negative rate.

(C) TRUE. Precision = $\text{TPR} \cdot \pi / (\text{TPR} \cdot \pi + \text{FPR}(1 - \pi))$.

(D) TRUE. FPR stays small when negatives are abundant, hiding many false positives.

Q.15 — Answer: (C) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

Key features of the plotted curve: S-shaped, range (0, 1), value 0.5 at $z = 0$.

Hence $g = \text{Sigmoid } \sigma(z) = 1/(1+e^{-z})$.

$\sigma'(z) = \sigma(z)(1 - \sigma(z))$: Bell-shaped with peak 0.25 at $z = 0$. – does not match.

$\tanh(z)$: S-shaped, range $(-1, 1)$, passes through the origin. – does not match.

Softplus: $\ln(1 + e^z)$: Smooth, always positive, value $\ln 2 \approx 0.69$ at $z = 0$, $\sim z$ for large z . – does not match.

Q.16 — Answer: (B) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Target AND truth table for $(x_1, x_2) = (0,0), (0,1), (1,0), (1,1)$: 0, 0, 0, 1. For each option compute $a = w_1 x_1 + w_2 x_2 + b$ at the four inputs and output 1 iff $a > 0$.

(A) FALSE. $a = -3, -3, -1, -1 \rightarrow$ outputs 0, 0, 0, 0 – constant 0.

(B) TRUE. $a = -2.5, -0.5, -0.5, 1.5 \rightarrow$ outputs 0, 0, 0, 1 – AND.

(C) FALSE. $a = 0.5, -1, 2.5, 1 \rightarrow$ outputs 1, 0, 1, 1 – x_1 OR (NOT x_2).

(D) FALSE. $a = 1, 4, 2.5, 5.5 \rightarrow$ outputs 1, 1, 1, 1 – constant 1.

Q.17 — Answer: 3.1621 to 3.2021 [NAT, 2M · Regression & Model Selection > MLE & MAP view]

$\bar{x} = 4.8, \bar{y} = 18.8$.

i	1	2	3	4	5
$x_i - \bar{x}$	-3.8	-2.8	0.2	2.2	4.2
$y_i - \bar{y}$	-5.8	-0.8	0.2	4.2	2.2

$(x_i - \bar{x})(y_i - \bar{y})$	22.04	2.24	0.04	9.24	9.24
----------------------------------	-------	------	------	------	------

$$S_{xx} = \sum (x_i - \bar{x})^2 = 44.8, S_{xy} = 42.8.$$

$$\beta_1 = S_{xy}/S_{xx} = 0.9554, \beta_0 = \bar{y} - \beta_1 \bar{x} = 14.2143.$$

$$SSE = S_{yy} - S_{xy}^2/S_{xx} = 56.8 - 40.8893 = 15.9107.$$

Maximising the Gaussian likelihood over σ^2 gives $\hat{\sigma}_{ML}^2 = SSE/n = 15.9107/5 = 3.1821$.

Answer: **3.18**.

Q.18 — Answer: 6.99 to 7.01 [NAT, 2M · Unsupervised Learning · Hierarchical clustering]

Use complete linkage: $d(A, B) = \max_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: {P3} + {P5} at height **2**. Updated: $d(\{P3, P5\}, \{P1\}) = 15$, $d(\{P3, P5\}, \{P2\}) = 9$, $d(\{P3, P5\}, \{P4\}) = 12$.

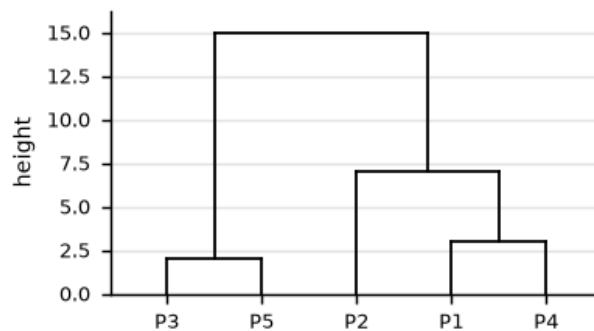
Merge 2: {P1} + {P4} at height **3**. Updated: $d(\{P1, P4\}, \{P2\}) = 7$, $d(\{P1, P4\}, \{P3, P5\}) = 15$.

Merge 3: {P2} + {P1, P4} at height **7**. Updated: $d(\{P1, P2, P4\}, \{P3, P5\}) = 15$.

Merge 4: {P3, P5} + {P1, P2, P4} at height **15**.

The third merge occurs at height **7**.

Dendrogram:



Q.19 — Answer: 0.7926 to 0.8126 [NAT, 2M · Probabilistic & Instance-based Classifiers · Linear discriminant analysis]

With equal variances, $\ln[\pi_1 p(x|C_1)/(\pi_0 p(x|C_0))] = (\mu_1 - \mu_0)x/\sigma^2 - (\mu_1^2 - \mu_0^2)/(2\sigma^2) + \ln(\pi_1/\pi_0)$ — linear in x (LDA).

Slope $a = (\mu_1 - \mu_0)/\sigma^2 = 1.5$; intercept $c = -(\mu_1^2 - \mu_0^2)/(2\sigma^2) + \ln(\pi_1/\pi_0) = -3.0973$.

$P(C_1|x) = \sigma(a x + c) = \sigma(1.4027) = \mathbf{0.8}$.

Q.20 — Answer: (D) [MCQ, 2M · Unsupervised Learning · Distance measures]

Compute the measure from $q = (1, 2)$ to every point (cosine column shows similarity, larger = more similar):

Point	coordinates	Manhattan	Euclidean	Chebyshev	cosine similarity
A	(0, 9)	8	7.071	7	0.894
B	(7, 5)	9	6.708	6	0.884
C	(9, 8)	14	10	8	0.928
D	(5, 8)	10	7.211	6	0.995

Under cosine similarity the most similar to q is **point D**.

Note how the answer depends on the measure: Manhattan (L_1) distance $\rightarrow$ A; Euclidean (L_2) distance $\rightarrow$ B; Chebyshev (L_∞) distance $\rightarrow$ B; cosine similarity $\rightarrow$ D. Cosine similarity looks only at the angle from the origin, L_1 sums coordinate gaps, L_∞ takes the largest gap.

Q.21 — Answer: 0.0931 to 0.0992 [NAT, 2M · Regression & Model Selection · Ridge & Lasso]

$\beta_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y$ with $\lambda = 3$.

$$X^T X = \begin{bmatrix} 8 & 6 \\ 6 & 5 \end{bmatrix}$$

$$X^T X + \lambda I = \begin{bmatrix} 11 & 6 \\ 6 & 8 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 6 & 8 \\ -10 & -5 \end{bmatrix}$$

$$\det = 11 \times 8 - (6)^2 = 52; (X^T X + \lambda I)^{-1} = (1/52)[[8, -6], [-6, 11]].$$

$$\beta = (-0.9615, 0.0962).$$

Answer: **0.096**.

Q.22 — Answer: -0.019 to -0.015 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

For the average cross-entropy, $\partial L / \partial w_j = (1/n) \sum (\hat{p}_i - y_i) x_{ij}$ and $\partial L / \partial b = (1/n) \sum (\hat{p}_i - y_i)$, with $\hat{p}_i = \sigma(z_i)$.

i	z_i	$\hat{p}_i$	$\hat{p}_i - y_i$
1	0	0.5	-0.5
2	-0.5	0.3775	0.3775
3	-0.5	0.3775	0.3775

$$\text{Gradient: } \partial L / \partial w_1 = 0.2517, \partial L / \partial w_2 = 0.5034, \partial L / \partial b = 0.085.$$

$$\text{Update } \theta \leftarrow \theta - \eta \nabla L: w_1 = 0.5 - 0.2 \times (0.2517) = 0.4497, w_2 = -0.5 - 0.2 \times (0.5034) = -0.6007, b = 0 - 0.2 \times (0.085) = -0.017.$$

Required: $b = -0.017$.

Q.23 — Answer: (B), (C) [MSQ, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

TP = 30, FN = 58, FP = 64, TN = 43.

(A) FALSE. $P = 0.3191$, $R = 0.3409$.

(B) TRUE. specificity (TNR) = $TN / (TN + FP) = 0.4019$.

(C) TRUE. F1-score = $2PR / (P + R) = 0.3297$.

(D) FALSE. false positive rate = $FP / (FP + TN) = 0.5981$, not 0.68.

Q.24 — Answer: (B), (D) [MSQ, 2M · SVM & Decision Trees > SVM: soft margin & hinge loss]

$$\xi_i = \max(0, 1 - y_i f(x_i)).$$

Point	$y f(x)$	ξ
P1	-2	3
P2	-2.5	3.5
P3	3	0
P4	1.5	0
P5	-0.5	1.5
P6	0.5	0.5

(A) FALSE. $y f = 3 \Rightarrow \xi = 0$.

(B) TRUE. $y f = 0.5 \Rightarrow \xi = 0.5$.

(C) FALSE. $y f = -0.5 \Rightarrow \xi = 1.5$.

(D) TRUE. Count of points with $y f < 1$.

Q.25 — Answer: 0.5477 to 0.5677 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Sort the values: 37(N), 44(N), 47(N), 65(Y), 66(Y), 72(N), 80(Y), 83(Y), 98(Y).

Parent [5 Yes, 4 No], $H = 0.9911$. The optimal threshold always lies where the class changes between neighbours (Fayyad-Irani), giving 3 candidates:

Threshold	Left ($\leq t$) [Yes, No]	Right [Yes, No]	IG
56	[0, 3]	[5, 1]	0.5577
69	[2, 3]	[3, 1]	0.0911
76	[2, 4]	[3, 0]	0.3789

Best: $t = 56$ with $IG = 0.5577$. Answer: **0.56**.

Q.26 — Answer: (B), (C), (D) [MSQ, 2M · SVM & Decision Trees > SVM: dual & support vectors]

P3 and P6 (same class) lie on one margin line and P4 on the other; the optimal boundary is parallel to the line through P3, P6, midway to P4. Support vectors: P3, P4, P6.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (0, 1)$, $b = -2$; $\|w\| = 1$, margin $2/\|w\| = 2$.

Point	P1	P2	P3	P4	P5	P6	P7
$y(w^T x + b)$	3	2	1	1	2	1	2

Support vectors are exactly the points with $y_i(w^T x_i + b) = 1$ ($\alpha_i > 0$): **P3, P4, P6**.

Dual check: $\alpha = (\alpha_3 = 1/3, \alpha_4 = 1/2, \alpha_6 = 1/6)$ satisfy $\sum \alpha_i y_i = 0$ and $\sum \alpha_i y_i x_i = w$.

Q.27 — Answer: 0.5348 to 0.5408 [NAT, 2M · Regression & Model Selection > Bias-variance trade-off]

Mean prediction $\bar{f} = (1/6)\sum f_m = 12.7333$.

Bias² = $(\bar{f} - f(x_0))^2 = (12.7333 - 12)^2 = 0.5378$.

Variance = $(1/6)\sum (f_m - \bar{f})^2 = 0.4056$.

Expected test error = Bias² + Variance + $\sigma^2 = 0.5378 + 0.4056 + 1.5 = 2.4433$.

Answer: **0.538**.

Q.28 — Answer: (B) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$p = \text{softmax}((-1, 0, 2)) = (0.042, 0.1142, 0.8438)$; $y = (0, 0, 1)$.

$\partial L / \partial z = p - y = \mathbf{(0.04, 0.11, -0.16)}$.

Distractors: $y - p = (-0.04, -0.11, 0.16)$ (wrong sign); p alone = $(0.04, 0.11, 0.84)$ (forgot the target); $(p - y) \odot p \odot (1 - p) = (0, 0.01, -0.02)$ (spurious σ' -type factor, which cancels for cross-entropy); $-y/p = (0, 0, -1.19)$ is $\partial L / \partial p$, not $\partial L / \partial z$.

Q.29 — Answer: 0.3233 to 0.3433 [NAT, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Classify each point by its nearest neighbour(s) among the remaining points:

x	true	3 nearest (excluding itself)	LOO prediction	
2	0	4(0), 10(1), 14(1)	1	wrong
4	0	2(0), 10(1), 14(1)	1	wrong
10	1	14(1), 4(0), 16(1)	1	
14	1	16(1), 17(0), 10(1)	1	
16	1	17(0), 14(1), 10(1)	1	
17	0	16(1), 14(1), 10(1)	1	wrong
25	1	26(1), 30(1), 17(0)	1	
26	1	25(1), 30(1), 17(0)	1	
30	1	26(1), 25(1), 17(0)	1	

Misclassified: 3 of 9. Error rate = $3/9 = \mathbf{0.33}$.

Note: the *training* error of 1-NN is 0, but the LOOCV error is not — a point cannot be its own neighbour.

Q.30 — Answer: (A), (B), (C), (D) [MSQ, 2M · Unsupervised Learning > PCA & dimensionality reduction]

(A) TRUE. The matrix is singular.

(B) TRUE. All eigenvalues scale by c^2 .

(C) TRUE. $S \rightarrow RSR^T$ is a similarity transform.

(D) TRUE. $S = X^T X / (n - 1) = V \Sigma^2 V^T / (n - 1)$.

Q.31 — Answer: 7.19 to 7.21 [NAT, 2M · Unsupervised Learning > k-means]

Iteration 1 - assignment (nearest initial centroid, squared Euclidean distance):

Point	$d^2(\cdot, C1)$	$d^2(\cdot, C2)$	cluster
P1 (1, 5)	90	26	C2
P2 (4, 10)	100	4	C2

P3 (0, 7)	125	13	C2
P4 (9, 6)	17	65	C1
P5 (1, 4)	85	37	C2
P6 (1, 10)	145	1	C2
P7 (6, 1)	17	97	C1

Update: each centroid becomes the mean of its assigned points: $C1 \rightarrow (7.5, 3.5)$; $C2 \rightarrow (1.4, 7.2)$.

Iteration 2 - assignment with the updated centroids:

Point	$d^2(\cdot, C1)$	$d^2(\cdot, C2)$	cluster
P1 (1, 5)	44.5	5	C2
P2 (4, 10)	54.5	14.6	C2
P3 (0, 7)	68.5	2	C2
P4 (9, 6)	8.5	59.2	C1
P5 (1, 4)	42.5	10.4	C2
P6 (1, 10)	84.5	8	C2
P7 (6, 1)	8.5	59.6	C1

Update: $C1 \rightarrow (7.5, 3.5)$; $C2 \rightarrow (1.4, 7.2)$. Required coordinate = **7.2**.

Note: each step of Lloyd's algorithm can only decrease (or keep) J , which is why it converges.

Q.32 — Answer: 0.623 to 0.627 [NAT, 2M · Regression & Model Selection > Gradient descent]

Hessian of J : $\nabla^2 J = (2/n)X^T X$. Eigenvalues of $X^T X$: $9 \pm 7 = 16, 2$ (eigenvectors $(1, 1), (1, -1)$).

So the Hessian eigenvalues are $L = 3.2$ and $\mu = 0.4$.

GD error evolves as $w_{t+1} - w^* = (I - \eta \nabla^2 J)(w_t - w^*)$, so it converges iff $|1 - \eta \lambda| < 1$ for every Hessian eigenvalue λ .

Need $\eta < 2/L = 2/3.2 = 0.625$.

Answer: **0.625**.

Q.33 — Answer: 13 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Rule: keep track of the best (lowest) validation loss; an epoch that does not strictly improve on it increments a counter, which is reset on improvement; stop when the counter reaches patience $P = 3$.

Trace – 1: 1.294 (new best); 2: 1.124 (new best); 3: 0.889 (new best); 4: 0.777 (new best); 5: 0.710 (new best); 6: 0.616 (new best); 7: 0.560 (new best); 8: 0.545 (new best); 9: 0.552 (no improvement, count = 1); 10: 0.528 (new best); 11: 0.567 (no improvement, count = 1); 12: 0.586 (no improvement, count = 2); 13: 0.620 (no improvement, count = 3).

Training stops after epoch 13; the best validation loss 0.528 occurred at epoch 10, whose weights are restored.

Answer: **13**. The widening train/validation gap after this point is the signature of overfitting.

Q.34 — Answer: 0.0657 to 0.067 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Activation functions]

Forward: $a_1 = 2 \cdot 1 + -0.5 = 1.5$, $h_1 = 0.8176$; $a_2 = -3 \cdot 0.8176 + 1 = -1.4527$, $h_2 = 0.1896$; $a_3 = -2 \cdot 0.1896 + 0 = -0.3792$, $y = 0.4063$.

Local derivatives $\sigma'(a)$: $\sigma'(a) = \sigma(a)(1 - \sigma(a))$: $\sigma'(a_1) = 0.1491$, $\sigma'(a_2) = 0.1536$, $\sigma'(a_3) = 0.2412$.

Chain rule: $\partial y / \partial x = \sigma'(a_3) \cdot w_3 \cdot \sigma'(a_2) \cdot w_2 \cdot \sigma'(a_1) \cdot w_1 = \mathbf{0.0663}$.

Each layer contributes a factor $\sigma' \leq 0.25$ – the product shrinks with depth (vanishing gradient).

Q.35 — Answer: 77.728 to 77.828 [NAT, 2M · SVM & Decision Trees > Decision trees: structure & traversal]

Route each record from the root to a leaf:

#	Predicted	Actual	Correct?
1	Yes	Yes	yes
2	Yes	No	no
3	No	No	yes

4	No	No	yes
5	Yes	Yes	yes
6	Yes	No	no
7	Yes	Yes	yes
8	Yes	Yes	yes
9	Yes	Yes	yes

Misclassified: 2 of 9 (accuracy 77.78%). Answer: **77.8**.

Mock Test 32

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	4	10	21

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Neural Networks (MLP / Feed-forward)	7	11
Probabilistic & Instance-based Classifiers	7	11
Regression & Model Selection	7	11
Unsupervised Learning	7	11
SVM & Decision Trees	7	11

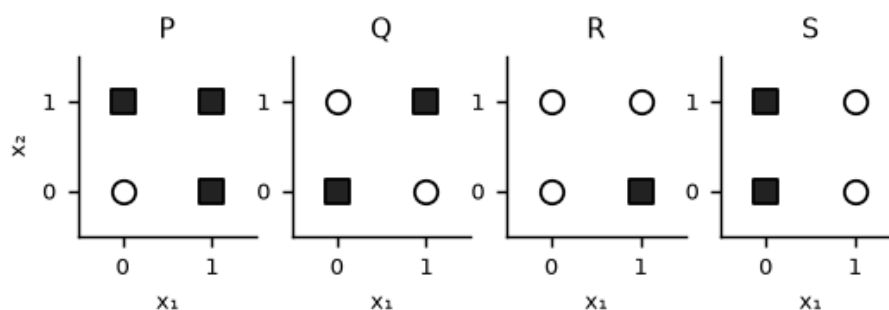
Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MCQ]

Each panel shows the four binary inputs (x_1, x_2) labelled as class 1 (filled square) or class 0 (open circle). Which dataset can NOT be classified perfectly by a single perceptron (one linear threshold unit)?



(A) Dataset P

(B) Dataset Q

(C) Dataset R

(D) Dataset S

Q.2 [1 mark · NAT (numerical answer)]

A naive Bayes classifier has $d = 4$ binary features and $K = 2$ classes. The number of independent parameters (class priors plus class-conditional feature probabilities) to be estimated is _____ (Answer in integer)

Q.3 [1 mark · MSQ (one or more correct)]

Linear regression is trained by minimising the squared-error loss. Which of the following statements about gradient-based training is/are CORRECT?

- (A) The MSE loss of linear regression can have local minima that are not global minima.
- (B) Gradient descent on the MSE loss converges for every learning rate $\eta > 0$.
- (C) Stochastic gradient descent updates the parameters using the gradient on one (or a few) randomly chosen example(s).
- (D) With a constant learning rate, SGD iterates typically keep fluctuating in a neighbourhood of the minimiser.

Q.4 [1 mark · NAT (numerical answer)]

The value of the softplus function $\ln(1 + e^z)$ at $z = -2$ is _____ (rounded off to three decimal places)

Q.5 [1 mark · NAT (numerical answer)]

For $\mathbf{x} = (-1, 7, 6)$ and $\mathbf{y} = (6, 4, -1)$, the cosine similarity between $\mathbf{x}$ and $\mathbf{y}$ is _____ (rounded off to two decimal places)

Q.6 [1 mark · NAT (numerical answer)]

How many distinct monomials of total degree **exactly** 2 can be formed from 3 variables (e.g. for $p = 2, d = 2: x_1^2, x_1x_2, x_2^2$)? _____ (Answer in integer)

Q.7 [1 mark · NAT (numerical answer)]

A binary logistic regression model is given by $P(y = 1 | \mathbf{x}) = \sigma(w_0 + w_1x_1 + w_2x_2)$ with $w_0 = -1.5, w_1 = 1, w_2 = -0.5$, where $\sigma(z) = 1/(1 + e^{-z})$. For the input $\mathbf{x} = (x_1, x_2) = (0, 2)$, the value of $P(y = 1 | \mathbf{x})$ is _____ (rounded off to two decimal places)

Q.8 [1 mark · NAT (numerical answer)]

A node of a classification tree contains 26 training records: 12 of class C_1 , 3 of class C_2 , 11 of class C_3 . The Gini index of this node is _____ (rounded off to two decimal places)

Q.9 [1 mark · MSQ (one or more correct)]

Which of the following statements about decision-tree learning is/are CORRECT?

- (A) A tree with axis-parallel splits predicts a constant within each axis-aligned rectangular region.
- (B) A decision tree is a linear classifier.
- (C) Pruning a subtree can increase the training error of the tree.
- (D) Cost-complexity pruning minimises $R(T) + \alpha|T|$, where $|T|$ is the number of leaves.

Q.10 [1 mark · MCQ]

For a fitted linear regression model $\hat{y} = \mathbf{w}^T \mathbf{x} + b$, the output for a new input $\mathbf{x}$ is:

- (A) a probability in $[0, 1]$
- (B) a real number
- (C) a vector of class probabilities summing to 1
- (D) a class label in $\{0, 1\}$

Q.11 [1 mark · NAT (numerical answer)]

The 4 two-dimensional observations below have zero mean in each feature. Then the total variance (trace of the sample covariance matrix, divisor $n - 1$) is _____ (rounded off to two decimal places)

Point	x1	x2	x3	x4
x1	4	4	-4	-4
x2	-1	4	-1	-2

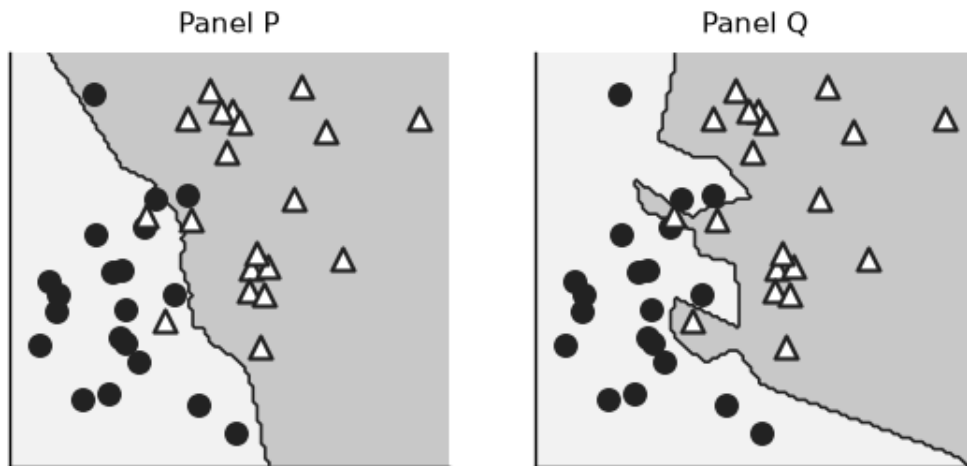
Q.12 [1 mark · MSQ (one or more correct)]

Which of the following statements about k-means clustering is/are CORRECT?

- (A) Each iteration of Lloyd's k-means algorithm cannot increase the within-cluster sum of squares.
- (B) k-means works best when clusters are roughly spherical and of similar size.
- (C) Finding the globally optimal k-means partition is NP-hard in general.
- (D) In the update step, a cluster's new centroid is the arithmetic mean of the points assigned to it.

Q.13 [1 mark · MCQ]

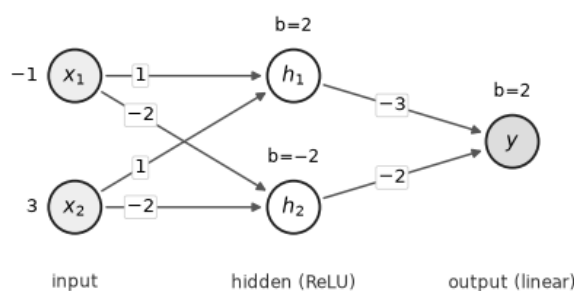
The two panels show the decision regions of k-NN classifiers trained on the same 2-D data set, one with $k = 1$ and the other with $k = 15$. Which of the following statements is CORRECT?



- (A) Panel Q uses $k = 1$; it has lower bias and higher variance
- (B) Panel P uses $k = 15$; it has zero training error
- (C) Panel Q uses $k = 1$; it has higher bias and lower variance
- (D) Panel P uses $k = 1$; it has lower bias and higher variance

Q.14 [1 mark · NAT (numerical answer)]

For the ReLU network shown (edge labels are weights, 'b' denotes a bias) and input $x = (-1, 3)$, the output y is ____ (Answer in integer)

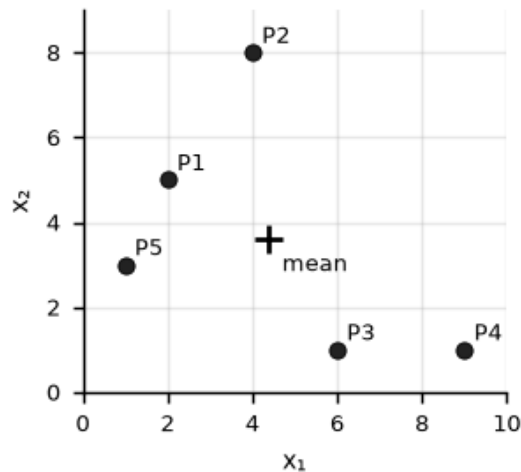
**Q.15 [1 mark · NAT (numerical answer)]**

A binary decision tree must have at least 90 leaves to separate 90 distinct classes. The minimum possible depth of such a tree is ____ (Answer in integer)

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

PCA is applied to the five 2-D points below (data are mean-centered first; sample covariance uses divisor $n - 1$). Taking the unit-length first principal direction with a **positive first component**, the score (projection) of P1 on PC1 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5
x_1	2	4	6	9	1
x_2	5	8	1	1	3

**Q.17 [2 marks · NAT (numerical answer)]**

Consider the model $y = wx + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$, $\sigma^2 = 1$, and $w \sim N(0, \tau^2)$ with $\tau^2 = 0.5$. Given the four observations below, the MAP estimate of w is _____ (rounded off to three decimal places)

x	3	-2	4	4
y	4	2	1	-1

Q.18 [2 marks · MCQ]

Ridge regression is tuned by 10-fold cross-validation. For each λ the table gives the mean CV MSE and its standard error (SE) across folds. (Larger λ means a simpler, more regularised model.) Using the **one-standard-error rule** (choose the most regularised model whose CV error is within one standard error of the minimum CV error), the selected value is

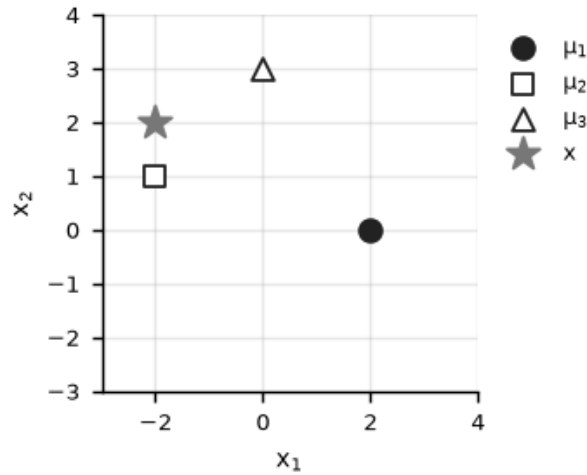
λ	0.01	0.1	0.3	1	3	10
CV MSE	5.32	4.93	5.27	5.44	7.06	7.98
SE	0.33	0.46	0.37	0.41	0.31	0.28

(A) $\lambda = 0.3$ **(B)** $\lambda = 0.01$ **(C)** $\lambda = 3$ **(D)** $\lambda = 0.1$

Q.19 [2 marks · NAT (numerical answer)]

An LDA classifier for 3 classes has shared covariance matrix Σ (below), class means $\mu_1 = (2, 0)^T$; $\mu_2 = (-2, 1)^T$; $\mu_3 = (0, 3)^T$, and priors $\pi_1 = 0.6$, $\pi_2 = 0.2$, $\pi_3 = 0.2$. The linear discriminant is $\delta_k(\mathbf{x}) = \mathbf{x}^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln \pi_k$ (natural log). For $\mathbf{x} = (-2, 2)^T$, the index $k \in \{1, 2, 3\}$ of the class predicted by LDA is _____. (Answer in integer)

$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

**Q.20 [2 marks · MSQ (one or more correct)]**

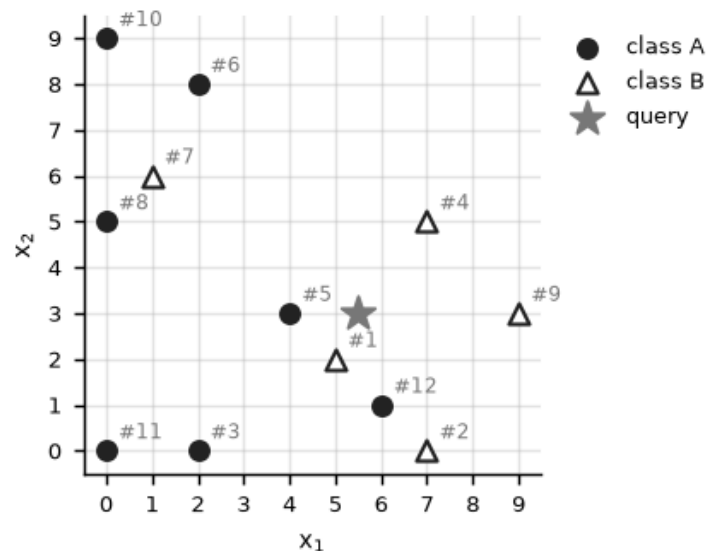
Let σ be the logistic sigmoid, and $'$ denote the derivative. At $z = 0.5$ (and $-z = -0.5$), which of the following is/are CORRECT? ($\text{softplus}(z) = \ln(1 + e^z)$.)

- (A) $|\tanh(0.5)| > |0.5|$ (B) $\sigma(0.5) + \sigma(-0.5) = 1$
 (C) $\tanh(0.5) = 2\sigma(1) - 1$ (D) $\sigma'(0.5) = \sigma(0.5)^2 = 0.387$

Q.21 [2 marks · MSQ (one or more correct)]

The figure and table show 12 labelled training points (class A: filled circles, class B: open triangles) and a query point (star) at (5.5, 3). A k-NN classifier with **Euclidean** distance and majority vote is used. Which of the following statements is/are CORRECT?

#	1	2	3	4	5	6	7	8	9	10	11	12
x_1	5	7	2	7	4	2	1	0	9	0	0	6
x_2	2	0	0	5	3	8	6	5	3	9	0	1
class	B	B	A	B	A	A	B	A	B	A	A	A



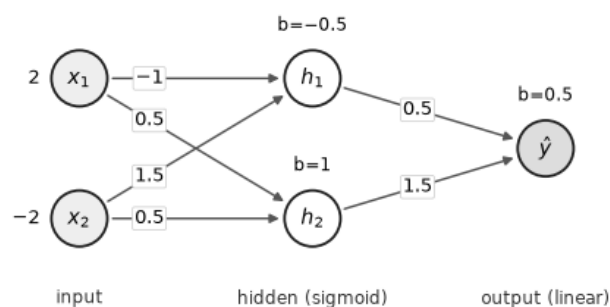
- (A) With $k = 1$, the query is classified as class A.
- (B) With $k = 5$, the query is classified as class A.
- (C) With $k = 7$, the query is classified as class A.
- (D) With $k = 3$, the query is classified as class A.

Q.22 [2 marks · NAT (numerical answer)]

A logistic regression classifier has $P(y = 1 | \mathbf{x}) = \sigma(2 + (2)x_1 + (3)x_2)$. An input is labelled $y = 1$ if and only if $P(y = 1 | \mathbf{x}) \geq 0.9$. For this rule, the perpendicular distance of the decision boundary from the origin is _____ (rounded off to two decimal places)

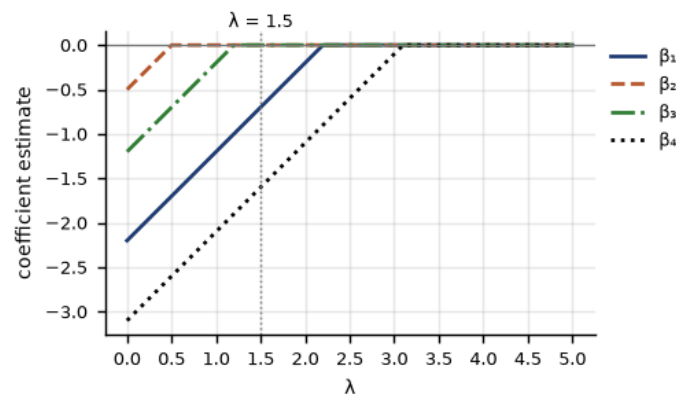
Q.23 [2 marks · NAT (numerical answer)]

The feed-forward network below has sigmoid hidden units and a linear output unit. The number on each edge is its weight and 'b' above a node is that node's bias. For input $\mathbf{x} = (2, -2)$ (values shown left of the inputs), the activation of hidden unit h_2 is _____ (rounded off to three decimal places)



Q.24 [2 marks · MSQ (one or more correct)]

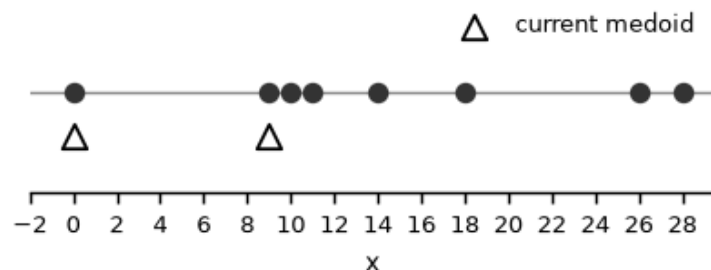
For a design with orthonormal columns ($X^T X = I$), the figure shows the coefficient paths $\beta_i(\lambda)$ of a penalised least-squares estimator that minimises $\frac{1}{2}\|y - X\beta\|^2 + P_\lambda(\beta)$, where the penalty P_λ is either the ridge penalty $(\lambda/2)\|\beta\|_2^2$ or the lasso penalty $\lambda\|\beta\|_1$. Which of the following statements is/are CORRECT?



- (A) At $\lambda = 0$ all coefficients equal zero.
- (B) No coefficient becomes exactly zero for any finite λ .
- (C) At $\lambda = 0$ the coefficients equal the least-squares estimates.
- (D) As λ increases, the training residual sum of squares decreases.

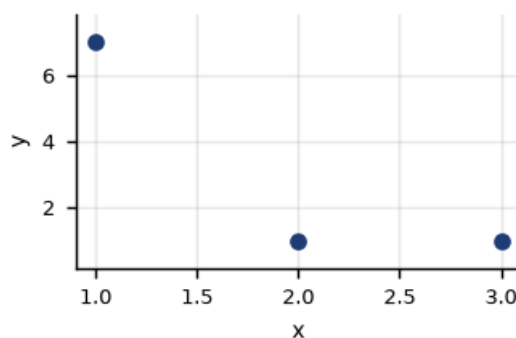
Q.25 [2 marks · NAT (numerical answer)]

PAM (k-medoids, $k = 2$, absolute-difference dissimilarity) is applied to the 1-D data $\{0, 9, 10, 11, 14, 18, 26, 28\}$. The current medoids are 0 and 9. The change in total cost, (cost after swap) – (cost before swap), when medoid 0 is swapped with non-medoid 10 is _____ (Answer in integer)

**Q.26 [2 marks · NAT (numerical answer)]**

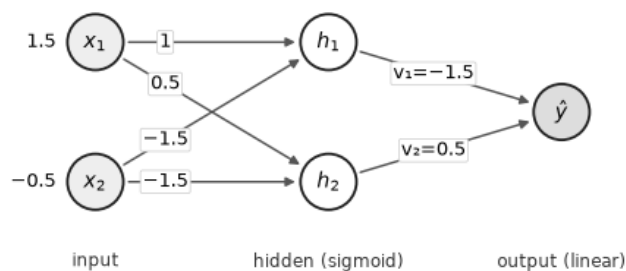
A linear model $\hat{y} = w_0 + w_1 x$ is trained on the 3 points below by batch gradient descent on the loss $J(w) = (1/2n)\sum (w_0 + w_1 x_i - y_i)^2$, starting from $(w_0, w_1) = (0, 1)$ with learning rate $\eta = 0.01$. Then the value of w_0 after one iteration is _____ (rounded off to three decimal places)

x	1	2	3
y	7	1	1



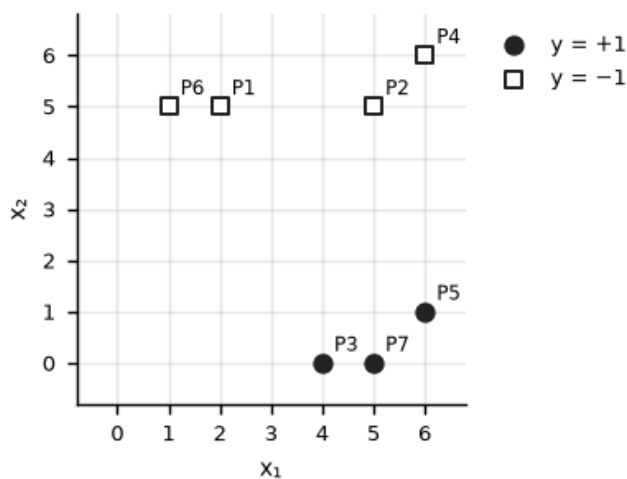
Q.27 [2 marks • NAT (numerical answer)]

In the network below, hidden units are sigmoid, the output is linear ($\hat{y} = v_1 h_1 + v_2 h_2$), all biases are 0, and the loss is $L = \frac{1}{2}(\hat{y} - y)^2$. For input $x = (1.5, -0.5)$ and target $y = 0.5$, the value of $\partial L / \partial b_1$ (bias of hidden unit h_1 , currently 0) is _____ (rounded off to three decimal places)

**Q.28 [2 marks • NAT (numerical answer)]**

The labelled training points shown below are linearly separable. A hard-margin linear SVM is trained on them. Compute the width of the margin (distance between the two margin hyperplanes). (rounded off to two decimal places)

Point	x_1	x_2	y
P1	2	5	-1
P2	5	5	-1
P3	4	0	+1
P4	6	6	-1
P5	6	1	+1
P6	1	5	-1
P7	5	0	+1

**Q.29 [2 marks • MSQ (one or more correct)]**

Each option shows a 2×2 matrix $[[K(x_1, x_1), K(x_1, x_2)], [K(x_2, x_1), K(x_2, x_2)]]$. Which of them can be the Gram matrix of some valid (Mercer) kernel $\tilde{K}$ on two points x_1, x_2 ?

- (A) $[[-4, 0], [0, 7]]$
 (B) $[[8, 0], [2, 7]]$
 (C) $[[1, -1], [-1, 6]]$
 (D) $[[7, 5], [5, 8]]$

Q.30 [2 marks · NAT (numerical answer)]

In C4.5, the attribute Branch splits a node of 29 records as tabulated (class Yes/No counts in each branch). Compute the gain ratio of Branch (all logarithms base 2). (rounded off to two decimal places)

Branch	#Yes	#No
v1	6	1
v2	7	0
v3	0	6
v4	6	3

Q.31 [2 marks · NAT (numerical answer)]

Agglomerative clustering with **complete linkage** is applied to the distance matrix below. The cophenetic distance between P4 and P5 (the dendrogram height at which they are first joined) is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5	P6
P1	0	1	2	4	7	10
P2		0	3	5	8	11
P3			0	2	5	8
P4				0	7	6
P5					0	9
P6						0

Q.32 [2 marks · MSQ (one or more correct)]

Consider k-means (Lloyd's algorithm) and k-medoids (PAM). Which of the following statements is/are CORRECT?

- (A) For 1-D data and a single cluster, the k-medoids representative under absolute distance is a median data point.
- (B) For a 1-D cluster with an odd number of points, the medoid under absolute distance is the mean.
- (C) k-medoids (PAM) can be run using only a pairwise dissimilarity matrix, without coordinates.
- (D) If the centroids do not change between two successive iterations, the assignments will not change in any later iteration either.

Q.33 [2 marks · MSQ (one or more correct)]

A node with class counts [40+, 50−] can be split by either of two binary splits S1 or S2 (table). For each impurity measure, the “gain” is the parent impurity minus the weighted child impurity. Which of the following statements is/are CORRECT?

Split	Left child [+ , −]	Right child [+ , −]
S1	[15+, 31−]	[25+, 19−]
S2	[10+, 44−]	[30+, 6−]

- (A) Information gain (entropy) prefers S2 over S1.
- (B) The information gain of S2 is 0.32 bits (to two decimals).
- (C) Misclassification-error reduction prefers S2 over S1.
- (D) Gini gain prefers S2 over S1.

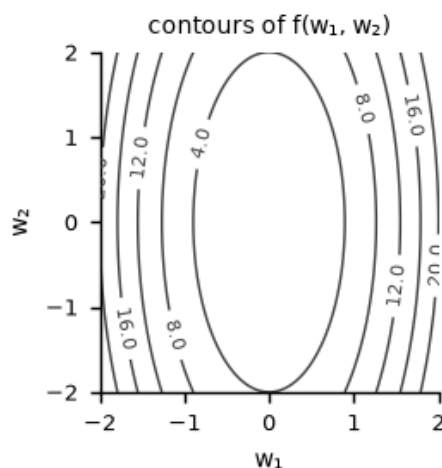
Q.34 [2 marks · NAT (numerical answer)]

The actual and predicted labels of a binary classifier (1 = positive) on 15 test examples are given below. The F_β -score with $\beta = 0.5$, $F_\beta = (1 + \beta^2)PR/(\beta^2P + R)$, is _____ (rounded off to two decimal places)

#	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
actual y	0	0	0	0	0	0	1	0	0	0	1	1	1	1	1
predicted $\hat{y}$	1	0	1	1	0	1	1	0	0	0	0	0	1	0	1

Q.35 [2 marks · MSQ (one or more correct)]

Gradient descent $w \leftarrow w - \eta \nabla f(w)$ is applied to $f(w_1, w_2) = \frac{1}{2}(10w_1^2 + 2w_2^2)$ from a generic starting point (both coordinates non-zero). For which of the following learning rates does GD **diverge**?



- (A) $\eta = 0.22$
(C) $\eta = 0.05$

- (B) $\eta = 0.3$
(D) $\eta = 0.03$

— END OF QUESTION PAPER —

Mock Test 32 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MCQ	1	(B)	19	NAT	2	2
2	NAT	1	9	20	MSQ	2	(B), (C)
3	MSQ	1	(C), (D)	21	MSQ	2	(D)
4	NAT	1	0.1249 to 0.1289	22	NAT	2	0.0447 to 0.0647
5	NAT	1	0.227 to 0.247	23	NAT	2	0.7291 to 0.7331
6	NAT	1	6	24	MSQ	2	(C)
7	NAT	1	0.0659 to 0.0859	25	NAT	2	3
8	NAT	1	0.5847 to 0.6047	26	NAT	2	0.008 to 0.012
9	MSQ	1	(A), (C), (D)	27	NAT	2	0.1854 to 0.1894
10	MCQ	1	(B)	28	NAT	2	4.1131 to 4.1331
11	NAT	1	28.6567 to 28.6767	29	MSQ	2	(C), (D)
12	MSQ	1	(A), (B), (C), (D)	30	NAT	2	0.2428 to 0.2628
13	MCQ	1	(A)	31	NAT	2	7.99 to 8.01
14	NAT	1	-10	32	MSQ	2	(A), (C), (D)
15	NAT	1	7	33	MSQ	2	(A), (B), (C), (D)
16	NAT	2	-2.7437 to -2.7237	34	NAT	2	0.4312 to 0.4512
17	NAT	2	0.1682 to 0.1722	35	MSQ	2	(A), (B)
18	MCQ	2	(A)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Neural Networks (MLP / Feed-forward)					11
Probabilistic & Instance-based Classifiers					11
Regression & Model Selection					11
Unsupervised Learning					11
SVM & Decision Trees					11
Total					55

Mock Test 32 — Detailed Solutions

Q.1 — Answer: (B) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

A single perceptron can classify a dataset perfectly iff the two classes are linearly separable.

Dataset P: labels for (0,0),(0,1),(1,0),(1,1) = 0, 1, 1, 1 → OR – linearly separable.

Dataset Q: labels for (0,0),(0,1),(1,0),(1,1) = 1, 0, 0, 1 → XNOR – **not** linearly separable (diagonal pattern).

Dataset R: labels for (0,0),(0,1),(1,0),(1,1) = 0, 0, 1, 0 → x_1 AND (NOT x_2) – linearly separable.

Dataset S: labels for (0,0),(0,1),(1,0),(1,1) = 1, 1, 0, 0 → NOT x_1 – linearly separable.

Answer: **Dataset Q**. Of the 14 non-constant Boolean functions of two inputs, only XOR and XNOR are not linearly separable.

Q.2 — Answer: 9 [NAT, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Priors: $K - 1 = 1$. Each binary feature needs 1 parameter $P(x_j=1|c)$ per class: $Kd = 8$.

Total = $8 + 1 = 9$.

Q.3 — Answer: (C), (D) [MSQ, 1M · Regression & Model Selection > Gradient descent]

(A) **FALSE**. It is convex; every local minimum is global.

(B) **FALSE**. It diverges if η is too large.

(C) **TRUE**. Definition of SGD / mini-batch GD.

(D) **TRUE**. Gradient noise does not vanish; decaying step sizes are needed for exact convergence.

Q.4 — Answer: 0.1249 to 0.1289 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

$\ln(1 + e^{-2}) = \ln(1 + 0.1353) = \mathbf{0.1269}$. (Its derivative is $\sigma(z) = 0.1192$.)

Q.5 — Answer: 0.227 to 0.247 [NAT, 1M · Unsupervised Learning > Distance measures]

$x \cdot y = 16$, $\|x\| = \sqrt{86}$, $\|y\| = \sqrt{53}$; $\cos \theta = x \cdot y / (\|x\| \|y\|) = 0.237 = \mathbf{0.24}$.

For reference: $L_1 = 17$, $L_2 = 10.34$, $L_\infty = 7$ (always $L_\infty \leq L_2 \leq L_1$).

Q.6 — Answer: 6 [NAT, 1M · Regression & Model Selection > Basis functions & feature scaling]

Multisets of size d from p variables: $C(p + d - 1, d) = C(4, 2) = \mathbf{6}$.

Q.7 — Answer: 0.0659 to 0.0859 [NAT, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

$z = w_0 + w_1 x_1 + w_2 x_2 = -1.5 + (1)(0) + (-0.5)(2) = -2.5$ (this is the log-odds).

$P(y = 1 | \mathbf{x}) = \sigma(z) = 1/(1 + e^{2.5}) = 0.0759$.

Answer: **0.08**.

Q.8 — Answer: 0.5847 to 0.6047 [NAT, 1M · SVM & Decision Trees > Decision trees: impurity measures]

Gini = $1 - \sum p_k^2 = 1 - [(12/26)^2 + (3/26)^2 + (11/26)^2] = \mathbf{0.595}$.

(Maximum possible with 3 classes: $1 - 1/3 = 0.667$.)

Q.9 — Answer: (A), (C), (D) [MSQ, 1M · SVM & Decision Trees > Decision trees: overfitting & pruning]

(A) **TRUE**. Each leaf corresponds to a hyper-rectangle.

(B) **FALSE**. Its boundary is piecewise axis-parallel, generally non-linear.

(C) **TRUE**. Pruning trades training fit for simplicity.

(D) **TRUE**. CART pruning criterion.

Q.10 — Answer: (B) [MCQ, 1M · Regression & Model Selection > Regression vs classification]

Linear regression outputs an unbounded real-valued prediction.

Hence the answer is **a real number**.

Q.11 — Answer: 28.6567 to 28.6767 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

The data are already centered (column means are 0), so $S = X^T X / (n - 1)$.

$$\Sigma_{X_1^2} = 64, \Sigma_{X_2^2} = 22, \Sigma_{X_1X_2} = 24.$$

$$S = [[21.3333, 8], [8, 7.3333]].$$

Required value = **28.67**. (Dividing by n instead of $n - 1$ would give 21.5.)

Q.12 — Answer: (A), (B), (C), (D) [MSQ, 1M · Unsupervised Learning > k-means]

(A) TRUE. Both the assignment step and the mean-update step minimize the objective with the other block fixed.

(B) TRUE. Its objective implicitly assumes isotropic, equal-variance clusters.

(C) TRUE. Hence the use of the Lloyd heuristic.

(D) TRUE. The mean minimizes the sum of squared Euclidean distances to the points.

Q.13 — Answer: (A) [MCQ, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

The jagged boundary with small islands around individual points (Panel Q) is produced by $k = 1$, which fits every training point (zero training error): low bias, high variance.

The smooth boundary (Panel P) comes from $k = 15$: averaging many neighbours raises bias and lowers variance; it misclassifies some training points, so its training error is not zero.

Answer: **Panel Q uses $k = 1$; it has lower bias and higher variance.**

Q.14 — Answer: -10 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Forward pass]

$a = Wx + b = (4, -6)$; $h = \text{ReLU}(a) = (4, 0)$ - a negative pre-activation is clipped to 0.

$$y = -3 \cdot 4 + (-2) \cdot 0 + (2) = -10.$$

A common slip is to forget the ReLU and use the raw pre-activation (which would give 2).

Q.15 — Answer: 7 [NAT, 1M · SVM & Decision Trees > Decision trees: structure & traversal]

Depth d allows at most 2^d leaves, so we need $2^d \geq 90 \Rightarrow d \geq \log_2 90 = 6.492 \Rightarrow d = 7$.

Q.16 — Answer: -2.7437 to -2.7237 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

Mean $\bar{x} = (4.4, 3.6)$. Centered data rows: P1: $(-2.4, 1.4)$, P2: $(-0.4, 4.4)$, P3: $(1.6, -2.6)$, P4: $(4.6, -2.6)$, P5: $(-3.4, -0.6)$.

Sample covariance $S = (1/(n - 1)) \Sigma(x - \bar{x})(x - \bar{x})^T = [[10.3, -4.8], [-4.8, 8.8]]$.

Eigenvalues: $\lambda_1 = 14.4082$, $\lambda_2 = 4.6918$; unit PC1 direction $v_1 = (0.7597, -0.6502)$.

$$\text{Score } z_1 = v_1 \cdot (x - \bar{x}) = 0.7597 \cdot (-2.4) + (-0.6502) \cdot 1.4 = -2.7337.$$

Answer: **-2.73** (the sign depends on the orientation of v_1 ; with the stated convention it is -2.73).

Q.17 — Answer: 0.1682 to 0.1722 [NAT, 2M · Regression & Model Selection > MLE & MAP view]

$$-\ln p(w \mid \text{data}) = (1/(2\sigma^2)) \Sigma(y_i - wx_i)^2 + w^2/(2\tau^2) + \text{const.}$$

Multiplying by $2\sigma^2$: $\Sigma(y_i - wx_i)^2 + (\sigma^2/\tau^2)w^2$ — ridge with $\lambda = \sigma^2/\tau^2 = 2$.

$$\Sigma x_i^2 = 45, \Sigma x_i y_i = 8; \hat{w}_{\text{MAP}} = \Sigma xy / (\Sigma x^2 + \lambda) = 8/(45 + 2) = 0.1702.$$

Answer: **0.17**.

Q.18 — Answer: (A) [MCQ, 2M · Regression & Model Selection > Cross-validation]

Minimum CV MSE = 4.93 at $\lambda = 0.1$, with SE = 0.46.

$$\text{One-SE threshold} = 4.93 + 0.46 = 5.39.$$

Models within the threshold: $\lambda = 0.01$ (5.32), $\lambda = 0.1$ (4.93), $\lambda = 0.3$ (5.27).

The one-SE rule picks the largest such λ (simplest model): $\lambda = 0.3$; the minimum-CV rule picks $\lambda = 0.1$.

Answer: **$\lambda = 0.3$** .

Q.19 — Answer: 2 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$$\Sigma^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 0.5 \end{bmatrix}$$

$$k = 1: \Sigma^{-1}\mu_1 = (2, 0); \mathbf{x}^T \Sigma^{-1}\mu = -4; \frac{1}{2}\mu^T \Sigma^{-1}\mu = 2; \ln \pi = -0.5108 \Rightarrow \delta_1 = -6.5108.$$

$$k = 2: \Sigma^{-1}\mu_2 = (-2, 0.5); \mathbf{x}^T \Sigma^{-1}\mu = 5; \frac{1}{2}\mu^T \Sigma^{-1}\mu = 2.25; \ln \pi = -1.6094 \Rightarrow \delta_2 = 1.1406.$$

$$k = 3: \Sigma^{-1}\mu_3 = (0, 1.5); \mathbf{x}^T \Sigma^{-1}\mu = 3; \frac{1}{2}\mu^T \Sigma^{-1}\mu = 2.25; \ln \pi = -1.6094 \Rightarrow \delta_3 = -0.8594.$$

The largest discriminant is $\delta_2 \Rightarrow$ predicted class **2**.

Q.20 — Answer: (B), (C) [MSQ, 2M · Neural Networks (MLP / Feed-forward) · Activation functions]

$\sigma(0.5) = 0.6225$, $\tanh(0.5) = 0.4621$.

(A) **FALSE**. $|\tanh z| < |z|$ for $z \neq 0$.

(B) **TRUE**. $\sigma(-z) = 1 - \sigma(z)$.

(C) **TRUE**. Both equal 0.4621.

(D) **FALSE**. $\sigma' = \sigma(1-\sigma) = 0.235$.

Q.21 — Answer: (D) [MSQ, 2M · Probabilistic & Instance-based Classifiers · k-nearest neighbours]

Distances from the query, sorted:

rank	#	(x_1, x_2)	distance	class
1	1	(5, 2)	$\sqrt{1.25} = 1.118$	B
2	5	(4, 3)	$\sqrt{2.25} = 1.5$	A
3	12	(6, 1)	$\sqrt{4.25} = 2.062$	A
4	4	(7, 5)	$\sqrt{6.25} = 2.5$	B
5	2	(7, 0)	$\sqrt{11.25} = 3.354$	B
6	9	(9, 3)	$\sqrt{12.25} = 3.5$	B
7	3	(2, 0)	$\sqrt{21.25} = 4.61$	A
8	7	(1, 6)	$\sqrt{29.25} = 5.408$	B

(A) **FALSE**. 1 nearest: #1 $\rightarrow$ 0 A vs 1 B $\Rightarrow$ class B.

(B) **FALSE**. 5 nearest: #1, #5, #12, #4, #2 $\rightarrow$ 2 A vs 3 B $\Rightarrow$ class B.

(C) **FALSE**. 7 nearest: #1, #5, #12, #4, #2, #9, #3 $\rightarrow$ 3 A vs 4 B $\Rightarrow$ class B.

(D) **TRUE**. 3 nearest: #1, #5, #12 $\rightarrow$ 2 A vs 1 B $\Rightarrow$ class A.

Q.22 — Answer: 0.0447 to 0.0647 [NAT, 2M · Probabilistic & Instance-based Classifiers · Logistic regression]

$P(y=1|x) \geq 0.9 \Leftrightarrow z \geq c$, where $c = \ln(0.9/0.1) = 2.1972$.

Decision boundary: $2 + (2)x_1 + (3)x_2 = 2.1972$ (a straight line; changing the threshold only shifts it parallel to the 0.5 boundary).

distance = $|w_0 - c|/\|(w_1, w_2)\| = |2 - 2.1972|/\sqrt{(13)} = \mathbf{0.05}$.

Q.23 — Answer: 0.7291 to 0.7331 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Forward pass]

Hidden pre-activations: $a_1 = -1 \cdot 2 + (1.5) \cdot -2 + (-0.5) = -5.5$; $a_2 = 0.5 \cdot 2 + (0.5) \cdot -2 + (1) = 1$.

Hidden activations ($\sigma(a) = 1/(1+e^{-a})$): $h_1 = 0.0041$, $h_2 = 0.7311$.

Required: $h_2 = \mathbf{0.7311}$.

Q.24 — Answer: (C) [MSQ, 2M · Regression & Model Selection · Ridge & Lasso]

With $X^T X = I$ the ridge path is $b/(1 + \lambda)$ and the lasso path is $\text{sign}(b)(|b| - \lambda)_+$. The plot shows paths reaching zero at finite $\lambda \Rightarrow$ lasso; here it is **lasso**.

(A) **FALSE**. At $\lambda = 0$ they are the OLS estimates.

(B) **FALSE**. Under lasso $\beta_j = 0$ for $\lambda \geq |b_j|$.

(C) **TRUE**. Penalty vanishes at $\lambda = 0$.

(D) **FALSE**. Training RSS is minimised at $\lambda = 0$ and increases with λ .

Q.25 — Answer: 3 [NAT, 2M · Unsupervised Learning · k-medoids]

PAM cost = Σ over all points of the absolute distance to the nearest medoid.

Current medoids {0, 9}: distances 0 $\rightarrow$ 0, 9 $\rightarrow$ 0, 10 $\rightarrow$ 1, 11 $\rightarrow$ 2, 14 $\rightarrow$ 5, 18 $\rightarrow$ 9, 26 $\rightarrow$ 17, 28 $\rightarrow$ 19; cost = **53**.

After swap, medoids {9, 10}: distances 0 $\rightarrow$ 9, 9 $\rightarrow$ 0, 10 $\rightarrow$ 0, 11 $\rightarrow$ 1, 14 $\rightarrow$ 4, 18 $\rightarrow$ 8, 26 $\rightarrow$ 16, 28 $\rightarrow$ 18; cost = 56.

Change = 56 - 53 = **3** (positive $\rightarrow$ PAM rejects the swap).

Q.26 — Answer: 0.008 to 0.012 [NAT, 2M · Regression & Model Selection · Gradient descent]

$\partial J/\partial w_0 = (1/n)\Sigma r_i$, $\partial J/\partial w_1 = (1/n)\Sigma r_i x_i$, with residual $r_i = w_0 + w_1 x_i - y_i$.

Step 1: $w = (0, 1)$, $r = (-6, 1, 2)$, $\nabla J = (-1, 0.6667) \Rightarrow w \leftarrow w - 0.01 \nabla J = (0.01, 0.9933)$.

Answer: **0.01**.

Q.27 — Answer: 0.1854 to 0.1894 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

Forward: $a = Wx = (2.25, 1.5)$, $h = \sigma(a) = (0.9047, 0.8176)$, $\hat{y} = v \cdot h = -0.9482$.

Output error $\delta_o = \partial L / \partial \hat{y} = \hat{y} - y = -1.4482$ (linear output, $L = \frac{1}{2}(\hat{y} - y)^2$).

Hidden delta $\delta_1 = \delta_o \cdot v_1 \cdot h_1(1 - h_1) = -1.4482 \times -1.5 \times 0.0863 = 0.1874$.

$\partial L / \partial b_1 = \delta_1 = \mathbf{0.1874}$.

Q.28 — Answer: 4.1131 to 4.1331 [NAT, 2M · SVM & Decision Trees > SVM: margin & geometry]

The closest pair of opposite-class points is $P2 = (5, 5)$ and $P5 = (6, 1)$. Their perpendicular bisector separates the classes and no other point comes closer to it, so Hence $w \parallel (x_+ - x_-)$ and the support vectors are $P2, P5$.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (2/17, -8/17)$, $b = 13/17$; $\|w\| = 0.4851$, margin $2/\|w\| = 4.1231$.

Point	P1	P2	P3	P4	P5	P6	P7
$y(w^T x + b)$	1.353	1	1.235	1.353	1	1.471	1.353

All values are ≥ 1 (feasible), with equality exactly at the support vectors.

Margin width = $2/\|w\| = \mathbf{4.12}$.

Q.29 — Answer: (C), (D) [MSQ, 2M · SVM & Decision Trees > SVM: kernels]

A Gram matrix of a valid kernel must be symmetric and positive semidefinite. For a symmetric 2×2 matrix this is equivalent to: both diagonal entries ≥ 0 and determinant ≥ 0 .

(A) **FALSE**. a diagonal entry $K(x, x) = \|\phi(x)\|^2$ is negative — impossible.

(B) **FALSE**. not symmetric, but $K(x, z) = K(z, x)$ for every kernel.

(C) **TRUE**. symmetric, diagonal ≥ 0 , $\det = 5 > 0 \Rightarrow$ PD (negative off-diagonal entries are fine).

(D) **TRUE**. symmetric, diagonal entries ≥ 0 and $\det = 31 > 0 \Rightarrow$ positive definite.

Q.30 — Answer: 0.2428 to 0.2628 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [19 Yes, 10 No]: $H = 0.9294$.

Child entropies: $v1: 0.5917$, $v2: 0$, $v3: 0$, $v4: 0.9183$.

$IG = 0.9294 - \sum (n_v/29) H_v = 0.5016$.

$\text{SplitInfo} = -\sum (n_v/N) \log_2(n_v/N) = -(7/29)\log_2(7/29) - (7/29)\log_2(7/29) - (6/29)\log_2(6/29) - (9/29)\log_2(9/29) = 1.9841$.

Gain ratio = $IG / \text{SplitInfo} = 0.5016/1.9841 = \mathbf{0.25}$.

C4.5 uses the gain ratio to penalise attributes that split the data into many small branches.

Q.31 — Answer: 7.99 to 8.01 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Use complete linkage: $d(A, B) = \max_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P1\} + \{P2\}$ at height **1**. Updated: $d(\{P1, P2\}, \{P3\}) = 3$, $d(\{P1, P2\}, \{P4\}) = 5$, $d(\{P1, P2\}, \{P5\}) = 8$, $d(\{P1, P2\}, \{P6\}) = 11$.

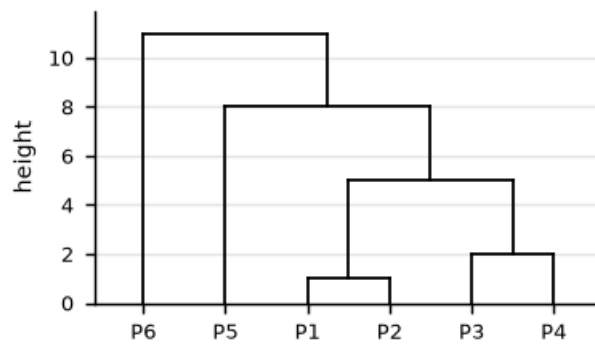
Merge 2: $\{P3\} + \{P4\}$ at height **2**. Updated: $d(\{P3, P4\}, \{P5\}) = 7$, $d(\{P3, P4\}, \{P6\}) = 8$, $d(\{P3, P4\}, \{P1, P2\}) = 5$.

Merge 3: $\{P1, P2\} + \{P3, P4\}$ at height **5**. Updated: $d(\{P1, P2, P3, P4\}, \{P5\}) = 8$, $d(\{P1, P2, P3, P4\}, \{P6\}) = 11$.

Merge 4: $\{P5\} + \{P1, P2, P3, P4\}$ at height **8**. Updated: $d(\{P1, P2, P3, P4, P5\}, \{P6\}) = 11$.

Merge 5: $\{P6\} + \{P1, P2, P3, P4, P5\}$ at height **11**.

$P4$ and $P5$ first lie in the same cluster at the merge of height 8, so their cophenetic distance is **8** (their original distance was 7).



Q.32 — Answer: (A), (C), (D) [MSQ, 2M · Unsupervised Learning > k-means]

(A) TRUE. The median minimizes the sum of absolute deviations.

(B) FALSE. It is the median, which is a data point.

(C) TRUE. Medoids are data points; no means are needed.

(D) TRUE. Same centroids give the same assignments; the algorithm is at a fixed point.

Q.33 — Answer: (A), (B), (C), (D) [MSQ, 2M · SVM & Decision Trees > Decision trees: impurity measures]

Parent [40+, 50-]: $H = 0.9911$, $Gini = 0.4938$, $error = 0.4444$.

S1: children $H = 0.9109, 0.9865$; $Gini = 0.4395, 0.4907$; $error = 0.3261, 0.4318$. Gains: $IG = 0.0432$, $Gini = 0.0293$, $error = 0.0667$.

S2: children $H = 0.6913, 0.65$; $Gini = 0.3018, 0.2778$; $error = 0.1852, 0.1667$. Gains: $IG = 0.3163$, $Gini = 0.2016$, $error = 0.2667$.

(A) TRUE. S1: 0.0432, S2: 0.3163.

(B) TRUE. $IG(S2) = 0.3163$.

(C) TRUE. S1: 0.0667, S2: 0.2667.

(D) TRUE. S1: 0.0293, S2: 0.2016.

Q.34 — Answer: 0.4312 to 0.4512 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Counting: $TP = 3$, $FP = 4$, $FN = 3$.

$P = 3/7 = 0.4286$, $R = 3/6 = 0.5$.

$F_{0.5} = (1 + 0.25) \times 0.4286 \times 0.5 / (0.25 \times 0.4286 + 0.5) = \mathbf{0.44}$.

$\beta < 1$ weights precision more heavily (F1 here would be 0.462).

Q.35 — Answer: (A), (B) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$\nabla f = (10w_1, 2w_2) = w_i \leftarrow (1 - \eta a_i) w_i$. GD converges (from every start) iff $|1 - \eta a_i| < 1$ for both i , i.e. $0 < \eta < 2/\max(a_i) = 2/10 = 0.2$. The steep direction ($a = 10$) limits the learning rate.

(A) TRUE. $\eta = 0.22$: $|1 - \eta \cdot 10| = 1.2$, $|1 - \eta \cdot 2| = 0.56 \rightarrow$ diverges.

(B) TRUE. $\eta = 0.3$: $|1 - \eta \cdot 10| = 2$, $|1 - \eta \cdot 2| = 0.4 \rightarrow$ diverges.

(C) FALSE. $\eta = 0.05$: $|1 - \eta \cdot 10| = 0.5$, $|1 - \eta \cdot 2| = 0.9 \rightarrow$ converges.

(D) FALSE. $\eta = 0.03$: $|1 - \eta \cdot 10| = 0.7$, $|1 - \eta \cdot 2| = 0.94 \rightarrow$ converges.

Mock Test 33

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	6	6	23

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Probabilistic & Instance-based Classifiers	7	11
Regression & Model Selection	7	11
Neural Networks (MLP / Feed-forward)	7	11
Unsupervised Learning	7	11
SVM & Decision Trees	7	11

Start time: _____ End time: _____ Score: _____ / 55

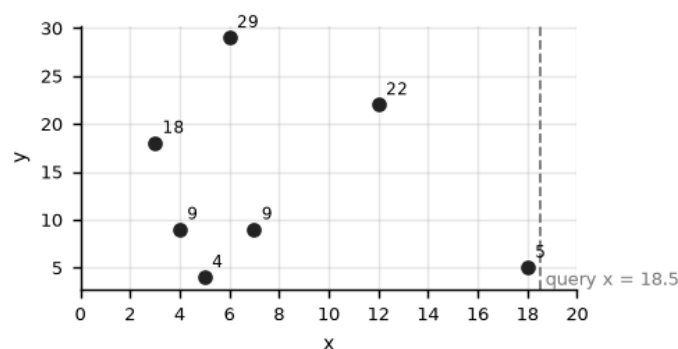
Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

The training data for a k-NN regressor are given below (also plotted). Using $k = 3$ and absolute distance $|x - x_i|$, the predicted value (unweighted average of neighbours) at $x = 18.5$ is _____ (rounded off to two decimal places)

x	3	4	5	6	7	12	18
y	18	9	4	29	9	22	5



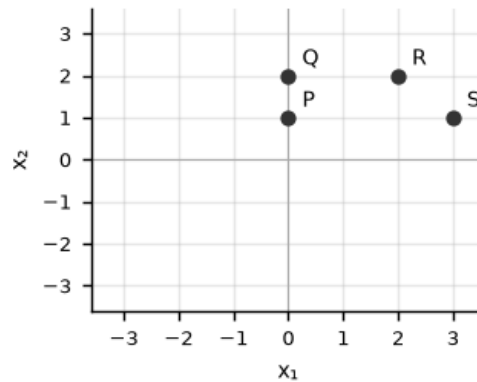
Q.2 [1 mark · MSQ (one or more correct)]

Consider ridge regression (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_2^2$) and the lasso (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_1$) with $\lambda \geq 0$. Which of the following statements is/are CORRECT?

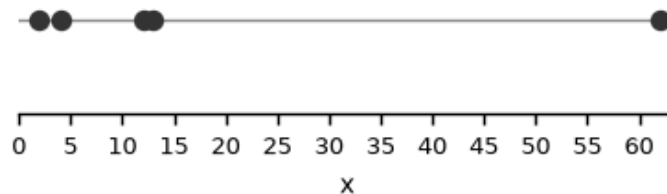
- (A) Ridge solutions are invariant to rescaling the individual predictors.
- (B) The lasso constraint region $\sum |\beta_j| \leq t$ in two dimensions is a diamond (square rotated by 45°).
- (C) The training error of ridge regression decreases as λ increases.
- (D) The lasso objective $\|y - X\beta\|^2 + \lambda\|\beta\|_1$ is convex in β .

Q.3 [1 mark · MSQ (one or more correct)]

A perceptron with weights $w = (-3, 1)$ and bias $b = 0$ outputs $\hat{y} = +1$ if $w \cdot x + b \geq 0$ and $\hat{y} = -1$ otherwise. Which of the points shown below are classified as $\hat{y} = +1$?

**(A)** Point P = (0, 1)**(B)** Point Q = (0, 2)**(C)** Point R = (2, 2)**(D)** Point S = (3, 1)**Q.4 [1 mark · NAT (numerical answer)]**

A cluster contains the 1-D values $\{2, 4, 12, 13, 62\}$. Treating them as one cluster, the absolute difference between the k-means centroid (mean) and the k-medoids representative (absolute-difference dissimilarity) is _____ (rounded off to two decimal places)

**Q.5 [1 mark · MSQ (one or more correct)]**

Regarding the bias-variance trade-off in supervised regression with squared-error loss, which of the following statements is/are CORRECT?

(A) Test error plotted against flexibility is typically U-shaped.**(B)** The squared bias term can be negative when the model overestimates f .**(C)** For nested least-squares polynomial fits, training MSE never increases as the degree increases.**(D)** Underfitting is characterised by low training error and high test error.**Q.6 [1 mark · NAT (numerical answer)]**

A linear regression model with an intercept uses 3 numeric features and a categorical feature *payment mode* with 5 levels (Cash, Card, UPI, Wallet, NetBanking); a categorical feature *season* with 4 levels (Winter, Summer, Monsoon, Autumn). Each categorical feature is encoded by dummy (indicator) variables using a reference level, so that the design matrix has full column rank. The model is fitted to $n = 70$ observations. The residual degrees of freedom used in the unbiased estimate $\hat{\sigma}^2 = \text{SSE}/(\text{degrees of freedom})$ is _____ (Answer in integer)

Q.7 [1 mark · MCQ]

A soft-margin linear SVM has learned $f(x) = x_1 + 0.5x_2 + 3$. For the training example $x = (0, 1)$ with label $y = -1$, which statement is CORRECT?

- (A) Correctly classified but strictly inside the margin ($0 < \xi < 1$)
- (B) Exactly on the decision boundary ($\xi = 1$)
- (C) Correctly classified, on or outside the margin ($\xi = 0$)
- (D) Misclassified ($\xi > 1$)

Q.8 [1 mark · NAT (numerical answer)]

In a disease screening test, a false negative costs 19 units and a false positive costs 10 units (correct decisions cost 0). The model outputs $p = P(\text{positive} \mid x)$, i.e. the probability that the patient has the disease. The Bayes-optimal rule predicts positive iff p exceeds the threshold _____ (rounded off to two decimal places)

Q.9 [1 mark · NAT (numerical answer)]

The RBF (Gaussian) kernel is $K(x, z) = \exp(-\gamma\|x - z\|^2)$ with $\gamma = 0.5$. For $x = (2, 1, 2)$ and $z = (0, 1, 2)$, the value of $K(x, z)$ is _____ (rounded off to three decimal places)

Q.10 [1 mark · NAT (numerical answer)]

A node contains 96 records spread equally over 32 classes (3 records per class). Its entropy (in bits) is _____ (rounded off to two decimal places)

Q.11 [1 mark · MSQ (one or more correct)]

Which of the following statements about activation functions is/are CORRECT? (σ denotes the logistic sigmoid.)

- (A) Softmax is invariant to multiplying all logits by the same positive constant.
- (B) ReLU is not differentiable at $z = 0$; in practice a subgradient value (0 or 1) is used there.
- (C) $\sigma(0) = 0$ for the logistic sigmoid.
- (D) As the temperature $T \rightarrow 0^+$, $\text{softmax}(z/T)$ approaches the uniform distribution.

Q.12 [1 mark · NAT (numerical answer)]

A training run on $N = 1000$ examples with batch size $B = 16$ (last partial batch kept) performed 945 parameter updates in total. The number of epochs completed is _____ (Answer in integer)

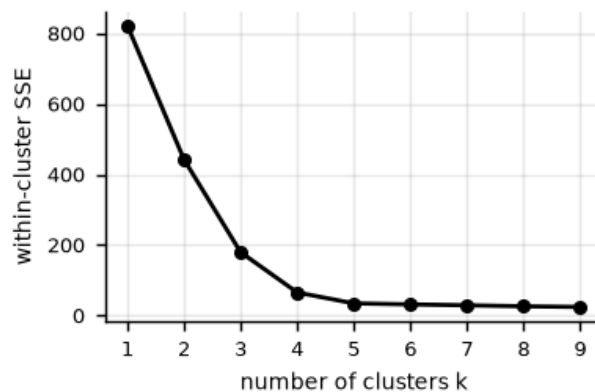
Q.13 [1 mark · MCQ]

In a logistic regression model for the probability of passing the exam, the coefficient of *hours of study* is $\beta = -0.7$. Holding all other features fixed, increasing *hours of study* by 1 unit multiplies the **odds** of passing the exam by (approximately)

- (A) 0.5
- (B) 0.3
- (C) 0.33
- (D) -0.7

Q.14 [1 mark · MCQ]

k-means is run for $k = 1, 2, \dots, 9$ on a dataset and the final within-cluster SSE is plotted below. Using the elbow method, the most appropriate number of clusters is



(A) 9

(B) 5

(C) 4

(D) 6

Q.15 [1 mark · NAT (numerical answer)]

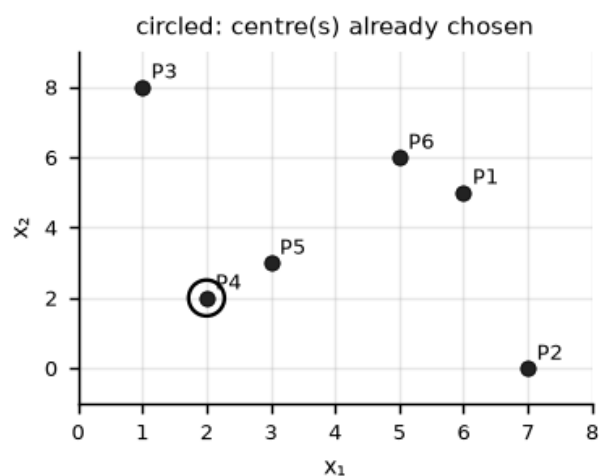
A clustering algorithm groups labelled data into three clusters; the contingency table of cluster versus true class counts is given below. The purity of this clustering is _____ (rounded off to two decimal places)

	Class P	Class Q	Class R
Cluster 1	13	11	2
Cluster 2	4	3	16
Cluster 3	9	7	4

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

k-means++ initialization (squared Euclidean seeding) is applied to the six points below. P4 has already been chosen as centre. The probability that P2 is selected as the second centre is _____ (rounded off to three decimal places)

Point	P1	P2	P3	P4	P5	P6
x ₁	6	7	1	2	3	5
x ₂	5	0	8	2	3	6



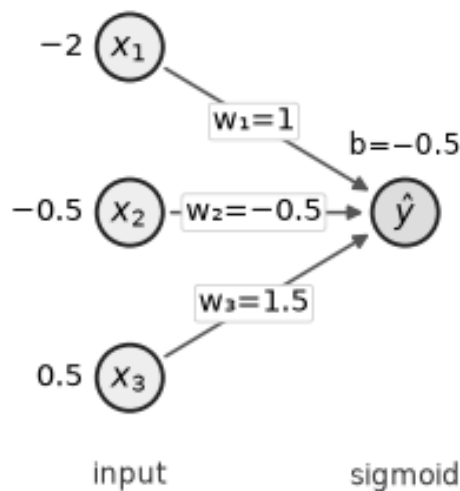
Q.17 [2 marks · NAT (numerical answer)]

For a two-class problem, the class means are $\mu_0 = (-1, -1)^T$ and $\mu_1 = (4, -2)^T$ and the pooled within-class scatter matrix S_W is given below. Fisher's linear discriminant uses $\mathbf{w} \propto S_W^{-1}(\mu_1 - \mu_0)$. Then the ratio w_2/w_1 of the components of the Fisher discriminant direction $\mathbf{w}$ is _____ (rounded off to two decimal places)

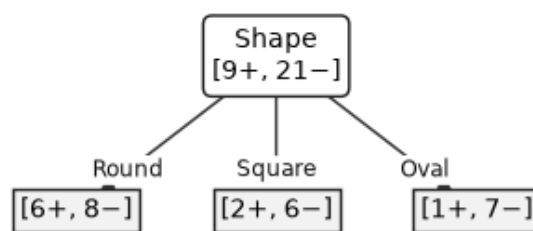
$$S_W = \begin{bmatrix} 8 & 4 \\ 4 & 12 \end{bmatrix}$$

Q.18 [2 marks · NAT (numerical answer)]

A single sigmoid neuron (shown below) receives $\mathbf{x} = (-2, -0.5, 0.5)$ and has target $y = 1$. The loss is $L = \frac{1}{2}(\hat{y} - y)^2$. The value of the updated weight w_1 after one gradient-descent step with $\eta = 1$ is _____ (rounded off to three decimal places)

**Q.19 [2 marks · NAT (numerical answer)]**

A node with 30 training records (class counts [positive+, negative-]) is split on the attribute Shape as shown. Compute the information gain (in bits) of splitting on Shape. (rounded off to two decimal places)

**Q.20 [2 marks · MSQ (one or more correct)]**

Consider agglomerative hierarchical clustering on n points with a dissimilarity matrix (no ties). Which of the following statements is/are CORRECT?

- (A) The Lance-Williams formula updates inter-cluster distances after a merge without going back to the raw points.
- (B) Cutting a dendrogram (with distinct merge heights) at all possible heights gives exactly n different clusterings, with 1, 2, ..., n clusters.
- (C) Single linkage is robust to a few noise points lying between two clusters.
- (D) Cophenetic distances from single, complete or average linkage satisfy the ultrametric inequality $d(x, z) \leq \max(d(x, y), d(y, z))$.

Q.21 [2 marks · NAT (numerical answer)]

A 2-2-1 network uses the identity (linear) activation in its hidden layer: $\hat{y} = W_2(W_1x + b_1) + b_2$, with the parameters below. The network is equivalent to a single linear model $\hat{y} = \tilde{w}_1x_1 + \tilde{w}_2x_2 + \tilde{b}$. Then the coefficient $\tilde{w}_2$ is _____ (Answer in integer)

$$W_1 = \begin{bmatrix} -2 & 1 \\ 2 & -2 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 1 & -2 \end{bmatrix}$$

$$b_2 = -1$$

Q.22 [2 marks · MCQ]

For $n = 9$ paired observations: $\sum x = 75$, $\sum y = 118$, $\sum x^2 = 779$, $\sum y^2 = 1852$, $\sum xy = 1182$. The least-squares regression line of $\mathbf{x}$ on $\mathbf{y}$ is (coefficients rounded to two decimals)

(A) $\hat{x} = -1.83 + 0.78y$

(B) $\hat{x} = -0.21 + 0.65y$

(C) $\hat{x} = 16.88 + 0.65y$

(D) $\hat{x} = -0.03 + 0.64y$

Q.23 [2 marks · NAT (numerical answer)]

Ridge regression without intercept minimises $\|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2$ with $\lambda = 3$, for the data below ($\beta = (\beta_1, \beta_2)^T$). The value of β_2 is _____ (rounded off to three decimal places)

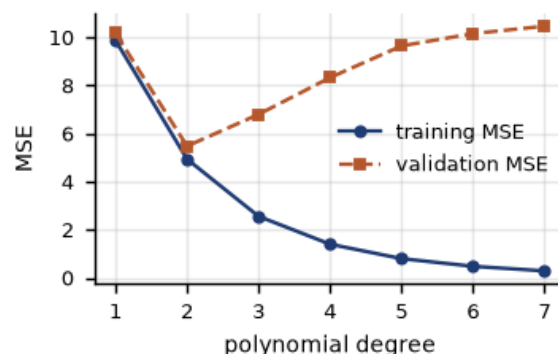
$$X = \begin{bmatrix} 2 & -2 \\ 1 & 2 \\ 2 & -1 \\ 1 & 0 \end{bmatrix}$$

$$y = \begin{bmatrix} -2 \\ -1 \\ 7 \\ 7 \end{bmatrix}$$

Q.24 [2 marks · MSQ (one or more correct)]

Polynomial regression models of degree 1 to 7 are fitted by least squares on a training set and evaluated on a separate validation set. The resulting mean squared errors are given below. Which of the following statements is/are CORRECT?

Degree	1	2	3	4	5	6	7
Training MSE	9.88	4.94	2.56	1.4	0.8	0.48	0.29
Validation MSE	10.18	5.48	6.8	8.33	9.65	10.16	10.47

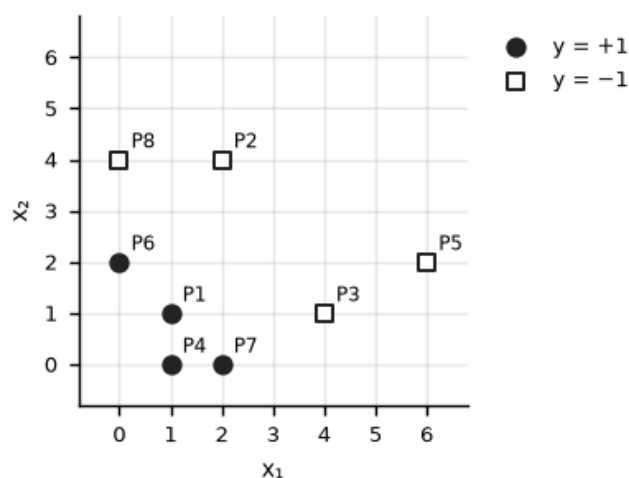


- (A) The degree-1 model underfits: its training and validation errors are both high.
- (B) The degree-7 model overfits: it has the lowest training MSE but its validation MSE exceeds the minimum.
- (C) Based on the validation MSE, degree 2 should be selected.
- (D) The degree-7 model has higher variance than the degree-1 model.

Q.25 [2 marks • MCQ]

Four candidate linear decision boundaries are proposed for the labelled training points shown. Which boundary separates the two classes with the LARGEST geometric margin (minimum distance from the boundary to any training point, all points correctly classified)?

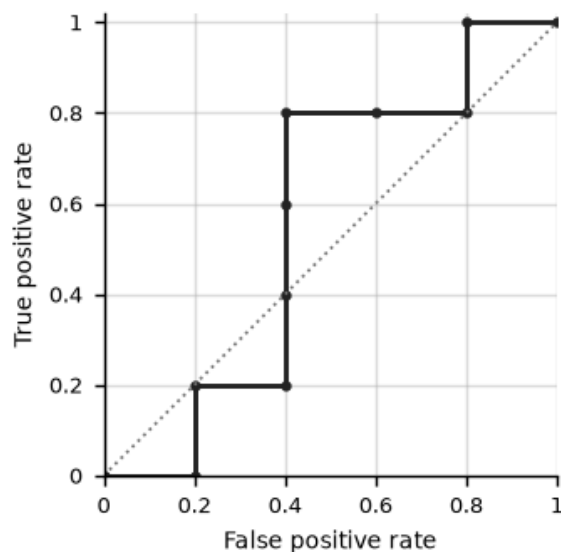
Point	x_1	x_2	y
P1	1	1	+1
P2	2	4	-1
P3	4	1	-1
P4	1	0	+1
P5	6	2	-1
P6	0	2	+1
P7	2	0	+1
P8	0	4	-1



- (A) $x_1 + x_2 - 2.5 = 0$
 (B) $x_1 + x_2 - 3 = 0$
 (C) $x_1 + x_2 - 3.5 = 0$
 (D) $x_1 - 2x_2 + 7 = 0$

Q.26 [2 marks • NAT (numerical answer)]

The ROC curve below was obtained from a test set with 5 positives and 5 negatives (all scores distinct, so the curve moves in steps of 1/5 vertically and 1/5 horizontally). From the figure, the area under this ROC curve (AUC) is _____ (rounded off to two decimal places)



Q.27 [2 marks · NAT (numerical answer)]

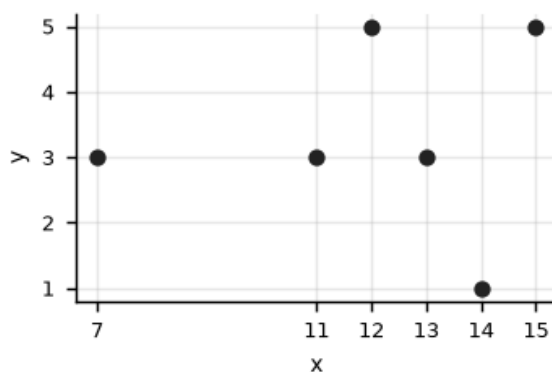
Two clusterings U and V of 8 points are given below (cluster names are arbitrary). Then the number of point pairs placed in the same cluster by **both** clusterings is _____ (Answer in integer)

Point	P1	P2	P3	P4	P5	P6	P7	P8
Clustering U	α	β	β	β	β	β	β	α
Clustering V	1	1	3	2	2	3	2	1

Q.28 [2 marks · NAT (numerical answer)]

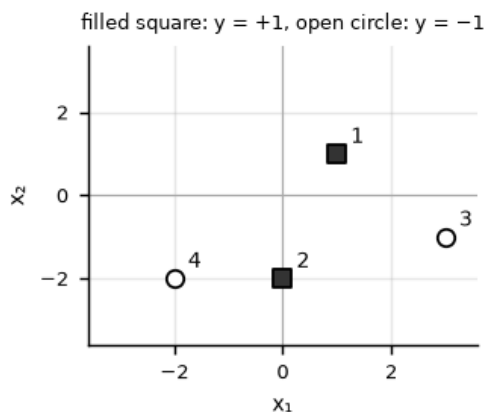
A regression tree (squared-error criterion) is fitted to the 6 points below. The root split " $x \leq t$ " is chosen among midpoints of consecutive x values to minimise the total SSE of the two children (each child predicts its mean). Find the prediction of the resulting depth-1 regression tree (stump) at $x = 6$. (rounded off to two decimal places)

x	7	11	12	13	14	15
y	3	3	5	3	1	5

**Q.29 [2 marks · NAT (numerical answer)]**

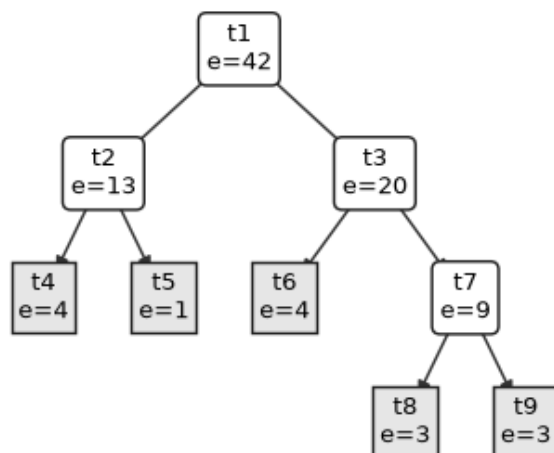
A perceptron with parameters $w \in \mathbb{R}^2$, b is trained on the data below, starting from $w = (0, 0)$, $b = 0$, with learning rate $\eta = 0.5$. Examples are presented once in the order 1, 2, 3, 4 (one epoch). An update $w \leftarrow w + \eta y x$, $b \leftarrow b + \eta y$ is made whenever $y(w \cdot x + b) \leq 0$. After the epoch, the weight w_1 is _____ (Answer in integer)

i	x_1	x_2	y
1	1	1	1
2	0	-2	1
3	3	-1	-1
4	-2	-2	-1



Q.30 [2 marks · NAT (numerical answer)]

A classification tree was grown on $N = 100$ training records. Each node shows $e(t)$, the number of training records that would be misclassified if t were a leaf; $R(t) = e(t)/N$. The smallest value of α at which cost-complexity pruning first prunes the tree (the weakest link) is _____ (rounded off to three decimal places)

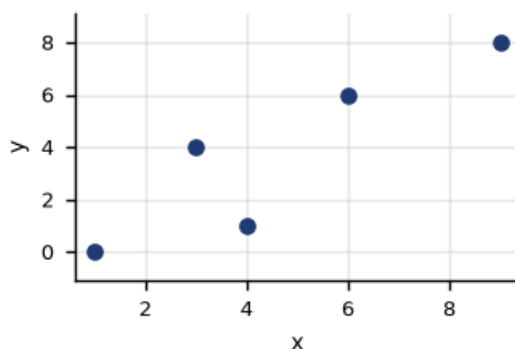
**Q.31 [2 marks · NAT (numerical answer)]**

Gradient descent with (heavy-ball) momentum minimises $f(w) = (0.5/2)w^2$ using $v_t = \beta v_{t-1} + \nabla f(w_{t-1})$ and $w_t = w_{t-1} - \eta v_t$, with $w_0 = 3$, $v_0 = 0$, $\eta = 0.05$, $\beta = 0.8$. The value of w_2 is _____ (rounded off to three decimal places)

Q.32 [2 marks · NAT (numerical answer)]

Assume $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with ε_i i.i.d. $N(0, \sigma^2)$. For the 5 observations below, with β estimated by least squares (= maximum likelihood), the maximised log-likelihood $\ln L(\beta, \hat{\sigma}^2)$ (natural log, with $\hat{\sigma}^2$ the maximum-likelihood estimate) is _____ (rounded off to two decimal places)

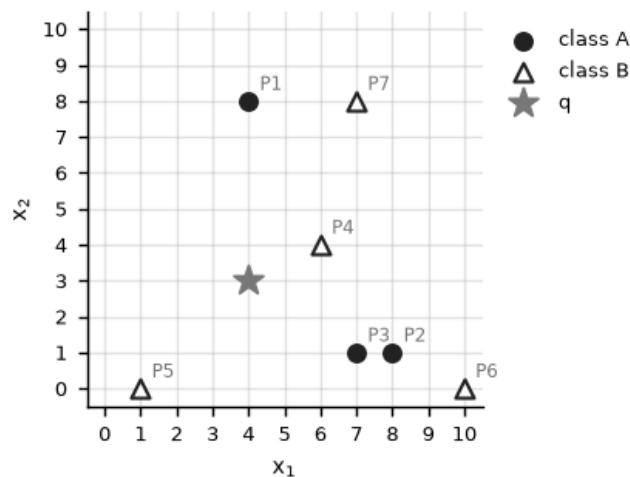
x	1	3	4	6	9
y	0	4	1	6	8



Q.33 [2 marks · MCQ]

Consider the training set below. A 3-NN classifier (majority vote) is applied to the query point $\mathbf{q} = (4, 3)$, once with Euclidean distance and once with Manhattan distance. The predicted classes are

Point	x_1	x_2	class
P1	4	8	A
P2	8	1	A
P3	7	1	A
P4	6	4	B
P5	1	0	B
P6	10	0	B
P7	7	8	B



- (A) Euclidean: A; Manhattan: A
(C) Euclidean: A; Manhattan: B

- (B) Euclidean: B; Manhattan: A
(D) Euclidean: B; Manhattan: B

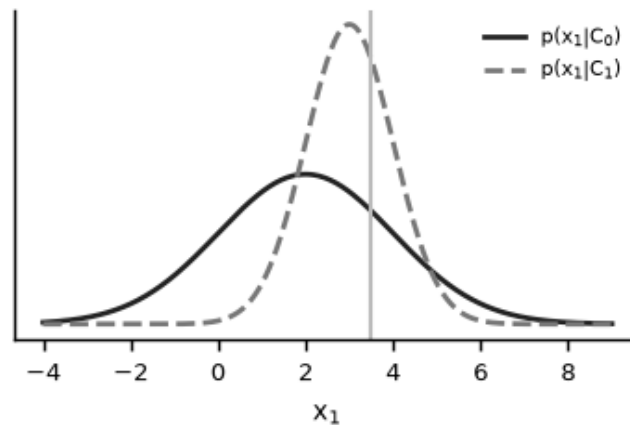
Q.34 [2 marks · NAT (numerical answer)]

A centered data matrix X with $n = 51$ rows (samples) and $d = 3$ columns has singular values 16, 12, 6. Using the sample covariance with divisor $n - 1$, the total variance (trace of the sample covariance matrix) is _____ (rounded off to two decimal places)

Q.35 [2 marks · NAT (numerical answer)]

A Gaussian naive Bayes classifier with feature x_1 has the class priors and per-class Gaussian parameters below (σ^2 = variance). For the input $x = 3.5$, the posterior probability $P(C_1 | x)$ is _____ (rounded off to two decimal places)

class	prior	$\mu(x_1)$	$\sigma^2(x_1)$
C_0	0.4	2	4
C_1	0.6	3	1



— END OF QUESTION PAPER —

Mock Test 33 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	11.99 to 12.01	19	NAT	2	0.0502 to 0.0702
2	MSQ	1	(B), (D)	20	MSQ	2	(A), (B), (D)
3	MSQ	1	(A), (B)	21	NAT	2	5
4	NAT	1	6.59 to 6.61	22	MCQ	2	(B)
5	MSQ	1	(A), (C)	23	NAT	2	-0.0101 to -0.0041
6	NAT	1	59	24	MSQ	2	(A), (B), (C), (D)
7	MCQ	1	(D)	25	MCQ	2	(B)
8	NAT	1	0.3348 to 0.3548	26	NAT	2	0.55 to 0.57
9	NAT	1	0.1333 to 0.1373	27	NAT	2	5
10	NAT	1	4.99 to 5.01	28	NAT	2	2.99 to 3.01
11	MSQ	1	(B)	29	NAT	2	0
12	NAT	1	15	30	NAT	2	0.0285 to 0.0315
13	MCQ	1	(A)	31	NAT	2	2.7899 to 2.7939
14	MCQ	1	(B)	32	NAT	2	-8.5563 to -8.4542
15	NAT	1	0.5407 to 0.5607	33	MCQ	2	(B)
16	NAT	2	0.2438 to 0.2478	34	NAT	2	8.71 to 8.73
17	NAT	2	-0.4475 to -0.4275	35	NAT	2	0.7681 to 0.7881
18	NAT	2	0.7541 to 0.7581				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Probabilistic & Instance-based Classifiers					11
Regression & Model Selection					11
Neural Networks (MLP / Feed-forward)					11
Unsupervised Learning					11

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
SVM & Decision Trees					11
Total					55

Mock Test 33 — Detailed Solutions

Q.1 — Answer: 11.99 to 12.01 [NAT, 1M · Probabilistic & Instance-based Classifiers › k-nearest neighbours]

Distances $|x_i - 18.5|$: $x=18$: 0.5, $x=12$: 6.5, $x=7$: 11.5, $x=6$: 12.5, $x=5$: 13.5, $x=4$: 14.5, $x=3$: 15.5.

The 3 nearest neighbours are $x = 18, 12, 7$ with $y = 5, 22, 9$.

$\hat{y} = (5 + 22 + 9)/3 = 12$.

Q.2 — Answer: (B), (D) [MSQ, 1M · Regression & Model Selection › Ridge & Lasso]

(A) FALSE. Rescaling x_j changes how strongly β_j is penalised.

(B) TRUE. Its vertices lie on the coordinate axes.

(C) FALSE. $\lambda = 0$ (OLS) minimises training RSS; training error is non-decreasing in λ .

(D) TRUE. Sum of two convex functions.

Q.3 — Answer: (A), (B) [MSQ, 1M · Neural Networks (MLP / Feed-forward) › Perceptron]

Net input $a = w \cdot x + b = -3x_1 + (1)x_2 + (0)$; output +1 iff $a \geq 0$.

(P) $a = -3 \cdot 0 + (1) \cdot 1 + (0) = 1 \rightarrow +1$ (selected).

(Q) $a = -3 \cdot 0 + (1) \cdot 2 + (0) = 2 \rightarrow +1$ (selected).

(R) $a = -3 \cdot 2 + (1) \cdot 2 + (0) = -4 \rightarrow -1$.

(S) $a = -3 \cdot 3 + (1) \cdot 1 + (0) = -8 \rightarrow -1$.

Answer: (A), (B).

Q.4 — Answer: 6.59 to 6.61 [NAT, 1M · Unsupervised Learning › k-medoids]

Mean = $93/5 = 18.6$ (pulled towards the outlier 62).

Medoid = data point minimizing $\sum |x_i - m|$. Costs: $m=2$: 83, $m=4$: 77, $m=12$: 69, $m=13$: 70, $m=62$: 217.

The minimum is at $m = 12$ (the median), cost 69.

Answer: 6.6. The medoid is barely affected by the outlier, unlike the mean.

Q.5 — Answer: (A), (C) [MSQ, 1M · Regression & Model Selection › Bias-variance trade-off]

(A) TRUE. Bias falls, variance rises.

(B) FALSE. A square is non-negative.

(C) TRUE. A higher-degree model contains the lower-degree one as a special case.

(D) FALSE. That describes overfitting; underfitting has high training error.

Q.6 — Answer: 59 [NAT, 1M · Regression & Model Selection › Multiple linear regression]

A categorical feature with k levels needs $k - 1$ dummies (one level is the reference absorbed by the intercept; using all k dummies with an intercept gives perfect multicollinearity).

Parameters = 1 (intercept) + 3 (numeric) + $(5 - 1) + (4 - 1) = 11$.

Residual degrees of freedom = $n - (\text{number of coefficients}) = 70 - 11 = 59$.

Answer: 59.

Q.7 — Answer: (D) [MCQ, 1M · SVM & Decision Trees › SVM: soft margin & hinge loss]

$y \cdot f(x) = (-1) \cdot (3.5) = -3.5$.

Slack $\xi = \max(0, 1 - y f(x)) = 4.5$.

Rules: $y f \geq 1 \Rightarrow \xi = 0$ (on/outside margin); $0 < y f < 1 \Rightarrow$ inside margin, still correct; $y f = 0 \Rightarrow$ on the boundary ($\xi = 1$); $y f < 0 \Rightarrow$ misclassified ($\xi > 1$).

Hence: **Misclassified ($\xi > 1$)**.

Q.8 — Answer: 0.3348 to 0.3548 [NAT, 1M · Probabilistic & Instance-based Classifiers › Bayes optimal classifier]

Expected cost of predicting positive = $(1 - p) \cdot C_{FP}$; of predicting negative = $p \cdot C_{FN}$.

Predict positive iff $p \cdot 19 > (1 - p) \cdot 10 \Leftrightarrow p > C_{FP} / (C_{FP} + C_{FN}) = 10/29 = 0.34$.

Q.9 — Answer: 0.1333 to 0.1373 [NAT, 1M · SVM & Decision Trees › SVM: kernels]

$\|x - z\|^2 = 4 + 0 + 0 = 4$.

$$K = \exp(-0.5 \times 4) = \exp(-2) = \mathbf{0.135}.$$

Note: $K \in (0, 1]$, $K(x, x) = 1$, and K decreases as the points move apart or as γ grows.

Q.10 — Answer: 4.99 to 5.01 [NAT, 1M · SVM & Decision Trees > Decision trees: impurity measures]

$p_k = 3/96 = 1/32$ for every class $\Rightarrow H = -32 \cdot (1/32) \log_2(1/32) = \log_2 32 = \mathbf{5}$ — the maximum possible for 32 classes.

Q.11 — Answer: (B) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

(A) **FALSE**. Scaling changes the sharpness (temperature); only additive shifts leave it unchanged.

(B) **TRUE**. Left and right derivatives are 0 and 1.

(C) **FALSE**. $\sigma(0) = 1/2$.

(D) **FALSE**. It approaches a one-hot vector at $\arg\max z$ (uniform is the $T \rightarrow \infty$ limit).

Q.12 — Answer: 15 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Updates per epoch = $\lceil 1000/16 \rceil = 63$; epochs = $945/63 = \mathbf{15}$.

Q.13 — Answer: (A) [MCQ, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

$\log\text{-odds} = \beta_0 + \beta x + \dots$, so $\Delta(\log\text{-odds}) = \beta \cdot \Delta x = -0.7 \times 1 = -0.7$.

Odds are multiplied by $e^{\beta \Delta x} = e^{-0.7} = \mathbf{0.5}$.

Distractors: $\beta \cdot \Delta x$ is the change in log-odds (not a multiplier); e^β ignores Δx ; $\sigma(\beta \Delta x)$ is a probability, not an odds ratio.

Q.14 — Answer: (B) [MCQ, 1M · Unsupervised Learning > k-means]

The SSE drops steeply up to $k = 5$ and then decreases only slowly (relative drops: $k=2$: 46%, $k=3$: 59%, $k=4$: 64%, $k=5$: 47%, $k=6$: 8%, $k=7$: 8%, $k=8$: 10%, $k=9$: 7%). The bend ('elbow') is at $\mathbf{k = 5}$. $k = 9$ gives the lowest SSE but SSE always (weakly) decreases with k , so minimizing it would pick the largest k ; it is not a model-selection criterion by itself.

Q.15 — Answer: 0.5407 to 0.5607 [NAT, 1M · Unsupervised Learning > Cluster evaluation]

Majority count in each cluster: 13, 16, 9 (sum 38); $N = 69$.

Purity = $(1/N) \sum_k \max_j n_{kj} = 38/69 = 0.5507$.

Misclassified = $N - \sum \max = 31$.

Answer: **0.55**.

Purity is an external measure (needs class labels) and is trivially 1 if each point forms its own cluster, so it cannot be used to choose k .

Q.16 — Answer: 0.2438 to 0.2478 [NAT, 2M · Unsupervised Learning > k-means]

$k\text{-means}++$ picks the next centre x with probability $D(x)^2 / \sum_{x'} D(x')^2$, where $D(x)$ is the distance from x to the **nearest** centre already chosen (chosen centres have $D = 0$).

Point	P1	P2	P3	P4	P5	P6
$D(x)^2$	25	29	37	0	2	25

$\sum D^2 = 118$, so $P(P2) = 29/118 = \mathbf{0.246}$.

Common mistakes: using D instead of D^2 (that is not $k\text{-means}++$), or using the distance to the *farthest*/first centre instead of the nearest chosen centre.

Q.17 — Answer: -0.4475 to -0.4275 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$\mu_1 - \mu_0 = (5, -1)^T$; $\det S_W = 80$.

$$S_W^{-1} = \begin{bmatrix} 0.15 & -0.05 \\ -0.05 & 0.1 \end{bmatrix}$$

$w = S_W^{-1}(\mu_1 - \mu_0) = (0.8, -0.35)^T$.

$w_2/w_1 = -0.35/0.8 = \mathbf{-0.44}$. (The direction, not the length, matters.)

Note: using $\mu_1 - \mu_0$ without S_W^{-1} is correct only when $S_W \propto I$.

Q.18 — Answer: 0.7541 to 0.7581 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

$$a = w \cdot x + b = -1.5; \hat{y} = \sigma(a) = 0.1824.$$

$$L = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial L / \partial \hat{y} = \hat{y} - y = -0.8176; \partial \hat{y} / \partial a = \hat{y}(1 - \hat{y}) = 0.1491.$$

$$\delta = \partial L / \partial a = (\hat{y} - y)\hat{y}(1 - \hat{y}) = -0.1219 (= \partial L / \partial b).$$

$$\partial L / \partial w_1 = \delta \cdot x_1 = -0.1219 \times -2 = 0.2439.$$

$$w_1 \leftarrow w_1 - \eta \partial L / \partial w_1 = 1 - 1 \times (0.2439) = \mathbf{0.7561}.$$

Forgetting the factor $\hat{y}(1 - \hat{y})$ (i.e. treating the output as linear) is the most common error.

Q.19 — Answer: 0.0502 to 0.0702 [NAT, 2M · SVM & Decision Trees · Decision trees: split selection]

$$\text{Parent [9+, 21-]: } H = -(9/30)\log_2(9/30) - (21/30)\log_2(21/30) = 0.8813.$$

$$\text{Shape = Round: [6+, 8-], } H = -(6/14)\log_2(6/14) - (8/14)\log_2(8/14) = 0.9852.$$

$$\text{Shape = Square: [2+, 6-], } H = -(2/8)\log_2(2/8) - (6/8)\log_2(6/8) = 0.8113.$$

$$\text{Shape = Oval: [1+, 7-], } H = -(1/8)\log_2(1/8) - (7/8)\log_2(7/8) = 0.5436.$$

$$\text{Weighted child entropy} = (14/30)(0.9852) + (8/30)(0.8113) + (8/30)(0.5436) = 0.8211.$$

$$IG = 0.8813 - 0.8211 = 0.0602.$$

Answer: **0.06**.

Q.20 — Answer: (A), (B), (D) [MSQ, 2M · Unsupervised Learning · Hierarchical clustering]

(A) TRUE. It covers single, complete, average, Ward and others.

(B) TRUE. One clustering per number of clusters.

(C) FALSE. Such bridge points cause chaining and merge the clusters.

(D) TRUE. Monotone dendrograms define ultrametrics.

Q.21 — Answer: 5 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Expressiveness]

With identity activations, $\hat{y} = W_2(W_1x + b_1) + b_2 = (W_2W_1)x + (W_2b_1 + b_2)$: a single affine map.

$$\tilde{w}^T = W_2W_1 = (-6, 5); \tilde{b} = W_2b_1 + b_2 = 0.$$

Answer: **5**.

Hence stacking linear layers adds no expressive power – a non-linear activation is essential.

Q.22 — Answer: (B) [MCQ, 2M · Regression & Model Selection · Simple linear regression]

$$\bar{x} = 8.3333, \bar{y} = 13.1111.$$

$$S_{xx} = \Sigma x^2 - (\Sigma x)^2/n = 154, S_{yy} = \Sigma y^2 - (\Sigma y)^2/n = 304.8889, S_{xy} = \Sigma xy - \Sigma x \Sigma y/n = 198.6667.$$

$$\text{Regress x on y: } b' = S_{xy}/S_{yy} = 0.6516; a' = \bar{x} - b'\bar{y} = -0.2099.$$

$$\text{Line: } \hat{\mathbf{x}} = \mathbf{-0.21 + 0.65y}.$$

A common error is to invert the y-on-x line (slope 1/b), which minimises the wrong (vertical) errors.

Q.23 — Answer: -0.0101 to -0.0041 [NAT, 2M · Regression & Model Selection · Ridge & Lasso]

$$\beta_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y \text{ with } \lambda = 3.$$

$$X^T X = \begin{bmatrix} 10 & -4 \\ -4 & 9 \end{bmatrix}$$

$$X^T X + \lambda I = \begin{bmatrix} 13 & -4 \\ -4 & 12 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 16 \\ -5 \end{bmatrix}$$

$$\det = 13 \times 12 - (-4)^2 = 140; (X^T X + \lambda I)^{-1} = (1/140)[[12, 4], [4, 13]].$$

$$\beta = (1.2286, -0.0071).$$

Answer: **-0.007**.

Q.24 — Answer: (A), (B), (C), (D) [MSQ, 2M · Regression & Model Selection · Bias-variance trade-off]

(A) TRUE. Training 9.88 and validation 10.18 are both large — high bias.

(B) TRUE. Train 0.29 (lowest), validation 10.47 > 5.48.

(C) TRUE. Validation MSE is minimum (5.48) at degree 2.

(D) TRUE. Higher-degree polynomials are more flexible → higher variance.

Q.25 — Answer: (B) [MCQ, 2M · SVM & Decision Trees > SVM: margin & geometry]

For $n^T x - c = 0$, the geometric margin is $\min_i y_i s(n^T x_i - c) / \|n\|$ with orientation $s = \pm 1$ chosen so that the positive class is on the positive side; a negative value means some point is misclassified.

Hyperplane	$\ n\ $	$\min_i$ distance (signed, best orientation)
$x_1 + x_2 - 2.5 = 0$	1.414	0.354
$x_1 + x_2 - 3 = 0$	1.414	0.707
$x_1 + x_2 - 3.5 = 0$	1.414	0.354
$x_1 - 2x_2 + 7 = 0$	2.236	-4.025 (does not separate)

The largest margin is 0.707, for $x_1 + x_2 - 3 = 0$. Note that a boundary can separate the data perfectly yet pass close to some point — only the minimum distance matters.

Q.26 — Answer: 0.55 to 0.57 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

The ROC curve is a staircase: each vertical step is a positive and each horizontal step a negative. Area = Σ over horizontal steps of (width $1/N$) $\times$ (current TPR): $0 + 0.2 + 0.8 + 0.8 + 1$, times $1/5$. AUC = **0.56**.

Q.27 — Answer: 5 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

Total pairs = $C(8,2) = 28$.

a = pairs together in both = 5; b = pairs separated in both = 10; pairs together in U only = 11; together in V only = 2.

Rand index = $(a + b) / C(n,2) = (5 + 10) / 28 = 0.5357$.

Answer: **5**. (Cluster names do not matter: only whether pairs are together.)

Q.28 — Answer: 2.99 to 3.01 [NAT, 2M · SVM & Decision Trees > Regression trees]

Parent mean $\hat{y} = 3.333$, $SSE_0 = \Sigma(y - \hat{y})^2 = 11.333$.

Evaluate every split:

Split t	Left mean	Right mean	$SSE_L + SSE_R$
9	3	3.4	11.2
11.5	3	3.5	11
12.5	3.667	3	10.667
13.5	3.5	3	11
14.5	3	5	8

Best split $t = 14.5$: left mean 3, right mean 5, $SSE = 8$, reduction = 3.333.

Answer: **3**.

Q.29 — Answer: 0 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Rule: if $y_i(w \cdot x_i + b) \leq 0$, then $w \leftarrow w + \eta y_i x_i$, $b \leftarrow b + \eta y_i$ ($\eta = 0.5$). Start $w = (0, 0)$, $b = 0$.

Example 1: $a = 0$, $y \cdot a = 0 \rightarrow$ **mistake**, update $\rightarrow w = (0.5, 0.5)$, $b = 0.5$.

Example 2: $a = -0.5$, $y \cdot a = -0.5 \rightarrow$ **mistake**, update $\rightarrow w = (0.5, -0.5)$, $b = 1$.

Example 3: $a = 3$, $y \cdot a = -3 \rightarrow$ **mistake**, update $\rightarrow w = (-1, 0)$, $b = 0.5$.

Example 4: $a = 2.5$, $y \cdot a = -2.5 \rightarrow$ **mistake**, update $\rightarrow w = (0, 1)$, $b = 0$.

Required value = **0**.

Note: a point with net input exactly 0 counts as a mistake ($y \cdot a \leq 0$) — this is why the first example always triggers an update when starting from zero.

Q.30 — Answer: 0.0285 to 0.0315 [NAT, 2M · SVM & Decision Trees > Decision trees: overfitting & pruning]

$R(t) = e(t)/N$ with $N = 100$. For an internal node t with subtree T_t : $g(t) = [R(t) - R(T_t)] / (|T_t| - 1)$, where $|T_t|$ = number of leaves.

Node	$R(t)$	$R(T_t)$	$ T_t $	$g(t)$
t1	0.42	0.15	5	0.0675

t2	0.13	0.05	2	0.08
t3	0.2	0.1	3	0.05
t7	0.09	0.06	2	0.03

The weakest link is t7 with the minimum $g = \mathbf{0.03}$; for $\alpha \geq$ this value, collapsing t7 into a leaf does not increase R_α .

Q.31 — Answer: 2.7899 to 2.7939 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

$f(w) = (0.5/2)w^2 \Rightarrow \nabla f(w) = 0.5w$. Update: $v_t = \beta v_{t-1} + \nabla f(w_{t-1})$, $w_t = w_{t-1} - \eta v_t$, with $v_0 = 0$, $\beta = 0.8$, $\eta = 0.05$.

$t = 1$: $g = 0.5 \times 3 = 1.5$; $v_1 = 0.8 \times 0 + 1.5 = 1.5$; $w_1 = 3 - 0.05 \times 1.5 = 2.925$.

$t = 2$: $g = 0.5 \times 2.925 = 1.4625$; $v_2 = 0.8 \times 1.5 + 1.4625 = 2.6625$; $w_2 = 2.925 - 0.05 \times 2.6625 = 2.7919$.

Answer: **2.7919**. (Plain gradient descent without momentum would give $w_2 = 2.8519$.)

Q.32 — Answer: -8.5563 to -8.4542 [NAT, 2M · Regression & Model Selection › MLE & MAP view]
 $\bar{x} = 4.6$, $\bar{y} = 3.8$.

i	1	2	3	4	5
$x_i - \bar{x}$	-3.6	-1.6	-0.6	1.4	4.4
$y_i - \bar{y}$	-3.8	0.2	-2.8	2.2	4.2
$(x_i - \bar{x})(y_i - \bar{y})$	13.68	-0.32	1.68	3.08	18.48

$S_{xx} = \sum (x_i - \bar{x})^2 = 37.2$, $S_{xy} = 36.6$.

$\beta_1 = S_{xy}/S_{xx} = 0.9839$, $\beta_0 = \bar{y} - \beta_1 \bar{x} = -0.7258$.

$SSE = S_{yy} - S_{xy}^2/S_{xx} = 44.8 - 36.0097 = 8.7903$.

$\ln L = -(n/2) \ln(2\pi\sigma^2) - SSE/(2\sigma^2)$. At $\hat{\sigma}^2 = SSE/n$: $\ln L = -(n/2)[\ln(2\pi\hat{\sigma}^2) + 1]$.

$\hat{\sigma}^2 = 1.7581 \Rightarrow \ln L = -(5/2)[\ln(2\pi \times 1.7581) + 1] = -8.5052$.

Answer: **-8.51**.

Q.33 — Answer: (B) [MCQ, 2M · Probabilistic & Instance-based Classifiers › k-nearest neighbours]
Distances from $\mathbf{q}$:

Point	class	Euclidean d	Manhattan d
P1	A	5	5
P2	A	4.472	6
P3	A	3.606	5
P4	B	2.236	3
P5	B	4.243	6
P6	B	6.708	9
P7	B	5.831	8

Euclidean 3 nearest: P4(B), P3(A), P5(B) $\Rightarrow$ **B**.

Manhattan 3 nearest: P4(B), P1(A), P3(A) $\Rightarrow$ **A**.

Answer: **Euclidean: B; Manhattan: A**. The two metrics disagree because L_1 penalises diagonal offsets more than L_2 , changing the neighbour set.

Q.34 — Answer: 8.71 to 8.73 [NAT, 2M · Unsupervised Learning › PCA & dimensionality reduction]
With $X = U\Sigma V^T$ (X centered, $n \times d$), the sample covariance is $S = X^T X / (n - 1) = V(\Sigma^2 / (n - 1))V^T$, so the PC directions are the columns of V and the eigenvalues are $\lambda_j = \sigma_j^2 / (n - 1)$.

$\lambda = 16^2/50 = 5.12$, $12^2/50 = 2.88$, $6^2/50 = 0.72$.

Cumulative % variance: 58.72, 91.74, 100 (note the $(n - 1)$ factor cancels in proportions: $PVE = \sigma_j^2 / \Sigma \sigma^2$).

Answer: **8.72**. (Using σ instead of σ^2 is the classic mistake.)

Q.35 — Answer: 0.7681 to 0.7881 [NAT, 2M · Probabilistic & Instance-based Classifiers › Naive Bayes]

$N(x; \mu, \sigma^2) = \exp(-(x-\mu)^2/(2\sigma^2))/\sqrt{2\pi\sigma^2}$. Naive Bayes multiplies the per-feature densities.

C_0 : $N(3.5; 2, 4) = 0.15057$; $\times$ prior $0.4 \Rightarrow 0.060227$.

C_1 : $N(3.5; 3, 1) = 0.35207$; $\times$ prior $0.6 \Rightarrow 0.211239$.

$P(C_1 | x) = 0.211239/(0.060227 + 0.211239) = \mathbf{0.78}$.

Mock Test 34

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	3	8	24

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Unsupervised Learning	7	11
Regression & Model Selection	7	11
SVM & Decision Trees	7	11
Probabilistic & Instance-based Classifiers	7	11
Neural Networks (MLP / Feed-forward)	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

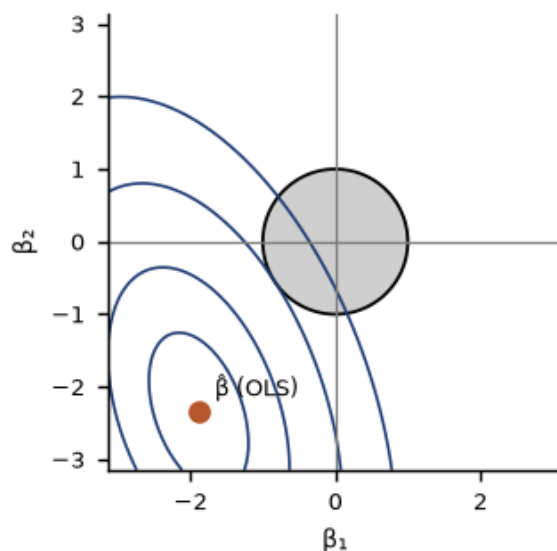
Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

A complete (binary) dendrogram over $n = 13$ points has how many nodes in total (leaves plus internal nodes)? (Answer in integer)

Q.2 [1 mark · MCQ]

For a two-predictor penalised least-squares problem, the figure shows the constraint region (shaded) of the penalty together with elliptical contours of the residual sum of squares centred at the least-squares estimate $\hat{\beta}$ (OLS). The penalised estimate (β_1, β_2) satisfies



- (A) $\beta_1 = 0$ exactly and $\beta_2 \neq 0$
- (B) $\beta_1 \neq 0$ and $\beta_2 \neq 0$ (both shrunk, neither exactly zero)
- (C) $\beta_1 = \beta_2 = 0$
- (D) $\beta_2 = 0$ exactly and $\beta_1 \neq 0$

Q.3 [1 mark · MSQ (one or more correct)]

Consider the soft-margin SVM: minimise $\frac{1}{2}\|w\|^2 + C \sum_i \xi_i$ subject to $y_i(w^T x_i + b) \geq 1 - \xi_i$, $\xi_i \geq 0$. Which of the following statements is/are CORRECT?

- (A) As $C \rightarrow \infty$, the soft-margin SVM approaches the hard-margin SVM on separable data.
- (B) Points with $0 < \alpha_i < C$ lie exactly on the margin ($\xi_i = 0$).
- (C) The soft-margin SVM has a solution even when the data are not linearly separable.
- (D) In the soft-margin dual the multipliers satisfy the box constraint $0 \leq \alpha_i \leq C$.

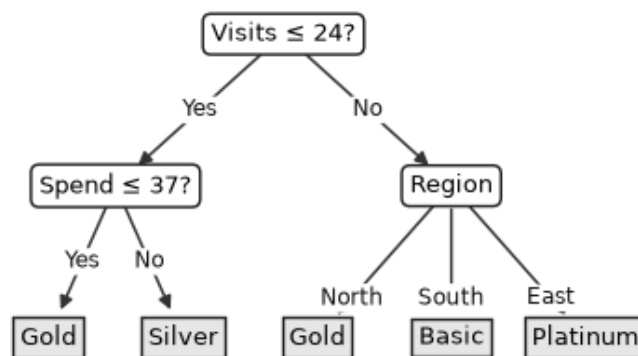
Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about principal component analysis (PCA) is/are CORRECT?

- (A) The proportion of variance explained by the first k PCs is $(\lambda_1 + \dots + \lambda_k)/(\lambda_1 + \dots + \lambda_d)$.
- (B) A scree plot shows the eigenvalues (or variance explained) in decreasing order against component number.
- (C) The principal component directions are eigenvectors of the sample covariance matrix.
- (D) The sample covariance matrix of n points has at most $\min(n - 1, d)$ non-zero eigenvalues.

Q.5 [1 mark · MCQ]

The decision tree below predicts the membership tier offered to a customer. Edges labelled Yes/No answer the test in the node. What is the prediction for the record shown in the table?



Visits	Spend	Tenure	Region
2	6	11	East

- (A) Gold
- (B) Basic
- (C) Platinum
- (D) Silver

Q.6 [1 mark · NAT (numerical answer)]

The homogeneous polynomial kernel $K(x, z) = (x^T z)^4$ on $x \in \mathbb{R}^3$ corresponds to a feature map of all (scaled) monomials of degree exactly 4. The number of distinct such features is _____. (Answer in integer)

Q.7 [1 mark · NAT (numerical answer)]

Lloyd's k-means is run on $n = 200$ points in $d = 3$ dimensions with $k = 6$ for $T = 9$ iterations. Counting one subtraction per coordinate per point-centroid pair in the assignment step, the total number of such subtractions is _____ (Answer in integer)

Q.8 [1 mark · MSQ (one or more correct)]

Which of the following statements about logistic regression is/are CORRECT?

- (A) Increasing x_j by one unit (others fixed) multiplies the odds of $y = 1$ by e^{β_j} .
- (B) For finite z , the sigmoid output $\sigma(z)$ can be exactly 0 or exactly 1.
- (C) The derivative of the sigmoid is $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.
- (D) Multiplying w and b by the same positive constant does not change the 0.5-threshold decision boundary.

Q.9 [1 mark · MSQ (one or more correct)]

Which of the following statements about naive Bayes classifiers is/are CORRECT?

- (A) With d binary features and K classes, naive Bayes needs $Kd + K - 1$ independent parameters.
- (B) Naive Bayes assumes the features are marginally (unconditionally) independent.
- (C) Naive Bayes posterior probabilities are often over-confident (close to 0 or 1) when features are correlated.
- (D) Fitting categorical naive Bayes requires iterative gradient-based optimisation.

Q.10 [1 mark · NAT (numerical answer)]

In a training set, the 4 examples of class Buys = Yes have the following counts for the feature Age. Using Laplace (add-one) smoothing, the naive Bayes estimate of $P(\text{Age} = \text{Young} \mid \text{Buys} = \text{Yes})$ is _____ (rounded off to three decimal places)

Age	Young	Middle	Senior
count in Buys = Yes	4	0	0

Q.11 [1 mark · MSQ (one or more correct)]

Which of the following statements about the representational power of feed-forward neural networks is/are CORRECT?

- (A) XOR can be computed by an MLP with one hidden layer containing two threshold units.
- (B) A single sigmoid neuron can represent XOR because the sigmoid is non-linear.
- (C) Some functions that a deep network represents with few units require exponentially many units in a shallow (one-hidden-layer) network.
- (D) A single sigmoid neuron (logistic regression) has a linear decision boundary.

Q.12 [1 mark · MSQ (one or more correct)]

Which of the following statements about cross-validation (CV) is/are CORRECT? (n denotes the number of observations.)

- (A) LOOCV is k -fold cross-validation with $k = n$.
- (B) The LOOCV estimate involves no randomness in the choice of splits.
- (C) k -fold CV of a single model configuration requires k model fits.
- (D) Nested CV gives a nearly unbiased estimate of generalisation error when hyperparameters are tuned by CV.

Q.13 [1 mark · NAT (numerical answer)]

A model has weights $w = (-1, -2, 0.5)$. The L2 penalty term $(\lambda/2)\|w\|^2$ with $\lambda = 1$ added to the loss equals _____ (rounded off to three decimal places)

Q.14 [1 mark · NAT (numerical answer)]

A fully connected MLP has layer sizes 50-128-16-10-10 (input-...-output), every layer having a bias vector. The number of bias parameters is _____ (Answer in integer)

Q.15 [1 mark · MSQ (one or more correct)]

Which of the following statements about basis-function (polynomial) regression and feature scaling is/are CORRECT?

- (A) Polynomial regression is non-linear in its parameters and therefore cannot be solved with the normal equations.
- (B) Gradient descent for linear regression usually converges faster when features are on comparable scales.
- (C) Min-max scaling maps the training values of a feature onto $[0, 1]$.
- (D) If all features are centred to mean zero, the OLS intercept equals $\bar{y}$.

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · MSQ (one or more correct)]**

A linear regression of house price (₹ lakh) on Area (area in hundreds of sq. ft), Furnished (1 if furnished, 0 otherwise) and zone $\in \{\text{North, South, East}\}$ encoded by dummies ZoneSouth and ZoneEast (reference level North) gives the fitted model

$$\hat{y} = 22 + 2.4 \cdot \text{Area} + 0.7 \cdot \text{Furnished} - 4.6 \cdot \text{ZoneSouth} - 2.7 \cdot \text{ZoneEast}$$

Which of the following is/are CORRECT?

- (A) Holding other variables fixed, the predicted response for zone South minus that for zone East is -1.9 .
- (B) The predicted house price for a furnished house in the East zone with area 7 hundred sq. ft is 36.8.
- (C) Increasing Area by 2 hundred sq. ft, other variables fixed, changes the prediction by 4.8.
- (D) Adding a third dummy for zone North while keeping the intercept would make $X^T X$ singular.

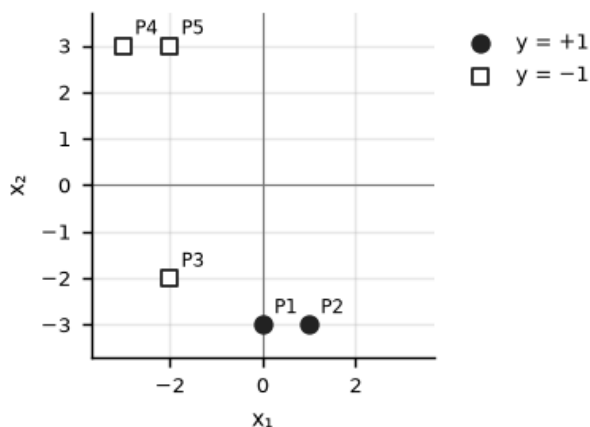
Q.17 [2 marks · NAT (numerical answer)]

$B = 10$ regression trees are averaged. Each tree's prediction has variance $\sigma^2 = 4$, and every pair of trees has correlation $\rho = 0.2$. The variance of the averaged prediction is _____ (rounded off to two decimal places)

Q.18 [2 marks · NAT (numerical answer)]

A perceptron with augmented weight vector $w = (w_0, w_1, w_2)$ (w_0 is the bias, $\tilde{x} = (1, x_1, x_2)$) starts from $w = (0, 0, 0)$ and uses learning rate $\eta = 1$. The points P1, P2, ... are presented once, in order; whenever $y \cdot w^T \tilde{x} \leq 0$ the update $w \leftarrow w + \eta y \tilde{x}$ is applied. Find the value of w_2 after this epoch. (Answer in integer)

Point	x_1	x_2	y
P1	0	-3	+1
P2	1	-3	+1
P3	-2	-2	-1
P4	-3	3	-1
P5	-2	3	-1

**Q.19 [2 marks · NAT (numerical answer)]**

The labelled data below are linearly separable through the origin by the unit vector $u = (1, 0)$ (i.e. $y_i = \text{sign}(u \cdot x_i)$). Take $\gamma = \min_i y_i(u \cdot x_i)$ and $R = \max_i \|x_i\|$. The perceptron convergence theorem (no bias, $w_0 = 0$, $\eta = 1$) bounds the number of mistakes by $(R/\gamma)^2$. The value of this bound is _____ (rounded off to two decimal places)

i	1	2	3	4	5
x_i	(2, 4)	(-2, 3)	(4, 3)	(-1, 4)	(1, 0)
y_i	1	-1	1	-1	1

Q.20 [2 marks · NAT (numerical answer)]

The eigenvalues of the sample covariance matrix (divisor $n - 1$) of a centered 6-dimensional dataset are 17, 11, 7, 5.5, 5, 2.5. Then the percentage of total variance **lost** by keeping only the top 2 principal components is _____ (rounded off to two decimal places)

Q.21 [2 marks · NAT (numerical answer)]

A naive Bayes classifier is trained on the 10 examples below, using maximum-likelihood estimates (no smoothing). For the new instance (Fever = Mild, Cough = Yes, Fatigue = No), the posterior probability $P(\text{Flu} = \text{Pos} \mid x)$ is _____ (rounded off to two decimal places)

#	Fever	Cough	Fatigue	Flu
1	High	Yes	No	Pos
2	Mild	Yes	Yes	Pos
3	High	Yes	Yes	Pos
4	Mild	Yes	Yes	Pos
5	Mild	No	No	Neg
6	High	No	Yes	Neg
7	Mild	Yes	No	Neg
8	None	No	No	Pos
9	High	Yes	Yes	Pos
10	Mild	No	Yes	Pos

Q.22 [2 marks · MCQ]

Six points have the pairwise distances below. Agglomerative clustering with **single linkage** is run and the dendrogram is cut to give exactly **2** clusters. The resulting clusters are

	P1	P2	P3	P4	P5	P6
P1	0	1	7	8	13	7
P2		0	8	9	14	6
P3			0	11	10	14
P4				0	5	3
P5					0	8
P6						0

(A) {P1, P2, P3, P4, P5}, {P6}

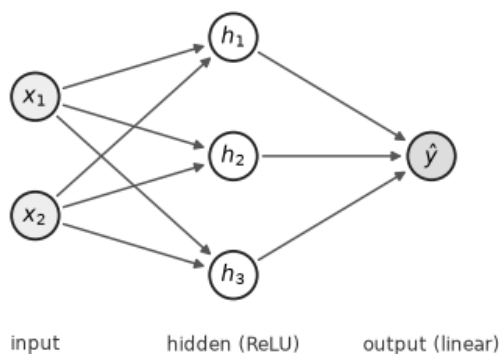
(C) {P1, P2, P4, P5, P6}, {P3}

(B) {P1, P2}, {P3}, {P4, P5, P6}

(D) {P1, P2, P3}, {P4, P5, P6}

Q.23 [2 marks · NAT (numerical answer)]

The 2-3-1 network below computes $\hat{y} = v \cdot \text{ReLU}(Wx + b) + c$ with W , b , v given below and $c = 0$. With $x = (2, 1)$, target $y = -2$ and loss $L = \frac{1}{2}(\hat{y} - y)^2$, the value of $\partial L / \partial x_2$ (gradient with respect to the input) is _____ (Answer in integer)



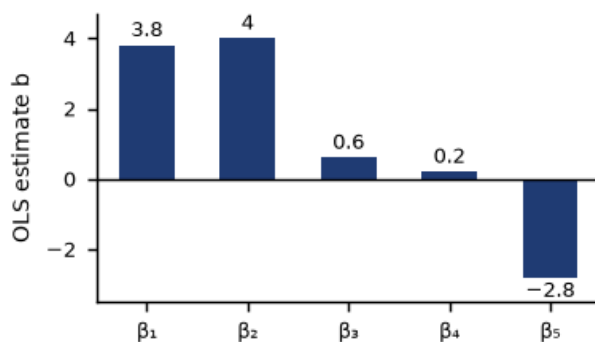
$$W = \begin{bmatrix} -2 & 1 \\ -2 & -2 \\ 0 & 0 \end{bmatrix}$$

$$b = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$v^T = \begin{bmatrix} 1 & -2 & -1 \end{bmatrix}$$

Q.24 [2 marks · NAT (numerical answer)]

In a regression problem the design matrix X has orthonormal columns ($X^T X = I$). The ordinary least-squares estimates $b = X^T y$ are shown below. Ridge minimises $\|y - X\beta\|^2 + \lambda \|\beta\|_2^2$ and the lasso minimises $\frac{1}{2} \|y - X\beta\|^2 + \lambda \|\beta\|_1$, each with $\lambda = 2$. Then the sum $\beta_1 + \dots + \beta_5$ of the lasso coefficients is _____ (rounded off to two decimal places)

**Q.25 [2 marks · NAT (numerical answer)]**

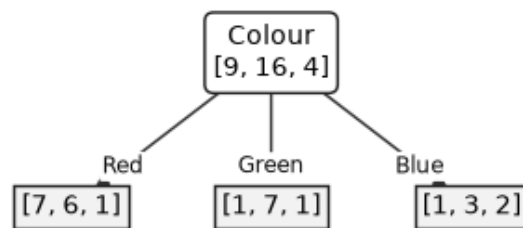
A softmax-regression output layer computes $z = Wx + b$, $p = \text{softmax}(z)$ with W and b below, and is trained with cross-entropy loss $L = -\ln p_y$. For $x = (2, 1)^T$ with true class $y = 2$, $\partial L / \partial b_3$ is _____ (rounded off to three decimal places)

$$W = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 1 \\ 0 & 1 \end{bmatrix}$$

$$b = \begin{bmatrix} 0 \\ -0.5 \\ 0 \end{bmatrix}$$

Q.26 [2 marks · NAT (numerical answer)]

A CART node with 29 training records (class counts [C1, C2, C3]) is split on Colour as shown. Compute the reduction in Gini index (Gini gain) obtained by splitting on Colour. (rounded off to two decimal places)

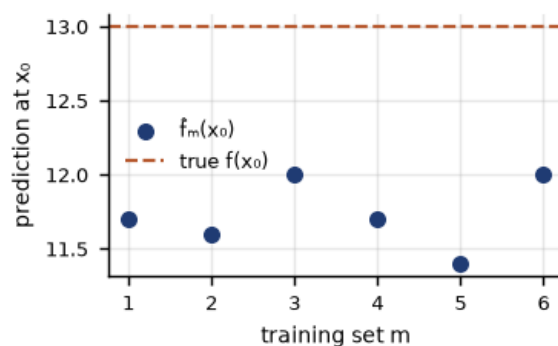
**Q.27 [2 marks · NAT (numerical answer)]**

A classifier for a rare condition has sensitivity (recall) 0.9 and specificity 0.97. It is deployed on a population where the prevalence of the positive class is 0.2. The expected negative predictive value in this population is _____ (rounded off to three decimal places)

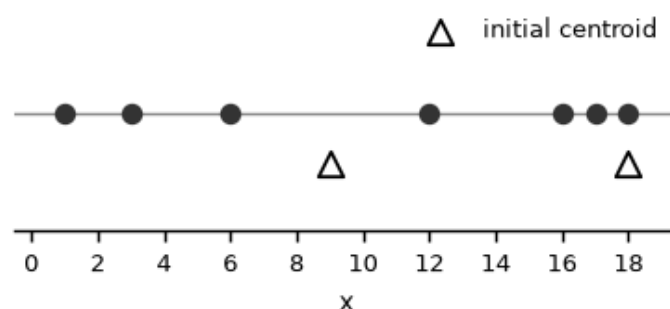
Q.28 [2 marks · NAT (numerical answer)]

Data are generated as $y = f(x) + \varepsilon$ with $E[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2 = 1$. At a test point x_0 the true value is $f(x_0) = 13$. A learning method is trained on 6 independent training sets, giving the predictions $\hat{f}_m(x_0)$ in the table (also plotted). Treating these as the full distribution of the method's prediction (variance with divisor 6), the expected squared prediction error $E[(y_0 - \hat{f}(x_0))^2]$ at x_0 is _____ (rounded off to three decimal places)

m	1	2	3	4	5	6
$\hat{f}_m(x_0)$	11.7	11.6	12	11.7	11.4	12

**Q.29 [2 marks · NAT (numerical answer)]**

The 1-D dataset $\{1, 3, 6, 12, 16, 17, 18\}$ is clustered with k-means ($k = 2$) using initial centroids 9 and 18 (shown as triangles). Lloyd's iterations (assign to nearest centroid, then recompute means) are run until convergence. Then the final within-cluster sum of squared errors (SSE) is _____ (rounded off to two decimal places)



Q.30 [2 marks · NAT (numerical answer)]

A 3-class softmax (multinomial logistic) regression model computes scores $\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}$ and $P(y = k | \mathbf{x}) = \frac{e^{z_k}}{\sum_j e^{z_j}}$, where row k of $\mathbf{W}$ holds the weights of class k . For the parameters below and $\mathbf{x} = (-1, 1)^T$, the predicted probability $P(y = 3 | \mathbf{x})$ is _____ (rounded off to two decimal places)

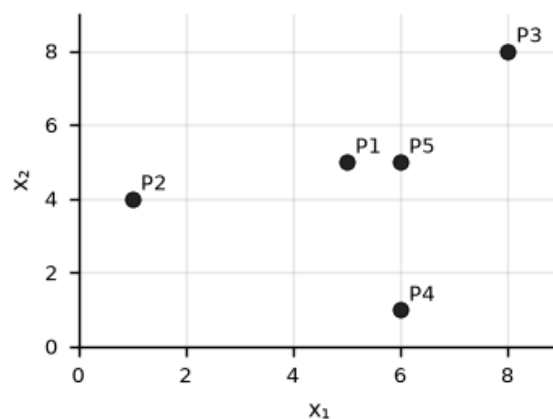
$$\mathbf{W} = \begin{bmatrix} 1 & -2 \\ 2 & 0 \\ 2 & -2 \end{bmatrix}$$

$$\mathbf{b} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Q.31 [2 marks · NAT (numerical answer)]

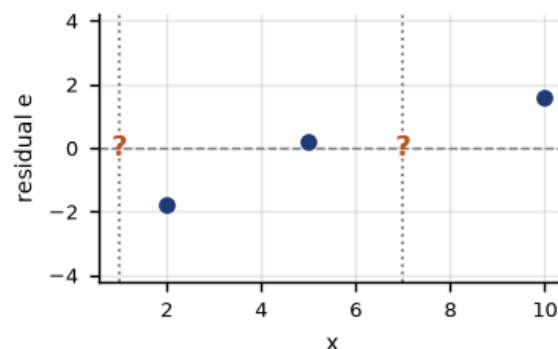
A cluster contains the 5 points below. Distances are Euclidean (L_2). The medoid is the point that minimizes the sum of distances to all points of the cluster. The minimum total distance (cost of the medoid) is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5
x_1	5	1	8	6	6
x_2	5	4	8	1	5

**Q.32 [2 marks · NAT (numerical answer)]**

A simple linear regression with intercept is fitted by least squares. The table gives the x -values and the residuals $e_i = y_i - \hat{y}_i$, but two residuals are missing. The fitted line is $\hat{y} = 3.2 + 1.7x$. The observed value y at $x = 1$ is _____ (rounded off to two decimal places)

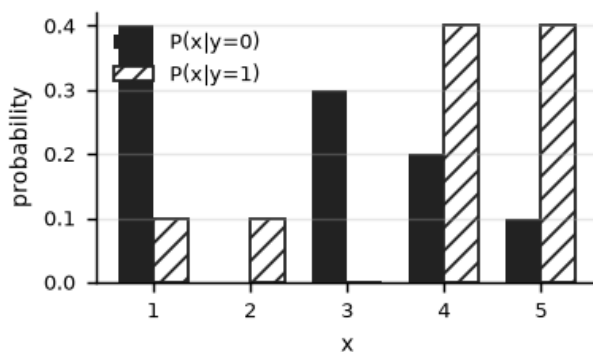
x_i	1	2	5	7	10
e_i	?	-1.8	0.2	?	1.6



Q.33 [2 marks · NAT (numerical answer)]

A discrete feature $x \in \{1, \dots, 5\}$ has the class-conditional distributions below (also plotted), with priors $P(y = 0) = 0.4$ and $P(y = 1) = 0.6$. Then the posterior probability $P(y = 1 \mid x = 3)$ is _____ (rounded off to three decimal places)

x	1	2	3	4	5
$P(x \mid y=0)$	0.4	0	0.3	0.2	0.1
$P(x \mid y=1)$	0.1	0.1	0	0.4	0.4

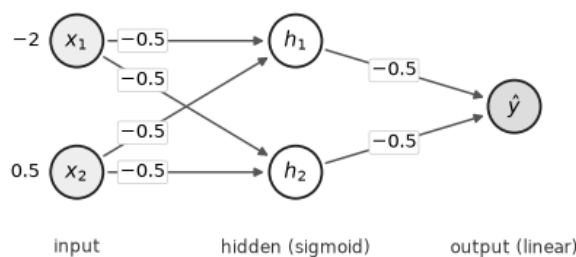
**Q.34 [2 marks · NAT (numerical answer)]**

In one AdaBoost round the current example weights and whether the chosen weak classifier h classifies each example correctly are given. Using $\alpha = \frac{1}{2} \ln((1 - \epsilon)/\epsilon)$ and the update $w_i \leftarrow w_i \cdot \exp(-\alpha y_i h(x_i))/Z$, find the normalisation constant $Z = \sum_i w_i \cdot \exp(-\alpha y_i h(x_i))$. (rounded off to three decimal places)

Example i	1	2	3	4	5	6	7
Weight w_i	2/23	5/23	1/23	1/23	4/23	5/23	5/23
$h(x_i)$ correct?	Yes	Yes	Yes	Yes	Yes	No	Yes

Q.35 [2 marks · NAT (numerical answer)]

In the 2-2-1 network below (sigmoid hidden units, linear output $\hat{y} = v_1 h_1 + v_2 h_2$, all biases 0), **every** weight is initialised to -0.5 . One gradient-descent step with $\eta = 0.1$ is taken on $L = \frac{1}{2}(\hat{y} - y)^2$ for the input $x = (-2, 0.5)$ with target $y = 2$. Then $w_{12} - w_{22}$ after the update (difference between the weights from x_2 into the two hidden units) is _____ (rounded off to 4 decimal places)



— END OF QUESTION PAPER —

Mock Test 34 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	25	19	NAT	2	24.95 to 25.05
2	MCQ	1	(B)	20	NAT	2	41.6567 to 41.6767
3	MSQ	1	(A), (B), (C), (D)	21	NAT	2	0.5694 to 0.5894
4	MSQ	1	(A), (B), (C), (D)	22	MCQ	2	(C)
5	MCQ	1	(A)	23	NAT	2	0
6	NAT	1	15	24	NAT	2	2.988 to 3.012
7	NAT	1	32400	25	NAT	2	0.5445 to 0.5485
8	MSQ	1	(A), (C), (D)	26	NAT	2	0.0579 to 0.0779
9	MSQ	1	(A), (C)	27	NAT	2	0.9729 to 0.9769
10	NAT	1	0.7123 to 0.7163	28	NAT	2	2.642 to 2.658
11	MSQ	1	(A), (C), (D)	29	NAT	2	33.4067 to 33.4267
12	MSQ	1	(A), (B), (C), (D)	30	NAT	2	0.032 to 0.052
13	NAT	1	2.623 to 2.627	31	NAT	2	13.4789 to 13.4989
14	NAT	1	164	32	NAT	2	7.0977 to 7.169
15	MSQ	1	(B), (C), (D)	33	NAT	2	−0.002 to 0.002
16	MSQ	2	(A), (B), (C), (D)	34	NAT	2	0.8229 to 0.8269
17	NAT	2	1.11 to 1.13	35	NAT	2	−0.0002 to 0.0002
18	NAT	2	−1				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Unsupervised Learning					11
Regression & Model Selection					11
SVM & Decision Trees					11
Probabilistic & Instance-based Classifiers					11
Neural Networks (MLP / Feed-forward)					11
Total					55

Mock Test 34 — Detailed Solutions

Q.1 — Answer: 25 [NAT, 1M · Unsupervised Learning > Hierarchical clustering]

n leaves + $(n - 1)$ internal (merge) nodes = $2n - 1 = 25$.

Q.2 — Answer: (B) [MCQ, 1M · Regression & Model Selection > Ridge & Lasso]

The shaded region is the constraint set of the ridge (L2: $\beta_1^2 + \beta_2^2 \leq t^2$, a disc). The penalised estimate is the first point at which the RSS contours (ellipses centred at β_{OLS} , which lies outside the region) touch the region.

From the figure the tangency point is approximately $(-0.84, -0.55)$.

The contour touches the boundary at a point that is not on an axis, so both coefficients are non-zero. For the disc this is the generic case — ridge shrinks but does not zero out.

Answer: **$\beta_1 \neq 0$ and $\beta_2 \neq 0$ (both shrunk, neither exactly zero)**. Both coefficients cannot be zero since the OLS point lies outside the region (the constraint is active but $t > 0$).

Q.3 — Answer: (A), (B), (C), (D) [MSQ, 1M · SVM & Decision Trees > SVM: soft margin & hinge loss]

(A) **TRUE**. Any slack becomes infinitely costly.

(B) **TRUE**. KKT: $\alpha_i < C \Rightarrow \xi_i = 0$, and $\alpha_i > 0 \Rightarrow y f = 1 - \xi$.

(C) **TRUE**. Slack variables make the problem always feasible.

(D) **TRUE**. C caps the influence of any single point.

Q.4 — Answer: (A), (B), (C), (D) [MSQ, 1M · Unsupervised Learning > PCA & dimensionality reduction]

(A) **TRUE**. Eigenvalues are variances along PCs.

(B) **TRUE**. Used to choose the number of PCs.

(C) **TRUE**. Standard result.

(D) **TRUE**. Centered data has rank $\leq n - 1$.

Q.5 — Answer: (A) [MCQ, 1M · SVM & Decision Trees > Decision trees: structure & traversal]

Follow the record from the root, taking the branch that matches each test:

Visits = $2 \leq 24 \rightarrow$ Yes.

Spend = $6 \leq 37 \rightarrow$ Yes.

(Only the tests on the actual path matter.) The record reaches the leaf **Gold**.

Note: " $\leq$ " includes equality, so a value equal to the threshold goes to the Yes branch.

Q.6 — Answer: 15 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

Monomials of degree exactly d in p variables = $C(p + d - 1, d)$ (stars and bars).

$C(6, 4) = 15$.

(Including all lower degrees, as for $(x^T z + 1)^4$, would give $C(7, 4) = 35$.)

Q.7 — Answer: 32400 [NAT, 1M · Unsupervised Learning > k-means]

Assignment cost per iteration = $n \cdot k \cdot d = 200 \cdot 6 \cdot 3 = 3600$; over T iterations **32400**. This is the $O(nkdT)$ complexity of k-means.

Q.8 — Answer: (A), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

(A) **TRUE**. The log-odds increase by β_j .

(B) **FALSE**. $0 < \sigma(z) < 1$ for every finite z .

(C) **TRUE**. Standard identity; maximal value 0.25 at $z = 0$.

(D) **TRUE**. The set $\{w^T x + b = 0\}$ is unchanged; only the confidence (steepness) changes.

Q.9 — Answer: (A), (C) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

(A) **TRUE**. Kd Bernoulli parameters + $K - 1$ priors.

(B) **FALSE**. The assumption is conditional independence given the class.

(C) **TRUE**. Correlated evidence is counted multiple times.

(D) **FALSE**. MLEs are closed-form frequencies.

Q.10 — Answer: 0.7123 to 0.7163 [NAT, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Add- α estimate: $(\text{count} + \alpha)/(N_c + \alpha V)$ with $V = 3$ possible values.

$$= (4 + 1)/(4 + 1 \times 3) = 5/7 = \mathbf{0.714}.$$

(The unsmoothed estimate would be $4/4$.)

Q.11 — Answer: (A), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Expressiveness]

(A) TRUE. E.g. $h_1 = \text{OR}$, $h_2 = \text{NAND}$, $y = \text{AND}(h_1, h_2)$.

(B) FALSE. Its decision boundary $w \cdot x + b = 0$ is still linear.

(C) TRUE. Depth-efficiency results (e.g. Telgarsky's sawtooth construction).

(D) TRUE. $\sigma(w \cdot x + b) = 0.5 \Leftrightarrow w \cdot x + b = 0$.

Q.12 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Cross-validation]

(A) TRUE. Each fold contains a single observation.

(B) TRUE. There is exactly one way to leave one out.

(C) TRUE. One fit per held-out fold.

(D) TRUE. The outer loop evaluates the whole tuning procedure on untouched data.

Q.13 — Answer: 2.623 to 2.627 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$$\|w\|^2 = 1 + 4 + 0.25 = 5.25; (\lambda/2)\|w\|^2 = 0.5 \times 5.25 = \mathbf{2.625}.$$

Q.14 — Answer: 164 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Parameter counting]

Fully connected layer with n_{in} inputs and n_{out} outputs: $n_{\text{in}} \cdot n_{\text{out}}$ weights + n_{out} biases.

Layer 1: $50 \times 128 = 6400$ weights, 128 biases.

Layer 2: $128 \times 16 = 2048$ weights, 16 biases.

Layer 3: $16 \times 10 = 160$ weights, 10 biases.

Layer 4: $10 \times 10 = 100$ weights, 10 biases.

Weights = 8708, biases = 164. Required: **164**.

Q.15 — Answer: (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Basis functions & feature scaling]

(A) FALSE. It is linear in the parameters.

(B) TRUE. Better conditioned Hessian.

(C) TRUE. $(x - \min)/(\max - \min)$.

(D) TRUE. $\beta_0 = \bar{y} - \sum \beta_j \bar{x}_j = \bar{y}$.

Q.16 — Answer: (A), (B), (C), (D) [MSQ, 2M · Regression & Model Selection > Multiple linear regression]

(A) TRUE. Difference of the two dummy coefficients: $-4.6 - (-2.7)$.

(B) TRUE. $\hat{y} = 22 + 2.4 \times 7 + (0.7) \times 1 + (-2.7) = 36.8$.

(C) TRUE. $2 \times 2.4 = 4.8$.

(D) TRUE. The three dummies would sum to the intercept column (dummy-variable trap).

Q.17 — Answer: 1.11 to 1.13 [NAT, 2M · SVM & Decision Trees > Ensembles: bagging & random forests]

$$\text{Var}((1/B)\Sigma f_p) = (1/B^2)[B\sigma^2 + B(B-1)\rho\sigma^2] = \rho\sigma^2 + (1-\rho)\sigma^2/B.$$

$$= 0.2 \times 4 + 0.8 \times 4/10 = 0.8 + 0.32 = \mathbf{1.12}.$$

Even with $B \rightarrow \infty$ the variance cannot fall below $\rho\sigma^2 = 0.8$ — which is why random forests try to reduce ρ by feature subsampling.

Q.18 — Answer: -1 [NAT, 2M · SVM & Decision Trees > Perceptron (linear separators)]

Trace the epoch (a point with $y \cdot w^T \tilde{x} \leq 0$, including 0, triggers an update):

Step	$\tilde{x} = (1, x_1, x_2)$	y	$w^T \tilde{x}$	$y \cdot w^T \tilde{x} \leq 0$?	w after step
1	(1, 0, -3)	+1	0	yes $\rightarrow$ update	(1, 0, -3)

2	(1, 1, -3)	+1	10	no	(1, 0, -3)
3	(1, -2, -2)	-1	7	yes → update	(0, 2, -1)
4	(1, -3, 3)	-1	-9	no	(0, 2, -1)
5	(1, -2, 3)	-1	-7	no	(0, 2, -1)

Final $w = (0, 2, -1)$, number of updates = 2. Answer: **-1**.

Q.19 — Answer: 24.95 to 25.05 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Unit separator $u = (1, 0)$. Margins $y_i(u \cdot x_i)$: 2, 2, 4, 1, 1 → $\gamma = \min = 1$.

Norms $\|x_i\|$: 4.472, 3.606, 5, 4.123, 1 → $R = \max = 5$.

Novikoff bound: number of mistakes $\leq (R/\gamma)^2 = (5/1)^2 = \mathbf{25}$.

Common mistakes: forgetting to normalise u (γ must be measured with $\|u\| = 1$), or using R/γ without squaring.

Q.20 — Answer: 41.6567 to 41.6767 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

For centered data, projecting onto the top- k PCs and reconstructing leaves the components along the discarded eigenvectors; their average squared length (divisor $n - 1$) is the sum of the discarded eigenvalues.

Discarded eigenvalues: 7, 5.5, 5, 2.5; sum = 20; total = 48.

Lost fraction = $20/48 = \mathbf{41.67\%}$.

Q.21 — Answer: 0.5694 to 0.5894 [NAT, 2M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Naive Bayes: $P(c | x) \propto P(c) \prod_j P(x_j | c)$.

Pos: $P(\text{Flu}=\text{Pos}) = 7/10$; $P(\text{Fever}=\text{Mild}|\text{Pos}) = 3/7$; $P(\text{Cough}=\text{Yes}|\text{Pos}) = 5/7$; $P(\text{Fatigue}=\text{No}|\text{Pos}) = 2/7$. Product = 0.06122.

Neg: $P(\text{Flu}=\text{Neg}) = 3/10$; $P(\text{Fever}=\text{Mild}|\text{Neg}) = 2/3$; $P(\text{Cough}=\text{Yes}|\text{Neg}) = 1/3$; $P(\text{Fatigue}=\text{No}|\text{Neg}) = 2/3$. Product = 0.04444.

$P(\text{Pos} | x) = 0.06122 / (0.06122 + 0.04444) = \mathbf{0.58}$.

Q.22 — Answer: (C) [MCQ, 2M · Unsupervised Learning > Hierarchical clustering]

Use single linkage: $d(A, B) = \min_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P1\} + \{P2\}$ at height **1**. Updated: $d(\{P1, P2\}, \{P3\}) = 7$, $d(\{P1, P2\}, \{P4\}) = 8$, $d(\{P1, P2\}, \{P5\}) = 13$, $d(\{P1, P2\}, \{P6\}) = 6$.

Merge 2: $\{P4\} + \{P6\}$ at height **3**. Updated: $d(\{P4, P6\}, \{P3\}) = 11$, $d(\{P4, P6\}, \{P5\}) = 5$, $d(\{P4, P6\}, \{P1, P2\}) = 6$.

Merge 3: $\{P5\} + \{P4, P6\}$ at height **5**. Updated: $d(\{P4, P5, P6\}, \{P3\}) = 10$, $d(\{P4, P5, P6\}, \{P1, P2\}) = 6$.

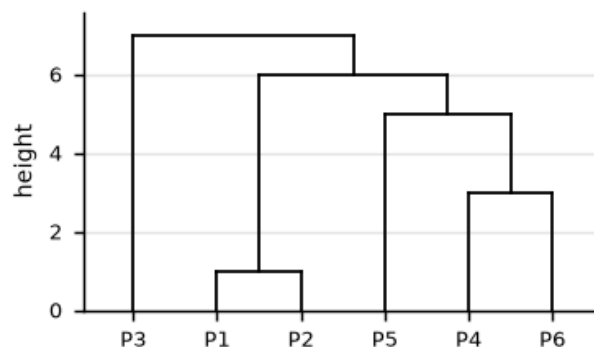
Merge 4: $\{P1, P2\} + \{P4, P5, P6\}$ at height **6**. Updated: $d(\{P1, P2, P4, P5, P6\}, \{P3\}) = 7$.

Merge 5: $\{P3\} + \{P1, P2, P4, P5, P6\}$ at height **7**.

Stopping when 2 clusters remain (i.e. after 4 merges): **$\{P1, P2, P4, P5, P6\}, \{P3\}$** .

$\{P1, P2, P3\}, \{P4, P5, P6\}$ is what *average* linkage would give.

$\{P1, P2\}, \{P3\}, \{P4, P5, P6\}$ has 3 clusters (cut at the wrong level).



Q.23 — Answer: 0 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

$a = Wx + b = (-4, -5, 1)$, $h = \text{ReLU}(a) = (0, 0, 1)$, $\hat{y} = v \cdot h + c = -1$; $\partial L / \partial \hat{y} = \hat{y} - y = 1$.

$\text{ReLU}'(a) = (0, 0, 1)$ (unit j is active iff $a_j > 0$).

$\partial L / \partial x_2 = (\hat{y} - y) \sum_j v_j \text{ReLU}'(a_j) W_{j2} = 1 \times (1 \cdot 0 \cdot 1 + -2 \cdot 0 \cdot -2 + -1 \cdot 1 \cdot 0) = 0$.

Q.24 — Answer: 2.988 to 3.012 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

With $X^T X = I$: $\beta_{\text{OLS}} = X^T y = b$.

Ridge: $\nabla[\|y - X\beta\|^2 + \lambda\|\beta\|^2] = 0 \Rightarrow (1 + \lambda)\beta = b \Rightarrow \beta_{\text{ridge}} = b/(1 + \lambda)$.

Lasso: the problem separates per coordinate, $\min \frac{1}{2}(\beta_j - b_j)^2 + \lambda|\beta_j| \Rightarrow \text{soft-thresholding } \beta_j = \text{sign}(b_j) \cdot \max(|b_j| - \lambda, 0)$.

j	1	2	3	4	5
b_j	3.8	4	0.6	0.2	-2.8
ridge	1.2667	1.3333	0.2	0.0667	-0.9333
lasso	1.8	2	0	0	-0.8

Answer: **3**.

Q.25 — Answer: 0.5445 to 0.5485 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$z = Wx + b = (0.5, -0.5, 1)$; $p = \text{softmax}(z) = (0.3315, 0.122, 0.5465)$.

$\delta = \partial L / \partial z = p - y = (0.3315, -0.878, 0.5465)$ (true class 2).

Since $z_k = \sum_j W_{kj} x_j + b_k$: $\partial L / \partial W_{kj} = (p_k - y_k) x_j$ and $\partial L / \partial b_k = p_k - y_k$ (gradient matrix = $(p - y)x^T$).

$\partial L / \partial b_3 = 0.5465$.

Q.26 — Answer: 0.0579 to 0.0779 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [9, 16, 4]: Gini = $1 - [(9/29)^2 + (16/29)^2 + (4/29)^2] = 0.5803$.

Colour = Red: [7, 6, 1], Gini = $1 - [(7/14)^2 + (6/14)^2 + (1/14)^2] = 0.5612$.

Colour = Green: [1, 7, 1], Gini = $1 - [(1/9)^2 + (7/9)^2 + (1/9)^2] = 0.3704$.

Colour = Blue: [1, 3, 2], Gini = $1 - [(1/6)^2 + (3/6)^2 + (2/6)^2] = 0.6111$.

Weighted Gini = $(14/29)(0.5612) + (9/29)(0.3704) + (6/29)(0.6111) = 0.5123$.

Gini gain = $0.5803 - 0.5123 = 0.0679$.

Answer: **0.07**.

Q.27 — Answer: 0.9729 to 0.9769 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Per unit population: $TP = Se \cdot \pi$, $FN = (1 - Se) \cdot \pi$, $FP = (1 - Sp)(1 - \pi)$, $TN = Sp(1 - \pi)$.

$TP = 0.18$, $FN = 0.02$, $FP = 0.024$, $TN = 0.776$.

Negative predictive value = $TN / (TN + FN) = 0.975$.

Q.28 — Answer: 2.642 to 2.658 [NAT, 2M · Regression & Model Selection > Bias-variance trade-off]

Mean prediction $\bar{f} = (1/6) \sum f_m = 11.7333$.

$\text{Bias}^2 = (\bar{f} - f(x_0))^2 = (11.7333 - 13)^2 = 1.6044$.

$\text{Variance} = (1/6) \sum (f_m - \bar{f})^2 = 0.0456$.

Expected test error = $\text{Bias}^2 + \text{Variance} + \sigma^2 = 1.6044 + 0.0456 + 1 = 2.65$.

Answer: **2.65**.

Q.29 — Answer: 33.4067 to 33.4267 [NAT, 2M · Unsupervised Learning > k-means]

Data: {1, 3, 6, 12, 16, 17, 18}; initial centroids $c_1 = 9$, $c_2 = 18$.

Iter	centroids used	cluster 1	cluster 2	new centroids
1	9, 18	{1, 3, 6, 12}	{16, 17, 18}	5.5, 17
2	5.5, 17	{1, 3, 6}	{12, 16, 17, 18}	3.33, 15.75
3	3.33, 15.75	{1, 3, 6}	{12, 16, 17, 18}	3.33, 15.75

In the last row the centroids do not change, so the algorithm has converged: final centroids 3.33 and 15.75.

$$\text{SSE} = (1 - 3.33)^2 + (3 - 3.33)^2 + (6 - 3.33)^2 + (12 - 15.75)^2 + (16 - 15.75)^2 + (17 - 15.75)^2 + (18 - 15.75)^2 = \mathbf{33.42}.$$

Q.30 — Answer: 0.032 to 0.052 [NAT, 2M · Probabilistic & Instance-based Classifiers · Logistic regression]

$$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b} = (-3, -1, -4).$$

$$\mathbf{e}^{\mathbf{z}} = (0.0498, 0.3679, 0.0183), \text{ sum} = 0.436.$$

$$\text{Softmax probabilities} = (0.1142, 0.8438, 0.042).$$

$$P(y = 3 | \mathbf{x}) = \mathbf{0.04}.$$

Q.31 — Answer: 13.4789 to 13.4989 [NAT, 2M · Unsupervised Learning · k-medoids]

Compute the Euclidean (L_2) distance matrix and its row sums:

	P1	P2	P3	P4	P5	Σ
P1	0	4.12	4.24	4.12	1	13.49
P2	4.12	0	8.06	5.83	5.1	23.12
P3	4.24	8.06	0	7.28	3.61	23.19
P4	4.12	5.83	7.28	0	4	21.23
P5	1	5.1	3.61	4	0	13.7

The smallest row sum is 13.49 for P1, so the medoid is P1.

Answer: **13.49**.

Unlike a k-means centroid, the medoid is always an actual data point.

Q.32 — Answer: 7.0977 to 7.169 [NAT, 2M · Regression & Model Selection · Simple linear regression]

For the least-squares fit with intercept the normal equations give $\Sigma e_i = 0$ and $\Sigma x_i e_i = 0$.

Known: $\Sigma e = 0$, $\Sigma xe = 13.4$. With unknowns $u = e$ at $x = 1$ and $v = e$ at $x = 7$:

$$u + v = 0, 1u + 7v = -13.4.$$

$$\text{Solving: } u = 2.2333, v = -2.2333.$$

$$y = \hat{y} + e = (3.2 + (1.7)(1)) + (2.2333) = 7.1333.$$

Answer: **7.13**.

Q.33 — Answer: -0.002 to 0.002 [NAT, 2M · Probabilistic & Instance-based Classifiers · Bayes optimal classifier]

$$P(y=1|x=3) = \pi_1 P(x|1) / (\pi_0 P(x|0) + \pi_1 P(x|1)) = 0 / (0.12 + 0) = \mathbf{0}.$$

Q.34 — Answer: 0.8229 to 0.8269 [NAT, 2M · SVM & Decision Trees · Ensembles: boosting]

$$\varepsilon = \sum_{\text{misclassified}} w_i = 0.2174; \alpha = \frac{1}{2} \ln(0.7826/0.2174) = 0.6405.$$

$$\text{Misclassified weights} \times e^\alpha = 1.8974; \text{correct weights} \times e^{-\alpha} = 0.527.$$

$$Z = \varepsilon e^\alpha + (1 - \varepsilon) e^{-\alpha} = 2\sqrt{\varepsilon(1 - \varepsilon)} = 0.8249.$$

$$\text{New weights: } 0.0556, 0.1389, 0.0278, 0.0278, 0.1111, 0.5, 0.1389.$$

Shortcut: after normalisation the misclassified set has total weight exactly $\frac{1}{2}$: $w_i' = w_i / (2\varepsilon)$ if misclassified, $w_i / (2(1 - \varepsilon))$ otherwise.

Answer: **0.825**.

Q.35 — Answer: -0.0002 to 0.0002 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Training & regularisation]

All weights = -0.5, biases 0 $\Rightarrow$ both hidden units get $a = -0.5 \times (-2 + 0.5) = 0.75$, $h_1 = h_2 = \sigma(a) = 0.6792$; $\hat{y} = 2 \times -0.5 \times 0.6792 = -0.6792$.

$$\hat{y} - y = -2.6792. \partial L / \partial v_j = (\hat{y} - y) h_j = -1.8196 \text{ (same for both } j).$$

$$\partial L / \partial w_{jk} = (\hat{y} - y) v_j h_j (1 - h_j) x_k = -2.6792 \times -0.5 \times 0.2179 \times 0.5 = 0.1459 \text{ (identical for } j = 1, 2).$$

Both hidden units receive identical gradients, so the difference stays **0**.

Because the two hidden units start identical and get identical updates, they remain identical forever (symmetry problem) - hence weights must be initialised randomly.

Mock Test 35

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	6	4	25

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
SVM & Decision Trees	7	11
Probabilistic & Instance-based Classifiers	7	11
Unsupervised Learning	7	11
Regression & Model Selection	7	11
Neural Networks (MLP / Feed-forward)	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

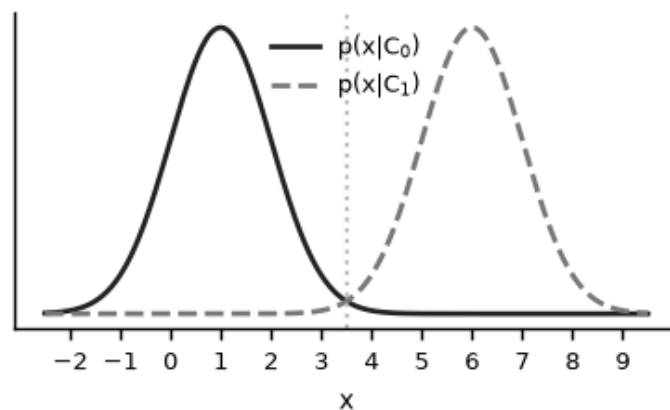
Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

A node of a classification tree contains 17 training records: 8 of class C_1 , 9 of class C_2 . The entropy (in bits) of this node is _____ (rounded off to two decimal places)

Q.2 [1 mark · MCQ]

Two classes have unit-variance Gaussian class-conditional densities with means 1 (C_0) and 6 (C_1), as plotted. With equal priors the Bayes decision threshold is the dotted line at $x = 3.5$. If the prior of C_1 is increased to 0.9, the threshold



- (A) It moves towards $x = 6$, enlarging the region assigned to C_1
- (B) It stays at $x = 3.5$ because the two variances are equal
- (C) It splits into two thresholds, giving a quadratic boundary
- (D) It moves towards $x = 1$, enlarging the region assigned to C_1

Q.3 [1 mark · NAT (numerical answer)]

A decision tree is built over 5 Boolean attributes, testing each attribute at most once on any root-to-leaf path. The maximum possible number of leaves is _____ (Answer in integer)

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about hierarchical clustering is/are CORRECT?

- (A) Single and complete linkage always produce the same dendrogram.
- (B) The cophenetic distance between two points always equals their original distance.
- (C) A merge made in agglomerative clustering can be undone in a later step.
- (D) Single-linkage clustering is closely related to the minimum spanning tree of the data.

Q.5 [1 mark · NAT (numerical answer)]

In a training set, the 9 examples of class Play = No have the following counts for the feature Outlook. Using Laplace (add-one) smoothing, the naive Bayes estimate of $P(\text{Outlook} = \text{Overcast} \mid \text{Play} = \text{No})$ is _____ (rounded off to three decimal places)

Outlook	Sunny	Overcast	Rain
count in Play = No	4	0	5

Q.6 [1 mark · NAT (numerical answer)]

The confusion matrix of a binary classifier on a test set is shown below (note which of rows/columns are actual and predicted). The recall (sensitivity, TPR) of the classifier is _____ (rounded off to two decimal places)

	Predicted +	Predicted –
Actual +	39	40
Actual –	20	5

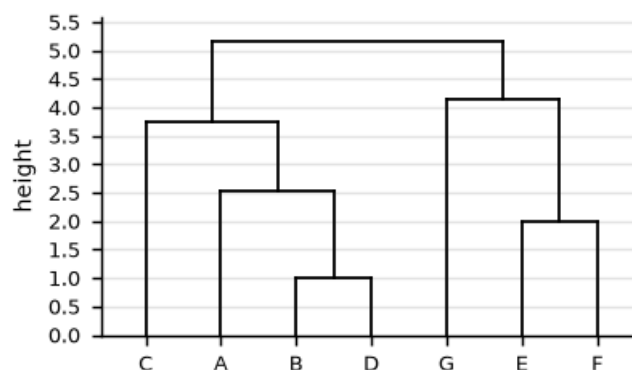
Q.7 [1 mark · MSQ (one or more correct)]

Which of the following statements about support vector machines is/are CORRECT?

- (A) Moving a non-support-vector point (keeping it strictly outside the margin) changes the SVM boundary.
- (B) Logistic regression is influenced by all training points, while the SVM solution depends only on support vectors.
- (C) Every training point contributes a non-zero term to w in the SVM solution.
- (D) In the SVM dual, the training data enter only through inner products $x_i^T x_j$.

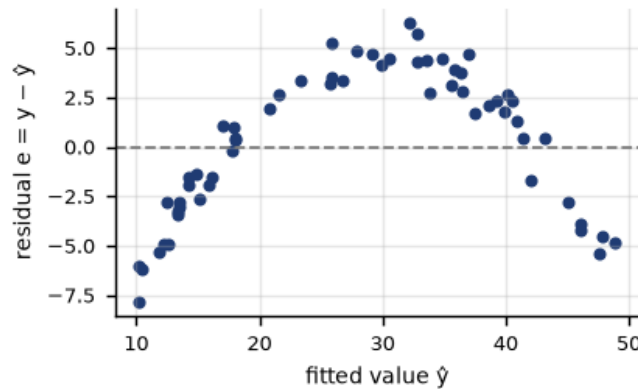
Q.8 [1 mark · NAT (numerical answer)]

The dendrogram below was obtained by agglomerative clustering of 7 points A-G. If the dendrogram is cut by a horizontal line at height 3.1, the number of clusters obtained is _____ (Answer in integer)



Q.9 [1 mark · MCQ]

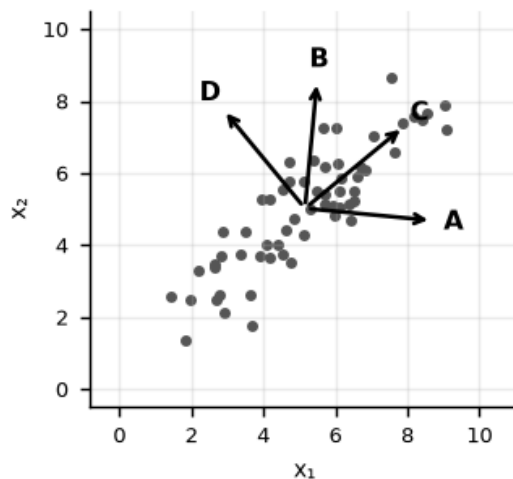
After fitting a linear regression model, the residuals are plotted against the fitted values as shown. The plot most strongly suggests:



- (A) A non-linear relationship not captured by the model (e.g. a missing x^2 term)
- (B) No evident violation of the linear-model assumptions
- (C) A single outlying observation, with the rest of the fit looking adequate
- (D) Non-constant error variance (heteroscedasticity)

Q.10 [1 mark · MCQ]

The scatter plot shows a 2-D dataset together with four candidate directions A-D drawn from the data mean. Which direction best represents the first principal component (PC1)?



- (A) Direction A
- (B) Direction B
- (C) Direction C
- (D) Direction D

Q.11 [1 mark · NAT (numerical answer)]

For logits $z = (0, 3, 0.5, 1.5)$, the softmax probability $p_2 = \exp(z_2) / \sum_j \exp(z_j)$ is _____ (rounded off to three decimal places)

Q.12 [1 mark · MCQ]

The 0-1 loss $L(y, \hat{y}) = 1[y \neq \hat{y}]$ is the natural loss for:

- (A) classification
- (B) ridge regression
- (C) clustering with k-means
- (D) principal component analysis

Q.13 [1 mark · NAT (numerical answer)]

Regressing predictor x_3 on all the other predictors gives $R^2 = 0.79$. The VIF of x_3 is _____ (rounded off to two decimal places)

Q.14 [1 mark · MSQ (one or more correct)]

Let $p = \text{softmax}(z)$ for logits $z \in \mathbb{R}^K$ and $L = -\ln p_y$ the cross-entropy loss for true class y . Which of the following is/are CORRECT?

- (A) For softmax outputs with cross-entropy loss and one-hot target y , $\partial L / \partial z = p - y$.
- (B) Adding the same constant to all logits leaves the softmax output and the cross-entropy loss unchanged.
- (C) The Jacobian of softmax is $\partial p_i / \partial z_j = p_i(\delta_{ij} - p_j)$.
- (D) For a confidently wrong prediction ($p_y \rightarrow 0$), the cross-entropy loss grows without bound.

Q.15 [1 mark · NAT (numerical answer)]

A layer of 1000 units uses dropout with keep probability $p = 0.9$. The expected number of units dropped (set to zero) in one forward pass during training is _____ (rounded off to two decimal places)

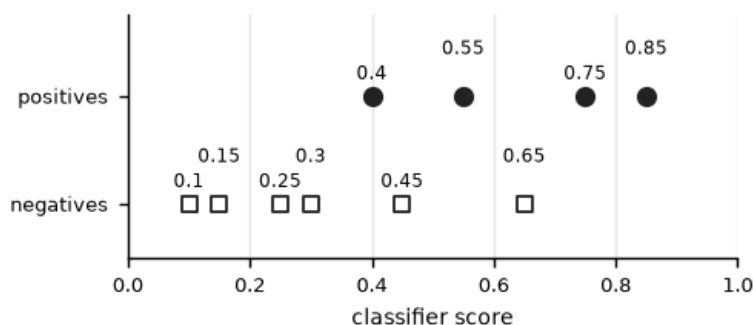
Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

For the kernel $K(x, z) = (x^T z + 1)^2$ on $\mathbb{R}^2$ with feature map ϕ , compute the cosine of the angle between $\phi(x)$ and $\phi(z)$, where $x = (-2, 2)$ and $z = (0, -1)$. (rounded off to two decimal places)

Q.17 [2 marks · NAT (numerical answer)]

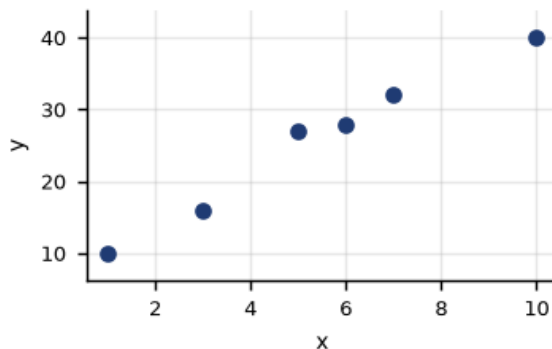
A classifier assigns the scores below to 10 test examples (4 positives, label 1; 6 negatives, label 0). The area under the ROC curve (AUC) is _____ (rounded off to two decimal places)

example	1	2	3	4	5	6	7	8	9	10
score	0.75	0.1	0.55	0.3	0.4	0.25	0.65	0.45	0.15	0.85
label	1	0	1	0	1	0	0	0	0	1

**Q.18 [2 marks · NAT (numerical answer)]**

A simple linear regression (with intercept) is fitted by least squares to the 6 points below. Using leave-one-out cross-validation, the leave-one-out residual $y_i - \hat{y}_{(-i)}(x_i)$ for the point (3, 16) — where $\hat{y}_{(-i)}$ is the line fitted without that point — is _____ (rounded off to two decimal places)

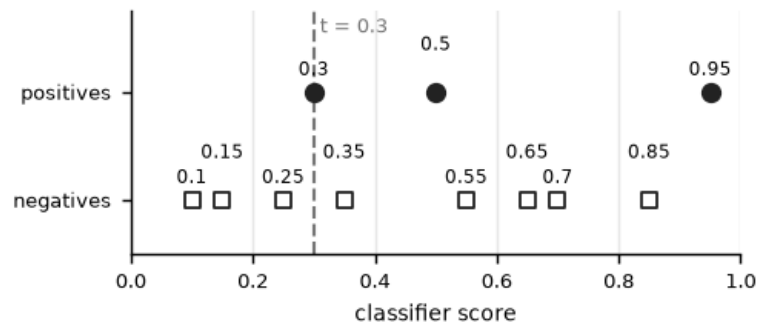
x	1	3	5	6	7	10
y	10	16	27	28	32	40



Q.19 [2 marks · NAT (numerical answer)]

A probabilistic classifier produces the scores below on 11 labelled examples (1 = positive). An example is predicted positive if its score is $\geq t$, with $t = 0.3$. The false positive rate at this threshold is _____ (rounded off to two decimal places)

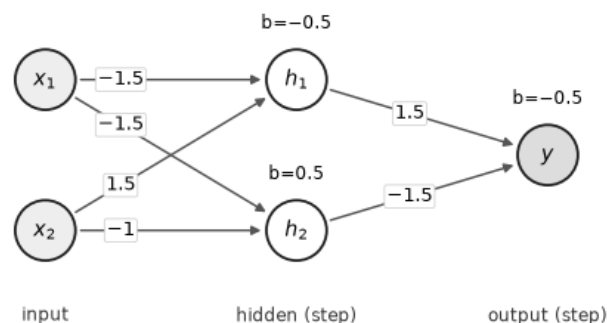
example	1	2	3	4	5	6	7	8	9	10	11
score	0.35	0.55	0.85	0.7	0.25	0.65	0.1	0.5	0.3	0.15	0.95
label	0	0	0	0	0	0	0	1	1	0	1

**Q.20 [2 marks · NAT (numerical answer)]**

The 2×2 matrix $X^T X$ of a (centred) design matrix X has eigenvalues 25, 9. For ridge regression with penalty $\lambda \|\beta\|_2^2$, the reduction in effective degrees of freedom relative to ordinary least squares, $p - df(\lambda)$, for $\lambda = 2$ is _____ (rounded off to two decimal places)

Q.21 [2 marks · MCQ]

In the network below every unit (hidden and output) is a threshold unit that outputs 1 if its weighted input plus bias is > 0 , and 0 otherwise. Edge labels are weights. For binary inputs $x_1, x_2 \in \{0, 1\}$, the network computes



- (A) (NOT x_1) AND x_2
(C) XOR

- (B) NOR
(D) OR

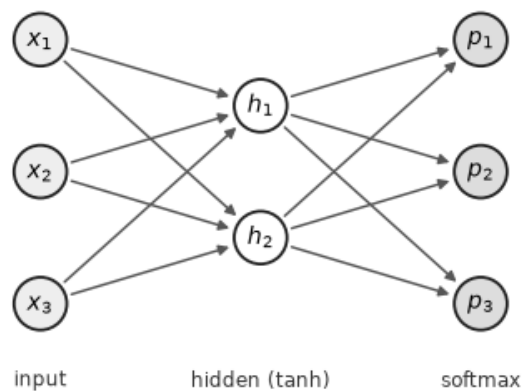
Q.22 [2 marks · NAT (numerical answer)]

Three customers are described by age and income. Age ranges over [20, 60] years and income over [10, 210] thousand rupees in the full dataset. On the **raw** (unscaled) features, the Euclidean distance from A to its nearest neighbour among B and C is _____ (rounded off to two decimal places)

Customer	Age (years)	Income (₹ thousand)
A	25	65
B	45	10
C	53	35

Q.23 [2 marks · NAT (numerical answer)]

A 3-2-3 network computes $h = \tanh(W_1 x + b_1)$, $z = W_2 h + b_2$, $p = \text{softmax}(z)$, with the parameters below. For $x = (1, 1, 1)^T$, the predicted probability p_2 of class 2 is _____ (rounded off to three decimal places)



$$W_1 = \begin{bmatrix} 0.5 & 1 & -0.5 \\ 1 & -1 & 0.5 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

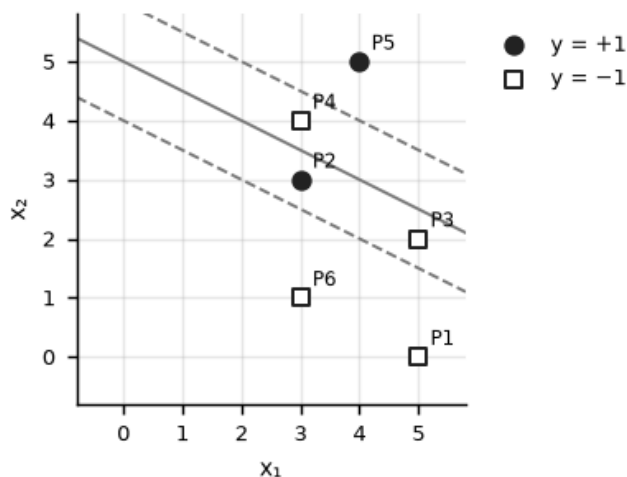
$$W_2 = \begin{bmatrix} -2 & 0 \\ 2 & -1 \\ 0 & 1 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 1 \\ 1 \\ -0.5 \end{bmatrix}$$

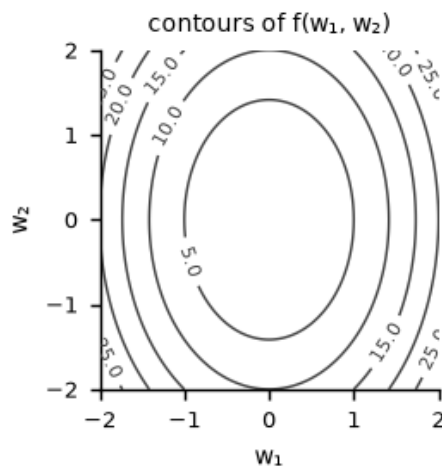
Q.24 [2 marks · NAT (numerical answer)]

A linear SVM with $f(x) = 0.5x_1 + x_2 - 5$ is evaluated on the training points in the table (also plotted, with the boundary $f = 0$ solid and $f = \pm 1$ dashed). Compute the average hinge loss $(1/n) \sum_i \max(0, 1 - y_i f(x_i))$. (rounded off to two decimal places)

Point	x_1	x_2	y
P1	5	0	-1
P2	3	3	+1
P3	5	2	-1
P4	3	4	-1
P5	4	5	+1
P6	3	1	-1

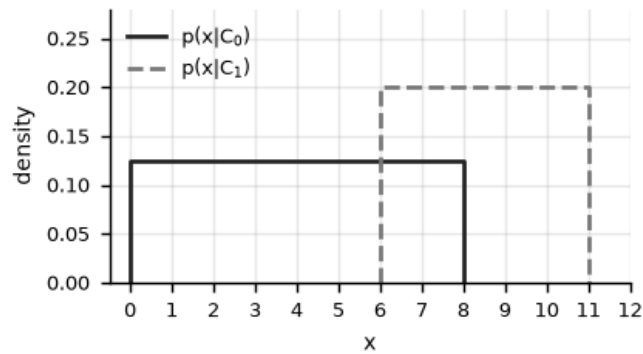
**Q.25 [2 marks · MSQ (one or more correct)]**

Gradient descent $w \leftarrow w - \eta \nabla f(w)$ is applied to $f(w_1, w_2) = \frac{1}{2}(10w_1^2 + 5w_2^2)$ from a generic starting point (both coordinates non-zero). For which of the following learning rates does GD **converge to the minimum**?

**(A)** $\eta = 0.19$ **(B)** $\eta = 0.12$ **(C)** $\eta = 0.3$ **(D)** $\eta = 0.03$

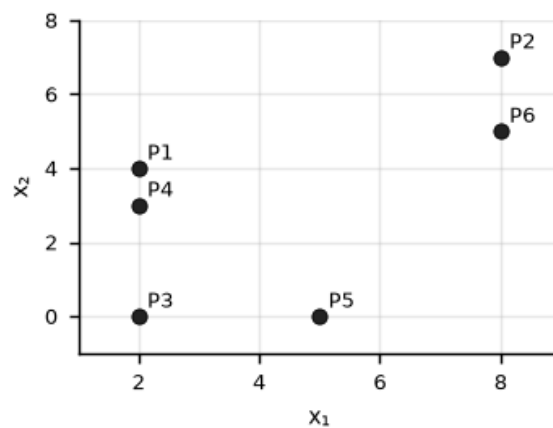
Q.26 [2 marks · NAT (numerical answer)]

For a binary problem with scalar x , $p(x | C_0)$ is uniform on $[0, 8]$ and $p(x | C_1)$ is uniform on $[6, 11]$, with priors $P(C_0) = 0.7$ and $P(C_1) = 0.3$. Then the error rate of the rule "predict C_0 if $x < 6.5$, else C_1 " is _____ (rounded off to three decimal places)

**Q.27 [2 marks · NAT (numerical answer)]**

Single-linkage agglomerative clustering with Manhattan distance is applied to the six points below. Then the height at which the **last** merge (forming a single cluster) occurs is _____ (Answer in integer)

Point	P1	P2	P3	P4	P5	P6
x_1	2	8	2	2	5	8
x_2	4	7	0	3	0	5



Q.28 [2 marks · MCQ]

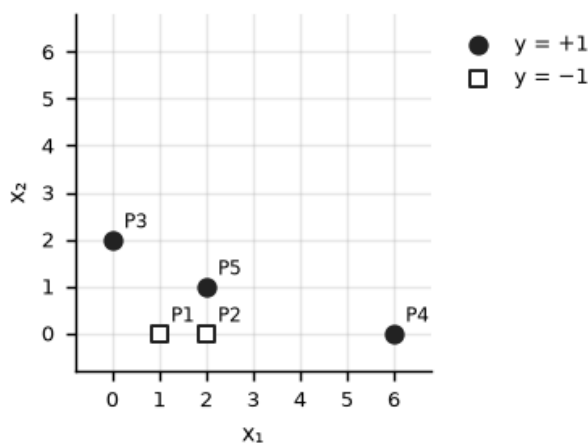
A decision tree is to be grown on the 12 training records below using Gini gain as the splitting criterion (multi-way split on each categorical attribute). Which attribute is chosen at the root?

#	Colour	Humidity	Size	Weekend	Class
1	Blue	Normal	Large	Yes	No
2	Green	High	Large	Yes	No
3	Green	Normal	Large	Yes	No
4	Blue	High	Large	Yes	Yes
5	Red	High	Large	Yes	No
6	Green	Normal	Large	No	Yes
7	Red	Normal	Large	No	Yes
8	Green	Normal	Small	Yes	No
9	Green	Normal	Small	Yes	No
10	Red	High	Large	Yes	No
11	Blue	High	Large	No	Yes
12	Red	High	Small	Yes	No

(A) Colour**(B)** Humidity**(C)** Size**(D)** Weekend**Q.29 [2 marks · NAT (numerical answer)]**

A hard-margin linear SVM is trained on the points below. Solving the dual problem gives the Lagrange multipliers α_i listed in the table. Using the KKT conditions, compute the weight component w_2 . (rounded off to two decimal places)

Point	x_1	x_2	y	α_i
P1	1	0	-1	0
P2	2	0	-1	2.125
P3	0	2	+1	0
P4	6	0	+1	0.125
P5	2	1	+1	2

**Q.30 [2 marks · NAT (numerical answer)]**

A 4-class classifier outputs logits $z = (-2, -1, 1, -0.5)$ and probabilities $p = \text{softmax}(z)$. The true class is 4, and the loss is the cross-entropy $L = -\ln p_4$. Then $\partial L / \partial z_3$ is _____ (rounded off to three decimal places)

Q.31 [2 marks · NAT (numerical answer)]

A one-feature logistic regression model $P(y = 1 | x) = \sigma(wx + b)$ has $w = 1$ and $b = 0$. It is evaluated on the 4 examples below. Using natural logarithms, the **average** binary cross-entropy loss $-[y \ln \hat{p} + (1 - y) \ln(1 - \hat{p})]$ over these examples is _____ (rounded off to two decimal places)

i	1	2	3	4
x_i	0	2	0	-1
y_i	1	1	1	0

Q.32 [2 marks · NAT (numerical answer)]

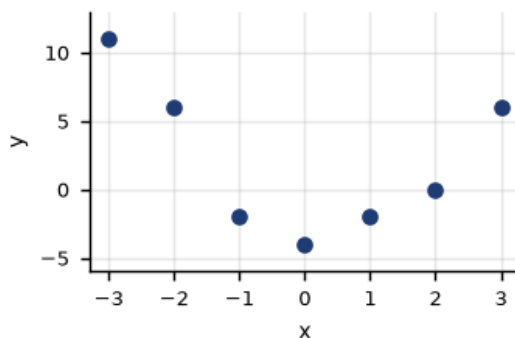
The sample covariance matrix of a two-feature dataset is S given below. In principal component analysis, the variance of the data along the second principal component is _____ (rounded off to two decimal places)

$$S = \begin{bmatrix} 11 & 5 \\ 5 & 4 \end{bmatrix}$$

Q.33 [2 marks · NAT (numerical answer)]

The quadratic model $y = a + bx + cx^2$ is fitted by least squares to the 7 points below (a polynomial / basis-function regression with basis $1, x, x^2$). Then the fitted value $\hat{y}$ at $x = 0.5$ is _____ (rounded off to two decimal places)

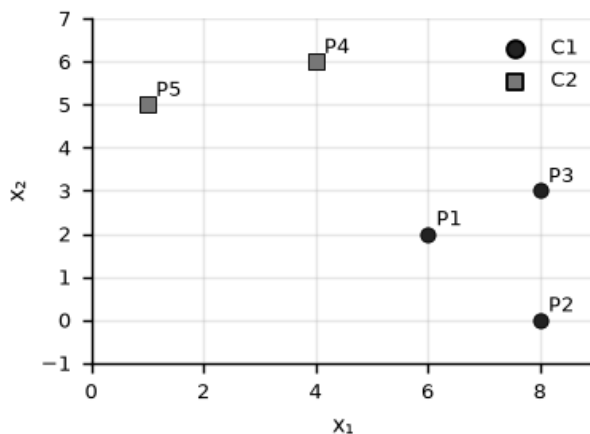
x	-3	-2	-1	0	1	2	3
y	11	6	-2	-4	-2	0	6



Q.34 [2 marks · NAT (numerical answer)]

A clustering of 5 points into 2 clusters is shown below. Using Euclidean distance, the silhouette coefficient $s(i) = (b(i) - a(i))/\max(a(i), b(i))$ of point P4 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5
x_1	6	8	8	4	1
x_2	2	0	3	6	5
cluster	C1	C1	C1	C2	C2

**Q.35 [2 marks · NAT (numerical answer)]**

Least-squares linear regression on $n = 2$ observations minimises $J(w) = (1/2n)\|Xw - y\|^2$ by batch gradient descent with constant learning rate η . The data give the matrix $X^T X$ below. Then the constant learning rate η that minimises the worst-case contraction factor $\max_i |1 - \eta \lambda_i|$ is _____ (rounded off to three decimal places)

$$X^T X = \begin{bmatrix} 9 & 7 \\ 7 & 9 \end{bmatrix}$$

— END OF QUESTION PAPER —

Mock Test 35 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	0.9875 to 1.0075	19	NAT	2	0.615 to 0.635
2	MCQ	1	(D)	20	NAT	2	0.2459 to 0.2659
3	NAT	1	32	21	MCQ	2	(A)
4	MSQ	1	(D)	22	NAT	2	41.0266 to 41.0466
5	NAT	1	0.0813 to 0.0853	23	NAT	2	0.8327 to 0.8367
6	NAT	1	0.4837 to 0.5037	24	NAT	2	0.5733 to 0.5933
7	MSQ	1	(B), (D)	25	MSQ	2	(A), (B), (D)
8	NAT	1	4	26	NAT	2	0.1593 to 0.1633
9	MCQ	1	(A)	27	NAT	2	7
10	MCQ	1	(C)	28	MCQ	2	(D)
11	NAT	1	0.736 to 0.74	29	NAT	2	1.99 to 2.01
12	MCQ	1	(A)	30	NAT	2	0.7081 to 0.7121
13	NAT	1	4.7381 to 4.7857	31	NAT	2	0.4466 to 0.4666
14	MSQ	1	(A), (B), (C), (D)	32	NAT	2	1.3867 to 1.4067
15	NAT	1	99.99 to 100.01	33	NAT	2	-3.4042 to -3.3636
16	NAT	2	0.0456 to 0.0656	34	NAT	2	0.4214 to 0.4414
17	NAT	2	0.865 to 0.885	35	NAT	2	0.2202 to 0.2242
18	NAT	2	-2.105 to -2.0633				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
SVM & Decision Trees					11
Probabilistic & Instance-based Classifiers					11
Unsupervised Learning					11
Regression & Model Selection					11
Neural Networks (MLP / Feed-forward)					11
Total					55

Mock Test 35 — Detailed Solutions

Q.1 — Answer: 0.9875 to 1.0075 [NAT, 1M · SVM & Decision Trees › Decision trees: impurity measures]

$$H = -\sum p_k \log_2 p_k = -(8/17)\log_2(8/17) - (9/17)\log_2(9/17) = \mathbf{0.998} \text{ bits.}$$

(Maximum possible with 2 classes: $\log_2 2 = 1$.)

Q.2 — Answer: (D) [MCQ, 1M · Probabilistic & Instance-based Classifiers › Linear discriminant analysis]

$$x^* = (\mu_0 + \mu_1)/2 + \sigma^2 \ln(\pi_0/\pi_1)/(\mu_1 - \mu_0) = 3.061.$$

A larger prior for C_1 means more of the x-axis is assigned to it, so the threshold moves away from μ_1 towards $\mu_0 = 1$. Answer: **It moves towards $x = 1$, enlarging the region assigned to C_1 .**

Equal variances keep the boundary a single point (linear); priors shift it but never make it quadratic.

Q.3 — Answer: 32 [NAT, 1M · SVM & Decision Trees › Decision trees: structure & traversal]

Depth is at most 5 (one test per attribute), so leaves $\leq 2^5 = \mathbf{32}$.

Q.4 — Answer: (D) [MSQ, 1M · Unsupervised Learning › Hierarchical clustering]

(A) **FALSE**. They generally differ.

(B) **FALSE**. It equals the merge height, which generally differs.

(C) **FALSE**. Merges are greedy and final.

(D) **TRUE**. Its merge heights are the MST edge weights.

Q.5 — Answer: 0.0813 to 0.0853 [NAT, 1M · Probabilistic & Instance-based Classifiers › Naive Bayes]

Add- α estimate: $(\text{count} + \alpha)/(N_c + \alpha V)$ with $V = 3$ possible values.

$$= (0 + 1)/(9 + 1 \times 3) = 1/12 = \mathbf{0.083}.$$

(The unsmoothed estimate would be $0/9$, i.e. zero, which would wipe out the whole posterior product.)

Q.6 — Answer: 0.4837 to 0.5037 [NAT, 1M · Probabilistic & Instance-based Classifiers › Evaluation metrics]

TP = 39, FN = 40, FP = 20, TN = 5 ($N = 104$).

Recall (sensitivity, TPR) = $TP/(TP + FN)$.

Value = **0.49**.

Watch the orientation: swapping FP and FN is the most common error (it swaps precision and recall).

Q.7 — Answer: (B), (D) [MSQ, 1M · SVM & Decision Trees › SVM: dual & support vectors]

(A) **FALSE**. Its constraint stays inactive, so the solution is unchanged.

(B) **TRUE**. Log-loss gradient is non-zero for every point.

(C) **FALSE**. Only support vectors have $\alpha_i > 0$.

(D) **TRUE**. This is what makes the kernel trick possible.

Q.8 — Answer: 4 [NAT, 1M · Unsupervised Learning › Hierarchical clustering]

Merges below the cut height 3.1: 3 (heights 1, 2, 2.53).

Each merge reduces the number of clusters by one: $7 - 3 = \mathbf{4}$. Equivalently, count the vertical lines crossed by the horizontal cut.

Q.9 — Answer: (A) [MCQ, 1M · Regression & Model Selection › Simple linear regression]

Residuals show a systematic U / $\cap$ pattern $\rightarrow$ the mean function is mis-specified; add non-linear terms or basis functions.

Answer: **A non-linear relationship not captured by the model (e.g. a missing x^2 term).**

Other patterns: funnel $\rightarrow$ heteroscedasticity; curvature $\rightarrow$ missing non-linearity; isolated point $\rightarrow$ outlier; random band $\rightarrow$ adequate fit.

Q.10 — Answer: (C) [MCQ, 1M · Unsupervised Learning › PCA & dimensionality reduction]

PC1 is the direction of maximum variance — along the long axis of the elongated cloud (direction C). PC2 must be orthogonal to PC1 and captures the remaining variance (direction D).

Answer: **Direction C**. Directions B and A are at 45° to the main axis and capture intermediate variance.

Q.11 — Answer: 0.736 to 0.74 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

exp values: 1, 20.0855, 1.6487, 4.4817; sum = 27.2159.

$p_2 = 20.0855/27.2159 = \mathbf{0.738}$.

Q.12 — Answer: (A) [MCQ, 1M · Regression & Model Selection > Regression vs classification]

0-1 loss counts mistakes in predicted labels.

Hence the answer is **classification**.

Q.13 — Answer: 4.7381 to 4.7857 [NAT, 1M · Regression & Model Selection > Multiple linear regression]

Variance inflation factor of predictor j : $VIF_j = 1/(1 - R_j^2)$, where R_j^2 is the R^2 from regressing x_j on all other predictors. It is the factor by which $\text{Var}(\beta_j)$ is inflated by collinearity; $\sqrt{VIF}$ inflates the standard error.

$VIF = 1/(1 - 0.79) = 4.7619$.

Answer: **4.76**.

Q.14 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

(A) TRUE. Standard result.

(B) TRUE. Shift invariance.

(C) TRUE. Standard derivative.

(D) TRUE. $-\ln p_y \rightarrow \infty$.

Q.15 — Answer: 99.99 to 100.01 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Each unit is dropped independently with probability $1 - p = 0.1$; expected number = $1000 \times 0.1 = \mathbf{100}$.

Q.16 — Answer: 0.0456 to 0.0656 [NAT, 2M · SVM & Decision Trees > SVM: kernels]

$\cos \theta = \varphi(x)^T \varphi(z) / (\|\varphi(x)\| \|\varphi(z)\|) = K(x, z) / \sqrt{K(x, x)K(z, z)}$.

$K(x, z) = 1$, $K(x, x) = 81$, $K(z, z) = 4$.

$\cos \theta = 1/\sqrt{81 \times 4} = \mathbf{0.06}$.

Q.17 — Answer: 0.865 to 0.885 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

AUC = fraction of (positive, negative) pairs in which the positive has the higher score (Mann-Whitney statistic).

For each positive, count negatives with lower score: $0.85 \rightarrow 6$; $0.75 \rightarrow 6$; $0.55 \rightarrow 5$; $0.4 \rightarrow 4$.

Concordant pairs = 21 out of $4 \times 6 = 24$.

$AUC = 21/24 = \mathbf{0.88}$.

(Equivalently, trace the ROC staircase by lowering the threshold and add the rectangle areas.)

Q.18 — Answer: -2.105 to -2.0633 [NAT, 2M · Regression & Model Selection > Cross-validation]

$\bar{x} = 5.333$, $\bar{y} = 25.5$.

i	1	2	3	4	5	6
$x_i - \bar{x}$	-4.33	-2.33	-0.33	0.67	1.67	4.67
$y_i - \bar{y}$	-15.5	-9.5	1.5	2.5	6.5	14.5
$(x_i - \bar{x})(y_i - \bar{y})$	67.17	22.17	-0.5	1.67	10.83	67.67

$S_{xx} = \sum (x_i - \bar{x})^2 = 49.333$, $S_{xy} = 169$.

$\beta_1 = S_{xy}/S_{xx} = 3.4257$, $\beta_0 = \bar{y} - \beta_1 \bar{x} = 7.2297$.

For a linear smoother, the leave-one-out residual is $e_{(i)} = e_i / (1 - h_{ii})$, with $h_{ii} = 1/n + (x_i - \bar{x})^2 / S_{xx}$.

x_i	1	3	5	6	7	10
e_i	-0.655	-1.507	2.642	0.216	0.791	-1.486
h_{ii}	0.547	0.277	0.169	0.176	0.223	0.608
$e_i/(1-h_{ii})$	-1.448	-2.084	3.179	0.262	1.017	-3.793

Answer: **-2.08** (verified by explicitly refitting without the point).

Q.19 — Answer: 0.615 to 0.635 [NAT, 2M · Probabilistic & Instance-based Classifiers · Evaluation metrics]

Predicted positive (score ≥ 0.3): examples 1, 2, 3, 4, 6, 8, 9, 11.

TP = 3, FP = 5, FN = 0, TN = 3.

False positive rate = $FP/(FP + TN) = 0.62$.

Raising t typically raises precision and lowers recall/FPR; lowering t does the opposite (threshold moving).

Q.20 — Answer: 0.2459 to 0.2659 [NAT, 2M · Regression & Model Selection · Ridge & Lasso]

Ridge fitted values $\hat{y} = H_\lambda y$ with $H_\lambda = X(X^T X + \lambda I)^{-1} X^T$. Writing $X^T X = V D V^T$ with eigenvalues d_j^2 gives $\text{trace}(H_\lambda) = \sum d_j^2 / (d_j^2 + \lambda)$ (effective degrees of freedom).

$25/(25 + 2) + 9/(9 + 2) = 1.7441$.

OLS has $df = \text{trace}(H) = p = 2$; reduction = $2 - 1.7441 = 0.2559$.

Answer: **0.26**.

Q.21 — Answer: (A) [MCQ, 2M · Neural Networks (MLP / Feed-forward) · Expressiveness]

Each unit outputs 1 iff its net input (weighted sum + bias) is > 0 . Evaluate on all four inputs:

x_1	x_2	h_1	h_2	y
0	0	0	1	0
0	1	1	0	1
1	0	0	0	0
1	1	0	0	0

h_1 computes (NOT x_1) AND x_2 , h_2 computes NOR; the output column (0,0)→(1,1) is 0, 1, 0, 0, i.e. **(NOT x_1) AND x_2** .

Q.22 — Answer: 41.0266 to 41.0466 [NAT, 2M · Unsupervised Learning · Distance measures]

Min-max scaling: $x' = (x - \min)/(max - \min)$; age range [20, 60], income range [10, 210].

Customer	age'	income'
A	0.125	0.275
B	0.625	0
C	0.825	0.125

Raw distances: $d(A,B) = 58.52$, $d(A,C) = 41.04 \rightarrow$ nearest is C (income dominates because of its large range).

Scaled distances: $d(A,B) = 0.5706$, $d(A,C) = 0.7159 \rightarrow$ nearest is B.

Answer: **41.04**. Scaling changes which neighbour is nearest — distance-based methods (k-means, k-NN, hierarchical clustering) should be run on standardized features.

Q.23 — Answer: 0.8327 to 0.8367 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Forward pass]

$a = W_1 x + b_1 = (1, 0.5)$; $h = \tanh(a) = (0.7616, 0.4621)$.

$z = W_2 h + b_2 = (-0.5232, 2.0611, -0.0379)$.

$\exp(z) = (0.5926, 7.8544, 0.9628)$, sum = 9.4098.

$p_2 = \exp(z_2) / \sum \exp(z_j) = 0.8347$. (All probabilities: (0.063, 0.835, 0.102).)

Q.24 — Answer: 0.5733 to 0.5933 [NAT, 2M · SVM & Decision Trees · SVM: soft margin & hinge loss]

Compute $f(x_i)$, the functional margin $y_i f(x_i)$ and the slack/hinge loss for each point:

Point	x	y	f(x)	$y f(x)$	$\xi = \max(0, 1 - y f)$
P1	(5, 0)	-1	-2.5	2.5	0
P2	(3, 3)	+1	-0.5	-0.5	1.5
P3	(5, 2)	-1	-0.5	0.5	0.5
P4	(3, 4)	-1	0.5	-0.5	1.5
P5	(4, 5)	+1	2	2	0
P6	(3, 1)	-1	-2.5	2.5	0

Average = $3.5/6 = \mathbf{0.58}$.

Points with $y f \geq 1$ contribute 0; points inside the margin contribute between 0 and 1; misclassified points contribute more than 1.

Q.25 — Answer: (A), (B), (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$\nabla f = (10w_1, 5w_2) = w_i \leftarrow (1 - \eta a_i) w_i$. GD converges (from every start) iff $|1 - \eta a_i| < 1$ for both i , i.e. $0 < \eta < 2/\max(a_i) = 2/10 = 0.2$. The steep direction ($a = 10$) limits the learning rate.

(A) TRUE. $\eta = 0.19$: $|1 - \eta \cdot 10| = 0.9$, $|1 - \eta \cdot 5| = 0.05 \rightarrow$ converges.

(B) TRUE. $\eta = 0.12$: $|1 - \eta \cdot 10| = 0.2$, $|1 - \eta \cdot 5| = 0.4 \rightarrow$ converges.

(C) FALSE. $\eta = 0.3$: $|1 - \eta \cdot 10| = 2$, $|1 - \eta \cdot 5| = 0.5 \rightarrow$ diverges.

(D) TRUE. $\eta = 0.03$: $|1 - \eta \cdot 10| = 0.7$, $|1 - \eta \cdot 5| = 0.85 \rightarrow$ converges.

Q.26 — Answer: 0.1593 to 0.1633 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

$P(\text{error}) = \pi_0 P(x \geq 6.5 | C_0) + \pi_1 P(x < 6.5 | C_1)$.

$= 0.7 \times (8 - 6.5)/8 + 0.3 \times (6.5 - 6)/5 = 0.1312 + 0.03 = \mathbf{0.161}$.

(For comparison, the Bayes error is 0.12: the Bayes rule assigns the whole overlap to the class with the larger $\pi_k p(x|C_k)$.)

Q.27 — Answer: 7 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Single-linkage merge heights are exactly the edge weights of a minimum spanning tree (Kruskal's algorithm adds the shortest edge joining two different components = single-linkage merge).

MST edges (Prim's algorithm): P1-P4 (1); P4-P3 (3); P3-P5 (3); P1-P6 (7); P6-P2 (2).

Sorted merge heights: 1, 2, 3, 3, 7.

The last single-linkage merge happens at the largest MST edge: 7.

Q.28 — Answer: (D) [MCQ, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent: 4 Yes, 8 No $\Rightarrow$ Gini = 0.4444.

Colour — Red: [1+, 3-], Green: [1+, 4-], Blue: [2+, 1-]; gain = 0.075.

Humidity — High: [2+, 4-], Normal: [2+, 4-]; gain = 0.

Size — Small: [0+, 3-], Large: [4+, 5-]; gain = 0.0741.

Weekend — Yes: [1+, 8-], No: [3+, 0-]; gain = 0.2963.

The largest Gini gain is for **Weekend**, so it is chosen at the root.

Q.29 — Answer: 1.99 to 2.01 [NAT, 2M · SVM & Decision Trees > SVM: dual & support vectors]

$w = \sum \alpha_i y_i x_i$ over points with $\alpha_i > 0$:

$w = (2.125)(-1)(2, 0) + (0.125)(+1)(6, 0) + (2)(+1)(2, 1) = (1/2, 2)$.

b from any support vector, e.g. P2: $y = w^T x + b = b = -1 - (1) = -2$.

Check: $\sum \alpha_i y_i = 0$ ($= 0$, as required).

Answer: **2**.

Q.30 — Answer: 0.7081 to 0.7121 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$p = \text{softmax}(z)$: $\exp(z) = (0.1353, 0.3679, 2.7183, 0.6065)$, $\text{sum} = 3.828$, so $p = (0.0354, 0.0961, 0.7101, 0.1584)$.

For $L = -\ln p_y$ with softmax, $\partial L / \partial z = p - y$ (y one-hot, true class 4): $\partial L / \partial z = (0.0354, 0.0961, 0.7101, -0.8416)$.

$$\partial L / \partial z_3 = p_3 - y_3 = 0.7101 - 0 = \mathbf{0.7101}.$$

$$\text{Note } \sum_k \partial L / \partial z_k = \sum p_k - 1 = 0.$$

Q.31 — Answer: 0.4466 to 0.4666 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

$$z_i = 1x_i + 0; \hat{p}_i = \sigma(z_i): 0.5, 0.8808, 0.5, 0.2689.$$

Per-example loss $L_i = -\ln \hat{p}_i$ if $y_i = 1$ and $-\ln(1 - \hat{p}_i)$ if $y_i = 0$:

i	y	$\hat{p}$	L_i
1	1	0.5	0.6931
2	1	0.8808	0.1269
3	1	0.5	0.6931
4	0	0.2689	0.3133

$$\sum L_i = 1.8265; \text{average} = 1.8265/4 = \mathbf{0.46}.$$

Common mistakes: using $\ln \hat{p}$ for $y = 0$ examples, or $\log_{10}$ instead of $\ln$.

Q.32 — Answer: 1.3867 to 1.4067 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

Eigenvalues of $[[a, b], [b, c]]$: $\lambda = (a + c)/2 \pm \sqrt{((a - c)/2)^2 + b^2}$.

$$\lambda = 7.5 \pm \sqrt{(12.25 + 25)} = 7.5 \pm 6.1033 \rightarrow \lambda_1 = 13.6033, \lambda_2 = 1.3967 \text{ (check: } \lambda_1 + \lambda_2 = \text{trace} = 15, \lambda_1 \lambda_2 = \text{det} = 19).$$

$$\text{Eigenvector for } \lambda_1: (S - \lambda_1 I)v = 0 \rightarrow (-2.6033)v_1 + (5)v_2 = 0 \rightarrow v \propto (1, 0.5207).$$

$$\text{PVE(PC1)} = \lambda_1 / (\lambda_1 + \lambda_2) = 90.69\%.$$

Answer: **1.4**.

Q.33 — Answer: -3.4042 to -3.3636 [NAT, 2M · Regression & Model Selection > Basis functions & feature scaling]

Design columns 1, x , x^2 . Because the x -values are symmetric about 0, $\sum x = \sum x^3 = 0$, so the normal equations decouple:

$$\sum x^2 = 28, \sum x^4 = 196, \sum y = 15, \sum xy = -27, \sum x^2 y = 173.$$

$$b = \sum xy / \sum x^2 = -27/28 = -0.9643.$$

$$[n, \sum x^2, \sum x^4][a; c] = [\sum y; \sum x^2 y] = 7a + 28c = 15, 28a + 196c = 173.$$

$$\text{Solving: } c = (n\sum x^2 y - \sum x^2 \sum y) / (n\sum x^4 - (\sum x^2)^2) = 1.3452, a = (\sum y - c\sum x^2) / n = -3.2381.$$

$$\hat{y}(0.5) = -3.2381 + (-0.9643)(0.5) + (1.3452)(0.5)^2 = -3.3839.$$

Answer: **-3.38**.

Q.34 — Answer: 0.4214 to 0.4414 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

a(i) = mean Euclidean distance from P_4 to the other members of its own cluster C_2 : $d(P_4, P_5) = 3.162 \rightarrow a = 3.1623$.

b(i) = smallest mean distance to the points of another cluster: C_1 : 5.5611 $\rightarrow b = 5.5611$.

$$s(i) = (b - a) / \max(a, b) = (5.5611 - 3.1623) / 5.5611 = \mathbf{0.43}.$$

s ranges in $[-1, 1]$; values near 1 mean well clustered, negative values suggest a wrong assignment.

Q.35 — Answer: 0.2202 to 0.2242 [NAT, 2M · Regression & Model Selection > Gradient descent]

Hessian of J : $\nabla^2 J = (1/n)X^T X$. Eigenvalues of $X^T X$: $9 \pm 7 = 16, 2$ (eigenvectors $(1, 1), (1, -1)$).

So the Hessian eigenvalues are $L = 8$ and $\mu = 1$.

GD error evolves as $w_{t+1} - w^* = (I - \eta \nabla^2 J)(w_t - w^*)$, so it converges iff $|1 - \eta \lambda| < 1$ for every Hessian eigenvalue λ .

$$\text{Balance } |1 - \eta L| = |1 - \eta \mu| \Rightarrow \eta^* = 2/(L + \mu) = 2/(8 + 1) = 0.2222.$$

Answer: **0.222**.

Mock Test 36

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	3	15	17

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Unsupervised Learning	7	11
SVM & Decision Trees	7	11
Probabilistic & Instance-based Classifiers	7	11
Regression & Model Selection	7	11
Neural Networks (MLP / Feed-forward)	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

For $x = (0, -2, 0, 7, 2)$ and $y = (3, 5, 4, 5, 7)$, the Minkowski distance with $p = 3$ between x and y is _____ (rounded off to two decimal places)

Q.2 [1 mark · MSQ (one or more correct)]

Which of the following statements about principal component analysis (PCA) is/are CORRECT?

- (A) PCA is an unsupervised, linear dimensionality-reduction technique.
- (B) PCA on the uncentered data matrix gives the same directions as PCA on centered data for every dataset.
- (C) A few linear principal components can unfold a nonlinear manifold such as a Swiss roll.
- (D) The variance of the projected data along the j -th PC equals the j -th largest covariance eigenvalue.

Q.3 [1 mark · MSQ (one or more correct)]

Which of the following statements about kernel functions (Mercer kernels) is/are CORRECT?

- (A) The linear kernel $k(x, z) = x^T z$ is valid.
- (B) If k_1 and k_2 are valid kernels, the product $k_1 \cdot k_2$ is a valid kernel.
- (C) $k(x, z) = f(x)f(z)$ is valid for any real-valued f .
- (D) $(x^T z + c)^d$ with $c \geq 0$ and positive integer d is a valid kernel.

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about Bayes-optimal classification and generative vs. discriminative models is/are CORRECT?

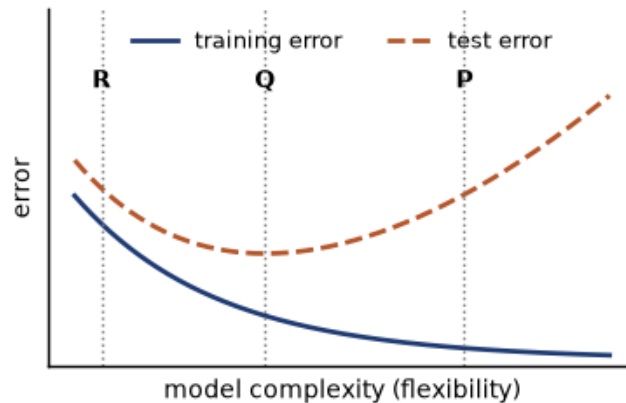
- (A) Naive Bayes, LDA and QDA are generative; logistic regression is discriminative.
- (B) The Bayes classifier can be computed knowing only the class priors.
- (C) For binary classification with 0-1 loss, the Bayes error at x equals $\min(P(y=0|x), P(y=1|x))$.
- (D) Given enough training data, any classifier can reach zero test error for every distribution.

Q.5 [1 mark · NAT (numerical answer)]

The distance of the hyperplane $2x_1 + 3x_2 + 6x_3 - 9 = 0$ from the origin is _____ (rounded off to two decimal places)

Q.6 [1 mark · MCQ]

The figure shows training and test error as a function of model complexity, with three complexity levels marked P, Q and R. Which marked level corresponds to underfitting (high bias, low variance)?



- (A) None of P, Q, R (B) Q
(C) R (D) P

Q.7 [1 mark · MSQ (one or more correct)]

Linear regression is trained by minimising the squared-error loss. Which of the following statements about gradient-based training is/are CORRECT?

- (A)** Standardising features changes the minimum training MSE achievable by OLS with an intercept.
- (B)** The MSE loss of linear regression can have local minima that are not global minima.
- (C)** One epoch of SGD with batch size 1 on n examples performs n parameter updates.
- (D)** Stochastic gradient descent updates the parameters using the gradient on one (or a few) randomly chosen example(s).

Q.8 [1 mark · NAT (numerical answer)]

For $K = 4$ classes and $d = 3$ binary features, a Bayes classifier that models the FULL joint class-conditional distribution $P(x_1, \dots, x_d | c)$ (no independence assumption) needs how many independent parameters, including the class priors? (Answer in integer)

Q.9 [1 mark · MSQ (one or more correct)]

A simple linear regression $\hat{y} = \beta_0 + \beta_1 x$ is fitted by ordinary least squares (with intercept) to data (x_i, y_i) , $i = 1, \dots, n$, with residuals $e_i = y_i - \hat{y}_i$ and sample correlation r . Which of the following statements is/are CORRECT?

- (A)** If $r = 0$, the least-squares line is horizontal at height $\bar{y}$.
(B) If $r = 0$ the least-squares line is $\hat{y} = 0$.
(C) $SST = SSR + SSE$ for the least-squares fit with intercept.
(D) $\sum \hat{y}_i e_i = 0$ for the least-squares fit.

Q.10 [1 mark · NAT (numerical answer)]

A sigmoid-output classifier gives the probabilities $p = P(y = 1 | x)$ below. The mean binary cross-entropy $-[y \ln p + (1 - y) \ln(1 - p)]$ over the 4 examples is _____ (rounded off to three decimal places)

example	$p = P(y = 1)$	y
1	0.4	0
2	0.8	1
3	0.3	0
4	0.9	1

Q.11 [1 mark · MSQ (one or more correct)]

Which of the following statements about initialisation and regularisation of neural networks is/are CORRECT?

- (A) Dropout is applied at test time with freshly sampled random masks, exactly as during training.
- (B) With inverted dropout (keep probability p), retained activations are scaled by $1/p$ during training, so no scaling is needed at test time.
- (C) Initialising all weights of an MLP to the same value makes the hidden units of a layer compute identical functions and receive identical updates, so they stay identical.
- (D) Early stopping acts as a regulariser.

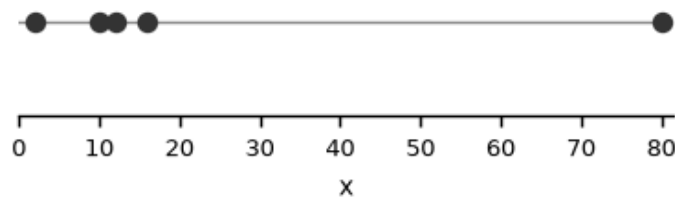
Q.12 [1 mark · NAT (numerical answer)]

A hard-margin SVM has exactly three support vectors with the optimal dual variables shown. The component w_2 of the optimal weight vector is _____ (rounded off to two decimal places)

i	x_i	y_i	α_i
1	(4, 0)	-1	0.5
2	(-2, 2)	+1	1.5
3	(-1, 4)	-1	1

Q.13 [1 mark · NAT (numerical answer)]

A cluster contains the 1-D values $\{2, 10, 12, 16, 80\}$. Treating them as one cluster, the medoid of this cluster (absolute-difference dissimilarity) is _____ (rounded off to two decimal places)

**Q.14 [1 mark · MSQ (one or more correct)]**

Which of the following statements about logistic regression is/are CORRECT?

- (A) Softmax regression with $K = 2$ classes is equivalent to binary logistic regression.
- (B) Increasing x_j by one unit increases $P(y = 1 | x)$ by exactly β_j .
- (C) The odds ratio for a one-unit increase in x_j equals β_j .
- (D) Changing the probability threshold from 0.5 to 0.8 moves the logistic-regression decision boundary to a parallel hyperplane.

Q.15 [1 mark · MSQ (one or more correct)]

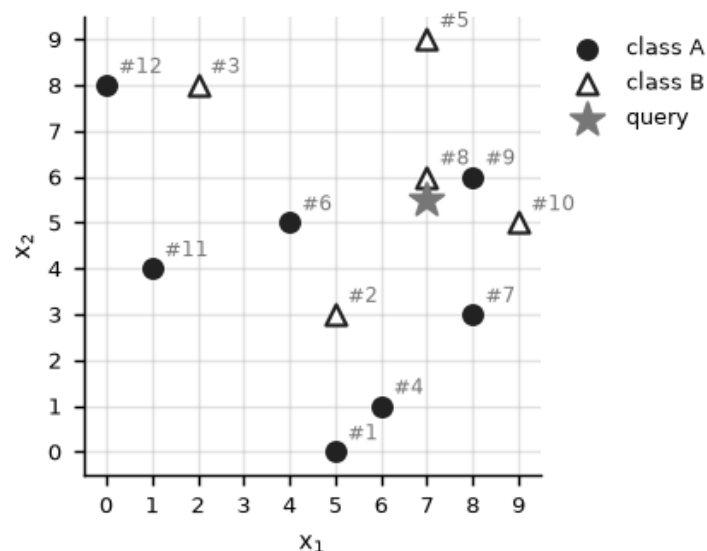
Which of the following statements about the perceptron is/are CORRECT?

- (A) When the data are not linearly separable, the basic perceptron algorithm does not settle on a fixed weight vector.
- (B) The weight vector w is normal (perpendicular) to the perceptron's decision boundary.
- (C) The perceptron mistake bound $(R/\gamma)^2$ does not depend explicitly on the input dimension d .
- (D) A single perceptron can implement AND, OR and NAND on binary inputs.

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · MSQ (one or more correct)]**

The figure and table show 12 labelled training points (class A: filled circles, class B: open triangles) and a query point (star) at (7, 5.5). A k-NN classifier with **Euclidean** distance and majority vote is used. Which of the following statements is/are CORRECT?

#	1	2	3	4	5	6	7	8	9	10	11	12
x_1	5	5	2	6	7	4	8	7	8	9	1	0
x_2	0	3	8	1	9	5	3	6	6	5	4	8
class	A	B	B	A	B	A	A	B	A	B	A	A

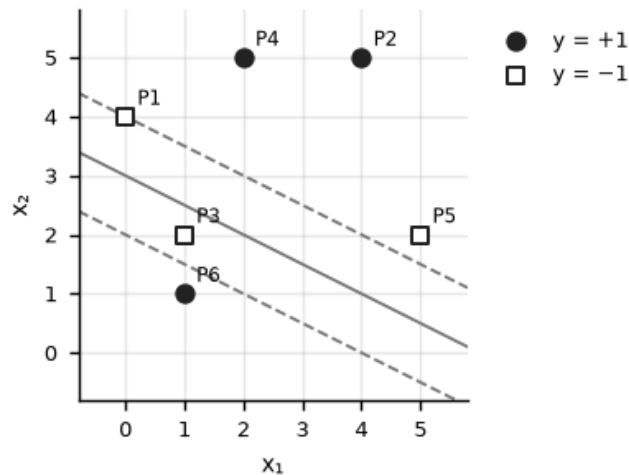


- (A) With $k = 7$, the query is classified as class A.
- (B) With $k = 1$, the query is classified as class A.
- (C) With $k = 3$, the query is classified as class A.
- (D) With $k = 5$, the query is classified as class A.

Q.17 [2 marks · MSQ (one or more correct)]

A soft-margin linear SVM has learned $f(x) = 0.5x_1 + x_2 - 3$. The training points are listed below and plotted (boundary solid, margins $f = \pm 1$ dashed). With ξ_i the optimal slack of point i , which of the following statements is/are CORRECT?

Point	x_1	x_2	y
P1	0	4	-1
P2	4	5	+1
P3	1	2	-1
P4	2	5	+1
P5	5	2	-1
P6	1	1	+1



- (A) P2 is misclassified ($\xi_2 > 1$).
- (B) P1 is correctly classified but lies strictly inside the margin.
- (C) P4 has slack $\xi_4 = 0$.
- (D) P2 has slack $\xi_2 = 0$.

Q.18 [2 marks · MSQ (one or more correct)]

Which of the following statements about PCA, SVD and dimensionality reduction is/are CORRECT?

- (A) PCA always preserves the class separability present in the original feature space.
- (B) In high dimensions, most of the volume of a hypercube is concentrated near its centre.
- (C) Adding a constant to every value of one feature changes the principal components.
- (D) Fisher LDA for C classes yields at most $C - 1$ discriminant directions.

Q.19 [2 marks · NAT (numerical answer)]

Agglomerative clustering with **average linkage** is applied to the distance matrix below. The cophenetic distance between P1 and P3 (the dendrogram height at which they are first joined) is ____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5	P6
P1	0	9	3	9	2	8
P2		0	10	6	7	5
P3			0	12	5	7
P4				0	7	7
P5					0	6
P6						0

Q.20 [2 marks · NAT (numerical answer)]

Consider the model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ (first column of X is the intercept column), fitted by least squares using $\beta = (X^T X)^{-1} X^T y$ with the design matrix X and response y below. Then the predicted response at $(x_1, x_2) = (-1.5, 1)$ is _____ (rounded off to two decimal places)

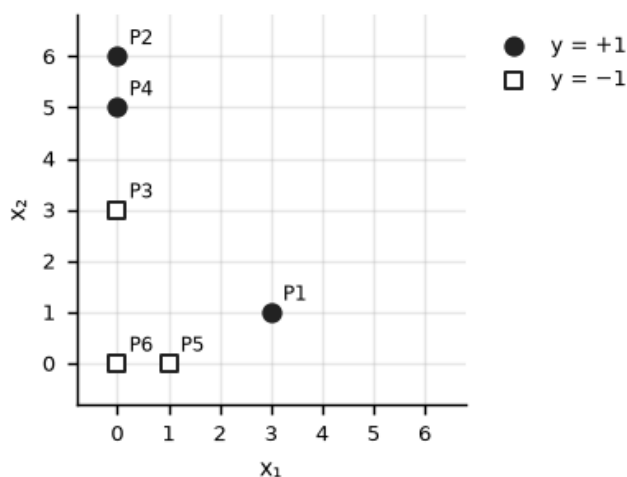
$$X = \begin{bmatrix} 1 & -1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$y = \begin{bmatrix} 5 \\ 6 \\ 2 \\ 7 \end{bmatrix}$$

Q.21 [2 marks · MSQ (one or more correct)]

A hard-margin linear SVM is trained on the 6 labelled points shown. Which of the following points is/are support vectors of the resulting maximum-margin classifier?

Point	x_1	x_2	y
P1	3	1	+1
P2	0	6	+1
P3	0	3	-1
P4	0	5	+1
P5	1	0	-1
P6	0	0	-1



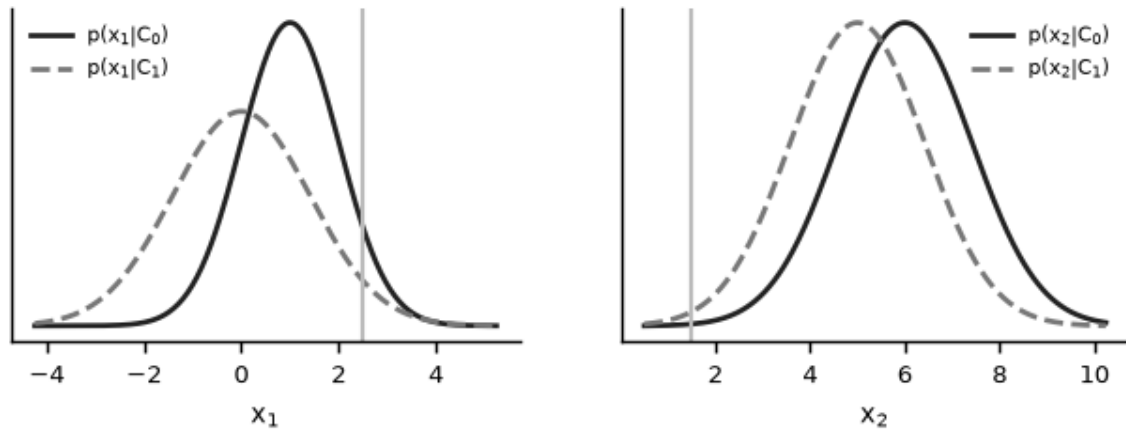
- (A) P1
(C) P3

- (B) P2
(D) P4

Q.22 [2 marks · NAT (numerical answer)]

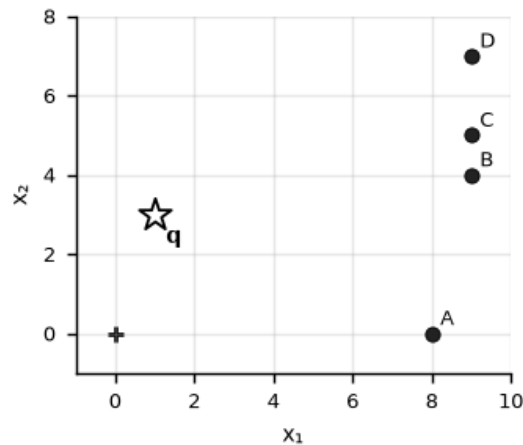
A Gaussian naive Bayes classifier with features x_1, x_2 has the class priors and per-class Gaussian parameters below (σ^2 = variance). For the input $x = (2.5, 1.5)$, the posterior probability $P(C_1 | x)$ is _____ (rounded off to two decimal places)

class	prior	$\mu(x_1)$	$\sigma^2(x_1)$	$\mu(x_2)$	$\sigma^2(x_2)$
C_0	0.7	1	1	6	2
C_1	0.3	0	2	5	2

**Q.23 [2 marks · MCQ]**

A query point $q = (1, 3)$ and four points A-D are shown below. Using **Manhattan (L_1) distance**, which point is nearest to q ?

Point	A	B	C	D
x_1	8	9	9	9
x_2	0	4	5	7

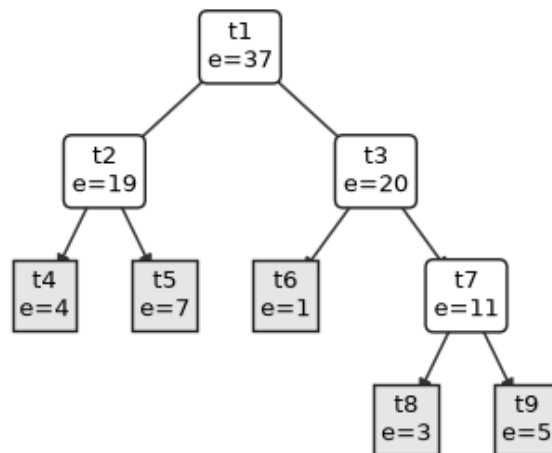


- (A) Point A
(C) Point C

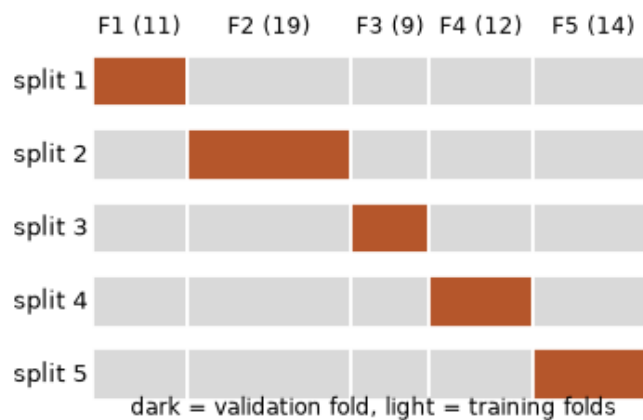
- (B) Point B
(D) Point D

Q.24 [2 marks · NAT (numerical answer)]

A classification tree was grown on $N = 100$ training records. Each node shows $e(t)$, the number of training records that would be misclassified if t were a leaf; $R(t) = e(t)/N$. The smallest value of α at which cost-complexity pruning first prunes the tree (the weakest link) is _____ (rounded off to three decimal places)

**Q.25 [2 marks · NAT (numerical answer)]**

A regression model is evaluated by 5-fold cross-validation on $n = 65$ observations. The folds (F1 ... F5) have unequal sizes, as shown in the diagram and table. The k -fold CV estimate is defined as the mean squared error over all n out-of-fold predictions (i.e. pooled over observations). Its value is _____ (rounded off to two decimal places)



Fold	F1	F2	F3	F4	F5
size	11	19	9	12	14
validation MSE	6.2	8.2	1.7	3.7	4.5

Q.26 [2 marks · MSQ (one or more correct)]

A threshold unit outputs 1 if $w_1x_1 + w_2x_2 + b > 0$ and 0 otherwise, for binary inputs $x_1, x_2 \in \{0, 1\}$. Which of the following parameter settings make the unit implement the Boolean **OR** function?

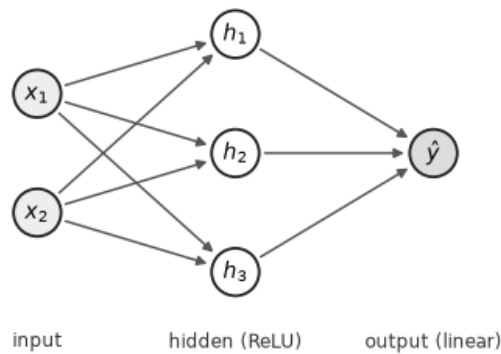
- (A) $w_1 = 2, w_2 = -2.5, b = -1.5$
- (B) $w_1 = 0, w_2 = -2.5, b = 1.5$
- (C) $w_1 = 3, w_2 = 1.5, b = -1$
- (D) $w_1 = 3, w_2 = 2.5, b = -2$

Q.27 [2 marks · NAT (numerical answer)]

A simple linear regression fitted by ordinary least squares gives $\hat{y} = 1.1 + 3.8x$. The response y is in $^{\circ}\text{C}$. If y is converted to $^{\circ}\text{F}$ ($y' = 1.8y + 32$) and the model is refitted, the new intercept is _____ (rounded off to two decimal places)

Q.28 [2 marks · NAT (numerical answer)]

The 2-3-1 network below computes $\hat{y} = v \cdot \text{ReLU}(Wx + b) + c$ with W, b, v given below and $c = 0$. With $x = (-1, 2)$, target $y = 2$ and loss $L = \frac{1}{2}(\hat{y} - y)^2$, the value of $\partial L / \partial v_2$ is _____ (Answer in integer)



$$W = \begin{bmatrix} 1 & -2 \\ 2 & 0 \\ 2 & 2 \end{bmatrix}$$

$$b = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$v^T = \begin{bmatrix} -2 & 2 & -1 \end{bmatrix}$$

Q.29 [2 marks · MSQ (one or more correct)]

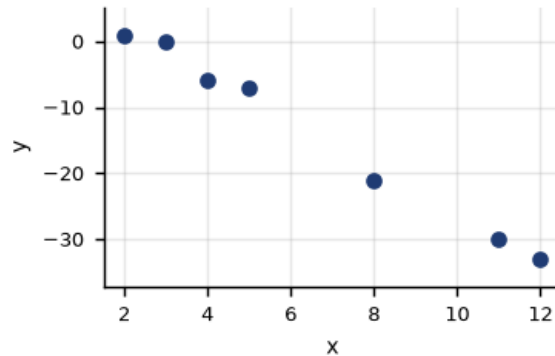
Which of the following statements about loss functions and optimisation for training neural networks is/are CORRECT?

- (A) Increasing the batch size reduces the variance of the gradient estimate.
- (B) With momentum β ($v \leftarrow \beta v + g$, $w \leftarrow w - \eta v$) and a constant gradient g , the step size approaches $\eta g / (1 - \beta)$.
- (C) The number of parameter updates in one epoch of mini-batch training is $\lceil N/B \rceil$ (keeping the last partial batch).
- (D) Decaying the learning rate during training can help SGD settle closer to a minimum.

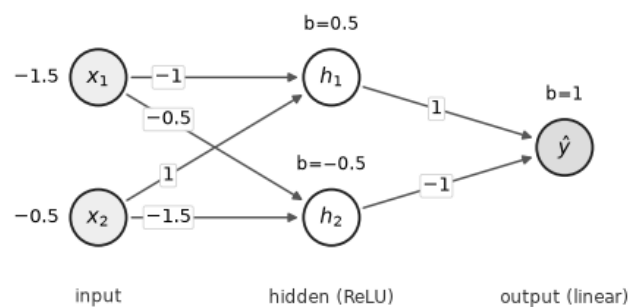
Q.30 [2 marks · NAT (numerical answer)]

The 7 observations shown in the table are plotted below. For the simple linear regression of y on x (with intercept), the intercept β_0 of the least-squares line $\hat{y} = \beta_0 + \beta_1 x$ is _____ (rounded off to two decimal places)

x	2	3	4	5	8	11	12
y	1	0	-6	-7	-21	-30	-33

**Q.31 [2 marks · NAT (numerical answer)]**

The feed-forward network below has ReLU hidden units and a linear output unit. The number on each edge is its weight and 'b' above a node is that node's bias. For input $x = (-1.5, -0.5)$ (values shown left of the inputs), the network output $\hat{y}$ is _____ (rounded off to three decimal places)

**Q.32 [2 marks · MSQ (one or more correct)]**

Which of the following statements about LDA, QDA and logistic regression is/are CORRECT?

- (A) Fisher's LDA directions coincide with the directions of maximum total variance found by PCA.
- (B) LDA and logistic regression always produce identical coefficient estimates.
- (C) Changing the class priors in LDA shifts the decision boundary but does not change its orientation.
- (D) Fisher's two-class direction is always $w \propto \mu_1 - \mu_0$, regardless of S_W .

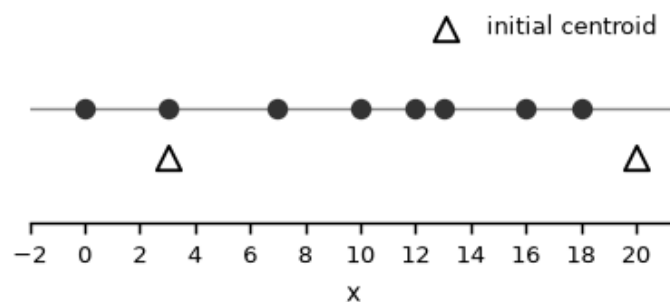
Q.33 [2 marks · MCQ]

A decision tree is to be grown on the 12 training records below using Gini gain as the splitting criterion (multi-way split on each categorical attribute). Which attribute is chosen at the root?

#	Shape	Colour	Wind	Temp	Class
1	Square	Green	Strong	Cool	Yes
2	Square	Blue	Strong	Hot	No
3	Square	Blue	Weak	Cool	No
4	Square	Green	Strong	Hot	No
5	Square	Green	Strong	Hot	No
6	Square	Green	Strong	Mild	Yes
7	Square	Blue	Strong	Mild	No
8	Square	Red	Weak	Hot	No
9	Square	Green	Weak	Cool	No
10	Square	Green	Strong	Hot	No
11	Round	Red	Weak	Mild	Yes
12	Square	Blue	Weak	Cool	No

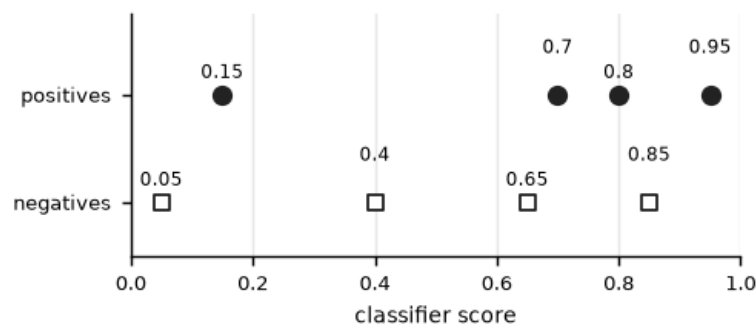
(A) Shape**(B)** Colour**(C)** Wind**(D)** Temp**Q.34 [2 marks · NAT (numerical answer)]**

The 1-D dataset $\{0, 3, 7, 10, 12, 13, 16, 18\}$ is clustered with k-means ($k = 2$) using initial centroids 3 and 20 (shown as triangles). Lloyd's iterations (assign to nearest centroid, then recompute means) are run until convergence. Then the number of iterations in which at least one centroid **changes** its value is _____ (Answer in integer)

**Q.35 [2 marks · NAT (numerical answer)]**

A classifier assigns the scores below to 8 test examples (4 positives, label 1; 4 negatives, label 0). The area under the ROC curve (AUC) is _____ (rounded off to two decimal places)

example	1	2	3	4	5	6	7	8
score	0.65	0.95	0.4	0.05	0.7	0.8	0.85	0.15
label	0	1	0	0	1	1	0	1



— END OF QUESTION PAPER —

Mock Test 36 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	8.2668 to 8.2868	19	NAT	2	3.99 to 4.01
2	MSQ	1	(A), (D)	20	NAT	2	7.2138 to 7.2862
3	MSQ	1	(A), (B), (C), (D)	21	MSQ	2	(A), (C), (D)
4	MSQ	1	(A), (C)	22	NAT	2	0.5811 to 0.6011
5	NAT	1	1.2757 to 1.2957	23	MCQ	2	(B)
6	MCQ	1	(C)	24	NAT	2	0.0285 to 0.0315
7	MSQ	1	(C), (D)	25	NAT	2	5.3238 to 5.3438
8	NAT	1	31	26	MSQ	2	(C), (D)
9	MSQ	1	(A), (C), (D)	27	NAT	2	33.912 to 34.048
10	NAT	1	0.297 to 0.301	28	NAT	2	0
11	MSQ	1	(B), (C), (D)	29	MSQ	2	(A), (B), (C), (D)
12	NAT	1	−1.01 to −0.99	30	NAT	2	9.1416 to 9.2334
13	NAT	1	11.99 to 12.01	31	NAT	2	1.498 to 1.502
14	MSQ	1	(A), (D)	32	MSQ	2	(C)
15	MSQ	1	(A), (B), (C), (D)	33	MCQ	2	(D)
16	MSQ	2	(D)	34	NAT	2	2
17	MSQ	2	(C), (D)	35	NAT	2	0.6775 to 0.6975
18	MSQ	2	(D)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Unsupervised Learning					11
SVM & Decision Trees					11
Probabilistic & Instance-based Classifiers					11
Regression & Model Selection					11
Neural Networks (MLP / Feed-forward)					11
Total					55

Mock Test 36 — Detailed Solutions

Q.1 — Answer: 8.2668 to 8.2868 [NAT, 1M · Unsupervised Learning › Distance measures]

$$(\sum |x_i - y_i|^3)^{1/3} = (567)^{1/3} = \mathbf{8.28}.$$

For reference: $L_1 = 21$, $L_2 = 10.15$, $L_\infty = 7$ (always $L_\infty \leq L_2 \leq L_1$).

Q.2 — Answer: (A), (D) [MSQ, 1M · Unsupervised Learning › PCA & dimensionality reduction]

(A) **TRUE**. It uses no labels and linear projections.

(B) **FALSE**. Without centering, the first direction tends to point towards the mean.

(C) **FALSE**. PCA is linear; kernel PCA or manifold learning is needed.

(D) **TRUE**. $v^T S v = \lambda$ for a unit eigenvector v .

Q.3 — Answer: (A), (B), (C), (D) [MSQ, 1M · SVM & Decision Trees › SVM: kernels]

(A) **TRUE**. $\varphi(x) = x$.

(B) **TRUE**. Schur product theorem.

(C) **TRUE**. Rank-one PSD Gram matrix.

(D) **TRUE**. Products and sums of valid kernels.

Q.4 — Answer: (A), (C) [MSQ, 1M · Probabilistic & Instance-based Classifiers › Bayes optimal classifier]

(A) **TRUE**. Standard classification.

(B) **FALSE**. It needs the class-conditional densities too.

(C) **TRUE**. The Bayes rule errs with the probability of the less probable class.

(D) **FALSE**. Error cannot go below the Bayes error, which can be positive.

Q.5 — Answer: 1.2757 to 1.2957 [NAT, 1M · SVM & Decision Trees › SVM: margin & geometry]

$$\text{Distance of origin} = |w^T 0 + b| / \|w\| = |b| / \|w\| = 9/7 = \mathbf{1.29}.$$

Q.6 — Answer: (C) [MCQ, 1M · Regression & Model Selection › Bias-variance trade-off]

At R: both training and test errors are high (left side) → underfitting.

At Q: the test error is at its minimum → best trade-off.

At P: training error is very low but test error has risen well above its minimum (right side) → overfitting.

Answer: **R**.

Q.7 — Answer: (C), (D) [MSQ, 1M · Regression & Model Selection › Gradient descent]

(A) **FALSE**. Affine rescaling of features does not change the span of the design, hence not the fitted values.

(B) **FALSE**. It is convex; every local minimum is global.

(C) **TRUE**. One update per example.

(D) **TRUE**. Definition of SGD / mini-batch GD.

Q.8 — Answer: 31 [NAT, 1M · Probabilistic & Instance-based Classifiers › Naive Bayes]

Each class-conditional joint over $2^3 = 8$ configurations needs $2^3 - 1 = 7$ parameters; for K classes: 28.

Priors: $K - 1 = 3$. Total = **31**.

Naive Bayes would need only $Kd + K - 1 = 15$ — linear rather than exponential in d .

Q.9 — Answer: (A), (C), (D) [MSQ, 1M · Regression & Model Selection › Simple linear regression]

(A) **TRUE**. $\beta_1 = 0$ and $\beta_0 = \bar{y}$.

(B) **FALSE**. It is $\hat{y} = \bar{y}$.

(C) **TRUE**. Cross term $2\sum(\hat{y}_i - \bar{y})e_i$ vanishes by the normal equations.

(D) **TRUE**. $\hat{y}$ is a linear combination of 1 and x , both orthogonal to e .

Q.10 — Answer: 0.297 to 0.301 [NAT, 1M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

Example 1: $y = 0 \rightarrow -\ln(1 - 0.4) = 0.5108$.

Example 2: $y = 1 \rightarrow -\ln 0.8 = 0.2231$.

Example 3: $y = 0 \rightarrow -\ln(1 - 0.3) = 0.3567$.

Example 4: $y = 1 \rightarrow -\ln 0.9 = 0.1054$.

Mean = **0.299**.

Q.11 — Answer: (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

(A) **FALSE**. Dropout is disabled at test time (deterministic prediction).

(B) **TRUE**. $E[m \cdot h/p] = h$.

(C) **TRUE**. The symmetry is never broken by GD.

(D) **TRUE**. It limits how far the weights move from their initialisation.

Q.12 — Answer: -1.01 to -0.99 [NAT, 1M · SVM & Decision Trees > SVM: dual & support vectors]

$w = \sum_i \alpha_i y_i x_i$ (sum over support vectors).

$w_2 = (0.5)(-1)(0) + (1.5)(+1)(2) + (1)(-1)(4) = -1$.

(Full vector $w = (-4, -1)$; note $\sum \alpha_i y_i = 0$ holds.)

Q.13 — Answer: 11.99 to 12.01 [NAT, 1M · Unsupervised Learning > k-medoids]

Mean = $120/5 = 24$ (pulled towards the outlier 80).

Medoid = data point minimizing $\sum |x_i - m|$. Costs: $m=2$: 110, $m=10$: 86, $m=12$: 84, $m=16$: 88, $m=80$: 280.

The minimum is at $m = 12$ (the median), cost 84.

Answer: **12**. The medoid is barely affected by the outlier, unlike the mean.

Q.14 — Answer: (A), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

(A) **TRUE**. $P(1) = e^{z_1}/(e^{z_1} + e^{z_2}) = \sigma(z_1 - z_2)$.

(B) **FALSE**. β_j is the change in log-odds; the change in probability depends on x .

(C) **FALSE**. It equals e^{β_j} .

(D) **TRUE**. The boundary becomes $w^T x + b = \ln 4$, same normal vector w .

Q.15 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

(A) **TRUE**. Some example is always misclassified, so updates never stop.

(B) **TRUE**. For any two points on the boundary, $w \cdot (x - x') = 0$.

(C) **TRUE**. It depends only on the radius R of the data and the margin γ .

(D) **TRUE**. All three are linearly separable, e.g. AND: $w = (1, 1)$, $b = -1.5$.

Q.16 — Answer: (D) [MSQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Distances from the query, sorted:

rank	#	(x_1, x_2)	distance	class
1	8	(7, 6)	$\sqrt{0.25} = 0.5$	B
2	9	(8, 6)	$\sqrt{1.25} = 1.118$	A
3	10	(9, 5)	$\sqrt{4.25} = 2.062$	B
4	7	(8, 3)	$\sqrt{7.25} = 2.693$	A
5	6	(4, 5)	$\sqrt{9.25} = 3.041$	A
6	2	(5, 3)	$\sqrt{10.25} = 3.202$	B
7	5	(7, 9)	$\sqrt{12.25} = 3.5$	B
8	4	(6, 1)	$\sqrt{21.25} = 4.61$	A

(A) **FALSE**. 7 nearest: #8, #9, #10, #7, #6, #2, #5 $\rightarrow 3$ A vs 4 B $\Rightarrow$ class B.

(B) **FALSE**. 1 nearest: #8 $\rightarrow 0$ A vs 1 B $\Rightarrow$ class B.

(C) **FALSE**. 3 nearest: #8, #9, #10 $\rightarrow 1$ A vs 2 B $\Rightarrow$ class B.

(D) **TRUE**. 5 nearest: #8, #9, #10, #7, #6 $\rightarrow 3$ A vs 2 B $\Rightarrow$ class A.

Q.17 — Answer: (C), (D) [MSQ, 2M · SVM & Decision Trees > SVM: soft margin & hinge loss]

$$\xi_i = \max(0, 1 - y_i f(x_i)).$$

Point	$y f(x)$	ξ
P1	-1	2
P2	4	0
P3	0.5	0.5
P4	3	0
P5	-1.5	2.5
P6	-1.5	2.5

(A) FALSE. $y f = 4 \Rightarrow \xi = 0$.

(B) FALSE. $y f = -1 \Rightarrow \xi = 2$.

(C) TRUE. $y f = 3 \Rightarrow \xi = 0$.

(D) TRUE. $y f = 4 \Rightarrow \xi = 0$.

Q.18 — Answer: (D) [MSQ, 2M · Unsupervised Learning › PCA & dimensionality reduction]

(A) FALSE. Discarded low-variance directions may be the discriminative ones.

(B) FALSE. It concentrates near the boundary.

(C) FALSE. Centering removes the shift.

(D) TRUE. The between-class scatter has rank $\leq C - 1$.

Q.19 — Answer: 3.99 to 4.01 [NAT, 2M · Unsupervised Learning › Hierarchical clustering]

Use average linkage (UPGMA): $d(A, B) = (1/|A||B|) \sum_{a \in A} \sum_{b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: {P1} + {P5} at height 2. Updated: $d(\{P1, P5\}, \{P2\}) = 8$, $d(\{P1, P5\}, \{P3\}) = 4$, $d(\{P1, P5\}, \{P4\}) = 8$, $d(\{P1, P5\}, \{P6\}) = 7$.

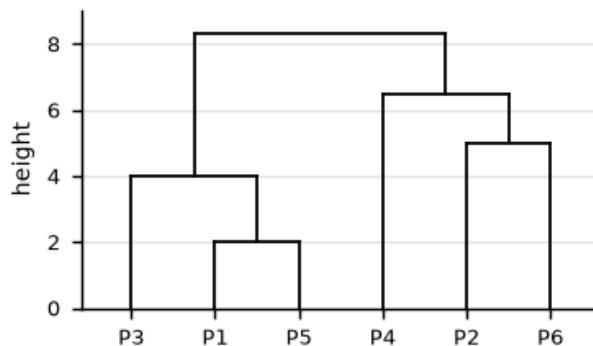
Merge 2: {P3} + {P1, P5} at height 4. Updated: $d(\{P1, P3, P5\}, \{P2\}) = 8.67$, $d(\{P1, P3, P5\}, \{P4\}) = 9.33$, $d(\{P1, P3, P5\}, \{P6\}) = 7$.

Merge 3: {P2} + {P6} at height 5. Updated: $d(\{P2, P6\}, \{P4\}) = 6.5$, $d(\{P2, P6\}, \{P1, P3, P5\}) = 7.83$.

Merge 4: {P4} + {P2, P6} at height 6.5. Updated: $d(\{P2, P4, P6\}, \{P1, P3, P5\}) = 8.33$.

Merge 5: {P1, P3, P5} + {P2, P4, P6} at height 8.33.

P1 and P3 first lie in the same cluster at the merge of height 4, so their cophenetic distance is 4 (their original distance was 3).



Q.20 — Answer: 7.2138 to 7.2862 [NAT, 2M · Regression & Model Selection › Multiple linear regression]

The columns of X are mutually orthogonal, so $X^T X$ is diagonal and each coefficient is $\beta_j = (x_j^T y) / (x_j^T x_j)$.

$$X^T X = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 20 \\ -2 \\ 6 \end{bmatrix}$$

$$\beta = (20/4, -2/4, 6/4) = (5, -0.5, 1.5).$$

$$\hat{y} = 5 + (-0.5)(-1.5) + (1.5)(1) = 7.25.$$

Answer: **7.25**.

Q.21 — Answer: (A), (C), (D) [MSQ, 2M · SVM & Decision Trees · SVM: dual & support vectors]

P1 and P4 (same class) lie on one margin line and P3 on the other; the optimal boundary is parallel to the line through P1, P4, midway to P3. Support vectors: P1, P3, P4.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (4/3, 1)$, $b = -4$; $\|w\| = 1.6667$, margin $2/\|w\| = 1.2$.

Point	P1	P2	P3	P4	P5	P6
$y(w^T x + b)$	1	2	1	1	2.667	4

Support vectors are exactly the points with $y_i(w^T x_i + b) = 1$ ($\alpha_i > 0$): **P1, P3, P4**.

Dual check: $\alpha = (\alpha_1 = 4/9, \alpha_3 = 25/18, \alpha_4 = 17/18)$ satisfy $\sum \alpha_i y_i = 0$ and $\sum \alpha_i y_i x_i = w$.

Q.22 — Answer: 0.5811 to 0.6011 [NAT, 2M · Probabilistic & Instance-based Classifiers · Naive Bayes]

$N(x; \mu, \sigma^2) = \exp(-(x-\mu)^2/(2\sigma^2))/\sqrt{(2\pi\sigma^2)}$. Naive Bayes multiplies the per-feature densities.

$$C_0: N(2.5; 1, 1) = 0.12952 \times N(1.5; 6, 2) = 0.00179; \times \text{prior } 0.7 \Rightarrow 0.000162.$$

$$C_1: N(2.5; 0, 2) = 0.05913 \times N(1.5; 5, 2) = 0.01319; \times \text{prior } 0.3 \Rightarrow 0.000234.$$

$$P(C_1 | x) = 0.000234/(0.000162 + 0.000234) = \mathbf{0.59}.$$

Q.23 — Answer: (B) [MCQ, 2M · Unsupervised Learning · Distance measures]

Compute the measure from $q = (1, 3)$ to every point (cosine column shows similarity, larger = more similar):

Point	coordinates	Manhattan	Euclidean	Chebyshev	cosine similarity
A	(8, 0)	10	7.616	7	0.316
B	(9, 4)	9	8.062	8	0.674
C	(9, 5)	10	8.246	8	0.737
D	(9, 7)	12	8.944	8	0.832

Under Manhattan (L_1) distance the nearest to q is **point B**.

Note how the answer depends on the measure: Manhattan (L_1) distance $\rightarrow$ B; Euclidean (L_2) distance $\rightarrow$ A; Chebyshev (L_∞) distance $\rightarrow$ A; cosine similarity $\rightarrow$ D. Cosine similarity looks only at the angle from the origin, L_1 sums coordinate gaps, L_∞ takes the largest gap.

Q.24 — Answer: 0.0285 to 0.0315 [NAT, 2M · SVM & Decision Trees · Decision trees: overfitting & pruning]

$R(t) = e(t)/N$ with $N = 100$. For an internal node t with subtree T_t : $g(t) = [R(t) - R(T_t)]/(|T_t| - 1)$, where $|T_t|$ = number of leaves.

Node	$R(t)$	$R(T_t)$	$ T_t $	$g(t)$
t1	0.37	0.2	5	0.0425
t2	0.19	0.11	2	0.08
t3	0.2	0.09	3	0.055
t7	0.11	0.08	2	0.03

The weakest link is t7 with the minimum $g = \mathbf{0.03}$; for $\alpha \geq$ this value, collapsing t7 into a leaf does not increase R_α .

Q.25 — Answer: 5.3238 to 5.3438 [NAT, 2M · Regression & Model Selection · Cross-validation]

Pooled CV MSE = $\sum_k n_k \cdot \text{MSE}_k / n$ (a size-weighted average, not the plain average of fold MSEs, since folds are unequal).

$$\sum n_k \text{MSE}_k = 11 \times 6.2 + 19 \times 8.2 + 9 \times 1.7 + 12 \times 3.7 + 14 \times 4.5 = 346.7; n = 65.$$

$$\text{CV} = 346.7/65 = 5.3338 \text{ (plain average of fold MSEs} = 4.86 \text{ — a tempting but wrong value).}$$

Answer: **5.33**.

Q.26 — Answer: (C), (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) · Perceptron]

Target OR truth table for $(x_1, x_2) = (0,0), (0,1), (1,0), (1,1)$: 0, 1, 1, 1. For each option compute $a = w_1x_1 + w_2x_2 + b$ at the four inputs and output 1 iff $a > 0$.

(A) **FALSE**. $a = -1.5, -4, 0.5, -2 \rightarrow$ outputs 0, 0, 1, 0 - x_1 AND (NOT x_2).

(B) **FALSE**. $a = 1.5, -1, 1.5, -1 \rightarrow$ outputs 1, 0, 1, 0 - NOT x_2 .

(C) **TRUE**. $a = -1, 0.5, 2, 3.5 \rightarrow$ outputs 0, 1, 1, 1 - OR.

(D) **TRUE**. $a = -2, 0.5, 1, 3.5 \rightarrow$ outputs 0, 1, 1, 1 - OR.

Q.27 — Answer: 33.912 to 34.048 [NAT, 2M · Regression & Model Selection > Basis functions & feature scaling]

The OLS coefficients transform linearly with y : $\beta'_0 = 1.8\beta_0 + 32$, $\beta'_1 = 1.8\beta_1$.

$\beta'_0 = 1.8 \times 1.1 + 32 = 33.98$; $\beta'_1 = 1.8 \times 3.8 = 6.84$.

Answer: **33.98**.

Q.28 — Answer: 0 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

$a = Wx + b = (-4, -3, 3)$, $h = \text{ReLU}(a) = (0, 0, 3)$, $\hat{y} = v \cdot h + c = -3$; $\partial L / \partial \hat{y} = \hat{y} - y = -5$.

$\text{ReLU}'(a) = (0, 0, 1)$ (unit j is active iff $a_j > 0$).

$\partial L / \partial v_2 = (\hat{y} - y) \cdot h_2 = -5 \times 0 = 0$.

Q.29 — Answer: (A), (B), (C), (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

(A) **TRUE**. Variance $\propto 1/B$.

(B) **TRUE**. Geometric series $1 + \beta + \beta^2 + \dots$

(C) **TRUE**. One update per mini-batch.

(D) **TRUE**. Reduces the noise floor of SGD.

Q.30 — Answer: 9.1416 to 9.2334 [NAT, 2M · Regression & Model Selection > Simple linear regression]

$\bar{x} = 6.429$, $\bar{y} = -13.714$.

i	1	2	3	4	5	6	7
$x_i - \bar{x}$	-4.43	-3.43	-2.43	-1.43	1.57	4.57	5.57
$y_i - \bar{y}$	14.71	13.71	7.71	6.71	-7.29	-16.29	-19.29
$(x_i - \bar{x})(y_i - \bar{y})$	-65.16	-47.02	-18.73	-9.59	-11.45	-74.45	-107.45

$S_{xx} = \sum (x_i - \bar{x})^2 = 93.714$, $S_{xy} = -333.857$.

$\beta_1 = S_{xy} / S_{xx} = -3.5625$, $\beta_0 = \bar{y} - \beta_1 \bar{x} = 9.1875$.

Answer: **9.19**.

Q.31 — Answer: 1.498 to 1.502 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

Hidden pre-activations: $a_1 = -1 \cdot -1.5 + (1) \cdot -0.5 + (0.5) = 1.5$; $a_2 = -0.5 \cdot -1.5 + (-1.5) \cdot -0.5 + (-0.5) = 1$.

Hidden activations ($\text{ReLU}(a) = \max(0, a)$): $h_1 = 1.5$, $h_2 = 1$.

Output pre-activation: $o = 1 \cdot 1.5 + (-1) \cdot 1 + (1) = 1.5$.

$\hat{y} = o = 1.5$.

Q.32 — Answer: (C) [MSQ, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

(A) **FALSE**. PCA ignores labels; LDA uses them.

(B) **FALSE**. Different estimation criteria.

(C) **TRUE**. Priors only enter the constant term $\ln \pi_k$.

(D) **FALSE**. Only when $S_W \propto I$.

Q.33 — Answer: (D) [MCQ, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent: 3 Yes, 9 No $\Rightarrow$ Gini = 0.375.

Shape — Round: [1+, 0-], Square: [2+, 9-]; gain = 0.1023.

Colour — Red: [1+, 1-], Green: [2+, 4-], Blue: [0+, 4-]; gain = 0.0694.

Wind — Weak: [1+, 4-], Strong: [2+, 5-]; gain = 0.0036.

Temp — Hot: [0+, 5−], Mild: [2+, 1−], Cool: [1+, 3−]; gain = 0.1389.

The largest Gini gain is for **Temp**, so it is chosen at the root.

Q.34 — Answer: 2 [NAT, 2M · Unsupervised Learning › k-means]

Data: {0, 3, 7, 10, 12, 13, 16, 18}; initial centroids $c_1 = 3$, $c_2 = 20$.

Iter	centroids used	cluster 1	cluster 2	new centroids
1	3, 20	{0, 3, 7, 10}	{12, 13, 16, 18}	5, 14.75
2	5, 14.75	{0, 3, 7}	{10, 12, 13, 16, 18}	3.33, 13.8
3	3.33, 13.8	{0, 3, 7}	{10, 12, 13, 16, 18}	3.33, 13.8

In the last row the centroids do not change, so the algorithm has converged: final centroids 3.33 and 13.8.

Centroids changed in 2 iteration(s); the final iteration only confirms convergence. Answer **2**.

Q.35 — Answer: 0.6775 to 0.6975 [NAT, 2M · Probabilistic & Instance-based Classifiers › Evaluation metrics]

AUC = fraction of (positive, negative) pairs in which the positive has the higher score (Mann-Whitney statistic).

For each positive, count negatives with lower score: $0.95 \rightarrow 4$; $0.8 \rightarrow 3$; $0.7 \rightarrow 3$; $0.15 \rightarrow 1$.

Concordant pairs = 11 out of $4 \times 4 = 16$.

AUC = $11/16 = \mathbf{0.69}$.

(Equivalently, trace the ROC staircase by lowering the threshold and add the rectangle areas.)

Mock Test 37

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	5	10	20

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Unsupervised Learning	7	11
Neural Networks (MLP / Feed-forward)	7	11
SVM & Decision Trees	7	11
Probabilistic & Instance-based Classifiers	7	11
Regression & Model Selection	7	11

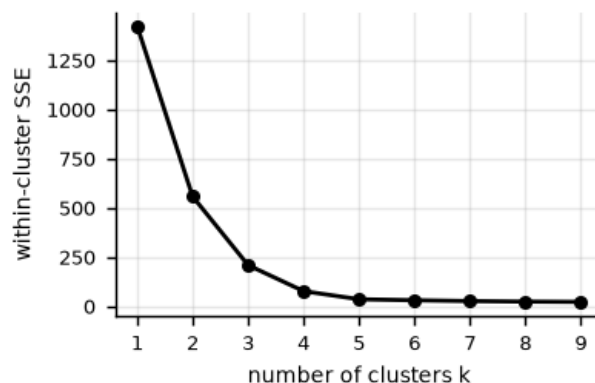
Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MCQ]

k-means is run for $k = 1, 2, \dots, 9$ on a dataset and the final within-cluster SSE is plotted below. Using the elbow method, the most appropriate number of clusters is



- (A) 9 (B) 4
(C) 5 (D) 6

Q.2 [1 mark · MSQ (one or more correct)]

Which of the following statements about initialisation and regularisation of neural networks is/are CORRECT?

- (A) With penalty $(\lambda/2)\|w\|^2$ and SGD, each step first shrinks the weights by the factor $(1 - \eta\lambda)$ (weight decay).
 (B) With inverted dropout (keep probability p), retained activations are scaled by $1/p$ during training, so no scaling is needed at test time.
 (C) Training loss decreasing while validation loss increases is a sign of overfitting.
 (D) Increasing the L2 penalty λ generally increases bias and reduces variance.

Q.3 [1 mark · NAT (numerical answer)]

In AdaBoost with $\varepsilon_t = 0.25$, before normalisation each misclassified example's weight is multiplied by e^{α_t} , where $\alpha_t = \frac{1}{2} \ln((1 - \varepsilon_t)/\varepsilon_t)$. This multiplying factor is _____ (rounded off to two decimal places)

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about bagging, random forests and boosting is/are CORRECT?

- (A) AdaBoost trains its weak learners independently and in parallel.
- (B) Bagging mainly reduces the variance of the base learner.
- (C) Bagged trees can be trained in parallel, whereas boosting is inherently sequential.
- (D) Bagging is mainly used to reduce the bias of high-bias learners such as decision stumps.

Q.5 [1 mark · MSQ (one or more correct)]

Let $p = \text{softmax}(z)$ for logits $z \in \mathbb{R}^K$ and $L = -\ln p_y$ the cross-entropy loss for true class y . Which of the following is/are CORRECT?

- (A) For a confidently wrong prediction ($p_y \rightarrow 0$), the cross-entropy loss grows without bound.
- (B) For softmax + cross-entropy, $\partial L / \partial z = y - p$.
- (C) Subtracting $\max(z)$ from all logits before exponentiating is a standard trick for numerical stability.
- (D) The softmax Jacobian is diagonal, so each p_i depends only on z_i .

Q.6 [1 mark · MSQ (one or more correct)]

Which of the following statements about the k-nearest-neighbour method is/are CORRECT?

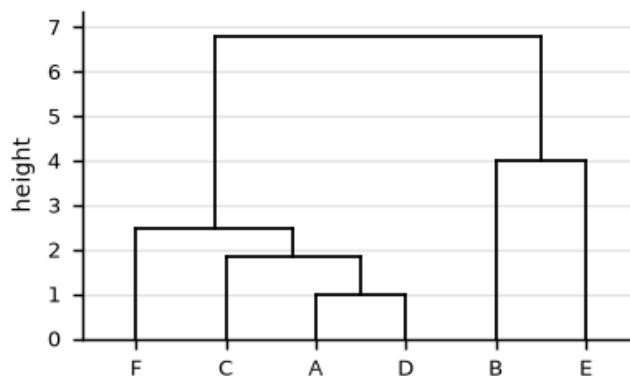
- (A) Unweighted k-NN regression with $k = n$ (all training points) predicts the training mean of y for every query.
- (B) The training error of 1-NN is zero when no two identical inputs carry different labels.
- (C) In high dimensions, distances between random points tend to concentrate, making nearest neighbours less informative.
- (D) In distance-weighted k-NN with weights $1/d$, farther neighbours get larger weights.

Q.7 [1 mark · NAT (numerical answer)]

In a fraud detector, a false negative costs 6 units and a false positive costs 9 units (correct decisions cost 0). The model outputs $p = P(\text{positive} | x)$, i.e. the probability that the transaction is fraudulent. The Bayes-optimal rule predicts positive iff p exceeds the threshold _____ (rounded off to two decimal places)

Q.8 [1 mark · NAT (numerical answer)]

The dendrogram below was obtained by agglomerative clustering of 6 points A-F. If the dendrogram is cut by a horizontal line at height 5.4, the number of clusters obtained is _____ (Answer in integer)



Q.9 [1 mark · NAT (numerical answer)]

A hard-margin linear SVM trained on a separable data set returns the canonical solution $w = (3, 4)$, $b = 1$ (so that $\min_i y_i(w^T x_i + b) = 1$). The width of the margin, i.e. the distance between the hyperplanes $w^T x + b = +1$ and $w^T x + b = -1$, is _____ (rounded off to two decimal places)

Q.10 [1 mark · MSQ (one or more correct)]

Let X be an $n \times p$ design matrix of full column rank p ($p < n$), and let $H = X(X^T X)^{-1} X^T$ be the hat matrix of least-squares regression of y on X . Which of the following statements is/are CORRECT?

- (A) The residual vector is $e = (I - H)y$.
- (B) The vector of fitted values is $\hat{y} = Hy$.
- (C) The average leverage $(1/n)\sum h_{ii}$ equals p/n .
- (D) $\text{trace}(H) = n$.

Q.11 [1 mark · MSQ (one or more correct)]

Which of the following tasks is/are most naturally formulated as **regression** problems?

- (A) Deciding whether an X-ray shows pneumonia or not
- (B) Estimating the fuel efficiency (km per litre) of a car from engine specifications
- (C) Recognising the species of an iris flower from petal measurements
- (D) Predicting the annual salary of an employee from experience and education

Q.12 [1 mark · NAT (numerical answer)]

For a data set, $\bar{x} = 13$, $\bar{y} = 57$, $s_x = 4$, $s_y = 11$ and the sample correlation between x and y is $r = 0.5$. The intercept of the least-squares regression line of y on x is _____ (rounded off to two decimal places)

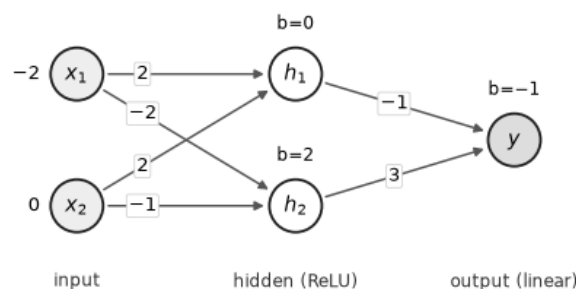
Q.13 [1 mark · NAT (numerical answer)]

Two shopping baskets are encoded as binary vectors (1 = item bought). The Jaccard similarity between basket A and basket B is _____ (rounded off to two decimal places)

Item	milk	bread	eggs	butter	jam	tea	rice	salt
Basket A	1	0	0	0	0	0	1	1
Basket B	1	0	0	1	0	0	0	0

Q.14 [1 mark · NAT (numerical answer)]

For the ReLU network shown (edge labels are weights, 'b' denotes a bias) and input $x = (-2, 0)$, the output y is _____ (Answer in integer)

**Q.15 [1 mark · MCQ]**

A fitted logistic regression model reports that each one-unit increase in *hours of study* multiplies the odds of passing the exam by 2 (other features fixed). The coefficient of *hours of study* in the model is

- (A) 0.88
- (B) 0.5
- (C) 1
- (D) 0.69

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

The sample covariance matrix of a two-feature dataset is S given below. In principal component analysis, the variance of the data along the second principal component is _____ (rounded off to two decimal places)

$$S = \begin{bmatrix} 10 & -6 \\ -6 & 12 \end{bmatrix}$$

Q.17 [2 marks · MSQ (one or more correct)]

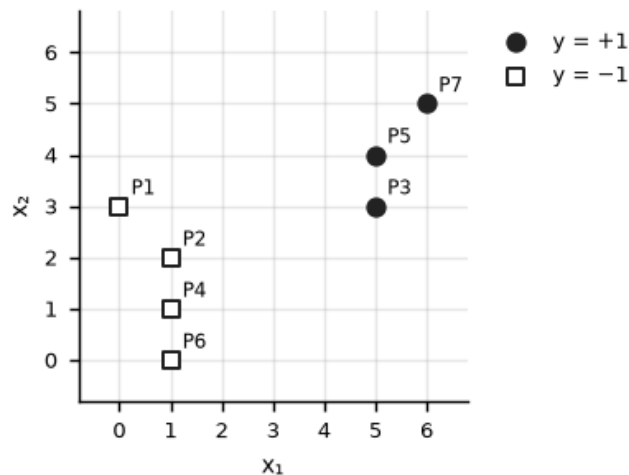
Let σ be the logistic sigmoid, and $'$ denote the derivative. At $z = -2$ (and $-z = 2$), which of the following is/are CORRECT? ($\text{softplus}(z) = \ln(1 + e^z)$.)

- (A) $\text{ReLU}'(-2) = 1$ (B) $\text{softplus}(-2) > \text{ReLU}(-2)$
 (C) $\sigma(-2) + \sigma(2) = 1$ (D) $\sigma'(-2) = \sigma(-2)^2 = 0.014$

Q.18 [2 marks · MCQ]

Four candidate linear decision boundaries are proposed for the labelled training points shown. Which boundary separates the two classes with the LARGEST geometric margin (minimum distance from the boundary to any training point, all points correctly classified)?

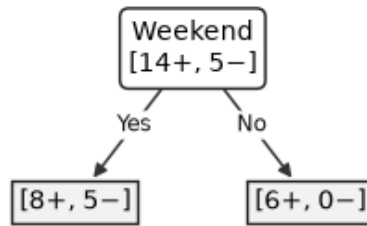
Point	x_1	x_2	y
P1	0	3	-1
P2	1	2	-1
P3	5	3	+1
P4	1	1	-1
P5	5	4	+1
P6	1	0	-1
P7	6	5	+1



- (A) $2x_1 + x_2 - 12.5 = 0$
 (B) $2x_1 - x_2 - 4 = 0$
 (C) $x_1 + x_2 - 4.5 = 0$
 (D) $2x_1 + x_2 - 9.5 = 0$

Q.19 [2 marks · NAT (numerical answer)]

A CART node with 19 training records (class counts [positive+, negative−]) is split on Weekend as shown. Compute the reduction in Gini index (Gini gain) obtained by splitting on Weekend. (rounded off to two decimal places)

**Q.20 [2 marks · MSQ (one or more correct)]**

For the confusion matrix below, which of the following statements is/are CORRECT? (values rounded to 2 decimals)

	Actual +	Actual −
Predicted +	77	47
Predicted −	60	62

- (A) The recall (sensitivity, TPR) is 0.62.
- (B) The false positive rate is 0.38.
- (C) Precision is greater than recall.
- (D) The balanced accuracy is 0.57.

Q.21 [2 marks · MCQ]

A naive Bayes classifier is trained on the 11 examples below. The new instance (Age = Young, Student = No, Rating = Excellent) is classified (i) with maximum-likelihood (unsmoothed) estimates and (ii) with Laplace (add-one) smoothing of the class-conditional probabilities (class priors unsmoothed). The predicted classes are

#	Age	Student	Rating	Buys
1	Young	Yes	Fair	Yes
2	Middle	Yes	Fair	Yes
3	Middle	Yes	Fair	Yes
4	Young	No	Excellent	Yes
5	Young	Yes	Fair	Yes
6	Middle	Yes	Excellent	No
7	Middle	No	Excellent	No
8	Middle	Yes	Fair	Yes
9	Young	Yes	Fair	Yes
10	Middle	Yes	Fair	Yes
11	Senior	No	Excellent	No

- (A) Without smoothing: No; with Laplace smoothing: Yes
- (B) Without smoothing: No; with Laplace smoothing: No
- (C) Without smoothing: Yes; with Laplace smoothing: Yes
- (D) Without smoothing: Yes; with Laplace smoothing: No

Q.22 [2 marks • NAT (numerical answer)]

Agglomerative hierarchical clustering with **average linkage** is applied to 5 points whose pairwise distance matrix is given below (upper triangle shown). The height (linkage distance) at which the **third** merge takes place is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5
P1	0	8	7	1	11
P2		0	7	7	3
P3			0	6	8
P4				0	10
P5					0

Q.23 [2 marks • NAT (numerical answer)]

A linear model $\hat{y} = w_1 x_1 + w_2 x_2 + b$ is trained by stochastic gradient descent (batch size 1) on the per-example loss $\frac{1}{2}(\hat{y} - y)^2$ with learning rate $\eta = 0.1$, starting from $w = (1, -1)$, $b = 1$. The two examples below are processed once, in the given order. After both updates, the value of the model's prediction for $x = (0, 1)$ is _____ (rounded off to three decimal places)

example	x_1	x_2	y
1	3	1	-1
2	3	0	-1

Q.24 [2 marks • NAT (numerical answer)]

Three customers are described by age and income. Age ranges over [20, 60] years and income over [10, 210] thousand rupees in the full dataset. After min-max scaling of each feature to [0, 1] (using the stated ranges), the Euclidean distance from A to its **nearest** neighbour among B and C is _____ (rounded off to two decimal places)

Customer	Age (years)	Income (₹ thousand)
A	44	40
B	44	95
C	59	35

Q.25 [2 marks • MCQ]

A network computes $\hat{y} = W_2 \text{ReLU}(W_1 x + b_1) + b_2$ with the parameters below. For $x = (1, -2, -1)^T$, the output $\hat{y}$ is

$$W_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & -2 \\ 2 & 0 & 0 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 0 & 2 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

(A) $\hat{y} = (1, 4)^T$

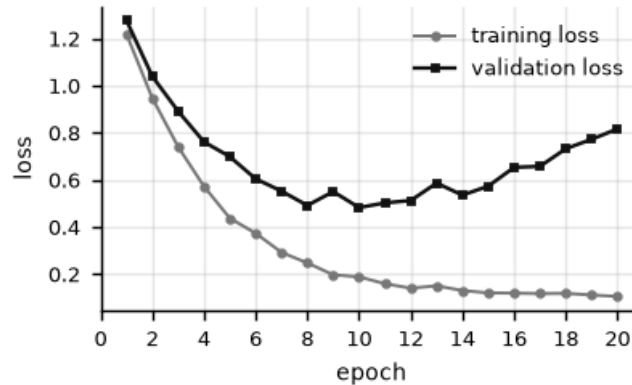
(C) $\hat{y} = (1, 2)^T$

(B) $\hat{y} = (-3, 4)^T$

(D) $\hat{y} = (5, 0)^T$

Q.26 [2 marks · NAT (numerical answer)]

A network is trained with early stopping on validation loss using patience $P = 4$: training stops as soon as the validation loss has failed to improve on its best value for 4 consecutive epochs, and the weights from the best epoch are restored. Using the validation losses below (also plotted), the epoch whose weights are returned (restored) as the final model is _____ (Answer in integer)



epoch	1	2	3	4	5	6	7	8	9	10
val. loss	1.278	1.040	0.892	0.762	0.699	0.605	0.553	0.492	0.551	0.483

epoch	11	12	13	14	15	16	17	18	19	20
val. loss	0.503	0.513	0.587	0.536	0.574	0.654	0.658	0.733	0.773	0.817

Q.27 [2 marks · NAT (numerical answer)]

A regression with intercept and $p = 4$ predictors on $n = 23$ observations has $R^2 = 0.7$. Its adjusted R^2 is _____ (rounded off to three decimal places)

Q.28 [2 marks · MSQ (one or more correct)]

A node with class counts $[20+, 20-]$ can be split by either of two binary splits S1 or S2 (table). For each impurity measure, the “gain” is the parent impurity minus the weighted child impurity. Which of the following statements is/are CORRECT?

Split	Left child $[+, -]$	Right child $[+, -]$
S1	$[14+, 11-]$	$[6+, 9-]$
S2	$[3+, 8-]$	$[17+, 12-]$

- (A) Information gain (entropy) prefers S2 over S1.
- (B) Misclassification-error reduction prefers S2 over S1.
- (C) The information gain of S2 is 0.06 bits (to two decimals).
- (D) Gini gain prefers S2 over S1.

Q.29 [2 marks · MSQ (one or more correct)]

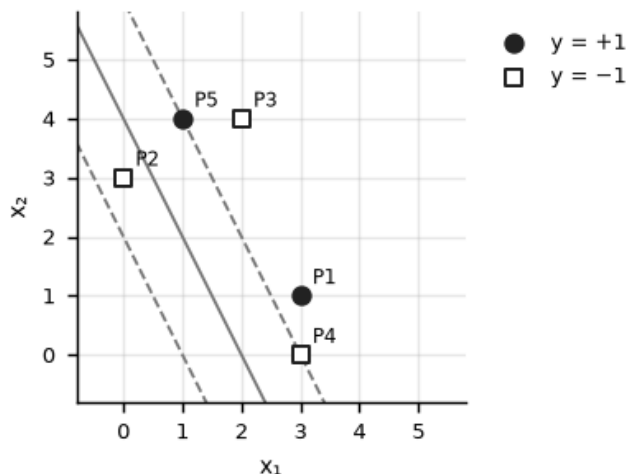
Consider k-means (Lloyd's algorithm) and k-medoids (PAM). Which of the following statements is/are CORRECT?

- (A) As k-means lowers the within-cluster SS, the between-cluster SS also decreases.
- (B) For mean centroids, total sum of squares = within-cluster SS + between-cluster SS.
- (C) Multiplying one feature by 10 never changes the k-means partition.
- (D) The number of ways to split n points into 2 non-empty clusters is $n(n-1)/2$.

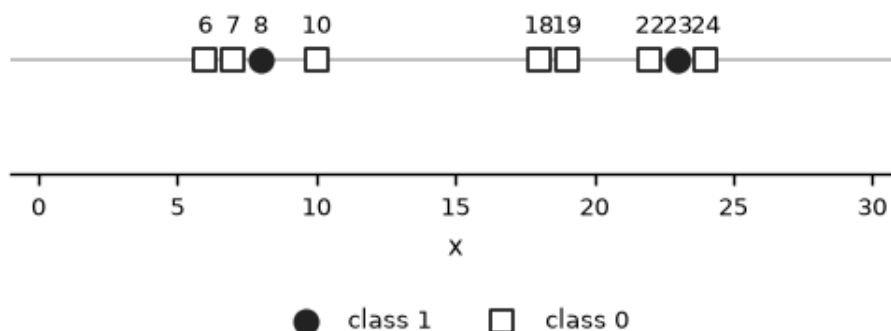
Q.30 [2 marks · NAT (numerical answer)]

A linear SVM with $f(x) = x_1 + 0.5x_2 - 2$ is evaluated on the training points in the table (also plotted, with the boundary $f = 0$ solid and $f = \pm 1$ dashed). Compute the value of the soft-margin primal objective $\frac{1}{2}\|w\|^2 + C \sum_i \xi_i$ with $C = 1$ (using the optimal slacks for this (w, b)). (rounded off to two decimal places)

Point	x_1	x_2	y
P1	3	1	+1
P2	0	3	-1
P3	2	4	-1
P4	3	0	-1
P5	1	4	+1

**Q.31 [2 marks · NAT (numerical answer)]**

9 one-dimensional points with binary labels are shown on the number line (filled circle = class 1, open square = class 0). A 3-NN classifier (absolute distance, majority vote) is evaluated by leave-one-out cross-validation: each point is classified using all the *other* points. The leave-one-out cross-validation error rate (as a fraction) is _____ (rounded off to two decimal places)



Q.32 [2 marks · NAT (numerical answer)]

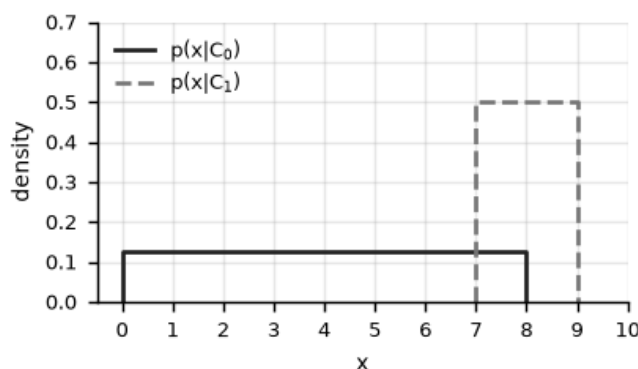
A softmax-regression output layer computes $z = Wx + b$, $p = \text{softmax}(z)$ with W and b below, and is trained with cross-entropy loss $L = -\ln p_y$. For $x = (2, -1)^T$ with true class $y = 2$, $\partial L / \partial W_{11}$ is _____ (rounded off to three decimal places)

$$W = \begin{bmatrix} 1 & -1 \\ 0 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

$$b = \begin{bmatrix} 0.5 \\ 0.5 \\ 0 \end{bmatrix}$$

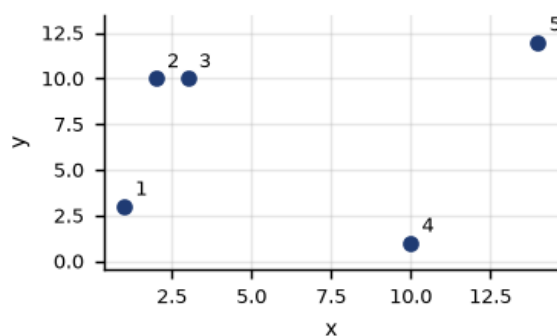
Q.33 [2 marks · NAT (numerical answer)]

For a binary problem with scalar x , $p(x | C_0)$ is uniform on $[0, 8]$ and $p(x | C_1)$ is uniform on $[7, 9]$, with priors $P(C_0) = 0.6$ and $P(C_1) = 0.4$. Then the error rate of the rule "predict C_0 if $x < 7.5$, else C_1 " is _____ (rounded off to three decimal places)

**Q.34 [2 marks · NAT (numerical answer)]**

Consider the 5 training points (x_i, y_i) shown below. The model always predicts the mean of the training targets (ignoring x). Its leave-one-out cross-validation MSE is _____ (rounded off to two decimal places)

x	1	2	3	10	14
y	3	10	10	1	12

**Q.35 [2 marks · NAT (numerical answer)]**

$X_1, \dots, X_n$ are i.i.d. with mean $\mu = 1$ and variance $\sigma^2 = 9$, with $n = 25$. Consider the shrinkage estimator $\hat{\mu} = c \cdot \bar{x}$ of μ , where $\bar{x}$ is the sample mean. Using $\text{MSE} = \text{Bias}^2 + \text{Variance}$, $\text{MSE}(\bar{x}) - \text{MSE}(0.9 \cdot \bar{x})$ is (a negative value means shrinkage is worse) _____ (rounded off to three decimal places)

— END OF QUESTION PAPER —

Mock Test 37 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MCQ	1	(C)	19	NAT	2	0.0539 to 0.0739
2	MSQ	1	(A), (B), (C), (D)	20	MSQ	2	(C), (D)
3	NAT	1	1.7221 to 1.7421	21	MCQ	2	(D)
4	MSQ	1	(B), (C)	22	NAT	2	6.49 to 6.51
5	MSQ	1	(A), (C)	23	NAT	2	−0.9027 to −0.8973
6	MSQ	1	(A), (B), (C)	24	NAT	2	0.265 to 0.285
7	NAT	1	0.59 to 0.61	25	MCQ	2	(C)
8	NAT	1	2	26	NAT	2	10
9	NAT	1	0.39 to 0.41	27	NAT	2	0.6313 to 0.6353
10	MSQ	1	(A), (B), (C)	28	MSQ	2	(A), (B), (C), (D)
11	MSQ	1	(B), (D)	29	MSQ	2	(B)
12	NAT	1	38.9294 to 39.3206	30	NAT	2	6.115 to 6.135
13	NAT	1	0.24 to 0.26	31	NAT	2	0.2122 to 0.2322
14	NAT	1	17	32	NAT	2	1.8349 to 1.8389
15	MCQ	1	(D)	33	NAT	2	0.1355 to 0.1395
16	NAT	2	4.9072 to 4.9272	34	NAT	2	29.4769 to 29.7731
17	MSQ	2	(B), (C)	35	NAT	2	0.0554 to 0.0614
18	MCQ	2	(D)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Unsupervised Learning					11
Neural Networks (MLP / Feed-forward)					11
SVM & Decision Trees					11
Probabilistic & Instance-based Classifiers					11
Regression & Model Selection					11
Total					55

Mock Test 37 — Detailed Solutions

Q.1 — Answer: (C) [MCQ, 1M · Unsupervised Learning › k-means]

The SSE drops steeply up to $k = 5$ and then decreases only slowly (relative drops: $k=2$: 61%, $k=3$: 62%, $k=4$: 63%, $k=5$: 53%, $k=6$: 14%, $k=7$: 13%, $k=8$: 10%, $k=9$: 6%). The bend ('elbow') is at $k = 5$.

$k = 9$ gives the lowest SSE but SSE always (weakly) decreases with k , so minimizing it would pick the largest k ; it is not a model-selection criterion by itself.

Q.2 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

(A) TRUE. $w \leftarrow w - \eta(\nabla L + \lambda w) = (1 - \eta\lambda)w - \eta\nabla L$.

(B) TRUE. $E[m \cdot h/p] = h$.

(C) TRUE. Generalisation gap grows.

(D) TRUE. Stronger shrinkage.

Q.3 — Answer: 1.7221 to 1.7421 [NAT, 1M · SVM & Decision Trees › Ensembles: boosting]

$e^\alpha = e^{\frac{1}{2} \ln((1-\epsilon)/\epsilon)} = \sqrt{((1-\epsilon)/\epsilon)} = \sqrt{3} = 1.73$.

Correctly classified examples are multiplied by $e^{-\alpha} = \sqrt{(\epsilon/(1-\epsilon))}$.

$\epsilon < 0.5 \Rightarrow \alpha > 0$; $\epsilon = 0.5 \Rightarrow \alpha = 0$ (no better than chance).

Q.4 — Answer: (B), (C) [MSQ, 1M · SVM & Decision Trees › Ensembles: bagging & random forests]

(A) FALSE. It is sequential.

(B) TRUE. Averaging bootstrap models smooths fluctuations.

(C) TRUE. Boosting needs the previous round's weights.

(D) FALSE. Bagging reduces variance; boosting reduces bias.

Q.5 — Answer: (A), (C) [MSQ, 1M · Neural Networks (MLP / Feed-forward) › Softmax + cross-entropy]

(A) TRUE. $-\ln p_y \rightarrow \infty$.

(B) FALSE. Sign is reversed: it is $p - y$.

(C) TRUE. Avoids overflow without changing the output.

(D) FALSE. Off-diagonal terms $-p_i p_j$ are non-zero.

Q.6 — Answer: (A), (B), (C) [MSQ, 1M · Probabilistic & Instance-based Classifiers › k-nearest neighbours]

(A) TRUE. All points are neighbours of every query.

(B) TRUE. Each training point is its own nearest neighbour.

(C) TRUE. The ratio of farthest to nearest distance approaches 1.

(D) FALSE. Weights decrease with distance.

Q.7 — Answer: 0.59 to 0.61 [NAT, 1M · Probabilistic & Instance-based Classifiers › Bayes optimal classifier]

Expected cost of predicting positive = $(1-p) \cdot C_{FP}$; of predicting negative = $p \cdot C_{FN}$.

Predict positive iff $p \cdot 6 > (1-p) \cdot 9 \Leftrightarrow p > C_{FP}/(C_{FP} + C_{FN}) = 9/15 = 0.6$.

Q.8 — Answer: 2 [NAT, 1M · Unsupervised Learning › Hierarchical clustering]

Merges below the cut height 5.4: 4 (heights 1, 1.83, 2.47, 4).

Each merge reduces the number of clusters by one: $6 - 4 = 2$. Equivalently, count the vertical lines crossed by the horizontal cut.

Q.9 — Answer: 0.39 to 0.41 [NAT, 1M · SVM & Decision Trees › SVM: margin & geometry]

Margin width = $2/\|w\|$. $\|w\| = \sqrt{9 + 16} = 5$.

Width = $2/5 = 0.4$.

Common mistakes: $1/\|w\|$ (that is the distance from the boundary to one margin hyperplane) or $2/\|w\|^2$ — the bias b plays no role.

Q.10 — Answer: (A), (B), (C) [MSQ, 1M · Regression & Model Selection › Multiple linear regression]**(A) TRUE.** $e = y - \hat{y} = (I - H)y$.**(B) TRUE.** $\hat{y} = X\beta = X(X^T X)^{-1} X^T y$.**(C) TRUE.** $\sum h_{ii} = \text{trace}(H) = p$.**(D) FALSE.** $\text{trace}(H) = p$.**Q.11 — Answer: (B), (D)** [MSQ, 1M · Regression & Model Selection › Regression vs classification]

A supervised task is *regression* when the target y is real-valued (loss such as squared error), and *classification* when y takes values in a finite set of labels (loss such as cross-entropy or 0-1 loss).

(A) FALSE. The output (pneumonia / normal) is a discrete class label → classification.**(B) TRUE.** The output (km per litre) is a continuous real-valued quantity → regression.**(C) FALSE.** The output (species label) is a discrete class label → classification.**(D) TRUE.** The output (salary) is a continuous real-valued quantity → regression.**Q.12 — Answer: 38.9294 to 39.3206** [NAT, 1M · Regression & Model Selection › Simple linear regression]

$$\beta_1 = r \cdot s_y / s_x = 0.5 \times 11/4 = 1.375.$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x} = 57 - (1.375)(13) = 39.125.$$

Answer: **39.12**.**Q.13 — Answer: 0.24 to 0.26** [NAT, 1M · Unsupervised Learning › Distance measures]Counts: $f_{11} = 1, f_{10} = 2, f_{01} = 1, f_{00} = 4$.

$$\text{Jaccard } J = f_{11} / (f_{11} + f_{10} + f_{01}) = 1/4 = 0.25 \text{ (0-0 matches are ignored).}$$

$$\text{SMC} = (f_{11} + f_{00}) / m = 5/8 = 0.625.$$

Required Jaccard similarity = **0.25**.**Q.14 — Answer: 17** [NAT, 1M · Neural Networks (MLP / Feed-forward) › Forward pass] $a = Wx + b = (-4, 6)$; $h = \text{ReLU}(a) = (0, 6)$ - a negative pre-activation is clipped to 0.

$$y = -1 \cdot 0 + (3) \cdot 6 + (-1) = 17.$$

A common slip is to forget the ReLU and use the raw pre-activation (which would give 21).

Q.15 — Answer: (D) [MCQ, 1M · Probabilistic & Instance-based Classifiers › Logistic regression]Odds ratio for a unit increase = $e^\beta = 2 \Rightarrow \beta = \ln 2 = 0.69$.Distractors: the odds ratio itself, $\log_{10}$, or $1/R$ are common slips; the model is linear in the *natural* log-odds.**Q.16 — Answer: 4.9072 to 4.9272** [NAT, 2M · Unsupervised Learning › PCA & dimensionality reduction]Eigenvalues of $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$: $\lambda = (a + c)/2 \pm \sqrt{((a - c)/2)^2 + b^2}$.

$$\lambda = 11 \pm \sqrt{(1 + 36)} = 11 \pm 6.0828 \rightarrow \lambda_1 = 17.0828, \lambda_2 = 4.9172 \text{ (check: } \lambda_1 + \lambda_2 = \text{trace} = 22, \lambda_1 \lambda_2 = \det = 84).$$

$$\text{Eigenvector for } \lambda_1: (S - \lambda_1 I)v = 0 \rightarrow (-7.0828)v_1 + (-6)v_2 = 0 \rightarrow v \propto (1, -1.1805).$$

$$\text{PVE(PC1)} = \lambda_1 / (\lambda_1 + \lambda_2) = 77.65\%.$$

Answer: **4.92**.**Q.17 — Answer: (B), (C)** [MSQ, 2M · Neural Networks (MLP / Feed-forward) › Activation functions]

$$\sigma(-2) = 0.1192, \tanh(-2) = -0.964.$$

(A) FALSE. $\text{ReLU}' = 1$ for $z > 0$ and 0 for $z < 0$.**(B) TRUE.** $\text{softplus} = 0.1269 > 0$; $\ln(1 + e^z) > \max(0, z)$ always.**(C) TRUE.** $\sigma(-z) = 1 - \sigma(z)$.**(D) FALSE.** $\sigma' = \sigma(1 - \sigma) = 0.105$.**Q.18 — Answer: (D)** [MCQ, 2M · SVM & Decision Trees › SVM: margin & geometry]

For $n^T x - c = 0$, the geometric margin is $\min_i y_i s(n^T x_i - c) / \|n\|$ with orientation $s = \pm 1$ chosen so that the positive class is on the positive side; a negative value means some point is misclassified.

Hyperplane	$\ n\ $	$\min_i$ distance (signed, best orientation)
$2x_1 + x_2 - 12.5 = 0$	2.236	0.224
$2x_1 - x_2 - 4 = 0$	2.236	0.894
$x_1 + x_2 - 4.5 = 0$	1.414	1.061
$2x_1 + x_2 - 9.5 = 0$	2.236	1.565

The largest margin is 1.565, for $2x_1 + x_2 - 9.5 = 0$. Note that a boundary can separate the data perfectly yet pass close to some point — only the minimum distance matters.

Q.19 — Answer: 0.0539 to 0.0739 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [14+, 5-]: Gini = $1 - [(14/19)^2 + (5/19)^2] = 0.3878$.

Weekend = Yes: [8+, 5-], Gini = $1 - [(8/13)^2 + (5/13)^2] = 0.4734$.

Weekend = No: [6+, 0-], Gini = $1 - [(6/6)^2 + (0/6)^2] = 0$.

Weighted Gini = $(13/19)(0.4734) + (6/19)(0) = 0.3239$.

Gini gain = $0.3878 - 0.3239 = 0.0639$.

Answer: **0.06**.

Q.20 — Answer: (C), (D) [MSQ, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

TP = 77, FN = 60, FP = 47, TN = 62.

(A) **FALSE**. recall (sensitivity, TPR) = $TP/(TP + FN) = 0.562$, not 0.62.

(B) **FALSE**. false positive rate = $FP/(FP + TN) = 0.4312$, not 0.38.

(C) **TRUE**. P = 0.621, R = 0.562.

(D) **TRUE**. balanced accuracy = $(TPR + TNR)/2 = 0.5654$.

Q.21 — Answer: (D) [MCQ, 2M · Probabilistic & Instance-based Classifiers > Naive Bayes]

(i) **No smoothing.**

Yes: $P(\text{Buys}=\text{Yes}) = 8/11$; $P(\text{Age}=\text{Young}|\text{Yes}) = 4/8$; $P(\text{Student}=\text{No}|\text{Yes}) = 1/8$;

$P(\text{Rating}=\text{Excellent}|\text{Yes}) = 1/8 \Rightarrow \text{score} = 0.00568$.

No: $P(\text{Buys}=\text{No}) = 3/11$; $P(\text{Age}=\text{Young}|\text{No}) = 0/3$; $P(\text{Student}=\text{No}|\text{No}) = 2/3$;

$P(\text{Rating}=\text{Excellent}|\text{No}) = 3/3 \Rightarrow \text{score} = 0$.

A zero count makes the whole product for No equal to 0 (zero-frequency problem), so the prediction is Yes.

(ii) **Laplace smoothing.**

Yes: $P(\text{Buys}=\text{Yes}) = 8/11$; $P(\text{Age}=\text{Young}|\text{Yes}) = (4+1)/(8+3) = 5/11$; $P(\text{Student}=\text{No}|\text{Yes}) = (1+1)/(8+2) = 1/5$; $P(\text{Rating}=\text{Excellent}|\text{Yes}) = (1+1)/(8+2) = 1/5 \Rightarrow \text{score} = 0.01322$.

No: $P(\text{Buys}=\text{No}) = 3/11$; $P(\text{Age}=\text{Young}|\text{No}) = (0+1)/(3+3) = 1/6$; $P(\text{Student}=\text{No}|\text{No}) = (2+1)/(3+2) = 3/5$; $P(\text{Rating}=\text{Excellent}|\text{No}) = (3+1)/(3+2) = 4/5 \Rightarrow \text{score} = 0.02182$.

Larger score: No. Answer: **Without smoothing: Yes; with Laplace smoothing: No.**

Q.22 — Answer: 6.49 to 6.51 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Use average linkage (UPGMA): $d(A,B) = (1/|A||B|) \sum_{a \in A} \sum_{b \in B} d(a,b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: {P1} + {P4} at height **1**. Updated: $d(\{P1, P4\}, \{P2\}) = 7.5$, $d(\{P1, P4\}, \{P3\}) = 6.5$, $d(\{P1, P4\}, \{P5\}) = 10.5$.

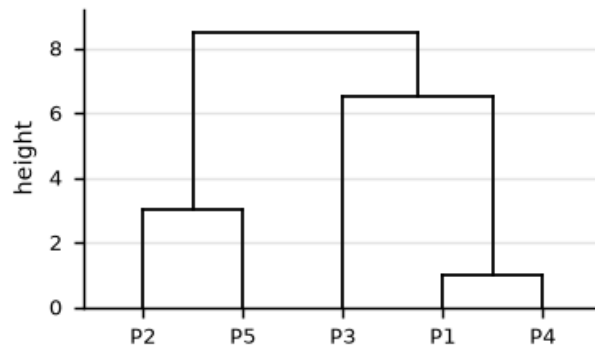
Merge 2: {P2} + {P5} at height **3**. Updated: $d(\{P2, P5\}, \{P3\}) = 7.5$, $d(\{P2, P5\}, \{P1, P4\}) = 9$.

Merge 3: {P3} + {P1, P4} at height **6.5**. Updated: $d(\{P1, P3, P4\}, \{P2, P5\}) = 8.5$.

Merge 4: {P2, P5} + {P1, P3, P4} at height **8.5**.

The third merge occurs at height **6.5**.

Dendrogram:



Q.23 — Answer: -0.9027 to -0.8973 [NAT, 2M · Regression & Model Selection > Gradient descent]

Per-example loss $\mathcal{L} = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial \mathcal{L} / \partial w = (\hat{y} - y)x$, $\partial \mathcal{L} / \partial b = (\hat{y} - y)$. Update after each example.

Example 1: $\hat{y} = 3$, error $\hat{y} - y = 4$; $\nabla_w = (12, 4)$, $\nabla_b = 4 \Rightarrow w = (-0.2, -1.4)$, $b = 0.6$.

Example 2: $\hat{y} = 0$, error $\hat{y} - y = 1$; $\nabla_w = (3, 0)$, $\nabla_b = 1 \Rightarrow w = (-0.5, -1.4)$, $b = 0.5$.

$\hat{y} = -0.5 \times 0 + -1.4 \times 1 + 0.5 = -0.9$.

Answer: **-0.9**.

Q.24 — Answer: 0.265 to 0.285 [NAT, 2M · Unsupervised Learning > Distance measures]

Min-max scaling: $x' = (x - \min) / (\max - \min)$; age range [20, 60], income range [10, 210].

Customer	age'	income'
A	0.6	0.15
B	0.6	0.425
C	0.975	0.125

Raw distances: $d(A,B) = 55$, $d(A,C) = 15.81 \rightarrow$ nearest is C (income dominates because of its larger range).

Scaled distances: $d(A,B) = 0.275$, $d(A,C) = 0.3758 \rightarrow$ nearest is B.

Answer: **0.28**. Scaling changes which neighbour is nearest — distance-based methods (k-means, k-NN, hierarchical clustering) should be run on standardized features.

Q.25 — Answer: (C) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

$a = W_1 x + b_1 = (2, -2, 3)^T$; $h = \text{ReLU}(a) = (2, 0, 3)^T$.

$\hat{y} = W_2 h + b_2 = (1, 2)^T$.

Distractors: skipping the ReLU gives $(-3, 4)^T$; using W_1^T instead of W_1 gives $(1, 4)^T$; dropping the biases gives $(0, 2)^T$; using $|a|$ instead of ReLU gives $(5, 0)^T$.

Q.26 — Answer: 10 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Rule: keep track of the best (lowest) validation loss; an epoch that does not strictly improve on it increments a counter, which is reset on improvement; stop when the counter reaches patience $P = 4$.

Trace - 1: 1.278 (new best); 2: 1.040 (new best); 3: 0.892 (new best); 4: 0.762 (new best); 5: 0.699 (new best); 6: 0.605 (new best); 7: 0.553 (new best); 8: 0.492 (new best); 9: 0.551 (no improvement, count = 1); 10: 0.483 (new best); 11: 0.503 (no improvement, count = 1); 12: 0.513 (no improvement, count = 2); 13: 0.587 (no improvement, count = 3); 14: 0.536 (no improvement, count = 4).

Training stops after epoch 14; the best validation loss 0.483 occurred at epoch 10, whose weights are restored.

Answer: **10**. The widening train/validation gap after this point is the signature of overfitting.

Q.27 — Answer: 0.6313 to 0.6353 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

Adjusted $R^2 = 1 - [\text{SSE}/(n - p - 1)] / [\text{SST}/(n - 1)] = 1 - (1 - R^2)(n - 1)/(n - p - 1)$, where p is the number of predictors (excluding the intercept).

$= 1 - (1 - 0.7) \times 22/18 = 0.6333$.

Answer: **0.633**.

Q.28 — Answer: (A), (B), (C), (D) [MSQ, 2M · SVM & Decision Trees > Decision trees: impurity measures]

Parent [20+, 20-]: $H = 1$, Gini = 0.5, error = 0.5.

S1: children $H = 0.9896, 0.971$; Gini = 0.4928, 0.48; error = 0.44, 0.4. Gains: IG = 0.0174, Gini = 0.012, error = 0.075.

S2: children $H = 0.8454, 0.9784$; Gini = 0.3967, 0.4851; error = 0.2727, 0.4138. Gains: IG = 0.0582, Gini = 0.0392, error = 0.125.

(A) TRUE. S1: 0.0174, S2: 0.0582.

(B) TRUE. S1: 0.075, S2: 0.125.

(C) TRUE. IG(S2) = 0.0582.

(D) TRUE. S1: 0.012, S2: 0.0392.

Q.29 — Answer: (B) [MSQ, 2M · Unsupervised Learning > k-means]

(A) FALSE. TSS is fixed, so BSS increases.

(B) TRUE. The standard ANOVA decomposition; reducing WSS increases BSS.

(C) FALSE. Non-uniform scaling changes distances and can change the partition.

(D) FALSE. It is $2^{n-1} - 1$.

Q.30 — Answer: 6.115 to 6.135 [NAT, 2M · SVM & Decision Trees > SVM: soft margin & hinge loss]

Compute $f(x_i)$, the functional margin $y_i f(x_i)$ and the slack/hinge loss for each point:

Point	x	y	$f(x)$	$y f(x)$	$\xi = \max(0, 1 - y f)$
P1	(3, 1)	+1	1.5	1.5	0
P2	(0, 3)	-1	-0.5	0.5	0.5
P3	(2, 4)	-1	2	-2	3
P4	(3, 0)	-1	1	-1	2
P5	(1, 4)	+1	1	1	0

$\frac{1}{2}\|w\|^2 = 0.625$; objective = $0.625 + 1 \times 5.5 = \mathbf{6.12}$.

Points with $y f \geq 1$ contribute 0; points inside the margin contribute between 0 and 1; misclassified points contribute more than 1.

Q.31 — Answer: 0.2122 to 0.2322 [NAT, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Classify each point by its nearest neighbour(s) among the remaining points:

x	true	3 nearest (excluding itself)	LOO prediction	
6	0	7(0), 8(1), 10(0)	0	
7	0	6(0), 8(1), 10(0)	0	
8	1	7(0), 6(0), 10(0)	0	wrong
10	0	8(1), 7(0), 6(0)	0	
18	0	19(0), 22(0), 23(1)	0	
19	0	18(0), 22(0), 23(1)	0	
22	0	23(1), 24(0), 19(0)	0	
23	1	22(0), 24(0), 19(0)	0	wrong
24	0	23(1), 22(0), 19(0)	0	

Misclassified: 2 of 9. Error rate = $2/9 = \mathbf{0.22}$.

Note: the *training* error of 1-NN is 0, but the LOOCV error is not — a point cannot be its own neighbour.

Q.32 — Answer: 1.8349 to 1.8389 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$z = Wx + b = (3.5, 1, -1.5)$; $p = \text{softmax}(z) = (0.9184, 0.0754, 0.0062)$.

$\delta = \partial L / \partial z = p - y = (0.9184, -0.9246, 0.0062)$ (true class 2).

Since $z_k = \sum_j W_{kj} x_j + b_k$: $\partial L / \partial W_{kj} = (p_k - y_k) x_j$ and $\partial L / \partial b_k = p_k - y_k$ (gradient matrix = $(p - y)x^T$).
 $\partial L / \partial W_{11} = 0.9184 \times 2 = \mathbf{1.8368}$.

Q.33 — Answer: 0.1355 to 0.1395 [NAT, 2M · Probabilistic & Instance-based Classifiers › Bayes optimal classifier]

$$P(\text{error}) = \pi_0 P(x \geq 7.5 | C_0) + \pi_1 P(x < 7.5 | C_1).$$

$$= 0.6 \times (8 - 7.5) / 8 + 0.4 \times (7.5 - 7) / 2 = 0.0375 + 0.1 = \mathbf{0.138}.$$

(For comparison, the Bayes error is 0.075: the Bayes rule assigns the whole overlap to the class with the larger $\pi_k p(x|C_k)$.)

Q.34 — Answer: 29.4769 to 29.7731 [NAT, 2M · Regression & Model Selection › Cross-validation]

Leaving out point i , the prediction is $(\sum y - y_i) / (n - 1)$ with $\sum y = 36$.

y_i	3	10	10	1	12
prediction	8.25	6.5	6.5	8.75	6
error	-5.25	3.5	3.5	-7.75	6

(Shortcut: error = $n(y_i - \bar{y}) / (n - 1)$, $\bar{y} = 7.2$.) LOOCV MSE = mean of squared errors = 29.625.

Answer: **29.62**.

Q.35 — Answer: 0.0554 to 0.0614 [NAT, 2M · Regression & Model Selection › Bias-variance trade-off]

$$E[c \bar{x}] = c\mu = \text{Bias} = (c - 1)\mu; \text{Var}(c \bar{x}) = c^2 \sigma^2 / n. \text{MSE}(c) = (c - 1)^2 \mu^2 + c^2 \sigma^2 / n.$$

Here $\mu = 1$, $\sigma^2 / n = 9/25 = 0.36$.

$$\text{MSE}(\bar{x}) = \sigma^2 / n = 0.36 \text{ (unbiased)}. \text{MSE}(0.9\bar{x}) = 0.3016. \text{Difference} = 0.0584.$$

Answer: **0.058**.

Mock Test 38

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	6	8	21

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Neural Networks (MLP / Feed-forward)	7	11
Probabilistic & Instance-based Classifiers	7	11
SVM & Decision Trees	7	11
Unsupervised Learning	7	11
Regression & Model Selection	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

Pure (single-example) SGD, i.e. batch size 1, is run for $E = 5$ epochs on $N = 50000$ examples. The number of parameter updates is _____ (Answer in integer)

Q.2 [1 mark · MSQ (one or more correct)]

Which of the following statements about naive Bayes classifiers is/are CORRECT?

- (A) If a feature is duplicated, naive Bayes counts its evidence twice.
- (B) The number of parameters of naive Bayes grows exponentially with the number of features.
- (C) With Laplace (add-one) smoothing, a feature value with zero count in a class still gets probability 0.
- (D) Naive Bayes cannot be applied to problems with more than two classes.

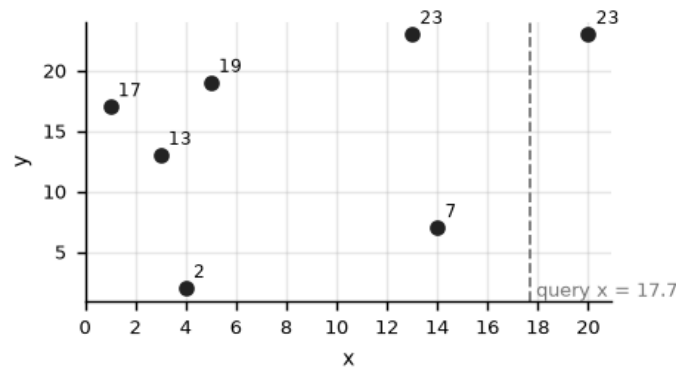
Q.3 [1 mark · NAT (numerical answer)]

Consider the polynomial kernel $K(\mathbf{x}, \mathbf{z}) = (\mathbf{x}^T \mathbf{z} + 1)^2$ with $\mathbf{x} = (-1, 3)$ and $\mathbf{z} = (2, 0)$. The value of $K(\mathbf{x}, \mathbf{z})$ is _____ (Answer in integer)

Q.4 [1 mark · NAT (numerical answer)]

The training data for a k-NN regressor are given below (also plotted). Using $k = 4$ and absolute distance $|x - x_i|$, the predicted value (unweighted average of neighbours) at $x = 17.7$ is _____ (rounded off to two decimal places)

x	1	3	4	5	13	14	20
y	17	13	2	19	23	7	23

**Q.5 [1 mark · MSQ (one or more correct)]**

Which of the following statements about backpropagation is/are CORRECT?

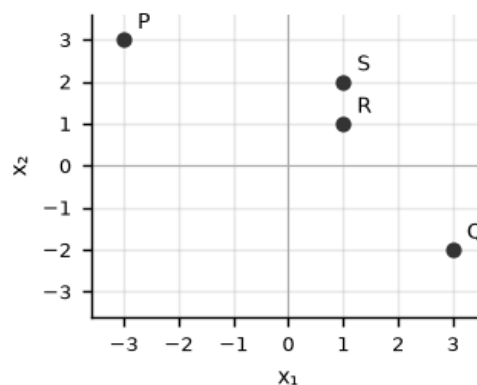
- (A) Backpropagation can only be used with sigmoid activation functions.
- (B) For a hidden unit j , $\delta_j = f'(a_j) \sum_k w_{kj} \delta_k$, summing over the units k that j feeds into.
- (C) Hidden-layer deltas are computed before the output-layer deltas.
- (D) The gradient with respect to a bias b_j equals δ_j .

Q.6 [1 mark · NAT (numerical answer)]

Lloyd's k-means is run on $n = 1500$ points in $d = 2$ dimensions with $k = 6$ for $T = 15$ iterations. Counting one subtraction per coordinate per point-centroid pair in the assignment step, the total number of such subtractions is _____ (Answer in integer)

Q.7 [1 mark · MSQ (one or more correct)]

A perceptron with weights $w = (-2, 1)$ and bias $b = 4$ outputs $\hat{y} = +1$ if $w \cdot x + b \geq 0$ and $\hat{y} = -1$ otherwise. Which of the points shown below are classified as $\hat{y} = +1$?



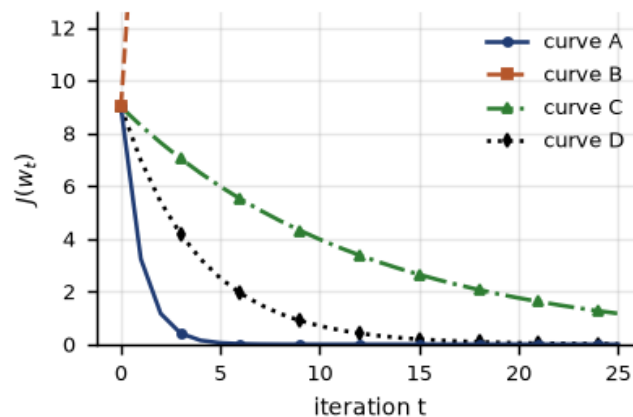
- (A) Point P = (-3, 3)
- (B) Point Q = (3, -2)
- (C) Point R = (1, 1)
- (D) Point S = (1, 2)

Q.8 [1 mark · NAT (numerical answer)]

A regularisation parameter is tuned over 8 candidate values using 3-fold cross-validation. After selecting the best value, the model is refitted once on the full training data. The total number of times the model is fitted is _____ (Answer in integer)

Q.9 [1 mark · MCQ]

Gradient descent $w_{t+1} = w_t - \eta J'(w_t)$ is run on $J(w) = (w - 5)^2$ from $w_0 = 8$ with four learning rates $\eta \in \{0.02, 0.06, 0.2, 1.25\}$. The loss curves are shown below (curves that leave the plotted range are cut off). Which curve corresponds to $\eta = 1.25$?



- (A) Curve C
(C) Curve A

- (B) Curve B
(D) Curve D

Q.10 [1 mark · MCQ]

Cost-complexity pruning of a classification tree produced the nested subtrees $T1 \supset T2 \supset T3 \supset T4$ below. For complexity parameter $\alpha = 0.002$, which subtree minimises $R_\alpha(T) = R(T) + \alpha|T|$?

Subtree	T1	T2	T3	T4
Leaves $ T $	13	12	7	5
Training error $R(T)$	0.045	0.09	0.105	0.145

- (A) T1
(C) T3

- (B) T2
(D) T4

Q.11 [1 mark · NAT (numerical answer)]

A clustering algorithm groups labelled data into three clusters; the contingency table of cluster versus true class counts is given below. If every point in a cluster is labelled with the cluster's majority class, the number of misclassified points is ____ (Answer in integer)

	Class P	Class Q	Class R
Cluster 1	10	25	2
Cluster 2	6	4	26
Cluster 3	5	10	0

Q.12 [1 mark · MSQ (one or more correct)]

Which of the following statements about hierarchical clustering is/are CORRECT?

- (A) Naive agglomerative clustering runs in $O(n)$ time.
 (B) Agglomerative clustering starts with each point as its own cluster and repeatedly merges the two closest clusters.
 (C) Cutting a dendrogram at a greater height produces more clusters.
 (D) Storing the full distance matrix for agglomerative clustering needs $O(n^2)$ memory.

Q.13 [1 mark · NAT (numerical answer)]

A 3-class softmax classifier produces the logits $z_1 = 0$, $z_2 = -2$, $z_3 = -1$ for an input $\mathbf{x}$. The predicted probability of class 3, $P(y = 3 | \mathbf{x})$, is ____ (rounded off to two decimal places)

Q.14 [1 mark · MCQ]

If the noise in $y = w^T x + \varepsilon$ is Laplace distributed (instead of Gaussian), the MLE of w minimises:

- (A) $\|w\|_1$
- (B) $\sum \max(0, 1 - y_i w^T x_i)$
- (C) $\sum \log(1 + e^{-y_i w^T x_i})$
- (D) $\sum |y_i - w^T x_i|$

Q.15 [1 mark · NAT (numerical answer)]

A bootstrap sample of size 11 is drawn with replacement from 11 distinct training points. The expected number of distinct points appearing in the sample is _____ (rounded off to two decimal places)

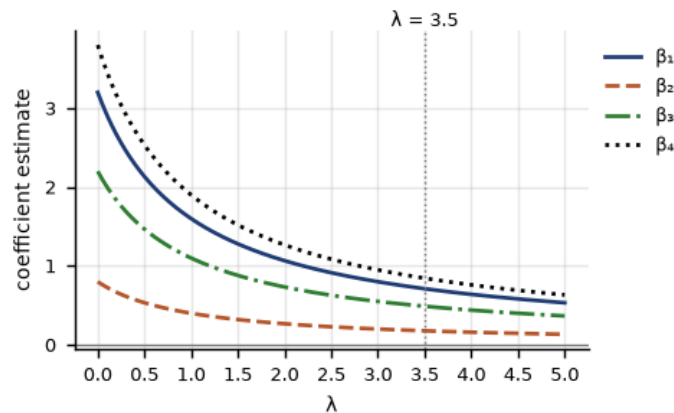
Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

For a two-class problem, the class means are $\mu_0 = (0, 3)^T$ and $\mu_1 = (1, 3)^T$ and the pooled within-class scatter matrix S_W is given below. Fisher's linear discriminant uses $w \propto S_W^{-1}(\mu_1 - \mu_0)$. Then the ratio w_2/w_1 of the components of the Fisher discriminant direction w is _____ (rounded off to two decimal places)

$$S_W = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

Q.17 [2 marks · MSQ (one or more correct)]

For a design with orthonormal columns ($X^T X = I$), the figure shows the coefficient paths $\beta_i(\lambda)$ of a penalised least-squares estimator that minimises $\frac{1}{2}\|y - X\beta\|^2 + P_\lambda(\beta)$, where the penalty P_λ is either the ridge penalty $(\lambda/2)\|\beta\|_2^2$ or the lasso penalty $\lambda\|\beta\|_1$. Which of the following statements is/are CORRECT?



- (A) At $\lambda = 3.5$, every coefficient equals its least-squares value divided by 4.5.
- (B) The plotted paths are those of the lasso (L1 penalty).
- (C) No coefficient becomes exactly zero for any finite λ .
- (D) At $\lambda = 3.5$, at least one coefficient is exactly zero.

Q.18 [2 marks · MCQ]

For logits $z = (0, 2, -1)$, softmax probabilities p and true class 3, the gradient of the cross-entropy loss $L = -\ln p_3$ with respect to z (rounded to two decimals) is

- (A) $(-0.89, -0.16, -0.96)$
- (B) $(0.11, 0.84, 0.04)$
- (C) $(0, 0, -23.8)$
- (D) $(0.11, 0.84, -0.96)$

Q.19 [2 marks · NAT (numerical answer)]

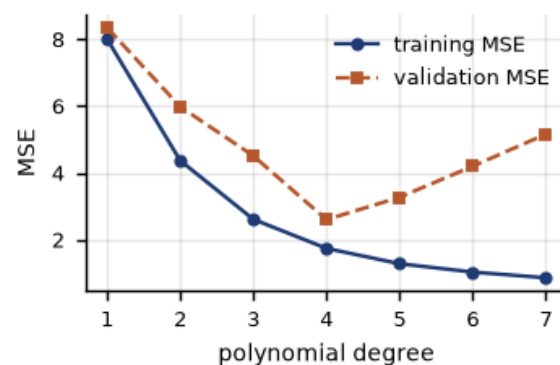
In C4.5, the attribute Customer-type splits a node of 11 records as tabulated (class Yes/No counts in each branch). Compute the split information of Customer-type (all logarithms base 2). (rounded off to two decimal places)

Customer-type	#Yes	#No
v1	1	5
v2	4	1

Q.20 [2 marks · MSQ (one or more correct)]

Polynomial regression models of degree 1 to 7 are fitted by least squares on a training set and evaluated on a separate validation set. The resulting mean squared errors are given below. Which of the following statements is/are CORRECT?

Degree	1	2	3	4	5	6	7
Training MSE	8.01	4.37	2.61	1.74	1.28	1.03	0.86
Validation MSE	8.35	5.98	4.51	2.6	3.27	4.2	5.16

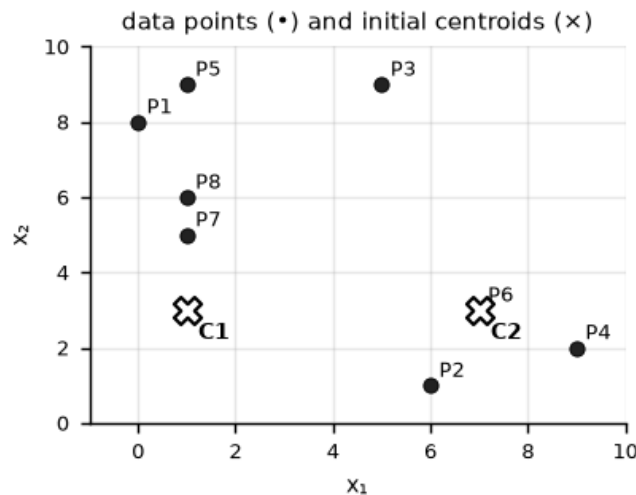


- (A) Based on the validation MSE, degree 4 should be selected.
- (B) Based on the validation MSE, degree 6 should be selected.
- (C) The generalisation gap (validation MSE – training MSE) is largest for degree 7.
- (D) The generalisation gap (validation MSE – training MSE) is largest for degree 5.

Q.21 [2 marks · NAT (numerical answer)]

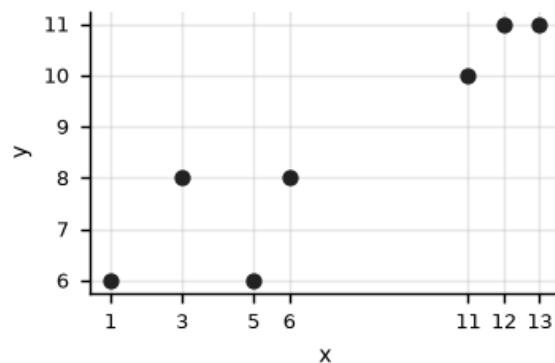
k-means (Lloyd's algorithm, Euclidean distance) with $k = 2$ is run on the 8 points below, starting from the initial centroids $C1 = (1, 3)$, $C2 = (7, 3)$. After **one** iteration (assignment step followed by update step), the x_1 -coordinate of the updated centroid of cluster $C2$ is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5	P6	P7	P8
x_1	0	6	5	9	1	7	1	1
x_2	8	1	9	2	9	3	5	6

**Q.22 [2 marks · NAT (numerical answer)]**

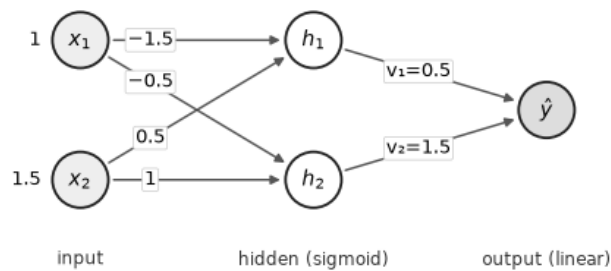
A regression tree (squared-error criterion) is fitted to the 7 points below. The root split " $x \leq t$ " is chosen among midpoints of consecutive x values to minimise the total SSE of the two children (each child predicts its mean). Find the prediction of the resulting depth-1 regression tree (stump) at $x = 10$. (rounded off to two decimal places)

x	1	3	5	6	11	12	13
y	6	8	6	8	10	11	11

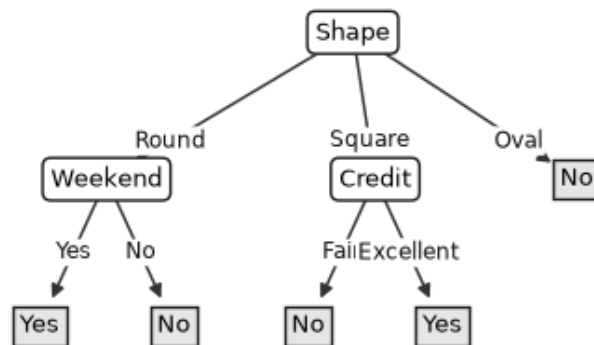


Q.23 [2 marks · NAT (numerical answer)]

In the network below, hidden units are sigmoid, the output is linear ($\hat{y} = v_1 h_1 + v_2 h_2$), all biases are 0, and the loss is $L = \frac{1}{2}(\hat{y} - y)^2$. For input $x = (1, 1.5)$ and target $y = 1$, the value of $\partial L / \partial w_{12}$ (the weight from x_2 to h_1) is _____ (rounded off to 4 decimal places)

**Q.24 [2 marks · NAT (numerical answer)]**

The decision tree shown is applied to the 10 labelled records in the table (attributes not tested on a path are ignored). Find the number of records misclassified by the tree. (Answer in integer)

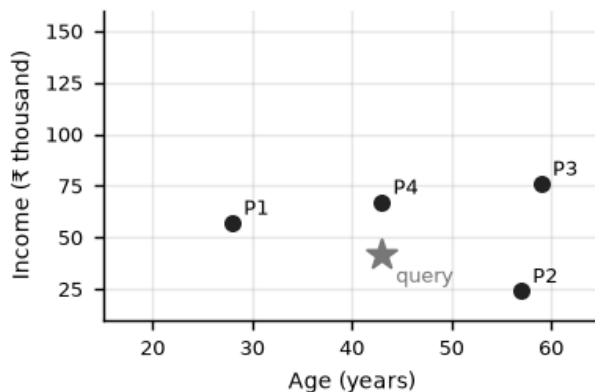


#	Shape	Weekend	Credit	Actual
1	Oval	No	Fair	Yes
2	Oval	No	Excellent	No
3	Oval	Yes	Excellent	Yes
4	Oval	Yes	Fair	No
5	Round	No	Excellent	No
6	Oval	Yes	Fair	No
7	Round	No	Excellent	Yes
8	Square	Yes	Excellent	No
9	Oval	No	Excellent	No
10	Oval	Yes	Excellent	No

Q.25 [2 marks · MCQ]

A 1-NN classifier uses the two features below. Before computing Euclidean distances, each feature is min-max scaled to $[0, 1]$ using the minimum and maximum of that feature over the four training points P1-P4 (the query is scaled with the same constants). The nearest neighbour of the query after scaling is

Point	Age (years)	Monthly income (₹ thousand)
P1	28	57
P2	57	24
P3	59	76
P4	43	67
Query	43	42



- (A) P1
(C) P3

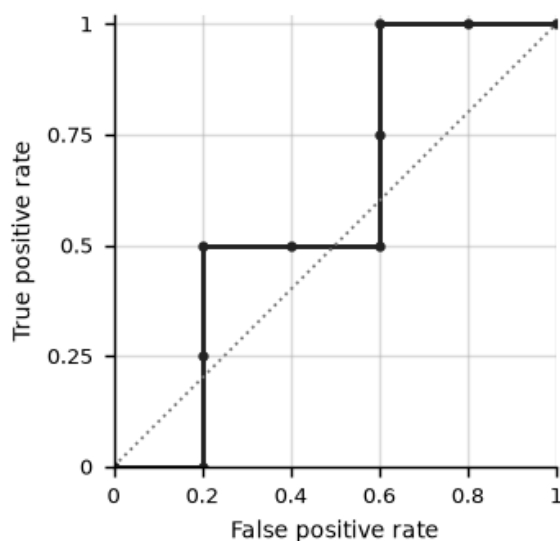
- (B) P2
(D) P4

Q.26 [2 marks · NAT (numerical answer)]

A network with one hidden layer maps 50 inputs to 1 output; all layers are fully connected and have biases. It has exactly 3173 trainable parameters. The number of hidden units is _____ (Answer in integer)

Q.27 [2 marks · NAT (numerical answer)]

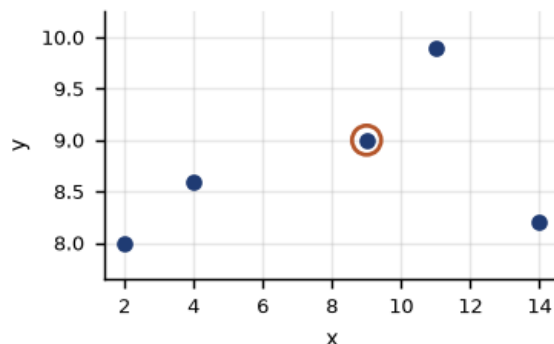
The ROC curve below was obtained from a test set with 4 positives and 5 negatives (all scores distinct, so the curve moves in steps of $1/4$ vertically and $1/5$ horizontally). From the figure, the area under this ROC curve (AUC) is _____ (rounded off to two decimal places)



Q.28 [2 marks · NAT (numerical answer)]

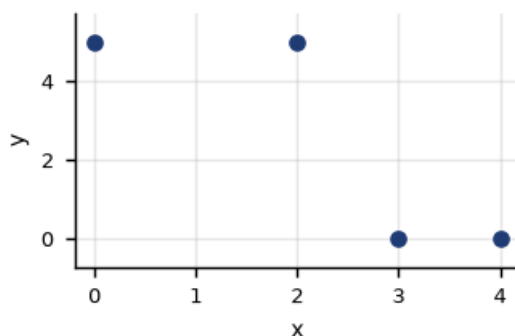
A simple linear regression (with intercept) is fitted by least squares to the 5 points shown. With $H = X(X^T X)^{-1} X^T$, if the errors have variance $\sigma^2 = 5$, the variance of the residual e_i of the observation with $x = 9$ (circled) is _____ (rounded off to three decimal places)

x	2	4	9	11	14
y	8	8.6	9	9.9	8.2

**Q.29 [2 marks · NAT (numerical answer)]**

A linear model $\hat{y} = w_0 + w_1 x$ is trained on the 4 points below by batch gradient descent on the loss $J(w) = (1/2n) \sum (w_0 + w_1 x_i - y_i)^2$, starting from $(w_0, w_1) = (0, 0)$ with learning rate $\eta = 0.05$. Then the value of w_1 after one iteration is _____ (rounded off to three decimal places)

x	0	2	3	4
y	5	5	0	0

**Q.30 [2 marks · NAT (numerical answer)]**

The eigenvalues of the sample covariance matrix (divisor $n - 1$) of a centered 7-dimensional dataset are 15, 12, 8.5, 3.5, 3, 1.5, 0.5. Then the mean squared reconstruction error $(1/(n - 1)) \sum_i \|x_i - \hat{x}_i\|^2$, when each point is projected onto the top 2 principal components, is _____ (rounded off to two decimal places)

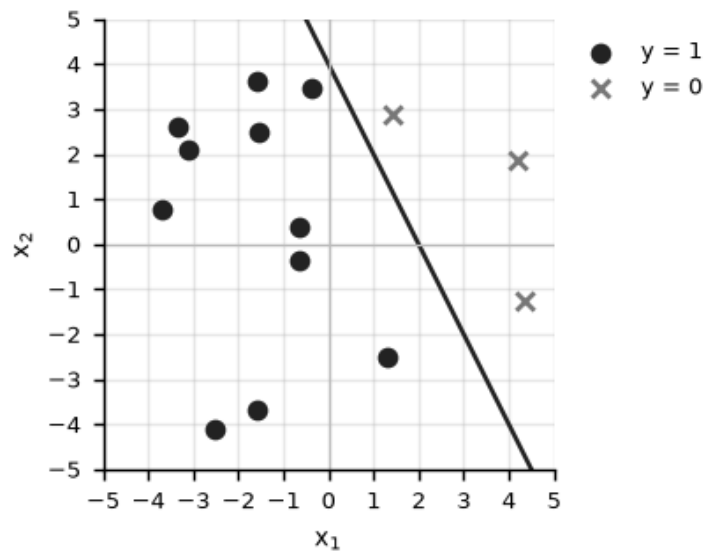
Q.31 [2 marks · MSQ (one or more correct)]

Consider agglomerative hierarchical clustering on n points with a dissimilarity matrix (no ties). Which of the following statements is/are CORRECT?

- (A) Single linkage is robust to a few noise points lying between two clusters.
- (B) Cophenetic distances from single, complete or average linkage satisfy the ultrametric inequality $d(x, z) \leq \max(d(x, y), d(y, z))$.
- (C) Replacing every dissimilarity by its square never changes the average-linkage dendrogram topology.
- (D) Complete-linkage merge heights equal the edge weights of a minimum spanning tree.

Q.32 [2 marks · MCQ]

A logistic regression model $P(y = 1 | \mathbf{x}) = \sigma(w_0 + w_1x_1 + w_2x_2)$ perfectly classifies the training points shown below using threshold 0.5. Its decision boundary is the solid line (it crosses the axes at grid points). Which of the following parameter vectors is consistent with the figure?



- (A) $(w_0, w_1, w_2) = (4, -1, -2)$
 (B) $(w_0, w_1, w_2) = (4, 2, -1)$
 (C) $(w_0, w_1, w_2) = (4, 1, 2)$
 (D) $(w_0, w_1, w_2) = (4, -2, -1)$

Q.33 [2 marks · NAT (numerical answer)]

The labelled data below are linearly separable through the origin by the unit vector $\mathbf{u} = (-3, 4)/5$ (i.e. $y_i = \text{sign}(\mathbf{u} \cdot \mathbf{x}_i)$). Take $\gamma = \min_i y_i(\mathbf{u} \cdot \mathbf{x}_i)$ and $R = \max_i \|\mathbf{x}_i\|$. The perceptron convergence theorem (no bias, $w_0 = 0$, $\eta = 1$) bounds the number of mistakes by $(R/\gamma)^2$. The value of this bound is _____ (rounded off to two decimal places)

i	1	2	3	4	5
$\mathbf{x}_i$	$(-2, -2)$	$(3, 3)$	$(4, -2)$	$(1, -2)$	$(-1, -3)$
y_i	-1	1	-1	-1	-1

Q.34 [2 marks · MSQ (one or more correct)]

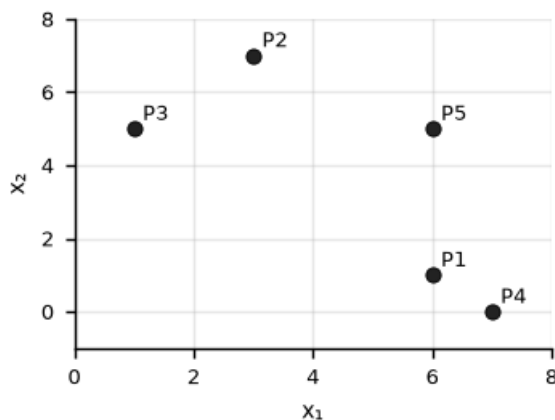
Each option shows a 2×2 matrix $[[K(x_1, x_1), K(x_1, x_2)], [K(x_2, x_1), K(x_2, x_2)]]$. Which of them can be the Gram matrix of some valid (Mercer) kernel K on two points x_1, x_2 ?

- (A) $[[9, 0], [0, 1]]$
 (B) $[[4, 1], [1, 1]]$
 (C) $[[9, 8], [8, 7]]$
 (D) $[[2, 1], [1, 5]]$

Q.35 [2 marks · NAT (numerical answer)]

A cluster contains the 5 points below. Distances are Manhattan (L_1). The medoid is the point that minimizes the sum of distances to all points of the cluster. The minimum total distance (cost of the medoid) is _____ (Answer in integer)

Point	P1	P2	P3	P4	P5
x_1	6	3	1	7	6
x_2	1	7	5	0	5



— END OF QUESTION PAPER —

Mock Test 38 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	250000	19	NAT	2	0.984 to 1.004
2	MSQ	1	(A)	20	MSQ	2	(A), (C)
3	NAT	1	1	21	NAT	2	6.74 to 6.76
4	NAT	1	17.99 to 18.01	22	NAT	2	10.6567 to 10.6767
5	MSQ	1	(B), (D)	23	NAT	2	0.0412 to 0.0428
6	NAT	1	270000	24	NAT	2	4
7	MSQ	1	(A), (C), (D)	25	MCQ	2	(D)
8	NAT	1	25	26	NAT	2	61
9	MCQ	1	(B)	27	NAT	2	0.59 to 0.61
10	MCQ	1	(A)	28	NAT	2	3.9371 to 3.9608
11	NAT	1	27	29	NAT	2	0.123 to 0.127
12	MSQ	1	(B), (D)	30	NAT	2	16.99 to 17.01
13	NAT	1	0.2347 to 0.2547	31	MSQ	2	(B)
14	MCQ	1	(D)	32	MCQ	2	(D)
15	NAT	1	7.1346 to 7.1546	33	NAT	2	124.75 to 125.25
16	NAT	2	-0.01 to 0.01	34	MSQ	2	(A), (B), (D)
17	MSQ	2	(A), (C)	35	NAT	2	20
18	MCQ	2	(D)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Neural Networks (MLP / Feed-forward)					11
Probabilistic & Instance-based Classifiers					11
SVM & Decision Trees					11
Unsupervised Learning					11

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Regression & Model Selection					11
Total					55

Mock Test 38 — Detailed Solutions

Q.1 — Answer: 250000 [NAT, 1M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

Batch size 1 $\Rightarrow$ N updates per epoch; total = $50000 \times 5 = 250000$. (Full-batch gradient descent would make only 5.)

Q.2 — Answer: (A) [MSQ, 1M · Probabilistic & Instance-based Classifiers › Naive Bayes]

(A) TRUE. The duplicated factor appears twice in the product, overstating confidence.

(B) FALSE. That is the full joint model; naive Bayes is linear in d.

(C) FALSE. It gets $1/(n_c + V) > 0$.

(D) FALSE. It handles any K.

Q.3 — Answer: 1 [NAT, 1M · SVM & Decision Trees › SVM: kernels]

$$x^T z = (-1)(2) + (3)(0) = -2.$$

$$K(x, z) = (-2 + 1)^2 = -1^2 = 1.$$

The kernel equals the inner product $\phi(x)^T \phi(z)$ of the explicit (all-monomials-up-to-degree-d) feature maps, without forming ϕ .

Q.4 — Answer: 17.99 to 18.01 [NAT, 1M · Probabilistic & Instance-based Classifiers › k-nearest neighbours]

Distances $|x_i - 17.7|$: x=20: 2.3, x=14: 3.7, x=13: 4.7, x=5: 12.7, x=4: 13.7, x=3: 14.7, x=1: 16.7.

The 4 nearest neighbours are x = 20, 14, 13, 5 with y = 23, 7, 23, 19.

$$\hat{y} = (23 + 7 + 23 + 19)/4 = 18.$$

Q.5 — Answer: (B), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) › Backpropagation]

(A) FALSE. Any (sub)differentiable activation works.

(B) TRUE. Standard backprop recursion.

(C) FALSE. Order is output $\rightarrow$ input.

(D) TRUE. a_j depends on b_j with coefficient 1.

Q.6 — Answer: 270000 [NAT, 1M · Unsupervised Learning › k-means]

Assignment cost per iteration = $n \cdot k \cdot d = 1500 \cdot 6 \cdot 2 = 18000$; over T iterations **270000**. This is the $O(nkdT)$ complexity of k-means.

Q.7 — Answer: (A), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) › Perceptron]

Net input $a = w \cdot x + b = -2x_1 + (1)x_2 + (4)$; output +1 iff $a \geq 0$.

(P) $a = -2 \cdot -3 + (1) \cdot 3 + (4) = 13 \rightarrow +1$ (selected).

(Q) $a = -2 \cdot 3 + (1) \cdot -2 + (4) = -4 \rightarrow -1$.

(R) $a = -2 \cdot 1 + (1) \cdot 1 + (4) = 3 \rightarrow +1$ (selected).

(S) $a = -2 \cdot 1 + (1) \cdot 2 + (4) = 4 \rightarrow +1$ (selected).

Answer: **(A), (C), (D)**.

Q.8 — Answer: 25 [NAT, 1M · Regression & Model Selection › Cross-validation]

Each candidate needs $k = 3$ fits: $8 \times 3 = 24$. Plus 1 final refit: **25**.

Q.9 — Answer: (B) [MCQ, 1M · Regression & Model Selection › Gradient descent]

$J(w) = (w - 5)^2$ has $J'' = 2$. GD: $w_{t+1} - 5 = (1 - 2\eta)(w_t - 5)$, so $J(w_t) = (1 - 2\eta)^{2t} J(w_0)$.

Ratios $(1 - 2\eta)^2$ per step: $\eta = 0.02 \rightarrow 0.9216$ (curve C); $\eta = 0.06 \rightarrow 0.7744$ (curve D); $\eta = 0.2 \rightarrow 0.36$ (curve A); $\eta = 1.25 \rightarrow 2.25$ (curve B).

Ratio $> 1 \Rightarrow$ divergence ($\eta > 1 = 2/J''$); among convergent rates (all < 0.5 here), a smaller η gives a ratio closer to 1, i.e. slower decrease.

Hence $\eta = 1.25$ corresponds to **Curve B**.

Q.10 — Answer: (A) [MCQ, 1M · SVM & Decision Trees › Decision trees: overfitting & pruning]

Subtree	$R(T) + \alpha T $
T1	$0.045 + 0.002 \times 13 = 0.071$

T2	$0.09 + 0.002 \times 12 = 0.114$
T3	$0.105 + 0.002 \times 7 = 0.119$
T4	$0.145 + 0.002 \times 5 = 0.155$

Minimum at **T1**. Larger α penalises leaves more and favours smaller trees; $\alpha = 0$ would pick the largest tree T1.

Q.11 — Answer: 27 [NAT, 1M · Unsupervised Learning > Cluster evaluation]

Majority count in each cluster: 25, 26, 10 (sum 61); $N = 88$.

Purity = $(1/N) \sum_k \max_j n_{kj} = 61/88 = 0.6932$.

Misclassified = $N - \sum \max = 27$.

Answer: **27**.

Purity is an external measure (needs class labels) and is trivially 1 if each point forms its own cluster, so it cannot be used to choose k .

Q.12 — Answer: (B), (D) [MSQ, 1M · Unsupervised Learning > Hierarchical clustering]

(A) **FALSE**. The naive algorithm is $O(n^3)$ ($O(n^2 \log n)$ with heaps).

(B) **TRUE**. Bottom-up procedure.

(C) **FALSE**. Higher cuts give fewer clusters.

(D) **TRUE**. There are $n(n-1)/2$ pairs.

Q.13 — Answer: 0.2347 to 0.2547 [NAT, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

softmax: $P(y = k | \mathbf{x}) = e^{z_k} / \sum_j e^{z_j}$.

e^z values: 1, 0.1353, 0.3679; sum = 1.5032.

$P(y = 3 | \mathbf{x}) = 0.3679/1.5032 = \mathbf{0.24}$.

Q.14 — Answer: (D) [MCQ, 1M · Regression & Model Selection > MLE & MAP view]

Laplace density $\propto \exp(-|\epsilon|/b)$; $-\log\text{-likelihood} \propto \sum |\text{residual}|$.

Answer: $\sum |y_i - \mathbf{w}^s \mathbf{x}_i|$.

Q.15 — Answer: 7.1346 to 7.1546 [NAT, 1M · SVM & Decision Trees > Ensembles: bagging & random forests]

By linearity: $n \cdot P(\text{point appears}) = 11 \cdot [1 - (1 - 1/11)^{11}] = \mathbf{7.145}$.

Q.16 — Answer: -0.01 to 0.01 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$\mu_1 - \mu_0 = (1, 0)^T$; $\det S_W = 2$.

$$S_W^{-1} = \begin{bmatrix} 0.5 & 0 \\ 0 & 1 \end{bmatrix}$$

$\mathbf{w} = S_W^{-1}(\mu_1 - \mu_0) = (0.5, 0)^T$.

$w_2/w_1 = 0/0.5 = \mathbf{0}$. (The direction, not the length, matters.)

Note: using $\mu_1 - \mu_0$ without S_W^{-1} is correct only when $S_W \propto I$.

Q.17 — Answer: (A), (C) [MSQ, 2M · Regression & Model Selection > Ridge & Lasso]

With $X^T X = I$ the ridge path is $b/(1 + \lambda)$ and the lasso path is $\text{sign}(b)(|b| - \lambda)_+$. The plot shows paths reaching zero at finite $\lambda \Leftrightarrow$ lasso; here it is **ridge**.

(A) **TRUE**. $\beta(\lambda) = b/(1 + \lambda)$.

(B) **FALSE**. No path reaches zero at a finite λ .

(C) **TRUE**. $b/(1 + \lambda) \neq 0$ whenever $b \neq 0$.

(D) **FALSE**. Ridge gives no exact zeros.

Q.18 — Answer: (D) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$\mathbf{p} = \text{softmax}((0, 2, -1)) = (0.1142, 0.8438, 0.042)$; $\mathbf{y} = (0, 0, 1)$.

$\partial L / \partial \mathbf{z} = \mathbf{p} - \mathbf{y} = \mathbf{(0.11, 0.84, -0.96)}$.

Distractors: $\mathbf{y} - \mathbf{p} = (-0.11, -0.84, 0.96)$ (wrong sign); $\mathbf{p}$ alone = $(0.11, 0.84, 0.04)$ (forgot the target); $(\mathbf{p} - \mathbf{y}) \odot \mathbf{p} \odot (1 - \mathbf{p}) = (0.01, 0.11, -0.04)$ (spurious σ' -type factor, which cancels for

cross-entropy); $-y/p = (0, 0, -23.8)$ is $\partial L/\partial p$, not $\partial L/\partial z$.

Q.19 — Answer: 0.984 to 1.004 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [5 Yes, 6 No]: $H = 0.994$.

Child entropies: $v_1: 0.65$, $v_2: 0.7219$.

$IG = 0.994 - \sum (n_v/11)H_v = 0.3113$.

SplitInfo = $-\sum (n_v/N) \log_2(n_v/N) = -(6/11)\log_2(6/11) - (5/11)\log_2(5/11) = 0.994$.

Answer: **0.99**.

C4.5 uses the gain ratio to penalise attributes that split the data into many small branches.

Q.20 — Answer: (A), (C) [MSQ, 2M · Regression & Model Selection > Bias-variance trade-off]

(A) TRUE. Validation MSE is minimum (2.6) at degree 4.

(B) FALSE. The minimum validation MSE is at degree 4, not 6.

(C) TRUE. Gaps: 0.34, 1.61, 1.9, 0.86, 1.99, 3.17, 4.3.

(D) FALSE. Gaps: 0.34, 1.61, 1.9, 0.86, 1.99, 3.17, 4.3; the largest is at degree 7.

Q.21 — Answer: 6.74 to 6.76 [NAT, 2M · Unsupervised Learning > k-means]

Iteration 1 - assignment (nearest initial centroid, squared Euclidean distance):

Point	$d^2(\cdot, C1)$	$d^2(\cdot, C2)$	cluster
P1 (0, 8)	26	74	C1
P2 (6, 1)	29	5	C2
P3 (5, 9)	52	40	C2
P4 (9, 2)	65	5	C2
P5 (1, 9)	36	72	C1
P6 (7, 3)	36	0	C2
P7 (1, 5)	4	40	C1
P8 (1, 6)	9	45	C1

Update: each centroid becomes the mean of its assigned points: $C1 \rightarrow (0.75, 7)$; $C2 \rightarrow (6.75, 3.75)$.

Required coordinate = **6.75**.

Note: each step of Lloyd's algorithm can only decrease (or keep) J , which is why it converges.

Q.22 — Answer: 10.6567 to 10.6767 [NAT, 2M · SVM & Decision Trees > Regression trees]

Parent mean $\bar{y} = 8.571$, $SSE_0 = \sum (y - \bar{y})^2 = 27.714$.

Evaluate every split:

Split t	Left mean	Right mean	$SSE_L + SSE_R$
2	6	9	20
4	7	9.2	20.8
5.5	6.667	10	8.667
8.5	7	10.667	4.667
11.5	7.6	11	11.2
12.5	8.167	11	20.833

Best split $t = 8.5$: left mean 7, right mean 10.667, $SSE = 4.667$, reduction = 23.048.

Answer: **10.67**.

Q.23 — Answer: 0.0412 to 0.0428 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

Forward: $a = Wx = (-0.75, 1)$, $h = \sigma(a) = (0.3208, 0.7311)$, $\hat{y} = v \cdot h = 1.257$.

Output error $\delta_o = \partial L/\partial \hat{y} = \hat{y} - y = 0.257$ (linear output, $L = \frac{1}{2}(\hat{y} - y)^2$).

Hidden delta $\delta_1 = \delta_o \cdot v_1 \cdot h_1(1 - h_1) = 0.257 \times 0.5 \times 0.2179 = 0.028$.

$\partial L/\partial w_{12} = \delta_1 \cdot x_2 = 0.028 \times 1.5 = \mathbf{0.042}$.

Q.24 — Answer: 4 [NAT, 2M · SVM & Decision Trees > Decision trees: structure & traversal]

Route each record from the root to a leaf:

#	Predicted	Actual	Correct?
1	No	Yes	no
2	No	No	yes
3	No	Yes	no
4	No	No	yes
5	No	No	yes
6	No	No	yes
7	No	Yes	no
8	Yes	No	no
9	No	No	yes
10	No	No	yes

Misclassified: 4 of 10 (accuracy 60%). Answer: **4**.

Q.25 — Answer: (D) [MCQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Ranges: age [28, 59], income [24, 76]. Scaled query = (0.484, 0.346).

Point	scaled age	scaled income	scaled distance	raw distance
P1	0	0.635	0.563	21.21
P2	0.935	0	0.569	22.8
P3	1	1	0.833	37.58
P4	0.484	0.827	0.481	25

Nearest after scaling: **P4**.

Without scaling the nearest point would be P1, because income differences (tens of thousands) dominate the age differences — exactly why k-NN needs feature scaling.

Q.26 — Answer: 61 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Parameter counting]

$P = (d + 1)h + (h + 1)k = h(d + k + 1) + k \Rightarrow h = (P - k)/(d + k + 1) = (3173 - 1)/(50 + 1 + 1) = 61$.

Q.27 — Answer: 0.59 to 0.61 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

The ROC curve is a staircase: each vertical step is a positive and each horizontal step a negative.

Area = Σ over horizontal steps of (width $1/N$) $\times$ (current TPR): $0 + 0.5 + 0.5 + 1 + 1$, times $1/5$.

AUC = **0.6**.

Q.28 — Answer: 3.9371 to 3.9608 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

For simple linear regression with intercept, the leverage (diagonal of the hat matrix) is $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$.

$\bar{x} = 8$, $S_{xx} = 98$.

$h = 0.2102$; $\text{Var}(e_i) = \sigma^2(1 - h_{ii}) = 5 \times 0.7898 = 3.949$.

Answer: **3.949**.

Q.29 — Answer: 0.123 to 0.127 [NAT, 2M · Regression & Model Selection > Gradient descent]

$\partial J/\partial w_0 = (1/n)\Sigma r_i$, $\partial J/\partial w_1 = (1/n)\Sigma r_i x_i$, with residual $r_i = w_0 + w_1 x_i - y_i$.

Step 1: $w = (0, 0)$, $r = (-5, -5, 0, 0)$, $\nabla J = (-2.5, -2.5) \Rightarrow w \leftarrow w - 0.05 \nabla J = (0.125, 0.125)$.

Answer: **0.125**.

Q.30 — Answer: 16.99 to 17.01 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

For centered data, projecting onto the top-k PCs and reconstructing leaves the components along the discarded eigenvectors; their average squared length (divisor $n - 1$) is the sum of the discarded eigenvalues.

Discarded eigenvalues: 8.5, 3.5, 3, 1.5, 0.5; sum = 17; total = 44.

Answer: **17**. (A common mistake is to add the *retained* eigenvalues 27.)

Q.31 — Answer: (B) [MSQ, 2M · Unsupervised Learning › Hierarchical clustering]

(A) **FALSE**. Such bridge points cause chaining and merge the clusters.

(B) **TRUE**. Monotone dendrograms define ultrametrics.

(C) **FALSE**. Average linkage depends on actual values, not only on ranks.

(D) **FALSE**. That holds for single linkage.

Q.32 — Answer: (D) [MCQ, 2M · Probabilistic & Instance-based Classifiers › Logistic regression]

The line passes through (2, 0) and (0, 4): $x_1/2 + x_2/4 = 1$, i.e. $4x_1 + 2x_2 - 8 = 0$. Hence (w_0, w_1, w_2) must be a non-zero multiple of $(-8, 4, 2)$.

Orientation: the class $y = 1$ region must have $w_0 + w_1x_1 + w_2x_2 > 0$. At the origin the $y = 1$ side has sign $+$ ($w_0 = 4$), which matches the figure (the origin lies on the $y = 1$ side).

So the answer is $(w_0, w_1, w_2) = (4, -2, -1)$ (any positive multiple gives the same classifier).

Other options either describe a different line (swapped/negated coefficients move the intercepts) or the same line with the sign flipped, which would label every training point incorrectly.

Q.33 — Answer: 124.75 to 125.25 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Perceptron]

Unit separator $u = (-3, 4)/5$. Margins $y_i(u \cdot x_i)$: 0.4, 0.6, 4, 2.2, 1.8 $\rightarrow \gamma = \min = 0.4$.

Norms $\|x_i\|$: 2.828, 4.243, 4.472, 2.236, 3.162 $\rightarrow R = \max = 4.4721$.

Novikoff bound: number of mistakes $\leq (R/\gamma)^2 = (4.4721/0.4)^2 = 125$.

Common mistakes: forgetting to normalise u (γ must be measured with $\|u\| = 1$), or using R/γ without squaring.

Q.34 — Answer: (A), (B), (D) [MSQ, 2M · SVM & Decision Trees › SVM: kernels]

A Gram matrix of a valid kernel must be symmetric and positive semidefinite. For a symmetric 2×2 matrix this is equivalent to: both diagonal entries ≥ 0 and determinant ≥ 0 .

(A) **TRUE**. symmetric, diagonal entries ≥ 0 and $\det = 9 > 0 \Rightarrow$ positive definite.

(B) **TRUE**. symmetric, diagonal entries ≥ 0 and $\det = 3 > 0 \Rightarrow$ positive definite.

(C) **FALSE**. $\det = -1 < 0 \Rightarrow$ one negative eigenvalue, not PSD.

(D) **TRUE**. symmetric, diagonal entries ≥ 0 and $\det = 9 > 0 \Rightarrow$ positive definite.

Q.35 — Answer: 20 [NAT, 2M · Unsupervised Learning › k-medoids]

Compute the Manhattan (L_1) distance matrix and its row sums:

	P1	P2	P3	P4	P5	Σ
P1	0	9	9	2	4	24
P2	9	0	4	11	5	29
P3	9	4	0	11	5	29
P4	2	11	11	0	6	30
P5	4	5	5	6	0	20

The smallest row sum is 20 for P5, so the medoid is P5.

Answer: **20**.

Unlike a k-means centroid, the medoid is always an actual data point.

Mock Test 39

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	8	4	23

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Neural Networks (MLP / Feed-forward)	7	11
Regression & Model Selection	7	11
Unsupervised Learning	7	11
SVM & Decision Trees	7	11
Probabilistic & Instance-based Classifiers	7	11

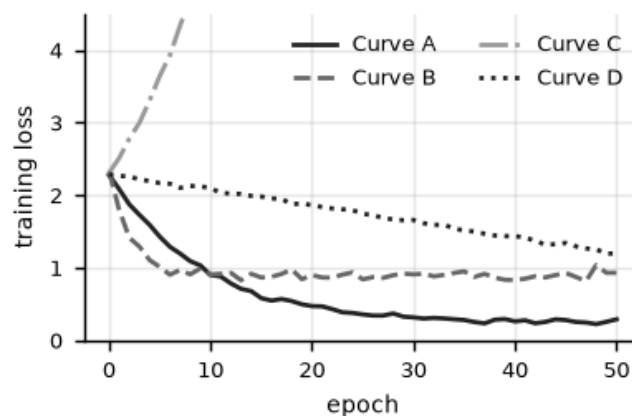
Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MCQ]

The training-loss curves below were obtained with four different learning rates (everything else fixed). Which curve corresponds to a learning rate that is somewhat too high (fast initial drop, then the loss plateaus at a high value)?



- (A) Curve A
(C) Curve C

- (B) Curve B
(D) Curve D

Q.2 [1 mark · MSQ (one or more correct)]

Which of the following statements about cross-validation (CV) is/are CORRECT? (n denotes the number of observations.)

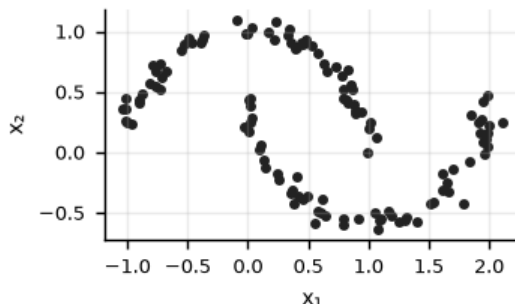
- (A) Stratified k-fold CV keeps the class proportions in each fold close to those of the full data.
(B) In nested CV, the inner loop estimates generalisation error and the outer loop tunes hyperparameters.
(C) Standardising the whole data set before splitting into folds causes no information leakage.
(D) LOOCV is k-fold cross-validation with $k = n$.

Q.3 [1 mark · NAT (numerical answer)]

Agglomerative clustering of 41 points is stopped when 2 clusters remain. The number of merges performed is _____ (Answer in integer)

Q.4 [1 mark · MCQ]

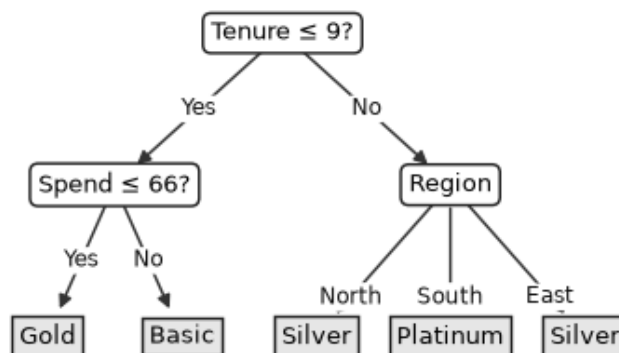
k-means with $k = 2$ fails to separate the two interleaving half-moons shown below, even with many random restarts. The main reason is that k-means



- (A) cannot be applied to two-dimensional data
- (B) produces convex clusters separated by linear (Voronoi) boundaries
- (C) converges only when the data are standardized
- (D) requires class labels to place the centroids

Q.5 [1 mark · MCQ]

The decision tree below predicts the membership tier offered to a customer. Edges labelled Yes/No answer the test in the node. What is the prediction for the record shown in the table?

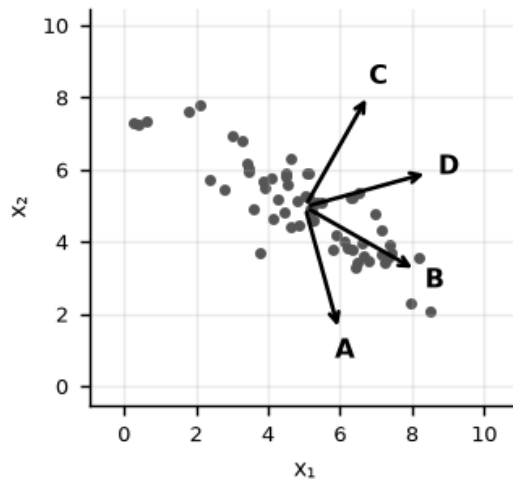


Tenure	Spend	Visits	Region
15	65	17	South

- (A) Silver
- (B) Gold
- (C) Platinum
- (D) Basic

Q.6 [1 mark · MCQ]

The scatter plot shows a 2-D dataset together with four candidate directions A-D drawn from the data mean. Which direction best represents the second principal component (PC2)?



- (A) Direction A (B) Direction B
(C) Direction C (D) Direction D

Q.7 [1 mark · NAT (numerical answer)]

Fisher's multi-class LDA is applied to data with $K = 8$ classes and $p = 3$ features. The maximum number of discriminant directions (non-zero generalised eigenvalues of $S_W^{-1}S_B$) is ____
(Answer in integer)

Q.8 [1 mark · NAT (numerical answer)]

A fully connected MLP has layer sizes 784-10-100-512-1 (input...-output), every layer having a bias vector. The total number of trainable parameters (weights and biases) is ____ (Answer in integer)

Q.9 [1 mark · MSQ (one or more correct)]

Which of the following statements about activation functions is/are CORRECT? (σ denotes the logistic sigmoid.)

- (A) $d/dz \tanh(z) = 1 + \tanh^2(z)$.
 (B) Dividing the logits by a temperature $T > 1$ makes the softmax distribution more uniform.
 (C) ReLU suffers from vanishing gradients for large positive inputs.
 (D) $\sigma'(z) = \sigma(z)(1 - \sigma(z))$, whose maximum value is 0.25 (at $z = 0$).

Q.10 [1 mark · NAT (numerical answer)]

Points are drawn uniformly from the unit hypercube $[0, 1]^{10}$. The fraction of points lying within distance 0.1 of the boundary (i.e. outside the inner cube $[0.1, 0.9]^{10}$) is _____ (rounded off to two decimal places)

Q.11 [1 mark · NAT (numerical answer)]

For a hard-margin linear SVM, the support vectors $\mathbf{x}_+ = (2, 2)$ (class +1) and $\mathbf{x}_- = (-2, 0)$ (class -1) lie on the hyperplanes $\mathbf{w}^T \mathbf{x} + b = +1$ and $\mathbf{w}^T \mathbf{x} + b = -1$ respectively, and the segment joining them is perpendicular to the decision boundary. The value of $\|\mathbf{w}\|$ is _____ (rounded off to two decimal places)

Q.12 [1 mark · MSQ (one or more correct)]

Which of the following statements about classifier evaluation is/are CORRECT?

- (A)** A classifier that assigns random scores has an expected ROC AUC of 0.5.
(B) Macro-averaging gives every class equal weight regardless of its number of examples.
(C) The F1-score is the harmonic mean of precision and recall.
(D) Raising the decision threshold can only increase the false positive rate.

Q.13 [1 mark · NAT (numerical answer)]

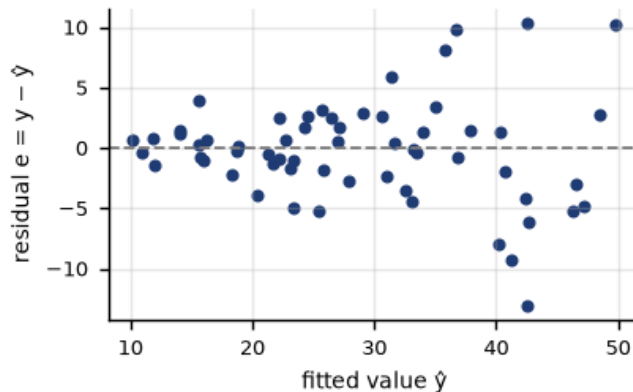
A full polynomial regression of degree 4 in 3 input variables $x_1, \dots, x_3$ includes every monomial in these variables of total degree at most 4 (including the constant term and all interaction terms). The number of regression coefficients is _____ (Answer in integer)

Q.14 [1 mark · NAT (numerical answer)]

Consider the polynomial kernel $K(x, z) = (x^T z)^2$ with $x = (1, -2, 3)$ and $z = (1, -2, 3)$. The value of $K(x, x)$ is _____ (Answer in integer)

Q.15 [1 mark · MCQ]

After fitting a linear regression model, the residuals are plotted against the fitted values as shown. The plot most strongly suggests:



- (A) Non-constant error variance (heteroscedasticity)
- (B) A non-linear relationship not captured by the model (e.g. a missing x^2 term)
- (C) A single outlying observation, with the rest of the fit looking adequate
- (D) No evident violation of the linear-model assumptions

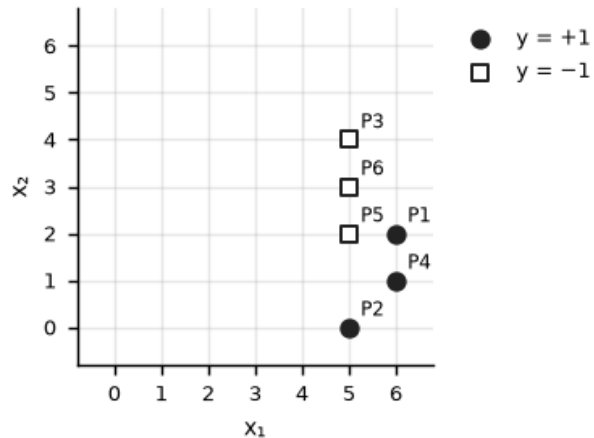
Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

A centered data matrix X with $n = 9$ rows (samples) and $d = 4$ columns has singular values 17, 12, 6, 2. Using the sample covariance with divisor $n - 1$, the total variance (trace of the sample covariance matrix) is _____ (rounded off to two decimal places)

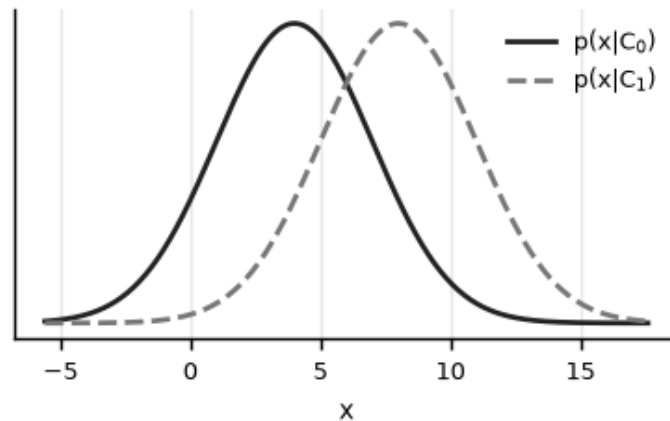
Q.17 [2 marks · NAT (numerical answer)]

A hard-margin linear SVM is trained on the points below. Solving the dual problem gives the Lagrange multipliers α_i listed in the table. Using the KKT conditions, compute the weight component w_2 . (rounded off to two decimal places)

Point	x_1	x_2	y	α_i
P1	6	2	+1	2
P2	5	0	+1	0.5
P3	5	4	-1	0
P4	6	1	+1	0
P5	5	2	-1	2.5
P6	5	3	-1	0

**Q.18 [2 marks · NAT (numerical answer)]**

In a two-class problem with one feature, $p(x | C_0) = N(4, 9)$ and $p(x | C_1) = N(8, 9)$ (second argument = variance), with priors $P(C_0) = 0.8$ and $P(C_1) = 0.2$. Under 0-1 loss, the posterior probability $P(C_1 | x = 7)$ is _____ (rounded off to two decimal places)



Class-conditional densities (not weighted by priors).

Q.19 [2 marks · NAT (numerical answer)]

A naive Bayes classifier is trained on the 11 examples below, using **Laplace (add-one) smoothing** for the class-conditional feature probabilities (the class prior is not smoothed). For the new instance (Fever = None, Cough = No, Fatigue = Yes), the posterior probability $P(\text{Flu} = \text{Pos} \mid x)$ is _____ (rounded off to two decimal places)

#	Fever	Cough	Fatigue	Flu
1	High	Yes	Yes	Neg
2	High	Yes	Yes	Pos
3	High	No	Yes	Neg
4	High	Yes	Yes	Pos
5	High	No	No	Pos
6	High	Yes	Yes	Neg
7	Mild	No	Yes	Pos
8	High	Yes	No	Pos
9	High	Yes	Yes	Pos
10	High	Yes	No	Neg
11	None	No	Yes	Neg

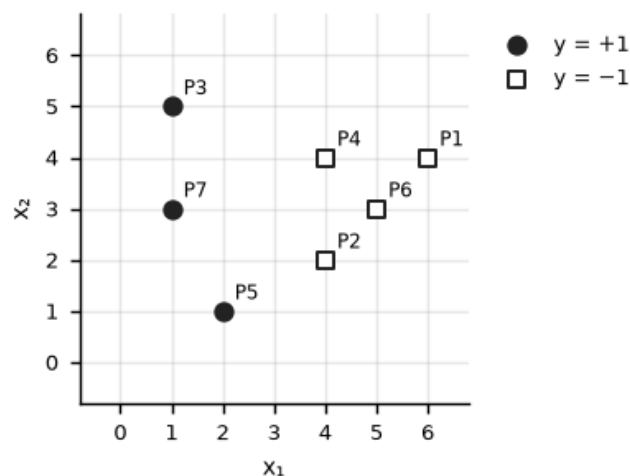
Q.20 [2 marks · NAT (numerical answer)]

Gradient descent with (heavy-ball) momentum minimises $f(w) = (1/2)w^2$ using $v_t = \beta v_{t-1} + \nabla f(w_{t-1})$ and $w_t = w_{t-1} - \eta v_t$, with $w_0 = 4$, $v_0 = 0$, $\eta = 0.1$, $\beta = 0.8$. The value of v_2 is _____ (rounded off to three decimal places)

Q.21 [2 marks · NAT (numerical answer)]

The labelled training points shown below are linearly separable. A hard-margin linear SVM is trained on them. Compute the width of the margin (distance between the two margin hyperplanes). (rounded off to two decimal places)

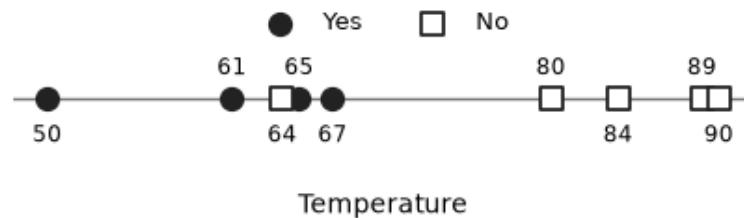
Point	x_1	x_2	y
P1	6	4	-1
P2	4	2	-1
P3	1	5	+1
P4	4	4	-1
P5	2	1	+1
P6	5	3	-1
P7	1	3	+1



Q.22 [2 marks · NAT (numerical answer)]

The continuous attribute Temperature and the class label of 9 training records are given below (unsorted). A binary split “Temperature $\leq t$ ” is chosen to maximise information gain, with t restricted to midpoints between consecutive sorted values. Find the threshold value t chosen for the split “Temperature $\leq t$ ”. (rounded off to one decimal place)

Temperature	89	65	61	80	67	50	90	64	84
Class	No	Yes	Yes	No	Yes	Yes	No	No	No

**Q.23 [2 marks · NAT (numerical answer)]**

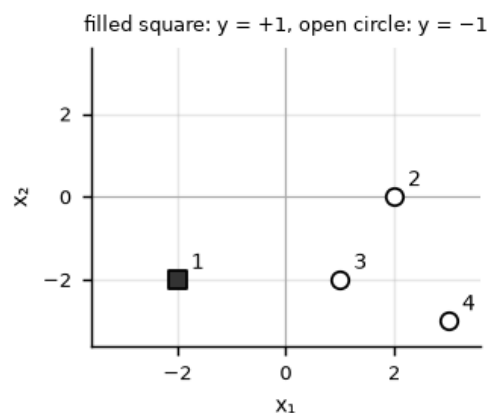
In one AdaBoost round the current example weights and whether the chosen weak classifier h classifies each example correctly are given. Using $\alpha = \frac{1}{2} \ln((1 - \epsilon)/\epsilon)$ and the update $w_i \leftarrow w_i \cdot \exp(-\alpha y_i h(x_i))/Z$, find the normalised weight of example 5 for the next round. (rounded off to three decimal places)

Example i	1	2	3	4	5	6	7	8
Weight w_i	1/9	2/27	4/27	1/27	5/27	4/27	4/27	4/27
$h(x_i)$ correct?	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes

Q.24 [2 marks · NAT (numerical answer)]

A perceptron with parameters $w \in \mathbb{R}^2$, b is trained on the data below, starting from $w = (0, 0)$, $b = 0$, with learning rate $\eta = 1$. Examples are presented once in the order 1, 2, 3, 4 (one epoch). An update $w \leftarrow w + \eta y x$, $b \leftarrow b + \eta y$ is made whenever $y(w \cdot x + b) \leq 0$. After the epoch, the net input $w \cdot x + b$ for the test point $x = (-1, 0)$ using the final parameters is _____. (Answer in integer)

i	x_1	x_2	y
1	-2	-2	1
2	2	0	-1
3	1	-2	-1
4	3	-3	-1



Q.25 [2 marks · NAT (numerical answer)]

For an input x , a classifier's posterior probabilities are $P(C_1|x) = 0.3$, $P(C_2|x) = 0.1$, $P(C_3|x) = 0.6$. The loss of taking action α_i (deciding C_i) when the true class is C_j is given below. Then the conditional risk of the action chosen by the MAP rule (i.e. choosing the class with highest posterior) is _____ (rounded off to two decimal places)

loss $L(\alpha_i, C_j)$	true C_1	true C_2	true C_3
α_1 : decide C_1	0	9	9
α_2 : decide C_2	1	0	2
α_3 : decide C_3	9	9	0

Q.26 [2 marks · MSQ (one or more correct)]

A linear regression of house price (₹ lakh) on Area (area in hundreds of sq. ft), Furnished (1 if furnished, 0 otherwise) and zone $\in \{\text{North, South, East}\}$ encoded by dummies ZoneSouth and ZoneEast (reference level North) gives the fitted model

$$\hat{y} = 19 + 1.3 \cdot \text{Area} + 1.3 \cdot \text{Furnished} - 5.2 \cdot \text{ZoneSouth} + 2.4 \cdot \text{ZoneEast}$$

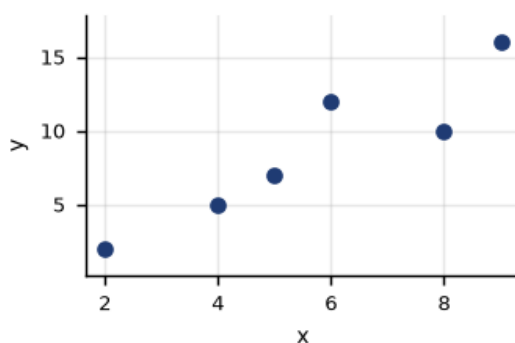
Which of the following is/are CORRECT?

- (A) Adding a third dummy for zone North while keeping the intercept would make $X^T X$ singular.
- (B) The predicted house price for a unfurnished house in the South zone with area 8 hundred sq. ft is 24.2.
- (C) The coefficient of ZoneSouth is the expected difference in response between zone South and zone North, holding the other variables fixed.
- (D) Increasing Area by 3 hundred sq. ft, other variables fixed, changes the prediction by 3.9.

Q.27 [2 marks · NAT (numerical answer)]

Assume $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with ε_i i.i.d. $N(0, \sigma^2)$. For the 6 observations below, with β estimated by least squares (= maximum likelihood), the unbiased estimate $s^2 = \text{SSE}/(n - 2)$ of σ^2 is _____ (rounded off to two decimal places)

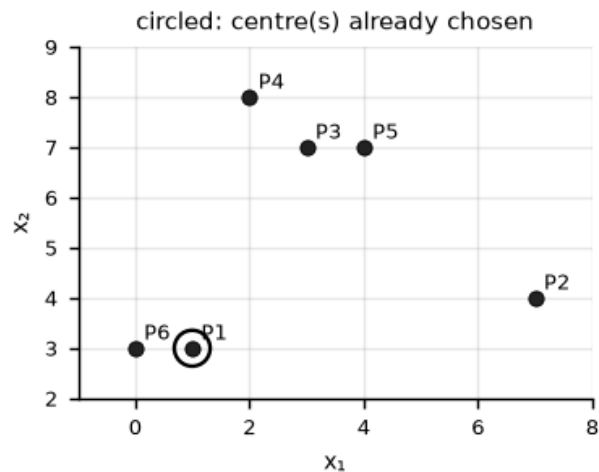
x	2	4	5	6	8	9
y	2	5	7	12	10	16



Q.28 [2 marks · NAT (numerical answer)]

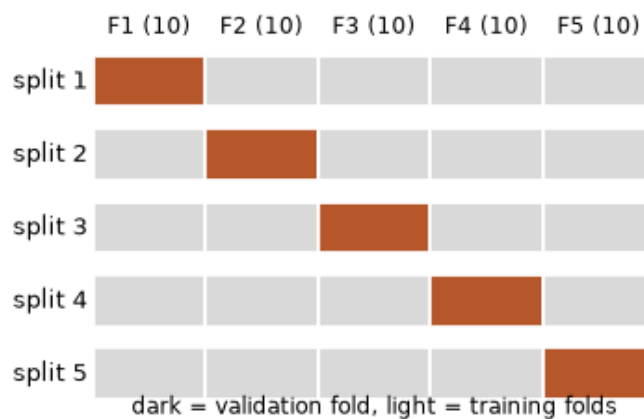
k-means++ initialization (squared Euclidean seeding) is applied to the six points below. P1 has already been chosen as centre. The probability that P5 is selected as the second centre is _____ (rounded off to three decimal places)

Point	P1	P2	P3	P4	P5	P6
x_1	1	7	3	2	4	0
x_2	3	4	7	8	7	3

**Q.29 [2 marks · NAT (numerical answer)]**

Ridge regression is tuned by 5-fold cross-validation (equal fold sizes) over $\lambda \in \{0.01, 0.5, 5\}$. The validation MSE on each fold is shown. The CV estimate (mean over folds) of the λ that would be selected is _____ (rounded off to two decimal places)

λ	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
0.01	5	4.8	4.5	2.1	2.3
0.5	4.6	4.1	5.8	4.6	4.8
5	5.3	4.9	5	2.6	2.1

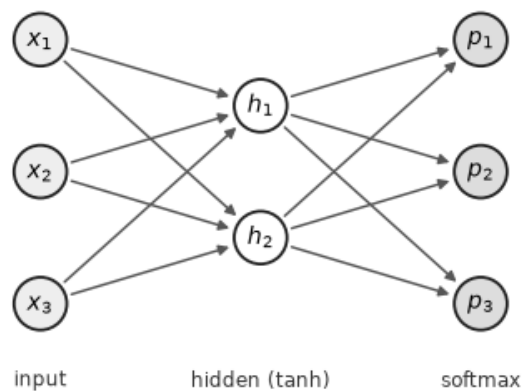
**Q.30 [2 marks · NAT (numerical answer)]**

Two clusterings U and V of 7 points are given below (cluster names are arbitrary). Then the Rand index between the two clusterings is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5	P6	P7
Clustering U	β	β	α	α	β	β	γ
Clustering V	2	2	1	3	3	2	3

Q.31 [2 marks · NAT (numerical answer)]

A 3-2-3 network computes $h = \tanh(W_1 x + b_1)$, $z = W_2 h + b_2$, $p = \text{softmax}(z)$, with the parameters below. For $x = (2, 1, 2)^T$, the predicted probability p_1 of class 1 is _____ (rounded off to three decimal places)



$$W_1 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & -1 & 0.5 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 1 \\ -0.5 \end{bmatrix}$$

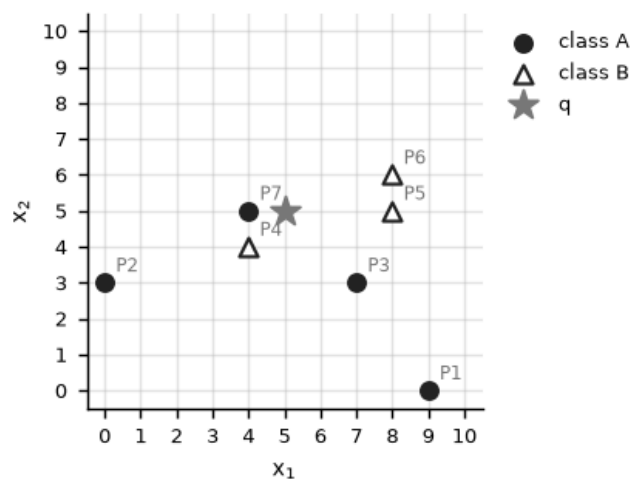
$$W_2 = \begin{bmatrix} 0 & -1 \\ -1 & 2 \\ -2 & 1 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} -0.5 \\ -0.5 \\ -0.5 \end{bmatrix}$$

Q.32 [2 marks · MCQ]

Consider the training set below. A 3-NN classifier (majority vote) is applied to the query point $q = (5, 5)$, once with Euclidean distance and once with Manhattan distance. The predicted classes are

Point	x_1	x_2	class
P1	9	0	A
P2	0	3	A
P3	7	3	A
P4	4	4	B
P5	8	5	B
P6	8	6	B
P7	4	5	A



- (A) Euclidean: A; Manhattan: B
(C) Euclidean: B; Manhattan: B

- (B) Euclidean: A; Manhattan: A
(D) Euclidean: B; Manhattan: A

Q.33 [2 marks · MCQ]

For $n = 6$ paired observations: $\Sigma x = 40$, $\Sigma y = -18$, $\Sigma x^2 = 292$, $\Sigma y^2 = 100$, $\Sigma xy = -144$. The least-squares regression line of x on y is (coefficients rounded to two decimals)

- (A) $\hat{x} = 5.1 - 0.52y$
(C) $\hat{x} = 2.35 - 1.44y$

- (B) $\hat{x} = 3.5 - 1.06y$
(D) $\hat{x} = 8.23 - 0.52y$

Q.34 [2 marks · MCQ]

Six points have the pairwise distances below. Agglomerative clustering with **complete linkage** is run and the dendrogram is cut to give exactly **3** clusters. The resulting clusters are

	P1	P2	P3	P4	P5	P6
P1	0	7	7	1	2	12
P2		0	14	8	5	9
P3			0	6	9	7
P4				0	3	11
P5					0	12
P6						0

- (A) $\{P1, P4, P5\}, \{P2\}, \{P3, P6\}$
(C) $\{P1, P2, P4, P5\}, \{P3, P6\}$

- (B) $\{P1, P2, P4, P5\}, \{P3\}, \{P6\}$
(D) $\{P1, P4, P5\}, \{P2\}, \{P3\}, \{P6\}$

Q.35 [2 marks · NAT (numerical answer)]

A logistic-regression neuron $p = \sigma(w_1x_1 + w_2x_2 + b)$ has $w = (0.5, 0)$, $b = 1$ and is trained with binary cross-entropy loss. After one stochastic gradient step on **example 1 only** with learning rate $\eta = 0.1$, the new value of w_1 is _____ (rounded off to three decimal places)

example	x_1	x_2	y
1	-2	1	0
2	1.5	-1.5	1

— END OF QUESTION PAPER —

Mock Test 39 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MCQ	1	(B)	19	NAT	2	0.2799 to 0.2999
2	MSQ	1	(A), (D)	20	NAT	2	6.798 to 6.802
3	NAT	1	39	21	NAT	2	2.1728 to 2.1928
4	MCQ	1	(B)	22	NAT	2	73.49 to 73.51
5	MCQ	1	(C)	23	NAT	2	0.498 to 0.502
6	MCQ	1	(C)	24	NAT	2	3
7	NAT	1	3	25	NAT	2	3.59 to 3.61
8	NAT	1	61175	26	MSQ	2	(A), (B), (C), (D)
9	MSQ	1	(B), (D)	27	NAT	2	4.5053 to 4.5597
10	NAT	1	0.8826 to 0.9026	28	NAT	2	0.2274 to 0.2314
11	NAT	1	0.4372 to 0.4572	29	NAT	2	3.73 to 3.75
12	MSQ	1	(A), (B), (C)	30	NAT	2	0.6567 to 0.6767
13	NAT	1	35	31	NAT	2	0.1333 to 0.1373
14	NAT	1	196	32	MCQ	2	(A)
15	MCQ	1	(A)	33	MCQ	2	(A)
16	NAT	2	59.115 to 59.135	34	MCQ	2	(A)
17	NAT	2	-1.01 to -0.99	35	NAT	2	0.598 to 0.602
18	NAT	2	0.2705 to 0.2905				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Neural Networks (MLP / Feed-forward)					11
Regression & Model Selection					11
Unsupervised Learning					11
SVM & Decision Trees					11
Probabilistic & Instance-based Classifiers					11
Total					55

Mock Test 39 — Detailed Solutions

Q.1 — Answer: (B) [MCQ, 1M · Neural Networks (MLP / Feed-forward) · Training & regularisation]

Typical signatures: far-too-high $\eta \rightarrow$ loss blows up; somewhat-too-high $\eta \rightarrow$ quick drop then plateau at a high loss (bouncing around the minimum); too-low $\eta \rightarrow$ slow, nearly linear decrease; good $\eta \rightarrow$ fast decrease to the lowest loss.

Curve A: a well-chosen learning rate.

Curve B: a learning rate that is somewhat too high (fast initial drop, then the loss plateaus at a high value).

Curve C: a learning rate that is far too high (training diverges).

Curve D: a learning rate that is too low (very slow, almost linear decrease).

Hence the answer is **Curve B**.

Q.2 — Answer: (A), (D) [MSQ, 1M · Regression & Model Selection · Cross-validation]

(A) **TRUE**. That is its purpose for classification.

(B) **FALSE**. It is the reverse: inner loop tunes, outer loop estimates performance.

(C) **FALSE**. Validation-fold statistics influence the training transformation.

(D) **TRUE**. Each fold contains a single observation.

Q.3 — Answer: 39 [NAT, 1M · Unsupervised Learning · Hierarchical clustering]

$41 - 2 = 39$ merges.

Q.4 — Answer: (B) [MCQ, 1M · Unsupervised Learning · k-means]

Points are assigned to the nearest centroid, so each k-means cluster is a convex Voronoi cell and two clusters are separated by the perpendicular bisector of their centroids. Non-convex shapes such as rings or moons cannot be separated by such a boundary.

k-means converges on raw data, works in any dimension, and is unsupervised (no labels).

Q.5 — Answer: (C) [MCQ, 1M · SVM & Decision Trees · Decision trees: structure & traversal]

Follow the record from the root, taking the branch that matches each test:

Tenure = 15 > 9 $\rightarrow$ No.

Region = South $\rightarrow$ branch South.

(Only the tests on the actual path matter.) The record reaches the leaf **Platinum**.

Note: " $\leq$ " includes equality, so a value equal to the threshold goes to the Yes branch.

Q.6 — Answer: (C) [MCQ, 1M · Unsupervised Learning · PCA & dimensionality reduction]

PC1 is the direction of maximum variance — along the long axis of the elongated cloud (direction B). PC2 must be orthogonal to PC1 and captures the remaining variance (direction C).

Answer: **Direction C**. Directions D and A are at 45° to the main axis and capture intermediate variance.

Q.7 — Answer: 3 [NAT, 1M · Probabilistic & Instance-based Classifiers · Linear discriminant analysis]

S_B is a sum of K outer products of $(\mu_k - \mu)$ which are linearly dependent (weighted sum is 0), so $\text{rank}(S_B) \leq K - 1 = 7$; also $\leq p = 3$.

Maximum = $\min(K - 1, p) = 3$.

Q.8 — Answer: 61175 [NAT, 1M · Neural Networks (MLP / Feed-forward) · Parameter counting]

Fully connected layer with n_{in} inputs and n_{out} outputs: $n_{\text{in}} \cdot n_{\text{out}}$ weights + n_{out} biases.

Layer 1: $784 \times 10 = 7840$ weights, 10 biases.

Layer 2: $10 \times 100 = 1000$ weights, 100 biases.

Layer 3: $100 \times 512 = 51200$ weights, 512 biases.

Layer 4: $512 \times 1 = 512$ weights, 1 biases.

Weights = 60552, biases = 623. Required: **61175**.

Q.9 — Answer: (B), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) · Activation functions]

(A) **FALSE**. It is $1 - \tanh^2(z)$.

- (B) **TRUE**. Differences between logits shrink.
 (C) **FALSE**. Its gradient is exactly 1 for $z > 0$.
 (D) **TRUE**. $\sigma(1-\sigma)$ is maximised at $\sigma = 1/2$.

Q.10 — Answer: 0.8826 to 0.9026 [NAT, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Inner cube volume = $(1 - 2 \times 0.1)^{10} = 0.8^{10} = 0.1074$.

Fraction near the boundary = $1 - 0.1074 = \mathbf{0.89}$.

In high dimensions most points lie near the boundary, far from each other — a symptom of the curse of dimensionality that hurts k-NN.

Q.11 — Answer: 0.4372 to 0.4572 [NAT, 1M · SVM & Decision Trees > SVM: margin & geometry]

Since $x_+ - x_- = (4, 2)$ is parallel to w , the margin width equals $\|x_+ - x_-\| = 4.4721$.

Margin = $2/\|w\| \Rightarrow \|w\| = 2/4.4721 = \mathbf{0.45}$.

(Check: $w^T(x_+ - x_-) = 2$ and $w \parallel (x_+ - x_-)$ give the same result.)

Q.12 — Answer: (A), (B), (C) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

(A) **TRUE**. Its ROC curve is the diagonal.

(B) **TRUE**. Simple mean over classes.

(C) **TRUE**. $F1 = 2PR/(P + R)$.

(D) **FALSE**. FPR can only stay the same or fall.

Q.13 — Answer: 35 [NAT, 1M · Regression & Model Selection > Basis functions & feature scaling]

Number of monomials of degree $\leq d$ in p variables = $C(p + d, d) = C(7, 4) = \mathbf{35}$.

Q.14 — Answer: 196 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

$x^T x = (1)(1) + (-2)(-2) + (3)(3) = 14$.

$K(x, x) = (14 + 0)^2 = 14^2 = \mathbf{196}$.

The kernel equals the inner product $\phi(x)^T \phi(z)$ of the explicit (all-monomials-up-to-degree- d) feature maps, without forming ϕ .

Q.15 — Answer: (A) [MCQ, 1M · Regression & Model Selection > Simple linear regression]

The spread of residuals grows with the fitted value (funnel shape) $\rightarrow$ the constant-variance assumption is violated; a transformation of y (e.g. log) or weighted least squares may help.

Answer: **Non-constant error variance (heteroscedasticity)**.

Other patterns: funnel $\rightarrow$ heteroscedasticity; curvature $\rightarrow$ missing non-linearity; isolated point $\rightarrow$ outlier; random band $\rightarrow$ adequate fit.

Q.16 — Answer: 59.115 to 59.135 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

With $X = U\Sigma V^T$ (X centered, $n \times d$), the sample covariance is $S = X^T X / (n - 1) = V(\Sigma^2 / (n - 1))V^T$, so the PC directions are the columns of V and the eigenvalues are $\lambda_j = \sigma_j^2 / (n - 1)$.

$\lambda = 17^2/8 = 36.125$, $12^2/8 = 18$, $6^2/8 = 4.5$, $2^2/8 = 0.5$.

Cumulative % variance: 61.1, 91.54, 99.15, 100 (note the $(n - 1)$ factor cancels in proportions: $PVE = \sigma_j^2 / \Sigma \sigma^2$).

Answer: **59.12**. (Using σ instead of σ^2 is the classic mistake.)

Q.17 — Answer: -1.01 to -0.99 [NAT, 2M · SVM & Decision Trees > SVM: dual & support vectors]

$w = \sum \alpha_i y_i x_i$ over points with $\alpha_i > 0$:

$w = (2)(+1)(6, 2) + (0.5)(+1)(5, 0) + (2.5)(-1)(5, 2) = (2, -1)$.

b from any support vector, e.g. P1: $y = w^T x + b \Rightarrow b = +1 - (10) = -9$.

Check: $\sum \alpha_i y_i = 0$ ($= 0$, as required).

Answer: **-1**.

Q.18 — Answer: 0.2705 to 0.2905 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

With equal variances, $\ln[\pi_1 p(x|C_1)/(\pi_0 p(x|C_0))] = (\mu_1 - \mu_0)x/\sigma^2 - (\mu_1^2 - \mu_0^2)/(2\sigma^2) + \ln(\pi_1/\pi_0)$ — linear in x (LDA).

Slope $a = (\mu_1 - \mu_0)/\sigma^2 = 0.4444$; intercept $c = -(\mu_1^2 - \mu_0^2)/(2\sigma^2) + \ln(\pi_1/\pi_0) = -4.053$.

$P(C_1|x) = \sigma(ax + c) = \sigma(-0.9418) = \mathbf{0.28}$.

Q.19 — Answer: 0.2799 to 0.2999 [NAT, 2M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Naive Bayes: $P(c|x) \propto P(c) \prod_j P(x_j|c)$. With Laplace smoothing, $P(x_j = v|c) = (\text{count} + 1)/(n_c + V_j)$, V_j = number of values of feature j .

Pos: $P(\text{Flu}=\text{Pos}) = 6/11$; $P(\text{Fever}=\text{None}|\text{Pos}) = (0+1)/(6+3) = 1/9$; $P(\text{Cough}=\text{No}|\text{Pos}) = (2+1)/(6+2) = 3/8$; $P(\text{Fatigue}=\text{Yes}|\text{Pos}) = (4+1)/(6+2) = 5/8$. Product = 0.0142.

Neg: $P(\text{Flu}=\text{Neg}) = 5/11$; $P(\text{Fever}=\text{None}|\text{Neg}) = (1+1)/(5+3) = 1/4$; $P(\text{Cough}=\text{No}|\text{Neg}) = (2+1)/(5+2) = 3/7$; $P(\text{Fatigue}=\text{Yes}|\text{Neg}) = (4+1)/(5+2) = 5/7$. Product = 0.03479.

$P(\text{Pos}|x) = 0.0142/(0.0142 + 0.03479) = \mathbf{0.29}$.

Q.20 — Answer: 6.798 to 6.802 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$f(w) = (1/2)w^2 = \nabla f(w) = 1w$. Update: $v_t = \beta v_{t-1} + \nabla f(w_{t-1})$, $w_t = w_{t-1} - \eta v_t$, with $v_0 = 0$, $\beta = 0.8$, $\eta = 0.1$.

$t = 1$: $g = 1 \times 4 = 4$; $v_1 = 0.8 \times 0 + 4 = 4$; $w_1 = 4 - 0.1 \times 4 = 3.6$.

$t = 2$: $g = 1 \times 3.6 = 3.6$; $v_2 = 0.8 \times 4 + 3.6 = 6.8$; $w_2 = 3.6 - 0.1 \times 6.8 = 2.92$.

Answer: **6.8**.

Q.21 — Answer: 2.1728 to 2.1928 [NAT, 2M · SVM & Decision Trees > SVM: margin & geometry]

P3 and P5 (same class) lie on one margin line and P2 on the other; the optimal boundary is parallel to the line through P3, P5, midway to P2. Support vectors: P2, P3, P5.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (-8/9, -2/9)$, $b = 3$; $\|w\| = 0.9162$, margin $2/\|w\| = \mathbf{2.1828}$.

Point	P1	P2	P3	P4	P5	P6	P7
$y(w^T x + b)$	3.222	1	1	1.444	1	2.111	1.444

All values are ≥ 1 (feasible), with equality exactly at the support vectors.

Margin width = $2/\|w\| = \mathbf{2.18}$.

Q.22 — Answer: 73.49 to 73.51 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Sort the values: 50(Y), 61(Y), 64(N), 65(Y), 67(Y), 80(N), 84(N), 89(N), 90(N).

Parent [4 Yes, 5 No], $H = 0.9911$. The optimal threshold always lies where the class changes between neighbours (Fayyad-Irani), giving 3 candidates:

Threshold	Left ($\leq t$) [Yes, No]	Right [Yes, No]	IG
62.5	[2, 0]	[2, 5]	0.3198
64.5	[2, 1]	[2, 4]	0.0728
73.5	[4, 1]	[0, 4]	0.59

Best: $t = 73.5$ with IG = 0.59. Answer: **73.5**.

Q.23 — Answer: 0.498 to 0.502 [NAT, 2M · SVM & Decision Trees > Ensembles: boosting]

$\varepsilon = \sum_{\text{misclassified}} w_i = 0.1852$; $\alpha = \frac{1}{2} \ln(0.8148/0.1852) = 0.7408$.

Misclassified weights $\times e^\alpha = 2.0976$; correct weights $\times e^{-\alpha} = 0.4767$.

$Z = \varepsilon e^\alpha + (1 - \varepsilon) e^{-\alpha} = 2\sqrt{\varepsilon(1 - \varepsilon)} = 0.7769$.

New weights: 0.0682, 0.0455, 0.0909, 0.0227, 0.5, 0.0909, 0.0909, 0.0909.

Shortcut: after normalisation the misclassified set has total weight exactly $\frac{1}{2}$: $w_i' = w_i/(2\varepsilon)$ if misclassified, $w_i/(2(1 - \varepsilon))$ otherwise.

Answer: **0.5**.

Q.24 — Answer: 3 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Rule: if $y_i(w \cdot x_i + b) \leq 0$, then $w \leftarrow w + \eta y_i x_i$, $b \leftarrow b + \eta y_i$ ($\eta = 1$). Start $w = (0, 0)$, $b = 0$.

Example 1: $a = 0, y \cdot a = 0 \rightarrow$ **mistake**, update $\rightarrow w = (-2, -2), b = 1$.

Example 2: $a = -3, y \cdot a = 3 \rightarrow$ correct, no change.

Example 3: $a = 3, y \cdot a = -3 \rightarrow$ **mistake**, update $\rightarrow w = (-3, 0), b = 0$.

Example 4: $a = -9, y \cdot a = 9 \rightarrow$ correct, no change.

Test: $w \cdot x + b = -3 \cdot -1 + 0 \cdot 0 + 0 = 3$.

Note: a point with net input exactly 0 counts as a mistake ($y \cdot a \leq 0$) – this is why the first example always triggers an update when starting from zero.

Q.25 — Answer: 3.59 to 3.61 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

Conditional risk $R(\alpha_i|x) = \sum_j L(\alpha_i, C_j)P(C_j|x)$:

$$R(\alpha_1) = 0 \times 0.3 + 9 \times 0.1 + 9 \times 0.6 = 6.3$$

$$R(\alpha_2) = 1 \times 0.3 + 0 \times 0.1 + 2 \times 0.6 = 1.5$$

$$R(\alpha_3) = 9 \times 0.3 + 9 \times 0.1 + 0 \times 0.6 = 3.6$$

Minimum risk: α_2 with $R = 1.5$. The MAP rule picks C_3 (risk 3.6) — different, because the losses are asymmetric.

Answer: **3.6**.

Q.26 — Answer: (A), (B), (C), (D) [MSQ, 2M · Regression & Model Selection > Multiple linear regression]

(A) **TRUE**. The three dummies would sum to the intercept column (dummy-variable trap).

(B) **TRUE**. $\hat{y} = 19 + 1.3 \times 8 + (1.3) \times 0 + (-5.2) = 24.2$.

(C) **TRUE**. North is the reference level (all dummies 0).

(D) **TRUE**. $3 \times 1.3 = 3.9$.

Q.27 — Answer: 4.5053 to 4.5597 [NAT, 2M · Regression & Model Selection > MLE & MAP view]

$\bar{x} = 5.667, \bar{y} = 8.667$.

i	1	2	3	4	5	6
$x_i - \bar{x}$	-3.67	-1.67	-0.67	0.33	2.33	3.33
$y_i - \bar{y}$	-6.67	-3.67	-1.67	3.33	1.33	7.33
$(x_i - \bar{x})(y_i - \bar{y})$	24.44	6.11	1.11	1.11	3.11	24.44

$$S_{xx} = \sum (x_i - \bar{x})^2 = 33.333, S_{xy} = 60.333.$$

$$\beta_1 = S_{xy}/S_{xx} = 1.81, \beta_0 = \bar{y} - \beta_1 \bar{x} = -1.59.$$

$$SSE = S_{yy} - S_{xy}^2/S_{xx} = 127.3333 - 109.2033 = 18.13.$$

Two parameters are estimated, so $s^2 = SSE/(n - 2) = 18.13/4 = 4.5325$.

Answer: **4.53**.

Q.28 — Answer: 0.2274 to 0.2314 [NAT, 2M · Unsupervised Learning > k-means]

k-means++ picks the next centre x with probability $D(x)^2 / \sum_{x'} D(x')^2$, where $D(x)$ is the distance from x to the **nearest** centre already chosen (chosen centres have $D = 0$).

Point	P1	P2	P3	P4	P5	P6
$D(x)^2$	0	37	20	26	25	1

$$\sum D^2 = 109, \text{ so } P(P5) = 25/109 = \mathbf{0.229}.$$

Common mistakes: using D instead of D^2 (that is not k-means++), or using the distance to the *farthest*/first centre instead of the nearest chosen centre.

Q.29 — Answer: 3.73 to 3.75 [NAT, 2M · Regression & Model Selection > Cross-validation]

$CV(\lambda)$ = average of the fold MSEs: $\lambda = 0.01$: 3.74; $\lambda = 0.5$: 4.78; $\lambda = 5$: 3.98.

Smallest CV error at $\lambda = 0.01$: $CV = 3.74$.

Answer: **3.74**.

Q.30 — Answer: 0.6567 to 0.6767 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

Total pairs = $C(7, 2) = 21$.

a = pairs together in both = 3; b = pairs separated in both = 11; pairs together in U only = 4; together in V only = 3.

Rand index = $(a + b)/C(n,2) = (3 + 11)/21 = 0.6667$.

Answer: **0.67**. (Cluster names do not matter: only whether pairs are together.)

Q.31 — Answer: 0.1333 to 0.1373 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Forward pass]

$a = W_1x + b_1 = (4, 1.5)$; $h = \tanh(a) = (0.9993, 0.9051)$.

$z = W_2h + b_2 = (-1.4051, 0.311, -1.5935)$.

$\exp(z) = (0.2453, 1.3647, 0.2032)$, sum = 1.8133.

$p_1 = \exp(z_1)/\sum \exp(z_j) = \mathbf{0.1353}$. (All probabilities: (0.135, 0.753, 0.112).)

Q.32 — Answer: (A) [MCQ, 2M · Probabilistic & Instance-based Classifiers › k-nearest neighbours]
Distances from q :

Point	class	Euclidean d	Manhattan d
P1	A	6.403	9
P2	A	5.385	7
P3	A	2.828	4
P4	B	1.414	2
P5	B	3	3
P6	B	3.162	4
P7	A	1	1

Euclidean 3 nearest: P7(A), P4(B), P3(A) = **A**.

Manhattan 3 nearest: P7(A), P4(B), P5(B) = **B**.

Answer: **Euclidean: A; Manhattan: B**. The two metrics disagree because L_1 penalises diagonal offsets more than L_2 , changing the neighbour set.

Q.33 — Answer: (A) [MCQ, 2M · Regression & Model Selection › Simple linear regression]

$\bar{x} = 6.6667$, $\bar{y} = -3$.

$S_{xx} = \sum x^2 - (\sum x)^2/n = 25.3333$, $S_{yy} = \sum y^2 - (\sum y)^2/n = 46$, $S_{xy} = \sum xy - \sum x \sum y/n = -24$.

Regress x on y : $b' = S_{xy}/S_{yy} = -0.5217$; $a' = \bar{x} - b'\bar{y} = 5.1014$.

Line: $\hat{x} = \mathbf{5.1 - 0.52y}$.

A common error is to invert the y -on- x line (slope $1/b$), which minimises the wrong (vertical) errors.

Q.34 — Answer: (A) [MCQ, 2M · Unsupervised Learning › Hierarchical clustering]

Use complete linkage: $d(A,B) = \max_{a \in A, b \in B} d(a,b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P1\} + \{P4\}$ at height **1**. Updated: $d(\{P1, P4\}, \{P2\}) = 8$, $d(\{P1, P4\}, \{P3\}) = 7$, $d(\{P1, P4\}, \{P5\}) = 3$, $d(\{P1, P4\}, \{P6\}) = 12$.

Merge 2: $\{P5\} + \{P1, P4\}$ at height **3**. Updated: $d(\{P1, P4, P5\}, \{P2\}) = 8$, $d(\{P1, P4, P5\}, \{P3\}) = 9$, $d(\{P1, P4, P5\}, \{P6\}) = 12$.

Merge 3: $\{P3\} + \{P6\}$ at height **7**. Updated: $d(\{P3, P6\}, \{P2\}) = 14$, $d(\{P3, P6\}, \{P1, P4, P5\}) = 12$.

Merge 4: $\{P2\} + \{P1, P4, P5\}$ at height **8**. Updated: $d(\{P1, P2, P4, P5\}, \{P3, P6\}) = 14$.

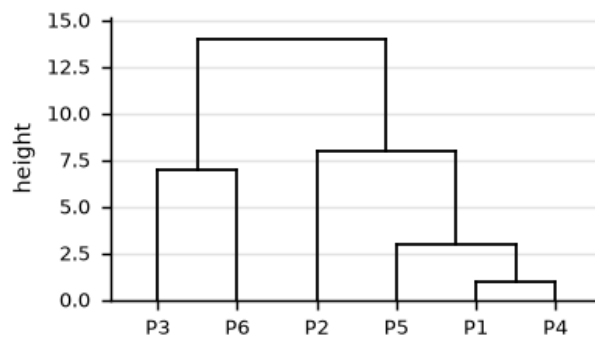
Merge 5: $\{P3, P6\} + \{P1, P2, P4, P5\}$ at height **14**.

Stopping when 3 clusters remain (i.e. after 3 merges): **$\{P1, P4, P5\}$, $\{P2\}$, $\{P3, P6\}$** .

$\{P1, P2, P4, P5\}$, $\{P3\}$, $\{P6\}$ is what *single* linkage would give.

$\{P1, P2, P4, P5\}$, $\{P3, P6\}$ has 2 clusters (cut at the wrong level).

$\{P1, P4, P5\}$, $\{P2\}$, $\{P3\}$, $\{P6\}$ has 4 clusters (cut at the wrong level).



Q.35 — Answer: 0.598 to 0.602 [NAT, 2M · Neural Networks (MLP / Feed-forward) >

Backpropagation]

For σ output with binary cross-entropy $L = -[y \ln p + (1 - y) \ln(1 - p)]$, $\partial L / \partial a = p - y$, so $\partial L / \partial w_j = (p - y)x_j$ and $\partial L / \partial b = p - y$.

Example 1: $a = 0$, $p = 0.5$, $p - y = 0.5$.

Gradient for w_1 : -1 .

$w_1 \leftarrow 0.5 - 0.1 \times (-1) = \mathbf{0.6}$.

Note there is no σ' factor: it cancels with the derivative of the log loss.

Mock Test 40

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	4	5	26

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Unsupervised Learning	7	11
SVM & Decision Trees	7	11
Neural Networks (MLP / Feed-forward)	7	11
Regression & Model Selection	7	11
Probabilistic & Instance-based Classifiers	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MSQ (one or more correct)]

Which of the following statements about clustering evaluation measures is/are CORRECT?

- (A) A negative silhouette suggests that the point may be assigned to the wrong cluster.
- (B) A silhouette value of -1 indicates a perfectly clustered point.
- (C) The Rand index requires the two clusterings to have the same number of clusters.
- (D) The Rand index is the fraction of point pairs on which two clusterings agree (both together or both apart).

Q.2 [1 mark · NAT (numerical answer)]

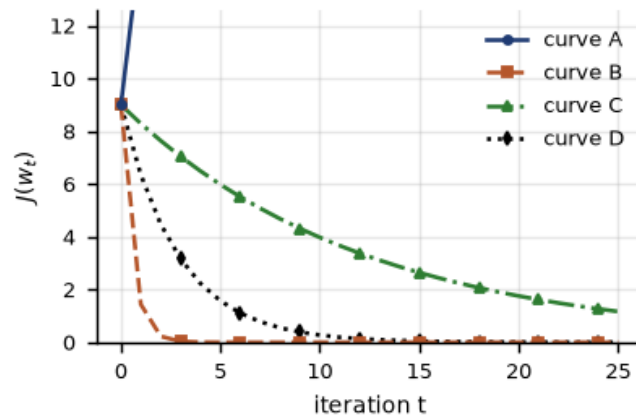
The distance of the hyperplane $x_1 + 3x_2 - 9 = 0$ from the origin is _____ (rounded off to two decimal places)

Q.3 [1 mark · NAT (numerical answer)]

A layer of 40 units uses dropout with keep probability $p = 0.5$. The expected number of units dropped (set to zero) in one forward pass during training is _____ (rounded off to two decimal places)

Q.4 [1 mark · MCQ]

Gradient descent $w_{t+1} = w_t - \eta J'(w_t)$ is run on $J(w) = (w - 4)^2$ from $w_0 = 1$ with four learning rates $\eta \in \{0.02, 0.08, 0.3, 1.15\}$. The loss curves are shown below (curves that leave the plotted range are cut off). Which curve corresponds to $\eta = 0.3$?



- (A) Curve C
(C) Curve A

- (B) Curve D
(D) Curve B

Q.5 [1 mark · MSQ (one or more correct)]

Which of the following statements about kernel functions (Mercer kernels) is/are CORRECT?

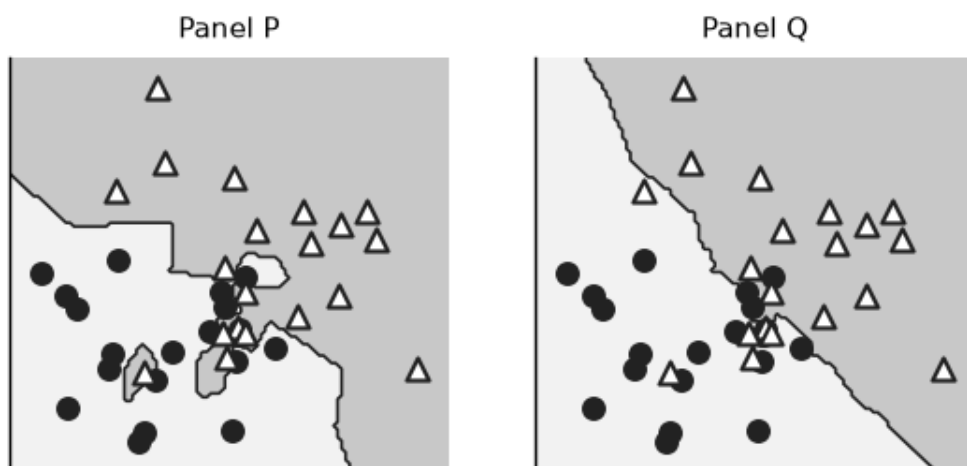
- (A) $\exp(-\gamma\|x - z\|^2)$ with $\gamma > 0$ is a valid kernel.
(B) $k(x, z) = f(x)f(z)$ is valid for any real-valued f .
(C) For the RBF kernel, $k(x, x) = 1$ for every x .
(D) Every valid kernel yields a symmetric PSD Gram matrix for any finite set of points.

Q.6 [1 mark · NAT (numerical answer)]

A binary logistic regression model is given by $\ln[P(y=1|\mathbf{x}) / P(y=0|\mathbf{x})] = -1 + (2)x_1 + (-0.5)x_2$. For the input $\mathbf{x} = (x_1, x_2) = (2, 3)$, the value of $P(y = 0 | \mathbf{x})$ is _____ (rounded off to two decimal places)

Q.7 [1 mark · MCQ]

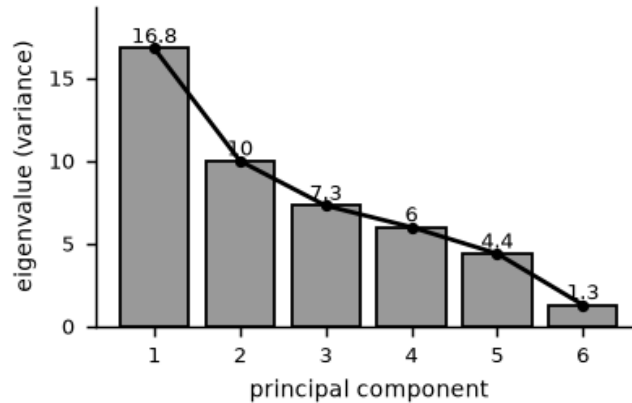
The two panels show the decision regions of k -NN classifiers trained on the same 2-D data set, one with $k = 1$ and the other with $k = 21$. Which of the following statements is CORRECT?



- (A) Panel P uses $k = 1$; it has higher bias and lower variance
(B) Panel Q uses $k = 1$; it has higher bias and lower variance
(C) Panel P uses $k = 1$; it has lower bias and higher variance
(D) Panel Q uses $k = 21$; it has zero training error

Q.8 [1 mark · NAT (numerical answer)]

The scree plot below shows the eigenvalues of the covariance matrix of a 6-dimensional dataset. The percentage of total variance explained by the first two principal components together is _____ (rounded off to one decimal place)

**Q.9 [1 mark · NAT (numerical answer)]**

A bootstrap sample of size 9 is drawn with replacement from 9 distinct training points. The expected number of distinct points appearing in the sample is _____ (rounded off to two decimal places)

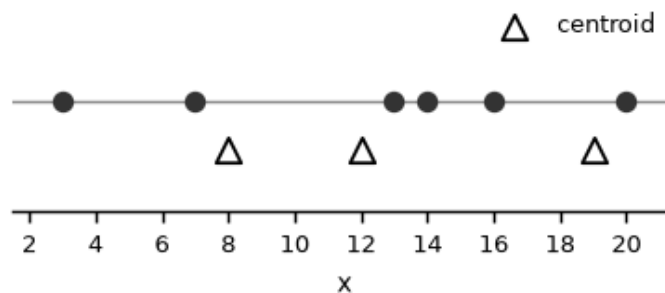
Q.10 [1 mark · MSQ (one or more correct)]

Which of the following tasks is/are most naturally formulated as **classification** problems?

- (A) Estimating the fuel efficiency (km per litre) of a car from engine specifications
- (B) Predicting whether a student will pass or fail a course
- (C) Predicting the number of minutes a user will spend on an app tomorrow
- (D) Recognising which of 26 letters a typed character image shows

Q.11 [1 mark · NAT (numerical answer)]

One-dimensional data $\{3, 7, 13, 14, 16, 20\}$ and $k = 3$ centroids at 8, 12, 19 are given. Each point is assigned to its nearest centroid (centroids are **not** updated). The resulting within-cluster sum of squared errors $\sum (x_i - c_{(i)})^2$ is _____ (rounded off to two decimal places)

**Q.12 [1 mark · NAT (numerical answer)]**

The value of the derivative of $\tanh(z)$ at $z = -2$ is _____ (rounded off to 4 decimal places)

Q.13 [1 mark · MSQ (one or more correct)]

Which of the following statements about the perceptron is/are CORRECT?

- (A) The perceptron always finds the maximum-margin separating hyperplane.
- (B) The decision boundary of a perceptron, $w \cdot x + b = 0$, is a hyperplane.
- (C) When the data are not linearly separable, the basic perceptron algorithm does not settle on a fixed weight vector.
- (D) In the perceptron learning rule (labels ± 1), the weights change only on misclassified examples.

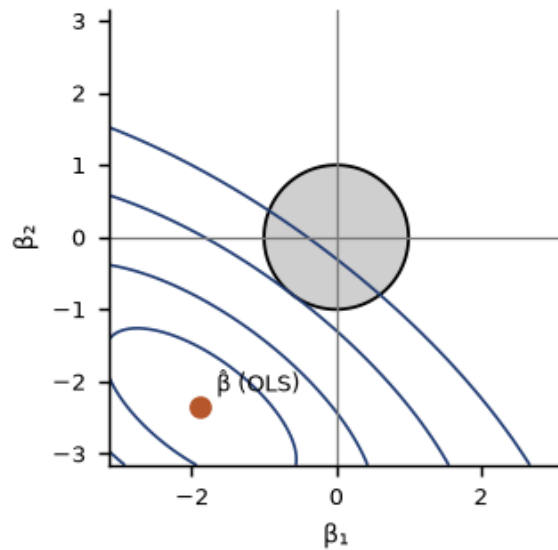
Q.14 [1 mark · MSQ (one or more correct)]

Which of the following statements about Bayes-optimal classification and generative vs. discriminative models is/are CORRECT?

- (A) For binary classification under 0-1 loss, the Bayes error at x equals $\max(P(y=0|x), P(y=1|x))$.
- (B) Collecting more training data (with the same features) cannot reduce the Bayes error rate.
- (C) A generative model can be used to sample synthetic input vectors x .
- (D) Adding a new informative feature can reduce the Bayes error rate.

Q.15 [1 mark · MCQ]

For a two-predictor penalised least-squares problem, the figure shows the constraint region (shaded) of the penalty together with elliptical contours of the residual sum of squares centred at the least-squares estimate β (OLS). The penalised estimate (β_1, β_2) satisfies



- (A) $\beta_2 = 0$ exactly and $\beta_1 \neq 0$
- (B) $\beta_1 = \beta_2 = 0$
- (C) $\beta_1 \neq 0$ and $\beta_2 \neq 0$ (both shrunk, neither exactly zero)
- (D) $\beta_1 = 0$ exactly and $\beta_2 \neq 0$

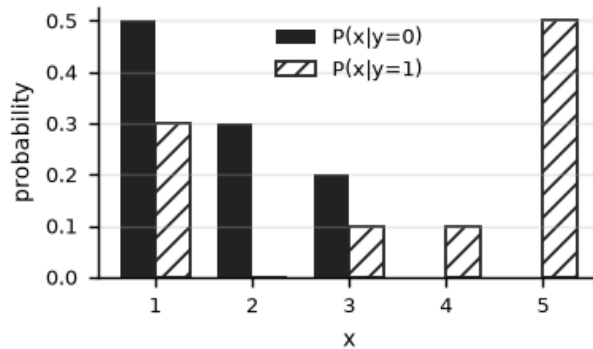
Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

Let φ be the feature map of the kernel $K(x, z) = (x^T z + 1)^2$ on $\mathbb{R}^2$, so that $K(x, z) = \varphi(x)^T \varphi(z)$. For $x = (2, -2)$ and $z = (0, -1)$, the squared distance $\|\varphi(x) - \varphi(z)\|^2$ in feature space is _____. (Answer in integer)

Q.17 [2 marks · NAT (numerical answer)]

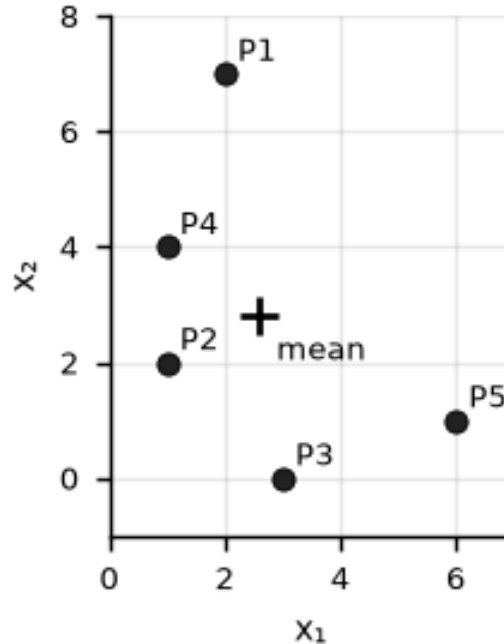
A discrete feature $x \in \{1, \dots, 5\}$ has the class-conditional distributions below (also plotted), with priors $P(y = 0) = 0.6$ and $P(y = 1) = 0.4$. Then the posterior probability $P(y = 1 \mid x = 3)$ is _____ (rounded off to three decimal places)

x	1	2	3	4	5
$P(x \mid y=0)$	0.5	0.3	0.2	0	0
$P(x \mid y=1)$	0.3	0	0.1	0.1	0.5

**Q.18 [2 marks · NAT (numerical answer)]**

PCA is applied to the five 2-D points below (data are mean-centered first; sample covariance uses divisor $n - 1$). The sample variance (divisor $n - 1$) of the PC1 scores of the five points is _____ (rounded off to two decimal places)

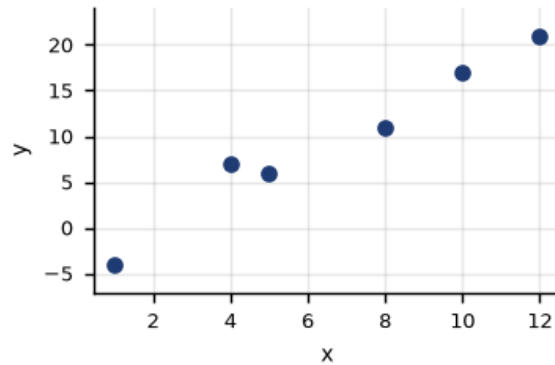
Point	P1	P2	P3	P4	P5
x_1	2	1	3	1	6
x_2	7	2	0	4	1



Q.19 [2 marks · NAT (numerical answer)]

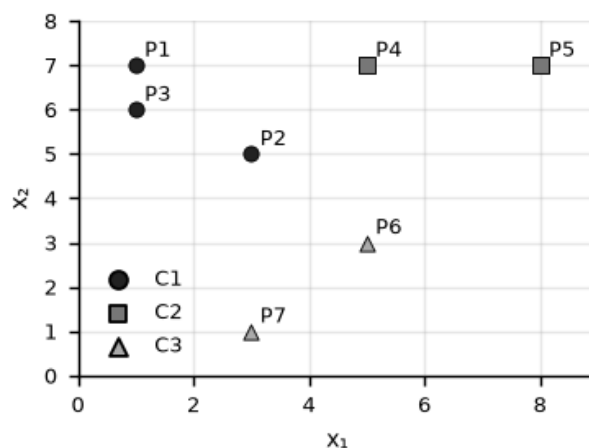
The 6 observations shown in the table are plotted below. For the simple linear regression of y on x (with intercept), the slope β_1 of the least-squares line $\hat{y} = \beta_0 + \beta_1 x$ is _____ (rounded off to two decimal places)

x	1	4	5	8	10	12
y	-4	7	6	11	17	21

**Q.20 [2 marks · NAT (numerical answer)]**

A clustering of 7 points into 3 clusters is shown below. Using Manhattan distance, the silhouette coefficient $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$ of point P3 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5	P6	P7
x_1	1	3	1	5	8	5	3
x_2	7	5	6	7	7	3	1
cluster	C1	C1	C1	C2	C2	C3	C3

**Q.21 [2 marks · MCQ]**

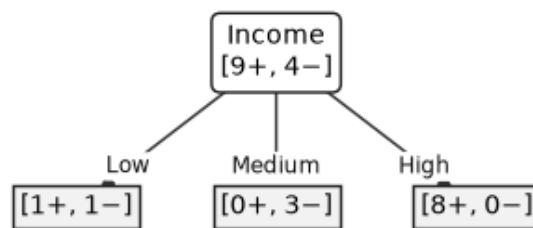
Ridge regression is tuned by 10-fold cross-validation. For each λ the table gives the mean CV MSE and its standard error (SE) across folds. (Larger λ means a simpler, more regularised model.) Using the **one-standard-error rule** (choose the most regularised model whose CV error is within one standard error of the minimum CV error), the selected value is

λ	0.01	0.1	0.3	1	3	10
CV MSE	5.39	4.94	4.72	4.9	5.76	6.31
SE	0.26	0.25	0.44	0.16	0.43	0.19

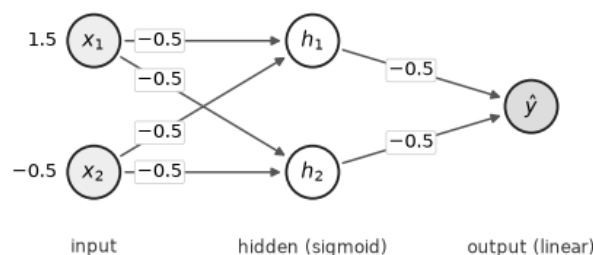
(A) $\lambda = 0.01$ **(B)** $\lambda = 0.1$ **(C)** $\lambda = 1$ **(D)** $\lambda = 10$

Q.22 [2 marks · NAT (numerical answer)]

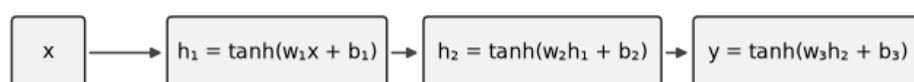
A node with 13 training records (class counts [positive+, negative−]) is split on the attribute Income as shown. Compute the information gain (in bits) of splitting on Income. (rounded off to two decimal places)

**Q.23 [2 marks · NAT (numerical answer)]**

In the 2-2-1 network below (sigmoid hidden units, linear output $\hat{y} = v_1 h_1 + v_2 h_2$, all biases 0), **every** weight is initialised to -0.5 . One gradient-descent step with $\eta = 1$ is taken on $L = \frac{1}{2}(\hat{y} - y)^2$ for the input $x = (1.5, -0.5)$ with target $y = 2$. Then the updated value of w_{22} (weight from x_2 to h_2) is _____ (rounded off to three decimal places)

**Q.24 [2 marks · NAT (numerical answer)]**

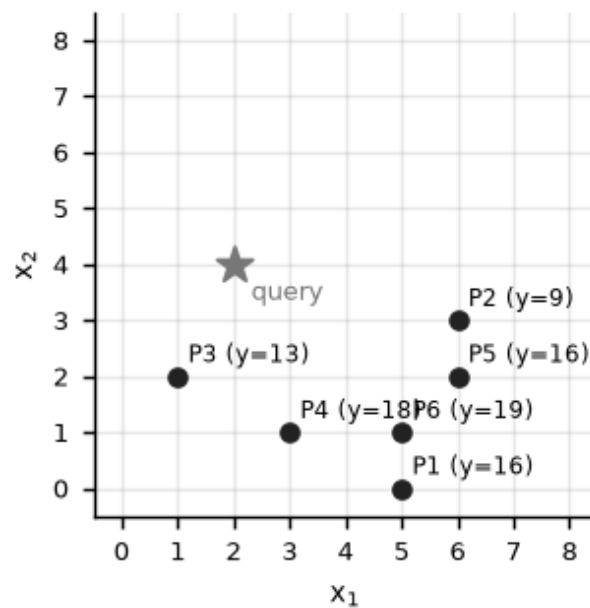
Consider the scalar chain shown below with weights $w_1 = 1$, $w_2 = 1$, $w_3 = 3$, biases $b_1 = -0.5$, $b_2 = 1$, $b_3 = -0.5$ and activation $\tanh$ at every stage. For input $x = -0.5$, the value of $\partial y / \partial w_1$ is _____ (rounded off to 4 decimal places)



Q.25 [2 marks · NAT (numerical answer)]

A distance-weighted k-NN regressor with $k = 3$ predicts $\hat{y} = \sum w_i y_i / \sum w_i$ over the 3 nearest neighbours, with weights $w_i = 1/d_i^2$. For the training data below, the prediction at $\mathbf{x} = (2, 4)$ (Euclidean distance) is _____ (rounded off to two decimal places)

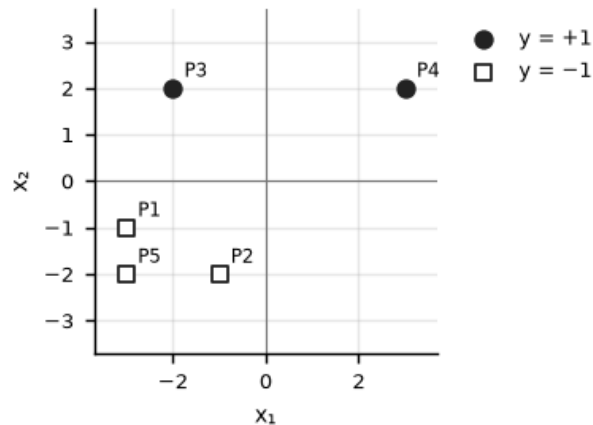
Point	x_1	x_2	y
P1	5	0	16
P2	6	3	9
P3	1	2	13
P4	3	1	18
P5	6	2	16
P6	5	1	19



Q.26 [2 marks · NAT (numerical answer)]

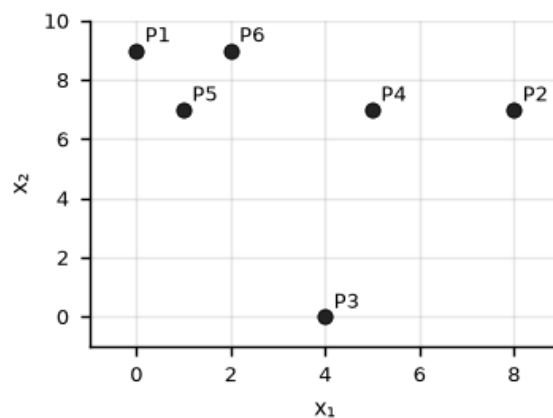
A perceptron with augmented weight vector $w = (w_0, w_1, w_2)$ (w_0 is the bias, $\tilde{x} = (1, x_1, x_2)$) starts from $w = (-1, 0, 0)$ and uses learning rate $\eta = 0.5$. The points P1, P2, ... are presented once, in order; whenever $y \cdot w^T \tilde{x} \leq 0$ the update $w \leftarrow w + \eta y \tilde{x}$ is applied. Find the value of w_2 after this epoch. (rounded off to one decimal place)

Point	x_1	x_2	y
P1	-3	-1	-1
P2	-1	-2	-1
P3	-2	2	+1
P4	3	2	+1
P5	-3	-2	-1

**Q.27 [2 marks · NAT (numerical answer)]**

Single-linkage agglomerative clustering with Manhattan distance is applied to the six points below. Then the height at which the **last** merge (forming a single cluster) occurs is _____ (Answer in integer)

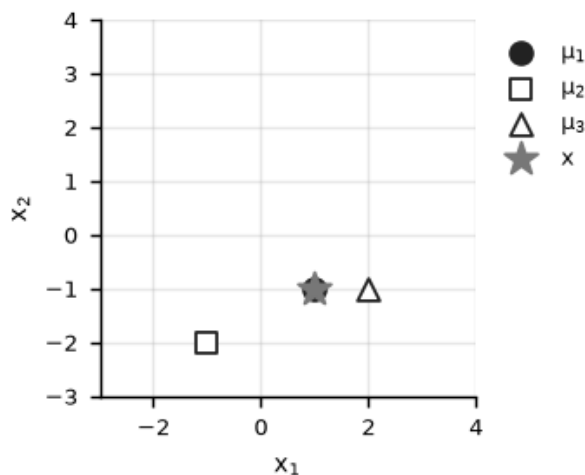
Point	P1	P2	P3	P4	P5	P6
x_1	0	8	4	5	1	2
x_2	9	7	0	7	7	9



Q.28 [2 marks · NAT (numerical answer)]

An LDA classifier for 3 classes has shared covariance matrix Σ (below), class means $\mu_1 = (1, -1)^T$; $\mu_2 = (-1, -2)^T$; $\mu_3 = (2, -1)^T$, and priors $\pi_1 = 0.6$, $\pi_2 = 0.2$, $\pi_3 = 0.2$. The linear discriminant is $\delta_k(\mathbf{x}) = \mathbf{x}^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln \pi_k$ (natural log). For $\mathbf{x} = (1, -1)^T$, the index $k \in \{1, 2, 3\}$ of the class predicted by LDA is ____ (Answer in integer)

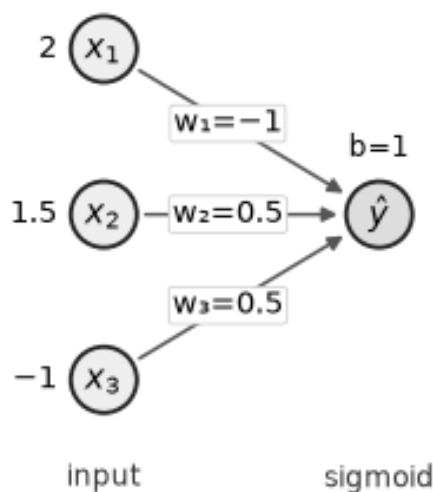
$$\Sigma = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

**Q.29 [2 marks · NAT (numerical answer)]**

$X_1, \dots, X_n$ are i.i.d. with mean $\mu = 1$ and variance $\sigma^2 = 16$, with $n = 16$. Consider the shrinkage estimator $\hat{\mu} = c \cdot \bar{x}$ of μ , where $\bar{x}$ is the sample mean. Using $\text{MSE} = \text{Bias}^2 + \text{Variance}$, the mean squared error of $\hat{\mu} = 0.7 \cdot \bar{x}$ is ____ (rounded off to three decimal places)

Q.30 [2 marks · NAT (numerical answer)]

A single sigmoid neuron (shown below) receives $\mathbf{x} = (2, 1.5, -1)$ and has target $y = 1$. The loss is $L = \frac{1}{2}(\hat{y} - y)^2$. The value of $\partial L / \partial b$ is ____ (rounded off to three decimal places)

**Q.31 [2 marks · NAT (numerical answer)]**

An ensemble combines 5 binary classifiers by majority vote. Each classifier is correct with probability 0.8, independently of the others. Compute the probability that the majority vote is correct. (rounded off to three decimal places)

Q.32 [2 marks · NAT (numerical answer)]

A 3-class softmax (multinomial logistic) regression model computes scores $\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}$ and $P(y = k | \mathbf{x}) = \frac{e^{z_k}}{\sum_j e^{z_j}}$, where row k of $\mathbf{W}$ holds the weights of class k . For the parameters below and $\mathbf{x} = (0, -1)^T$, the cross-entropy loss $-\ln P(y = 1 | \mathbf{x})$ incurred if the true class is 1 (natural log) is _____ (rounded off to two decimal places)

$$\mathbf{W} = \begin{bmatrix} 0 & 0 \\ 0 & -2 \\ -1 & -2 \end{bmatrix}$$

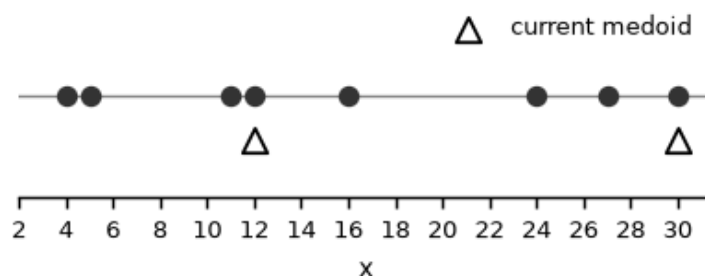
$$\mathbf{b} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Q.33 [2 marks · NAT (numerical answer)]

A 4-class classifier outputs logits $\mathbf{z} = (-1.5, 2, 1.5, 0)$ and probabilities $\mathbf{p} = \text{softmax}(\mathbf{z})$. The true class is 3, and the loss is the cross-entropy $L = -\ln p_3$. Then the Euclidean norm $\|\partial L / \partial \mathbf{z}\|$ of the gradient with respect to the logit vector is _____ (rounded off to three decimal places)

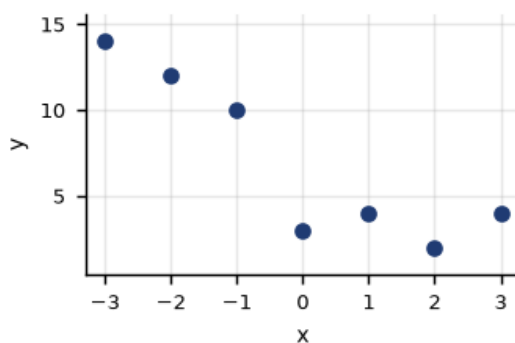
Q.34 [2 marks · NAT (numerical answer)]

PAM (k-medoids, $k = 2$, absolute-difference dissimilarity) is applied to the 1-D data $\{4, 5, 11, 12, 16, 24, 27, 30\}$. The current medoids are 12 and 30. The total cost $\sum_i \min_m |x_i - m|$ of the current configuration is _____ (Answer in integer)

**Q.35 [2 marks · NAT (numerical answer)]**

The quadratic model $y = a + bx + cx^2$ is fitted by least squares to the 7 points below (a polynomial / basis-function regression with basis $1, x, x^2$). Then the fitted value $\hat{y}$ at $x = 0.5$ is _____ (rounded off to two decimal places)

x	-3	-2	-1	0	1	2	3
y	14	12	10	3	4	2	4



— END OF QUESTION PAPER —

Mock Test 40 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MSQ	1	(A), (D)	19	NAT	2	2.108 to 2.148
2	NAT	1	2.836 to 2.856	20	NAT	2	0.6823 to 0.7023
3	NAT	1	19.99 to 20.01	21	MCQ	2	(C)
4	MCQ	1	(D)	22	NAT	2	0.7266 to 0.7466
5	MSQ	1	(A), (B), (C), (D)	23	NAT	2	−0.3623 to −0.3583
6	NAT	1	0.1724 to 0.1924	24	NAT	2	−0.5775 to −0.5661
7	MCQ	1	(C)	25	NAT	2	13.7277 to 13.7477
8	NAT	1	58.315 to 58.715	26	NAT	2	1.99 to 2.01
9	NAT	1	5.872 to 5.892	27	NAT	2	8
10	MSQ	1	(B), (D)	28	NAT	2	1
11	NAT	1	40.99 to 41.01	29	NAT	2	0.577 to 0.583
12	NAT	1	0.0692 to 0.0721	30	NAT	2	−0.15 to −0.146
13	MSQ	1	(B), (C), (D)	31	NAT	2	0.9401 to 0.9441
14	MSQ	1	(B), (C), (D)	32	NAT	2	4.3166 to 4.3366
15	MCQ	1	(C)	33	NAT	2	0.8682 to 0.8722
16	NAT	2	67	34	NAT	2	29
17	NAT	2	0.248 to 0.252	35	NAT	2	4.3665 to 4.4192
18	NAT	2	9.0964 to 9.1164				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Unsupervised Learning					11
SVM & Decision Trees					11
Neural Networks (MLP / Feed-forward)					11
Regression & Model Selection					11
Probabilistic & Instance-based Classifiers					11
Total					55

Mock Test 40 — Detailed Solutions

Q.1 — Answer: (A), (D) [MSQ, 1M · Unsupervised Learning > Cluster evaluation]

(A) TRUE. It is on average closer to another cluster ($b < a$).

(B) FALSE. +1 is ideal; -1 is worst.

(C) FALSE. It compares pairs only.

(D) TRUE. $RI = (a + b)/C(n, 2)$.

Q.2 — Answer: 2.836 to 2.856 [NAT, 1M · SVM & Decision Trees > SVM: margin & geometry]

Distance of origin = $|w^T 0 + b|/\|w\| = |b|/\|w\| = 9/3.1623 = 2.85$.

Q.3 — Answer: 19.99 to 20.01 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Each unit is dropped independently with probability $1 - p = 0.5$; expected number = $40 \times 0.5 = 20$.

Q.4 — Answer: (D) [MCQ, 1M · Regression & Model Selection > Gradient descent]

$J(w) = (w - 4)^2$ has $J' = 2$. GD: $w_{t+1} - 4 = (1 - 2\eta)(w_t - 4)$, so $J(w_t) = (1 - 2\eta)^{2t} J(w_0)$.

Ratios $(1 - 2\eta)^2$ per step: $\eta = 0.02 \rightarrow 0.9216$ (curve C); $\eta = 0.08 \rightarrow 0.7056$ (curve D); $\eta = 0.3 \rightarrow 0.16$ (curve B); $\eta = 1.15 \rightarrow 1.69$ (curve A).

Ratio $> 1 \Rightarrow$ divergence ($\eta > 1 = 2/J'$); among convergent rates (all < 0.5 here), a smaller η gives a ratio closer to 1, i.e. slower decrease.

Hence $\eta = 0.3$ corresponds to **Curve B**.

Q.5 — Answer: (A), (B), (C), (D) [MSQ, 1M · SVM & Decision Trees > SVM: kernels]

(A) TRUE. The Gaussian/RBF kernel.

(B) TRUE. Rank-one PSD Gram matrix.

(C) TRUE. $\exp(0) = 1$.

(D) TRUE. Mercer's condition.

Q.6 — Answer: 0.1724 to 0.1924 [NAT, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

$z = w_0 + w_1 x_1 + w_2 x_2 = -1 + (2)(2) + (-0.5)(3) = 1.5$ (this is the log-odds).

$P(y = 1 | \mathbf{x}) = \sigma(z) = 1/(1 + e^{-1.5}) = 0.8176$.

$P(y = 0 | \mathbf{x}) = 1 - \sigma(z) = \sigma(-z) = 0.18$.

Q.7 — Answer: (C) [MCQ, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

The jagged boundary with small islands around individual points (Panel P) is produced by $k = 1$, which fits every training point (zero training error): low bias, high variance.

The smooth boundary (Panel Q) comes from $k = 21$: averaging many neighbours raises bias and lowers variance; it misclassifies some training points, so its training error is not zero.

Answer: **Panel P uses $k = 1$; it has lower bias and higher variance.**

Q.8 — Answer: 58.315 to 58.715 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

Total variance = $\Sigma \lambda = 45.8$. Cumulative proportion of variance explained:

k	1	2	3	4	5	6
λ_k	16.8	10	7.3	6	4.4	1.3
cumulative %	36.68	58.52	74.45	87.55	97.16	100

$(\lambda_1 + \lambda_2)/\Sigma \lambda = 26.8/45.8 = 58.5\%$.

Q.9 — Answer: 5.872 to 5.892 [NAT, 1M · SVM & Decision Trees > Ensembles: bagging & random forests]

By linearity: $n \cdot P(\text{point appears}) = 9 \cdot [1 - (1 - 1/9)^9] = 5.882$.

Q.10 — Answer: (B), (D) [MSQ, 1M · Regression & Model Selection > Regression vs classification]

A supervised task is *regression* when the target y is real-valued (loss such as squared error), and *classification* when y takes values in a finite set of labels (loss such as cross-entropy or 0-1 loss).

- (A) **FALSE**. The output (km per litre) is a continuous real-valued quantity → regression.
 (B) **TRUE**. The output (pass / fail) is a discrete class label → classification.
 (C) **FALSE**. The output (time in minutes) is a continuous real-valued quantity → regression.
 (D) **TRUE**. The output (one of 26 letters) is a discrete class label → classification.

Q.11 — Answer: 40.99 to 41.01 [NAT, 1M · Unsupervised Learning > k-means]

Assignment (nearest centroid): $c = 8$: {3, 7}; $c = 12$: {13, 14}; $c = 19$: {16, 20}.

$$\text{SSE} = (3-8)^2 + (7-8)^2 + (13-12)^2 + (14-12)^2 + (16-19)^2 + (20-19)^2 = \mathbf{41}.$$

Q.12 — Answer: 0.0692 to 0.0721 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

$$\tanh(-2) = -0.964; d/dz \tanh(z) = 1 - \tanh^2(z) = 1 - 0.9293 = \mathbf{0.0707}.$$

Q.13 — Answer: (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

- (A) **FALSE**. It stops at any separating hyperplane; max margin is the SVM objective.
 (B) **TRUE**. It is an affine set of dimension $d-1$.
 (C) **TRUE**. Some example is always misclassified, so updates never stop.
 (D) **TRUE**. Update $w \leftarrow w + \eta yx$ is applied only when $y(w \cdot x + b) \leq 0$.

Q.14 — Answer: (B), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

- (A) **FALSE**. It is the minimum of the two posteriors.
 (B) **TRUE**. It is a property of the true distribution.
 (C) **TRUE**. It models the distribution of x .
 (D) **TRUE**. The classes may overlap less in the larger space.

Q.15 — Answer: (C) [MCQ, 1M · Regression & Model Selection > Ridge & Lasso]

The shaded region is the constraint set of the ridge (L2: $\beta_1^2 + \beta_2^2 \leq t^2$, a disc). The penalised estimate is the first point at which the RSS contours (ellipses centred at β_{OLS} , which lies outside the region) touch the region.

From the figure the tangency point is approximately $(-0.61, -0.8)$.

The contour touches the boundary at a point that is not on an axis, so both coefficients are non-zero. For the disc this is the generic case — ridge shrinks but does not zero out.

Answer: **$\beta_1 \neq 0$ and $\beta_2 \neq 0$ (both shrunk, neither exactly zero)**. Both coefficients cannot be zero since the OLS point lies outside the region (the constraint is active but $t > 0$).

Q.16 — Answer: 67 [NAT, 2M · SVM & Decision Trees > SVM: kernels]

$$\|\varphi(x) - \varphi(z)\|^2 = K(x, x) + K(z, z) - 2K(x, z).$$

$$K(x, x) = (8 + 1)^2 = 81, K(z, z) = (1 + 1)^2 = 4, K(x, z) = (2 + 1)^2 = 9.$$

$$\text{Squared distance} = 81 + 4 - 2(9) = \mathbf{67}.$$

Q.17 — Answer: 0.248 to 0.252 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

$$P(y=1|x=3) = \pi_1 P(x|1) / (\pi_0 P(x|0) + \pi_1 P(x|1)) = 0.04 / (0.12 + 0.04) = \mathbf{0.25}.$$

Q.18 — Answer: 9.0964 to 9.1164 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

Mean $\bar{x} = (2.6, 2.8)$. Centered data rows: P1: $(-0.6, 4.2)$, P2: $(-1.6, -0.8)$, P3: $(0.4, -2.8)$, P4: $(-1.6, 1.2)$, P5: $(3.4, -1.8)$.

$$\text{Sample covariance } S = (1/(n-1)) \sum (x - \bar{x})(x - \bar{x})^T = [[4.3, -2.6], [-2.6, 7.7]].$$

$$\text{Eigenvalues: } \lambda_1 = 9.1064, \lambda_2 = 2.8936; \text{ unit PC1 direction } v_1 = (0.4758, -0.8796).$$

$$\text{Score } z_1 = v_1 \cdot (x - \bar{x}) = 0.4758 \cdot -0.6 + -0.8796 \cdot 4.2 = -3.9796.$$

The variance of the PC1 scores equals the top eigenvalue: **9.11**.

Q.19 — Answer: 2.108 to 2.148 [NAT, 2M · Regression & Model Selection > Simple linear regression]

$$\bar{x} = 6.667, \bar{y} = 9.667.$$

i	1	2	3	4	5	6
$x_i - \bar{x}$	-5.67	-2.67	-1.67	1.33	3.33	5.33
$y_i - \bar{y}$	-13.67	-2.67	-3.67	1.33	7.33	11.33
$(x_i - \bar{x})(y_i - \bar{y})$	77.44	7.11	6.11	1.78	24.44	60.44

$$S_{xx} = \sum (x_i - \bar{x})^2 = 83.333, S_{xy} = 177.333.$$

$$\beta_1 = S_{xy}/S_{xx} = 2.128, \beta_0 = \bar{y} - \beta_1 \bar{x} = -4.52.$$

Answer: **2.13**.

Q.20 — Answer: 0.6823 to 0.7023 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

a(i) = mean Manhattan distance from P3 to the other members of its own cluster C1: $d(P3, P1) = 1$, $d(P3, P2) = 3 \rightarrow a = 2$.

b(i) = smallest mean distance to the points of another cluster: C2: 6.5; C3: 7 $\rightarrow b = 6.5$.

$$s(i) = (b - a)/\max(a, b) = (6.5 - 2)/6.5 = \mathbf{0.69}.$$

s ranges in $[-1, 1]$; values near 1 mean well clustered, negative values suggest a wrong assignment.

Q.21 — Answer: (C) [MCQ, 2M · Regression & Model Selection > Cross-validation]

Minimum CV MSE = 4.72 at $\lambda = 0.3$, with SE = 0.44.

$$\text{One-SE threshold} = 4.72 + 0.44 = 5.16.$$

Models within the threshold: $\lambda = 0.1$ (4.94), $\lambda = 0.3$ (4.72), $\lambda = 1$ (4.9).

The one-SE rule picks the largest such λ (simplest model): $\lambda = 1$; the minimum-CV rule picks $\lambda = 0.3$.

Answer: $\lambda = 1$.

Q.22 — Answer: 0.7266 to 0.7466 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

$$\text{Parent } [9+, 4-]: H = -(9/13)\log_2(9/13) - (4/13)\log_2(4/13) = 0.8905.$$

$$\text{Income = Low: } [1+, 1-], H = -(1/2)\log_2(1/2) - (1/2)\log_2(1/2) = 1.$$

$$\text{Income = Medium: } [0+, 3-], H = 0 \text{ (pure node)} = 0.$$

$$\text{Income = High: } [8+, 0-], H = 0 \text{ (pure node)} = 0.$$

$$\text{Weighted child entropy} = (2/13)(1) + (3/13)(0) + (8/13)(0) = 0.1538.$$

$$IG = 0.8905 - 0.1538 = 0.7366.$$

Answer: **0.74**.

Q.23 — Answer: -0.3623 to -0.3583 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

All weights = -0.5, biases 0 $\Rightarrow$ both hidden units get $a = -0.5 \times (1.5 + -0.5) = -0.5$, $h_1 = h_2 = \sigma(a) = 0.3775$; $\hat{y} = 2 \times -0.5 \times 0.3775 = -0.3775$.

$$\hat{y} - y = -2.3775. \partial L / \partial v_j = (\hat{y} - y)h_j = -0.8976 \text{ (same for both } j).$$

$$\partial L / \partial w_{jk} = (\hat{y} - y) v_j h_j (1 - h_j) x_k = -2.3775 \times -0.5 \times 0.235 \times -0.5 = -0.1397 \text{ (identical for } j = 1, 2).$$

$$w_{22} \leftarrow -0.5 - 1 \times (-0.1397) = \mathbf{-0.3603}.$$

Because the two hidden units start identical and get identical updates, they remain identical forever (symmetry problem) - hence weights must be initialised randomly.

Q.24 — Answer: -0.5775 to -0.5661 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Activation functions]

Forward: $a_1 = 1 \cdot -0.5 + -0.5 = -1$, $h_1 = -0.7616$; $a_2 = 1 \cdot -0.7616 + 1 = 0.2384$, $h_2 = 0.234$; $a_3 = 3 \cdot 0.234 + -0.5 = 0.202$, $y = 0.1993$.

Local derivatives $\tanh'(a)$: $\tanh'(a) = 1 - \tanh^2(a)$: $\tanh'(a_1) = 0.42$, $\tanh'(a_2) = 0.9452$, $\tanh'(a_3) = 0.9603$.

$$\text{Chain rule: } \partial y / \partial w_1 = \tanh'(a_3) \cdot w_3 \cdot \tanh'(a_2) \cdot w_2 \cdot \tanh'(a_1) \cdot x = \mathbf{-0.5718}.$$

Each layer contributes a factor $\tanh' \leq 1$ - the product shrinks with depth (vanishing gradient).

Q.25 — Answer: 13.7277 to 13.7477 [NAT, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Three nearest neighbours: P3 ($d^2=5$, $y=13$), P4 ($d^2=10$, $y=18$), P2 ($d^2=17$, $y=9$).

Weights $1/d^2$: 0.2, 0.1, 0.0588; $\Sigma w = 0.3588$.

$\Sigma w_i y_i = 4.9294 \Rightarrow \hat{y} = 4.9294/0.3588 = \mathbf{13.74}$.

(The unweighted 3-NN average would be 13.33; weighting pulls $\hat{y}$ towards the closest point.)

Q.26 — Answer: 1.99 to 2.01 [NAT, 2M · SVM & Decision Trees > Perceptron (linear separators)]

Trace the epoch (a point with $y \cdot w^T \tilde{x} \leq 0$, including 0, triggers an update):

Step	$\tilde{x} = (1, x_1, x_2)$	y	$w^T \tilde{x}$	$y \cdot w^T \tilde{x} \leq 0$?	w after step
1	(1, -3, -1)	-1	-1	no	(-1, 0, 0)
2	(1, -1, -2)	-1	-1	no	(-1, 0, 0)
3	(1, -2, 2)	+1	-1	yes → update	(-0.5, -1, 1)
4	(1, 3, 2)	+1	-1.5	yes → update	(0, 0.5, 2)
5	(1, -3, -2)	-1	-5.5	no	(0, 0.5, 2)

Final $w = (0, 0.5, 2)$, number of updates = 2. Answer: **2**.

Q.27 — Answer: 8 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Single-linkage merge heights are exactly the edge weights of a minimum spanning tree (Kruskal's algorithm adds the shortest edge joining two different components = single-linkage merge).

MST edges (Prim's algorithm): P1-P6 (2); P1-P5 (3); P5-P4 (4); P4-P2 (3); P4-P3 (8).

Sorted merge heights: 2, 3, 3, 4, 8.

The last single-linkage merge happens at the largest MST edge: **8**.

Q.28 — Answer: 1 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$$\Sigma^{-1} = \begin{bmatrix} 0.667 & 0.333 \\ 0.333 & 0.667 \end{bmatrix}$$

$k = 1$: $\Sigma^{-1}\mu_1 = (0.3333, -0.3333)$; $\mathbf{x}^T \Sigma^{-1}\mu = 0.6667$; $\frac{1}{2}\mu^T \Sigma^{-1}\mu = 0.3333$; $\ln \pi = -0.5108 \Rightarrow \delta_1 = -0.1775$.

$k = 2$: $\Sigma^{-1}\mu_2 = (-1.3333, -1.6667)$; $\mathbf{x}^T \Sigma^{-1}\mu = 0.3333$; $\frac{1}{2}\mu^T \Sigma^{-1}\mu = 2.3333$; $\ln \pi = -1.6094 \Rightarrow \delta_2 = -3.6094$.

$k = 3$: $\Sigma^{-1}\mu_3 = (1, 0)$; $\mathbf{x}^T \Sigma^{-1}\mu = 1$; $\frac{1}{2}\mu^T \Sigma^{-1}\mu = 1$; $\ln \pi = -1.6094 \Rightarrow \delta_3 = -1.6094$.

The largest discriminant is $\delta_1 \Rightarrow$ predicted class **1**.

Q.29 — Answer: 0.577 to 0.583 [NAT, 2M · Regression & Model Selection > Bias-variance trade-off]

$E[c \bar{x}] = c\mu \Rightarrow \text{Bias} = (c - 1)\mu$; $\text{Var}(c \bar{x}) = c^2\sigma^2/n$. $\text{MSE}(c) = (c - 1)^2\mu^2 + c^2\sigma^2/n$.

Here $\mu = 1$, $\sigma^2/n = 16/16 = 1$.

$\text{MSE} = (-0.3)^2(1)^2 + (0.7)^2(1) = 0.09 + 0.49 = 0.58$.

Answer: **0.58**.

Q.30 — Answer: -0.15 to -0.146 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

$a = w \cdot x + b = -0.75$; $\hat{y} = \sigma(a) = 0.3208$.

$L = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial L / \partial \hat{y} = \hat{y} - y = -0.6792$; $\partial \hat{y} / \partial a = \hat{y}(1 - \hat{y}) = 0.2179$.

$\delta = \partial L / \partial a = (\hat{y} - y)\hat{y}(1 - \hat{y}) = -0.148 (= \partial L / \partial b)$.

Answer: **-0.148**.

Forgetting the factor $\hat{y}(1 - \hat{y})$ (i.e. treating the output as linear) is the most common error.

Q.31 — Answer: 0.9401 to 0.9441 [NAT, 2M · SVM & Decision Trees > Ensembles: bagging & random forests]

The vote is correct when at least 3 of 5 classifiers are correct (Binomial(5, 0.8)).

$P(\text{correct}) = C(5,3)(0.8)^3(0.2)^2 + C(5,4)(0.8)^4(0.2)^1 + C(5,5)(0.8)^5(0.2)^0 = 0.9421$.

Answer: **0.942**. Independence is crucial — correlated classifiers gain much less.

Q.32 — Answer: 4.3166 to 4.3366 [NAT, 2M · Probabilistic & Instance-based Classifiers › Logistic regression]

$$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b} = (-1, 3, 2).$$

$$e^{\mathbf{z}} = (0.3679, 20.0855, 7.3891), \text{ sum} = 27.8425.$$

$$\text{Softmax probabilities} = (0.0132, 0.7214, 0.2654).$$

$$\text{Loss} = -\ln(0.0132) = \mathbf{4.33}.$$

Q.33 — Answer: 0.8682 to 0.8722 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Softmax + cross-entropy]

$$\mathbf{p} = \text{softmax}(\mathbf{z}): \exp(\mathbf{z}) = (0.2231, 7.3891, 4.4817, 1), \text{ sum} = 13.0939, \text{ so } \mathbf{p} = (0.017, 0.5643, 0.3423, 0.0764).$$

$$\text{For } L = -\ln p_y \text{ with softmax, } \partial L / \partial \mathbf{z} = \mathbf{p} - \mathbf{y} \text{ (y one-hot, true class 3): } \partial L / \partial \mathbf{z} = (0.017, 0.5643, -0.6577, 0.0764).$$

$$\|\mathbf{p} - \mathbf{y}\| = \sqrt{(0.0003 + 0.3185 + 0.4326 + 0.0058)} = \mathbf{0.8702}.$$

$$\text{Note } \sum_k \partial L / \partial z_k = \sum p_k - 1 = 0.$$

Q.34 — Answer: 29 [NAT, 2M · Unsupervised Learning › k-medoids]

PAM cost = Σ over all points of the absolute distance to the nearest medoid.

Current medoids {12, 30}: distances 4→8, 5→7, 11→1, 12→0, 16→4, 24→6, 27→3, 30→0; cost = **29**.

Q.35 — Answer: 4.3665 to 4.4192 [NAT, 2M · Regression & Model Selection › Basis functions & feature scaling]

Design columns 1, x, x^2 . Because the x-values are symmetric about 0, $\Sigma x = \Sigma x^3 = 0$, so the normal equations decouple:

$$\Sigma x^2 = 28, \Sigma x^4 = 196, \Sigma y = 49, \Sigma xy = -56, \Sigma x^2 y = 232.$$

$$b = \Sigma xy / \Sigma x^2 = -56/28 = -2.$$

$$[n, \Sigma x^2; \Sigma x^2, \Sigma x^4][a; c] = [\Sigma y; \Sigma x^2 y] = 7a + 28c = 49, 28a + 196c = 232.$$

$$\text{Solving: } c = (n\Sigma x^2 y - \Sigma x^2 \Sigma y) / (n\Sigma x^4 - (\Sigma x^2)^2) = 0.4286, a = (\Sigma y - c\Sigma x^2) / n = 5.2857.$$

$$\hat{y}(0.5) = 5.2857 + (-2)(0.5) + (0.4286)(0.5)^2 = 4.3929.$$

Answer: **4.39**.

Mock Test 41

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	5	11	19

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Neural Networks (MLP / Feed-forward)	7	11
Probabilistic & Instance-based Classifiers	7	11
Regression & Model Selection	7	11
SVM & Decision Trees	7	11
Unsupervised Learning	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

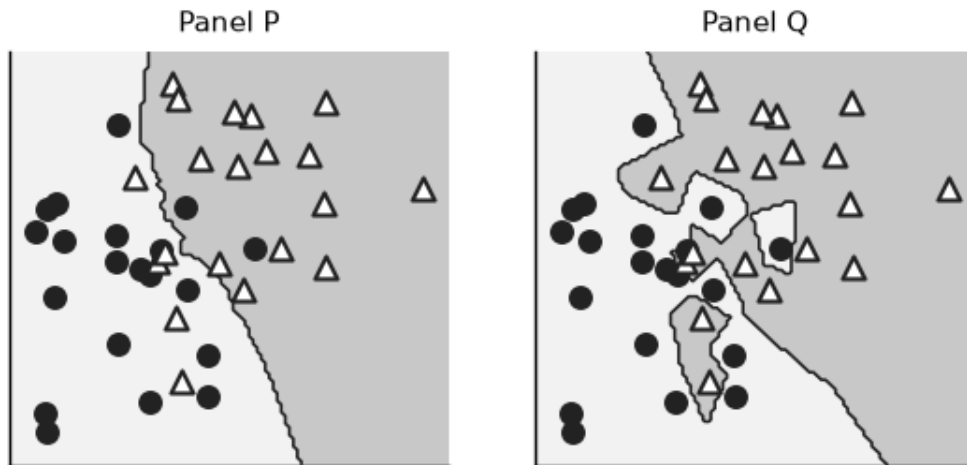
A weight $w = 2$ is trained on the regularised objective $L(w) + (\lambda/2)w^2$ with $\lambda = 0.5$. The data-loss gradient is $\partial L/\partial w = 0$. After one gradient-descent step with $\eta = 0.2$, the new weight is _____ (rounded off to three decimal places)

Q.2 [1 mark · NAT (numerical answer)]

Fisher's multi-class LDA is applied to data with $K = 5$ classes and $p = 7$ features. The maximum number of discriminant directions (non-zero generalised eigenvalues of $S_W^{-1}S_B$) is _____ (Answer in integer)

Q.3 [1 mark · MCQ]

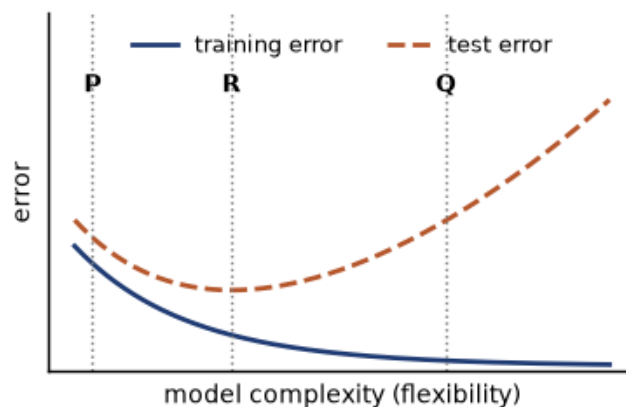
The two panels show the decision regions of k-NN classifiers trained on the same 2-D data set, one with $k = 1$ and the other with $k = 21$. Which of the following statements is CORRECT?



- (A) Panel P uses $k = 1$; it has higher bias and lower variance
- (B) Panel P uses $k = 21$; it has zero training error
- (C) Panel P uses $k = 1$; it has lower bias and higher variance
- (D) Panel Q uses $k = 1$; it has lower bias and higher variance

Q.4 [1 mark · MCQ]

The figure shows training and test error as a function of model complexity, with three complexity levels marked P, Q and R. Which marked level corresponds to overfitting (low bias, high variance)?



- (A) None of P, Q, R
- (B) R
- (C) Q
- (D) P

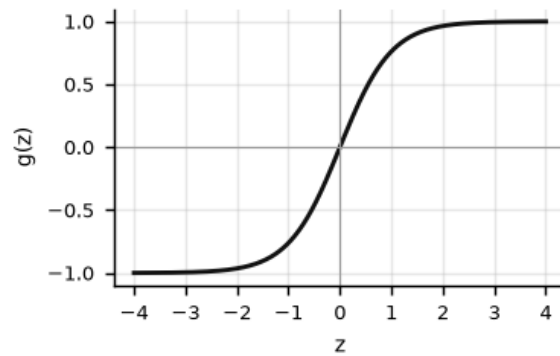
Q.5 [1 mark · MSQ (one or more correct)]

Consider the soft-margin SVM: minimise $\frac{1}{2}\|w\|^2 + C \sum_i \xi_i$ subject to $y_i(w^T x_i + b) \geq 1 - \xi_i$, $\xi_i \geq 0$. Which of the following statements is/are CORRECT?

- (A) In the soft-margin dual the multipliers satisfy the box constraint $0 \leq \alpha_i \leq C$.
- (B) Increasing C increases the regularisation strength.
- (C) A point lying exactly on the margin ($y f(x) = 1$) incurs a positive hinge loss.
- (D) The soft-margin SVM is infeasible if the data are not linearly separable.

Q.6 [1 mark · MCQ]

The figure shows the graph of a function $g(z)$ used in neural networks. Which function is it?



- (A) ELU with $\alpha = 1$
- (B) Sigmoid $\sigma(z) = 1/(1+e^{-z})$
- (C) $\tanh(z)$
- (D) $1 - \tanh^2(z)$

Q.7 [1 mark · MSQ (one or more correct)]

Which of the following statements about distance / similarity measures is/are CORRECT?

- (A) Minkowski 'distance' with $0 < p < 1$ violates the triangle inequality.
- (B) Jaccard similarity counts 0-0 matches as agreements.
- (C) Manhattan distance is invariant to rotations of the coordinate axes.
- (D) Cosine distance $1 - \cos \theta$ does not satisfy the triangle inequality in general.

Q.8 [1 mark · MSQ (one or more correct)]

Which of the following statements about basis-function (polynomial) regression and feature scaling is/are CORRECT?

- (A) Multiplying a feature by 10 changes the OLS predictions (model with intercept).
- (B) Ridge regression solutions are invariant to rescaling individual features.
- (C) Polynomial regression $y = w_0 + w_1x + w_2x^2$ is linear in the parameters, so it can be fitted by ordinary least squares.
- (D) Test data should be standardised with the test set's own mean and standard deviation.

Q.9 [1 mark · MSQ (one or more correct)]

Let X be an $n \times p$ design matrix of full column rank p ($p < n$), and let $H = X(X^T X)^{-1} X^T$ be the hat matrix of least-squares regression of y on X . Which of the following statements is/are CORRECT?

- (A) $H^2 = 2H$.
- (B) The vector of fitted values is $\hat{y} = Hy$.
- (C) H is an orthogonal matrix ($H^T H = I$).
- (D) Each diagonal element (leverage) satisfies $0 \leq h_{ii} \leq 1$.

Q.10 [1 mark · MSQ (one or more correct)]

Which of the following statements about the representational power of feed-forward neural networks is/are CORRECT?

- (A) A network with a single hidden layer of two ReLU units can approximate any continuous function on $[0, 1]$ to arbitrary accuracy.
- (B) The universal approximation theorem also holds with ReLU activations.
- (C) A ReLU network with finitely many units can represent a discontinuous step function exactly.
- (D) The universal approximation theorem fails for sigmoid activations; it holds only for ReLU.

Q.11 [1 mark · MSQ (one or more correct)]

Which of the following statements about k-means clustering is/are CORRECT?

- (A) With Euclidean distance, the boundary between two k-means clusters is the perpendicular bisector (hyperplane) of their centroids.
- (B) k-means works best when clusters are roughly spherical and of similar size.
- (C) Each k-means cluster region (Voronoi cell) is convex.
- (D) The elbow method plots the within-cluster SSE against k and picks the k where the decrease levels off.

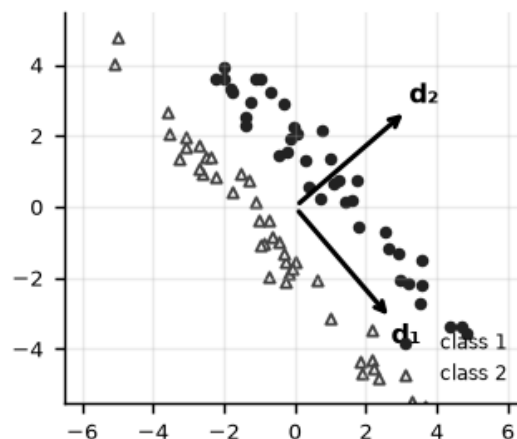
Q.12 [1 mark · NAT (numerical answer)]

A hard-margin SVM has exactly three support vectors with the optimal dual variables shown. The component w_1 of the optimal weight vector is _____ (rounded off to two decimal places)

i	x_i	y_i	α_i
1	(0, -2)	-1	1
2	(-1, 2)	+1	1.5
3	(2, -2)	-1	0.5

Q.13 [1 mark · MCQ]

Two labelled classes are shown below, together with two orthogonal directions d_1 and d_2 . A single direction is to be selected by (i) PCA (first principal component, ignoring labels) and (ii) Fisher's LDA. Which statement is correct?



- (A) PCA (1st PC) chooses d_2 ; LDA chooses d_1
- (B) PCA (1st PC) chooses d_1 ; LDA chooses d_2
- (C) Both PCA and LDA choose d_2
- (D) Both PCA and LDA choose d_1

Q.14 [1 mark · NAT (numerical answer)]

A 4-class softmax classifier produces the logits $z_1 = -1$, $z_2 = 1$, $z_3 = -1$, $z_4 = 1$ for an input $\mathbf{x}$. The predicted probability of class 2, $P(y = 2 | \mathbf{x})$, is _____ (rounded off to two decimal places)

Q.15 [1 mark · NAT (numerical answer)]

The maximum number of leaf nodes in a binary decision tree of depth 5 (root at depth 0) is _____ (Answer in integer)

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

A 2-2-1 network uses the identity (linear) activation in its hidden layer: $\hat{y} = W_2(W_1x + b_1) + b_2$, with the parameters below. The network is equivalent to a single linear model $\hat{y} = \tilde{w}_1x_1 + \tilde{w}_2x_2 + \tilde{b}$. Then the intercept $\tilde{b}$ is _____ (Answer in integer)

$$W_1 = \begin{bmatrix} -2 & 0 \\ -1 & 2 \end{bmatrix}$$

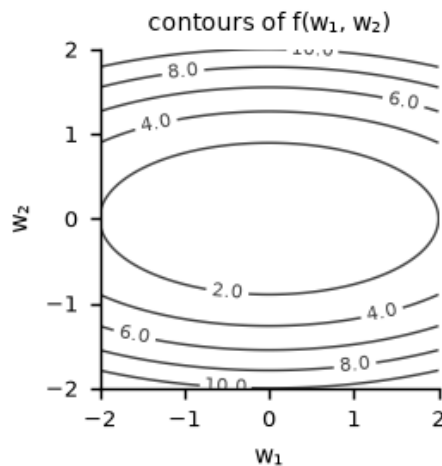
$$b_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 1 & -2 \end{bmatrix}$$

$$b_2 = 2$$

Q.17 [2 marks · MSQ (one or more correct)]

Gradient descent $w \leftarrow w - \eta \nabla f(w)$ is applied to $f(w_1, w_2) = \frac{1}{2}(1w_1^2 + 5w_2^2)$ from a generic starting point (both coordinates non-zero). For which of the following learning rates does GD **diverge**?



(A) $\eta = 0.52$

(B) $\eta = 0.38$

(C) $\eta = 0.1$

(D) $\eta = 0.06$

Q.18 [2 marks · MSQ (one or more correct)]

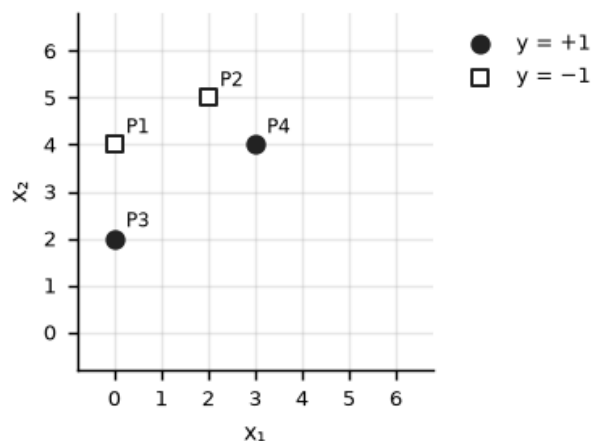
Which of the following statements about PCA, SVD and dimensionality reduction is/are CORRECT?

- (A) Kernel PCA can capture nonlinear structure by performing PCA in a feature space.
- (B) To capture a fraction r of uniform data in a d -dimensional unit hypercube, a sub-cube needs edge length $r^{1/d}$, which approaches 1 as d grows.
- (C) For a covariance matrix $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$ with $b > 0$, PC1 is along $(1, 1)/\sqrt{2}$.
- (D) Fisher LDA for C classes yields at most $C - 1$ discriminant directions.

Q.19 [2 marks · NAT (numerical answer)]

A hard-margin linear SVM is trained on the points below. Solving the dual problem gives the Lagrange multipliers α_i listed in the table. Using the KKT conditions, compute the weight component w_1 . (rounded off to two decimal places)

Point	x_1	x_2	y	α_i
P1	0	4	-1	0
P2	2	5	-1	1.04
P3	0	2	+1	0.08
P4	3	4	+1	0.96

**Q.20 [2 marks · NAT (numerical answer)]**

A logistic regression classifier has $P(y = 1 \mid \mathbf{x}) = \sigma(-1 + (1)x_1 + (3)x_2)$. An input is labelled $y = 1$ if and only if $P(y = 1 \mid \mathbf{x}) \geq 0.8$. For this rule, the perpendicular distance of the decision boundary from the origin is _____ (rounded off to two decimal places)

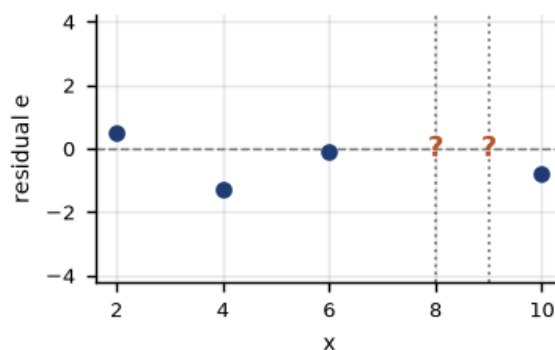
Q.21 [2 marks · NAT (numerical answer)]

The 3×3 matrix $X^T X$ of a (centred) design matrix X has eigenvalues 25, 20, 5. For ridge regression with penalty $\lambda \|\beta\|_2^2$, $df(5) - df(20)$ is _____ (rounded off to two decimal places)

Q.22 [2 marks · NAT (numerical answer)]

A simple linear regression with intercept is fitted by least squares. The table gives the x -values and the residuals $e_i = y_i - \hat{y}_i$, but two residuals are missing. The missing residual at $x = 8$ is _____ (rounded off to two decimal places)

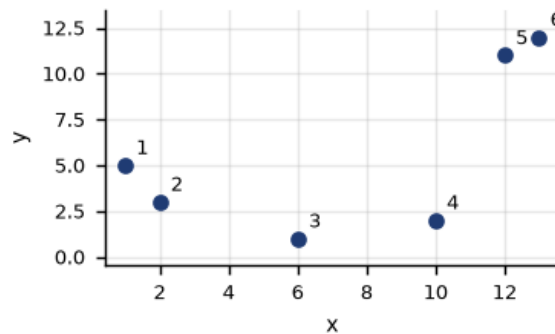
x_i	2	4	6	8	9	10
e_i	0.5	-1.3	-0.1	?	?	-0.8



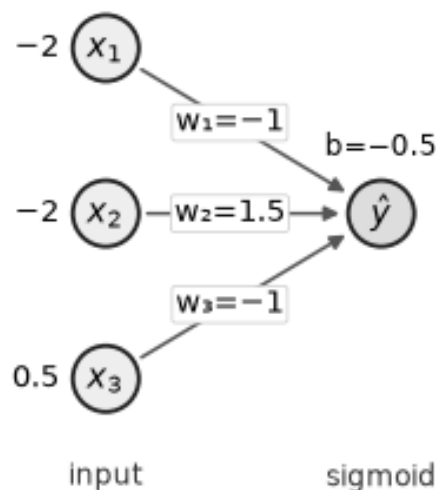
Q.23 [2 marks · NAT (numerical answer)]

Consider the 6 training points (x_i, y_i) shown below. Two-fold cross-validation is used with fold 1 = {points 1, 3, 5, ...} and fold 2 = {points 2, 4, ...} (points numbered in increasing x). The model predicts the mean of the training-fold targets. The pooled 2-fold CV MSE (over all n points) is _____ (rounded off to two decimal places)

x	1	2	6	10	12	13
y	5	3	1	2	11	12

**Q.24 [2 marks · NAT (numerical answer)]**

A single sigmoid neuron (shown below) receives $x = (-2, -2, 0.5)$ and has target $y = 1$. The loss is $L = \frac{1}{2}(\hat{y} - y)^2$. The value of $\partial L / \partial w_1$ is _____ (rounded off to three decimal places)

**Q.25 [2 marks · NAT (numerical answer)]**

Three customers are described by age and income. Age ranges over [20, 60] years and income over [10, 210] thousand rupees in the full dataset. On the **raw** (unscaled) features, the Euclidean distance from A to its nearest neighbour among B and C is _____ (rounded off to two decimal places)

Customer	Age (years)	Income (₹ thousand)
A	54	110
B	42	195
C	25	155

Q.26 [2 marks • NAT (numerical answer)]

In C4.5, the attribute Region splits a node of 12 records as tabulated (class Yes/No counts in each branch). Compute the gain ratio of Region (all logarithms base 2). (rounded off to two decimal places)

Region	#Yes	#No
v1	4	1
v2	1	6

Q.27 [2 marks • MCQ]

A network computes $\hat{y} = W_2 \text{ReLU}(W_1 x + b_1) + b_2$ with the parameters below. For $x = (-2, 1, 2)^T$, the output $\hat{y}$ is

$$W_1 = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & -2 \\ -1 & -1 & 1 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 0 & -1 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

(A) $\hat{y} = (7, -3)^T$

(B) $\hat{y} = (1, -2)^T$

(C) $\hat{y} = (5, -2)^T$

(D) $\hat{y} = (6, -3)^T$

Q.28 [2 marks • MSQ (one or more correct)]

Each option shows a 2×2 matrix $[[K(x_1, x_1), K(x_1, x_2)], [K(x_2, x_1), K(x_2, x_2)]]$. Which of them can be the Gram matrix of some valid (Mercer) kernel $\tilde{K}$ on two points x_1, x_2 ?

(A) $[[6, 0], [0, 7]]$

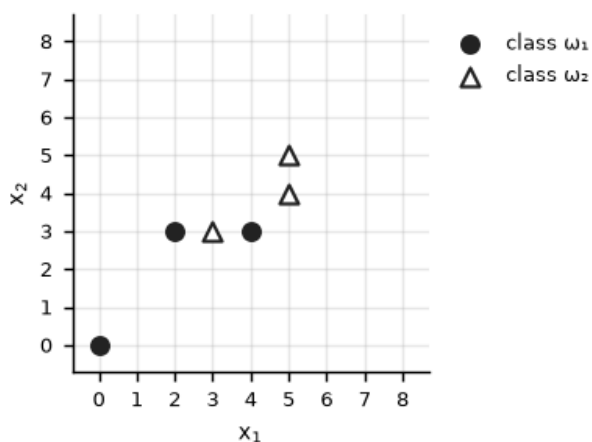
(B) $[[-1, 0], [0, 7]]$

(C) $[[7, 0], [1, 11]]$

(D) $[[3, -2], [-2, 5]]$

Q.29 [2 marks • NAT (numerical answer)]

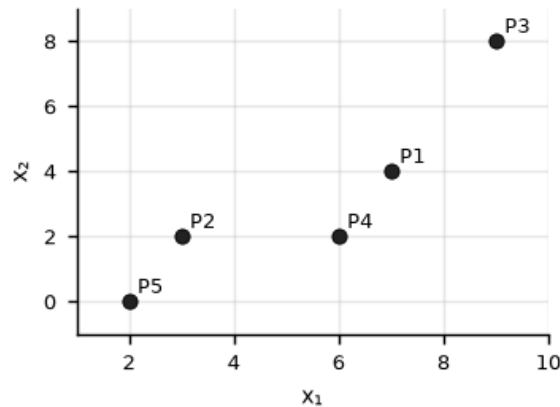
Class ω_1 has training points $(0, 0)$, $(2, 3)$, $(4, 3)$ and class ω_2 has points $(5, 4)$, $(3, 3)$, $(5, 5)$. Let $\mathbf{m}_1, \mathbf{m}_2$ be the class means and $S_W = \sum_k \sum_{x \in \omega_k} (x - \mathbf{m}_k)(x - \mathbf{m}_k)^T$ the within-class scatter matrix (no division by n). Then $(\mathbf{m}_2 - \mathbf{m}_1)^T S_W^{-1} (\mathbf{m}_2 - \mathbf{m}_1)$ is _____ (rounded off to two decimal places)



Q.30 [2 marks · NAT (numerical answer)]

A cluster contains the 5 points below. Distances are Euclidean (L_2). The medoid is the point that minimizes the sum of distances to all points of the cluster. The minimum total distance (cost of the medoid) is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5
x_1	7	3	9	6	2
x_2	4	2	8	2	0

**Q.31 [2 marks · NAT (numerical answer)]**

Least-squares linear regression on $n = 10$ observations minimises $J(w) = (1/n)\|Xw - y\|^2$ by batch gradient descent with constant learning rate η . The data give the matrix $X^T X$ below. Then the constant learning rate η that minimises the worst-case contraction factor $\max_i |1 - \eta \lambda_i|$ is _____ (rounded off to three decimal places)

$$X^T X = \begin{bmatrix} 12 & 4 \\ 4 & 12 \end{bmatrix}$$

Q.32 [2 marks · MSQ (one or more correct)]

Consider k-means (Lloyd's algorithm) and k-medoids (PAM). Which of the following statements is/are CORRECT?

- (A) For 1-D data and a single cluster, the k-medoids representative under absolute distance is a median data point.
- (B) k-medoids (PAM) can be run using only a pairwise dissimilarity matrix, without coordinates.
- (C) The medoid of a cluster is the member that minimizes the sum of dissimilarities to the other members.
- (D) The number of ways to split n points into 2 non-empty clusters is $n(n-1)/2$.

Q.33 [2 marks · NAT (numerical answer)]

A classifier for a rare condition has sensitivity (recall) 0.9 and specificity 0.99. It is deployed on a population where the prevalence of the positive class is 0.05. The expected accuracy in this population is _____ (rounded off to three decimal places)

Q.34 [2 marks · NAT (numerical answer)]

The predicted probabilities $\hat{p}_i = P(y_i = 1 | x_i)$ of a binary classifier on 4 examples, and their true labels, are given below. Using natural logarithms, the **average** binary cross-entropy loss $-[y \ln \hat{p} + (1 - y) \ln(1 - \hat{p})]$ over these examples is _____ (rounded off to two decimal places)

i	1	2	3	4
y_i	1	0	1	1
$\hat{p}_i = P(y_i=1)$	0.25	0.5	0.8	0.15

Q.35 [2 marks · MSQ (one or more correct)]

A node with class counts [50+, 60–] can be split by either of two binary splits S1 or S2 (table). For each impurity measure, the “gain” is the parent impurity minus the weighted child impurity. Which of the following statements is/are CORRECT?

Split	Left child [+ , –]	Right child [+ , –]
S1	[46+, 39–]	[4+, 21–]
S2	[9+, 28–]	[41+, 32–]

- (A) Misclassification-error reduction prefers S2 over S1.
 (B) Information gain (entropy) prefers S2 over S1.
 (C) Information gain (entropy) prefers S1 over S2.
 (D) Gini gain prefers S2 over S1.

— END OF QUESTION PAPER —

Mock Test 41 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	1.798 to 1.802	19	NAT	2	0.79 to 0.81
2	NAT	1	4	20	NAT	2	0.7446 to 0.7646
3	MCQ	1	(D)	21	NAT	2	0.8678 to 0.8878
4	MCQ	1	(C)	22	NAT	2	2.4875 to 2.5125
5	MSQ	1	(A)	23	NAT	2	18.4628 to 18.6483
6	MCQ	1	(C)	24	NAT	2	0.183 to 0.187
7	MSQ	1	(A), (D)	25	NAT	2	53.525 to 53.545
8	MSQ	1	(C)	26	NAT	2	0.3308 to 0.3508
9	MSQ	1	(B), (D)	27	MCQ	2	(A)
10	MSQ	1	(B)	28	MSQ	2	(A), (D)
11	MSQ	1	(A), (B), (C), (D)	29	NAT	2	0.5317 to 0.5517
12	NAT	1	–2.51 to –2.49	30	NAT	2	16.4064 to 16.4264
13	MCQ	1	(B)	31	NAT	2	0.4147 to 0.4187
14	NAT	1	0.4304 to 0.4504	32	MSQ	2	(A), (B), (C)
15	NAT	1	32	33	NAT	2	0.9835 to 0.9875
16	NAT	2	–3	34	NAT	2	1.0399 to 1.0599
17	MSQ	2	(A)	35	MSQ	2	(A), (C)
18	MSQ	2	(A), (B), (C), (D)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Neural Networks (MLP / Feed-forward)					11
Probabilistic & Instance-based Classifiers					11
Regression & Model Selection					11
SVM & Decision Trees					11
Unsupervised Learning					11
Total					55

Mock Test 41 — Detailed Solutions

Q.1 — Answer: 1.798 to 1.802 [NAT, 1M · Neural Networks (MLP / Feed-forward) › Training & regularisation]

$$\nabla = \partial L / \partial w + \lambda w = 0 + 0.5 \times 2 = 1.$$

$$w \leftarrow 2 - 0.2 \times 1 = \mathbf{1.8} \text{ (equivalently } (1 - \eta \lambda)w - \eta \partial L / \partial w \text{).}$$

Q.2 — Answer: 4 [NAT, 1M · Probabilistic & Instance-based Classifiers › Linear discriminant analysis]

S_B is a sum of K outer products of $(\mu_k - \mu)$ which are linearly dependent (weighted sum is 0), so $\text{rank}(S_B) \leq K - 1 = 4$; also $\leq p = 7$.

$$\text{Maximum} = \min(K - 1, p) = \mathbf{4}.$$

Q.3 — Answer: (D) [MCQ, 1M · Probabilistic & Instance-based Classifiers › k-nearest neighbours]

The jagged boundary with small islands around individual points (Panel Q) is produced by $k = 1$, which fits every training point (zero training error): low bias, high variance.

The smooth boundary (Panel P) comes from $k = 21$: averaging many neighbours raises bias and lowers variance; it misclassifies some training points, so its training error is not zero.

Answer: **Panel Q uses $k = 1$; it has lower bias and higher variance.**

Q.4 — Answer: (C) [MCQ, 1M · Regression & Model Selection › Bias-variance trade-off]

At P: both training and test errors are high (left side) → underfitting.

At R: the test error is at its minimum → best trade-off.

At Q: training error is very low but test error has risen well above its minimum (right side) → overfitting.

Answer: **Q.**

Q.5 — Answer: (A) [MSQ, 1M · SVM & Decision Trees › SVM: soft margin & hinge loss]

(A) TRUE. C caps the influence of any single point.

(B) FALSE. Larger C means weaker regularisation.

(C) FALSE. Hinge loss is 0 there.

(D) FALSE. Slack makes it always feasible.

Q.6 — Answer: (C) [MCQ, 1M · Neural Networks (MLP / Feed-forward) › Activation functions]

Key features of the plotted curve: S-shaped, range $(-1, 1)$, passes through the origin.

Hence $g = \mathbf{\tanh(z)}$.

ELU with $\alpha = 1$: Smoothly saturates to -1 for $z \rightarrow -\infty$, linear for $z > 0$, passes through origin. – does not match.

Sigmoid $\sigma(z) = 1/(1+e^{-z})$: S-shaped, range $(0, 1)$, value 0.5 at $z = 0$. – does not match.

$1 - \tanh^2(z)$: Bell-shaped with peak 1 at $z = 0$. – does not match.

Q.7 — Answer: (A), (D) [MSQ, 1M · Unsupervised Learning › Distance measures]

(A) TRUE. e.g. $p = 0.5$: $d((0,0),(1,1)) = 4$, which exceeds $d((0,0),(1,0)) + d((1,0),(1,1)) = 2$.

(B) FALSE. That is the simple matching coefficient.

(C) FALSE. Only L_2 is rotation-invariant.

(D) TRUE. So it is not a true metric.

Q.8 — Answer: (C) [MSQ, 1M · Regression & Model Selection › Basis functions & feature scaling]

(A) FALSE. The coefficient simply becomes one-tenth; predictions are unchanged.

(B) FALSE. They are not.

(C) TRUE. Treat 1, x , x^2 as columns of the design matrix.

(D) FALSE. Use the training statistics.

Q.9 — Answer: (B), (D) [MSQ, 1M · Regression & Model Selection › Multiple linear regression]

(A) FALSE. H is idempotent: $H^2 = H$.

(B) TRUE. $\hat{y} = X\beta = X(X^T X)^{-1} X^T y$.

(C) FALSE. $H^T H = H \neq I$ since H is singular.

(D) TRUE. $h_{ii} = \sum_j h_{ij}^2 \geq h_{ii}^2$ gives $0 \leq h_{ii} \leq 1$.

Q.10 — Answer: (B) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Expressiveness]

(A) FALSE. Width must grow with the required accuracy.

(B) TRUE. ReLU is non-polynomial, so the Leshno et al. theorem applies.

(C) FALSE. ReLU networks compute continuous functions.

(D) FALSE. The original result (Cybenko) was proved for sigmoids.

Q.11 — Answer: (A), (B), (C), (D) [MSQ, 1M · Unsupervised Learning > k-means]

(A) TRUE. Points equidistant from two centroids form that hyperplane.

(B) TRUE. Its objective implicitly assumes isotropic, equal-variance clusters.

(C) TRUE. A Voronoi cell is an intersection of half-spaces.

(D) TRUE. That bend is the 'elbow'.

Q.12 — Answer: -2.51 to -2.49 [NAT, 1M · SVM & Decision Trees > SVM: dual & support vectors]

$w = \sum_i \alpha_i y_i x_i$ (sum over support vectors).

$w_1 = (1)(-1)(0) + (1.5)(+1)(-1) + (0.5)(-1)(2) = -2.5$.

(Full vector $w = (-2.5, 6)$; note $\sum \alpha_i y_i = 0$ holds.)

Q.13 — Answer: (B) [MCQ, 1M · Unsupervised Learning > PCA & dimensionality reduction]

PCA is unsupervised: it picks the direction of maximum total variance, which is along the long axis of the clouds (d_1).

LDA is supervised: it maximizes between-class separation relative to within-class scatter, so it picks the direction across which the class means differ (d_2).

Answer: **PCA (1st PC) chooses d_1 ; LDA chooses d_2** . Projecting on the PCA direction would mix the two classes completely.

Q.14 — Answer: 0.4304 to 0.4504 [NAT, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

softmax: $P(y = k | \mathbf{x}) = e^{z_k} / \sum_j e^{z_j}$.

e^z values: 0.3679, 2.7183, 0.3679, 2.7183; sum = 6.1723.

$P(y = 2 | \mathbf{x}) = 2.7183/6.1723 = \mathbf{0.44}$.

Q.15 — Answer: 32 [NAT, 1M · SVM & Decision Trees > Decision trees: structure & traversal]

Each level at most doubles the nodes: leaves $\leq 2^d = 2^5 = \mathbf{32}$.

Q.16 — Answer: -3 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Expressiveness]

With identity activations, $\hat{y} = W_2(W_1x + b_1) + b_2 = (W_2W_1)x + (W_2b_1 + b_2)$: a single affine map.

$\tilde{w}^T = W_2W_1 = (0, -4)$; $\tilde{b} = W_2b_1 + b_2 = -3$.

Answer: **-3**.

Hence stacking linear layers adds no expressive power – a non-linear activation is essential.

Q.17 — Answer: (A) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$\nabla f = (1w_1, 5w_2) = w_i \leftarrow (1 - \eta a_i) w_i$. GD converges (from every start) iff $|1 - \eta a_i| < 1$ for both i , i.e. $0 < \eta < 2/\max(a_i) = 2/5 = 0.4$. The steep direction ($a = 5$) limits the learning rate.

(A) TRUE. $\eta = 0.52$: $|1 - \eta \cdot 1| = 0.48$, $|1 - \eta \cdot 5| = 1.6 \rightarrow$ diverges.

(B) FALSE. $\eta = 0.38$: $|1 - \eta \cdot 1| = 0.62$, $|1 - \eta \cdot 5| = 0.9 \rightarrow$ converges.

(C) FALSE. $\eta = 0.1$: $|1 - \eta \cdot 1| = 0.9$, $|1 - \eta \cdot 5| = 0.5 \rightarrow$ converges.

(D) FALSE. $\eta = 0.06$: $|1 - \eta \cdot 1| = 0.94$, $|1 - \eta \cdot 5| = 0.7 \rightarrow$ converges.

Q.18 — Answer: (A), (B), (C), (D) [MSQ, 2M · Unsupervised Learning > PCA & dimensionality reduction]

(A) TRUE. Via the kernel trick.

(B) TRUE. Neighbourhoods stop being local.

(C) TRUE. Eigenvalue $a + b$.

(D) TRUE. The between-class scatter has rank $\leq C - 1$.

Q.19 — Answer: 0.79 to 0.81 [NAT, 2M · SVM & Decision Trees > SVM: dual & support vectors]

$w = \sum \alpha_i y_i x_i$ over points with $\alpha_i > 0$:

$$w = (1.04)(-1)(2, 5) + (0.08)(+1)(0, 2) + (0.96)(+1)(3, 4) = (4/5, -6/5).$$

b from any support vector, e.g. P2: $y = w^T x + b \Rightarrow b = -1 - (-4.4) = 17/5$.

Check: $\sum \alpha_i y_i = 0$ ($= 0$, as required).

Answer: **0.8**.

Q.20 — Answer: 0.7446 to 0.7646 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

$P(y=1|x) \geq 0.8 \Leftrightarrow z \geq c$, where $c = \ln(0.8/0.2) = 1.3863$.

Decision boundary: $-1 + (1)x_1 + (3)x_2 = 1.3863$ (a straight line; changing the threshold only shifts it parallel to the 0.5 boundary).

$$\text{distance} = |w_0 - c| / \|(w_1, w_2)\| = |-1 - 1.3863| / \sqrt{10} = \mathbf{0.75}.$$

Q.21 — Answer: 0.8678 to 0.8878 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

Ridge fitted values $\hat{y} = H_\lambda y$ with $H_\lambda = X(X^T X + \lambda I)^{-1} X^T$. Writing $X^T X = V D V^T$ with eigenvalues d_j^2 gives $\text{trace}(H_\lambda) = \sum d_j^2 / (d_j^2 + \lambda)$ (effective degrees of freedom).

$$25/(25 + 5) + 20/(20 + 5) + 5/(5 + 5) = 2.1333.$$

$$25/(25 + 20) + 20/(20 + 20) + 5/(5 + 20) = 1.2556.$$

$$\text{Difference} = 2.1333 - 1.2556 = 0.8778 \text{ (df decreases as } \lambda \text{ grows)}.$$

Answer: **0.88**.

Q.22 — Answer: 2.4875 to 2.5125 [NAT, 2M · Regression & Model Selection > Simple linear regression]

For the least-squares fit with intercept the normal equations give $\sum e_i = 0$ and $\sum x_i e_i = 0$.

Known: $\sum e = -1.7$, $\sum x e = -12.8$. With unknowns $u = e$ at $x = 8$ and $v = e$ at $x = 9$:

$$u + v = 1.7, 8u + 9v = 12.8.$$

Solving: $u = 2.5$, $v = -0.8$.

Answer: **2.5**.

Q.23 — Answer: 18.4628 to 18.6483 [NAT, 2M · Regression & Model Selection > Cross-validation]

Fold 1 validated with mean of fold-2 targets = 5.6667; fold 2 validated with mean of fold-1 targets = 5.6667.

$$\sum \text{squared errors} = 111.3333; \text{pooled MSE} = 111.3333/6 = 18.5556.$$

Answer: **18.56**.

Q.24 — Answer: 0.183 to 0.187 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

$$a = w \cdot x + b = -2; \hat{y} = \sigma(a) = 0.1192.$$

$$L = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial L / \partial \hat{y} = \hat{y} - y = -0.8808; \partial \hat{y} / \partial a = \hat{y}(1 - \hat{y}) = 0.105.$$

$$\delta = \partial L / \partial a = (\hat{y} - y)\hat{y}(1 - \hat{y}) = -0.0925 (= \partial L / \partial b).$$

$$\partial L / \partial w_1 = \delta \cdot x_1 = -0.0925 \times -2 = 0.185.$$

Answer: **0.185**.

Forgetting the factor $\hat{y}(1 - \hat{y})$ (i.e. treating the output as linear) is the most common error.

Q.25 — Answer: 53.525 to 53.545 [NAT, 2M · Unsupervised Learning > Distance measures]

Min-max scaling: $x' = (x - \min) / (\max - \min)$; age range [20, 60], income range [10, 210].

Customer	age'	income'
A	0.85	0.5
B	0.55	0.925
C	0.125	0.725

Raw distances: $d(A, B) = 85.84$, $d(A, C) = 53.54 \rightarrow$ nearest is C (income dominates because of its larger range).

Scaled distances: $d(A, B) = 0.5202$, $d(A, C) = 0.7591 \rightarrow$ nearest is B.

Answer: **53.54**. Scaling changes which neighbour is nearest — distance-based methods (k-means, k-NN, hierarchical clustering) should be run on standardized features.

Q.26 — Answer: 0.3308 to 0.3508 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [5 Yes, 7 No]: $H = 0.9799$.

Child entropies: $v_1: 0.7219$, $v_2: 0.5917$.

$IG = 0.9799 - \sum (n_v/12)H_v = 0.3339$.

$SplitInfo = -\sum (n_v/N) \log_2(n_v/N) = -(5/12)\log_2(5/12) - (7/12)\log_2(7/12) = 0.9799$.

Gain ratio = $IG / SplitInfo = 0.3339/0.9799 = \mathbf{0.34}$.

C4.5 uses the gain ratio to penalise attributes that split the data into many small branches.

Q.27 — Answer: (A) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

$a = W_1x + b_1 = (-1, -5, 3)^T$; $h = \text{ReLU}(a) = (0, 0, 3)^T$.

$\hat{y} = W_2h + b_2 = \mathbf{(7, -3)^s}$.

Distractors: skipping the ReLU gives $(1, -2)^T$; using W_1^T instead of W_1 gives $(5, -2)^T$; dropping the biases gives $(6, -3)^T$; using $|a|$ instead of ReLU gives $(13, -4)^T$.

Q.28 — Answer: (A), (D) [MSQ, 2M · SVM & Decision Trees > SVM: kernels]

A Gram matrix of a valid kernel must be symmetric and positive semidefinite. For a symmetric 2×2 matrix this is equivalent to: both diagonal entries ≥ 0 and determinant ≥ 0 .

(A) **TRUE**. symmetric, diagonal entries ≥ 0 and $\det = 42 > 0 \Rightarrow$ positive definite.

(B) **FALSE**. a diagonal entry $K(x, x) = \|\phi(x)\|^2$ is negative — impossible.

(C) **FALSE**. not symmetric, but $K(x, z) = K(z, x)$ for every kernel.

(D) **TRUE**. symmetric, diagonal ≥ 0 , $\det = 11 > 0 \Rightarrow$ PD (negative off-diagonal entries are fine).

Q.29 — Answer: 0.5317 to 0.5517 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$\mathbf{m}_1 = (2, 2)$, $\mathbf{m}_2 = (4.333, 4)$.

$$S_1 = \begin{bmatrix} 8 & 6 \\ 6 & 6 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 2.667 & 2 \\ 2 & 2 \end{bmatrix}$$

$$S_W = \begin{bmatrix} 10.667 & 8 \\ 8 & 8 \end{bmatrix}$$

$\mathbf{m}_2 - \mathbf{m}_1 = (2.333, 2)$.

$\mathbf{w} = S_W^{-1}(\mathbf{m}_2 - \mathbf{m}_1) = (0.125, 0.125)$.

$(\mathbf{m}_2 - \mathbf{m}_1)^T \mathbf{w} = \mathbf{0.54}$.

Q.30 — Answer: 16.4064 to 16.4264 [NAT, 2M · Unsupervised Learning > k-medoids]

Compute the Euclidean (L_2) distance matrix and its row sums:

	P1	P2	P3	P4	P5	Σ
P1	0	4.47	4.47	2.24	6.4	17.58
P2	4.47	0	8.49	3	2.24	18.19
P3	4.47	8.49	0	6.71	10.63	30.3
P4	2.24	3	6.71	0	4.47	16.42
P5	6.4	2.24	10.63	4.47	0	23.74

The smallest row sum is 16.42 for P4, so the medoid is P4.

Answer: **16.42**.

Unlike a k-means centroid, the medoid is always an actual data point.

Q.31 — Answer: 0.4147 to 0.4187 [NAT, 2M · Regression & Model Selection > Gradient descent]

Hessian of J : $\nabla^2 J = (2/n)X^T X$. Eigenvalues of $X^T X$: $12 \pm 4 = 16, 8$ (eigenvectors $(1, 1)$, $(1, -1)$).

So the Hessian eigenvalues are $L = 3.2$ and $\mu = 1.6$.

GD error evolves as $w_{t+1} - w^* = (I - \eta \nabla^2 J)(w_t - w^*)$, so it converges iff $|1 - \eta \lambda| < 1$ for every Hessian eigenvalue λ .

Balance $|1 - \eta L| = |1 - \eta \mu| \Rightarrow \eta^* = 2/(L + \mu) = 2/(3.2 + 1.6) = 0.4167$.

Answer: **0.417**.

Q.32 — Answer: (A), (B), (C) [MSQ, 2M · Unsupervised Learning > k-means]

(A) **TRUE**. The median minimizes the sum of absolute deviations.

(B) **TRUE**. Medoids are data points; no means are needed.

(C) **TRUE**. This is the definition.

(D) **FALSE**. It is $2^{n-1} - 1$.

Q.33 — Answer: 0.9835 to 0.9875 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Per unit population: $TP = Se \cdot \pi$, $FN = (1 - Se) \cdot \pi$, $FP = (1 - Sp)(1 - \pi)$, $TN = Sp(1 - \pi)$.

$TP = 0.045$, $FN = 0.005$, $FP = 0.0095$, $TN = 0.9405$.

Accuracy = $TP + TN = \mathbf{0.986}$.

Note that the trivial 'always negative' classifier would have accuracy 0.95 — accuracy is misleading under imbalance.

Q.34 — Answer: 1.0399 to 1.0599 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

Per-example loss $L_i = -\ln \hat{p}_i$ if $y_i = 1$ and $-\ln(1 - \hat{p}_i)$ if $y_i = 0$:

i	y	$\hat{p}$	L_i
1	1	0.25	1.3863
2	0	0.5	0.6931
3	1	0.8	0.2231
4	1	0.15	1.8971

$\sum L_i = 4.1997$; average = $4.1997/4 = \mathbf{1.05}$.

Common mistakes: using $\ln \hat{p}$ for $y = 0$ examples, or $\log_{10}$ instead of $\ln$.

Q.35 — Answer: (A), (C) [MSQ, 2M · SVM & Decision Trees > Decision trees: impurity measures]

Parent [50+, 60-]: $H = 0.994$, $Gini = 0.4959$, error = 0.4545.

S1: children $H = 0.9951, 0.6343$; $Gini = 0.4966, 0.2688$; error = 0.4588, 0.16. Gains: $IG = 0.0809$, $Gini = 0.051$, error = 0.0636.

S2: children $H = 0.8004, 0.989$; $Gini = 0.3682, 0.4924$; error = 0.2432, 0.4384. Gains: $IG = 0.0685$, $Gini = 0.0453$, error = 0.0818.

(A) **TRUE**. S1: 0.0636, S2: 0.0818.

(B) **FALSE**. S1: 0.0809, S2: 0.0685.

(C) **TRUE**. S1: 0.0809, S2: 0.0685.

(D) **FALSE**. S1: 0.051, S2: 0.0453.

Mock Test 42

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	8	2	25

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Neural Networks (MLP / Feed-forward)	7	11
Probabilistic & Instance-based Classifiers	7	11
Unsupervised Learning	7	11
SVM & Decision Trees	7	11
Regression & Model Selection	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

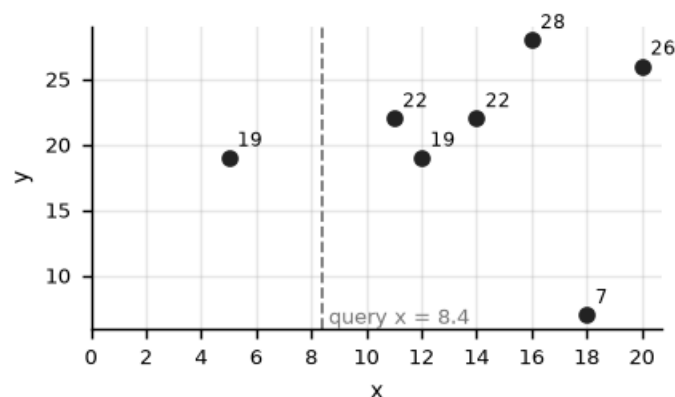
Q.1 [1 mark · NAT (numerical answer)]

For logits $z = (-0.5, 2, 1.5)$, the temperature-scaled softmax $p_k = \exp(z_k/T) / \sum_j \exp(z_j/T)$ with $T = 2$ gives $p_3 = \underline{\hspace{2cm}}$ (rounded off to three decimal places)

Q.2 [1 mark · NAT (numerical answer)]

The training data for a k-NN regressor are given below (also plotted). Using $k = 2$ and absolute distance $|x - x_i|$, the predicted value (unweighted average of neighbours) at $x = 8.4$ is _____ (rounded off to two decimal places)

x	5	11	12	14	16	18	20
y	19	22	19	22	28	7	26



Q.3 [1 mark · NAT (numerical answer)]

The 4 two-dimensional observations below have zero mean in each feature. Then the sample variance of x_1 (divisor $n - 1$) is _____ (rounded off to two decimal places)

Point	x_1	x_2	x_3	x_4
x_1	4	-4	2	-2
x_2	-1	-4	-2	7

Q.4 [1 mark · NAT (numerical answer)]

In a training set, the 8 examples of class Approve = Yes have the following counts for the feature *Income*. Using Laplace (add-one) smoothing, the naive Bayes estimate of $P(\text{Income} = \text{Low} \mid \text{Approve} = \text{Yes})$ is _____ (rounded off to three decimal places)

Income	Low	Medium	High
count in Approve = Yes	0	4	4

Q.5 [1 mark · NAT (numerical answer)]

The homogeneous polynomial kernel $K(x, z) = (x^T z)^3$ on $x \in \mathbb{R}^4$ corresponds to a feature map of all (scaled) monomials of degree exactly 3. The number of distinct such features is _____ (Answer in integer)

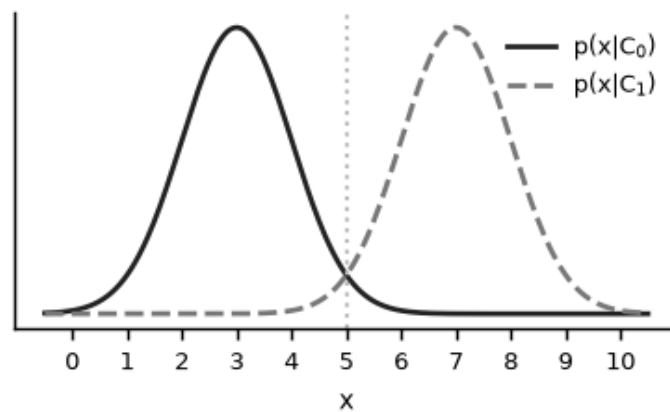
Q.6 [1 mark · MCQ]

A soft-margin linear SVM has learned $f(x) = 0.5x_1 + 0.5x_2 - 3$. For the training example $x = (-1, 4)$ with label $y = +1$, which statement is CORRECT?

- (A) Exactly on the decision boundary ($\xi = 1$)
- (B) Correctly classified but strictly inside the margin ($0 < \xi < 1$)
- (C) Misclassified ($\xi > 1$)
- (D) Correctly classified, on or outside the margin ($\xi = 0$)

Q.7 [1 mark · MCQ]

Two classes have unit-variance Gaussian class-conditional densities with means 3 (C_0) and 7 (C_1), as plotted. With equal priors the Bayes decision threshold is the dotted line at $x = 5$. If the prior of C_0 is increased to 0.8, the threshold



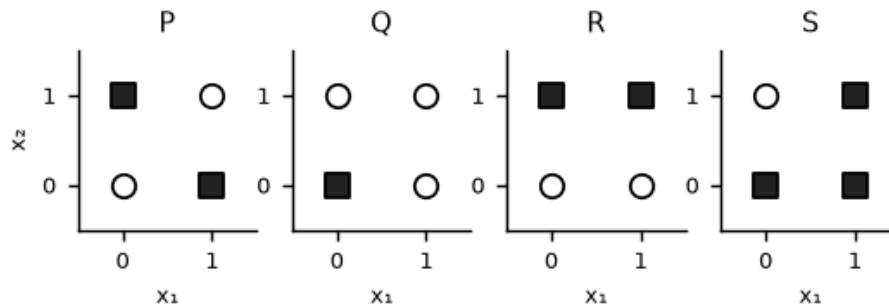
- (A) It stays at $x = 5$ because the two variances are equal
- (B) It moves towards $x = 3$, enlarging the region assigned to C_0
- (C) It moves towards $x = 3$, shrinking the region assigned to C_0
- (D) It moves towards $x = 7$, enlarging the region assigned to C_0

Q.8 [1 mark · NAT (numerical answer)]

Data are uniformly distributed in the d -dimensional unit hypercube with $d = 50$. A sub-cube is to capture a fraction $r = 0.01$ of the data. The required edge length of the sub-cube is _____ (rounded off to two decimal places)

Q.9 [1 mark · MCQ]

Each panel shows the four binary inputs (x_1, x_2) labelled as class 1 (filled square) or class 0 (open circle). Which dataset can NOT be classified perfectly by a single perceptron (one linear threshold unit)?



- (A) Dataset P
(B) Dataset Q
(C) Dataset R
(D) Dataset S

Q.10 [1 mark · NAT (numerical answer)]

The number of ways to partition 5 distinct points into **at most** 3 non-empty (unlabeled) clusters is ____ (Answer in integer)

Q.11 [1 mark · NAT (numerical answer)]

Repeated 10-fold cross-validation with 3 repetitions (different random partitions) is used to compare 4 models. Ignoring any final refit, the number of model fits is ____ (Answer in integer)

Q.12 [1 mark · NAT (numerical answer)]

In AdaBoost, a weak learner has weighted training error $\varepsilon = 0.29$. Its vote weight $\alpha = \frac{1}{2} \ln((1 - \varepsilon)/\varepsilon)$ is ____ (rounded off to two decimal places)

Q.13 [1 mark · MSQ (one or more correct)]

A simple linear regression $\hat{y} = \beta_0 + \beta_1 x$ is fitted by ordinary least squares (with intercept) to data (x_i, y_i) , $i = 1, \dots, n$, with residuals $e_i = y_i - \hat{y}_i$ and sample correlation r . Which of the following statements is/are CORRECT?

- (A) For the least-squares line, $\sum x_i e_i = 0$.
(B) The least-squares slope equals $r \cdot s_y / s_x$, where r is the sample correlation.
(C) For the least-squares line (with intercept), the residuals sum to zero: $\sum e_i = 0$.
(D) The least-squares line passes through the point $(\bar{x}, \bar{y})$.

Q.14 [1 mark · NAT (numerical answer)]

A 3-class softmax classifier gives the predicted probabilities below for a mini-batch of 3 examples. The mean categorical cross-entropy loss (natural log) over the mini-batch is ____ (rounded off to three decimal places)

example	p(class 1)	p(class 2)	p(class 3)	true class
1	0.2	0.4	0.4	3
2	0.1	0.8	0.1	3
3	0.3	0.6	0.1	3

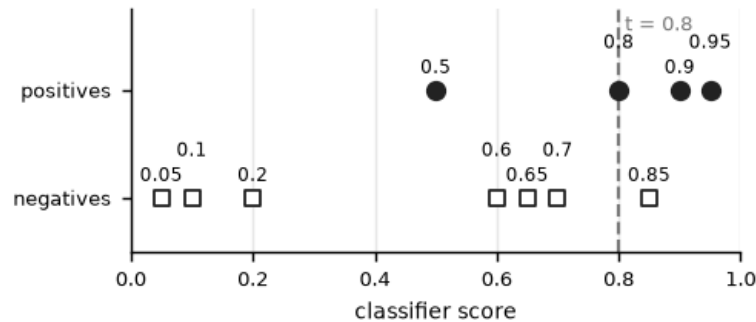
Q.15 [1 mark · NAT (numerical answer)]

Regressing x_2 on the remaining predictors gives $R^2 = 0.52$. By what factor is the standard error of β_2 inflated relative to the case of x_2 being uncorrelated with the other predictors? ____ (rounded off to two decimal places)

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

A probabilistic classifier produces the scores below on 11 labelled examples (1 = positive). An example is predicted positive if its score is $\geq t$, with $t = 0.8$. The precision at this threshold is _____ (rounded off to two decimal places)

example	1	2	3	4	5	6	7	8	9	10	11
score	0.9	0.1	0.95	0.5	0.6	0.65	0.85	0.8	0.05	0.7	0.2
label	1	0	1	1	0	0	0	1	0	0	0

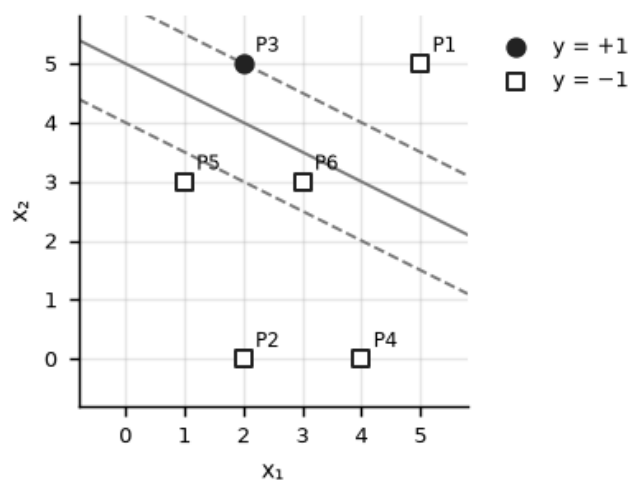
**Q.17 [2 marks · NAT (numerical answer)]**

On $n = 16$ observations, model A (intercept + 1 predictor) has $R^2 = 0.71$, and model B (intercept + 2 predictors, containing those of A) has $R^2 = 0.78$. The value of (adjusted R^2 of B) – (adjusted R^2 of A) is _____ (rounded off to three decimal places)

Q.18 [2 marks · NAT (numerical answer)]

A linear SVM with $f(x) = 0.5x_1 + x_2 - 5$ is evaluated on the training points in the table (also plotted, with the boundary $f = 0$ solid and $f = \pm 1$ dashed). Compute the total hinge loss $\sum_i \max(0, 1 - y_i f(x_i))$. (rounded off to two decimal places)

Point	x_1	x_2	y
P1	5	5	-1
P2	2	0	-1
P3	2	5	+1
P4	4	0	-1
P5	1	3	-1
P6	3	3	-1



Q.19 [2 marks · NAT (numerical answer)]

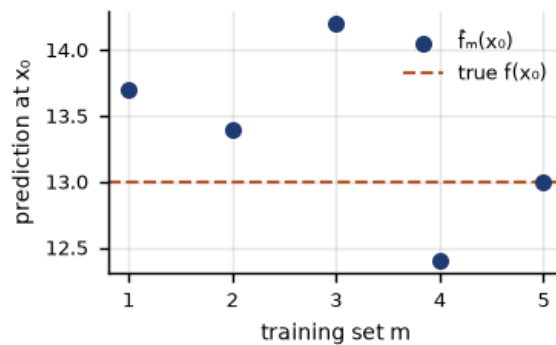
The confusion matrix of a 3-class classifier is given below (rows = actual class, columns = predicted class). The weighted-average (by class support) recall is _____ (rounded off to two decimal places)

actual \ predicted	A	B	C
A	11	3	5
B	4	20	4
C	5	2	8

Q.20 [2 marks · NAT (numerical answer)]

Data are generated as $y = f(x) + \varepsilon$ with $E[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2 = 0.25$. At a test point x_0 the true value is $f(x_0) = 13$. A learning method is trained on 5 independent training sets, giving the predictions $\hat{f}_m(x_0)$ in the table (also plotted). Treating these as the full distribution of the method's prediction (variance with divisor 5), the expected squared prediction error $E[(y_0 - \hat{f}(x_0))^2]$ at x_0 is _____ (rounded off to three decimal places)

m	1	2	3	4	5
$\hat{f}_m(x_0)$	13.7	13.4	14.2	12.4	13

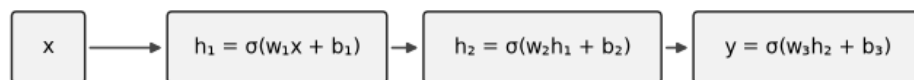
**Q.21 [2 marks · NAT (numerical answer)]**

The actual and predicted labels of a binary classifier (1 = positive) on 14 test examples are given below. The F_β -score with $\beta = 0.5$, $F_\beta = (1 + \beta^2)PR/(\beta^2P + R)$, is _____ (rounded off to two decimal places)

#	1	2	3	4	5	6	7	8	9	10	11	12	13	14
actual y	0	1	1	1	1	1	0	0	0	1	1	1	1	0
predicted $\hat{y}$	0	1	0	1	0	1	0	1	0	1	0	1	1	1

Q.22 [2 marks · NAT (numerical answer)]

Consider the scalar chain shown below with weights $w_1 = -2$, $w_2 = 3$, $w_3 = 2$, biases $b_1 = 1$, $b_2 = -1$, $b_3 = 1$ and activation the logistic sigmoid σ at every stage. For input $x = 1$, the value of $\partial y / \partial x$ is _____ (rounded off to 4 decimal places)

**Q.23 [2 marks · NAT (numerical answer)]**

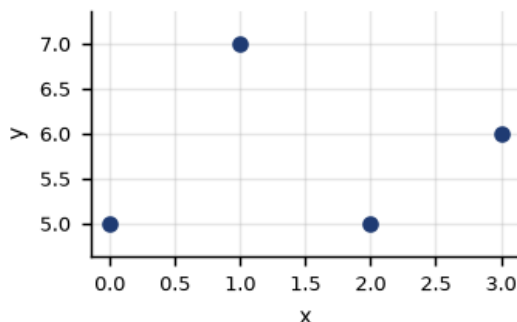
The sample covariance matrix of a two-feature dataset is S given below. In principal component analysis, the variance of the data along the second principal component is _____ (rounded off to two decimal places)

$$S = \begin{bmatrix} 6 & 4 \\ 4 & 4 \end{bmatrix}$$

Q.24 [2 marks · NAT (numerical answer)]

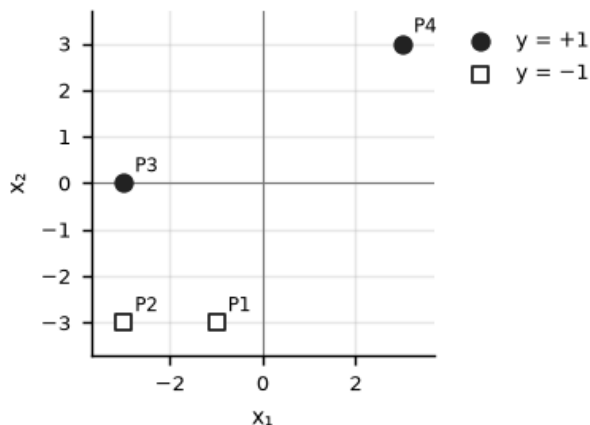
A linear model $\hat{y} = w_0 + w_1 x$ is trained on the 4 points below by batch gradient descent on the loss $J(w) = (1/2n)\sum(w_0 + w_1 x_i - y_i)^2$, starting from $(w_0, w_1) = (0, 1)$ with learning rate $\eta = 0.01$. Then the value of w_0 after one iteration is _____ (rounded off to three decimal places)

x	0	1	2	3
y	5	7	5	6

**Q.25 [2 marks · NAT (numerical answer)]**

A perceptron with augmented weight vector $w = (w_0, w_1, w_2)$ (w_0 is the bias, $\tilde{x} = (1, x_1, x_2)$) starts from $w = (-1, 0, 0)$ and uses learning rate $\eta = 1$. The points P1, P2, ... are presented once, in order; whenever $y \cdot w^T \tilde{x} \leq 0$ the update $w \leftarrow w + \eta y \tilde{x}$ is applied. Find the value of w_1 after this epoch. (Answer in integer)

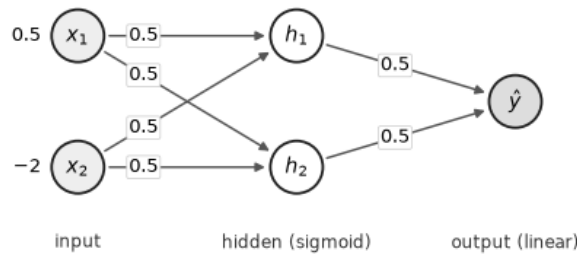
Point	x_1	x_2	y
P1	-1	-3	-1
P2	-3	-3	-1
P3	-3	0	+1
P4	3	3	+1

**Q.26 [2 marks · NAT (numerical answer)]**

A simple linear regression fitted by ordinary least squares gives $\hat{y} = 4.5 - 2.7x$. The sample standard deviations are $s_x = 12$ and $s_y = 25$. If both x and y are z-score standardised and the regression is refitted, the new slope (the standardised coefficient) is _____ (rounded off to two decimal places)

Q.27 [2 marks · NAT (numerical answer)]

In the 2-2-1 network below (sigmoid hidden units, linear output $\hat{y} = v_1 h_1 + v_2 h_2$, all biases 0), **every** weight is initialised to 0.5. One gradient-descent step with $\eta = 0.1$ is taken on $L = \frac{1}{2}(\hat{y} - y)^2$ for the input $x = (0.5, -2)$ with target $y = 0$. Then the updated value of w_{21} (weight from x_1 to h_2) is _____ (rounded off to three decimal places)

**Q.28 [2 marks · NAT (numerical answer)]**

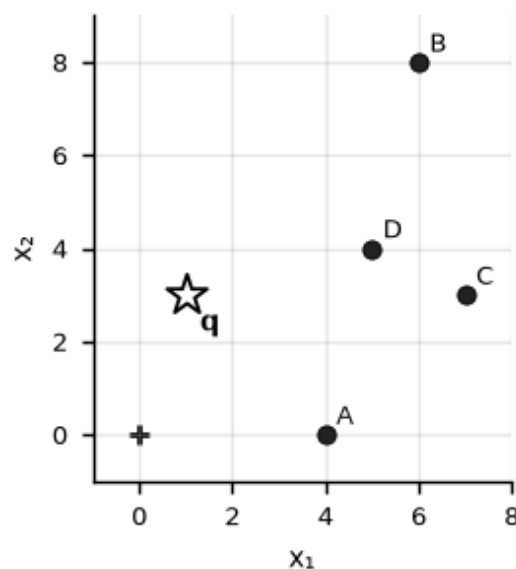
Logistic regression $P(y = 1 | \mathbf{x}) = \sigma(w_1 x + b)$ is trained by **batch** gradient descent on the **average** cross-entropy loss over the 2 examples below, with learning rate $\eta = 1$. Starting from $w_1 = 0.5$, $b = -0.5$, the value of b after ONE gradient-descent update is _____ (rounded off to three decimal places)

i	x	y
1	-1	0
2	-1	0

Q.29 [2 marks · MCQ]

A query point $q = (1, 3)$ and four points A-D are shown below. Using **Manhattan (L_1) distance**, which point is nearest to q ?

Point	A	B	C	D
x_1	4	6	7	5
x_2	0	8	3	4

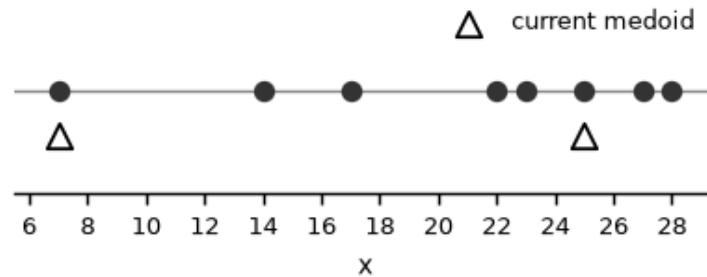


- (A) Point A
(C) Point C

- (B) Point B
(D) Point D

Q.30 [2 marks · NAT (numerical answer)]

PAM (k-medoids, $k = 2$, absolute-difference dissimilarity) is applied to the 1-D data $\{7, 14, 17, 22, 23, 25, 27, 28\}$. The current medoids are 7 and 25. The total cost $\sum_i \min_m |x_i - m|$ of the current configuration is _____ (Answer in integer)

**Q.31 [2 marks · MCQ]**

For logits $z = (1, -1, 0)$, softmax probabilities p and true class 3, the gradient of the cross-entropy loss $L = -\ln p_3$ with respect to z (rounded to two decimals) is

- (A) $(0, 0, -4.09)$ (B) $(0.15, 0.01, -0.14)$
 (C) $(0.67, 0.09, -0.76)$ (D) $(-0.67, -0.09, 0.76)$

Q.32 [2 marks · MSQ (one or more correct)]

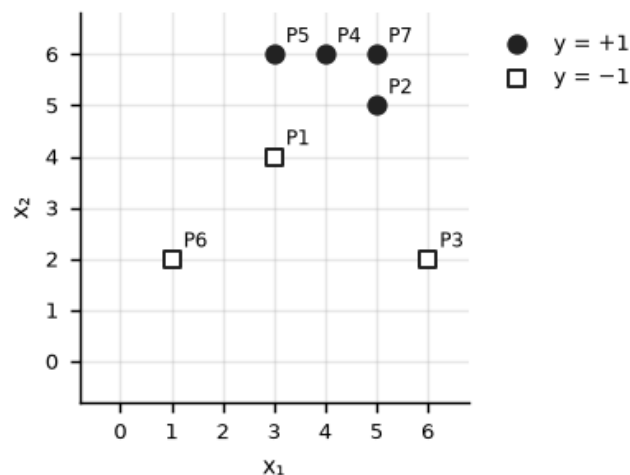
Consider agglomerative hierarchical clustering on n points with a dissimilarity matrix (no ties). Which of the following statements is/are CORRECT?

- (A) Cophenetic distances from single, complete or average linkage satisfy the ultrametric inequality $d(x, z) \leq \max(d(x, y), d(y, z))$.
 (B) For any two points, the complete-linkage cophenetic distance is $\geq$ their original distance.
 (C) Replacing every dissimilarity by its square does not change the single-linkage merge order.
 (D) Clusterings obtained by cutting one dendrogram at different heights are nested.

Q.33 [2 marks · MCQ]

Four candidate linear decision boundaries are proposed for the labelled training points shown. Which boundary separates the two classes with the LARGEST geometric margin (minimum distance from the boundary to any training point, all points correctly classified)?

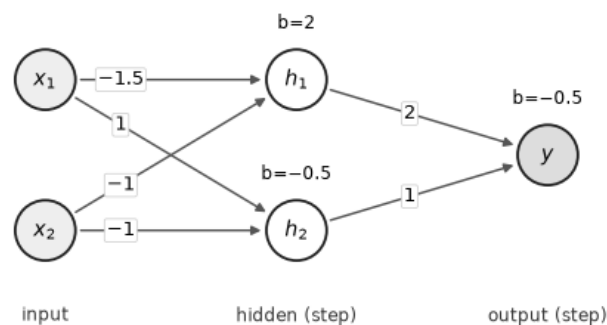
Point	x_1	x_2	y
P1	3	4	-1
P2	5	5	+1
P3	6	2	-1
P4	4	6	+1
P5	3	6	+1
P6	1	2	-1
P7	5	6	+1



- (A) $x_1 + 2x_2 - 13 = 0$
 (B) $x_1 + 2x_2 - 14 = 0$
 (C) $x_1 + x_2 - 8.5 = 0$
 (D) $x_1 - 2x_2 - 2.5 = 0$

Q.34 [2 marks · MCQ]

In the network below every unit (hidden and output) is a threshold unit that outputs 1 if its weighted input plus bias is > 0 , and 0 otherwise. Edge labels are weights. For binary inputs $x_1, x_2 \in \{0, 1\}$, the network computes



- (A) NAND
 (B) XOR
 (C) AND
 (D) OR

Q.35 [2 marks · MCQ]

A decision tree is to be grown on the 10 training records below using Gini gain as the splitting criterion (multi-way split on each categorical attribute). Which attribute is chosen at the root?

#	Wind	Shape	Colour	Age	Class
1	Weak	Square	Green	Young	Yes
2	Weak	Oval	Red	Middle	No
3	Weak	Square	Blue	Young	Yes
4	Weak	Round	Blue	Senior	Yes
5	Strong	Square	Blue	Young	No
6	Weak	Square	Red	Senior	No
7	Weak	Oval	Green	Senior	No
8	Weak	Square	Red	Middle	No
9	Strong	Round	Red	Middle	No
10	Weak	Round	Blue	Senior	Yes

(A) Wind**(B)** Shape**(C)** Colour**(D)** Age

— END OF QUESTION PAPER —

Mock Test 42 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	0.3751 to 0.3791	19	NAT	2	0.619 to 0.639
2	NAT	1	20.49 to 20.51	20	NAT	2	0.737 to 0.743
3	NAT	1	13.3233 to 13.3433	21	NAT	2	0.7217 to 0.7417
4	NAT	1	0.0889 to 0.0929	22	NAT	2	−0.0666 to −0.0653
5	NAT	1	20	23	NAT	2	0.8669 to 0.8869
6	MCQ	1	(C)	24	NAT	2	0.0405 to 0.0445
7	MCQ	1	(D)	25	NAT	2	0
8	NAT	1	0.902 to 0.922	26	NAT	2	−1.306 to −1.286
9	MCQ	1	(A)	27	NAT	2	0.4963 to 0.5002
10	NAT	1	41	28	NAT	2	−0.7709 to −0.7669
11	NAT	1	120	29	MCQ	2	(D)
12	NAT	1	0.4377 to 0.4577	30	NAT	2	25
13	MSQ	1	(A), (B), (C), (D)	31	MCQ	2	(C)
14	NAT	1	1.8385 to 1.8425	32	MSQ	2	(A), (B), (C), (D)
15	NAT	1	1.4334 to 1.4534	33	MCQ	2	(A)
16	NAT	2	0.74 to 0.76	34	MCQ	2	(A)
17	NAT	2	0.0549 to 0.0589	35	MCQ	2	(C)
18	NAT	2	3.99 to 4.01				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Neural Networks (MLP / Feed-forward)					11
Probabilistic & Instance-based Classifiers					11
Unsupervised Learning					11
SVM & Decision Trees					11
Regression & Model Selection					11
Total					55

Mock Test 42 — Detailed Solutions

Q.1 — Answer: 0.3751 to 0.3791 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

Scaled logits $z/T = (-0.25, 1, 0.75)$.

exp values: 0.7788, 2.7183, 2.117; sum = 5.6141.

$p_3 = 2.117/5.6141 = \mathbf{0.3771}$.

Q.2 — Answer: 20.49 to 20.51 [NAT, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Distances $|x_i - 8.4|$: $x=11$: 2.6, $x=5$: 3.4, $x=12$: 3.6, $x=14$: 5.6, $x=16$: 7.6, $x=18$: 9.6, $x=20$: 11.6.

The 2 nearest neighbours are $x = 11, 5$ with $y = 22, 19$.

$\hat{y} = (22 + 19)/2 = \mathbf{20.5}$.

Q.3 — Answer: 13.3233 to 13.3433 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

The data are already centered (column means are 0), so $S = X^T X / (n - 1)$.

$\Sigma x_1^2 = 40$, $\Sigma x_2^2 = 70$, $\Sigma x_1 x_2 = -6$.

$S = [[13.3333, -2], [-2, 23.3333]]$.

Required value = **13.33**. (Dividing by n instead of $n - 1$ would give 10.)

Q.4 — Answer: 0.0889 to 0.0929 [NAT, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Add- α estimate: $(\text{count} + \alpha) / (N_c + \alpha V)$ with $V = 3$ possible values.

$= (0 + 1) / (8 + 1 \times 3) = 1/11 = \mathbf{0.091}$.

(The unsmoothed estimate would be 0/8, i.e. zero, which would wipe out the whole posterior product.)

Q.5 — Answer: 20 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

Monomials of degree exactly d in p variables = $C(p + d - 1, d)$ (stars and bars).

$C(6, 3) = \mathbf{20}$.

(Including all lower degrees, as for $(x^T z + 1)^3$, would give $C(7, 3) = 35$.)

Q.6 — Answer: (C) [MCQ, 1M · SVM & Decision Trees > SVM: soft margin & hinge loss]

$y \cdot f(x) = (+1) \cdot (-1.5) = -1.5$.

Slack $\xi = \max(0, 1 - y f(x)) = 2.5$.

Rules: $y f \geq 1 \Rightarrow \xi = 0$ (on/outside margin); $0 < y f < 1 \Rightarrow$ inside margin, still correct; $y f = 0 \Rightarrow$ on the boundary ($\xi = 1$); $y f < 0 \Rightarrow$ misclassified ($\xi > 1$).

Hence: **Misclassified ($\xi > 1$)**.

Q.7 — Answer: (D) [MCQ, 1M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$x^* = (\mu_0 + \mu_1)/2 + \sigma^2 \ln(\pi_0/\pi_1)/(\mu_1 - \mu_0) = 5.347$.

A larger prior for C_0 means more of the x -axis is assigned to it, so the threshold moves away from μ_0 towards $\mu_1 = 7$. Answer: **It moves towards $x = 7$, enlarging the region assigned to C_0 .**

Equal variances keep the boundary a single point (linear); priors shift it but never make it quadratic.

Q.8 — Answer: 0.902 to 0.922 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

Volume of sub-cube = $e^d = r \Rightarrow e = r^{1/d} = 0.01^{1/50} = \mathbf{0.91}$.

Even to capture a small fraction of data, the 'local' neighbourhood must span most of each feature's range — the curse of dimensionality.

Q.9 — Answer: (A) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

A single perceptron can classify a dataset perfectly iff the two classes are linearly separable.

Dataset P: labels for $(0,0), (0,1), (1,0), (1,1) = 0, 1, 1, 0 \rightarrow$ XOR — **not** linearly separable (diagonal pattern).

Dataset Q: labels for (0,0),(0,1),(1,0),(1,1) = 1, 0, 0, 0 → NOR - linearly separable.

Dataset R: labels for (0,0),(0,1),(1,0),(1,1) = 0, 1, 0, 1 → x_2 (identity of x_2) - linearly separable.

Dataset S: labels for (0,0),(0,1),(1,0),(1,1) = 1, 0, 1, 1 → x_1 OR (NOT x_2) - linearly separable.

Answer: **Dataset P**. Of the 14 non-constant Boolean functions of two inputs, only XOR and XNOR are not linearly separable.

Q.10 — Answer: 41 [NAT, 1M · Unsupervised Learning > k-means]

Sum Stirling numbers of the second kind: $S(5,1) = 1 + S(5,2) = 15 + S(5,3) = 25 \rightarrow$ total **41**.

Q.11 — Answer: 120 [NAT, 1M · Regression & Model Selection > Cross-validation]

3 repetitions $\times$ 10 folds $\times$ 4 models = **120**.

Q.12 — Answer: 0.4377 to 0.4577 [NAT, 1M · SVM & Decision Trees > Ensembles: boosting]

$\alpha = \frac{1}{2} \ln(0.71/0.29) = \frac{1}{2} \ln(2.4483) = \mathbf{0.45}$.

$\varepsilon < 0.5 \Rightarrow \alpha > 0$; $\varepsilon = 0.5 \Rightarrow \alpha = 0$ (no better than chance).

Q.13 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Simple linear regression]

(A) TRUE. This is the normal equation for β_1 .

(B) TRUE. $\beta_1 = S_{xy}/S_{xx} = r\sqrt{(S_{yy}/S_{xx})}$.

(C) TRUE. The normal equation for β_0 is exactly $\sum(y_i - \beta_0 - \beta_1 x_i) = 0$.

(D) TRUE. $\beta_0 = \bar{y} - \beta_1 \bar{x} = \hat{y}(\bar{x}) = \bar{y}$.

Q.14 — Answer: 1.8385 to 1.8425 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

CE for one example = $-\ln p_{\text{true class}}$.

Example 1: $-\ln 0.4 = 0.9163$.

Example 2: $-\ln 0.1 = 2.3026$.

Example 3: $-\ln 0.1 = 2.3026$.

Mean = **1.8405**.

Q.15 — Answer: 1.4334 to 1.4534 [NAT, 1M · Regression & Model Selection > Multiple linear regression]

Variance inflation factor of predictor j: $VIF_j = 1/(1 - R_j^2)$, where R_j^2 is the R^2 from regressing x_j on all other predictors. It is the factor by which $\text{Var}(\beta_j)$ is inflated by collinearity; $\sqrt{VIF}$ inflates the standard error.

$VIF = 1/(1 - 0.52) = 2.0833$; SE inflation = $\sqrt{VIF} = 1.4434$.

Answer: **1.44**.

Q.16 — Answer: 0.74 to 0.76 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Predicted positive (score ≥ 0.8): examples 1, 3, 7, 8.

TP = 3, FP = 1, FN = 1, TN = 6.

Precision = $TP/(TP + FP) = \mathbf{0.75}$.

Raising t typically raises precision and lowers recall/FPR; lowering t does the opposite (threshold moving).

Q.17 — Answer: 0.0549 to 0.0589 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

Adjusted $R^2 = 1 - [SSE/(n - p - 1)]/[SST/(n - 1)] = 1 - (1 - R^2)(n - 1)/(n - p - 1)$, where p is the number of predictors (excluding the intercept).

A: $1 - (1 - 0.71)(15)/(14) = 0.6893$.

B: $1 - (1 - 0.78)(15)/(13) = 0.7462$.

Difference = 0.0569.

Answer: **0.057**.

Q.18 — Answer: 3.99 to 4.01 [NAT, 2M · SVM & Decision Trees > SVM: soft margin & hinge loss]

Compute $f(x_i)$, the functional margin $y_i f(x_i)$ and the slack/hinge loss for each point:

Point	x	y	f(x)	$y f(x)$	$\xi = \max(0, 1 - y f)$
P1	(5, 5)	-1	2.5	-2.5	3.5
P2	(2, 0)	-1	-4	4	0
P3	(2, 5)	+1	1	1	0
P4	(4, 0)	-1	-3	3	0
P5	(1, 3)	-1	-1.5	1.5	0
P6	(3, 3)	-1	-0.5	0.5	0.5

$$\sum \xi_i = 4.$$

Points with $y f \geq 1$ contribute 0; points inside the margin contribute between 0 and 1; misclassified points contribute more than 1.

Q.19 — Answer: 0.619 to 0.639 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

class	TP	predicted total	actual total	precision	recall	F1
A	11	20	19	0.55	0.5789	0.5641
B	20	25	28	0.8	0.7143	0.7547
C	8	17	15	0.4706	0.5333	0.5

Precision of class $k = C_{kk}/(\text{column sum})$; recall = $C_{kk}/(\text{row sum})$.

Weighted recall = $\sum(\text{support}_k \times \text{recall}_k)/N = \sum C_{kk}/N = \text{accuracy} = \mathbf{0.63}$.

Q.20 — Answer: 0.737 to 0.743 [NAT, 2M · Regression & Model Selection > Bias-variance trade-off]

Mean prediction $\bar{f} = (1/5)\sum f_m = 13.34$.

Bias² = $(\bar{f} - f(x_0))^2 = (13.34 - 13)^2 = 0.1156$.

Variance = $(1/5)\sum (f_m - \bar{f})^2 = 0.3744$.

Expected test error = Bias² + Variance + $\sigma^2 = 0.1156 + 0.3744 + 0.25 = 0.74$.

Answer: **0.74**.

Q.21 — Answer: 0.7217 to 0.7417 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Counting: TP = 6, FP = 2, FN = 3.

$P = 6/8 = 0.75$, $R = 6/9 = 0.6667$.

$F_{0.5} = (1 + 0.25) \times 0.75 \times 0.6667 / (0.25 \times 0.75 + 0.6667) = \mathbf{0.73}$.

$\beta < 1$ weights precision more heavily (F1 here would be 0.706).

Q.22 — Answer: -0.0666 to -0.0653 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Activation functions]

Forward: $a_1 = -2 \cdot 1 + 1 = -1$, $h_1 = 0.2689$; $a_2 = 3 \cdot 0.2689 + -1 = -0.1932$, $h_2 = 0.4519$; $a_3 = 2 \cdot 0.4519 + 1 = 1.9037$, $y = 0.8703$.

Local derivatives $\sigma'(a)$: $\sigma'(a) = \sigma(a)(1 - \sigma(a))$: $\sigma'(a_1) = 0.1966$, $\sigma'(a_2) = 0.2477$, $\sigma'(a_3) = 0.1129$.

Chain rule: $\partial y / \partial x = \sigma'(a_3) \cdot w_3 \cdot \sigma'(a_2) \cdot w_2 \cdot \sigma'(a_1) \cdot w_1 = \mathbf{-0.066}$.

Each layer contributes a factor $\sigma' \leq 0.25$ - the product shrinks with depth (vanishing gradient).

Q.23 — Answer: 0.8669 to 0.8869 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

Eigenvalues of $[[a, b], [b, c]]$: $\lambda = (a + c)/2 \pm \sqrt{((a - c)/2)^2 + b^2}$.

$\lambda = 5 \pm \sqrt{(1 + 16)} = 5 \pm 4.1231 \rightarrow \lambda_1 = 9.1231$, $\lambda_2 = 0.8769$ (check: $\lambda_1 + \lambda_2 = \text{trace} = 10$, $\lambda_1 \lambda_2 = \det = 8$).

Eigenvector for λ_1 : $(S - \lambda_1 I)v = 0 \rightarrow (-3.1231)v_1 + (4)v_2 = 0 \rightarrow v \propto (1, 0.7808)$.

PVE(PC1) = $\lambda_1/(\lambda_1 + \lambda_2) = 91.23\%$.

Answer: **0.88**.

Q.24 — Answer: 0.0405 to 0.0445 [NAT, 2M · Regression & Model Selection > Gradient descent]

$\partial J/\partial w_0 = (1/n)\sum r_i$, $\partial J/\partial w_1 = (1/n)\sum r_i x_i$, with residual $r_i = w_0 + w_1 x_i - y_i$.

Step 1: $w = (0, 1)$, $r = (-5, -6, -3, -3)$, $\nabla J = (-4.25, -5.25) = w \leftarrow w - 0.01\nabla J = (0.0425, 1.0525)$.

Answer: **0.043**.

Q.25 — Answer: 0 [NAT, 2M · SVM & Decision Trees > Perceptron (linear separators)]

Trace the epoch (a point with $y \cdot w^T \tilde{x} \leq 0$, including 0, triggers an update):

Step	$\tilde{x} = (1, x_1, x_2)$	y	$w^T \tilde{x}$	$y \cdot w^T \tilde{x} \leq 0$?	w after step
1	(1, -1, -3)	-1	-1	no	(-1, 0, 0)
2	(1, -3, -3)	-1	-1	no	(-1, 0, 0)
3	(1, -3, 0)	+1	-1	yes → update	(0, -3, 0)
4	(1, 3, 3)	+1	-9	yes → update	(1, 0, 3)

Final $w = (1, 0, 3)$, number of updates = 2. Answer: **0**.

Q.26 — Answer: -1.306 to -1.286 [NAT, 2M · Regression & Model Selection > Basis functions & feature scaling]

Standardising y divides the slope by s_y ; standardising x multiplies it by s_x : $\beta^* = \beta_1 s_x / s_y = -2.7 \times 12/25 = -1.296$ (this equals the correlation r in simple regression).

Answer: **-1.3**.

Q.27 — Answer: 0.4963 to 0.5002 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

All weights = 0.5, biases 0 = both hidden units get $a = 0.5 \times (0.5 + -2) = -0.75$, $h_1 = h_2 = \sigma(a) = 0.3208$; $\hat{y} = 2 \times 0.5 \times 0.3208 = 0.3208$.

$\hat{y} - y = 0.3208$. $\partial L/\partial v_j = (\hat{y} - y)h_j = 0.1029$ (same for both j).

$\partial L/\partial w_{jk} = (\hat{y} - y) v_j h_j (1 - h_j) x_k = 0.3208 \times 0.5 \times 0.2179 \times 0.5 = 0.0175$ (identical for $j = 1, 2$).

$w_{21} \leftarrow 0.5 - 0.1 \times (0.0175) = \mathbf{0.4983}$.

Because the two hidden units start identical and get identical updates, they remain identical forever (symmetry problem) - hence weights must be initialised randomly.

Q.28 — Answer: -0.7709 to -0.7669 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

For the average cross-entropy, $\partial L/\partial w_j = (1/n)\sum (\hat{p}_i - y_i)x_{ij}$ and $\partial L/\partial b = (1/n)\sum (\hat{p}_i - y_i)$, with $\hat{p}_i = \sigma(z_i)$.

i	z_i	$\hat{p}_i$	$\hat{p}_i - y_i$
1	-1	0.2689	0.2689
2	-1	0.2689	0.2689

Gradient: $\partial L/\partial w_1 = -0.2689$, $\partial L/\partial b = 0.2689$.

Update $\theta \leftarrow \theta - \eta \nabla L$: $w_1 = 0.5 - 1 \times (-0.2689) = 0.7689$, $b = -0.5 - 1 \times (0.2689) = -0.7689$.

Required: $b = \mathbf{-0.769}$.

Q.29 — Answer: (D) [MCQ, 2M · Unsupervised Learning > Distance measures]

Compute the measure from $q = (1, 3)$ to every point (cosine column shows similarity, larger = more similar):

Point	coordinates	Manhattan	Euclidean	Chebyshev	cosine similarity
A	(4, 0)	6	4.243	3	0.316
B	(6, 8)	10	7.071	5	0.949
C	(7, 3)	6	6	6	0.664
D	(5, 4)	5	4.123	4	0.84

Under Manhattan (L_1) distance the nearest to q is **point D**.

Note how the answer depends on the measure: Manhattan (L_1) distance → D; Euclidean (L_2) distance → D; Chebyshev (L_∞) distance → A; cosine similarity → B. Cosine similarity looks only at the angle from the origin, L_1 sums coordinate gaps, L_∞ takes the largest gap.

Q.30 — Answer: 25 [NAT, 2M · Unsupervised Learning > k-medoids]

PAM cost = Σ over all points of the absolute distance to the nearest medoid.

Current medoids {7, 25}: distances 7→0, 14→7, 17→8, 22→3, 23→2, 25→0, 27→2, 28→3; cost = **25**.

Q.31 — Answer: (C) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$p = \text{softmax}((1, -1, 0)) = (0.6652, 0.09, 0.2447)$; $y = (0, 0, 1)$.

$\partial L / \partial z = p - y = \mathbf{(0.67, 0.09, -0.76)}$.

Distractors: $y - p = (-0.67, -0.09, 0.76)$ (wrong sign); p alone = $(0.67, 0.09, 0.24)$ (forgot the target); $(p - y) \odot p \odot (1 - p) = (0.15, 0.01, -0.14)$ (spurious σ' -type factor, which cancels for cross-entropy); $-y/p = (0, 0, -4.09)$ is $\partial L / \partial p$, not $\partial L / \partial z$.

Q.32 — Answer: (A), (B), (C), (D) [MSQ, 2M · Unsupervised Learning > Hierarchical clustering]

(A) TRUE. Monotone dendrograms define ultrametrics.

(B) TRUE. The merge height is a maximum over cross pairs that includes this pair.

(C) TRUE. Single (and complete) linkage depend only on the ranks of the distances.

(D) TRUE. A higher cut only merges lower-level clusters.

Q.33 — Answer: (A) [MCQ, 2M · SVM & Decision Trees > SVM: margin & geometry]

For $n^T x - c = 0$, the geometric margin is $\min_i y_i s(n^T x_i - c) / \|n\|$ with orientation $s = \pm 1$ chosen so that the positive class is on the positive side; a negative value means some point is misclassified.

Hyperplane	$\ n\ $	$\min_i$ distance (signed, best orientation)
$x_1 + 2x_2 - 13 = 0$	2.236	0.894
$x_1 + 2x_2 - 14 = 0$	2.236	0.447
$x_1 + x_2 - 8.5 = 0$	1.414	0.354
$x_1 - 2x_2 - 2.5 = 0$	2.236	-3.354 (does not separate)

The largest margin is 0.894, for $\mathbf{x_1 + 2x_2 - 13 = 0}$. Note that a boundary can separate the data perfectly yet pass close to some point — only the minimum distance matters.

Q.34 — Answer: (A) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Expressiveness]

Each unit outputs 1 iff its net input (weighted sum + bias) is > 0 . Evaluate on all four inputs:

x_1	x_2	h_1	h_2	y
0	0	1	0	1
0	1	1	0	1
1	0	1	1	1
1	1	0	0	0

h_1 computes NAND, h_2 computes x_1 AND (NOT x_2); the output column (0,0)→(1,1) is 1, 1, 1, 0, i.e. **NAND**.

Q.35 — Answer: (C) [MCQ, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent: 4 Yes, 6 No $\Rightarrow$ Gini = 0.48.

Wind — Weak: [4+, 4-], Strong: [0+, 2-]; gain = 0.08.

Shape — Round: [2+, 1-], Square: [2+, 3-], Oval: [0+, 2-]; gain = 0.1067.

Colour — Red: [0+, 4-], Green: [1+, 1-], Blue: [3+, 1-]; gain = 0.23.

Age — Young: [2+, 1-], Middle: [0+, 3-], Senior: [2+, 2-]; gain = 0.1467.

The largest Gini gain is for **Colour**, so it is chosen at the root.

Mock Test 43

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	3	12	20

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Unsupervised Learning	7	11
Neural Networks (MLP / Feed-forward)	7	11
Probabilistic & Instance-based Classifiers	7	11
SVM & Decision Trees	7	11
Regression & Model Selection	7	11

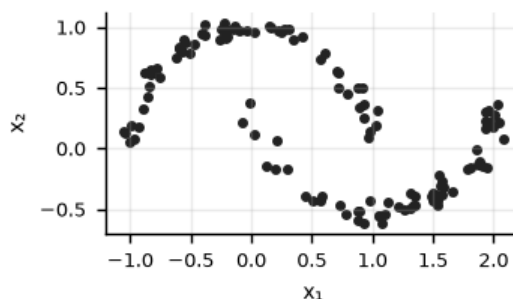
Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MCQ]

k-means with $k = 2$ fails to separate the two interleaving half-moons shown below, even with many random restarts. The main reason is that k-means



- (A) requires class labels to place the centroids
- (B) converges only when the data are standardized
- (C) produces convex clusters separated by linear (Voronoi) boundaries
- (D) cannot be applied to two-dimensional data

Q.2 [1 mark · NAT (numerical answer)]

A model has weights $w = (1, 0.5, -1.5)$. The L2 penalty term $(\lambda/2)\|w\|^2$ with $\lambda = 0.5$ added to the loss equals _____ (rounded off to three decimal places)

Q.3 [1 mark · NAT (numerical answer)]

A naive Bayes classifier has $d = 8$ binary features and $K = 2$ classes. The number of independent parameters (class priors plus class-conditional feature probabilities) to be estimated is _____ (Answer in integer)

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about activation functions is/are CORRECT? (σ denotes the logistic sigmoid.)

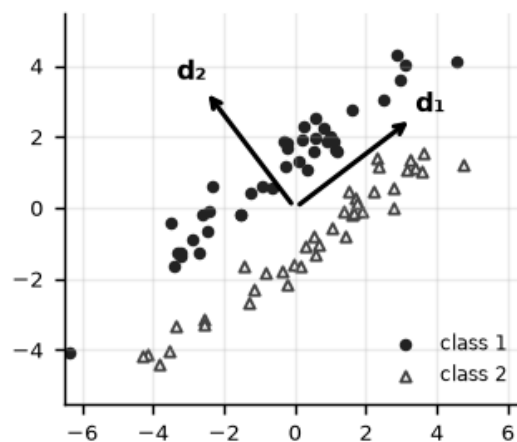
- (A) ReLU is not differentiable at $z = 0$; in practice a subgradient value (0 or 1) is used there.
- (B) For $z > 0$ the derivative of ReLU equals 1, so ReLU does not saturate for positive inputs.
- (C) Dividing the logits by a temperature $T > 1$ makes the softmax distribution more uniform.
- (D) The output of ReLU lies in $[0, \infty)$.

Q.5 [1 mark · NAT (numerical answer)]

The Gaussian kernel is $K(x, z) = \exp(-\|x - z\|^2 / (2\sigma^2))$ with $\sigma = 3$. For $x = (2, -2)$ and $z = (2, -1)$, the value of $K(x, z)$ is _____ (rounded off to three decimal places)

Q.6 [1 mark · MCQ]

Two labelled classes are shown below, together with two orthogonal directions d_1 and d_2 . A single direction is to be selected by (i) PCA (first principal component, ignoring labels) and (ii) Fisher's LDA. Which statement is correct?



- (A) Both PCA and LDA choose d_2
- (B) PCA (1st PC) chooses d_2 ; LDA chooses d_1
- (C) Both PCA and LDA choose d_1
- (D) PCA (1st PC) chooses d_1 ; LDA chooses d_2

Q.7 [1 mark · MSQ (one or more correct)]

Linear regression is trained by minimising the squared-error loss. Which of the following statements about gradient-based training is/are CORRECT?

- (A) Standardising features usually speeds up gradient descent for linear regression.
- (B) The Hessian of $\|Xw - y\|^2$ is $2X^T X$.
- (C) For $J(w) = (1/n)\|Xw - y\|^2$, the gradient descent update is $w \leftarrow w - \eta(2/n)X^T(Xw - y)$.
- (D) One epoch of SGD with batch size 1 on n examples performs n parameter updates.

Q.8 [1 mark · NAT (numerical answer)]

The maximum possible entropy (in bits) of a decision-tree node for a 20-class problem is _____ (rounded off to two decimal places)

Q.9 [1 mark · MSQ (one or more correct)]

Which of the following statements about classifier evaluation is/are CORRECT?

- (A) Under heavy class imbalance, precision-recall curves are usually more informative than ROC curves.
- (B) At fixed TPR and FPR, precision does not depend on the class prevalence.
- (C) The ROC curve plots precision against recall.
- (D) Recall equals $TP / (TP + FP)$.

Q.10 [1 mark · MSQ (one or more correct)]

Consider ridge regression (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_2^2$) and the lasso (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_1$) with $\lambda \geq 0$. Which of the following statements is/are CORRECT?

- (A) Elastic net combines L1 and L2 penalties.
- (B) The lasso penalty is $\lambda \sum \beta_j^2$.
- (C) As $\lambda \rightarrow \infty$, all (penalised) ridge coefficients shrink towards 0.
- (D) The ridge objective $\|y - X\beta\|^2 + \lambda\|\beta\|^2$ ($\lambda > 0$) is non-convex.

Q.11 [1 mark · MSQ (one or more correct)]

Which of the following statements about distance / similarity measures is/are CORRECT?

- (A) Cosine similarity changes when both vectors are multiplied by the same positive constant.
- (B) The Chebyshev distance is always $\geq$ the Euclidean distance.
- (C) The simple matching coefficient and the Jaccard coefficient are always equal for binary vectors.
- (D) Jaccard distance $1 - J(A, B)$ is a metric on finite sets.

Q.12 [1 mark · NAT (numerical answer)]

A binary logistic regression model is given by $\ln[P(y=1|\mathbf{x}) / P(y=0|\mathbf{x})] = -3 + (-0.5)x_1 + (2)x_2$. For the input $\mathbf{x} = (x_1, x_2) = (1, 2)$, the value of $P(y = 0 | \mathbf{x})$ is _____ (rounded off to two decimal places)

Q.13 [1 mark · MSQ (one or more correct)]

Which of the following statements about support vector machines is/are CORRECT?

- (A) In the hard-margin SVM, $\alpha_i > 0$ only for points lying on the hyperplanes $w^T x + b = \pm 1$.
- (B) Logistic regression is influenced by all training points, while the SVM solution depends only on support vectors.
- (C) A hard-margin linear SVM has a feasible solution only if the training data are linearly separable.
- (D) The kernel trick computes inner products in a feature space without explicitly constructing $\phi(x)$.

Q.14 [1 mark · MSQ (one or more correct)]

Which of the following statements about backpropagation is/are CORRECT?

- (A) The gradient of a sum of per-example losses equals the sum of the per-example gradients.
- (B) For a sigmoid output with binary cross-entropy (or softmax with cross-entropy), the error at the output pre-activation is $\hat{y} - y$.
- (C) The gradient with respect to a bias b_j equals δ_j .
- (D) If a ReLU unit's pre-activation is negative for an input, no gradient flows through that unit for that input.

Q.15 [1 mark · NAT (numerical answer)]

A linear regression model with an intercept uses 5 numeric features and a categorical feature *region* with 3 levels (North, South, East). Each categorical feature is encoded by dummy (indicator) variables using a reference level, so that the design matrix has full column rank. The total number of regression coefficients (including the intercept) to be estimated is _____ (Answer in integer)

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

A logistic-regression neuron $p = \sigma(w_1x_1 + w_2x_2 + b)$ has $w = (1, 1)$, $b = 1$ and is trained with binary cross-entropy loss. After one stochastic gradient step on **example 1 only** with learning rate $\eta = 1$, the new value of w_1 is _____ (rounded off to three decimal places)

example	x_1	x_2	y
1	-2	-1	1
2	1	-1	1

Q.17 [2 marks · MSQ (one or more correct)]

Let σ be the logistic sigmoid, and $'$ denote the derivative. At $z = -0.5$ (and $-z = 0.5$), which of the following is/are CORRECT? ($\text{softplus}(z) = \ln(1 + e^z)$.)

(A) $\sigma'(-0.5) \geq 0.25$

(B) $\sigma(-0.5) + \sigma(0.5) = 1$

(C) $\sigma'(-0.5) = \sigma'(0.5)$

(D) $\text{softplus}(-0.5) > \text{ReLU}(-0.5)$

Q.18 [2 marks · NAT (numerical answer)]

Ridge regression without intercept minimises $\|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2$ with $\lambda = 4$, for the data below ($\beta = (\beta_1, \beta_2)^T$). The value of the ridge prediction at $x = (0, -1)$ is _____ (rounded off to three decimal places)

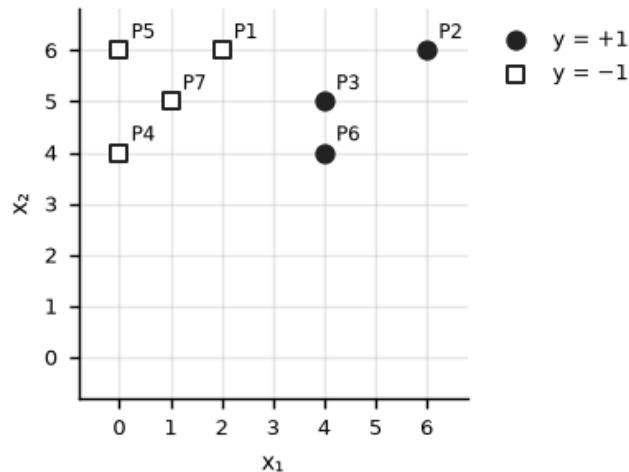
$$X = \begin{bmatrix} -2 & 1 \\ -2 & 3 \\ 3 & 3 \end{bmatrix}$$

$$y = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

Q.19 [2 marks · MSQ (one or more correct)]

A hard-margin linear SVM is trained on the 7 labelled points shown. Which of the following points is/are support vectors of the resulting maximum-margin classifier?

Point	x ₁	x ₂	y
P1	2	6	−1
P2	6	6	+1
P3	4	5	+1
P4	0	4	−1
P5	0	6	−1
P6	4	4	+1
P7	1	5	−1

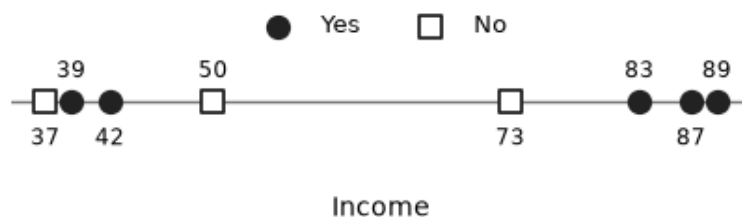


- (A)** P1 **(B)** P3
(C) P6 **(D)** P7

Q.20 [2 marks · NAT (numerical answer)]

The continuous attribute Income and the class label of 8 training records are given below (unsorted). A binary split “Income $\leq t$ ” is chosen to maximise information gain, with t restricted to midpoints between consecutive sorted values. Find the threshold value t chosen for the split “Income $\leq t$ ”. (rounded off to one decimal place)

Income	83	39	73	89	37	42	87	50
Class	Yes	Yes	No	Yes	No	Yes	Yes	No



Q.21 [2 marks · NAT (numerical answer)]

The labelled data below are linearly separable through the origin by the unit vector $u = (-3, 4)/5$ (i.e. $y_i = \text{sign}(u \cdot x_i)$). Take $\gamma = \min_i y_i(u \cdot x_i)$ and $R = \max_i \|x_i\|$. The perceptron convergence theorem (no bias, $w_0 = 0$, $\eta = 1$) bounds the number of mistakes by $(R/\gamma)^2$. The value of this bound is _____ (rounded off to two decimal places)

i	1	2	3	4	5
x_i	$(-2, 3)$	$(1, -1)$	$(0, 4)$	$(1, 3)$	$(2, 4)$
y_i	1	-1	1	1	1

Q.22 [2 marks · NAT (numerical answer)]

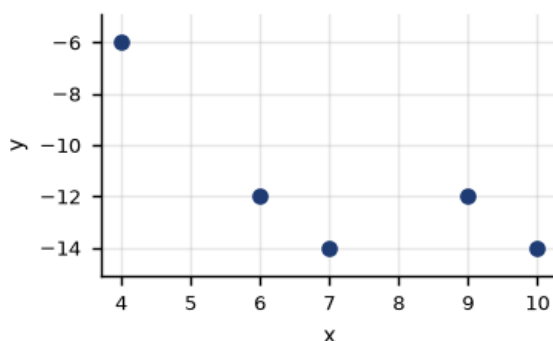
Logistic regression $P(y = 1 \mid \mathbf{x}) = \sigma(w_1x_1 + w_2x_2 + b)$ is trained by **batch** gradient descent on the **average** cross-entropy loss over the 3 examples below, with learning rate $\eta = 0.2$. Starting from $w_1 = 1$, $w_2 = -0.5$, $b = 0.5$, the value of w_1 after ONE gradient-descent update is _____ (rounded off to three decimal places)

i	x_1	x_2	y
1	0	3	1
2	0	1	1
3	2	1	1

Q.23 [2 marks · NAT (numerical answer)]

A simple linear regression (with intercept) is fitted by least squares to the 5 points below. Using leave-one-out cross-validation, the leave-one-out residual $y_i - \hat{y}_{(-i)}(x_i)$ for the point $(6, -12)$ — where $\hat{y}_{(-i)}$ is the line fitted without that point — is _____ (rounded off to two decimal places)

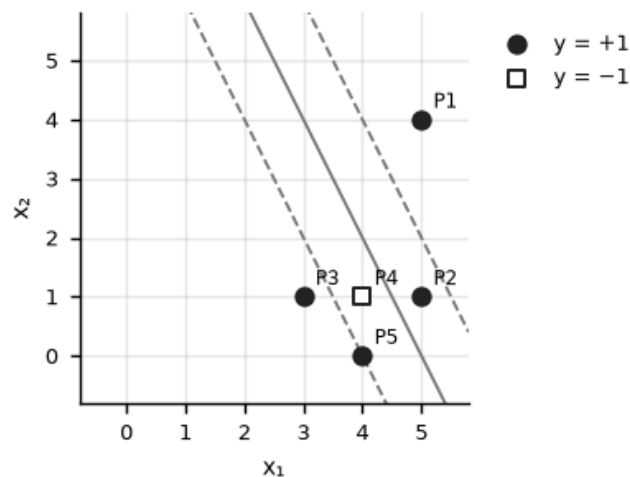
x	4	6	7	9	10
y	-6	-12	-14	-12	-14



Q.24 [2 marks · MSQ (one or more correct)]

A soft-margin linear SVM has learned $f(x) = x_1 + 0.5x_2 - 5$. The training points are listed below and plotted (boundary solid, margins $f = \pm 1$ dashed). With ξ_i the optimal slack of point i , which of the following statements is/are CORRECT?

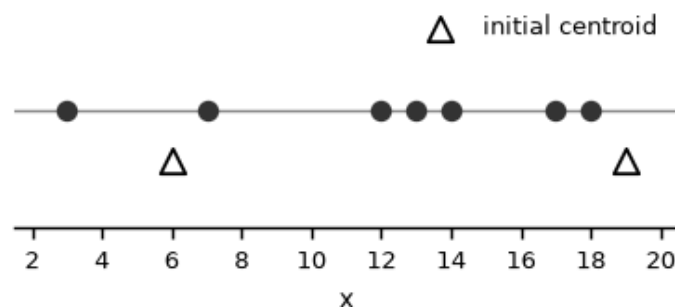
Point	x_1	x_2	y
P1	5	4	+1
P2	5	1	+1
P3	3	1	+1
P4	4	1	-1
P5	4	0	+1



- (A) P3 is misclassified ($\xi_3 > 1$).
- (B) Exactly 5 of the points have $\xi_i > 0$.
- (C) P1 has slack $\xi_1 = 0$.
- (D) $\sum \xi_i = 5.5$.

Q.25 [2 marks · NAT (numerical answer)]

The 1-D dataset $\{3, 7, 12, 13, 14, 17, 18\}$ is clustered with k-means ($k = 2$) using initial centroids 6 and 19 (shown as triangles). Lloyd's iterations (assign to nearest centroid, then recompute means) are run until convergence. Then the final within-cluster sum of squared errors (SSE) is _____ (rounded off to two decimal places)

**Q.26 [2 marks · NAT (numerical answer)]**

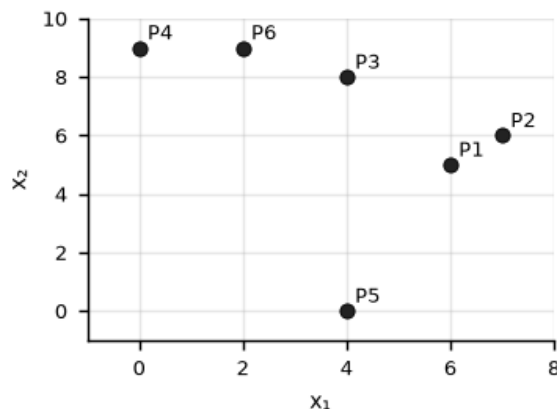
Consider the model $y = wx + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$, $\sigma^2 = 2$, and $w \sim N(0, \tau^2)$ with $\tau^2 = 0.5$. Given the four observations below, the MAP estimate of w is _____ (rounded off to three decimal places)

x	0	1	-1	-1
y	3	4	0	1

Q.27 [2 marks · NAT (numerical answer)]

Single-linkage agglomerative clustering with Manhattan distance is applied to the six points below. Then the height at which the **last** merge (forming a single cluster) occurs is _____. (Answer in integer)

Point	P1	P2	P3	P4	P5	P6
x_1	6	7	4	0	4	2
x_2	5	6	8	9	0	9

**Q.28 [2 marks · MSQ (one or more correct)]**

Which of the following statements about loss functions and optimisation for training neural networks is/are CORRECT?

- (A) For classification with a sigmoid output, MSE loss is convex in the weights and is therefore preferred over cross-entropy.
- (B) Gradient descent with a fixed learning rate decreases the training loss at every step regardless of η .
- (C) Momentum accumulates an exponentially decaying average of past gradients, which damps oscillations across steep directions.
- (D) Mini-batch gradient descent with batch size $B = N$ is the same as single-example SGD.

Q.29 [2 marks · NAT (numerical answer)]

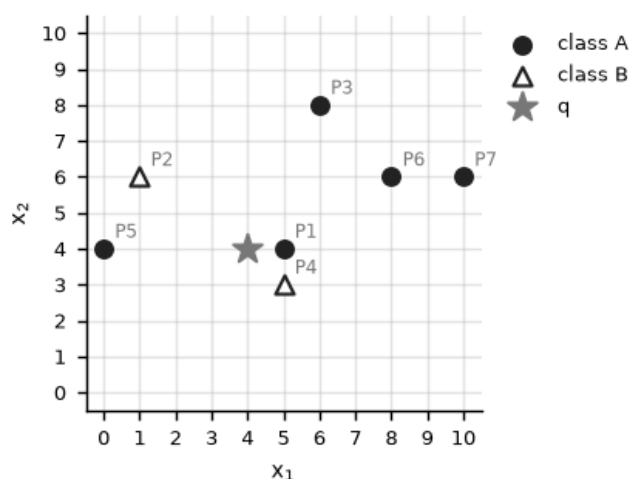
Two clusterings U and V of 6 points are given below (cluster names are arbitrary). Then the number of point pairs placed in the same cluster by **both** clusterings is _____ (Answer in integer)

Point	P1	P2	P3	P4	P5	P6
Clustering U	β	α	α	β	α	β
Clustering V	2	1	3	2	1	2

Q.30 [2 marks · MCQ]

Consider the training set below. A 3-NN classifier (majority vote) is applied to the query point $\mathbf{q} = (4, 4)$, once with Euclidean distance and once with Manhattan distance. The predicted classes are

Point	x_1	x_2	class
P1	5	4	A
P2	1	6	B
P3	6	8	A
P4	5	3	B
P5	0	4	A
P6	8	6	A
P7	10	6	A



- (A) Euclidean: A; Manhattan: B
(C) Euclidean: B; Manhattan: B

- (B) Euclidean: B; Manhattan: A
(D) Euclidean: A; Manhattan: A

Q.31 [2 marks · NAT (numerical answer)]

For a two-class problem, the class means are $\mu_0 = (-2, -2)^T$ and $\mu_1 = (-1, 2)^T$ and the pooled within-class scatter matrix S_W is given below. Fisher's linear discriminant uses $\mathbf{w} \propto S_W^{-1}(\mu_1 - \mu_0)$. Then the separation of the projected class means, $\mathbf{w}^T(\mu_1 - \mu_0)$, with $\mathbf{w} = S_W^{-1}(\mu_1 - \mu_0)$ is _____ (rounded off to two decimal places)

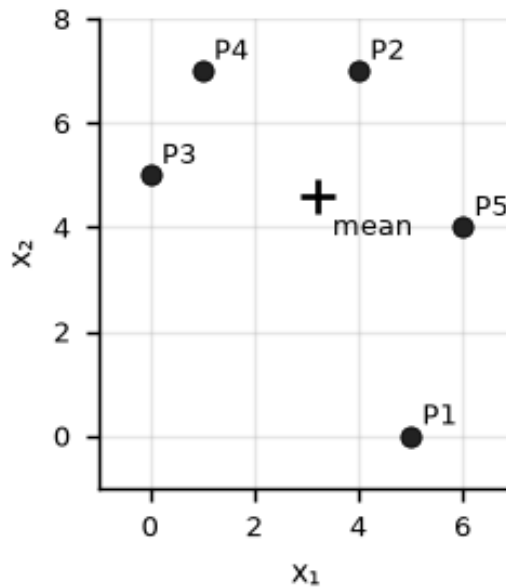
$S_W =$

4	2
2	4

Q.32 [2 marks · NAT (numerical answer)]

PCA is applied to the five 2-D points below (data are mean-centered first; sample covariance uses divisor $n - 1$). If each point is reconstructed from its PC1 score only ($\hat{x} = \bar{x} + z_1 v_1$), the squared reconstruction error $\|x - \hat{x}\|^2$ of P2 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5
x_1	5	4	0	1	6
x_2	0	7	5	7	4

**Q.33 [2 marks · NAT (numerical answer)]**

A linear model $\hat{y} = w_1 x_1 + w_2 x_2 + b$ is trained by stochastic gradient descent (batch size 1) on the per-example loss $\frac{1}{2}(\hat{y} - y)^2$ with learning rate $\eta = 0.05$, starting from $w = (1, 0)$, $b = 1$. The two examples below are processed once, in the given order. After both updates, the value of w_2 is _____ (rounded off to three decimal places)

example	x_1	x_2	y
1	-1	-2	6
2	-1	3	3

Q.34 [2 marks · NAT (numerical answer)]

For the RBF kernel $K(x, z) = \exp(-\gamma \|x - z\|^2)$ with $\gamma = 1$, let ϕ be its (infinite-dimensional) feature map. For $x = (1, 0)$ and $z = (-1, 2)$, the Euclidean distance $\|\phi(x) - \phi(z)\|$ in feature space is _____ (rounded off to two decimal places)

Q.35 [2 marks · MSQ (one or more correct)]

For the confusion matrix below, which of the following statements is/are CORRECT? (values rounded to 2 decimals)

	Actual +	Actual -
Predicted +	22	76
Predicted -	32	45

- (A) The F1-score is less than the arithmetic mean of precision and recall.
- (B) The specificity (TNR) is 0.37.
- (C) The negative predictive value is 0.58.
- (D) The recall (sensitivity, TPR) is 0.22.

— END OF QUESTION PAPER —

Mock Test 43 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MCQ	1	(C)	19	MSQ	2	(A), (B)
2	NAT	1	0.873 to 0.877	20	NAT	2	77.99 to 78.01
3	NAT	1	17	21	NAT	2	10.1837 to 10.2245
4	MSQ	1	(A), (B), (C), (D)	22	NAT	2	1.0139 to 1.0179
5	NAT	1	0.944 to 0.948	23	NAT	2	−2.3086 to −2.2629
6	MCQ	1	(D)	24	MSQ	2	(A), (C), (D)
7	MSQ	1	(A), (B), (C), (D)	25	NAT	2	34.79 to 34.81
8	NAT	1	4.3119 to 4.3319	26	NAT	2	0.4266 to 0.4306
9	MSQ	1	(A)	27	NAT	2	7
10	MSQ	1	(A), (C)	28	MSQ	2	(C)
11	MSQ	1	(D)	29	NAT	2	4
12	NAT	1	0.3675 to 0.3875	30	MCQ	2	(B)
13	MSQ	1	(A), (B), (C), (D)	31	NAT	2	4.3233 to 4.3433
14	MSQ	1	(A), (B), (C), (D)	32	NAT	2	4.5174 to 4.5374
15	NAT	1	8	33	NAT	2	0.028 to 0.032
16	NAT	2	−0.7636 to −0.7596	34	NAT	2	1.404 to 1.424
17	MSQ	2	(B), (C), (D)	35	MSQ	2	(A), (B), (C)
18	NAT	2	−0.6275 to −0.6215				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Unsupervised Learning					11
Neural Networks (MLP / Feed-forward)					11
Probabilistic & Instance-based Classifiers					11
SVM & Decision Trees					11
Regression & Model Selection					11
Total					55

Mock Test 43 — Detailed Solutions

Q.1 — Answer: (C) [MCQ, 1M · Unsupervised Learning > k-means]

Points are assigned to the nearest centroid, so each k-means cluster is a convex Voronoi cell and two clusters are separated by the perpendicular bisector of their centroids. Non-convex shapes such as rings or moons cannot be separated by such a boundary.

k-means converges on raw data, works in any dimension, and is unsupervised (no labels).

Q.2 — Answer: 0.873 to 0.877 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$$\|w\|^2 = 1 + 0.25 + 2.25 = 3.5; (\lambda/2)\|w\|^2 = 0.25 \times 3.5 = \mathbf{0.875}.$$

Q.3 — Answer: 17 [NAT, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Priors: $K - 1 = 1$. Each binary feature needs 1 parameter $P(x_j=1|c)$ per class: $Kd = 16$.

Total = $16 + 1 = \mathbf{17}$.

Q.4 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

(A) TRUE. Left and right derivatives are 0 and 1.

(B) TRUE. Gradient passes unchanged through active ReLUs.

(C) TRUE. Differences between logits shrink.

(D) TRUE. $\max(0, z)$ is never negative and is unbounded above.

Q.5 — Answer: 0.944 to 0.948 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

$$\|x - z\|^2 = 1; 2\sigma^2 = 18.$$

$$K = \exp(-1/18) = \mathbf{0.946}.$$

Note: $K \in (0, 1]$, $K(x, x) = 1$, and K decreases as the points move apart or as γ grows.

Q.6 — Answer: (D) [MCQ, 1M · Unsupervised Learning > PCA & dimensionality reduction]

PCA is unsupervised: it picks the direction of maximum total variance, which is along the long axis of the clouds (d_1).

LDA is supervised: it maximizes between-class separation relative to within-class scatter, so it picks the direction across which the class means differ (d_2).

Answer: **PCA (1st PC) chooses d_1 ; LDA chooses d_2** . Projecting on the PCA direction would mix the two classes completely.

Q.7 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Gradient descent]

(A) TRUE. It reduces the condition number of $X^T X$, making contours rounder.

(B) TRUE. Differentiate $2X^T(Xw - y)$ once more.

(C) TRUE. $\nabla J = (2/n)X^T(Xw - y)$.

(D) TRUE. One update per example.

Q.8 — Answer: 4.3119 to 4.3319 [NAT, 1M · SVM & Decision Trees > Decision trees: impurity measures]

Entropy is maximised by the uniform distribution $p_k = 1/20$: $H_{\max} = \log_2 20 = \mathbf{4.32}$ bits.

Q.9 — Answer: (A) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

(A) TRUE. FPR stays small when negatives are abundant, hiding many false positives.

(B) FALSE. It does.

(C) FALSE. It plots TPR against FPR.

(D) FALSE. That is precision.

Q.10 — Answer: (A), (C) [MSQ, 1M · Regression & Model Selection > Ridge & Lasso]

(A) TRUE. Penalty $\lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2$.

(B) FALSE. That is the ridge (L2) penalty.

(C) TRUE. The penalty dominates the RSS.

(D) FALSE. It is strictly convex.

Q.11 — Answer: (D) [MSQ, 1M · Unsupervised Learning > Distance measures]

- (A) FALSE. Scaling does not change the angle.
 (B) FALSE. $L_\infty \leq L_2$.
 (C) FALSE. They differ whenever there are 0-0 matches.
 (D) TRUE. It satisfies the triangle inequality.

Q.12 — Answer: 0.3675 to 0.3875 [NAT, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

$z = w_0 + w_1x_1 + w_2x_2 = -3 + (-0.5)(1) + (2)(2) = 0.5$ (this is the log-odds).
 $P(y = 1 | \mathbf{x}) = \sigma(z) = 1/(1 + e^{-0.5}) = 0.6225$.
 $P(y = 0 | \mathbf{x}) = 1 - \sigma(z) = \sigma(-z) = \mathbf{0.38}$.

Q.13 — Answer: (A), (B), (C), (D) [MSQ, 1M · SVM & Decision Trees > SVM: dual & support vectors]

- (A) TRUE. Complementary slackness: $\alpha_i[y_i(w^Tx_i + b) - 1] = 0$.
 (B) TRUE. Log-loss gradient is non-zero for every point.
 (C) TRUE. Otherwise the constraints $y_i(w^Tx_i + b) \geq 1$ cannot all hold.
 (D) TRUE. $K(x, z) = \phi(x)^T\phi(z)$ is evaluated directly.

Q.14 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Backpropagation]

- (A) TRUE. Differentiation is linear.
 (B) TRUE. The σ' factor cancels with the log-loss derivative.
 (C) TRUE. a_j depends on b_j with coefficient 1.
 (D) TRUE. $\text{ReLU}'(a) = 0$ for $a < 0$.

Q.15 — Answer: 8 [NAT, 1M · Regression & Model Selection > Multiple linear regression]

A categorical feature with k levels needs $k - 1$ dummies (one level is the reference absorbed by the intercept; using all k dummies with an intercept gives perfect multicollinearity).
 Parameters = 1 (intercept) + 5 (numeric) + (3 - 1) = 8.
 Answer: **8**.

Q.16 — Answer: -0.7636 to -0.7596 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

For σ output with binary cross-entropy $L = -[y \ln p + (1 - y) \ln(1 - p)]$, $\partial L / \partial a = p - y$, so $\partial L / \partial w_j = (p - y)x_j$ and $\partial L / \partial b = p - y$.
 Example 1: $a = -2$, $p = 0.1192$, $p - y = -0.8808$.
 Gradient for w_1 : 1.7616.
 $w_1 \leftarrow 1 - 1 \times (1.7616) = \mathbf{-0.7616}$.
 Note there is no σ' factor: it cancels with the derivative of the log loss.

Q.17 — Answer: (B), (C), (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Activation functions]

$\sigma(-0.5) = 0.3775$, $\tanh(-0.5) = -0.4621$.
 (A) FALSE. $\sigma'(-0.5) = 0.235 < 0.25$ (max only at 0).
 (B) TRUE. $\sigma(-z) = 1 - \sigma(z)$.
 (C) TRUE. σ' is an even function.
 (D) TRUE. $\text{softplus} = 0.4741 > 0$; $\ln(1 + e^z) > \max(0, z)$ always.

Q.18 — Answer: -0.6275 to -0.6215 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

$\beta_{\text{ridge}} = (X^TX + \lambda I)^{-1}X^Ty$ with $\lambda = 4$.

$$X^TX = \begin{bmatrix} 17 & 1 \\ 1 & 19 \end{bmatrix}$$

$$X^TX + \lambda I = \begin{bmatrix} 21 & 1 \\ 1 & 23 \end{bmatrix}$$

$$X^Ty = \begin{bmatrix} -7 \end{bmatrix}$$

$$\boxed{14}$$

$$\det = 21 \times 23 - (1)^2 = 482; (X^T X + \lambda I)^{-1} = (1/482)[[23, -1], [-1, 21]].$$

$$\beta = (-0.3631, 0.6245).$$

$$\hat{y} = 0 \times -0.3631 + -1 \times 0.6245 = -0.6245.$$

Answer: **-0.624**.

Q.19 — Answer: (A), (B) [MSQ, 2M · SVM & Decision Trees > SVM: dual & support vectors]

The closest pair of opposite-class points is $P1 = (2, 6)$ and $P3 = (4, 5)$, Their perpendicular bisector separates the classes and no other point comes closer to it, so Hence $w \parallel (x_+ - x_-)$ and the support vectors are $P1, P3$.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (4/5, -2/5)$, $b = -1/5$; $\|w\| = 0.8944$, margin $2/\|w\| = 2.2361$.

Point	P1	P2	P3	P4	P5	P6	P7
$y(w^T x + b)$	1	2.2	1	1.8	2.6	1.4	1.4

Support vectors are exactly the points with $y_i(w^T x_i + b) = 1$ ($\alpha_i > 0$): **P1, P3**.

Dual check: $\alpha = (\alpha_1 = 2/5, \alpha_3 = 2/5)$ satisfy $\sum \alpha_i y_i = 0$ and $\sum \alpha_i y_i x_i = w$.

Q.20 — Answer: 77.99 to 78.01 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Sort the values: 37(N), 39(Y), 42(Y), 50(N), 73(N), 83(Y), 87(Y), 89(Y).

Parent [5 Yes, 3 No], $H = 0.9544$. The optimal threshold always lies where the class changes between neighbours (Fayyad-Irani), giving 3 candidates:

Threshold	Left ($\leq t$) [Yes, No]	Right [Yes, No]	IG
38	[0, 1]	[5, 2]	0.1992
46	[2, 1]	[3, 2]	0.0032
78	[2, 3]	[3, 0]	0.3476

Best: $t = 78$ with $IG = 0.3476$. Answer: **78**.

Q.21 — Answer: 10.1837 to 10.2245 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Unit separator $u = (-3, 4)/5$. Margins $y_i(u \cdot x_i)$: 3.6, 1.4, 3.2, 1.8, 2 $\rightarrow \gamma = \min = 1.4$.

Norms $\|x_i\|$: 3.606, 1.414, 4, 3.162, 4.472 $\rightarrow R = \max = 4.4721$.

Novikoff bound: number of mistakes $\leq (R/\gamma)^2 = (4.4721/1.4)^2 = \mathbf{10.2}$.

Common mistakes: forgetting to normalise u (γ must be measured with $\|u\| = 1$), or using R/γ without squaring.

Q.22 — Answer: 1.0139 to 1.0179 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

For the average cross-entropy, $\partial L / \partial w_j = (1/n) \sum (\hat{p}_i - y_i) x_{ij}$ and $\partial L / \partial b = (1/n) \sum (\hat{p}_i - y_i)$, with $\hat{p}_i = \sigma(z_i)$.

i	z_i	$\hat{p}_i$	$\hat{p}_i - y_i$
1	-1	0.2689	-0.7311
2	0	0.5	-0.5
3	2	0.8808	-0.1192

Gradient: $\partial L / \partial w_1 = -0.0795$, $\partial L / \partial w_2 = -0.9375$, $\partial L / \partial b = -0.4501$.

Update $\theta \leftarrow \theta - \eta \nabla L$: $w_1 = 1 - 0.2 \times (-0.0795) = 1.0159$, $w_2 = -0.5 - 0.2 \times (-0.9375) = -0.3125$, $b = 0.5 - 0.2 \times (-0.4501) = 0.59$.

Required: $w_1 = \mathbf{1.016}$.

Q.23 — Answer: -2.3086 to -2.2629 [NAT, 2M · Regression & Model Selection > Cross-validation]

$\bar{x} = 7.2$, $\bar{y} = -11.6$.

i	1	2	3	4	5
$x_i - \bar{x}$	-3.2	-1.2	-0.2	1.8	2.8
$y_i - \bar{y}$	5.6	-0.4	-2.4	-0.4	-2.4

$(x_i - \bar{x})(y_i - \bar{y})$	-17.92	0.48	0.48	-0.72	-6.72
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$$S_{xx} = \sum (x_i - \bar{x})^2 = 22.8, S_{xy} = -24.4.$$

$$\beta_1 = S_{xy}/S_{xx} = -1.0702, \beta_0 = \bar{y} - \beta_1 \bar{x} = -3.8947.$$

For a linear smoother, the leave-one-out residual is $e_{(i)} = e_i / (1 - h_{ii})$, with $h_{ii} = 1/n + (x_i - \bar{x})^2 / S_{xx}$.

x_i	4	6	7	9	10
e_i	2.175	-1.684	-2.614	1.526	0.596
h_{ii}	0.649	0.263	0.202	0.342	0.544
$e_i / (1 - h_{ii})$	6.2	-2.286	-3.275	2.32	1.308

Answer: **-2.29** (verified by explicitly refitting without the point).

Q.24 — Answer: (A), (C), (D) [MSQ, 2M · SVM & Decision Trees > SVM: soft margin & hinge loss]

$$\xi_i = \max(0, 1 - y_i f(x_i)).$$

Point	$y f(x)$	ξ
P1	2	0
P2	0.5	0.5
P3	-1.5	2.5
P4	0.5	0.5
P5	-1	2

(A) TRUE. $y f = -1.5 \Rightarrow \xi = 2.5$.

(B) FALSE. The correct count is 4.

(C) TRUE. $y f = 2 \Rightarrow \xi = 0$.

(D) TRUE. Sum of the individual slacks.

Q.25 — Answer: 34.79 to 34.81 [NAT, 2M · Unsupervised Learning > k-means]

Data: {3, 7, 12, 13, 14, 17, 18}; initial centroids $c_1 = 6, c_2 = 19$.

Iter	centroids used	cluster 1	cluster 2	new centroids
1	6, 19	{3, 7, 12}	{13, 14, 17, 18}	7.33, 15.5
2	7.33, 15.5	{3, 7}	{12, 13, 14, 17, 18}	5, 14.8
3	5, 14.8	{3, 7}	{12, 13, 14, 17, 18}	5, 14.8

In the last row the centroids do not change, so the algorithm has converged: final centroids 5 and 14.8.

$$SSE = (3 - 5)^2 + (7 - 5)^2 + (12 - 14.8)^2 + (13 - 14.8)^2 + (14 - 14.8)^2 + (17 - 14.8)^2 + (18 - 14.8)^2 = \mathbf{34.8}.$$

Q.26 — Answer: 0.4266 to 0.4306 [NAT, 2M · Regression & Model Selection > MLE & MAP view]

$$-\ln p(w | \text{data}) = (1/(2\sigma^2)) \sum (y_i - wx_i)^2 + w^2/(2\tau^2) + \text{const.}$$

Multiplying by $2\sigma^2$: $\sum (y_i - wx_i)^2 + (\sigma^2/\tau^2)w^2$ — ridge with $\lambda = \sigma^2/\tau^2 = 4$.

$$\sum x_i^2 = 3, \sum x_i y_i = 3; \hat{w}_{\text{MAP}} = \sum xy / (\sum x^2 + \lambda) = 3/(3 + 4) = 0.4286.$$

Answer: **0.429**.

Q.27 — Answer: 7 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Single-linkage merge heights are exactly the edge weights of a minimum spanning tree (Kruskal's algorithm adds the shortest edge joining two different components = single-linkage merge).

MST edges (Prim's algorithm): P1-P2 (2); P1-P3 (5); P3-P6 (3); P6-P4 (2); P1-P5 (7).

Sorted merge heights: 2, 2, 3, 5, 7.

The last single-linkage merge happens at the largest MST edge: 7.

Q.28 — Answer: (C) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

(A) FALSE. $MSE \cdot \sigma$ is non-convex; CE is the standard choice.

(B) FALSE. Large η can increase the loss.

(C) TRUE. Opposite-sign components cancel.

(D) FALSE. $B = N$ is full-batch gradient descent.

Q.29 — Answer: 4 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

Total pairs = $C(6,2) = 15$.

a = pairs together in both = 4; b = pairs separated in both = 9; pairs together in U only = 2; together in V only = 0.

Rand index = $(a + b)/C(n,2) = (4 + 9)/15 = 0.8667$.

Answer: **4**. (Cluster names do not matter: only whether pairs are together.)

Q.30 — Answer: (B) [MCQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Distances from q :

Point	class	Euclidean d	Manhattan d
P1	A	1	1
P2	B	3.606	5
P3	A	4.472	6
P4	B	1.414	2
P5	A	4	4
P6	A	4.472	6
P7	A	6.325	8

Euclidean 3 nearest: P1(A), P4(B), P2(B) $\Rightarrow$ **B**.

Manhattan 3 nearest: P1(A), P4(B), P5(A) $\Rightarrow$ **A**.

Answer: **Euclidean: B; Manhattan: A**. The two metrics disagree because L_1 penalises diagonal offsets more than L_2 , changing the neighbour set.

Q.31 — Answer: 4.3233 to 4.3433 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$\mu_1 - \mu_0 = (1, 4)^T$; $\det S_W = 12$.

$$S_W^{-1} = \begin{bmatrix} 0.3333 & -0.1667 \\ -0.1667 & 0.3333 \end{bmatrix}$$

$w = S_W^{-1}(\mu_1 - \mu_0) = (-0.3333, 1.1667)^T$.

$w^T(\mu_1 - \mu_0) = (\mu_1 - \mu_0)^T S_W^{-1}(\mu_1 - \mu_0) = \mathbf{4.33}$ (always ≥ 0 since S_W is positive definite).

Note: using $\mu_1 - \mu_0$ without S_W^{-1} is correct only when $S_W \propto I$.

Q.32 — Answer: 4.5174 to 4.5374 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

Mean $\bar{x} = (3.2, 4.6)$. Centered data rows: P1: (1.8, -4.6), P2: (0.8, 2.4), P3: (-3.2, 0.4), P4: (-2.2, 2.4), P5: (2.8, -0.6).

Sample covariance $S = (1/(n-1)) \sum (x - \bar{x})(x - \bar{x})^T = [[6.7, -3.65], [-3.65, 8.3]]$.

Eigenvalues: $\lambda_1 = 11.2366$, $\lambda_2 = 3.7634$; unit PC1 direction $v_1 = (0.6269, -0.7791)$.

Score $z_1 = v_1 \cdot (x - \bar{x}) = 0.6269 \cdot 0.8 + -0.7791 \cdot 2.4 = -1.3684$.

$\|x - \bar{x}\|^2 = 6.4$; error = $\|x - \bar{x}\|^2 - z_1^2 = 6.4 - 1.8726 = \mathbf{4.53}$ (Pythagoras: the residual is the PC2 component).

Q.33 — Answer: 0.028 to 0.032 [NAT, 2M · Regression & Model Selection > Gradient descent]

Per-example loss $\square = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial \square / \partial w = (\hat{y} - y)x$, $\partial \square / \partial b = (\hat{y} - y)$. Update after each example.

Example 1: $\hat{y} = 0$, error $\hat{y} - y = -6$; $\nabla_w = (6, 12)$, $\nabla_b = -6 \Rightarrow w = (0.7, -0.6)$, $b = 1.3$.

Example 2: $\hat{y} = -1.2$, error $\hat{y} - y = -4.2$; $\nabla_w = (4.2, -12.6)$, $\nabla_b = -4.2 \Rightarrow w = (0.49, 0.03)$, $b = 1.51$.

Answer: **0.03**.

Q.34 — Answer: 1.404 to 1.424 [NAT, 2M · SVM & Decision Trees > SVM: kernels]

$\|\varphi(x) - \varphi(z)\|^2 = K(x, x) + K(z, z) - 2K(x, z) = 2 - 2K(x, z)$ for the RBF kernel.

$\|x - z\|^2 = 8$, $K(x, z) = \exp(-8) = 0.0003$.

Distance = $\sqrt{2 - 2 \times 0.0003} = \mathbf{1.41}$ (always below $\sqrt{2} \approx 1.414$).

Q.35 — Answer: (A), (B), (C) [MSQ, 2M · Probabilistic & Instance-based Classifiers › Evaluation metrics]

TP = 22, FN = 32, FP = 76, TN = 45.

(A) TRUE. The harmonic mean is $\leq$ the arithmetic mean, with equality only if $P = R$ (not the case here).

(B) TRUE. specificity (TNR) = $TN/(TN + FP) = 0.3719$.

(C) TRUE. negative predictive value = $TN/(TN + FN) = 0.5844$.

(D) FALSE. recall (sensitivity, TPR) = $TP/(TP + FN) = 0.4074$, not 0.22.

Mock Test 44

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	6	7	22

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Regression & Model Selection	7	11
Unsupervised Learning	7	11
SVM & Decision Trees	7	11
Neural Networks (MLP / Feed-forward)	7	11
Probabilistic & Instance-based Classifiers	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MSQ (one or more correct)]

Regarding the bias-variance trade-off in supervised regression with squared-error loss, which of the following statements is/are CORRECT?

- (A) Increasing model flexibility typically decreases bias.
- (B) k-NN regression with $k = 1$ has low bias and high variance.
- (C) The bias of $\hat{f}$ at x_0 is $E[\hat{f}(x_0)] - f(x_0)$, the expectation being over training sets.
- (D) Increasing model flexibility typically increases variance.

Q.2 [1 mark · MSQ (one or more correct)]

Which of the following statements about clustering evaluation measures is/are CORRECT?

- (A) A silhouette close to 1 means the point is much closer to its own cluster than to the nearest other cluster.
- (B) Purity is an external validation measure: it needs ground-truth class labels.
- (C) The average silhouette width can be used to choose k by picking the k with the largest value.
- (D) Purity is an internal measure that needs no class labels.

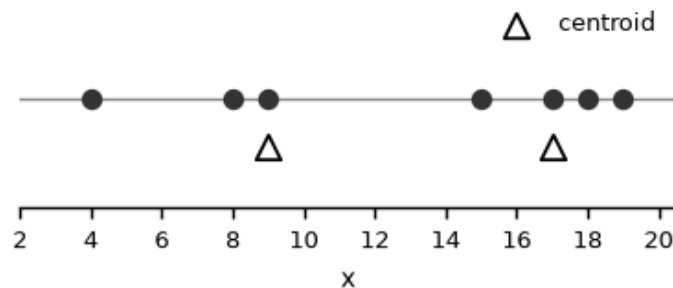
Q.3 [1 mark · MSQ (one or more correct)]

Which of the following statements about bagging, random forests and boosting is/are CORRECT?

- (A) A random forest with $m = p$ (all features considered at each split) is equivalent to bagged trees.
- (B) In AdaBoost, the weights of misclassified examples increase for the next round.
- (C) Bagging mainly reduces the variance of the base learner.
- (D) A weak learner with weighted error 0.3 receives a negative α in AdaBoost.

Q.4 [1 mark · NAT (numerical answer)]

One-dimensional data $\{4, 8, 9, 15, 17, 18, 19\}$ and $k = 2$ centroids at 9, 17 are given. Each point is assigned to its nearest centroid (centroids are **not** updated). The resulting within-cluster sum of squared errors $\sum (x_i - c_{c(i)})^2$ is _____ (rounded off to two decimal places)

**Q.5 [1 mark · NAT (numerical answer)]**

Inverted dropout with keep probability $p = 0.9$ is applied during training. A hidden unit with activation $h = 4$ that is **kept** passes on the value _____ (rounded off to two decimal places)

Q.6 [1 mark · NAT (numerical answer)]

In a spam filter, a false negative costs 5 units and a false positive costs 2 units (correct decisions cost 0). The model outputs $p = P(\text{positive} | x)$, i.e. the probability that the email is spam. The Bayes-optimal rule predicts positive iff p exceeds the threshold _____ (rounded off to two decimal places)

Q.7 [1 mark · NAT (numerical answer)]

For points uniform in the unit hypercube $[0, 1]^{100}$, the fraction of the volume lying within distance 0.01 of the boundary (i.e. outside the inner cube $[0.01, 0.99]^{100}$) is _____ (rounded off to two decimal places)

Q.8 [1 mark · MCQ]

Which loss function is the most robust to a few large outliers in the target of a regression problem?

- (A) Squared error divided by n
- (B) Squared error $(y - \hat{y})^2$
- (C) Exponential error $e^{|y - \hat{y}|}$
- (D) Absolute error $|y - \hat{y}|$

Q.9 [1 mark · NAT (numerical answer)]

For a data set, $\bar{x} = 30$, $\bar{y} = 113$, $s_x = 3$, $s_y = 10$ and the sample correlation between x and y is $r = -0.45$. Using the least-squares regression line of y on x , the predicted value of y at $x = 25$ is _____ (rounded off to two decimal places)

Q.10 [1 mark · NAT (numerical answer)]

A node of a classification tree contains 19 training records: 12 of class C_1 , 7 of class C_2 . The Gini index of this node is _____ (rounded off to two decimal places)

Q.11 [1 mark · MCQ]

Cost-complexity pruning of a classification tree produced the nested subtrees $T1 \supset T2 \supset T3 \supset T4$ below. For complexity parameter $\alpha = 0.02$, which subtree minimises $R_\alpha(T) = R(T) + \alpha|T|$?

Subtree	T1	T2	T3	T4
Leaves $ T $	14	12	8	4
Training error $R(T)$	0.035	0.065	0.095	0.105

- (A) T1
- (B) T2
- (C) T3
- (D) T4

Q.12 [1 mark · MSQ (one or more correct)]

Which of the following statements about the perceptron is/are CORRECT?

- (A) A two-layer network of threshold units can compute XOR.
- (B) The decision boundary of a perceptron, $w \cdot x + b = 0$, is a hyperplane.
- (C) The perceptron is not trained by gradient descent on the 0-1 loss, because that loss has zero gradient almost everywhere.
- (D) Each mistake of the perceptron increases $w \cdot w^*$ by at least γ , where w^* is a unit-norm separator with margin γ ($\eta = 1$).

Q.13 [1 mark · MSQ (one or more correct)]

Which of the following statements about logistic regression is/are CORRECT?

- (A) $\sigma(0) = 0$.
- (B) The decision boundaries between pairs of classes in softmax (multinomial logistic) regression are linear in x .
- (C) A negative coefficient β_j means the predicted probability of $y = 1$ decreases as x_j increases (others fixed).
- (D) Adding an L2 penalty $\lambda \|w\|^2$ ($\lambda > 0$) keeps the logistic-regression weights finite even on separable data.

Q.14 [1 mark · NAT (numerical answer)]

For logits $z = (-0.5, 0, 3, 1)$, the softmax probability $p_1 = \exp(z_1) / \sum_j \exp(z_j)$ is _____ (rounded off to 4 decimal places)

Q.15 [1 mark · MSQ (one or more correct)]

Which of the following statements about naive Bayes classifiers is/are CORRECT?

- (A) Gaussian naive Bayes corresponds to Gaussian class-conditionals with diagonal covariance matrices.
- (B) If a feature is duplicated, naive Bayes counts its evidence twice.
- (C) Naive Bayes is a generative classifier: it models $P(x | y)$ and $P(y)$.
- (D) Naive Bayes can classify accurately even when the independence assumption is violated, since only the arg max of the posterior matters.

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

Consider the model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ (first column of X is the intercept column), fitted by least squares using $\beta = (X^T X)^{-1} X^T y$ with the design matrix X and response y below. Then the least-squares estimate of β_0 is _____ (rounded off to two decimal places)

$$X = \begin{bmatrix} 1 & -1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$y = \begin{bmatrix} 10 \\ 9 \\ 6 \\ 5 \\ 6 \end{bmatrix}$$

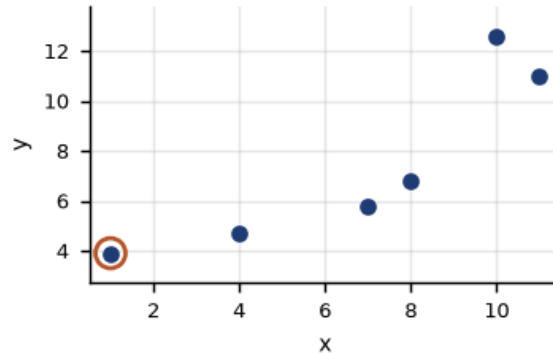
Q.17 [2 marks · NAT (numerical answer)]

An ensemble combines 5 binary classifiers by majority vote. Each classifier is correct with probability 0.65, independently of the others. Compute the probability that the majority vote is correct. (rounded off to three decimal places)

Q.18 [2 marks • NAT (numerical answer)]

A simple linear regression (with intercept) is fitted by least squares to the 6 points shown. With $H = X(X^T X)^{-1} X^T$, if the errors have variance $\sigma^2 = 1$, the variance of the residual e_i of the observation with $x = 1$ (circled) is _____ (rounded off to three decimal places)

x	1	4	7	8	10	11
y	3.9	4.7	5.8	6.8	12.6	11

**Q.19 [2 marks • NAT (numerical answer)]**

An autoencoder with layer sizes 256-64-256 (fully connected, separate encoder and decoder weights, biases in both layers) has _____ trainable parameters. (Answer in integer)

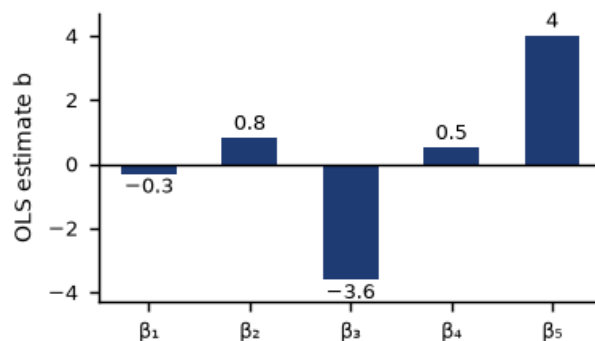
Q.20 [2 marks • MSQ (one or more correct)]

A threshold unit outputs 1 if $w_1 x_1 + w_2 x_2 + b > 0$ and 0 otherwise, for binary inputs $x_1, x_2 \in \{0, 1\}$. Which of the following parameter settings make the unit implement the Boolean **NOR** function?

- (A) $w_1 = -2.5, w_2 = -3, b = 0.5$
- (B) $w_1 = -1, w_2 = -2.5, b = 0.5$
- (C) $w_1 = -3, w_2 = -1, b = -2.5$
- (D) $w_1 = 0, w_2 = -2.5, b = 2$

Q.21 [2 marks • NAT (numerical answer)]

In a regression problem the design matrix X has orthonormal columns ($X^T X = I$). The ordinary least-squares estimates $b = X^T y$ are shown below. Ridge minimises $\|y - X\beta\|^2 + \lambda \|\beta\|_2^2$ and the lasso minimises $\frac{1}{2} \|y - X\beta\|^2 + \lambda \|\beta\|_1$, each with $\lambda = 1.5$. Then $\|\beta_{\text{lasso}}\|_1$ is _____ (rounded off to two decimal places)

**Q.22 [2 marks • MCQ]**

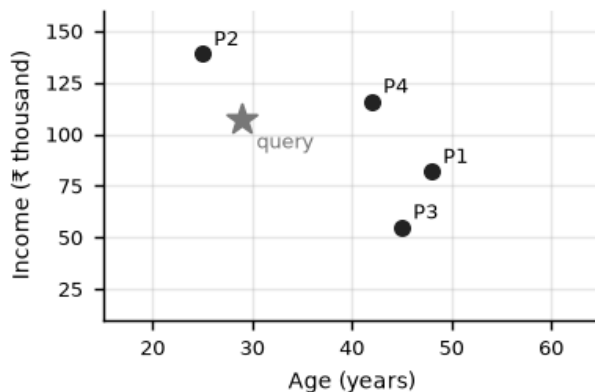
For $n = 9$ paired observations: $\Sigma x = 77, \Sigma y = 92, \Sigma x^2 = 817, \Sigma y^2 = 1078, \Sigma xy = 900$. The sample correlation coefficient r between x and y is

- (A) 0.765
- (B) 0.713
- (C) 0.586
- (D) -0.765

Q.23 [2 marks · MCQ]

A 1-NN classifier uses the two features below. Before computing Euclidean distances, each feature is min-max scaled to $[0, 1]$ using the minimum and maximum of that feature over the four training points P1-P4 (the query is scaled with the same constants). The nearest neighbour of the query after scaling is

Point	Age (years)	Monthly income (₹ thousand)
P1	48	82
P2	25	139
P3	45	55
P4	42	116
Query	29	107



- (A) P1
(C) P3

- (B) P2
(D) P4

Q.24 [2 marks · NAT (numerical answer)]

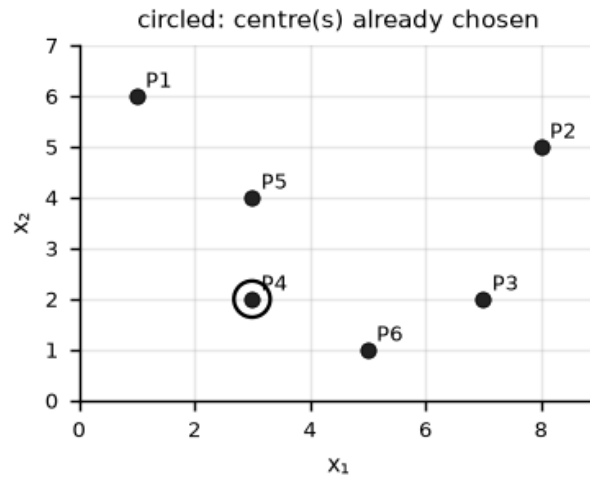
For an input x , a classifier's posterior probabilities are $P(C_1|x) = 0.5$, $P(C_2|x) = 0.3$, $P(C_3|x) = 0.2$. The loss of taking action α_i (deciding C_i) when the true class is C_j is given below. Then the index i of the Bayes-optimal (minimum-risk) action α_i is ____ (Answer in integer)

loss $L(\alpha_i, C_j)$	true C_1	true C_2	true C_3
α_1 : decide C_1	0	10	4
α_2 : decide C_2	3	0	9
α_3 : decide C_3	8	9	0

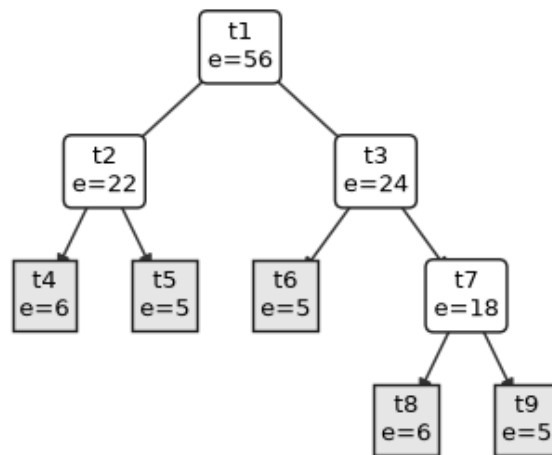
Q.25 [2 marks · NAT (numerical answer)]

k-means++ initialization (squared Euclidean seeding) is applied to the six points below. P4 has already been chosen as centre. The probability that P6 is selected as the second centre is _____ (rounded off to three decimal places)

Point	P1	P2	P3	P4	P5	P6
x_1	1	8	7	3	3	5
x_2	6	5	2	2	4	1

**Q.26 [2 marks · NAT (numerical answer)]**

A classification tree was grown on $N = 100$ training records. Each node shows $e(t)$, the number of training records that would be misclassified if t were a leaf; $R(t) = e(t)/N$. For $\alpha = 0.01$, the cost-complexity $R_\alpha(T) = R(T) + \alpha|T|$ of the full tree T is _____ (rounded off to three decimal places)



Q.27 [2 marks · MCQ]

Six points have the pairwise distances below. Agglomerative clustering with **single linkage** is run and the dendrogram is cut to give exactly **2** clusters. The resulting clusters are

	P1	P2	P3	P4	P5	P6
P1	0	7	9	4	7	6
P2		0	2	3	10	11
P3			0	5	12	13
P4				0	7	8
P5					0	1
P6						0

(A) {P1, P2, P3, P4}, {P5, P6}**(B)** {P1, P4, P5, P6}, {P2, P3}**(C)** {P1, P2, P4, P5}, {P3, P6}**(D)** {P1}, {P2, P3, P4}, {P5, P6}**Q.28 [2 marks · NAT (numerical answer)]**

In one AdaBoost round the current example weights and whether the chosen weak classifier h classifies each example correctly are given. Using $\alpha = \frac{1}{2} \ln((1 - \epsilon)/\epsilon)$ and the update $w_i \leftarrow w_i \cdot \exp(-\alpha y_i h(x_i))/Z$, find the normalisation constant $Z = \sum_i w_i \cdot \exp(-\alpha y_i h(x_i))$. (rounded off to three decimal places)

Example i	1	2	3	4	5	6
Weight w_i	1/6	1/6	1/6	1/6	1/6	1/6
$h(x_i)$ correct?	No	Yes	Yes	Yes	Yes	Yes

Q.29 [2 marks · MCQ]

A naive Bayes classifier is trained on the 10 examples below. The new instance (Outlook = Rain, Temp = Hot, Wind = Strong) is classified (i) with maximum-likelihood (unsmoothed) estimates and (ii) with Laplace (add-one) smoothing of the class-conditional probabilities (class priors unsmoothed). The predicted classes are

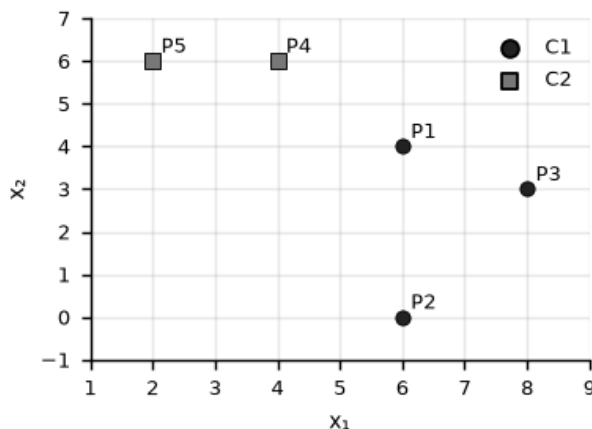
#	Outlook	Temp	Wind	Play
1	Rain	Hot	Weak	No
2	Rain	Cool	Strong	No
3	Rain	Mild	Weak	Yes
4	Sunny	Mild	Weak	Yes
5	Rain	Mild	Strong	No
6	Rain	Cool	Weak	No
7	Sunny	Mild	Weak	Yes
8	Sunny	Cool	Strong	Yes
9	Rain	Mild	Weak	No
10	Rain	Mild	Strong	No

(A) Without smoothing: No; with Laplace smoothing: No**(B)** Without smoothing: Yes; with Laplace smoothing: No**(C)** Without smoothing: No; with Laplace smoothing: Yes**(D)** Without smoothing: Yes; with Laplace smoothing: Yes

Q.30 [2 marks · NAT (numerical answer)]

A clustering of 5 points into 2 clusters is shown below. Using Manhattan distance, the silhouette coefficient $s(i) = (b(i) - a(i))/\max(a(i), b(i))$ of point P5 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5
x_1	6	6	8	4	2
x_2	4	0	3	6	6
cluster	C1	C1	C1	C2	C2

**Q.31 [2 marks · NAT (numerical answer)]**

A centered data matrix X with $n = 26$ rows (samples) and $d = 3$ columns has singular values 25, 11, 2. Using the sample covariance with divisor $n - 1$, the minimum number of principal components needed to retain at least 90% of the variance is _____ (Answer in integer)

Q.32 [2 marks · NAT (numerical answer)]

A 2-2-1 network uses the identity (linear) activation in its hidden layer: $\hat{y} = W_2(W_1x + b_1) + b_2$, with the parameters below. The network is equivalent to a single linear model $\hat{y} = \tilde{w}_1x_1 + \tilde{w}_2x_2 + \tilde{b}$. Then the intercept $\tilde{b}$ is _____ (Answer in integer)

$$W_1 = \begin{bmatrix} 1 & -2 \\ 1 & 0 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

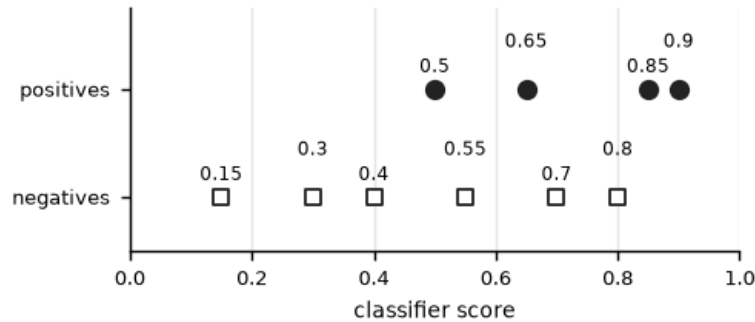
$$W_2 = \begin{bmatrix} 2 & -2 \end{bmatrix}$$

$$b_2 = -2$$

Q.33 [2 marks · NAT (numerical answer)]

A classifier assigns the scores below to 10 test examples (4 positives, label 1; 6 negatives, label 0). The area under the ROC curve (AUC) is _____ (rounded off to two decimal places)

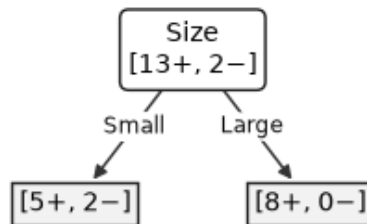
example	1	2	3	4	5	6	7	8	9	10
score	0.5	0.3	0.15	0.8	0.4	0.85	0.55	0.9	0.7	0.65
label	1	0	0	0	0	1	0	1	0	1

**Q.34 [2 marks · NAT (numerical answer)]**

Gradient descent with (heavy-ball) momentum minimises $f(w) = (1/2)w^2$ using $v_t = \beta v_{t-1} + \nabla f(w_{t-1})$ and $w_t = w_{t-1} - \eta v_t$, with $w_0 = 1$, $v_0 = 0$, $\eta = 0.25$, $\beta = 0.5$. The value of v_2 is _____ (rounded off to three decimal places)

Q.35 [2 marks · NAT (numerical answer)]

A node with 15 training records (class counts [positive+, negative−]) is split on the attribute Size as shown. Compute the information gain (in bits) of splitting on Size. (rounded off to two decimal places)



— END OF QUESTION PAPER —

Mock Test 44 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MSQ	1	(A), (B), (C), (D)	19	NAT	2	33088
2	MSQ	1	(A), (B), (C)	20	MSQ	2	(A), (B)
3	MSQ	1	(A), (B), (C)	21	NAT	2	4.5816 to 4.6184
4	NAT	1	34.99 to 35.01	22	MCQ	2	(A)
5	NAT	1	4.4344 to 4.4544	23	MCQ	2	(B)
6	NAT	1	0.2757 to 0.2957	24	NAT	2	2
7	NAT	1	0.8574 to 0.8774	25	NAT	2	0.0613 to 0.0653
8	MCQ	1	(D)	26	NAT	2	0.3185 to 0.3215
9	NAT	1	119.8975 to 121.1025	27	MCQ	2	(A)
10	NAT	1	0.4554 to 0.4754	28	NAT	2	0.7434 to 0.7474
11	MCQ	1	(D)	29	MCQ	2	(A)
12	MSQ	1	(A), (B), (C), (D)	30	NAT	2	0.75 to 0.77
13	MSQ	1	(B), (C), (D)	31	NAT	2	2
14	NAT	1	0.0244 to 0.0253	32	NAT	2	0
15	MSQ	1	(A), (B), (C), (D)	33	NAT	2	0.7817 to 0.8017
16	NAT	2	7.164 to 7.236	34	NAT	2	1.248 to 1.252
17	NAT	2	0.7628 to 0.7668	35	NAT	2	0.1537 to 0.1737
18	NAT	2	0.3509 to 0.3549				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Regression & Model Selection					11
Unsupervised Learning					11
SVM & Decision Trees					11
Neural Networks (MLP / Feed-forward)					11
Probabilistic & Instance-based Classifiers					11
Total					55

Mock Test 44 — Detailed Solutions

Q.1 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Bias-variance trade-off]

- (A) **TRUE**. Flexible models can approximate the true f better.
- (B) **TRUE**. It reproduces each training point.
- (C) **TRUE**. Definition of bias.
- (D) **TRUE**. Flexible fits change more with the training sample.

Q.2 — Answer: (A), (B), (C) [MSQ, 1M · Unsupervised Learning > Cluster evaluation]

- (A) **TRUE**. $a \sqcup b$ gives $s \approx 1$.
- (B) **TRUE**. Majority class per cluster.
- (C) **TRUE**. A common model-selection heuristic.
- (D) **FALSE**. It needs ground-truth labels.

Q.3 — Answer: (A), (B), (C) [MSQ, 1M · SVM & Decision Trees > Ensembles: bagging & random forests]

- (A) **TRUE**. No feature subsampling.
- (B) **TRUE**. Multiplied by e^α .
- (C) **TRUE**. Averaging bootstrap models smooths fluctuations.
- (D) **FALSE**. $\alpha = \frac{1}{2} \ln(0.7/0.3) > 0$.

Q.4 — Answer: 34.99 to 35.01 [NAT, 1M · Unsupervised Learning > k-means]

Assignment (nearest centroid): $c = 9$: {4, 8, 9}; $c = 17$: {15, 17, 18, 19}.

$$\text{SSE} = (4-9)^2 + (8-9)^2 + (9-9)^2 + (15-17)^2 + (17-17)^2 + (18-17)^2 + (19-17)^2 = \mathbf{35}.$$

Q.5 — Answer: 4.4344 to 4.4544 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Inverted dropout multiplies kept activations by $1/p$ so that $E[\text{output}] = h$: $4/0.9 = \mathbf{4.4444}$. No scaling is applied at test time.

Q.6 — Answer: 0.2757 to 0.2957 [NAT, 1M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

Expected cost of predicting positive = $(1 - p) \cdot C_{\text{FP}}$; of predicting negative = $p \cdot C_{\text{FN}}$.

Predict positive iff $p \cdot 5 > (1 - p) \cdot 2 \Leftrightarrow p > C_{\text{FP}} / (C_{\text{FP}} + C_{\text{FN}}) = 2/7 = \mathbf{0.29}$.

Q.7 — Answer: 0.8574 to 0.8774 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

Inner cube volume = $(1 - 2\varepsilon)^d = 0.98^{100} = 0.1326$.

Fraction near boundary = $1 - 0.1326 = \mathbf{0.87}$. In high dimensions most of the volume lies near the boundary.

Q.8 — Answer: (D) [MCQ, 1M · Regression & Model Selection > Regression vs classification]

Absolute error grows linearly in the residual, so outliers have far less influence than with squared (or faster-growing) losses.

Hence the answer is **Absolute error** $|y - \hat{y}|$.

Q.9 — Answer: 119.8975 to 121.1025 [NAT, 1M · Regression & Model Selection > Simple linear regression]

$$\beta_1 = r \cdot s_y / s_x = -0.45 \times 10/3 = -1.5.$$

$$\hat{y} = \bar{y} + \beta_1(x_0 - \bar{x}) = 113 + (-1.5)(-5) = 120.5 \text{ (the line passes through } (\bar{x}, \bar{y}) \text{)}.$$

Answer: **120.5**.

Q.10 — Answer: 0.4554 to 0.4754 [NAT, 1M · SVM & Decision Trees > Decision trees: impurity measures]

$$\text{Gini} = 1 - \sum p_k^2 = 1 - [(12/19)^2 + (7/19)^2] = \mathbf{0.465}.$$

(Maximum possible with 2 classes: $1 - 1/2 = 0.5$.)

Q.11 — Answer: (D) [MCQ, 1M · SVM & Decision Trees > Decision trees: overfitting & pruning]

Subtree	$R(T) + \alpha T $
T1	$0.035 + 0.02 \times 14 = 0.315$
T2	$0.065 + 0.02 \times 12 = 0.305$
T3	$0.095 + 0.02 \times 8 = 0.255$
T4	$0.105 + 0.02 \times 4 = 0.185$

Minimum at **T4**. Larger α penalises leaves more and favours smaller trees; $\alpha = 0$ would pick the largest tree T1.

Q.12 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

(A) **TRUE**. E.g. $XOR = AND(OR(x_1, x_2), NAND(x_1, x_2))$.

(B) **TRUE**. It is an affine set of dimension $d-1$.

(C) **TRUE**. The 0-1 loss is piecewise constant in w .

(D) **TRUE**. Key step of the convergence proof: $w \cdot w^*$ grows linearly in the number of mistakes.

Q.13 — Answer: (B), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

(A) **FALSE**. $\sigma(0) = 1/(1+1) = 0.5$.

(B) **TRUE**. Boundary between k and l : $(w_k - w_l)^T x + (b_k - b_l) = 0$.

(C) **TRUE**. σ is increasing and the log-odds decrease with x_j .

(D) **TRUE**. The penalty grows with $\|w\|$ and makes the objective strictly convex and coercive.

Q.14 — Answer: 0.0244 to 0.0253 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

exp values: 0.6065, 1, 20.0855, 2.7183; sum = 24.4103.

$p_1 = 0.6065/24.4103 = \mathbf{0.0248}$.

Q.15 — Answer: (A), (B), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

(A) **TRUE**. Independence of Gaussian features given the class $\Leftrightarrow$ diagonal covariance.

(B) **TRUE**. The duplicated factor appears twice in the product, overstating confidence.

(C) **TRUE**. Classification then uses Bayes' rule.

(D) **TRUE**. Probability estimates may be poor while the ranking of classes is still right.

Q.16 — Answer: 7.164 to 7.236 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

The columns of X are mutually orthogonal, so $X^T X$ is diagonal and each coefficient is $\beta_j = (x_j^T y)/(x_j^T x_j)$.

$$X^T X = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 36 \\ -8 \\ -2 \end{bmatrix}$$

$\beta = (36/5, -8/4, -2/4) = (7.2, -2, -0.5)$.

Answer: **7.2**.

Q.17 — Answer: 0.7628 to 0.7668 [NAT, 2M · SVM & Decision Trees > Ensembles: bagging & random forests]

The vote is correct when at least 3 of 5 classifiers are correct (Binomial(5, 0.65)).

$P(\text{correct}) = C(5,3)(0.65)^3(0.35)^2 + C(5,4)(0.65)^4(0.35)^1 + C(5,5)(0.65)^5(0.35)^0 = 0.7648$.

Answer: **0.765**. Independence is crucial — correlated classifiers gain much less.

Q.18 — Answer: 0.3509 to 0.3549 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

For simple linear regression with intercept, the leverage (diagonal of the hat matrix) is $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$.

$\bar{x} = 6.8333$, $S_{xx} = 70.8333$.

$h = 0.6471$; $\text{Var}(e_i) = \sigma^2(1 - h_{ii}) = 1 \times 0.3529 = 0.3529$.

Answer: **0.353**.

Q.19 — Answer: 33088 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Parameter counting]

Encoder: $256 \times 64 + 64 = 16448$; decoder: $64 \times 256 + 256 = 16640$. Total = **33088**.

Q.20 — Answer: (A), (B) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Target NOR truth table for $(x_1, x_2) = (0,0), (0,1), (1,0), (1,1)$: 1, 0, 0, 0. For each option compute $a = w_1x_1 + w_2x_2 + b$ at the four inputs and output 1 iff $a > 0$.

(A) TRUE. $a = 0.5, -2.5, -2, -5 \rightarrow$ outputs 1, 0, 0, 0 - NOR.

(B) TRUE. $a = 0.5, -2, -0.5, -3 \rightarrow$ outputs 1, 0, 0, 0 - NOR.

(C) FALSE. $a = -2.5, -3.5, -5.5, -6.5 \rightarrow$ outputs 0, 0, 0, 0 - constant 0.

(D) FALSE. $a = 2, -0.5, 2, -0.5 \rightarrow$ outputs 1, 0, 1, 0 - NOT x_2 .

Q.21 — Answer: 4.5816 to 4.6184 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

With $X^T X = I$: $\beta_{OLS} = X^T y = b$.

Ridge: $\nabla[\|y - X\beta\|^2 + \lambda\|\beta\|^2] = 0 \Rightarrow (1 + \lambda)\beta = b \Rightarrow \beta_{\text{ridge}} = b/(1 + \lambda)$.

Lasso: the problem separates per coordinate, $\min \frac{1}{2}(\beta_j - b_j)^2 + \lambda|\beta_j| \Rightarrow$ soft-thresholding $\beta_j = \text{sign}(b_j) \cdot \max(|b_j| - \lambda, 0)$.

j	1	2	3	4	5
b_j	-0.3	0.8	-3.6	0.5	4
ridge	-0.12	0.32	-1.44	0.2	1.6
lasso	0	0	-2.1	0	2.5

Answer: **4.6**.

Q.22 — Answer: (A) [MCQ, 2M · Regression & Model Selection > Simple linear regression]

$\bar{x} = 8.5556$, $\bar{y} = 10.2222$.

$S_{xx} = \Sigma x^2 - (\Sigma x)^2/n = 158.2222$, $S_{yy} = \Sigma y^2 - (\Sigma y)^2/n = 137.5556$, $S_{xy} = \Sigma xy - \Sigma x \Sigma y/n = 112.8889$.

$r = S_{xy}/\sqrt{S_{xx}S_{yy}} = 112.8889/\sqrt{158.2222 \times 137.5556} = \mathbf{0.765}$.

($r^2 = 0.586$ would be R^2 ; the slopes $b_{yx} = 0.713$, $b_{xy} = 0.821$ satisfy $b_{yx}b_{xy} = r^2$.)

Q.23 — Answer: (B) [MCQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Ranges: age [25, 48], income [55, 139]. Scaled query = (0.174, 0.619).

Point	scaled age	scaled income	scaled distance	raw distance
P1	1	0.321	0.878	31.4
P2	0	1	0.419	32.25
P3	0.87	0	0.931	54.41
P4	0.739	0.726	0.575	15.81

Nearest after scaling: **P2**.

Without scaling the nearest point would be P4, because income differences (tens of thousands) dominate the age differences — exactly why k-NN needs feature scaling.

Q.24 — Answer: 2 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

Conditional risk $R(\alpha_i|x) = \sum_j L(\alpha_i, C_j)P(C_j|x)$:

$R(\alpha_1) = 0 \times 0.5 + 10 \times 0.3 + 4 \times 0.2 = 3.8$

$R(\alpha_2) = 3 \times 0.5 + 0 \times 0.3 + 9 \times 0.2 = 3.3$

$R(\alpha_3) = 8 \times 0.5 + 9 \times 0.3 + 0 \times 0.2 = 6.7$

Minimum risk: α_2 with $R = 3.3$. The MAP rule picks C_1 (risk 3.8) — different, because the losses are asymmetric.

Answer: **2**.

Q.25 — Answer: 0.0613 to 0.0653 [NAT, 2M · Unsupervised Learning > k-means]

k-means++ picks the next centre x with probability $D(x)^2 / \sum_{x'} D(x')^2$, where $D(x)$ is the distance from x to the **nearest** centre already chosen (chosen centres have $D = 0$).

Point	P1	P2	P3	P4	P5	P6
$D(x)^2$	20	34	16	0	4	5

$\sum D^2 = 79$, so $P(P6) = 5/79 = \mathbf{0.063}$.

Common mistakes: using D instead of D^2 (that is not k-means++), or using the distance to the *farthest*/first centre instead of the nearest chosen centre.

Q.26 — Answer: 0.3185 to 0.3215 [NAT, 2M · SVM & Decision Trees > Decision trees: overfitting & pruning]

$R(T) = \text{sum of leaf errors} / N = (6 + 5 + 5 + 6 + 5)/100 = 0.27$; $|T| = 5$ leaves.

$R_\alpha(T) = 0.27 + 0.01 \times 5 = \mathbf{0.32}$.

Q.27 — Answer: (A) [MCQ, 2M · Unsupervised Learning > Hierarchical clustering]

Use single linkage: $d(A, B) = \min_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P5\} + \{P6\}$ at height **1**. Updated: $d(\{P5, P6\}, \{P1\}) = 6$, $d(\{P5, P6\}, \{P2\}) = 10$, $d(\{P5, P6\}, \{P3\}) = 12$, $d(\{P5, P6\}, \{P4\}) = 7$.

Merge 2: $\{P2\} + \{P3\}$ at height **2**. Updated: $d(\{P2, P3\}, \{P1\}) = 7$, $d(\{P2, P3\}, \{P4\}) = 3$, $d(\{P2, P3\}, \{P5, P6\}) = 10$.

Merge 3: $\{P4\} + \{P2, P3\}$ at height **3**. Updated: $d(\{P2, P3, P4\}, \{P1\}) = 4$, $d(\{P2, P3, P4\}, \{P5, P6\}) = 7$.

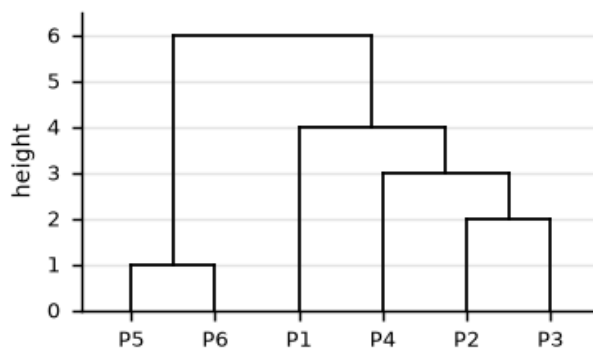
Merge 4: $\{P1\} + \{P2, P3, P4\}$ at height **4**. Updated: $d(\{P1, P2, P3, P4\}, \{P5, P6\}) = 6$.

Merge 5: $\{P5, P6\} + \{P1, P2, P3, P4\}$ at height **6**.

Stopping when 2 clusters remain (i.e. after 4 merges): **$\{P1, P2, P3, P4\}, \{P5, P6\}$** .

$\{P1, P4, P5, P6\}, \{P2, P3\}$ is what *complete* linkage would give.

$\{P1\}, \{P2, P3, P4\}, \{P5, P6\}$ has 3 clusters (cut at the wrong level).

**Q.28 — Answer: 0.7434 to 0.7474** [NAT, 2M · SVM & Decision Trees > Ensembles: boosting]

$\varepsilon = \sum_{\text{misclassified}} w_i = 0.1667$; $\alpha = \frac{1}{2} \ln(0.8333/0.1667) = 0.8047$.

Misclassified weights $\times e^\alpha = 2.2361$; correct weights $\times e^{-\alpha} = 0.4472$.

$Z = \varepsilon e^\alpha + (1 - \varepsilon) e^{-\alpha} = 2\sqrt{\varepsilon(1 - \varepsilon)} = 0.7454$.

New weights: 0.5, 0.1, 0.1, 0.1, 0.1, 0.1.

Shortcut: after normalisation the misclassified set has total weight exactly $\frac{1}{2}$: $w_i' = w_i/(2\varepsilon)$ if misclassified, $w_i/(2(1 - \varepsilon))$ otherwise.

Answer: **0.745**.

Q.29 — Answer: (A) [MCQ, 2M · Probabilistic & Instance-based Classifiers > Naive Bayes]

(i) No smoothing.

Yes: $P(\text{Play}=\text{Yes}) = 4/10$; $P(\text{Outlook}=\text{Rain}|\text{Yes}) = 1/4$; $P(\text{Temp}=\text{Hot}|\text{Yes}) = 0/4$;

$P(\text{Wind}=\text{Strong}|\text{Yes}) = 1/4 = \text{score} = 0$.

No: $P(\text{Play}=\text{No}) = 6/10$; $P(\text{Outlook}=\text{Rain}|\text{No}) = 6/6$; $P(\text{Temp}=\text{Hot}|\text{No}) = 1/6$; $P(\text{Wind}=\text{Strong}|\text{No}) = 3/6 \Rightarrow \text{score} = 0.05$.

A zero count makes the whole product for Yes equal to 0 (zero-frequency problem), so the prediction is No.

(ii) Laplace smoothing.

Yes: $P(\text{Play}=\text{Yes}) = 4/10$; $P(\text{Outlook}=\text{Rain}|\text{Yes}) = (1+1)/(4+3) = 2/7$; $P(\text{Temp}=\text{Hot}|\text{Yes}) = (0+1)/(4+3) = 1/7$; $P(\text{Wind}=\text{Strong}|\text{Yes}) = (1+1)/(4+2) = 1/3 \Rightarrow \text{score} = 0.00544$.

No: $P(\text{Play}=\text{No}) = 6/10$; $P(\text{Outlook}=\text{Rain}|\text{No}) = (6+1)/(6+3) = 7/9$; $P(\text{Temp}=\text{Hot}|\text{No}) = (1+1)/(6+3) = 2/9$; $P(\text{Wind}=\text{Strong}|\text{No}) = (3+1)/(6+2) = 1/2 \Rightarrow \text{score} = 0.05185$.

Larger score: No. Answer: **Without smoothing: No; with Laplace smoothing: No.**

Q.30 — Answer: 0.75 to 0.77 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

a(i) = mean Manhattan distance from P5 to the other members of its own cluster C2: $d(P5, P4) = 2 \rightarrow a = 2$.

b(i) = smallest mean distance to the points of another cluster: C1: 8.3333 $\rightarrow b = 8.3333$.

s(i) = $(b - a)/\max(a, b) = (8.3333 - 2)/8.3333 = 0.76$.

s ranges in $[-1, 1]$; values near 1 mean well clustered, negative values suggest a wrong assignment.

Q.31 — Answer: 2 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

With $X = U\Sigma V^T$ (X centered, $n \times d$), the sample covariance is $S = X^T X / (n - 1) = V(\Sigma^2 / (n - 1))V^T$, so the PC directions are the columns of V and the eigenvalues are $\lambda_j = \sigma_j^2 / (n - 1)$.

$\lambda = 25^2/25 = 25$, $11^2/25 = 4.84$, $2^2/25 = 0.16$.

Cumulative % variance: 83.33, 99.47, 100 (note the $(n - 1)$ factor cancels in proportions: $\text{PVE} = \sigma_j^2 / \Sigma \sigma^2$).

Answer: **2**. (Using σ instead of σ^2 is the classic mistake.)

Q.32 — Answer: 0 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Expressiveness]

With identity activations, $\hat{y} = W_2(W_1x + b_1) + b_2 = (W_2W_1)x + (W_2b_1 + b_2)$: a single affine map.

$\tilde{w}^T = W_2W_1 = (0, -4)$; $\tilde{b} = W_2b_1 + b_2 = 0$.

Answer: **0**.

Hence stacking linear layers adds no expressive power – a non-linear activation is essential.

Q.33 — Answer: 0.7817 to 0.8017 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

AUC = fraction of (positive, negative) pairs in which the positive has the higher score (Mann-Whitney statistic).

For each positive, count negatives with lower score: $0.9 \rightarrow 6$; $0.85 \rightarrow 6$; $0.65 \rightarrow 4$; $0.5 \rightarrow 3$.

Concordant pairs = 19 out of $4 \times 6 = 24$.

AUC = $19/24 = 0.79$.

(Equivalently, trace the ROC staircase by lowering the threshold and add the rectangle areas.)

Q.34 — Answer: 1.248 to 1.252 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$f(w) = (1/2)w^2 \Rightarrow \nabla f(w) = 1w$. Update: $v_t = \beta v_{t-1} + \nabla f(w_{t-1})$, $w_t = w_{t-1} - \eta v_t$, with $v_0 = 0$, $\beta = 0.5$, $\eta = 0.25$.

$t = 1$: $g = 1 \times 1 = 1$; $v_1 = 0.5 \times 0 + 1 = 1$; $w_1 = 1 - 0.25 \times 1 = 0.75$.

$t = 2$: $g = 1 \times 0.75 = 0.75$; $v_2 = 0.5 \times 1 + 0.75 = 1.25$; $w_2 = 0.75 - 0.25 \times 1.25 = 0.4375$.

Answer: **1.25**.

Q.35 — Answer: 0.1537 to 0.1737 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent $[13+, 2-]$: $H = -(13/15)\log_2(13/15) - (2/15)\log_2(2/15) = 0.5665$.

Size = Small: $[5+, 2-]$, $H = -(5/7)\log_2(5/7) - (2/7)\log_2(2/7) = 0.8631$.

Size = Large: $[8+, 0-]$, $H = 0$ (pure node) = 0.

Weighted child entropy = $(7/15)(0.8631) + (8/15)(0) = 0.4028$.

IG = $0.5665 - 0.4028 = 0.1637$.

Answer: **0.16**.

Mock Test 45

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	6	7	22

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Regression & Model Selection	7	11
Probabilistic & Instance-based Classifiers	7	11
Neural Networks (MLP / Feed-forward)	7	11
Unsupervised Learning	7	11
SVM & Decision Trees	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MSQ (one or more correct)]

Let X be an $n \times p$ design matrix of full column rank p ($p < n$), and let $H = X(X^T X)^{-1} X^T$ be the hat matrix of least-squares regression of y on X . Which of the following statements is/are CORRECT?

- (A) H is an orthogonal matrix ($H^T H = I$).
- (B) $\text{rank}(H) = n$.
- (C) $HX = X$.
- (D) $\text{rank}(H) = p$.

Q.2 [1 mark · NAT (numerical answer)]

The confusion matrix of a binary classifier on a test set is shown below (note which of rows/columns are actual and predicted). The balanced accuracy of the classifier is _____ (rounded off to two decimal places)

	Actual +	Actual -
Predicted +	37	26
Predicted -	53	23

Q.3 [1 mark · NAT (numerical answer)]

The value of the softplus function $\ln(1 + e^z)$ at $z = 1.5$ is _____ (rounded off to three decimal places)

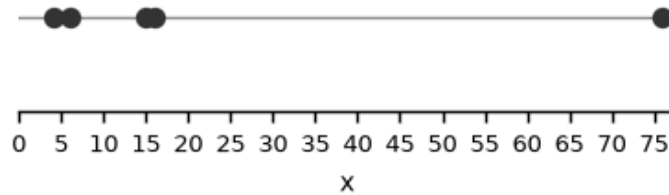
Q.4 [1 mark · MCQ]

Assume $y_i = w^T x_i + \varepsilon_i$ with ε_i i.i.d. $N(0, \sigma^2)$. The maximum-likelihood estimate of w is obtained by minimising:

- (A) $\max_i |y_i - w^T x_i|$
- (B) $\sum (y_i - w^T x_i)^2 + \lambda \|w\|^2$
- (C) $\sum (y_i - w^T x_i)^2$
- (D) $\sum (y_i - w^T x_i)^4$

Q.5 [1 mark · NAT (numerical answer)]

A cluster contains the 1-D values $\{4, 6, 15, 16, 76\}$. Treating them as one cluster, the medoid of this cluster (absolute-difference dissimilarity) is _____ (rounded off to two decimal places)

**Q.6 [1 mark · MCQ]**

Predicting the number of goals (0, 1, 2, ...) a football team will score is most naturally a:

- (A) clustering problem
- (B) binary classification problem
- (C) regression problem with a count-valued target (e.g. Poisson regression)
- (D) unsupervised density-estimation problem

Q.7 [1 mark · MSQ (one or more correct)]

Let $p = \text{softmax}(z)$ for logits $z \in \mathbb{R}^K$ and $L = -\ln p_y$ the cross-entropy loss for true class y . Which of the following is/are CORRECT?

- (A) For softmax + cross-entropy, the gradient of a wrong-class logit is negative.
- (B) For a confidently wrong prediction ($p_y \rightarrow 0$), the cross-entropy loss grows without bound.
- (C) For softmax + cross-entropy, $\partial L / \partial z_k = (p_k - y_k) p_k (1 - p_k)$.
- (D) The Jacobian of softmax is $\partial p_i / \partial z_j = p_i (\delta_{ij} - p_j)$.

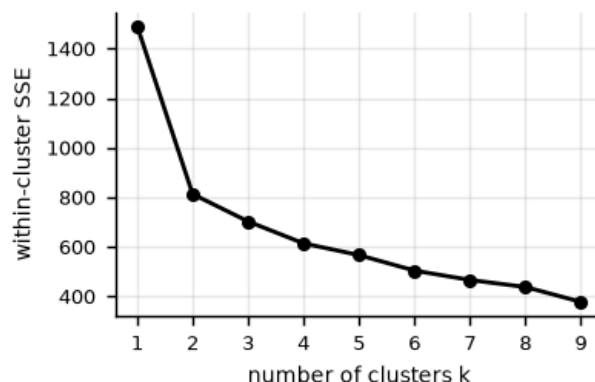
Q.8 [1 mark · MSQ (one or more correct)]

Which of the following statements about kernel functions (Mercer kernels) is/are CORRECT?

- (A) If k is a valid (non-zero) kernel, $-k$ is also valid.
- (B) The polynomial kernel $(x^T z + 1)^d$ corresponds to an infinite-dimensional feature space.
- (C) If k_1 and k_2 are valid kernels, $k_1 + k_2$ is a valid kernel.
- (D) If k is valid, $f(x)k(x, z)f(z)$ is valid for any real function f .

Q.9 [1 mark · MCQ]

k-means is run for $k = 1, 2, \dots, 9$ on a dataset and the final within-cluster SSE is plotted below. Using the elbow method, the most appropriate number of clusters is



- (A) 2
- (B) 3
- (C) 9
- (D) 4

Q.10 [1 mark · NAT (numerical answer)]

Points are drawn uniformly from the unit hypercube $[0, 1]^5$. The fraction of points lying within distance 0.02 of the boundary (i.e. outside the inner cube $[0.02, 0.98]^5$) is _____ (rounded off to two decimal places)

Q.11 [1 mark · MSQ (one or more correct)]

Which of the following statements about decision-tree learning is/are CORRECT?

- (A) Pruning a subtree can increase the training error of the tree.
- (B) A binary tree of depth d has at most 2^d leaves.
- (C) With squared-error loss, a regression-tree leaf predicts the median of its training responses.
- (D) A tree with finitely many axis-parallel splits can represent the diagonal boundary $x_1 = x_2$ exactly on continuous inputs.

Q.12 [1 mark · MSQ (one or more correct)]

Which of the following statements about initialisation and regularisation of neural networks is/are CORRECT?

- (A) With inverted dropout (keep probability p), retained activations are scaled by $1/p$ during training, so no scaling is needed at test time.
- (B) With penalty $(\lambda/2)\|w\|^2$ and SGD, each step first shrinks the weights by the factor $(1 - \eta\lambda)$ (weight decay).
- (C) Data augmentation can reduce overfitting.
- (D) Increasing the L2 penalty λ generally increases bias and reduces variance.

Q.13 [1 mark · MCQ]

In a logistic regression model for the probability of having the disease, the coefficient of *age (years)* is $\beta = 0.7$. Holding all other features fixed, increasing *age (years)* by 3 units multiplies the **odds** of having the disease by (approximately)

- (A) 6.04
- (B) 8.17
- (C) 2.1
- (D) 3.1

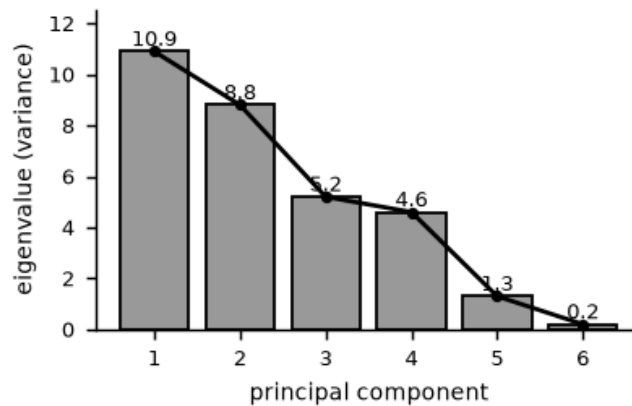
Q.14 [1 mark · NAT (numerical answer)]

A hard-margin SVM has exactly three support vectors with the optimal dual variables shown. The component w_2 of the optimal weight vector is _____ (rounded off to two decimal places)

i	x_i	y_i	α_i
1	(3, 1)	+1	0.25
2	(-2, 2)	+1	0.25
3	(-1, 4)	-1	0.5

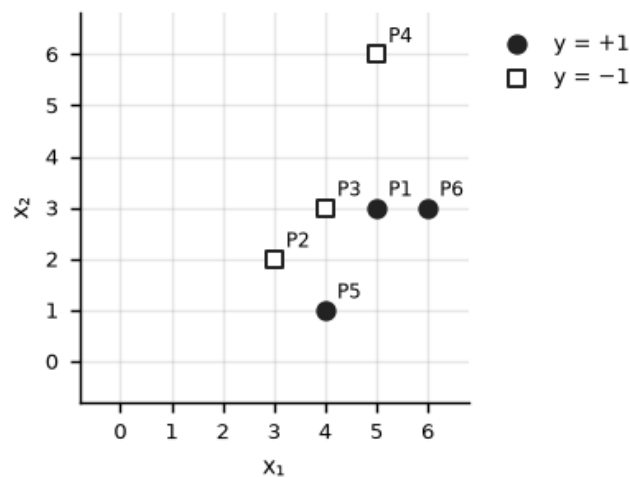
Q.15 [1 mark · NAT (numerical answer)]

The scree plot below shows the eigenvalues of the covariance matrix of a 6-dimensional dataset. The percentage of total variance explained by the first two principal components together is _____ (rounded off to one decimal place)

**Q.16 - Q.35 : Two-mark questions****Q.16 [2 marks · NAT (numerical answer)]**

The labelled training points shown below are linearly separable. A hard-margin linear SVM is trained on them. Compute $\|w\|$ for the canonical solution (support vectors satisfy $y_i(w^T x_i + b) = 1$). (rounded off to two decimal places)

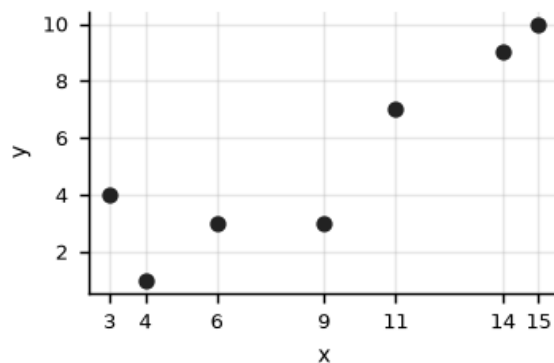
Point	x_1	x_2	y
P1	5	3	+1
P2	3	2	-1
P3	4	3	-1
P4	5	6	-1
P5	4	1	+1
P6	6	3	+1



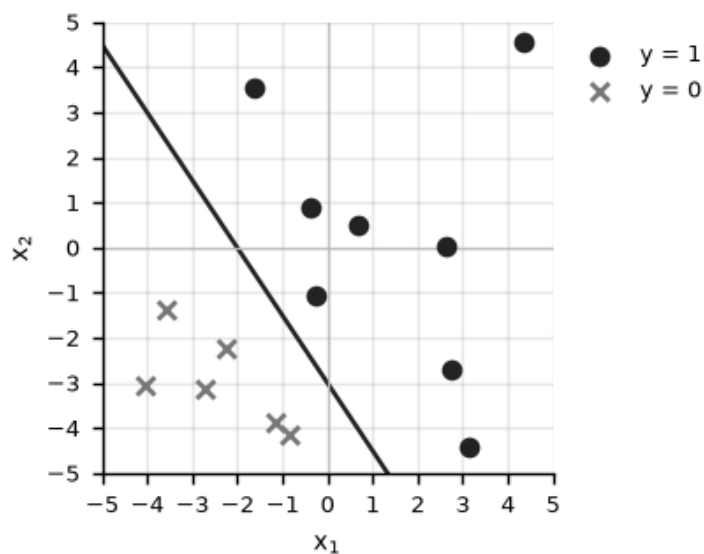
Q.17 [2 marks · NAT (numerical answer)]

A regression tree (squared-error criterion) is fitted to the 7 points below. The root split “ $x \leq t$ ” is chosen among midpoints of consecutive x values to minimise the total SSE of the two children (each child predicts its mean). Find the total SSE of the two child nodes after the best split. (rounded off to two decimal places)

x	3	4	6	9	11	14	15
y	4	1	3	3	7	9	10

**Q.18 [2 marks · MCQ]**

A logistic regression model $P(y = 1 | \mathbf{x}) = \sigma(w_0 + w_1x_1 + w_2x_2)$ perfectly classifies the training points shown below using threshold 0.5. Its decision boundary is the solid line (it crosses the axes at grid points). Which of the following parameter vectors is consistent with the figure?

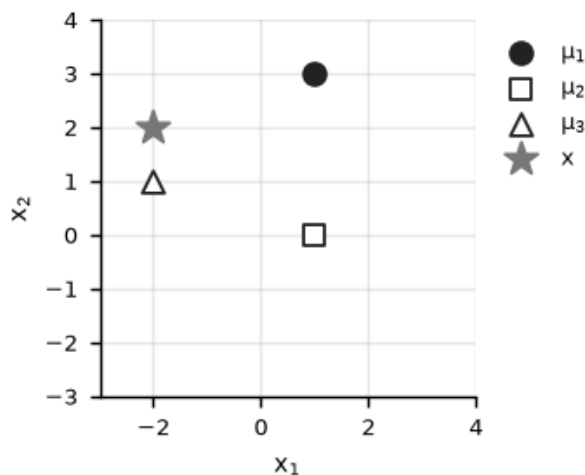


- (A) $(w_0, w_1, w_2) = (-12, 4, 6)$
 (B) $(w_0, w_1, w_2) = (-12, 6, 4)$
 (C) $(w_0, w_1, w_2) = (12, 6, -4)$
 (D) $(w_0, w_1, w_2) = (12, 6, 4)$

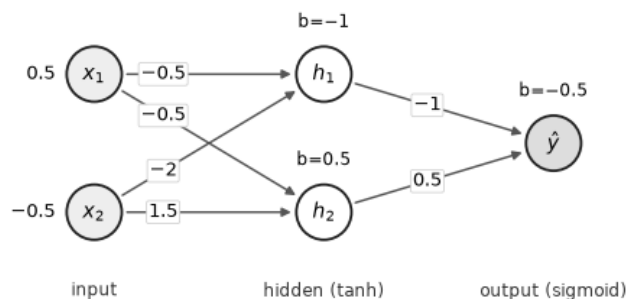
Q.19 [2 marks · NAT (numerical answer)]

An LDA classifier for 3 classes has shared covariance matrix Σ (below), class means $\mu_1 = (1, 3)^T$; $\mu_2 = (1, 0)^T$; $\mu_3 = (-2, 1)^T$, and priors $\pi_1 = 1/3$, $\pi_2 = 1/3$, $\pi_3 = 1/3$. The linear discriminant is $\delta_k(\mathbf{x}) = \mathbf{x}^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln \pi_k$ (natural log). For $\mathbf{x} = (-2, 2)^T$, the value of the discriminant function $\delta_1(\mathbf{x})$ is _____ (rounded off to two decimal places)

$$\Sigma = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$$

**Q.20 [2 marks · NAT (numerical answer)]**

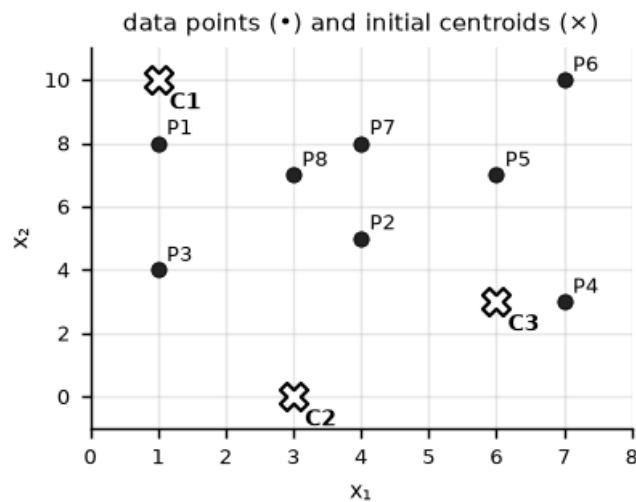
The feed-forward network below has tanh hidden units and a sigmoid output unit. The number on each edge is its weight and 'b' above a node is that node's bias. For input $\mathbf{x} = (0.5, -0.5)$ (values shown left of the inputs), the network output $\hat{y}$ is _____ (rounded off to three decimal places)



Q.21 [2 marks · NAT (numerical answer)]

k-means (Lloyd's algorithm, Euclidean distance) with $k = 3$ is run on the 8 points below, starting from the initial centroids $C1 = (1, 10)$, $C2 = (3, 0)$, $C3 = (6, 3)$. After **one** iteration (assignment step followed by update step), the within-cluster sum of squared Euclidean distances $J = \sum_i \|x_i - \mu_{c(i)}\|^2$, computed with the *updated* centroids and the assignments of that iteration, is _____ (rounded off to two decimal places)

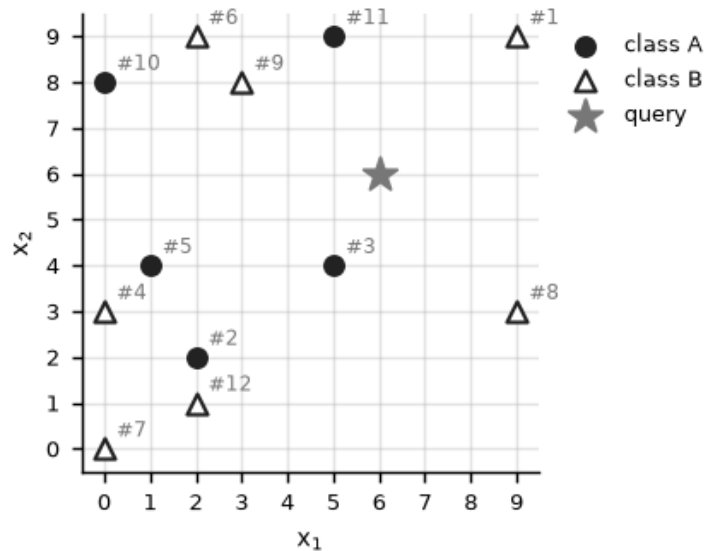
Point	P1	P2	P3	P4	P5	P6	P7	P8
x_1	1	4	1	7	6	7	4	3
x_2	8	5	4	3	7	10	8	7



Q.22 [2 marks · MSQ (one or more correct)]

The figure and table show 12 labelled training points (class A: filled circles, class B: open triangles) and a query point (star) at (6, 6). A k-NN classifier with **Euclidean** distance and majority vote is used. Which of the following statements is/are CORRECT?

#	1	2	3	4	5	6	7	8	9	10	11	12
x_1	9	2	5	0	1	2	0	9	3	0	5	2
x_2	9	2	4	3	4	9	0	3	8	8	9	1
class	B	A	A	B	A	B	B	B	B	A	A	B



- (A) With $k = 7$, the query is classified as class A.
 (B) With $k = 5$, the query is classified as class A.
 (C) With $k = 3$, the query is classified as class A.
 (D) With $k = 1$, the query is classified as class A.

Q.23 [2 marks · NAT (numerical answer)]

A simple linear regression fitted by ordinary least squares gives $\hat{y} = 10.4 - 3x$. The feature x was measured in metres. If x is re-expressed in centimetres (so $x' = 100x$) and the model is refitted, the new slope is _____ (rounded off to 4 decimal places)

Q.24 [2 marks · MCQ]

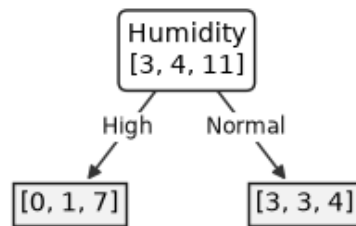
Ridge regression is tuned by 10-fold cross-validation. For each λ the table gives the mean CV MSE and its standard error (SE) across folds. (Larger λ means a simpler, more regularised model.) Using the **one-standard-error rule** (choose the most regularised model whose CV error is within one standard error of the minimum CV error), the selected value is

λ	0.01	0.1	0.3	1	3	10
CV MSE	3.89	3.48	3.19	3.29	3.58	4.01
SE	0.21	0.47	0.16	0.32	0.49	0.22

- (A) $\lambda = 0.1$ (B) $\lambda = 10$
 (C) $\lambda = 1$ (D) $\lambda = 0.3$

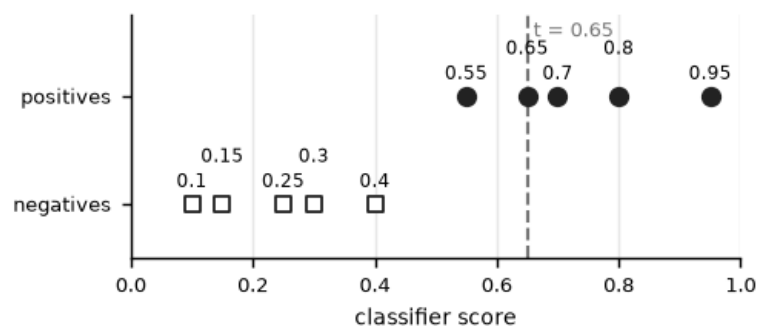
Q.25 [2 marks · NAT (numerical answer)]

A CART node with 18 training records (class counts [C1, C2, C3]) is split on Humidity as shown. Compute the reduction in Gini index (Gini gain) obtained by splitting on Humidity. (rounded off to two decimal places)

**Q.26 [2 marks · NAT (numerical answer)]**

A probabilistic classifier produces the scores below on 10 labelled examples (1 = positive). An example is predicted positive if its score is $\geq t$, with $t = 0.65$. The false positive rate at this threshold is _____ (rounded off to two decimal places)

example	1	2	3	4	5	6	7	8	9	10
score	0.95	0.55	0.25	0.4	0.3	0.65	0.7	0.1	0.8	0.15
label	1	1	0	0	0	1	1	0	1	0

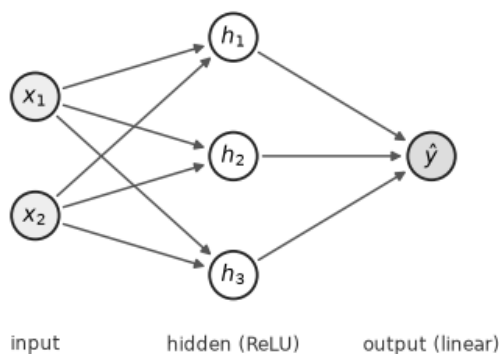
**Q.27 [2 marks · NAT (numerical answer)]**

Agglomerative hierarchical clustering with **single linkage** is applied to 5 points whose pairwise distance matrix is given below (upper triangle shown). The height (linkage distance) at which the **second** merge takes place is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5
P1	0	2	11	11	10
P2		0	11	11	10
P3			0	4	5
P4				0	9
P5					0

Q.28 [2 marks · NAT (numerical answer)]

The 2-3-1 network below computes $\hat{y} = v \cdot \text{ReLU}(Wx + b) + c$ with W , b , v given below and $c = 0$. With $x = (2, -1)$, target $y = -1$ and loss $L = \frac{1}{2}(\hat{y} - y)^2$, the value of $\partial L / \partial W_{31}$ is _____ (Answer in integer)



$$W = \begin{bmatrix} 1 & 0 \\ -1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$b = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$v^T = \begin{bmatrix} 2 & 1 & 1 \end{bmatrix}$$

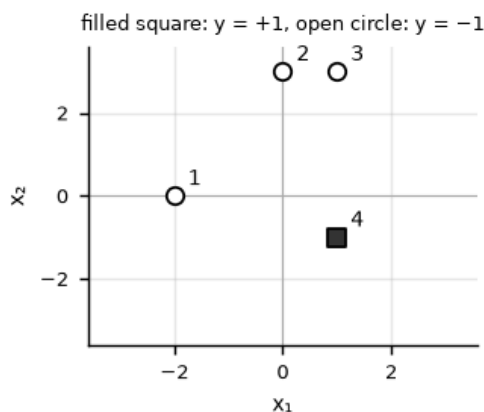
Q.29 [2 marks · NAT (numerical answer)]

The eigenvalues of the sample covariance matrix (divisor $n - 1$) of a centered 5-dimensional dataset are 12.5, 9, 5.5, 3, 1.5. Then with $n = 51$ samples, the total squared reconstruction error $\sum_i \|x_i - \hat{x}_i\|^2$ when keeping the top 2 principal components is _____ (rounded off to two decimal places)

Q.30 [2 marks · NAT (numerical answer)]

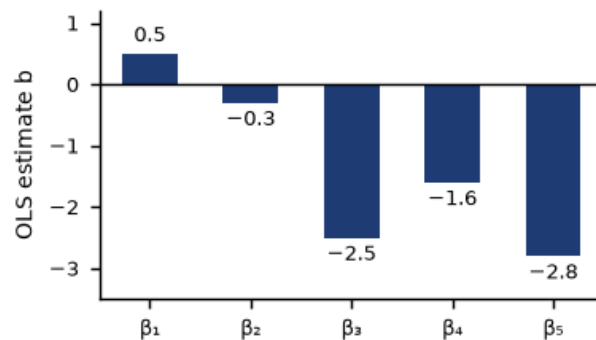
A perceptron with parameters $w \in \mathbb{R}^2$, b is trained on the data below, starting from $w = (0, 0)$, $b = 0$, with learning rate $\eta = 1$. Examples are presented once in the order 1, 2, 3, 4 (one epoch). An update $w \leftarrow w + \eta y x$, $b \leftarrow b + \eta y$ is made whenever $y(w \cdot x + b) \leq 0$. After the epoch, the weight w_1 is _____ (Answer in integer)

i	x_1	x_2	y
1	-2	0	-1
2	0	3	-1
3	1	3	-1
4	1	-1	1

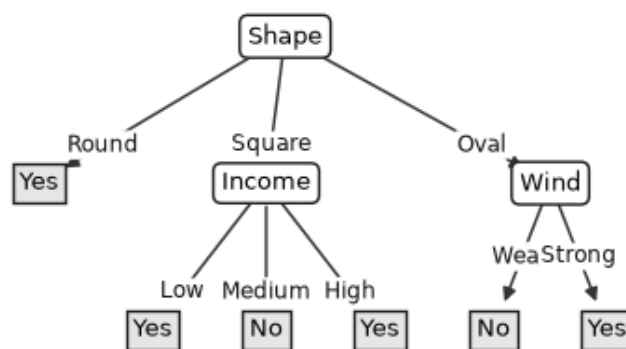


Q.31 [2 marks · NAT (numerical answer)]

In a regression problem the design matrix X has orthonormal columns ($X^T X = I$). The ordinary least-squares estimates $b = X^T y$ are shown below. Ridge minimises $\|y - X\beta\|^2 + \lambda\|\beta\|_2^2$ and the lasso minimises $\frac{1}{2}\|y - X\beta\|^2 + \lambda\|\beta\|_1$, each with $\lambda = 1$. Then the number of non-zero coefficients in the lasso solution is _____ (Answer in integer)

**Q.32 [2 marks · NAT (numerical answer)]**

The decision tree shown is applied to the 9 labelled records in the table (attributes not tested on a path are ignored). Find the precision of the tree for the class “Yes”. (rounded off to two decimal places)

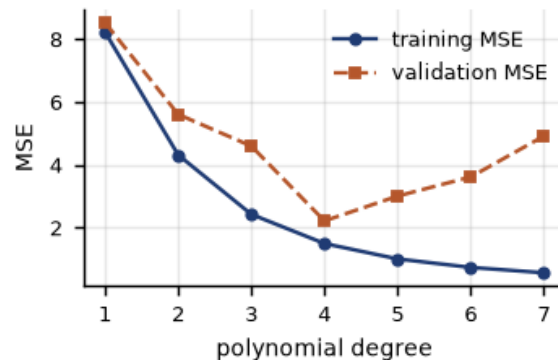


#	Shape	Income	Wind	Actual
1	Oval	Low	Strong	Yes
2	Square	Low	Strong	Yes
3	Square	Low	Weak	Yes
4	Oval	High	Strong	No
5	Round	High	Weak	No
6	Oval	Low	Weak	No
7	Oval	Low	Strong	Yes
8	Round	Medium	Weak	No
9	Oval	Medium	Strong	Yes

Q.33 [2 marks · MSQ (one or more correct)]

Polynomial regression models of degree 1 to 7 are fitted by least squares on a training set and evaluated on a separate validation set. The resulting mean squared errors are given below. Which of the following statements is/are CORRECT?

Degree	1	2	3	4	5	6	7
Training MSE	8.23	4.32	2.43	1.5	1.01	0.74	0.57
Validation MSE	8.53	5.61	4.6	2.22	3.01	3.62	4.92



- (A) Based on the validation MSE, degree 4 should be selected.
- (B) The generalisation gap (validation MSE – training MSE) is largest for degree 1.
- (C) The degree-7 model overfits: it has the lowest training MSE but its validation MSE exceeds the minimum.
- (D) The generalisation gap (validation MSE – training MSE) is largest for degree 7.

Q.34 [2 marks · NAT (numerical answer)]

Agglomerative clustering with **complete linkage** is applied to the distance matrix below. The cophenetic distance between P3 and P4 (the dendrogram height at which they are first joined) is ____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5	P6
P1	0	9	8	5	6	12
P2		0	11	4	7	5
P3			0	7	4	6
P4				0	3	7
P5					0	6
P6						0

Q.35 [2 marks · NAT (numerical answer)]

A softmax-regression output layer computes $z = Wx + b$, $p = \text{softmax}(z)$ with W and b below, and is trained with cross-entropy loss $L = -\ln p_y$. For $x = (-2, -2)^T$ with true class $y = 1$, the updated W_{22} after one gradient-descent step with $\eta = 1$ is ____ (rounded off to three decimal places)

$$W = \begin{bmatrix} 1 & -1 \\ 0.5 & -1 \\ 0.5 & 0.5 \end{bmatrix}$$

$$b = \begin{bmatrix} -0.5 \\ 0 \\ 0 \end{bmatrix}$$

— END OF QUESTION PAPER —

Mock Test 45 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MSQ	1	(C), (D)	19	NAT	2	0.1914 to 0.2114
2	NAT	1	0.4302 to 0.4502	20	NAT	2	0.3788 to 0.3828
3	NAT	1	1.6994 to 1.7034	21	NAT	2	36.1567 to 36.1767
4	MCQ	1	(C)	22	MSQ	2	(C), (D)
5	NAT	1	14.99 to 15.01	23	NAT	2	−0.0301 to −0.0299
6	MCQ	1	(C)	24	MCQ	2	(C)
7	MSQ	1	(B), (D)	25	NAT	2	0.0755 to 0.0955
8	MSQ	1	(C), (D)	26	NAT	2	−0.01 to 0.01
9	MCQ	1	(A)	27	NAT	2	3.99 to 4.01
10	NAT	1	0.1746 to 0.1946	28	NAT	2	0
11	MSQ	1	(A), (B)	29	NAT	2	499.99 to 500.01
12	MSQ	1	(A), (B), (C), (D)	30	NAT	2	1
13	MCQ	1	(B)	31	NAT	2	3
14	NAT	1	−1.26 to −1.24	32	NAT	2	0.615 to 0.635
15	NAT	1	63.348 to 63.748	33	MSQ	2	(A), (C), (D)
16	NAT	2	2.2261 to 2.2461	34	NAT	2	7.99 to 8.01
17	NAT	2	9.4067 to 9.4267	35	NAT	2	0.5692 to 0.5732
18	MCQ	2	(D)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Regression & Model Selection					11
Probabilistic & Instance-based Classifiers					11
Neural Networks (MLP / Feed-forward)					11
Unsupervised Learning					11
SVM & Decision Trees					11
Total					55

Mock Test 45 — Detailed Solutions

Q.1 — Answer: (C), (D) [MSQ, 1M · Regression & Model Selection > Multiple linear regression]

(A) FALSE. $H^T H = H \neq I$ since H is singular.

(B) FALSE. $\text{rank}(H) = p < n$.

(C) TRUE. $X(X^T X)^{-1} X^T X = X$ — H leaves the column space of X unchanged.

(D) TRUE. For an idempotent matrix $\text{rank} = \text{trace} = p$.

Q.2 — Answer: 0.4302 to 0.4502 [NAT, 1M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

TP = 37, FN = 53, FP = 26, TN = 23 (N = 139).

Balanced accuracy = $(\text{TPR} + \text{TNR})/2$.

TPR = $37/90 = 0.4111$, TNR = $23/49 = 0.4694$; balanced accuracy = **0.44**.

Watch the orientation: swapping FP and FN is the most common error (it swaps precision and recall).

Q.3 — Answer: 1.6994 to 1.7034 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

$\ln(1 + e^{1.5}) = \ln(1 + 4.4817) = \mathbf{1.7014}$. (Its derivative is $\sigma(z) = 0.8176$.)

Q.4 — Answer: (C) [MCQ, 1M · Regression & Model Selection > MLE & MAP view]

$-\log L = (n/2)\log(2\pi\sigma^2) + (1/2\sigma^2)\sum(y_i - w^T x_i)^2$; only the sum of squares depends on w .

Answer: $\sum(y_i - w^T x_i)^2$.

Q.5 — Answer: 14.99 to 15.01 [NAT, 1M · Unsupervised Learning > k-medoids]

Mean = $117/5 = 23.4$ (pulled towards the outlier 76).

Medoid = data point minimizing $\sum |x_i - m|$. Costs: $m=4$: 97, $m=6$: 91, $m=15$: 82, $m=16$: 83, $m=76$: 263.

The minimum is at $m = 15$ (the median), cost 82.

Answer: **15**. The medoid is barely affected by the outlier, unlike the mean.

Q.6 — Answer: (C) [MCQ, 1M · Regression & Model Selection > Regression vs classification]

The target is a numeric count, so it is a (count) regression problem.

Hence the answer is **regression problem with a count-valued target (e.g. Poisson regression)**.

Q.7 — Answer: (B), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

(A) FALSE. It is $p_k > 0$, so GD lowers wrong-class logits.

(B) TRUE. $-\ln p_y \rightarrow \infty$.

(C) FALSE. The extra factor would appear only for an MSE-type loss on a sigmoid; it cancels here.

(D) TRUE. Standard derivative.

Q.8 — Answer: (C), (D) [MSQ, 1M · SVM & Decision Trees > SVM: kernels]

(A) FALSE. $-K$ has a negative diagonal entry somewhere.

(B) FALSE. Its feature space has dimension $C(p + d, d)$.

(C) TRUE. Sum of PSD Gram matrices is PSD.

(D) TRUE. Gram matrix $D K D$ with D diagonal.

Q.9 — Answer: (A) [MCQ, 1M · Unsupervised Learning > k-means]

The SSE drops steeply up to $k = 2$ and then decreases only slowly (relative drops: $k=2$: 45%, $k=3$: 14%, $k=4$: 13%, $k=5$: 8%, $k=6$: 11%, $k=7$: 8%, $k=8$: 6%, $k=9$: 14%). The bend ('elbow') is at **$k = 2$** .

$k = 9$ gives the lowest SSE but SSE always (weakly) decreases with k , so minimizing it would pick the largest k ; it is not a model-selection criterion by itself.

Q.10 — Answer: 0.1746 to 0.1946 [NAT, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Inner cube volume = $(1 - 2 \times 0.02)^5 = 0.96^5 = 0.8154$.

Fraction near the boundary = $1 - 0.8154 = \mathbf{0.18}$.

In high dimensions most points lie near the boundary, far from each other — a symptom of the curse of dimensionality that hurts k-NN.

Q.11 — Answer: (A), (B) [MSQ, 1M · SVM & Decision Trees > Decision trees: overfitting & pruning]

(A) TRUE. Pruning trades training fit for simplicity.

(B) TRUE. Doubling per level.

(C) FALSE. The mean minimises SSE (median minimises absolute error).

(D) FALSE. Only a staircase approximation is possible.

Q.12 — Answer: (A), (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

(A) TRUE. $E[m \cdot h/p] = h$.

(B) TRUE. $w \leftarrow w - \eta(\nabla L + \lambda w) = (1 - \eta\lambda)w - \eta\nabla L$.

(C) TRUE. It effectively enlarges the training set.

(D) TRUE. Stronger shrinkage.

Q.13 — Answer: (B) [MCQ, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

$\log\text{-odds} = \beta_0 + \beta x + \dots$, so $\Delta(\log\text{-odds}) = \beta \cdot \Delta x = 0.7 \times 3 = 2.1$.

Odds are multiplied by $e^{\beta \Delta x} = e^{2.1} = \mathbf{8.17}$.

Distractors: $\beta \cdot \Delta x$ is the change in log-odds (not a multiplier); e^β ignores Δx ; $\sigma(\beta \Delta x)$ is a probability, not an odds ratio.

Q.14 — Answer: -1.26 to -1.24 [NAT, 1M · SVM & Decision Trees > SVM: dual & support vectors]

$w = \sum_i \alpha_i y_i x_i$ (sum over support vectors).

$w_2 = (0.25)(+1)(1) + (0.25)(+1)(2) + (0.5)(-1)(4) = \mathbf{-1.25}$.

(Full vector $w = (0.75, -1.25)$; note $\sum \alpha_i y_i = 0$ holds.)

Q.15 — Answer: 63.348 to 63.748 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

Total variance = $\sum \lambda = 31$. Cumulative proportion of variance explained:

k	1	2	3	4	5	6
λ_k	10.9	8.8	5.2	4.6	1.3	0.2
cumulative %	35.16	63.55	80.32	95.16	99.35	100

$(\lambda_1 + \lambda_2)/\sum \lambda = 19.7/31 = \mathbf{63.5\%}$.

Q.16 — Answer: 2.2261 to 2.2461 [NAT, 2M · SVM & Decision Trees > SVM: margin & geometry]

P1 and P5 (same class) lie on one margin line and P3 on the other; the optimal boundary is parallel to the line through P1, P5, midway to P3. Support vectors: P1, P3, P5.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (2, -1)$, $b = -6$; $\|w\| = 2.2361$, margin $2/\|w\| = 0.8944$.

Point	P1	P2	P3	P4	P5	P6
$y(w^T x + b)$	1	2	1	2	1	3

All values are ≥ 1 (feasible), with equality exactly at the support vectors.

$\|w\| = \mathbf{2.24}$.

Q.17 — Answer: 9.4067 to 9.4267 [NAT, 2M · SVM & Decision Trees > Regression trees]

Parent mean $\bar{y} = 5.286$, $SSE_0 = \sum (y - \bar{y})^2 = 69.429$.

Evaluate every split:

Split t	Left mean	Right mean	$SSE_L + SSE_R$
3.5	4	5.5	67.5
5	2.5	6.4	47.7
7.5	2.667	7.25	33.417
10	2.75	8.667	9.417

12.5	3.6	9.5	19.7
14.5	4.5	10	43.5

Best split $t = 10$: left mean 2.75, right mean 8.667, SSE = 9.417, reduction = 60.012.

Answer: **9.42**.

Q.18 — Answer: (D) [MCQ, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

The line passes through $(-2, 0)$ and $(0, -3)$: $x_1/-2 + x_2/-3 = 1$, i.e. $-3x_1 - 2x_2 - 6 = 0$. Hence (w_0, w_1, w_2) must be a non-zero multiple of $(-6, -3, -2)$.

Orientation: the class $y = 1$ region must have $w_0 + w_1x_1 + w_2x_2 > 0$. At the origin the $y = 1$ side has sign $+$ ($w_0 = 12$), which matches the figure (the origin lies on the $y = 1$ side).

So the answer is $(w_0, w_1, w_2) = (12, 6, 4)$ (any positive multiple gives the same classifier).

Other options either describe a different line (swapped/negated coefficients move the intercepts) or the same line with the sign flipped, which would label every training point incorrectly.

Q.19 — Answer: 0.1914 to 0.2114 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$$\Sigma^{-1} = \begin{bmatrix} 0.4 & -0.2 \\ -0.2 & 0.6 \end{bmatrix}$$

$k = 1$: $\Sigma^{-1}\mu_1 = (-0.2, 1.6)$; $\mathbf{x}^T\Sigma^{-1}\mu = 3.6$; $\frac{1}{2}\mu^T\Sigma^{-1}\mu = 2.3$; $\ln \pi = -1.0986 \Rightarrow \delta_1 = 0.2014$.

$k = 2$: $\Sigma^{-1}\mu_2 = (0.4, -0.2)$; $\mathbf{x}^T\Sigma^{-1}\mu = -1.2$; $\frac{1}{2}\mu^T\Sigma^{-1}\mu = 0.2$; $\ln \pi = -1.0986 \Rightarrow \delta_2 = -2.4986$.

$k = 3$: $\Sigma^{-1}\mu_3 = (-1, 1)$; $\mathbf{x}^T\Sigma^{-1}\mu = 4$; $\frac{1}{2}\mu^T\Sigma^{-1}\mu = 1.5$; $\ln \pi = -1.0986 \Rightarrow \delta_3 = 1.4014$.

Required $\delta_3(\mathbf{x}) = \mathbf{0.2}$.

Q.20 — Answer: 0.3788 to 0.3828 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

Hidden pre-activations: $a_1 = -0.5 \cdot 0.5 + (-2) \cdot -0.5 + (-1) = -0.25$; $a_2 = -0.5 \cdot 0.5 + (1.5) \cdot -0.5 + (0.5) = -0.5$.

Hidden activations ($\tanh(a)$): $h_1 = -0.2449$, $h_2 = -0.4621$.

Output pre-activation: $o = -1 \cdot -0.2449 + (0.5) \cdot -0.4621 + (-0.5) = -0.4861$.

$\hat{y} = \sigma(o) = \mathbf{0.381}$.

Q.21 — Answer: 36.1567 to 36.1767 [NAT, 2M · Unsupervised Learning > k-means]

Iteration 1 - assignment (nearest initial centroid, squared Euclidean distance):

Point	$d^2(\cdot, C1)$	$d^2(\cdot, C2)$	$d^2(\cdot, C3)$	cluster
P1 (1, 8)	4	68	50	C1
P2 (4, 5)	34	26	8	C3
P3 (1, 4)	36	20	26	C2
P4 (7, 3)	85	25	1	C3
P5 (6, 7)	34	58	16	C3
P6 (7, 10)	36	116	50	C1
P7 (4, 8)	13	65	29	C1
P8 (3, 7)	13	49	25	C1

Update: each centroid becomes the mean of its assigned points: $C1 \rightarrow (3.75, 8.25)$; $C2 \rightarrow (1, 4)$; $C3 \rightarrow (5.67, 5)$.

$J = \sum \|x_i - \mu_{c(i)}\|^2$ with the new centroids = **36.17**.

Note: each step of Lloyd's algorithm can only decrease (or keep) J , which is why it converges.

Q.22 — Answer: (C), (D) [MSQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Distances from the query, sorted:

rank	#	(x_1, x_2)	distance	class
1	3	(5, 4)	$\sqrt{5} = 2.236$	A
2	11	(5, 9)	$\sqrt{10} = 3.162$	A

3	9	(3, 8)	$\sqrt{13} = 3.606$	B
4	1	(9, 9)	$\sqrt{18} = 4.243$	B
5	8	(9, 3)	$\sqrt{18} = 4.243$	B
6	6	(2, 9)	$\sqrt{25} = 5$	B
7	5	(1, 4)	$\sqrt{29} = 5.385$	A
8	2	(2, 2)	$\sqrt{32} = 5.657$	A

(A) FALSE. 7 nearest: #3, #11, #9, #1, #8, #6, #5 $\rightarrow$ 3 A vs 4 B $\Rightarrow$ class B.

(B) FALSE. 5 nearest: #3, #11, #9, #1, #8 $\rightarrow$ 2 A vs 3 B $\Rightarrow$ class B.

(C) TRUE. 3 nearest: #3, #11, #9 $\rightarrow$ 2 A vs 1 B $\Rightarrow$ class A.

(D) TRUE. 1 nearest: #3 $\rightarrow$ 1 A vs 0 B $\Rightarrow$ class A.

Q.23 — Answer: -0.0301 to -0.0299 [NAT, 2M · Regression & Model Selection > Basis functions & feature scaling]

$x = x'/100 \Rightarrow \hat{y} = \beta_0 + (\beta_1/100)x'$. Fitted values are unchanged; the slope becomes $-3/100 = -0.03$.

Answer: **-0.03**.

Q.24 — Answer: (C) [MCQ, 2M · Regression & Model Selection > Cross-validation]

Minimum CV MSE = 3.19 at $\lambda = 0.3$, with SE = 0.16.

One-SE threshold = $3.19 + 0.16 = 3.35$.

Models within the threshold: $\lambda = 0.3$ (3.19), $\lambda = 1$ (3.29).

The one-SE rule picks the largest such λ (simplest model): $\lambda = 1$; the minimum-CV rule picks $\lambda = 0.3$.

Answer: **$\lambda = 1$** .

Q.25 — Answer: 0.0755 to 0.0955 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [3, 4, 11]: Gini = $1 - [(3/18)^2 + (4/18)^2 + (11/18)^2] = 0.5494$.

Humidity = High: [0, 1, 7], Gini = $1 - [(0/8)^2 + (1/8)^2 + (7/8)^2] = 0.2188$.

Humidity = Normal: [3, 3, 4], Gini = $1 - [(3/10)^2 + (3/10)^2 + (4/10)^2] = 0.66$.

Weighted Gini = $(8/18)(0.2188) + (10/18)(0.66) = 0.4639$.

Gini gain = $0.5494 - 0.4639 = 0.0855$.

Answer: **0.09**.

Q.26 — Answer: -0.01 to 0.01 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Predicted positive (score ≥ 0.65): examples 1, 6, 7, 9.

TP = 4, FP = 0, FN = 1, TN = 5.

False positive rate = $FP/(FP + TN) = 0$.

Raising t typically raises precision and lowers recall/FPR; lowering t does the opposite (threshold moving).

Q.27 — Answer: 3.99 to 4.01 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Use single linkage: $d(A, B) = \min_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: {P1} + {P2} at height **2**. Updated: $d(\{P1, P2\}, \{P3\}) = 11$, $d(\{P1, P2\}, \{P4\}) = 11$, $d(\{P1, P2\}, \{P5\}) = 10$.

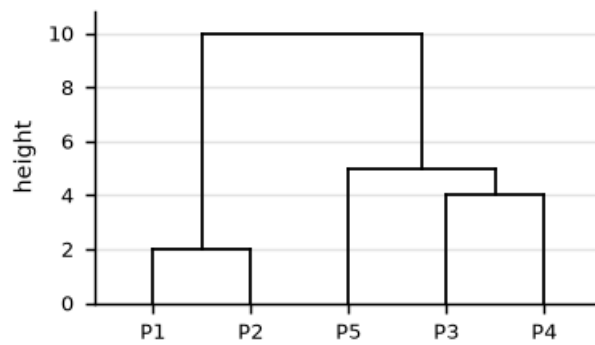
Merge 2: {P3} + {P4} at height **4**. Updated: $d(\{P3, P4\}, \{P5\}) = 5$, $d(\{P3, P4\}, \{P1, P2\}) = 11$.

Merge 3: {P5} + {P3, P4} at height **5**. Updated: $d(\{P3, P4, P5\}, \{P1, P2\}) = 10$.

Merge 4: {P1, P2} + {P3, P4, P5} at height **10**.

The second merge occurs at height **4**.

Dendrogram:



Q.28 — Answer: 0 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Backpropagation]

$a = Wx + b = (3, -2, -2)$, $h = \text{ReLU}(a) = (3, 0, 0)$, $\hat{y} = v \cdot h + c = 6$; $\partial L / \partial \hat{y} = \hat{y} - y = 7$.

$\text{ReLU}'(a) = (1, 0, 0)$ (unit j is active iff $a_j > 0$).

$\partial L / \partial W_{31} = (\hat{y} - y) \cdot v_3 \cdot \text{ReLU}'(a_3) \cdot x_1 = 7 \times 1 \times 0 \times 2 = 0$. The unit is inactive, so no gradient flows to its incoming weights.

Q.29 — Answer: 499.99 to 500.01 [NAT, 2M · Unsupervised Learning › PCA & dimensionality reduction]

For centered data, projecting onto the top- k PCs and reconstructing leaves the components along the discarded eigenvectors; their average squared length (divisor $n - 1$) is the sum of the discarded eigenvalues.

Discarded eigenvalues: 5.5, 3, 1.5; sum = 10; total = 31.5.

Total error = $(n - 1) \times 10 = 50 \times 10 = 500$.

Q.30 — Answer: 1 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Perceptron]

Rule: if $y_i(w \cdot x_i + b) \leq 0$, then $w \leftarrow w + \eta y_i x_i$, $b \leftarrow b + \eta y_i$ ($\eta = 1$). Start $w = (0, 0)$, $b = 0$.

Example 1: $a = 0$, $y \cdot a = 0 \rightarrow$ **mistake**, update $\rightarrow w = (2, 0)$, $b = -1$.

Example 2: $a = -1$, $y \cdot a = 1 \rightarrow$ correct, no change.

Example 3: $a = 1$, $y \cdot a = -1 \rightarrow$ **mistake**, update $\rightarrow w = (1, -3)$, $b = -2$.

Example 4: $a = 2$, $y \cdot a = 2 \rightarrow$ correct, no change.

Required value = 1.

Note: a point with net input exactly 0 counts as a mistake ($y \cdot a \leq 0$) – this is why the first example always triggers an update when starting from zero.

Q.31 — Answer: 3 [NAT, 2M · Regression & Model Selection › Ridge & Lasso]

With $X^T X = I$: $\beta_{\text{OLS}} = X^T y = b$.

Ridge: $\nabla [\|y - X\beta\|^2 + \lambda \|\beta\|^2] = 0 \Rightarrow (1 + \lambda)\beta = b \Rightarrow \beta_{\text{ridge}} = b / (1 + \lambda)$.

Lasso: the problem separates per coordinate, $\min \frac{1}{2}(\beta_j - b_j)^2 + \lambda |\beta_j| \Rightarrow$ soft-thresholding $\beta_j = \text{sign}(b_j) \cdot \max(|b_j| - \lambda, 0)$.

j	1	2	3	4	5
b_j	0.5	-0.3	-2.5	-1.6	-2.8
ridge	0.25	-0.15	-1.25	-0.8	-1.4
lasso	0	0	-1.5	-0.6	-1.8

Answer: 3.

Q.32 — Answer: 0.615 to 0.635 [NAT, 2M · SVM & Decision Trees › Decision trees: structure & traversal]

Route each record from the root to a leaf:

#	Predicted	Actual	Correct?
1	Yes	Yes	yes
2	Yes	Yes	yes
3	Yes	Yes	yes
4	Yes	No	no

5	Yes	No	no
6	No	No	yes
7	Yes	Yes	yes
8	Yes	No	no
9	Yes	Yes	yes

Misclassified: 3 of 9 (accuracy 66.67%). Answer: **0.62**.

Q.33 — Answer: (A), (C), (D) [MSQ, 2M · Regression & Model Selection › Bias-variance trade-off]

(A) TRUE. Validation MSE is minimum (2.22) at degree 4.

(B) FALSE. Gaps: 0.3, 1.29, 2.17, 0.72, 2, 2.88, 4.35; the largest is at degree 7.

(C) TRUE. Train 0.57 (lowest), validation 4.92 > 2.22.

(D) TRUE. Gaps: 0.3, 1.29, 2.17, 0.72, 2, 2.88, 4.35.

Q.34 — Answer: 7.99 to 8.01 [NAT, 2M · Unsupervised Learning › Hierarchical clustering]

Use complete linkage: $d(A, B) = \max_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: {P4} + {P5} at height **3**. Updated: $d(\{P4, P5\}, \{P1\}) = 6$, $d(\{P4, P5\}, \{P2\}) = 7$, $d(\{P4, P5\}, \{P3\}) = 7$, $d(\{P4, P5\}, \{P6\}) = 7$.

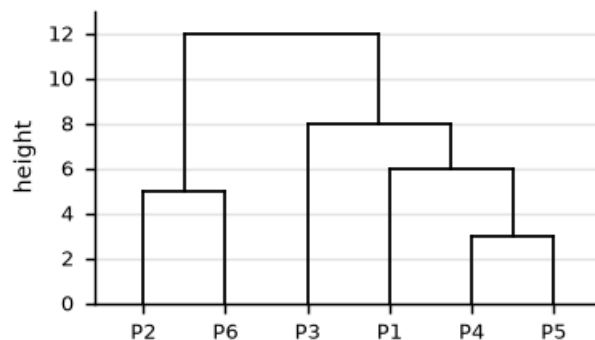
Merge 2: {P2} + {P6} at height **5**. Updated: $d(\{P2, P6\}, \{P1\}) = 12$, $d(\{P2, P6\}, \{P3\}) = 11$, $d(\{P2, P6\}, \{P4, P5\}) = 7$.

Merge 3: {P1} + {P4, P5} at height **6**. Updated: $d(\{P1, P4, P5\}, \{P3\}) = 8$, $d(\{P1, P4, P5\}, \{P2, P6\}) = 12$.

Merge 4: {P3} + {P1, P4, P5} at height **8**. Updated: $d(\{P1, P3, P4, P5\}, \{P2, P6\}) = 12$.

Merge 5: {P2, P6} + {P1, P3, P4, P5} at height **12**.

P3 and P4 first lie in the same cluster at the merge of height 8, so their cophenetic distance is **8** (their original distance was 7).



Q.35 — Answer: 0.5692 to 0.5732 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Softmax + cross-entropy]

$z = Wx + b = (-0.5, 1, -2)$; $p = \text{softmax}(z) = (0.1753, 0.7856, 0.0391)$.

$\delta = \partial L / \partial z = p - y = (-0.8247, 0.7856, 0.0391)$ (true class 1).

Since $z_k = \sum_j W_{kj} x_j + b_k$: $\partial L / \partial W_{kj} = (p_k - y_k) x_j$ and $\partial L / \partial b_k = p_k - y_k$ (gradient matrix = $(p - y)x^T$).

$W_{22} \leftarrow -1 - 1 \times 0.7856 \times -2 = \mathbf{0.5712}$.

Mock Test 46

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	6	10	19

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Probabilistic & Instance-based Classifiers	7	11
Unsupervised Learning	7	11
Regression & Model Selection	7	11
Neural Networks (MLP / Feed-forward)	7	11
SVM & Decision Trees	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MSQ (one or more correct)]

Which of the following statements about the k-nearest-neighbour method is/are CORRECT?

- (A) k-NN is a lazy (instance-based) learner: training essentially consists of storing the data.
- (B) Prediction with brute-force k-NN takes time independent of the training-set size.
- (C) As $n \rightarrow \infty$, the error rate of 1-NN is at most twice the Bayes error rate (binary case).
- (D) Increasing k generally increases bias and decreases variance of the k-NN classifier.

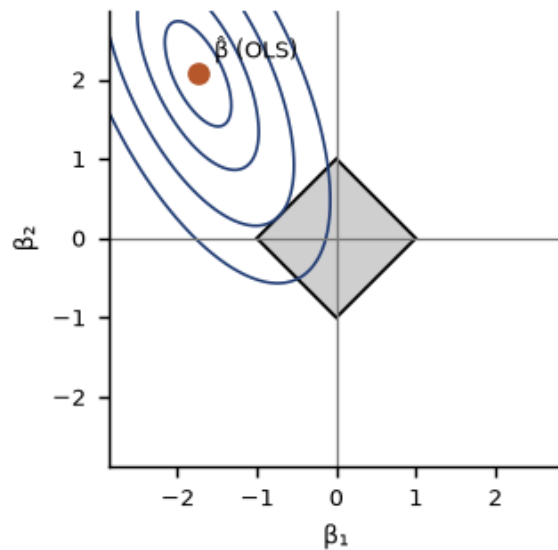
Q.2 [1 mark · MSQ (one or more correct)]

Which of the following statements about hierarchical clustering is/are CORRECT?

- (A) Hierarchical clustering requires the number of clusters before it can be run.
- (B) Average linkage defines the distance between clusters as the distance between their medoids.
- (C) Storing the full distance matrix for agglomerative clustering needs $O(n^2)$ memory.
- (D) Complete linkage is particularly prone to the chaining effect.

Q.3 [1 mark · MCQ]

For a two-predictor penalised least-squares problem, the figure shows the constraint region (shaded) of the penalty together with elliptical contours of the residual sum of squares centred at the least-squares estimate $\hat{\beta}$ (OLS). The penalised estimate (β_1, β_2) satisfies



- (A) $\beta_1 \neq 0$ and $\beta_2 \neq 0$ (both shrunk, neither exactly zero)
- (B) $\beta_1 = \beta_2 = 0$
- (C) $\beta_1 = 0$ exactly and $\beta_2 \neq 0$
- (D) $\beta_2 = 0$ exactly and $\beta_1 \neq 0$

Q.4 [1 mark · NAT (numerical answer)]

A Gaussian naive Bayes classifier has $d = 6$ continuous features and $K = 3$ classes, with a separate mean and variance for every (feature, class) pair. The number of independent parameters, including class priors, is _____ (Answer in integer)

Q.5 [1 mark · NAT (numerical answer)]

A fully connected MLP has layer sizes 100-64-100-256-2 (input-...-output), every layer having a bias vector. The number of bias parameters is _____ (Answer in integer)

Q.6 [1 mark · MCQ]

Cost-complexity pruning of a classification tree produced the nested subtrees $T1 \supset T2 \supset T3 \supset T4$ below. For complexity parameter $\alpha = 0.03$, which subtree minimises $R_\alpha(T) = R(T) + \alpha|T|$?

Subtree	T1	T2	T3	T4
Leaves $ T $	15	13	12	6
Training error $R(T)$	0.115	0.125	0.155	0.165

- (A) T1
- (B) T2
- (C) T3
- (D) T4

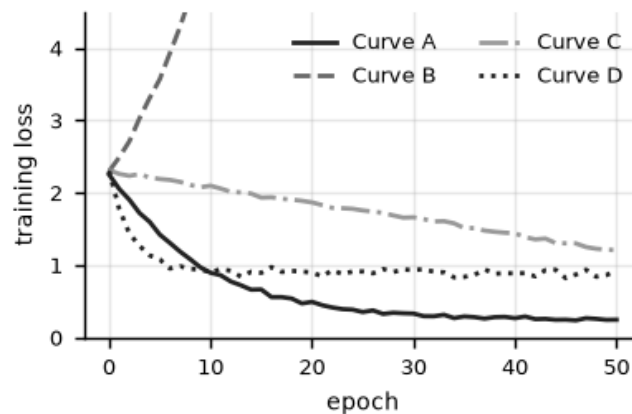
Q.7 [1 mark · MCQ]

A soft-margin linear SVM has learned $f(x) = 2x_1 + x_2 - 1$. For the training example $x = (2, -3)$ with label $y = +1$, which statement is CORRECT?

- (A) Correctly classified but strictly inside the margin ($0 < \xi < 1$)
- (B) Exactly on the decision boundary ($\xi = 1$)
- (C) Correctly classified, on or outside the margin ($\xi = 0$)
- (D) Misclassified ($\xi > 1$)

Q.8 [1 mark · MCQ]

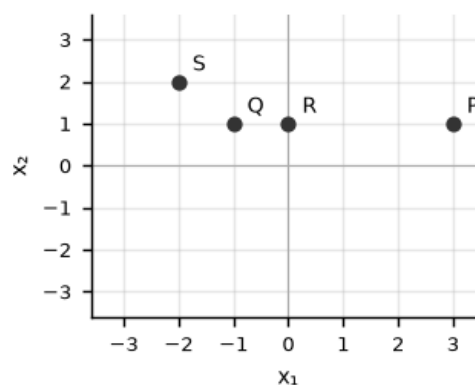
The training-loss curves below were obtained with four different learning rates (everything else fixed). Which curve corresponds to a learning rate that is too low (very slow, almost linear decrease)?

**(A)** Curve A**(B)** Curve B**(C)** Curve C**(D)** Curve D**Q.9 [1 mark · NAT (numerical answer)]**

The Gaussian kernel is $K(x, z) = \exp(-\|x - z\|^2 / (2\sigma^2))$ with $\sigma = 1.5$. For $x = (0, 2, 0)$ and $z = (2, 0, 2)$, the value of $K(x, z)$ is _____ (rounded off to three decimal places)

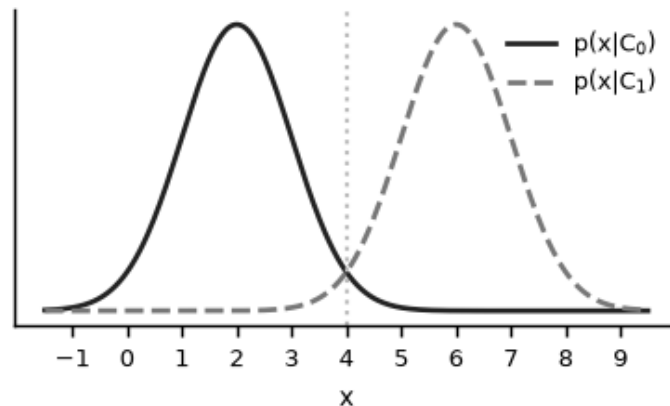
Q.10 [1 mark · MSQ (one or more correct)]

A perceptron with weights $w = (-1, 3)$ and bias $b = -1$ outputs $\hat{y} = +1$ if $w \cdot x + b \geq 0$ and $\hat{y} = -1$ otherwise. Which of the points shown below are classified as $\hat{y} = +1$?

**(A)** Point P = (3, 1)**(B)** Point Q = (-1, 1)**(C)** Point R = (0, 1)**(D)** Point S = (-2, 2)

Q.11 [1 mark · MCQ]

Two classes have unit-variance Gaussian class-conditional densities with means 2 (C_0) and 6 (C_1), as plotted. With equal priors the Bayes decision threshold is the dotted line at $x = 4$. If the prior of C_1 is increased to 0.7, the threshold



- (A) It moves towards $x = 6$, shrinking the region assigned to C_1
- (B) It moves towards $x = 2$, enlarging the region assigned to C_1
- (C) It moves towards $x = 6$, enlarging the region assigned to C_1
- (D) It splits into two thresholds, giving a quadratic boundary

Q.12 [1 mark · MSQ (one or more correct)]

Which of the following statements about cross-validation (CV) is/are CORRECT? (n denotes the number of observations.)

- (A) LOOCV estimates of test error have low bias but can have relatively high variance.
- (B) The minimum CV error over many candidate models is an unbiased estimate of the chosen model's test error.
- (C) For least-squares linear regression, LOOCV error can be computed from a single fit using the leverages h_{ii} .
- (D) Nested CV gives a nearly unbiased estimate of generalisation error when hyperparameters are tuned by CV.

Q.13 [1 mark · NAT (numerical answer)]

k -means with $k = 2$ searches over partitions of the data into k non-empty, unlabeled clusters. The number of distinct such partitions of $n = 5$ points is ____ (Answer in integer)

Q.14 [1 mark · MSQ (one or more correct)]

Regarding the bias-variance trade-off in supervised regression with squared-error loss, which of the following statements is/are CORRECT?

- (A) Regularisation trades a small increase in bias for a reduction in variance.
- (B) For a fixed model class, more training data tends to reduce variance.
- (C) For nested least-squares polynomial fits, training MSE never increases as the degree increases.
- (D) Increasing model flexibility typically increases variance.

Q.15 [1 mark · NAT (numerical answer)]

Two shopping baskets are encoded as binary vectors (1 = item bought). The Jaccard distance between basket A and basket B is ____ (rounded off to two decimal places)

Item	milk	bread	eggs	butter	jam	tea	rice	salt	sugar
Basket A	1	0	1	0	1	1	0	0	1
Basket B	1	1	0	0	0	1	1	0	1

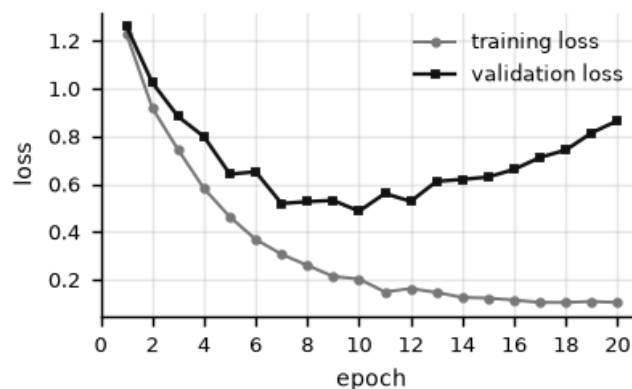
Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · MSQ (one or more correct)]**

Which of the following statements about LDA, QDA and logistic regression is/are CORRECT?

- (A) LDA and logistic regression always produce identical coefficient estimates.
- (B) Fisher's LDA with K classes can always find K discriminant directions.
- (C) Changing the class priors in LDA rotates the decision boundary.
- (D) LDA models each class-conditional density as Gaussian with a covariance matrix Σ common to all classes.

Q.17 [2 marks · NAT (numerical answer)]

A network is trained with early stopping on validation loss using patience $P = 4$: training stops as soon as the validation loss has failed to improve on its best value for 4 consecutive epochs, and the weights from the best epoch are restored. Using the validation losses below (also plotted), the epoch whose weights are returned (restored) as the final model is _____ (Answer in integer)



epoch	1	2	3	4	5	6	7	8	9	10
val. loss	1.261	1.025	0.883	0.799	0.642	0.652	0.519	0.527	0.532	0.487

epoch	11	12	13	14	15	16	17	18	19	20
val. loss	0.561	0.528	0.612	0.620	0.630	0.662	0.712	0.743	0.815	0.866

Q.18 [2 marks · MSQ (one or more correct)]

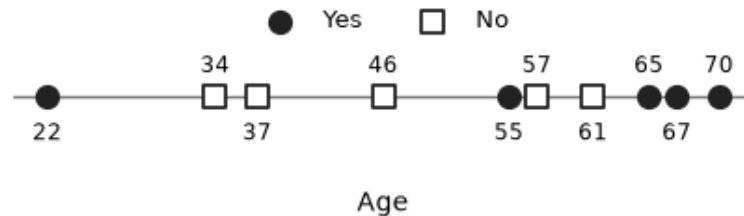
Each option shows a 2×2 matrix $[[K(x_1, x_1), K(x_1, x_2)], [K(x_2, x_1), K(x_2, x_2)]]$. Which of them can be the Gram matrix of some valid (Mercer) kernel $\tilde{K}$ on two points x_1, x_2 ?

- (A) $\begin{bmatrix} -2 & 1 \\ 1 & 6 \end{bmatrix}$
- (B) $\begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix}$
- (C) $\begin{bmatrix} 9 & 0 \\ 1 & 10 \end{bmatrix}$
- (D) $\begin{bmatrix} 8 & 2 \\ 3 & 5 \end{bmatrix}$

Q.19 [2 marks · NAT (numerical answer)]

The continuous attribute Age and the class label of 10 training records are given below (unsorted). A binary split “Age $\leq t$ ” is chosen to maximise information gain, with t restricted to midpoints between consecutive sorted values. Find the number of candidate thresholds that need to be evaluated (midpoints between consecutive sorted values at which the class label changes). (Answer in integer)

Age	37	22	70	65	67	34	61	57	46	55
Class	No	Yes	Yes	Yes	Yes	No	No	No	No	Yes

**Q.20 [2 marks · MSQ (one or more correct)]**

Consider k-means (Lloyd's algorithm) and k-medoids (PAM). Which of the following statements is/are CORRECT?

- (A) k-means is a limiting case of EM for a Gaussian mixture with equal spherical covariances $\sigma^2 I$ as $\sigma \rightarrow 0$.
- (B) Lloyd's algorithm can cycle forever between two partitions with different SSE values.
- (C) k-medoids requires the data to lie in a Euclidean space so that means can be computed.
- (D) During Lloyd's iterations a cluster can become empty; implementations handle it, e.g. by re-seeding.

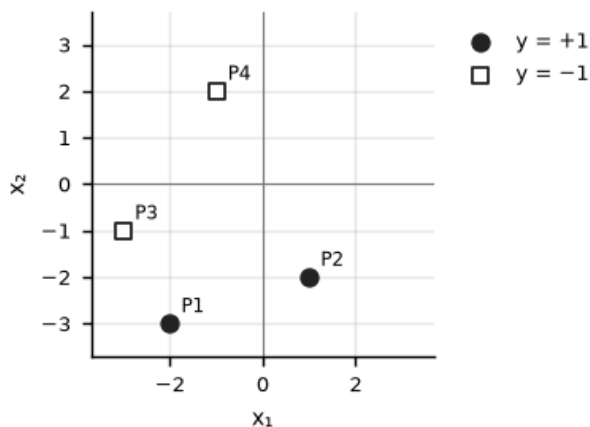
Q.21 [2 marks · NAT (numerical answer)]

$X_1, \dots, X_n$ are i.i.d. with mean $\mu = 2$ and variance $\sigma^2 = 4$, with $n = 10$. Consider the shrinkage estimator $\hat{\mu} = c \cdot \bar{x}$ of μ , where $\bar{x}$ is the sample mean. Using $\text{MSE} = \text{Bias}^2 + \text{Variance}$, the value of c that minimises the MSE of $\hat{\mu} = c \cdot \bar{x}$ is _____ (rounded off to three decimal places)

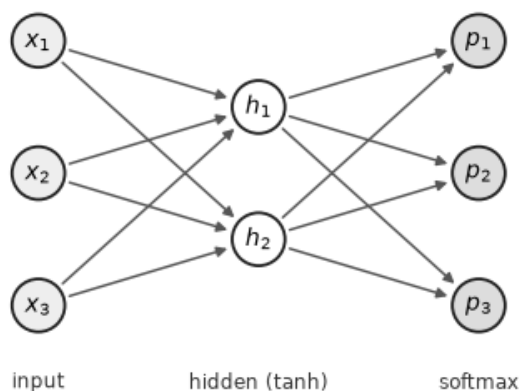
Q.22 [2 marks · NAT (numerical answer)]

A perceptron with augmented weight vector $w = (w_0, w_1, w_2)$ (w_0 is the bias, $\tilde{x} = (1, x_1, x_2)$) starts from $w = (0, 0, 0)$ and uses learning rate $\eta = 1$. The points P1, P2, ... are presented once, in order; whenever $y \cdot w^T \tilde{x} \leq 0$ the update $w \leftarrow w + \eta y \tilde{x}$ is applied. Find the number of weight updates made during this epoch. (Answer in integer)

Point	x_1	x_2	y
P1	-2	-3	+1
P2	1	-2	+1
P3	-3	-1	-1
P4	-1	2	-1

**Q.23 [2 marks · NAT (numerical answer)]**

A 3-2-3 network computes $h = \tanh(W_1 x + b_1)$, $z = W_2 h + b_2$, $p = \text{softmax}(z)$, with the parameters below. For $x = (-1, 1, 1)^T$, the predicted probability p_3 of class 3 is _____ (rounded off to three decimal places)



$$W_1 = \begin{bmatrix} -0.5 & -0.5 & 0 \\ 1 & 0.5 & 0 \end{bmatrix}$$

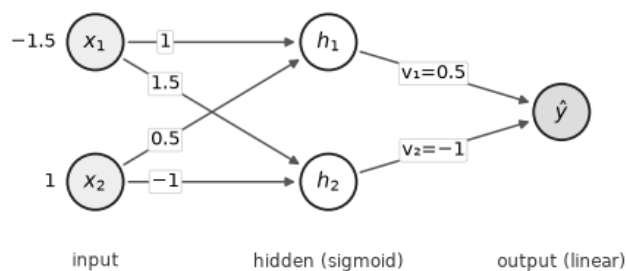
$$b_1 = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} -2 & -1 \\ -1 & -1 \\ -2 & -2 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} -1 \\ -0.5 \\ -1 \end{bmatrix}$$

Q.24 [2 marks · NAT (numerical answer)]

In the network below, hidden units are sigmoid, the output is linear ($\hat{y} = v_1 h_1 + v_2 h_2$), all biases are 0, and the loss is $L = \frac{1}{2}(\hat{y} - y)^2$. For input $x = (-1.5, 1)$ and target $y = -0.5$, the value of $\partial L / \partial v_1$ is _____ (rounded off to three decimal places)

**Q.25 [2 marks · NAT (numerical answer)]**

The sample covariance matrix of a two-feature dataset is S given below. In principal component analysis, the variance of the data along the second principal component is _____ (rounded off to two decimal places)

$$S = \begin{bmatrix} 10 & -3 \\ -3 & 10 \end{bmatrix}$$

Q.26 [2 marks · NAT (numerical answer)]

A 4-class classifier outputs logits $z = (-1.5, 1.5, -1, -0.5)$ and probabilities $p = \text{softmax}(z)$. The true class is 3, and the loss is the cross-entropy $L = -\ln p_3$. Then the Euclidean norm $\|\partial L / \partial z\|$ of the gradient with respect to the logit vector is _____ (rounded off to three decimal places)

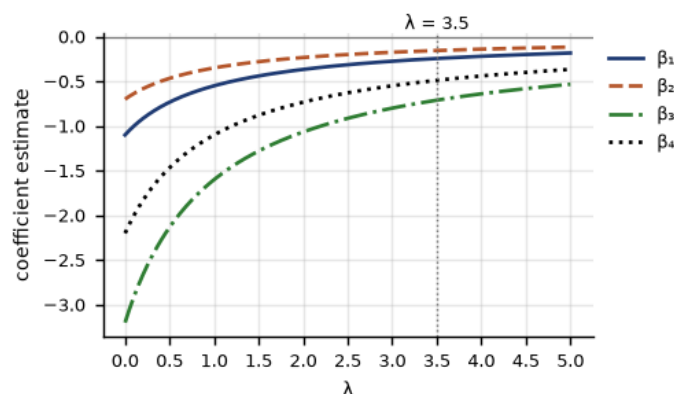
Q.27 [2 marks · NAT (numerical answer)]

The confusion matrix of a 3-class classifier is given below (rows = actual class, columns = predicted class). The weighted-average (by class support) recall is _____ (rounded off to two decimal places)

actual \ predicted	A	B	C
A	13	6	4
B	5	9	1
C	5	1	10

Q.28 [2 marks · MSQ (one or more correct)]

For a design with orthonormal columns ($X^T X = I$), the figure shows the coefficient paths $\beta_i(\lambda)$ of a penalised least-squares estimator that minimises $\frac{1}{2}\|y - X\beta\|^2 + P_\lambda(\beta)$, where the penalty P_λ is either the ridge penalty $(\lambda/2)\|\beta\|_2^2$ or the lasso penalty $\lambda\|\beta\|_1$. Which of the following statements is/are CORRECT?



- (A) As λ increases, the training residual sum of squares decreases.
- (B) At $\lambda = 3.5$, every coefficient equals its least-squares value divided by 4.5.
- (C) The plotted paths are those of ridge regression (L2 penalty).
- (D) At $\lambda = 3.5$, at least one coefficient is exactly zero.

Q.29 [2 marks · NAT (numerical answer)]

Agglomerative clustering with **average linkage** is applied to the distance matrix below. The cophenetic distance between P4 and P5 (the dendrogram height at which they are first joined) is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5
P1	0	4	8	13	3
P2		0	8	9	1
P3			0	11	9
P4				0	10
P5					0

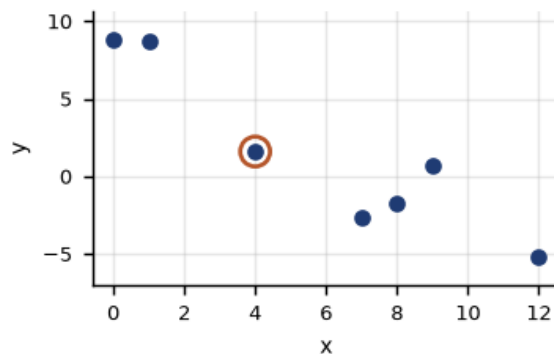
Q.30 [2 marks · NAT (numerical answer)]

A logistic regression classifier has $P(y = 1 | \mathbf{x}) = \sigma(3 + (-2)x_1 + (3)x_2)$. An input is labelled $y = 1$ if and only if $P(y = 1 | \mathbf{x}) \geq 0.5$. For this rule, the x_1 -intercept of the decision boundary (the value of x_1 where it meets $x_2 = 0$) is _____ (rounded off to two decimal places)

Q.31 [2 marks · NAT (numerical answer)]

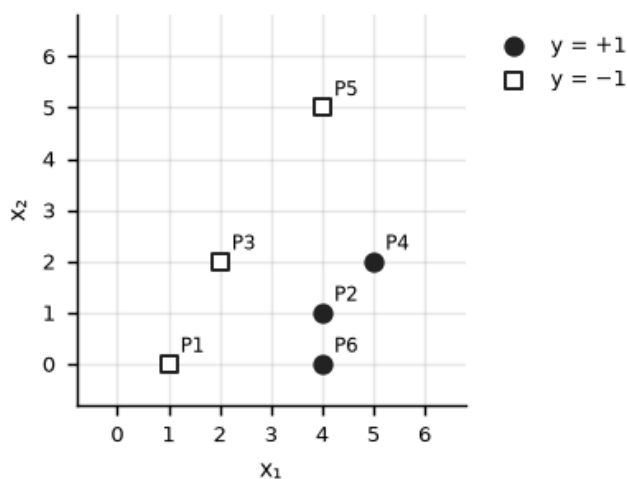
A simple linear regression (with intercept) is fitted by least squares to the 7 points shown. With $H = X(X^T X)^{-1} X^T$, if the errors have variance $\sigma^2 = 9$, the variance of the residual e_i of the observation with $x = 4$ (circled) is _____ (rounded off to three decimal places)

x	0	1	4	7	8	9	12
y	8.8	8.7	1.6	-2.6	-1.7	0.7	-5.2

**Q.32 [2 marks · MSQ (one or more correct)]**

A hard-margin linear SVM is trained on the 6 labelled points shown. Which of the following points is/are support vectors of the resulting maximum-margin classifier?

Point	x_1	x_2	y
P1	1	0	-1
P2	4	1	+1
P3	2	2	-1
P4	5	2	+1
P5	4	5	-1
P6	4	0	+1



(A) P1

(B) P2

(C) P3

(D) P5

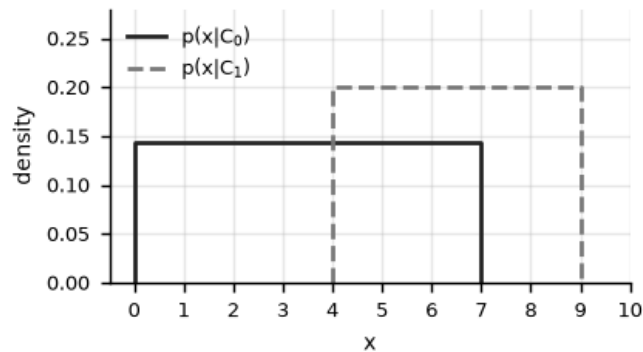
Q.33 [2 marks · NAT (numerical answer)]

Consider the model $y = wx + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$, $\sigma^2 = 4$, and $w \sim N(0, \tau^2)$ with $\tau^2 = 2$. Given the four observations below, the MAP estimate of w is _____ (rounded off to three decimal places)

x	2	3	-1	-2
y	-2	-2	1	-1

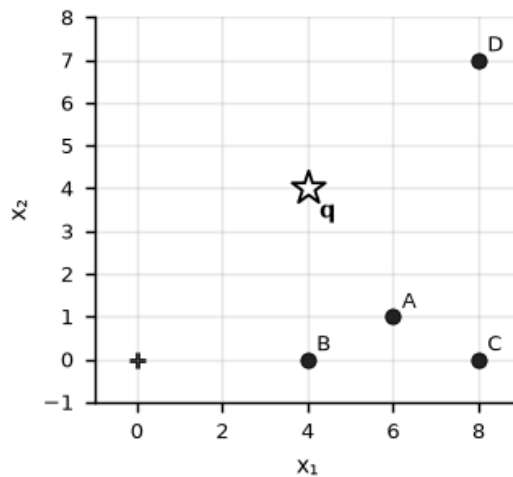
Q.34 [2 marks · NAT (numerical answer)]

For a binary problem with scalar x , $p(x | C_0)$ is uniform on $[0, 7]$ and $p(x | C_1)$ is uniform on $[4, 9]$, with priors $P(C_0) = 0.5$ and $P(C_1) = 0.5$. Then the Bayes (minimum achievable) error rate is _____ (rounded off to three decimal places)

**Q.35 [2 marks · MCQ]**

A query point $q = (4, 4)$ and four points A-D are shown below. Using **Euclidean (L_2) distance**, which point is nearest to q ?

Point	A	B	C	D
x_1	6	4	8	8
x_2	1	0	0	7



- (A) Point A
(C) Point C

- (B) Point B
(D) Point D

— END OF QUESTION PAPER —

Mock Test 46 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MSQ	1	(A), (C), (D)	19	NAT	2	4
2	MSQ	1	(C)	20	MSQ	2	(A), (D)
3	MCQ	1	(A)	21	NAT	2	0.9061 to 0.9121
4	NAT	1	38	22	NAT	2	2
5	NAT	1	422	23	NAT	2	0.1463 to 0.1503
6	MCQ	1	(D)	24	NAT	2	0.1586 to 0.1626
7	MCQ	1	(B)	25	NAT	2	6.99 to 7.01
8	MCQ	1	(C)	26	NAT	2	1.2269 to 1.2309
9	NAT	1	0.0675 to 0.0715	27	NAT	2	0.5826 to 0.6026
10	MSQ	1	(B), (C), (D)	28	MSQ	2	(B), (C)
11	MCQ	1	(B)	29	NAT	2	10.74 to 10.76
12	MSQ	1	(A), (C), (D)	30	NAT	2	1.49 to 1.51
13	NAT	1	15	31	NAT	2	7.4217 to 7.4664
14	MSQ	1	(A), (B), (C), (D)	32	MSQ	2	(A), (B), (D)
15	NAT	1	0.5614 to 0.5814	33	NAT	2	−0.452 to −0.448
16	MSQ	2	(D)	34	NAT	2	0.2123 to 0.2163
17	NAT	2	10	35	MCQ	2	(A)
18	MSQ	2	(B)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Probabilistic & Instance-based Classifiers					11
Unsupervised Learning					11
Regression & Model Selection					11
Neural Networks (MLP / Feed-forward)					11
SVM & Decision Trees					11
Total					55

Mock Test 46 — Detailed Solutions

Q.1 — Answer: (A), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

- (A) **TRUE**. All computation is deferred to prediction time.
- (B) **FALSE**. It is $O(nd)$ per query.
- (C) **TRUE**. Cover-Hart bound: $R^* \leq R_{1NN} \leq 2R^*(1 - R^*)$.
- (D) **TRUE**. Averaging more neighbours smooths the decision boundary.

Q.2 — Answer: (C) [MSQ, 1M · Unsupervised Learning > Hierarchical clustering]

- (A) **FALSE**. The dendrogram can be cut at any level afterwards.
- (B) **FALSE**. It averages all pairwise distances.
- (C) **TRUE**. There are $n(n-1)/2$ pairs.
- (D) **FALSE**. Chaining is typical of single linkage.

Q.3 — Answer: (A) [MCQ, 1M · Regression & Model Selection > Ridge & Lasso]

The shaded region is the constraint set of the lasso (L1: $|\beta_1| + |\beta_2| \leq t$, a diamond). The penalised estimate is the first point at which the RSS contours (ellipses centred at β_{OLS} , which lies outside the region) touch the region.

From the figure the tangency point is approximately $(-0.73, 0.27)$.

The contour touches the boundary at a point that is not on an axis, so both coefficients are non-zero. Here the lasso tangency is on an edge of the diamond.

Answer: **$\beta_1 \neq 0$ and $\beta_2 \neq 0$ (both shrunk, neither exactly zero)**. Both coefficients cannot be zero since the OLS point lies outside the region (the constraint is active but $t > 0$).

Q.4 — Answer: 38 [NAT, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

Means: $K_d = 18$; variances: $K_d = 18$; priors: $K - 1 = 2$.

Total = $2K_d + K - 1 = 38$.

Q.5 — Answer: 422 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Parameter counting]

Fully connected layer with n_{in} inputs and n_{out} outputs: $n_{in} \cdot n_{out}$ weights + n_{out} biases.

Layer 1: $100 \times 64 = 6400$ weights, 64 biases.

Layer 2: $64 \times 100 = 6400$ weights, 100 biases.

Layer 3: $100 \times 256 = 25600$ weights, 256 biases.

Layer 4: $256 \times 2 = 512$ weights, 2 biases.

Weights = 38912, biases = 422. Required: **422**.

Q.6 — Answer: (D) [MCQ, 1M · SVM & Decision Trees > Decision trees: overfitting & pruning]

Subtree	$R(T) + \alpha T $
T1	$0.115 + 0.03 \times 15 = 0.565$
T2	$0.125 + 0.03 \times 13 = 0.515$
T3	$0.155 + 0.03 \times 12 = 0.515$
T4	$0.165 + 0.03 \times 6 = 0.345$

Minimum at **T4**. Larger α penalises leaves more and favours smaller trees; $\alpha = 0$ would pick the largest tree T1.

Q.7 — Answer: (B) [MCQ, 1M · SVM & Decision Trees > SVM: soft margin & hinge loss]

$y \cdot f(x) = (+1) \cdot (0) = 0$.

Slack $\xi = \max(0, 1 - y f(x)) = 1$.

Rules: $y f \geq 1 \Rightarrow \xi = 0$ (on/outside margin); $0 < y f < 1 \Rightarrow$ inside margin, still correct; $y f = 0 \Rightarrow$ on the boundary ($\xi = 1$); $y f < 0 \Rightarrow$ misclassified ($\xi > 1$).

Hence: **Exactly on the decision boundary ($\xi = 1$)**.

Q.8 — Answer: (C) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Typical signatures: far-too-high $\eta \rightarrow$ loss blows up; somewhat-too-high $\eta \rightarrow$ quick drop then plateau at a high loss (bouncing around the minimum); too-low $\eta \rightarrow$ slow, nearly linear decrease; good $\eta \rightarrow$

fast decrease to the lowest loss.

Curve A: a well-chosen learning rate.

Curve B: a learning rate that is far too high (training diverges).

Curve C: a learning rate that is too low (very slow, almost linear decrease).

Curve D: a learning rate that is somewhat too high (fast initial drop, then the loss plateaus at a high value).

Hence the answer is **Curve C**.

Q.9 — Answer: 0.0675 to 0.0715 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

$$\|x - z\|^2 = 12; 2\sigma^2 = 4.5.$$

$$K = \exp(-12/4.5) = \mathbf{0.069}.$$

Note: $K \in (0, 1]$, $K(x, x) = 1$, and K decreases as the points move apart or as γ grows.

Q.10 — Answer: (B), (C), (D) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

Net input $a = w \cdot x + b = -1x_1 + (3)x_2 + (-1)$; output +1 iff $a \geq 0$.

$$(P) a = -1 \cdot 3 + (3) \cdot 1 + (-1) = -1 \rightarrow -1.$$

$$(Q) a = -1 \cdot -1 + (3) \cdot 1 + (-1) = 3 \rightarrow +1 \text{ (selected)}.$$

$$(R) a = -1 \cdot 0 + (3) \cdot 1 + (-1) = 2 \rightarrow +1 \text{ (selected)}.$$

$$(S) a = -1 \cdot -2 + (3) \cdot 2 + (-1) = 7 \rightarrow +1 \text{ (selected)}.$$

Answer: **(B), (C), (D)**.

Q.11 — Answer: (B) [MCQ, 1M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$$x^* = (\mu_0 + \mu_1)/2 + \sigma^2 \ln(\pi_0/\pi_1)/(\mu_1 - \mu_0) = 3.788.$$

A larger prior for C_1 means more of the x-axis is assigned to it, so the threshold moves away from μ_1 towards $\mu_0 = 2$. Answer: **It moves towards $x = 2$, enlarging the region assigned to C_1 .**

Equal variances keep the boundary a single point (linear); priors shift it but never make it quadratic.

Q.12 — Answer: (A), (C), (D) [MSQ, 1M · Regression & Model Selection > Cross-validation]

(A) TRUE. The n training sets overlap almost completely, so the fold errors are highly correlated.

(B) FALSE. Selecting the minimum makes it optimistically biased; use a separate test set or nested CV.

$$\mathbf{(C) TRUE.} CV_{(n)} = (1/n) \sum_i (e_i / (1 - h_{ii}))^2.$$

(D) TRUE. The outer loop evaluates the whole tuning procedure on untouched data.

Q.13 — Answer: 15 [NAT, 1M · Unsupervised Learning > k-means]

This is the Stirling number of the second kind $S(n, k)$, with $S(n, k) = k \cdot S(n-1, k) + S(n-1, k-1)$.

$$S(5, 2) = \mathbf{15.} \text{ (For } k = 2 \text{ it equals } 2^{n-1} - 1 = 15.)$$

Q.14 — Answer: (A), (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Bias-variance trade-off]

(A) TRUE. Shrinkage stabilises estimates.

(B) TRUE. Estimates stabilise as n grows.

(C) TRUE. A higher-degree model contains the lower-degree one as a special case.

(D) TRUE. Flexible fits change more with the training sample.

Q.15 — Answer: 0.5614 to 0.5814 [NAT, 1M · Unsupervised Learning > Distance measures]

$$\text{Counts: } f_{11} = 3, f_{10} = 2, f_{01} = 2, f_{00} = 2.$$

$$\text{Jaccard } J = f_{11} / (f_{11} + f_{10} + f_{01}) = 3/7 = 0.4286 \text{ (0-0 matches are ignored).}$$

$$\text{SMC} = (f_{11} + f_{00}) / m = 5/9 = 0.5556.$$

$$\text{Required Jaccard distance} = \mathbf{0.57}.$$

Q.16 — Answer: (D) [MSQ, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

(A) FALSE. Different estimation criteria.

- (B) FALSE. At most $K - 1$.
 (C) FALSE. Priors only shift it.
 (D) TRUE. That is the defining assumption.

Q.17 — Answer: 10 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Training & regularisation]

Rule: keep track of the best (lowest) validation loss; an epoch that does not strictly improve on it increments a counter, which is reset on improvement; stop when the counter reaches patience $P = 4$.

Trace - 1: 1.261 (new best); 2: 1.025 (new best); 3: 0.883 (new best); 4: 0.799 (new best); 5: 0.642 (new best); 6: 0.652 (no improvement, count = 1); 7: 0.519 (new best); 8: 0.527 (no improvement, count = 1); 9: 0.532 (no improvement, count = 2); 10: 0.487 (new best); 11: 0.561 (no improvement, count = 1); 12: 0.528 (no improvement, count = 2); 13: 0.612 (no improvement, count = 3); 14: 0.620 (no improvement, count = 4).

Training stops after epoch 14; the best validation loss 0.487 occurred at epoch 10, whose weights are restored.

Answer: **10**. The widening train/validation gap after this point is the signature of overfitting.

Q.18 — Answer: (B) [MSQ, 2M · SVM & Decision Trees · SVM: kernels]

A Gram matrix of a valid kernel must be symmetric and positive semidefinite. For a symmetric 2×2 matrix this is equivalent to: both diagonal entries ≥ 0 and determinant ≥ 0 .

- (A) FALSE. a diagonal entry $K(x, x) = \|\phi(x)\|^2$ is negative — impossible.
 (B) TRUE. symmetric, diagonal entries ≥ 0 and $\det = 7 > 0 \Rightarrow$ positive definite.
 (C) FALSE. not symmetric, but $K(x, z) = K(z, x)$ for every kernel.
 (D) FALSE. not symmetric, but $K(x, z) = K(z, x)$ for every kernel.

Q.19 — Answer: 4 [NAT, 2M · SVM & Decision Trees · Decision trees: split selection]

Sort the values: 22(Y), 34(N), 37(N), 46(N), 55(Y), 57(N), 61(N), 65(Y), 67(Y), 70(Y).

Parent [5 Yes, 5 No], $H = 1$. The optimal threshold always lies where the class changes between neighbours (Fayyad-Irani), giving 4 candidates:

Threshold	Left ($\leq t$) [Yes, No]	Right [Yes, No]	IG
28	[1, 0]	[4, 5]	0.108
50.5	[1, 3]	[4, 2]	0.1245
56	[2, 3]	[3, 2]	0.029
63	[2, 5]	[3, 0]	0.3958

Best: $t = 63$ with $IG = 0.3958$. Answer: **4**.

Q.20 — Answer: (A), (D) [MSQ, 2M · Unsupervised Learning · k-means]

- (A) TRUE. The responsibilities become hard 0/1 assignments.
 (B) FALSE. J strictly decreases whenever the partition changes, so it cannot return to a worse partition.
 (C) FALSE. k-medoids never computes means; any dissimilarity works.
 (D) TRUE. Nothing in the algorithm prevents a centroid from losing all its points.

Q.21 — Answer: 0.9061 to 0.9121 [NAT, 2M · Regression & Model Selection · Bias-variance trade-off]

$E[c \bar{x}] = c\mu \Rightarrow \text{Bias} = (c - 1)\mu$; $\text{Var}(c \bar{x}) = c^2\sigma^2/n$. $\text{MSE}(c) = (c - 1)^2\mu^2 + c^2\sigma^2/n$.

Here $\mu = 2$, $\sigma^2/n = 4/10 = 0.4$.

$d\text{MSE}/dc = 2(c - 1)\mu^2 + 2c\sigma^2/n = 0 \Rightarrow c^* = \mu^2/(\mu^2 + \sigma^2/n) = 4/(4 + 0.4) = 0.9091$.

$c^* < 1$: accepting some bias reduces variance enough to lower MSE.

Answer: **0.909**.

Q.22 — Answer: 2 [NAT, 2M · SVM & Decision Trees · Perceptron (linear separators)]

Trace the epoch (a point with $y \cdot w^T \tilde{x} \leq 0$, including 0, triggers an update):

Step	$\tilde{x} = (1, x_1, x_2)$	y	$w^T \tilde{x}$	$y \cdot w^T \tilde{x} \leq 0$?	w after step
------	-----------------------------	-----	-----------------	----------------------------------	----------------

1	(1, -2, -3)	+1	0	yes → update	(1, -2, -3)
2	(1, 1, -2)	+1	5	no	(1, -2, -3)
3	(1, -3, -1)	-1	10	yes → update	(0, 1, -2)
4	(1, -1, 2)	-1	-5	no	(0, 1, -2)

Final $w = (0, 1, -2)$, number of updates = 2. Answer: **2**.

Q.23 — Answer: 0.1463 to 0.1503 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Forward pass]

$a = W_1 x + b_1 = (0.5, 0.5)$; $h = \tanh(a) = (0.4621, 0.4621)$.

$z = W_2 h + b_2 = (-2.3864, -1.4242, -2.8485)$.

$\exp(z) = (0.092, 0.2407, 0.0579)$, sum = 0.3906.

$p_3 = \exp(z_3) / \sum \exp(z_j) = \mathbf{0.1483}$. (All probabilities: (0.235, 0.616, 0.148).)

Q.24 — Answer: 0.1586 to 0.1626 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Backpropagation]

Forward: $a = Wx = (-1, -3.25)$, $h = \sigma(a) = (0.2689, 0.0373)$, $\hat{y} = v \cdot h = 0.0971$.

Output error $\delta_o = \partial L / \partial \hat{y} = \hat{y} - y = 0.5971$ (linear output, $L = \frac{1}{2}(\hat{y} - y)^2$).

$\partial L / \partial v_1 = \delta_o \cdot h_1 = 0.5971 \times 0.2689 = \mathbf{0.1606}$.

Q.25 — Answer: 6.99 to 7.01 [NAT, 2M · Unsupervised Learning › PCA & dimensionality reduction]

Eigenvalues of $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$: $\lambda = (a + c)/2 \pm \sqrt{((a - c)/2)^2 + b^2}$.

$\lambda = 10 \pm \sqrt{(0 + 9)} = 10 \pm 3 \rightarrow \lambda_1 = 13, \lambda_2 = 7$ (check: $\lambda_1 + \lambda_2 = \text{trace} = 20$, $\lambda_1 \lambda_2 = \det = 91$).

Eigenvector for λ_1 : $(S - \lambda_1 I)v = 0 \rightarrow (-3)v_1 + (-3)v_2 = 0 \rightarrow v \propto (1, -1)$.

PVE(PC1) = $\lambda_1 / (\lambda_1 + \lambda_2) = 65\%$.

Answer: **7**.

Q.26 — Answer: 1.2269 to 1.2309 [NAT, 2M · Neural Networks (MLP / Feed-forward) › Softmax + cross-entropy]

$p = \text{softmax}(z)$: $\exp(z) = (0.2231, 4.4817, 0.3679, 0.6065)$, sum = 5.6792, so $p = (0.0393, 0.7891, 0.0648, 0.1068)$.

For $L = -\ln p_y$ with softmax, $\partial L / \partial z = p - y$ (y one-hot, true class 3): $\partial L / \partial z = (0.0393, 0.7891, -0.9352, 0.1068)$.

$\|p - y\| = \sqrt{0.0015 + 0.6227 + 0.8746 + 0.0114} = \mathbf{1.229}$.

Note $\sum_k \partial L / \partial z_k = \sum p_k - 1 = 0$.

Q.27 — Answer: 0.5826 to 0.6026 [NAT, 2M · Probabilistic & Instance-based Classifiers › Evaluation metrics]

class	TP	predicted total	actual total	precision	recall	F1
A	13	23	23	0.5652	0.5652	0.5652
B	9	16	15	0.5625	0.6	0.5806
C	10	15	16	0.6667	0.625	0.6452

Precision of class $k = C_{kk} / (\text{column sum})$; recall = $C_{kk} / (\text{row sum})$.

Weighted recall = $\sum (\text{support}_k \times \text{recall}_k) / N = \sum C_{kk} / N = \text{accuracy} = \mathbf{0.59}$.

Q.28 — Answer: (B), (C) [MSQ, 2M · Regression & Model Selection › Ridge & Lasso]

With $X^T X = I$ the ridge path is $b / (1 + \lambda)$ and the lasso path is $\text{sign}(b)(|b| - \lambda)_+$. The plot shows paths reaching zero at finite $\lambda \Rightarrow$ lasso; here it is **ridge**.

(A) **FALSE**. Training RSS is minimised at $\lambda = 0$ and increases with λ .

(B) **TRUE**. $\beta(\lambda) = b / (1 + \lambda)$.

(C) **TRUE**. Smooth decay $b / (1 + \lambda)$, no exact zeros.

(D) **FALSE**. Ridge gives no exact zeros.

Q.29 — Answer: 10.74 to 10.76 [NAT, 2M · Unsupervised Learning › Hierarchical clustering]

Use average linkage (UPGMA): $d(A, B) = (1/|A||B|) \sum_{a \in A} \sum_{b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

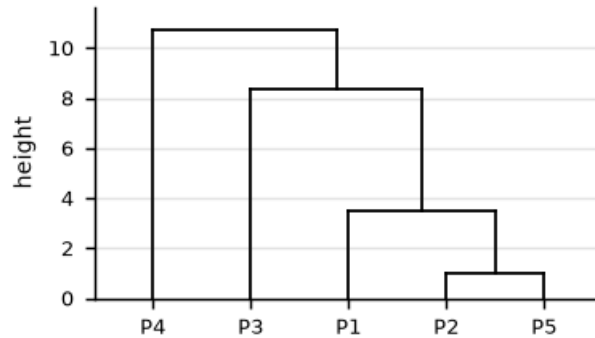
Merge 1: {P2} + {P5} at height **1**. Updated: $d(\{P2, P5\}, \{P1\}) = 3.5$, $d(\{P2, P5\}, \{P3\}) = 8.5$, $d(\{P2, P5\}, \{P4\}) = 9.5$.

Merge 2: {P1} + {P2, P5} at height **3.5**. Updated: $d(\{P1, P2, P5\}, \{P3\}) = 8.33$, $d(\{P1, P2, P5\}, \{P4\}) = 10.67$.

Merge 3: {P3} + {P1, P2, P5} at height **8.33**. Updated: $d(\{P1, P2, P3, P5\}, \{P4\}) = 10.75$.

Merge 4: {P4} + {P1, P2, P3, P5} at height **10.75**.

P4 and P5 first lie in the same cluster at the merge of height 10.75, so their cophenetic distance is **10.75** (their original distance was 10).



Q.30 — Answer: 1.49 to 1.51 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

$P(y=1|x) \geq 0.5 \Leftrightarrow z \geq c$, where $c = \ln(0.5/0.5) = 0$.

Here $c = 0$ because the threshold is 0.5.

Decision boundary: $3 + (-2)x_1 + (3)x_2 = 0$ (a straight line; changing the threshold only shifts it parallel to the 0.5 boundary).

$x_1 = (c - w_0)/w_1 = (0 - (3))/(-2) = 1.5$.

Q.31 — Answer: 7.4217 to 7.4664 [NAT, 2M · Regression & Model Selection > Multiple linear regression]

For simple linear regression with intercept, the leverage (diagonal of the hat matrix) is $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$.

$\bar{x} = 5.8571$, $S_{xx} = 114.8571$.

$h = 0.1729$; $\text{Var}(e_i) = \sigma^2(1 - h_{ii}) = 9 \times 0.8271 = 7.444$.

Answer: **7.444**.

Q.32 — Answer: (A), (B), (D) [MSQ, 2M · SVM & Decision Trees > SVM: dual & support vectors]

P1 and P5 (same class) lie on one margin line and P2 on the other; the optimal boundary is parallel to the line through P1, P5, midway to P2. Support vectors: P1, P2, P5.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (5/6, -1/2)$, $b = -11/6$; $\|w\| = 0.9718$, margin $2/\|w\| = 2.058$.

Point	P1	P2	P3	P4	P5	P6
$y(w^T x + b)$	1	1	1.167	1.333	1	1.5

Support vectors are exactly the points with $y_i(w^T x_i + b) = 1$ ($\alpha_i > 0$): **P1, P2, P5**.

Dual check: $\alpha = (\alpha_1 = 5/18, \alpha_2 = 17/36, \alpha_5 = 7/36)$ satisfy $\sum \alpha_i y_i = 0$ and $\sum \alpha_i y_i x_i = w$.

Q.33 — Answer: -0.452 to -0.448 [NAT, 2M · Regression & Model Selection > MLE & MAP view]

$-\ln p(w | \text{data}) = (1/(2\sigma^2))\sum (y_i - wx_i)^2 + w^2/(2\tau^2) + \text{const.}$

Multiplying by $2\sigma^2$: $\sum (y_i - wx_i)^2 + (\sigma^2/\tau^2)w^2$ — ridge with $\lambda = \sigma^2/\tau^2 = 2$.

$\sum x_i^2 = 18$, $\sum x_i y_i = -9$; $\hat{w}_{\text{MAP}} = \sum xy / (\sum x^2 + \lambda) = -9/(18 + 2) = -0.45$.

Answer: **-0.45**.

Q.34 — Answer: 0.2123 to 0.2163 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

$\pi_0 p(x|C_0) = 0.5/7 = 0.0714$ on $[0, 7]$; $\pi_1 p(x|C_1) = 0.5/5 = 0.1$ on $[4, 9]$.

Outside the overlap $[4, 7]$ only one class has non-zero density $\Rightarrow$ no error. Inside it the Bayes rule chooses C_1 (larger weighted density) and errs with density 0.0714.

Bayes error = (overlap length) $\times$ min = $3 \times 0.0714 = \mathbf{0.214}$.

Q.35 — Answer: (A) [MCQ, 2M · Unsupervised Learning › Distance measures]

Compute the measure from $q = (4, 4)$ to every point (cosine column shows similarity, larger = more similar):

Point	coordinates	Manhattan	Euclidean	Chebyshev	cosine similarity
A	(6, 1)	5	3.606	3	0.814
B	(4, 0)	4	4	4	0.707
C	(8, 0)	8	5.657	4	0.707
D	(8, 7)	7	5	4	0.998

Under Euclidean (L_2) distance the nearest to q is **point A**.

Note how the answer depends on the measure: Manhattan (L_1) distance $\rightarrow$ B; Euclidean (L_2) distance $\rightarrow$ A; Chebyshev (L_∞) distance $\rightarrow$ A; cosine similarity $\rightarrow$ D. Cosine similarity looks only at the angle from the origin, L_1 sums coordinate gaps, L_∞ takes the largest gap.

Mock Test 47

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	6	9	20

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Unsupervised Learning	7	11
SVM & Decision Trees	7	11
Neural Networks (MLP / Feed-forward)	7	11
Probabilistic & Instance-based Classifiers	7	11
Regression & Model Selection	7	11

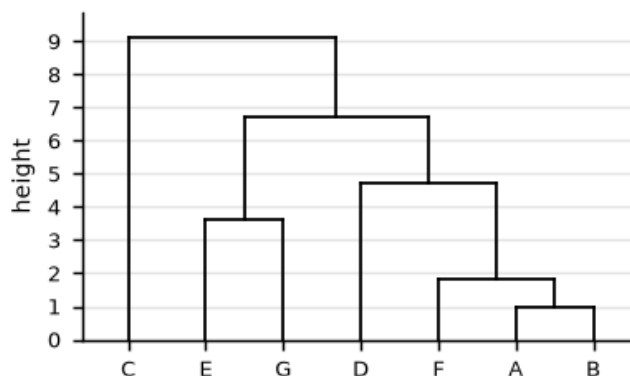
Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

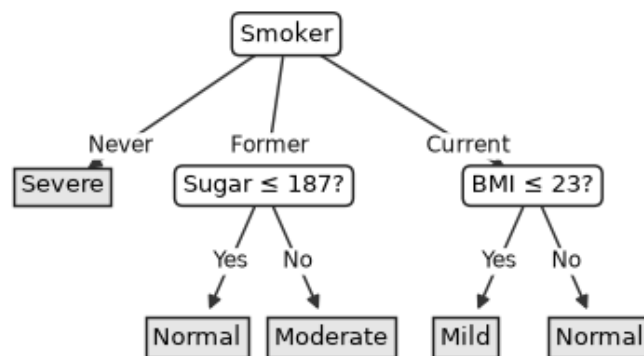
Q.1 [1 mark · NAT (numerical answer)]

The dendrogram below was obtained by agglomerative clustering of 7 points A-G. If it is cut to obtain exactly 3 clusters, the number of points in the cluster that contains point D is _____ (Answer in integer)



Q.2 [1 mark · MCQ]

The decision tree below predicts the diagnosis category of a patient. Edges labelled Yes/No answer the test in the node. What is the prediction for the record shown in the table?

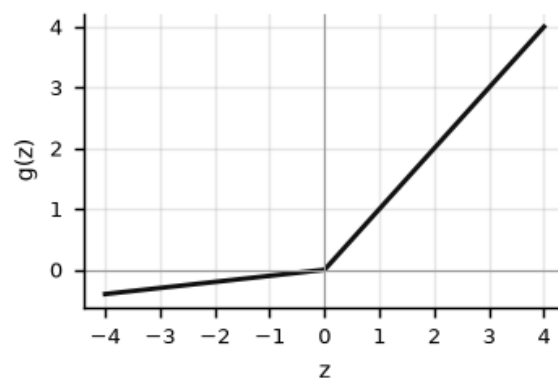


Sugar	BMI	BP	Smoker
186	19	94	Former

- (A) Normal (B) Severe
(C) Mild (D) Moderate

Q.3 [1 mark · MCQ]

The figure shows the graph of a function $g(z)$ used in neural networks. Which function is it?



- (A) ELU with $\alpha = 1$
(B) Leaky ReLU with $\alpha = 0.1$
(C) ReLU: $\max(0, z)$
(D) Softplus: $\ln(1 + e^z)$

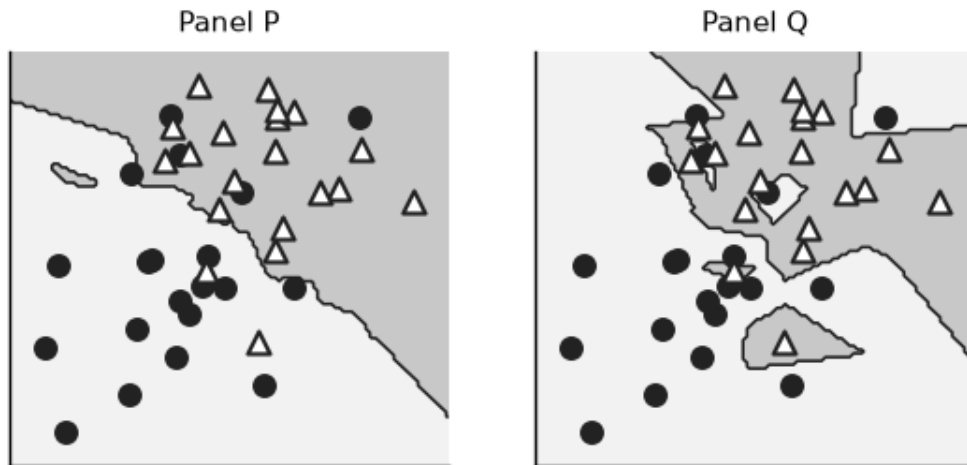
Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about the representational power of feed-forward neural networks is/are CORRECT?

- (A) A ReLU network computes a continuous piecewise-linear function of its input.
(B) Logistic regression has a non-linear decision boundary because the sigmoid is non-linear.
(C) A network with a single hidden layer of two ReLU units can approximate any continuous function on $[0, 1]$ to arbitrary accuracy.
(D) A ReLU network with finitely many units can represent a discontinuous step function exactly.

Q.5 [1 mark · MCQ]

The two panels show the decision regions of k-NN classifiers trained on the same 2-D data set, one with $k = 1$ and the other with $k = 15$. Which of the following statements is CORRECT?



- (A) Panel P uses $k = 15$; it has zero training error
- (B) Panel P uses $k = 1$; it has higher bias and lower variance
- (C) Panel Q uses $k = 1$; it has higher bias and lower variance
- (D) Panel Q uses $k = 1$; it has lower bias and higher variance

Q.6 [1 mark · NAT (numerical answer)]

In AdaBoost, a weak learner has weighted training error $\varepsilon = 0.28$. Its vote weight $\alpha = \frac{1}{2} \ln((1 - \varepsilon)/\varepsilon)$ is _____ (rounded off to two decimal places)

Q.7 [1 mark · NAT (numerical answer)]

A sigmoid-output classifier gives the probabilities $p = P(y = 1 | x)$ below. The mean binary cross-entropy $-[y \ln p + (1 - y) \ln(1 - p)]$ over the 4 examples is _____ (rounded off to three decimal places)

example	$p = P(y = 1)$	y
1	0.4	0
2	0.6	0
3	0.6	0
4	0.7	1

Q.8 [1 mark · MSQ (one or more correct)]

Which of the following statements about decision-tree learning is/are CORRECT?

- (A) The information gain of any split is non-negative.
- (B) A larger α in cost-complexity pruning yields a smaller (or equal-size) tree.
- (C) A decision tree's training predictions are unchanged by a strictly increasing transformation of a feature (e.g. $x \rightarrow \log x$).
- (D) Post-pruning grows a full tree and then removes subtrees that do not help validation performance.

Q.9 [1 mark · NAT (numerical answer)]

QDA is fitted with $K = 2$ classes and $p = 10$ features. Counting class priors, class means and the class-specific covariance matrices, the number of independent parameters is _____ (Answer in integer)

Q.10 [1 mark · MSQ (one or more correct)]

Which of the following statements about naive Bayes classifiers is/are CORRECT?

- (A) If a feature is duplicated, naive Bayes counts its evidence twice.
- (B) Laplace smoothing is used mainly to reduce the number of parameters of naive Bayes.
- (C) Naive Bayes is a discriminative classifier that models $P(y | x)$ directly.
- (D) The Laplace-smoothed estimate $(n_{v,c} + 1)/(n_c + V)$ does not depend on the number V of possible feature values.

Q.11 [1 mark · MSQ (one or more correct)]

Which of the following statements about principal component analysis (PCA) is/are CORRECT?

- (A) PCA is an unsupervised, linear dimensionality-reduction technique.
- (B) Scores on different principal components are uncorrelated.
- (C) The variance of the projected data along the j -th PC equals the j -th largest covariance eigenvalue.
- (D) A scree plot shows the eigenvalues (or variance explained) in decreasing order against component number.

Q.12 [1 mark · NAT (numerical answer)]

For a data set, $\bar{x} = 14$, $\bar{y} = 31$, $s_x = 4$, $s_y = 7$ and the sample correlation between x and y is $r = 0.9$. The slope of the least-squares regression line of y on x is _____ (rounded off to two decimal places)

Q.13 [1 mark · MSQ (one or more correct)]

Linear regression is trained by minimising the squared-error loss. Which of the following statements about gradient-based training is/are CORRECT?

- (A) Gradient descent on the MSE loss converges for every learning rate $\eta > 0$.
- (B) Batch gradient descent and the normal equations converge to the same minimiser when $X^T X$ is invertible.
- (C) A larger learning rate always gives faster convergence.
- (D) One epoch of SGD with batch size 1 on n examples performs n parameter updates.

Q.14 [1 mark · NAT (numerical answer)]

For $x = (3, 7, 4)$ and $y = (-2, 3, 2)$, the Chebyshev (L_∞) distance between x and y is _____ (rounded off to two decimal places)

Q.15 [1 mark · NAT (numerical answer)]

Regressing predictor x_3 on all the other predictors gives $R^2 = 0.8$. The VIF of x_3 is _____ (rounded off to two decimal places)

Q.16 - Q.35 : Two-mark questions

Q.16 [2 marks · MCQ]

A network computes $\hat{y} = W_2 \text{ReLU}(W_1 x + b_1) + b_2$ with the parameters below. For $x = (2, 0, 2)^T$, the output $\hat{y}$ is

$$W_1 = \begin{bmatrix} -2 & 0 & 0 \\ 1 & -1 & 2 \\ 2 & 2 & 2 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 0 & 2 & 0 \\ 1 & -1 & 1 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

(A) $\hat{y} = (12, 2)^T$

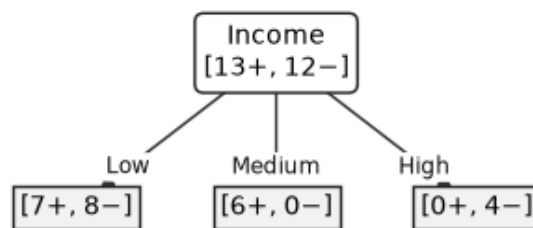
(B) $\hat{y} = (11, 2)^T$

(C) $\hat{y} = (7, 0)^T$

(D) $\hat{y} = (11, -3)^T$

Q.17 [2 marks · NAT (numerical answer)]

A node with 25 training records (class counts [positive+, negative−]) is split on the attribute Income as shown. Compute the information gain (in bits) of splitting on Income. (rounded off to two decimal places)

**Q.18 [2 marks · MSQ (one or more correct)]**

A linear regression of annual salary (₹ lakh) on Exp (years of experience), Female (1 if the employee is female, 0 otherwise) and city $\in \{A, B, C\}$ encoded by dummies CityB and CityC (reference level A) gives the fitted model

$$\hat{y} = 28 + 0.8 \cdot \text{Exp} + 2.3 \cdot \text{Female} - 1.1 \cdot \text{CityB} - 5.1 \cdot \text{CityC}$$

Which of the following is/are CORRECT?

(A) The predicted annual salary for a female employee in city B with 2 years of experience is 31.9.

(B) The intercept 28 is the predicted response when Female = 1, Exp = 0, in city C.

(C) Increasing Exp by 3 years, other variables fixed, changes the prediction by 2.4.

(D) Holding other variables fixed, the predicted response for city B minus that for city C is 4.

Q.19 [2 marks · NAT (numerical answer)]

A 3-class softmax (multinomial logistic) regression model computes scores $\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}$ and $P(y = k | \mathbf{x}) = \frac{e^{z_k}}{\sum_j e^{z_j}}$, where row k of $\mathbf{W}$ holds the weights of class k . For the parameters below and $\mathbf{x} = (1, 0)^T$, the cross-entropy loss $-\ln P(y = 3 | \mathbf{x})$ incurred if the true class is 3 (natural log) is _____ (rounded off to two decimal places)

$$\mathbf{W} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \\ -1 & -1 \end{bmatrix}$$

$$\mathbf{b} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

Q.20 [2 marks · MSQ (one or more correct)]

Let σ be the logistic sigmoid, and $'$ denote the derivative. At $z = 2$ (and $-z = -2$), which of the following is/are CORRECT? ($\text{softplus}(z) = \ln(1 + e^z)$.)

- (A) $|\tanh(2)| > |2|$ (B) $\tanh(2) + \tanh(-2) = 1$
 (C) $\sigma'(2) \geq 0.25$ (D) $\tanh'(2) = 1 - \tanh^2(2) = 0.071$

Q.21 [2 marks · NAT (numerical answer)]

Ridge regression without intercept minimises $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_2^2$ with $\lambda = 4$, for the data below ($\boldsymbol{\beta} = (\beta_1, \beta_2)^T$). The value of β_2 is _____ (rounded off to three decimal places)

$$\mathbf{X} = \begin{bmatrix} 2 & 2 \\ 3 & 0 \\ 3 & 0 \\ 3 & 0 \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} 7 \\ 7 \\ 2 \\ 2 \end{bmatrix}$$

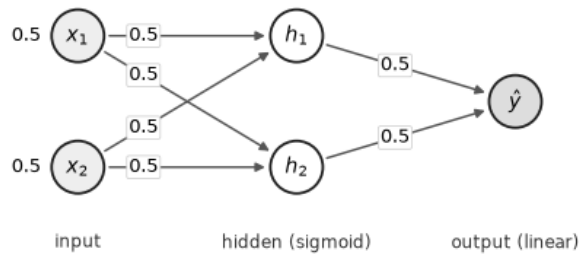
Q.22 [2 marks · NAT (numerical answer)]

A naive Bayes classifier is trained on the 11 examples below, using maximum-likelihood estimates (no smoothing). For the new instance (Income = Medium, Credit = Good, Employed = Yes), the posterior probability $P(\text{Approve} = \text{Yes} | \mathbf{x})$ is _____ (rounded off to two decimal places)

#	Income	Credit	Employed	Approve
1	High	Good	No	Yes
2	Medium	Good	No	No
3	Low	Good	Yes	No
4	Medium	Good	Yes	No
5	Low	Good	No	Yes
6	Medium	Good	Yes	Yes
7	Low	Good	No	Yes
8	Low	Good	No	Yes
9	High	Good	No	Yes
10	High	Good	Yes	No
11	Medium	Good	No	No

Q.23 [2 marks · NAT (numerical answer)]

In the 2-2-1 network below (sigmoid hidden units, linear output $\hat{y} = v_1 h_1 + v_2 h_2$, all biases 0), **every** weight is initialised to 0.5. One gradient-descent step with $\eta = 0.1$ is taken on $L = \frac{1}{2}(\hat{y} - y)^2$ for the input $x = (0.5, 0.5)$ with target $y = -1$. Then the updated value of w_{11} (weight from x_1 to h_1) is _____ (rounded off to three decimal places)

**Q.24 [2 marks · NAT (numerical answer)]**

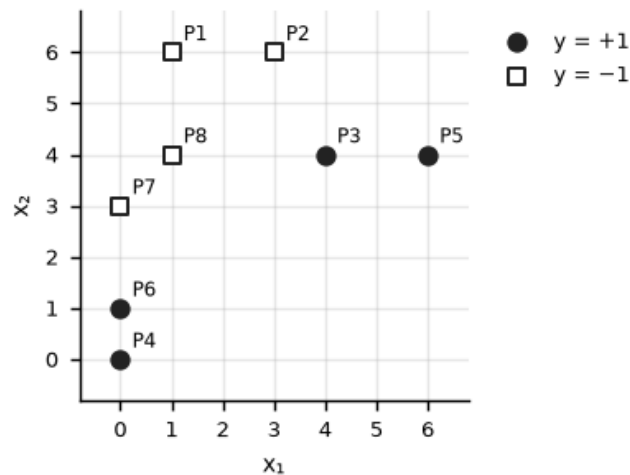
Two clusterings U and V of 7 points are given below (cluster names are arbitrary). Then the number of point pairs placed in the same cluster by **both** clusterings is _____ (Answer in integer)

Point	P1	P2	P3	P4	P5	P6	P7
Clustering U	β	β	α	β	α	α	β
Clustering V	2	1	2	2	1	1	1

Q.25 [2 marks • MCQ]

Four candidate linear decision boundaries are proposed for the labelled training points shown. Which boundary separates the two classes with the LARGEST geometric margin (minimum distance from the boundary to any training point, all points correctly classified)?

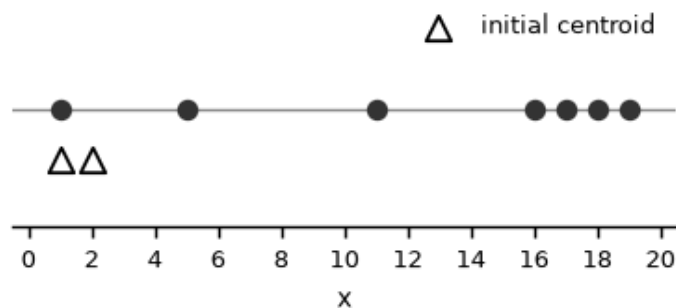
Point	x_1	x_2	y
P1	1	6	-1
P2	3	6	-1
P3	4	4	+1
P4	0	0	+1
P5	6	4	+1
P6	0	1	+1
P7	0	3	-1
P8	1	4	-1



- (A) $x_1 - x_2 + 1.5 = 0$
 (B) $x_1 - x_2 + 2 = 0$
 (C) $x_1 - 2x_2 + 4.5 = 0$
 (D) $x_1 - 2x_2 + 5.5 = 0$

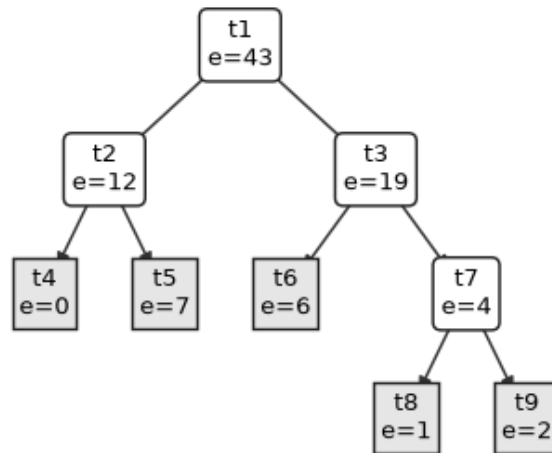
Q.26 [2 marks • NAT (numerical answer)]

The 1-D dataset $\{1, 5, 11, 16, 17, 18, 19\}$ is clustered with k-means ($k = 2$) using initial centroids 1 and 2 (shown as triangles). Lloyd's iterations (assign to nearest centroid, then recompute means) are run until convergence. Then the number of iterations in which at least one centroid **changes** its value is _____ (Answer in integer)



Q.27 [2 marks · NAT (numerical answer)]

A classification tree was grown on $N = 100$ training records. Each node shows $e(t)$, the number of training records that would be misclassified if t were a leaf; $R(t) = e(t)/N$. For $\alpha = 0.01$, the cost-complexity $R_\alpha(T) = R(T) + \alpha|T|$ of the full tree T is _____ (rounded off to three decimal places)

**Q.28 [2 marks · NAT (numerical answer)]**

A classifier for a rare condition has sensitivity (recall) 0.85 and specificity 0.99. It is deployed on a population where the prevalence of the positive class is 0.01. The expected accuracy in this population is _____ (rounded off to three decimal places)

Q.29 [2 marks · MSQ (one or more correct)]

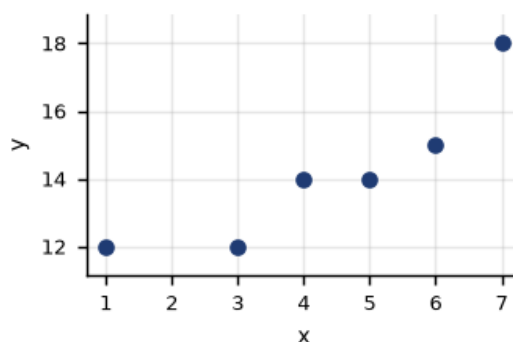
Which of the following statements about PCA, SVD and dimensionality reduction is/are CORRECT?

- (A) Rotating the data by an orthogonal matrix rotates the PC directions but leaves the eigenvalues unchanged.
- (B) PCA on standardized features is equivalent to eigen-decomposition of the correlation matrix.
- (C) The mean squared reconstruction error (divisor $n - 1$) when keeping k PCs equals the sum of the discarded eigenvalues.
- (D) If $X = U\Sigma V^T$, the PC scores are given by $V\Sigma$.

Q.30 [2 marks · NAT (numerical answer)]

Assume $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with ε_i i.i.d. $N(0, \sigma^2)$. For the 6 observations below, with β estimated by least squares (= maximum likelihood), the maximised log-likelihood $\ln L(\beta, \hat{\sigma}^2)$ (natural log, with $\hat{\sigma}^2$ the maximum-likelihood estimate) is _____ (rounded off to two decimal places)

x	1	3	4	5	6	7
y	12	12	14	14	15	18



Q.31 [2 marks · MSQ (one or more correct)]

A node with class counts [50+, 30−] can be split by either of two binary splits S1 or S2 (table). For each impurity measure, the “gain” is the parent impurity minus the weighted child impurity. Which of the following statements is/are CORRECT?

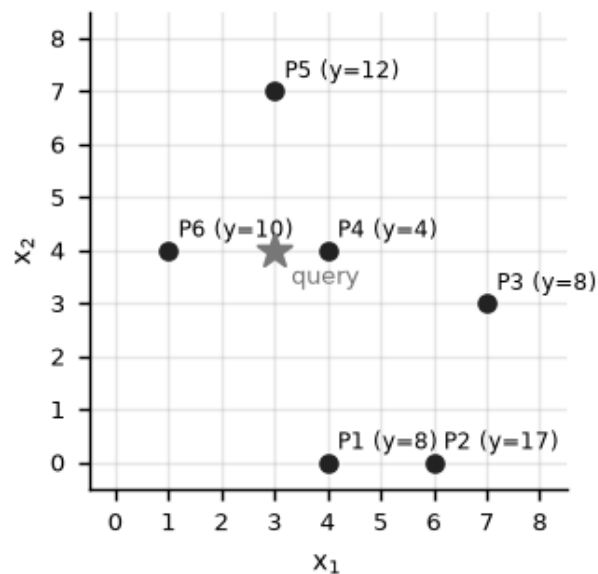
Split	Left child [+ , −]	Right child [+ , −]
S1	[21+, 18−]	[29+, 12−]
S2	[8+, 20−]	[42+, 10−]

- (A) Gini gain prefers S2 over S1.
- (B) Information gain (entropy) prefers S2 over S1.
- (C) The information gain of S2 is 0.19 bits (to two decimals).
- (D) Misclassification-error reduction prefers S2 over S1.

Q.32 [2 marks · NAT (numerical answer)]

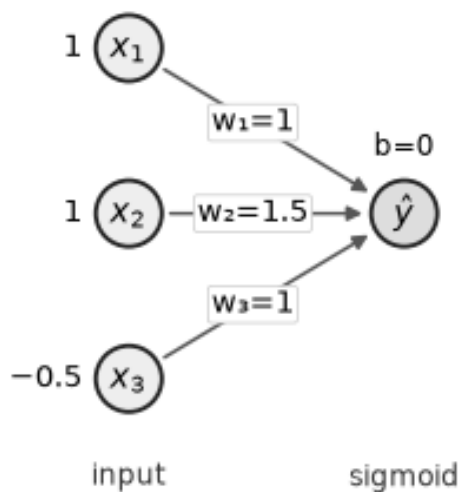
A distance-weighted k-NN regressor with $k = 3$ predicts $\hat{y} = \sum w_i y_i / \sum w_i$ over the 3 nearest neighbours, with weights $w_i = 1/d_i^2$. For the training data below, the prediction at $\mathbf{x} = (3, 4)$ (Euclidean distance) is _____ (rounded off to two decimal places)

Point	x_1	x_2	y
P1	4	0	8
P2	6	0	17
P3	7	3	8
P4	4	4	4
P5	3	7	12
P6	1	4	10



Q.33 [2 marks • NAT (numerical answer)]

A single sigmoid neuron (shown below) receives $x = (1, 1, -0.5)$ and has target $y = 1$. The loss is $L = \frac{1}{2}(\hat{y} - y)^2$. The value of $\partial L / \partial w_2$ is _____ (rounded off to 4 decimal places)

**Q.34 [2 marks • NAT (numerical answer)]**

Least-squares linear regression on $n = 2$ observations minimises $J(w) = (1/n)\|Xw - y\|^2$ by batch gradient descent with constant learning rate η . The data give the matrix $X^T X$ below. Then the constant learning rate η that minimises the worst-case contraction factor $\max_i |1 - \eta \lambda_i|$ is _____ (rounded off to three decimal places)

$$X^T X = \begin{bmatrix} 9 & 3 \\ 3 & 9 \end{bmatrix}$$

Q.35 [2 marks • MCQ]

Six points have the pairwise distances below. Agglomerative clustering with **complete linkage** is run and the dendrogram is cut to give exactly **2** clusters. The resulting clusters are

	P1	P2	P3	P4	P5	P6
P1	0	4	8	6	4	1
P2		0	6	2	8	5
P3			0	8	6	9
P4				0	10	7
P5					0	5
P6						0

(A) {P1, P5, P6}, {P2, P4}, {P3}

(B) {P1, P5, P6}, {P2, P3, P4}

(C) {P1, P4, P6}, {P2, P3, P5}

(D) {P1, P2, P4, P5, P6}, {P3}

— END OF QUESTION PAPER —

Mock Test 47 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	4	19	NAT	2	1.339 to 1.359
2	MCQ	1	(A)	20	MSQ	2	(D)
3	MCQ	1	(B)	21	NAT	2	1.1405 to 1.1474
4	MSQ	1	(A)	22	NAT	2	0.0747 to 0.0947
5	MCQ	1	(D)	23	NAT	2	0.4885 to 0.4925
6	NAT	1	0.4622 to 0.4822	24	NAT	2	3
7	NAT	1	0.673 to 0.677	25	MCQ	2	(B)
8	MSQ	1	(A), (B), (C), (D)	26	NAT	2	2
9	NAT	1	131	27	NAT	2	0.2085 to 0.2115
10	MSQ	1	(A)	28	NAT	2	0.9866 to 0.9906
11	MSQ	1	(A), (B), (C), (D)	29	MSQ	2	(A), (B), (C)
12	NAT	1	1.565 to 1.585	30	NAT	2	−7.8369 to −7.7434
13	MSQ	1	(B), (D)	31	MSQ	2	(A), (B), (C), (D)
14	NAT	1	4.99 to 5.01	32	NAT	2	5.7451 to 5.7651
15	NAT	1	4.975 to 5.025	33	NAT	2	−0.0128 to −0.0123
16	MCQ	2	(B)	34	NAT	2	0.1091 to 0.1131
17	NAT	2	0.3908 to 0.4108	35	MCQ	2	(B)
18	MSQ	2	(C), (D)				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Unsupervised Learning					11
SVM & Decision Trees					11
Neural Networks (MLP / Feed-forward)					11
Probabilistic & Instance-based Classifiers					11
Regression & Model Selection					11
Total					55

Mock Test 47 — Detailed Solutions

Q.1 — Answer: 4 [NAT, 1M · Unsupervised Learning > Hierarchical clustering]

Undo the top 2 merge(s) (equivalently cut just below height 6.72). Clusters: {A, B, D, F}, {C}, {E, G}.

D lies in a cluster of size 4.

Q.2 — Answer: (A) [MCQ, 1M · SVM & Decision Trees > Decision trees: structure & traversal]

Follow the record from the root, taking the branch that matches each test:

Smoker = Former → branch Former.

Sugar = $186 \leq 187 \rightarrow$ Yes.

(Only the tests on the actual path matter.) The record reaches the leaf **Normal**.

Note: " $\leq$ " includes equality, so a value equal to the threshold goes to the Yes branch.

Q.3 — Answer: (B) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

Key features of the plotted curve: Small negative linear part (slope 0.1) for $z < 0$, slope 1 for $z > 0$.

Hence g = **Leaky ReLU with $\alpha = 0.1$** .

ELU with $\alpha = 1$: Smoothly saturates to -1 for $z \rightarrow -\infty$, linear for $z > 0$, passes through origin. – does not match.

ReLU: $\max(0, z)$: Exactly 0 for $z \leq 0$, slope 1 for $z > 0$, kink at origin. – does not match.

Softplus: $\ln(1 + e^z)$: Smooth, always positive, value $\ln 2 \approx 0.69$ at $z = 0$, $\sim z$ for large z . – does not match.

Q.4 — Answer: (A) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Expressiveness]

(A) **TRUE**. Composition of affine maps and $\max(0, \cdot)$.

(B) **FALSE**. Thresholding $\sigma(w \cdot x + b)$ at 0.5 gives the hyperplane $w \cdot x + b = 0$.

(C) **FALSE**. Width must grow with the required accuracy.

(D) **FALSE**. ReLU networks compute continuous functions.

Q.5 — Answer: (D) [MCQ, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

The jagged boundary with small islands around individual points (Panel Q) is produced by $k = 1$, which fits every training point (zero training error): low bias, high variance.

The smooth boundary (Panel P) comes from $k = 15$: averaging many neighbours raises bias and lowers variance; it misclassifies some training points, so its training error is not zero.

Answer: **Panel Q uses $k = 1$; it has lower bias and higher variance.**

Q.6 — Answer: 0.4622 to 0.4822 [NAT, 1M · SVM & Decision Trees > Ensembles: boosting]

$\alpha = \frac{1}{2} \ln(0.72/0.28) = \frac{1}{2} \ln(2.5714) = 0.47$.

$\varepsilon < 0.5 \Rightarrow \alpha > 0$; $\varepsilon = 0.5 \Rightarrow \alpha = 0$ (no better than chance).

Q.7 — Answer: 0.673 to 0.677 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Example 1: $y = 0 \rightarrow -\ln(1 - 0.4) = 0.5108$.

Example 2: $y = 0 \rightarrow -\ln(1 - 0.6) = 0.9163$.

Example 3: $y = 0 \rightarrow -\ln(1 - 0.6) = 0.9163$.

Example 4: $y = 1 \rightarrow -\ln 0.7 = 0.3567$.

Mean = **0.675**.

Q.8 — Answer: (A), (B), (C), (D) [MSQ, 1M · SVM & Decision Trees > Decision trees: overfitting & pruning]

(A) **TRUE**. Conditioning cannot increase entropy on average.

(B) **TRUE**. Each leaf costs more.

(C) **TRUE**. Splits depend only on the ordering of values.

(D) **TRUE**. E.g. reduced-error or cost-complexity pruning.

Q.9 — Answer: 131 [NAT, 1M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

Priors $K - 1 = 1$; means $K_p = 20$; K covariances $K_p(p+1)/2 = 110$.
Total = **131**.

Q.10 — Answer: (A) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Naive Bayes]

- (A) **TRUE**. The duplicated factor appears twice in the product, overstating confidence.
(B) **FALSE**. It avoids zero probability estimates; the parameter count is unchanged.
(C) **FALSE**. It is generative.
(D) **FALSE**. V appears in the denominator.

Q.11 — Answer: (A), (B), (C), (D) [MSQ, 1M · Unsupervised Learning > PCA & dimensionality reduction]

- (A) **TRUE**. It uses no labels and linear projections.
(B) **TRUE**. The covariance of the scores is diagonal.
(C) **TRUE**. $v^T S v = \lambda$ for a unit eigenvector v .
(D) **TRUE**. Used to choose the number of PCs.

Q.12 — Answer: 1.565 to 1.585 [NAT, 1M · Regression & Model Selection > Simple linear regression]

$\beta_1 = r \cdot s_y / s_x = 0.9 \times 7/4 = 1.575$.
Answer: **1.57**.

Q.13 — Answer: (B), (D) [MSQ, 1M · Regression & Model Selection > Gradient descent]

- (A) **FALSE**. It diverges if η is too large.
(B) **TRUE**. The least-squares minimiser is unique in that case.
(C) **FALSE**. Beyond $2/\lambda_{\max}$ it diverges.
(D) **TRUE**. One update per example.

Q.14 — Answer: 4.99 to 5.01 [NAT, 1M · Unsupervised Learning > Distance measures]

$\max |x_i - y_i| = \max(5, 4, 2) = 5$.
For reference: $L_1 = 11$, $L_2 = 6.71$, $L_\infty = 5$ (always $L_\infty \leq L_2 \leq L_1$).

Q.15 — Answer: 4.975 to 5.025 [NAT, 1M · Regression & Model Selection > Multiple linear regression]

Variance inflation factor of predictor j : $VIF_j = 1/(1 - R_j^2)$, where R_j^2 is the R^2 from regressing x_j on all other predictors. It is the factor by which $\text{Var}(\beta_j)$ is inflated by collinearity; $\sqrt{VIF}$ inflates the standard error.

$VIF = 1/(1 - 0.8) = 5$.
Answer: **5**.

Q.16 — Answer: (B) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

$a = W_1 x + b_1 = (-5, 5, 8)^T$; $h = \text{ReLU}(a) = (0, 5, 8)^T$.
 $\hat{y} = W_2 h + b_2 = (11, 2)^S$.

Distractors: skipping the ReLU gives $(11, -3)^T$; using W_1^T instead of W_1 gives $(7, 0)^T$; dropping the biases gives $(12, 2)^T$; using $|a|$ instead of ReLU gives $(11, 7)^T$.

Q.17 — Answer: 0.3908 to 0.4108 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent $[13+, 12-]$: $H = -(13/25)\log_2(13/25) - (12/25)\log_2(12/25) = 0.9988$.
Income = Low: $[7+, 8-]$, $H = -(7/15)\log_2(7/15) - (8/15)\log_2(8/15) = 0.9968$.
Income = Medium: $[6+, 0-]$, $H = 0$ (pure node) = 0.
Income = High: $[0+, 4-]$, $H = 0$ (pure node) = 0.
Weighted child entropy = $(15/25)(0.9968) + (6/25)(0) + (4/25)(0) = 0.5981$.
 $IG = 0.9988 - 0.5981 = 0.4008$.
Answer: **0.4**.

Q.18 — Answer: (C), (D) [MSQ, 2M · Regression & Model Selection > Multiple linear regression]

- (A) **FALSE**. This omits the city dummy; the correct value is 30.8.

(B) FALSE. The intercept corresponds to Female = 0, Exp = 0 and the reference level A.

(C) TRUE. $3 \times 0.8 = 2.4$.

(D) TRUE. Difference of the two dummy coefficients: $-1.1 - (-5.1)$.

Q.19 — Answer: 1.339 to 1.359 [NAT, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

$$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b} = (1, -2, 0).$$

$$e^{\mathbf{z}} = (2.7183, 0.1353, 1), \text{ sum} = 3.8536.$$

$$\text{Softmax probabilities} = (0.7054, 0.0351, 0.2595).$$

$$\text{Loss} = -\ln(0.2595) = \mathbf{1.35}.$$

Q.20 — Answer: (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Activation functions]

$$\sigma(2) = 0.8808, \tanh(2) = 0.964.$$

(A) FALSE. $|\tanh z| < |z|$ for $z \neq 0$.

(B) FALSE. tanh is odd: the sum is 0.

(C) FALSE. $\sigma'(2) = 0.105 < 0.25$ (max only at 0).

(D) TRUE. tanh = 0.964.

Q.21 — Answer: 1.1405 to 1.1474 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

$$\beta_{\text{ridge}} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y} \text{ with } \lambda = 4.$$

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 31 & 4 \\ 4 & 4 \end{bmatrix}$$

$$\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I} = \begin{bmatrix} 35 & 4 \\ 4 & 8 \end{bmatrix}$$

$$\mathbf{X}^T\mathbf{y} = \begin{bmatrix} 47 \\ 14 \end{bmatrix}$$

$$\det = 35 \times 8 - (4)^2 = 264; (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1} = (1/264)[[8, -4], [-4, 35]].$$

$$\beta = (1.2121, 1.1439).$$

Answer: **1.144**.

Q.22 — Answer: 0.0747 to 0.0947 [NAT, 2M · Probabilistic & Instance-based Classifiers > Naive Bayes]

$$\text{Naive Bayes: } P(c | \mathbf{x}) \propto P(c) \prod_j P(x_j | c).$$

$$\text{Yes: } P(\text{Approve}=\text{Yes}) = 6/11; P(\text{Income}=\text{Medium}|\text{Yes}) = 1/6; P(\text{Credit}=\text{Good}|\text{Yes}) = 6/6; \\ P(\text{Employed}=\text{Yes}|\text{Yes}) = 1/6. \text{ Product} = 0.01515.$$

$$\text{No: } P(\text{Approve}=\text{No}) = 5/11; P(\text{Income}=\text{Medium}|\text{No}) = 3/5; P(\text{Credit}=\text{Good}|\text{No}) = 5/5; \\ P(\text{Employed}=\text{Yes}|\text{No}) = 3/5. \text{ Product} = 0.16364.$$

$$P(\text{Yes} | \mathbf{x}) = 0.01515 / (0.01515 + 0.16364) = \mathbf{0.08}.$$

Q.23 — Answer: 0.4885 to 0.4925 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$$\text{All weights} = 0.5, \text{ biases } 0 \Rightarrow \text{both hidden units get } a = 0.5 \times (0.5 + 0.5) = 0.5, h_1 = h_2 = \sigma(a) = 0.6225; \hat{y} = 2 \times 0.5 \times 0.6225 = 0.6225.$$

$$\hat{y} - y = 1.6225. \partial L / \partial v_j = (\hat{y} - y)h_j = 1.0099 \text{ (same for both } j).$$

$$\partial L / \partial w_{jk} = (\hat{y} - y) v_j h_j (1 - h_j) x_k = 1.6225 \times 0.5 \times 0.235 \times 0.5 = 0.0953 \text{ (identical for } j = 1, 2).$$

$$w_{11} \leftarrow 0.5 - 0.1 \times (0.0953) = \mathbf{0.4905}.$$

Because the two hidden units start identical and get identical updates, they remain identical forever (symmetry problem) – hence weights must be initialised randomly.

Q.24 — Answer: 3 [NAT, 2M · Unsupervised Learning > Cluster evaluation]

$$\text{Total pairs} = C(7,2) = 21.$$

a = pairs together in both = 3; b = pairs separated in both = 6; pairs together in U only = 6; together in V only = 6.

$$\text{Rand index} = (a + b) / C(n,2) = (3 + 6) / 21 = 0.4286.$$

Answer: **3**. (Cluster names do not matter: only whether pairs are together.)

Q.25 — Answer: (B) [MCQ, 2M · SVM & Decision Trees > SVM: margin & geometry]

For $n^T x - c = 0$, the geometric margin is $\min_i y_i s(n^T x_i - c)/\|n\|$ with orientation $s = \pm 1$ chosen so that the positive class is on the positive side; a negative value means some point is misclassified.

Hyperplane	$\ n\ $	$\min_i$ distance (signed, best orientation)
$x_1 - x_2 + 1.5 = 0$	1.414	0.354
$x_1 - x_2 + 2 = 0$	1.414	0.707
$x_1 - 2x_2 + 4.5 = 0$	2.236	0.224
$x_1 - 2x_2 + 5.5 = 0$	2.236	0.224

The largest margin is 0.707, for $x_1 - x_2 + 2 = 0$. Note that a boundary can separate the data perfectly yet pass close to some point — only the minimum distance matters.

Q.26 — Answer: 2 [NAT, 2M · Unsupervised Learning > k-means]

Data: {1, 5, 11, 16, 17, 18, 19}; initial centroids $c_1 = 1$, $c_2 = 2$.

Iter	centroids used	cluster 1	cluster 2	new centroids
1	1, 2	{1}	{5, 11, 16, 17, 18, 19}	1, 14.33
2	1, 14.33	{1, 5}	{11, 16, 17, 18, 19}	3, 16.2
3	3, 16.2	{1, 5}	{11, 16, 17, 18, 19}	3, 16.2

In the last row the centroids do not change, so the algorithm has converged: final centroids 3 and 16.2.

Centroids changed in 2 iteration(s); the final iteration only confirms convergence. Answer **2**.

Q.27 — Answer: 0.2085 to 0.2115 [NAT, 2M · SVM & Decision Trees > Decision trees: overfitting & pruning]

$R(T)$ = sum of leaf errors / $N = (0 + 7 + 6 + 1 + 2)/100 = 0.16$; $|T| = 5$ leaves.

$R_\alpha(T) = 0.16 + 0.01 \times 5 = \mathbf{0.21}$.

Q.28 — Answer: 0.9866 to 0.9906 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Per unit population: $TP = Se \cdot \pi$, $FN = (1 - Se) \cdot \pi$, $FP = (1 - Sp)(1 - \pi)$, $TN = Sp(1 - \pi)$.

$TP = 0.0085$, $FN = 0.0015$, $FP = 0.0099$, $TN = 0.9801$.

Accuracy = $TP + TN = \mathbf{0.989}$.

Note that the trivial 'always negative' classifier would have accuracy 0.99 — accuracy is misleading under imbalance.

Q.29 — Answer: (A), (B), (C) [MSQ, 2M · Unsupervised Learning > PCA & dimensionality reduction]

(A) **TRUE**. $S \rightarrow RSR^T$ is a similarity transform.

(B) **TRUE**. The covariance of z-scores is the correlation matrix.

(C) **TRUE**. The residual lies in the span of the discarded eigenvectors.

(D) **FALSE**. Scores are $U\Sigma = XV$.

Q.30 — Answer: -7.8369 to -7.7434 [NAT, 2M · Regression & Model Selection > MLE & MAP view]

$\bar{x} = 4.333$, $\bar{y} = 14.167$.

i	1	2	3	4	5	6
$x_i - \bar{x}$	-3.33	-1.33	-0.33	0.67	1.67	2.67
$y_i - \bar{y}$	-2.17	-2.17	-0.17	-0.17	0.83	3.83
$(x_i - \bar{x})(y_i - \bar{y})$	7.22	2.89	0.06	-0.11	1.39	10.22

$S_{xx} = \sum (x_i - \bar{x})^2 = 23.333$, $S_{xy} = 21.667$.

$\beta_1 = S_{xy}/S_{xx} = 0.9286$, $\beta_0 = \bar{y} - \beta_1 \bar{x} = 10.1429$.

$SSE = S_{yy} - S_{xy}^2/S_{xx} = 24.8333 - 20.119 = 4.7143$.

$\ln L = -(n/2) \ln(2\pi\sigma^2) - SSE/(2\sigma^2)$. At $\hat{\sigma}^2 = SSE/n$: $\ln L = -(n/2)[\ln(2\pi\hat{\sigma}^2) + 1]$.

$\hat{\sigma}^2 = 0.7857 \Rightarrow \ln L = -(6/2)[\ln(2\pi \times 0.7857) + 1] = -7.7901$.

Answer: **-7.79**.

Q.31 — Answer: (A), (B), (C), (D) [MSQ, 2M · SVM & Decision Trees > Decision trees: impurity measures]

Parent [50+, 30-]: $H = 0.9544$, $Gini = 0.4688$, $error = 0.375$.

S1: children $H = 0.9957$, 0.8722 ; $Gini = 0.497$, 0.414 ; $error = 0.4615$, 0.2927 . Gains: $IG = 0.022$, $Gini = 0.0142$, $error = 0$.

S2: children $H = 0.8631$, 0.7063 ; $Gini = 0.4082$, 0.3107 ; $error = 0.2857$, 0.1923 . Gains: $IG = 0.1933$, $Gini = 0.124$, $error = 0.15$.

(A) TRUE. S1: 0.0142, S2: 0.124.

(B) TRUE. S1: 0.022, S2: 0.1933.

(C) TRUE. $IG(S2) = 0.1933$.

(D) TRUE. S1: 0, S2: 0.15.

Q.32 — Answer: 5.7451 to 5.7651 [NAT, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Three nearest neighbours: P4 ($d^2=1$, $y=4$), P6 ($d^2=4$, $y=10$), P5 ($d^2=9$, $y=12$).

Weights $1/d^2$: 1, 0.25, 0.1111; $\Sigma w = 1.3611$.

$\Sigma w_i y_i = 7.8333 \Rightarrow \hat{y} = 7.8333/1.3611 = \mathbf{5.76}$.

(The unweighted 3-NN average would be 8.67; weighting pulls $\hat{y}$ towards the closest point.)

Q.33 — Answer: -0.0128 to -0.0123 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

$a = w \cdot x + b = 2$; $\hat{y} = \sigma(a) = 0.8808$.

$L = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial L / \partial \hat{y} = \hat{y} - y = -0.1192$; $\partial \hat{y} / \partial a = \hat{y}(1 - \hat{y}) = 0.105$.

$\delta = \partial L / \partial a = (\hat{y} - y)\hat{y}(1 - \hat{y}) = -0.0125 (= \partial L / \partial b)$.

$\partial L / \partial w_2 = \delta \cdot x_2 = -0.0125 \times 1 = -0.0125$.

Answer: **-0.0125**.

Forgetting the factor $\hat{y}(1 - \hat{y})$ (i.e. treating the output as linear) is the most common error.

Q.34 — Answer: 0.1091 to 0.1131 [NAT, 2M · Regression & Model Selection > Gradient descent]

Hessian of J : $\nabla^2 J = (2/n)X^T X$. Eigenvalues of $X^T X$: $9 \pm 3 = 12, 6$ (eigenvectors $(1, 1), (1, -1)$).

So the Hessian eigenvalues are $L = 12$ and $\mu = 6$.

GD error evolves as $w_{t+1} - w^* = (I - \eta \nabla^2 J)(w_t - w^*)$, so it converges iff $|1 - \eta \lambda| < 1$ for every Hessian eigenvalue λ .

Balance $|1 - \eta L| = |1 - \eta \mu| \Rightarrow \eta^* = 2/(L + \mu) = 2/(12 + 6) = 0.1111$.

Answer: **0.111**.

Q.35 — Answer: (B) [MCQ, 2M · Unsupervised Learning > Hierarchical clustering]

Use complete linkage: $d(A, B) = \max_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: $\{P1\} + \{P6\}$ at height **1**. Updated: $d(\{P1, P6\}, \{P2\}) = 5$, $d(\{P1, P6\}, \{P3\}) = 9$, $d(\{P1, P6\}, \{P4\}) = 7$, $d(\{P1, P6\}, \{P5\}) = 5$.

Merge 2: $\{P2\} + \{P4\}$ at height **2**. Updated: $d(\{P2, P4\}, \{P3\}) = 8$, $d(\{P2, P4\}, \{P5\}) = 10$, $d(\{P2, P4\}, \{P1, P6\}) = 7$.

Merge 3: $\{P5\} + \{P1, P6\}$ at height **5**. Updated: $d(\{P1, P5, P6\}, \{P3\}) = 9$, $d(\{P1, P5, P6\}, \{P2, P4\}) = 10$.

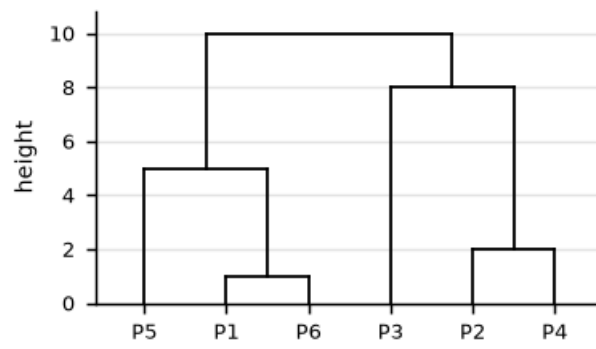
Merge 4: $\{P3\} + \{P2, P4\}$ at height **8**. Updated: $d(\{P2, P3, P4\}, \{P1, P5, P6\}) = 10$.

Merge 5: $\{P1, P5, P6\} + \{P2, P3, P4\}$ at height **10**.

Stopping when 2 clusters remain (i.e. after 4 merges): **$\{P1, P5, P6\}, \{P2, P3, P4\}$** .

$\{P1, P2, P4, P5, P6\}, \{P3\}$ is what *average* linkage would give.

$\{P1, P5, P6\}, \{P2, P4\}, \{P3\}$ has 3 clusters (cut at the wrong level).



Mock Test 48

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	5	4	26

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Unsupervised Learning	7	11
Regression & Model Selection	7	11
Probabilistic & Instance-based Classifiers	7	11
SVM & Decision Trees	7	11
Neural Networks (MLP / Feed-forward)	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · NAT (numerical answer)]

Agglomerative clustering of 31 points is stopped when 4 clusters remain. The number of merges performed is _____ (Answer in integer)

Q.2 [1 mark · NAT (numerical answer)]

A linear regression model with an intercept uses 1 numeric feature and a categorical feature *payment mode* with 4 levels (Cash, Card, UPI, Wallet). Each categorical feature is encoded by dummy (indicator) variables using a reference level, so that the design matrix has full column rank. The total number of regression coefficients (including the intercept) to be estimated is _____ (Answer in integer)

Q.3 [1 mark · MSQ (one or more correct)]

Which of the following statements about logistic regression is/are CORRECT?

- (A) There is no closed-form expression for the maximum-likelihood weights of logistic regression; they are found iteratively (gradient descent, Newton/IRLS).
- (B) Multiplying w and b by the same positive constant does not change the 0.5-threshold decision boundary.
- (C) The sigmoid satisfies $\sigma(-z) = 1 - \sigma(z)$.
- (D) Multiplying all weights and the bias by 2 shifts the 0.5-threshold decision boundary.

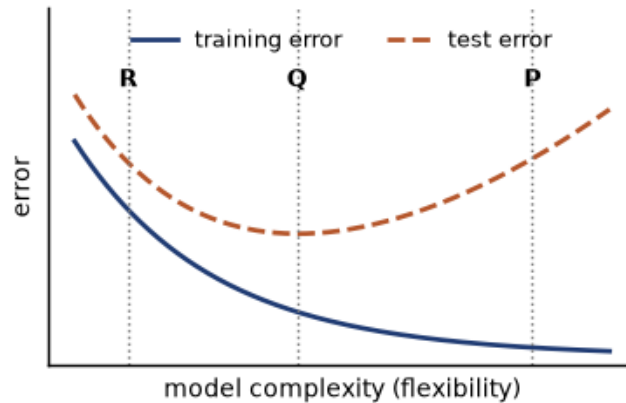
Q.4 [1 mark · NAT (numerical answer)]

The 5 two-dimensional observations below have zero mean in each feature. Then the correlation coefficient between x_1 and x_2 is _____ (rounded off to two decimal places)

Point	x_1	x_2	x_3	x_4	x_5
x_1	-3	-2	2	3	0
x_2	4	-4	-4	1	3

Q.5 [1 mark · MCQ]

The figure shows training and test error as a function of model complexity, with three complexity levels marked P, Q and R. Which marked level corresponds to the best generalisation (lowest expected test error)?



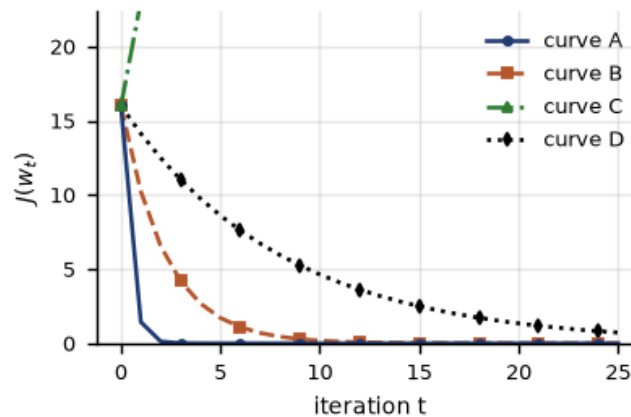
- (A) P (B) R
(C) Q (D) None of P, Q, R

Q.6 [1 mark · NAT (numerical answer)]

A hard-margin linear SVM trained on a separable data set returns the canonical solution $w = (5, 12)$, $b = -3$ (so that $\min_i y_i (w^T x_i + b) = 1$). The width of the margin, i.e. the distance between the hyperplanes $w^T x + b = +1$ and $w^T x + b = -1$, is _____ (rounded off to two decimal places)

Q.7 [1 mark · MCQ]

Gradient descent $w_{t+1} = w_t - \eta J'(w_t)$ is run on $J(w) = (w - 2)^2$ from $w_0 = -2$ with four learning rates $\eta \in \{0.03, 0.1, 0.35, 1.1\}$. The loss curves are shown below (curves that leave the plotted range are cut off). Which curve corresponds to $\eta = 1.1$?



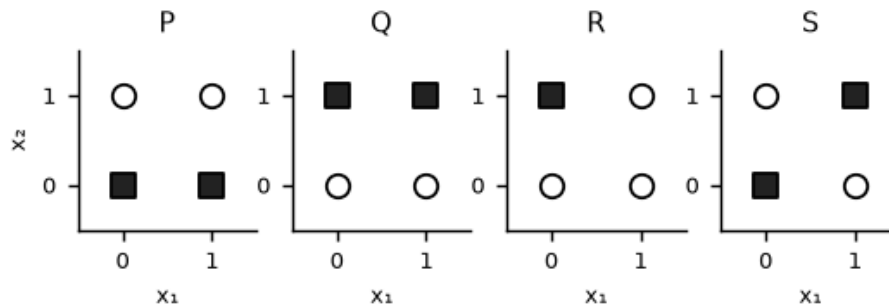
- (A) Curve A (B) Curve D
(C) Curve C (D) Curve B

Q.8 [1 mark · NAT (numerical answer)]

Lloyd's k-means is run on $n = 2000$ points in $d = 5$ dimensions with $k = 8$ for $T = 12$ iterations. Counting one subtraction per coordinate per point-centroid pair in the assignment step, the total number of such subtractions is _____ (Answer in integer)

Q.9 [1 mark · MCQ]

Each panel shows the four binary inputs (x_1, x_2) labelled as class 1 (filled square) or class 0 (open circle). Which dataset can NOT be classified perfectly by a single perceptron (one linear threshold unit)?



(A) Dataset P

(B) Dataset Q

(C) Dataset R

(D) Dataset S

Q.10 [1 mark · NAT (numerical answer)]

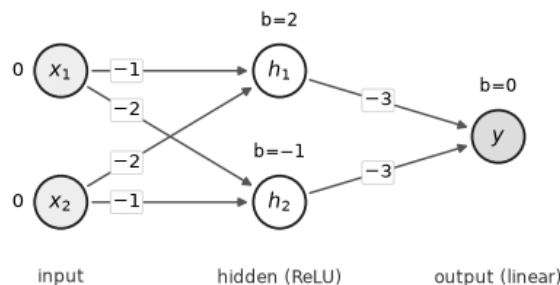
Mini-batch SGD is run on $N = 2020$ examples with batch size $B = 64$ for $E = 4$ epochs, and any incomplete final batch in an epoch is **dropped**. The total number of parameter updates is _____ (Answer in integer)

Q.11 [1 mark · NAT (numerical answer)]

The homogeneous polynomial kernel $K(x, z) = (x^T z)^4$ on $x \in \mathbb{R}^3$ corresponds to a feature map of all (scaled) monomials of degree exactly 4. The number of distinct such features is _____ (Answer in integer)

Q.12 [1 mark · NAT (numerical answer)]

For the ReLU network shown (edge labels are weights, 'b' denotes a bias) and input $x = (0, 0)$, the output y is _____ (Answer in integer)

**Q.13 [1 mark · NAT (numerical answer)]**

Points are drawn uniformly from the unit hypercube $[0, 1]^{50}$. The fraction of points lying within distance 0.05 of the boundary (i.e. outside the inner cube $[0.05, 0.95]^{50}$) is _____ (rounded off to two decimal places)

Q.14 [1 mark · NAT (numerical answer)]

In a disease screening test, a false negative costs 6 units and a false positive costs 4 units (correct decisions cost 0). For an input x the model gives $P(\text{positive} | x) = 0.4$. The expected cost of the Bayes-optimal decision for this x is _____ (rounded off to two decimal places)

Q.15 [1 mark · MSQ (one or more correct)]

Which of the following statements about bagging, random forests and boosting is/are CORRECT?

- (A) Averaging B independent predictors, each with variance σ^2 , gives variance σ^2/B .
- (B) Random forests reduce variance by making the individual trees more correlated.
- (C) Bagging draws bootstrap samples of size n without replacement.
- (D) Random forests decorrelate trees by considering a random subset of features at each split.

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

The labelled data below are linearly separable through the origin by the unit vector $u = (-3, 4)/5$ (i.e. $y_i = \text{sign}(u \cdot x_i)$). Take $\gamma = \min_i y_i(u \cdot x_i)$ and $R = \max_i \|x_i\|$. The perceptron convergence theorem (no bias, $w_0 = 0$, $\eta = 1$) bounds the number of mistakes by $(R/\gamma)^2$. The value of this bound is _____ (rounded off to two decimal places)

i	1	2	3	4	5
x_i	(3, -2)	(4, 1)	(-1, -4)	(-2, 3)	(1, 2)
y_i	-1	-1	-1	1	1

Q.17 [2 marks · MCQ]

A decision tree is to be grown on the 10 training records below using information gain as the splitting criterion (multi-way split on each categorical attribute). Which attribute is chosen at the root?

#	Size	Credit	Age	Colour	Class
1	Large	Fair	Senior	Red	Yes
2	Large	Excellent	Senior	Green	No
3	Large	Excellent	Young	Red	No
4	Small	Excellent	Young	Red	Yes
5	Small	Fair	Senior	Red	No
6	Large	Excellent	Middle	Blue	Yes
7	Small	Excellent	Middle	Red	No
8	Small	Excellent	Senior	Green	Yes
9	Small	Excellent	Young	Green	No
10	Small	Fair	Senior	Blue	Yes

- (A) Size
- (B) Credit
- (C) Age
- (D) Colour

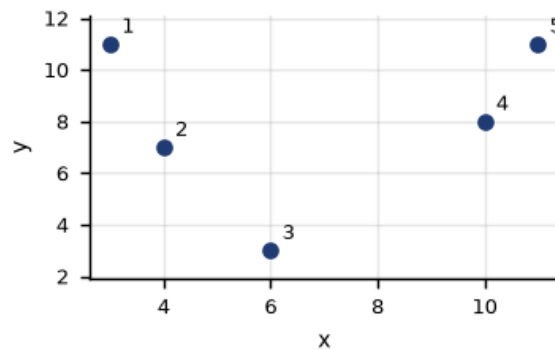
Q.18 [2 marks · NAT (numerical answer)]

The 2×2 matrix $X^T X$ of a (centred) design matrix X has eigenvalues 20, 5. For ridge regression with penalty $\lambda \|\beta\|_2^2$, the effective degrees of freedom $df(\lambda) = \text{trace}[X(X^T X + \lambda I)^{-1} X^T]$ for $\lambda = 1$ is _____ (rounded off to two decimal places)

Q.19 [2 marks · NAT (numerical answer)]

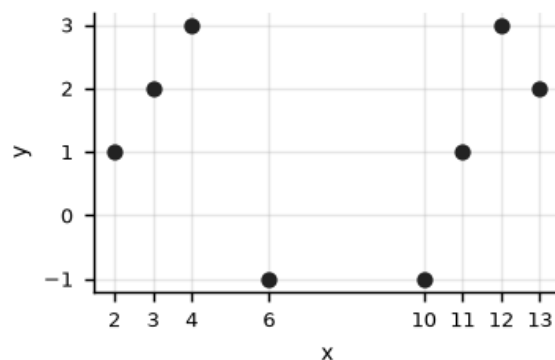
Consider the 5 training points (x_i, y_i) shown below. A 1-nearest-neighbour regressor (prediction = y-value of the closest training x , $|\cdot|$ distance) is evaluated by leave-one-out cross-validation. The LOOCV MSE is _____ (rounded off to two decimal places)

x	3	4	6	10	11
y	11	7	3	8	11

**Q.20 [2 marks · NAT (numerical answer)]**

A regression tree (squared-error criterion) is fitted to the 8 points below. The root split " $x \leq t$ " is chosen among midpoints of consecutive x values to minimise the total SSE of the two children (each child predicts its mean). Find the prediction of the resulting depth-1 regression tree (stump) at $x = 5$. (rounded off to two decimal places)

x	2	3	4	6	10	11	12	13
y	1	2	3	-1	-1	1	3	2

**Q.21 [2 marks · MSQ (one or more correct)]**

Consider agglomerative hierarchical clustering on n points with a dissimilarity matrix (no ties). Which of the following statements is/are CORRECT?

- (A) Single linkage tends to produce compact, roughly spherical clusters of equal diameter.
- (B) Complete-linkage merge heights equal the edge weights of a minimum spanning tree.
- (C) Replacing every dissimilarity by its square does not change the single-linkage merge order.
- (D) The single-linkage cophenetic distance between two points can exceed their original distance.

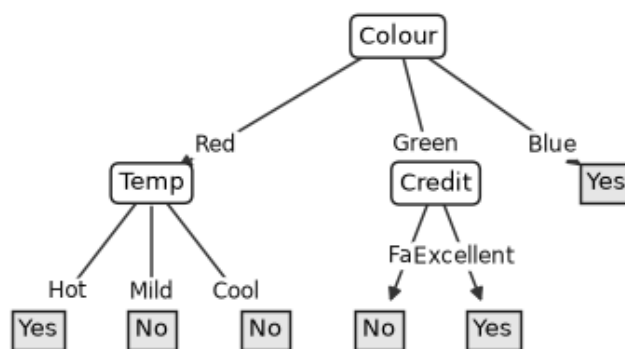
Q.22 [2 marks · NAT (numerical answer)]

The actual and predicted labels of a binary classifier (1 = positive) on 13 test examples are given below. The F_β -score with $\beta = 0.5$, $F_\beta = (1 + \beta^2)PR/(\beta^2P + R)$, is _____ (rounded off to two decimal places)

#	1	2	3	4	5	6	7	8	9	10	11	12	13
actual y	0	0	0	1	0	1	1	0	1	1	1	0	0
predicted $\hat{y}$	0	1	0	0	1	1	0	0	0	0	1	0	0

Q.23 [2 marks · NAT (numerical answer)]

The decision tree shown is applied to the 8 labelled records in the table (attributes not tested on a path are ignored). Find the number of records misclassified by the tree. (Answer in integer)



#	Colour	Temp	Credit	Actual
1	Green	Mild	Excellent	Yes
2	Red	Mild	Excellent	No
3	Red	Cool	Fair	No
4	Red	Hot	Excellent	Yes
5	Green	Hot	Excellent	Yes
6	Red	Mild	Excellent	Yes
7	Green	Hot	Fair	No
8	Blue	Hot	Excellent	Yes

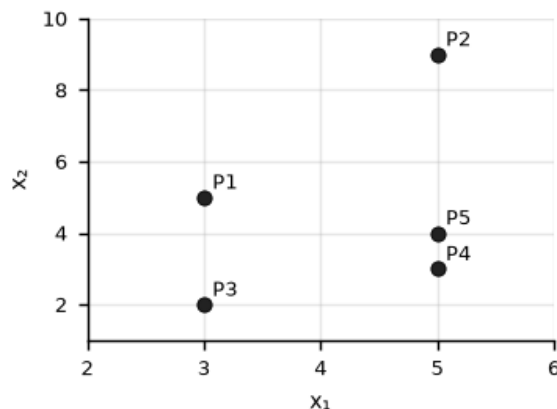
Q.24 [2 marks · NAT (numerical answer)]

The eigenvalues of the sample covariance matrix (divisor $n - 1$) of a centered 7-dimensional dataset are 18, 13.5, 11, 9.5, 8.5, 5, 2. Then with $n = 11$ samples, the total squared reconstruction error $\sum_i \|x_i - \hat{x}_i\|^2$ when keeping the top 2 principal components is _____ (rounded off to two decimal places)

Q.25 [2 marks · NAT (numerical answer)]

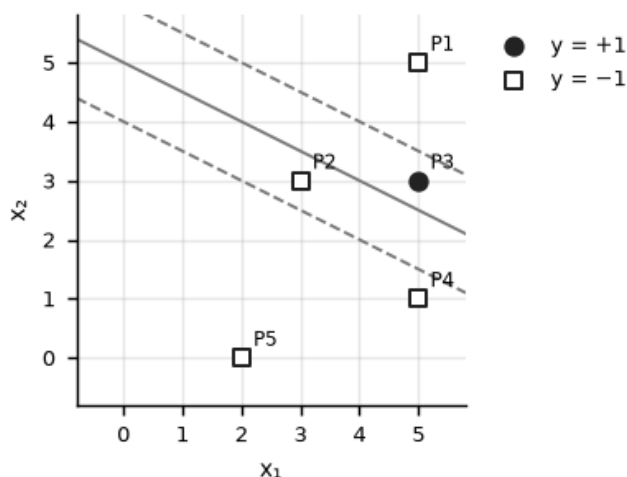
A cluster contains the 5 points below. Distances are Manhattan (L_1). If P3 were used as the cluster representative instead of the medoid, the total distance (cost) would increase by _____ (Answer in integer)

Point	P1	P2	P3	P4	P5
x_1	3	5	3	5	5
x_2	5	9	2	3	4

**Q.26 [2 marks · NAT (numerical answer)]**

A linear SVM with $f(x) = 0.5x_1 + x_2 - 5$ is evaluated on the training points in the table (also plotted, with the boundary $f = 0$ solid and $f = \pm 1$ dashed). Compute the average hinge loss $(1/n) \sum_i \max(0, 1 - y_i f(x_i))$. (rounded off to two decimal places)

Point	x_1	x_2	y
P1	5	5	-1
P2	3	3	-1
P3	5	3	+1
P4	5	1	-1
P5	2	0	-1

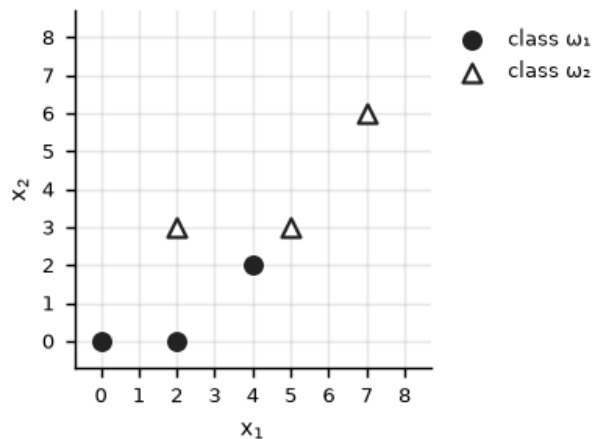
**Q.27 [2 marks · MCQ]**

For logits $z = (0, 1, -1)$, softmax probabilities p and true class 3, the gradient of the cross-entropy loss $L = -\ln p_3$ with respect to z (rounded to two decimals) is

(A) (0.24, 0.67, 0.09)**(B)** (-0.24, -0.67, 0.91)**(C)** (-0.76, -0.33, -0.91)**(D)** (0.24, 0.67, -0.91)

Q.28 [2 marks · NAT (numerical answer)]

Class ω_1 has training points (4, 2), (0, 0), (2, 0) and class ω_2 has points (2, 3), (7, 6), (5, 3). Let $\mathbf{m}_1, \mathbf{m}_2$ be the class means and $S_W = \sum_k \sum_{x \in \omega_k} (x - \mathbf{m}_k)(x - \mathbf{m}_k)^T$ the within-class scatter matrix (no division by n). Then $(\mathbf{m}_2 - \mathbf{m}_1)^T S_W^{-1} (\mathbf{m}_2 - \mathbf{m}_1)$ is _____ (rounded off to two decimal places)

**Q.29 [2 marks · MSQ (one or more correct)]**

Which of the following statements about loss functions and optimisation for training neural networks is/are CORRECT?

- (A) Mini-batch gradient descent with batch size $B = N$ is the same as single-example SGD.
- (B) Too small a learning rate leads to very slow convergence.
- (C) Minimising cross-entropy loss is equivalent to maximising the log-likelihood of the labels under the model.
- (D) Decaying the learning rate during training can help SGD settle closer to a minimum.

Q.30 [2 marks · NAT (numerical answer)]

A 2-2-1 network uses the identity (linear) activation in its hidden layer: $\hat{y} = W_2(W_1x + b_1) + b_2$, with the parameters below. The network is equivalent to a single linear model $\hat{y} = \tilde{w}_1x_1 + \tilde{w}_2x_2 + \tilde{b}$. Then the intercept $\tilde{b}$ is _____ (Answer in integer)

$$W_1 = \begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

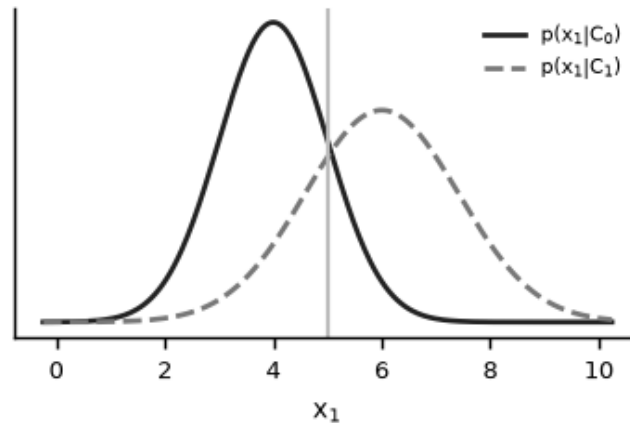
$$W_2 = \begin{bmatrix} 1 & -2 \end{bmatrix}$$

$$b_2 = -1$$

Q.31 [2 marks · NAT (numerical answer)]

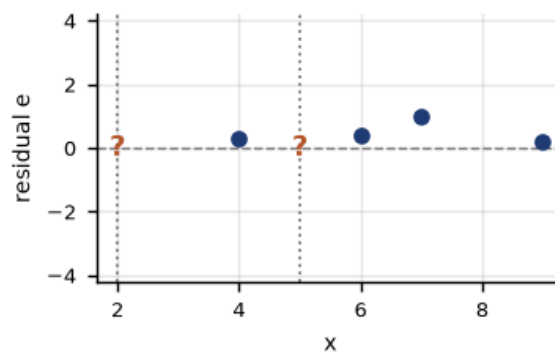
A Gaussian naive Bayes classifier with feature x_1 has the class priors and per-class Gaussian parameters below (σ^2 = variance). For the input $x = 5$, the posterior probability $P(C_1 | x)$ is _____ (rounded off to two decimal places)

class	prior	$\mu(x_1)$	$\sigma^2(x_1)$
C_0	0.7	4	1
C_1	0.3	6	2

**Q.32 [2 marks · NAT (numerical answer)]**

A simple linear regression with intercept is fitted by least squares. The table gives the x -values and the residuals $e_i = y_i - \hat{y}_i$, but two residuals are missing. The fitted line is $\hat{y} = 3 + 1.2x$. The observed value y at $x = 2$ is _____ (rounded off to two decimal places)

x_i	2	4	5	6	7	9
e_i	?	0.3	?	0.4	1	0.2

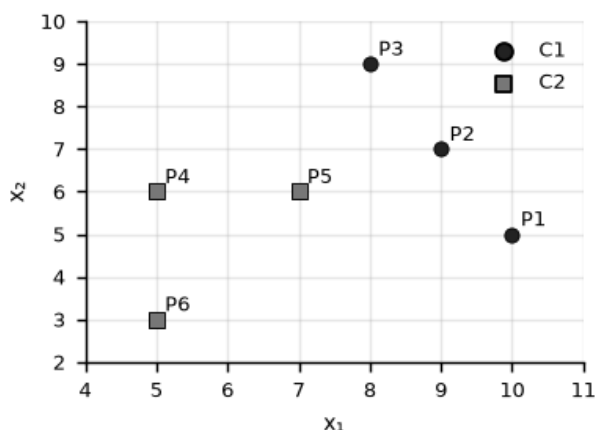
**Q.33 [2 marks · NAT (numerical answer)]**

A multiple regression with an intercept and $p = 4$ predictors is fitted to $n = 47$ observations, giving $SSE = 355$ and $SST = 798$. The adjusted R^2 is _____ (rounded off to three decimal places)

Q.34 [2 marks · NAT (numerical answer)]

A clustering of 6 points into 2 clusters is shown below. Using Euclidean distance, the silhouette coefficient $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$ of point P5 is _____ (rounded off to two decimal places)

Point	P1	P2	P3	P4	P5	P6
x_1	10	9	8	5	7	5
x_2	5	7	9	6	6	3
cluster	C1	C1	C1	C2	C2	C2

**Q.35 [2 marks · NAT (numerical answer)]**

A one-feature logistic regression model $P(y = 1 | x) = \sigma(wx + b)$ has $w = 1.5$ and $b = 0$. It is evaluated on the 5 examples below. Using natural logarithms, the **total** (summed) binary cross-entropy loss $-[y \ln \hat{p} + (1 - y) \ln(1 - \hat{p})]$ over these examples is _____ (rounded off to two decimal places)

i	1	2	3	4	5
x_i	-2	2	2	1	0
y_i	0	0	0	0	1

— END OF QUESTION PAPER —

Mock Test 48 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	NAT	1	27	19	NAT	2	13.134 to 13.266
2	NAT	1	5	20	NAT	2	0.8233 to 0.8433
3	MSQ	1	(A), (B), (C)	21	MSQ	2	(C)
4	NAT	1	-0.2418 to -0.2218	22	NAT	2	0.4445 to 0.4645
5	MCQ	1	(C)	23	NAT	2	1
6	NAT	1	0.1438 to 0.1638	24	NAT	2	359.99 to 360.01
7	MCQ	1	(C)	25	NAT	2	6
8	NAT	1	960000	26	NAT	2	0.89 to 0.91
9	MCQ	1	(D)	27	MCQ	2	(D)
10	NAT	1	124	28	NAT	2	1.6369 to 1.6569
11	NAT	1	15	29	MSQ	2	(B), (C), (D)
12	NAT	1	-6	30	NAT	2	1
13	NAT	1	0.9848 to 1.0048	31	NAT	2	0.2701 to 0.2901
14	NAT	1	2.39 to 2.41	32	NAT	2	6.3348 to 6.3985
15	MSQ	1	(A), (D)	33	NAT	2	0.5108 to 0.5148
16	NAT	2	16.966 to 17.034	34	NAT	2	0.0078 to 0.0278

Q	Type	M	Key	Q	Type	M	Key
17	MCQ	2	(D)	35	NAT	2	8.5303 to 8.5503
18	NAT	2	1.7757 to 1.7957				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Unsupervised Learning					11
Regression & Model Selection					11
Probabilistic & Instance-based Classifiers					11
SVM & Decision Trees					11
Neural Networks (MLP / Feed-forward)					11
Total					55

Mock Test 48 — Detailed Solutions

Q.1 — Answer: 27 [NAT, 1M · Unsupervised Learning > Hierarchical clustering]

$31 - 4 = 27$ merges.

Q.2 — Answer: 5 [NAT, 1M · Regression & Model Selection > Multiple linear regression]

A categorical feature with k levels needs $k - 1$ dummies (one level is the reference absorbed by the intercept; using all k dummies with an intercept gives perfect multicollinearity).

Parameters = 1 (intercept) + 1 (numeric) + $(4 - 1) = 5$.

Answer: 5.

Q.3 — Answer: (A), (B), (C) [MSQ, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

(A) TRUE. Setting the gradient $\Sigma(\sigma(w^T x_i) - y_i)x_i = 0$ gives non-linear equations.

(B) TRUE. The set $\{w^T x + b = 0\}$ is unchanged; only the confidence (steepness) changes.

(C) TRUE. $\sigma(-z) = 1/(1+e^z) = 1 - 1/(1+e^{-z})$.

(D) FALSE. $w^T x + b = 0 \Leftrightarrow 2w^T x + 2b = 0$; the boundary is the same.

Q.4 — Answer: -0.2418 to -0.2218 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

The data are already centered (column means are 0), so $S = X^T X / (n - 1)$.

$\Sigma x_1^2 = 26$, $\Sigma x_2^2 = 58$, $\Sigma x_1 x_2 = -9$.

$S = [[6.5, -2.25], [-2.25, 14.5]]$.

Required value = **-0.23**.

Q.5 — Answer: (C) [MCQ, 1M · Regression & Model Selection > Bias-variance trade-off]

At R: both training and test errors are high (left side) → underfitting.

At Q: the test error is at its minimum → best trade-off.

At P: training error is very low but test error has risen well above its minimum (right side) → overfitting.

Answer: Q.

Q.6 — Answer: 0.1438 to 0.1638 [NAT, 1M · SVM & Decision Trees > SVM: margin & geometry]

Margin width = $2/\|w\|$. $\|w\| = \sqrt{25 + 144} = 13$.

Width = $2/13 = \mathbf{0.15}$.

Common mistakes: $1/\|w\|$ (that is the distance from the boundary to one margin hyperplane) or $2/\|w\|^2$ — the bias b plays no role.

Q.7 — Answer: (C) [MCQ, 1M · Regression & Model Selection > Gradient descent]

$J(w) = (w - 2)^2$ has $J' = 2$. GD: $w_{t+1} - 2 = (1 - 2\eta)(w_t - 2)$, so $J(w_t) = (1 - 2\eta)^{2t} J(w_0)$.

Ratios $(1 - 2\eta)^2$ per step: $\eta = 0.03 \rightarrow 0.8836$ (curve D); $\eta = 0.1 \rightarrow 0.64$ (curve B); $\eta = 0.35 \rightarrow 0.09$ (curve A); $\eta = 1.1 \rightarrow 1.44$ (curve C).

Ratio $> 1 \Rightarrow$ divergence ($\eta > 1 = 2/J'$); among convergent rates (all < 0.5 here), a smaller η gives a ratio closer to 1, i.e. slower decrease.

Hence $\eta = 1.1$ corresponds to **Curve C**.

Q.8 — Answer: 960000 [NAT, 1M · Unsupervised Learning > k-means]

Assignment cost per iteration = $n \cdot k \cdot d = 2000 \cdot 8 \cdot 5 = 80000$; over T iterations **960000**. This is the $O(nkdT)$ complexity of k-means.

Q.9 — Answer: (D) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

A single perceptron can classify a dataset perfectly iff the two classes are linearly separable.

Dataset P: labels for $(0,0), (0,1), (1,0), (1,1) = 1, 0, 1, 0 \rightarrow$ NOT x_2 — linearly separable.

Dataset Q: labels for $(0,0), (0,1), (1,0), (1,1) = 0, 1, 0, 1 \rightarrow x_2$ (identity of x_2) — linearly separable.

Dataset R: labels for $(0,0), (0,1), (1,0), (1,1) = 0, 1, 0, 0 \rightarrow$ (NOT x_1) AND x_2 — linearly separable.

Dataset S: labels for $(0,0), (0,1), (1,0), (1,1) = 1, 0, 0, 1 \rightarrow$ XNOR — **not** linearly separable (diagonal pattern).

Answer: **Dataset S**. Of the 14 non-constant Boolean functions of two inputs, only XOR and XNOR are not linearly separable.

Q.10 — Answer: 124 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Updates per epoch = $\lceil N/B \rceil = \lceil 2020/64 \rceil = 31$.

Total = $31 \times 4 = 124$.

Q.11 — Answer: 15 [NAT, 1M · SVM & Decision Trees > SVM: kernels]

Monomials of degree exactly d in p variables = $C(p + d - 1, d)$ (stars and bars).

$C(6, 4) = 15$.

(Including all lower degrees, as for $(x^T z + 1)^4$, would give $C(7, 4) = 35$.)

Q.12 — Answer: -6 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Forward pass]

$a = Wx + b = (2, -1)$; $h = \text{ReLU}(a) = (2, 0)$ - a negative pre-activation is clipped to 0.

$y = -3 \cdot 2 + (-3) \cdot 0 + (0) = -6$.

A common slip is to forget the ReLU and use the raw pre-activation (which would give -3).

Q.13 — Answer: 0.9848 to 1.0048 [NAT, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Inner cube volume = $(1 - 2 \times 0.05)^{50} = 0.9^{50} = 0.0052$.

Fraction near the boundary = $1 - 0.0052 = 0.99$.

In high dimensions most points lie near the boundary, far from each other — a symptom of the curse of dimensionality that hurts k-NN.

Q.14 — Answer: 2.39 to 2.41 [NAT, 1M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

Predict positive: $(1 - 0.4) \times 4 = 2.4$. Predict negative: $0.4 \times 6 = 2.4$.

Optimal decision: positive; expected cost = **2.4**.

Q.15 — Answer: (A), (D) [MSQ, 1M · SVM & Decision Trees > Ensembles: bagging & random forests]

(A) **TRUE**. Variance of a mean.

(B) **FALSE**. They de-correlate them.

(C) **FALSE**. Without replacement it would reproduce the data.

(D) **TRUE**. Feature subsampling reduces correlation between trees.

Q.16 — Answer: 16.966 to 17.034 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Perceptron]

Unit separator $u = (-3, 4)/5$. Margins $y_i(u \cdot x_i)$: 3.4, 1.6, 2.6, 3.6, 1 $\rightarrow \gamma = \min = 1$.

Norms $\|x_i\|$: 3.606, 4.123, 4.123, 3.606, 2.236 $\rightarrow R = \max = 4.1231$.

Novikoff bound: number of mistakes $\leq (R/\gamma)^2 = (4.1231/1)^2 = 17$.

Common mistakes: forgetting to normalise u (γ must be measured with $\|u\| = 1$), or using R/γ without squaring.

Q.17 — Answer: (D) [MCQ, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent: 5 Yes, 5 No $\Rightarrow H = 1$.

Size — Small: [3+, 3-], Large: [2+, 2-]; gain = 0.

Credit — Fair: [2+, 1-], Excellent: [3+, 4-]; gain = 0.0349.

Age — Young: [1+, 2-], Middle: [1+, 1-], Senior: [3+, 2-]; gain = 0.039.

Colour — Red: [2+, 3-], Green: [1+, 2-], Blue: [2+, 0-]; gain = 0.239.

The largest information gain is for **Colour**, so it is chosen at the root.

Q.18 — Answer: 1.7757 to 1.7957 [NAT, 2M · Regression & Model Selection > Ridge & Lasso]

Ridge fitted values $\hat{y} = H_\lambda y$ with $H_\lambda = X(X^T X + \lambda I)^{-1} X^T$. Writing $X^T X = V D V^T$ with eigenvalues d_j^2 gives $\text{trace}(H_\lambda) = \sum d_j^2 / (d_j^2 + \lambda)$ (effective degrees of freedom).

$20/(20 + 1) + 5/(5 + 1) = 1.7857$.

Answer: **1.79**.

Q.19 — Answer: 13.134 to 13.266 [NAT, 2M · Regression & Model Selection > Cross-validation]

For each left-out point, the prediction is the y of its nearest remaining neighbour (no ties here).

x_i	3	4	6	10	11
nearest x	4	3	4	11	10
prediction	7	11	7	11	8
error	4	-4	-4	-3	3

$$\text{LOOCV MSE} = (1/5)\Sigma \text{error}^2 = 66/5 = 13.2.$$

Answer: **13.2**.

Q.20 — Answer: 0.8233 to 0.8433 [NAT, 2M · SVM & Decision Trees > Regression trees]

$$\text{Parent mean } \bar{y} = 1.25, \text{SSE}_0 = \Sigma(y - \bar{y})^2 = 17.5.$$

Evaluate every split:

Split t	Left mean	Right mean	$\text{SSE}_L + \text{SSE}_R$
2.5	1	1.286	17.429
3.5	1.5	1.167	17.333
5	2	0.8	14.8
8	1.25	1.25	17.5
10.5	0.8	2	14.8
11.5	0.833	2.5	13.333
12.5	1.143	2	16.857

Best split $t = 11.5$: left mean 0.833, right mean 2.5, $\text{SSE} = 13.333$, reduction = 4.167.

Answer: **0.83**.

Q.21 — Answer: (C) [MSQ, 2M · Unsupervised Learning > Hierarchical clustering]

(A) **FALSE**. That describes complete linkage; single linkage chains.

(B) **FALSE**. That holds for single linkage.

(C) **TRUE**. Single (and complete) linkage depend only on the ranks of the distances.

(D) **FALSE**. It is always $\leq$ the original distance.

Q.22 — Answer: 0.4445 to 0.4645 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

Counting: $\text{TP} = 2$, $\text{FP} = 2$, $\text{FN} = 4$.

$$P = 2/4 = 0.5, R = 2/6 = 0.3333.$$

$$F_{0.5} = (1 + 0.25) \times 0.5 \times 0.3333 / (0.25 \times 0.5 + 0.3333) = \mathbf{0.45}.$$

$\beta < 1$ weights precision more heavily (F1 here would be 0.4).

Q.23 — Answer: 1 [NAT, 2M · SVM & Decision Trees > Decision trees: structure & traversal]

Route each record from the root to a leaf:

#	Predicted	Actual	Correct?
1	Yes	Yes	yes
2	No	No	yes
3	No	No	yes
4	Yes	Yes	yes
5	Yes	Yes	yes
6	No	Yes	no
7	No	No	yes
8	Yes	Yes	yes

Misclassified: 1 of 8 (accuracy 87.5%). Answer: **1**.

Q.24 — Answer: 359.99 to 360.01 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

For centered data, projecting onto the top-k PCs and reconstructing leaves the components along the discarded eigenvectors; their average squared length (divisor $n - 1$) is the sum of the discarded eigenvalues.

Discarded eigenvalues: 11, 9.5, 8.5, 5, 2; sum = 36; total = 67.5.

Total error = $(n - 1) \times 36 = 10 \times 36 = \mathbf{360}$.

Q.25 — Answer: 6 [NAT, 2M · Unsupervised Learning > k-medoids]

Compute the Manhattan (L_1) distance matrix and its row sums:

	P1	P2	P3	P4	P5	Σ
P1	0	6	3	4	3	16
P2	6	0	9	6	5	26
P3	3	9	0	3	4	19
P4	4	6	3	0	1	14
P5	3	5	4	1	0	13

The smallest row sum is 13 for P5, so the medoid is P5.

Cost with P3: 19; increase = $19 - 13 = \mathbf{6}$.

Unlike a k-means centroid, the medoid is always an actual data point.

Q.26 — Answer: 0.89 to 0.91 [NAT, 2M · SVM & Decision Trees > SVM: soft margin & hinge loss]

Compute $f(x_i)$, the functional margin $y_i f(x_i)$ and the slack/hinge loss for each point:

Point	x	y	$f(x)$	$y f(x)$	$\xi = \max(0, 1 - y f)$
P1	(5, 5)	-1	2.5	-2.5	3.5
P2	(3, 3)	-1	-0.5	0.5	0.5
P3	(5, 3)	+1	0.5	0.5	0.5
P4	(5, 1)	-1	-1.5	1.5	0
P5	(2, 0)	-1	-4	4	0

Average = $4.5/5 = \mathbf{0.9}$.

Points with $y f \geq 1$ contribute 0; points inside the margin contribute between 0 and 1; misclassified points contribute more than 1.

Q.27 — Answer: (D) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Softmax + cross-entropy]

$p = \text{softmax}((0, 1, -1)) = (0.2447, 0.6652, 0.09)$; $y = (0, 0, 1)$.

$\partial L / \partial z = p - y = \mathbf{(0.24, 0.67, -0.91)}$.

Distractors: $y - p = (-0.24, -0.67, 0.91)$ (wrong sign); p alone = $(0.24, 0.67, 0.09)$ (forgot the target); $(p - y) \odot p \odot (1 - p) = (0.05, 0.15, -0.07)$ (spurious σ' -type factor, which cancels for cross-entropy); $-y/p = (0, 0, -11.11)$ is $\partial L / \partial p$, not $\partial L / \partial z$.

Q.28 — Answer: 1.6369 to 1.6569 [NAT, 2M · Probabilistic & Instance-based Classifiers > Linear discriminant analysis]

$\mathbf{m}_1 = (2, 0.667)$, $\mathbf{m}_2 = (4.667, 4)$.

$$S_1 = \begin{bmatrix} 8 & 4 \\ 4 & 2.667 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 12.667 & 7 \\ 7 & 6 \end{bmatrix}$$

$$S_W = \begin{bmatrix} 20.667 & 11 \\ 11 & 8.667 \end{bmatrix}$$

$\mathbf{m}_2 - \mathbf{m}_1 = (2.667, 3.333)$.

$\mathbf{w} = S_W^{-1}(\mathbf{m}_2 - \mathbf{m}_1) = (-0.2333, 0.6807)$.

$(\mathbf{m}_2 - \mathbf{m}_1)^T \mathbf{w} = \mathbf{1.65}$.

Q.29 — Answer: (B), (C), (D) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

(A) FALSE. $B = N$ is full-batch gradient descent.

(B) TRUE. Tiny steps.

(C) TRUE. CE = negative log-likelihood of a categorical/Bernoulli model.

(D) TRUE. Reduces the noise floor of SGD.

Q.30 — Answer: 1 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Expressiveness]

With identity activations, $\hat{y} = W_2(W_1x + b_1) + b_2 = (W_2W_1)x + (W_2b_1 + b_2)$: a single affine map.

$\tilde{w}^T = W_2W_1 = (5, -5)$; $\tilde{b} = W_2b_1 + b_2 = 1$.

Answer: **1**.

Hence stacking linear layers adds no expressive power – a non-linear activation is essential.

Q.31 — Answer: 0.2701 to 0.2901 [NAT, 2M · Probabilistic & Instance-based Classifiers · Naive Bayes]

$N(x; \mu, \sigma^2) = \exp(-(x-\mu)^2/(2\sigma^2))/\sqrt{(2\pi\sigma^2)}$. Naive Bayes multiplies the per-feature densities.

C_0 : $N(5; 4, 1) = 0.24197$; $\times$ prior $0.7 = 0.16938$.

C_1 : $N(5; 6, 2) = 0.2197$; $\times$ prior $0.3 = 0.065909$.

$P(C_1 | x) = 0.065909/(0.16938 + 0.065909) = \mathbf{0.28}$.

Q.32 — Answer: 6.3348 to 6.3985 [NAT, 2M · Regression & Model Selection · Simple linear regression]

For the least-squares fit with intercept the normal equations give $\Sigma e_i = 0$ and $\Sigma x_i e_i = 0$.

Known: $\Sigma e = 1.9$, $\Sigma x e = 12.4$. With unknowns $u = e$ at $x = 2$ and $v = e$ at $x = 5$:

$u + v = -1.9$, $2u + 5v = -12.4$.

Solving: $u = 0.9667$, $v = -2.8667$.

$y = \hat{y} + e = (3 + (1.2)(2)) + (0.9667) = 6.3667$.

Answer: **6.37**.

Q.33 — Answer: 0.5108 to 0.5148 [NAT, 2M · Regression & Model Selection · Multiple linear regression]

Adjusted $R^2 = 1 - [SSE/(n - p - 1)]/[SST/(n - 1)] = 1 - (1 - R^2)(n - 1)/(n - p - 1)$, where p is the number of predictors (excluding the intercept).

$= 1 - (355/42)/(798/46) = 1 - 8.4524/17.3478 = 0.5128$.

Answer: **0.513**.

Q.34 — Answer: 0.0078 to 0.0278 [NAT, 2M · Unsupervised Learning · Cluster evaluation]

a(i) = mean Euclidean distance from P_5 to the other members of its own cluster C_2 : $d(P_5, P_4) = 2$, $d(P_5, P_6) = 3.606 \rightarrow a = 2.8028$.

b(i) = smallest mean distance to the points of another cluster: C_1 : $2.8535 \rightarrow b = 2.8535$.

s(i) = $(b - a)/\max(a, b) = (2.8535 - 2.8028)/2.8535 = \mathbf{0.02}$.

s ranges in $[-1, 1]$; values near 1 mean well clustered, negative values suggest a wrong assignment.

Q.35 — Answer: 8.5303 to 8.5503 [NAT, 2M · Probabilistic & Instance-based Classifiers · Logistic regression]

$z_i = 1.5x_i + 0$; $\hat{p}_i = \sigma(z_i)$: 0.0474, 0.9526, 0.9526, 0.8176, 0.5.

Per-example loss $L_i = -\ln \hat{p}_i$ if $y_i = 1$ and $-\ln(1 - \hat{p}_i)$ if $y_i = 0$:

i	y	$\hat{p}$	L_i
1	0	0.0474	0.0486
2	0	0.9526	3.0486
3	0	0.9526	3.0486
4	0	0.8176	1.7014
5	1	0.5	0.6931

$\Sigma L_i = 8.5403 = \mathbf{8.54}$.

Common mistakes: using $\ln \hat{p}$ for $y = 0$ examples, or $\log_{10}$ instead of $\ln$.

Mock Test 49

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	2	9	24

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Neural Networks (MLP / Feed-forward)	7	11
SVM & Decision Trees	7	11
Regression & Model Selection	7	11
Probabilistic & Instance-based Classifiers	7	11
Unsupervised Learning	7	11

Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MSQ (one or more correct)]

Which of the following statements about the representational power of feed-forward neural networks is/are CORRECT?

- (A) A network with a single hidden layer of two ReLU units can approximate any continuous function on $[0, 1]$ to arbitrary accuracy.
- (B) An MLP of any depth with identity (linear) activations computes an affine function of its input.
- (C) Adding hidden layers with identity activations enlarges the set of functions the network can represent.
- (D) An MLP with 10 hidden layers and identity activations can represent XOR.

Q.2 [1 mark · NAT (numerical answer)]

A fully connected MLP has layer sizes 64-50-128-16-5 (input-...-output), every layer having a bias vector. The total number of trainable parameters (weights and biases) is _____ (Answer in integer)

Q.3 [1 mark · MSQ (one or more correct)]

Consider the soft-margin SVM: minimise $\frac{1}{2}\|w\|^2 + C \sum_i \xi_i$ subject to $y_i(w^T x_i + b) \geq 1 - \xi_i$, $\xi_i \geq 0$. Which of the following statements is/are CORRECT?

- (A) Points with $\alpha_i = C$ may lie inside the margin or be misclassified.
- (B) The hinge loss of every correctly classified point is zero.
- (C) The soft-margin SVM is infeasible if the data are not linearly separable.
- (D) Slack variables ξ_i can be negative for points far from the boundary.

Q.4 [1 mark · MSQ (one or more correct)]

Consider ridge regression (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_2^2$) and the lasso (minimise $\|y - X\beta\|^2 + \lambda\|\beta\|_1$) with $\lambda \geq 0$. Which of the following statements is/are CORRECT?

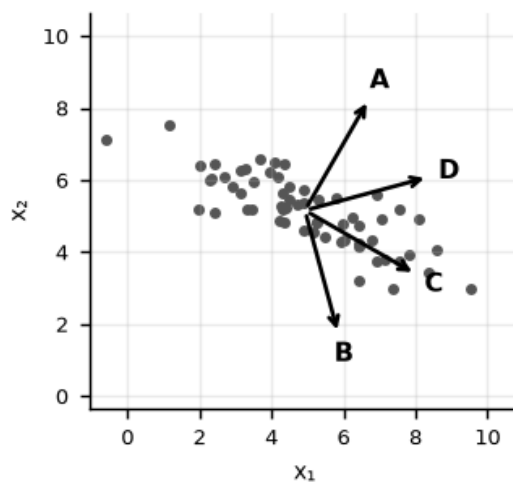
- (A) The lasso constraint region $\sum |\beta_j| \leq t$ in two dimensions is a diamond (square rotated by 45°).
- (B) The lasso has a closed-form solution for a general design matrix X , just like ridge.
- (C) The ridge penalty is $\lambda \sum |\beta_j|$.
- (D) Ridge regression requires $X^T X$ to be invertible.

Q.5 [1 mark · NAT (numerical answer)]

A 4-class softmax classifier produces the logits $z_1 = 1, z_2 = -2, z_3 = -1, z_4 = 2$ for an input $\mathbf{x}$. The predicted probability of class 2, $P(y = 2 | \mathbf{x})$, is _____ (rounded off to two decimal places)

Q.6 [1 mark · MCQ]

The scatter plot shows a 2-D dataset together with four candidate directions A-D drawn from the data mean. Which direction best represents the first principal component (PC1)?



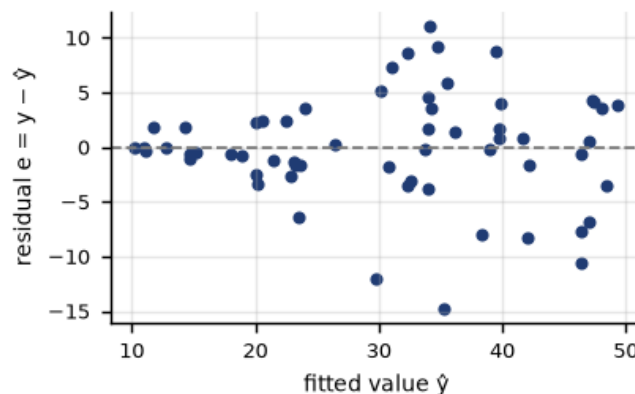
- (A) Direction A
- (B) Direction B
- (C) Direction C
- (D) Direction D

Q.7 [1 mark · NAT (numerical answer)]

Inverted dropout with keep probability $p = 0.5$ is applied during training. A hidden unit with activation $h = 3$ that is **kept** passes on the value _____ (rounded off to two decimal places)

Q.8 [1 mark · MCQ]

After fitting a linear regression model, the residuals are plotted against the fitted values as shown. The plot most strongly suggests:

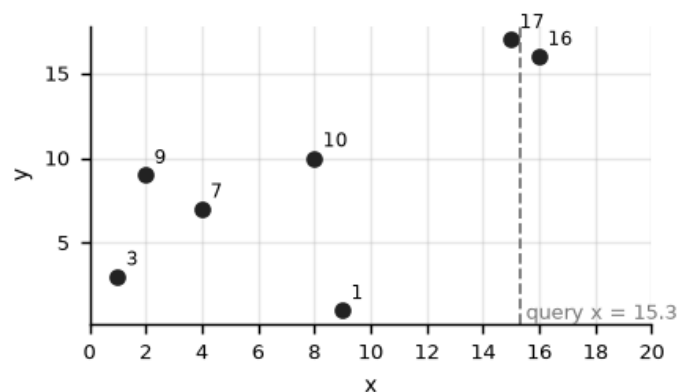


- (A) No evident violation of the linear-model assumptions
- (B) A non-linear relationship not captured by the model (e.g. a missing x^2 term)
- (C) Non-constant error variance (heteroscedasticity)
- (D) A single outlying observation, with the rest of the fit looking adequate

Q.9 [1 mark · NAT (numerical answer)]

The training data for a k-NN regressor are given below (also plotted). Using $k = 3$ and absolute distance $|x - x_i|$, the predicted value (unweighted average of neighbours) at $x = 15.3$ is _____ (rounded off to two decimal places)

x	1	2	4	8	9	15	16
y	3	9	7	10	1	17	16

**Q.10 [1 mark · MSQ (one or more correct)]**

Which of the following statements about k-means clustering is/are CORRECT?

- (A) k-means is sensitive to outliers.
- (B) k-means converges in a finite number of iterations.
- (C) k-means assigns each point to the centroid with the largest cosine similarity.
- (D) One iteration of Lloyd's algorithm costs $O(nkd)$ time.

Q.11 [1 mark · NAT (numerical answer)]

A node contains 6 records spread equally over 2 classes (3 records per class). Its entropy (in bits) is _____ (rounded off to two decimal places)

Q.12 [1 mark · MSQ (one or more correct)]

Which of the following tasks is/are most naturally formulated as **regression** problems?

- (A) Diagnosing whether a tumour is benign or malignant
- (B) Predicting the annual salary of an employee from experience and education
- (C) Predicting the blood group (A, B, AB or O) of a person from genetic markers
- (D) Estimating the age (in years) of a person from a photograph

Q.13 [1 mark · NAT (numerical answer)]

A clustering algorithm groups labelled data into three clusters; the contingency table of cluster versus true class counts is given below. The purity of this clustering is _____ (rounded off to two decimal places)

	Class P	Class Q	Class R
Cluster 1	0	11	25
Cluster 2	19	6	3
Cluster 3	15	11	6

Q.14 [1 mark · NAT (numerical answer)]

Consider the polynomial kernel $K(x, z) = (x^T z + 2)^2$ with $x = (-2, 3, -2)$ and $z = (3, 1, 3)$. The value of $K(x, z)$ is _____ (Answer in integer)

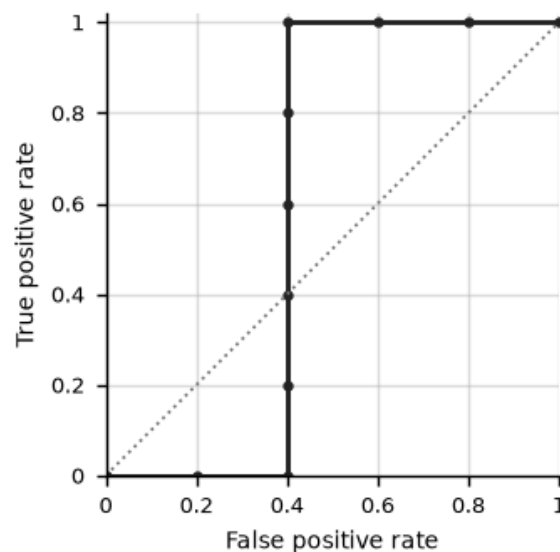
Q.15 [1 mark · MSQ (one or more correct)]

Which of the following statements about classifier evaluation is/are CORRECT?

- (A) Applying the transformation $s \rightarrow s^3$ to the scores changes the ROC AUC.
- (B) An AUC of 0.5 indicates a perfect classifier.
- (C) Balanced accuracy for binary classification is the mean of sensitivity and specificity.
- (D) Macro-averaging gives every class equal weight regardless of its number of examples.

Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

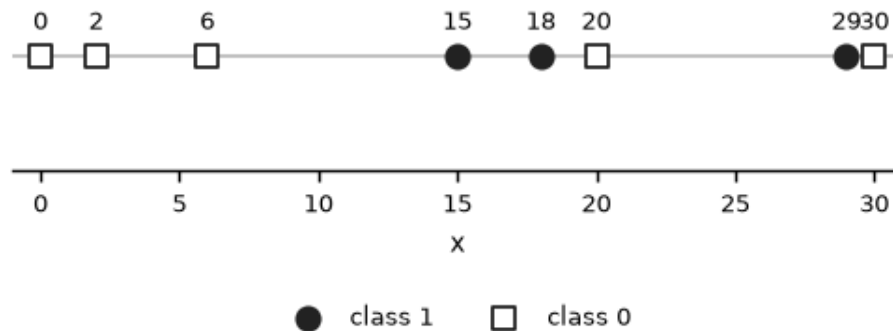
The ROC curve below was obtained from a test set with 5 positives and 5 negatives (all scores distinct, so the curve moves in steps of 1/5 vertically and 1/5 horizontally). From the figure, the highest TPR attainable at $FPR \leq 0.2$ is _____ (rounded off to two decimal places)

**Q.17 [2 marks · NAT (numerical answer)]**

$B = 10$ regression trees are averaged. Each tree's prediction has variance $\sigma^2 = 4$, and every pair of trees has correlation $\rho = 0.5$. The variance of the averaged prediction is _____ (rounded off to two decimal places)

Q.18 [2 marks · NAT (numerical answer)]

8 one-dimensional points with binary labels are shown on the number line (filled circle = class 1, open square = class 0). A 3-NN classifier (absolute distance, majority vote) is evaluated by leave-one-out cross-validation: each point is classified using all the *other* points. The number of training points misclassified in leave-one-out cross-validation is _____ (Answer in integer)

**Q.19 [2 marks · MSQ (one or more correct)]**

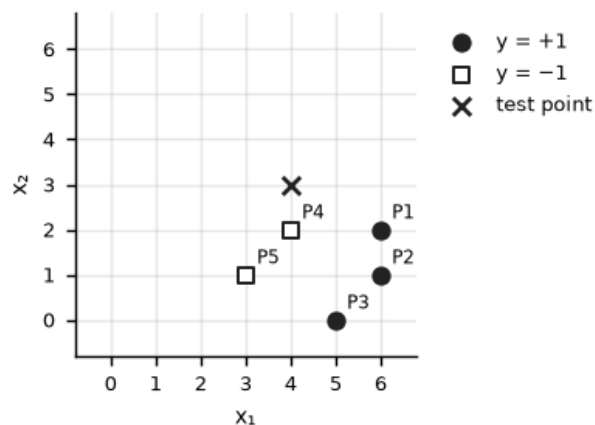
A threshold unit outputs 1 if $w_1x_1 + w_2x_2 + b > 0$ and 0 otherwise, for binary inputs $x_1, x_2 \in \{0, 1\}$. Which of the following parameter settings make the unit implement the Boolean **NOR** function?

- (A) $w_1 = -2, w_2 = -2, b = 1$
- (B) $w_1 = -1.5, w_2 = -3, b = 1$
- (C) $w_1 = -3, w_2 = 1.5, b = 2$
- (D) $w_1 = 3, w_2 = -3, b = -2$

Q.20 [2 marks · NAT (numerical answer)]

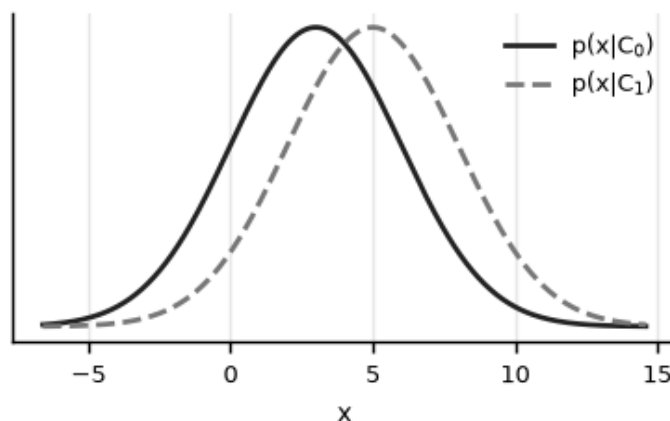
A hard-margin linear SVM is trained on the points below. Solving the dual problem gives the Lagrange multipliers α_i listed in the table. Using the KKT conditions, compute the decision value $f(x) = w^T x + b$ at $x = (4, 3)$. (rounded off to two decimal places)

Point	x_1	x_2	y	α_i
P1	6	2	+1	0.375
P2	6	1	+1	0
P3	5	0	+1	0.25
P4	4	2	-1	0.625
P5	3	1	-1	0



Q.21 [2 marks · NAT (numerical answer)]

In a two-class problem with one feature, $p(x | C_0) = N(3, 9)$ and $p(x | C_1) = N(5, 9)$ (second argument = variance), with priors $P(C_0) = 0.75$ and $P(C_1) = 0.25$. Under 0-1 loss, the posterior probability $P(C_1 | x = 3)$ is _____ (rounded off to two decimal places)



Class-conditional densities (not weighted by priors).

Q.22 [2 marks · NAT (numerical answer)]

Consider the model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ (first column of X is the intercept column), fitted by least squares using $\beta = (X^T X)^{-1} X^T y$ with the design matrix X and response y below. Then the least-squares estimate of β_1 is _____ (rounded off to two decimal places)

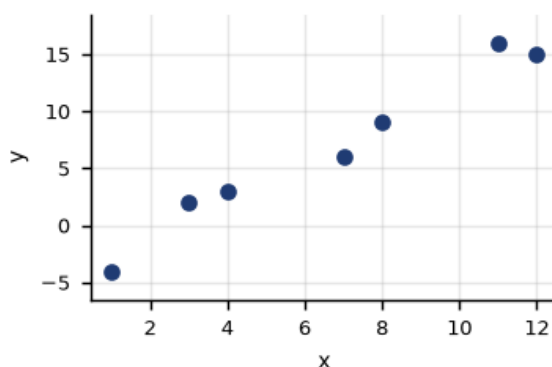
$$X = \begin{bmatrix} 1 & -2 & 2 \\ 1 & -1 & -1 \\ 1 & 0 & -2 \\ 1 & 1 & -1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$y = \begin{bmatrix} -7 \\ 4 \\ 5 \\ 5 \\ 3 \end{bmatrix}$$

Q.23 [2 marks · NAT (numerical answer)]

The 7 observations shown in the table are plotted below. For the simple linear regression of y on x (with intercept), the intercept β_0 of the least-squares line $\hat{y} = \beta_0 + \beta_1 x$ is _____ (rounded off to two decimal places)

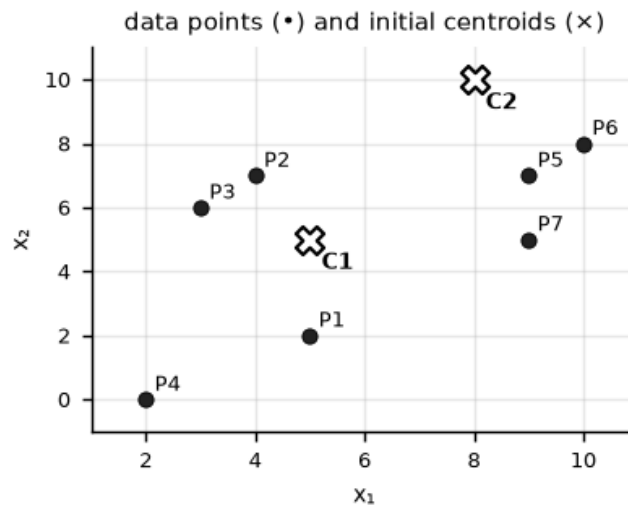
x	1	3	4	7	8	11	12
y	-4	2	3	6	9	16	15



Q.24 [2 marks · NAT (numerical answer)]

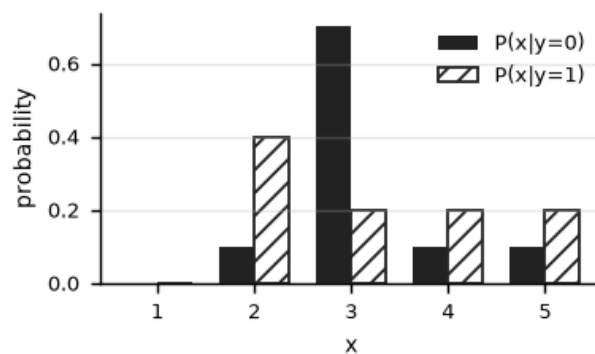
k-means (Lloyd's algorithm, Euclidean distance) with $k = 2$ is run on the 7 points below, starting from the initial centroids $C1 = (5, 5)$, $C2 = (8, 10)$. The number of points whose cluster membership **changes** in the second iteration (compared with the first-iteration assignment) is _____ (Answer in integer)

Point	P1	P2	P3	P4	P5	P6	P7
x_1	5	4	3	2	9	10	9
x_2	2	7	6	0	7	8	5

**Q.25 [2 marks · NAT (numerical answer)]**

A discrete feature $x \in \{1, \dots, 5\}$ has the class-conditional distributions below (also plotted), with priors $P(y = 0) = 0.7$ and $P(y = 1) = 0.3$. Then the Bayes error rate of the optimal classifier is _____ (rounded off to three decimal places)

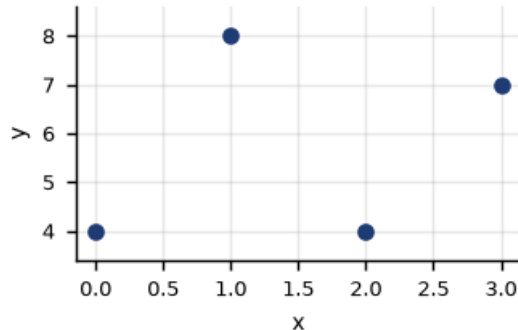
x	1	2	3	4	5
$P(x y=0)$	0	0.1	0.7	0.1	0.1
$P(x y=1)$	0	0.4	0.2	0.2	0.2



Q.26 [2 marks · NAT (numerical answer)]

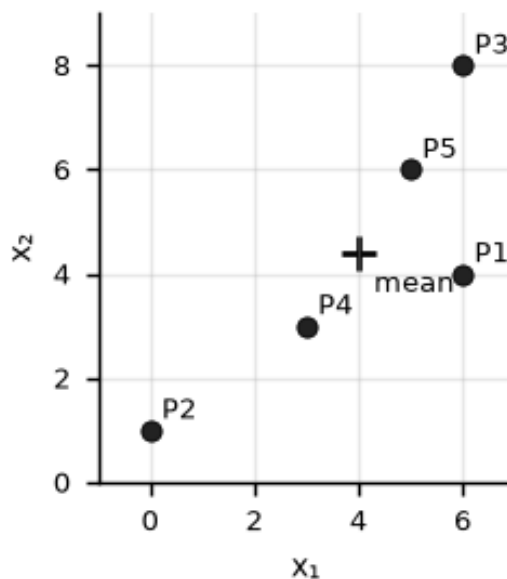
A linear model $\hat{y} = w_0 + w_1 x$ is trained on the 4 points below by batch gradient descent on the loss $J(w) = (1/2n)\sum(w_0 + w_1 x_i - y_i)^2$, starting from $(w_0, w_1) = (1, 0)$ with learning rate $\eta = 0.1$. Then the value of w_0 after one iteration is _____ (rounded off to three decimal places)

x	0	1	2	3
y	4	8	4	7

**Q.27 [2 marks · NAT (numerical answer)]**

PCA is applied to the five 2-D points below (data are mean-centered first; sample covariance uses divisor $n - 1$). Taking the unit-length first principal direction with a **positive first component**, the score (projection) of P4 on PC1 is _____ (rounded off to two decimal places)

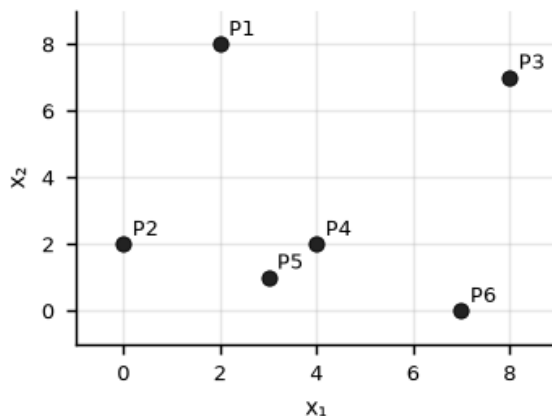
Point	P1	P2	P3	P4	P5
x_1	6	0	6	3	5
x_2	4	1	8	3	6



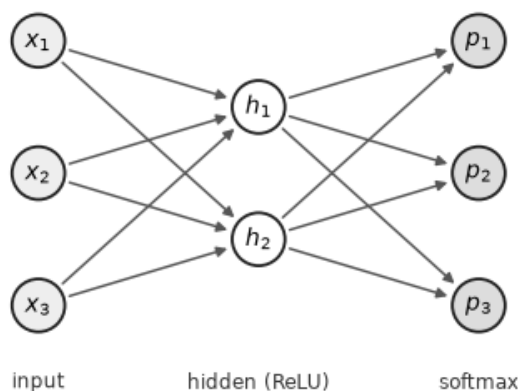
Q.28 [2 marks · NAT (numerical answer)]

Single-linkage agglomerative clustering with Manhattan distance is applied to the six points below. Then the height at which the **last** merge (forming a single cluster) occurs is ____ (Answer in integer)

Point	P1	P2	P3	P4	P5	P6
x_1	2	0	8	4	3	7
x_2	8	2	7	2	1	0

**Q.29 [2 marks · NAT (numerical answer)]**

A 3-2-3 network computes $h = \text{ReLU}(W_1 x + b_1)$, $z = W_2 h + b_2$, $p = \text{softmax}(z)$, with the parameters below. For $x = (1, 1, 0)^T$, the predicted probability p_1 of class 1 is ____ (rounded off to three decimal places)



$$W_1 = \begin{bmatrix} 0 & 0.5 & 0 \\ 0.5 & -0.5 & 0 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 1 \\ -0.5 \end{bmatrix}$$

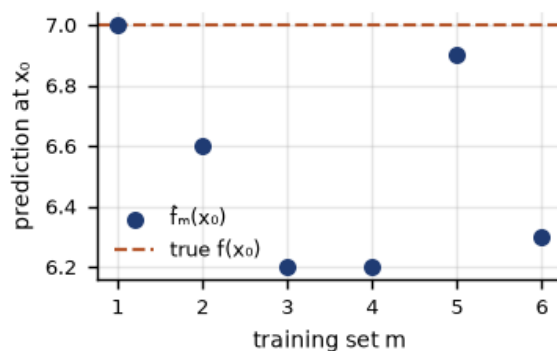
$$W_2 = \begin{bmatrix} 2 & -2 \\ -2 & 2 \\ -2 & 1 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} -1 \\ -1 \\ 0.5 \end{bmatrix}$$

Q.30 [2 marks · NAT (numerical answer)]

Data are generated as $y = f(x) + \varepsilon$ with $E[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2 = 1$. At a test point x_0 the true value is $f(x_0) = 7$. A learning method is trained on 6 independent training sets, giving the predictions $\hat{f}_m(x_0)$ in the table (also plotted). Treating these as the full distribution of the method's prediction (variance with divisor 6), the squared bias of the method at x_0 is _____ (rounded off to three decimal places)

m	1	2	3	4	5	6
$\hat{f}_m(x_0)$	7	6.6	6.2	6.2	6.9	6.3

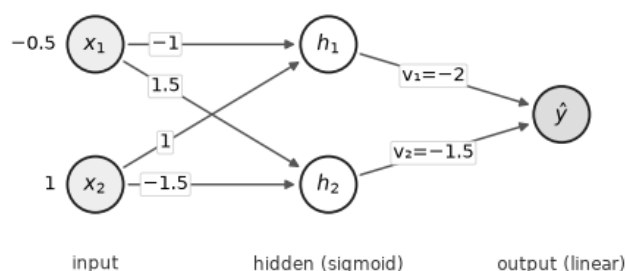
**Q.31 [2 marks · NAT (numerical answer)]**

Three customers are described by age and income. Age ranges over $[20, 60]$ years and income over $[10, 210]$ thousand rupees in the full dataset. On the **raw** (unscaled) features, the Euclidean distance from A to its nearest neighbour among B and C is _____ (rounded off to two decimal places)

Customer	Age (years)	Income (₹ thousand)
A	56	60
B	31	10
C	58	120

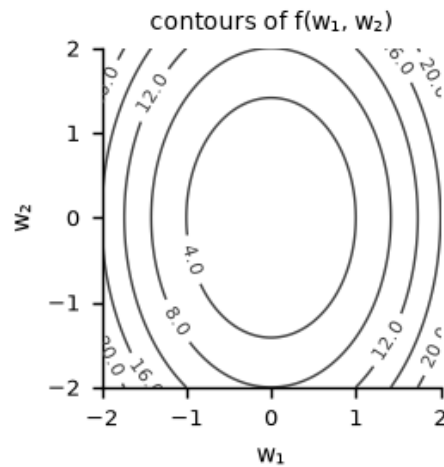
Q.32 [2 marks · NAT (numerical answer)]

In the network below, hidden units are sigmoid, the output is linear ($\hat{y} = v_1 h_1 + v_2 h_2$), all biases are 0, and the loss is $L = \frac{1}{2}(\hat{y} - y)^2$. For input $x = (-0.5, 1)$ and target $y = -1$, the value of $\partial L / \partial b_2$ (bias of hidden unit h_2 , currently 0) is _____ (rounded off to three decimal places)

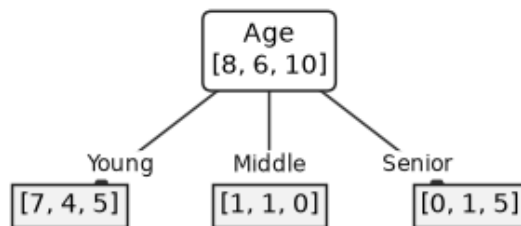


Q.33 [2 marks · MSQ (one or more correct)]

Gradient descent $w \leftarrow w - \eta \nabla f(w)$ is applied to $f(w_1, w_2) = \frac{1}{2}(8w_1^2 + 4w_2^2)$ from a generic starting point (both coordinates non-zero). For which of the following learning rates does GD converge to the minimum?

**(A)** $\eta = 0.1$ **(B)** $\eta = 0.0375$ **(C)** $\eta = 0.325$ **(D)** $\eta = 0.375$ **Q.34 [2 marks · NAT (numerical answer)]**

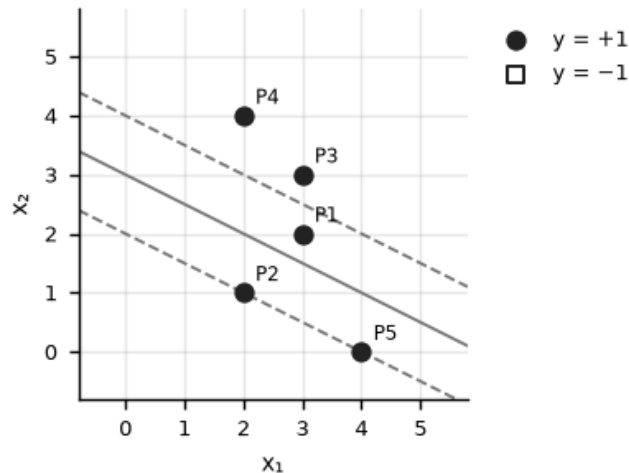
A CART node with 24 training records (class counts [C1, C2, C3]) is split on Age as shown. Compute the weighted Gini index of the split on Age. (rounded off to two decimal places)



Q.35 [2 marks · MSQ (one or more correct)]

A soft-margin linear SVM has learned $f(x) = 0.5x_1 + x_2 - 3$. The training points are listed below and plotted (boundary solid, margins $f = \pm 1$ dashed). With ξ_i the optimal slack of point i , which of the following statements is/are CORRECT?

Point	x_1	x_2	y
P1	3	2	+1
P2	2	1	+1
P3	3	3	+1
P4	2	4	+1
P5	4	0	+1



- (A) Exactly 3 of the points have $\xi_i > 0$.
 (B) P2 is misclassified ($\xi_2 > 1$).
 (C) P1 is correctly classified but lies strictly inside the margin.
 (D) $\sum \xi_i = 5$.

— END OF QUESTION PAPER —

Mock Test 49 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MSQ	1	(B)	19	MSQ	2	(A), (B)
2	NAT	1	11927	20	NAT	2	-1.51 to -1.49
3	MSQ	1	(A)	21	NAT	2	0.2007 to 0.2207
4	MSQ	1	(A)	22	NAT	2	2.08 to 2.12
5	NAT	1	0.0028 to 0.0228	23	NAT	2	-4.6241 to -4.5781
6	MCQ	1	(C)	24	NAT	2	1
7	NAT	1	5.99 to 6.01	25	NAT	2	0.248 to 0.252
8	MCQ	1	(C)	26	NAT	2	1.4706 to 1.4794
9	NAT	1	11.3233 to 11.3433	27	NAT	2	-1.7159 to -1.6959
10	MSQ	1	(A), (B), (D)	28	NAT	2	8
11	NAT	1	0.99 to 1.01	29	NAT	2	0.9846 to 0.9886
12	MSQ	1	(B), (D)	30	NAT	2	0.2148 to 0.2208
13	NAT	1	0.6046 to 0.6246	31	NAT	2	55.8917 to 55.9117
14	NAT	1	49	32	NAT	2	0.0987 to 0.1027
15	MSQ	1	(C), (D)	33	MSQ	2	(A), (B)
16	NAT	2	-0.01 to 0.01	34	NAT	2	0.5334 to 0.5534
17	NAT	2	2.19 to 2.21	35	MSQ	2	(A), (B), (C)
18	NAT	2	4				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Neural Networks (MLP / Feed-forward)					11
SVM & Decision Trees					11
Regression & Model Selection					11
Probabilistic & Instance-based Classifiers					11
Unsupervised Learning					11
Total					55

Mock Test 49 — Detailed Solutions

Q.1 — Answer: (B) [MSQ, 1M · Neural Networks (MLP / Feed-forward) > Expressiveness]

- (A) **FALSE**. Width must grow with the required accuracy.
 (B) **TRUE**. A composition of affine maps is affine.
 (C) **FALSE**. The composition remains affine.
 (D) **FALSE**. It is still a linear model; XOR is not linearly separable.

Q.2 — Answer: 11927 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Parameter counting]

Fully connected layer with n_{in} inputs and n_{out} outputs: $n_{in} \cdot n_{out}$ weights + n_{out} biases.

Layer 1: $64 \times 50 = 3200$ weights, 50 biases.

Layer 2: $50 \times 128 = 6400$ weights, 128 biases.

Layer 3: $128 \times 16 = 2048$ weights, 16 biases.

Layer 4: $16 \times 5 = 80$ weights, 5 biases.

Weights = 11728, biases = 199. Required: **11927**.

Q.3 — Answer: (A) [MSQ, 1M · SVM & Decision Trees > SVM: soft margin & hinge loss]

- (A) **TRUE**. Bound support vectors.
 (B) **FALSE**. Correct points inside the margin ($0 < y f < 1$) have positive loss.
 (C) **FALSE**. Slack makes it always feasible.
 (D) **FALSE**. $\xi_i \geq 0$ by definition.

Q.4 — Answer: (A) [MSQ, 1M · Regression & Model Selection > Ridge & Lasso]

- (A) **TRUE**. Its vertices lie on the coordinate axes.
 (B) **FALSE**. In general it requires iterative methods (coordinate descent, LARS).
 (C) **FALSE**. That is the lasso (L1) penalty; ridge uses $\lambda \Sigma \beta_j^2$.
 (D) **FALSE**. $X^T X + \lambda I$ is invertible for any $\lambda > 0$.

Q.5 — Answer: 0.0028 to 0.0228 [NAT, 1M · Probabilistic & Instance-based Classifiers > Logistic regression]

softmax: $P(y = k | \mathbf{x}) = e^{z_k} / \sum_j e^{z_j}$.

e^z values: 2.7183, 0.1353, 0.3679, 7.3891; sum = 10.6106.

$P(y = 2 | \mathbf{x}) = 0.1353/10.6106 = \mathbf{0.01}$.

Q.6 — Answer: (C) [MCQ, 1M · Unsupervised Learning > PCA & dimensionality reduction]

PC1 is the direction of maximum variance — along the long axis of the elongated cloud (direction C). PC2 must be orthogonal to PC1 and captures the remaining variance (direction A).

Answer: **Direction C**. Directions D and B are at 45° to the main axis and capture intermediate variance.

Q.7 — Answer: 5.99 to 6.01 [NAT, 1M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Inverted dropout multiplies kept activations by $1/p$ so that $E[\text{output}] = h$: $3/0.5 = \mathbf{6}$. No scaling is applied at test time.

Q.8 — Answer: (C) [MCQ, 1M · Regression & Model Selection > Simple linear regression]

The spread of residuals grows with the fitted value (funnel shape) → the constant-variance assumption is violated; a transformation of y (e.g. log) or weighted least squares may help.

Answer: **Non-constant error variance (heteroscedasticity)**.

Other patterns: funnel → heteroscedasticity; curvature → missing non-linearity; isolated point → outlier; random band → adequate fit.

Q.9 — Answer: 11.3233 to 11.3433 [NAT, 1M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Distances $|x_i - 15.3|$: $x=15$: 0.3, $x=16$: 0.7, $x=9$: 6.3, $x=8$: 7.3, $x=4$: 11.3, $x=2$: 13.3, $x=1$: 14.3.

The 3 nearest neighbours are $x = 15, 16, 9$ with $y = 17, 16, 1$.

$\hat{y} = (17 + 16 + 1)/3 = \mathbf{11.33}$.

Q.10 — Answer: (A), (B), (D) [MSQ, 1M · Unsupervised Learning · k-means]

(A) **TRUE**. Centroids are means, and squared distances give far-away points a large influence.

(B) **TRUE**. There are finitely many partitions and the objective never increases, so no partition repeats.

(C) **FALSE**. Standard k-means uses Euclidean distance.

(D) **TRUE**. $n \cdot k$ distances, each $O(d)$.

Q.11 — Answer: 0.99 to 1.01 [NAT, 1M · SVM & Decision Trees · Decision trees: impurity measures]

$p_k = 3/6 = 1/2$ for every class $\Rightarrow H = -2 \cdot (1/2) \log_2(1/2) = \log_2 2 = 1$ — the maximum possible for 2 classes.

Q.12 — Answer: (B), (D) [MSQ, 1M · Regression & Model Selection · Regression vs classification]

A supervised task is *regression* when the target y is real-valued (loss such as squared error), and *classification* when y takes values in a finite set of labels (loss such as cross-entropy or 0-1 loss).

(A) **FALSE**. The output (benign / malignant) is a discrete class label $\rightarrow$ classification.

(B) **TRUE**. The output (salary) is a continuous real-valued quantity $\rightarrow$ regression.

(C) **FALSE**. The output (one of 4 groups) is a discrete class label $\rightarrow$ classification.

(D) **TRUE**. The output (age in years) is a continuous real-valued quantity $\rightarrow$ regression.

Q.13 — Answer: 0.6046 to 0.6246 [NAT, 1M · Unsupervised Learning · Cluster evaluation]

Majority count in each cluster: 25, 19, 15 (sum 59); $N = 96$.

Purity $= (1/N) \sum_k \max_j n_{kj} = 59/96 = 0.6146$.

Misclassified $= N - \sum \max = 37$.

Answer: **0.61**.

Purity is an external measure (needs class labels) and is trivially 1 if each point forms its own cluster, so it cannot be used to choose k .

Q.14 — Answer: 49 [NAT, 1M · SVM & Decision Trees · SVM: kernels]

$x^T z = (-2)(3) + (3)(1) + (-2)(3) = -9$.

$K(x, z) = (-9 + 2)^2 = -7^2 = 49$.

The kernel equals the inner product $\phi(x)^T \phi(z)$ of the explicit (all-monomials-up-to-degree- d) feature maps, without forming ϕ .

Q.15 — Answer: (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers · Evaluation metrics]

(A) **FALSE**. Monotone transforms preserve the ranking.

(B) **FALSE**. 0.5 is chance level; 1 is perfect.

(C) **TRUE**. $(\text{TPR} + \text{TNR})/2$.

(D) **TRUE**. Simple mean over classes.

Q.16 — Answer: -0.01 to 0.01 [NAT, 2M · Probabilistic & Instance-based Classifiers · Evaluation metrics]

Among the operating points with $\text{FPR} \leq 0.2$, take the highest point of the curve (the top of the vertical segment at $\text{FPR} = 0.2$).

$\text{TPR} = 0$.

Q.17 — Answer: 2.19 to 2.21 [NAT, 2M · SVM & Decision Trees · Ensembles: bagging & random forests]

$\text{Var}((1/B) \sum_b f_b) = (1/B^2) [B\sigma^2 + B(B-1)\rho\sigma^2] = \rho\sigma^2 + (1-\rho)\sigma^2/B$
 $= 0.5 \times 4 + 0.5 \times 4/10 = 2 + 0.2 = 2.2$.

Even with $B \rightarrow \infty$ the variance cannot fall below $\rho\sigma^2 = 2$ — which is why random forests try to reduce ρ by feature subsampling.

Q.18 — Answer: 4 [NAT, 2M · Probabilistic & Instance-based Classifiers · k-nearest neighbours]

Classify each point by its nearest neighbour(s) among the remaining points:

x	true	3 nearest (excluding itself)	LOO prediction	
0	0	2(0), 6(0), 15(1)	0	
2	0	0(0), 6(0), 15(1)	0	
6	0	2(0), 0(0), 15(1)	0	
15	1	18(1), 20(0), 6(0)	0	wrong
18	1	20(0), 15(1), 29(1)	1	
20	0	18(1), 15(1), 29(1)	1	wrong
29	1	30(0), 20(0), 18(1)	0	wrong
30	0	29(1), 20(0), 18(1)	1	wrong

Misclassified: 4 of 8. Answer: **4**.

Note: the *training* error of 1-NN is 0, but the LOOCV error is not — a point cannot be its own neighbour.

Q.19 — Answer: (A), (B) [MSQ, 2M · Neural Networks (MLP / Feed-forward) · Perceptron]

Target NOR truth table for $(x_1, x_2) = (0,0), (0,1), (1,0), (1,1)$: 1, 0, 0, 0. For each option compute $a = w_1x_1 + w_2x_2 + b$ at the four inputs and output 1 iff $a > 0$.

(A) TRUE. $a = 1, -1, -1, -3 \rightarrow$ outputs 1, 0, 0, 0 - NOR.

(B) TRUE. $a = 1, -2, -0.5, -3.5 \rightarrow$ outputs 1, 0, 0, 0 - NOR.

(C) FALSE. $a = 2, 3.5, -1, 0.5 \rightarrow$ outputs 1, 1, 0, 1 - (NOT x_1) OR x_2 .

(D) FALSE. $a = -2, -5, 1, -2 \rightarrow$ outputs 0, 0, 1, 0 - x_1 AND (NOT x_2).

Q.20 — Answer: -1.51 to -1.49 [NAT, 2M · SVM & Decision Trees · SVM: dual & support vectors]

$w = \sum \alpha_i y_i x_i$ over points with $\alpha_i > 0$:

$w = (0.375)(+1)(6, 2) + (0.25)(+1)(5, 0) + (0.625)(-1)(4, 2) = (1, -1/2)$.

b from any support vector, e.g. P1: $y = w^T x + b \Rightarrow b = +1 - (5) = -4$.

Check: $\sum \alpha_i y_i = 0$ ($= 0$, as required).

$f((4, 3)) = (1)(4) + (-1/2)(3) - 4 = -1.5$.

Q.21 — Answer: 0.2007 to 0.2207 [NAT, 2M · Probabilistic & Instance-based Classifiers · Linear discriminant analysis]

With equal variances, $\ln[\pi_1 p(x|C_1)/(\pi_0 p(x|C_0))] = (\mu_1 - \mu_0)x/\sigma^2 - (\mu_1^2 - \mu_0^2)/(2\sigma^2) + \ln(\pi_1/\pi_0)$ — linear in x (LDA).

Slope $a = (\mu_1 - \mu_0)/\sigma^2 = 0.2222$; intercept $c = -(\mu_1^2 - \mu_0^2)/(2\sigma^2) + \ln(\pi_1/\pi_0) = -1.9875$.

$P(C_1|x) = \sigma(ax + c) = \sigma(-1.3208) = \mathbf{0.21}$.

Q.22 — Answer: 2.08 to 2.12 [NAT, 2M · Regression & Model Selection · Multiple linear regression]

The columns of X are mutually orthogonal, so $X^T X$ is diagonal and each coefficient is $\beta_j = (x_j^T y)/(x_j^T x_j)$.

$$X^T X = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 14 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 10 \\ 21 \\ -27 \end{bmatrix}$$

$\beta = (10/5, 21/10, -27/14) = (2, 2.1, -1.9286)$.

Answer: **2.1**.

Q.23 — Answer: -4.6241 to -4.5781 [NAT, 2M · Regression & Model Selection · Simple linear regression]

$\bar{x} = 6.571, \bar{y} = 6.714$.

i	1	2	3	4	5	6	7
$x_i - \bar{x}$	-5.57	-3.57	-2.57	0.43	1.43	4.43	5.43
$y_i - \bar{y}$	-10.71	-4.71	-3.71	-0.71	2.29	9.29	8.29

$(x_i - \bar{x})(y_i - \bar{y})$	59.69	16.84	9.55	-0.31	3.27	41.12	44.98
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$$S_{xx} = \sum (x_i - \bar{x})^2 = 101.714, S_{xy} = 175.143.$$

$$\beta_1 = S_{xy}/S_{xx} = 1.7219, \beta_0 = \bar{y} - \beta_1 \bar{x} = -4.6011.$$

Answer: **-4.6**.

Q.24 — Answer: 1 [NAT, 2M · Unsupervised Learning > k-means]

Iteration 1 - assignment (nearest initial centroid, squared Euclidean distance):

Point	$d^2(\cdot, C1)$	$d^2(\cdot, C2)$	cluster
P1 (5, 2)	9	73	C1
P2 (4, 7)	5	25	C1
P3 (3, 6)	5	41	C1
P4 (2, 0)	34	136	C1
P5 (9, 7)	20	10	C2
P6 (10, 8)	34	8	C2
P7 (9, 5)	16	26	C1

Update: each centroid becomes the mean of its assigned points: $C1 \rightarrow (4.6, 4)$; $C2 \rightarrow (9.5, 7.5)$.

Iteration 2 - assignment with the updated centroids:

Point	$d^2(\cdot, C1)$	$d^2(\cdot, C2)$	cluster
P1 (5, 2)	4.16	50.5	C1
P2 (4, 7)	9.36	30.5	C1
P3 (3, 6)	6.56	44.5	C1
P4 (2, 0)	22.76	112.5	C1
P5 (9, 7)	28.36	0.5	C2
P6 (10, 8)	45.16	0.5	C2
P7 (9, 5)	20.36	6.5	C2

Points that changed cluster: P7 $\rightarrow$ answer **1**.

Note: each step of Lloyd's algorithm can only decrease (or keep) J, which is why it converges.

Q.25 — Answer: 0.248 to 0.252 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

Bayes rule: choose the class with larger $\pi_k P(x|k)$. Bayes error = $\sum_x \min(\pi_0 P(x|0), \pi_1 P(x|1))$.

x	$\pi_0 P(x 0)$	$\pi_1 P(x 1)$	decision	min
1	0	0	tie	0
2	0.07	0.12	1	0.07
3	0.49	0.06	0	0.06
4	0.07	0.06	0	0.06
5	0.07	0.06	0	0.06

Bayes error = **0.25**.

Q.26 — Answer: 1.4706 to 1.4794 [NAT, 2M · Regression & Model Selection > Gradient descent]

$$\partial J / \partial w_0 = (1/n) \sum r_i, \partial J / \partial w_1 = (1/n) \sum r_i x_i, \text{ with residual } r_i = w_0 + w_1 x_i - y_i.$$

$$\text{Step 1: } w = (1, 0), r = (-3, -7, -3, -6), \nabla J = (-4.75, -7.75) = w \leftarrow w - 0.1 \nabla J = (1.475, 0.775).$$

Answer: **1.475**.

Q.27 — Answer: -1.7159 to -1.6959 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

Mean $\bar{x} = (4, 4.4)$. Centered data rows: P1: (2, -0.4), P2: (-4, -3.4), P3: (2, 3.6), P4: (-1, -1.4), P5: (1, 1.6).

$$\text{Sample covariance } S = (1/(n-1)) \sum (x - \bar{x})(x - \bar{x})^T = [[6.5, 5.75], [5.75, 7.3]].$$

$$\text{Eigenvalues: } \lambda_1 = 12.6639, \lambda_2 = 1.1361; \text{ unit PC1 direction } v_1 = (0.6821, 0.7312).$$

$$\text{Score } z_1 = v_1 \cdot (x - \bar{x}) = 0.6821 \cdot -1 + 0.7312 \cdot -1.4 = -1.7059.$$

Answer: **-1.71** (the sign depends on the orientation of v_1 ; with the stated convention it is -1.71).

Q.28 — Answer: 8 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Single-linkage merge heights are exactly the edge weights of a minimum spanning tree (Kruskal's algorithm adds the shortest edge joining two different components = single-linkage merge).

MST edges (Prim's algorithm): P1-P3 (7); P1-P2 (8); P2-P4 (4); P4-P5 (2); P4-P6 (5).

Sorted merge heights: 2, 4, 5, 7, 8.

The last single-linkage merge happens at the largest MST edge: **8**.

Q.29 — Answer: 0.9846 to 0.9886 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Forward pass]

$a = W_1x + b_1 = (1.5, -0.5)$; $h = \text{ReLU}(a) = (1.5, 0)$.

$z = W_2h + b_2 = (2, -4, -2.5)$.

$\exp(z) = (7.3891, 0.0183, 0.0821)$, $\text{sum} = 7.4895$.

$p_1 = \exp(z_1) / \sum \exp(z_j) = \mathbf{0.9866}$. (All probabilities: (0.987, 0.002, 0.011).)

Q.30 — Answer: 0.2148 to 0.2208 [NAT, 2M · Regression & Model Selection > Bias-variance trade-off]

Mean prediction $\bar{f} = (1/6)\sum f_m = 6.5333$.

$\text{Bias}^2 = (\bar{f} - f(x_0))^2 = (6.5333 - 7)^2 = 0.2178$.

$\text{Variance} = (1/6)\sum (f_m - \bar{f})^2 = 0.1056$.

Expected test error = $\text{Bias}^2 + \text{Variance} + \sigma^2 = 0.2178 + 0.1056 + 1 = 1.3233$.

Answer: **0.218**.

Q.31 — Answer: 55.8917 to 55.9117 [NAT, 2M · Unsupervised Learning > Distance measures]

Min-max scaling: $x' = (x - \min) / (\max - \min)$; age range [20, 60], income range [10, 210].

Customer	age'	income'
A	0.9	0.25
B	0.275	0
C	0.95	0.55

Raw distances: $d(A,B) = 55.9$, $d(A,C) = 60.03 \rightarrow$ nearest is B (income dominates because of its larger range).

Scaled distances: $d(A,B) = 0.6731$, $d(A,C) = 0.3041 \rightarrow$ nearest is C.

Answer: **55.9**. Scaling changes which neighbour is nearest — distance-based methods (k-means, k-NN, hierarchical clustering) should be run on standardized features.

Q.32 — Answer: 0.0987 to 0.1027 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Backpropagation]

Forward: $a = Wx = (1.5, -2.25)$, $h = \sigma(a) = (0.8176, 0.0953)$, $\hat{y} = v \cdot h = -1.7782$.

Output error $\delta_o = \partial L / \partial \hat{y} = \hat{y} - y = -0.7782$ (linear output, $L = \frac{1}{2}(\hat{y} - y)^2$).

Hidden delta $\delta_2 = \delta_o \cdot v_2 \cdot h_2(1 - h_2) = -0.7782 \times -1.5 \times 0.0863 = 0.1007$.

$\partial L / \partial b_2 = \delta_2 = \mathbf{0.1007}$.

Q.33 — Answer: (A), (B) [MSQ, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

$\nabla f = (8w_1, 4w_2) = w_i \leftarrow (1 - \eta a_i) w_i$. GD converges (from every start) iff $|1 - \eta a_i| < 1$ for both i , i.e. $0 < \eta < 2/\max(a_i) = 2/8 = 0.25$. The steep direction ($a = 8$) limits the learning rate.

(A) TRUE. $\eta = 0.1$: $|1 - \eta \cdot 8| = 0.2$, $|1 - \eta \cdot 4| = 0.6 \rightarrow$ converges.

(B) TRUE. $\eta = 0.0375$: $|1 - \eta \cdot 8| = 0.7$, $|1 - \eta \cdot 4| = 0.85 \rightarrow$ converges.

(C) FALSE. $\eta = 0.325$: $|1 - \eta \cdot 8| = 1.6$, $|1 - \eta \cdot 4| = 0.3 \rightarrow$ diverges.

(D) FALSE. $\eta = 0.375$: $|1 - \eta \cdot 8| = 2$, $|1 - \eta \cdot 4| = 0.5 \rightarrow$ diverges.

Q.34 — Answer: 0.5334 to 0.5534 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [8, 6, 10]: $\text{Gini} = 1 - [(8/24)^2 + (6/24)^2 + (10/24)^2] = 0.6528$.

Age = Young: [7, 4, 5], Gini = $1 - [(7/16)^2 + (4/16)^2 + (5/16)^2] = 0.6484$.

Age = Middle: [1, 1, 0], Gini = $1 - [(1/2)^2 + (1/2)^2 + (0/2)^2] = 0.5$.

Age = Senior: [0, 1, 5], Gini = $1 - [(0/6)^2 + (1/6)^2 + (5/6)^2] = 0.2778$.

Weighted Gini = $(16/24)(0.6484) + (2/24)(0.5) + (6/24)(0.2778) = 0.5434$.

Gini gain = $0.6528 - 0.5434 = 0.1094$.

Answer: **0.54**.

Q.35 — Answer: (A), (B), (C) [MSQ, 2M · SVM & Decision Trees › SVM: soft margin & hinge loss]

$\xi_i = \max(0, 1 - y_i f(x_i))$.

Point	$y f(x)$	ξ
P1	0.5	0.5
P2	-1	2
P3	1.5	0
P4	2	0
P5	-1	2

(A) TRUE. Count of points with $y f < 1$.

(B) TRUE. $y f = -1 \Rightarrow \xi = 2$.

(C) TRUE. $y f = 0.5 \Rightarrow \xi = 0.5$.

(D) FALSE. $\sum \xi_i = 4.5$.

Mock Test 50

Section 6 — Machine Learning | GATE DA pattern

Questions	Total marks	Duration	MCQ	MSQ	NAT
35	55	100 min	7	4	24

Q.1 - Q.15 carry **one mark** each; **Q.16 - Q.35** carry **two marks** each. For a wrong MCQ answer, $\frac{1}{3}$ mark (1-mark questions) or $\frac{2}{3}$ mark (2-mark questions) is deducted. MSQ and NAT questions carry **no negative marking**; MSQ requires choosing *all and only* the correct options (no partial credit). NAT answers are accepted within the stated range. A calculator is allowed (as in GATE, use the on-screen virtual calculator mindset: keep work tidy).

Topic area	Questions	Marks
Neural Networks (MLP / Feed-forward)	7	11
SVM & Decision Trees	7	11
Regression & Model Selection	7	11
Unsupervised Learning	7	11
Probabilistic & Instance-based Classifiers	7	11

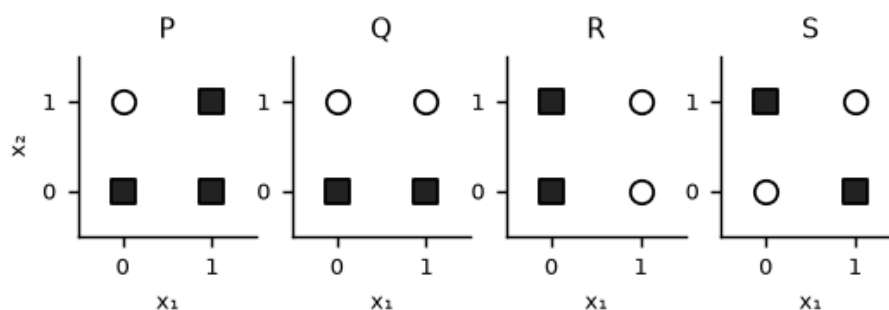
Start time: _____ End time: _____ Score: _____ / 55

Questions

Q.1 - Q.15 : One-mark questions

Q.1 [1 mark · MCQ]

Each panel shows the four binary inputs (x_1, x_2) labelled as class 1 (filled square) or class 0 (open circle). Which dataset can NOT be classified perfectly by a single perceptron (one linear threshold unit)?



- (A) Dataset P (B) Dataset Q
 (C) Dataset R (D) Dataset S

Q.2 [1 mark · MSQ (one or more correct)]

Which of the following statements about support vector machines is/are CORRECT?

- (A)** Without kernels, both the SVM and logistic regression produce linear (hyperplane) decision boundaries.
- (B)** Maximising the SVM margin is equivalent to maximising $\|w\|$.
- (C)** A kernel SVM predicts with $f(x) = \sum \alpha_i y_i K(x_i, x) + b$, where only support vectors contribute.
- (D)** The margin width of the canonical hard-margin SVM is $1/\|w\|^2$.

Q.3 [1 mark · NAT (numerical answer)]

A binary decision tree in which every internal node has exactly two children has 48 leaves. The number of internal (decision) nodes is _____ (Answer in integer)

Q.4 [1 mark · MSQ (one or more correct)]

Which of the following statements about basis-function (polynomial) regression and feature scaling is/are CORRECT?

- (A) Gaussian radial basis functions $\phi_j(x) = \exp(-(x - \mu_j)^2/(2s^2))$ can be used in a model that is linear in its weights.
- (B) Decision-tree splits are strongly affected by monotone rescaling of a feature.
- (C) k-NN regression predictions can change if one feature is rescaled.
- (D) Min-max scaling maps the training values of a feature onto $[0, 1]$.

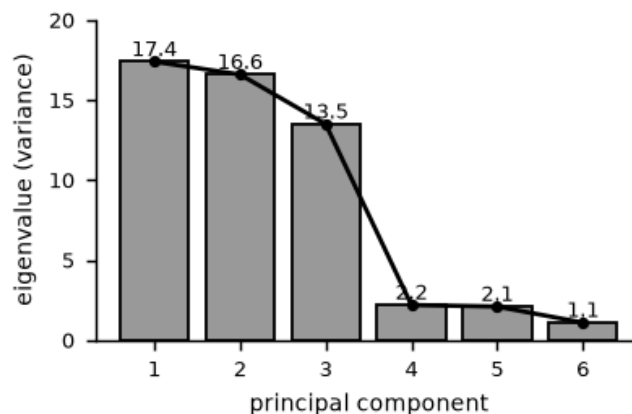
Q.5 [1 mark · MSQ (one or more correct)]

A simple linear regression $\hat{y} = \beta_0 + \beta_1 x$ is fitted by ordinary least squares (with intercept) to data (x_i, y_i) , $i = 1, \dots, n$, with residuals $e_i = y_i - \hat{y}_i$ and sample correlation r . Which of the following statements is/are CORRECT?

- (A) For the least-squares line, $\sum e_i^2 = 0$.
- (B) The product of the slope of the y-on-x line and the slope of the x-on-y line equals r^2 .
- (C) The least-squares slope equals $r \cdot s_y / s_x$, where r is the sample correlation.
- (D) For the least-squares line (with intercept), the residuals sum to zero: $\sum e_i = 0$.

Q.6 [1 mark · NAT (numerical answer)]

The scree plot below shows the eigenvalues of the covariance matrix of a 6-dimensional dataset. The minimum number of principal components needed to explain at least 85% of the total variance is ____ (Answer in integer)

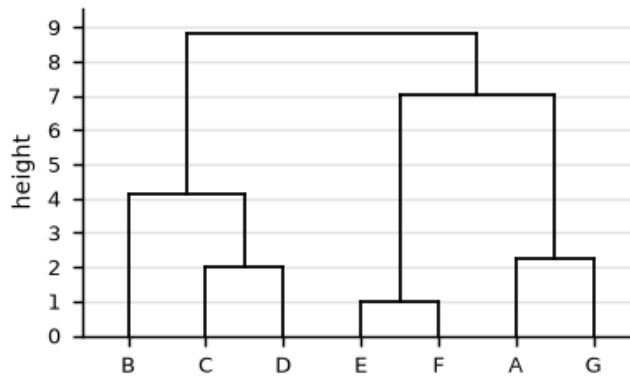
**Q.7 [1 mark · NAT (numerical answer)]**

The confusion matrix of a binary classifier on a test set is shown below (note which of rows/columns are actual and predicted). The specificity (TNR) of the classifier is ____ (rounded off to two decimal places)

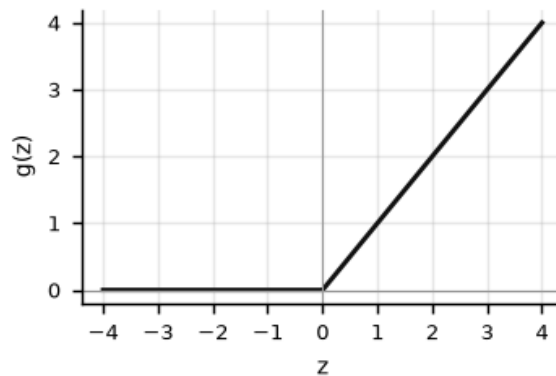
	Actual +	Actual -
Predicted +	44	17
Predicted -	10	52

Q.8 [1 mark · NAT (numerical answer)]

The dendrogram below was obtained by agglomerative clustering of 7 points A-G. If it is cut to obtain exactly 3 clusters, the number of points in the cluster that contains point D is _____ (Answer in integer)

**Q.9 [1 mark · MCQ]**

The figure shows the graph of a function $g(z)$ used in neural networks. Which function is it?



- (A) Softplus: $\ln(1 + e^z)$
- (B) ELU with $\alpha = 1$
- (C) ReLU: $\max(0, z)$
- (D) Leaky ReLU with $\alpha = 0.1$

Q.10 [1 mark · NAT (numerical answer)]

A bagged ensemble is trained on $N = 4100$ points. Using the large- N approximation, the expected number of training points that are out-of-bag for a single bootstrap sample of size N is _____ (Answer in integer)

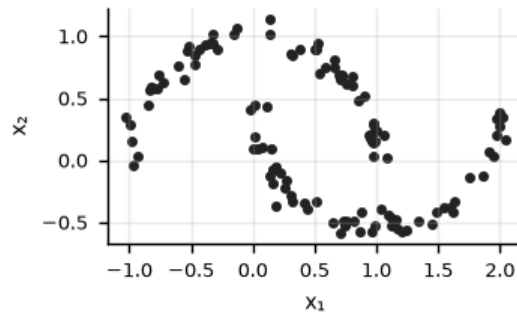
Q.11 [1 mark · MSQ (one or more correct)]

Which of the following statements about Bayes-optimal classification and generative vs. discriminative models is/are CORRECT?

- (A) When the misclassification costs are asymmetric, the Bayes-optimal rule need not pick the most probable class.
- (B) k-NN is a generative classifier because it uses the training data at prediction time.
- (C) The Bayes error rate is the lowest error rate achievable by any classifier for the given joint distribution of (x, y) .
- (D) Naive Bayes, LDA and QDA are generative; logistic regression is discriminative.

Q.12 [1 mark · MCQ]

The unlabeled data below visibly consists of two interleaving half-moons (well separated, dense). Which method is most likely to recover these two groups exactly?



- (A) Single-linkage agglomerative clustering, cut into 2 clusters
- (B) k-medoids (PAM) with $k = 2$
- (C) k-means with $k = 2$ (best of many restarts)
- (D) Complete-linkage agglomerative clustering, cut into 2 clusters

Q.13 [1 mark · MCQ]

A fitted logistic regression model reports that each one-unit increase in *number of links in an email* multiplies the odds of the email being spam by 3 (other features fixed). The coefficient of *number of links in an email* in the model is

- (A) 1.1
- (B) 0.48
- (C) 0.33
- (D) 0.95

Q.14 [1 mark · NAT (numerical answer)]

A regularisation parameter is tuned over 7 candidate values using 10-fold cross-validation. After selecting the best value, the model is refitted once on the full training data. The total number of times the model is fitted is _____ (Answer in integer)

Q.15 [1 mark · NAT (numerical answer)]

A network is trained with mini-batch SGD on $N = 50000$ examples with batch size $B = 128$ for $E = 19$ epochs. The last, smaller batch of each epoch is kept (not dropped). The total number of parameter updates is _____ (Answer in integer)

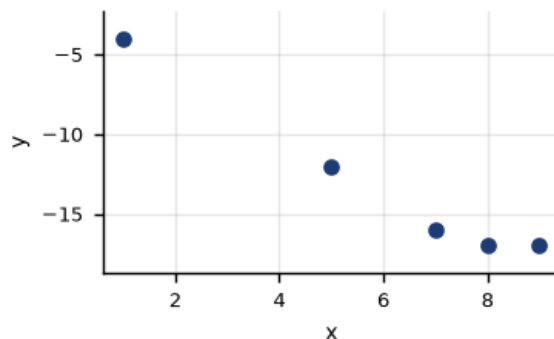
Q.16 - Q.35 : Two-mark questions**Q.16 [2 marks · NAT (numerical answer)]**

An autoencoder 50-64-50 uses tied weights: the decoder weight matrix is the transpose of the encoder weight matrix $W \in \mathbb{R}^{64 \times 50}$, while encoder and decoder have their own bias vectors. The number of trainable parameters is _____ (Answer in integer)

Q.17 [2 marks · NAT (numerical answer)]

A simple linear regression (with intercept) is fitted by least squares to the 5 points below. Using leave-one-out cross-validation, the LOOCV estimate of test MSE, $(1/n)\sum(y_i - \hat{y}_{(-i)}(x_i))^2$, is _____ (rounded off to two decimal places)

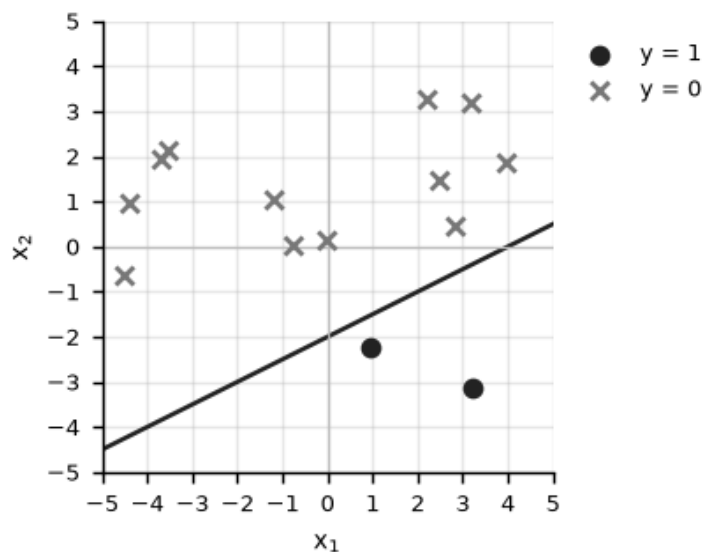
x	1	5	7	8	9
y	-4	-12	-16	-17	-17

**Q.18 [2 marks · NAT (numerical answer)]**

For the RBF kernel $K(x, z) = \exp(-\gamma\|x - z\|^2)$ with $\gamma = 1$, let ϕ be its (infinite-dimensional) feature map. For $x = (2, 2)$ and $z = (2, -2)$, the Euclidean distance $\|\phi(x) - \phi(z)\|$ in feature space is _____ (rounded off to two decimal places)

Q.19 [2 marks · MCQ]

A logistic regression model $P(y = 1 | \mathbf{x}) = \sigma(w_0 + w_1x_1 + w_2x_2)$ perfectly classifies the training points shown below using threshold 0.5. Its decision boundary is the solid line (it crosses the axes at grid points). Which of the following parameter vectors is consistent with the figure?



- (A) $(w_0, w_1, w_2) = (-4, 1, -2)$
- (B) $(w_0, w_1, w_2) = (-4, 1, 2)$
- (C) $(w_0, w_1, w_2) = (-4, -2, 1)$
- (D) $(w_0, w_1, w_2) = (4, -2, 1)$

Q.20 [2 marks · NAT (numerical answer)]

The confusion matrix of a 3-class classifier is given below (rows = actual class, columns = predicted class). The macro-averaged recall is _____ (rounded off to two decimal places)

actual \ predicted	A	B	C
A	20	5	4
B	5	5	5
C	5	2	9

Q.21 [2 marks · NAT (numerical answer)]

An AdaBoost ensemble has three weak classifiers with the weighted errors below, and their predictions on a test point x are listed. The ensemble score $F(x) = \sum \alpha_t h_t(x)$, with $\alpha_t = \frac{1}{2} \ln((1 - \varepsilon_t)/\varepsilon_t)$, is _____ (rounded off to two decimal places)

Round t	ε_t	$h_t(x)$
1	0.25	-1
2	0.15	-1
3	0.15	+1

Q.22 [2 marks · NAT (numerical answer)]

Agglomerative hierarchical clustering with **complete linkage** is applied to 5 points whose pairwise distance matrix is given below (upper triangle shown). The height (linkage distance) at which the **fourth** merge takes place is _____ (rounded off to two decimal places)

	P1	P2	P3	P4	P5
P1	0	13	1	8	12
P2		0	12	7	5
P3			0	7	11
P4				0	12
P5					0

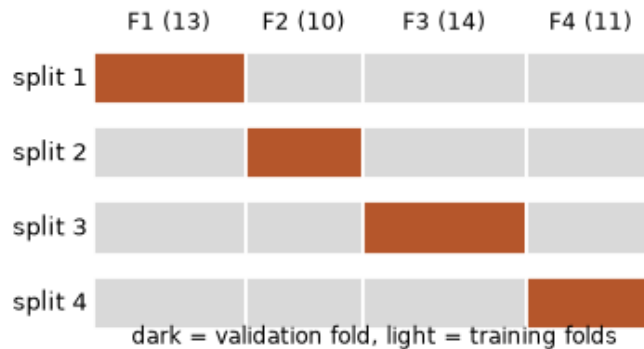
Q.23 [2 marks · NAT (numerical answer)]

In C4.5, the attribute Branch splits a node of 23 records as tabulated (class Yes/No counts in each branch). Compute the gain ratio of Branch (all logarithms base 2). (rounded off to two decimal places)

Branch	#Yes	#No
v1	8	0
v2	4	0
v3	2	1
v4	3	5

Q.24 [2 marks · NAT (numerical answer)]

A regression model is evaluated by 4-fold cross-validation on $n = 48$ observations. The folds (F1 ... F4) have unequal sizes, as shown in the diagram and table. The k-fold CV estimate is defined as the mean squared error over all n out-of-fold predictions (i.e. pooled over observations). Its value is _____ (rounded off to two decimal places)



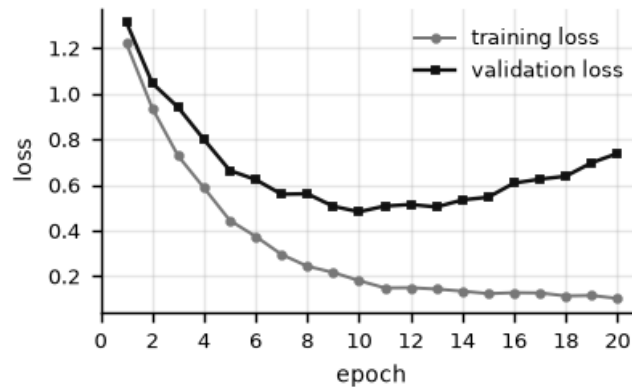
Fold	F1	F2	F3	F4
size	13	10	14	11
validation MSE	2.7	6.6	4.1	8.1

Q.25 [2 marks · NAT (numerical answer)]

A centered data matrix X with $n = 5$ rows (samples) and $d = 3$ columns has singular values 25, 22, 12. Using the sample covariance with divisor $n - 1$, the percentage of total variance explained by the first principal component is _____ (rounded off to two decimal places)

Q.26 [2 marks · NAT (numerical answer)]

A network is trained with early stopping on validation loss using patience $P = 2$: training stops as soon as the validation loss has failed to improve on its best value for 2 consecutive epochs, and the weights from the best epoch are restored. Using the validation losses below (also plotted), the epoch after which training is stopped is _____ (Answer in integer)



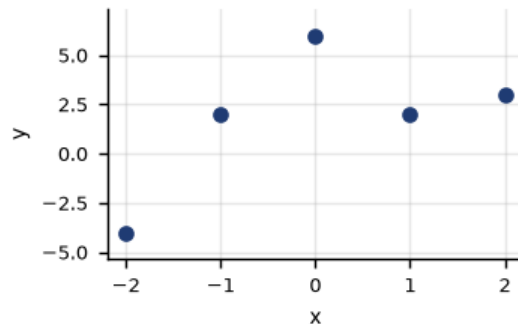
epoch	1	2	3	4	5	6	7	8	9	10
val. loss	1.314	1.047	0.941	0.800	0.664	0.624	0.560	0.562	0.507	0.483

epoch	11	12	13	14	15	16	17	18	19	20
val. loss	0.508	0.515	0.504	0.535	0.547	0.609	0.626	0.639	0.698	0.739

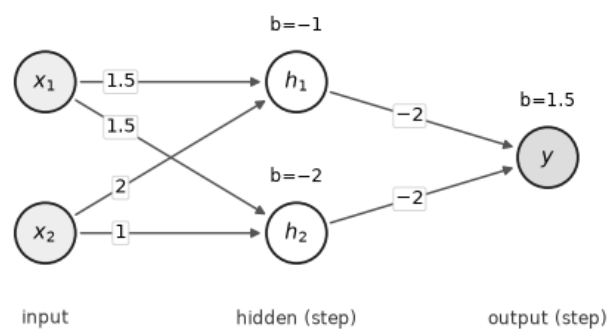
Q.27 [2 marks · NAT (numerical answer)]

The quadratic model $y = a + bx + cx^2$ is fitted by least squares to the 5 points below (a polynomial / basis-function regression with basis $1, x, x^2$). Then the coefficient c of x^2 is _____ (rounded off to two decimal places)

x	-2	-1	0	1	2
y	-4	2	6	2	3

**Q.28 [2 marks · MCQ]**

In the network below every unit (hidden and output) is a threshold unit that outputs 1 if its weighted input plus bias is > 0 , and 0 otherwise. Edge labels are weights. For binary inputs $x_1, x_2 \in \{0, 1\}$, the network computes

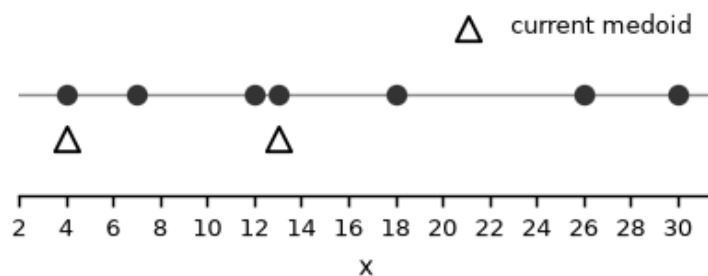
**(A)** (NOT x_1) AND x_2 **(B)** XNOR**(C)** NOR**(D)** x_1 AND (NOT x_2)**Q.29 [2 marks · NAT (numerical answer)]**

A linear model $\hat{y} = w_1x_1 + w_2x_2 + b$ is trained by stochastic gradient descent (batch size 1) on the per-example loss $\frac{1}{2}(\hat{y} - y)^2$ with learning rate $\eta = 0.05$, starting from $w = (-1, -1)$, $b = 1$. The two examples below are processed once, in the given order. After both updates, the value of w_2 is _____ (rounded off to three decimal places)

example	x_1	x_2	y
1	3	2	4
2	3	2	0

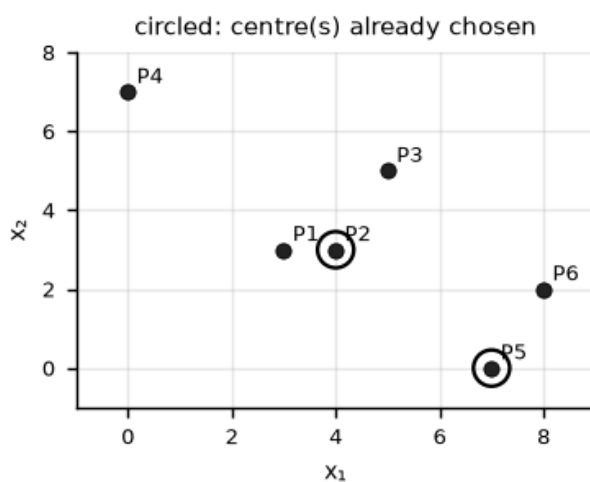
Q.30 [2 marks · NAT (numerical answer)]

PAM (k-medoids, $k = 2$, absolute-difference dissimilarity) is applied to the 1-D data $\{4, 7, 12, 13, 18, 26, 30\}$. The current medoids are 4 and 13. The total cost $\sum_i \min_m |x_i - m|$ of the current configuration is _____ (Answer in integer)

**Q.31 [2 marks · NAT (numerical answer)]**

k-means++ initialization (squared Euclidean seeding) is applied to the six points below. P2 and P5 have already been chosen as centres. The probability that P6 is selected as the third centre is _____ (rounded off to three decimal places)

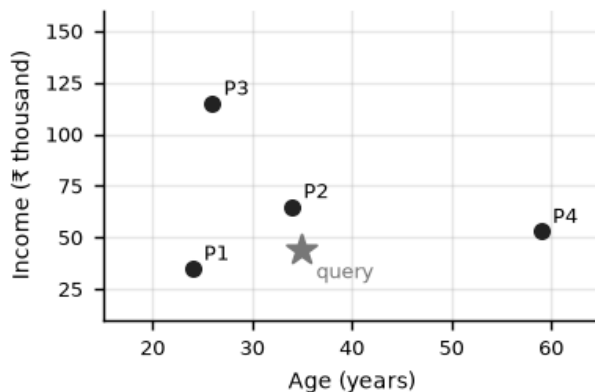
Point	P1	P2	P3	P4	P5	P6
x_1	3	4	5	0	7	8
x_2	3	3	5	7	0	2



Q.32 [2 marks · MCQ]

A 1-NN classifier uses the two features below. Before computing Euclidean distances, each feature is min-max scaled to $[0, 1]$ using the minimum and maximum of that feature over the four training points P1-P4 (the query is scaled with the same constants). The nearest neighbour of the query after scaling is

Point	Age (years)	Monthly income (₹ thousand)
P1	24	35
P2	34	65
P3	26	115
P4	59	53
Query	35	44

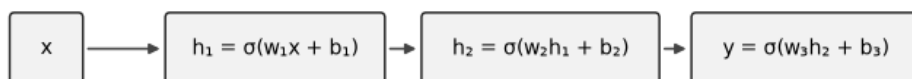


- (A) P1
(C) P3

- (B) P2
(D) P4

Q.33 [2 marks · NAT (numerical answer)]

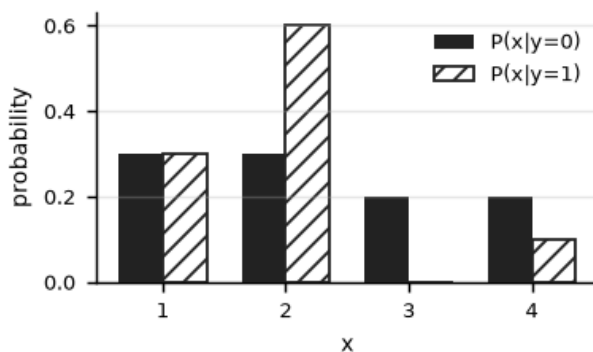
Consider the scalar chain shown below with weights $w_1 = -3$, $w_2 = 3$, $w_3 = -1$, biases $b_1 = -1$, $b_2 = 0$, $b_3 = 0.5$ and activation the logistic sigmoid σ at every stage. For input $x = -1$, the value of $\partial y / \partial w_1$ is _____ (rounded off to 4 decimal places)



Q.34 [2 marks · NAT (numerical answer)]

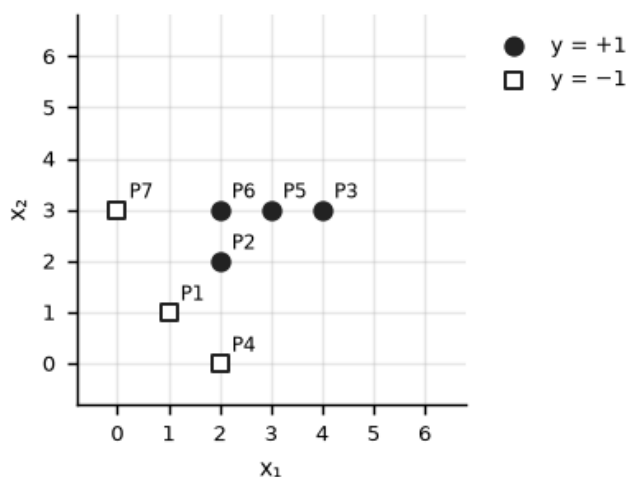
A discrete feature $x \in \{1, \dots, 4\}$ has the class-conditional distributions below (also plotted), with priors $P(y = 0) = 0.7$ and $P(y = 1) = 0.3$. Then the Bayes error rate of the optimal classifier is _____ (rounded off to three decimal places)

x	1	2	3	4
$P(x y=0)$	0.3	0.3	0.2	0.2
$P(x y=1)$	0.3	0.6	0	0.1

**Q.35 [2 marks · NAT (numerical answer)]**

The labelled training points shown below are linearly separable. A hard-margin linear SVM is trained on them. Compute the width of the margin (distance between the two margin hyperplanes). (rounded off to two decimal places)

Point	x_1	x_2	y
P1	1	1	-1
P2	2	2	+1
P3	4	3	+1
P4	2	0	-1
P5	3	3	+1
P6	2	3	+1
P7	0	3	-1



— END OF QUESTION PAPER —

Mock Test 50 — Answer Key

Q	Type	M	Key	Q	Type	M	Key
1	MCQ	1	(D)	19	MCQ	2	(A)
2	MSQ	1	(A), (C)	20	NAT	2	0.5185 to 0.5385
3	NAT	1	47	21	NAT	2	−0.5593 to −0.5393
4	MSQ	1	(A), (C), (D)	22	NAT	2	12.99 to 13.01
5	MSQ	1	(B), (C), (D)	23	NAT	2	0.1899 to 0.2099
6	NAT	1	3	24	NAT	2	5.1483 to 5.1683
7	NAT	1	0.7436 to 0.7636	25	NAT	2	49.8703 to 49.8903
8	NAT	1	3	26	NAT	2	12
9	MCQ	1	(C)	27	NAT	2	−1.3057 to −1.2657
10	NAT	1	1507 to 1510	28	MCQ	2	(C)
11	MSQ	1	(A), (C), (D)	29	NAT	2	−0.362 to −0.358
12	MCQ	1	(A)	30	NAT	2	39
13	MCQ	1	(A)	31	NAT	2	0.1143 to 0.1183
14	NAT	1	71	32	MCQ	2	(B)
15	NAT	1	7429	33	NAT	2	0.0044 to 0.005
16	NAT	2	3314	34	NAT	2	0.298 to 0.302
17	NAT	2	3.7911 to 3.8677	35	NAT	2	1.0994 to 1.1194
18	NAT	2	1.4042 to 1.4242				

Self-evaluation (fill after checking)

Topic area	Attempted	Correct	Wrong (MCQ)	Marks scored	Max
Neural Networks (MLP / Feed-forward)					11
SVM & Decision Trees					11
Regression & Model Selection					11
Unsupervised Learning					11
Probabilistic & Instance-based Classifiers					11
Total					55

Mock Test 50 — Detailed Solutions

Q.1 — Answer: (D) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Perceptron]

A single perceptron can classify a dataset perfectly iff the two classes are linearly separable.

Dataset P: labels for (0,0),(0,1),(1,0),(1,1) = 1, 0, 1, 1 $\rightarrow x_1$ OR (NOT x_2) – linearly separable.

Dataset Q: labels for (0,0),(0,1),(1,0),(1,1) = 1, 0, 1, 0 $\rightarrow$ NOT x_2 – linearly separable.

Dataset R: labels for (0,0),(0,1),(1,0),(1,1) = 1, 1, 0, 0 $\rightarrow$ NOT x_1 – linearly separable.

Dataset S: labels for (0,0),(0,1),(1,0),(1,1) = 0, 1, 1, 0 $\rightarrow$ XOR – **not** linearly separable (diagonal pattern).

Answer: **Dataset S**. Of the 14 non-constant Boolean functions of two inputs, only XOR and XNOR are not linearly separable.

Q.2 — Answer: (A), (C) [MSQ, 1M · SVM & Decision Trees > SVM: dual & support vectors]

(A) **TRUE**. Both threshold an affine score $w^T x + b$.

(B) **FALSE**. The margin $2/\|w\|$ is maximised by minimising $\|w\|$.

(C) **TRUE**. $\alpha_i = 0$ for all other points.

(D) **FALSE**. It is $2/\|w\|$.

Q.3 — Answer: 47 [NAT, 1M · SVM & Decision Trees > Decision trees: structure & traversal]

In a full binary tree, #internal = #leaves – 1 = 48 – 1 = **47**.

Q.4 — Answer: (A), (C), (D) [MSQ, 1M · Regression & Model Selection > Basis functions & feature scaling]

(A) **TRUE**. $y = \sum_j w_j \phi_j(x)$ is linear in w .

(B) **FALSE**. Trees depend only on the ordering of feature values.

(C) **TRUE**. Distances change.

(D) **TRUE**. $(x - \min)/(\max - \min)$.

Q.5 — Answer: (B), (C), (D) [MSQ, 1M · Regression & Model Selection > Simple linear regression]

(A) **FALSE**. Only for a perfect fit; in general $SSE > 0$.

(B) **TRUE**. $(S_{xy}/S_{xx})(S_{xy}/S_{yy}) = r^2$.

(C) **TRUE**. $\beta_1 = S_{xy}/S_{xx} = r \cdot \sqrt{(S_{yy}/S_{xx})}$.

(D) **TRUE**. The normal equation for β_0 is exactly $\sum (y_i - \beta_0 - \beta_1 x_i) = 0$.

Q.6 — Answer: 3 [NAT, 1M · Unsupervised Learning > PCA & dimensionality reduction]

Total variance = $\sum \lambda = 52.9$. Cumulative proportion of variance explained:

k	1	2	3	4	5	6
λ_k	17.4	16.6	13.5	2.2	2.1	1.1
cumulative %	32.89	64.27	89.79	93.95	97.92	100

The cumulative share first reaches 85% at $k = 3$.

Q.7 — Answer: 0.7436 to 0.7636 [NAT, 1M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

TP = 44, FN = 10, FP = 17, TN = 52 (N = 123).

Specificity (TNR) = $TN/(TN + FP)$.

Value = **0.75**.

Value = **0.75**.

Watch the orientation: swapping FP and FN is the most common error (it swaps precision and recall).

Q.8 — Answer: 3 [NAT, 1M · Unsupervised Learning > Hierarchical clustering]

Undo the top 2 merge(s) (equivalently cut just below height 7.04). Clusters: {A, G}, {B, C, D}, {E, F}.

D lies in a cluster of size **3**.

Q.9 — Answer: (C) [MCQ, 1M · Neural Networks (MLP / Feed-forward) > Activation functions]

Key features of the plotted curve: Exactly 0 for $z \leq 0$, slope 1 for $z > 0$, kink at origin.

Hence $g = \text{ReLU: } \max(0, z)$.

Softplus: $\ln(1 + e^z)$: Smooth, always positive, value $\ln 2 \approx 0.69$ at $z = 0$, $\sim z$ for large z . – does not match.

ELU with $\alpha = 1$: Smoothly saturates to -1 for $z \rightarrow -\infty$, linear for $z > 0$, passes through origin. – does not match.

Leaky ReLU with $\alpha = 0.1$: Small negative linear part (slope 0.1) for $z < 0$, slope 1 for $z > 0$. – does not match.

Q.10 — Answer: 1507 to 1510 [NAT, 1M · SVM & Decision Trees · Ensembles: bagging & random forests]

$P(\text{out-of-bag}) = (1 - 1/N)^N \approx e^{-1} \approx 0.3679$; expected count $\approx 4100 \times 0.3679 = \mathbf{1508}$.

Q.11 — Answer: (A), (C), (D) [MSQ, 1M · Probabilistic & Instance-based Classifiers · Bayes optimal classifier]

(A) **TRUE**. A costly error type shifts the decision threshold.

(B) **FALSE**. It does not model $P(x|y)$; it is a non-parametric discriminative method.

(C) **TRUE**. It is the error of the Bayes classifier.

(D) **TRUE**. Standard classification.

Q.12 — Answer: (A) [MCQ, 1M · Unsupervised Learning · k-means]

Each group is connected by chains of small gaps, while the gap between groups is larger. Single linkage merges via nearest neighbours (MST edges) and therefore follows these chains: cutting the largest MST edge separates the two interleaving half-moons.

k-means and k-medoids produce convex (Voronoi) clusters around representatives, so they split the shapes by a straight boundary. Complete linkage favours compact, small-diameter clusters and also cuts across the non-convex shapes.

Q.13 — Answer: (A) [MCQ, 1M · Probabilistic & Instance-based Classifiers · Logistic regression]

Odds ratio for a unit increase $= e^\beta = 3 \Rightarrow \beta = \ln 3 = \mathbf{1.1}$.

Distractors: the odds ratio itself, $\log_{10}$, or $1/R$ are common slips; the model is linear in the *natural* log-odds.

Q.14 — Answer: 71 [NAT, 1M · Regression & Model Selection · Cross-validation]

Each candidate needs $k = 10$ fits: $7 \times 10 = 70$. Plus 1 final refit: **71**.

Q.15 — Answer: 7429 [NAT, 1M · Neural Networks (MLP / Feed-forward) · Training & regularisation]

Updates per epoch $= \lceil N/B \rceil = \lceil 50000/128 \rceil = 391$ (390 full batches + 1 partial batch of 80).

Total $= 391 \times 19 = \mathbf{7429}$.

Q.16 — Answer: 3314 [NAT, 2M · Neural Networks (MLP / Feed-forward) · Parameter counting]

Shared W : $64 \times 50 = 3200$; encoder bias 64; decoder bias 50. Total $= \mathbf{3314}$.

(Without tying it would be $2 \times 3200 + 64 + 50 = 6514$.)

Q.17 — Answer: 3.7911 to 3.8677 [NAT, 2M · Regression & Model Selection · Cross-validation]

$\bar{x} = 6$, $\bar{y} = -13.2$.

i	1	2	3	4	5
$x_i - \bar{x}$	-5	-1	1	2	3
$y_i - \bar{y}$	9.2	1.2	-2.8	-3.8	-3.8
$(x_i - \bar{x})(y_i - \bar{y})$	-46	-1.2	-2.8	-7.6	-11.4

$S_{xx} = \sum (x_i - \bar{x})^2 = 40$, $S_{xy} = -69$.

$\beta_1 = S_{xy}/S_{xx} = -1.725$, $\beta_0 = \bar{y} - \beta_1 \bar{x} = -2.85$.

For a linear smoother, the leave-one-out residual is $e_{(i)} = e_i / (1 - h_{ii})$, with $h_{ii} = 1/n + (x_i - \bar{x})^2 / S_{xx}$.

x_i	1	5	7	8	9
e_i	0.575	-0.525	-1.075	-0.35	1.375
h_{ii}	0.825	0.225	0.225	0.3	0.425

$e_i/(1-h_{ii})$	3.286	-0.677	-1.387	-0.5	2.391
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$$CV_{(n)} = (1/5)\Sigma(e_i/(1-h_{ii}))^2 = 3.8294.$$

Answer: **3.83** (verified by explicitly refitting without the point).

Q.18 — Answer: 1.4042 to 1.4242 [NAT, 2M · SVM & Decision Trees > SVM: kernels]

$\|\varphi(x) - \varphi(z)\|^2 = K(x, x) + K(z, z) - 2K(x, z) = 2 - 2K(x, z)$ for the RBF kernel.

$\|x - z\|^2 = 16$, $K(x, z) = \exp(-16) = 0$.

Distance = $\sqrt{(2 - 2 \times 0)} = \mathbf{1.41}$ (always below $\sqrt{2} \approx 1.414$).

Q.19 — Answer: (A) [MCQ, 2M · Probabilistic & Instance-based Classifiers > Logistic regression]

The line passes through (4, 0) and (0, -2): $x_1/4 + x_2/-2 = 1$, i.e. $-2x_1 + 4x_2 - 8 = 0$. Hence (w_0, w_1, w_2) must be a non-zero multiple of (8, -2, 4).

Orientation: the class $y = 1$ region must have $w_0 + w_1x_1 + w_2x_2 > 0$. At the origin the $y = 1$ side has sign - ($w_0 = -4$), which matches the figure (the origin lies on the $y = 0$ side).

So the answer is $(w_0, w_1, w_2) = \mathbf{(-4, 1, -2)}$ (any positive multiple gives the same classifier).

Other options either describe a different line (swapped/negated coefficients move the intercepts) or the same line with the sign flipped, which would label every training point incorrectly.

Q.20 — Answer: 0.5185 to 0.5385 [NAT, 2M · Probabilistic & Instance-based Classifiers > Evaluation metrics]

class	TP	predicted total	actual total	precision	recall	F1
A	20	30	29	0.6667	0.6897	0.678
B	5	12	15	0.4167	0.3333	0.3704
C	9	18	16	0.5	0.5625	0.5294

Precision of class $k = C_{kk}/(\text{column sum})$; recall = $C_{kk}/(\text{row sum})$.

Macro recall = mean of per-class recalls = **0.53** (= balanced accuracy).

Q.21 — Answer: -0.5593 to -0.5393 [NAT, 2M · SVM & Decision Trees > Ensembles: boosting]

α_t : $\alpha_1 = 0.5493$, $\alpha_2 = 0.8673$, $\alpha_3 = 0.8673$.

$F(x) = -0.5493 - 0.8673 + 0.8673 = \mathbf{-0.55}$; prediction $\text{sign}(F) = -1$.

Note that a majority of weak votes need not win: a low-error learner carries a larger α .

Q.22 — Answer: 12.99 to 13.01 [NAT, 2M · Unsupervised Learning > Hierarchical clustering]

Use complete linkage: $d(A, B) = \max_{a \in A, b \in B} d(a, b)$. After each merge, recompute the distance of the new cluster to the others with this rule and merge the closest pair again.

Merge 1: {P1} + {P3} at height **1**. Updated: $d(\{P1, P3\}, \{P2\}) = 13$, $d(\{P1, P3\}, \{P4\}) = 8$, $d(\{P1, P3\}, \{P5\}) = 12$.

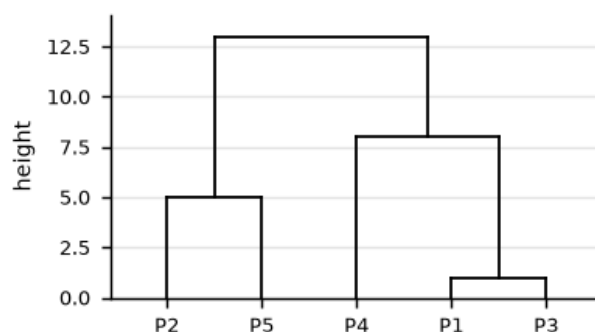
Merge 2: {P2} + {P5} at height **5**. Updated: $d(\{P2, P5\}, \{P4\}) = 12$, $d(\{P2, P5\}, \{P1, P3\}) = 13$.

Merge 3: {P4} + {P1, P3} at height **8**. Updated: $d(\{P1, P3, P4\}, \{P2, P5\}) = 13$.

Merge 4: {P2, P5} + {P1, P3, P4} at height **13**.

The fourth merge occurs at height **13**.

Dendrogram:



Q.23 — Answer: 0.1899 to 0.2099 [NAT, 2M · SVM & Decision Trees > Decision trees: split selection]

Parent [17 Yes, 6 No]: $H = 0.8281$.

Child entropies: $v_1: 0, v_2: 0, v_3: 0.9183, v_4: 0.9544$.

$IG = 0.8281 - \sum (n_v/23)H_v = 0.3763$.

$\text{SplitInfo} = -\sum (n_v/N) \log_2(n_v/N) = -(8/23)\log_2(8/23) - (4/23)\log_2(4/23) - (3/23)\log_2(3/23) - (8/23)\log_2(8/23) = 1.882$.

Gain ratio = $IG / \text{SplitInfo} = 0.3763/1.882 = 0.2$.

C4.5 uses the gain ratio to penalise attributes that split the data into many small branches.

Q.24 — Answer: 5.1483 to 5.1683 [NAT, 2M · Regression & Model Selection > Cross-validation]

Pooled CV MSE = $\sum_k n_k \cdot \text{MSE}_k / n$ (a size-weighted average, not the plain average of fold MSEs, since folds are unequal).

$\sum n_k \text{MSE}_k = 13 \times 2.7 + 10 \times 6.6 + 14 \times 4.1 + 11 \times 8.1 = 247.6; n = 48$.

CV = $247.6/48 = 5.1583$ (plain average of fold MSEs = 5.375 — a tempting but wrong value).

Answer: **5.16**.

Q.25 — Answer: 49.8703 to 49.8903 [NAT, 2M · Unsupervised Learning > PCA & dimensionality reduction]

With $X = U\Sigma V^T$ (X centered, $n \times d$), the sample covariance is $S = X^T X / (n - 1) = V(\Sigma^2 / (n - 1))V^T$, so the PC directions are the columns of V and the eigenvalues are $\lambda_j = \sigma_j^2 / (n - 1)$.

$\lambda = 25^2/4 = 156.25, 22^2/4 = 121, 12^2/4 = 36$.

Cumulative % variance: 49.88, 88.51, 100 (note the $(n - 1)$ factor cancels in proportions: $\text{PVE} = \sigma_j^2 / \Sigma \sigma^2$).

Answer: **49.88**. (Using σ instead of σ^2 is the classic mistake.)

Q.26 — Answer: 12 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Training & regularisation]

Rule: keep track of the best (lowest) validation loss; an epoch that does not strictly improve on it increments a counter, which is reset on improvement; stop when the counter reaches patience $P = 2$.

Trace - 1: 1.314 (new best); 2: 1.047 (new best); 3: 0.941 (new best); 4: 0.800 (new best); 5: 0.664 (new best); 6: 0.624 (new best); 7: 0.560 (new best); 8: 0.562 (no improvement, count = 1); 9: 0.507 (new best); 10: 0.483 (new best); 11: 0.508 (no improvement, count = 1); 12: 0.515 (no improvement, count = 2).

Training stops after epoch 12; the best validation loss 0.483 occurred at epoch 10, whose weights are restored.

Answer: **12**. The widening train/validation gap after this point is the signature of overfitting.

Q.27 — Answer: -1.3057 to -1.2657 [NAT, 2M · Regression & Model Selection > Basis functions & feature scaling]

Design columns 1, x , x^2 . Because the x -values are symmetric about 0, $\Sigma x = \Sigma x^3 = 0$, so the normal equations decouple:

$\Sigma x^2 = 10, \Sigma x^4 = 34, \Sigma y = 9, \Sigma xy = 14, \Sigma x^2 y = 0$.

$b = \Sigma xy / \Sigma x^2 = 14/10 = 1.4$.

$[n, \Sigma x^2; \Sigma x^2, \Sigma x^4][a; c] = [\Sigma y; \Sigma x^2 y] = 5a + 10c = 9, 10a + 34c = 0$.

Solving: $c = (n \Sigma x^2 y - \Sigma x^2 \Sigma y) / (n \Sigma x^4 - (\Sigma x^2)^2) = -1.2857, a = (\Sigma y - c \Sigma x^2) / n = 4.3714$.

Answer: **-1.29**.

Q.28 — Answer: (C) [MCQ, 2M · Neural Networks (MLP / Feed-forward) > Expressiveness]

Each unit outputs 1 iff its net input (weighted sum + bias) is > 0 . Evaluate on all four inputs:

x_1	x_2	h_1	h_2	y
0	0	0	0	1
0	1	1	0	0
1	0	1	0	0
1	1	1	1	0

h_1 computes OR, h_2 computes AND; the output column $(0,0) \rightarrow (1,1)$ is 1, 0, 0, 0, i.e. **NOR**.

Q.29 — Answer: -0.362 to -0.358 [NAT, 2M · Regression & Model Selection > Gradient descent]

Per-example loss $\mathcal{L} = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \partial \mathcal{L} / \partial w = (\hat{y} - y)x$, $\partial \mathcal{L} / \partial b = (\hat{y} - y)$. Update after each example.

Example 1: $\hat{y} = -4$, error $\hat{y} - y = -8$; $\nabla_w = (-24, -16)$, $\nabla_b = -8 \Rightarrow w = (0.2, -0.2)$, $b = 1.4$.

Example 2: $\hat{y} = 1.6$, error $\hat{y} - y = 1.6$; $\nabla_w = (4.8, 3.2)$, $\nabla_b = 1.6 \Rightarrow w = (-0.04, -0.36)$, $b = 1.32$.

Answer: **-0.36**.

Q.30 — Answer: 39 [NAT, 2M · Unsupervised Learning > k-medoids]

PAM cost = Σ over all points of the absolute distance to the nearest medoid.

Current medoids $\{4, 13\}$: distances $4 \rightarrow 0$, $7 \rightarrow 3$, $12 \rightarrow 1$, $13 \rightarrow 0$, $18 \rightarrow 5$, $26 \rightarrow 13$, $30 \rightarrow 17$; cost = **39**.

Q.31 — Answer: 0.1143 to 0.1183 [NAT, 2M · Unsupervised Learning > k-means]

k-means++ picks the next centre x with probability $D(x)^2 / \sum_{x'} D(x')^2$, where $D(x)$ is the distance from x to the **nearest** centre already chosen (chosen centres have $D = 0$).

Point	P1	P2	P3	P4	P5	P6
$D(x)^2$	1	0	5	32	0	5

$\Sigma D^2 = 43$, so $P(P6) = 5/43 = \mathbf{0.116}$.

Common mistakes: using D instead of D^2 (that is not k-means++), or using the distance to the *farthest*/first centre instead of the nearest chosen centre.

Q.32 — Answer: (B) [MCQ, 2M · Probabilistic & Instance-based Classifiers > k-nearest neighbours]

Ranges: age [24, 59], income [35, 115]. Scaled query = (0.314, 0.113).

Point	scaled age	scaled income	scaled distance	raw distance
P1	0	0	0.334	14.21
P2	0.286	0.375	0.264	21.02
P3	0.057	1	0.924	71.57
P4	1	0.225	0.695	25.63

Nearest after scaling: **P2**.

Without scaling the nearest point would be P1, because income differences (tens of thousands) dominate the age differences — exactly why k-NN needs feature scaling.

Q.33 — Answer: 0.0044 to 0.005 [NAT, 2M · Neural Networks (MLP / Feed-forward) > Activation functions]

Forward: $a_1 = -3 \cdot -1 + -1 = 2$, $h_1 = 0.8808$; $a_2 = 3 \cdot 0.8808 + 0 = 2.6424$, $h_2 = 0.9335$; $a_3 = -1 \cdot 0.9335 + 0.5 = -0.4335$, $y = 0.3933$.

Local derivatives $\sigma'(a)$: $\sigma'(a) = \sigma(a)(1 - \sigma(a))$: $\sigma'(a_1) = 0.105$, $\sigma'(a_2) = 0.062$, $\sigma'(a_3) = 0.2386$.

Chain rule: $\partial y / \partial w_1 = \sigma'(a_3) \cdot w_3 \cdot \sigma'(a_2) \cdot w_2 \cdot \sigma'(a_1) \cdot x = \mathbf{0.0047}$.

Each layer contributes a factor $\sigma' \leq 0.25$ — the product shrinks with depth (vanishing gradient).

Q.34 — Answer: 0.298 to 0.302 [NAT, 2M · Probabilistic & Instance-based Classifiers > Bayes optimal classifier]

Bayes rule: choose the class with larger $\pi_k P(x|k)$. Bayes error = $\sum_x \min(\pi_0 P(x|0), \pi_1 P(x|1))$.

x	$\pi_0 P(x 0)$	$\pi_1 P(x 1)$	decision	min
1	0.21	0.09	0	0.09
2	0.21	0.18	0	0.18
3	0.14	0	0	0
4	0.14	0.03	0	0.03

Bayes error = **0.3**.

Q.35 — Answer: 1.0994 to 1.1194 [NAT, 2M · SVM & Decision Trees > SVM: margin & geometry]

P4 and P7 (same class) lie on one margin line and P2 on the other; the optimal boundary is parallel to the line through P4, P7, midway to P2. Support vectors: P2, P4, P7.

Scaling so that $y_i(w^T x_i + b) = 1$ at the support vectors gives $w = (3/2, 1)$, $b = -4$; $\|w\| = 1.8028$, margin $2/\|w\| = 1.1094$.

Point	P1	P2	P3	P4	P5	P6	P7
$y(w^T x + b)$	1.5	1	5	1	3.5	2	1

All values are ≥ 1 (feasible), with equality exactly at the support vectors.

Margin width = $2/\|w\| = \mathbf{1.11}$.

Master Answer Key: all tests

MCQ/MSQ show option letters (for MSQ, every listed option must be chosen). NAT shows the exact value; the accepted range is given in each test's own answer key.

Mock Test 01

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	A,B,C, D	C	0.446	A,B,C	7	21	18848	3090.5	A,B,C, D	A,B,C, D	20625	B
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	B,C,D	0.214	C	0.19	A,B,C, D	0	1	A	0.867	7.5	0.45	0.529
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	C	C	A,D	B	D	0.208	B,D	1	C	0.2	B	

Mock Test 02

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	45	0.105	A	A,C,D	A,B,C, D	D	95	1	C	A	B,C,D	B
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	1.304	0.269	2	A,B,C, D	D	B,C,D	A,C,D	B	0	0.634	3	3.4
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	1.229	A,B,C, D	0.5	7	A,D	D	A	C	1.857	0.135	72.5	

Mock Test 03

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	1.506	32	A,B,C	1.286	C,D	0.938	4	A,C,D	A,C,D	0.549	A,B,D	2
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A,B,C	A,C	A	2	7.8	0.1	B,D	$-\frac{0.55}{4}$	0.872	D	0.491	A,B,C
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	B	6.083	0.25	B	6.47	C	0.5	-8.1	17.211	0.621	$-\frac{0.34}{6}$	

Mock Test 04

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	C	B	A	C	0.155	A,B,C, D	A	8.5	0.198	A	59	0.208
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A	B,D	-6	A,C,D	0.658	0.154	A,B,C, D	$-\frac{40.1}{6}$	0.098	4.472	0.444	26
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	B	$-\frac{1.38}{7}$	$-\frac{13.6}{17}$	B,C,D	0.25	2	0.193	0.414	16	$-\frac{0.03}{9}$	0.694	

Mock Test 05

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	C,D	C	22.333	C	A,B,C, D	D	C	0.083	0.471	D	0.912	15.6
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	205	0.49	216	0.559	9	0.273	$-\frac{0.00}{5}$	50	$-\frac{0.12}{4}$	0.197	$-\frac{0.05}{1}$	0.088
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	0.216	7.847	0.062	20.333	2.706	2	3	4.5	0.595	-6	1.377	

Mock Test 06

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	B,C	D	28.543	A,D	3.5	3	2.564	A	0.853	0.687	C,D	552
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	4	B	C	-3.4	4	B,C,D	A	1.554	$-\frac{0.14}{4}$	0.059	0.85	C
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	$-\frac{9.38}{6}$	1.438	8	$-\frac{1.84}{3}$	A,B,C	30.2	B,C,D	1.109	2	18	0.896	

Mock Test 07

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	D	A,B,C, D	0.64	B	A,B,C, D	90	B,C,D	41	C	16.75	127	A,B,C
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A,B,C, D	5	C	3.381	B	C	1.253	0	69.444	A	A	0.815
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	0.098	A,B,C, D	24	C	-3	11.121	0.565	0.8	D	1.528	6.5	

Mock Test 08

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	C,D	D	0.667	0.444	D	C	A	D	0.037	1	360	0.45
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	0.041	3.03	A,B,C, D	3.9	B	0.159	250	73	70	0.878	A,B,C	0.222
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	A	B,D	3	0.35	B,C	B	C	B	0.743	C	0.999	

Mock Test 09

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	$-\frac{3.76}{5}$	C	A	A	0.676	A	C	4	A	54.029	B	A,C
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	0.224	C	23	C	$-\frac{0.00}{5}$	8	4.186	B	A,B,C	15.75	7	0.943
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	$-\frac{0.45}{5}$	3	0.219	0.952	0.786	2.189	D	0.996	30.625	0	2	

Mock Test 10

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	89	1.225	B	A,B	56	0.398	0.675	13	A,C,D	A,D	0.7	0.667
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	0.181	B,C	D	15	B,D	0.213	98560	3	1.5	-0.02	0.959	$-\frac{2.37}{5}$
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	$-\frac{1.21}{4}$	2	A	1.052	0.32	B	20.556	0.438	0.438	A,C	90.881	

Mock Test 11

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	A,B,C, D	0.7	D	4.667	256	21	B	D	A,B,C, D	C	B,D	-0.57
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A,B,C, D	A	5	10	0.864	$-\frac{0.09}{2}$	30	$-\frac{0.56}{7}$	16.8	0.446	B	11
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	0.75	0.68	A	5.601	A	D	D	0.603	0.736	-32	26.39	

Mock Test 12

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	A	3274.5	A,C	B,D	A	14	A	0.235	8	2	8000	0.119
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	6240	23.333	6	A,B,C	2.697	0.86	0.714	C	0.6	0.25	1	4
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	0.05	-0.07 1	A	10	0.412	C	0.158	4.962	83.7	A,B,C, D	A	

Mock Test 13

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	484	A,B,C	C	A,B,C, D	0.127	A	A,B,C, D	0.9	A	0.8	A,B,D	B
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	32	A	A,B,C, D	D	0.21	9	1.707	-2.60 9	-3	2.441	0.331	1.311
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	2.5	0.7	C	1	0.976	0.083	12.831	B,C	A,B,D	40.311	A,B,C, D	

Mock Test 14

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	7	A,B,C, D	A,C	0.946	0.78	B	0.1	D	A	64	D	B
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A,B	1	B	A	A,B,C, D	-40	0	A,B,D	0.476	2.251	4.042	61
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	1.898	2	1.25	D	11.316	0	0.688	0.133	D	0.378	0.245	

Mock Test 15

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	D	7.6	125	21	B,C	A,B,C	B,C,D	47	A,B,C, D	0	B	11
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	B,D	A	D	4	5	3.5	A	0.353	D	4	0.157	1.061
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	A,B,C, D	0.708	0.562	1.118	B	14.128	B,C,D	3.986	27.119	24	5	

Mock Test 16

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	B	B,C,D	0.564	A,B,C, D	A	-0.74 4	D	0.368	0.8	0.381	C	0.096
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A	D	A,B,C, D	8	8	A,C,D	-2.17 7	25	-5	0	0.714	0.447
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	A,B	1.1	0.135	A,B,D	0.589	0.045	A,B,D	7.865	A,B,C, D	B,D	-0.07 2	

Mock Test 17

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	A,B,C, D	A,B,C, D	C	B,C	B	0.494	A	7	1	0.95	0.119	124
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	1.732	1.97	A,C	49	12.162	0	0.208	0	A	1.507	B	A,C,D
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	1.714	A	0.012	B	3.765	C	1.019	-22	0.067	-0.03	0.708	

Mock Test 18

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	21.75	B,C,D	A,B,C	A,D	10	-6	0.46	A,B,C, D	1.649	16	B	22.667
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	$-\frac{0.58}{6}$	3	71	A	A,B,D	1	3	8	2.333	C	B,C	C,D
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	$-\frac{0.40}{4}$	0.39	0.016	10	1.447	1.366	0.075	9.24	13.143	0.344	0.943	

Mock Test 19

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	A	0.182	4	A	0.3	3.991	A,B	25	3	A,B,C, D	A	13
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	$\frac{73500}{0}$	0.14	0.035	A,B,D	3.075	0.264	85.094	0.75	C	22.5	4.125	43
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	0.054	0.103	0.894	B,D	81	0.625	D	1.51	0.24	49.2	0.404	

Mock Test 20

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	A,B	1.225	0.353	C	27	A,D	90	C,D	D	84	B	A,C,D
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A,D	D	C	0.017	C	$-\frac{17.3}{82}$	$-\frac{1.13}{2}$	0.35	A,B,C, D	2.5	0.29	0.104
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	A,C,D	6.23	8	0.571	2	0.161	2.1	B	6	0.789	$-\frac{0.02}{9}$	

Mock Test 21

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	75	B	A	C	A,B,C, D	B	2.7	0.076	A,B	2.381	59.75	10
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	0.99	A,B,C, D	A	C,D	A	A	0	4.103	0.474	$-\frac{0.01}{5}$	0.036	A,B,C, D
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	1.378	3	C	0.75	10	2	0.6	15.851	0.21	A,B,C	A	

Mock Test 22

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	0.296	13.4	A,C,D	A,B,C	0.6	0.029	0.78	C	13	D	C,D	A,B
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	1.789	63	A,B,C, D	49.5	29	0.524	24.324	B	1.571	C,D	0.583	14.414
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	B,C,D	1	14.973	$-\frac{1.12}{2}$	4.5	A,D	10.85	-6	C	0.017	12	

Mock Test 23

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	1	A,B,C, D	21	D	B,C,D	B,C,D	0.794	D	35	B	A,B,C, D	A
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A	0.164	0.571	18.667	A	0.456	A,B,C, D	A,B,C, D	C	0.377	C	$-\frac{0.25}{3}$
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	0.571	0.186	0.289	3	0.13	44.096	0.044	1.4	1.44	-0.98	86016	

Mock Test 24

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	5	C	A,B,C, D	0.368	B	A,B,C, D	D	A,B,C, D	14.5	0.458	B,D	4
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A,B,C	-13	B,C	11.414	0.836	2.703	0.05	0.653	9	0.906	A	0.902
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	D	C	2.6	35.5	0.97	$-\frac{0.41}{8}$	C	0.14	9.8	0.603	0	

Mock Test 25

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	45.333	B,C	B	A,C,D	B	0.833	C	A,B,D	C	0.667	0.889	65
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A,C	64	0.871	4	18	0.08	A,C,D	4.8	D	0.101	2.732	4.151
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	2	0.018	B	0.357	2	0.104	15	1.423	4	A	0.823	

Mock Test 26

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	-4	C	0.135	A,B,C	D	A,C,D	6	A,C,D	C	D	0.696	A,B,C, D
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	B	A,B,C, D	54	0.288	1.165	B,D	1.145	A,B,C, D	2	1	4.4	A
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	3	A	0.78	$-\frac{0.03}{5}$	B,C,D	A,B,C, D	65.9	48	0.465	C	-2	

Mock Test 27

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	29927	B	A,B,C, D	13	14.333	A	A,C	A,B,C, D	53.701	882.5	B	B,C
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	104	45	D	2.341	0.804	D	$-\frac{1.00}{2}$	$-\frac{2.19}{7}$	A	44.1	0.14	D
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	0.7	1	4.817	1.795	B,C	A,B,C, D	A,C	B	2.141	-0.04	A,B,C, D	

Mock Test 28

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	B,C,D	A,B,C, D	1.105	0.25	B	B,C,D	7	B	1	D	10000 0	51300 0
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	0.881	33	0.523	-3.15	A,B,C, D	D	0.195	69.764	3.475	26	0.198	6
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	0.25	40.625	20.321	2	0.28	3	A,B,D	-4	A,B,C, D	0.8	2.306	

Mock Test 29

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	C	A	10000	B	B	54.598	2	0.474	A,B,C, D	A,B,C, D	B	71.72
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	35	A,B,C, D	320	0.5	6.831	0.01	0.431	1.504	B,C,D	0.46	1.571	10
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	C	0.62	32	-0.6	C	0.942	0.742	0.993	0.121	5	8.857	

Mock Test 30

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	0.056	0.75	56	A	C,D	3	A	A	A,B,D	1.6	63	1.57
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A	225	C	$-\frac{0.68}{9}$	A	0.832	0.04	2.571	5	B	31	1
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	7	4.4	3.883	A,B,C, D	0.08	6	0.681	3.5	1096	1800	A	

Mock Test 31

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	25	0.409	A,D	C	B	A,B,C, D	16000 0	B	2.887	60	A,B,C, D	1.8
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	C	A,B,C, D	C	B	3.182	7	0.803	D	0.096	$-\frac{0.01}{7}$	B,C	B,D
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	0.558	B,C,D	0.538	B	0.333	A,B,C, D	7.2	0.625	13	0.066	77.778	

Mock Test 32

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	B	9	C,D	0.127	0.237	6	0.076	0.595	A,C,D	B	28.667	A,B,C, D
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A	-10	7	$-\frac{2.73}{4}$	0.17	A	2	B,C	D	0.055	0.731	C
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	3	0.01	0.187	4.123	C,D	0.253	8	A,C,D	A,B,C, D	0.441	A,B	

Mock Test 33

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	12	B,D	A,B	6.6	A,C	59	D	0.345	0.135	5	B	15
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A	B	0.551	0.246	$-\frac{0.43}{8}$	0.756	0.06	A,B,D	5	B	$-\frac{0.00}{7}$	A,B,C, D
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	B	0.56	5	3	0	0.03	2.792	$-\frac{8.50}{5}$	B	8.72	0.778	

Mock Test 34

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	25	B	A,B,C, D	A,B,C, D	A	15	32400	A,C,D	A,C	0.714	A,C,D	A,B,C, D
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	2.625	164	B,C,D	A,B,C, D	1.12	-1	25	41.667	0.579	C	0	3
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	0.547	0.068	0.975	2.65	33.417	0.042	13.489	7.133	0	0.825	0	

Mock Test 35

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	0.998	D	32	D	0.083	0.494	B,D	4	A	C	0.738	A
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	4.762	A,B,C, D	100	0.056	0.875	$-\frac{2.08}{4}$	0.625	0.256	A	41.037	0.835	0.583
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	A,B,D	0.161	7	D	2	0.71	0.457	1.397	$-\frac{3.38}{4}$	0.431	0.222	

Mock Test 36

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	8.277	A,D	A,B,C, D	A,C	1.286	C	C,D	31	A,C,D	0.299	B,C,D	−1
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	12	A,D	A,B,C, D	D	C,D	D	4	7.25	A,C,D	0.591	B	0.03
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	5.334	C,D	33.98	0	A,B,C, D	9.188	1.5	C	D	2	0.688	

Mock Test 37

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	C	A,B,C, D	1.732	B,C	A,C	A,B,C	0.6	2	0.4	A,B,C	B,D	39.125
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	0.25	17	D	4.917	B,C	D	0.064	C,D	D	6.5	−0.9	0.275
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	C	10	0.633	A,B,C, D	B	6.125	0.222	1.837	0.138	29.625	0.058	

Mock Test 38

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	25000 0	A	1	18	B,D	27000 0	A,C,D	25	B	A	27	B,D
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	0.245	D	7.145	0	A,C	D	0.994	A,C	6.75	10.667	0.042	4
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	D	61	0.6	3.949	0.125	17	B	D	125	A,B,D	20	

Mock Test 39

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	B	A,D	39	B	C	C	3	61175	B,D	0.893	0.447	A,B,C
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	35	196	A	59.125	−1	0.28	0.29	6.8	2.183	73.5	0.5	3
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	3.6	A,B,C, D	4.533	0.229	3.74	0.667	0.135	A	A	A	0.6	

Mock Test 40

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	A,D	2.846	20	D	A,B,C, D	0.182	C	58.515	5.882	B,D	41	0.071
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	B,C,D	B,C,D	C	67	0.25	9.106	2.128	0.692	C	0.737	−0.36	−0.57 2
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	13.738	2	8	1	0.58	−0.14 8	0.942	4.327	0.87	29	4.393	

Mock Test 41

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	1.8	4	D	C	A	C	A,D	C	B,D	B	A,B,C, D	−2.5
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	B	0.44	32	−3	A	A,B,C, D	0.8	0.755	0.878	2.5	18.556	0.185
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	53.535	0.341	A	A,D	0.542	16.416	0.417	A,B,C	0.986	1.05	A,C	

Mock Test 42

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	0.377	20.5	13.333	0.091	20	C	D	0.912	A	41	120	0.448
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A,B,C, D	1.84	1.443	0.75	0.057	4	0.629	0.74	0.732	$-\frac{0.06}{6}$	0.877	0.042
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	0	$-\frac{1.29}{6}$	0.498	$-\frac{0.76}{9}$	D	25	C	A,B,C, D	A	A	C	

Mock Test 43

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	C	0.875	17	A,B,C, D	0.946	D	A,B,C, D	4.322	A	A,C	D	0.378
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A,B,C, D	A,B,C, D	8	$-\frac{0.76}{2}$	B,C,D	$-\frac{0.62}{4}$	A,B	78	10.204	1.016	$-\frac{2.28}{6}$	A,C,D
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	34.8	0.429	7	C	4	B	4.333	4.527	0.03	1.414	A,B,C	

Mock Test 44

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	A,B,C, D	A,B,C	A,B,C	35	4.444	0.286	0.867	D	120.5	0.465	D	A,B,C, D
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	B,C,D	0.025	A,B,C, D	7.2	0.765	0.353	33088	A,B	4.6	A	B	2
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	0.063	0.32	A	0.745	A	0.76	2	0	0.792	1.25	0.164	

Mock Test 45

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	C,D	0.44	1.701	C	15	C	B,D	C,D	A	0.185	A,B	A,B,C, D
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	B	-1.25	63.548	2.236	9.417	D	0.201	0.381	36.167	C,D	-0.03	C
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	0.085	0	4	0	500	1	3	0.625	A,C,D	8	0.571	

Mock Test 46

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	A,C,D	C	A	38	422	D	B	C	0.069	B,C,D	B	A,C,D
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	15	A,B,C, D	0.571	D	10	B	4	A,D	0.909	2	0.148	0.161
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	7	1.229	0.593	B,C	10.75	1.5	7.444	A,B,D	-0.45	0.214	A	

Mock Test 47

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	4	A	B	A	D	0.472	0.675	A,B,C, D	131	A	A,B,C, D	1.575
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	B,D	5	5	B	0.401	C,D	1.349	D	1.144	0.085	0.49	3
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	B	2	0.21	0.989	A,B,C	-7.79	A,B,C, D	5.755	$-\frac{0.01}{3}$	0.111	B	

Mock Test 48

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	27	5	A,B,C	$-\frac{0.23}{2}$	C	0.154	C	$\frac{96000}{0}$	D	124	15	-6
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	0.995	2.4	A,D	17	D	1.786	13.2	0.833	C	0.455	1	360
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	6	0.9	D	1.647	B,C,D	1	0.28	6.367	0.513	0.018	8.54	

Mock Test 49

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	B	11927	A	A	0.013	C	6	C	11.333	A,B,D	1	B,D
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	0.615	49	C,D	0	2.2	4	A,B	-1.5	0.211	2.1	$-\frac{4.60}{1}$	1
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	0.25	1.475	$-\frac{1.70}{6}$	8	0.987	0.218	55.902	0.101	A,B	0.543	A,B,C	

Mock Test 50

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	D	A,C	47	A,C,D	B,C,D	3	0.754	3	C	1508.5	A,C,D	A
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	A	71	7429	3314	3.829	1.414	A	0.528	$-\frac{0.54}{9}$	13	0.2	5.158
Q	25	26	27	28	29	30	31	32	33	34	35	
Key	49.88	12	$-\frac{1.28}{6}$	C	-0.36	39	0.116	B	0.005	0.3	1.109	